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(54) **ORGANIC ELECTROLUMINESCENT MATERIALS AND DEVICES**

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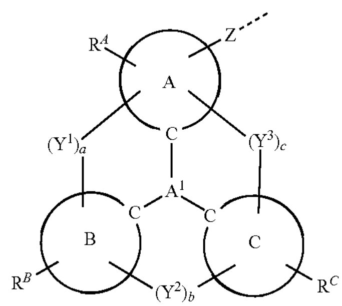
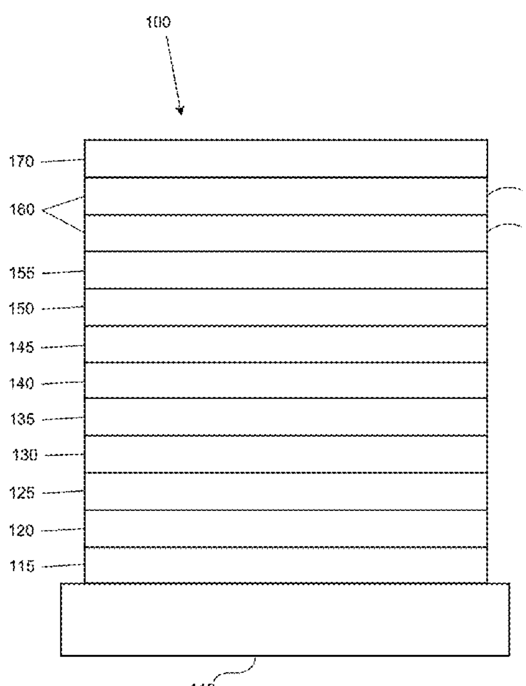
(57) **ABSTRACT**

A compound having a first ligand L_A comprising a structure of Formula I,



is provided. In Formula I, L_A is coordinated to a metal M selected from the group consisting of Ir, Rh, Re, Ru, Os, Pt, Pd, Ag, Au, and Cu; each of moiety A, moiety B, and moiety C is a monocyclic ring or a polycyclic fused ring system; A^1 is selected from the group consisting of B, N, P, P═O, P═S, Al, Ga, SiR", GeR", and SnR"; at least two of Y^1, Y^2, and Y^3 are present and are a direct bond or a linker; Z is a direct bond to metal M, or Z is a direct bond or an organic linker to a ring or a polycyclic fused ring system that is coordinated to M. Formulations, OLEDs, and consumer products containing the same are also provided.



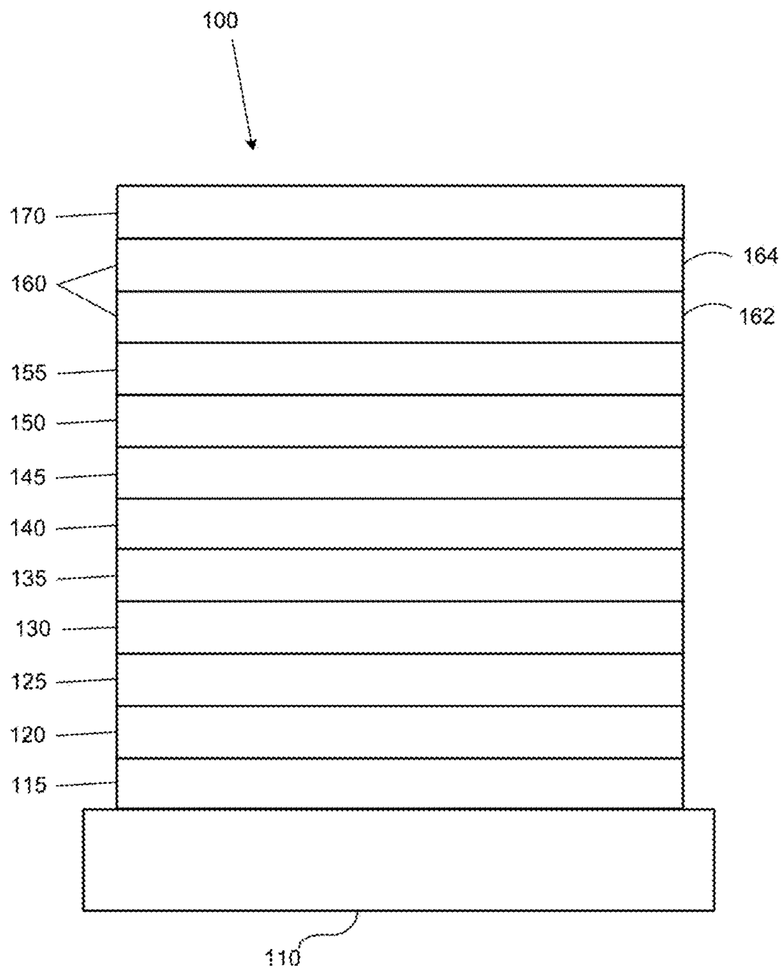

100
170
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FIG. 1

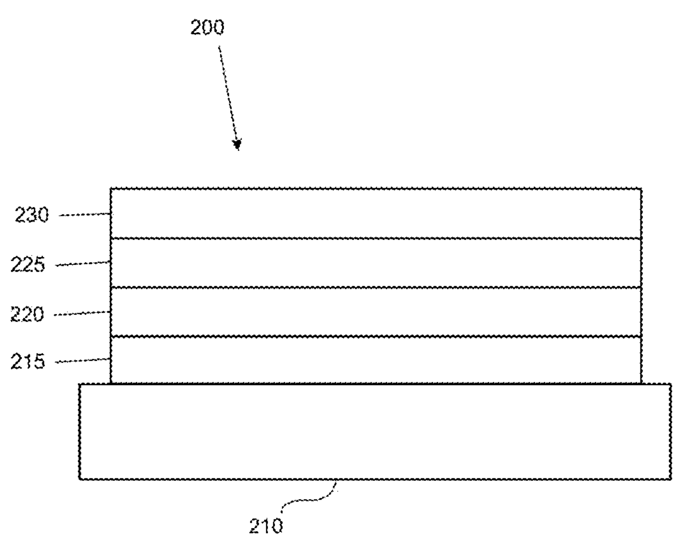

200
230
225
220
215
210

FIG. 2

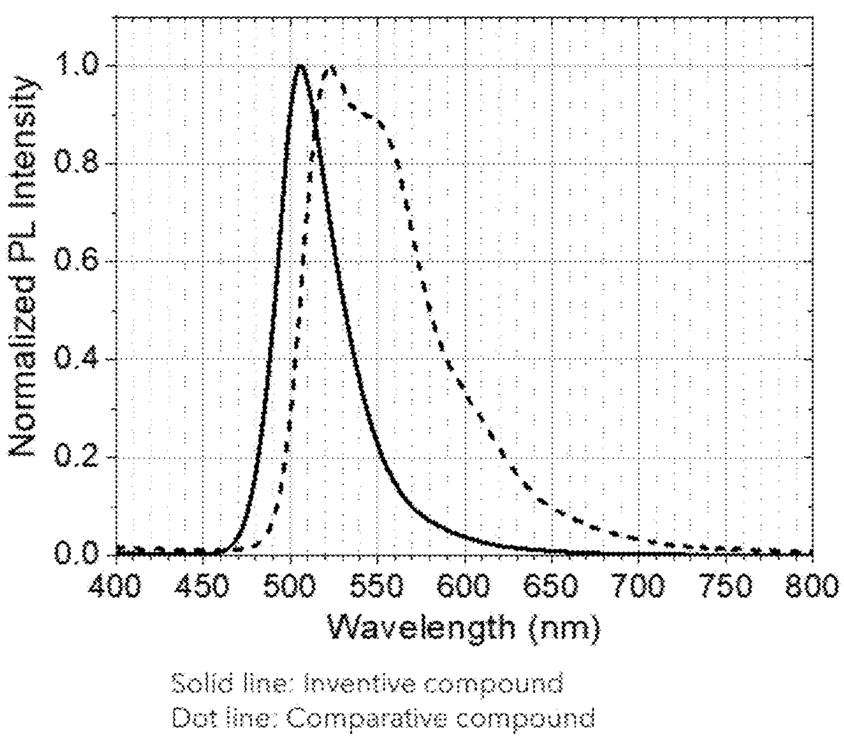

Normalized PL Intensity
1.0
0.8
0.6
0.4
0.2
0.0
400 450 500 550 600 650 700 750 800
Wavelength (nm)
Solid line: Inventive compound
Dot line: Comparative compound

FIG. 3

ORGANIC ELECTROLUMINESCENT MATERIALS AND DEVICES

CROSS-REFERENCES TO RELATED APPLICATIONS

[0001] This application claims priority under 35 U.S.C. § 119(e) to U.S. Provisional Application No. 63/563,019, filed on Mar. 8, 2024, the entire contents of which are incorporated herein by reference.

FIELD

[0002] The present disclosure generally relates to organic or metal coordination compounds and formulations and their various uses including as emitters, sensitizers, charge transporters, or exciton transporters in devices such as organic light emitting diodes and related electronic devices and consumer products.

BACKGROUND

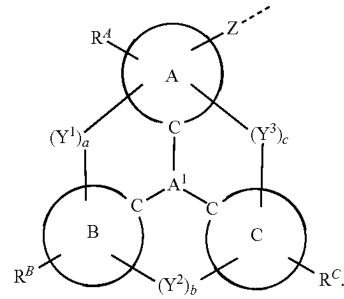
[0003] Opto-electronic devices that make use of organic materials are becoming increasingly desirable for various reasons. Many of the materials used to make such devices are relatively inexpensive, so organic opto-electronic devices have the potential for cost advantages over inorganic devices. In addition, the inherent properties of organic materials, such as their flexibility, may make them well suited for particular applications such as fabrication on a flexible substrate. Examples of organic opto-electronic devices include organic light emitting diodes/devices (OLEDs), organic phototransistors, organic photovoltaic cells, organic scintillators, and organic photodetectors. For OLEDs, the organic materials may have performance advantages over conventional materials.

[0004] OLEDs make use of thin organic films that emit light when voltage is applied across the device. OLEDs are becoming an increasingly interesting technology for use in applications such as displays, illumination, and backlighting.

[0005] One application for emissive molecules is a full color display. Industry standards for such a display call for pixels adapted to emit particular colors, referred to as "saturated" colors. In particular, these standards call for saturated red, green, and blue pixels. Alternatively, the OLED can be designed to emit white light. In conventional liquid crystal displays emission from a white backlight is filtered using absorption filters to produce red, green and blue emission. The same technique can also be used with OLEDs. The white OLED can be either a single emissive layer (EML) device or a stack structure. Color may be measured using CIE coordinates, which are well known to the art.

SUMMARY

[0006] In one aspect, the present disclosure provides a compound having a first ligand L_A comprising a structure of Formula



I,

In Formula I:

[0007] L_A is coordinated to a metal M selected from the group consisting of Ir, Rh, Re, Ru, Os, Pt, Pd, Ag, Au, and Cu; each of moiety A, moiety B, and moiety C is independently a monocyclic ring or a polycyclic fused ring system, wherein the monocyclic ring or each ring of the polycyclic fused ring system is independently a 5-membered to 10-membered carbocyclic or heterocyclic ring;

[0008] A^1 is selected from the group consisting of B, N, P, P=O, P=S, Al, Ga, SiR", GeR", and SnR";

[0009] each of a, b, and c is independently 0 or 1, where 0 represents absent and 1 represents present;

[0010] a+b+c is 2 or 3;

[0011] each of Y^1, Y^2, and Y^3 is independently selected from the group consisting of a direct bond, BR, BRR', NR, PR, P(O)R, O, S, Se, C=O, C=S, C=Se, C=NR, C=CRR', S=O, SO_2, CR, CRR', SiRR', and GeRR';

[0012] Z is a direct bond to metal M, or Z is a direct bond or an organic linker to a monocyclic ring or a polycyclic fused ring system that is coordinated to M;

[0013] each of R^A, R^B, and R^C independently represents mono to the maximum allowable substitution;

[0014] each R, R', R", R^A, R^B, and R^C is independently hydrogen or a substituent selected from the group consisting of deuterium, halogen, alkyl, cycloalkyl, heteroalkyl, heterocycloalkyl, arylalkyl, alkoxy, aryloxy, amino, silyl, germyl, boryl, alkenyl, cycloalkenyl, heteroalkenyl, alkynyl, aryl, heteroaryl, acyl, carboxylic acid, ether, ester, nitrile, isonitrile, sulfanyl, sulfinyl, sulfonyl, phosphino, selenyl, and combinations thereof;

[0015] M can be coordinated to other ligands;

[0016] L_A can be a monodentate, bidentate, tridentate, tetradentate, pentadentate ligand, or hexadentate ligand; and

[0017] wherein any two substituents can be joined or fused to form a ring.

[0018] In another aspect, the present disclosure provides a formulation comprising a compound having a first ligand L_A comprising a structure of Formula I as described herein.

[0019] In yet another aspect, the present disclosure provides an OLED having an organic layer comprising a compound having a first ligand L_A comprising a structure of Formula I as described herein.

[0020] In yet another aspect, the present disclosure provides a consumer product comprising an OLED with an

organic layer comprising a compound having a first ligand L_A comprising a structure of Formula I as described herein.

BRIEF DESCRIPTION OF THE DRAWINGS

[0021] FIG. **1** shows an organic light emitting device.

[0022] FIG. **2** shows an inverted organic light emitting device that does not have a separate electron transport layer.

[0023] FIG. **3** shows the PL spectra of an inventive compound and a comparative compound.

DETAILED DESCRIPTION

A. Terminology

[0024] Unless otherwise specified, the below terms used herein are defined as follows:

[0025] As used herein, "top" means furthest away from the substrate, while "bottom" means closest to the substrate. Where a first layer is described as "disposed over" a second layer, the first layer is disposed further away from substrate. There may be other layers between the first and second layer, unless it is specified that the first layer is "in contact with" the second layer. For example, a cathode may be described as "disposed over" an anode, even though there are various organic layers in between.

[0026] As used herein, "solution processable" means capable of being dissolved, dispersed, or transported in and/or deposited from a liquid medium, either in solution or suspension form.

[0027] As used herein, and as would be generally understood by one skilled in the art, a first "Highest Occupied Molecular Orbital" (HOMO) or "Lowest Unoccupied Molecular Orbital" (LUMO) energy level is "greater than" or "higher than" a second HOMO or LUMO energy level if the first energy level is closer to the vacuum energy level. Since ionization potentials (IP) are measured as a negative energy relative to a vacuum level, a higher HOMO energy level corresponds to an IP having a smaller absolute value (an IP that is less negative). Similarly, a higher LUMO energy level corresponds to an electron affinity (EA) having a smaller absolute value (an EA that is less negative). On a conventional energy level diagram, with the vacuum level at the top, the LUMO energy level of a material is higher than the HOMO energy level of the same material. A "higher" HOMO or LUMO energy level appears closer to the top of such a diagram than a "lower" HOMO or LUMO energy level.

[0028] As used herein, and as would be generally understood by one skilled in the art, a first work function is "greater than" or "higher than" a second work function if the first work function has a higher absolute value. Because work functions are generally measured as negative numbers relative to vacuum level, this means that a "higher" work function is more negative. On a conventional energy level diagram, with the vacuum level at the top, a "higher" work function is illustrated as further away from the vacuum level in the downward direction. Thus, the definitions of HOMO and LUMO energy levels follow a different convention than work functions.

[0029] Layers, materials, regions, and devices may be described herein in reference to the color of light they emit. In general, as used herein, an emissive region that is described as producing a specific color of light may include one or more emissive layers disposed over each other in a stack.

[0030] As used herein, a "NIR", "red", "green", "blue", "yellow" layer, material, region, or device refers to a layer, a material, a region, or a device that emits light in the wavelength range of about 700-1500 nm, 580-700 nm, 500-600 nm, 400-500 nm, 540-600 nm, respectively, or a layer, a material, a region, or a device that has a highest peak in its emission spectrum in the respective wavelength region. In some arrangements, separate regions, layers, materials, or devices may provide separate "deep blue" and "light blue" emissions. As used herein, the "deep blue" emission component refers to an emission having a peak emission wavelength that is at least about 4 nm less than the peak emission wavelength of the "light blue" emission component. Typically, a "light blue" emission component has a peak emission wavelength in the range of about 465-500 nm, and a "deep blue" emission component has a peak emission wavelength in the range of about 400-470 nm, though these ranges may vary for some configurations.

[0031] In some arrangements, a color altering layer that converts, modifies, or shifts the color of the light emitted by another layer to an emission having a different wavelength is provided. Such a color altering layer can be formulated to shift wavelength of the light emitted by the other layer by a defined amount, as measured by the difference in the wavelength of the emitted light and the wavelength of the resulting light. In general, there are two classes of color altering layers: color filters that modify a spectrum by removing light of unwanted wavelengths, and color changing layers that convert photons of higher energy to lower energy. For example, a "red" color filter can be present in order to filter an input light to remove light having a wavelength outside the range of about 580-700 nm. A component "of a color" refers to a component that, when activated or used, produces or otherwise emits light having a particular color as previously described. For example, a "first emissive region of a first color" and a "second emissive region of a second color different than the first color" describes two emissive regions that, when activated within a device, emit two different colors as previously described.

[0032] As used herein, emissive materials, layers, and regions may be distinguished from one another and from other structures based upon light initially generated by the material, layer or region, as opposed to light eventually emitted by the same or a different structure. The initial light generation typically is the result of an energy level change resulting in emission of a photon. For example, an organic emissive material may initially generate blue light, which may be converted by a color filter, quantum dot or other structure to red or green light, such that a complete emissive stack or sub-pixel emits the red or green light. In this case the initial emissive material, region, or layer may be referred to as a "blue" component, even though the sub-pixel is a "red" or "green" component.

[0033] In some cases, it may be preferable to describe the color of a component such as an emissive region, sub-pixel, color altering layer, or the like, in terms of 1931 CIE coordinates. For example, a yellow emissive material may have multiple peak emission wavelengths, one in or near an edge of the "green" region, and one within or near an edge of the "red" region as previously described. Accordingly, as used herein, each color term also corresponds to a shape in the 1931 CIE coordinate color space. The shape in 1931 CIE color space is constructed by following the locus between two color points and any additional interior points. For example, interior shape parameters for red, green, blue, and yellow may be defined as shown below:

Color	CIE Shape Parameters
Central Red	Locus: [0.6270, 0.3725]; [0.7347, 0.2653]; Interior: [0.5086, 0.2657]
Central Green	Locus: [0.0326, 0.3530]; [0.3731, 0.6245]; Interior: [0.2268, 0.3321
Central Blue	Locus: [0.1746, 0.0052]; [0.0326, 0.3530]; Interior: [0.2268, 0.3321]
Central Yellow	Locus: [0.3731, 0.6245]; [0.6270, 0.3725]; Interior: [0.3700, 0.4087]; [0.2886, 0.4572]

[0034] The terms "halo," "halogen," and "halide" are used interchangeably and refer to fluorine, chlorine, bromine, and iodine.

[0035] The term "acyl" refers to a substituted carbonyl group (—C(O)—R_s).

[0036] The term "ester" refers to a substituted oxycarbonyl (—O—C(O)—R_s or —C(O)—O—R_s) group.

[0037] The term "ether" refers to an —OR_s group.

[0038] The terms "sulfanyl" or "thio-ether" are used interchangeably and refer to a —SR_s group.

[0039] The term "selenyl" refers to a —SeR_s group.

[0040] The term "sulfinyl" refers to a —S(O)—R_s group.

[0041] The term "sulfonyl" refers to a —SO_2—R_s group.

[0042] The term "phosphino" refers to a group containing at least one phosphorus atom bonded to the relevant structure. Common examples of phosphino groups include, but are not limited to, groups such as a —$P(R_s)_2$ group or a —$PO(R_s)_2$ group, wherein each R_s can be same or different.

[0043] The term "silyl" refers to a group containing at least one silicon atom bonded to the relevant structure. Common examples of silyl groups include, but are not limited to, groups such as a —$Si(R_s)_3$ group, wherein each R_s can be same or different.

[0044] The term "germyl" refers to a group containing at least one germanium atom bonded to the relevant structure. Common examples of germyl groups include, but are not limited to, groups such as a —$Ge(R_s)_3$ group, wherein each R_s can be same or different.

[0045] The term "boryl" refers to a group containing at least one boron atom bonded to the relevant structure. Common examples of boryl groups include, but are not limited to, groups such as a —$B(R_s)_2$ group or its Lewis adduct —$B(R_s)_3$ group, wherein R can be same or different.

[0046] In each of the above, R_s can be hydrogen or a substituent selected from the group consisting of the general substituents as defined in this application. Preferred R_s is selected from the group consisting of deuterium, halogen, alkyl, cycloalkyl, heteroalkyl, heterocycloalkyl, arylalkyl, alkoxy, aryloxy, amino, silyl, alkenyl, cycloalkenyl, heteroalkenyl, alkynyl, aryl, heteroaryl, and combination thereof. More preferably R_s is selected from the group consisting of alkyl, cycloalkyl, aryl, heteroaryl, and combination thereof.

[0047] The term "alkyl" refers to and includes both straight and branched chain alkyl groups having an alkyl carbon atom bonded to the relevant structure. Preferred alkyl groups are those containing from one to fifteen carbon atoms, preferably one to nine carbon atoms, and includes methyl, ethyl, n-propyl, iso-propyl, n-butyl, sec-butyl, isobutyl, tert-butyl, n-pentyl, 2-methylbutyl, 3-methylbutyl, 2,2-dimethylpropyl, 1,3-dimethylpropyl, 1,1-dimethylpropyl, 2-ethylpropyl, 1,2-dimethylpropyl, n-hexyl, 2-methylpentyl, 3-methylpentyl, 2,2-dimethylbutyl, 2,3-dimethylbutyl, n-heptyl, 2-methylhexyl, 3-methylhexyl, 2,2-dimethylpentyl, 2,3-dimethylpentyl, 2,4-dimethylpentyl, 3,3-dimethylpentyl, 3-ethylpentyl, 2,2,3-trimethylbutyl, and the like. Additionally, the alkyl group can be further substituted.

[0048] The term "cycloalkyl" refers to and includes monocyclic, polycyclic, and spiro alkyl groups having a ring alkyl carbon atom bonded to the relevant structure. Preferred cycloalkyl groups are those containing 3 to 12 ring carbon atoms and includes cyclopropyl, cyclopentyl, cyclohexyl, bicyclo[3.1.1]heptyl, spiro[4.5]decyl, spiro[5.5]undecyl, adamantyl, and the like. Additionally, the cycloalkyl group can be further substituted.

[0049] The terms "heteroalkyl" or "heterocycloalkyl" refer to an alkyl or a cycloalkyl group, respectively, having at least one carbon atom replaced by a heteroatom. Optionally the at least one heteroatom is selected from O, S, N, P, B, Si, Ge and Se, preferably, O, S or N. Additionally, the heteroalkyl or heterocycloalkyl group can be further substituted.

[0050] The term "alkenyl" refers to and includes both straight and branched chain alkene groups. Alkenyl groups are essentially alkyl groups that include at least one carbon-carbon double bond in the alkyl chain with one carbon atom from the carbon-carbon double bond that is bonded to the relevant structure. Cycloalkenyl groups are essentially cycloalkyl groups that include at least one carbon-carbon double bond in the cycloalkyl ring. The term "heteroalkenyl" as used herein refers to an alkenyl group having at least one carbon atom replaced by a heteroatom. Optionally the at least one heteroatom is selected from O, S, N, P, B, Si, Ge, and Se, preferably, O, S, or N. Preferred alkenyl, cycloalkenyl, or heteroalkenyl groups are those containing two to fifteen carbon atoms. Additionally, the alkenyl, cycloalkenyl, or heteroalkenyl group can be further substituted.

[0051] The term "alkynyl" refers to and includes both straight and branched chain alkyne groups. Alkynyl groups are essentially alkyl groups that include at least one carbon-carbon triple bond in the alkyl chain with one carbon atom from the carbon-carbon triple bond that is bonded to the relevant structure. Preferred alkynyl groups are those containing two to fifteen carbon atoms. Additionally, the alkynyl group can be further substituted.

[0052] The terms "aralkyl" or "arylalkyl" are used interchangeably and refer to an aryl-substituted alkyl group having an alkyl carbon atom bonded to the relevant structure. Additionally, the aralkyl group can be further substituted.

[0053] The term "heterocyclic group" refers to and includes aromatic and non-aromatic cyclic groups containing at least one heteroatom. Optionally the at least one heteroatom is selected from O, S, Se, N, P, B, Si, Ge, and Se, preferably, O, S, N, or B. Hetero-aromatic cyclic groups may be used interchangeably with heteroaryl. Preferred hetero-non-aromatic cyclic groups are those containing 3 to 10 ring atoms, preferably those containing 3 to 7 ring atoms, which includes at least one hetero atom, and includes cyclic amines such as morpholino, piperidino, pyrrolidino, and the like, and cyclic ethers/thio-ethers, such as tetrahydrofuran, tetrahydropyran, tetrahydrothiophene, and the like. Additionally, the heterocyclic group can be further substituted or fused.

[0054] The term "aryl" refers to and includes both single-ring and polycyclic aromatic hydrocarbyl groups. The polycyclic rings may have two or more rings in which two carbons are common to two adjoining rings (the rings are "fused"). Preferred aryl groups are those containing six to thirty carbon atoms, preferably six to twenty-four carbon atoms, six to eighteen carbon atoms, and more preferably six to twelve carbon atoms. Especially preferred is an aryl group having six carbons, ten carbons, twelve carbons, fourteen carbons, or eighteen carbons. Suitable aryl groups include phenyl, biphenyl, triphenyl, triphenylene, tetraphenylene, naphthalene, anthracene, phenalene, phenanthrene, pyrene, chrysene, perylene, and azulene, preferably phenyl, biphenyl, triphenyl, triphenylene, and naphthalene. Additionally, the aryl group can be further substituted or fused, such as, without limitation, fluorene.

[0055] The term "heteroaryl" refers to and includes both single-ring aromatic groups and polycyclic aromatic ring systems that include at least one heteroatom. The heteroatoms include, but are not limited to O, S, Se, N, P, B, Si, Ge, and Se. In many instances, O, S, N, or B are the preferred heteroatoms. Hetero-single ring aromatic systems are preferably single rings with 5 or 6 ring atoms, and the ring can have from one to six heteroatoms. The hetero-polycyclic ring systems can have two or more aromatic rings in which two atoms are common to two adjoining rings (the rings are "fused") wherein at least one of the rings is a heteroaryl. The hetero-polycyclic aromatic ring systems can have from one to six heteroatoms per ring of the polycyclic aromatic ring system. Preferred heteroaryl groups are those containing three to thirty carbon atoms, preferably three to twenty-four carbon atoms, three to eighteen carbon atoms, and more preferably three to twelve carbon atoms. Suitable heteroaryl groups include dibenzothiophene, dibenzofuran, dibenzoselenophene, furan, thiophene, benzofuran, benzothiophene, benzoselenophene, carbazole, indolocarbazole, pyridylindole, pyrrolodipyridine, pyrazole, imidazole, triazole, oxazole, thiazole, oxadiazole, oxatriazole, dioxazole, thiadiazole, pyridine, pyridazine, pyrimidine, pyrazine, triazine, oxazine, oxathiazine, oxadiazine, indole, benzimidazole, indazole, indoxazine, benzoxazole, benzisoxazole, benzothiazole, quinoline, isoquinoline, cinnoline, quinazoline, quinoxaline, naphthyridine, phthalazine, pteridine, xanthene, acridine, phenazine, phenothiazine, phenoxazine, benzofuropyridine, furodipyridine, benzothienopyridine, thienodipyridine, benzoselenophenopyridine, selenophenodipyridine, azaborine, borazine, 5λ^2,9λ^2-diaza-13b-boranaphtho[2,3,4-de]anthracene, 52'-benzo[d]benzo[4,5]imidazo[3,2-a]imidazole, and 5,9-dioxa-13b-boranaphtho[3,2,1-de]anthracene; preferably dibenzothiophene, dibenzofuran, dibenzoselenophene, carbazole, indolocarbazole, imidazole, pyridine, triazine, benzimidazole, 5λ^2,9λ^2-diaza-13b-boranaphtho[2,3,4-de]anthracene, 52'-benzo[d]benzo[4,5]imidazo[3,2-a]imidazole, and 5,9-dioxa-13b-boranaphtho[3,2,1-de]anthracene. Additionally, the heteroaryl group can be further substituted or fused.

[0056] Of the aryl and heteroaryl groups listed above, the groups of triphenylene, naphthalene, anthracene, dibenzothiophene, dibenzofuran, dibenzoselenophene, carbazole, indolocarbazole, imidazole, pyridine, pyrazine, pyrimidine, triazine, benzimidazole, 5λ^2,9λ^2-diaza-13b-boranaphtho[2,3,4-de]anthracene, 5λ^2-benzo[d]benzo[4,5]imidazo[3,2-a]imidazole, 5,9-dioxa-13b-boranaphtho[3,2,1-de]anthracene, and the respective aza-analogs of each thereof are of particular interest.

[0057] In many instances, the General Substituents are selected from the group consisting of deuterium, halogen, alkyl, cycloalkyl, heteroalkyl, heterocycloalkyl, arylalkyl, alkoxy, aryloxy, amino, silyl, germyl, boryl, alkenyl, cycloalkenyl, heteroalkenyl, alkynyl, aryl, heteroaryl, acyl, carboxylic acid, ether, ester, nitrile, isonitrile, sulfanyl, selenyl, sulfinyl, sulfonyl, phosphino, and combinations thereof.

[0058] In some instances, the Preferred General Substituents are selected from the group consisting of deuterium, fluorine, alkyl, cycloalkyl, heteroalkyl, alkoxy, aryloxy, amino, silyl, germyl, boryl, alkenyl, cycloalkenyl, heteroalkenyl, aryl, heteroaryl, nitrile, isonitrile, sulfanyl, and combinations thereof.

[0059] In some instances, the More Preferred General Substituents are selected from the group consisting of deuterium, fluorine, alkyl, cycloalkyl, alkoxy, aryloxy, amino, silyl, germyl, boryl, aryl, heteroaryl, nitrile, sulfanyl, and combinations thereof.

[0060] In some instances, the Even More Preferred General Substituents are selected from the group consisting of deuterium, fluorine, alkyl, cycloalkyl, silyl, aryl, heteroaryl, nitrile, and combinations thereof.

[0061] In yet other instances, the Most Preferred General Substituents are selected from the group consisting of deuterium, alkyl, cycloalkyl, aryl, heteroaryl, and combinations thereof.

[0062] The terms "substituted" and "substitution" refer to a substituent other than H that is bonded to the relevant position, e.g., a carbon or nitrogen. For example, when R^1 represents mono-substitution, then one R^1 must be other than H (i.e., a substitution). Similarly, when R^1 represents di-substitution, then two of R^1 must be other than H. Similarly, when R^1 represents zero or no substitution, R^1, for example, can be a hydrogen for all available valencies of ring atoms, as in carbon atoms for benzene and the nitrogen atom in pyrrole, or simply represents nothing for ring atoms with fully filled valencies, e.g., the nitrogen atom in pyridine. The maximum number of substitutions possible in a ring structure will depend on the total number of available valencies in the ring atoms.

[0063] As used herein, "combinations thereof" indicates that one or more members of the applicable list are combined to form a known or chemically stable arrangement that one of ordinary skill in the art can envision from the applicable list. For example, an alkyl and deuterium can be combined to form a partial or fully deuterated alkyl group; a halogen and alkyl can be combined to form a halogenated alkyl substituent; and a halogen, alkyl, and aryl can be combined to form a halogenated arylalkyl. In one instance, the term substitution includes a combination of two to four of the listed groups. In another instance, the term substitution includes a combination of two to three groups. In yet another instance, the term substitution includes a combination of two groups. Preferred combinations of substituent groups are those that contain up to fifty atoms that are not hydrogen or deuterium, or those which include up to forty atoms that are not hydrogen or deuterium, or those that include up to thirty atoms that are not hydrogen or deuterium. In many instances, a preferred combination of substituent groups will include up to twenty atoms that are not hydrogen or deuterium.

[0064] The "aza" designation in the fragments described herein, i.e. aza-dibenzofuran, aza-dibenzothiophene, etc. means that one or more of the C—H groups in the respective aromatic ring can be replaced by a nitrogen atom, for example, and without any limitation, azatriphenylene encompasses both dibenzo[f,h]quinoxaline and dibenzo[f,h]quinoline. One of ordinary skill in the art can readily envision other nitrogen analogs of the aza-derivatives described above, and all such analogs are intended to be encompassed by the terms as set forth herein.

[0065] As used herein, "deuterium" refers to an isotope of hydrogen. Deuterated compounds can be readily prepared using methods known in the art. For example, U.S. Pat. No. 8,557,400, Patent Pub. No. WO 2006/095951, and U.S. Pat. Application Pub. No. US 2011/0037057, which are hereby incorporated by reference in their entireties, describe the making of deuterium-substituted organometallic complexes. Further reference is made to Ming Yan, et al., *Tetrahedron* 2015, 71, 1425-30 and Atzrodt et al., *Angew. Chem. Int. Ed.* (*Reviews*) 2007, 46, 7744-65, which are incorporated by reference in their entireties, describe the deuteration of the methylene hydrogens in benzyl amines and efficient pathways to replace aromatic ring hydrogens with deuterium, respectively.

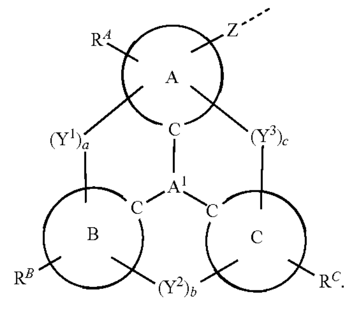
[0066] As used herein, any specifically listed substituent, such as, without limitation, methyl, phenyl, pyridyl, etc. includes undeuterated, partially deuterated, and fully deuterated versions thereof. Similarly, classes of substituents such as, without limitation, alkyl, aryl, cycloalkyl, heteroaryl, etc. also include undeuterated, partially deuterated, and fully deuterated versions thereof. Unless otherwise specified, atoms in chemical structures without valences fully filled by H or D should be considered to include undeuterated, partially deuterated, and fully deuterated versions thereof. For example, the chemical structure of C implies to include C_6H_6, C_6D_6, $C_6H_3D_3$, and any other partially deuterated variants thereof. Some common basic partially or fully deuterated group include, without limitation, CD_3, $CD_2C(CH_3)_3$, $C(CD_3)_3$, and C_6D_5.

[0067] It is to be understood that when a molecular fragment is described as being a substituent or otherwise attached to another moiety, its name may be written as if it were a fragment (e.g. phenyl, phenylene, naphthyl, dibenzofuryl) or as if it were the whole molecule (e.g. benzene, naphthalene, dibenzofuran). As used herein, these different ways of designating a substituent or attached fragment are considered to be equivalent.

[0068] In some instances, a pair of substituents in the molecule can be optionally joined or fused into a ring. The preferred ring is a five to nine-membered carbocyclic or heterocyclic ring, includes both instances where the portion of the ring formed by the pair of substituents is saturated and where the portion of the ring formed by the pair of substituents is unsaturated. In yet other instances, a pair of adjacent substituents can be optionally joined or fused into a ring. As used herein, "adjacent" means that the two substituents involved can be on the same ring next to each other, or on two neighboring rings having the two closest available substitutable positions, such as 2, 2' positions in a biphenyl, or 1, 8 position in a naphthalene.

B. The Compounds of the Present Disclosure

[0069] In one aspect, the present disclosure provides a compound having a first ligand L_A comprising a structure of Formula



I,

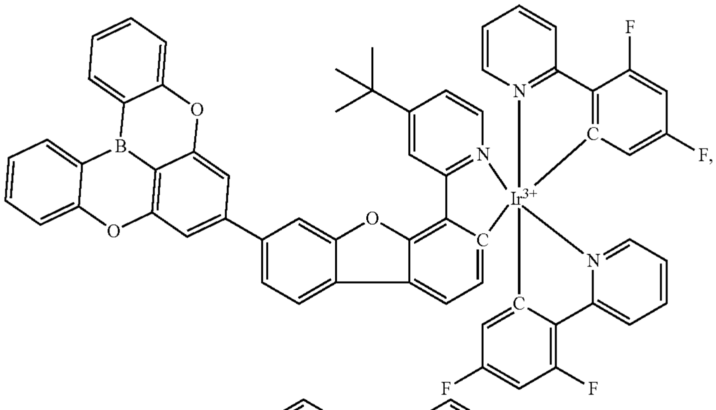
In Formula I:

- **[0070]** L_A is coordinated to a metal M selected from the group consisting of Ir, Rh, Re, Ru, Os, Pt, Pd, Ag, Au, and Cu;
- **[0071]** each of moiety A, moiety B, and moiety C is independently a monocyclic ring or a polycyclic fused ring system, wherein the monocyclic ring or each ring of the polycyclic fused ring system is independently a 5-membered to 10-membered carbocyclic or heterocyclic ring;
- **[0072]** A^1 is selected from the group consisting of B, N, P, P═O, P═S, Al, Ga, SiR", GeR", and SnR";
- **[0073]** each of a, b, and c is independently 0 or 1, where 0 represents absent and 1 represents present;
- **[0074]** a+b+c is 2 or 3;
- **[0075]** each of Y^1, Y^2, and Y^3 is independently selected from the group consisting of a direct bond, BR, BRR', NR, PR, P(O)R, O, S, Se, C═O, C═S, C═Se, C═NR, C═CRR', S═O, SO_2, CR, CRR', SiRR', and GeRR';
- **[0076]** Z is a direct bond to metal M, or Z is a direct bond or an organic linker to a monocyclic ring or a polycyclic fused ring system that is coordinated to M;
- **[0077]** each of R^A, R^B, and R^C independently represents mono to the maximum allowable substitution;
- **[0078]** each R, R', R", R^A, R^B, and R^C is independently hydrogen or a substituent selected from the group consisting of the General Substituents defined herein;
- **[0079]** M can be coordinated to other ligands;
- **[0080]** L_A can be a monodentate, bidentate, tridentate, tetradentate, pentadentate ligand, or hexadentate ligand; and
- **[0081]** wherein any two substituents can be joined or fused to form a ring.

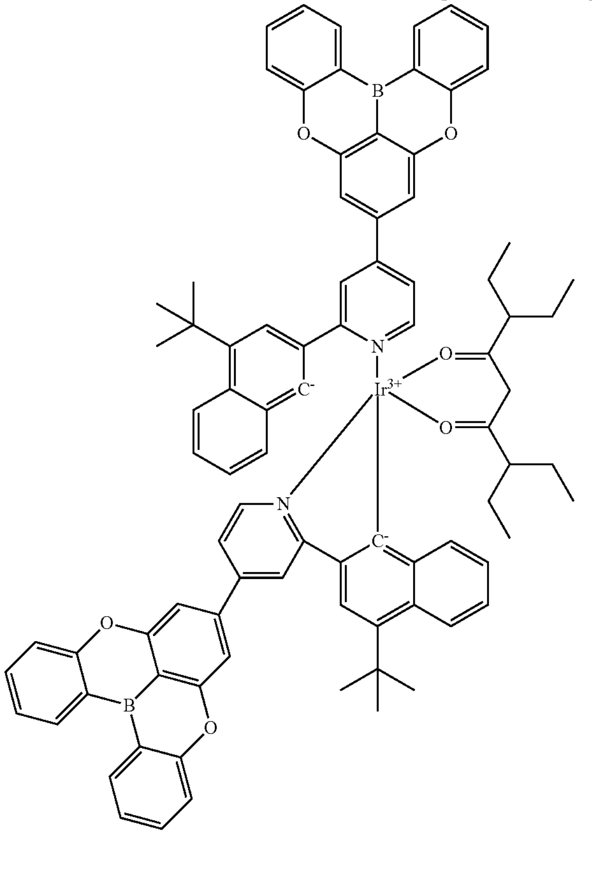
[0082] In some embodiments, if M is Pt, then L_A is a monodentate ligand.

[0083] In some embodiments, if Formula I is part of a bidentate ligand and ring A is a monocyclic ring, then Z is not a direct bond to the metal.

[0084] In some embodiments, the compound is not



or



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[0085] In some embodiments when Z is a direct bond to metal Ir, it is the one and only bond to the metal from Formula I. In some embodiments when Z is a direct bond to a monocyclic ring or a polycyclic fused ring system that is coordinated to Ir, no R^A is to join with any substituent on the monocyclic ring or on the polycyclic fused ring system to form a ring.

[0086] In some embodiments of Formula I, at least one R, R', R", R^A, R^B, or R^C is partially or fully deuterated. In some embodiments, at least one R^A is partially or fully deuterated. In some embodiments, at least one R^B is partially or fully deuterated. In some embodiments, at least one R^C is partially or fully deuterated. In some embodiments, at least one R or R' is partially or fully deuterated. In some embodiments, R" is partially or fully deuterated.

[0087] In some embodiments, each of moiety A, moiety B, and moiety C is independently a monocyclic ring or a polycyclic fused ring system, wherein the monocyclic ring or each ring of the polycyclic fused ring system is independently a 5-membered or 6-membered carbocyclic or heterocyclic ring.

[0088] In some embodiments, each of moiety A, moiety B, and moiety C is independently a monocyclic ring or a polycyclic fused ring system, wherein the monocyclic ring or each ring of the polycyclic fused ring system is independently a 5-membered or 6-membered aryl or heteroaryl ring.

[0089] In some embodiments, at least one R^A, R^B, or R^C is selected from the group consisting of the General Substituents defined herein. In some embodiments, at least one R^A is selected from the group consisting of the General Substituents defined herein. In some embodiments, at least one R^B is selected from the group consisting of the General Substituents defined herein. In some embodiments, at least one R^C is selected from the group consisting of the General Substituents defined herein. In some embodiments, at least one R^A, R^B, or R^C is selected from the group consisting of the Preferred General Substituents defined herein.

[0090] In some embodiments, each R, R', R", R^A, R^B, and R^C is independently hydrogen or a substituent selected from the group consisting of the Preferred General Substituents defined herein. In some embodiments, each R, R', R", R^A, R^B, and R^C is independently hydrogen or a substituent selected from the group consisting of the More Preferred General Substituents defined herein. In some embodiments, each R, R', R", R^A, R^B, and R^C is independently hydrogen or a substituent selected from the group consisting of the Even More Preferred General Substituents defined herein. In some embodiments, each R, R', R", R^A, R^B, and R^C is independently hydrogen or a substituent selected from the group consisting of the Most Preferred General Substituents defined herein.

[0091] In some embodiments, metal M is selected from the group consisting of Ru, Os, Rh, Ir, Pd, and Pt. In some embodiments, metal M is Ir. In some embodiments, metal M is Pt or Pd. In some embodiments, metal M is Pt. In some embodiments, metal M is Pd.

[0092] In some embodiments, each of moiety A, moiety B, and moiety C is independently selected from the group consisting of the following Cyclic Moiety List: benzene, pyridine, pyrimidine, pyridazine, pyrazine, triazine, imidazole, pyrazole, pyrrole, oxazole, furan, thiophene, thiazole, triazole, naphthalene, quinoline, isoquinoline, quinazoline, benzofuran, aza-benzofuran, benzoxazole, aza-benzoxazole, benzothiophene, aza-benzothiophene, benzothiazole, aza-benzothiazole, benzoselenophene, aza-benzoselenophene, indene, aza-indene, indole, aza-indole, benzimidazole, aza-benzimidazole, benzobenzimidazole, aza-benzobenzimidazole, carbazole, aza-carbazole, dibenzofuran, aza-dibenzofuran, phenanthro[3,2-b]benzofuran, dibenzothiophene, aza-dibenzothiophene, quinoxaline, phthalazine, phenanthrene, aza-phenanthrene, anthracene, aza-anthracene, phenanthridine, fluorene, and aza-fluorene. In some embodiments, the aza variant includes one N on a benzo ring.

[0093] In some embodiments, moiety A is a monocyclic ring. In some embodiments, moiety A is selected from the group consisting of benzene, pyridine, pyrimidine, pyridazine, pyrazine, triazine, imidazole, pyrazole, pyrrole, oxazole, furan, thiophene, thiazole, and triazole. In some embodiments, moiety A is benzene.

[0094] In some embodiments, moiety A is a polycyclic fused ring system. In some embodiments, moiety A is selected from the group consisting of naphthalene, quinoline, isoquinoline, quinazoline, benzofuran, aza-benzofuran, benzoxazole, aza-benzoxazole, benzothiophene, aza-benzothiophene, benzothiazole, aza-benzothiazole, benzoselenophene, aza-benzoselenophene, indene, aza-indene, indole, aza-indole, benzimidazole, aza-benzimidazole, carbazole, aza-carbazole, dibenzofuran, aza-dibenzofuran, dibenzothiophene, aza-dibenzothiophene, quinoxaline, phthalazine, phenanthrene, aza-phenanthrene, anthracene, aza-anthracene, phenanthridine, fluorene, and aza-fluorene. In some embodiments, moiety A is naphthalene.

[0095] In some embodiments, moiety B is a monocyclic ring. In some embodiments, moiety B is selected from the group consisting of benzene, pyridine, pyrimidine, pyridazine, pyrazine, triazine, imidazole, pyrazole, pyrrole, oxazole, furan, thiophene, thiazole, and triazole. In some embodiments, moiety B is benzene.

[0096] In some embodiments, moiety B is a polycyclic fused ring system. In some embodiments, moiety B is selected from the group consisting of naphthalene, quinoline, isoquinoline, quinazoline, benzofuran, aza-benzofuran, benzoxazole, aza-benzoxazole, benzothiophene, aza-benzothiophene, benzothiazole, aza-benzothiazole, benzoselenophene, aza-benzoselenophene, indene, aza-indene, indole, aza-indole, benzimidazole, aza-benzimidazole, carbazole, aza-carbazole, dibenzofuran, aza-dibenzofuran, dibenzothiophene, aza-dibenzothiophene, quinoxaline, phthalazine, phenanthrene, aza-phenanthrene, anthracene, aza-anthracene, phenanthridine, fluorene, and aza-fluorene. In some embodiments, moiety B is naphthalene.

[0097] In some embodiments, moiety C is a monocyclic ring. In some embodiments, moiety C is selected from the group consisting of benzene, pyridine, pyrimidine, pyridazine, pyrazine, triazine, imidazole, pyrazole, pyrrole, oxazole, furan, thiophene, thiazole, and triazole. In some embodiments, moiety C is benzene.

[0098] In some embodiments, moiety C is a polycyclic fused ring system. In some embodiments, moiety C is selected from the group consisting of naphthalene, quinoline, isoquinoline, quinazoline, benzofuran, aza-benzofuran, benzoxazole, aza-benzoxazole, benzothiophene, aza-benzothiophene, benzothiazole, aza-benzothiazole, benzoselenophene, aza-benzoselenophene, indene, aza-indene, indole, aza-indole, benzimidazole, aza-benzimidazole, carbazole, aza-carbazole, dibenzofuran, aza-dibenzofuran, dibenzothiophene, aza-dibenzothiophene, quinoxaline, phthalazine, phenanthrene, aza-phenanthrene, anthracene, aza-anthracene, phenanthridine, fluorene, and aza-fluorene. In some embodiments, moiety C is naphthalene.

[0099] In some embodiments, A^1 is selected from the group consisting of B, N, P, Al, and Ga. In some embodiments, A^1 is B.

[0100] In some embodiments, A^1 is selected from the group consisting of SiR", GeR", and SnR". In some embodiments, A^1 is selected from the group consisting of P=O and P=S.

[0101] In some embodiments, Z is a direct bond to metal M. In some embodiments, Z is a direct bond to a monocyclic ring or a polycyclic fused ring system that is coordinated to metal M.

[0102] In some embodiments, Z is an organic linker to a monocyclic ring or a polycyclic fused ring system that is coordinated to metal M.

[0103] In some embodiments, Z is an organic linker selected from the group consisting of BR, BRR', NR, PR, P(O)R, O, S, Se, C=O, C=S, C=Se, $C{=}NR^Z$, $C{=}CR^ZR^{Z'}$, S=O, SO_2, CR^Z, $CR^ZR^{Z'}$, SiR^ZR^{Z}", and $GeR^ZR^{Z'}$, alkyl, cycloalkyl, aryl, heteroaryl, and combinations thereof; and each of R^Z and $R^{Z'}$ is independently hydrogen or a substituent selected from the group consisting of the General Substituents defined herein.

[0104] In some embodiments, Z is selected from the group consisting of O, S, and Se. In some embodiments, Z is selected from the group consisting of BR, NR, and PR. In some embodiments, Z is selected from the group consisting of P(O)R, C═O, C═S, C═Se, C═NR', C═CRR', S═O, and SO_2. In some embodiments, Z is selected from the group consisting of BRR', CRR', SiRR', and GeRR'. In some embodiments, Z is CR. In some embodiments, Z is alkyl, cycloalkyl, aryl, heteroaryl, or a combination thereof. In some embodiments, Z is substituted or unsubstituted aryl. In some embodiments, Z is phenyl or biphenyl.

[0105] In some embodiments, a+b+c is 3.

[0106] In some embodiments, a+b+c is 2, and a is 0. In some embodiments, a+b+c is 2, and b is 0. In some embodiments, a+b+c is 2, and c is 0.

[0107] In some embodiments, Y^1 is a direct bond.

[0108] In some embodiments, Y^1 is selected from the group consisting of O, S, and Se. In some embodiments, Y^1 is selected from the group consisting of BR, NR, and PR. In some embodiments, Y^1 is BR. In some embodiments, Y^1 is NR. In some embodiments, Y^1 is PR. In some such embodiments, the R is aryl or heteroaryl. In some such embodiments, R is benzene. In some such embodiments, R is joined or fused with one R^B to form a ring. In some such embodiments, the ring is a 5-membered ring. In some such embodiments, the ring is a pyrrole ring.

[0109] In some embodiments, Y^1 is selected from the group consisting of P(O)R, C═O, C═S, C═Se, C═NR', C═CRR', S═O, and SO_2. In some embodiments, Y^1 is selected from the group consisting of BRR', CRR', SiRR', and GeRR'. In some embodiments, Y^1 is CR.

[0110] In some embodiments, Y^2 is a direct bond.

[0111] In some embodiments, Y^2 is selected from the group consisting of O, S, and Se. In some embodiments, Y^2 is selected from the group consisting of BR, NR, and PR. In some embodiments, Y^2 is BR. In some embodiments, Y^2 is NR. In some embodiments, Y^2 is PR. In some such embodiments, the R is aryl or heteroaryl. In some such embodiments, R is benzene. In some such embodiments, R is joined or fused with one R^B or R^C to form a ring. In some such embodiments, the ring is a 5-membered ring. In some such embodiments, the ring is a pyrrole ring.

[0112] In some embodiments, Y^2 is selected from the group consisting of P(O)R, C═O, C═S, C═Se, C═NR', C═CRR', S═O, and SO_2. In some embodiments, Y^2 is selected from the group consisting of BRR', CRR', SiRR', and GeRR'. In some embodiments, Y^2 is CR.

[0113] In some embodiments, Y^3 is a direct bond.

[0114] In some embodiments, Y^3 is selected from the group consisting of O, S, and Se. In some embodiments, Y^3 is selected from the group consisting of BR, NR, and PR. In some embodiments, Y^3 is BR. In some embodiments, Y^3 is NR. In some embodiments, Y^3 is PR. In some embodiments, R is aryl or heteroaryl. In some embodiments, R is benzene.

[0115] In some embodiments, R is joined or fused with one R^C to form a ring. In some such embodiments, the ring is a 5-membered ring. In some such embodiments, the ring is a pyrrole ring.

[0116] In some embodiments, R is joined or fused with one R^C to form a ring. In some such embodiments, the ring is a 6-membered ring. In some such embodiments, the ring comprises one B and one N.

[0117] In some embodiments, R is biphenyl and R^C is an N that is bonded to both phenyls of the biphenyl to form a pyrrole ring.

[0118] In some embodiments, Y^3 is selected from the group consisting of P(O)R, C═O, C═S, C═Se, C═NR', C═CRR', S═O, and SO_2. In some embodiments, Y^3 is selected from the group consisting of BRR', CRR', SiRR', and GeRR'. In some embodiments, Y^3 is CR.

[0119] In some embodiments, A^1 is B and at least two of Y^1 to Y^3 are NR.

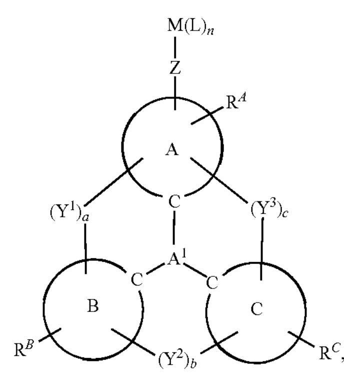
[0120] In some embodiments, A^1 is B and Y^1 and Y^3 are both NR. In some such embodiments, b=0.

[0121] In some embodiments, A^1 is B and each of Y^1 to Y^3 is NR.

[0122] In some embodiments when A^1 is N and two of Y^1, Y^2, or Y^3 are direct bonds, then at least one of R^A, R^B, or R^C is not H. In some such embodiments, at least one of R^A, R^B, or R^C is selected from the group consisting of the General Substituents defined herein. In some such embodiments, at least one R^A, R^B, or R^C is selected from the group consisting of the Preferred General Substituents defined herein. In some embodiments when A^1 is N and two of Y^1, Y^2, or Y^3 are direct bonds, then the remaining one is also present (a+b+c=3). In some embodiments when A^1 is N and one of Y^1, Y^2, or Y^3 is a direct bond, then at least one additional of Y^1, Y^2, or Y^3 is selected from the group consisting of BR, BRR', NR, PR, P(O)R, O, S, Se, C═O, C═S, C═Se, C═NR, C═CRR', S═O, SO_2, CR, CRR', SiRR', and GeRR'.

[0123] In some embodiments, Y^1 is NR, where the R is aryl bonded to R^B, and Y^3 is NR, where the R is aryl bonded to R^C.

[0124] In some embodiments, the compound comprises a structure of Formula IIA:



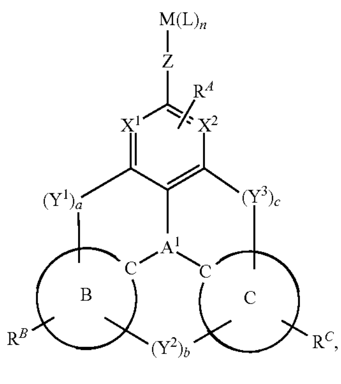
wherein:

[0125] n is an integer from 1 to 5;

[0126] each L is independently a monodentate, bidentate, tridentate, tetradentate, or pentadentate ligand; and

[0127] each L can be the same or different.

[0128] In some embodiments, the compound comprises a structure of Formula II:



wherein:

[0129] X^1 and X^2 are each independently C or N;

[0130] n is an integer from 1 to 5;

[0131] each L is independently a monodentate, bidentate, tridentate, tetradentate, or pentadentate ligand; and

[0132] each L can be the same or different.

[0133] In some embodiments of Formula II or Formula IIA, Z is a direct bond.

[0134] In some embodiments of Formula II, X^1 and X^2 are C.

[0135] In some embodiments of Formula II, X^1 is C and X^2 is N. In some embodiments of Formula II, X^1 is N and X^2 is C. In some embodiments of Formula II, X^1 and X^2 are N.

[0136] In some embodiments of Formula II or Formula IIA, n is 1. In some such embodiments, L is a tridentate ligand.

[0137] In some embodiments of Formula II or Formula IIA, n is 2. In some embodiments of Formula II, each L is a bidentate ligand.

[0138] In some embodiments of Formula II or Formula IIA, n is 3 or greater. In some embodiments of Formula II, n is 3, two Ls are monodentate, and one L is tridentate.

[0139] In some embodiments of Formula II or Formula IIA, L is not coordinated to the metal through a carbene carbon.

[0140] In some embodiments of Formula II or Formula IIA when M is Au, Ag, or Cu, L is not coordinated to the metal through a carbene carbon.

[0141] In some embodiments of Formula II or Formula IIA, each of Y^1, Y^2, and Y^3 is present (a+b+c=3).

[0142] In some embodiments of Formula II or Formula IIA when M is Au, Ag, or Cu, each of Y^1, Y^2, and Y^3 is present.

[0143] In some embodiments of Formula II or Formula IIA when one of Y^1, Y^2, or Y^3 is absent, at least one of the remaining two is not N or B.

[0144] In some embodiments of Formula II or Formula IIA when M is Au, Ag, or Cu and one of Y^1, Y^2, or Y^3 is absent, at least one of the remaining two is not N or B.

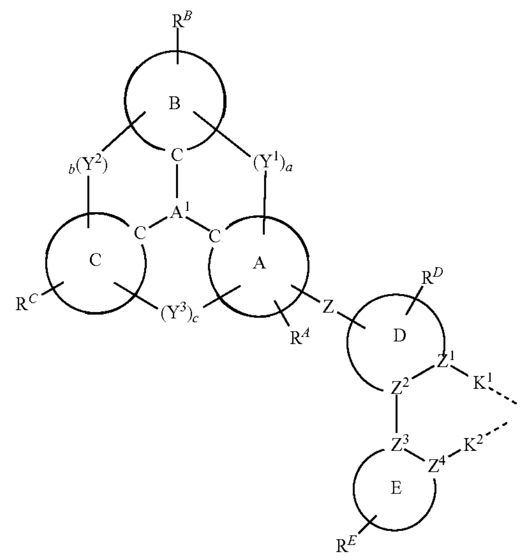
[0145] In some embodiments of Formula II or Formula IIA, two of Y^1, Y^2, or Y^3 are direct bonds.

[0146] In some embodiments of Formula II or Formula IIA when M is Au, Ag, or Cu, two of Y^1, Y^2, or Y^3 are direct bonds.

[0147] In some embodiments of Formula II or Formula IIA, at least one of moiety A, moiety B, or moiety C is a 5-membered ring.

[0148] In some embodiments of Formula II or Formula IIA when M is Au, Ag, or Cu, at least one of moiety A, moiety B, or moiety C is a 5-membered ring. In some such embodiments, moiety A is a 5-membered ring. In some such embodiments, moiety B is a 5-membered ring. In some such embodiments, moiety C is a 5-membered ring.

[0149] In some embodiments, L_A comprises a structure of Formula IIIA:



In Formula IIIA, each of moiety D and moiety E is independently a monocyclic ring or a polycyclic fused ring system, wherein the monocyclic ring or each ring of the polycyclic fused ring system is independently a 5-membered to 10-membered carbocyclic or heterocyclic ring;

[0150] each of Z^1 to Z^4 is independently C or N;

[0151] K^1 and K^2 are each independently a direct bond or selected from the group consisting of a direct bond, O, S, $N(R^\alpha)$, $P(R^\alpha)$, $B(R^\alpha)$, $C(R^\alpha)(R^\beta)$, and $Si(R^\alpha)(R^\beta)$;

[0152] R^D and R^E each independently represent mono to the maximum allowable substitution;

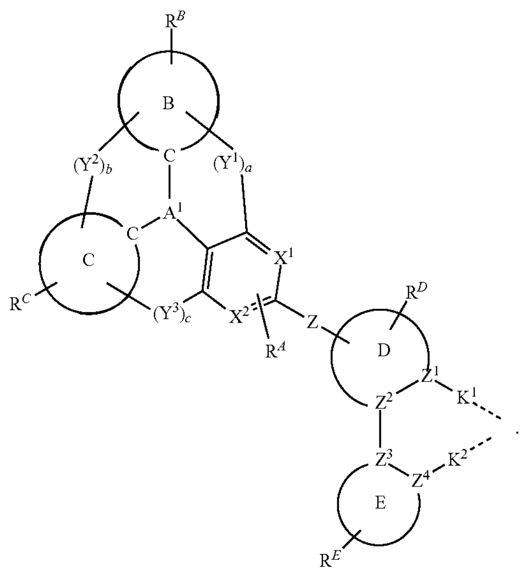
[0153] each R^α, R^β, R^D, and R^E is independently hydrogen or a substituent selected from the group consisting of the General Substituents defined herein;

[0154] the remaining variables are the same as previously defined;

[0155] any two substituents can be joined or fused to form a ring; and

[0156] L_A is coordinated to M through the indicated dashed lines.

[0157] In some embodiments, L_A comprises a structure of Formula III:



In Formula III,

[0158] each of moiety D and moiety E is independently a monocyclic ring or a polycyclic fused ring system, wherein the monocyclic ring or each ring of the polycyclic fused ring system is independently a 5-membered to 10-membered carbocyclic or heterocyclic ring;

[0159] each of X^1, X^2, and Z^1 to Z^4 is independently C or N;

[0160] K^1 and K^2 are each independently a direct bond or selected from the group consisting of a direct bond, O, S, $N(R^\alpha)$, $P(R^\alpha)$, $B(R^\alpha)$, $C(R^\alpha)(R^\beta)$, and $Si(R^\alpha)(R^\beta)$;

[0161] R^D and R^E each independently represent mono to the maximum allowable substitution;

[0162] each R^α, R^β, R^D, and R^E is independently hydrogen or a substituent selected from the group consisting of the General Substituents defined herein;

[0163] any two substituents can be joined or fused to form a ring; and

[0164] L_A is coordinated to M through the indicated dashed lines.

[0165] In some embodiments of Formula III or Formula IIIA, the first ligand L_A has a structure of Formula III or Formula IIIA. In some embodiments of Formula III or Formula IIIA, the first ligand L_A consists essentially of Formula III or Formula IIIA.

[0166] In some embodiments of Formula III or Formula IIIA, Z is a direct bond. In some embodiments of Formula III or Formula IIIA, Z is aryl. In some embodiments of Formula III or Formula IIIA, Z is phenyl.

[0167] In some embodiments of Formula III or Formula IIIA, at least one R^A, R^B, R_C, R^D, or R^E is selected from the group consisting of the General Substituents defined herein. In some embodiments, at least one R^A is selected from the group consisting of the General Substituents defined herein. In some embodiments, at least one R^B is selected from the group consisting of the General Substituents defined herein. In some embodiments, at least one R^C is selected from the group consisting of the General Substituents defined herein. In some embodiments, at least one R^D is selected from the group consisting of the General Substituents defined herein. In some embodiments, at least one R^E is selected from the group consisting of the General Substituents defined herein. In some embodiments, at least one R^A, R^B, R^C, R^D, or R^E is selected from the group consisting of the Preferred General Substituents defined herein.

[0168] In some embodiments of Formula III or Formula IIIA, at least one R, R', R", R^α, R^β, R^A, R^B, R^C, R^D, or R^E is partially or fully deuterated. In some embodiments, at least one R^A is partially or fully deuterated. In some embodiments, at least one R^B is partially or fully deuterated. In some embodiments, at least one R^C is partially or fully deuterated. In some embodiments, at least one R^D is partially or fully deuterated. In some embodiments, at least one R^E is partially or fully deuterated. In some embodiments, at least one R or R' is partially or fully deuterated. In some embodiments, R" is partially or fully deuterated. In some embodiments, at least one R^α or R^β is partially or fully deuterated.

[0169] In some embodiments of Formula III or Formula IIIA, each of moiety D and moiety E is independently a monocyclic ring or a polycyclic fused ring system, wherein the monocyclic ring or each ring of the polycyclic fused ring system is independently a 5-membered or 6-membered carbocyclic or heterocyclic ring. In some embodiments of Formula III, each of moiety D and moiety E is independently a monocyclic ring or a polycyclic fused ring system, wherein the monocyclic ring or each ring of the polycyclic fused ring system is independently a 5-membered or 6-membered aryl or heteroaryl ring.

[0170] In some embodiments of Formula III, X^1 and X^2 are C. In some embodiments of Formula III, X^1 is C and X^2 is N. In some embodiments of Formula III, X^1 is N and X^2 is C. In some embodiments of Formula III, X^1 and X^2 are N.

[0171] In some embodiments of Formula III or Formula IIIA, each of moiety D and moiety E is independently selected from the group consisting of the following Cyclic Moiety List: benzene, pyridine, pyrimidine, pyridazine, pyrazine, triazine, imidazole, imidazole derived carbene, pyrazole, pyrrole, oxazole, furan, thiophene, thiazole, triazole, naphthalene, quinoline, isoquinoline, quinazoline, benzofuran, aza-benzofuran, benzoxazole, aza-benzoxazole, benzothiophene, aza-benzothiophene, benzothiazole, aza-benzothiazole, benzoselenophene, aza-benzoselenophene, indene, aza-indene, indole, aza-indole, benzimidazole, aza-benzimidazole, benzimidazole derived carbene, aza-benzimidazole derived carbene, benzobenzimidazole, aza-benzobenzimidazole, carbazole, aza-carbazole, dibenzofuran, aza-dibenzofuran, phenanthro[3,2-b]benzofuran, dibenzothiophene, aza-dibenzothiophene, quinoxaline, phthalazine, phenanthrene, aza-phenanthrene, anthracene, aza-anthracene, phenanthridine, fluorene, and aza-fluorene. In some such embodiments, the aza variant includes one N on a benzo ring. In some such embodiments, the aza variant includes one N on a benzo ring and the N is bonded to the metal M.

[0172] In some embodiments of Formula III or Formula IIIA, moiety D is a monocyclic ring. In some embodiments of Formula III, moiety D is selected from the group consisting of benzene, pyridine, pyrimidine, pyridazine, pyrazine, triazine, imidazole, imidazole derived carbene, pyrazole, pyrrole, oxazole, furan, thiophene, thiazole, and triazole. In some embodiments of Formula III, moiety D is benzene, pyridine, or imidazole.

[0173] In some embodiments of Formula III or Formula IIIA, moiety D is a polycyclic fused ring system. In some embodiments of Formula III, moiety D is selected from the group consisting of naphthalene, quinoline, isoquinoline, quinazoline, benzofuran, aza-benzofuran, benzoxazole, aza-benzoxazole, benzothiophene, aza-benzothiophene, benzothiazole, aza-benzothiazole, benzoselenophene, aza-benzoselenophene, indene, aza-indene, indole, aza-indole, benzimidazole, aza-benzimidazole, benzimidazole derived carbene, aza-benzimidazole derived carbene, benzobenzimidazole, aza-benzobenzimidazole, carbazole, aza-carbazole, dibenzofuran, aza-dibenzofuran, phenanthro[3,2-b]benzofuran, dibenzothiophene, aza-dibenzothiophene, quinoxaline, phthalazine, phenanthrene, aza-phenanthrene, anthracene, aza-anthracene, phenanthridine, fluorene, and aza-fluorene. In some embodiments of Formula III, moiety D is naphthalene, quinoline, isoquinoline, or benzimidazole.

[0174] In some embodiments of Formula III or Formula IIIA, moiety E is a monocyclic ring. In some embodiments of Formula III, moiety E is selected from the group consisting of benzene, pyridine, pyrimidine, pyridazine, pyrazine, triazine, imidazole, imidazole derived carbene, pyrazole, pyrrole, oxazole, furan, thiophene, thiazole, and triazole. In some embodiments of Formula III, moiety E is benzene, pyridine, or imidazole.

[0175] In some embodiments of Formula III or Formula IIIA, moiety E is a polycyclic fused ring system. In some embodiments of Formula III, moiety E is selected from the group consisting of naphthalene, quinoline, isoquinoline, quinazoline, benzofuran, aza-benzofuran, benzoxazole, aza-benzoxazole, benzothiophene, aza-benzothiophene, benzothiazole, aza-benzothiazole, benzoselenophene, aza-benzoselenophene, indene, aza-indene, indole, aza-indole, benzimidazole, aza-benzimidazole, benzimidazole derived carbene, aza-benzimidazole derived carbene, benzobenzimidazole, aza-benzobenzimidazole, carbazole, aza-carbazole, dibenzofuran, aza-dibenzofuran, phenanthro[3,2-b]benzofuran, dibenzothiophene, aza-dibenzothiophene, quinoxaline, phthalazine, phenanthrene, aza-phenanthrene, anthracene, aza-anthracene, phenanthridine, fluorene, and aza-fluorene. In some embodiments of Formula III, moiety E is naphthalene, quinoline, isoquinoline, or benzimidazole.

[0176] In some embodiments of Formula III or Formula IIIA, moiety D is pyridine, imidazole, imidazole derived carbene, benzimidazole, quinoline, or isoquinoline, and moiety E is benzene or naphthalene.

[0177] In some embodiments of Formula III or Formula IIIA, moiety D is pyridine, imidazole, or benzimidazole and E is benzene or naphthalene. In some embodiments of Formula III or Formula IIIA, D is benzimidazole and E is benzene.

[0178] In some embodiments of Formula III or Formula IIIA, moiety E is pyridine, imidazole, imidazole derived carbene, benzimidazole, quinoline, or isoquinoline, and moiety D is benzene or naphthalene. In some embodiments of Formula III, moiety E is pyridine, imidazole, or benzimidazole and D is benzene or naphthalene. In some embodiments of Formula III, E is benzimidazole and D is benzene.

[0179] In some embodiments, each of moiety A, moiety B, moiety C, moiety D, and moiety E can independently be a polycyclic fused ring structure. In some embodiments, moiety A, moiety B, moiety C, moiety D, and moiety E can independently be a polycyclic fused ring structure comprising at least two fused rings. In some embodiments, the polycyclic fused ring structure has one 6-membered ring and one 5-membered ring. In some such embodiments, either the 5-membered ring or the 6-membered ring can coordinate to the metal. In some embodiments, the polycyclic fused ring structure has two 6-membered rings. In some embodiments, moiety A, moiety B, moiety C, moiety D, and moiety E can independently be selected from the group consisting of benzofuran, benzothiophene, benzoselenophene, naphthalene, and aza-variants thereof.

[0180] In some embodiments, moiety A, moiety B, moiety C, moiety D, and moiety E can independently be a polycyclic fused ring structure comprising at least three fused rings. In some embodiments, the polycyclic fused ring structure has two 6-membered rings and one 5-membered ring. In some such embodiments, the 5-membered ring is fused to the ring coordinated to metal M and the second 6-membered ring is fused to the 5-membered ring. In some embodiments, moiety A, moiety B, moiety C, moiety D, and moiety E can independently be selected from the group consisting of dibenzofuran, dibenzothiophene, dibenzoselenophene, and aza-variants thereof. In some such embodiments, moiety A, moiety B, moiety C, moiety D, and moiety E can independently be further substituted at the ortho- or meta-position of the O, S, or Se atom by a substituent selected from the group consisting of deuterium, fluorine, nitrile, alkyl, cycloalkyl, aryl, heteroaryl, and combinations thereof. In some such embodiments, the aza-variants contain exactly one N atom at the 6-position (ortho to the O, S, or Se) with a substituent at the 7-position (meta to the O, S, or Se).

[0181] In some embodiments, moiety A, moiety B, moiety C, moiety D, and moiety E can independently be a polycyclic fused ring structure comprising at least four fused rings. In some embodiments, the polycyclic fused ring structure comprises three 6-membered rings and one 5-membered ring. In some such embodiments, the 5-membered ring is fused to the ring coordinated to metal M, the second 6-membered ring is fused to the 5-membered ring, and the third 6-membered ring is fused to the second 6-membered ring. In some such embodiments, the third 6-membered ring is further substituted by a substituent selected from the group consisting of deuterium, fluorine, nitrile, alkyl, cycloalkyl, aryl, heteroaryl, and combinations thereof.

[0182] In some embodiments, moiety A, moiety B, moiety C, moiety D, and moiety E can independently be a polycyclic fused ring structure comprising at least five fused rings. In some embodiments, the polycyclic fused ring structure comprises four 6-membered rings and one 5-membered ring or three 6-membered rings and two 5-membered rings. In some embodiments comprising two 5-membered rings, the 5-membered rings are fused together. In some embodiments comprising two 5-membered rings, the 5-membered rings are separated by at least one 6-membered ring. In some embodiments with one 5-membered ring, the 5-membered ring is fused to the ring coordinated to metal M, the second 6-membered ring is fused to the 5-membered ring, the third 6-membered ring is fused to the second 6-membered ring, and the fourth 6-membered ring is fused to the third 6-membered ring.

[0183] In some embodiments, moiety A, moiety B, moiety C, moiety D, and moiety E can independently be an aza version of the polycyclic fused rings described above. In some such embodiments, moiety A, moiety B, moiety C, moiety D, and moiety E can independently contain exactly one aza N atom. In some such embodiments, moiety A, moiety B, moiety C, moiety D, and moiety E contains exactly two aza N atoms, which can be in one ring, or in two different rings. In some such embodiments, the ring having aza N atom is separated by at least two other rings from the metal M atom. In some such embodiments, the ring having aza N atom is separated by at least three other rings from the metal M atom. In some such embodiments, each of the ortho position of the aza N atom is substituted.

[0184] In some embodiments of Formula III or Formula IIIA, Z^1 is N and each of X^2 to X^4 are C. In some embodiments of Formula III or Formula IIIA, Z^4 is N and each of X^1 to X^3 is C.

[0185] In some embodiments of Formula III or Formula IIIA, K^1 is a direct bond.

[0186] In some embodiments of Formula III or Formula IIIA, K^1 is O or S. In some embodiments of Formula III, K^1 is O. In some embodiments of Formula III or Formula IIIA, K^1 is S.

[0187] In some embodiments of Formula III or Formula IIIA, K^1 is $N(R^{\alpha})$, $P(R^{\alpha})$, or $B(R^{\alpha})$. In some embodiments of Formula III, K^1 is $C(R^{\alpha})(R^{\beta})$ or $Si(R^{\alpha})(R^{\beta})$.

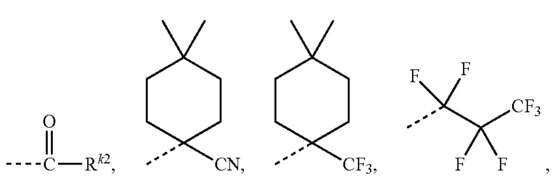
[0188] In some embodiments of Formula III or Formula IIIA, K^1 is K^2 is a direct bond. In some embodiments of Formula III, K^2 is O or S. In some embodiments of Formula III, K^2 is O. In some embodiments of Formula III, K^2 is S.

[0189] In some embodiments of Formula III or Formula IIIA, K^2 is $N(R^{\alpha})$, $P(R^{\alpha})$, or $B(R^{\alpha})$. In some embodiments of Formula III, K^2 is $C(R^{\alpha})(R^{\beta})$ or $Si(R^{\alpha})(R^{\beta})$.

[0190] In some embodiments of Formula III, K^2 is each of K^1 and K^2 is a direct bond.

[0191] In some embodiments of Formula III, K^2 is one of K^1 or K^2 is a direct bond and the other one of K^1 or K^2 is not a direct bond. In some such embodiments, one of K^1 or K^2 is O.

[0192] In some embodiments of Formula III, the first ligand L_A comprises an electron-withdrawing group selected from the group consisting of the following EWG1 LIST: F, CF_3, CN, $COCH_3$, CHO, $COCF_3$, COOMe, $COOCF_3$, NO_2, SF_3, SiF_3, PF_4, SF_5, OCF_3, SCF_3, $SeCF_3$, $SOCF_3$, $SeOCF_3$, SO_2F, SO_2CF_3, SeO_2CF_3, $OSeO_2CF_3$, OCN, SCN, SeCN, NC, $^{+}N(R^{k2})_3$, $(R^{k2})_2CCN$, $(R^{k2})_2CCF_3$, $CNC(CF_3)_2$, $BR^{k3}R^{k2}$, substituted or unsubstituted dibenzoborole, 1-substituted carbazole, 1,9-substituted carbazole, substituted or unsubstituted carbazole, substituted or unsubstituted pyridine, substituted or unsubstituted pyrimidine, substituted or unsubstituted pyrazine, substituted or unsubstituted pyridoxine, substituted or unsubstituted triazine, substituted or unsubstituted oxazole, substituted or unsubstituted benzoxazole, substituted or unsubstituted thiazole, substituted or unsubstituted benzothiazole, substituted or unsubstituted imidazole, substituted or unsubstituted benzimidazole, ketone, carboxylic acid, ester, nitrile, isonitrile, sulfinyl, sulfonyl, partially and fully fluorinated alkyl, partially and fully fluorinated aryl, partially and fully fluorinated heteroaryl, cyano-containing alkyl, cyano-containing aryl, cyano-containing heteroaryl, isocyanate,



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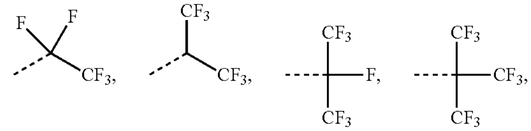

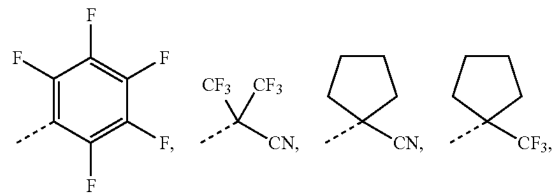

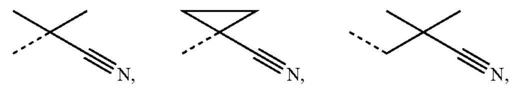

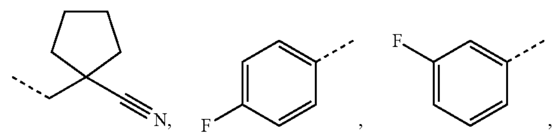

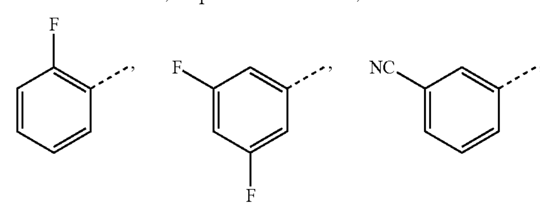

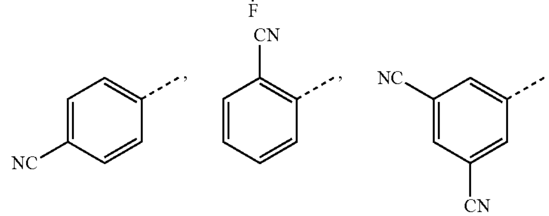

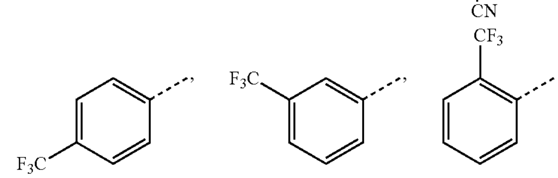

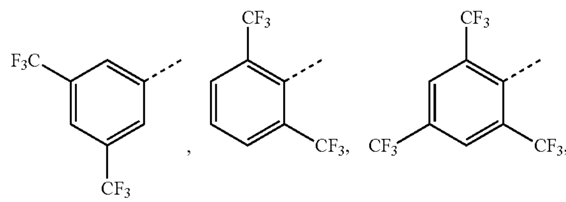

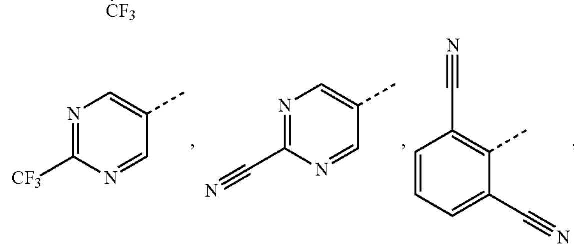

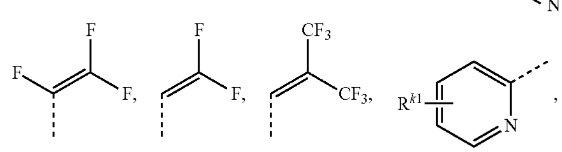

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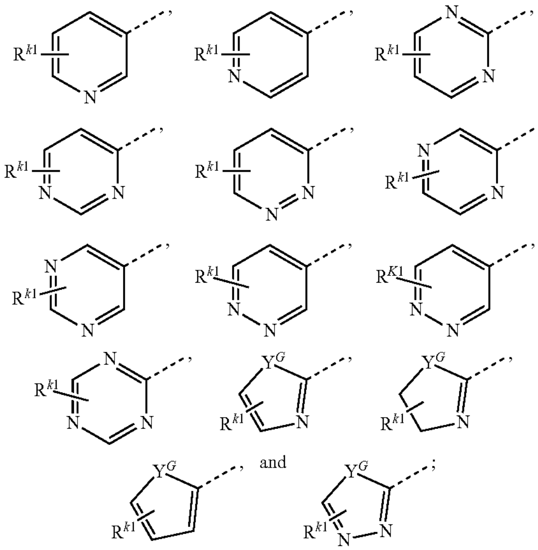

[0193] wherein each R^{k1} represents mono to the maximum allowable substitution, or no substitutions;

[0194] wherein Y^G is selected from the group consisting of BR_e, NR_e, PR_e, O, S, Se, C═O, S═O, SO_2, CR_eR_f, SiR_eR_f, and GeR_eR_f; and

[0195] wherein each of R^k, R^{k2}, R^{k3}, R_e, and R_f is independently a hydrogen, or a substituent selected from the group consisting of the General Substituents defined herein.

[0196] In some embodiments, the first ligand L_A comprises an electron-withdrawing group selected from the group consisting of the structures of the following EWG2 List:

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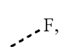

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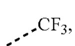

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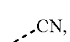

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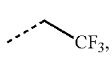

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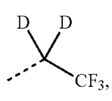

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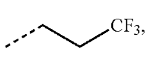

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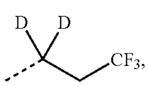

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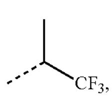

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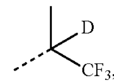

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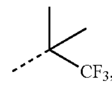

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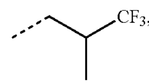

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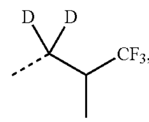

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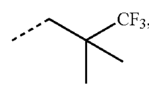

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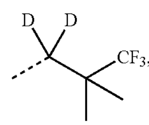

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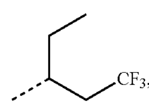

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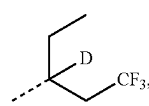

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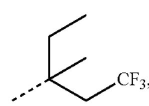

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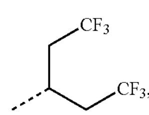

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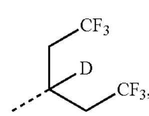

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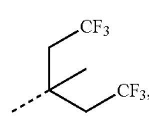

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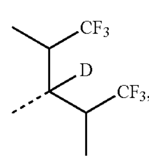

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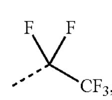

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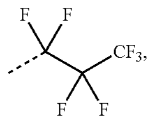

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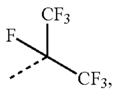

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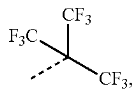

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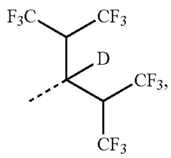

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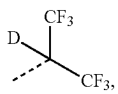

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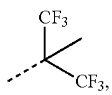

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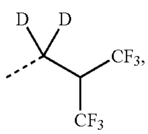

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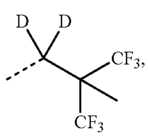

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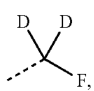

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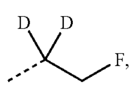

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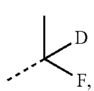

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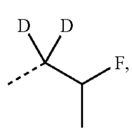

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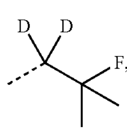

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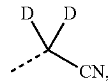

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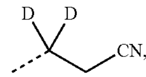

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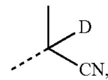

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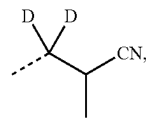

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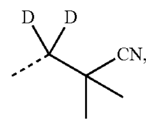

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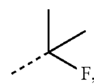

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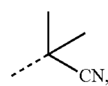

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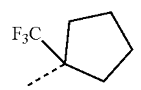

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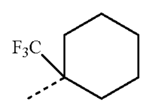

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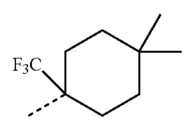

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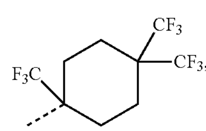

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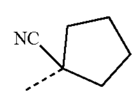

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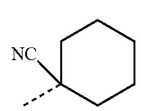

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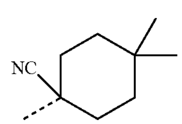

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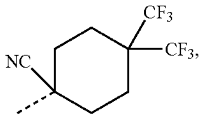

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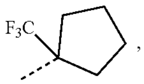

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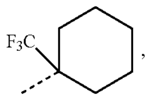

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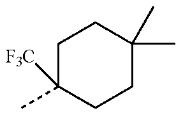

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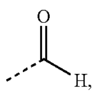

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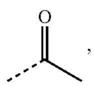

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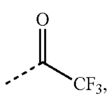

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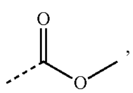

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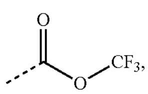

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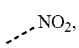

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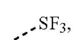

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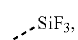

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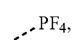

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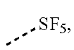

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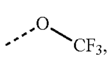

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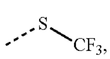

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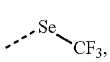

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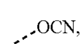

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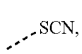

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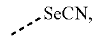

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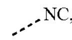

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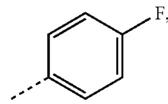

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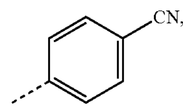

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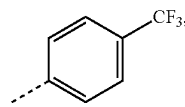

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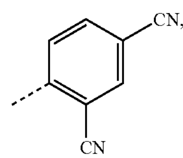

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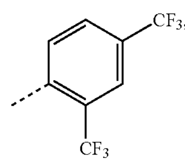

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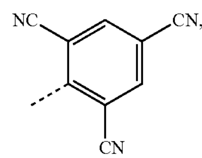

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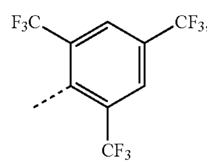

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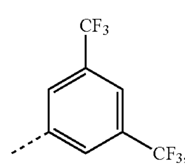

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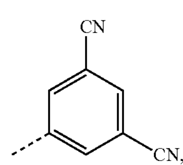

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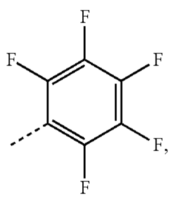

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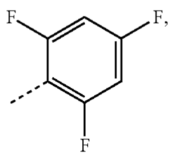

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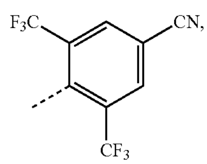

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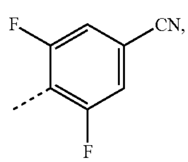

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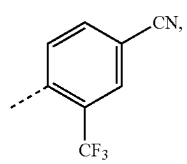

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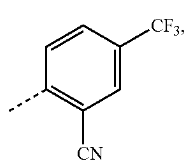

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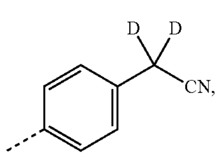

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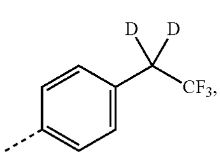

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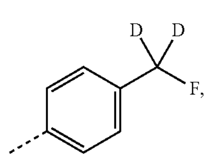

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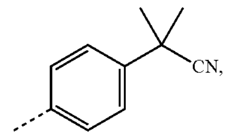

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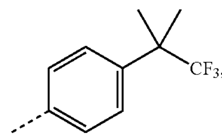

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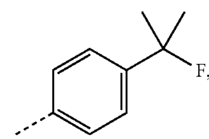

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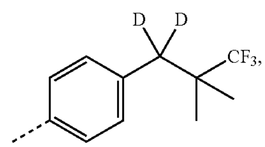

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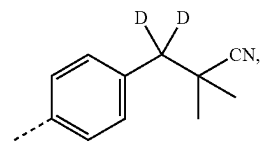

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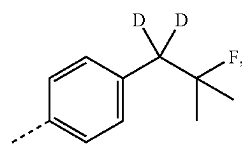

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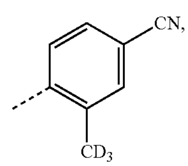

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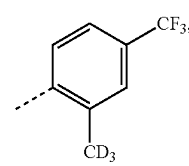

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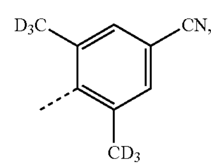

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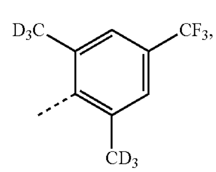

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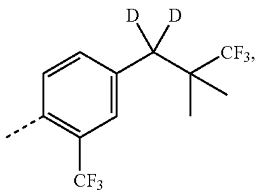

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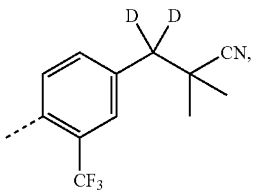

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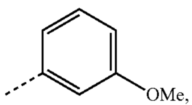

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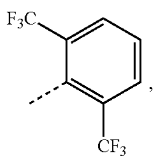

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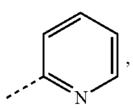

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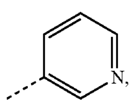

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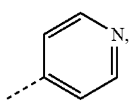

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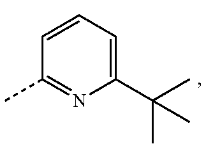

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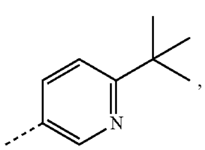

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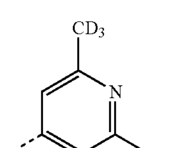

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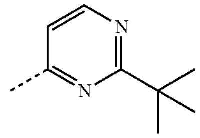

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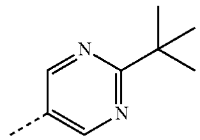

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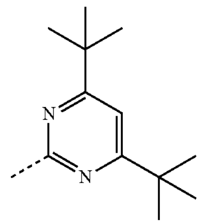

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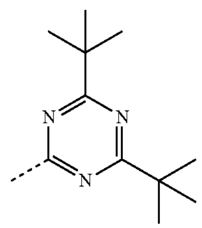

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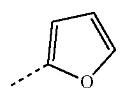

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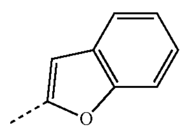

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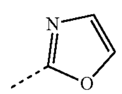

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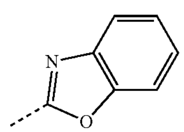

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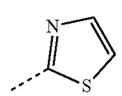

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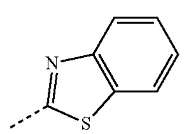

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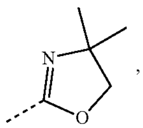

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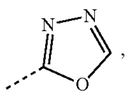

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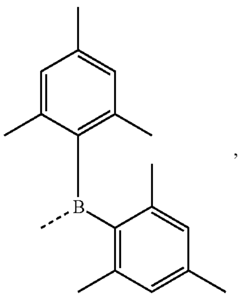

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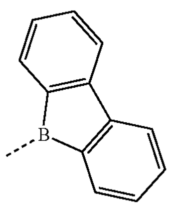

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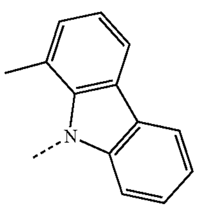

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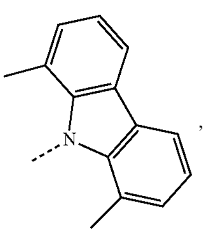

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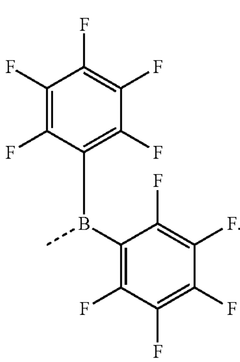
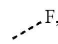
E125



[0197] In some embodiments, the first ligand L_A comprises an electron-withdrawing group selected from the group consisting of the structures of the following EWG3 LIST:

E1



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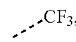
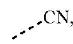


E3



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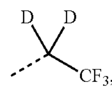

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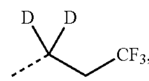

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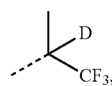

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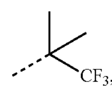

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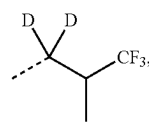

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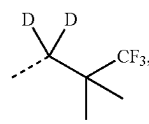

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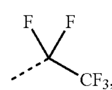

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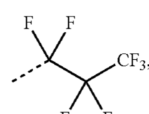

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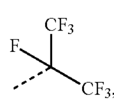

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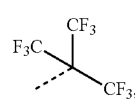

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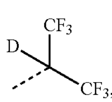

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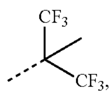

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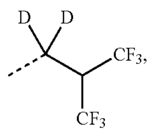

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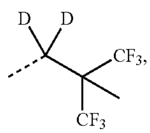

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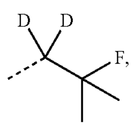

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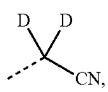

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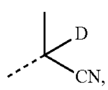

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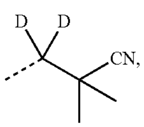

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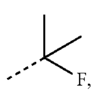

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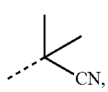

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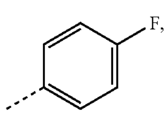

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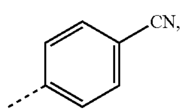

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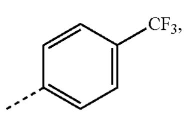

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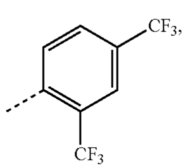

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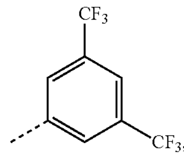

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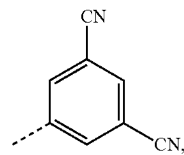

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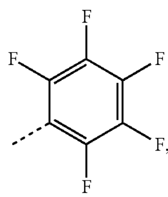

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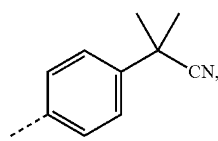

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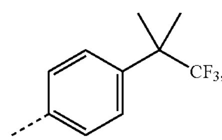

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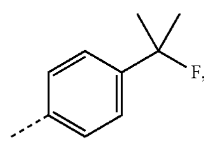

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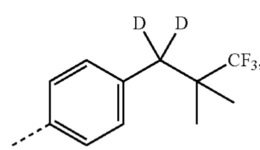

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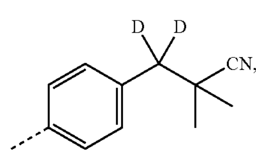

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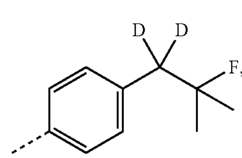

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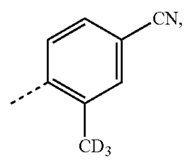

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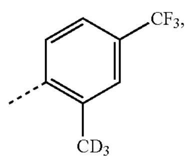

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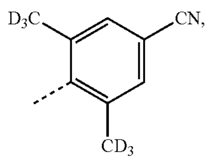

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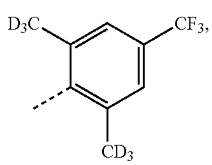

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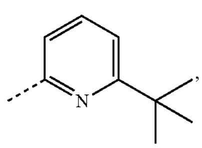

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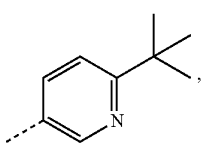

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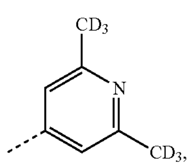

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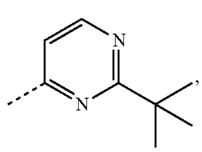

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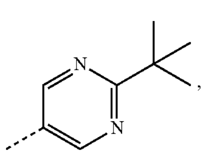

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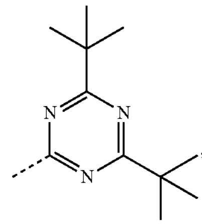

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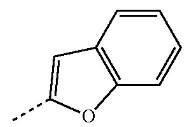

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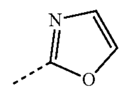
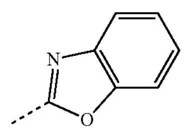
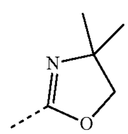


E116



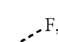
and

E119



[0198] In some embodiments, the first ligand L_A comprises an electron-withdrawing group selected from the group consisting of the structures of the following EWG4 LIST:

E1



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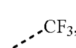

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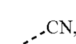

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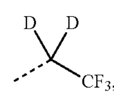
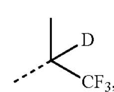


E9



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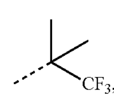

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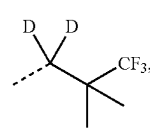

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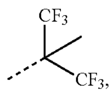

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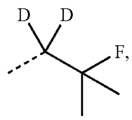

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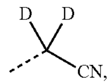

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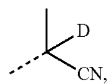

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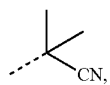

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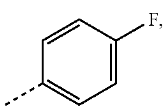

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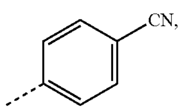

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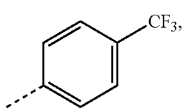

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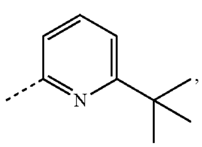

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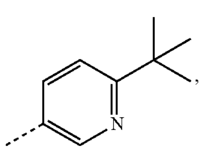

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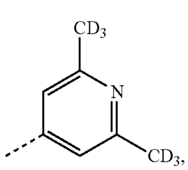

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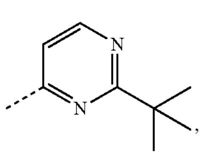

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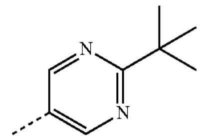
E110



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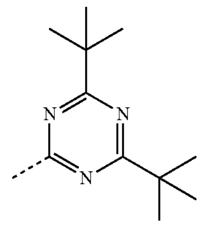

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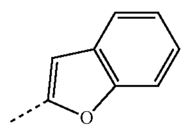

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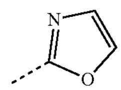

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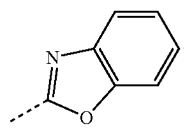

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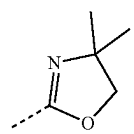
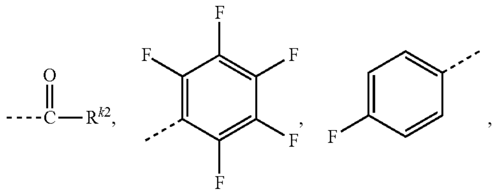
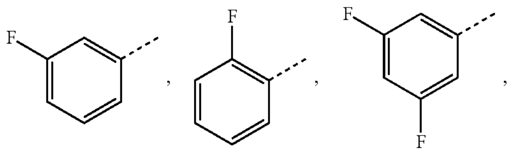
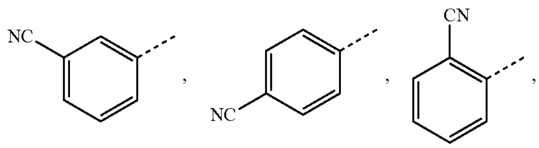
E119

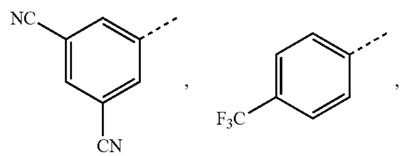


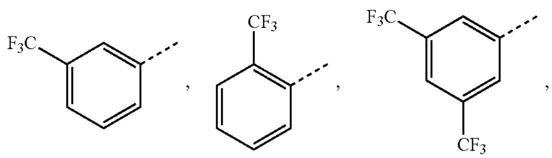
[0199] In some embodiments, the first ligand L_A comprises a π-electron deficient electron-withdrawing group selected from the group consisting of the structures of the following Pi-EWG LIST: CN, $COCH_3$, CHO, $COCF_3$, COOMe, $COOCF_3$, NO_2, SF_3, SiF_3, PF_4, SF_5, OCF_3, SCF_3, $SeCF_3$, $SOCF_3$, $SeOCF_3$, SO_2F, SO_2CF_3, SeO_2CF_3, $OSeO_2CF_3$, OCN, SCN, SeCN, NC, $^{+}N(R^{k2})_3$, $BR^{k2}Rk^3$, substituted or unsubstituted dibenzoborole, 1-substituted carbazole, 1,9-substituted carbazole, substituted or unsubstituted carbazole, substituted or unsubstituted pyridine, substituted or unsubstituted pyrimidine, substituted or unsubstituted pyrazine, substituted or unsubstituted pyridazine, substituted or unsubstituted triazine, substituted or unsubstituted oxazole, substituted or unsubstituted benzoxazole, substituted or unsubstituted thiazole, substituted or unsubstituted benzothiazole, substituted or unsubstituted imidazole, substituted or unsubstituted benzimidazole, ketone, carboxylic acid, ester, nitrile, isonitrile, sulfinyl, sulfonyl, partially and fully fluorinated aryl, partially and fully fluorinated heteroaryl, cyano-containing aryl, cyano-containing heteroaryl, isocyanate,

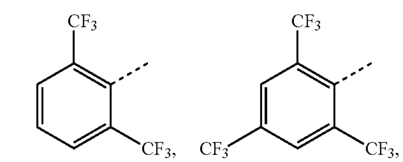


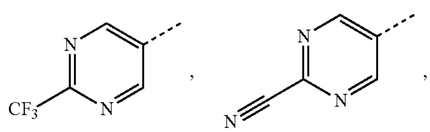


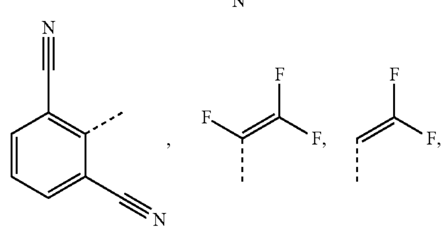


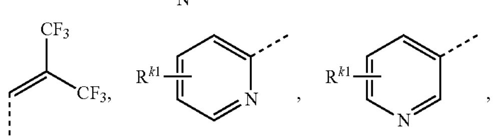


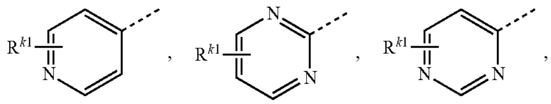


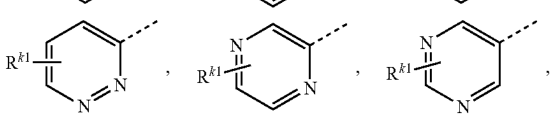












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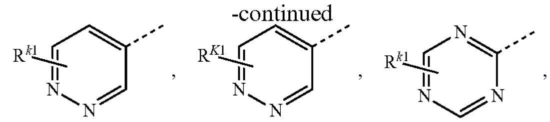
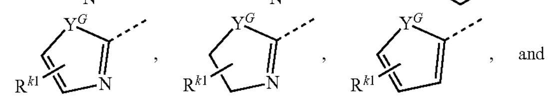
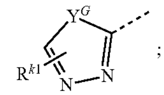






wherein the variables are the same as previously defined.

[0200] In some embodiments, the compound comprises an electron-withdrawing group from the EWG1 LIST as defined herein. In some embodiments, the compound comprises an electron-withdrawing group from the EWG2 LIST as defined herein. In some embodiments, the compound comprises an electron-withdrawing group from the EWG3 LIST as defined herein. In some embodiments, the compound comprises an electron-withdrawing group from the EWG4 LIST as defined herein. In some embodiments, the compound comprises an electron-withdrawing group from the Pi-EWG LIST as defined herein.

[0201] In some embodiments, at least one R^A is or comprises an electron-withdrawing group from the EWG1 LIST as defined herein. In some embodiments, at least one R^A is or comprises an electron-withdrawing group from the EWG2 LIST as defined herein. In some embodiments, at least one R^A is or comprises an electron-withdrawing group from the EWG3 LIST as defined herein. In some embodiments, at least one R^A is or comprises an electron-withdrawing group from the EWG4 LIST as defined herein. In some embodiments, at least one R^A is or comprises an electron-withdrawing group from the Pi-EWG LIST as defined herein.

[0202] In some embodiments, at least one R^B is or comprises an electron-withdrawing group from the EWG1 LIST as defined herein. In some embodiments, at least one R^B is or comprises an electron-withdrawing group from the EWG2 LIST as defined herein. In some embodiments, at least one R^B is or comprises an electron-withdrawing group from the EWG3 LIST as defined herein. In some embodiments, at least one R^B is or comprises an electron-withdrawing group from the EWG4 LIST as defined herein. In some embodiments, at least one R^B is or comprises an electron-withdrawing group from the Pi-EWG LIST as defined herein.

[0203] In some embodiments, at least one R^C is or comprises an electron-withdrawing group from the EWG1 LIST as defined herein. In some embodiments, at least one R^C is or comprises an electron-withdrawing group from the EWG2 LIST as defined herein. In some embodiments, at least one R^C is or comprises an electron-withdrawing group from the EWG3 LIST as defined herein. In some embodiments, at least one R^C is or comprises an electron-withdrawing group from the EWG4 LIST as defined herein. In some embodiments, at least one R^C is or comprises an electron-withdrawing group from the Pi-EWG LIST as defined herein.

[0204] In some embodiments, at least one R^D is or comprises an electron-withdrawing group from the EWG1 LIST as defined herein. In some embodiments, at least one R^D is or comprises an electron-withdrawing group from the EWG2 LIST as defined herein. In some embodiments, at least one R^D is or comprises an electron-withdrawing group from the EWG3 LIST as defined herein. In some embodiments, at least one R^D is or comprises an electron-withdrawing group from the EWG4 LIST as defined herein. In some embodiments, at least one R^D is or comprises an electron-withdrawing group from the Pi-EWG LIST as defined herein.

[0205] In some embodiments, at least one R^E is or comprises an electron-withdrawing group from the EWG1 LIST as defined herein. In some embodiments, at least one R^E is or comprises an electron-withdrawing group from the EWG2 LIST as defined herein. In some embodiments, at least one R^E is or comprises an electron-withdrawing group from the EWG3 LIST as defined herein. In some embodiments, at least one R^E is or comprises an electron-withdrawing group from the EWG4 LIST as defined herein. In some embodiments, at least one R^E is or comprises an electron-withdrawing group from the Pi-EWG LIST as defined herein.

[0206] In some embodiments, at least one R or R' is or comprises an electron-withdrawing group from the EWG1 LIST as defined herein. In some embodiments, at least one R or R' is or comprises an electron-withdrawing group from the EWG2 LIST as defined herein. In some embodiments, at least one R or R' is or comprises an electron-withdrawing group from the EWG3 LIST as defined herein. In some embodiments, at least one R or R' is or comprises an electron-withdrawing group from the EWG4 LIST as defined herein. In some embodiments, at least one R or R' is or comprises an electron-withdrawing group from the Pi-EWG LIST as defined herein.

[0207] In some embodiments, each R^A is hydrogen.

[0208] In some embodiments, at least one R^A is not hydrogen. In some embodiments, at least one R^A comprises at least one C atom. In some embodiments, at least one R^A comprises a substituent selected from the group consisting of alkyl, cycloalkyl, aryl, heteroaryl, and combinations thereof.

[0209] In some embodiments, each R^B is hydrogen.

[0210] In some embodiments, at least one R^B is not hydrogen. In some embodiments, at least one R^B comprises at least one C atom. In some embodiments, at least one R^B comprises a substituent selected from the group consisting of alkyl, cycloalkyl, aryl, heteroaryl, and combinations thereof. In some embodiments, at least one R^B comprises alkyl. In some embodiments, at least one R^B comprises aryl. In some embodiments, at least one R^B comprises alkyl comprising at least three C atoms.

[0211] In some embodiments, two R^B are joined or fused to form a 4- to 10-membered ring. In some such embodiments, the ring is a saturated ring, while the ring is an aromatic ring in other embodiments. In some embodiments, the ring is a 5- or 6-membered ring. In some embodiments, the ring is a 5-membered ring. In some embodiments, the ring is a 6-membered ring.

[0212] In some embodiments, each R^C is hydrogen.

[0213] In some embodiments, at least one R^C is not hydrogen. In some embodiments, at least one R^C comprises at least one C atom. In some embodiments, at least one R^C comprises a substituent selected from the group consisting of alkyl, cycloalkyl, aryl, heteroaryl, and combinations thereof. In some embodiments, at least one R^C comprises alkyl. In some embodiments, at least one R^C comprises aryl. In some embodiments, at least one R^C comprises alkyl comprising at least three C atoms.

[0214] In some embodiments, two R^C are joined or fused to form a 4- to 10-membered ring. In some such embodiments, the ring is a saturated ring, while the ring is an aromatic ring in other embodiments. In some embodiments, the ring is a 5- or 6-membered ring. In some embodiments, the ring is a 5-membered ring. In some embodiments, the ring is a 6-membered ring.

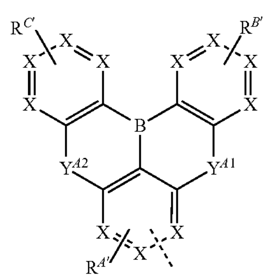
[0215] In some embodiments, each R^D is hydrogen.

[0216] In some embodiments, at least one R^D is not hydrogen. In some embodiments, at least one R^D comprises at least one C atom. In some embodiments, at least one R^D comprises a substituent selected from the group consisting of alkyl, cycloalkyl, aryl, heteroaryl, and combinations thereof. In some embodiments, at least one R^D comprises alkyl. In some embodiments, at least one R^D comprises aryl. In some embodiments, at least one R^D comprises alkyl comprising at least three C atoms.

[0217] In some embodiments, each R^E is hydrogen.

[0218] In some embodiments, at least one R^E is not hydrogen. In some embodiments, at least one R^E comprises at least one C atom. In some embodiments, at least one R^E comprises a substituent selected from the group consisting of alkyl, cycloalkyl, aryl, heteroaryl, and combinations thereof. In some embodiments, at least one R^E comprises alkyl. In some embodiments, at least one R^E comprises aryl. In some embodiments, at least one R^E comprises alkyl comprising at least three C atoms.

[0219] In some embodiments, the ligand L_A is selected from the group consisting of the structures of the following LIST 1: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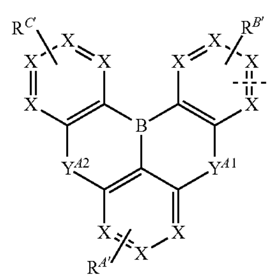
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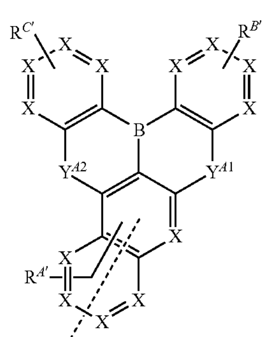
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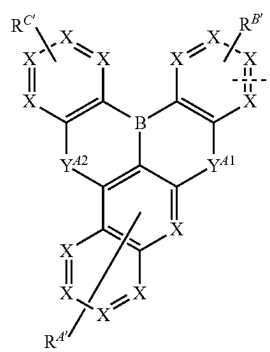
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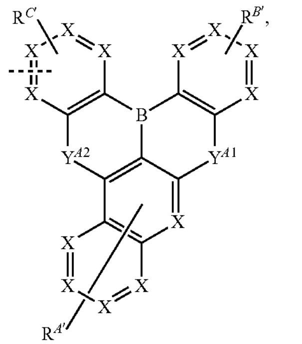

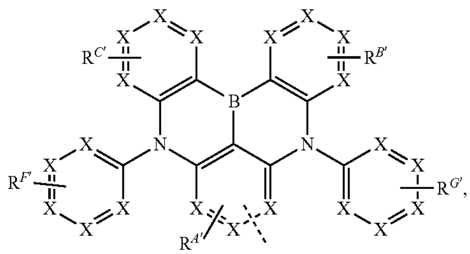

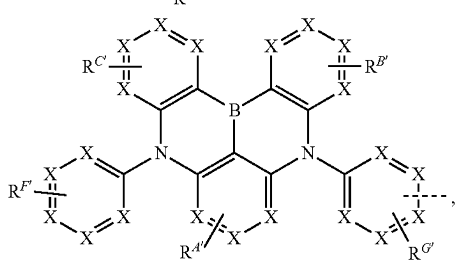

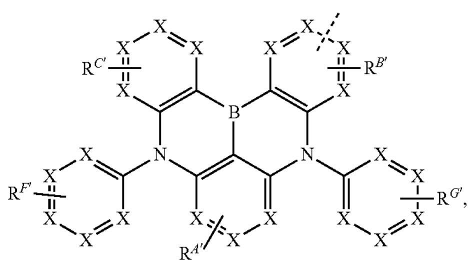

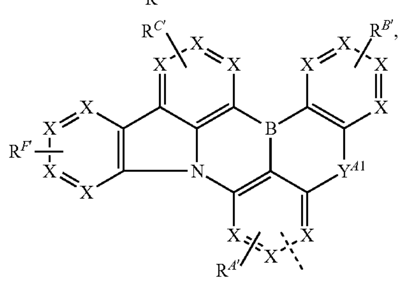

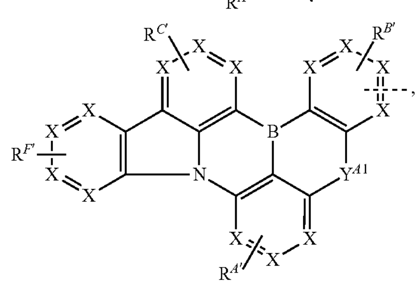

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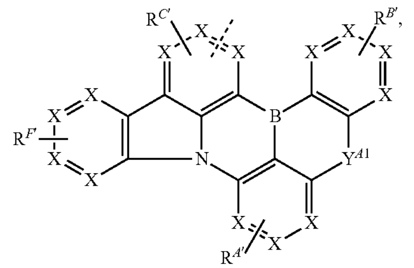

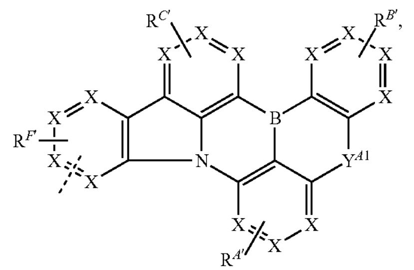

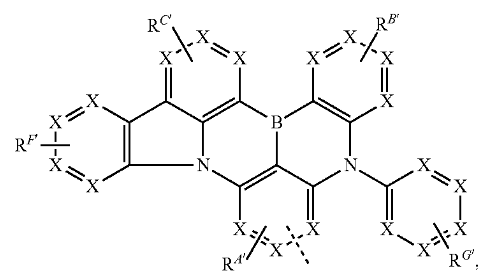

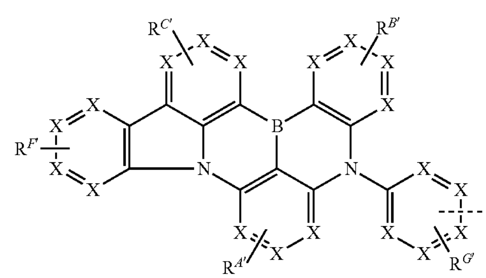

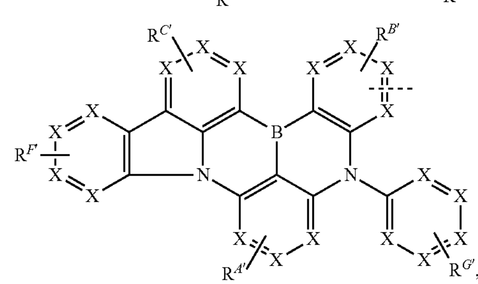

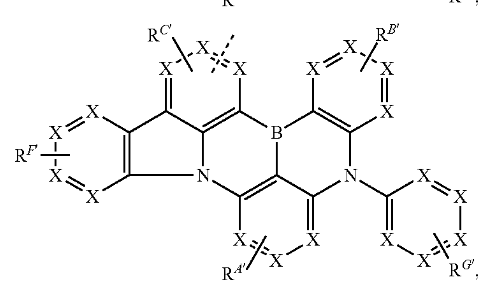

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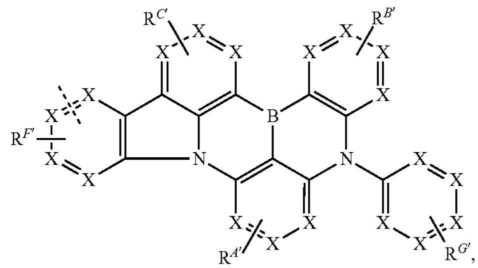

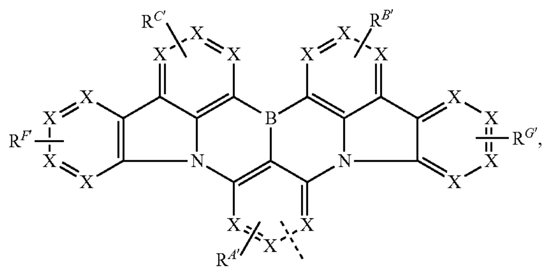

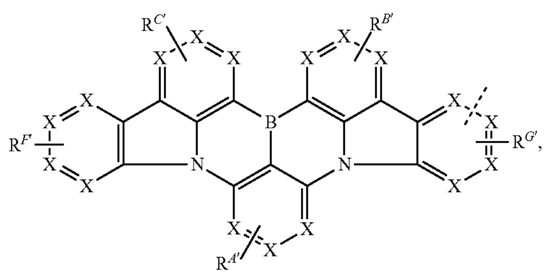

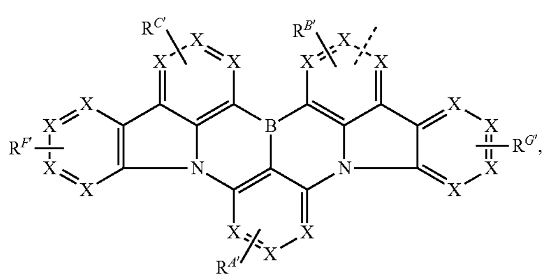

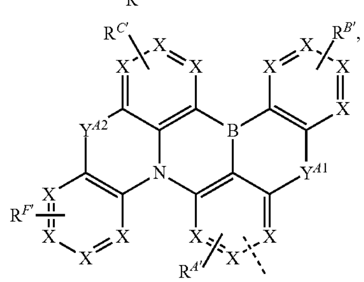

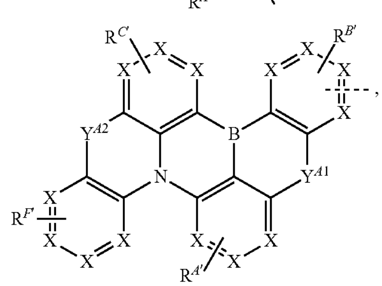

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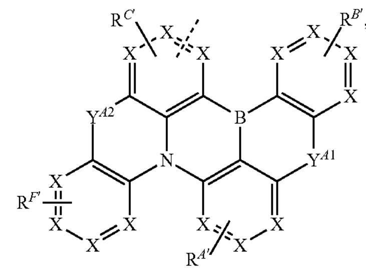

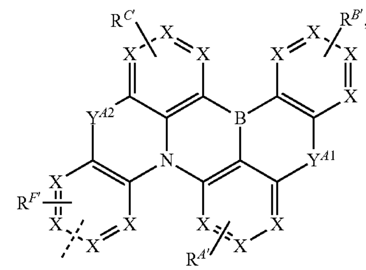

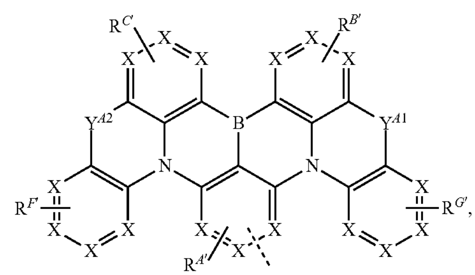

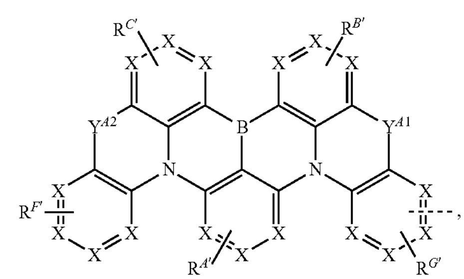

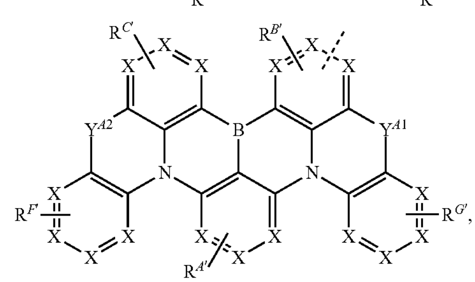

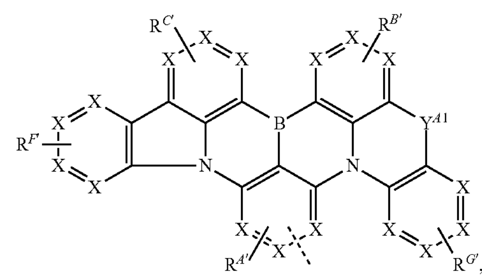

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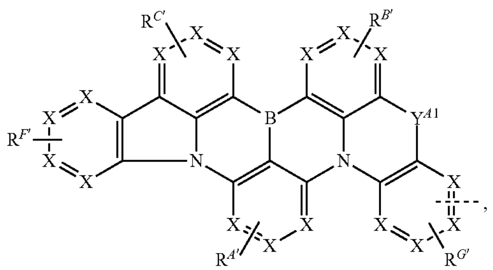

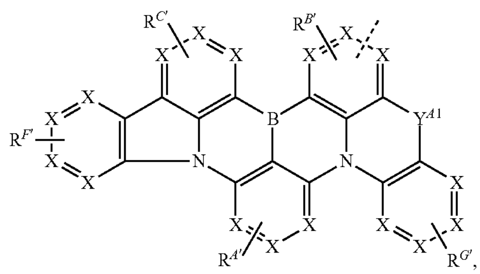

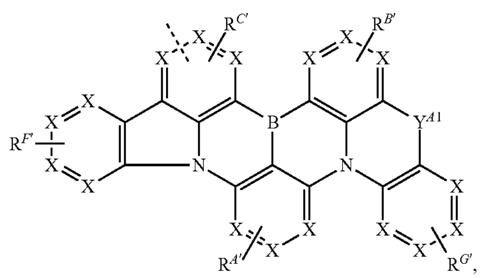

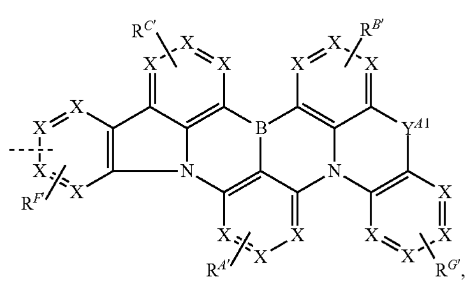

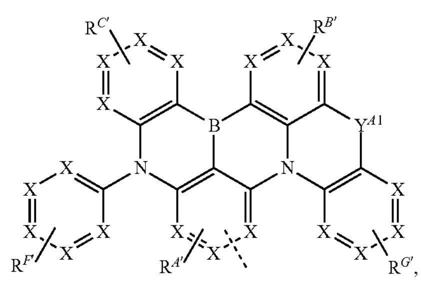

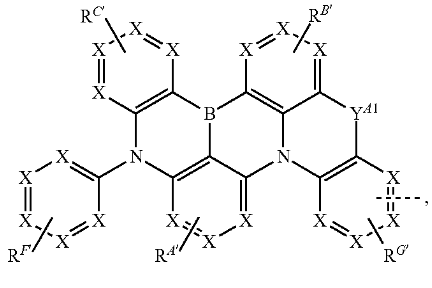

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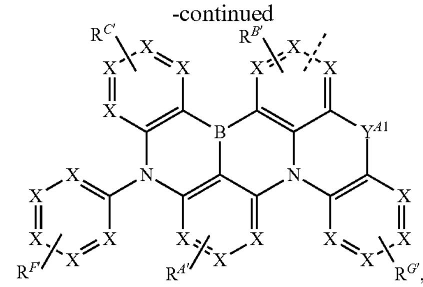

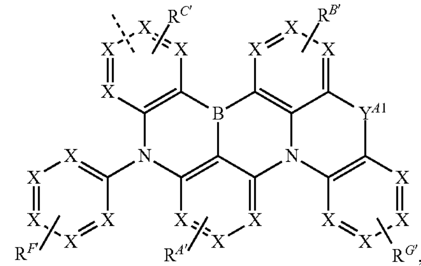

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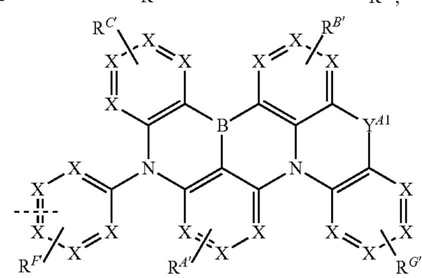


wherein

[0220] for each occurrence, X is independently C or N;

[0221] each of Y^{A1} and Y^{A2} is independently selected from the group consisting of a direct bond, BR, BRR', NR, PR, P(O)R, O, S, Se, C═O, C═S, C═Se, C═NR, C═CRR', S═O, SO_2, CR, CRR', SiRR', and GeRR';

[0222] - - - - represents chelation to the metal;

[0223] each of $R^{A'}$, $R^{B'}$, $R^{C'}$, $R^{F'}$, and $R^{G'}$ independently represents mono to the maximum allowable substitution; each R, R', $R^{A'}$, $R^{B'}$, $R^{C'}$, $R^{F'}$, and $R^{G'}$ is independently hydrogen or a substituent selected from the group consisting of the General Substituents defined herein; and any two substituents can be joined or fused to form a ring.

[0224] It should be understood that for each structure in LIST 1, X can be the same or different in each individual ring. In some embodiments, each X is C. In some embodiments, at least one X is N. In some embodiments, exactly one X is N.

[0225] Also, with respect to LIST 1, even though the ___ was drawn to show bonding from a particular ring to the metal for each structure in LIST 1, it should be understood that the bonding to the metal can be from any ring of the structure so long as it's the one and only bond to the metal for the particular structure of LIST 1.

[0226] In some embodiments, Formula I is selected from the group consisting of the structures of LIST 1. In such embodiments, the ___ was a direct bond or through an organic linker to a monocyclic ring or a polycyclic fused ring system that is coordinated to M. It should also be understood that the direct bond can be from any ring of the structure to the monocyclic ring or the polycyclic fused ring system or through the organic linker.

[0227] In some embodiments where ligand L_A is selected from LIST 1, at least one of R, R', $R^{A'}$, $R^{B'}$, $R^{C'}$, $R^{F'}$, or $R^{G'}$ is partially or fully deuterated. In some embodiments, at least one $R^{A'}$ is partially or fully deuterated. In some embodiments, at least one $R^{B'}$ is partially or fully deuterated. In some embodiments, at least one $R^{C'}$ is partially or fully deuterated. In some embodiments, at least one $R^{F'}$ is partially or fully deuterated. In some embodiments, at least one $R^{G'}$ is partially or fully deuterated. In some embodiments, at least one R or R' is partially or fully deuterated.

[0228] In some embodiments where ligand L_A is selected from LIST 1 or 3 (below), at least one $R^{A'}$ is or comprises an electron-withdrawing group from the EWG1 LIST as defined herein. In some embodiments, at least one $R^{A'}$ is or comprises an electron-withdrawing group from the EWG2 LIST as defined herein. In some embodiments, at least one $R^{A'}$ is or comprises an electron-withdrawing group from the EWG3 LIST as defined herein. In some embodiments, at least one $R^{A'}$ is or comprises an electron-withdrawing group from the EWG4 LIST as defined herein. In some embodiments, at least one $R^{A'}$ is or comprises an electron-withdrawing group from the Pi-EWG LIST as defined herein.

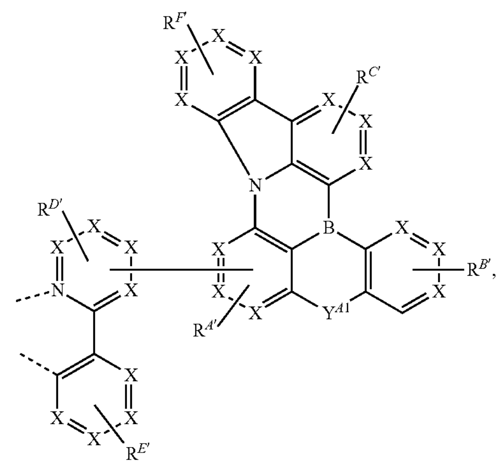
[0229] In some embodiments where ligand L_A is selected from LIST 1 or 3, at least one $R^{B'}$ is or comprises an electron-withdrawing group from the EWG1 LIST as defined herein. In some embodiments, at least one $R^{B'}$ is or comprises an electron-withdrawing group from the EWG2 LIST as defined herein. In some embodiments, at least one $R^{B'}$ is or comprises an electron-withdrawing group from the EWG3 LIST as defined herein. In some embodiments, at least one $R^{B'}$ is or comprises an electron-withdrawing group from the EWG4 LIST as defined herein. In some embodiments, at least one $R^{B'}$ is or comprises an electron-withdrawing group from the Pi-EWG LIST as defined herein.

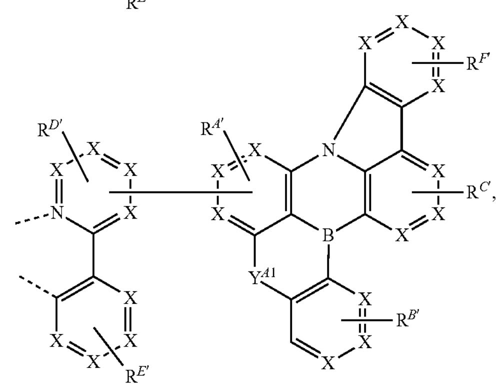
[0230] In some embodiments where ligand L_A is selected from LIST 1 or 3, at least one $R^{C'}$ is or comprises an electron-withdrawing group from the EWG1 LIST as defined herein. In some embodiments, at least one $R^{C'}$ is or comprises an electron-withdrawing group from the EWG2 LIST as defined herein. In some embodiments, at least one $R^{C'}$ is or comprises an electron-withdrawing group from the EWG3 LIST as defined herein. In some embodiments, at least one $R^{C'}$ is or comprises an electron-withdrawing group from the EWG4 LIST as defined herein. In some embodiments, at least one $R^{C'}$ is or comprises an electron-withdrawing group from the Pi-EWG LIST as defined herein.

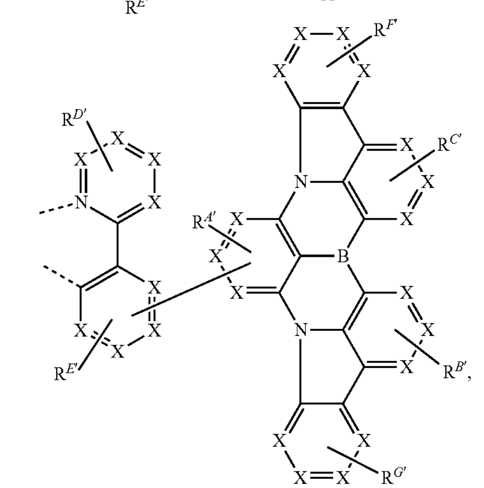
[0231] In some embodiments where ligand L_A is selected from LIST 1 or 3, at least one $R^{F'}$ is or comprises an electron-withdrawing group from the EWG1 LIST as defined herein. In some embodiments, at least one $R^{F'}$ is or comprises an electron-withdrawing group from the EWG2 LIST as defined herein. In some embodiments, at least one $R^{F'}$ is or comprises an electron-withdrawing group from the EWG3 LIST as defined herein. In some embodiments, at least one $R^{F'}$ is or comprises an electron-withdrawing group from the EWG4 LIST as defined herein. In some embodiments, at least one $R^{F'}$ is or comprises an electron-withdrawing group from the Pi-EWG LIST as defined herein.

[0232] In some embodiments where ligand L_A is selected from LIST 1 or 3, at least one $R^{G'}$ is or comprises an electron-withdrawing group from the EWG1 LIST as defined herein. In some embodiments, at least one $R^{G'}$ is or comprises an electron-withdrawing group from the EWG2 LIST as defined herein. In some embodiments, at least one $R^{G'}$ is or comprises an electron-withdrawing group from the EWG3 LIST as defined herein. In some embodiments, at least one $R^{G'}$ is or comprises an electron-withdrawing group from the EWG4 LIST as defined herein. In some embodiments, at least one $R^{G'}$ is or comprises an electron-withdrawing group from the Pi-EWG LIST as defined herein.

[0233] In some embodiments, the ligand L_A is selected from the group consisting of the structures of the following LIST 2:







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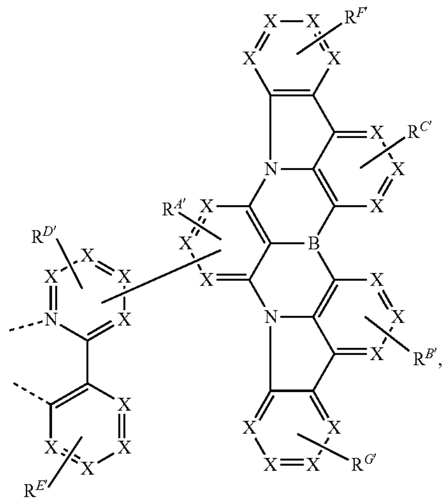

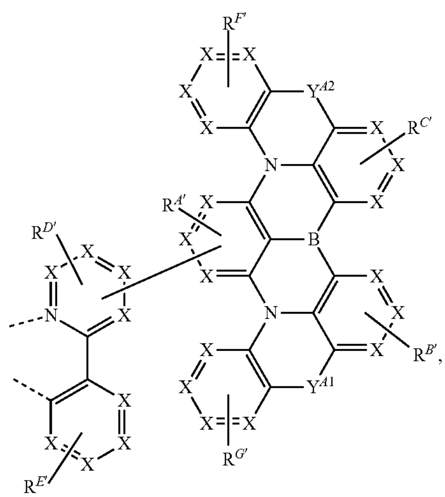

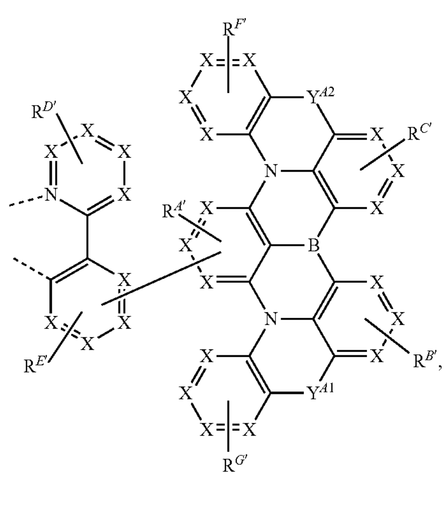

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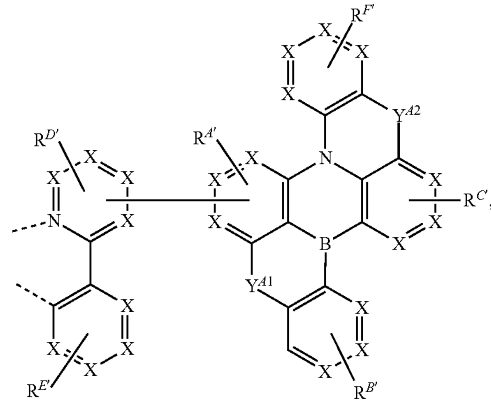

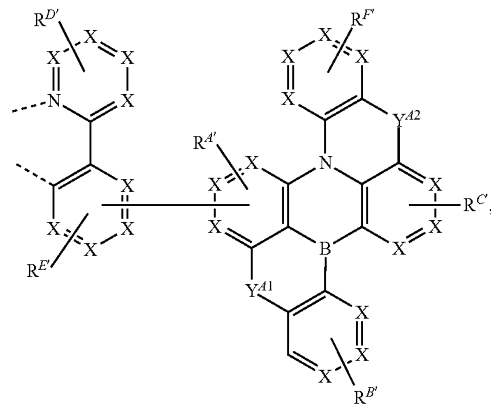

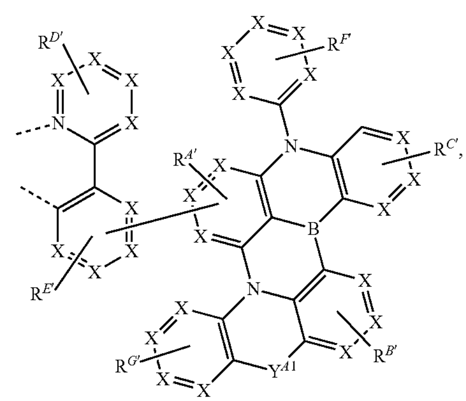

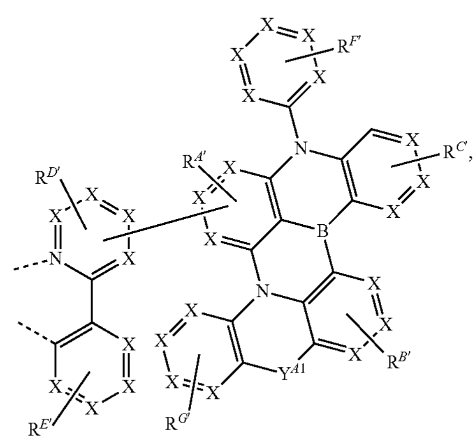

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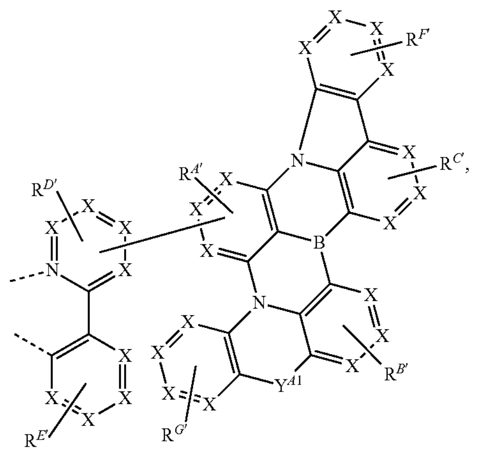

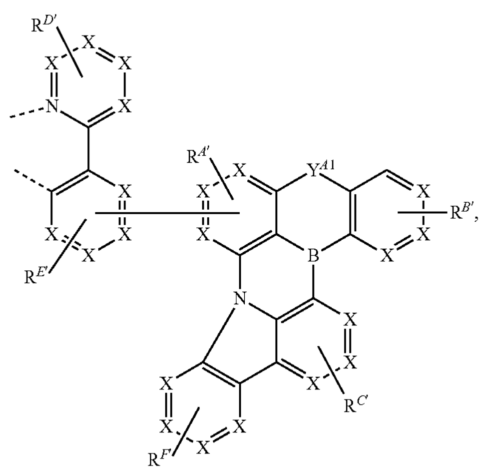

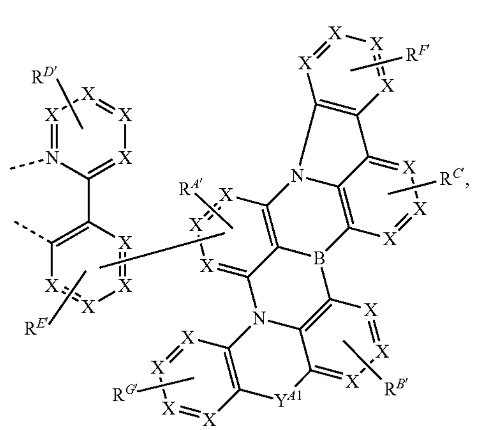

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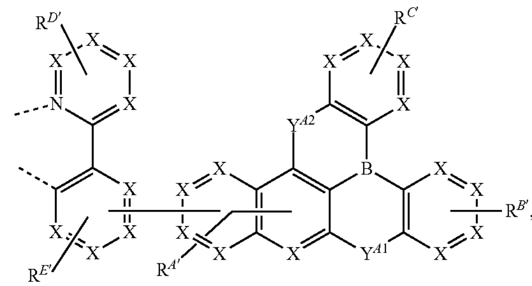

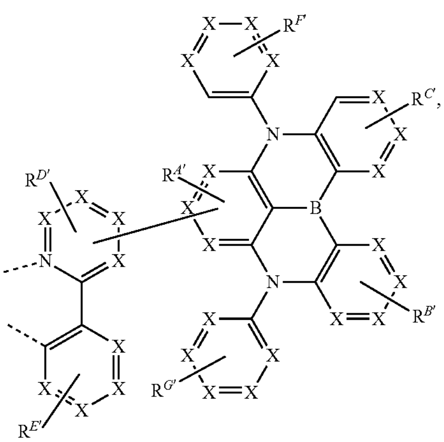

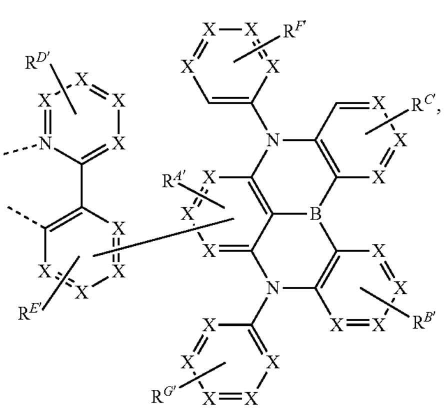

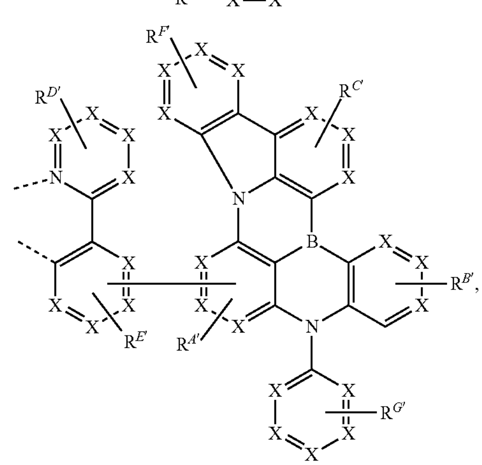

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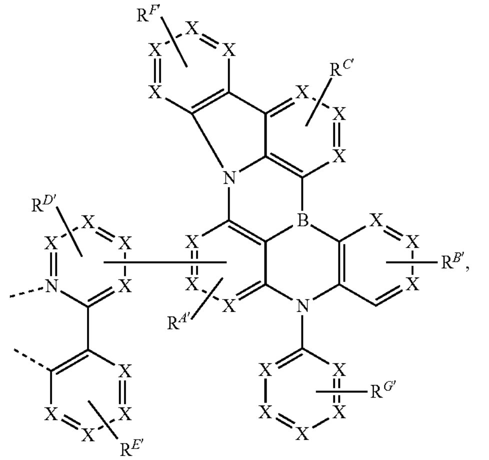

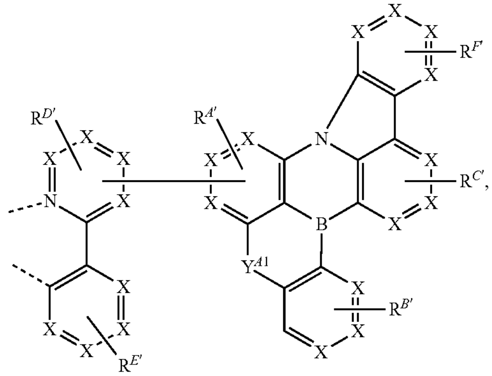

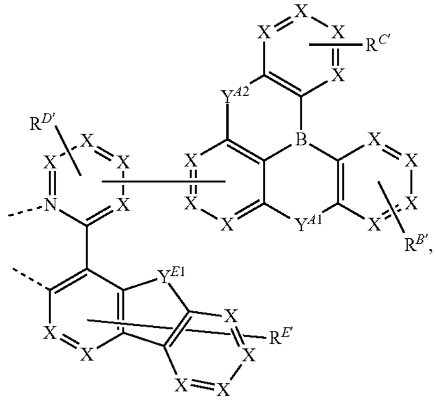

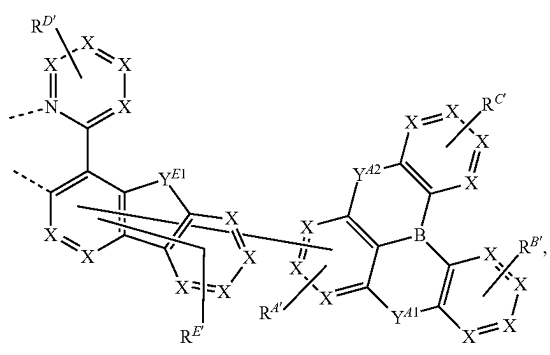

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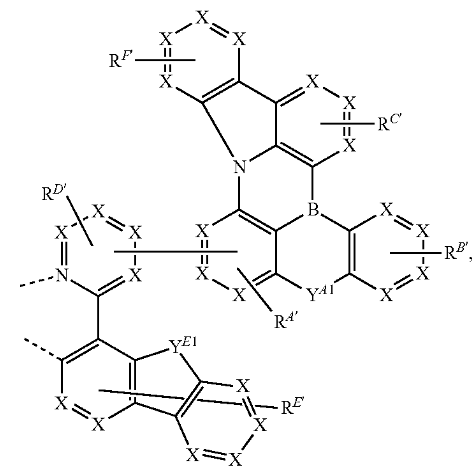

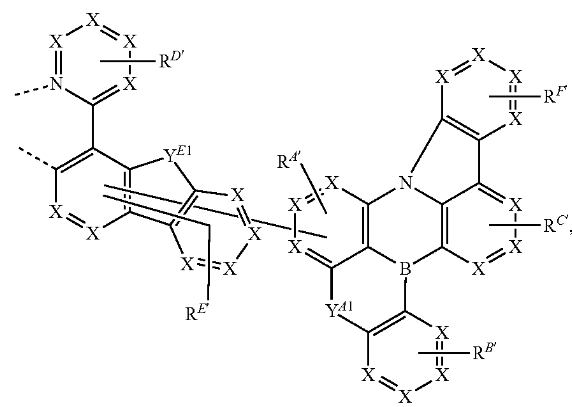

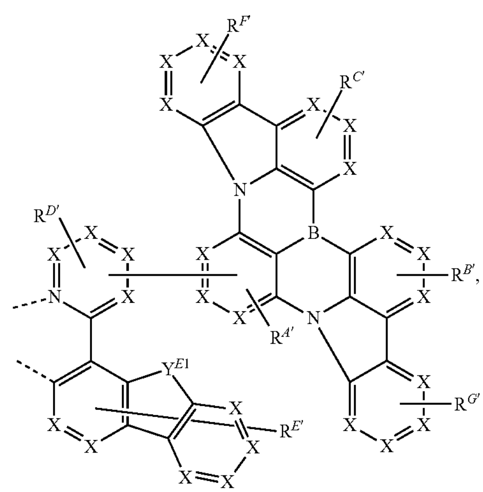

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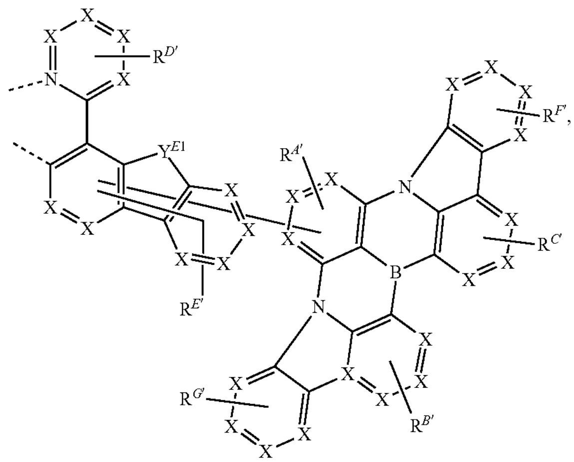

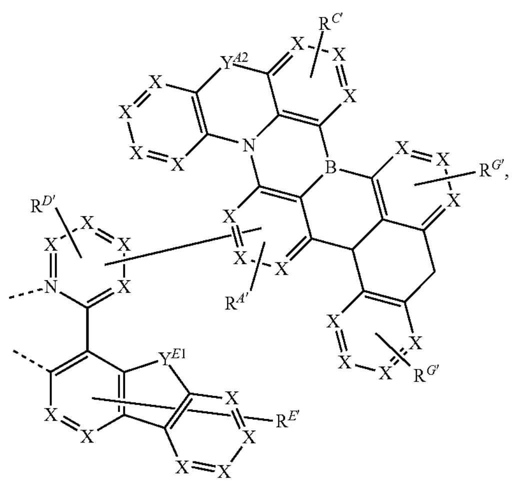

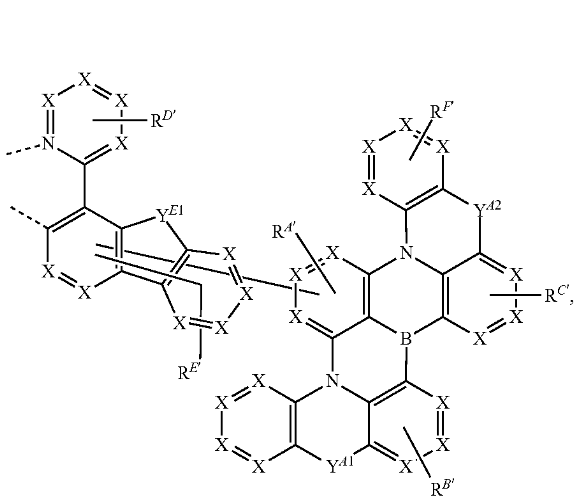

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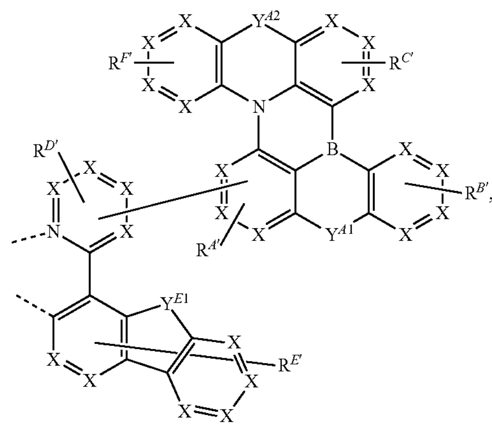

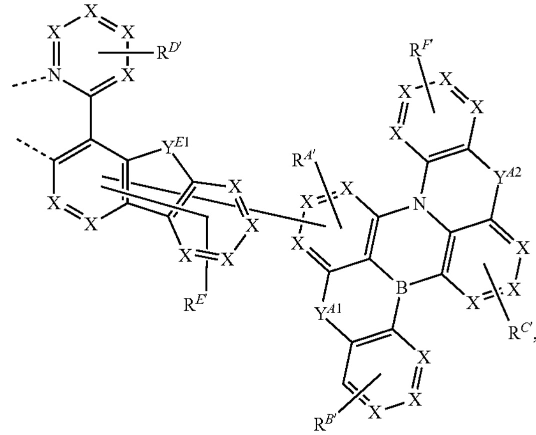

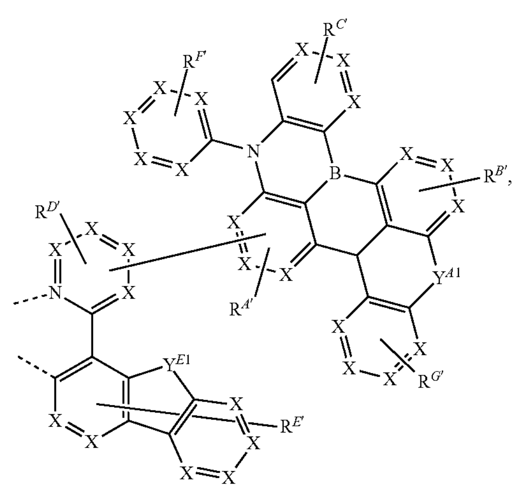

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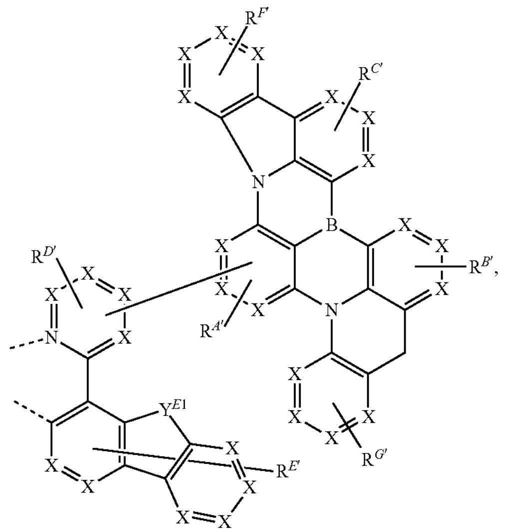

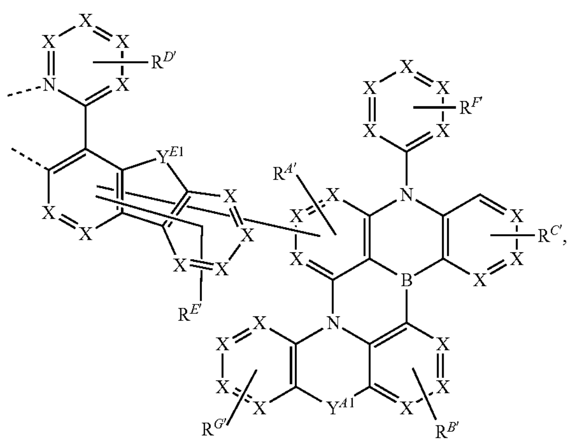

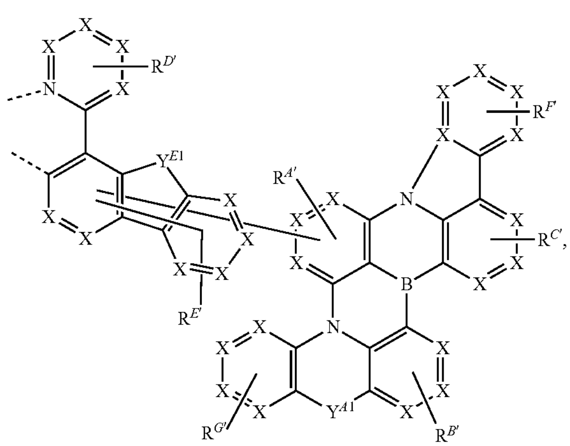

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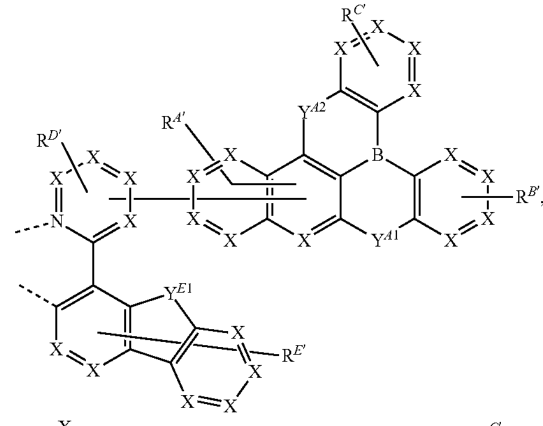

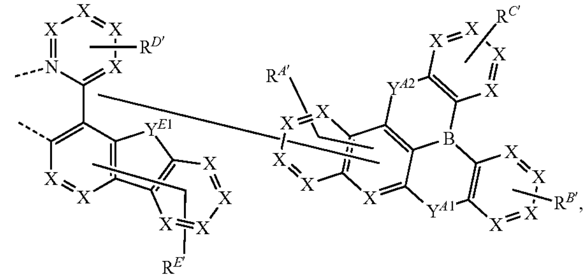

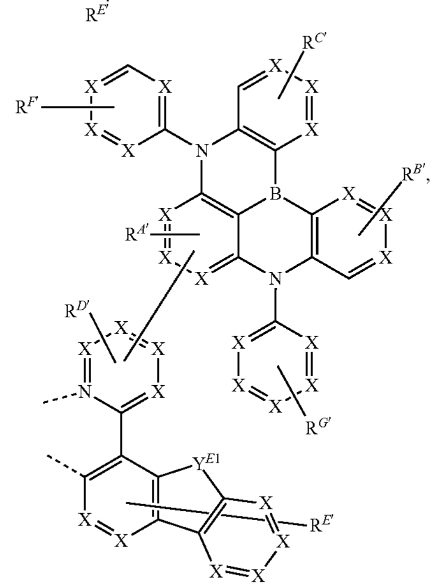

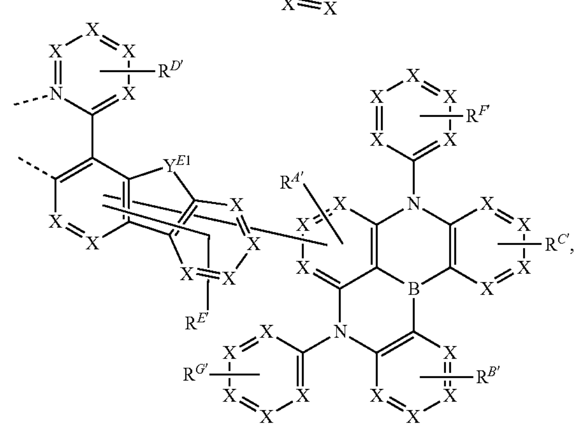

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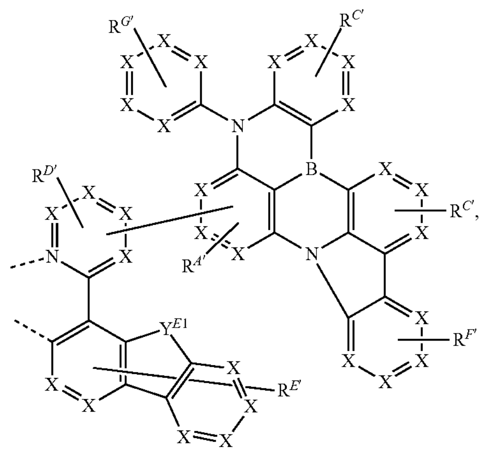

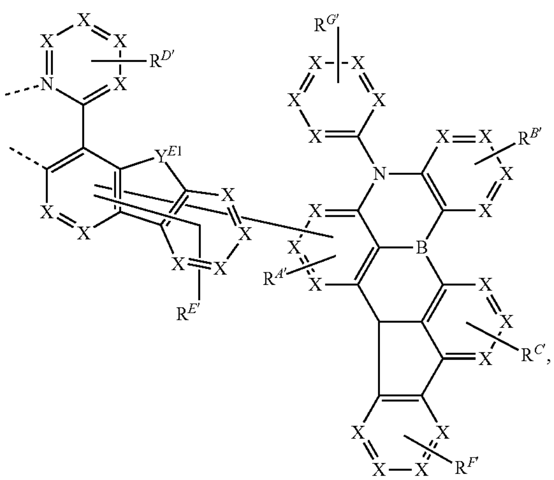

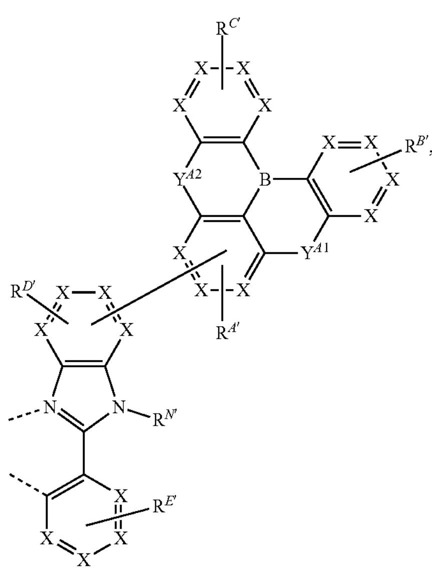

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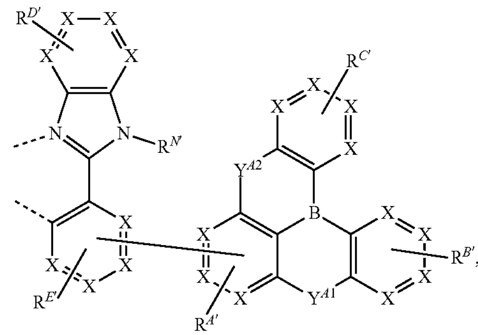

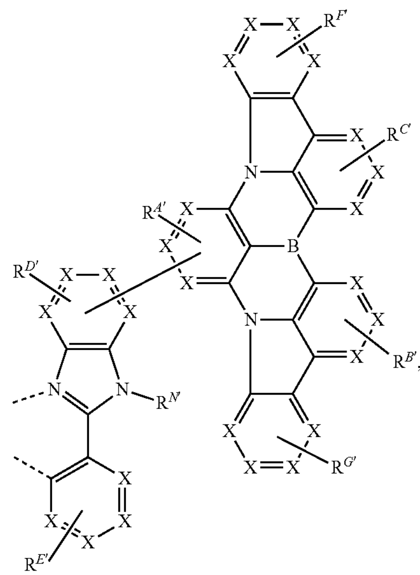

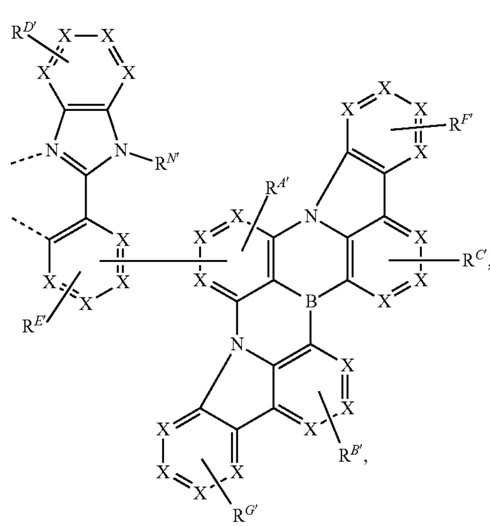

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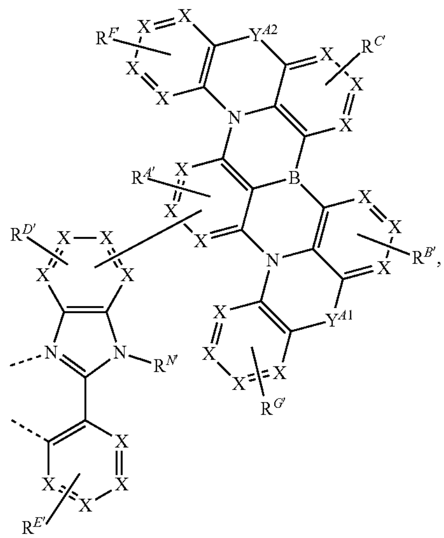

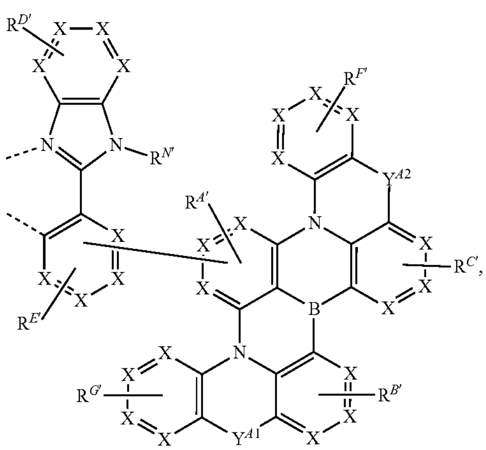

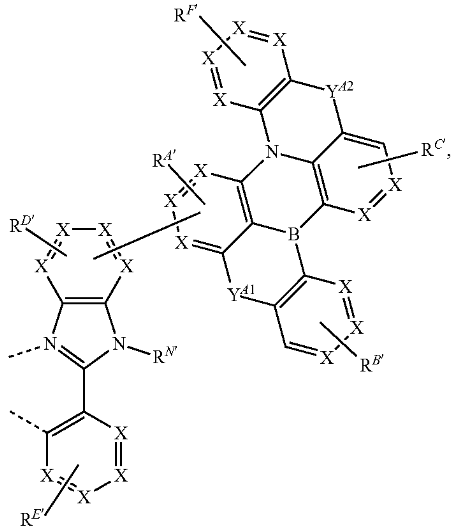

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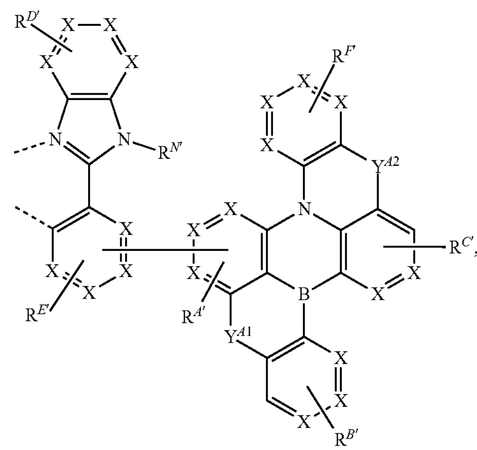

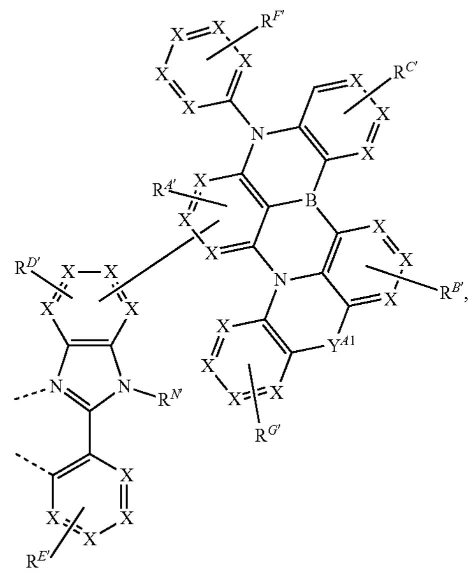

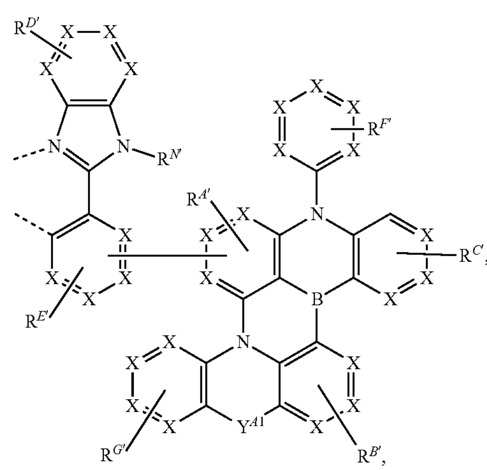

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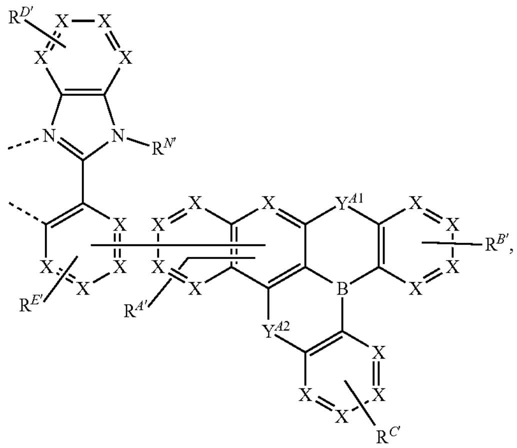

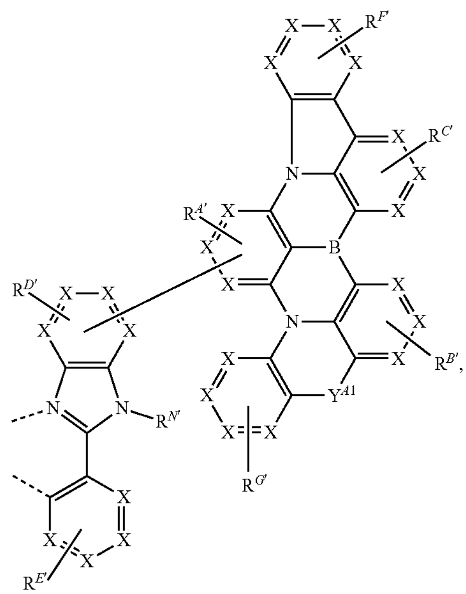

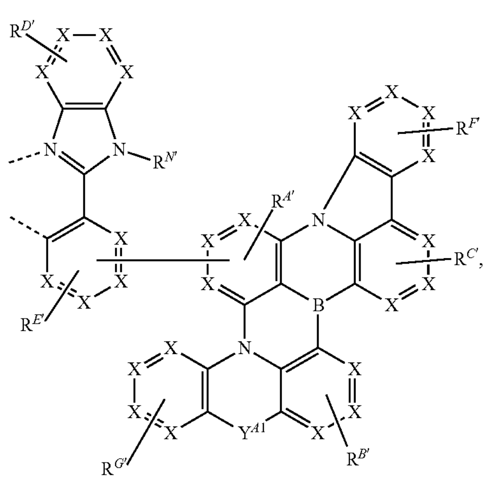

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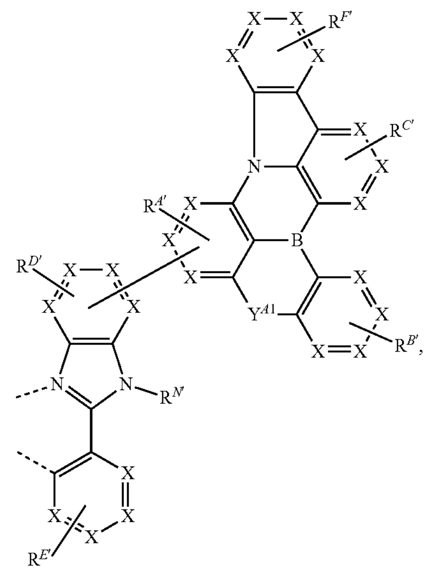

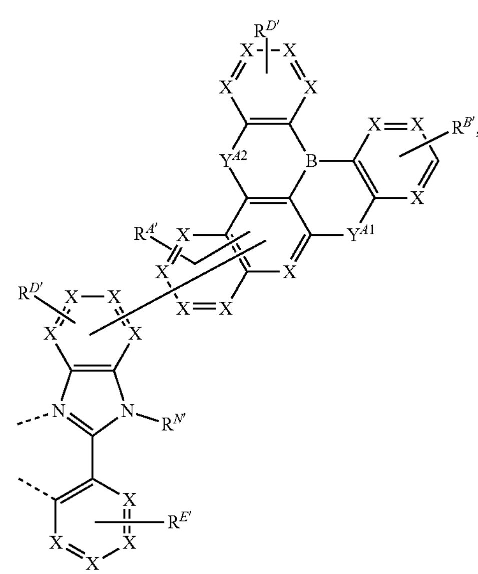

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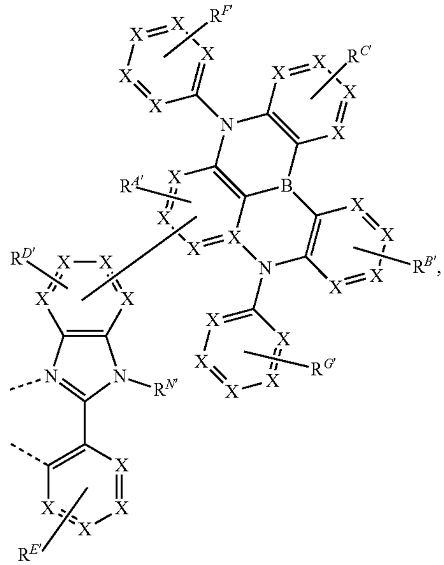

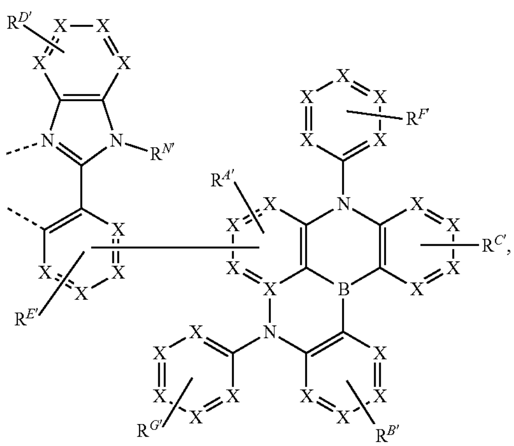

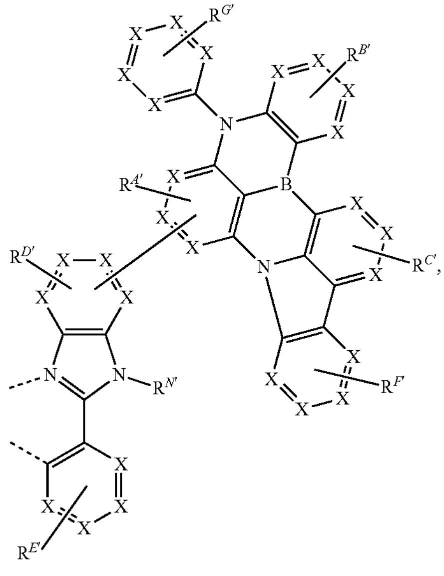

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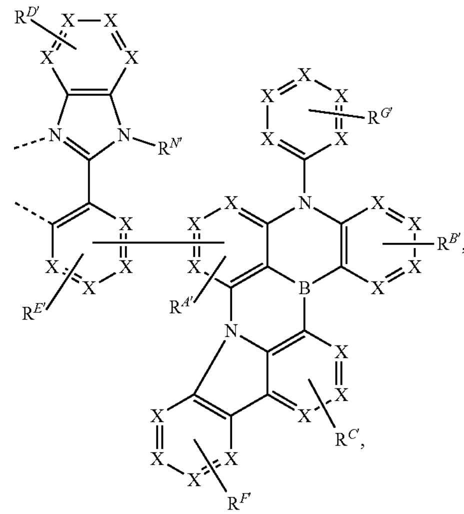

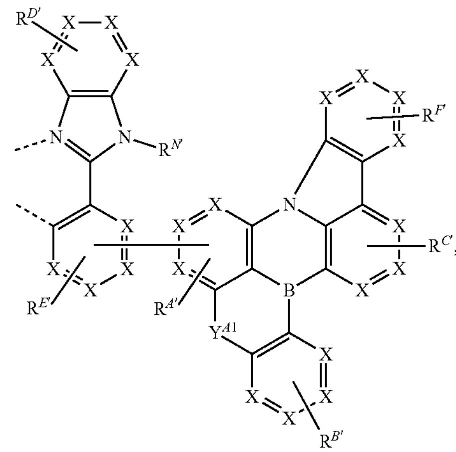

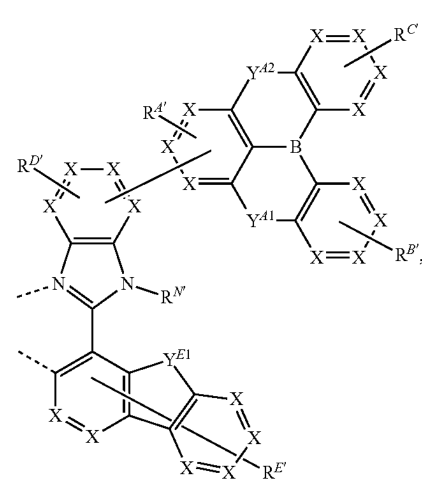

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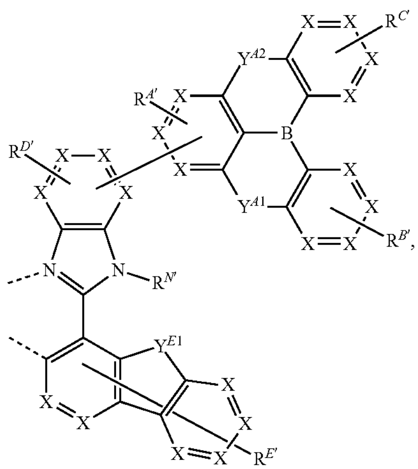

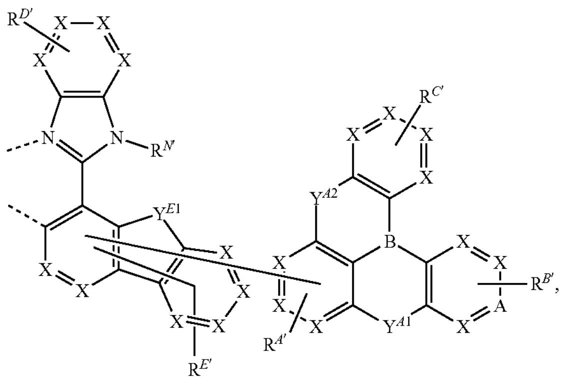

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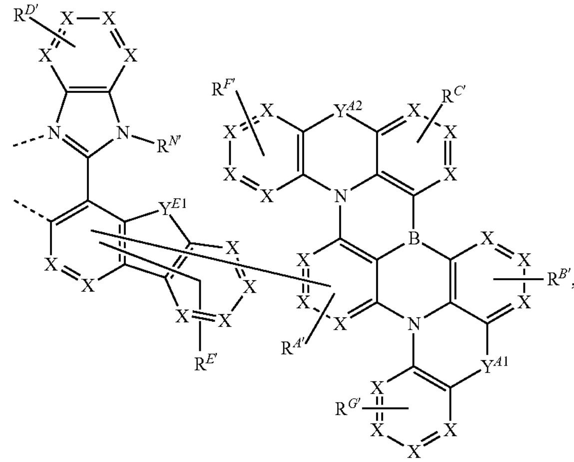

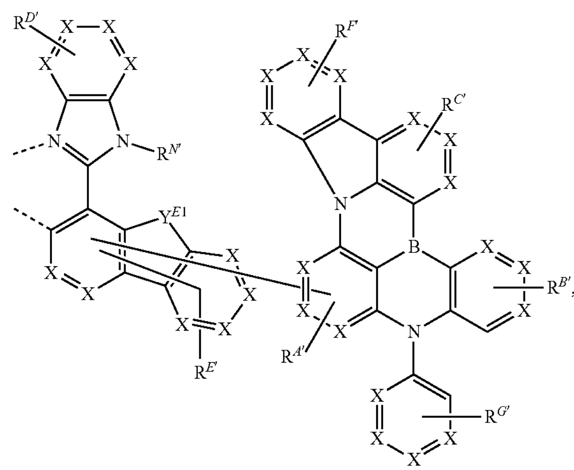

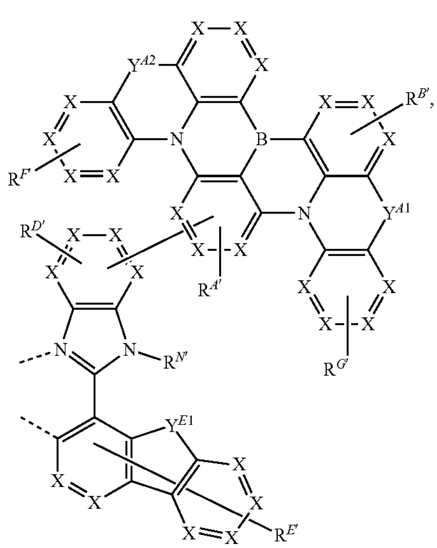

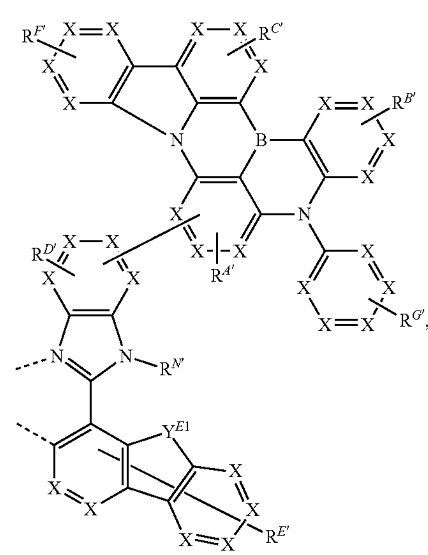

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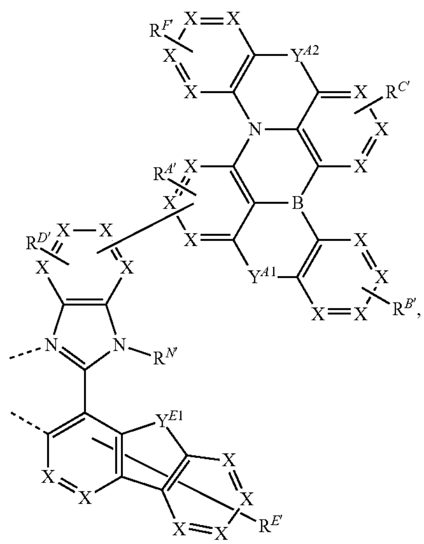

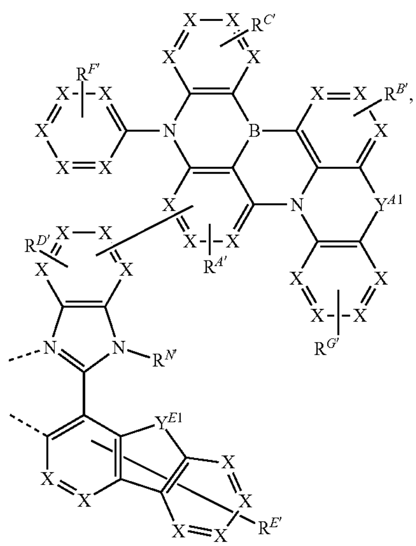

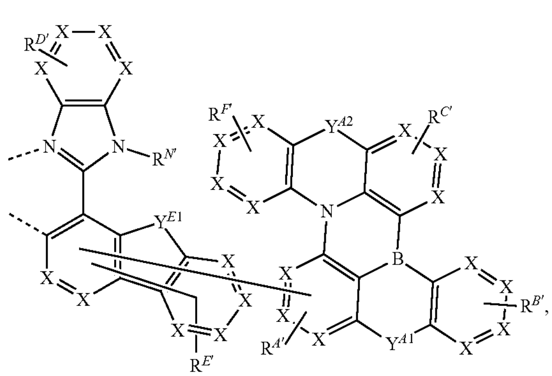

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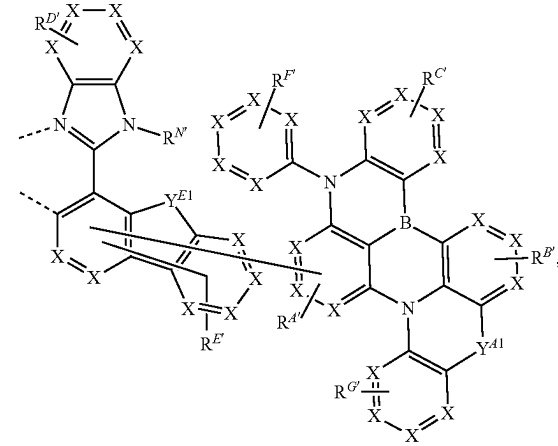

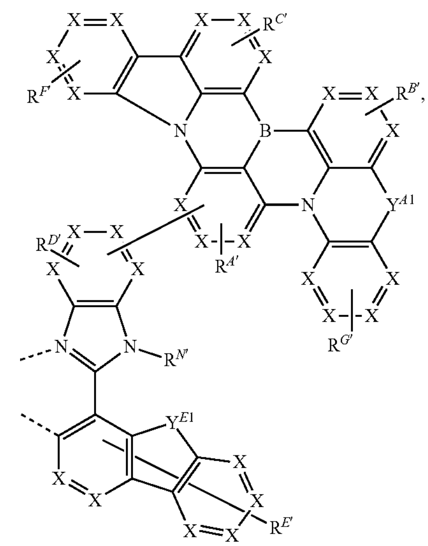

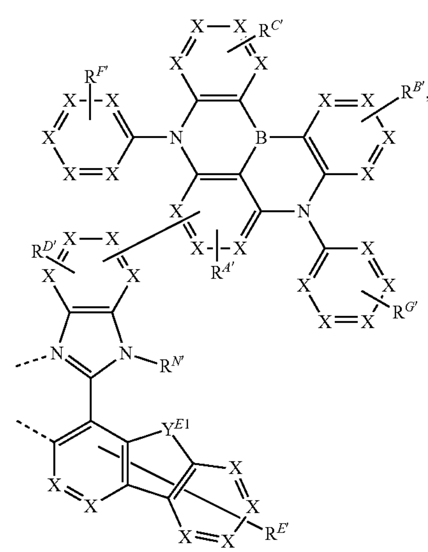

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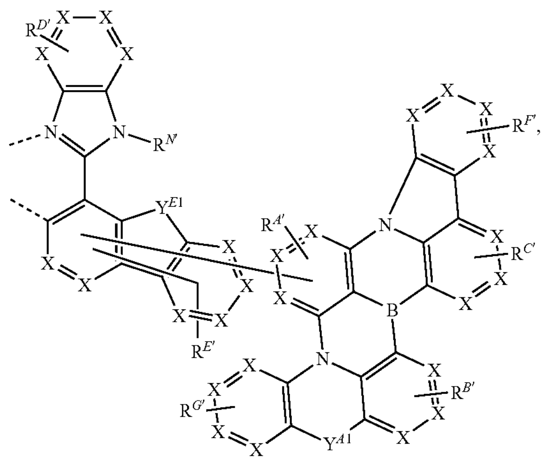

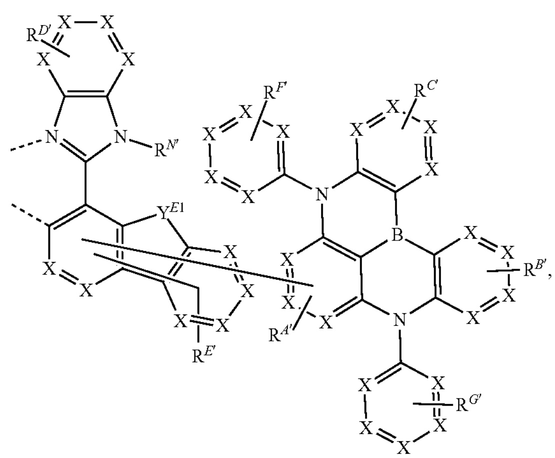

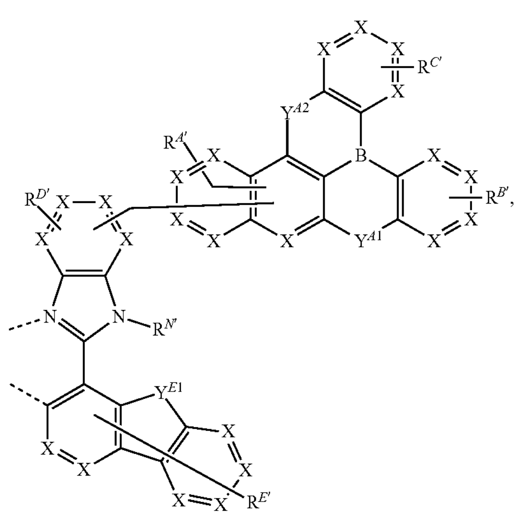

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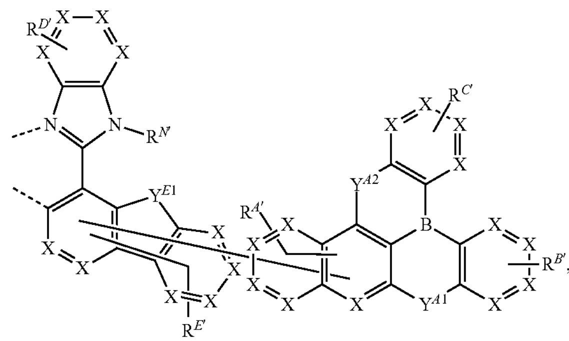

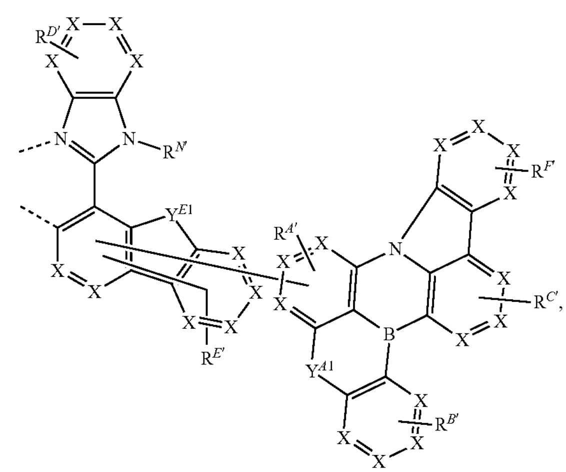

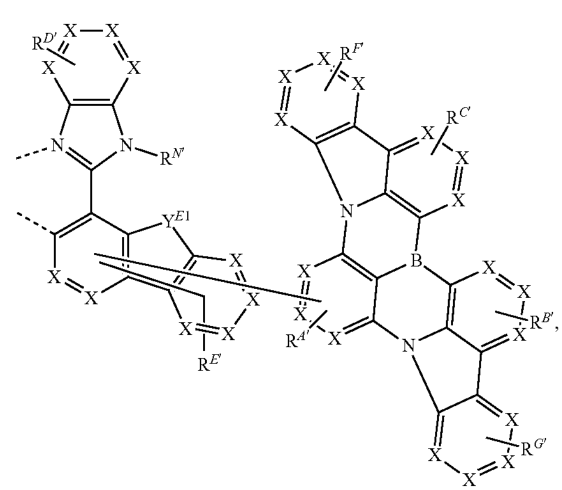

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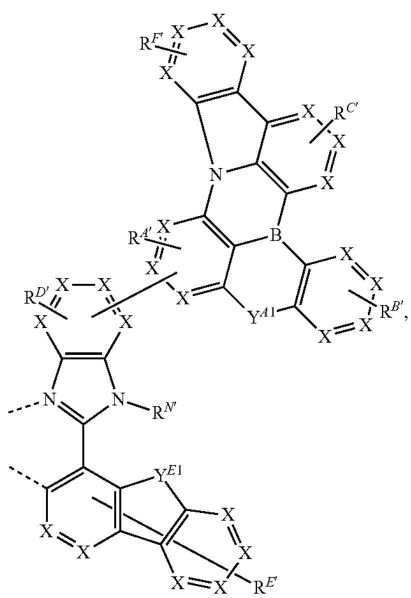

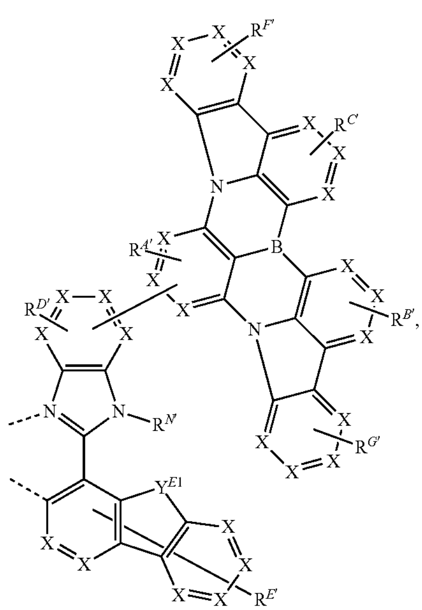

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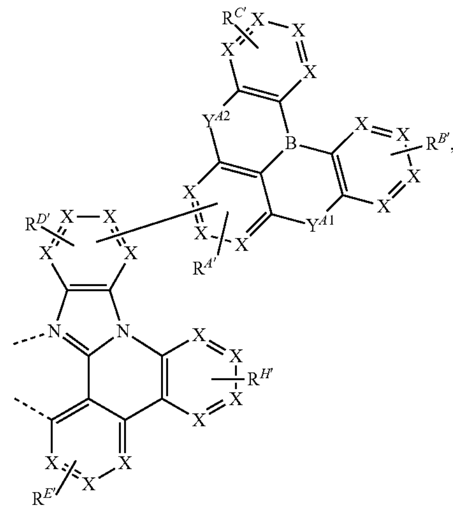

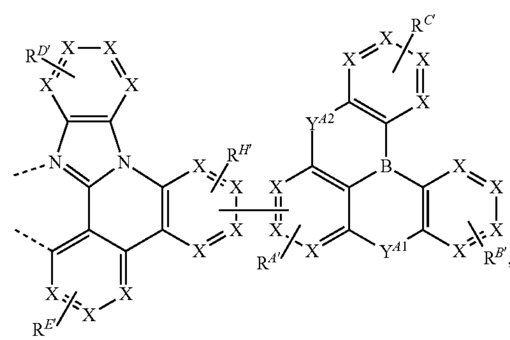

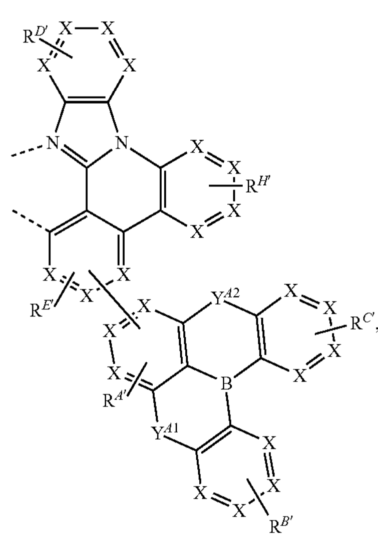

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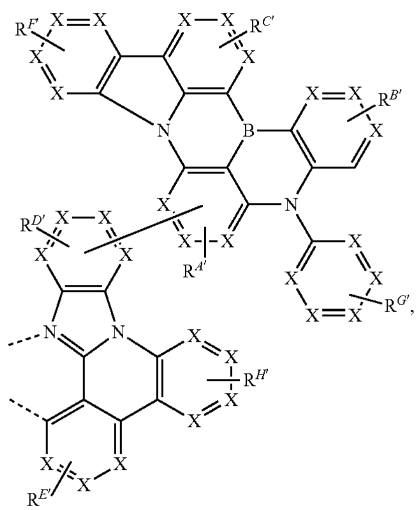

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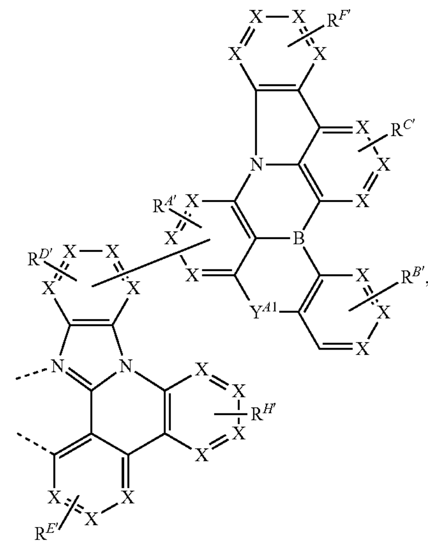

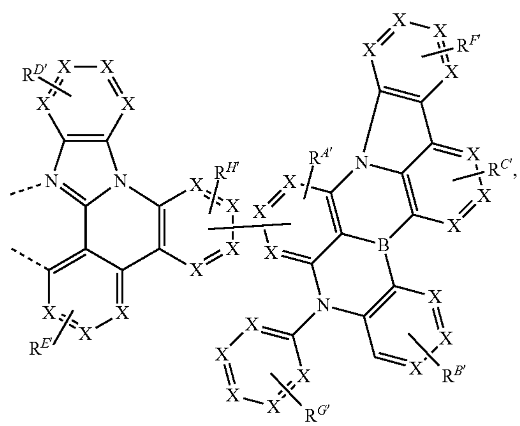

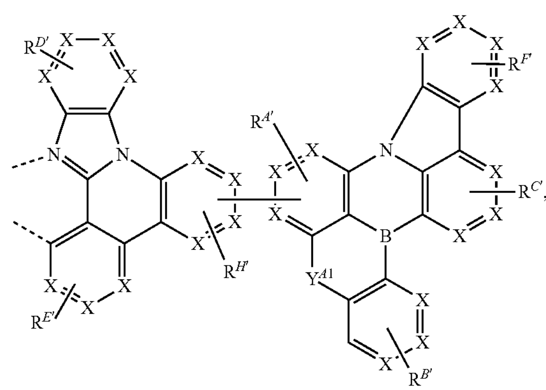

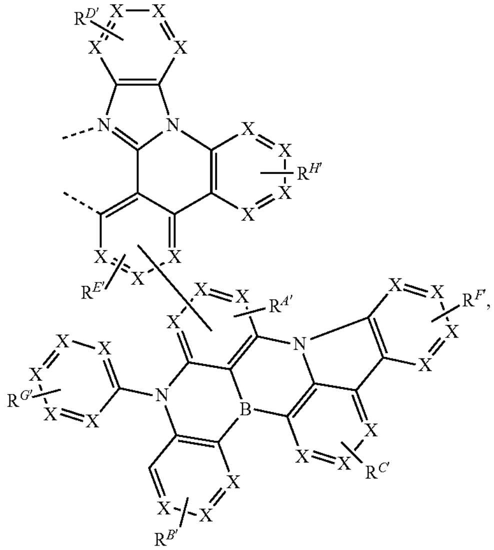

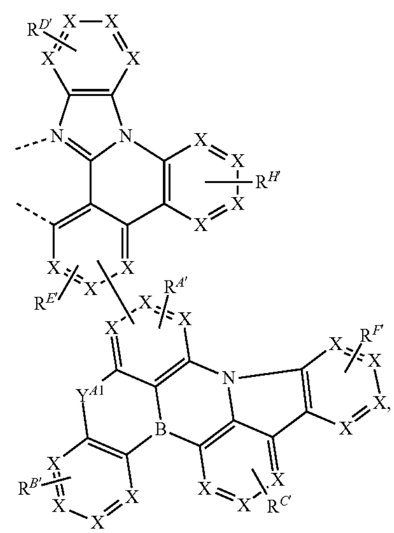

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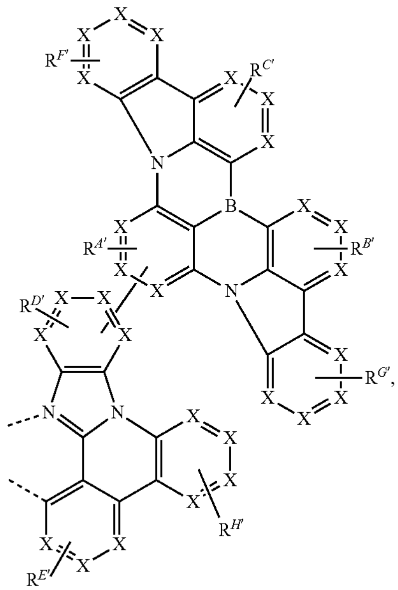

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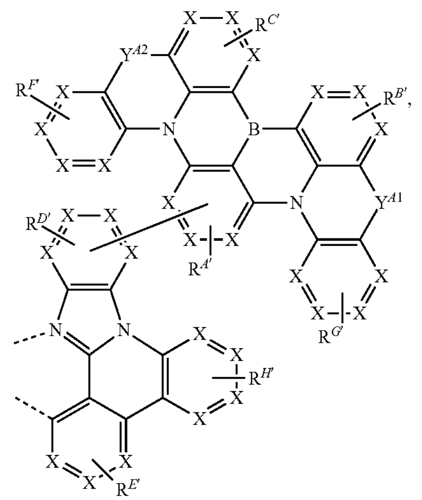

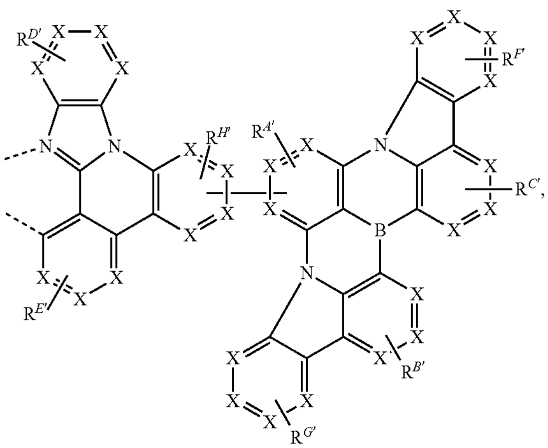

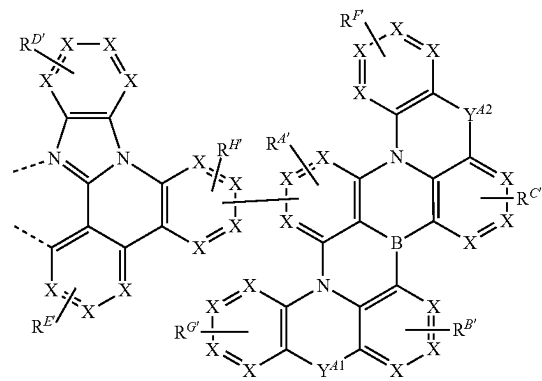

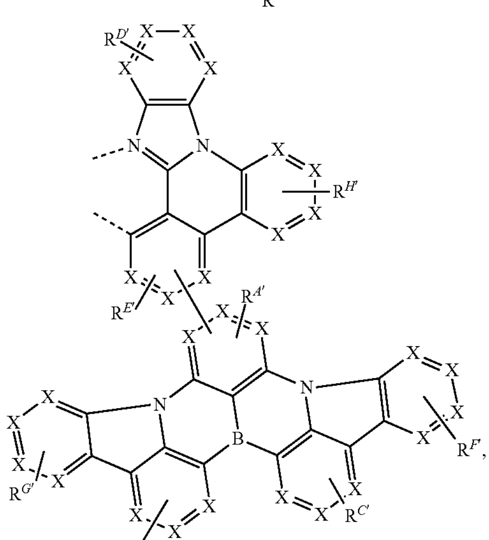

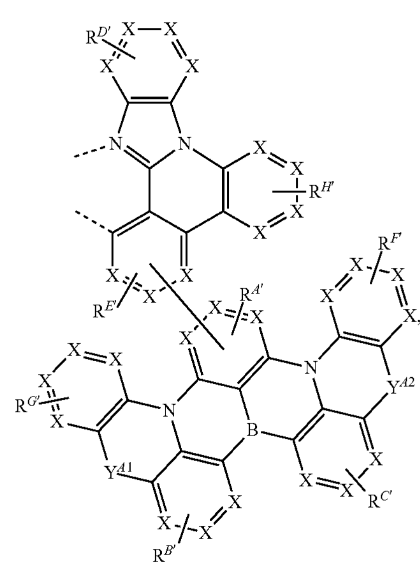

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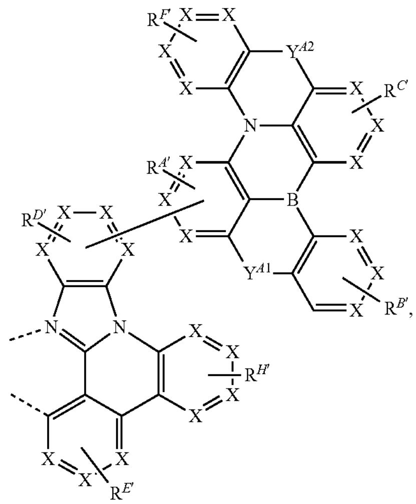

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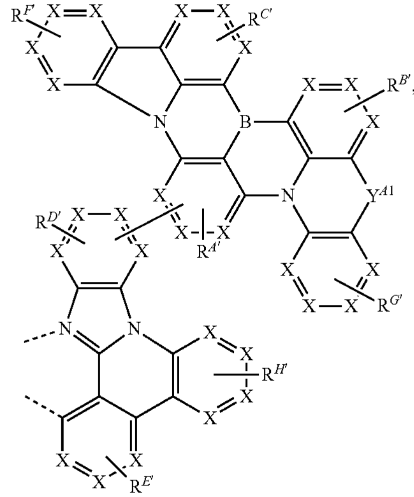

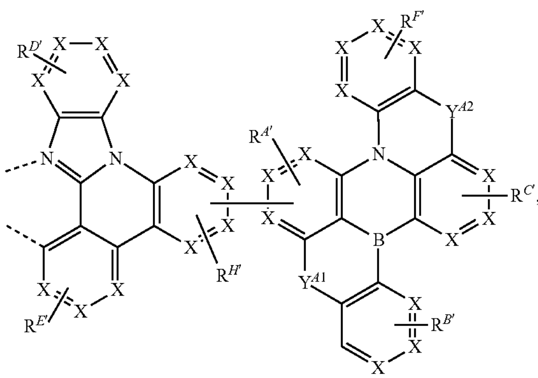

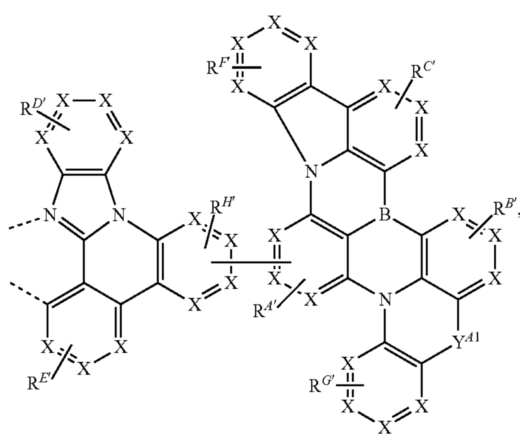

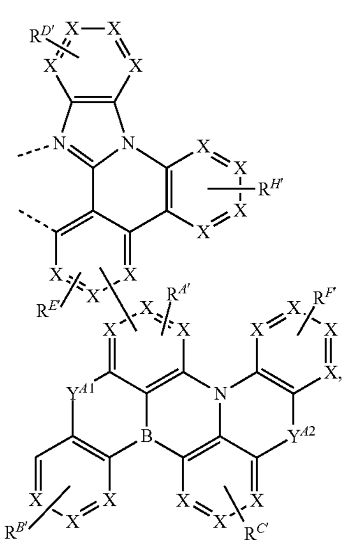

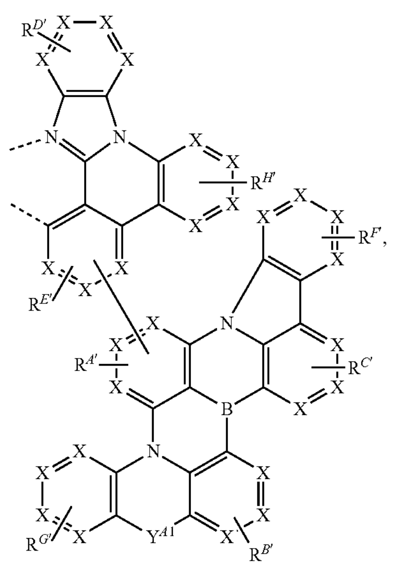

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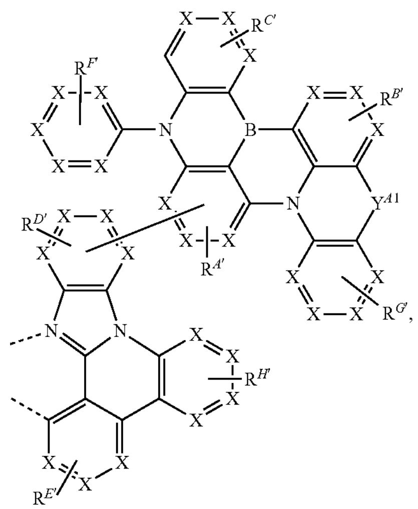

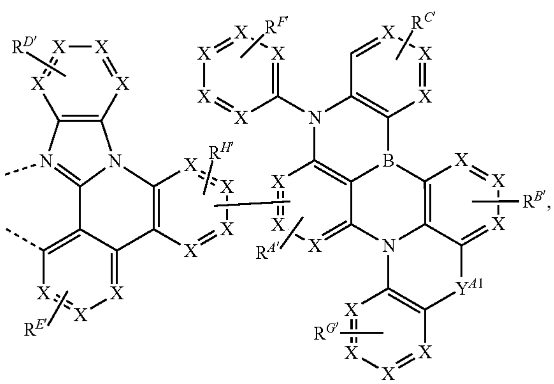

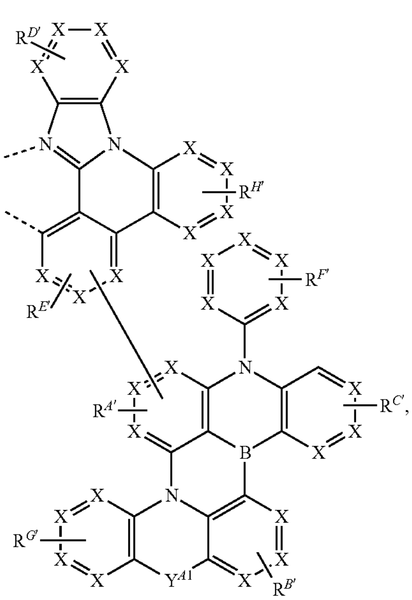

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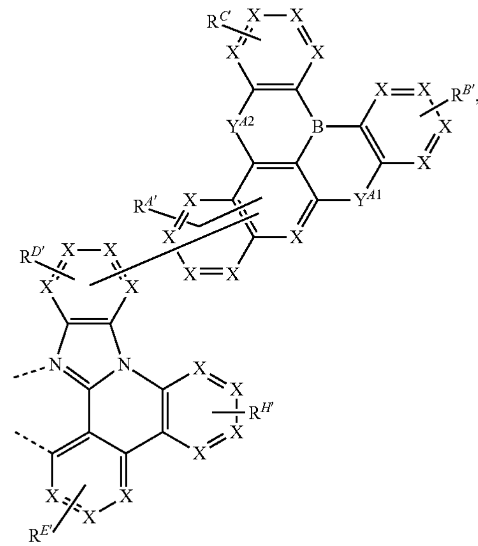

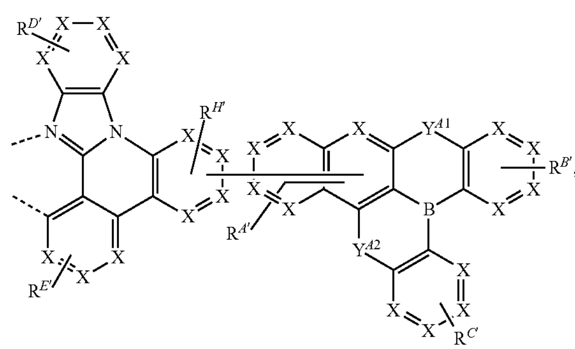

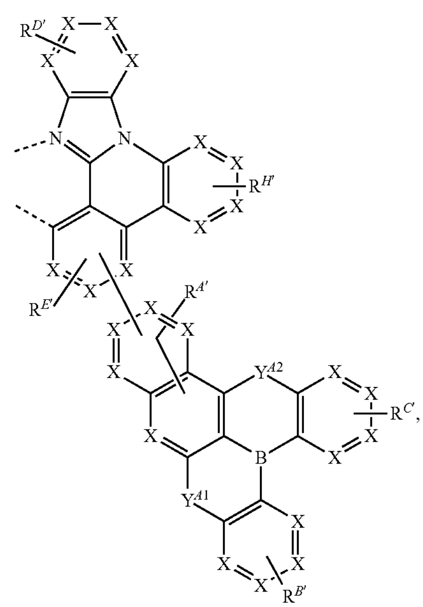

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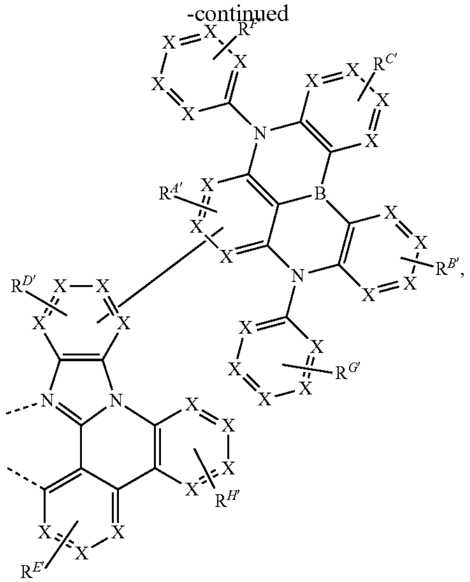

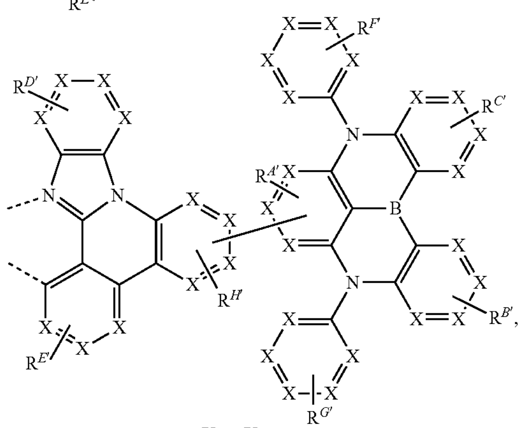

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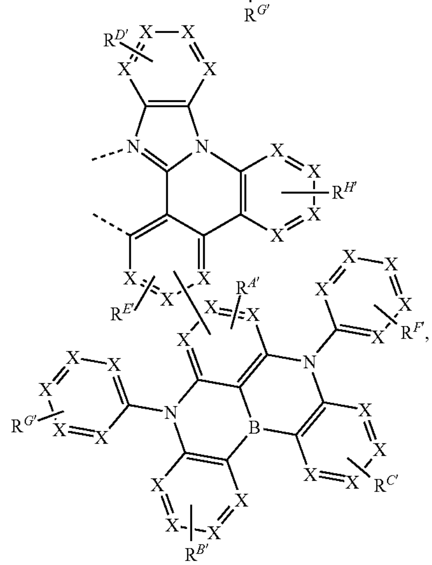


wherein

[0234] for each occurrence, X is independently C or N;

[0235] each of Y^{A1}, Y^{A2} and Y^{E1} is independently selected from the group consisting of a direct bond, BR, BRR', NR, PR, P(O)R, O, S, Se, C═O, C═S, C═Se, C═NR, C═CRR', S═O, SO_2, CR, CRR', SiRR', and GeRR';

[0236] each of the two - - - - represents chelation to the metal;

[0237] each of $R^{A'}$, $R^{B'}$, $R^{C'}$, $R^{D'}$, $R^{E'}$, $R^{F'}$, $R^{G'}$, and $R^{H'}$ independently represents mono to the maximum allowable substitution;

[0238] each R, R', $R^{A'}$, $R^{B'}$, $R^{C'}$, $R^{D'}$, $R^{E'}$, $R^{F'}$, $R^{G'}$, $R^{H'}$, and $R^{N'}$ is independently hydrogen or a substituent selected from the group consisting of the General Substituents defined herein; and

[0239] any two substituents may be optionally joined or fused to form a ring.

[0240] For each structure in LIST 2, X can be the same or different in each individual ring. In some embodiments, each X is C. In some embodiments, at least one X is N. In some embodiments, exactly one X is N.

[0241] Even though the Formula I moiety of each ligand L_A in LIST 2 was drawn to show attachment to one particular ring of the moiety chelated to metal M, it should be understood that in other embodiments Formula I may be attached to another ring of the moiety chelated to metal M. In some embodiments, the Formula I moiety is attached to the ring with $R^{D'}$ substituents. In some embodiments, the Formula I moiety is attached to the ring with $R^{E'}$ substituents. In some embodiments, the Formula I moiety is attached to the ring with $R^{H'}$ substituents.

[0242] In some embodiments where ligand L_A is selected from LIST 2, two $R^{D'}$ can be joined to form a fused ring. In some such embodiments, the fused ring may be benzene, pyridine, pyrimidine, pyridazine, pyrazine, triazine, imidazole, pyrazole, pyrrole, oxazole, furan, thiophene, or thiazole. In some such embodiments, the fused ring may be benzene.

[0243] In some embodiments where ligand L_A is selected from LIST 2, at least one $R^{A'}$, $R^{B'}$, $R^{C'}$, $R^{D'}$, $R^{E'}$ $R^{F'}$, $R^{G'}$, $R^{H'}$, or $R^{N'}$ is selected from the group consisting of the General Substituents defined herein. In some embodiments, at least one $R^{A'}$ is selected from the group consisting of the General Substituents defined herein. In some embodiments, at least one $R^{B'}$ is selected from the group consisting of the General Substituents defined herein. In some embodiments, at least one $R^{C'}$ is selected from the group consisting of the General Substituents defined herein. In some embodiments, at least one $R^{D'}$ is selected from the group consisting of the General Substituents defined herein. In some embodiments, at least one $R^{E'}$ is selected from the group consisting of the General Substituents defined herein. In some embodiments, at least one $R^{F'}$ is selected from the group consisting of the General Substituents defined herein. In some embodiments, at least one $R^{G'}$ is selected from the group consisting of the General Substituents defined herein. In some embodiments, at least one $R^{H'}$ is selected from the group consisting of the General Substituents defined herein. In some embodiments, at least one $R^{N'}$ is selected from the group consisting of the General Substituents defined herein. In some embodiments, at least one $R^{A'}$, $R^{B'}$, $R^{C'}$, $R^{D'}$, $R^{E'}$ $R^{F'}$, $R^{G'}$, $R^{H'}$, or $R^{N'}$ is selected from the group consisting of the Preferred General Substituents defined herein.

[0244] In some embodiments where ligand L_A is selected from LIST 2, at least one of R, R', $R^{A'}$, $R^{B'}$, $R^{C'}$, $R^{D'}$, $R^{E'}$ $R^{F'}$, $R^{G'}$, $R^{H'}$, or $R^{N'}$ is partially or fully deuterated. In some embodiments, at least one $R^{A'}$ is partially or fully deuterated.

In some embodiments, at least one $R^{B'}$ is partially or fully deuterated. In some embodiments, at least one $R^{C'}$ is partially or fully deuterated. In some embodiments, at least one $R^{D'}$ is partially or fully deuterated. In some embodiments, at least one $R^{E'}$ is partially or fully deuterated. In some embodiments, at least one $R^{F'}$ is partially or fully deuterated. In some embodiments, at least one $R^{G'}$ is partially or fully deuterated. In some embodiments, at least one $R^{H'}$ is partially or fully deuterated. In some embodiments, at least one $R^{N'}$ is partially or fully deuterated. In some embodiments, at least one R or R' is partially or fully deuterated.

[0245] In some embodiments where ligand L_A is selected from LIST 2 or 4 (below), at least one $R^{A'}$ is or comprises an electron-withdrawing group from the EWG1 LIST as defined herein. In some embodiments, at least one $R^{A'}$ is or comprises an electron-withdrawing group from the EWG2 LIST as defined herein. In some embodiments, at least one $R^{A'}$ is or comprises an electron-withdrawing group from the EWG3 LIST as defined herein. In some embodiments, at least one $R^{A'}$ is or comprises an electron-withdrawing group from the EWG4 LIST as defined herein. In some embodiments, at least one $R^{A'}$ is or comprises an electron-withdrawing group from the Pi-EWG LIST as defined herein.

[0246] In some embodiments where ligand L_A is selected from LIST 2 or 4, at least one $R^{B'}$ is or comprises an electron-withdrawing group from the EWG1 LIST as defined herein. In some embodiments, at least one $R^{B'}$ is or comprises an electron-withdrawing group from the EWG2 LIST as defined herein. In some embodiments, at least one $R^{B'}$ is or comprises an electron-withdrawing group from the EWG3 LIST as defined herein. In some embodiments, at least one $R^{B'}$ is or comprises an electron-withdrawing group from the EWG4 LIST as defined herein. In some embodiments, at least one $R^{B'}$ is or comprises an electron-withdrawing group from the Pi-EWG LIST as defined herein.

[0247] In some embodiments where ligand L_A is selected from LIST 2 or 4, at least one $R^{C'}$ is or comprises an electron-withdrawing group from the EWG1 LIST as defined herein. In some embodiments, at least one $R^{C'}$ is or comprises an electron-withdrawing group from the EWG2 LIST as defined herein. In some embodiments, at least one $R^{C'}$ is or comprises an electron-withdrawing group from the EWG3 LIST as defined herein. In some embodiments, at least one $R^{C'}$ is or comprises an electron-withdrawing group from the EWG4 LIST as defined herein. In some embodiments, at least one $R^{C'}$ is or comprises an electron-withdrawing group from the Pi-EWG LIST as defined herein.

[0248] In some embodiments where ligand L_A is selected from LIST 2 or 4, at least one $R^{D'}$ is or comprises an electron-withdrawing group from the EWG1 LIST as defined herein. In some embodiments, at least one $R^{D'}$ is or comprises an electron-withdrawing group from the EWG2 LIST as defined herein. In some embodiments, at least one $R^{D'}$ is or comprises an electron-withdrawing group from the EWG3 LIST as defined herein. In some embodiments, at least one $R^{D'}$ is or comprises an electron-withdrawing group from the EWG4 LIST as defined herein. In some embodiments, at least one $R^{D'}$ is or comprises an electron-withdrawing group from the Pi-EWG LIST as defined herein.

[0249] In some embodiments where ligand L_A is selected from LIST 2 or 4, at least one $R^{E'}$ is or comprises an electron-withdrawing group from the EWG1 LIST as defined herein. In some embodiments, at least one $R^{E'}$ is or comprises an electron-withdrawing group from the EWG2 LIST as defined herein. In some embodiments, at least one $R^{E'}$ is or comprises an electron-withdrawing group from the EWG3 LIST as defined herein. In some embodiments, at least one $R^{E'}$ is or comprises an electron-withdrawing group from the EWG4 LIST as defined herein. In some embodiments, at least one $R^{E'}$ is or comprises an electron-withdrawing group from the Pi-EWG LIST as defined herein.

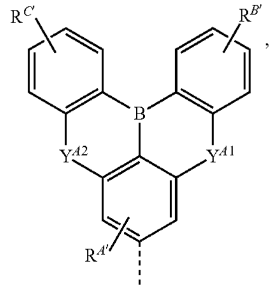
[0250] In some embodiments where ligand L_A is selected from LIST 2 or 4, at least one $R^{F'}$ is or comprises an electron-withdrawing group from the EWG1 LIST as defined herein. In some embodiments, at least one $R^{F'}$ is or comprises an electron-withdrawing group from the EWG2 LIST as defined herein. In some embodiments, at least one $R^{F'}$ is or comprises an electron-withdrawing group from the EWG3 LIST as defined herein. In some embodiments, at least one $R^{F'}$ is or comprises an electron-withdrawing group from the EWG4 LIST as defined herein. In some embodiments, at least one $R^{F'}$ is or comprises an electron-withdrawing group from the Pi-EWG LIST as defined herein.

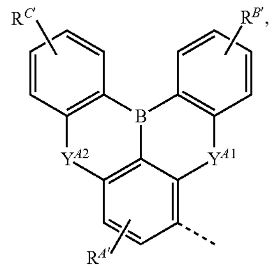
[0251] In some embodiments where ligand L_A is selected from LIST 2 or 4, at least one $R^{G'}$ is or comprises an electron-withdrawing group from the EWG1 LIST as defined herein. In some embodiments, at least one $R^{G'}$ is or comprises an electron-withdrawing group from the EWG2 LIST as defined herein. In some embodiments, at least one $R^{G'}$ is or comprises an electron-withdrawing group from the EWG3 LIST as defined herein. In some embodiments, at least one $R^{G'}$ is or comprises an electron-withdrawing group from the EWG4 LIST as defined herein. In some embodiments, at least one $R^{G'}$ is or comprises an electron-withdrawing group from the Pi-EWG LIST as defined herein.

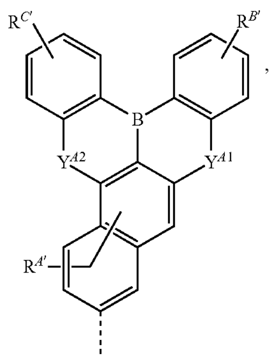
[0252] In some embodiments where ligand L_A is selected from LIST 2 or 4, at least one $R^{H'}$ is or comprises an electron-withdrawing group from the EWG1 LIST as defined herein. In some embodiments, at least one $R^{H'}$ is or comprises an electron-withdrawing group from the EWG2 LIST as defined herein. In some embodiments, at least one $R^{H'}$ is or comprises an electron-withdrawing group from the EWG3 LIST as defined herein. In some embodiments, at least one $R^{H'}$ is or comprises an electron-withdrawing group from the EWG4 LIST as defined herein. In some embodiments, at least one $R^{H'}$ is or comprises an electron-withdrawing group from the Pi-EWG LIST as defined herein.

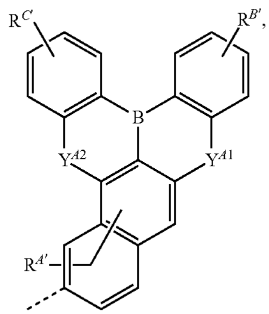
[0253] In some embodiments where ligand L_A is selected from LIST 2 or 4, $R^{N'}$ is or comprises an electron-withdrawing group from the EWG1 LIST as defined herein. In some embodiments, $R^{N'}$ is or comprises an electron-withdrawing group from the EWG2 LIST as defined herein. In some embodiments, $R^{N'}$ is or comprises an electron-withdrawing group from the EWG3 LIST as defined herein. In some embodiments, $R^{N'}$ is or comprises an electron-withdrawing group from the EWG4 LIST as defined herein. In some embodiments, $R^{N'}$ is or comprises an electron-withdrawing group from the Pi-EWG LIST as defined herein.

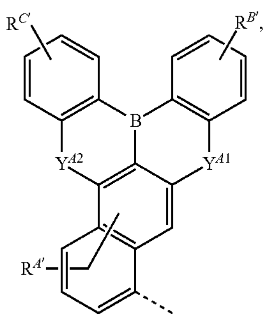
[0254] In some embodiments, the ligand L_A is selected from the group consisting of the structures of the following LIST 3:

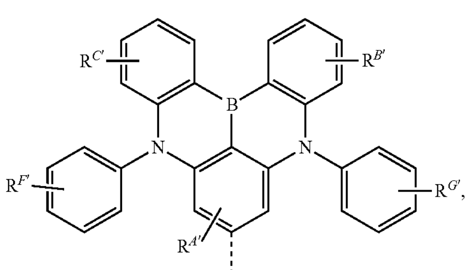


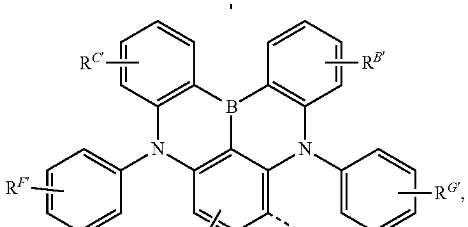












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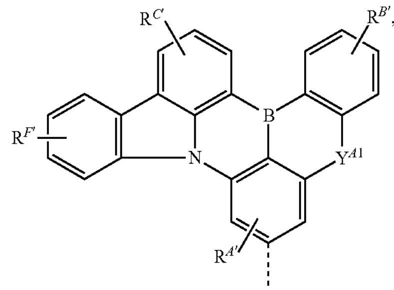

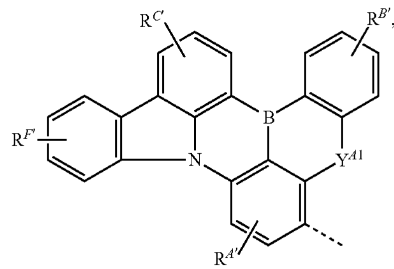

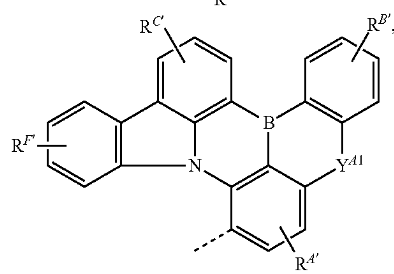

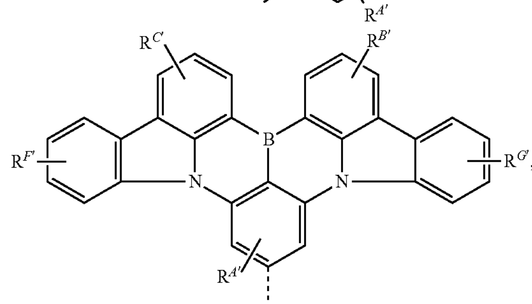

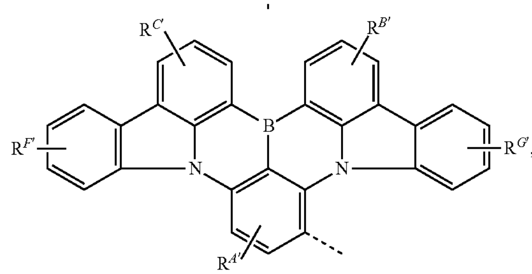

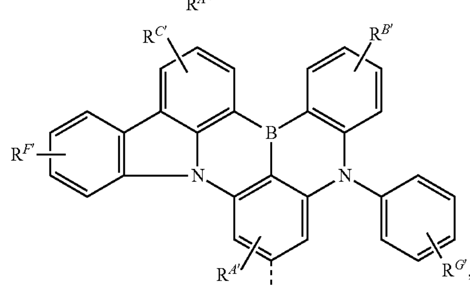

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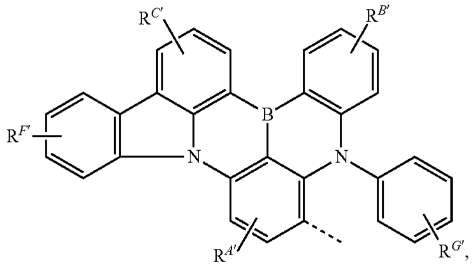

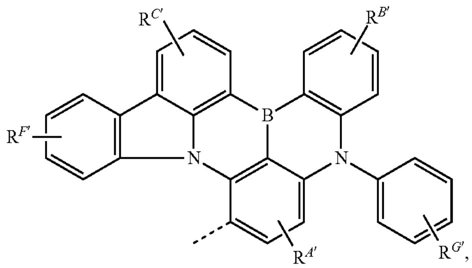

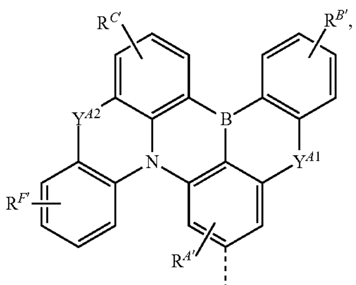

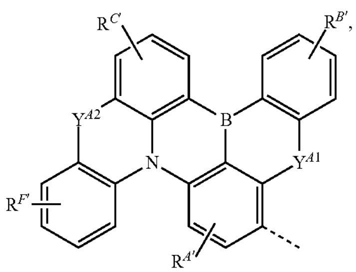

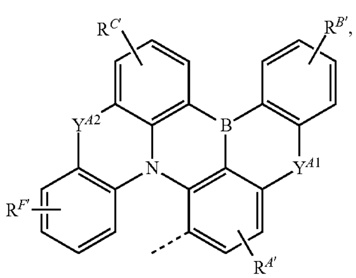

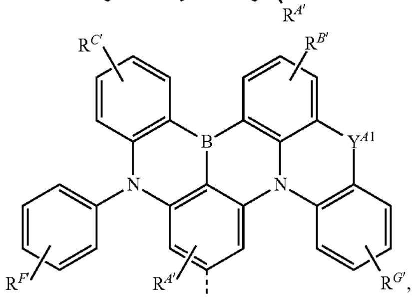

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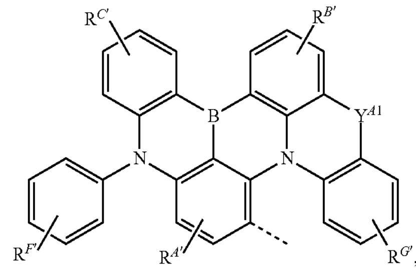

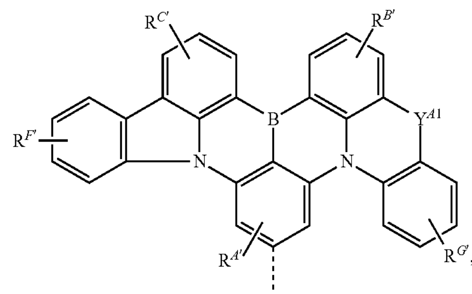

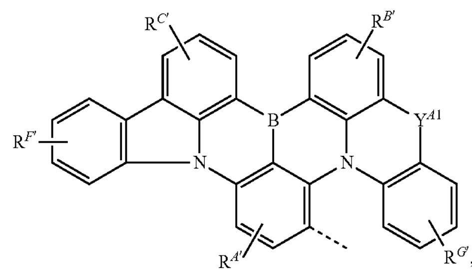

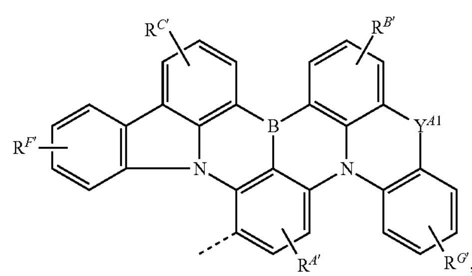

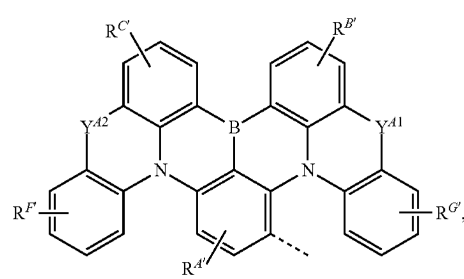

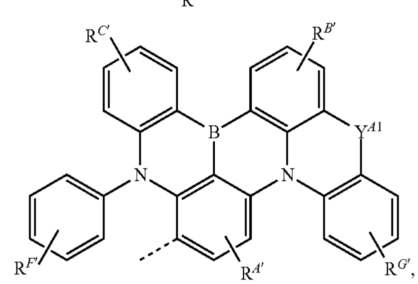

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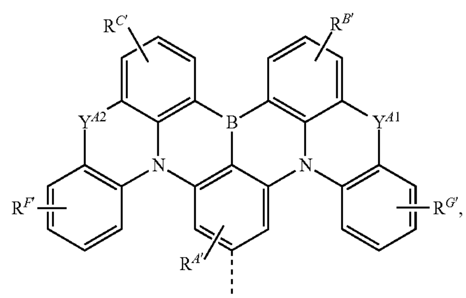


wherein:

[0255] each of Y^{A1} and Y^{A2} is independently selected from the group consisting of a direct bond, BR, BRR', NR, PR, P(O)R, O, S, Se, C═O, C═S, C═Se, C═NR, C═CRR', S═O, SO_2, CR, CRR', SiRR', and GeRR';

[0256] - - - - represents chelation to the metal;

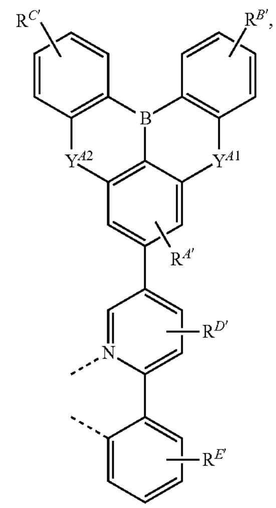
[0257] each of $R^{A'}$, $R^{B'}$, $R^{C'}$, $R^{F'}$, and $R^{G'}$ independently represents mono to the maximum allowable substitution;

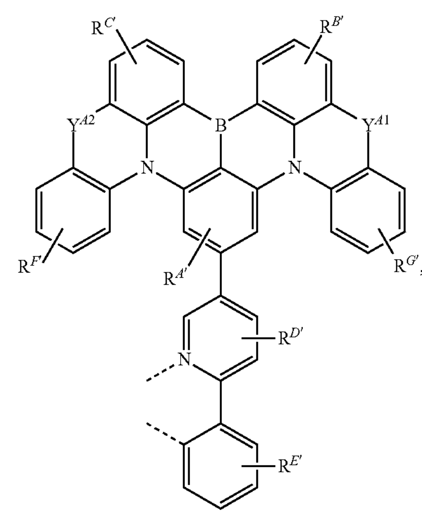
[0258] each R, R', R", $R^{A'}$, $R^{B'}$, $R^{C'}$, $R^{F'}$ and $R^{G'}$ is independently hydrogen or a substituent selected from the group consisting of the General Substituents defined herein; and any two substituents can be joined or fused to form a ring.

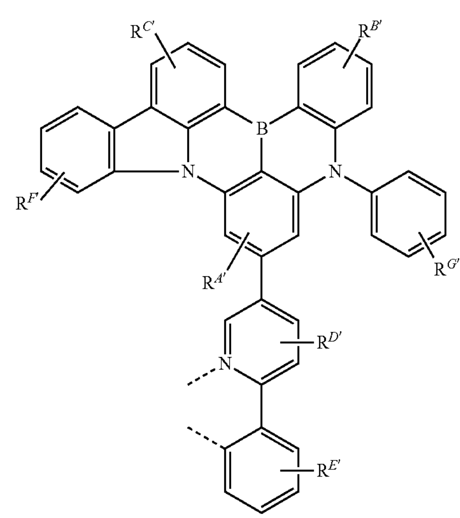
[0259] In some embodiments, Formula I is selected from the group consisting of the structures of LIST 3. In such embodiments, the - - - - - was a direct bond or through an organic linker to a monocyclic ring or polycyclic fused ring system that is coordinated to M. It should also be understood that the direct bond can be from any ring of the structure to the organic linker.

[0260] In some embodiments where ligand L_A is selected from LIST 3, at least one of R, R', $R^{A'}$, $R^{B'}$, $R^{C'}$, $R^{D'}$, $R^{E'}$ $R^{F'}$, $R^{G'}$, $R^{H'}$, or $R^{N'}$ is partially or fully deuterated. In some embodiments, at least one $R^{A'}$ is partially or fully deuterated. In some embodiments, at least one $R^{B'}$ is partially or fully deuterated. In some embodiments, at least one $R^{C'}$ is partially or fully deuterated. In some embodiments, at least one $R^{D'}$ is partially or fully deuterated. In some embodiments, at least one $R^{E'}$ is partially or fully deuterated. In some embodiments, at least one $R^{F'}$ is partially or fully deuterated. In some embodiments, at least one $R^{G'}$ is partially or fully deuterated. In some embodiments, at least one $R^{H'}$ is partially or fully deuterated. In some embodiments, at least one $R^{N'}$ is partially or fully deuterated. In some embodiments, at least one R or R' is partially or fully deuterated.

[0261] In some embodiments, the ligand L_A is selected from the group consisting of the structures of the following LIST 4:







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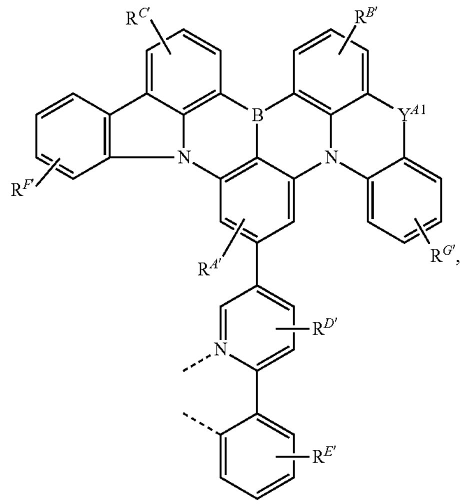

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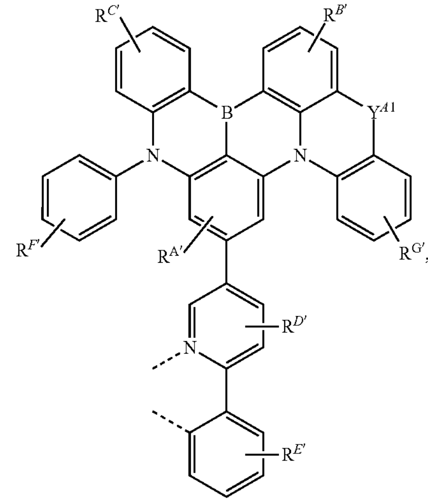

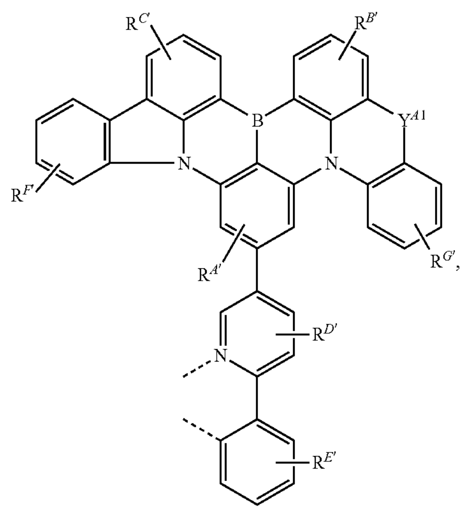

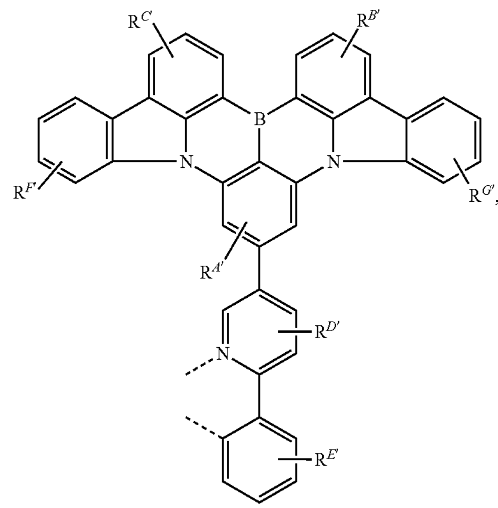

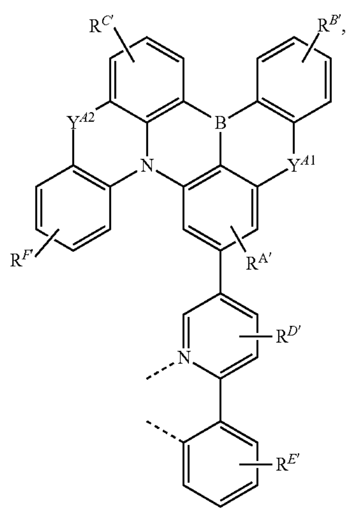

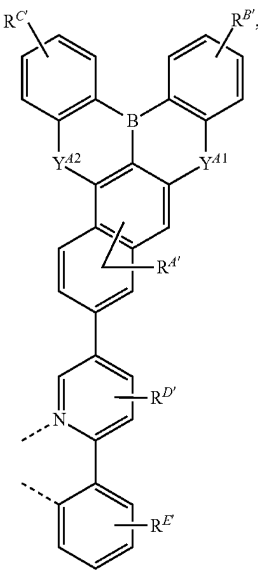

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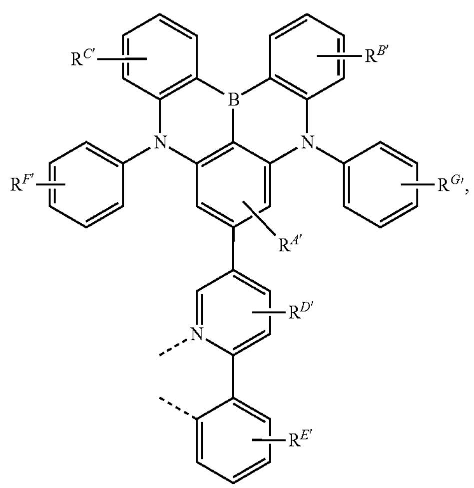

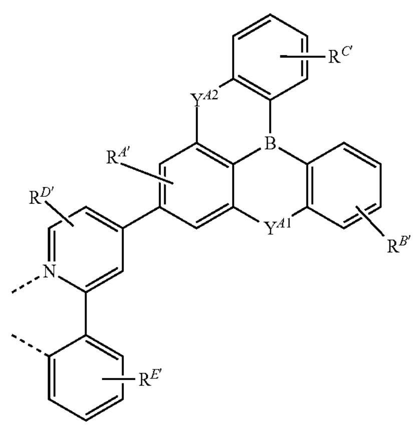

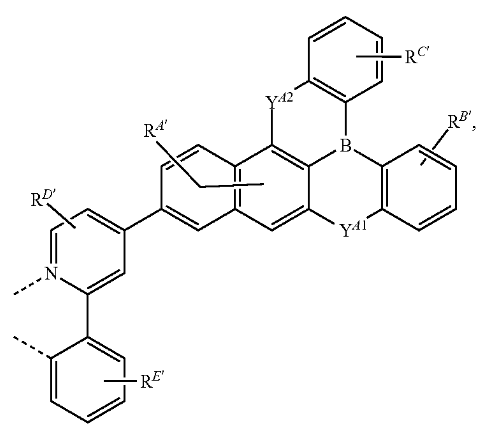

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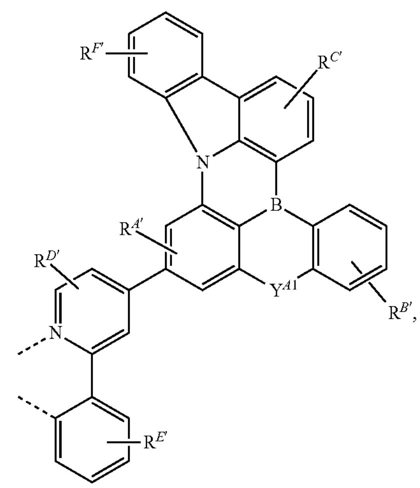

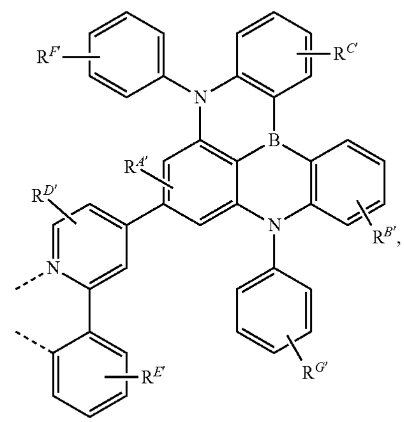

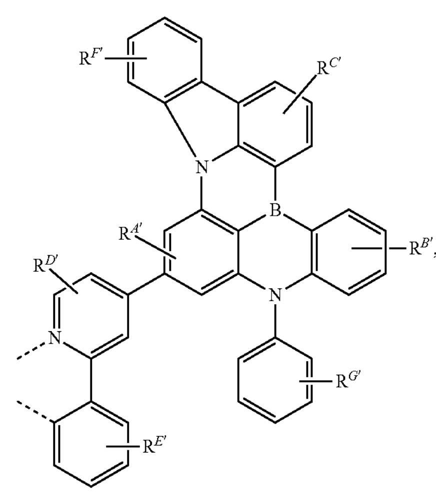

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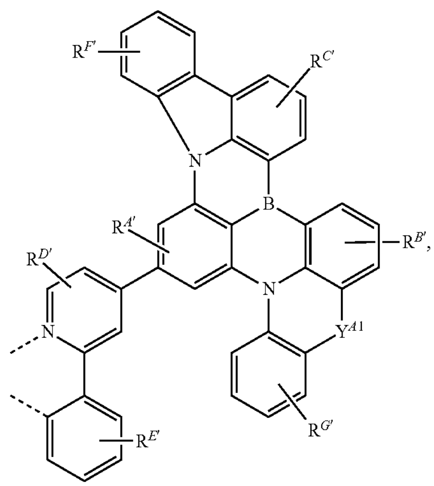

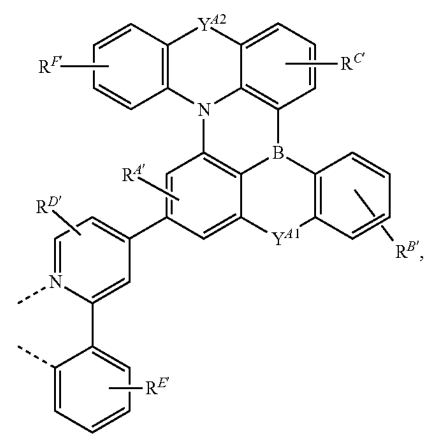

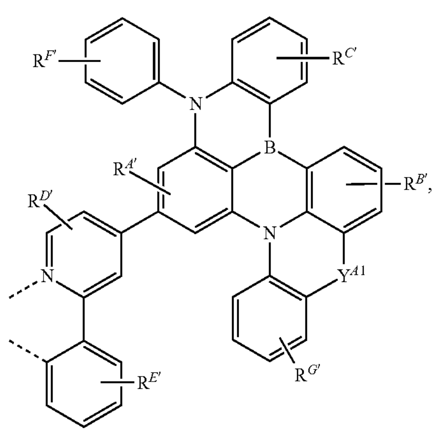

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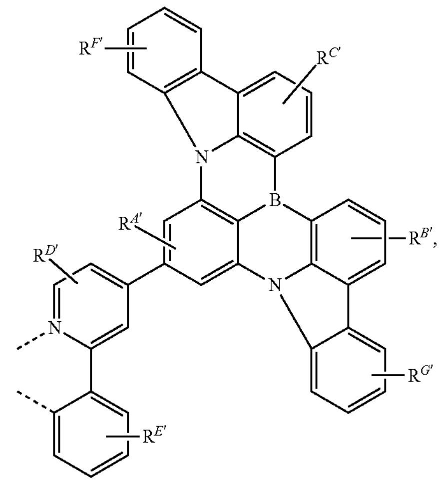

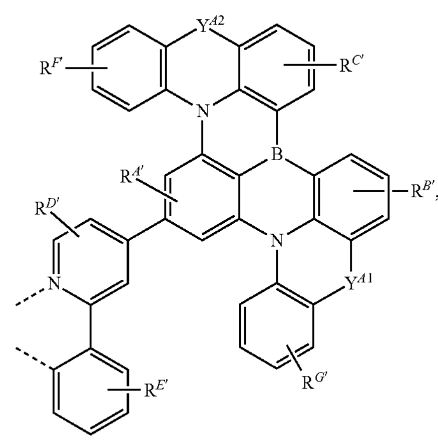

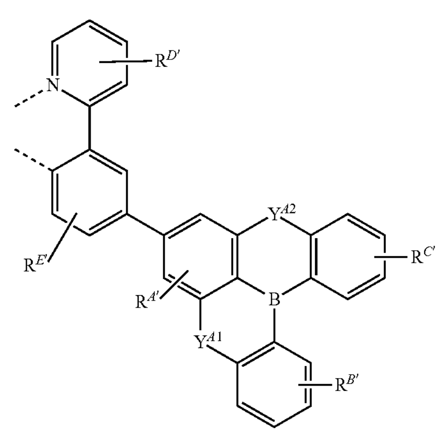

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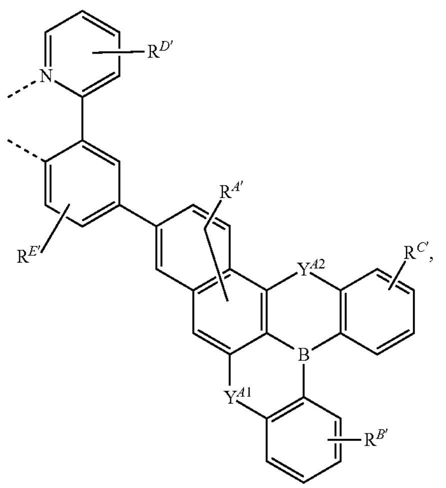

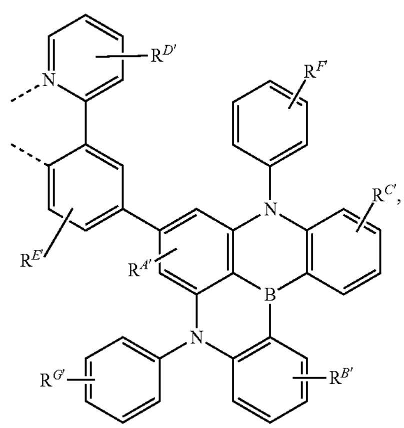

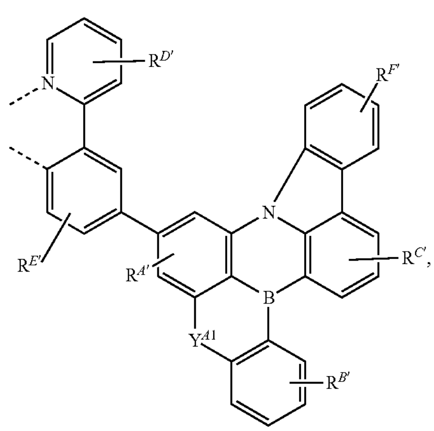

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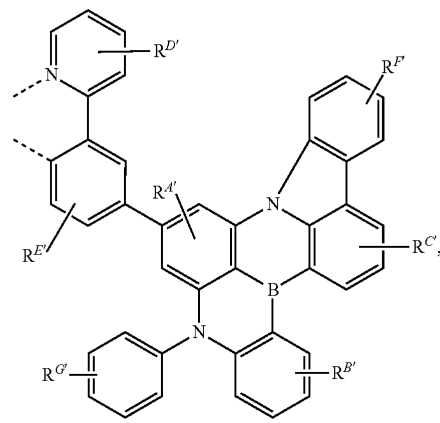

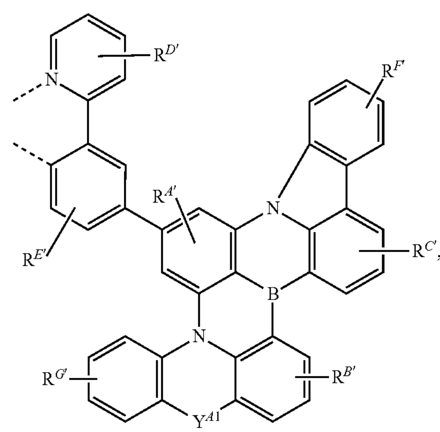

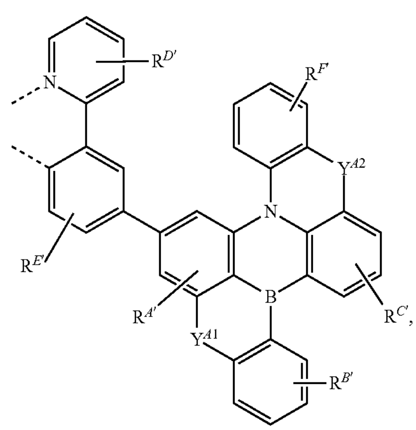

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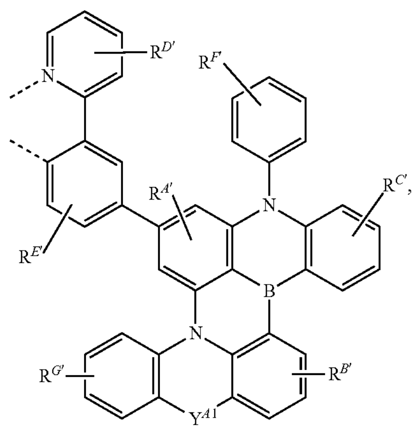

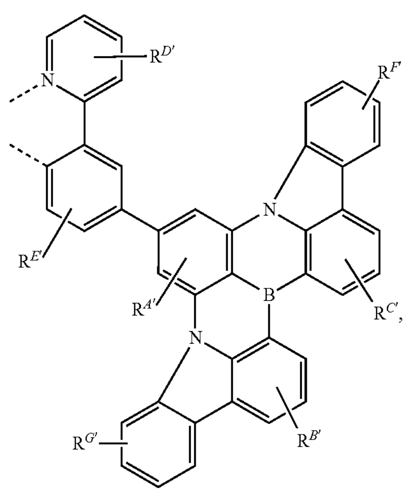

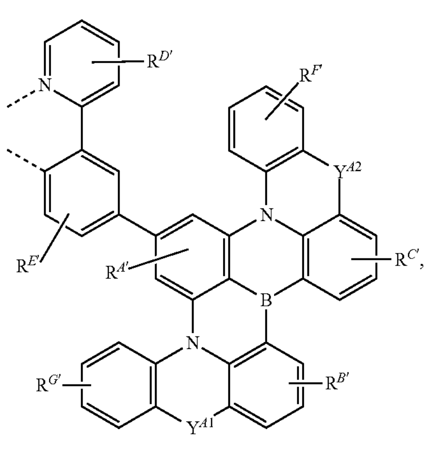

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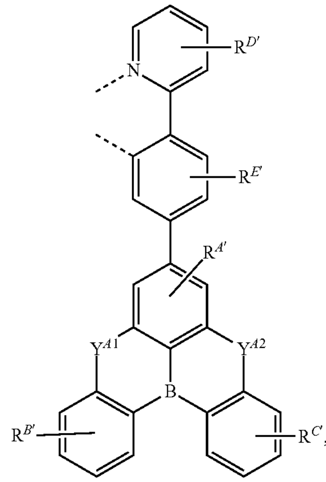

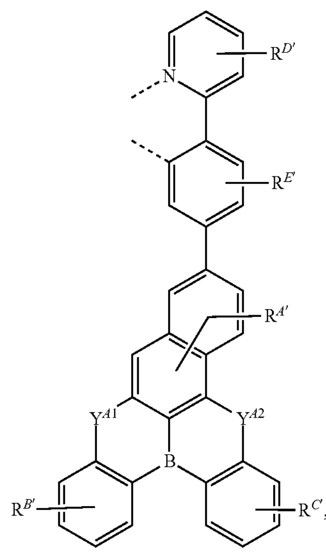

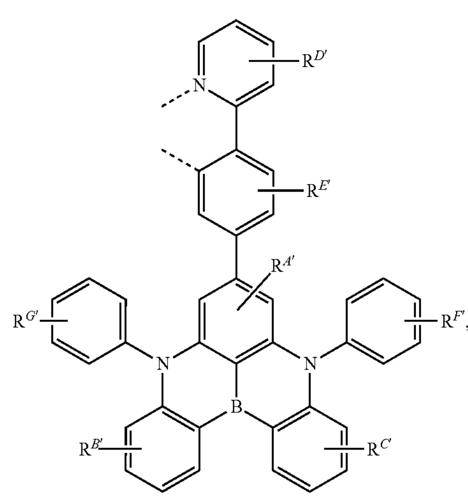

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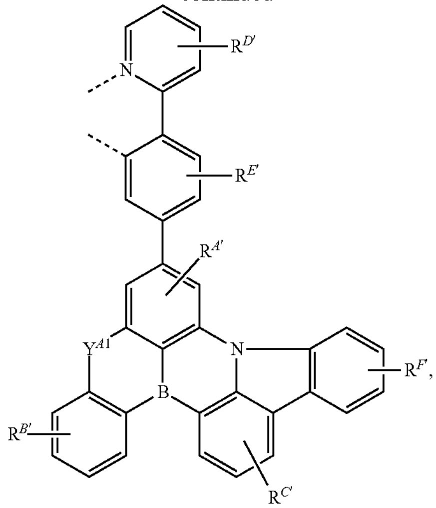

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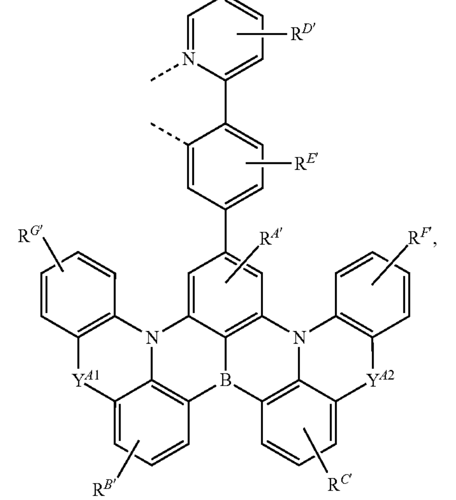

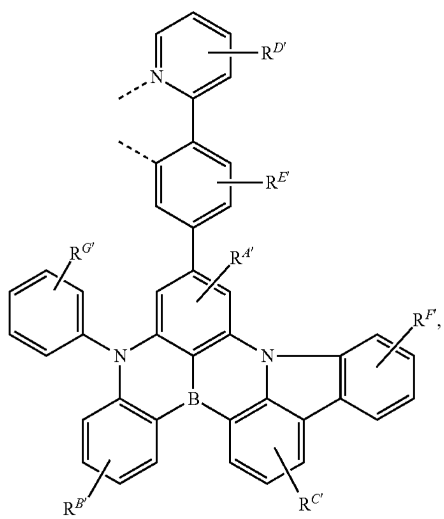

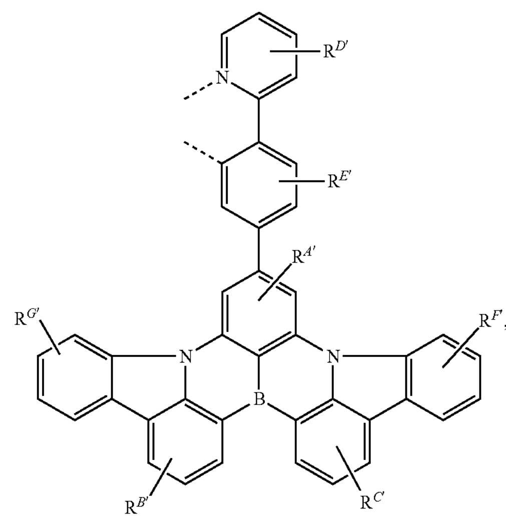

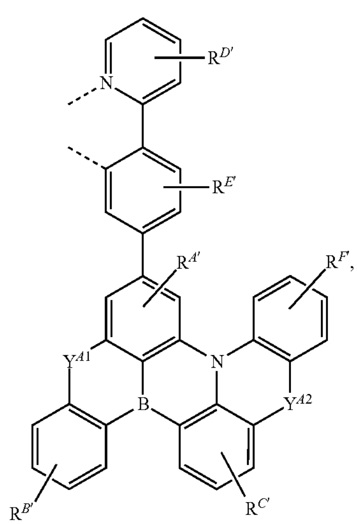

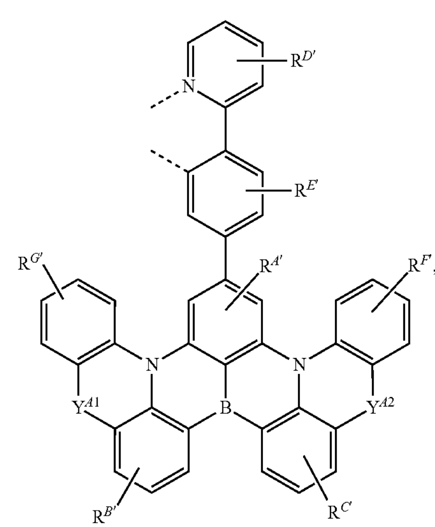

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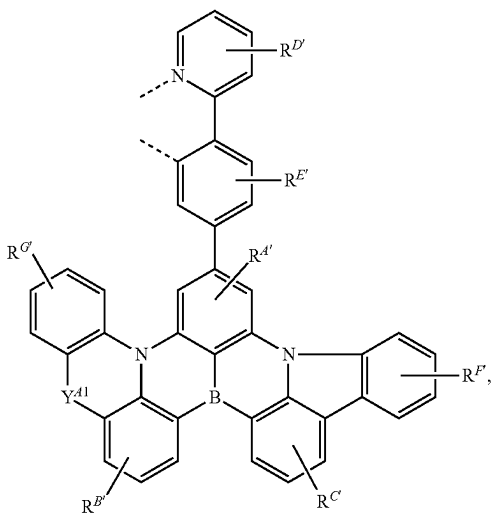

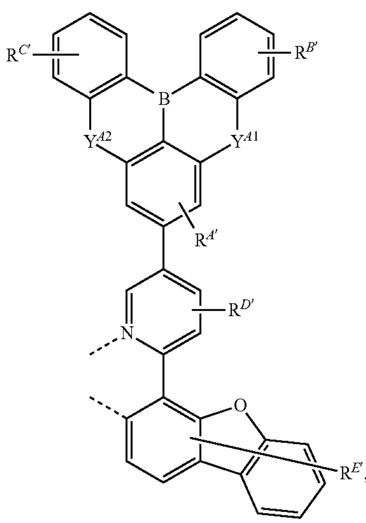

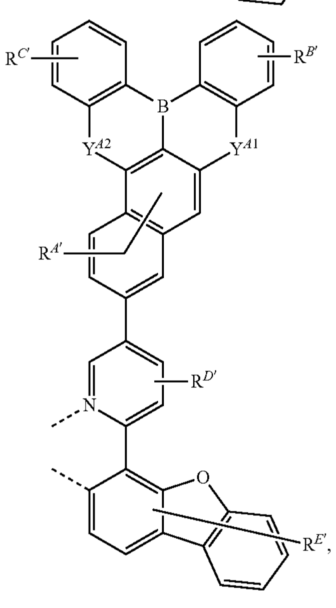

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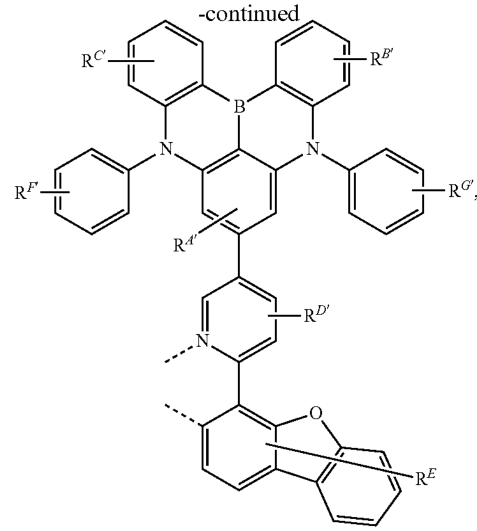

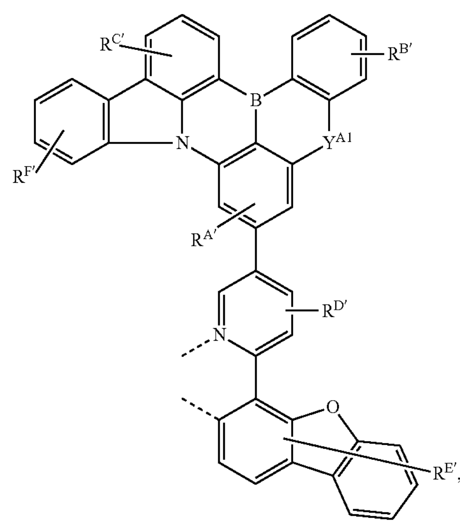

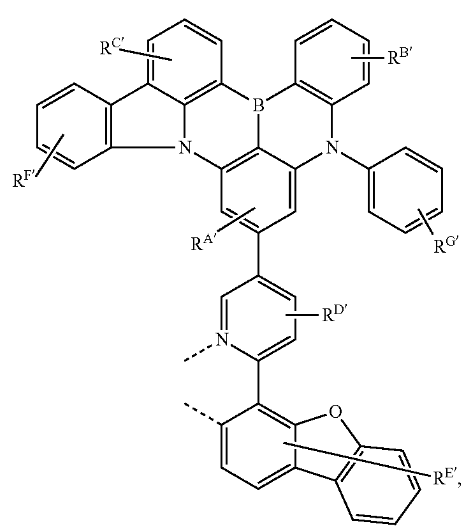

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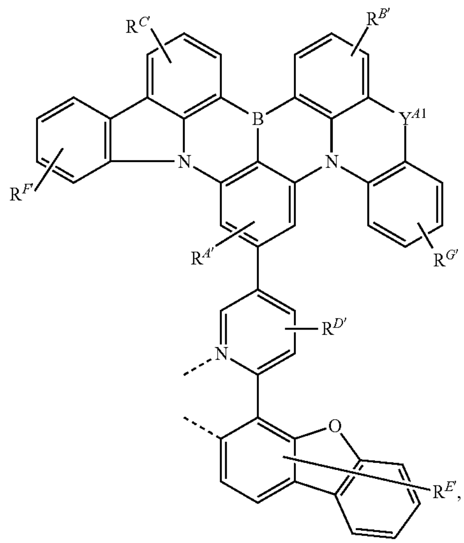

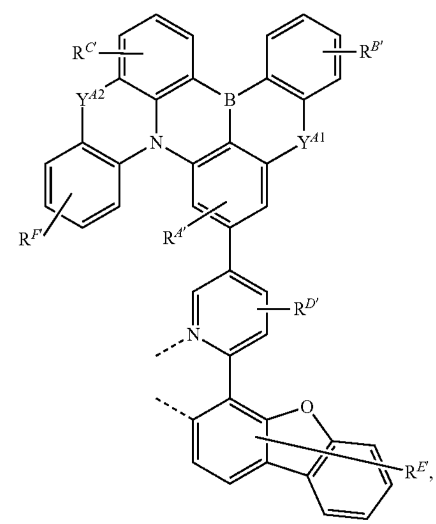

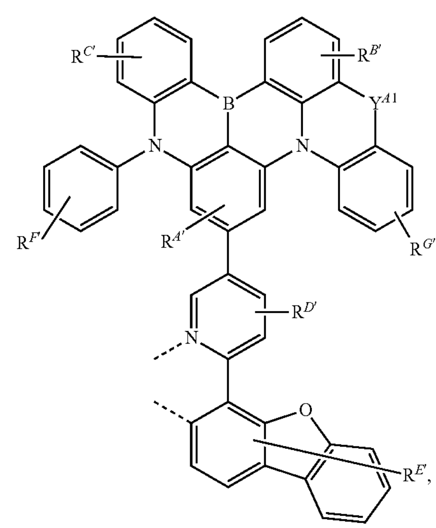

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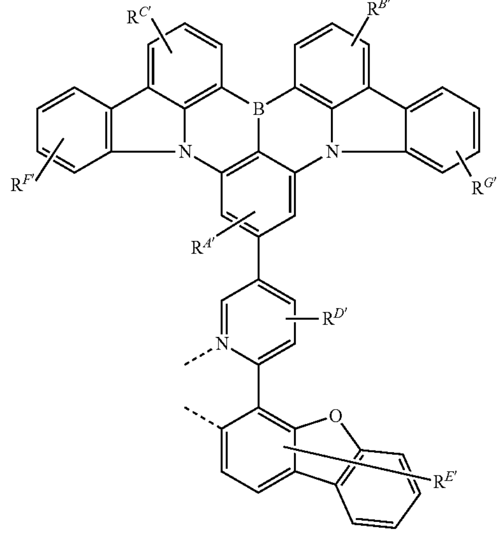

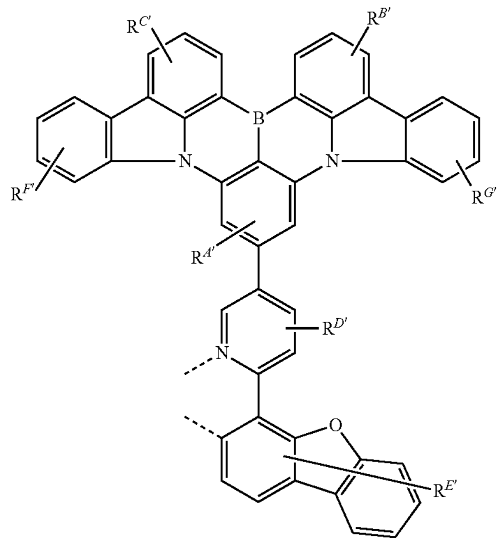

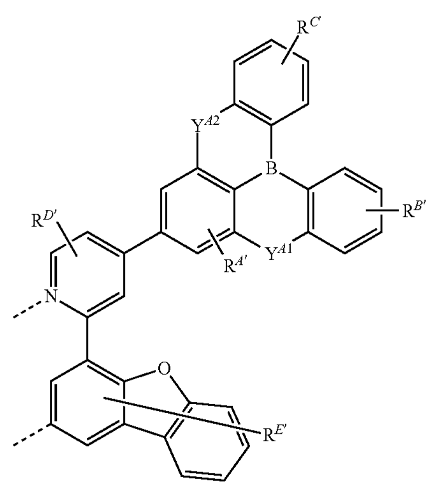

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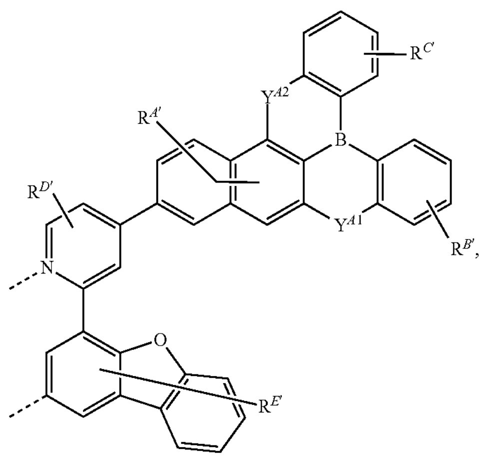

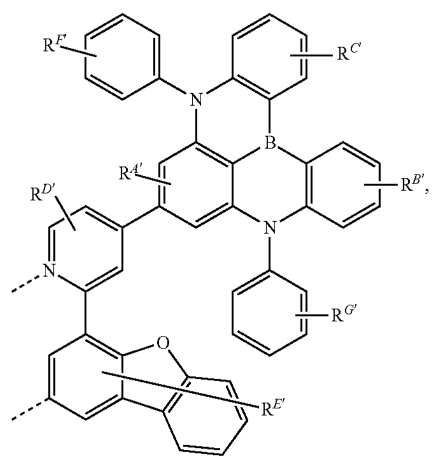

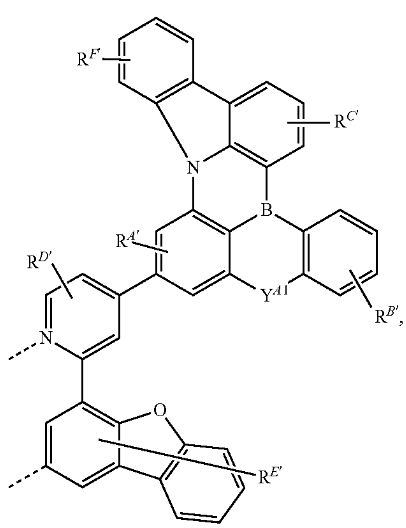

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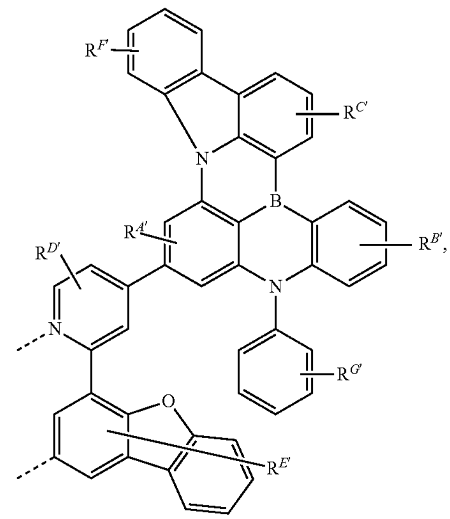

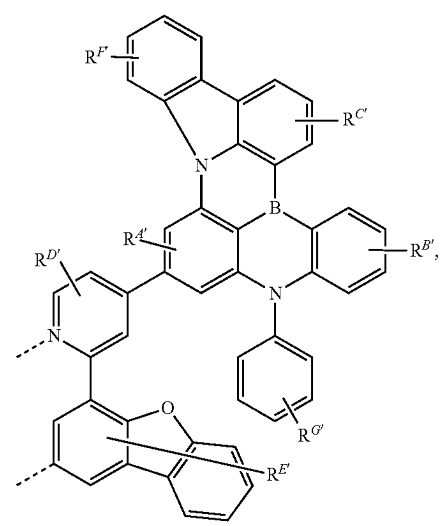

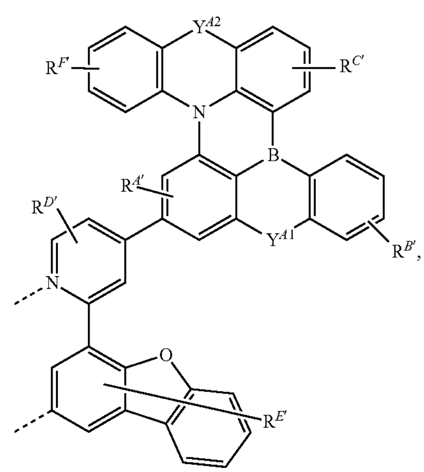

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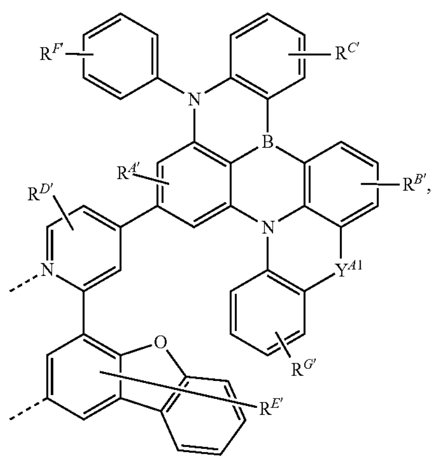

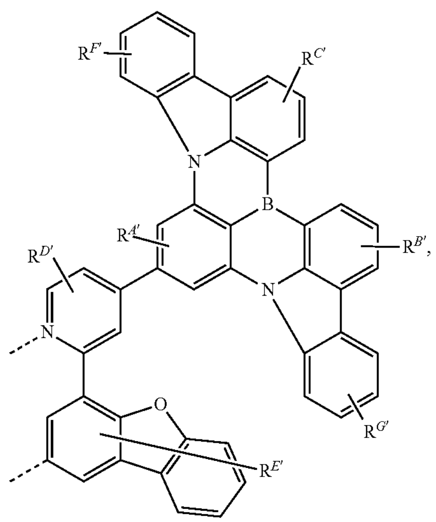

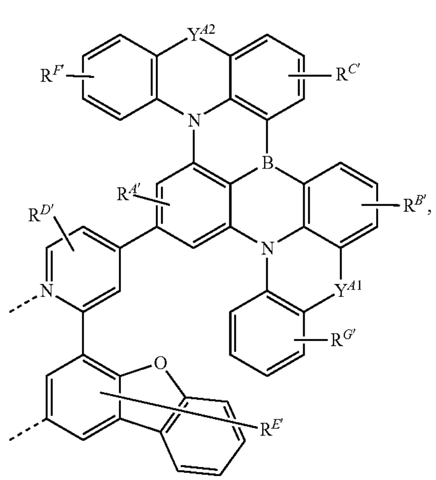

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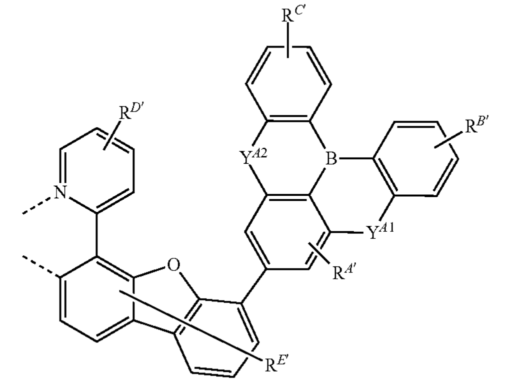

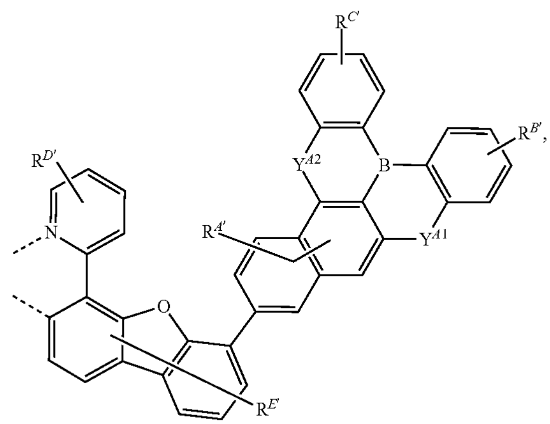

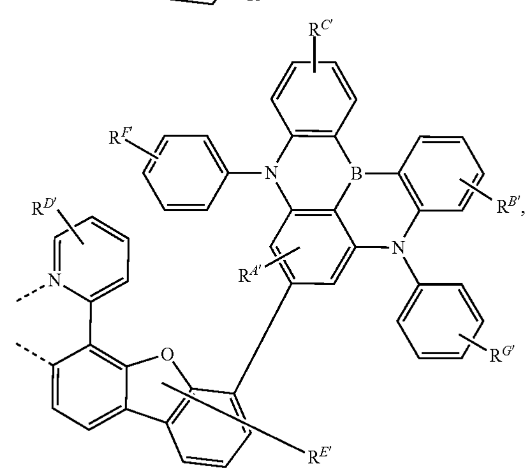

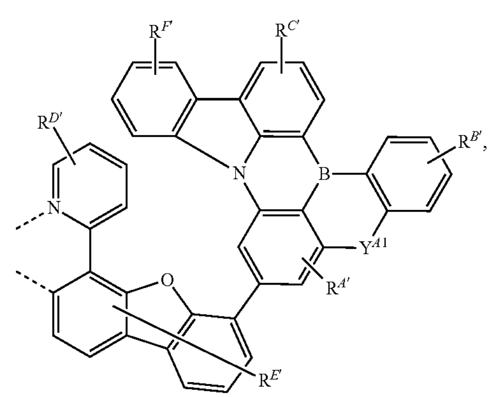

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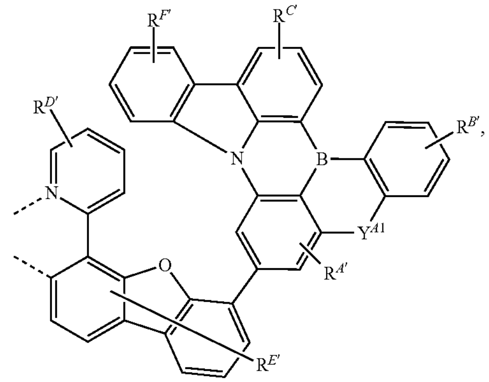

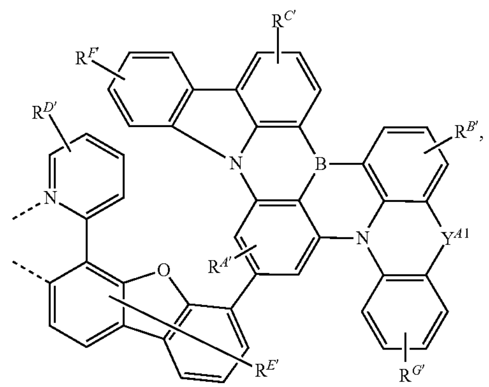

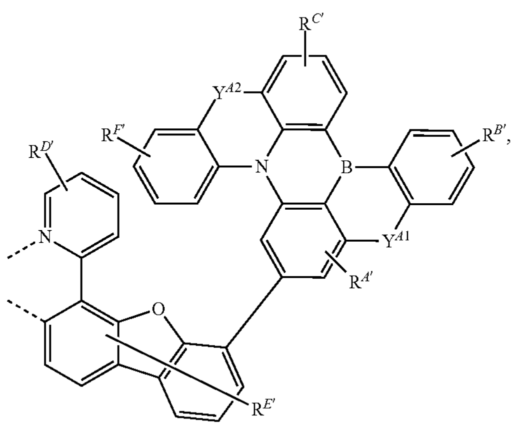

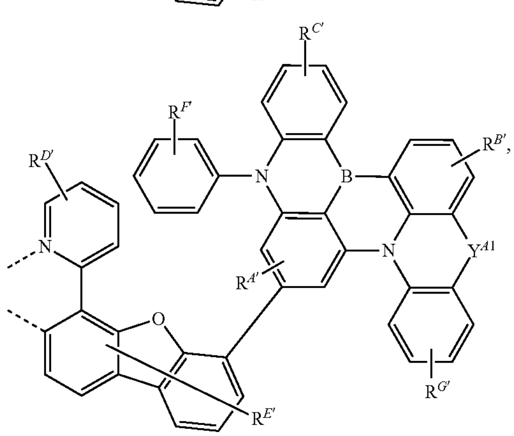

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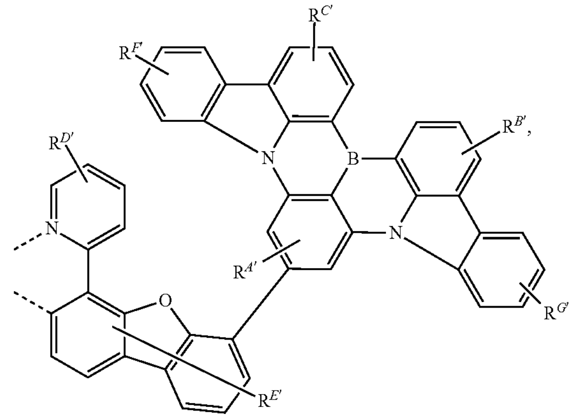

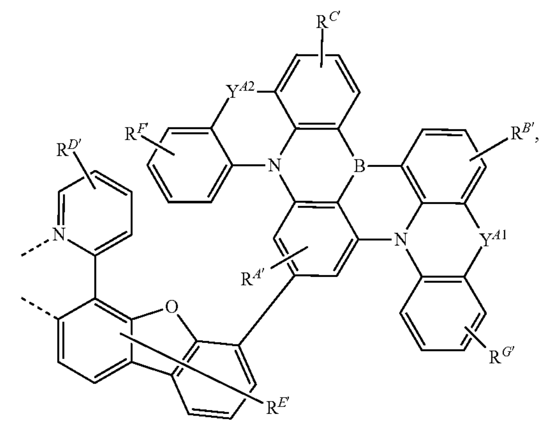

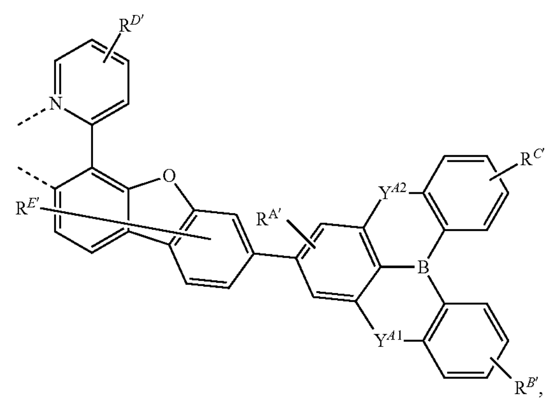

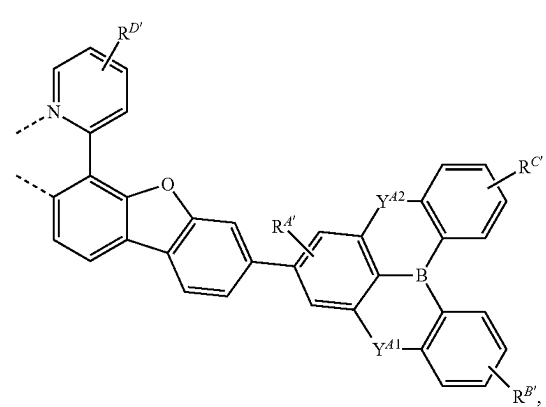

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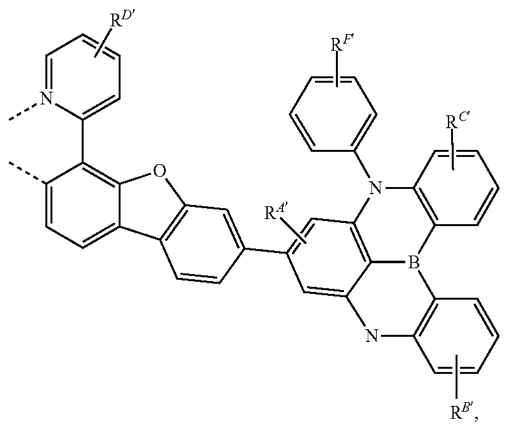

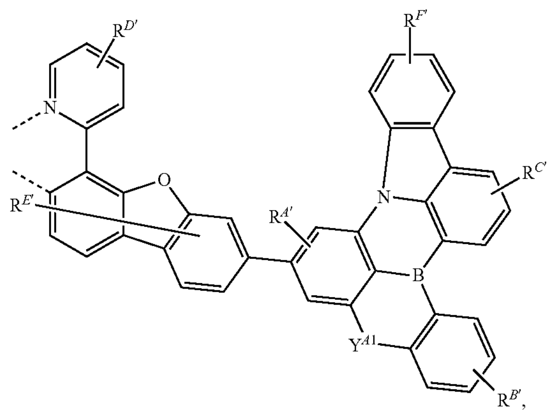

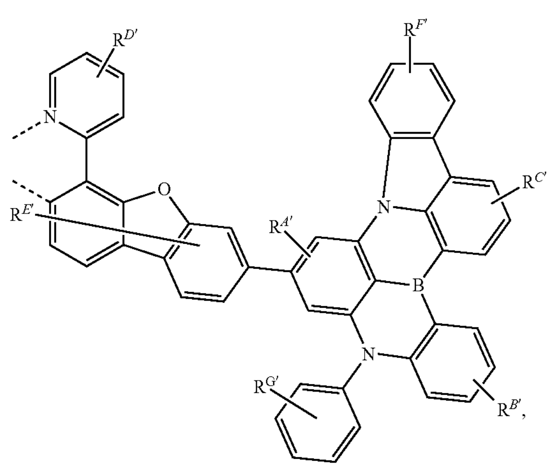

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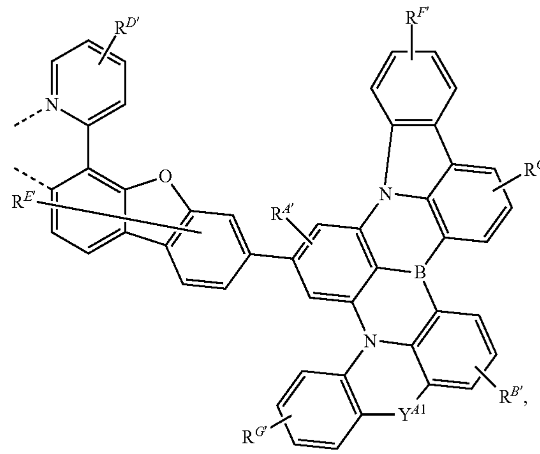

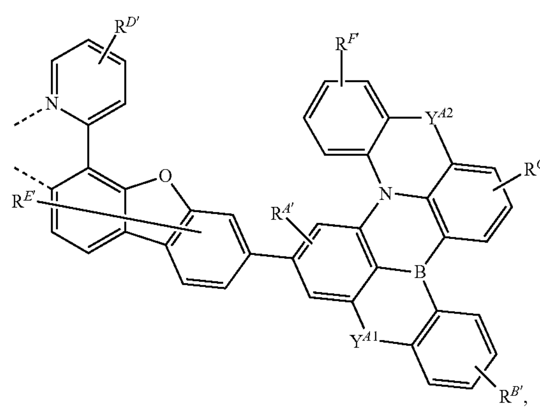

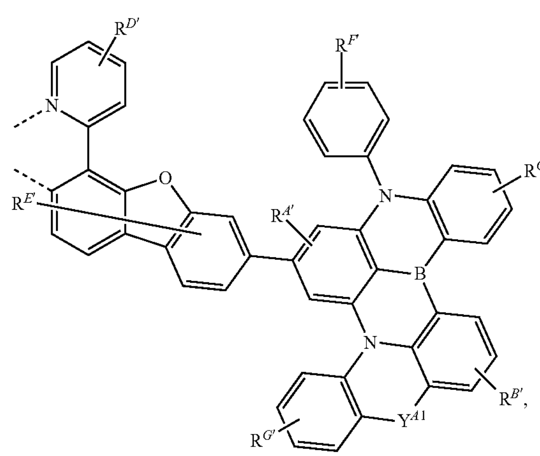

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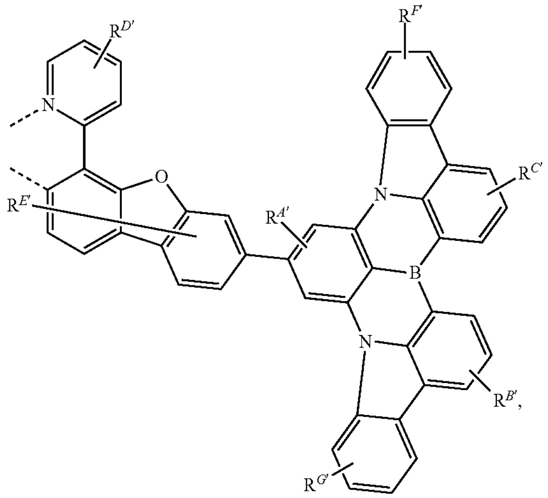

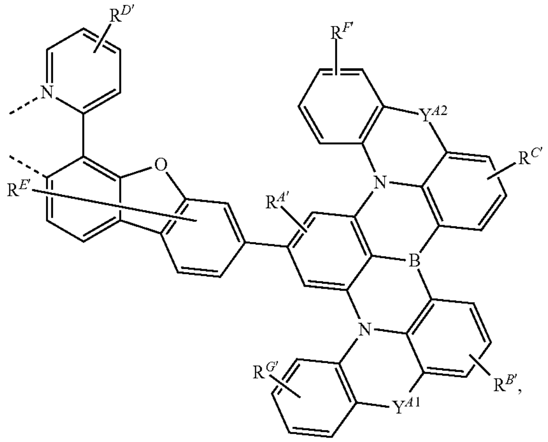

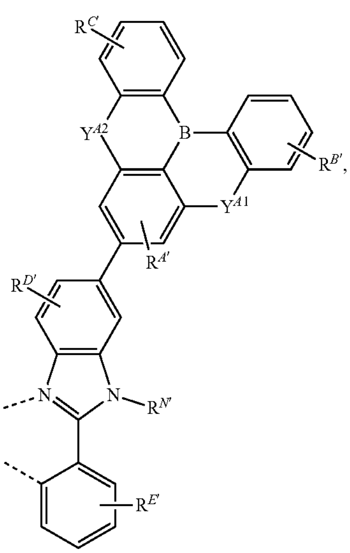

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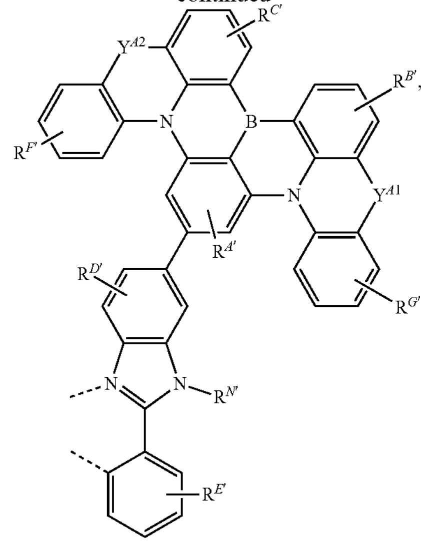

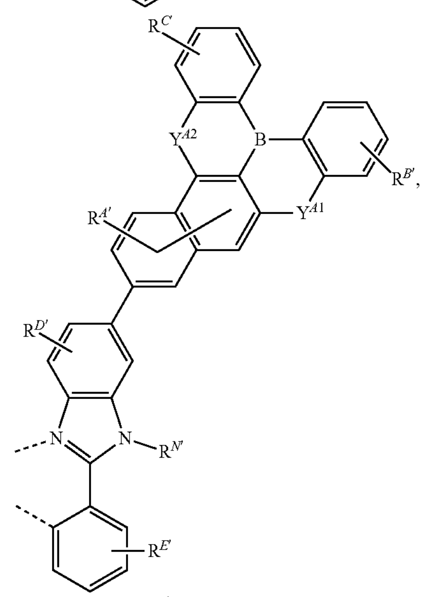

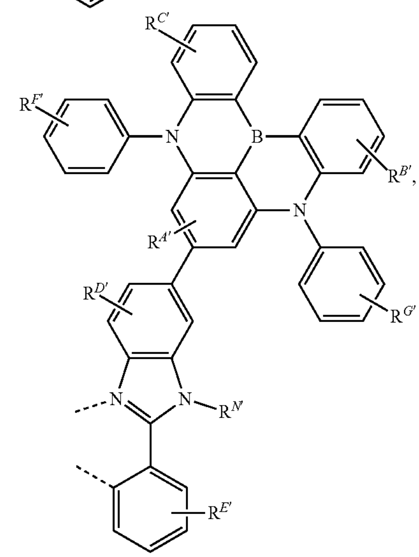

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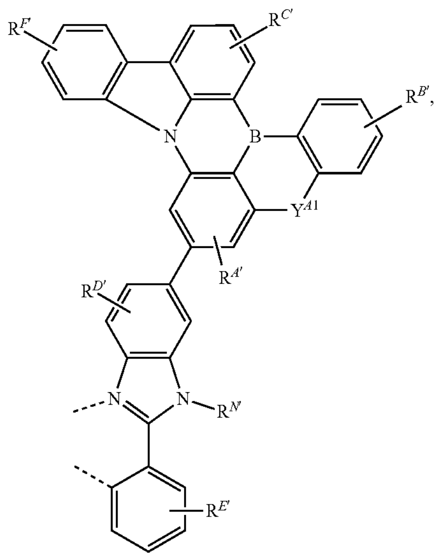

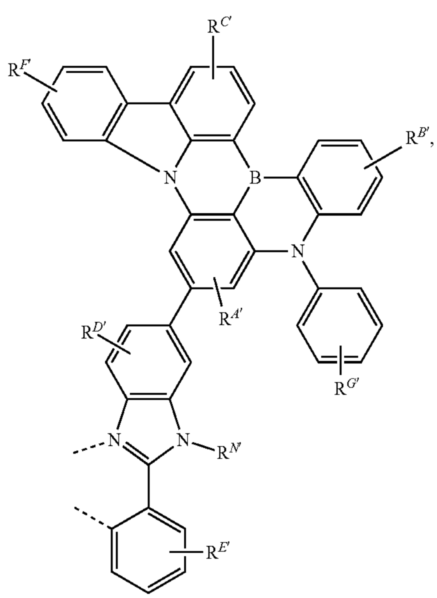

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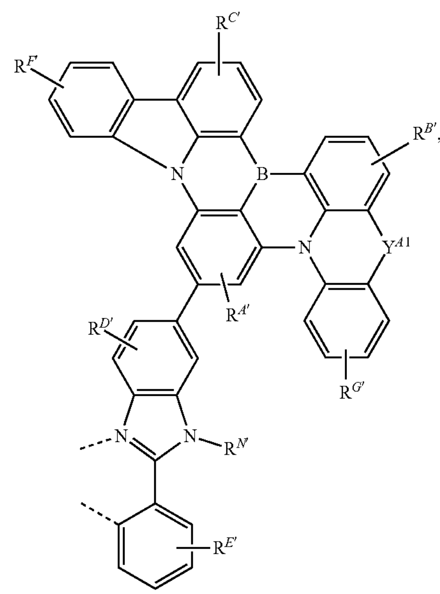

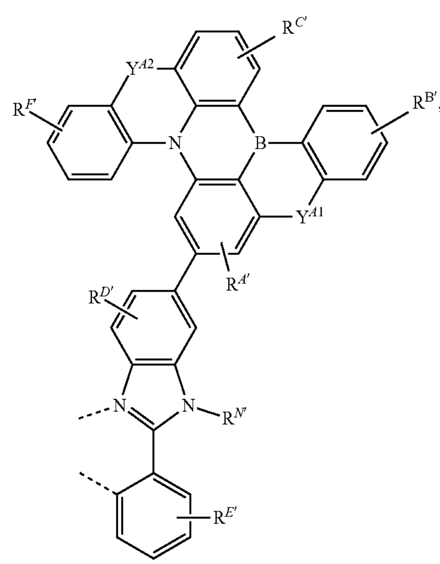

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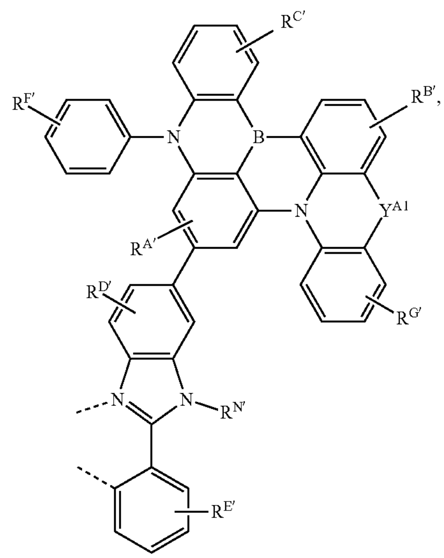

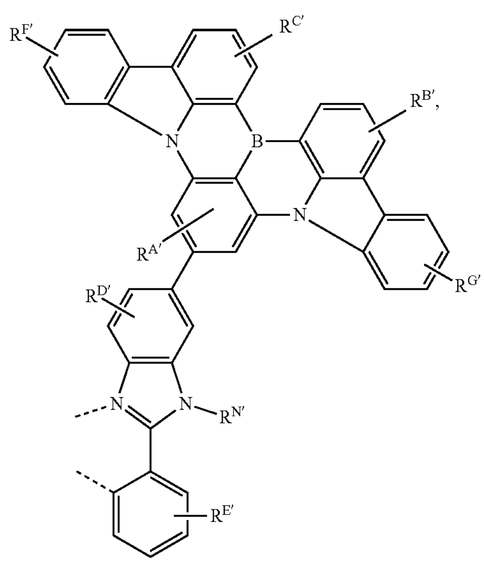

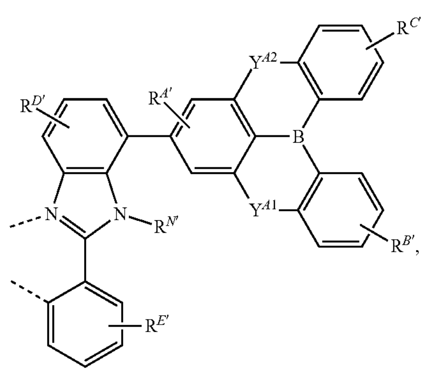

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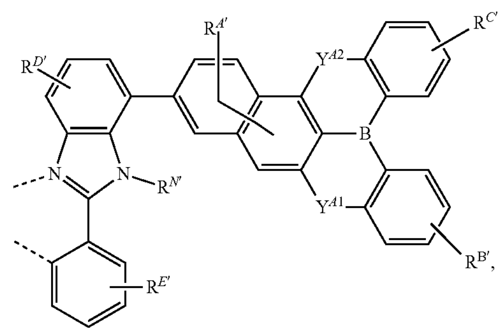

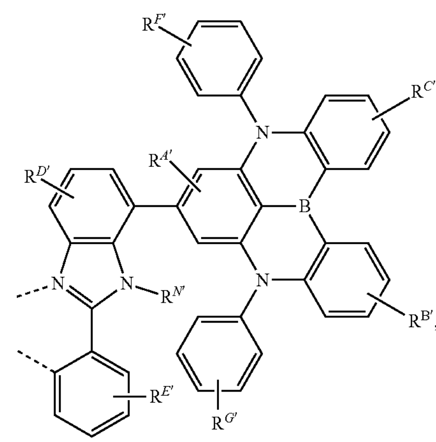

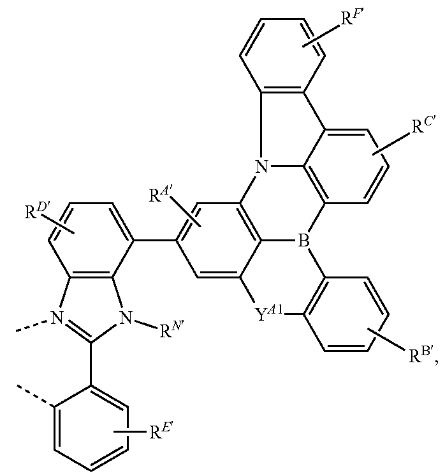

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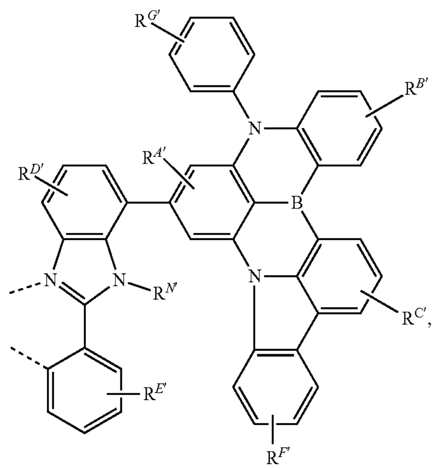

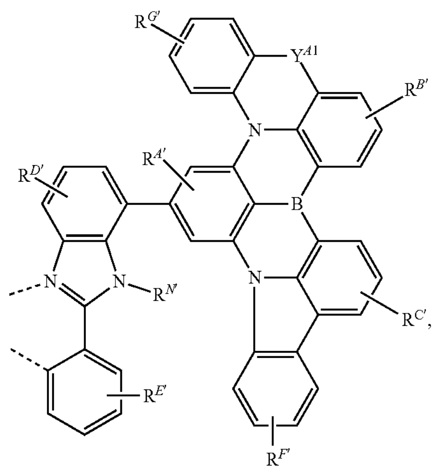

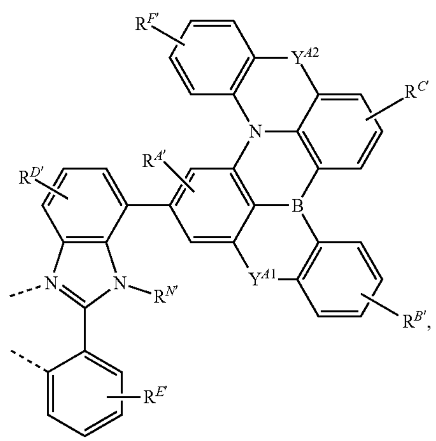

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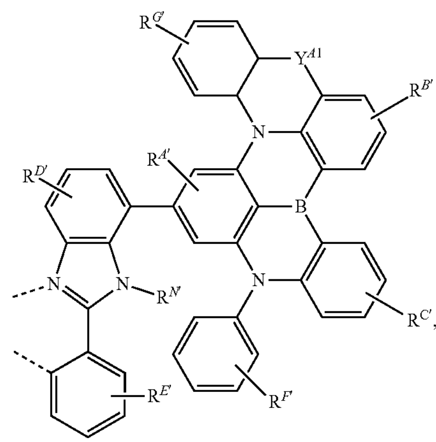

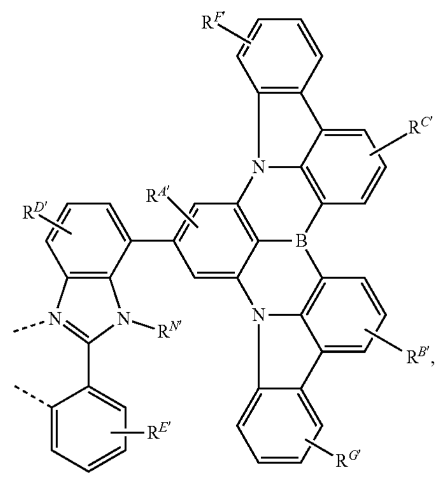

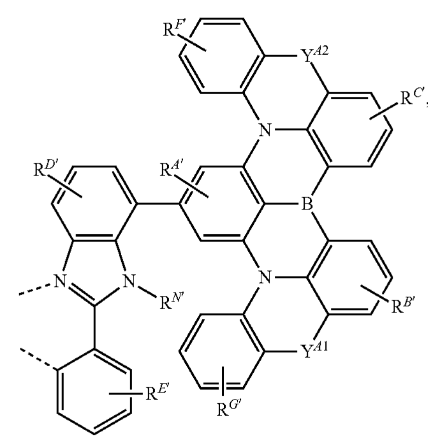

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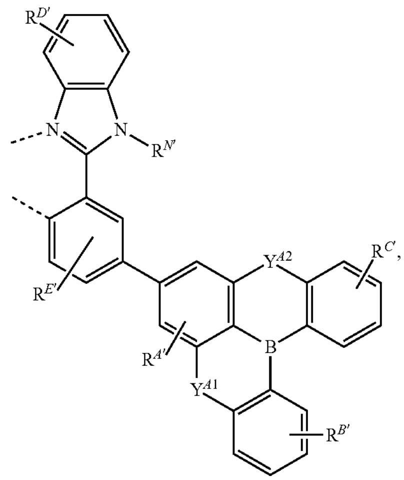

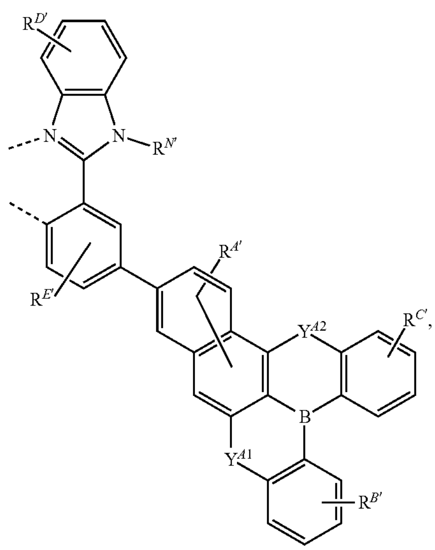

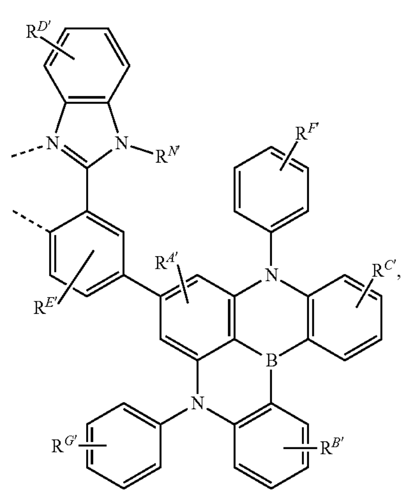

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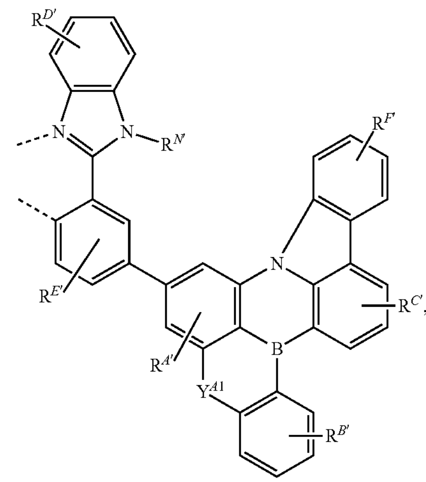

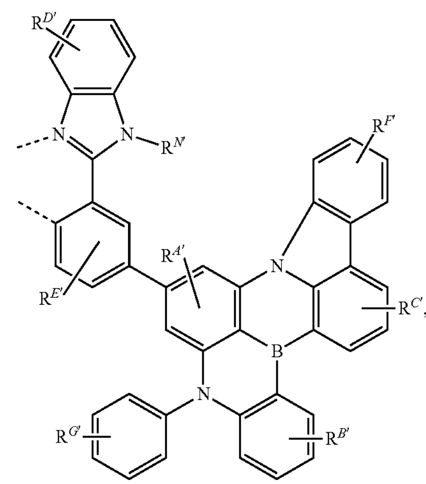

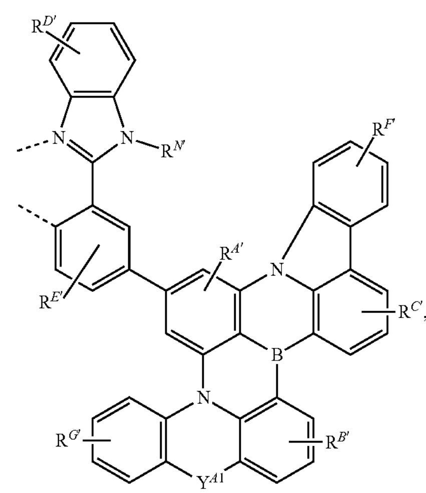

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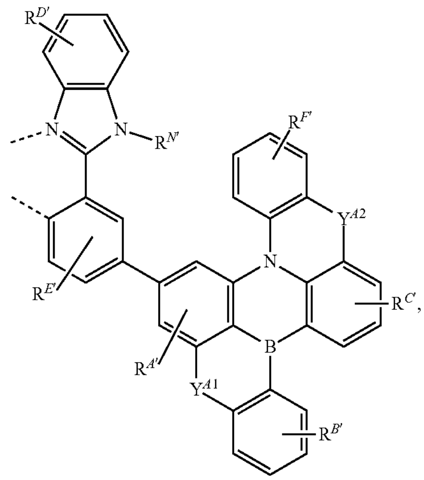

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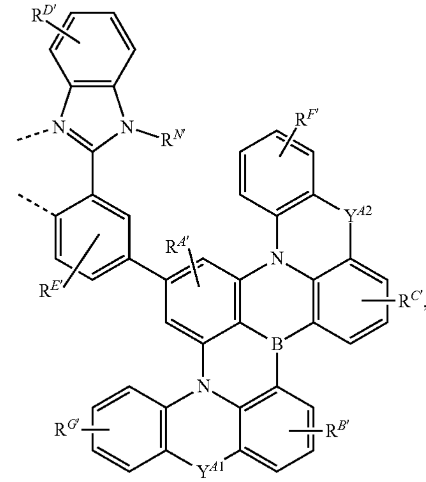

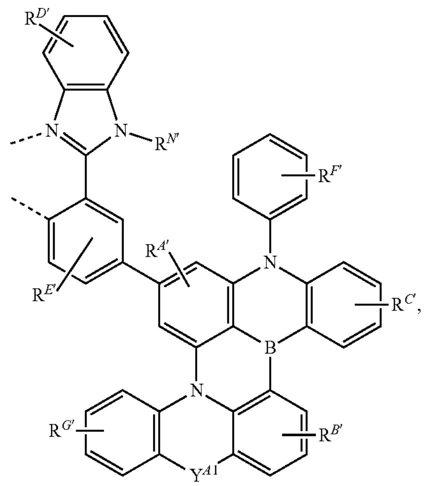

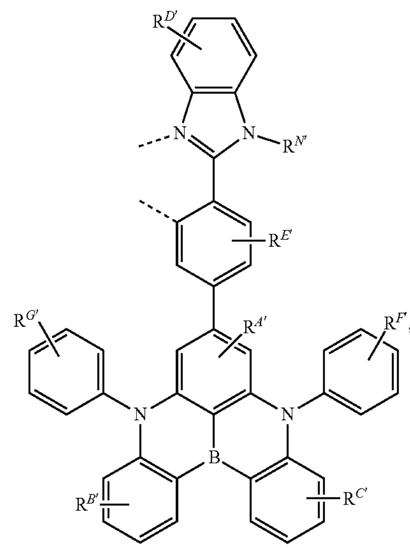

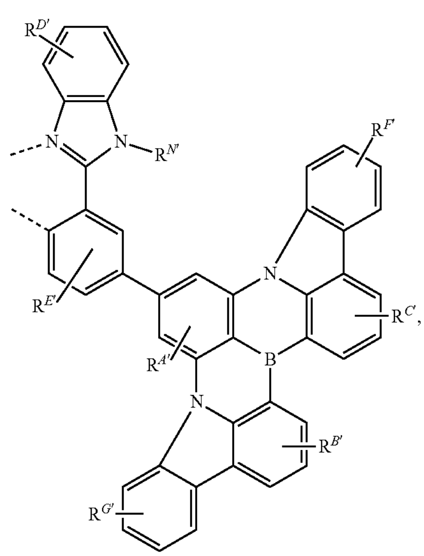

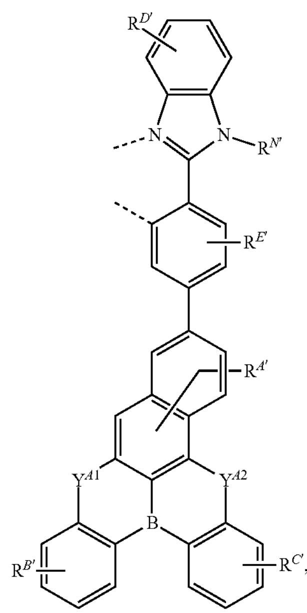

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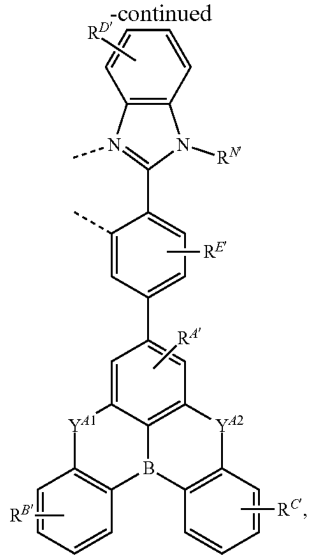

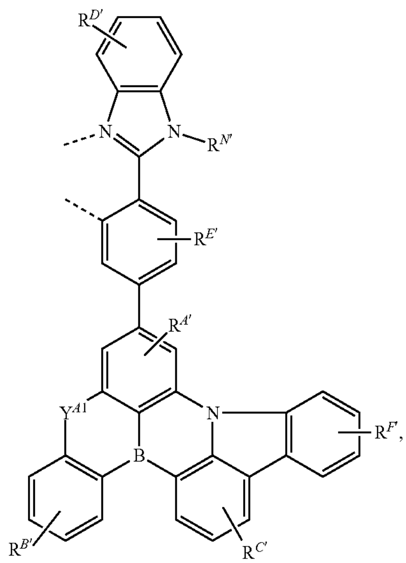

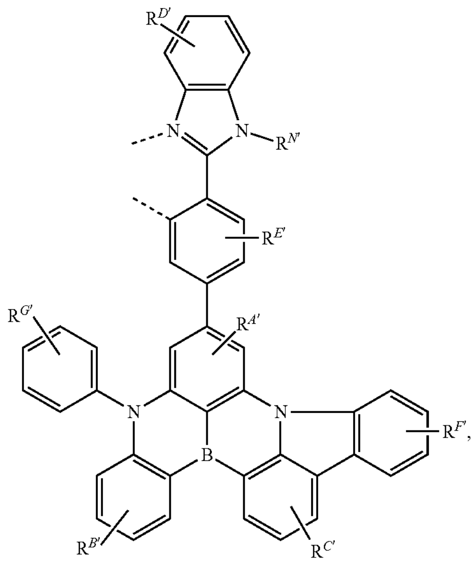

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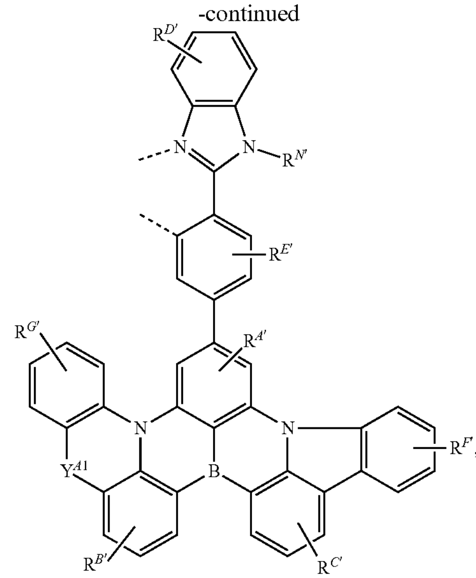

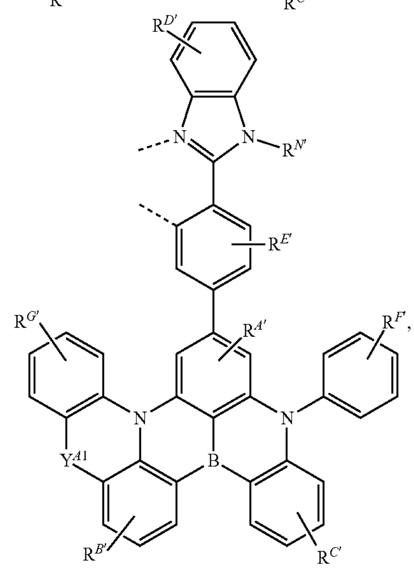

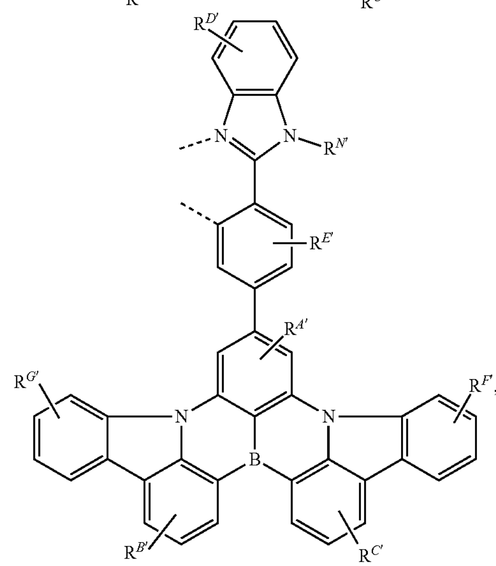

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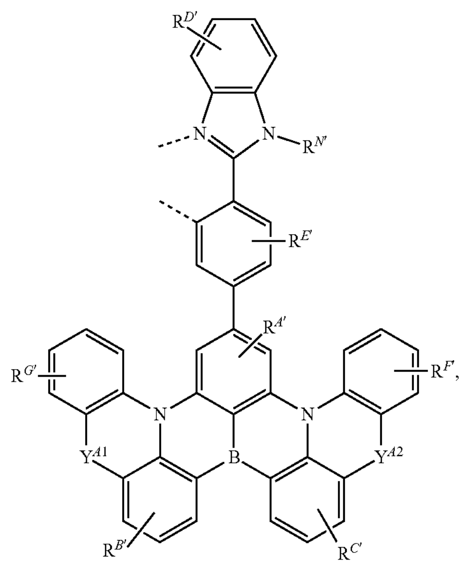

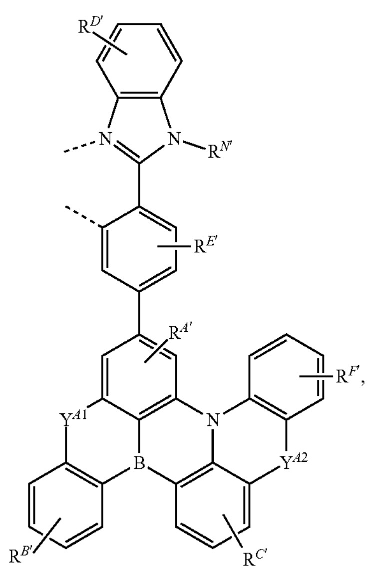

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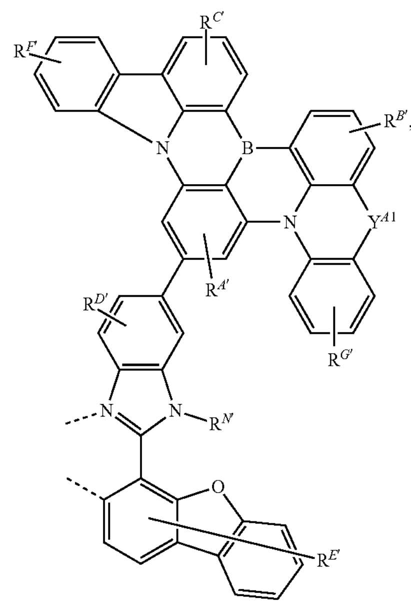

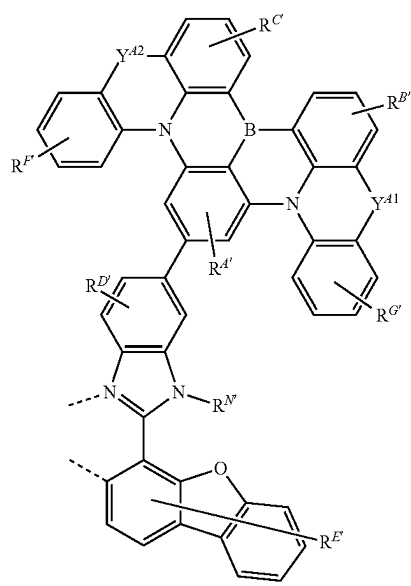

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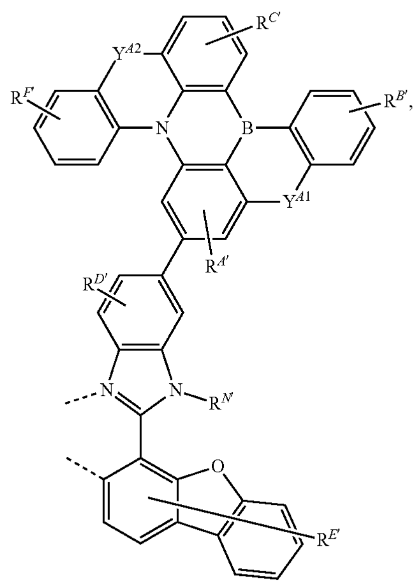

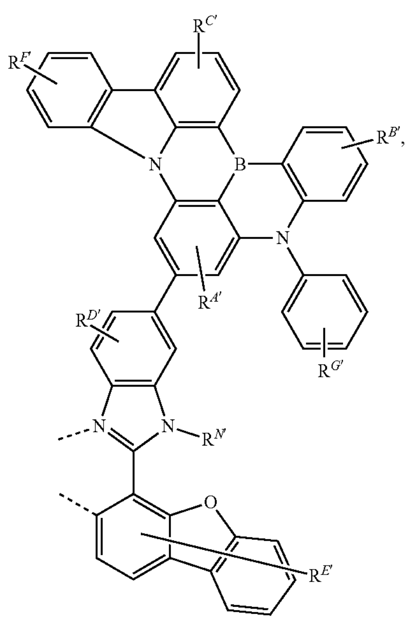

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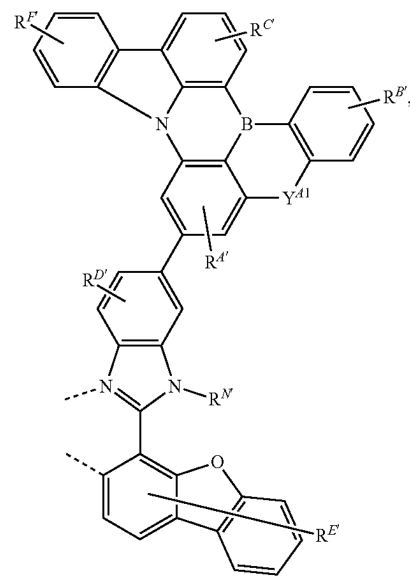

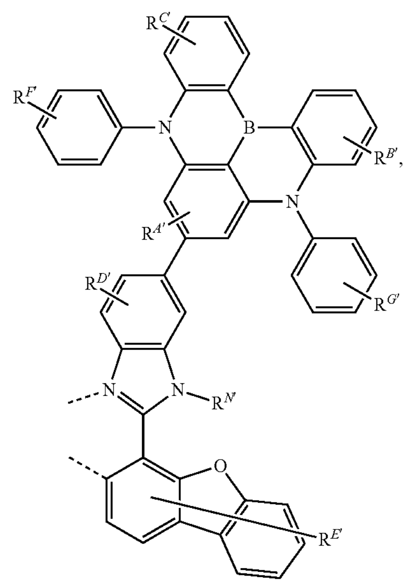

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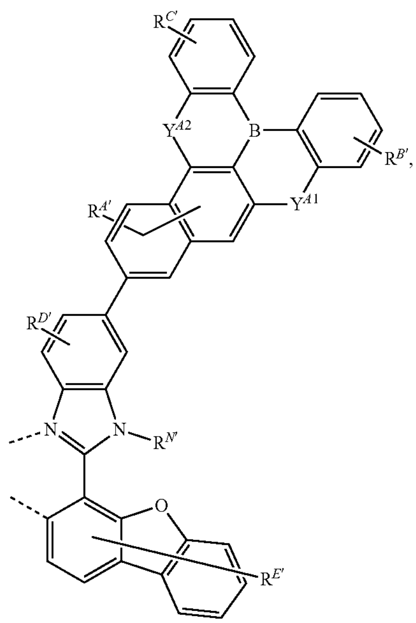

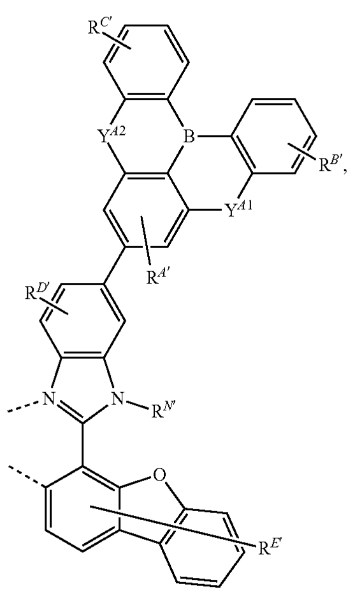

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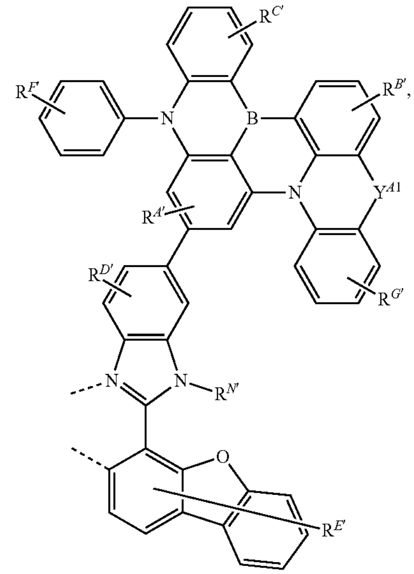

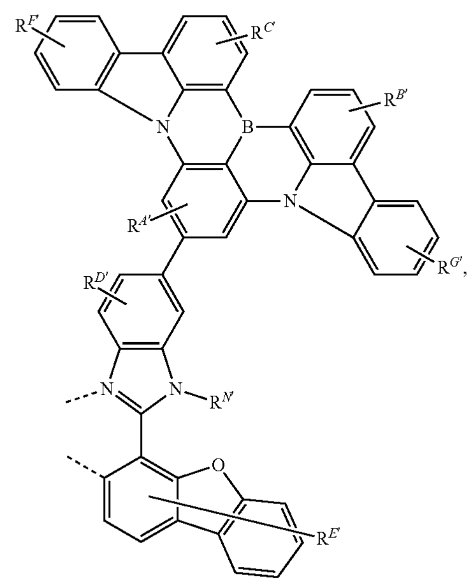

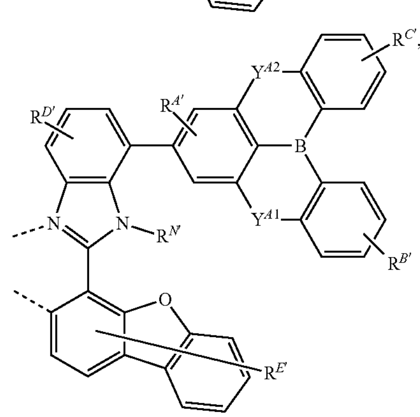

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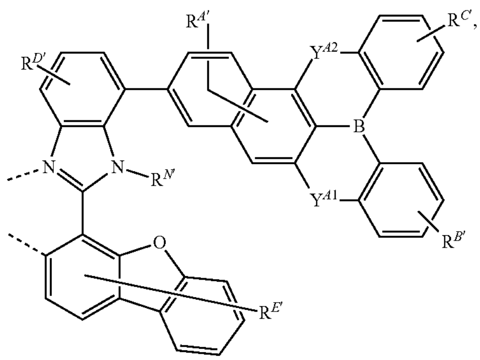

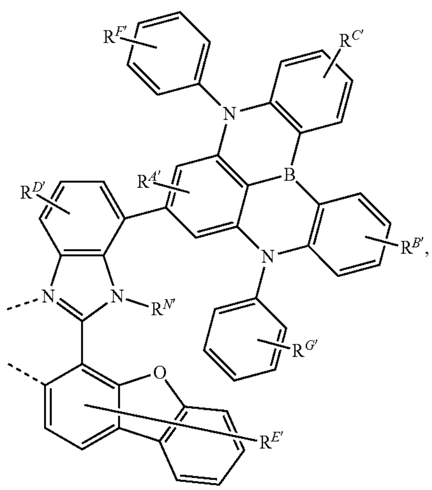

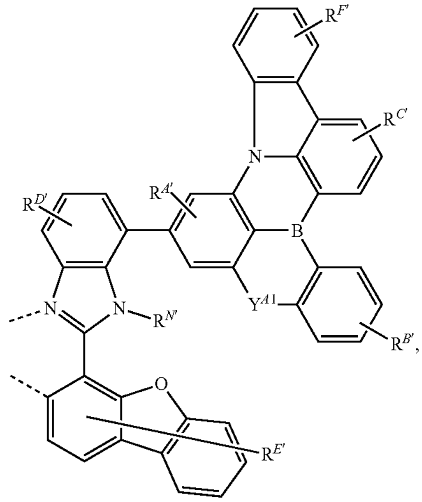

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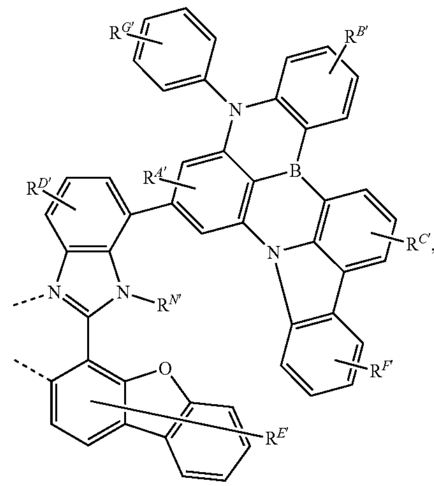

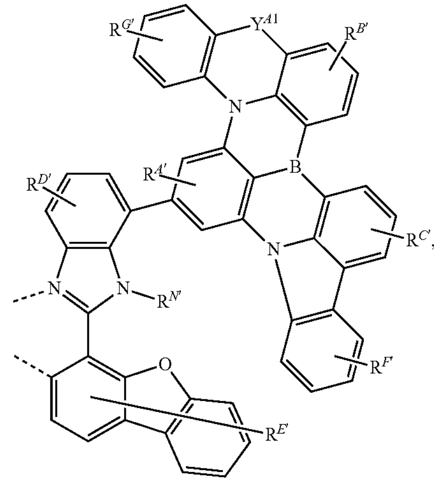

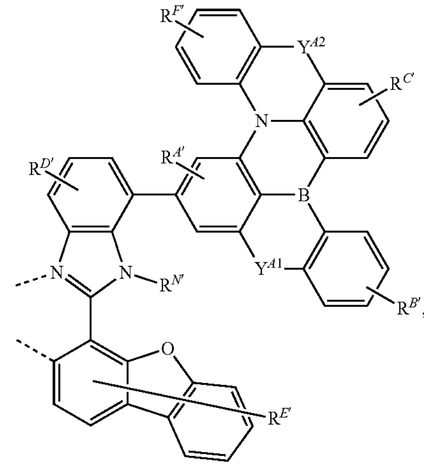

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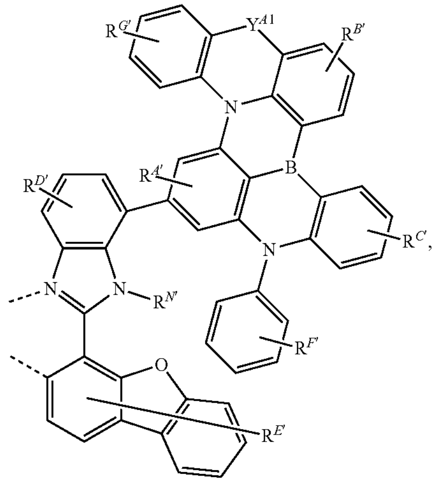

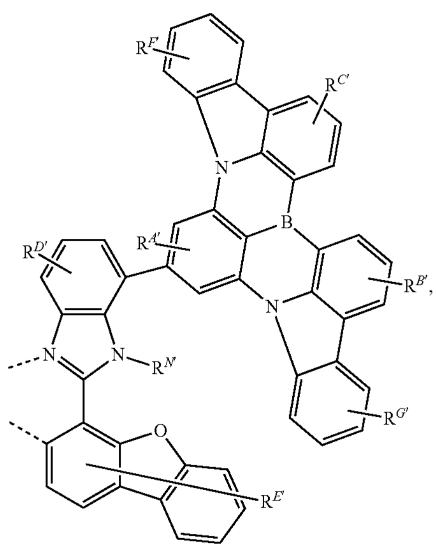

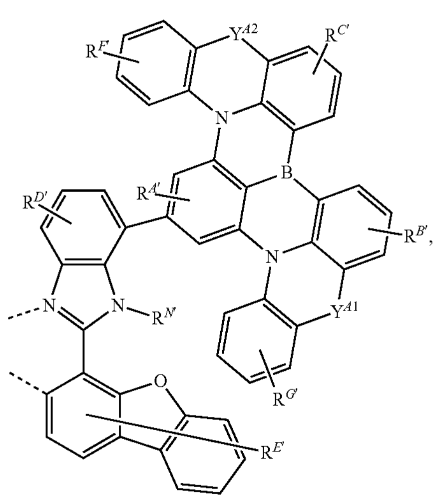

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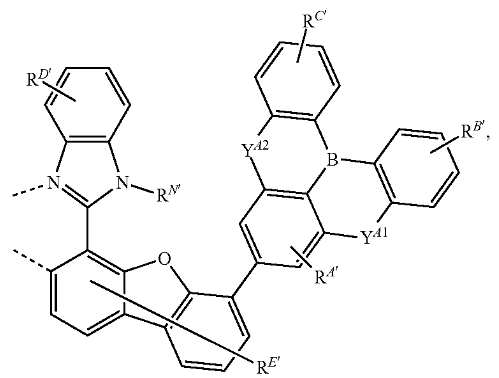

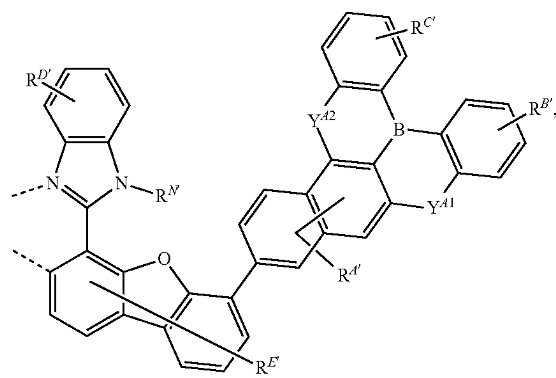

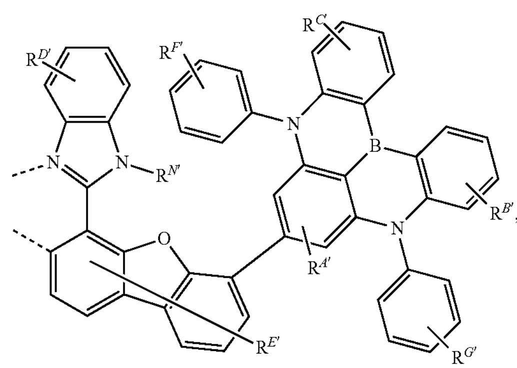

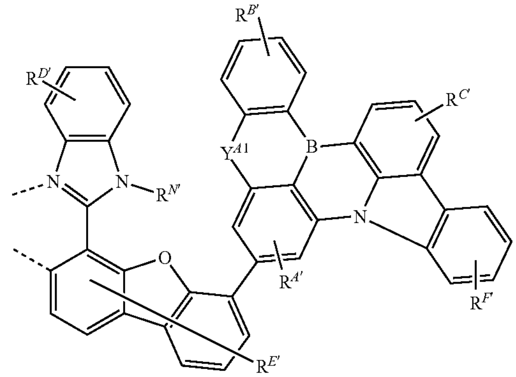

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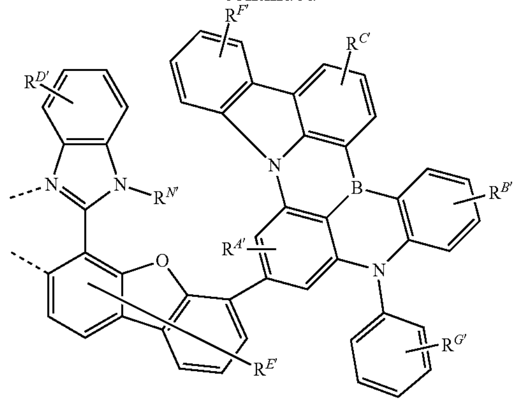

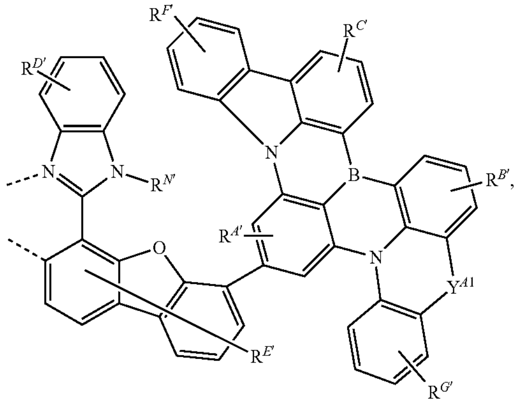

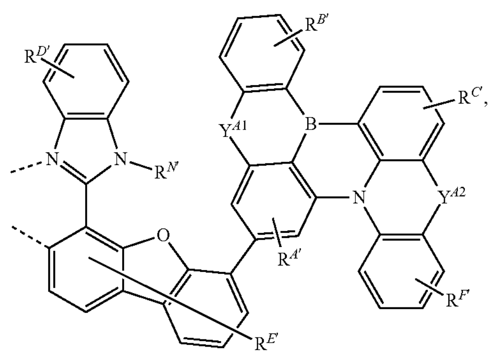

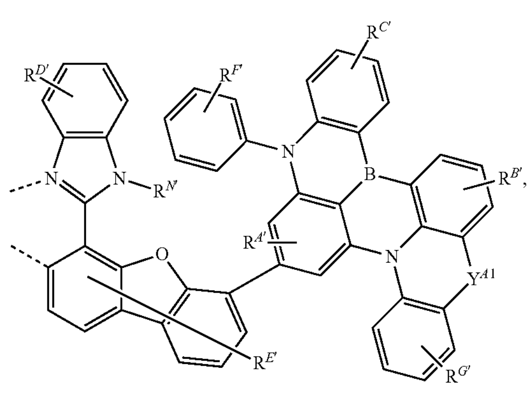

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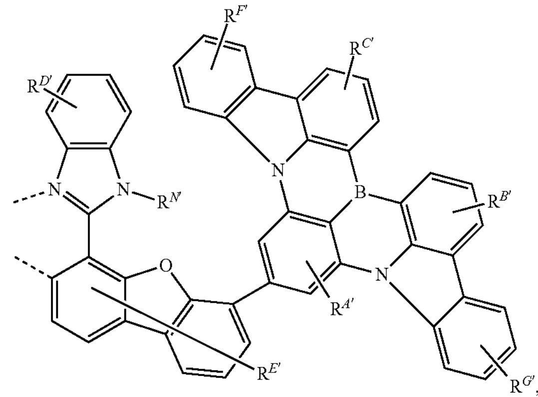

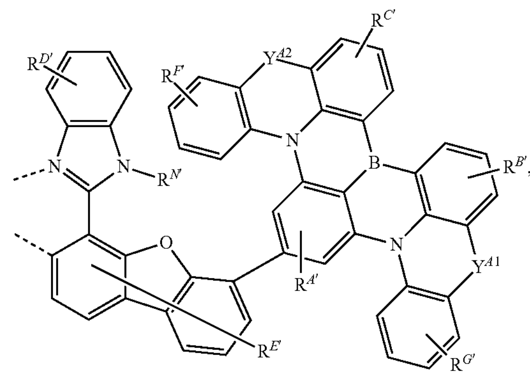

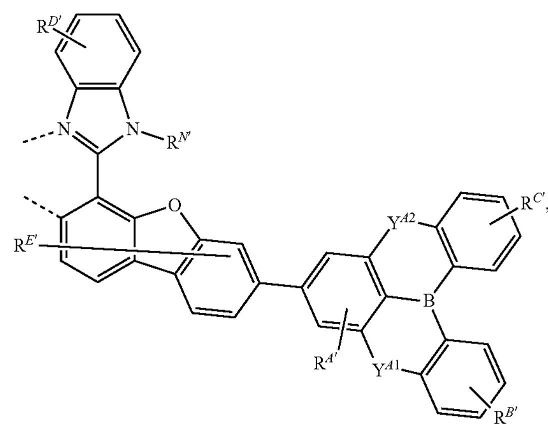

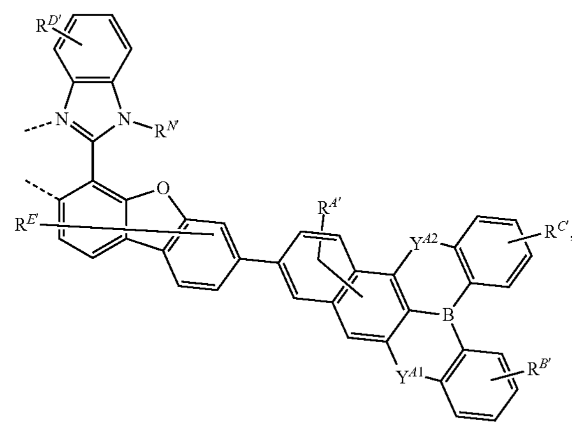

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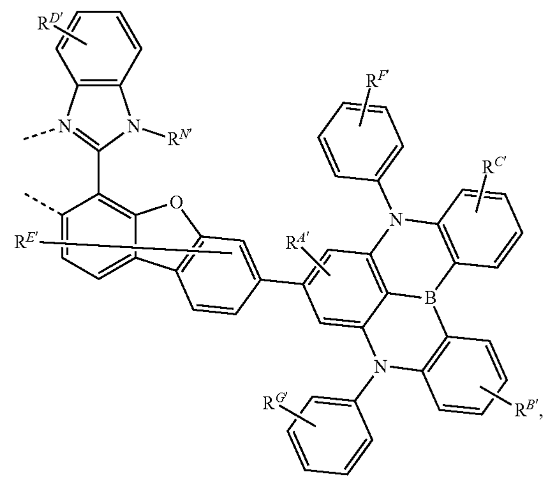

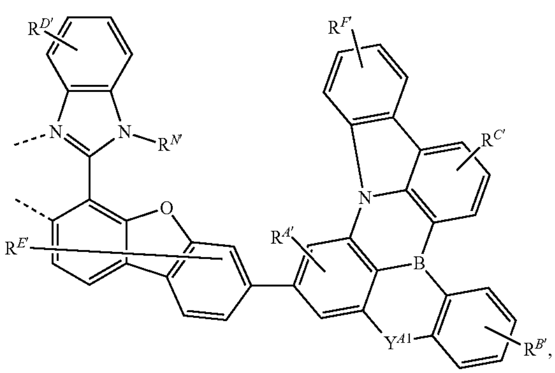

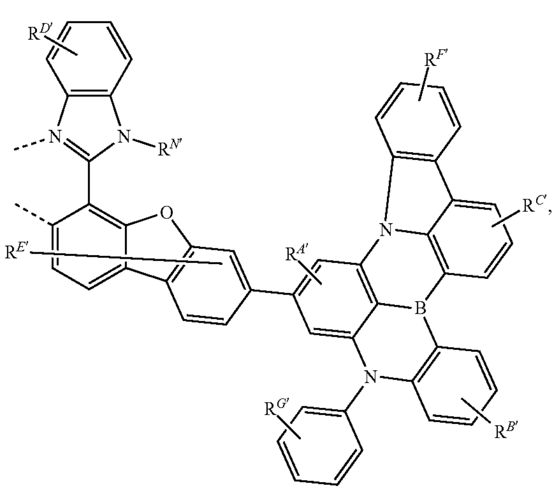

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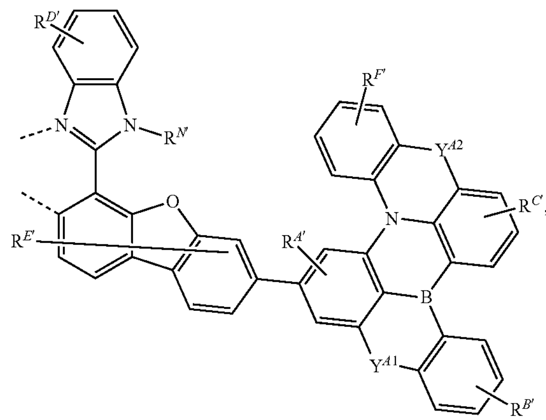

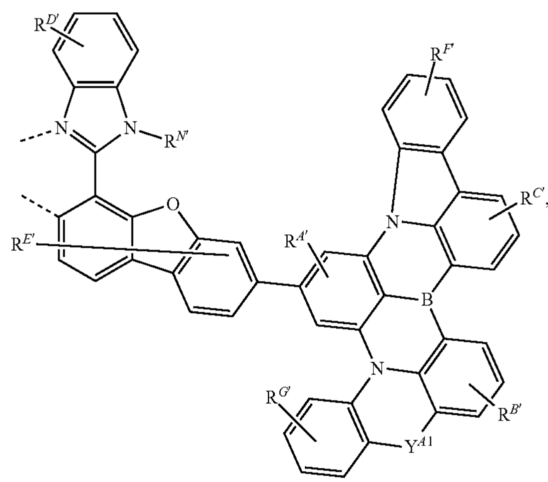

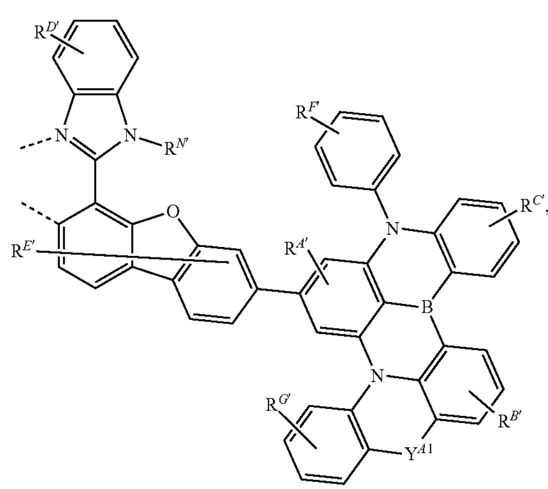

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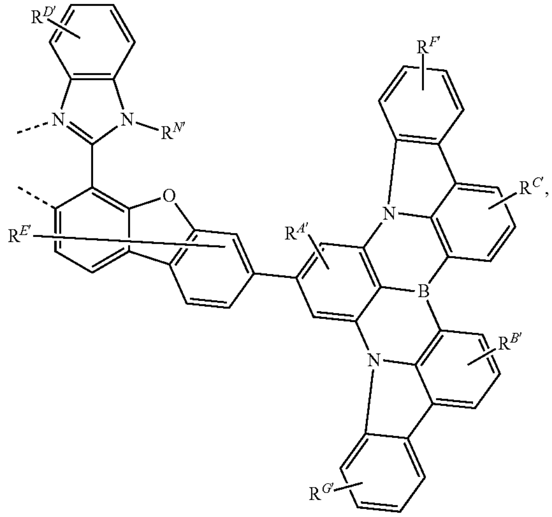

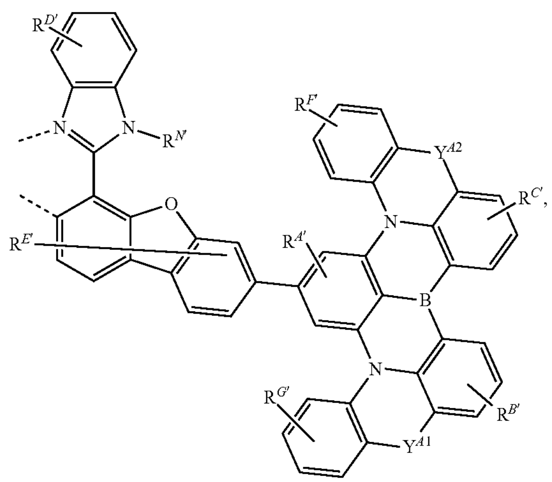

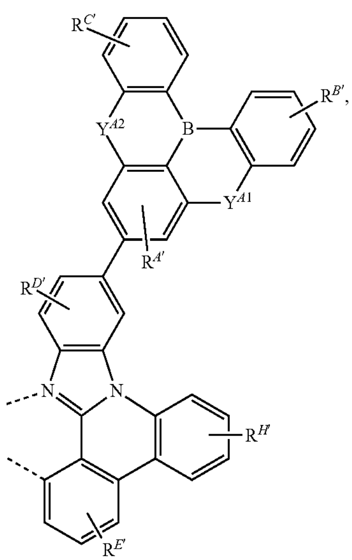

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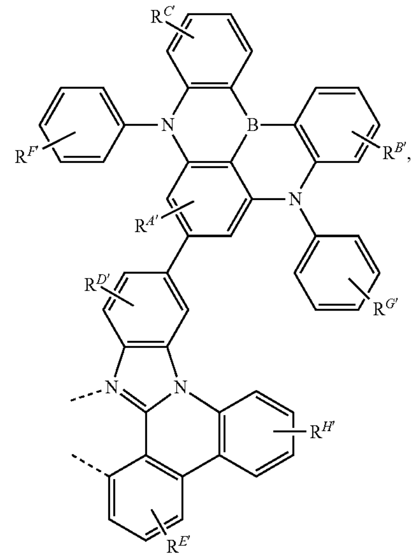

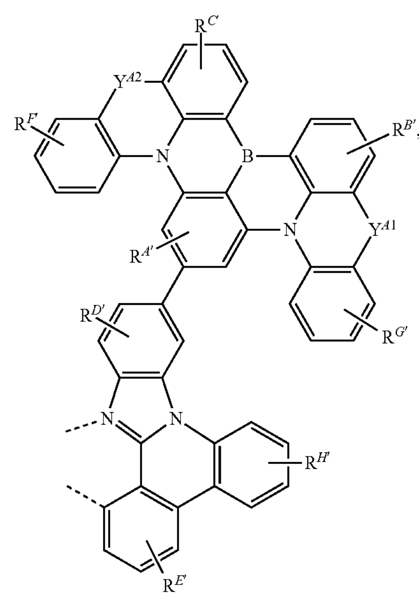

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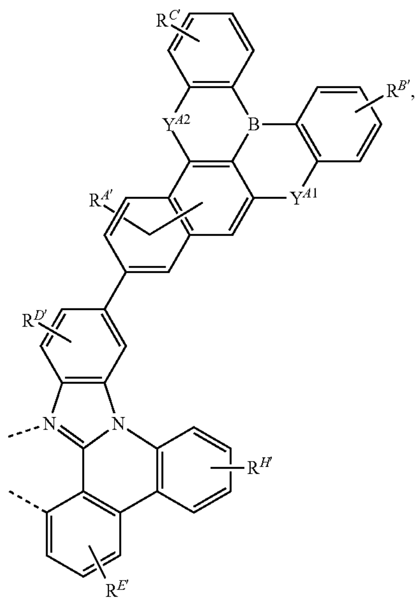

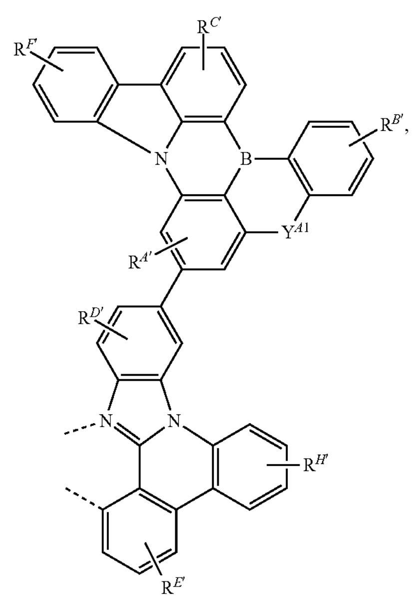

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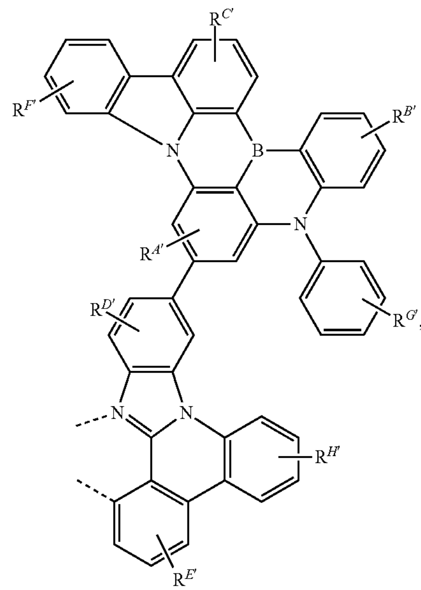

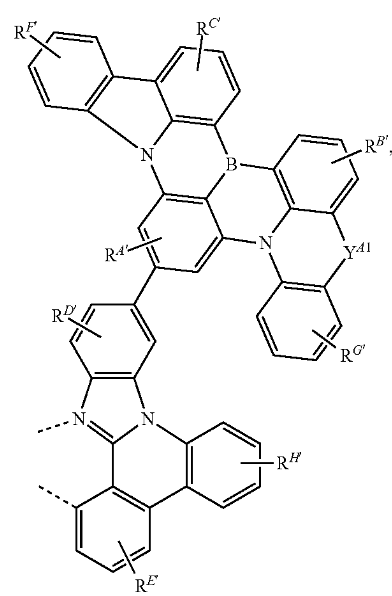

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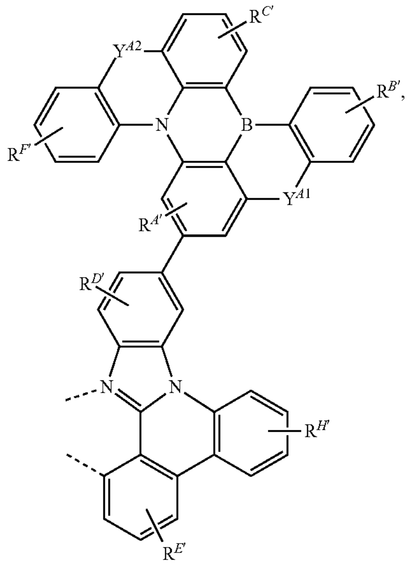

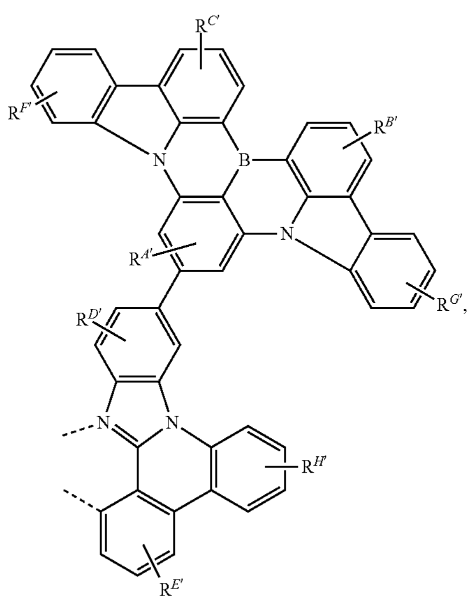

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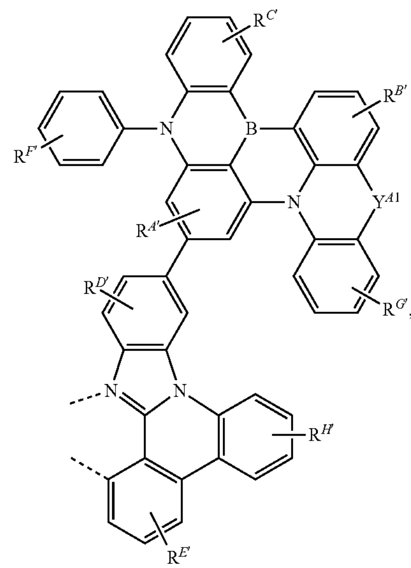

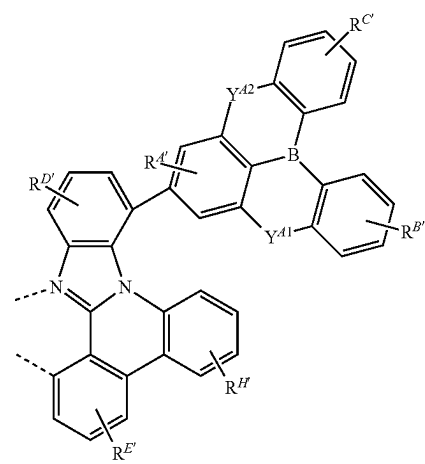

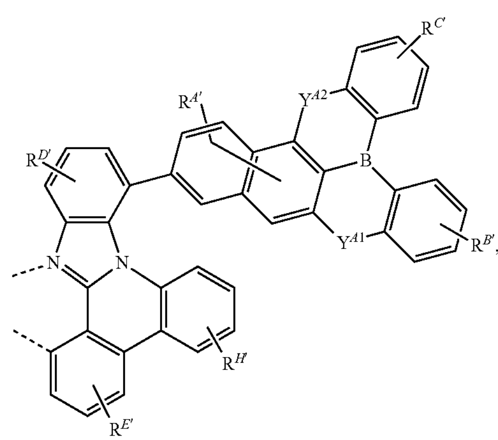

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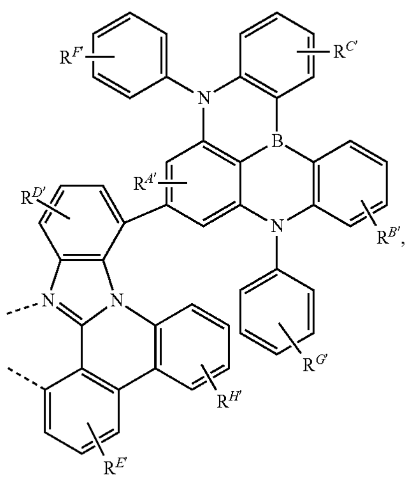

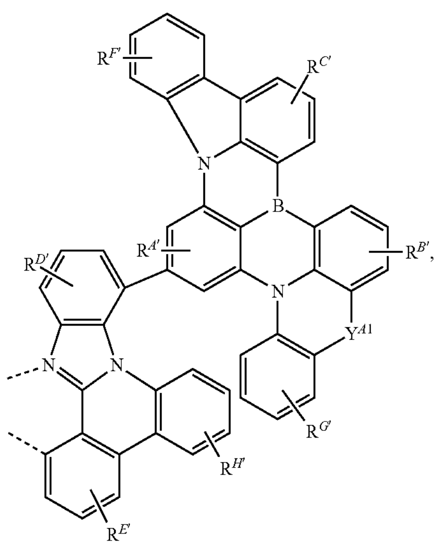

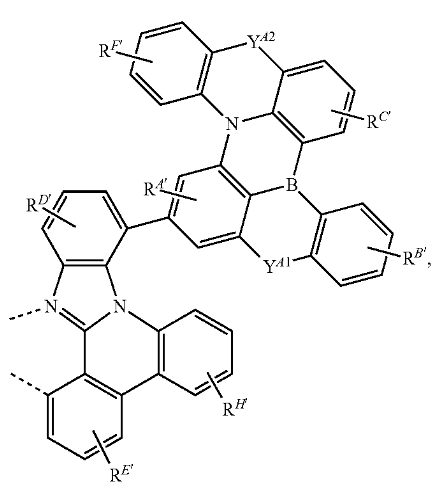

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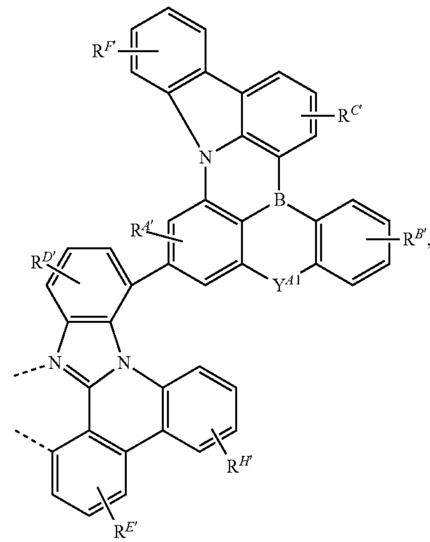

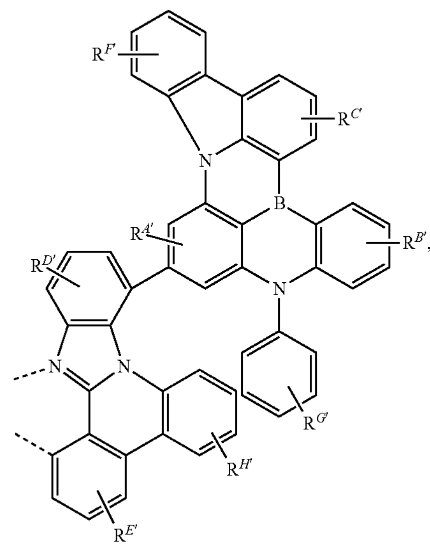

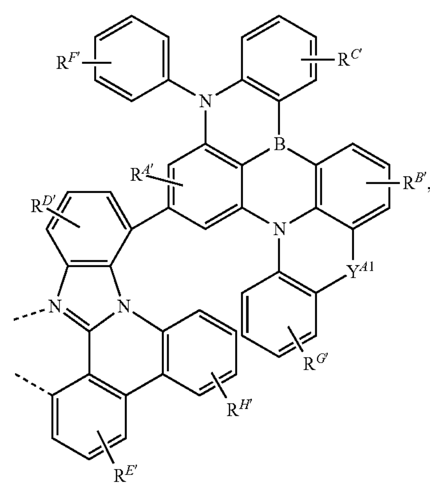

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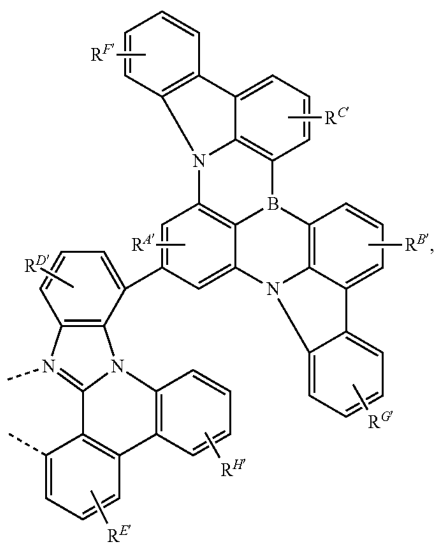

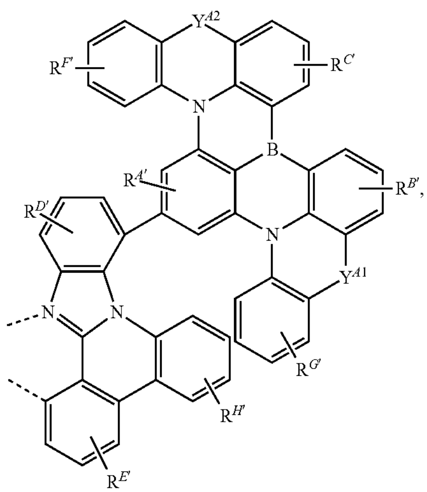

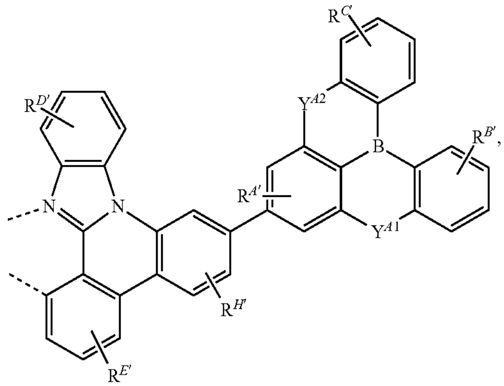

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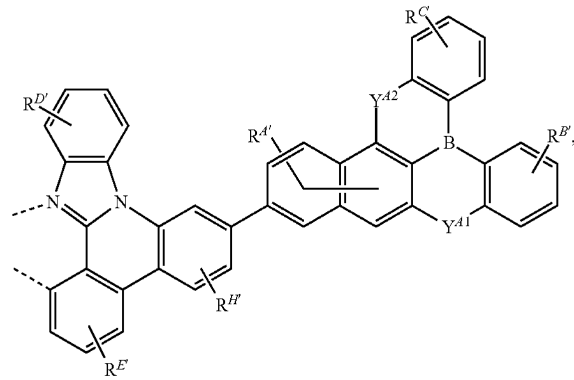

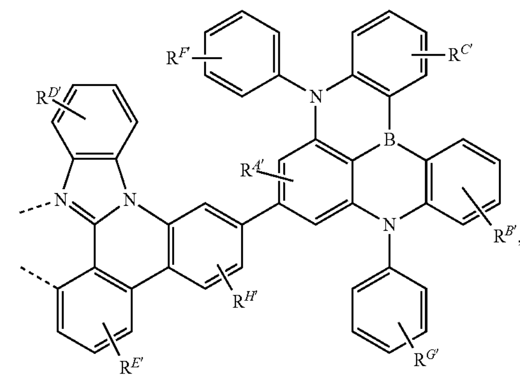

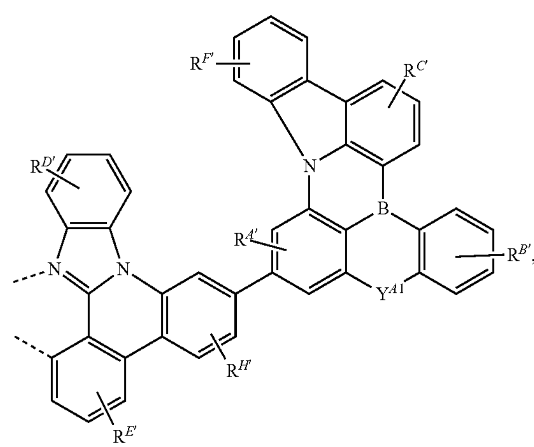

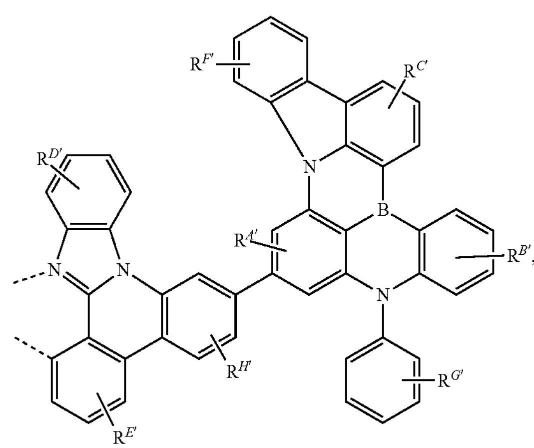

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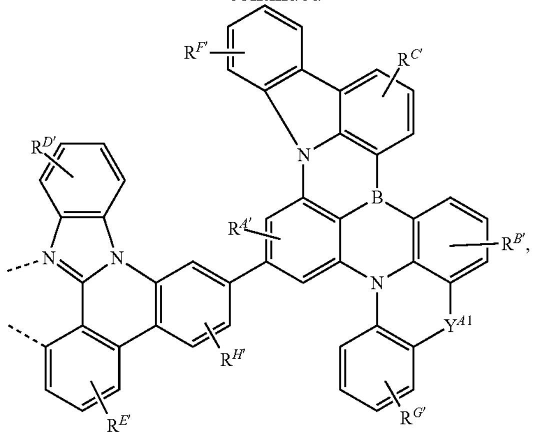

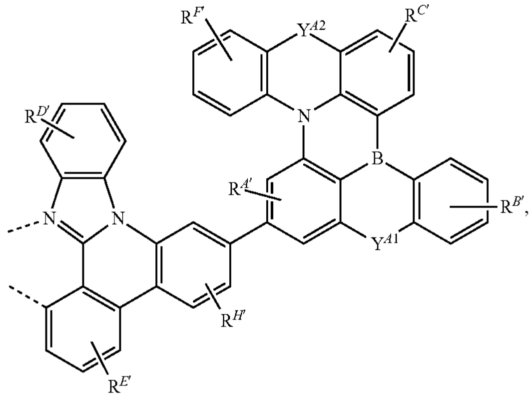

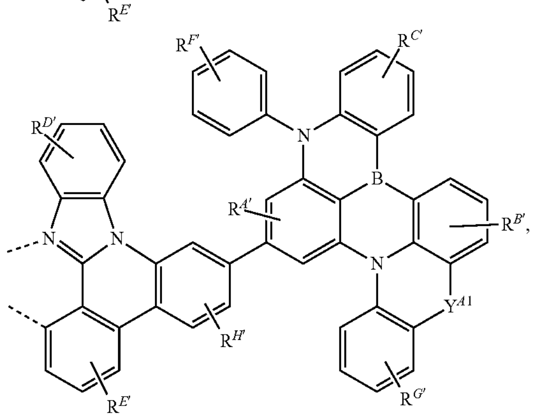

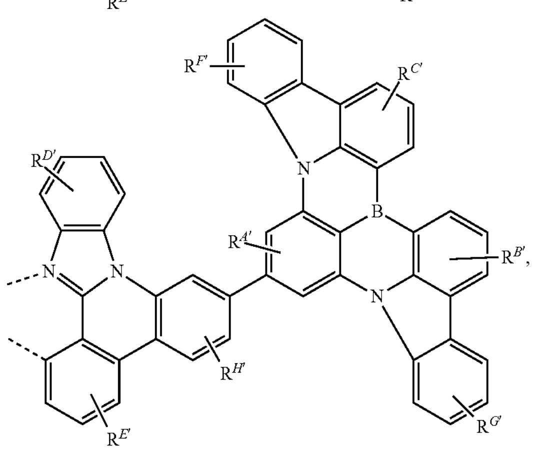

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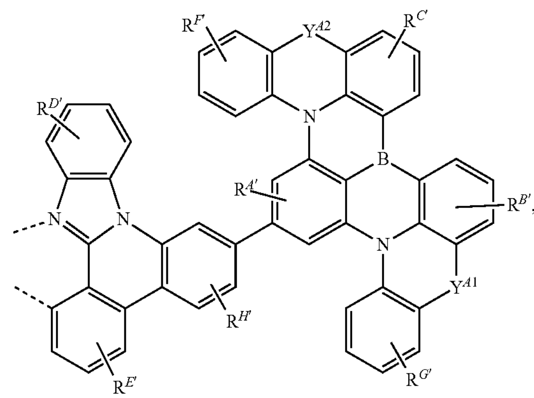

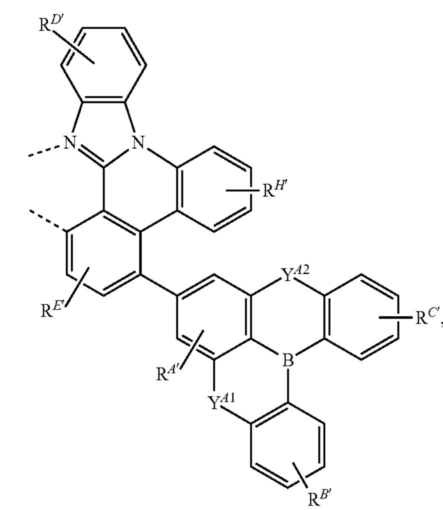

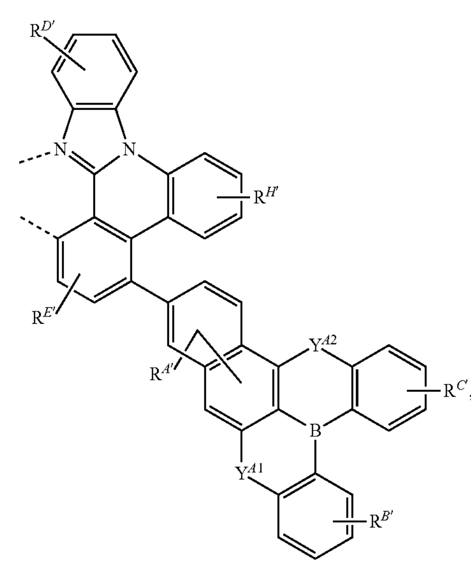

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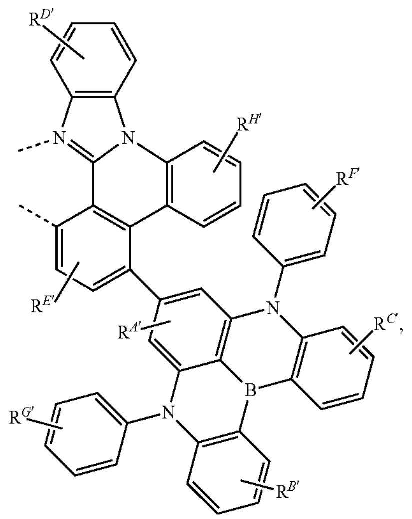

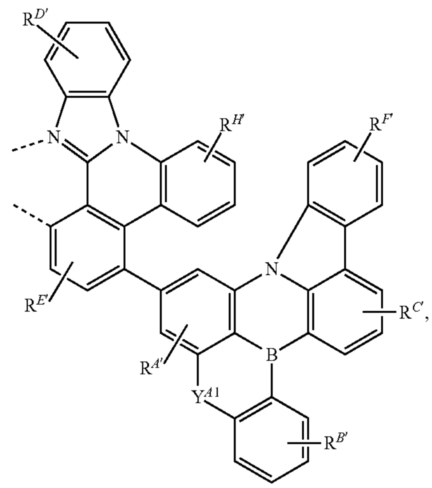

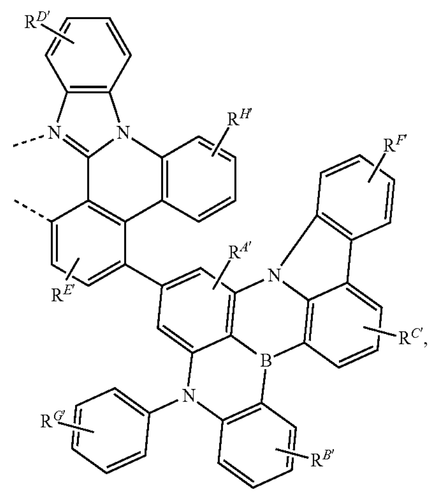

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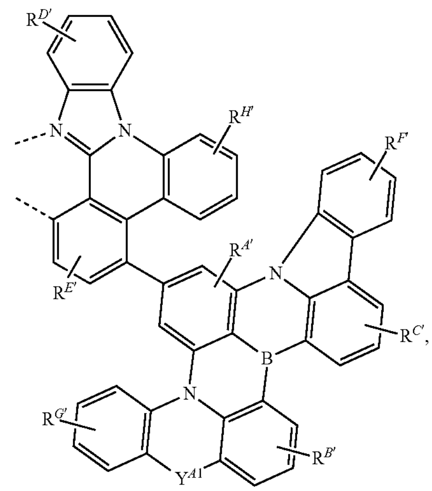

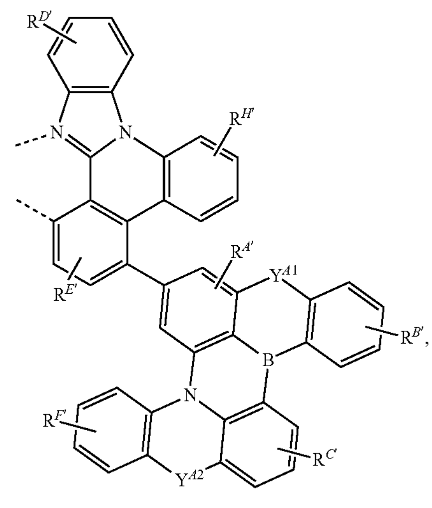

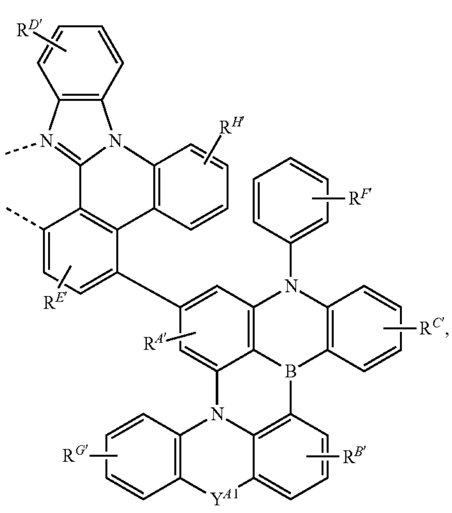

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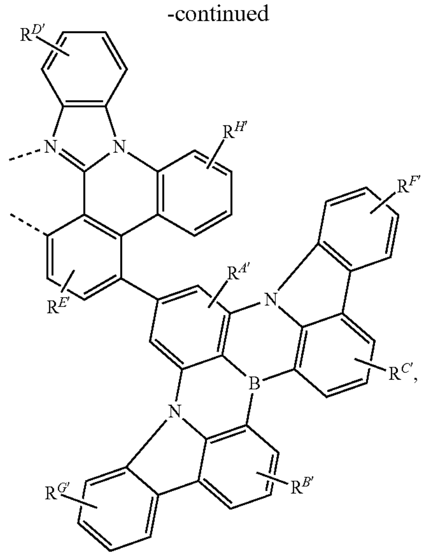

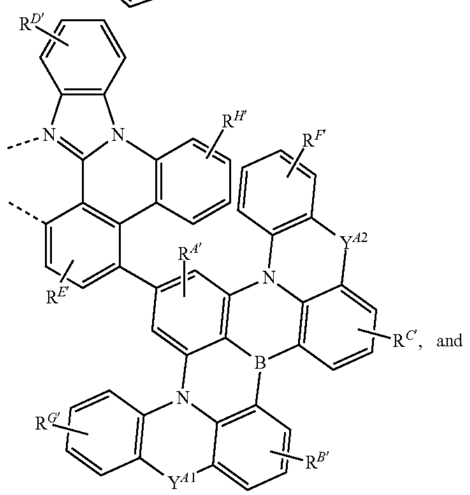

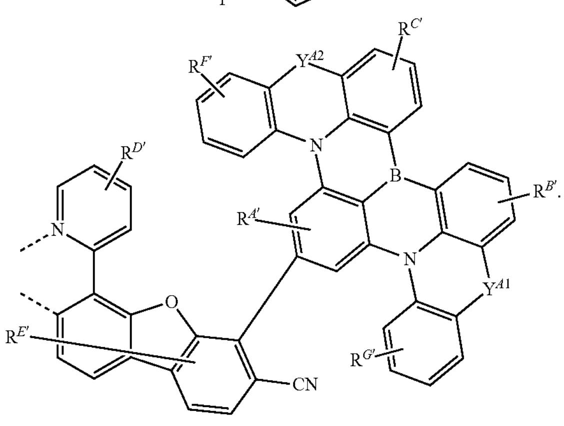

wherein

[0262] each of Y^{A1} and Y^{A2} is independently selected from the group consisting of a direct bond, BR, BRR', NR, PR, P(O)R, O, S, Se, C═O, C═S, C═Se, C═NR, C═CRR', S═O, SO, CR, CRR', SiRR', and GeRR';

[0263] each of the two - - - - represents chelation to the metal;

[0264] each of $R^{A'}$, $R^{B'}$, $R^{C'}$, $R^{D'}$, $R^{E'}$, $R^{F'}$, $R^{G'}$, and $R^{H'}$ independently represents mono to the maximum allowable substitution;

[0265] each R, R', $R^{A'}$, $R^{B'}$, $R^{C'}$, $R^{D'}$, $R^{E'}$, $R^{F'}$, $R^{G'}$, $R^{H'}$, and $R^{N'}$ is independently hydrogen or a substituent selected from the group consisting of the General Substituents; and

[0266] any two substituents may be optionally joined or fused to form a ring.

[0267] In some embodiments where ligand L_A is selected from LIST 4, two $R^{D'}$ can be joined to form a fused ring. In some such embodiments, the fused ring may be benzene, pyridine, pyrimidine, pyridazine, pyrazine, triazine, imidazole, pyrazole, pyrrole, oxazole, furan, thiophene, or thiazole. In some such embodiments, the fused ring may be benzene.

[0268] In some embodiments where ligand L_A is selected from LIST 4, at least one $R^{A'}$, $R^{B'}$, $R^{C'}$, $R^{D'}$, $R^{E'}$ $R^{F'}$, $R^{G'}$, $R^{H'}$, or $R^{N'}$ is selected from the group consisting of the General Substituents defined herein. In some embodiments, at least one $R^{A'}$ is selected from the group consisting of the General Substituents defined herein. In some embodiments, at least one $R^{B'}$ is selected from the group consisting of the General Substituents defined herein. In some embodiments, at least one $R^{C'}$ is selected from the group consisting of the General Substituents defined herein. In some embodiments, at least one $R^{D'}$ is selected from the group consisting of the General Substituents defined herein. In some embodiments, at least one $R^{E'}$ is selected from the group consisting of the General Substituents defined herein. In some embodiments, at least one $R^{F'}$ is selected from the group consisting of the General Substituents defined herein. In some embodiments, at least one $R^{G'}$ is selected from the group consisting of the General Substituents defined herein. In some embodiments, at least one $R^{H'}$ is selected from the group consisting of the General Substituents defined herein. In some embodiments, at least one $R^{N'}$ is selected from the group consisting of the General Substituents defined herein. In some embodiments, at least one $R^{A'}$, $R^{B'}$, $R^{C'}$, $R^{D'}$, $R^{E'}$ $R^{F'}$, $R^{G'}$, $R^{H'}$, or $R^{N'}$ is selected from the group consisting of the Preferred General Substituents defined herein.

[0269] In some embodiments where ligand L_A is selected from LIST 4, at least one of R, R', $R^{A'}$, $R^{B'}$, $R^{C'}$, $R^{D'}$, $R^{E'}$ $R^{F'}$, $R^{G'}$, $R^{H'}$, or $R^{N'}$ is partially or fully deuterated. In some embodiments, at least one $R^{A'}$ is partially or fully deuterated. In some embodiments, at least one $R^{B'}$ is partially or fully deuterated. In some embodiments, at least one $R^{C'}$ is partially or fully deuterated. In some embodiments, at least one $R^{D'}$ is partially or fully deuterated. In some embodiments, at least one $R^{E'}$ is partially or fully deuterated. In some embodiments, at least one $R^{F'}$ is partially or fully deuterated. In some embodiments, at least one $R^{G'}$ is partially or fully deuterated. In some embodiments, at least one $R^{H'}$ is partially or fully deuterated. In some embodiments, at least one $R^{N'}$ is partially or fully deuterated. In some embodiments, at least one R or R' is partially or fully deuterated.

[0270] In some embodiments, the ligand L_A is selected from L_{AAi}, wherein i is an integer from 1 to 15, and each L_{AAi} is defined in the following LIST 5:

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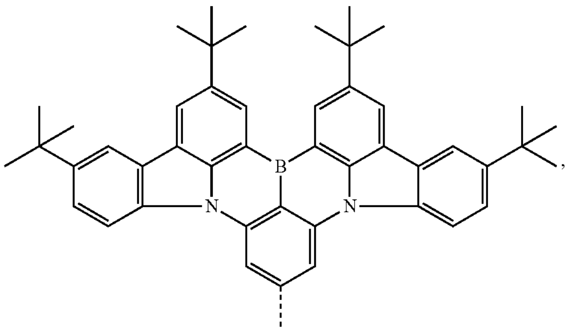
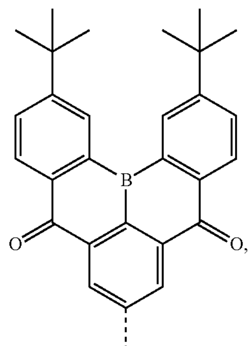
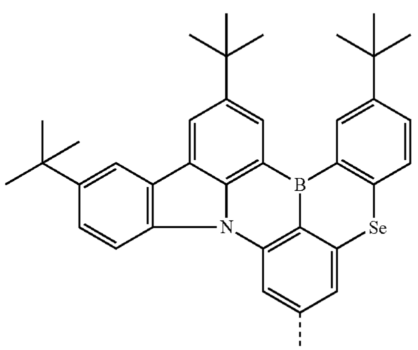
L_{AA1}



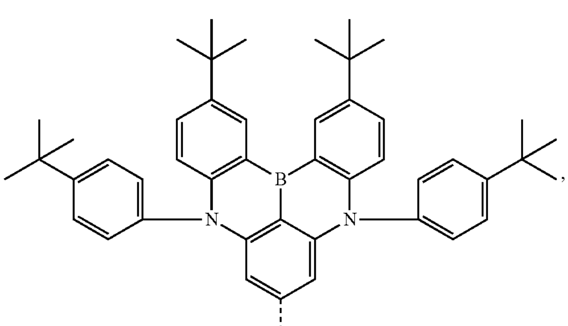
L_{AA2}



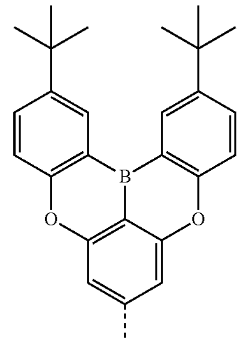
L_{AA3}



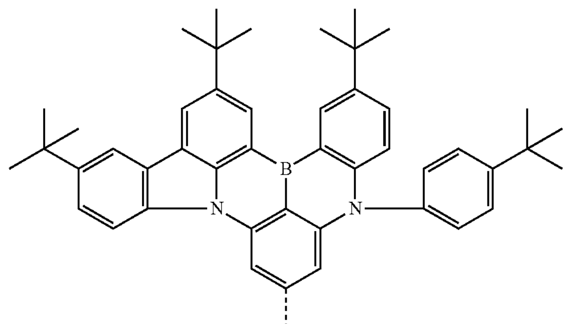
L_{AA4}



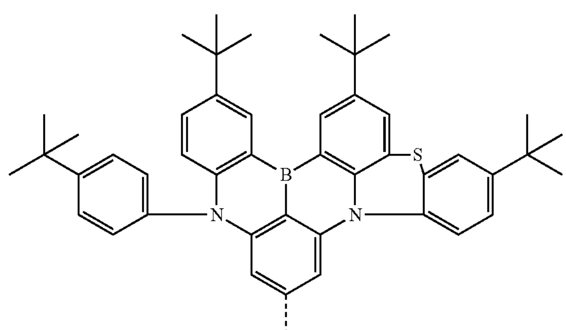
L_{AA5}



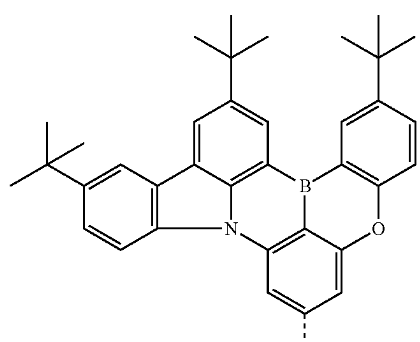
L_{AA6}



L_{AA7}



L_{AA8}



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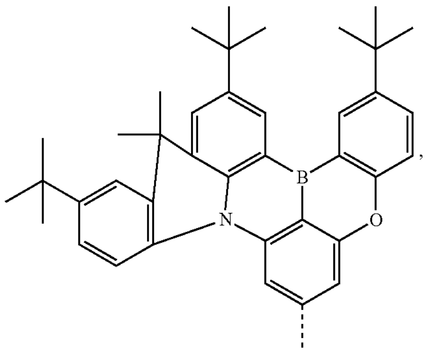
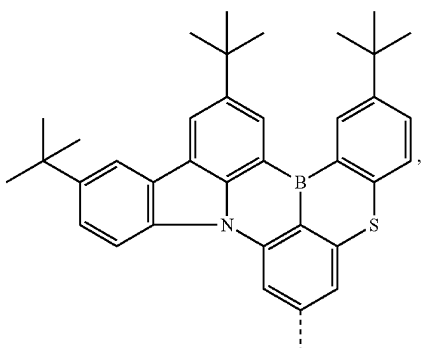
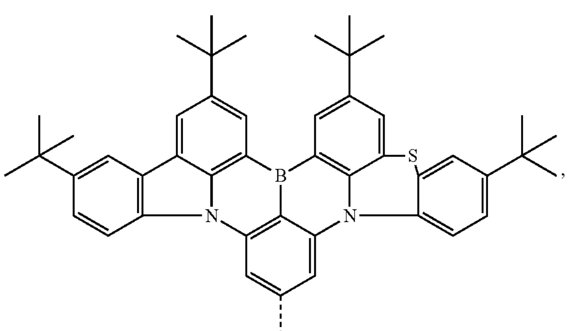
L_{AA9}



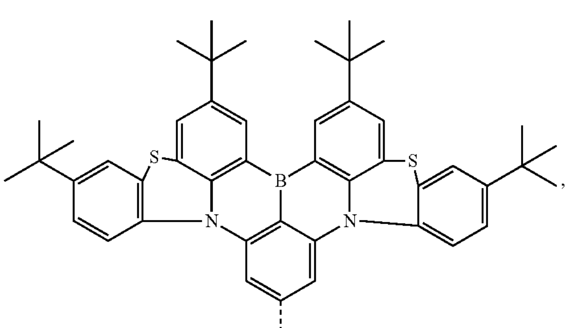
L_{AA10}



L_{AA11}



L_{AA12}



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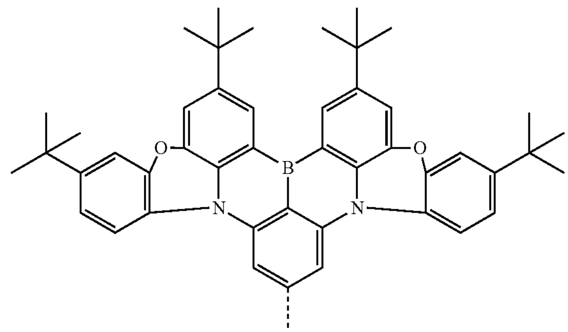
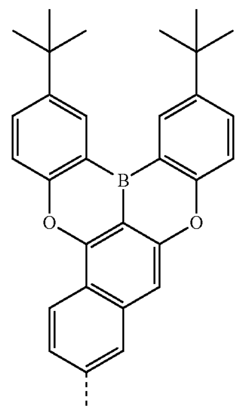
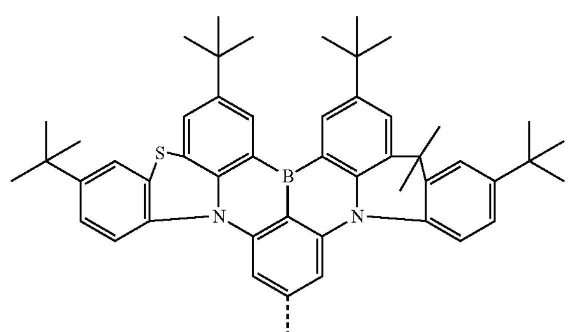
L_{AA13}



L_{AA14}

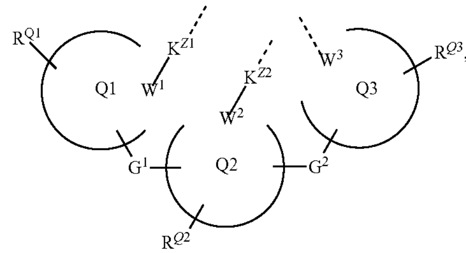


L_{AA15}



[0271] In some embodiments, the compound has a Formula $Ir(L_A)(L_{B'})(L_C)$ or $Pt(L_A)(L_{B'})$, wherein $L_{B'}$ is a tridentate ligand, L_C is a bidentate ligand.

[0272] In some embodiments, $L_{B'}$ may be a structure of Formula Z shown below:



wherein W^1, W^2, and W^3 are each independently C or N; each of moiety Q1, moiety Q2, and moiety Q3 is independently a monocyclic ring or a polycyclic fused ring system, wherein the monocyclic ring or each ring of the polycyclic fused ring system is independently a 5-membered to 10-membered carbocyclic or heterocyclic ring;

- **[0273]** each of K^{Z1} and K^{Z2} is independently a direct bond, S, or O;
- **[0274]** each of G^1, and G^2 is independently a direct bond or an organic linker;
- **[0275]** each of R^{Q1}, R^{Q2}, and R^{Q3} independently represents mono to the maximum allowable substitution;
- **[0276]** each of R^{Q1}, R^{Q2}, and R^{Q3} is independently hydrogen or a substituent selected from the group consisting of deuterium, halogen, alkyl, cycloalkyl, heteroalkyl, heterocycloalkyl, arylalkyl, alkoxy, aryloxy, amino, silyl, germyl, boryl, alkenyl, cycloalkenyl, heteroalkenyl, alkynyl, aryl, heteroaryl, acyl, carboxylic acid, ether, ester, nitrile, isonitrile, sulfanyl, sulfinyl, sulfonyl, phosphino, selenyl, and combinations thereof;
- **[0277]** the three dotted lines coordinate to the metal; and
- **[0278]** any two substituents may be optionally joined or fused to form a ring.

[0279] In some embodiments of Formula Z, each of K^{Z1} and R^{Z2} is a direct bond. In some embodiments, K^{Z1} is S. In some embodiments, K^{Z1} is O. In some embodiments, K^{Z2} is O.

[0280] In some embodiments of Formula Z, each of W^1, W^2, and W^3 is N. In some embodiments, each of W^1, W^2, and W^3 is C. In some embodiments, one of W^1, W^2, and W^3 is C, and the remaining two are N. In some embodiments, one of W^1, W^2, and W^3 is N, and the remaining two are C. In some embodiments, at least one of W^1, W^2, and W^3 is a carbene carbon,

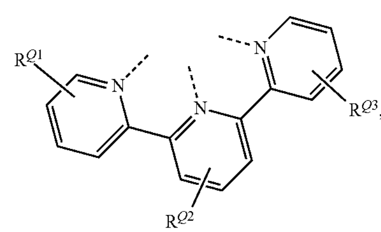
[0281] In some embodiments of Formula Z, both G^1 and G^2 are direct bonds. In some embodiments, one of G^1 and G^2 is a direct bond and the other is an organic linker. In some embodiments, both G^1 and G^2 are organic linkers. In such embodiments, the organic linker may be selected from the group consisting of BR, BRR', NR, PR, P(O)R, O, S, Se, C=O, C=S, C=Se, C=NR, C=CRR', S=O, SO_2, CR, CRR', SiRR', GeRR', alkylene, cycloalkyl, aryl, cycloalkylene, arylene, heteroarylene, and combinations thereof; wherein R and R' are the same as previously defined.

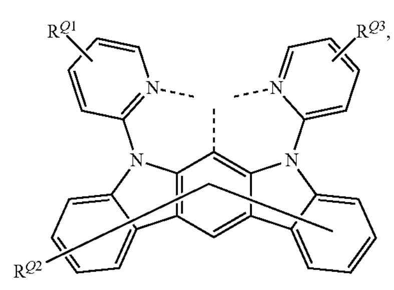
[0282] In some embodiments of Formula Z, the ring containing W^1, the ring containing W^2, and the ring containing W^3 are all 6-membered aromatic rings. In some embodiments, one of the ring containing W^1, the ring containing W^2, and the ring containing W^3 is a 6-membered aromatic ring, the remaining two are 5-membered aromatic rings. In some embodiments, one of the ring containing W^1, the ring containing W^2, and the ring containing W^3 is a 5-membered aromatic ring, the remaining two are 6-membered aromatic rings. In some embodiments, the ring containing W^1, the ring containing W^2, and the ring containing W^3 are all 5-membered aromatic rings. In some of the above embodiments, the 5-membered ring may be a imidazole ring containing a carbene carbon coordinated to the metal.

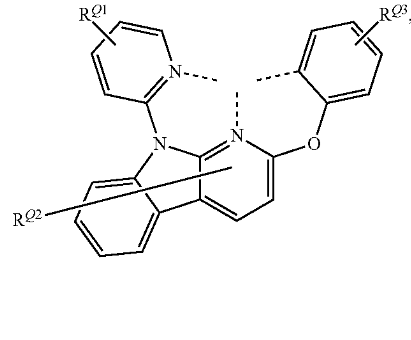
[0283] In some embodiments of Formula Z, each of moiety Q1, moiety Q2, and moiety Q3 may be independently benzene, pyridine, pyrimidine, pyridazine, pyrazine, triazine, imidazole, pyrazole, pyrrole, oxazole, furan, thiophene, thiazole, triazole, naphthalene, quinoline, isoquinoline, quinazoline, benzofuran, aza-benzofuran, benzoxazole, aza-benzoxazole, benzothiophene, aza-benzothiophene, benzothiazole, aza-benzothiazole, benzoselenophene, aza-benzoselenophene, indene, aza-indene, indole, aza-indole, benzimidazole, aza-benzimidazole, benzobenzimidazole, aza-benzobenzimidazole, carbazole, aza-carbazole, dibenzofuran, aza-dibenzofuran, phenanthro[3,2-b]benzofuran, dibenzothiophene, aza-dibenzothiophene, quinoxaline, phthalazine, phenanthrene, aza-phenanthrene, anthracene, aza-anthracene, phenanthridine, fluorene, or aza-fluorene,

[0284] In some embodiments of Formula Z, two R^{Q1} may be joined or fused to form a ring. In some embodiments, two R^{Q2} may be joined or fused to form a ring. In some embodiments, two R^{Q3} may be joined or fused to form a ring. In some embodiments, one R^{Q1} and one R^{Q2} may be joined or fused to form a ring. In some embodiments, one R^{Q2} and one R^{Q3} may be joined or fused to form a ring. In some embodiments, G^1 and R^{Q1} may be joined or fused to form a ring. In some embodiments, G^1 and R^{Q2} may be joined or fused to form a ring. In some embodiments, G^2 and R^{Q2} may be joined or fused to form a ring. In some embodiments, G^2 and R^{Q3} may be joined or fused to form a ring.

[0285] In some embodiments, $L_{B'}$ may be selected from the structures below (LIST 6a):







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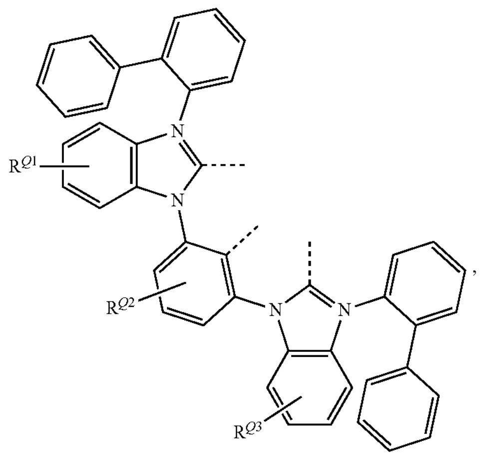

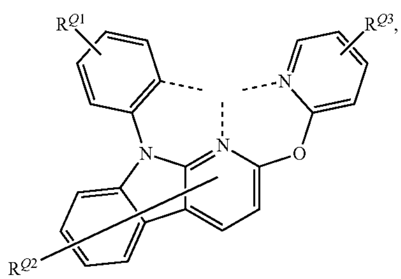

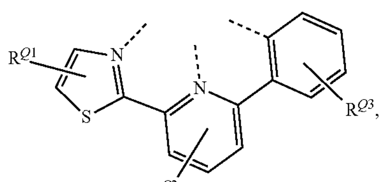

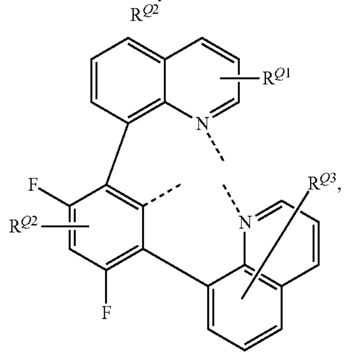

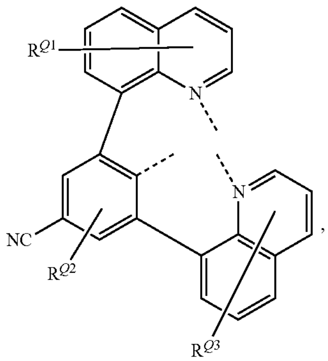

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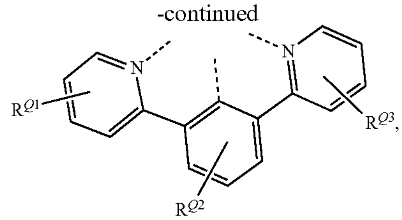

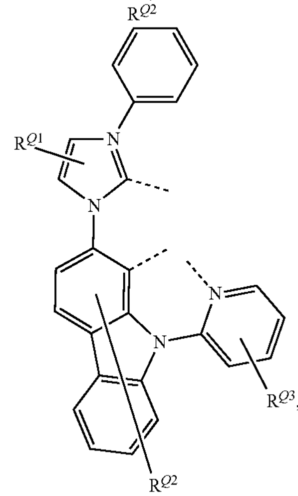

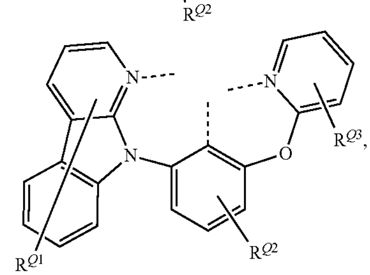

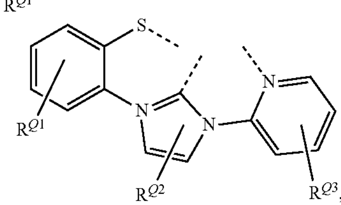

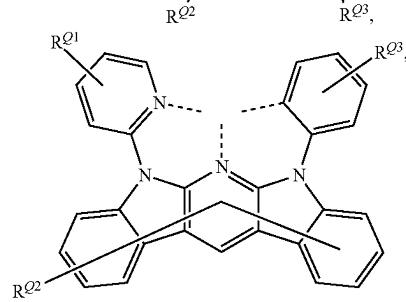

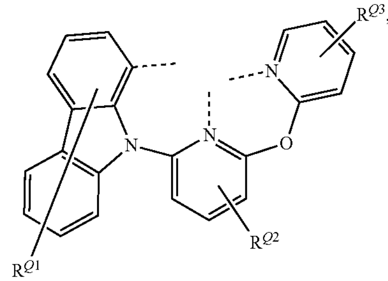

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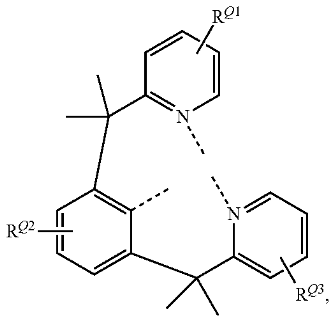

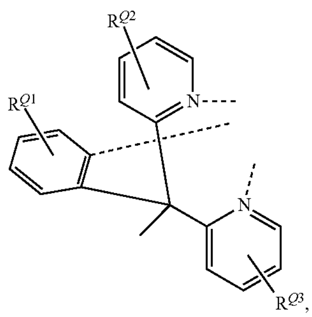

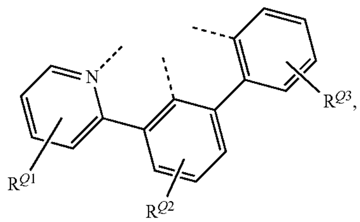

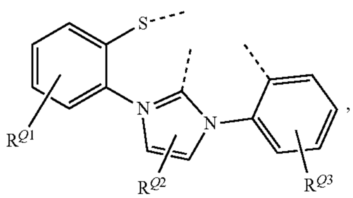

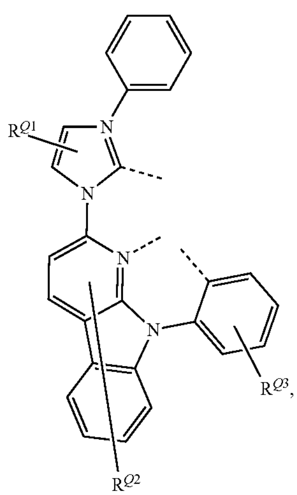

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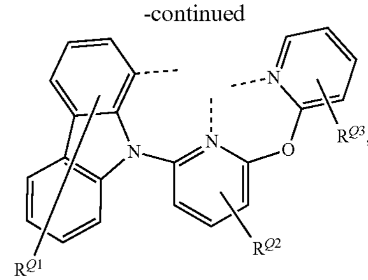

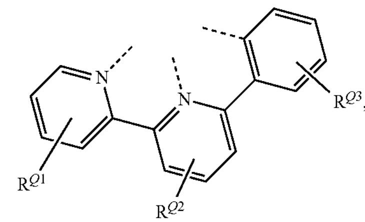

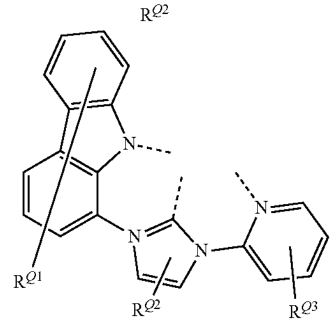

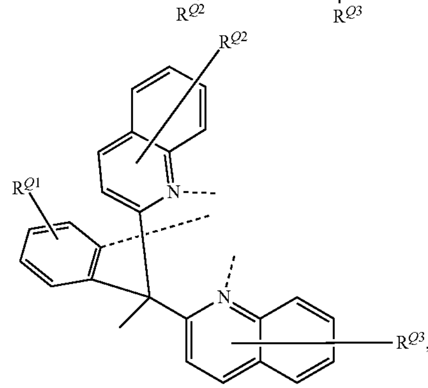

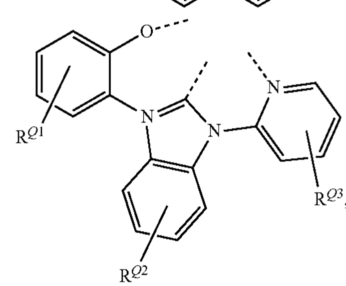

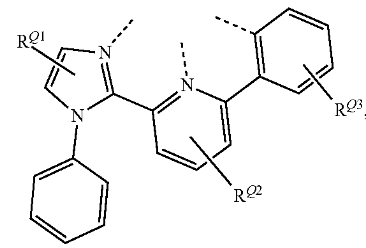

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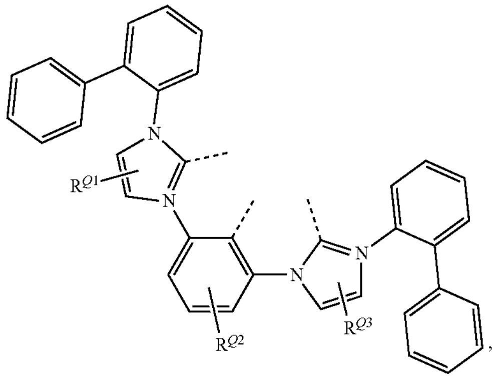

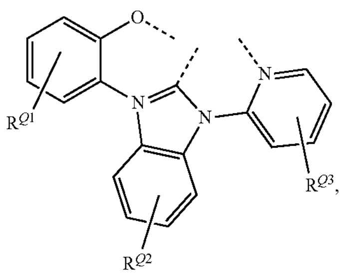

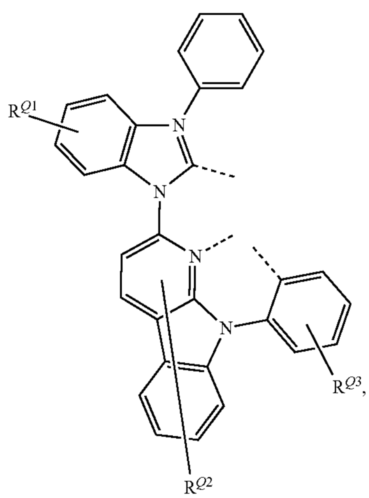

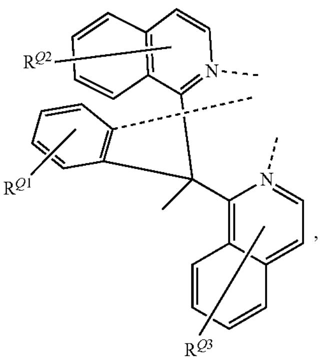

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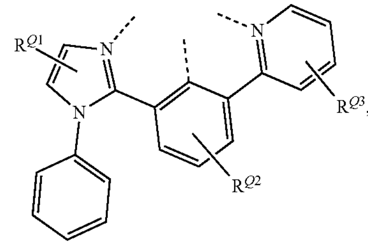

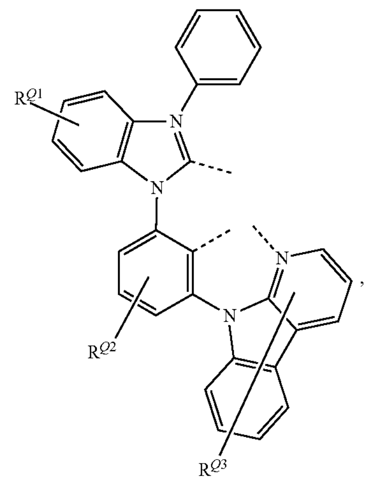

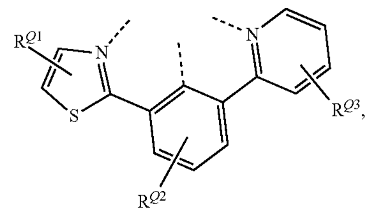

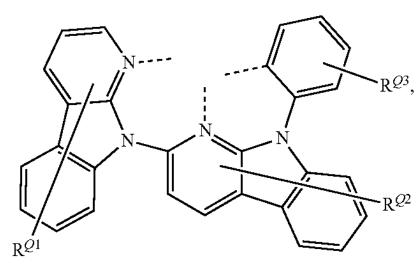

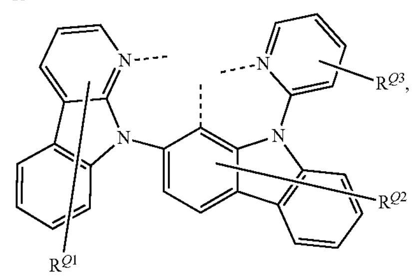

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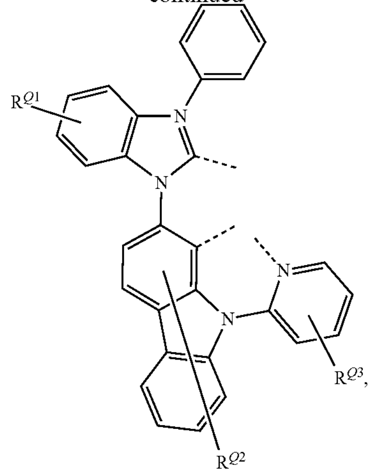

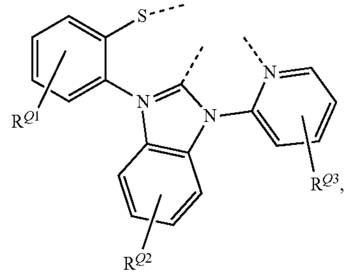

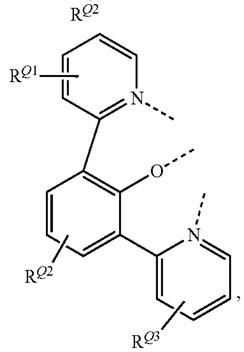

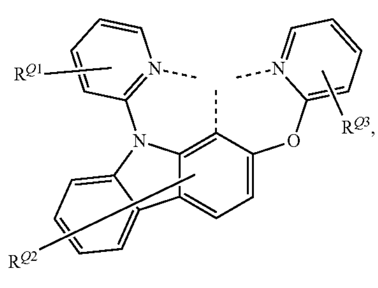

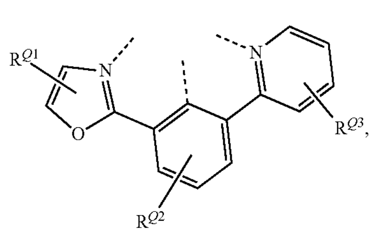

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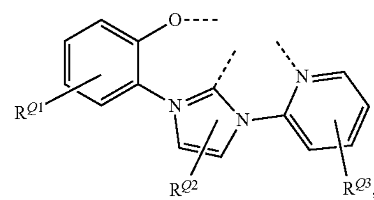

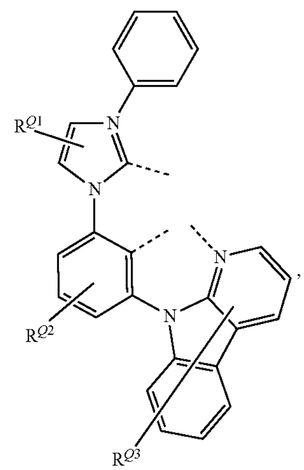

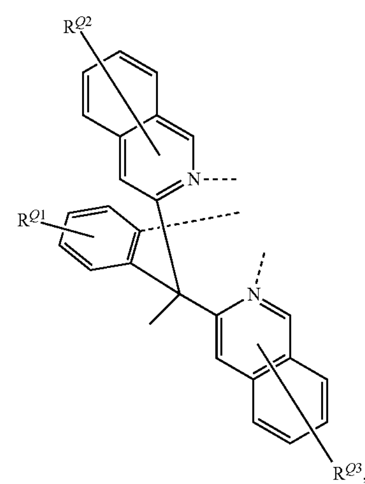

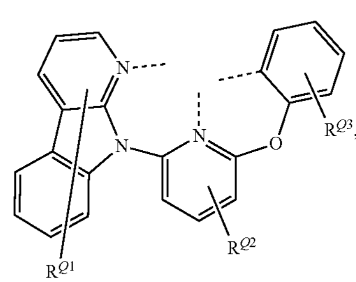

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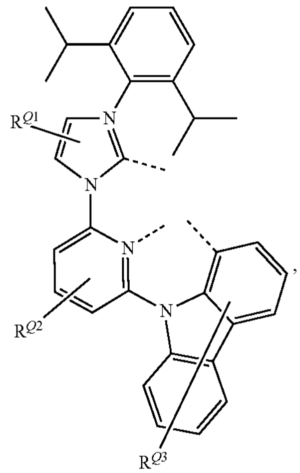

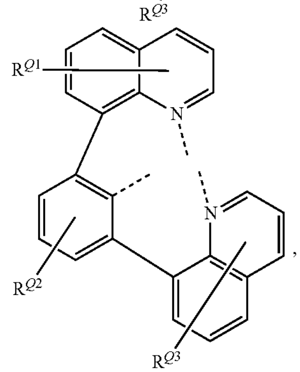

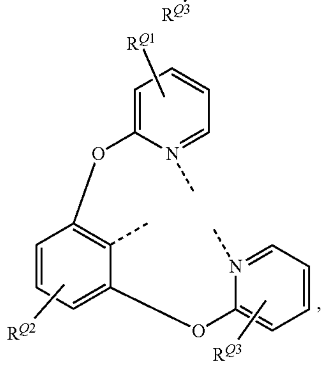

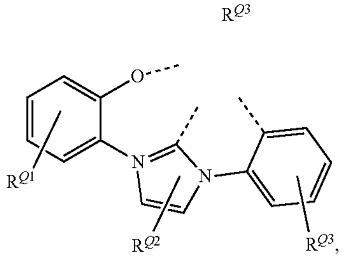

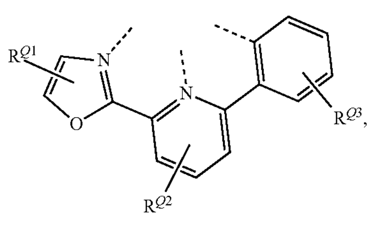

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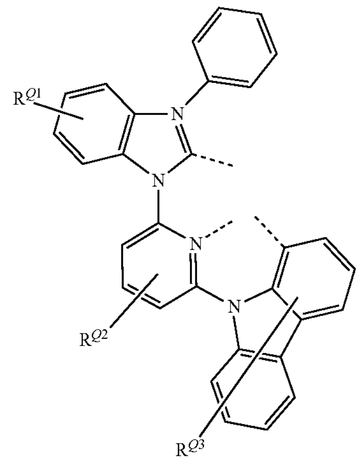

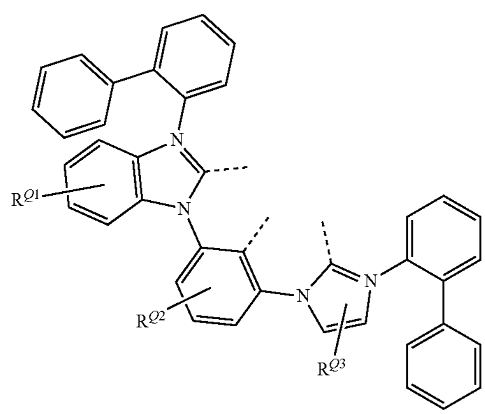

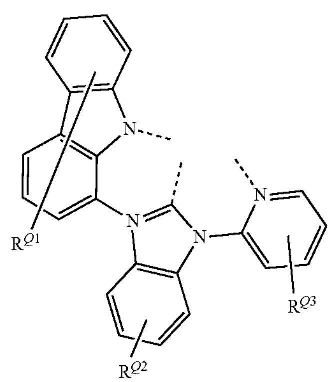

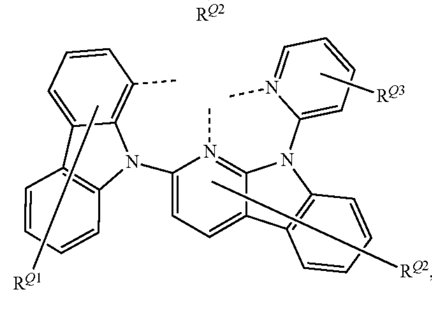

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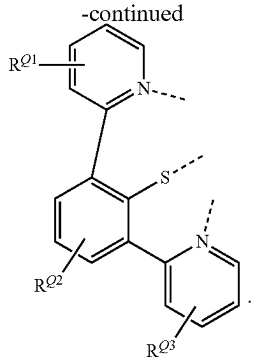
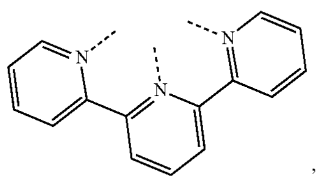


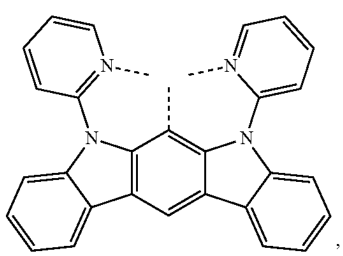
In some embodiments when a structure is selected from LIST 6a, at least one of R^{Q1}, R^{Q2} or R^{Q3} is selected from the group consisting of the General Substituents defined herein. In some embodiments when a structure is selected from LIST 6a, at least one of R^{Q1}, R^{Q2} or R^{Q3} is selected from the group consisting of the Preferred General Substituents defined herein.

[0286] In some embodiments, $L_{B'}$ is selected from $L_{BBi'}$, wherein i' is an integer from 1 to 53, and each $L_{BBi'}$ is defined in the following LIST 6:

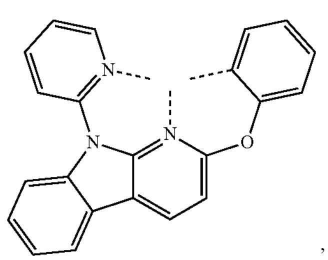
L_{BB1}



L_{BB2}



L_{BB3}



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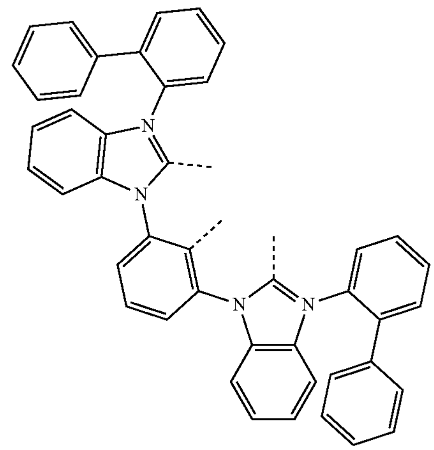
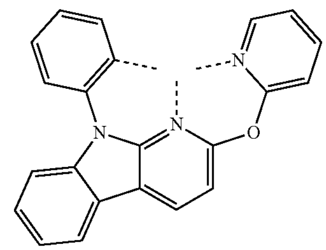
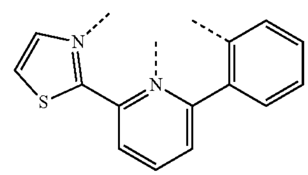
L_{BB4}



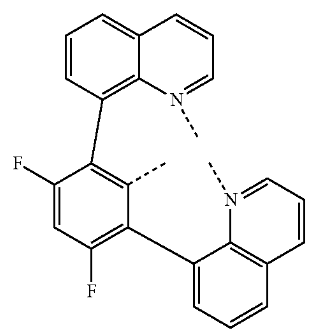
L_{BB5}



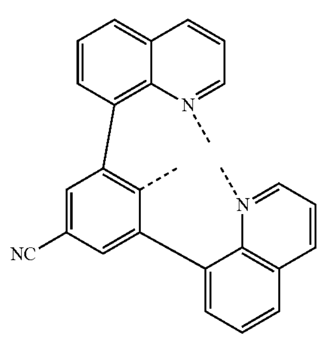
L_{BB6}



L_{BB7}



L_{BB8}



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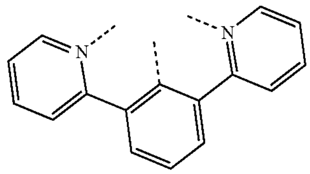
L_{BB9}



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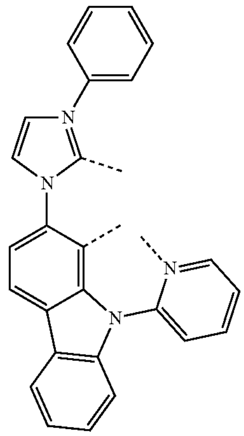
L_{BB10}



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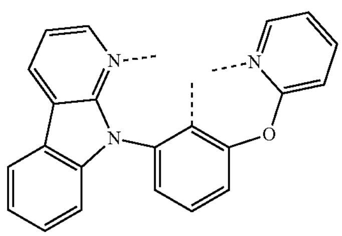
L_{BB11}



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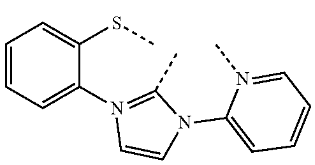
L_{BB12}



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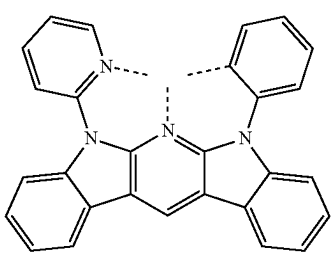
L_{BB13}



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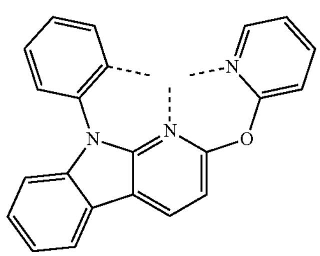
L_{BB14}



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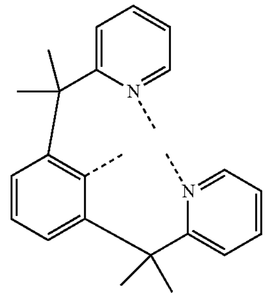
L_{BB15}



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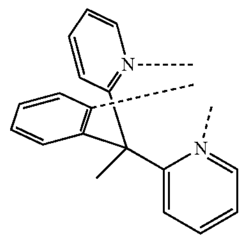
L_{BB16}



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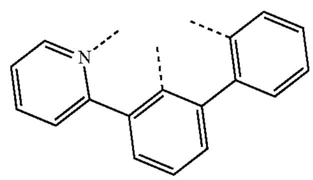
L_{BB17}



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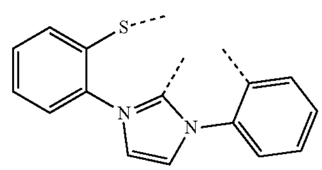
L_{BB18}



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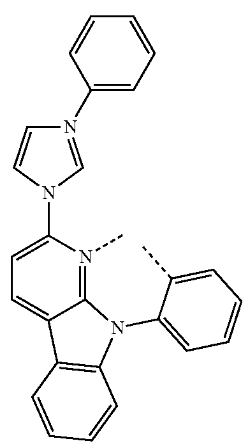
L_{BB19}



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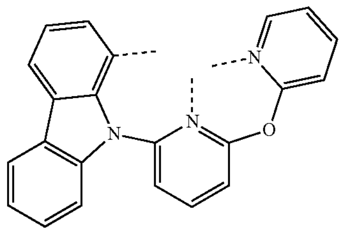
L_{BB20}



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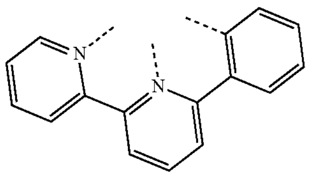
L_{BB21}



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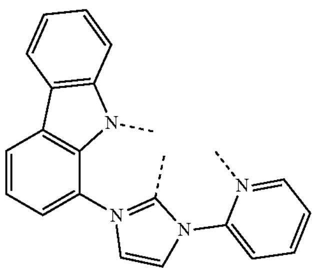
L_{BB22}



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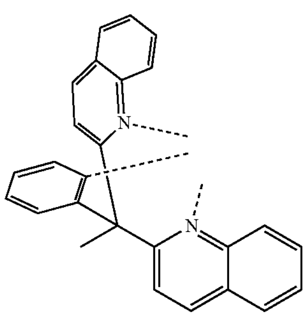
L_{BB23}



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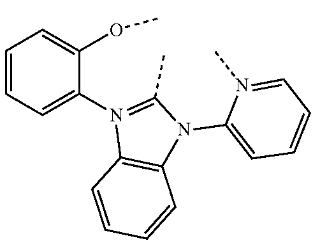
L_{BB24}



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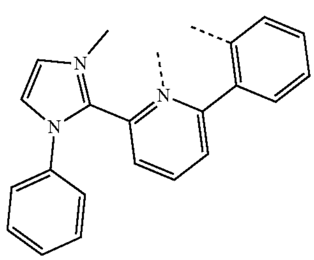
L_{BB25}



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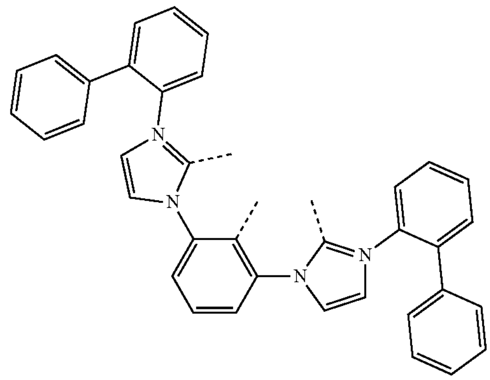
L_{BB26}



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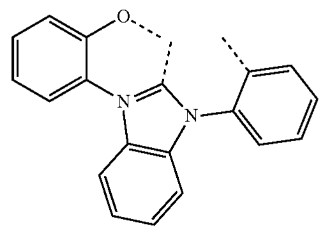
L_{BB27}



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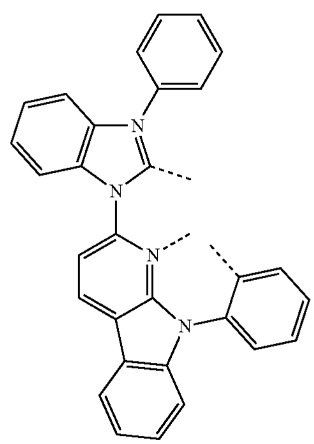
L_{BB28}



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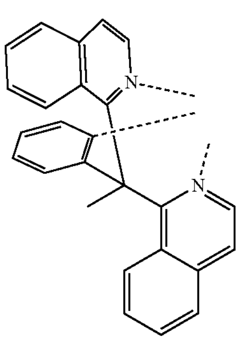
L_{BB29}



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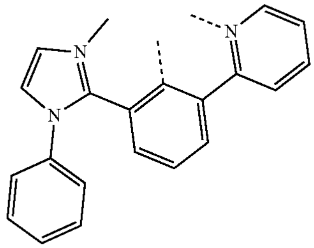
L_{BB30}



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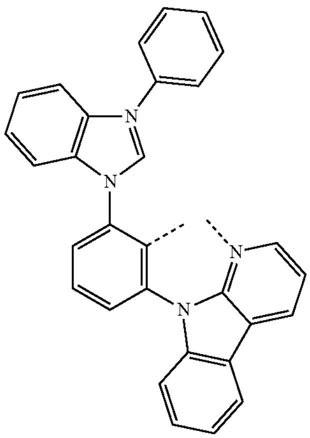
L_{BB31}



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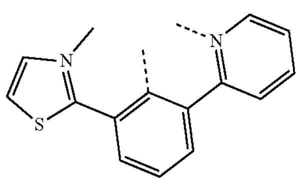
L_{BB32}



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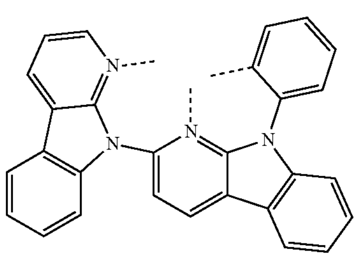
L_{BB33}



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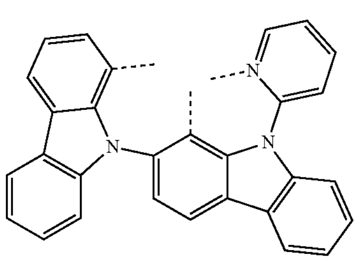
L_{BB34}



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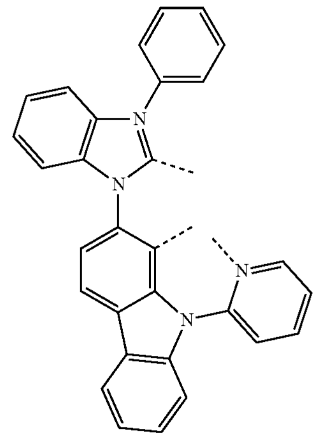
L_{BB35}



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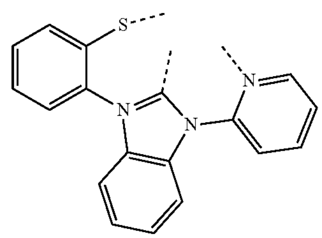
L_{BB36}



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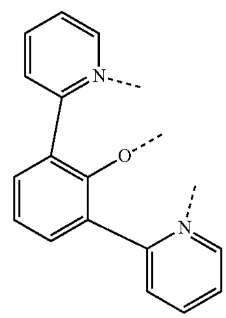
L_{BB37}



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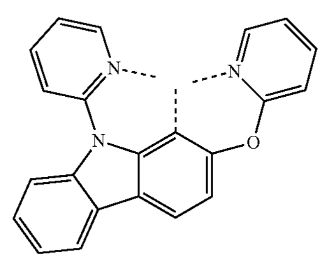
L_{BB38}



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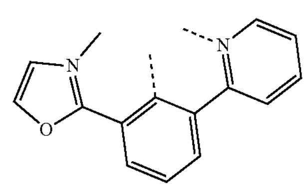
L_{BB39}



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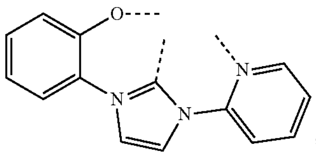
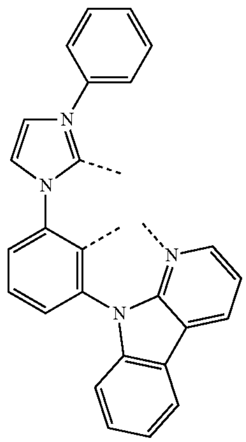
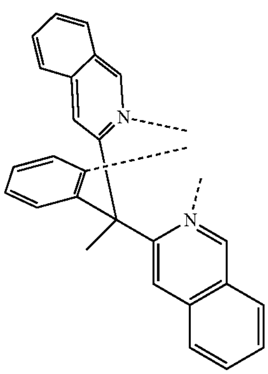
L_{BB40}



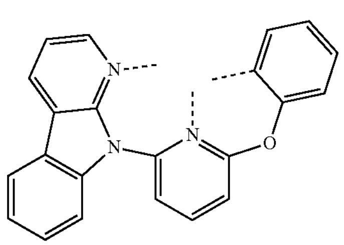
L_{BB41}



L_{BB42}



L_{BB43}



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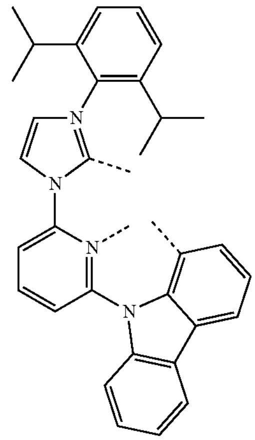
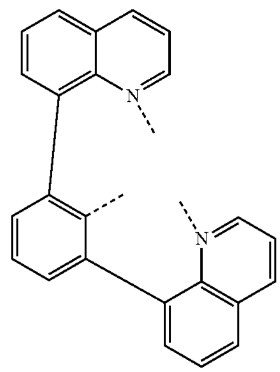
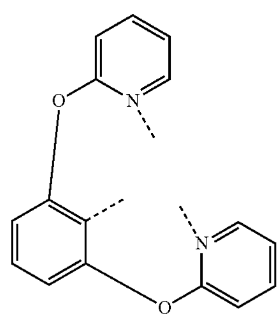
L_{BB44}



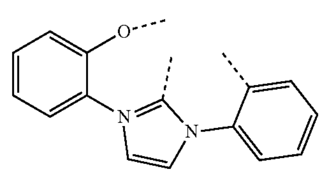
L_{BB45}



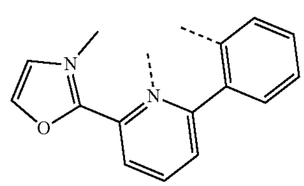
L_{BB46}



L_{BB47}



L_{BB48}



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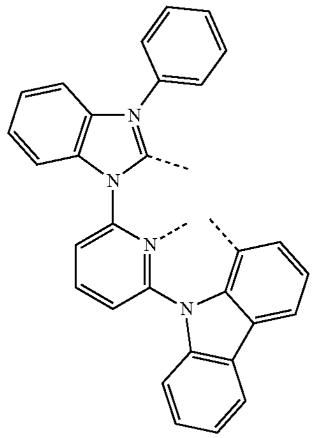
L_{BB49}



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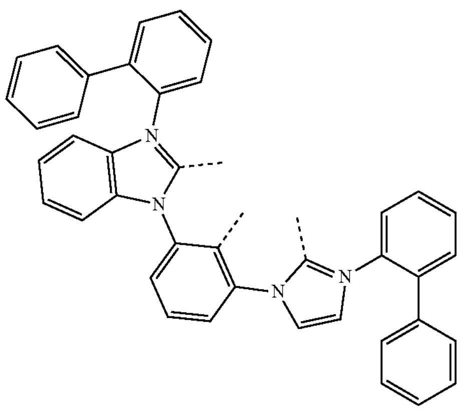
L_{BB50}



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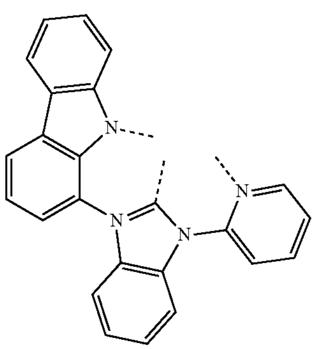
L_{BB51}



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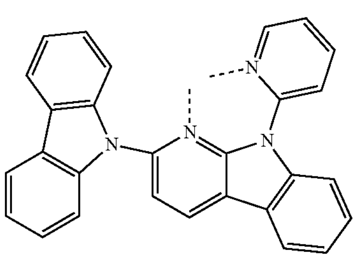
L_{BB52}



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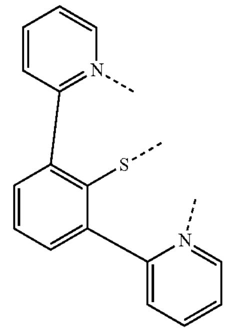
L_{BB53}



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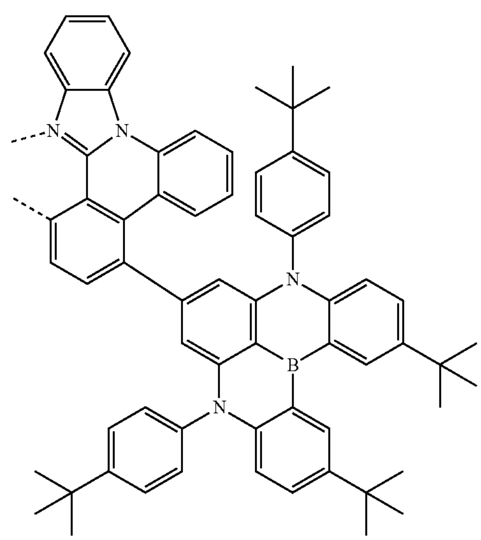
[0287] In some embodiments, the compound has a formula Pt(L_{AAi})($L_{BBi'}$) selected from compounds having the structure of Pt(L_{AA1})(L_{BB1}) to Pt(L_{AA15})(L_{BB53}). In some embodiments, the compound has a formula Pt(L_A)(L_B), wherein L_A is selected from the structures of LIST 1, LIST 3, or LIST 5 defined herein, and L_B is selected from LIST 6a or LIST 6 defined herein.

[0288] In some embodiments, the ligand L_A is Selected from $L_{Ai''}$, wherein C is an integer from 1 to 241, and each $L_{Ai''}$ as defined in the following LIST 7:

L_{A1}



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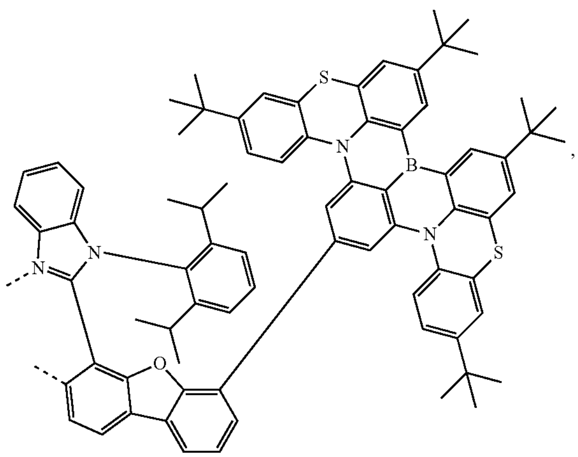
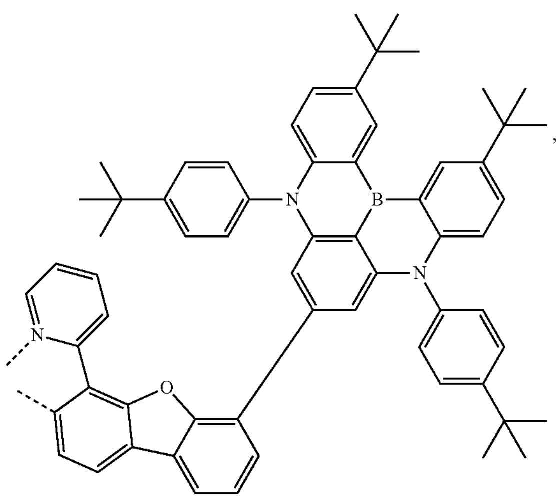
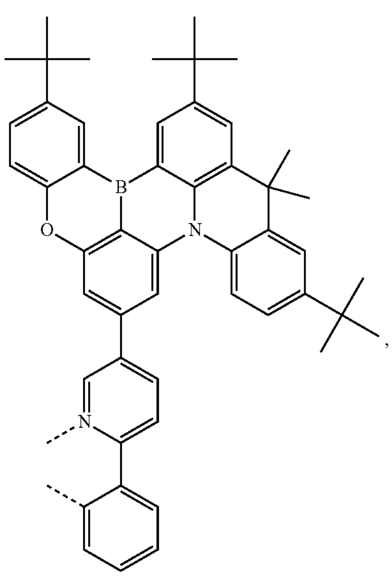
L_{A2}



L_{A3}



L_{A4}



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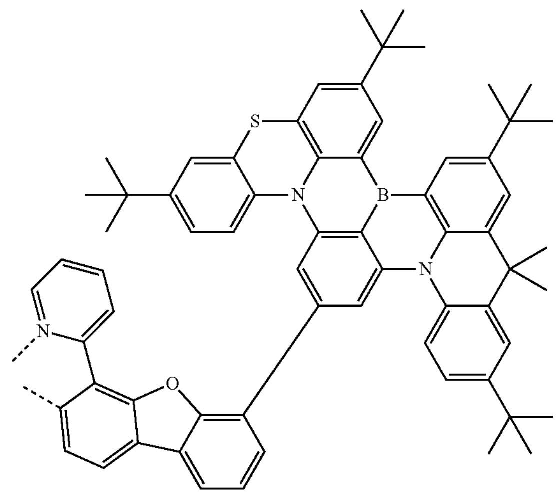
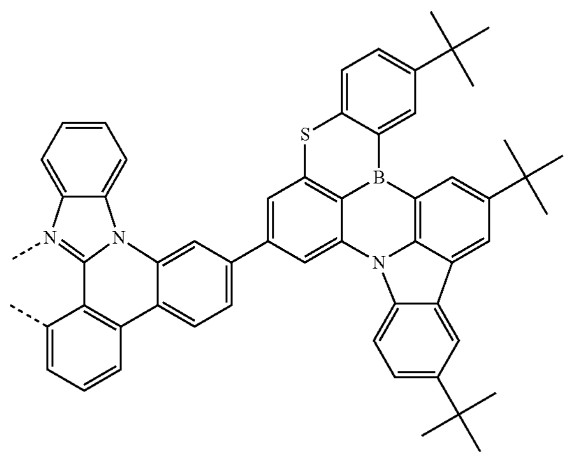
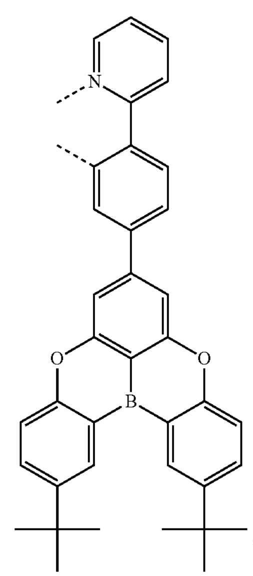
L_{A5}



L_{A6}



L_{A7}



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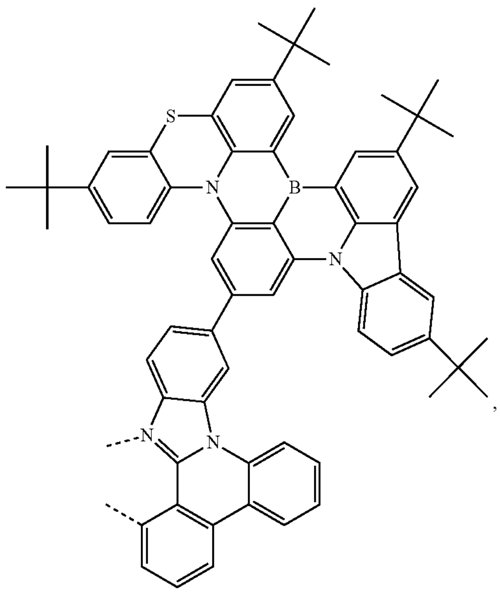
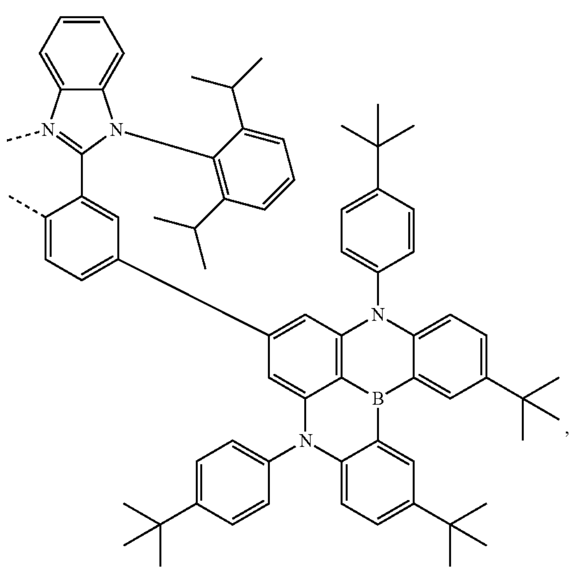
L_{A8}



L_{A9}



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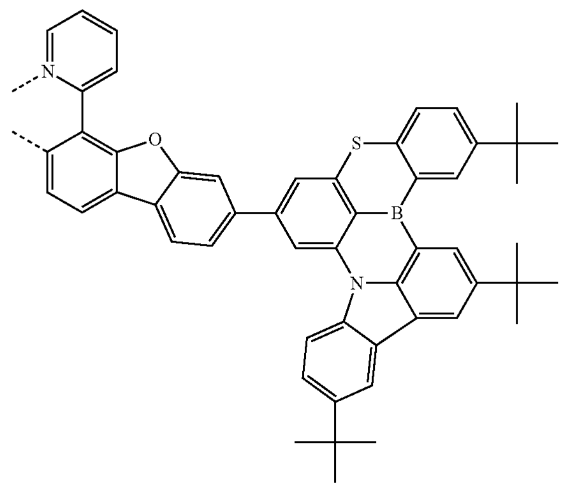
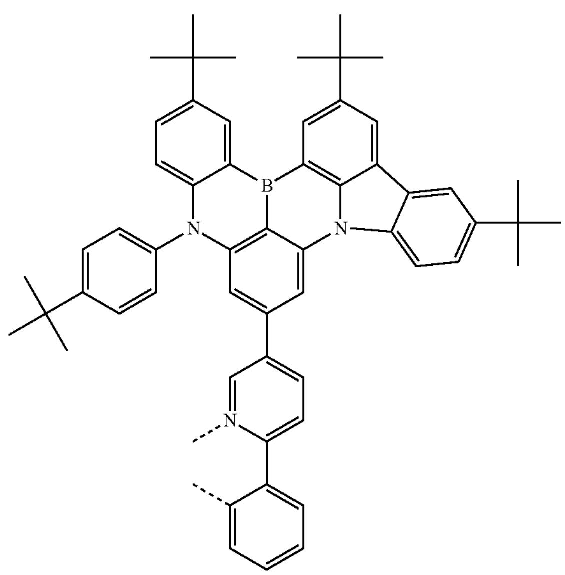
L_{A10}



L_{A11}



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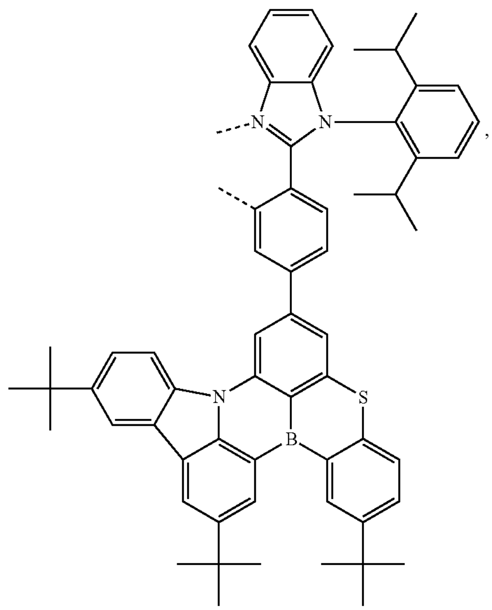
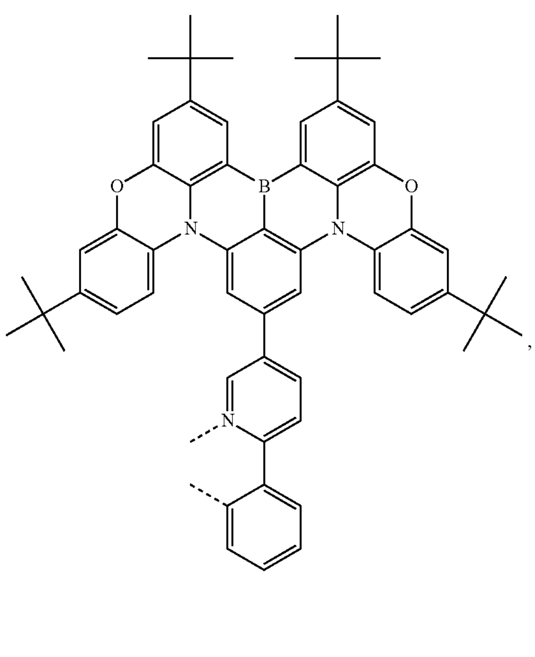
L_{A12}



L_{A13}



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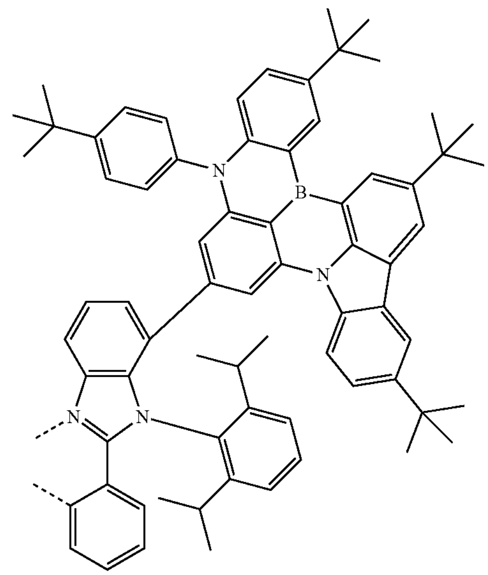
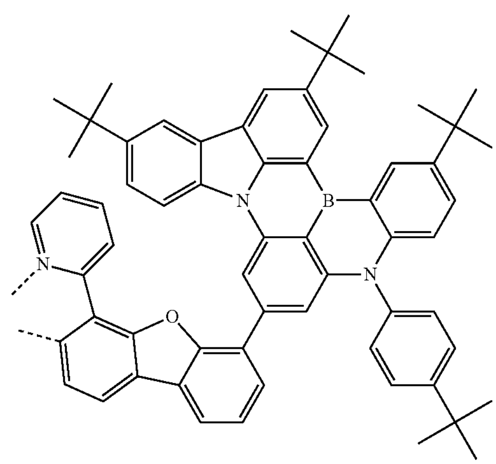
L_{A14}



L_{A15}



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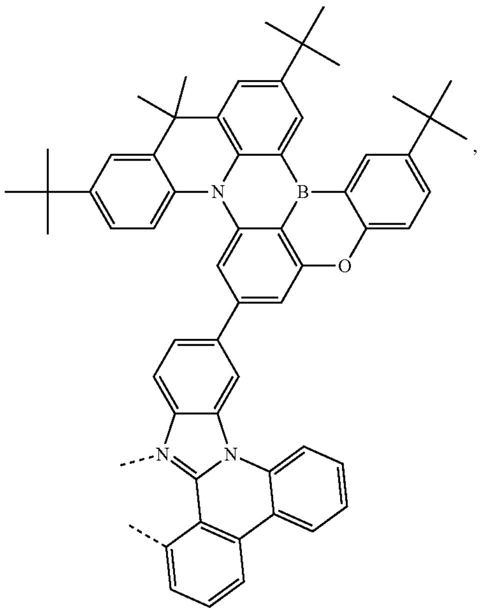
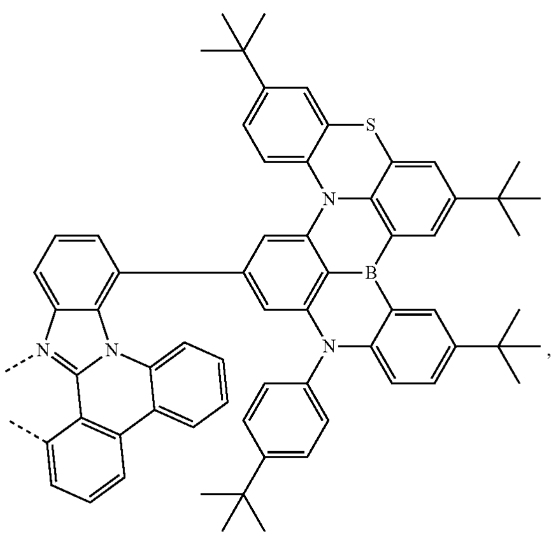
L_{A16}



L_{A17}



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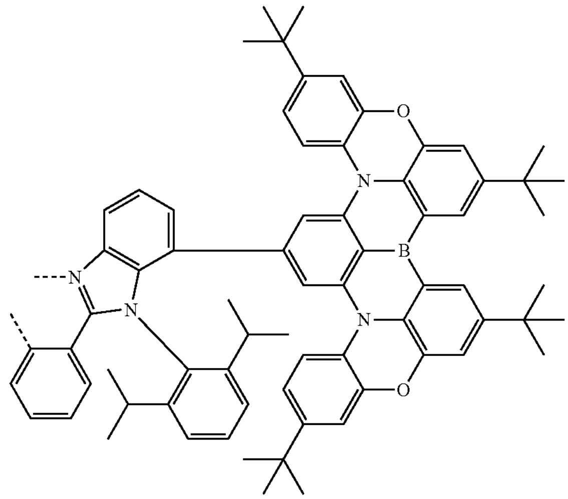
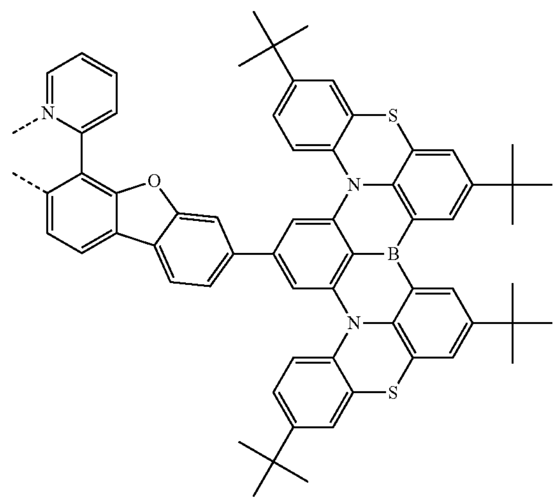
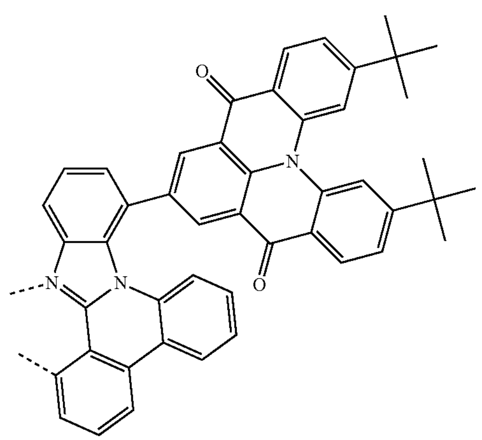
L_{A18}



L_{A19}



L_{A20}



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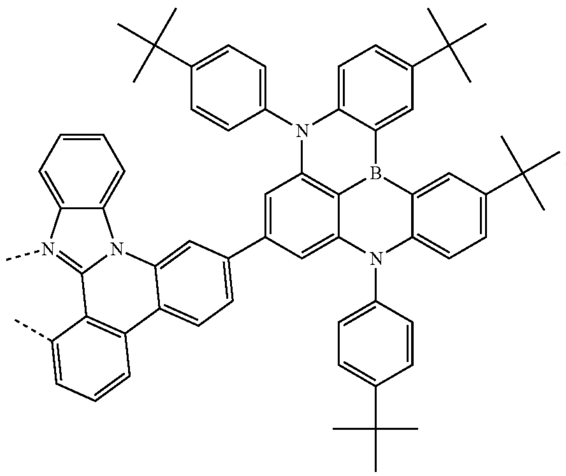
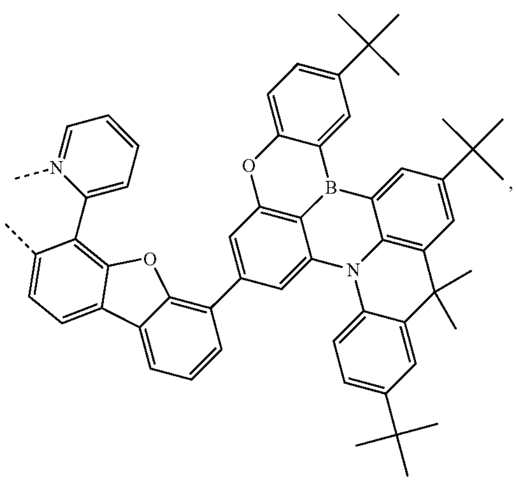
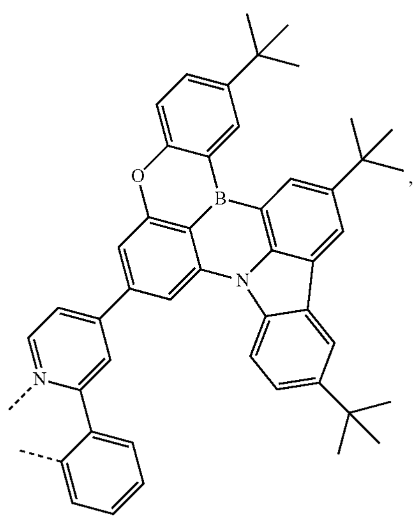
L_{A21}



L_{A22}



L_{A23}



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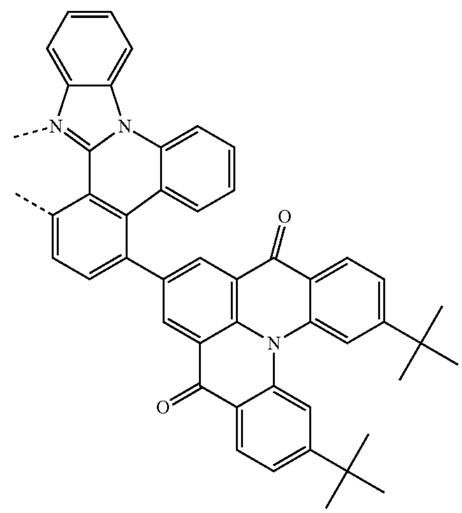
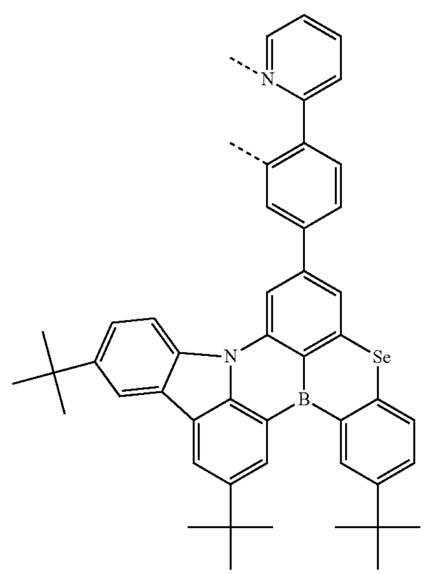
L_{A24}



L_{A25}



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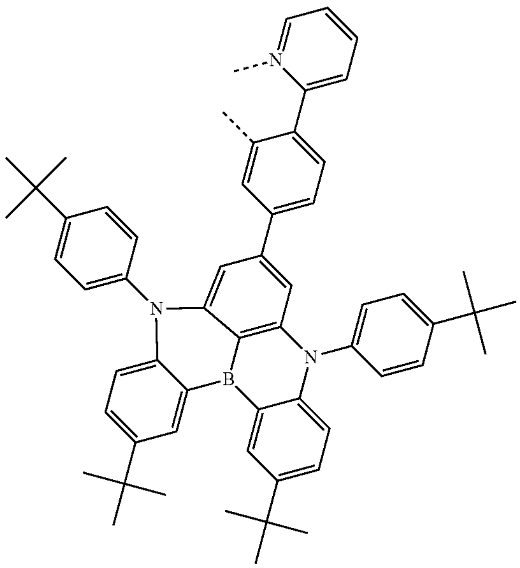
L_{A26}



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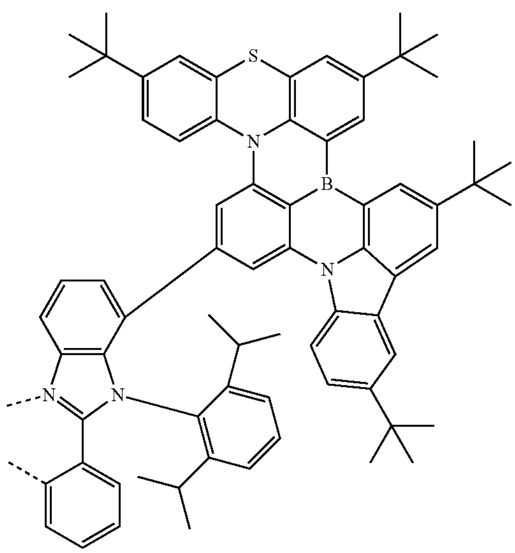
L_{A27}



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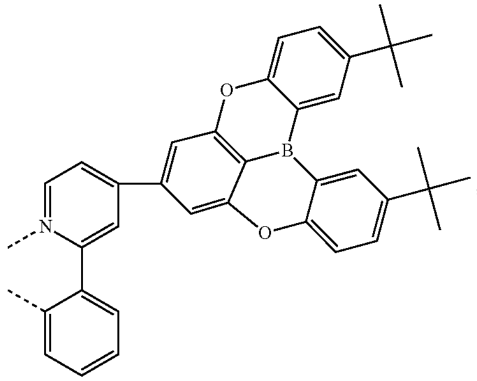
L_{A28}



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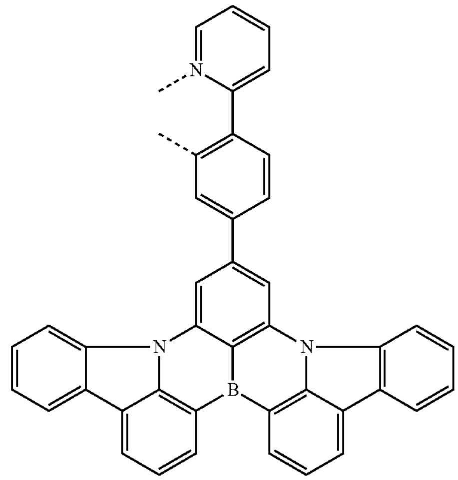
L_{A29}



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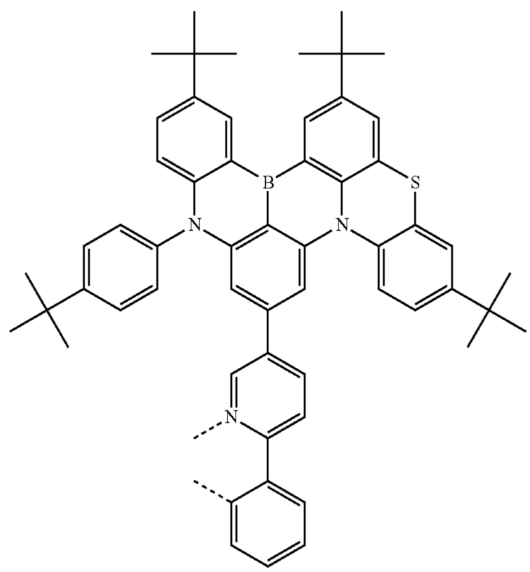
L_{A30}



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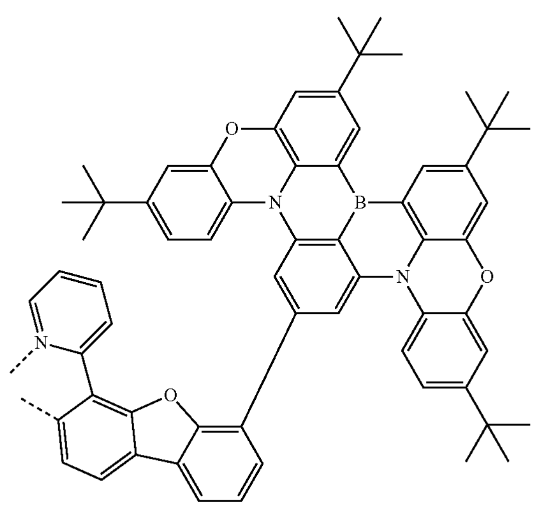
L_{A31}



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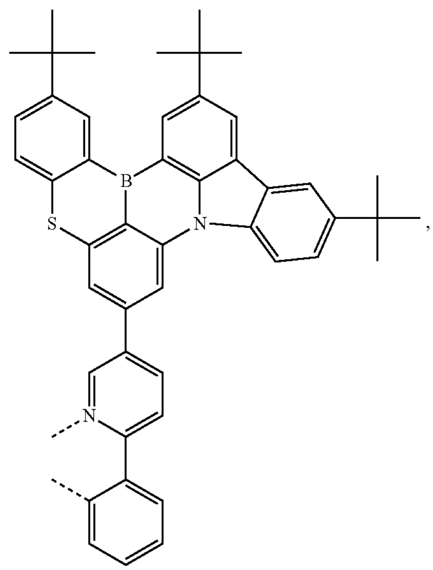
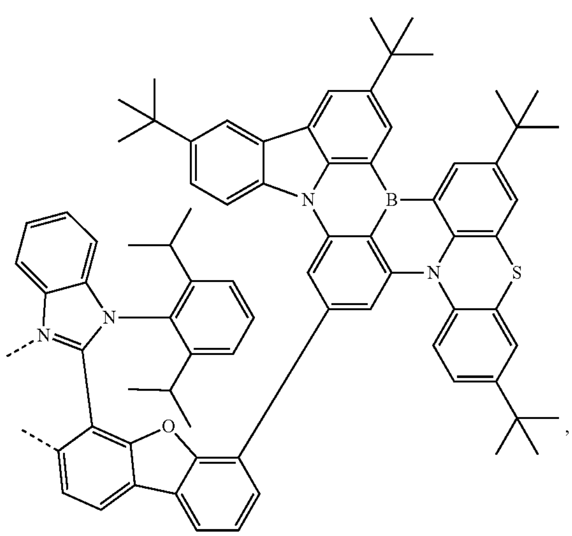
L_{A32}



L_{A33}



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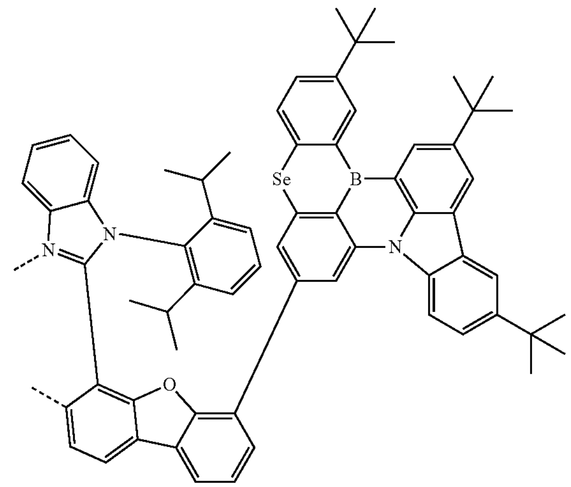
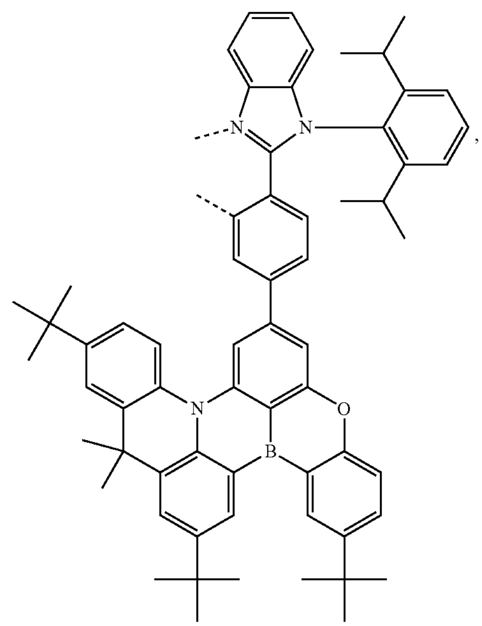
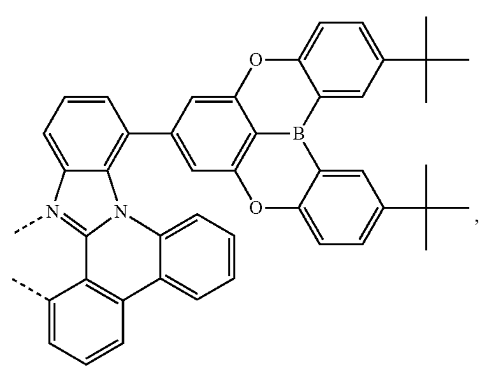
L_{A34}



L_{A35}



L_{A36}



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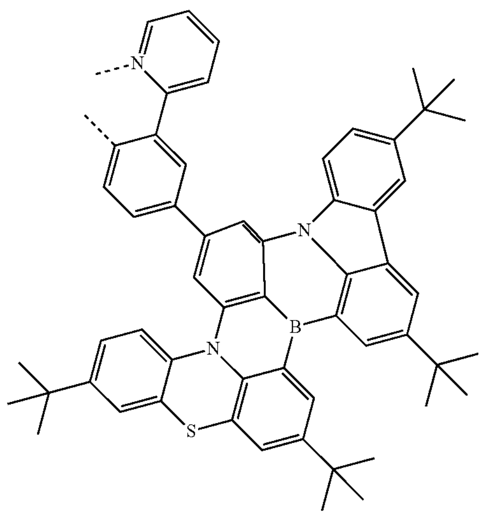
L_{A37}



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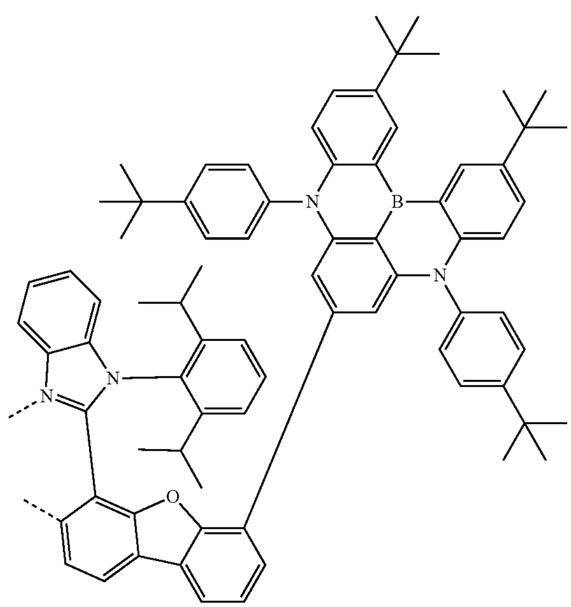
L_{A38}



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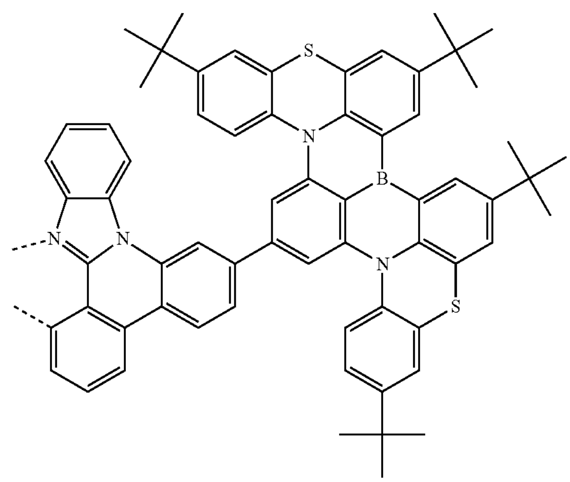
L_{A39}



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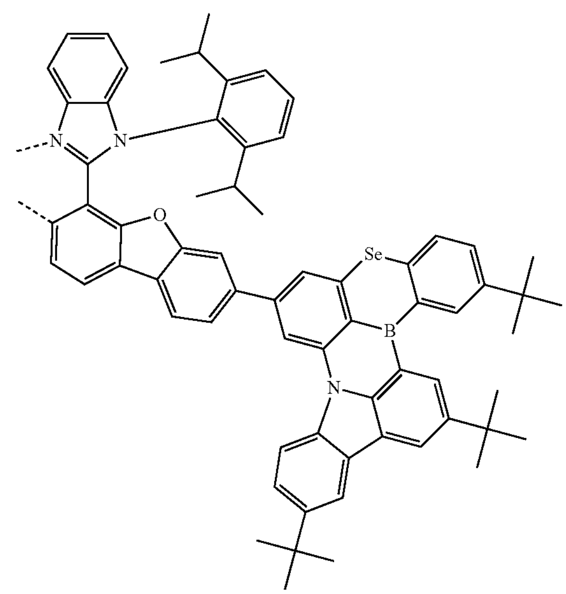
L_{A40}



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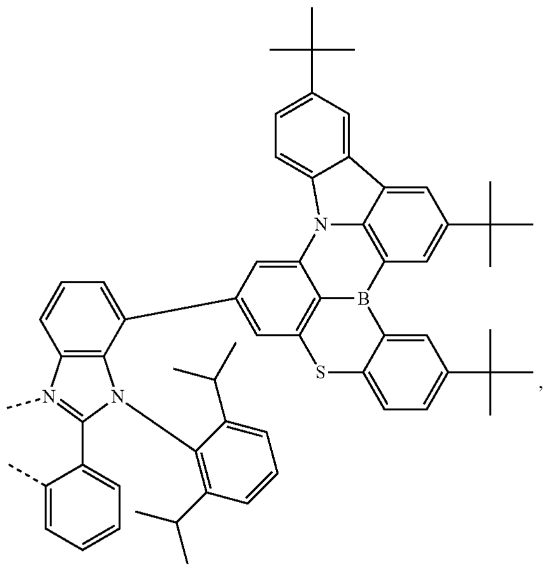

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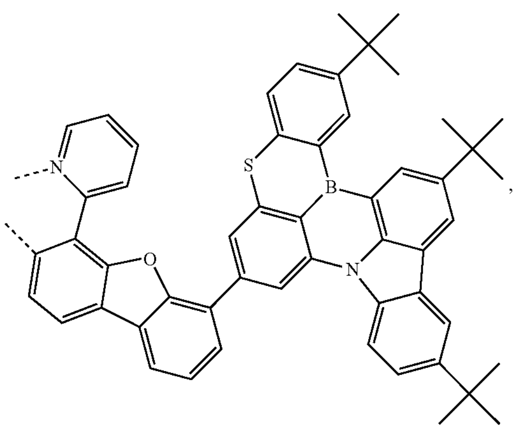

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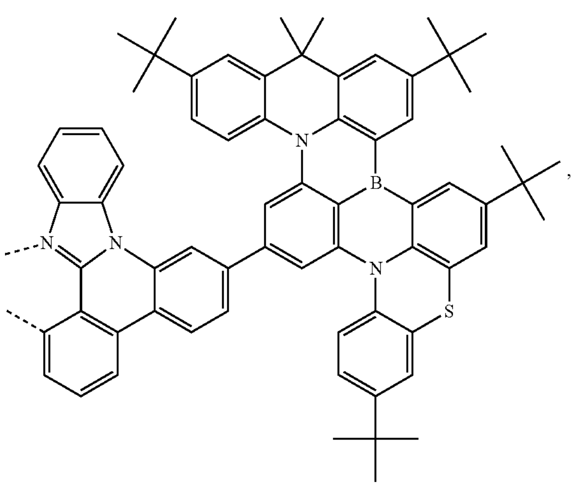

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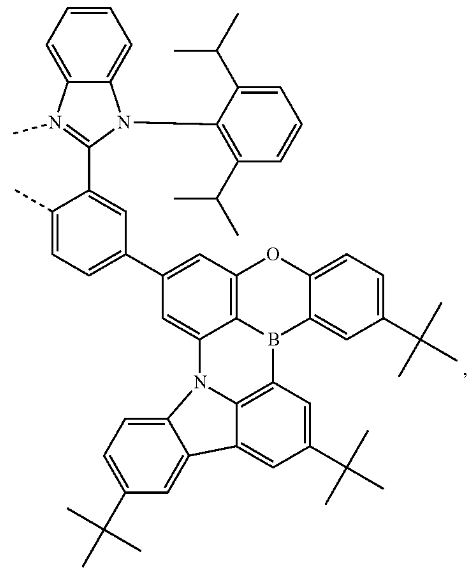

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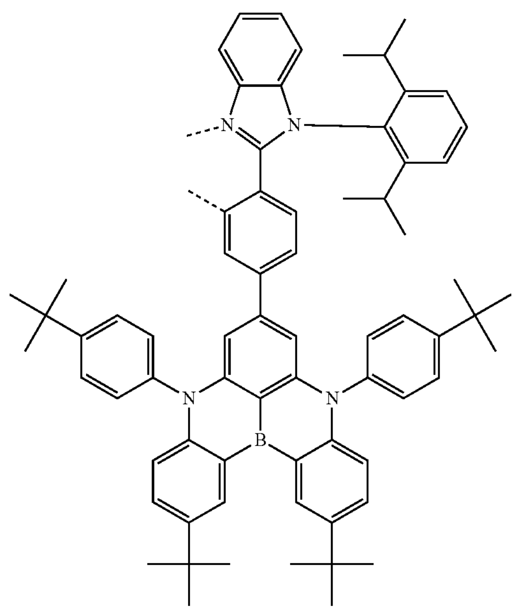

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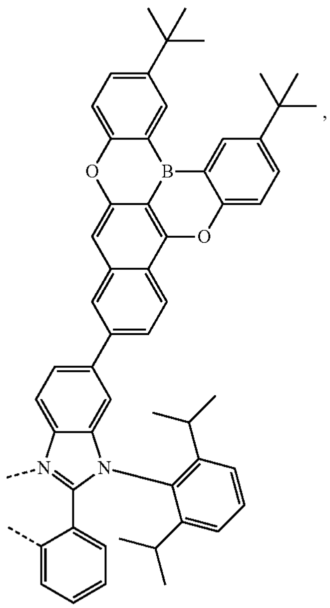

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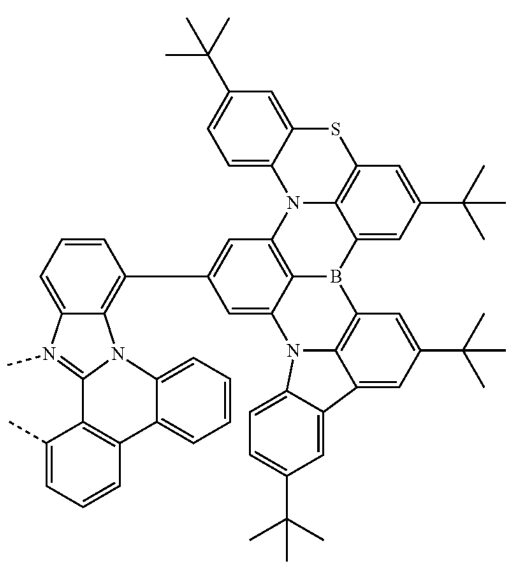

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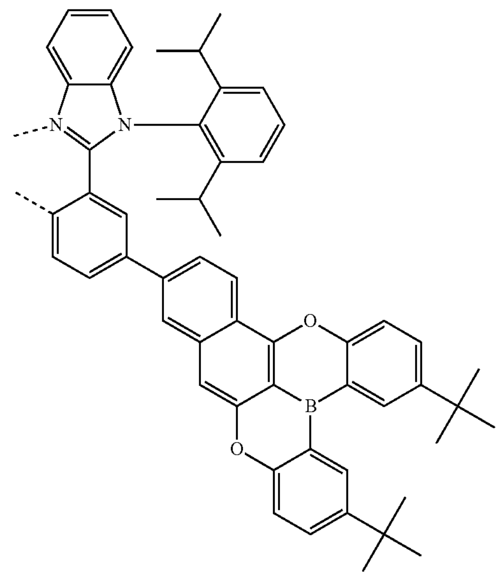

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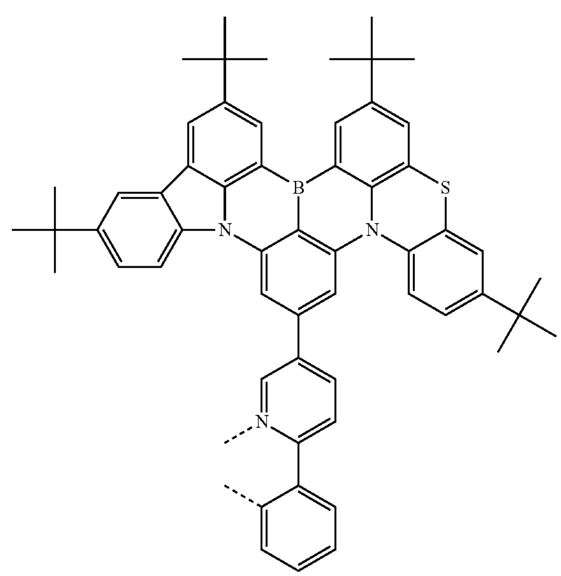

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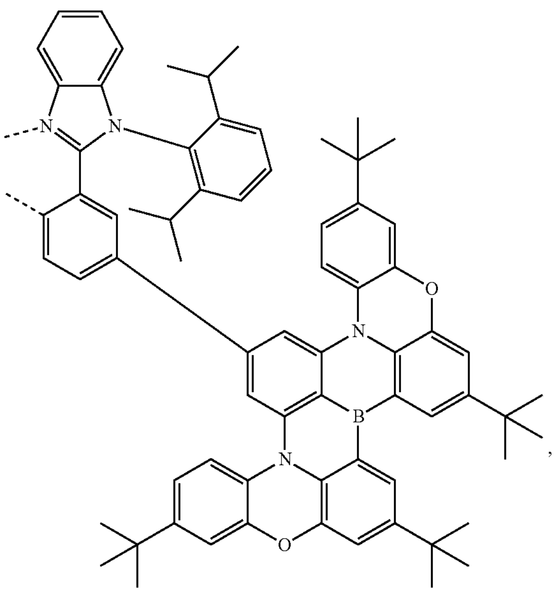

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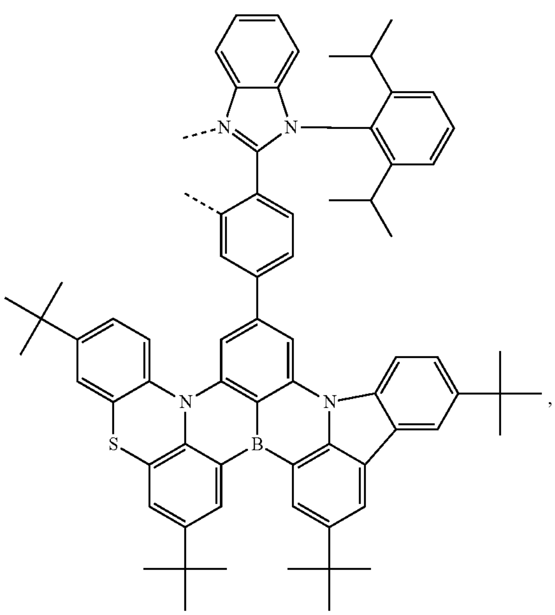

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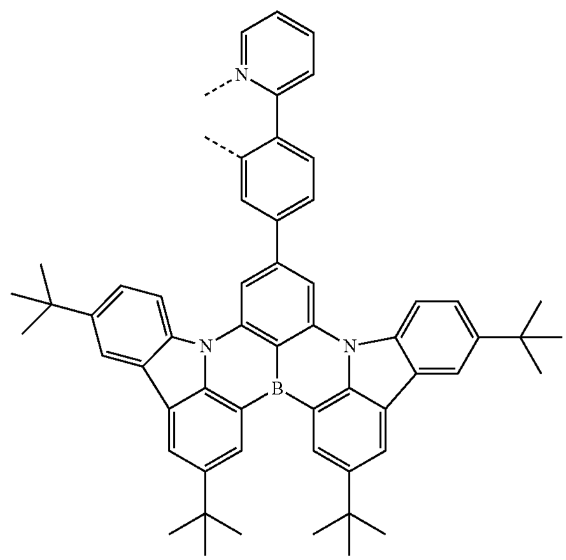

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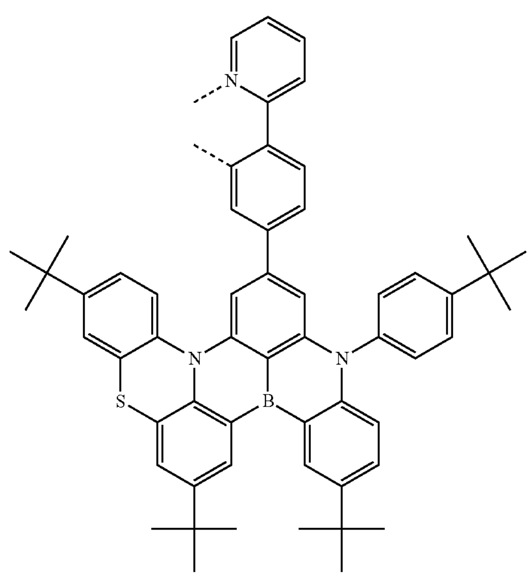

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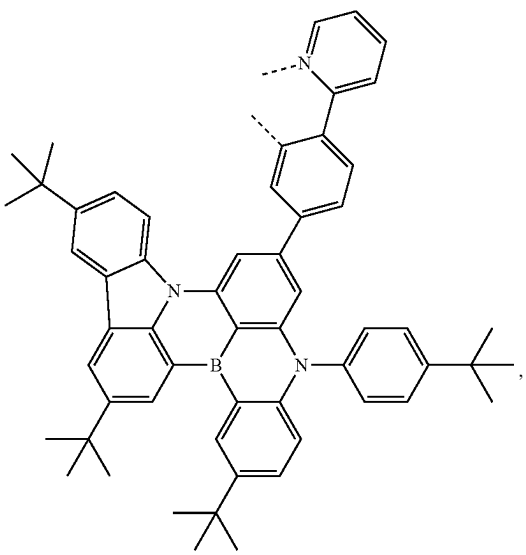

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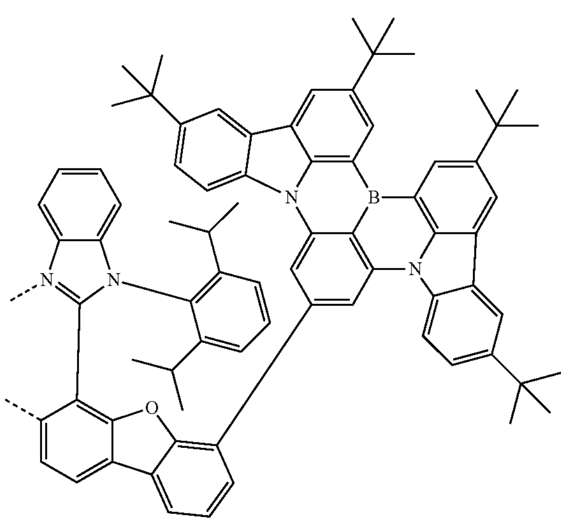

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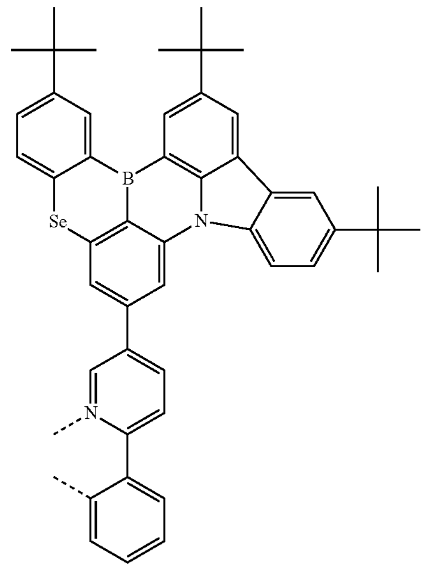

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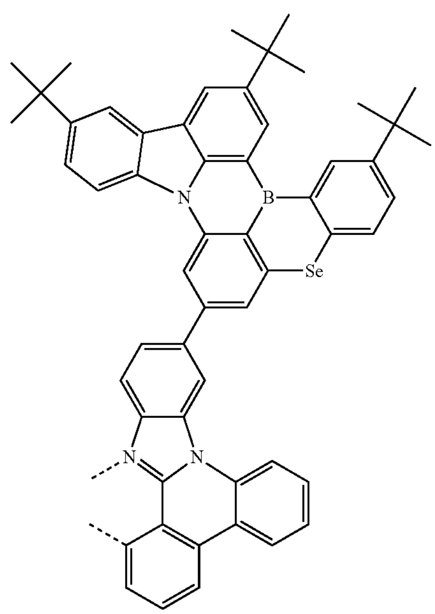

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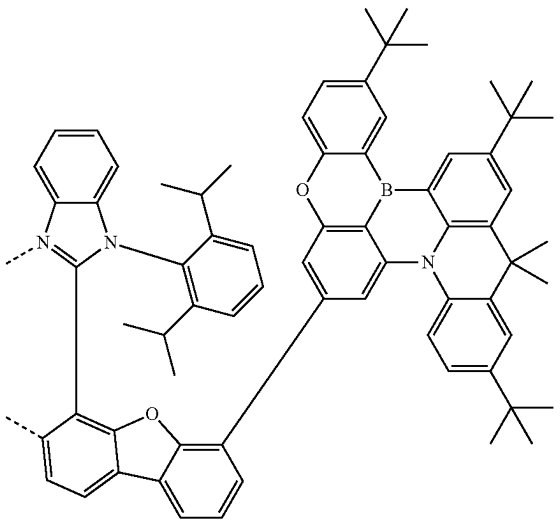

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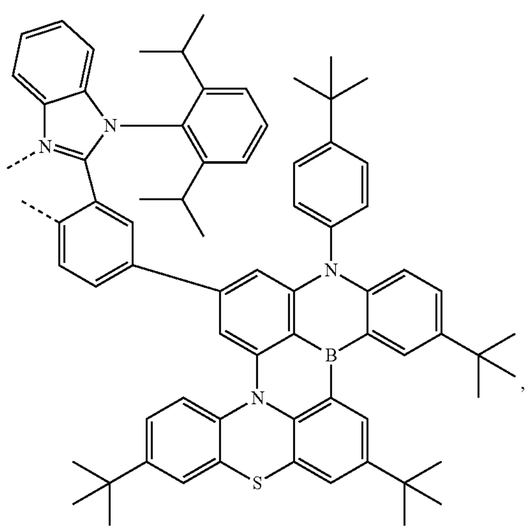

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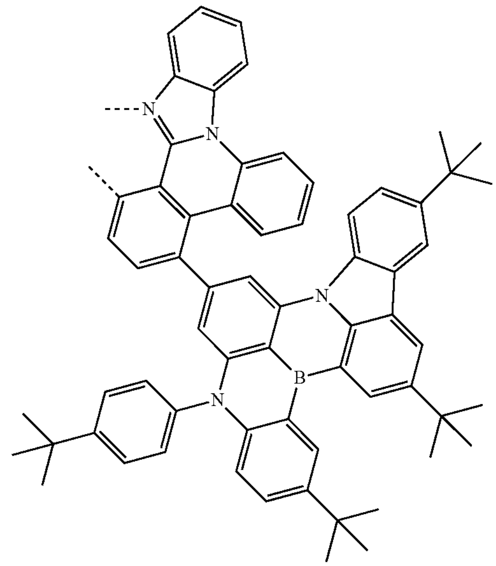

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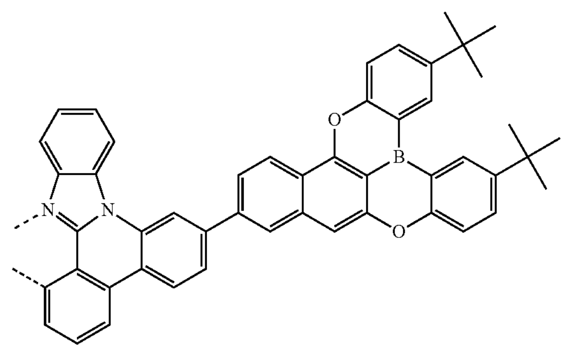

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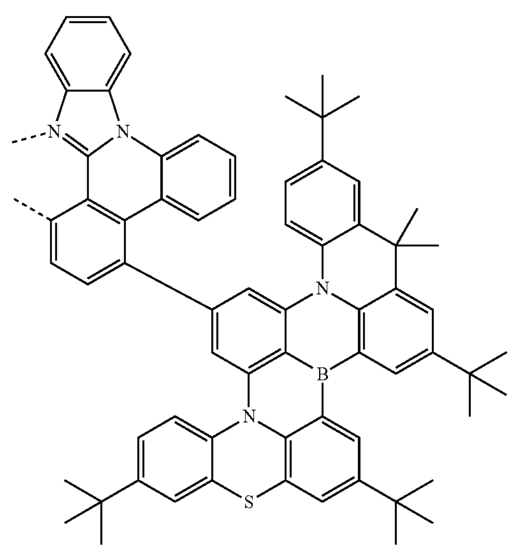

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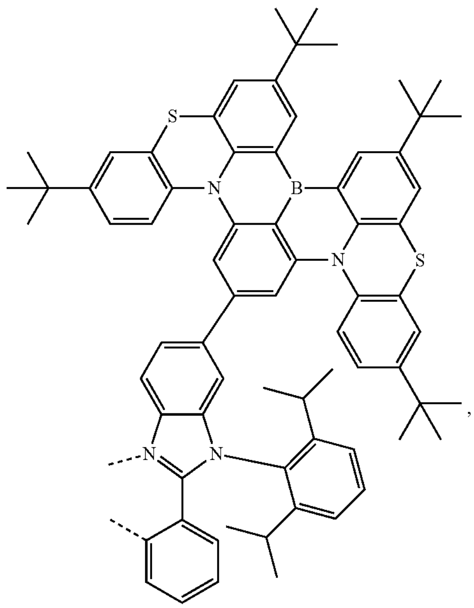

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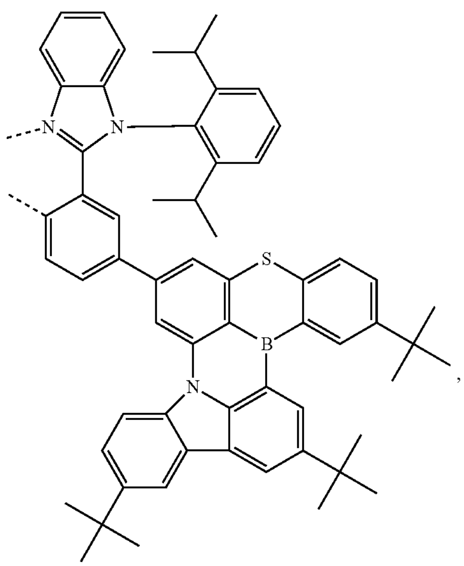

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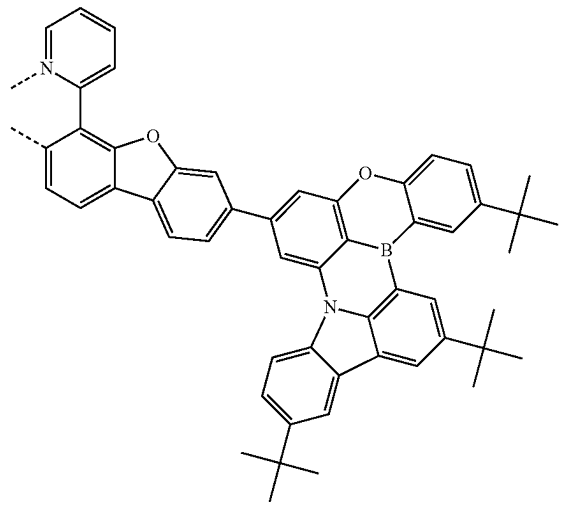

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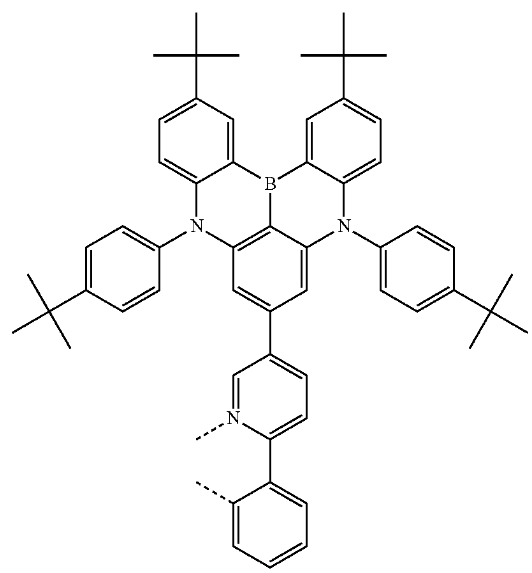

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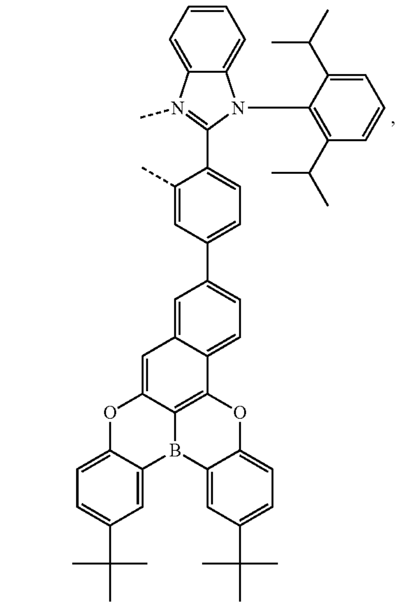

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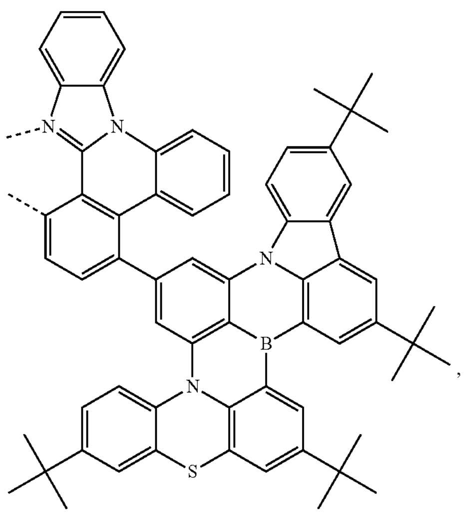

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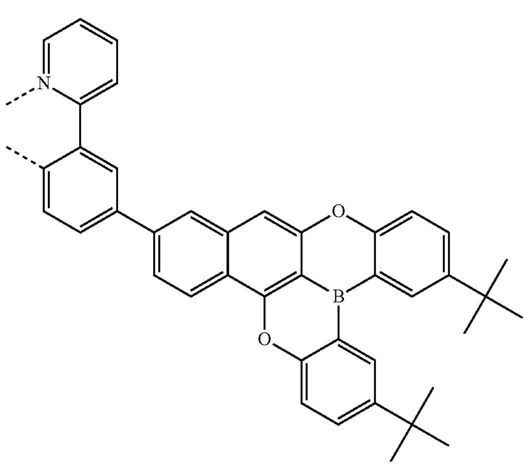

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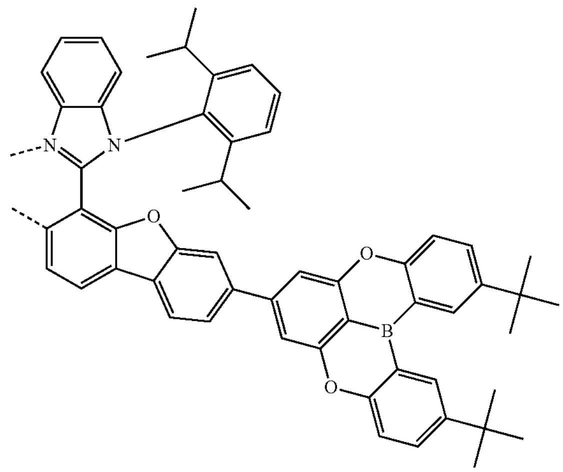

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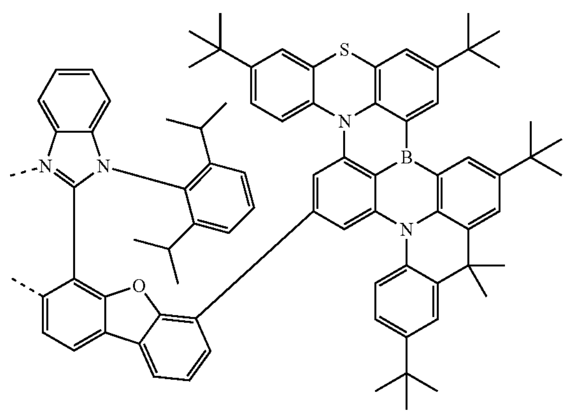

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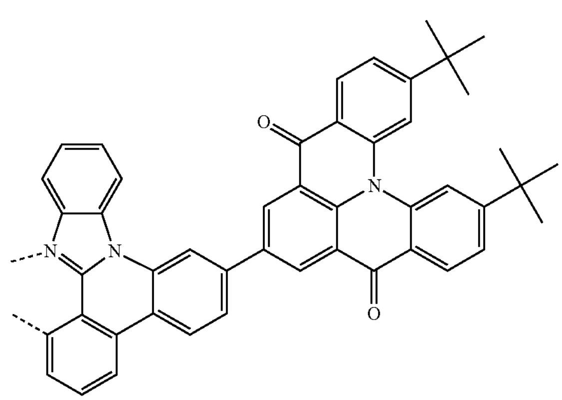

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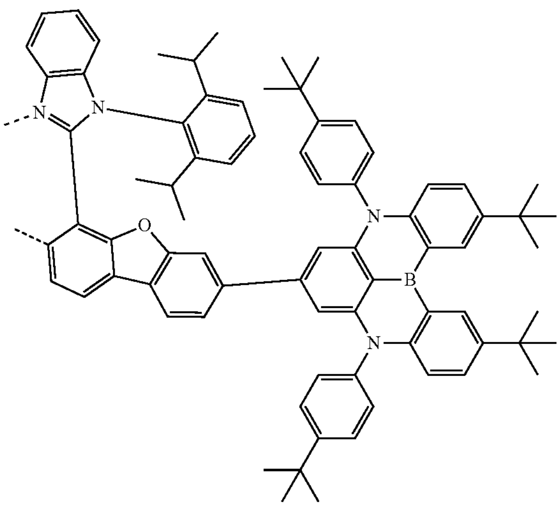

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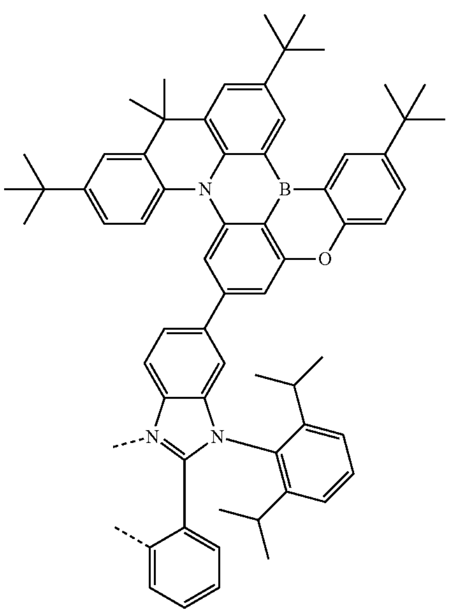

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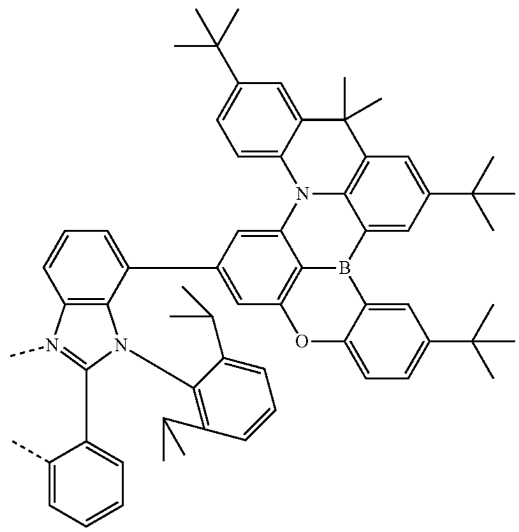

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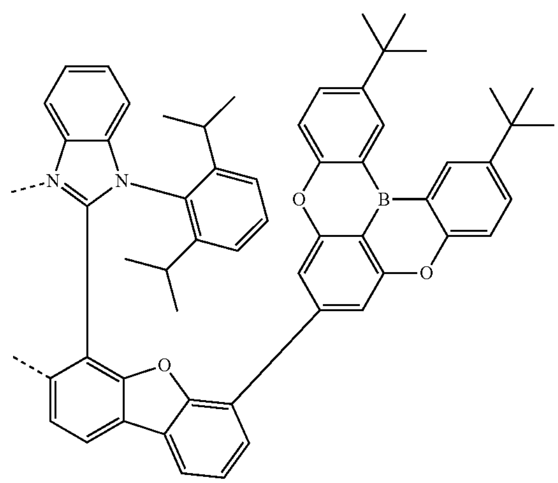

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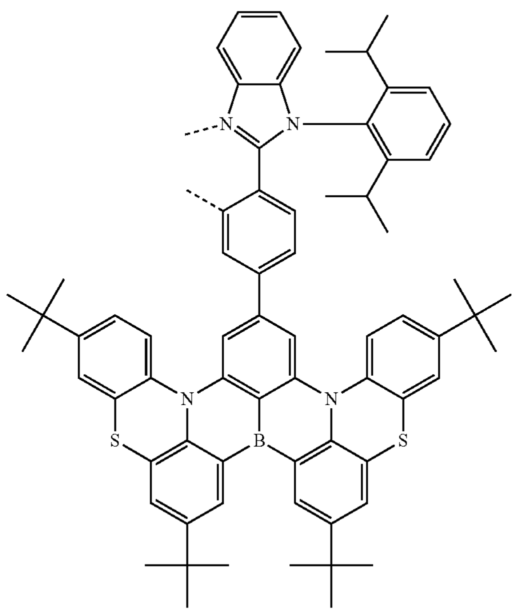

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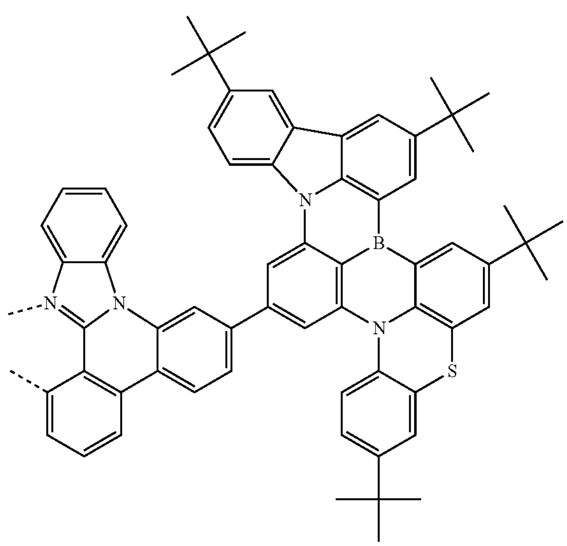

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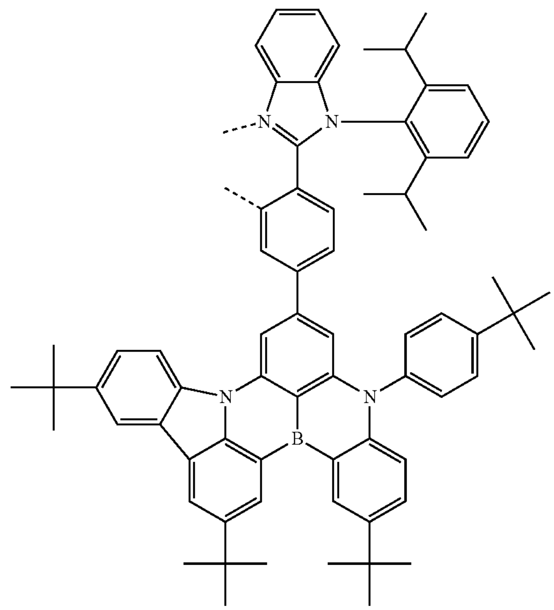

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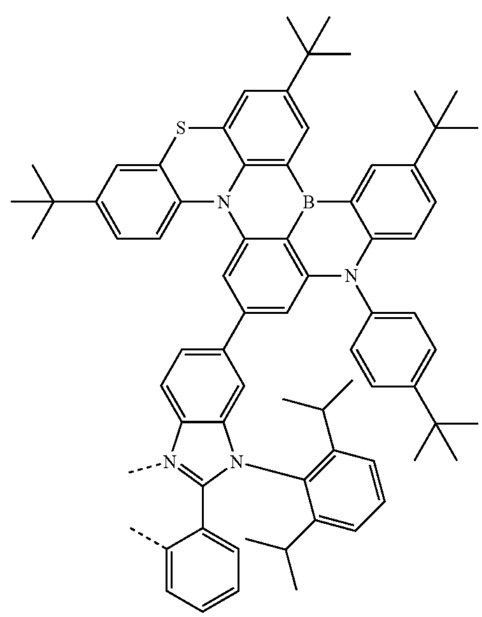

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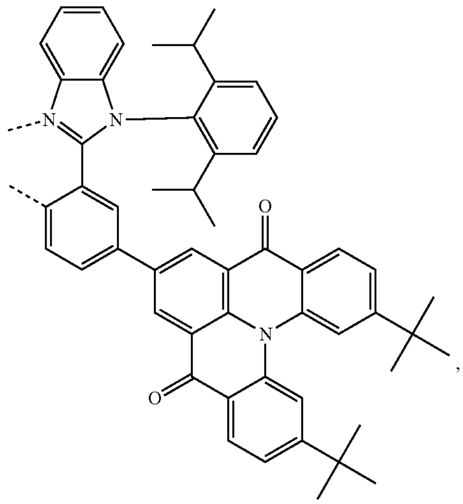

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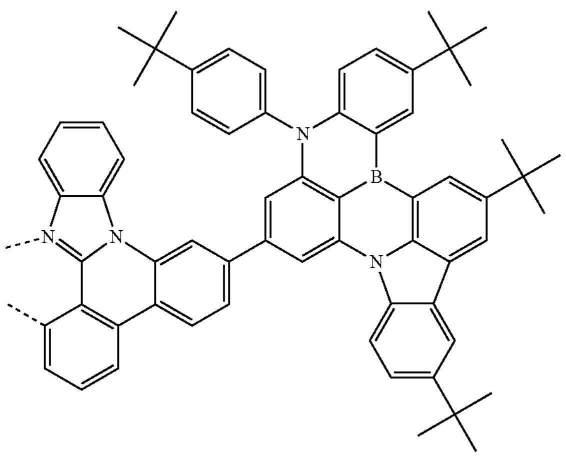

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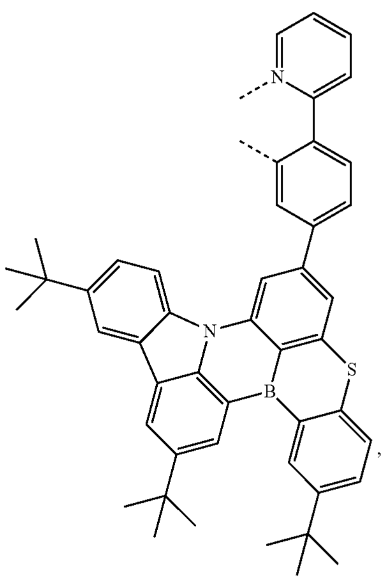

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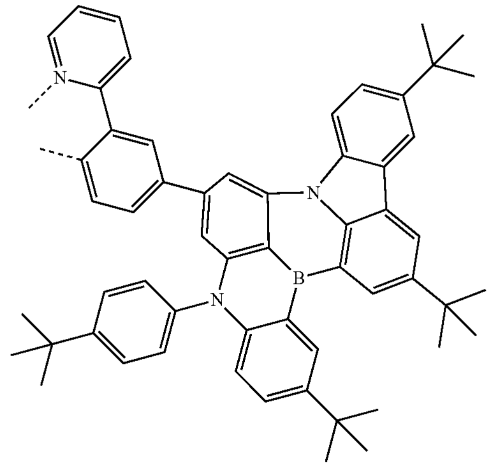

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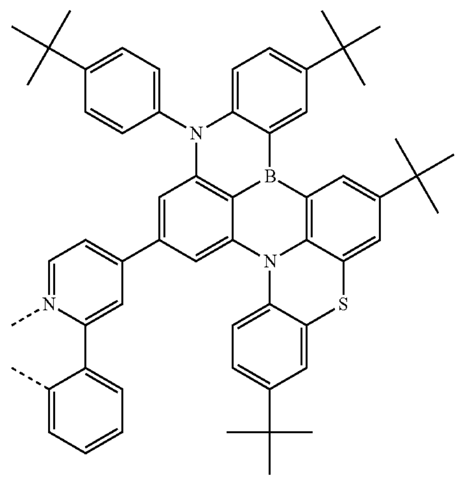

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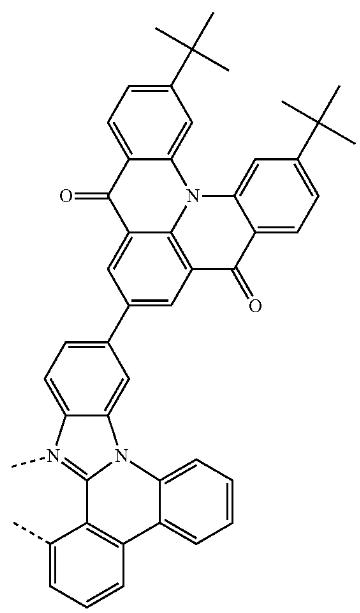

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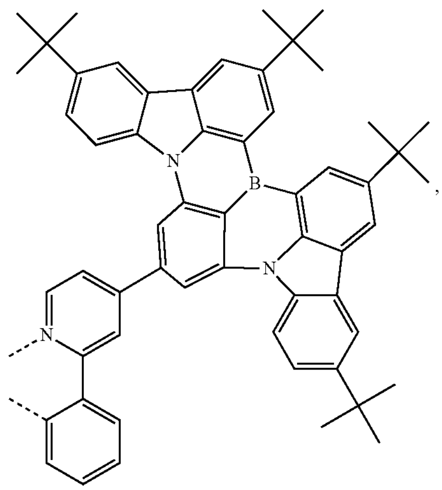

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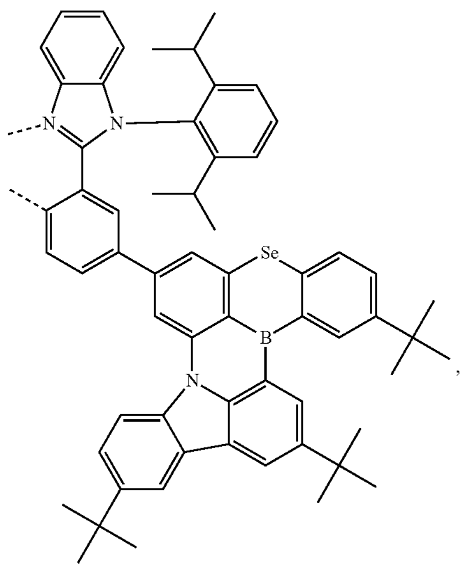

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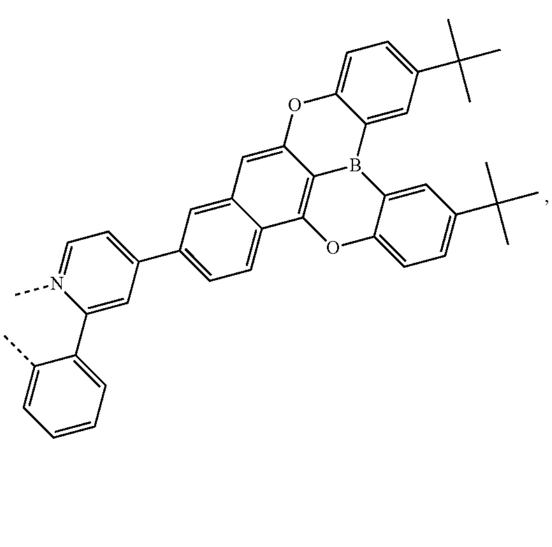

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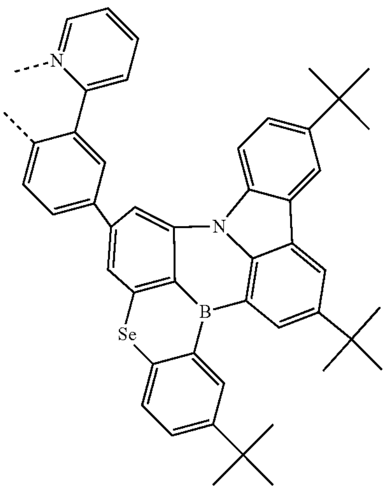

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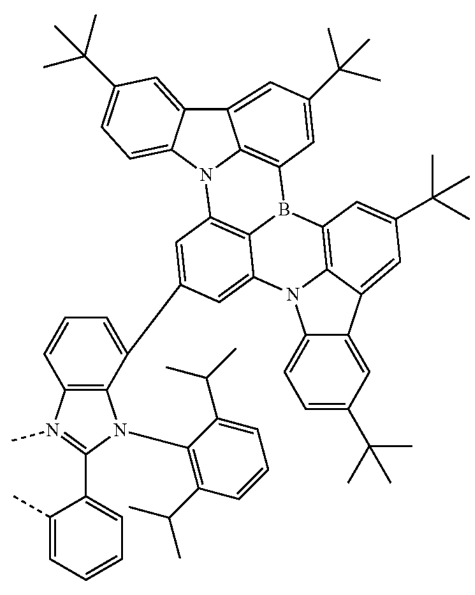

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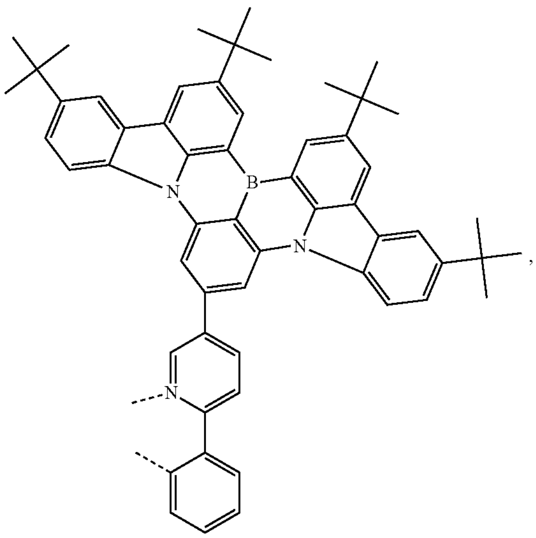

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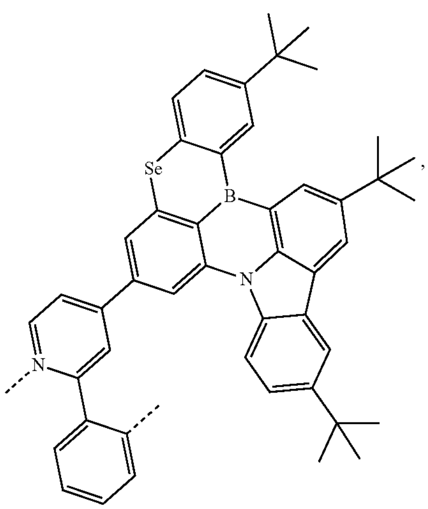

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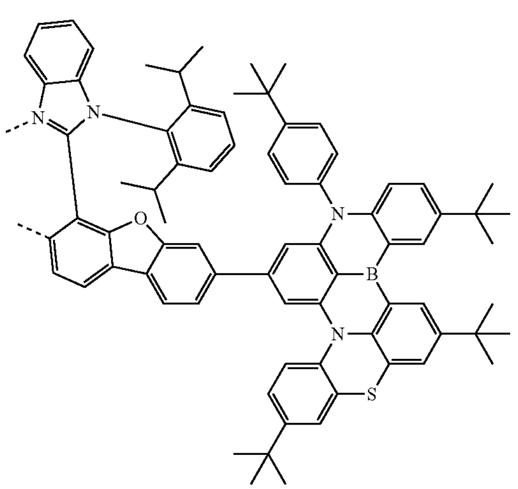

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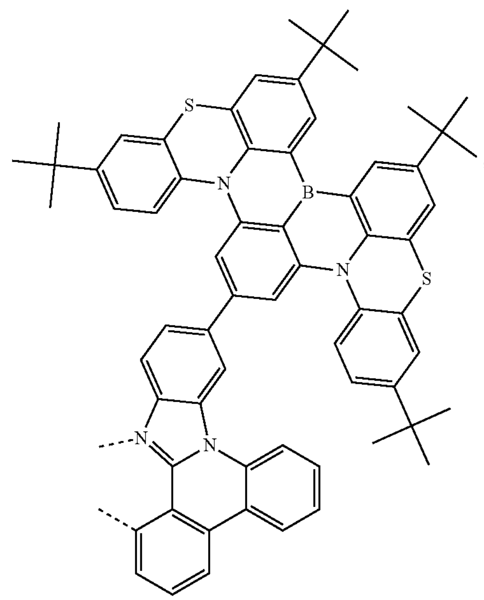

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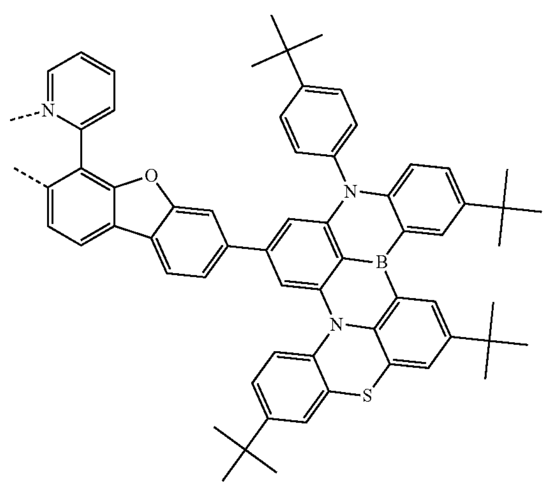

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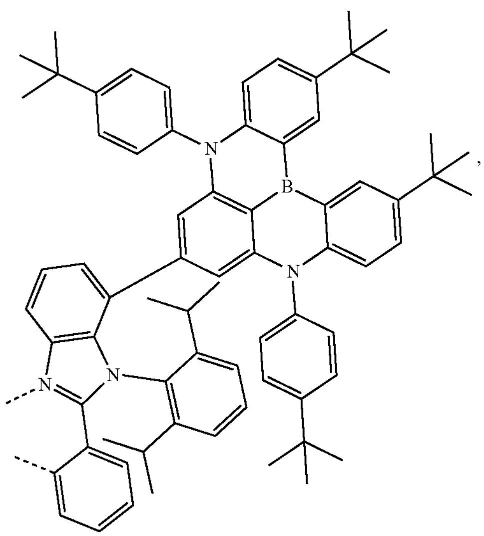

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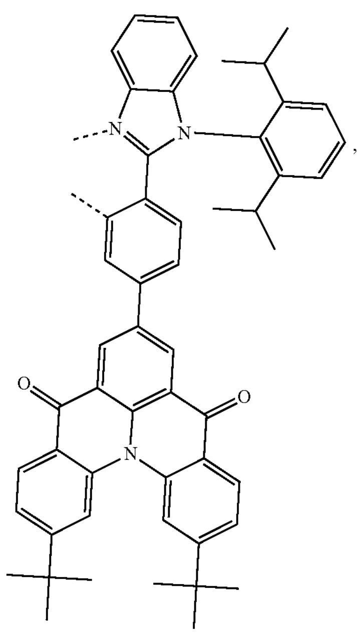

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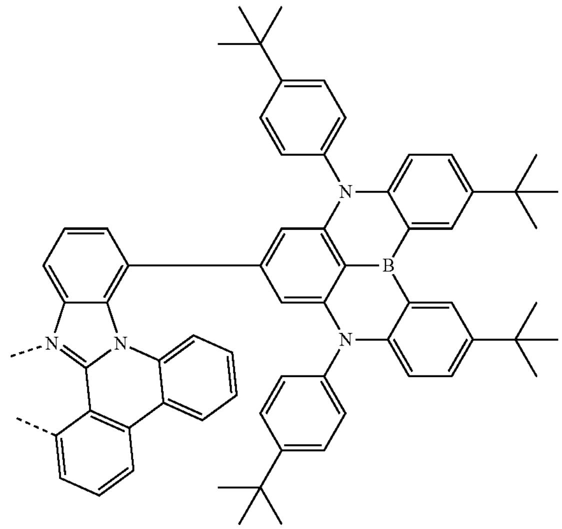

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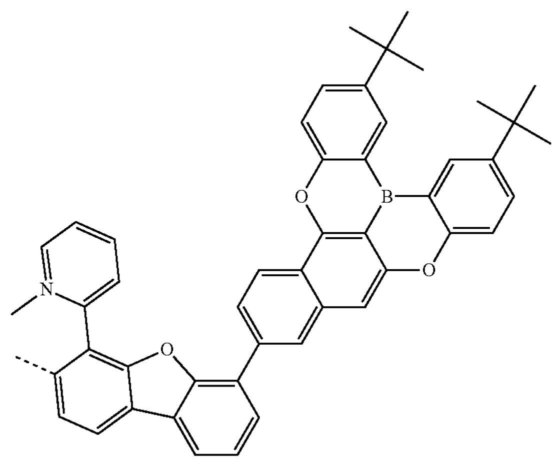

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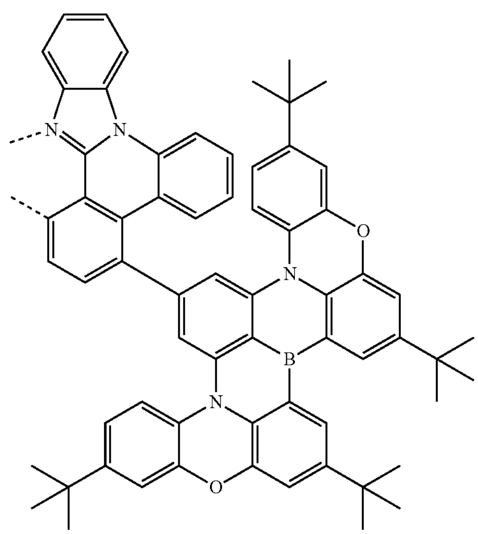

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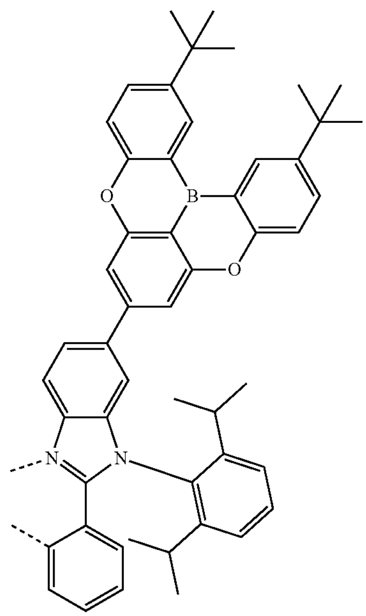

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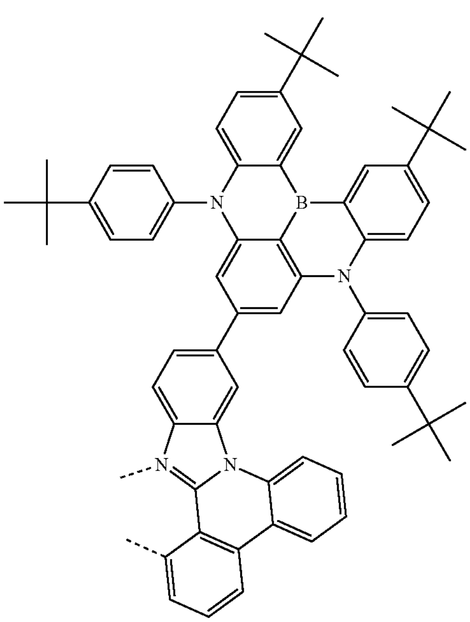

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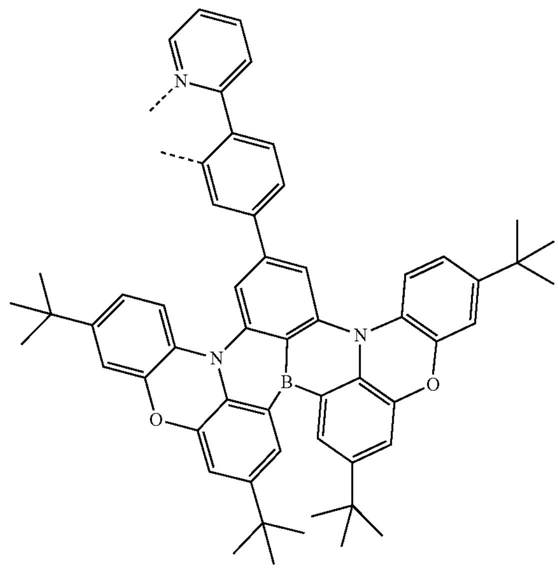

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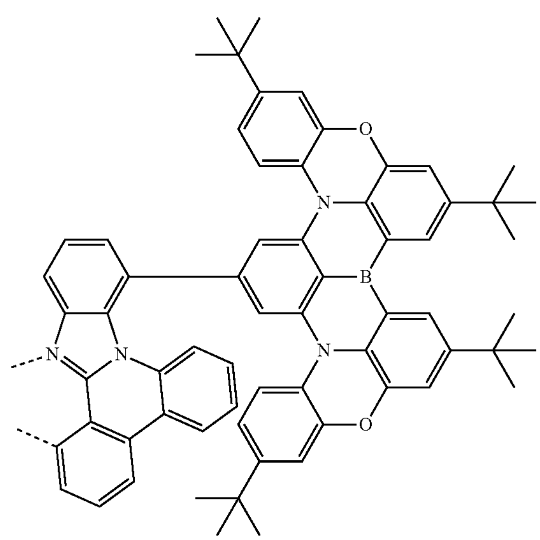

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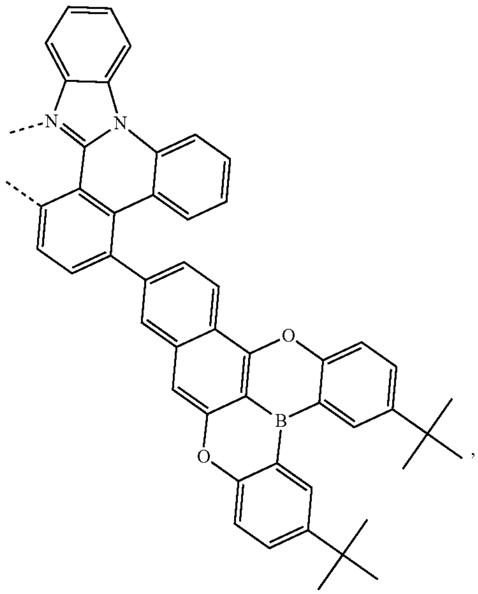

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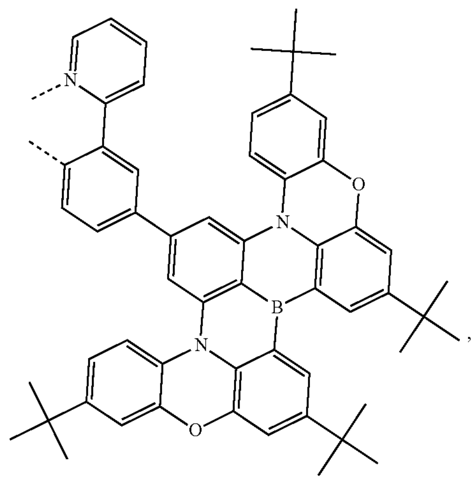

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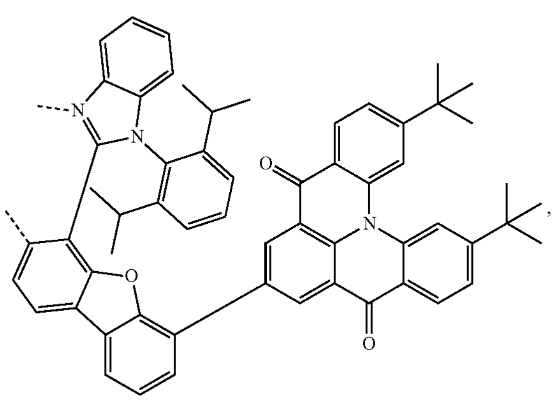

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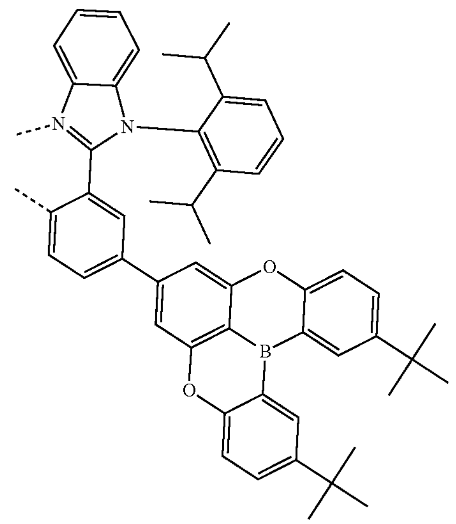

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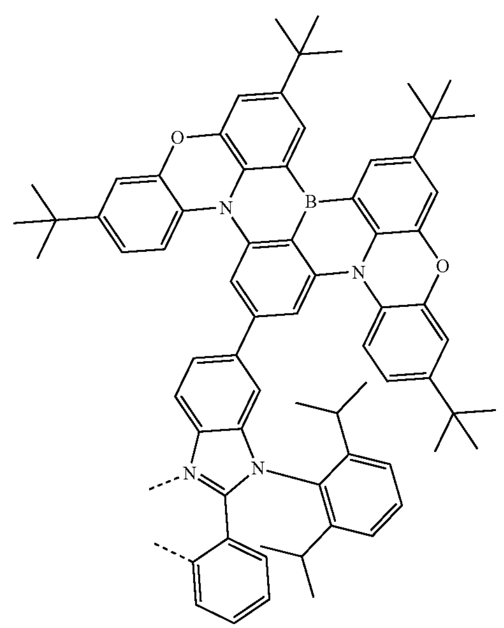

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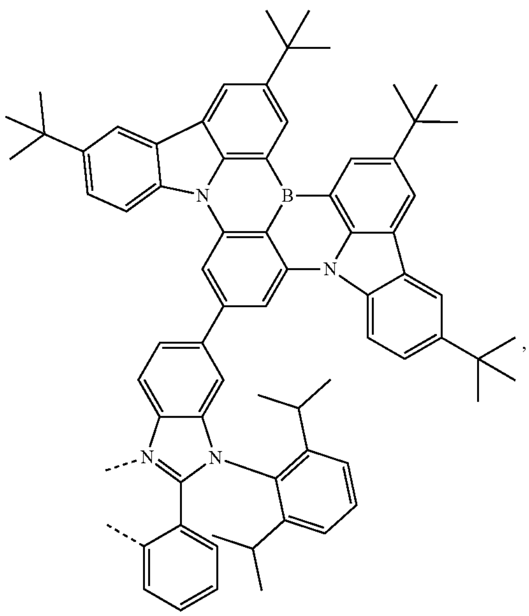

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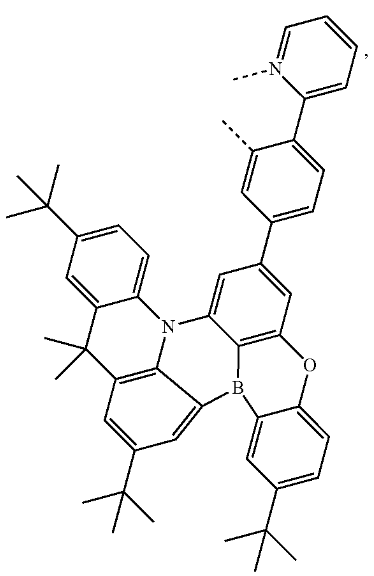

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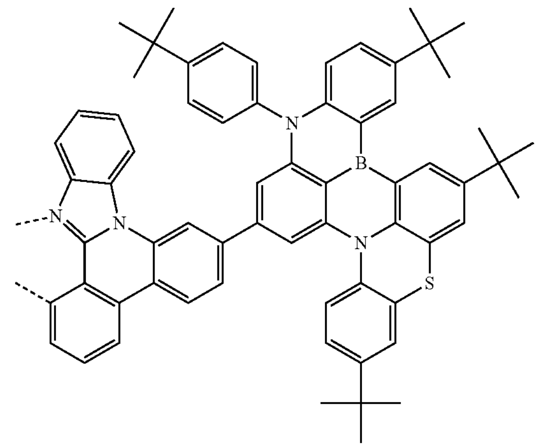

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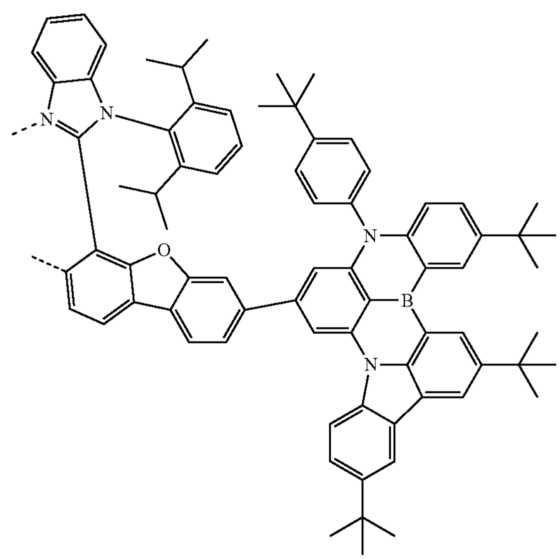

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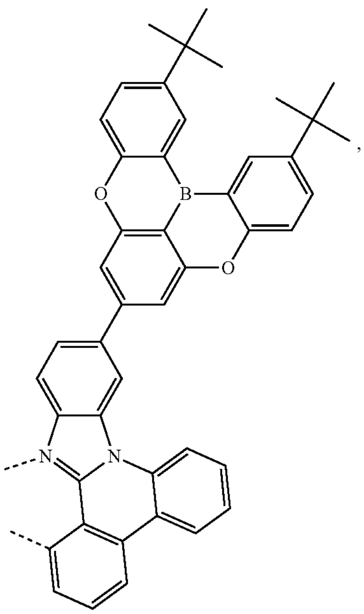

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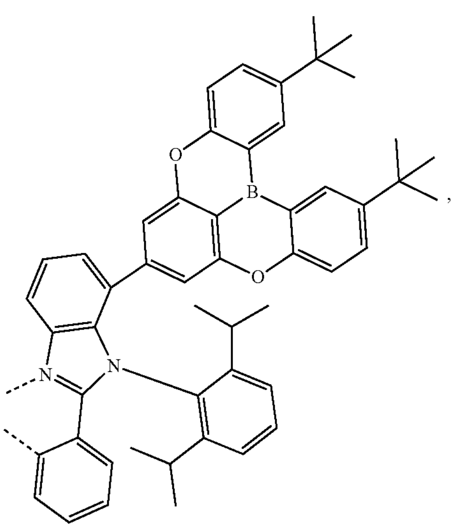

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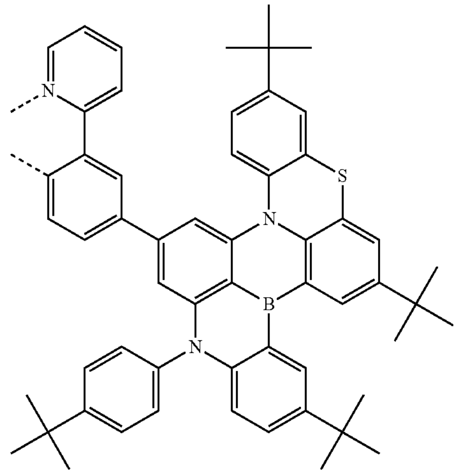

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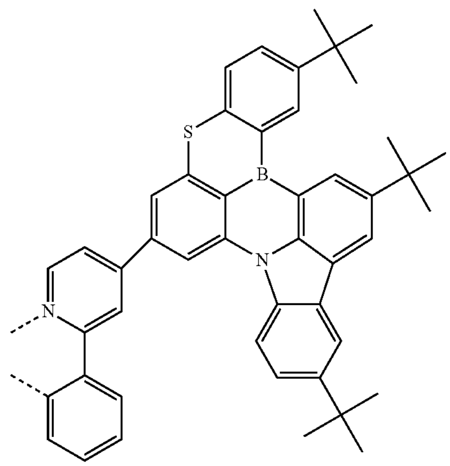

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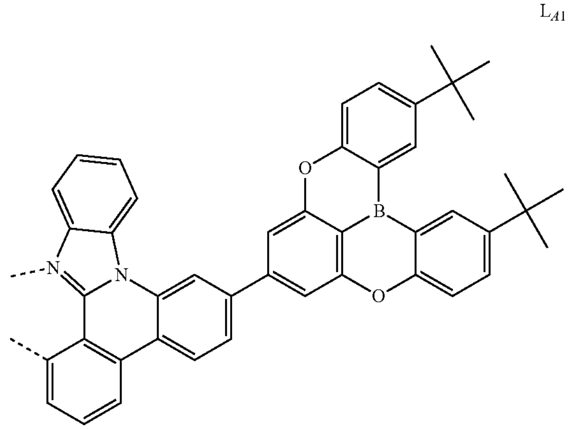

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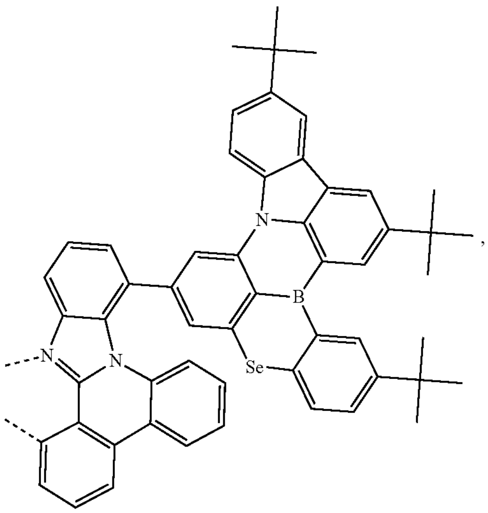

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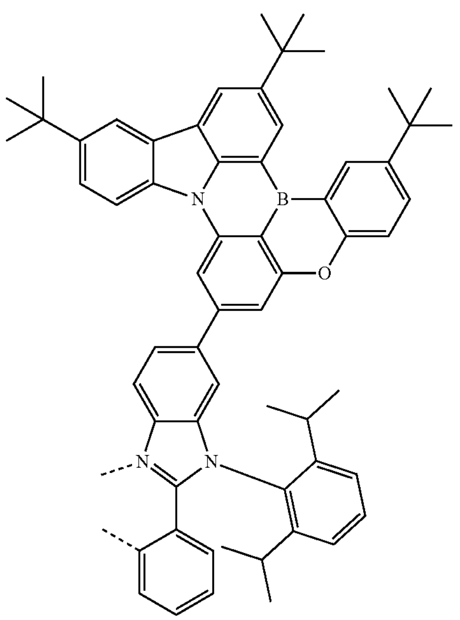

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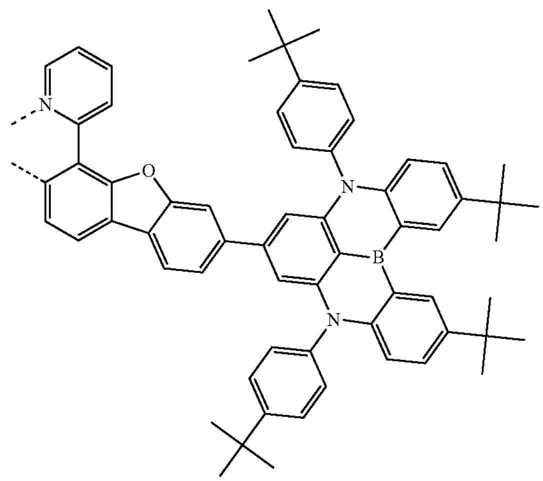

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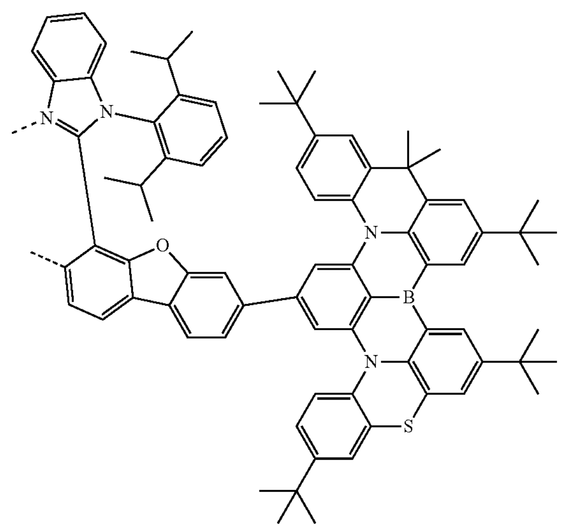

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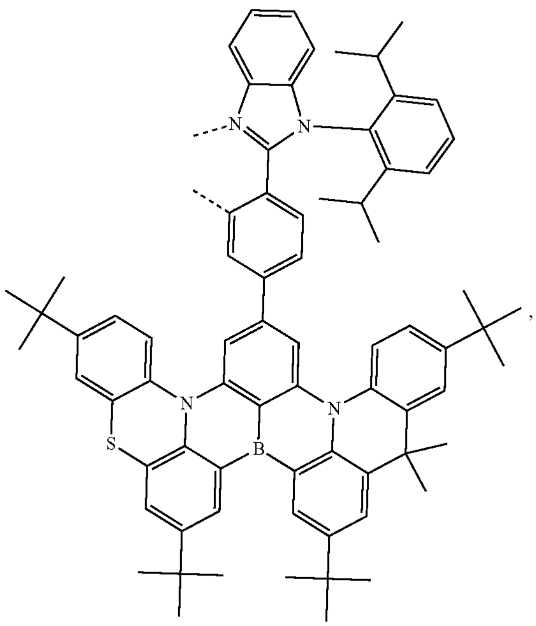

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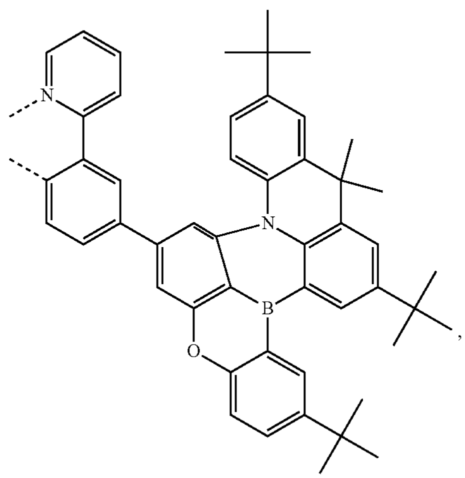

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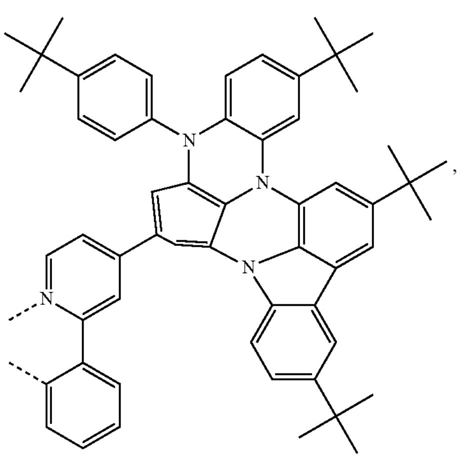

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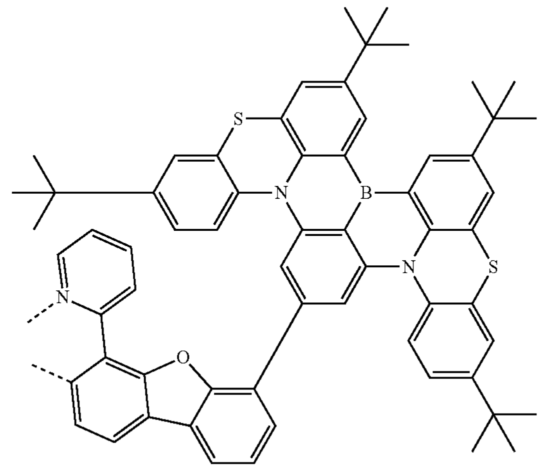

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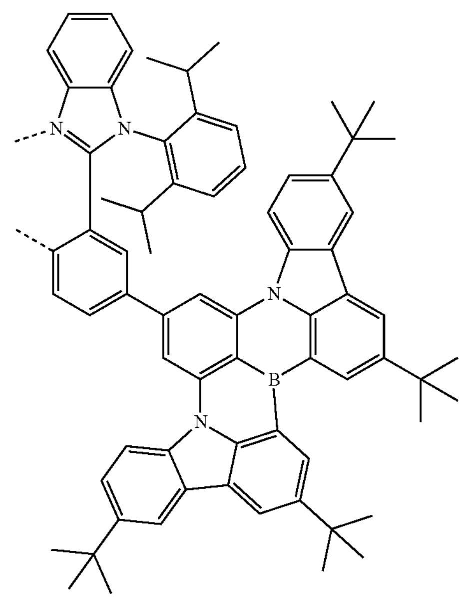

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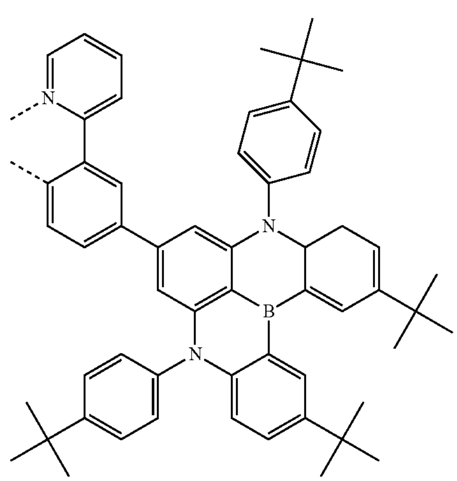

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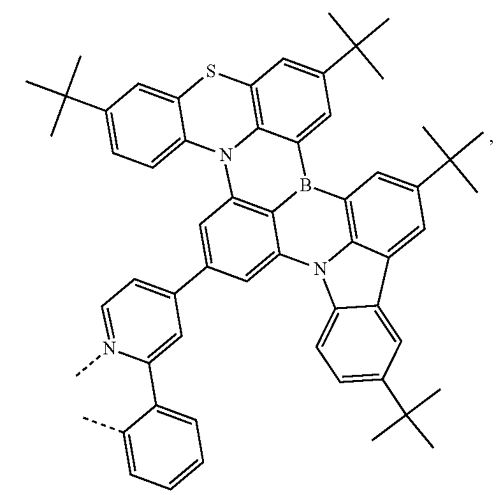

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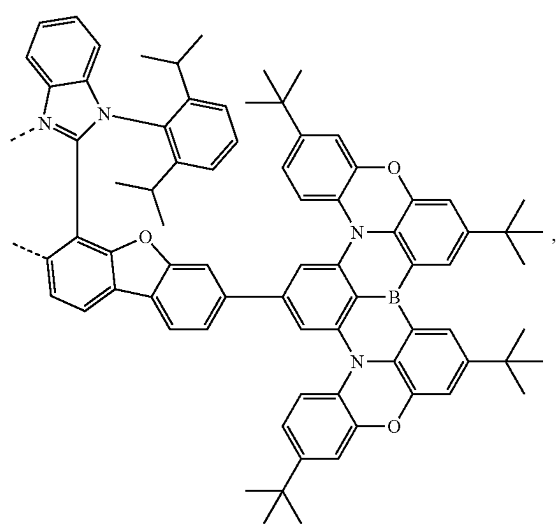

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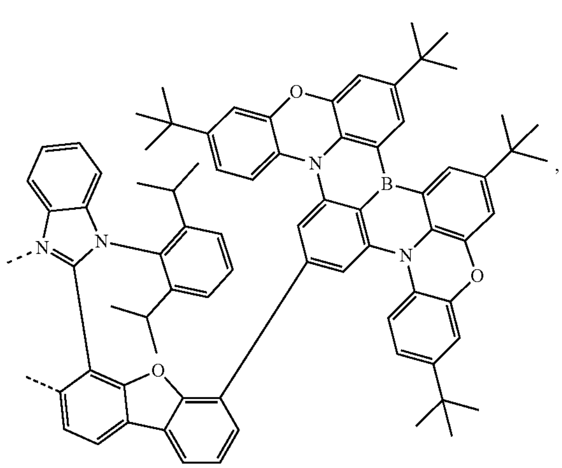

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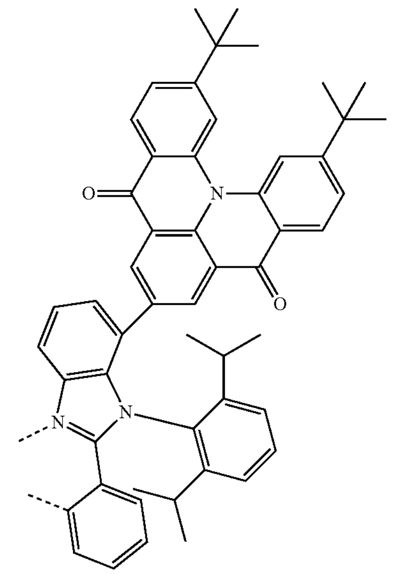

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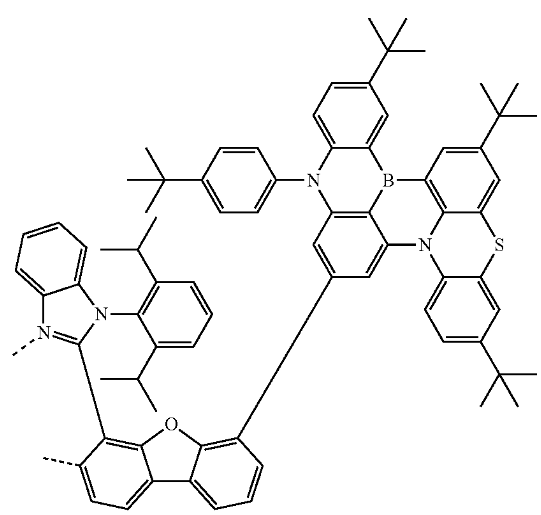

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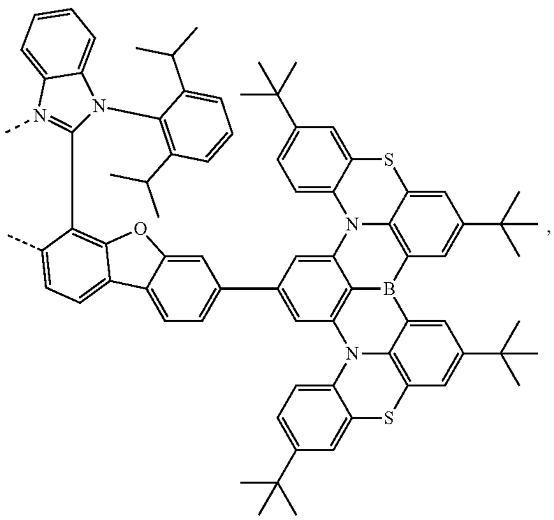

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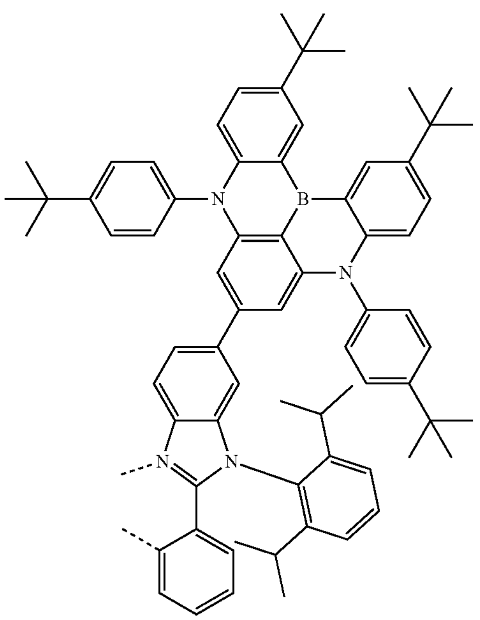

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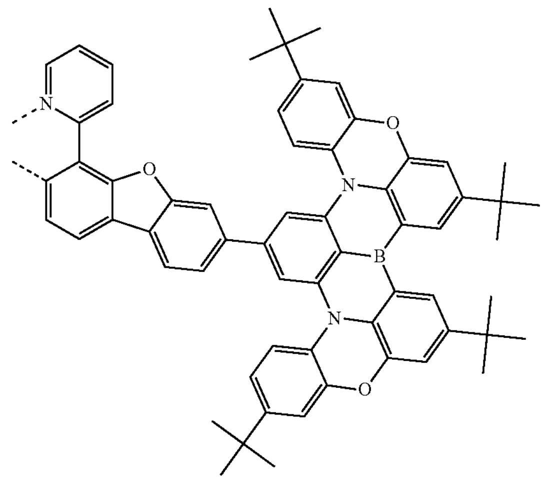

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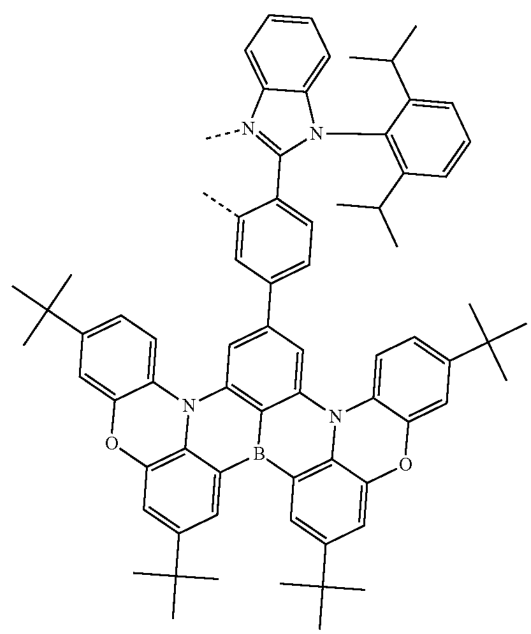

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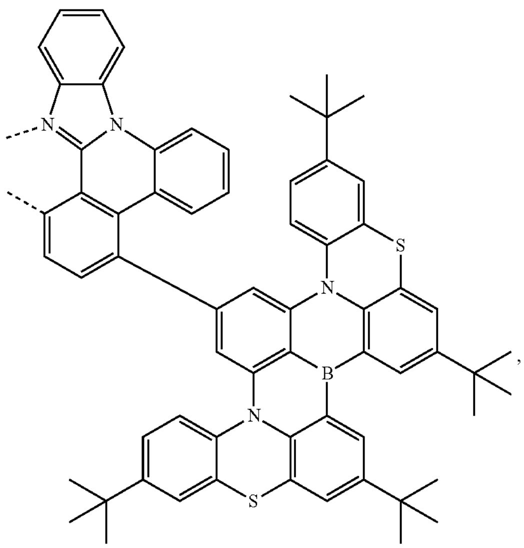

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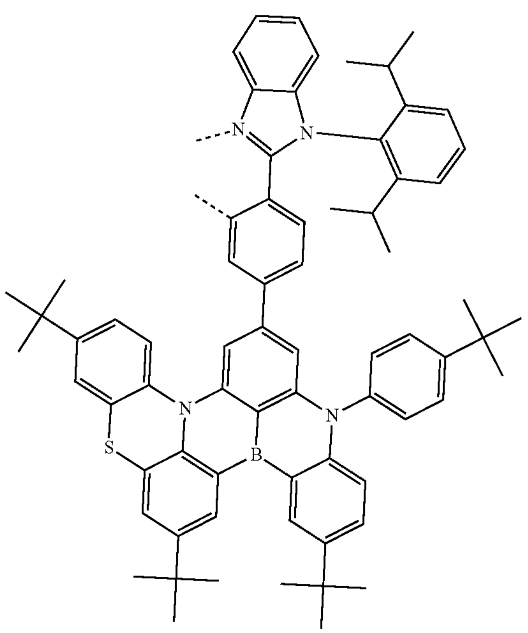

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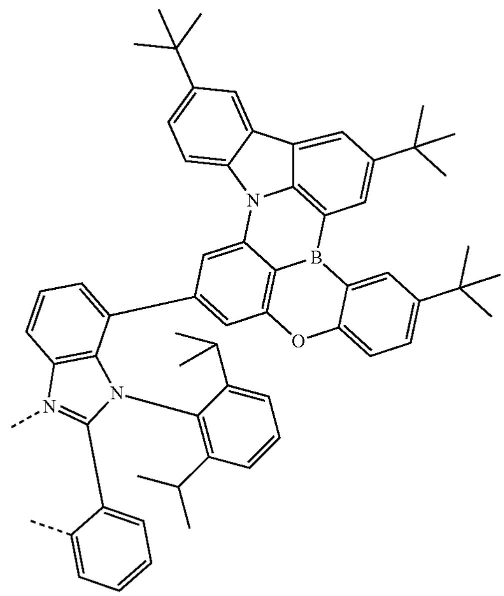

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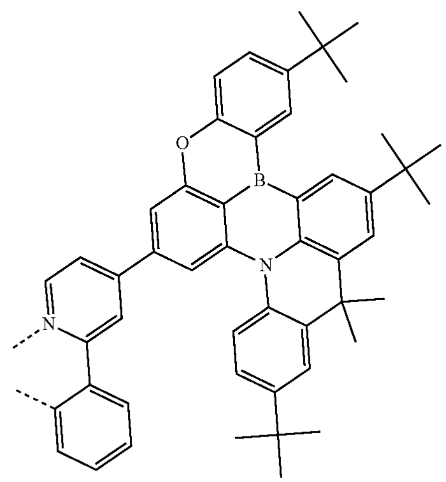

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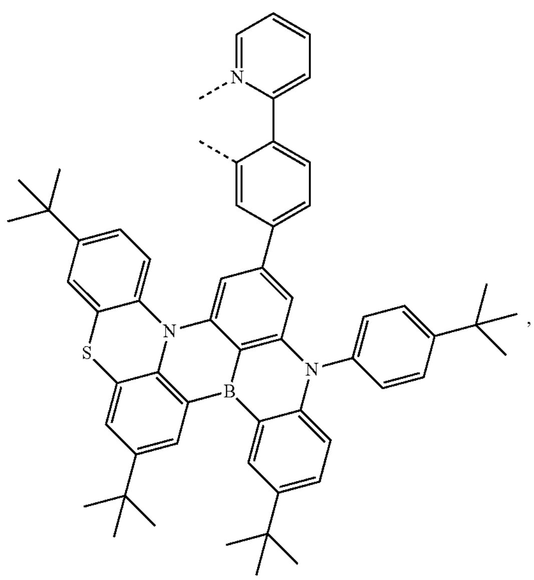

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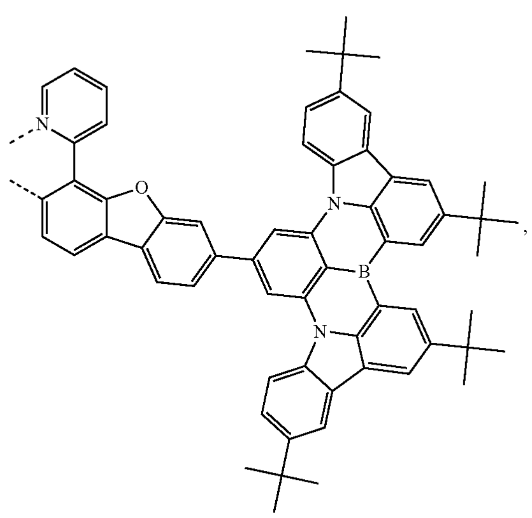

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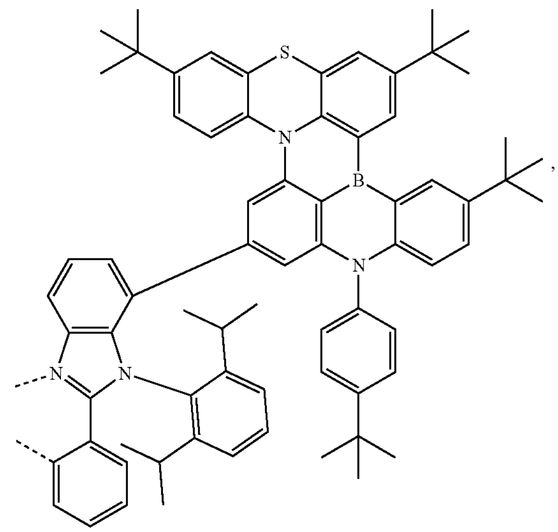

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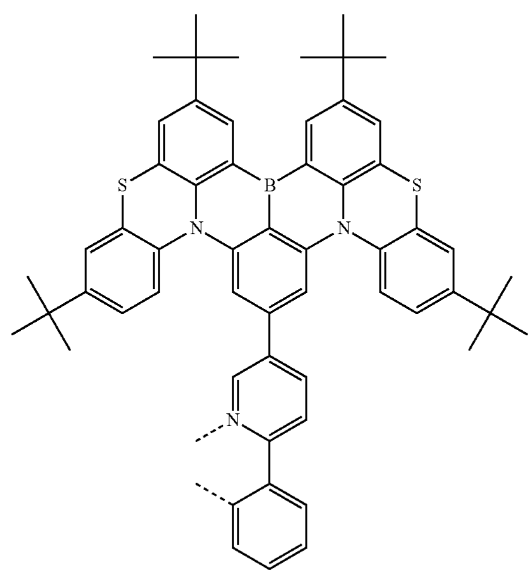

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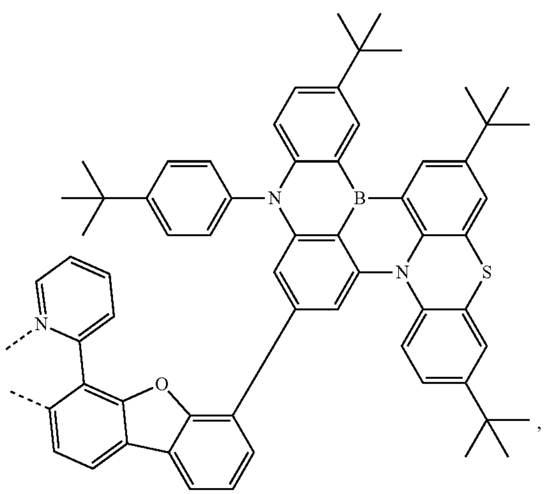

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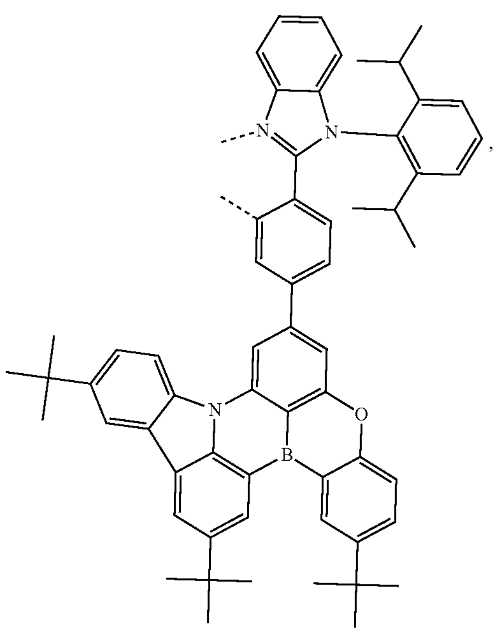

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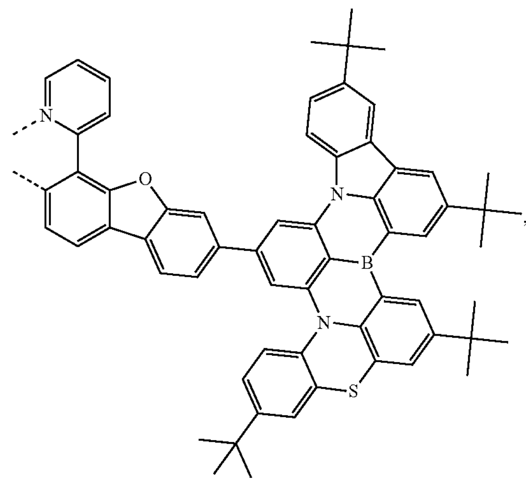

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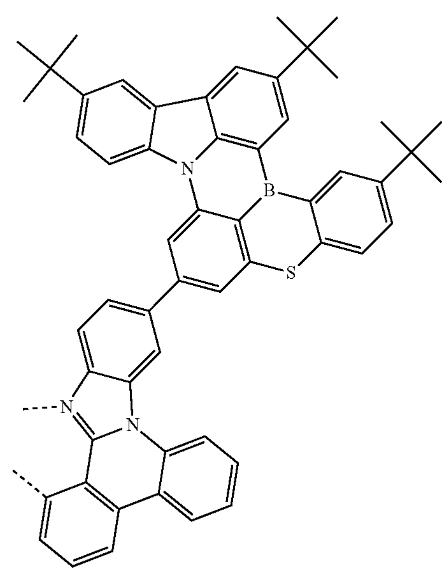

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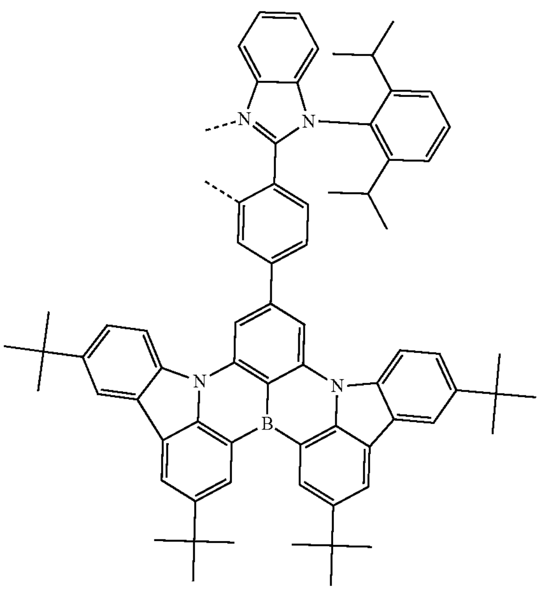

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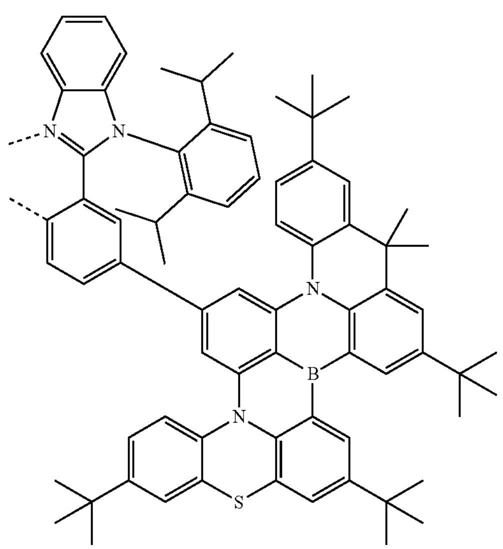

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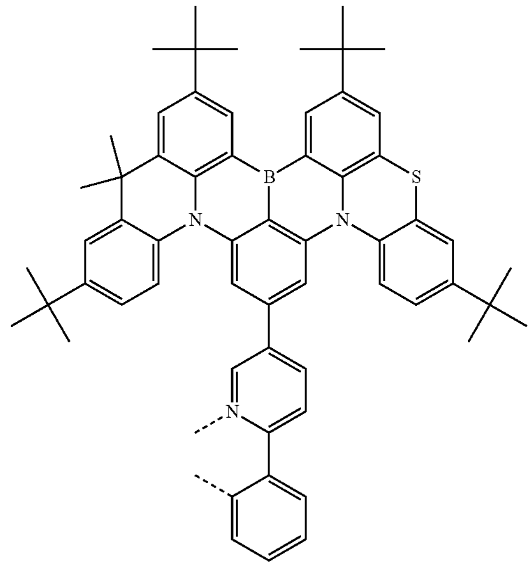

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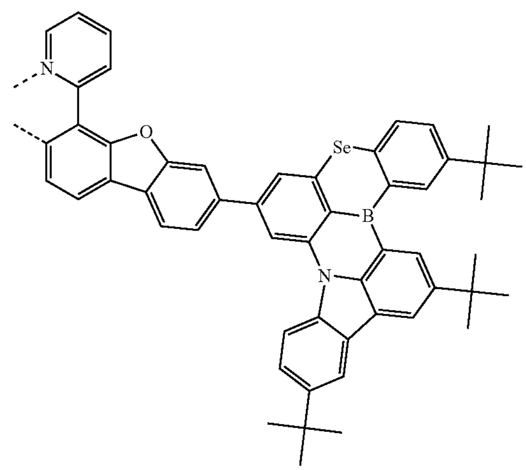

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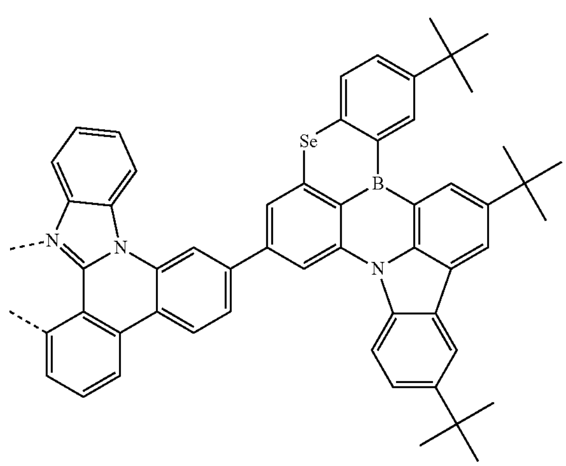

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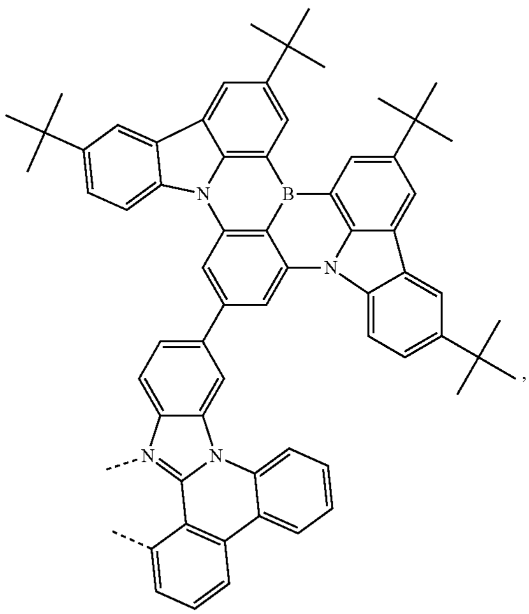

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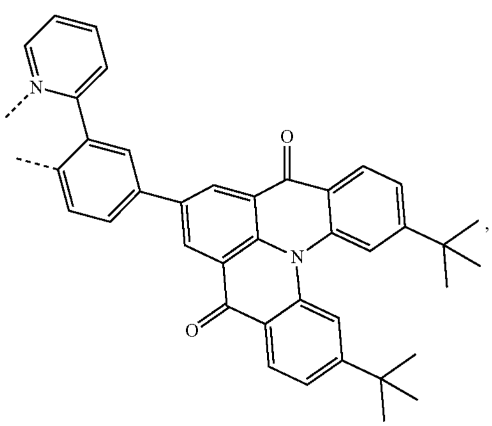

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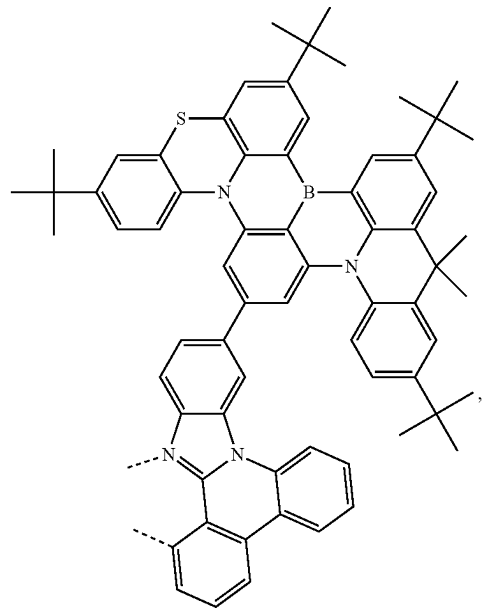

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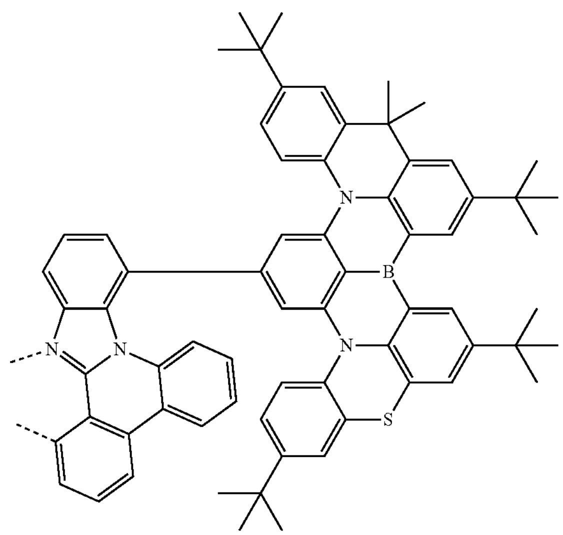

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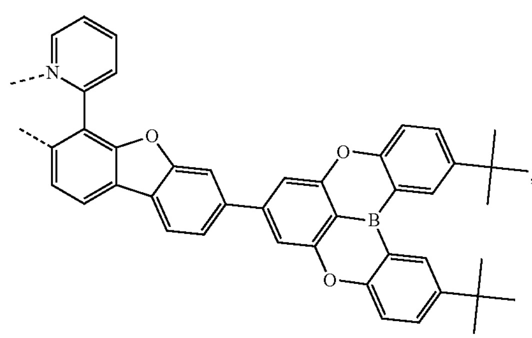

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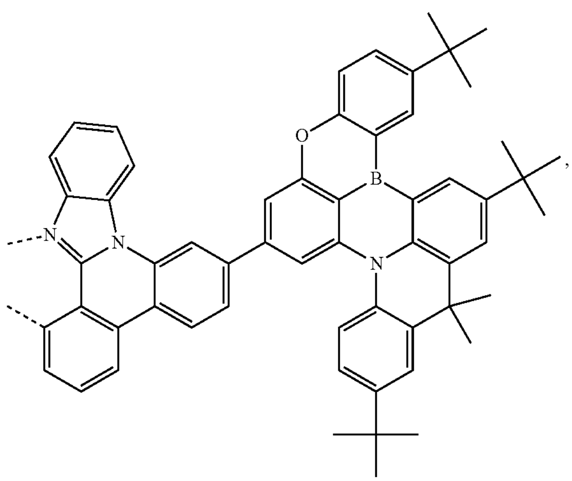

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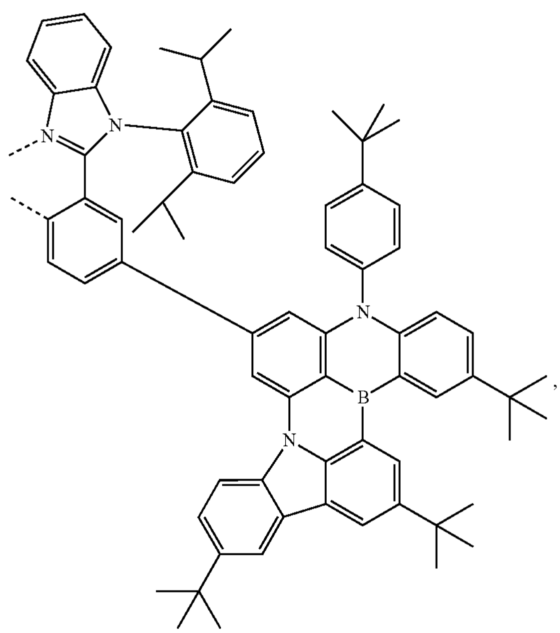

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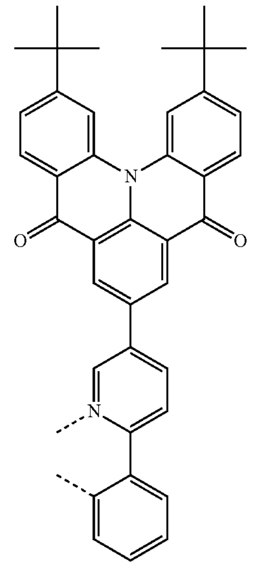

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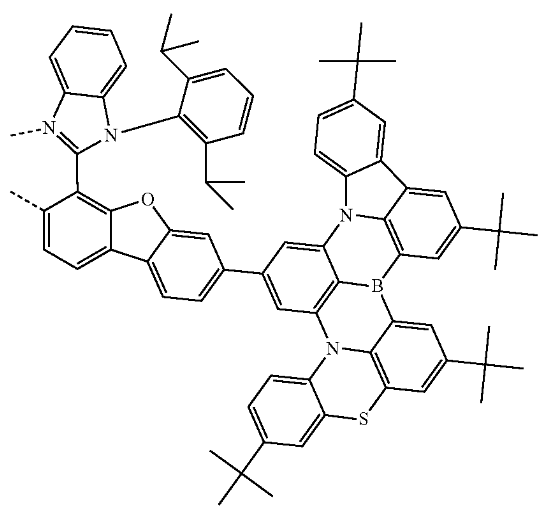

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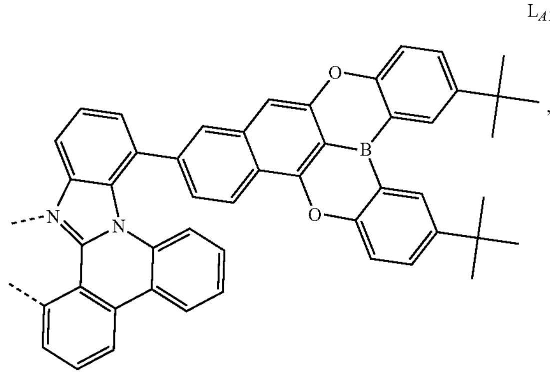

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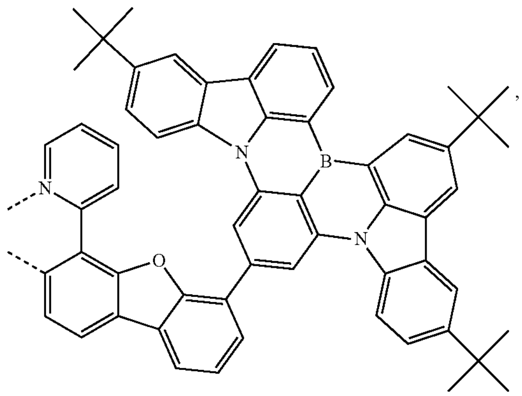

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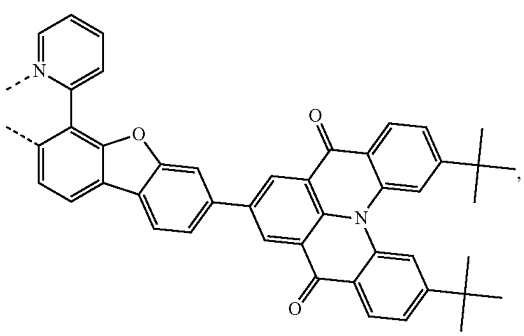

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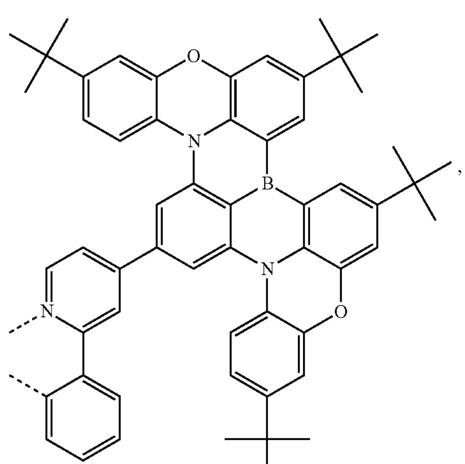

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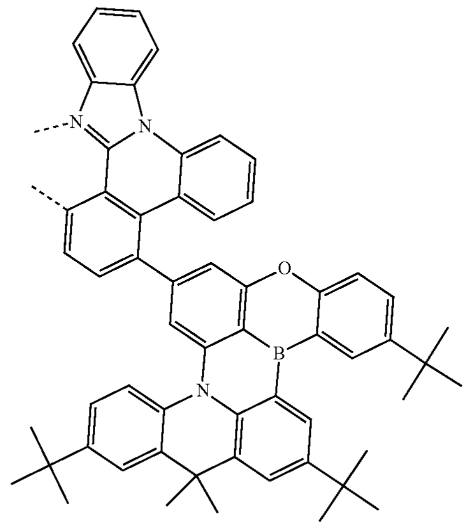

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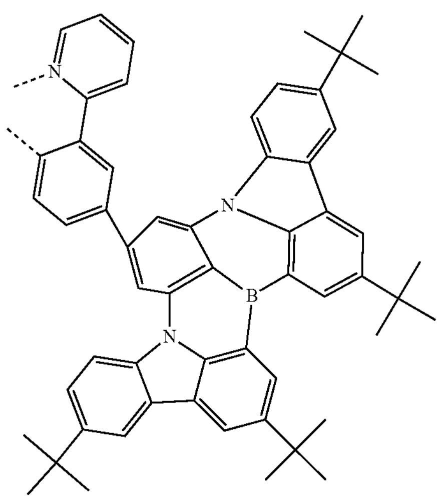

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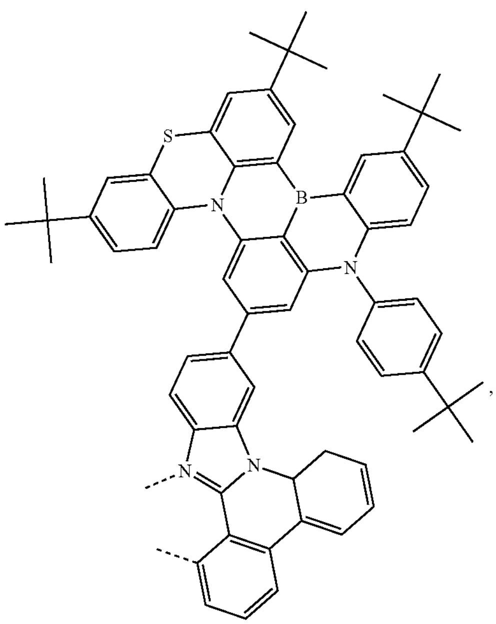

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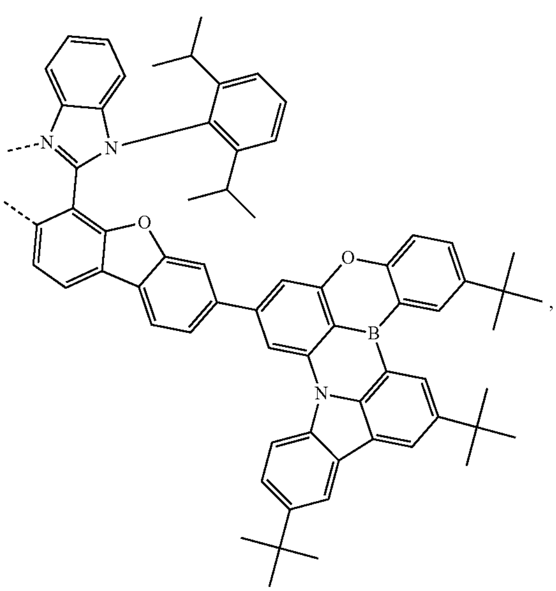

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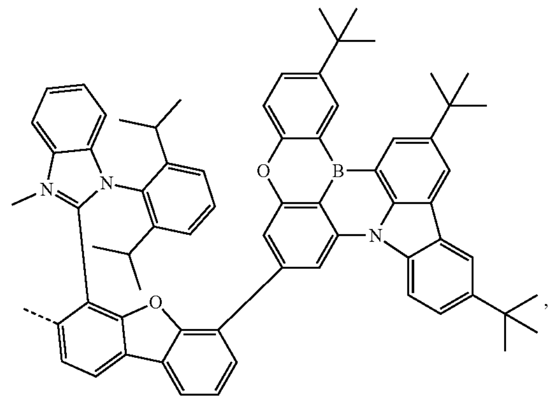

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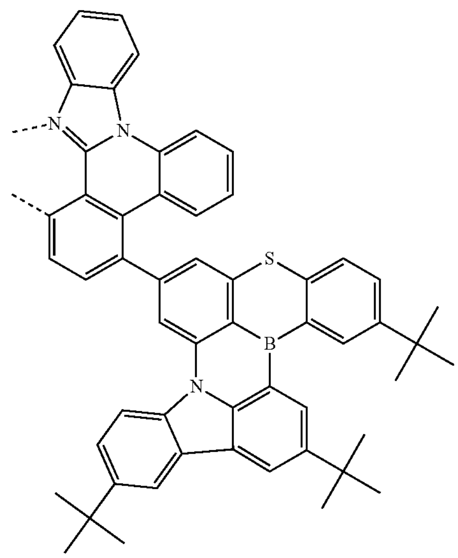

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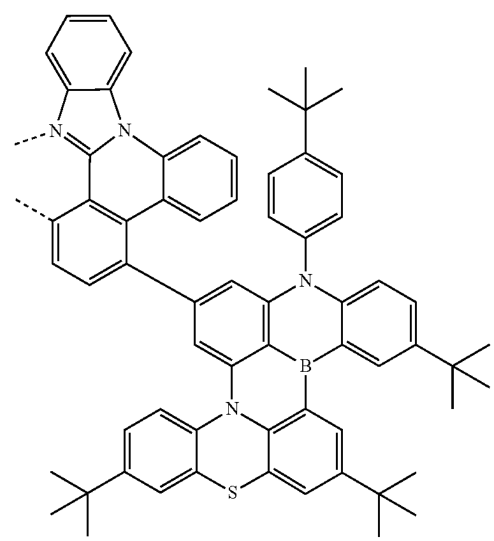

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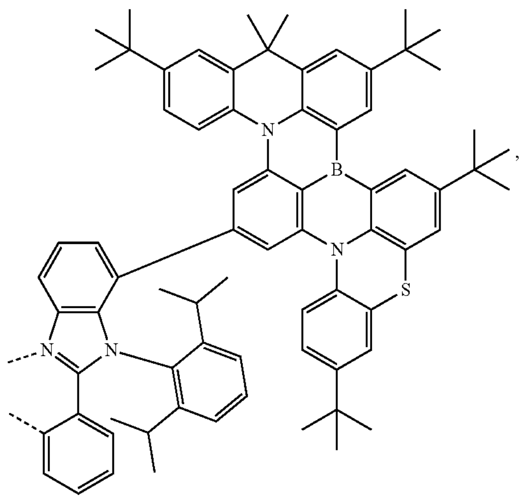

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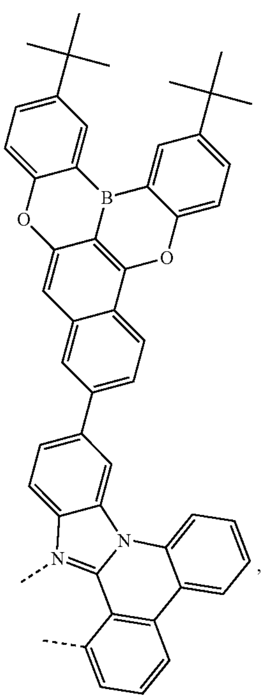

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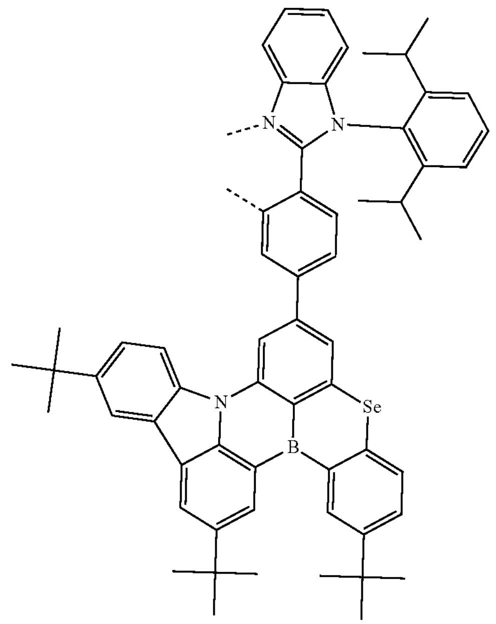

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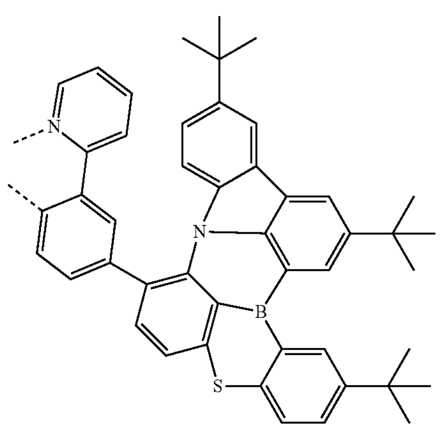

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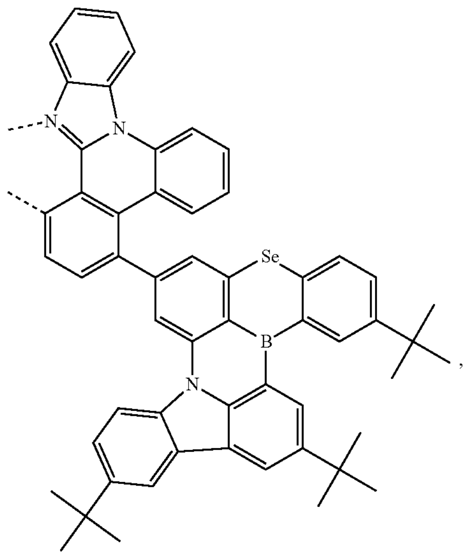

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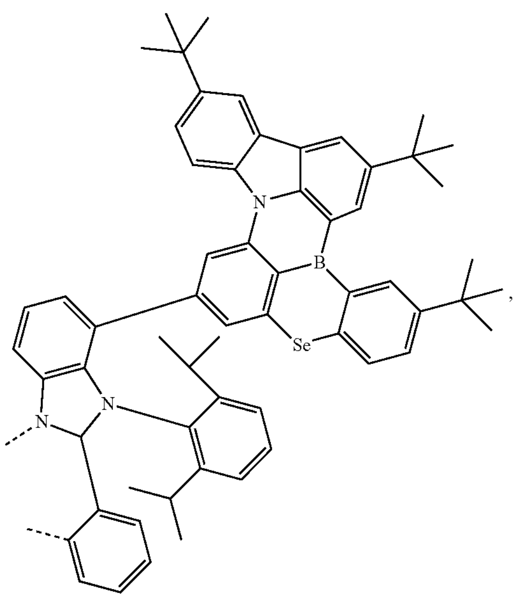

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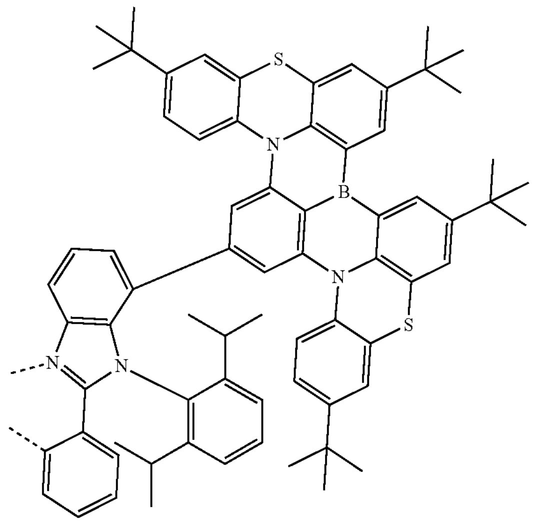

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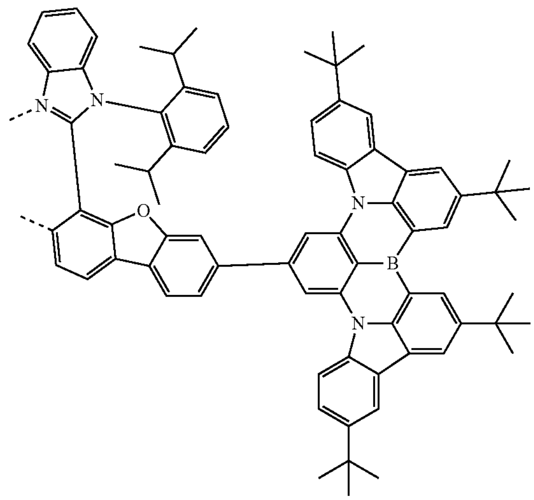

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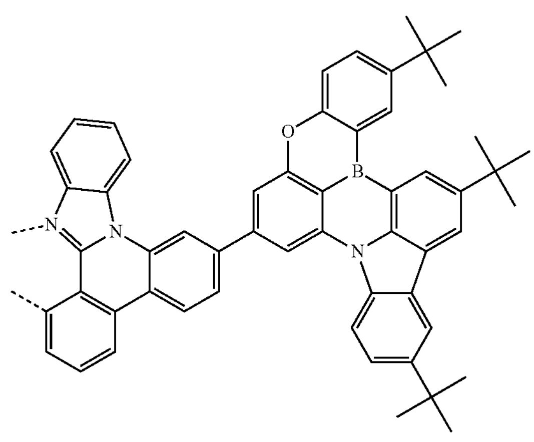

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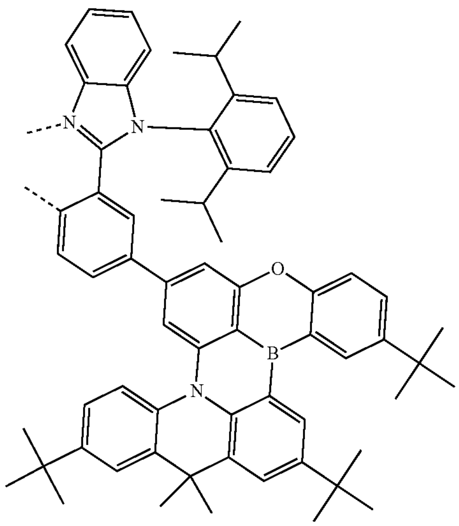

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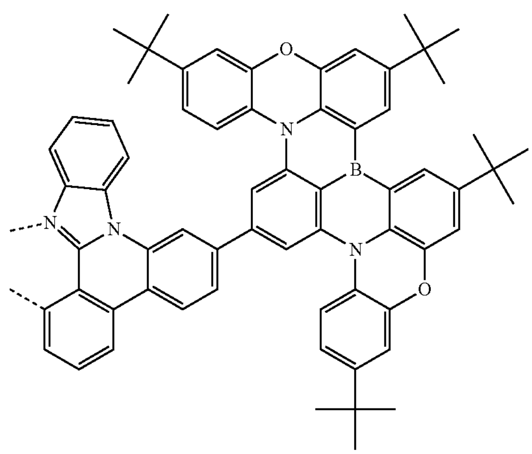

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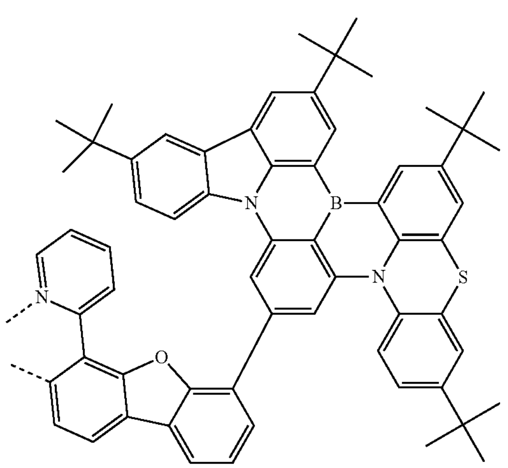

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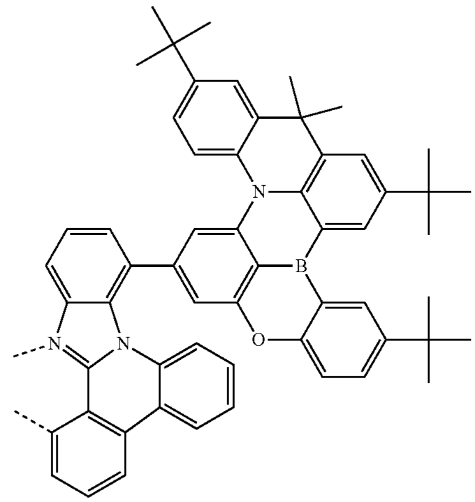

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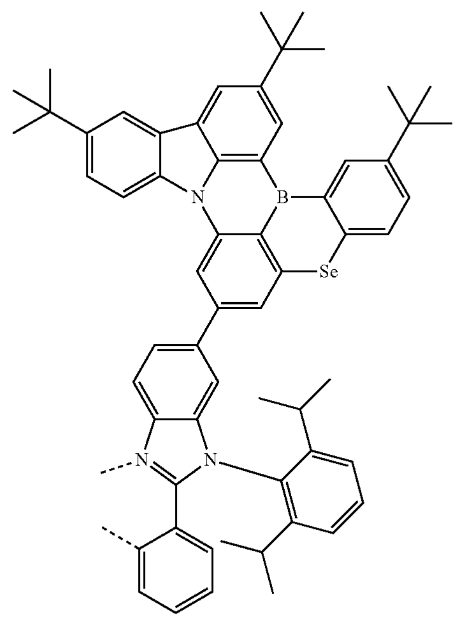

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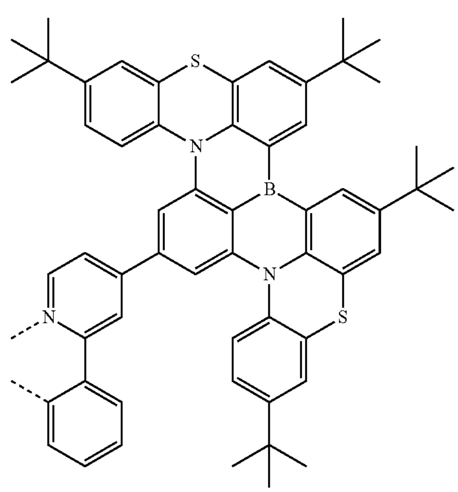

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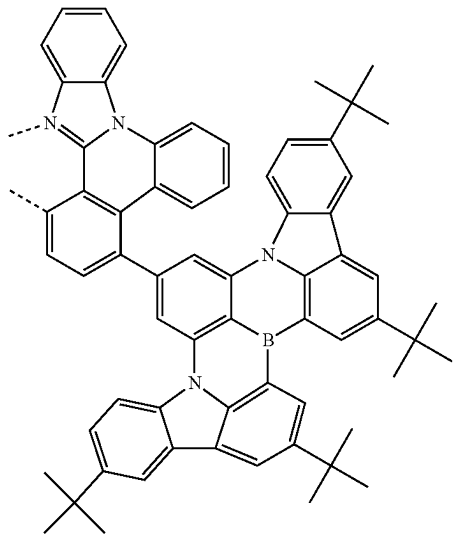

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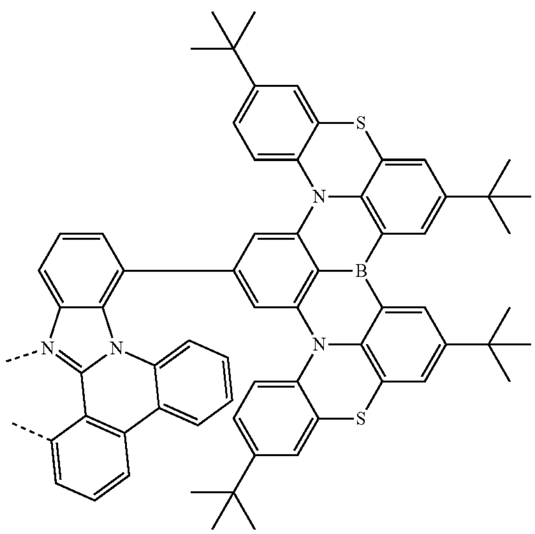

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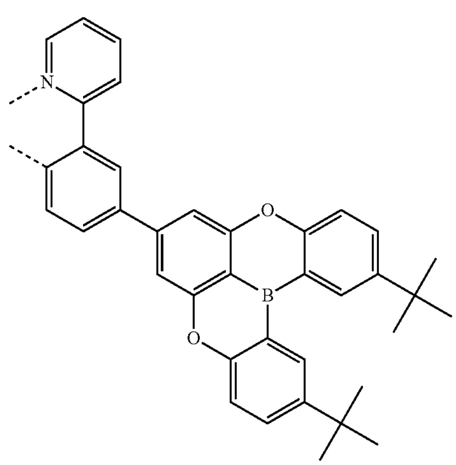

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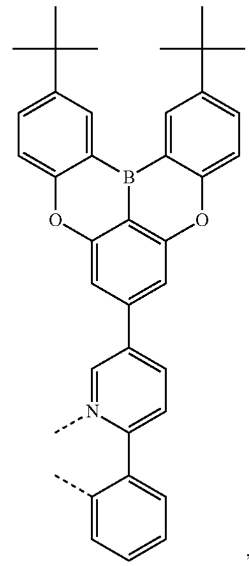

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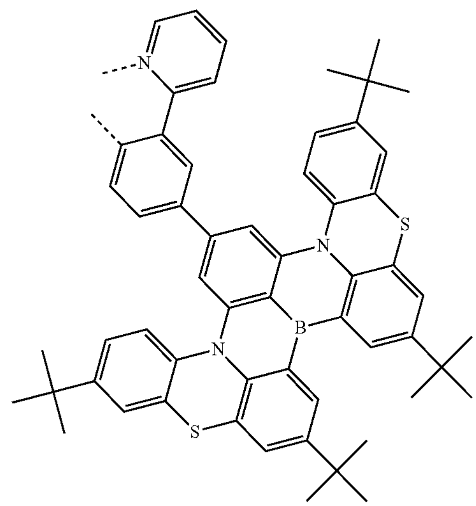

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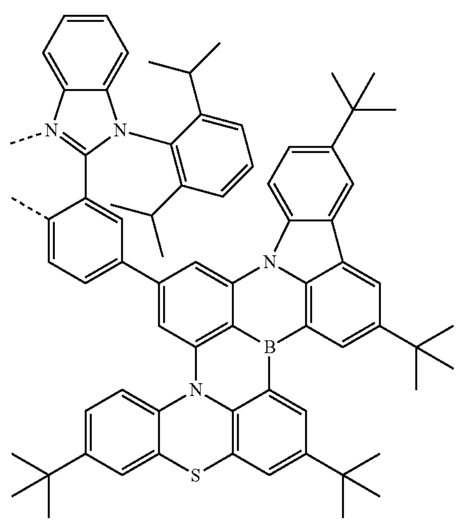

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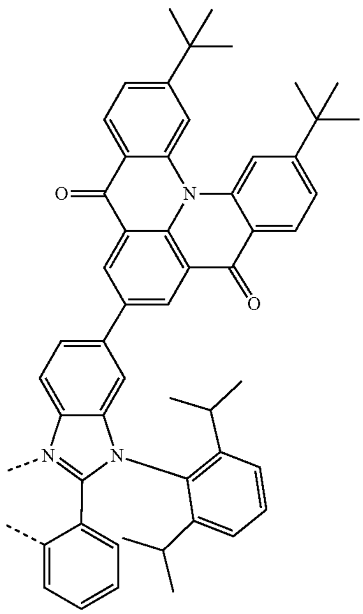

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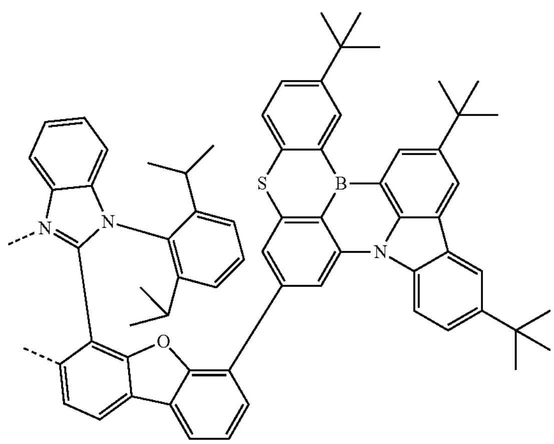

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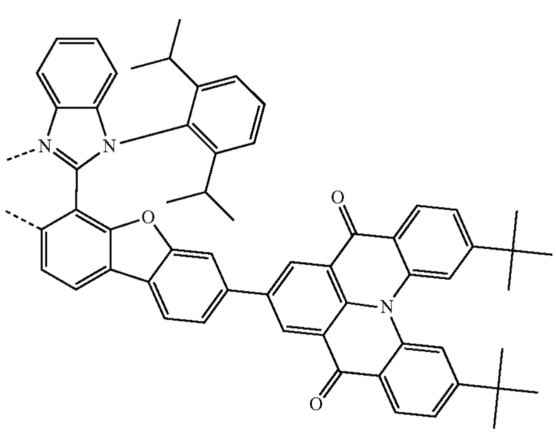

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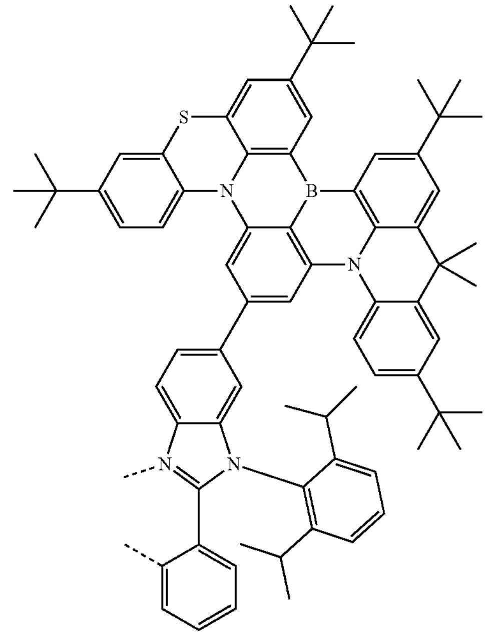

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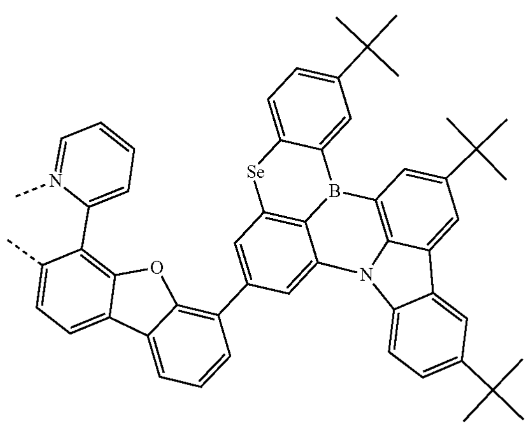

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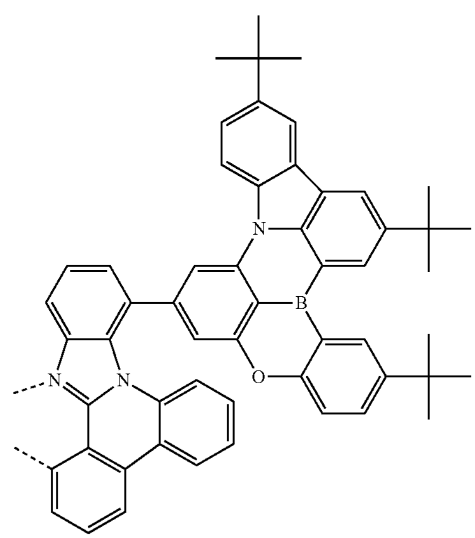

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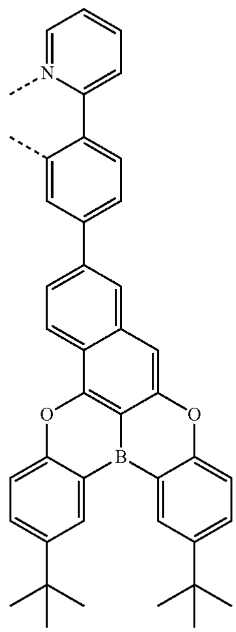

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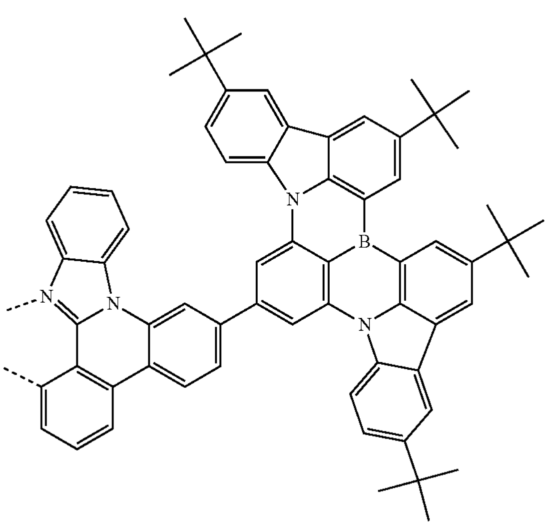

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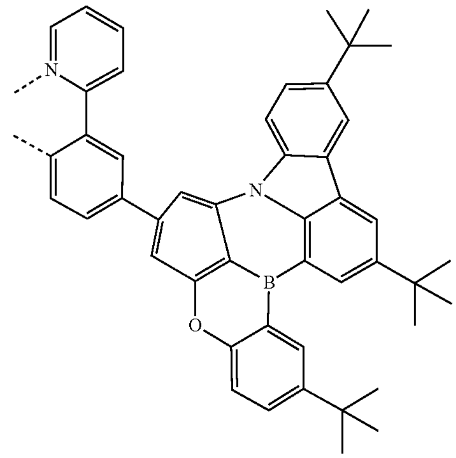

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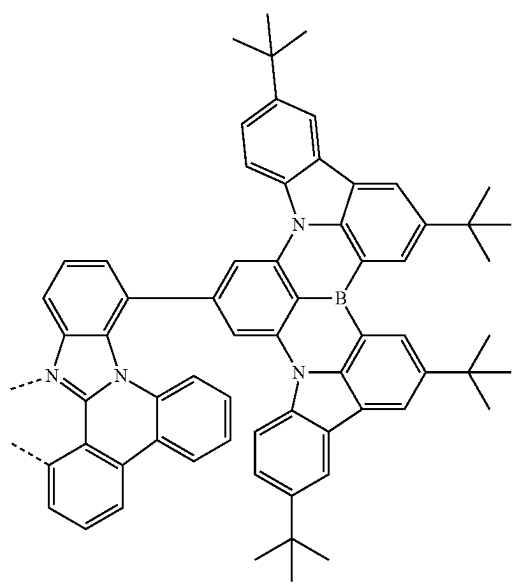

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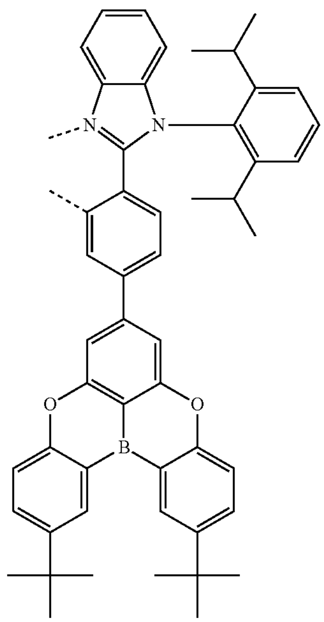

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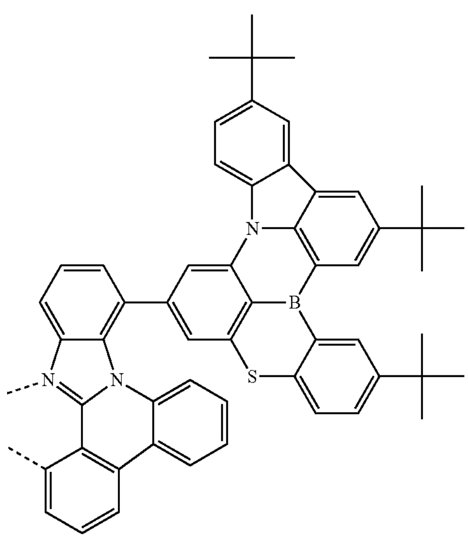

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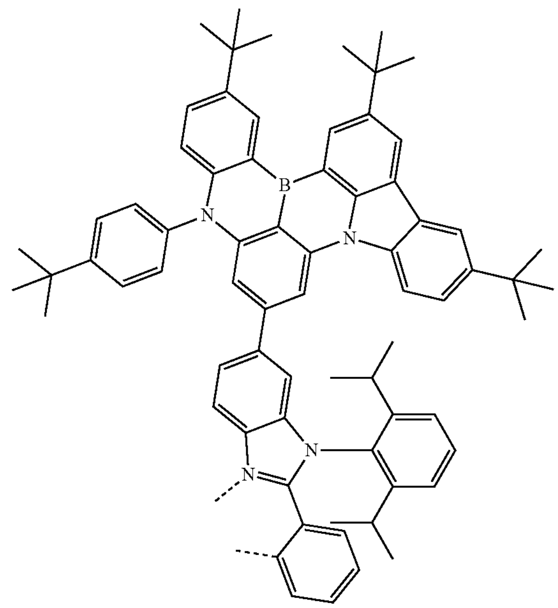

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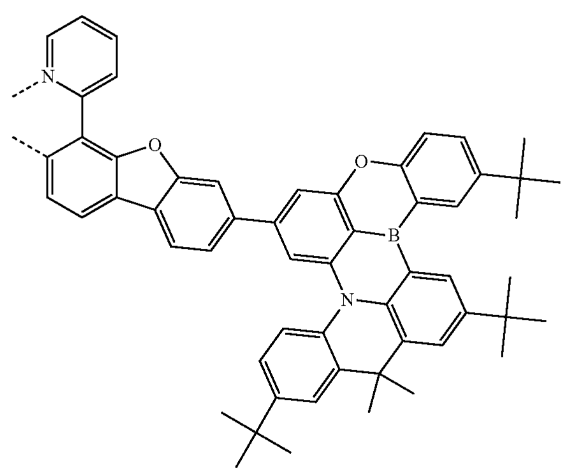

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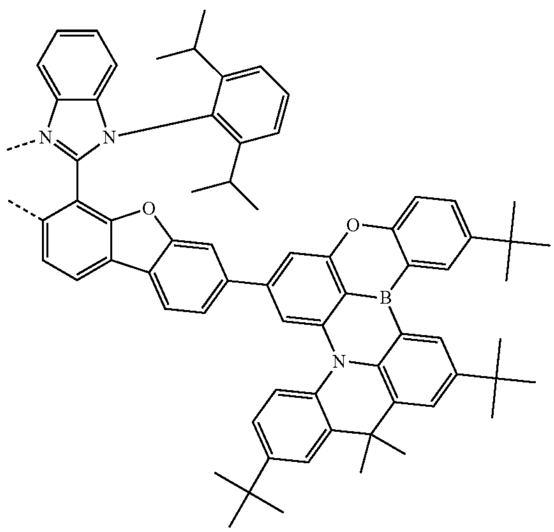

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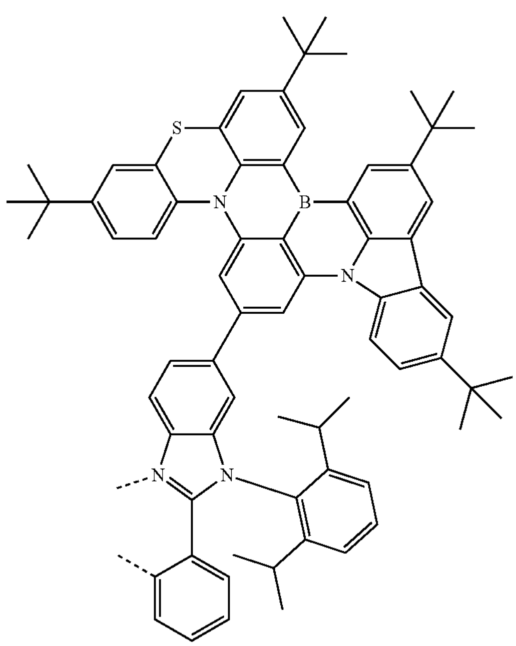

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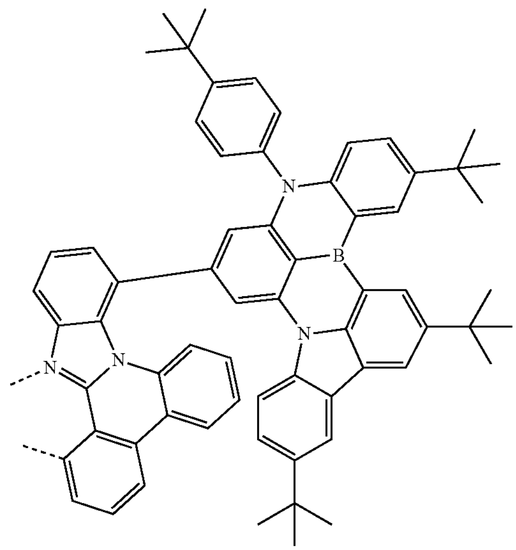

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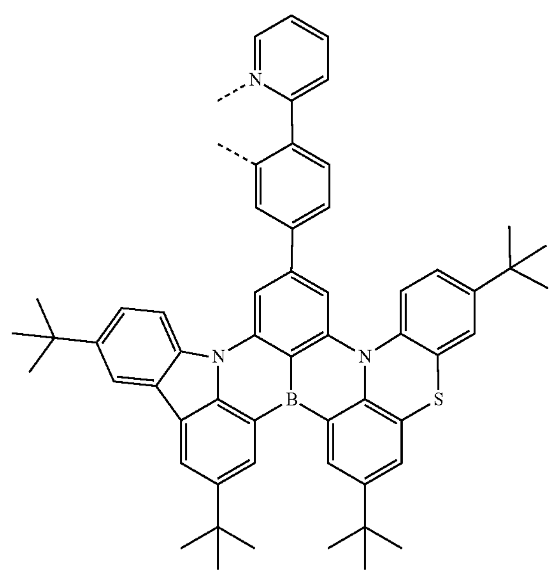

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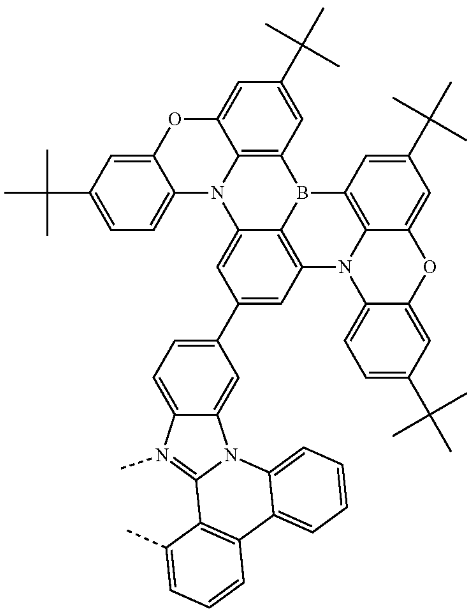

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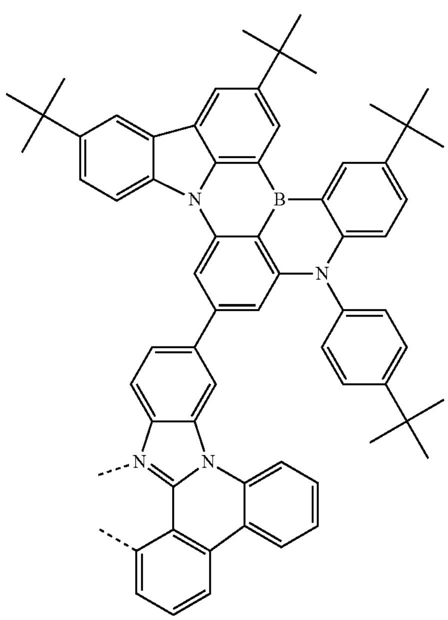

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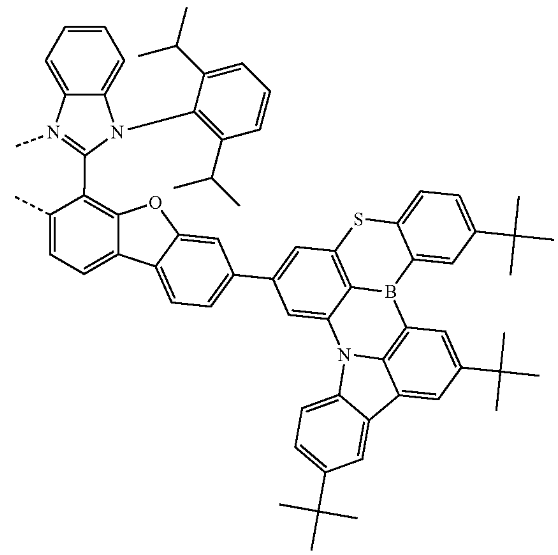

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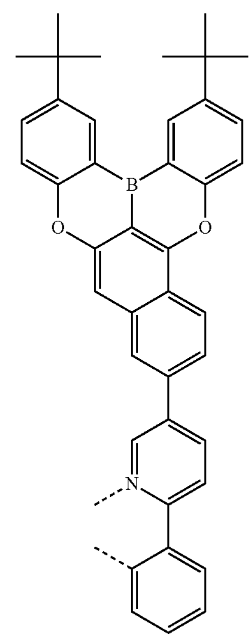

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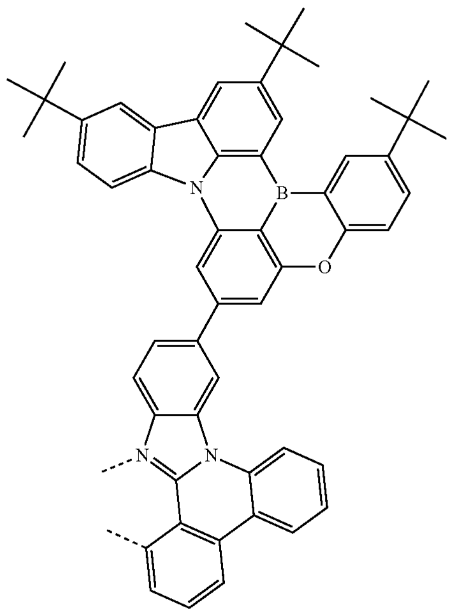

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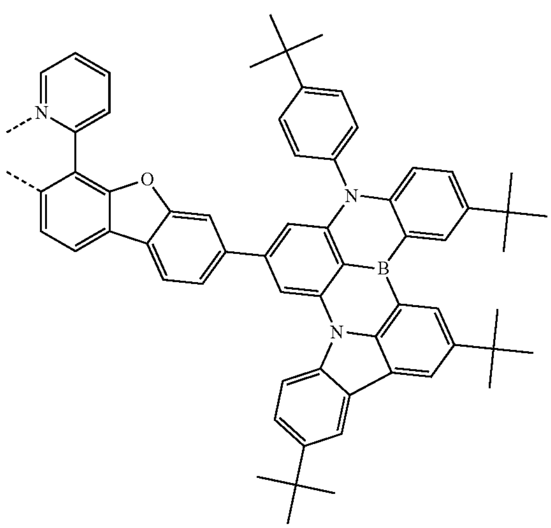

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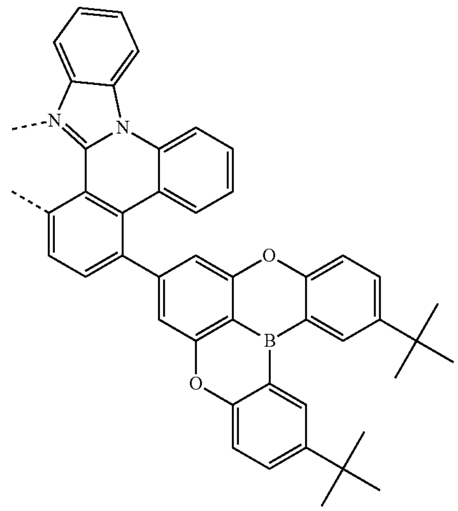

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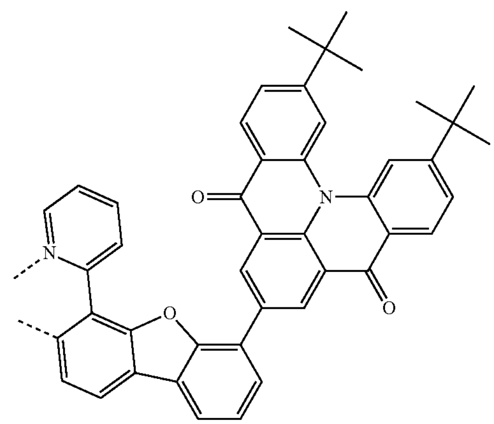

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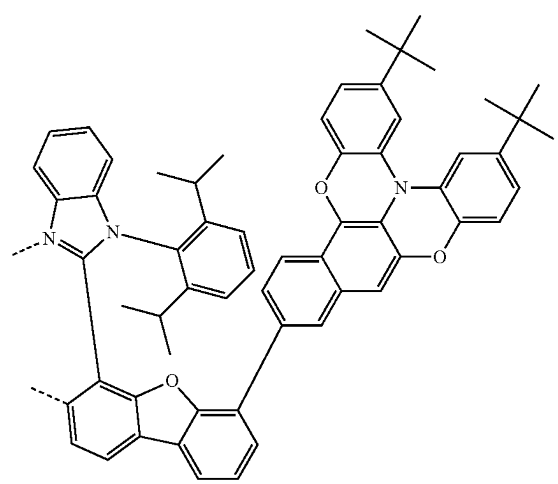

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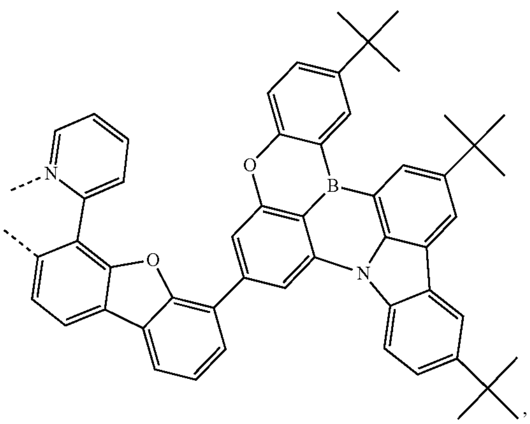

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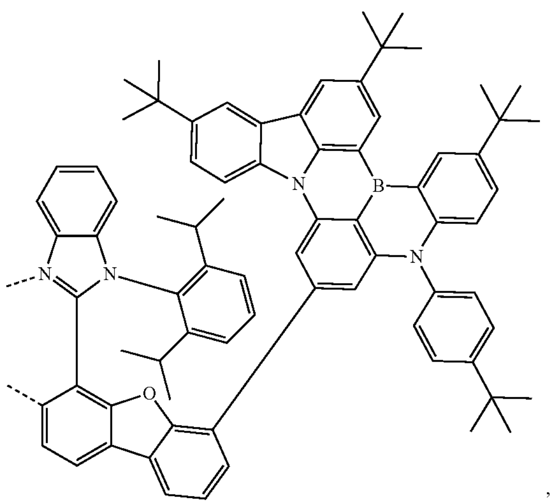

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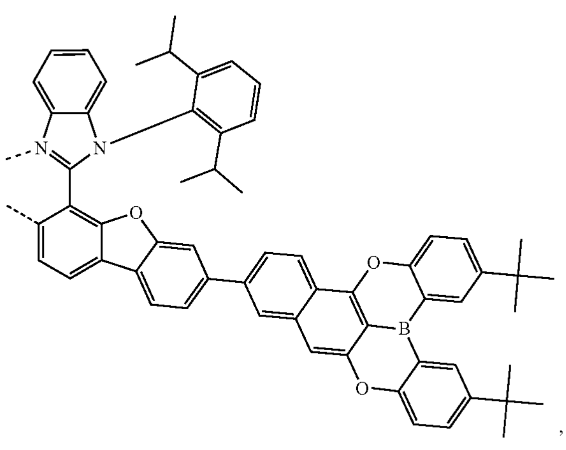

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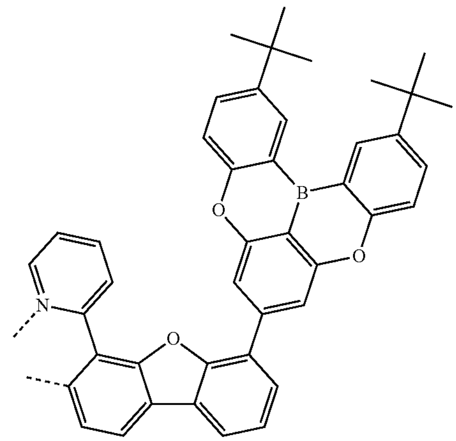

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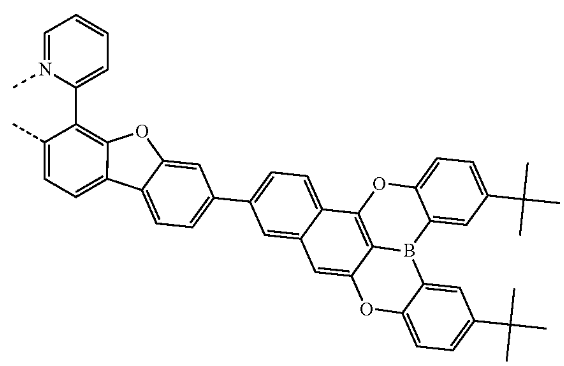

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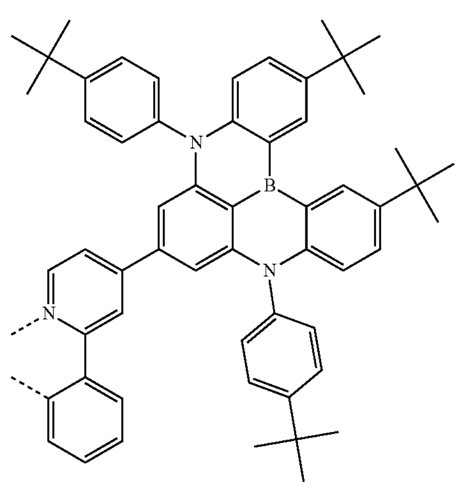

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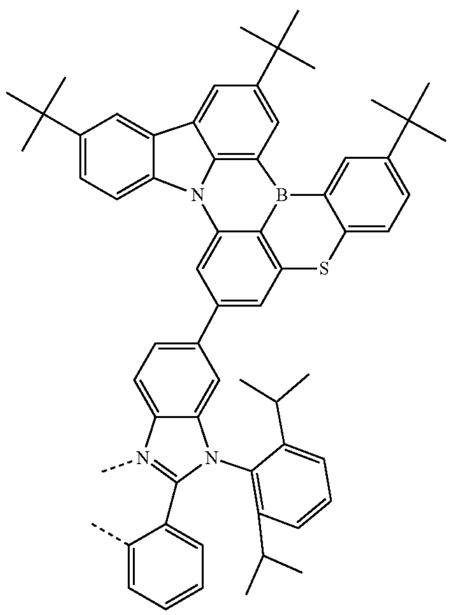

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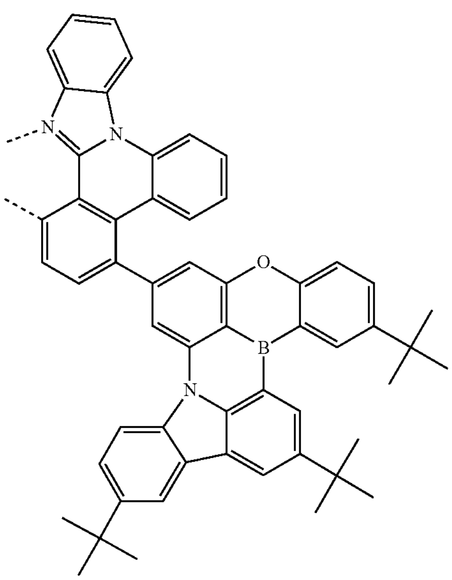

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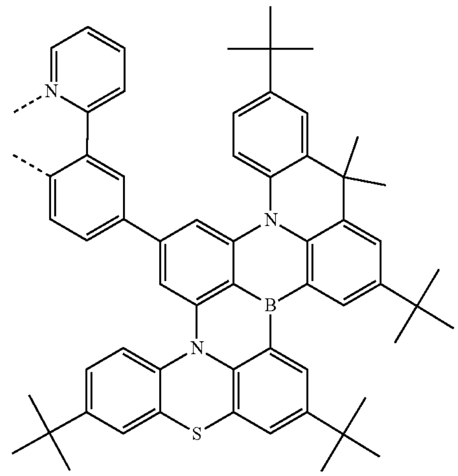

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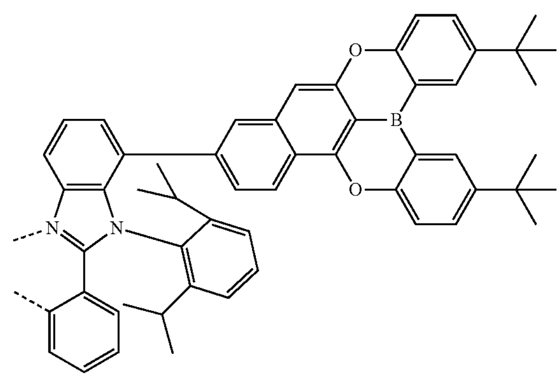

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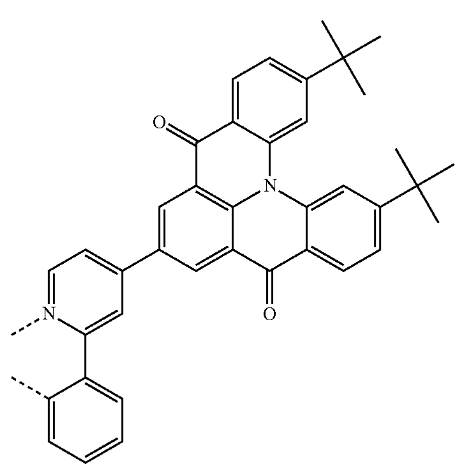

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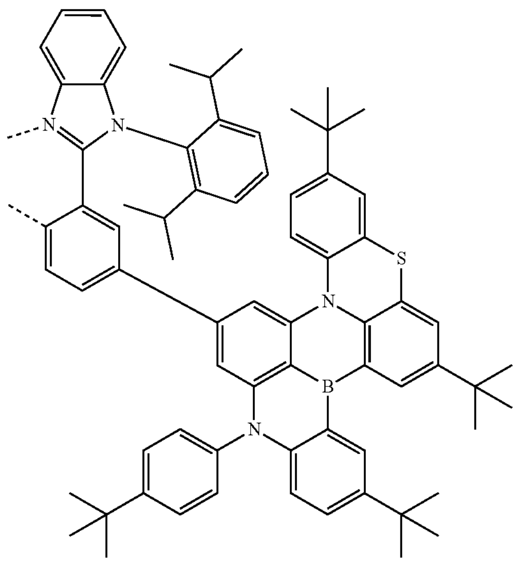

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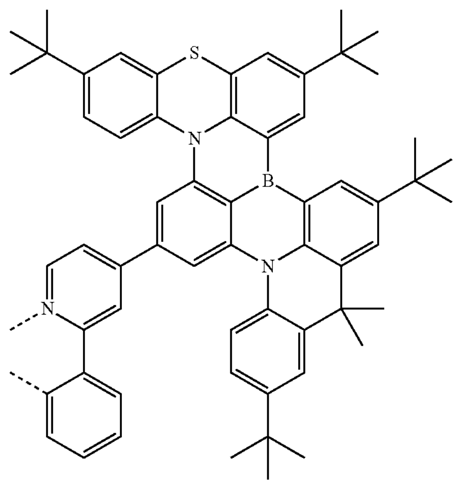

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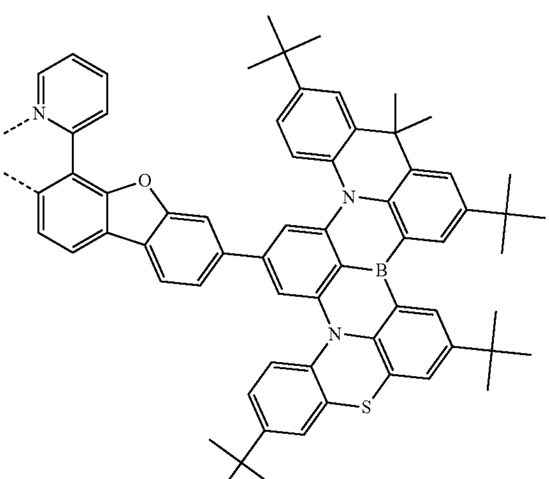

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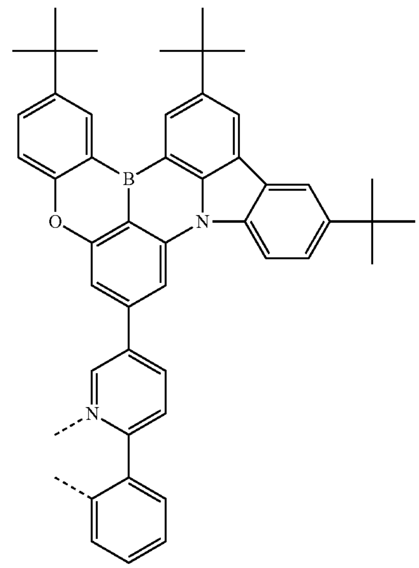

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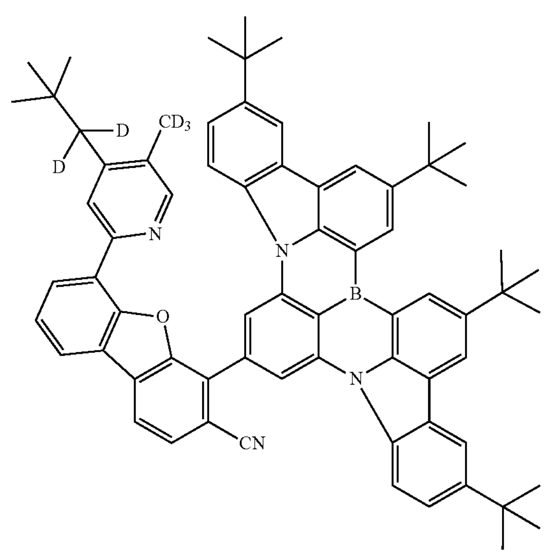

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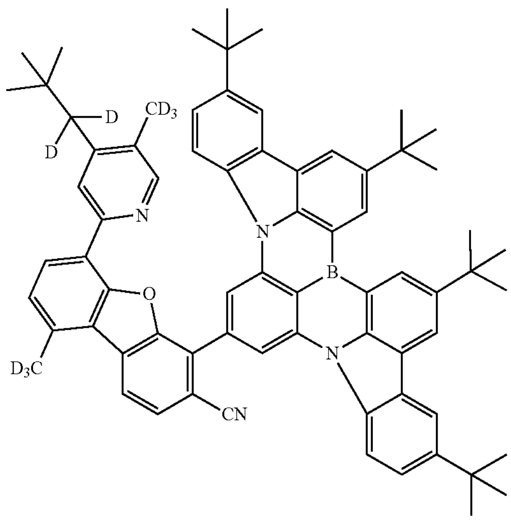

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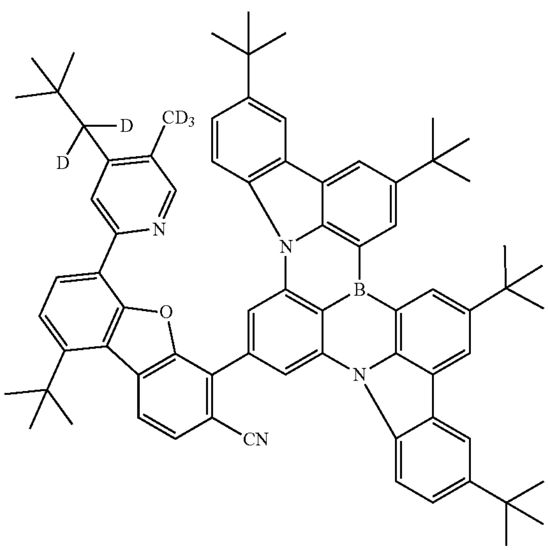
L_{A241}



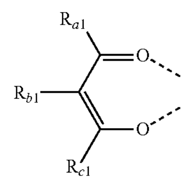
[0289] In some embodiments, the compound has a formula of $M(L_A)_p(L_B)_q(L_C)$, wherein L_B and L_C are each a bidentate ligand; and wherein p is 1, 2, or 3; q is 0, 1, or 2; r is 0, 1, or 2; and p+q+r is the oxidation state of the metal M.

[0290] In some embodiments, the compound has a formula selected from the group consisting of $Ir(L_A)_3$, $Ir(L_A)(L_B)_2$, $Ir(L_A)_2(L_B)$, $Ir(L_A)_2(L_C)$, and $Ir(L_A)(L_B)(L_C)$; and wherein L_A, L_B, and L_C are different from each other.

[0291] In some embodiments, L_B is a substituted or unsubstituted phenylpyridine, and L_C is a substituted or unsubstituted acetylacetonate.

[0292] In some embodiments, the compound has a formula of $Pt(L_A)(L_{B'})$. In some embodiments, L_A and $L_{B'}$ are connected to form a tetradentate ligand.

[0293] In some embodiments, L_B and L_C are each independently selected from the group consisting of the structures of the following LIST 8:



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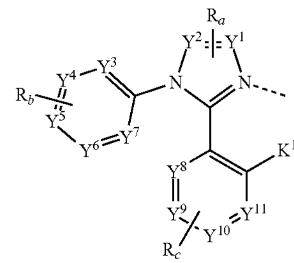

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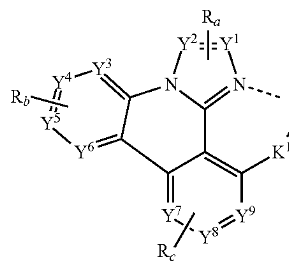

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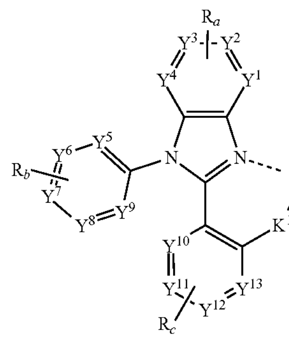

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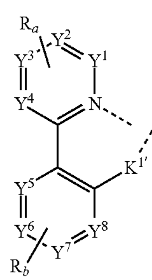

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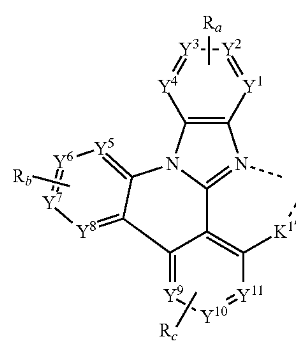

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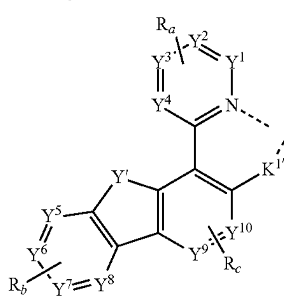

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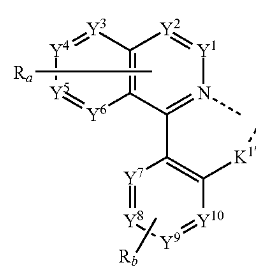

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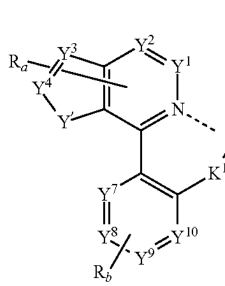

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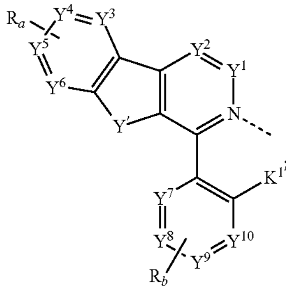

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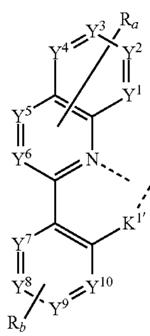

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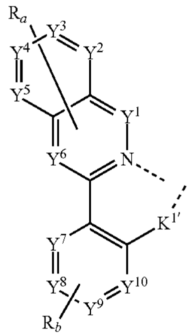

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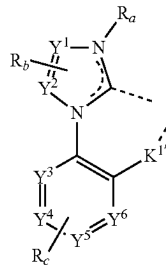

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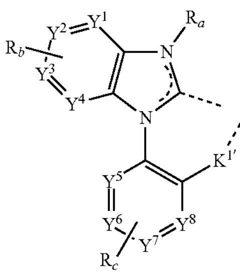

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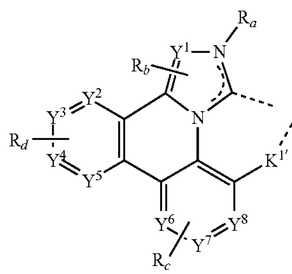

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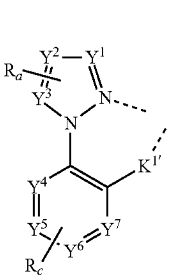

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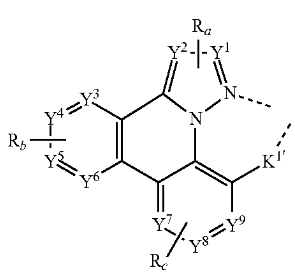

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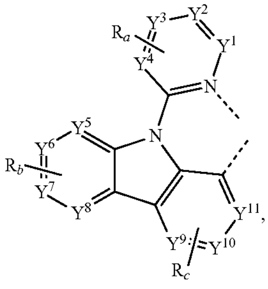

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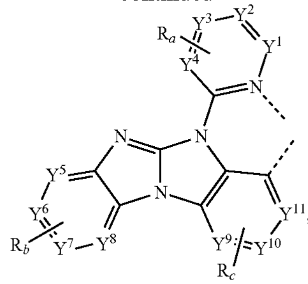

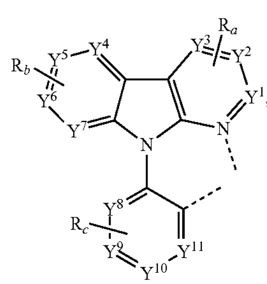

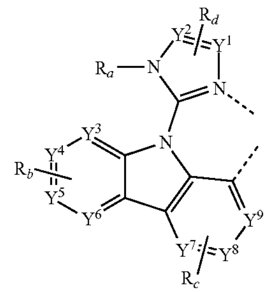

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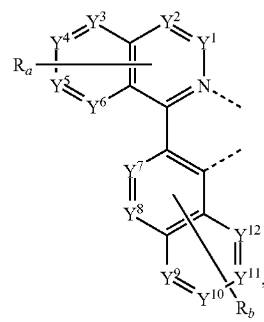

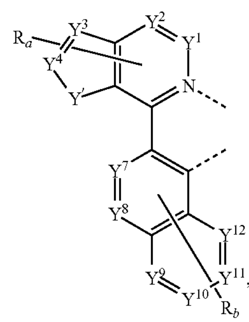

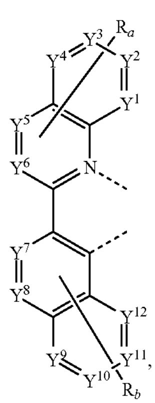

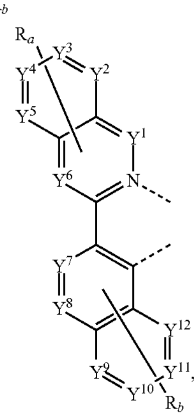

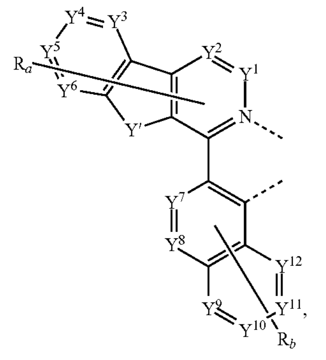

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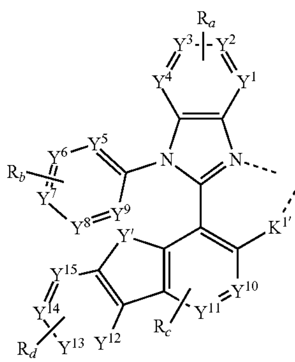

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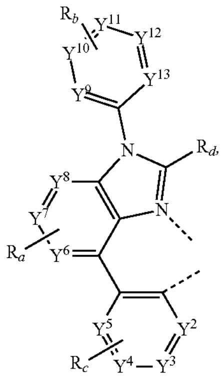

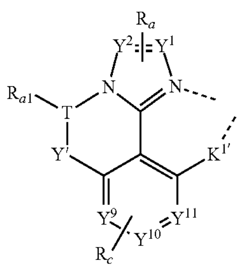

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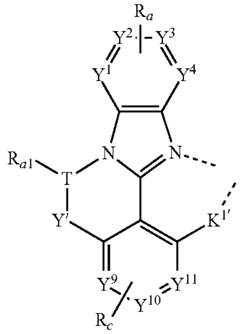

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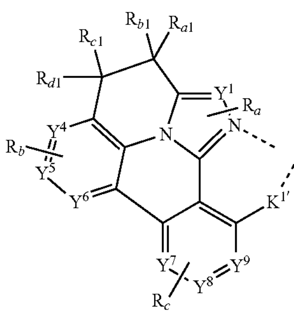

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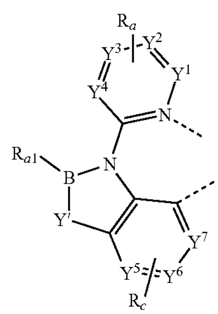

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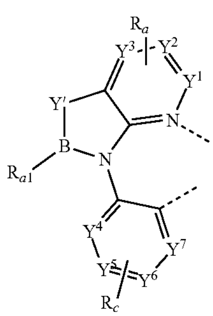

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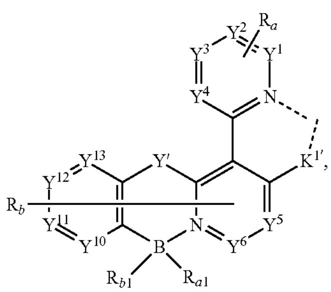

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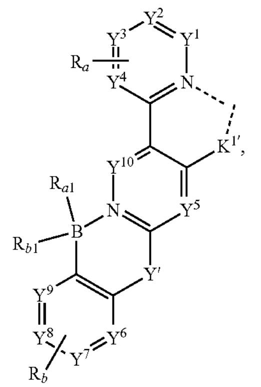


wherein:

[0294] T is selected from the group consisting of B, Al, Ga, and In;

[0295] $K^{1'}$ is selected from the group consisting of a single bond, O, S, NR_e, PR_e, BR_e, CR_eR_f, and SiR_eR_f;

[0296] each of Y^1 to Y^{13} is independently selected from the group consisting of C and N;

[0297] Y' is selected from the group consisting of BR_e, BR_eR_f, NR_e, PR_e, P(O)Re, O, S, Se, C═O, C═S, C═Se, C═NR_e, C═CR_eR_f, S═O, SO_2, CR_eR_f, SiR_eR_f, and GeR_eR_f;

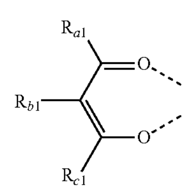
[0298] R_e and R_f can be fused or joined to form a ring;

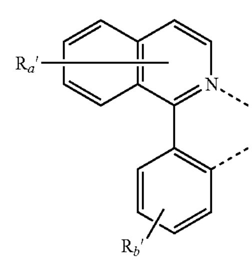
[0299] each R_a, R_b, R_c, and R_d independently represents from mono to the maximum allowed number of substitutions, or no substitution;

[0300] each of R_{a1}, R_{b1}, R_{c1}, R_{d1}, R_a, R_b, R_c, R_d, R_e, and R_f is independently a hydrogen or a subsituent selected from the group consisting of the General Substituents defined herein; and

[0301] any two substituents of R_{a1}, R_{b1}, R_{c1}, R_{d1}, R_a, R_b, R_c, and R_d can be fused or joined to form a ring or form a multidentate ligand.

[0302] In some embodiments, L_B and L_C are each independently selected from the group consisting of the structures of the following LIST 9:





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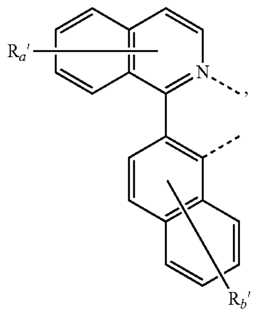

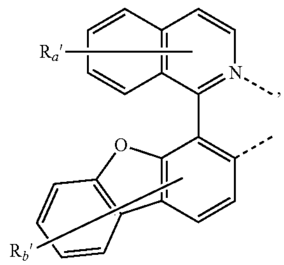

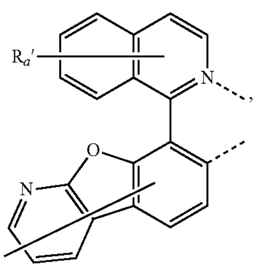

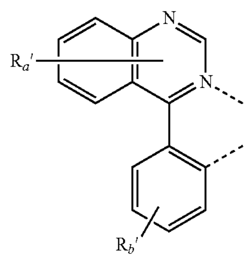

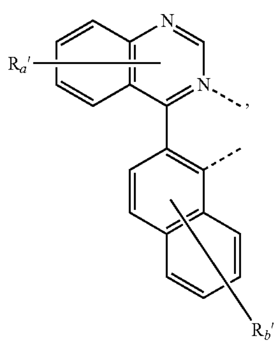

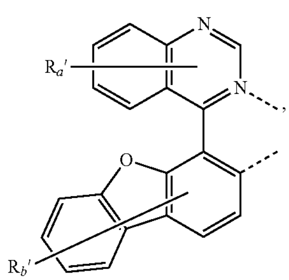

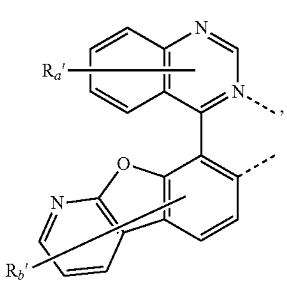

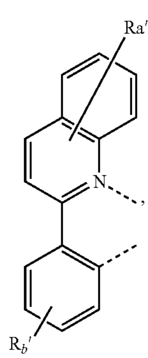

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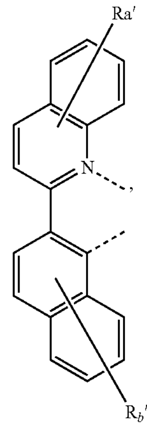

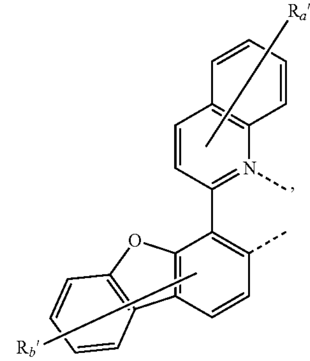

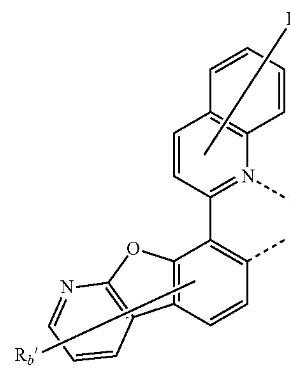

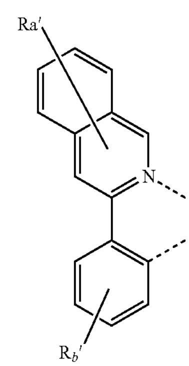

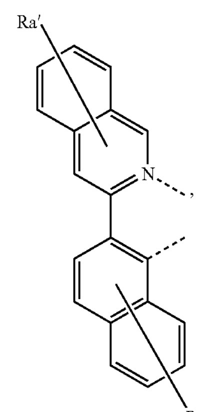

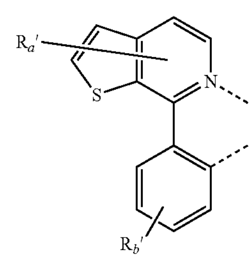

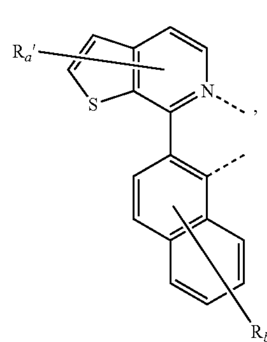

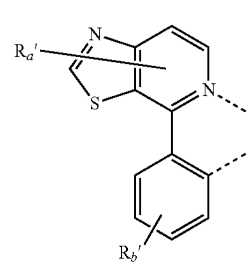

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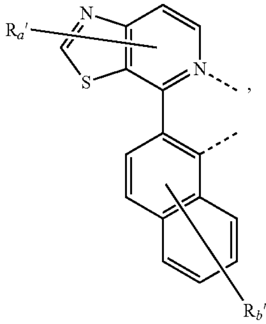

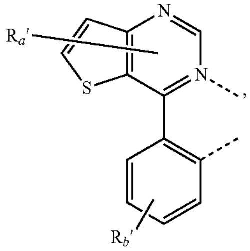

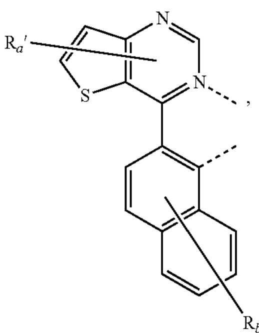

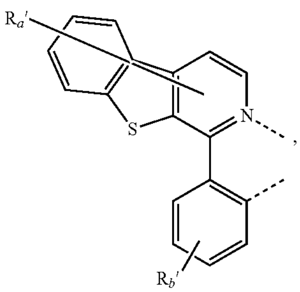

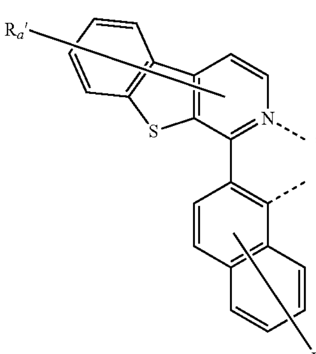

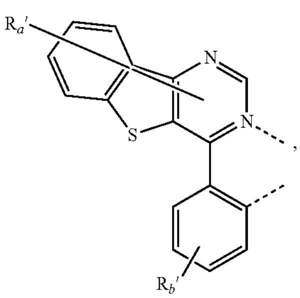

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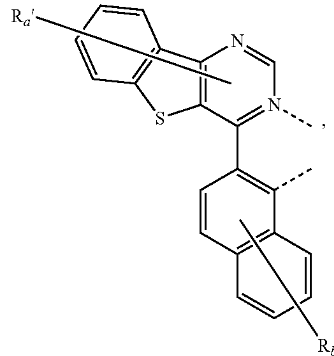

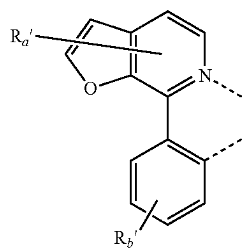

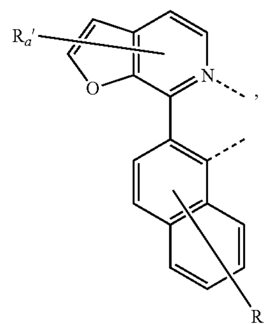

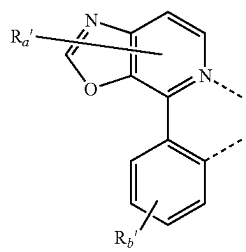

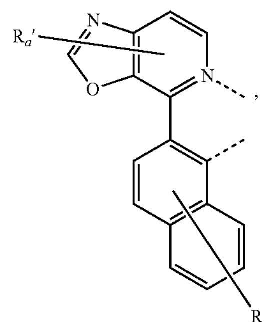

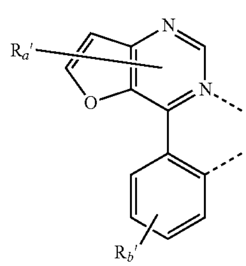

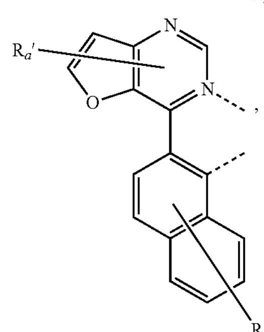

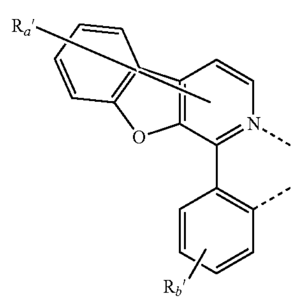

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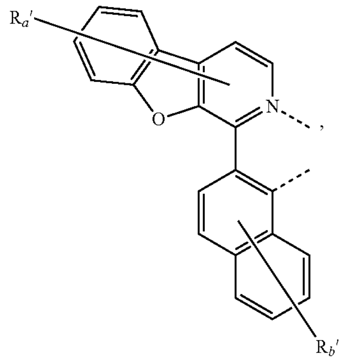

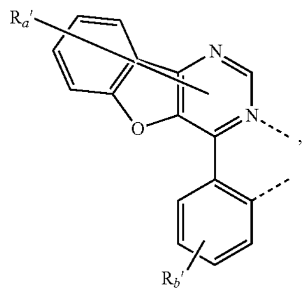

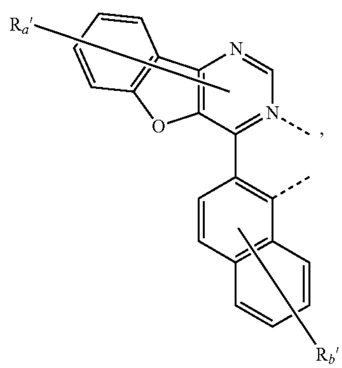

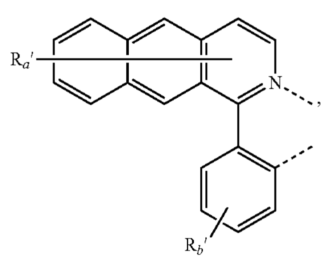

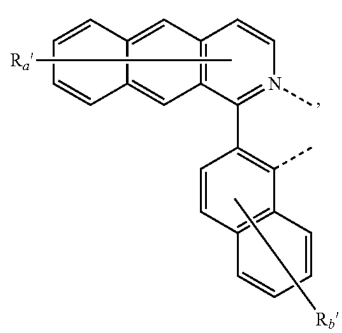

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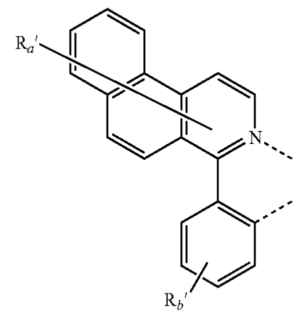

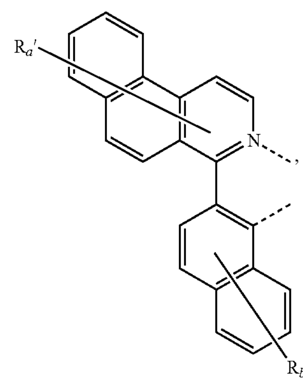

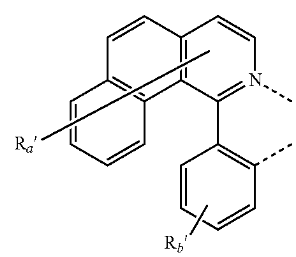

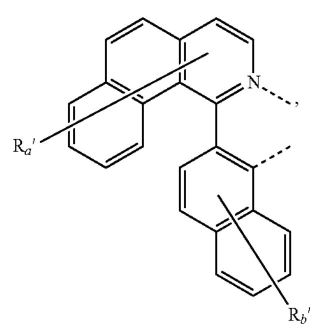

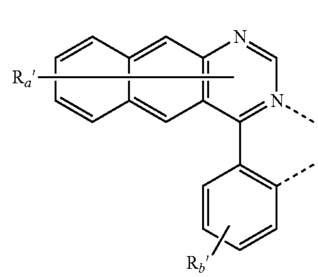

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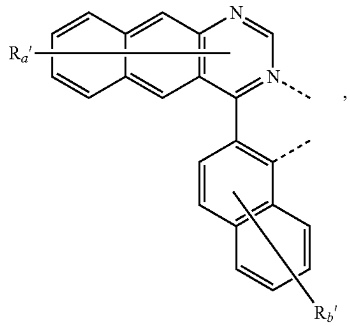

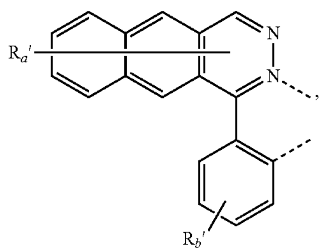

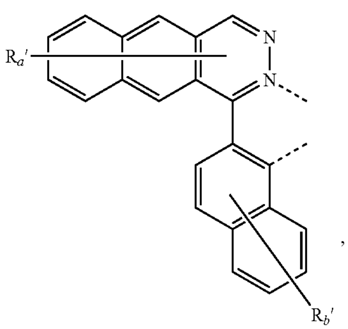

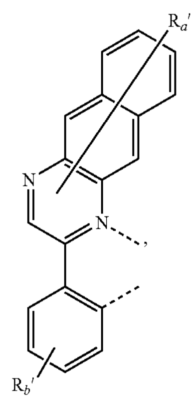

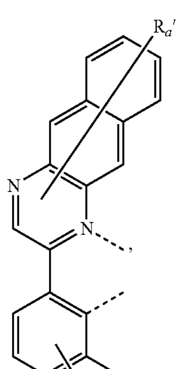

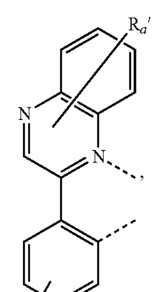

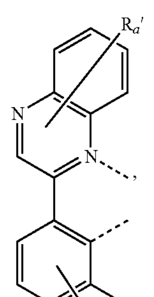

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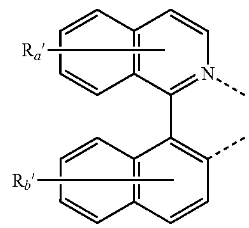

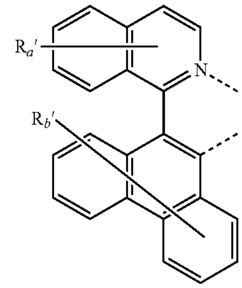

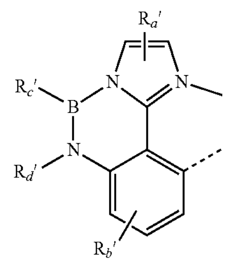

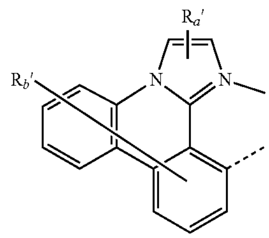

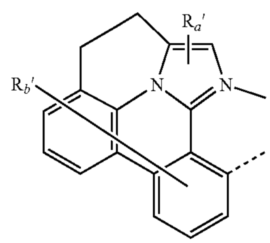

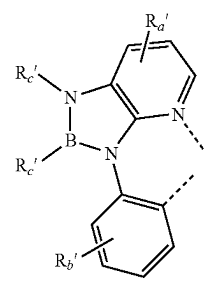

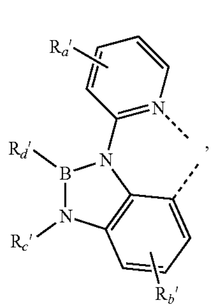

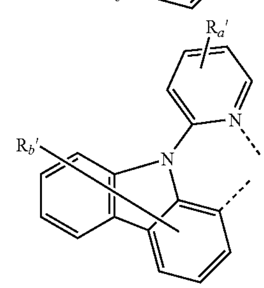

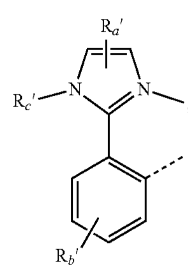

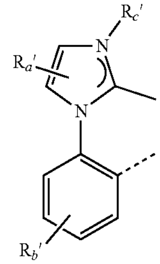

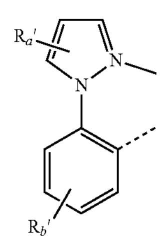

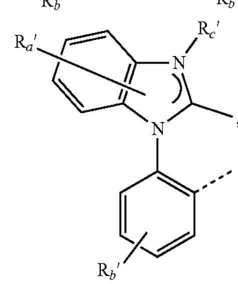

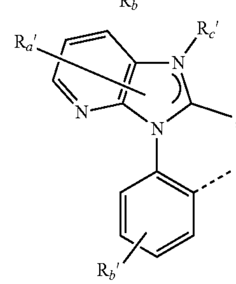

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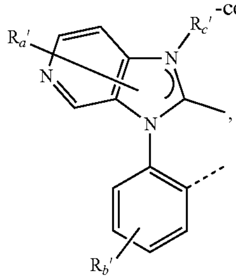

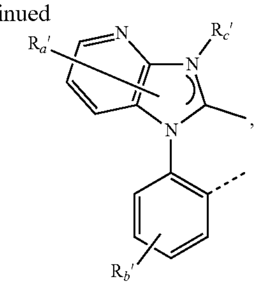

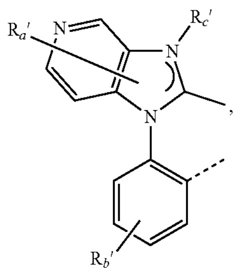

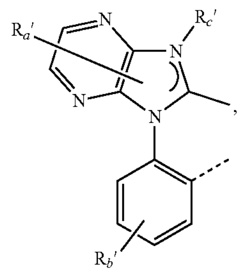

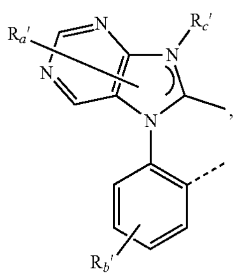

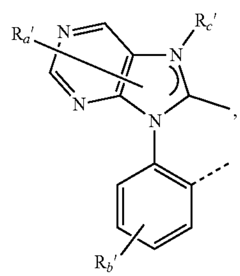

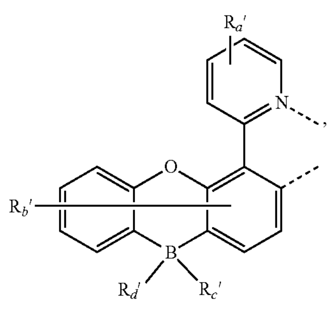

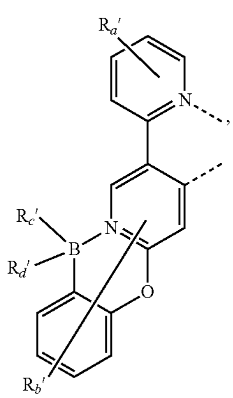

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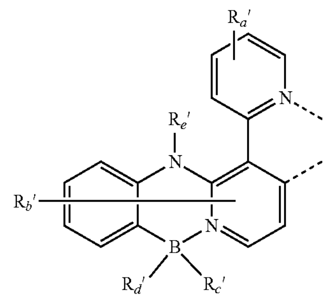

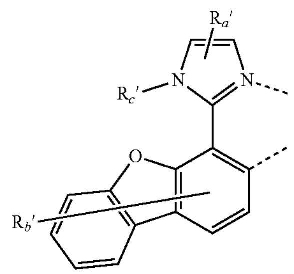

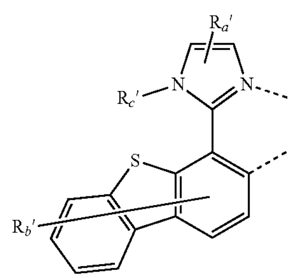

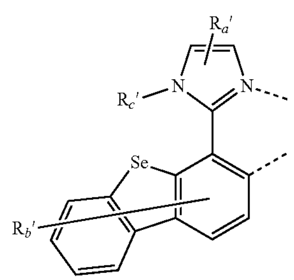

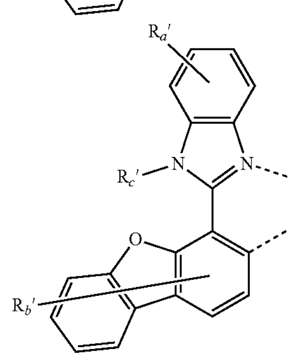

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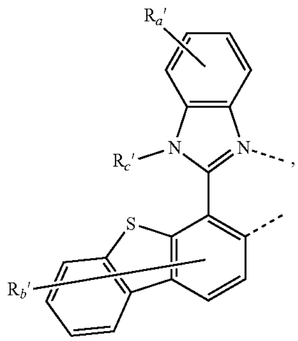

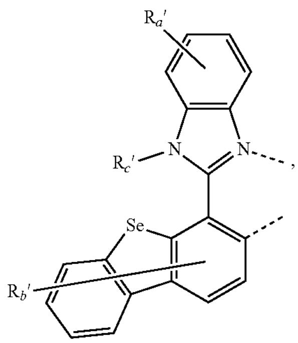

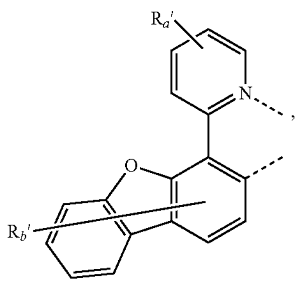

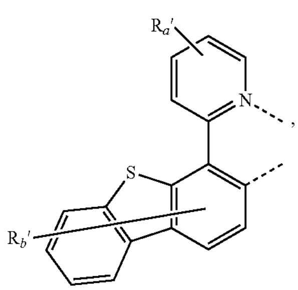

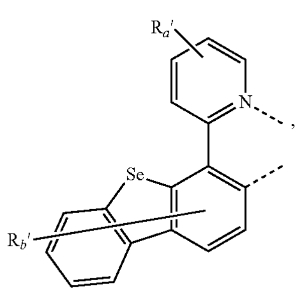

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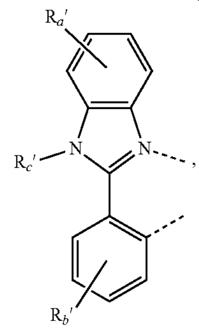

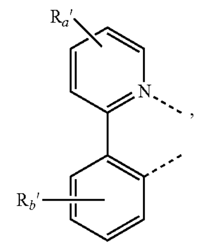

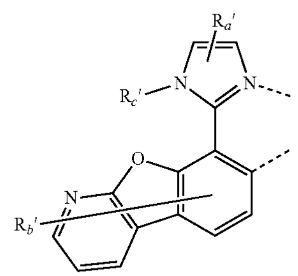

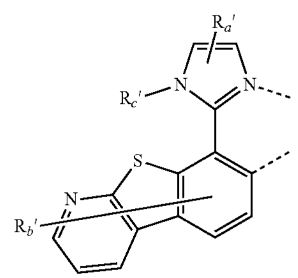

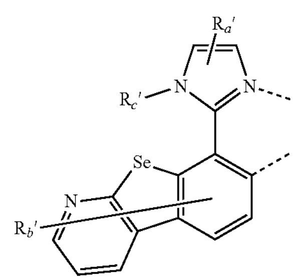

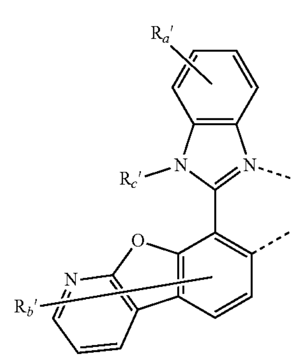

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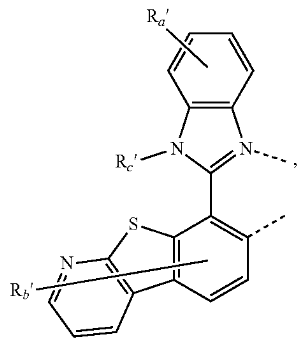

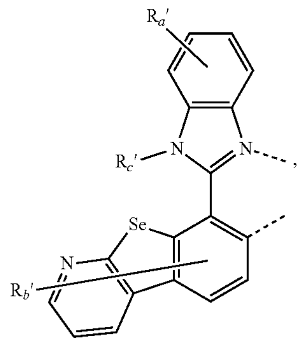

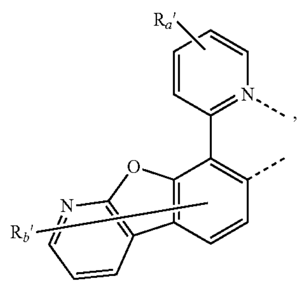

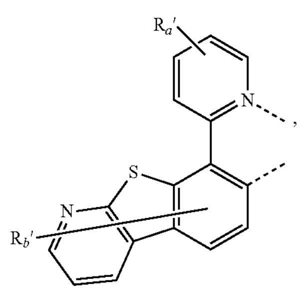

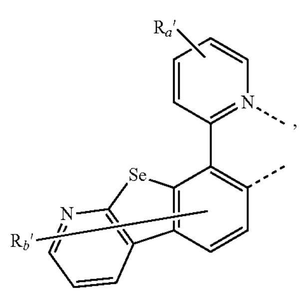

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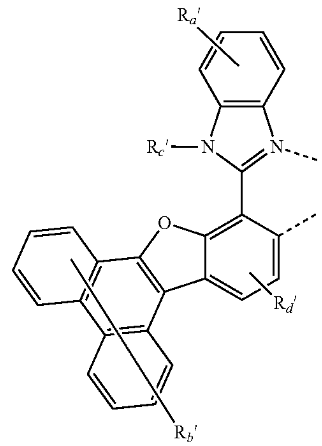

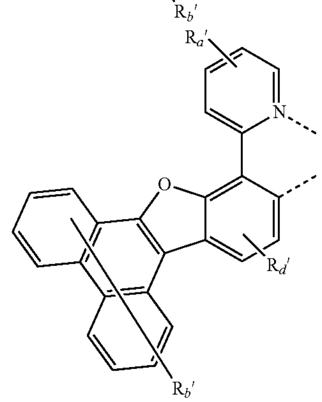

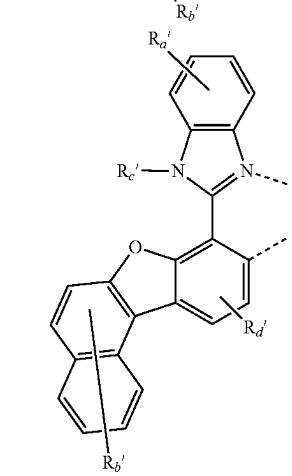

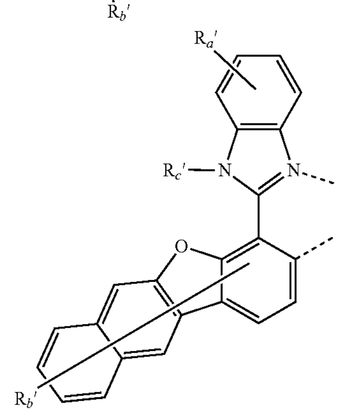

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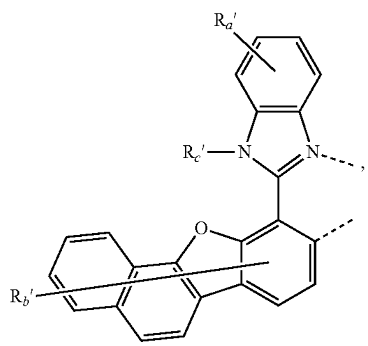

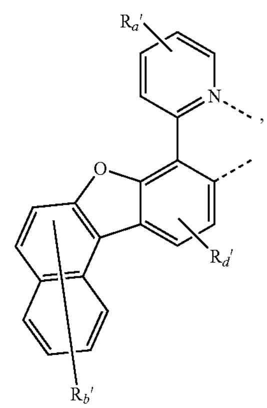

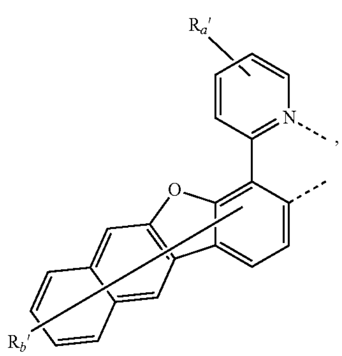

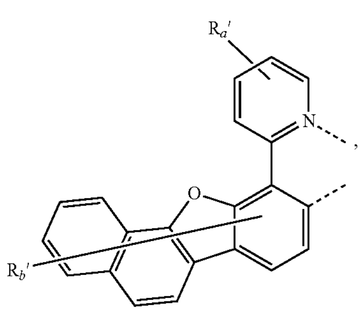

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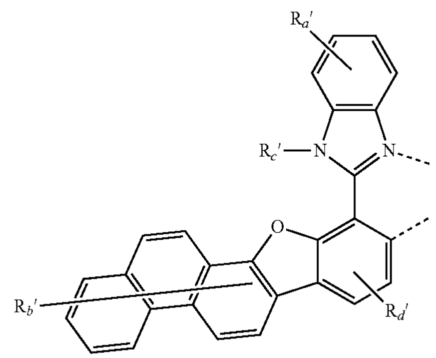

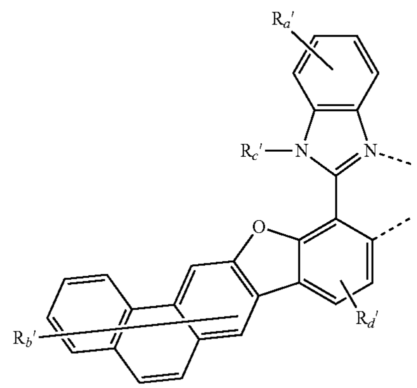

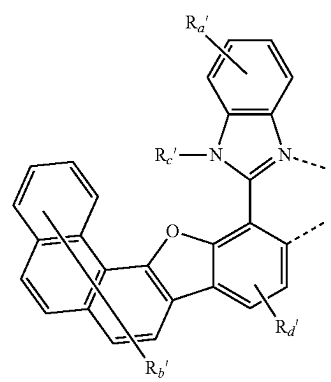

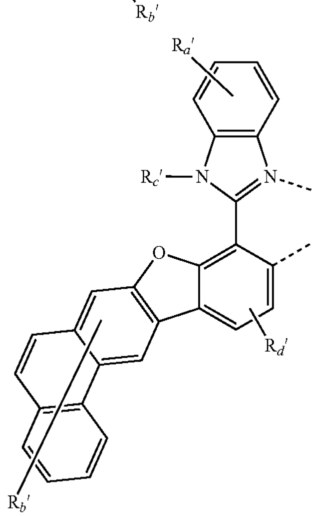

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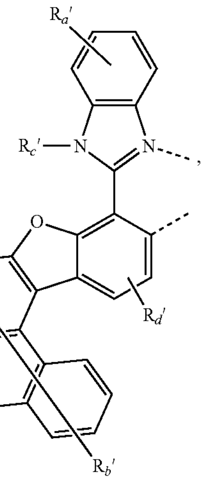

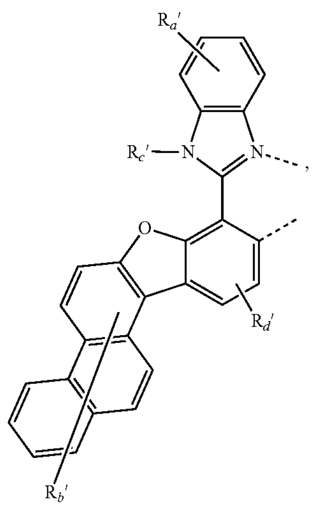

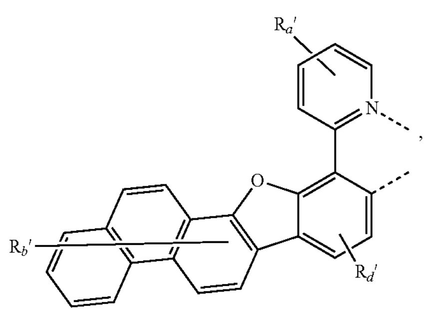

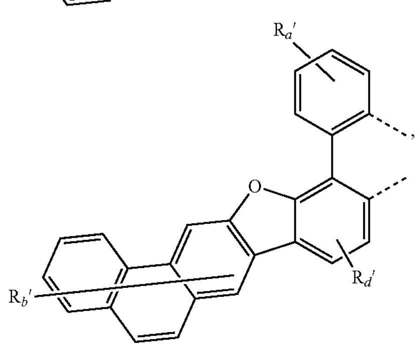

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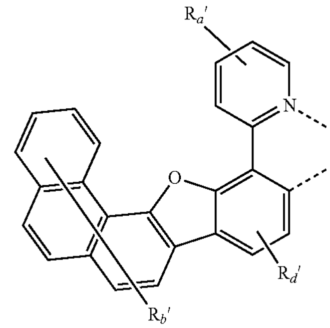

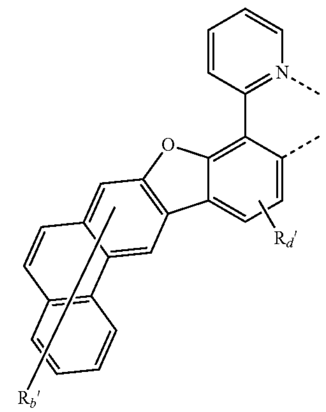

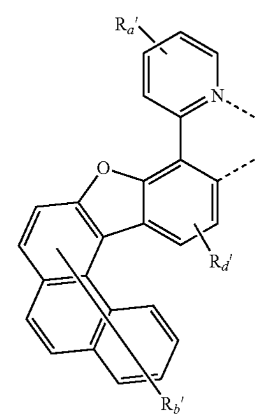

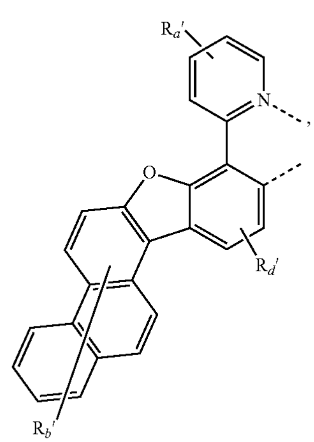

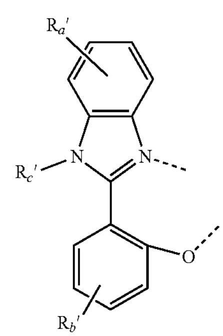

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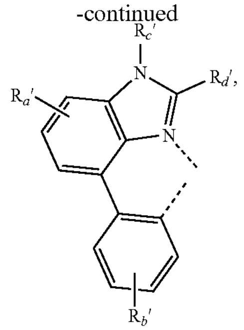

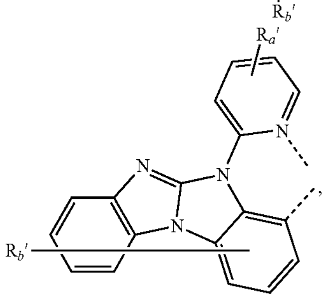

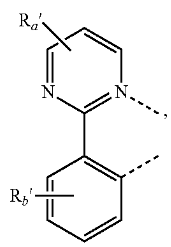

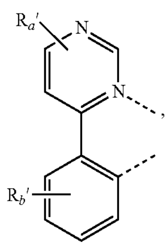

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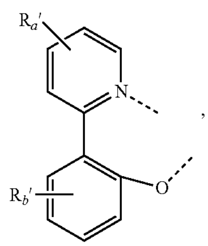


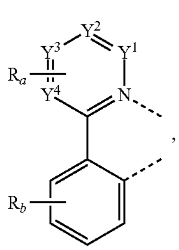
[0303] wherein:

[0304] R_a', R_b', R_c', R_d', and R_e' each independently represents zero, mono, or up to a maximum allowed number of substitution to its associated ring;

[0305] R_a', R_b', R_c', R_d', and R_e' each independently hydrogen or a substituent selected from the group consisting of the General Substituents defined herein; and

[0306] two substituents of R_a', R_b', R_c', R_d', and R_e' can be fused or joined to form a ring or form a multidentate ligand.

[0307] In some embodiments, L_B comprises a structure of



wherein the variables are the same as previously defined. In some embodiments, each of Y^1 to Y^4 is independently carbon. In some embodiments, at least one of Y^1 to Y^4 is N. In some embodiments, exactly one of Y^1 to Y^4 is N. In some embodiments, Y^1 is N. In some embodiments, Y^2 is N. In some embodiments, Y^3 is N. In some embodiments, Y^4 is N. In some embodiments, at least one of R_a is a tertiary alkyl, silyl or germyl. In some embodiments, at least one of R_a is a tertiary alkyl. In some embodiments, Y^3 is C and the R_a attached thereto is a tertiary alkyl, silyl or germyl. In some embodiments, Y^1 to Y^3 is C, Y^4 is N, and the R_a attached to Y^3 is a tertiary alkyl, silyl or germyl. In some embodiments, Y^1 to Y^3 is C, Y^4 is N, and the R_a attached to Y^2 is a tertiary alkyl, silyl or germyl. In some embodiments, at least one of R_b is a tertiary alkyl, silyl, or germyl. In some embodiments, the tertiary alkyl is tert-butyl. In some embodiments, at least one pair of R_a and R_b are joined or fused to form a ring.

[0308] In some embodiments, L_A can be selected from L_{Ai}, wherein i is an integer from 1 to 241; and L_B is selected from L_{Bk}, wherein k is an integer from 1 to 541, wherein:

[0309] when the compound has formula $Ir(L_{Ai})_3$, the compound is selected from the group consisting of $Ir(L_{A1})_3$ to $Ir(L_{A241})_3$;

[0310] when the compound has formula $Ir(L_{Ai})(L_{Bk})_2$, the compound is selected from the group consisting of $Ir(L_{A1})(L_{B1})_2$ to $Ir(L_{A241})(L_{B541})_2$;

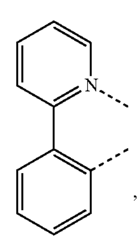
[0311] when the compound has formula $Ir(L_{Ai})_2(L_{Bk})$, the compound is selected from the group consisting of $Ir(L_{A1})_2(L_{B1})$ to $Ir(L_{A241})_2(L_{B541})$;

[0312] when the compound has formula $Ir(L_{Ai})_2(L_{Cj-I})$, j is an integer from 1 to 1416, wherein the compound is selected from the group consisting of $Ir(L_{A1})_2(L_{C1-I})$ to $Ir(L_{A241})_2(L_{C1416-I})$; and

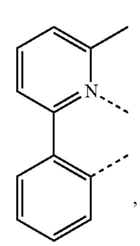
[0313] when the compound has formula $Ir(L_{Ai})_2(L_{Cj-II})$, j is an integer from 1 to 1416, wherein the compound is selected from the group consisting of $Ir(L_{A1})_2(L_{C1-II})$ to $Ir(L_{A241})_2(L_{C1416-II})$;

[0314] wherein k is an integer from 1 to 541, and each L_{Bk} has the structure defined in the following LIST 10:

L_{B1}



L_{B2}



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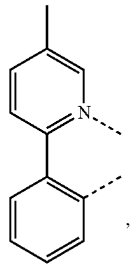
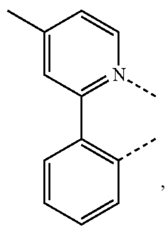
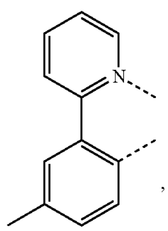
L_{B3}



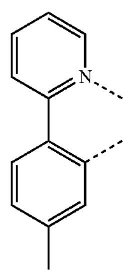
L_{B4}



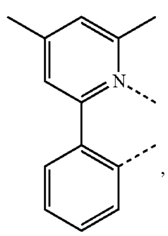
L_{B5}



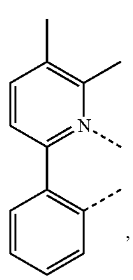
L_{B6}



L_{B7}



L_{B8}



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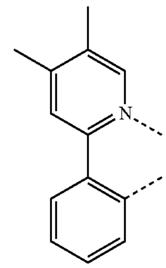
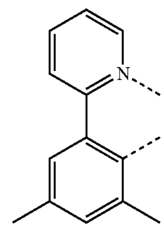
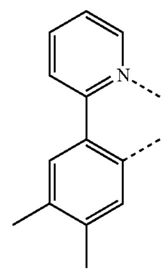
L_{B9}



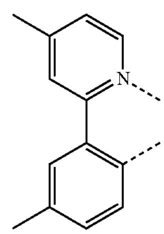
L_{B10}



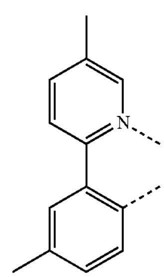
L_{B11}



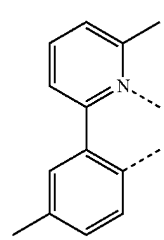
L_{B12}



L_{B13}



L_{B14}



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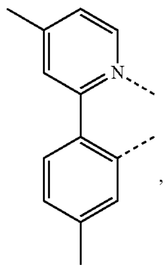
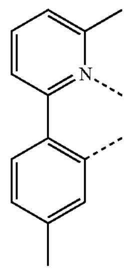
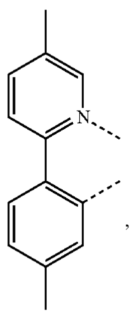
L_{B15}



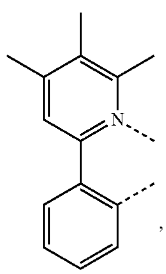
L_{B16}



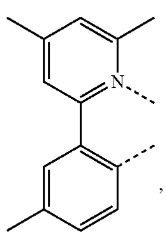
L_{B17}



L_{B18}



L_{B19}



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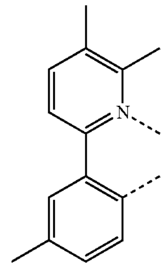
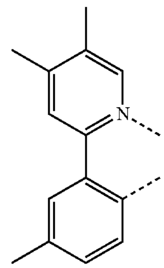
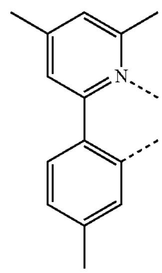
L_{B20}



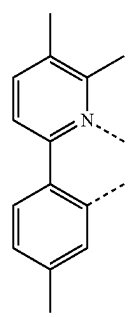
L_{B21}



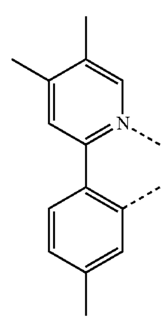
L_{B22}



L_{B23}



L_{B24}



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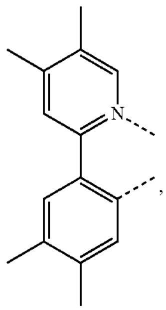
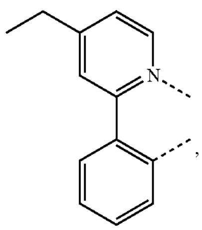
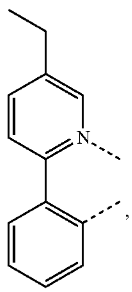
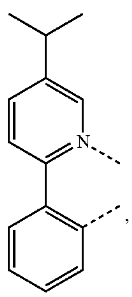
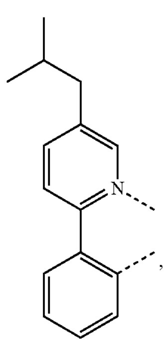

L_{B25}


L_{B26}


L_{B27}


L_{B28}


L_{B29}

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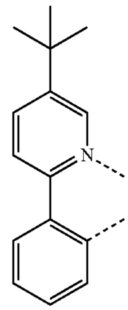
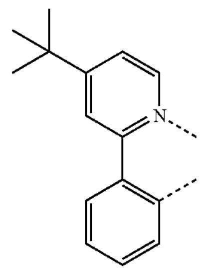
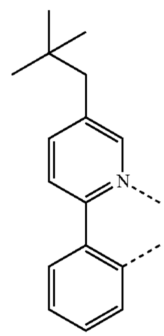
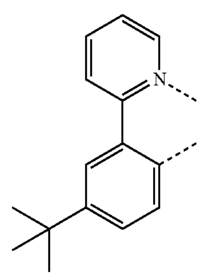
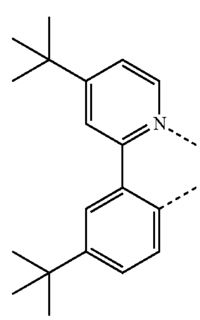

L_{B30}


L_{B31}


L_{B32}


L_{B33}


L_{B34}

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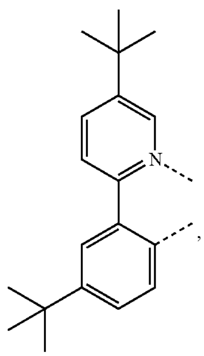
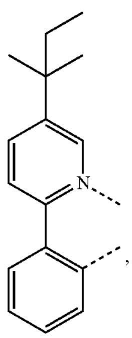
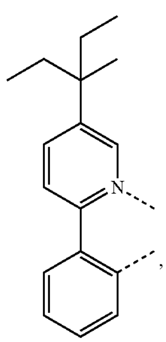
L$_{B35}$



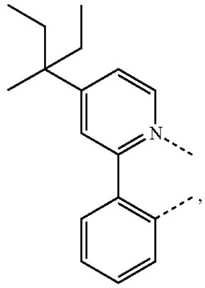
L$_{B36}$



L$_{B37}$



L$_{B38}$



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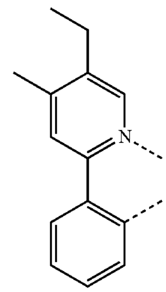
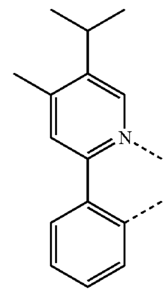
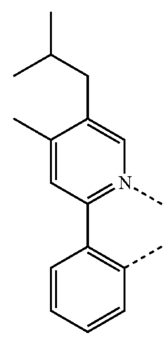
L$_{B39}$



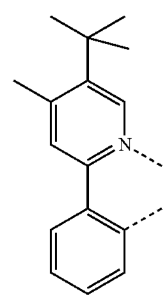
L$_{B40}$



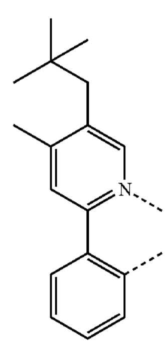
L$_{B41}$



L$_{B42}$



L$_{B43}$



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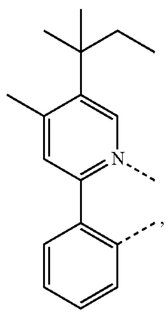
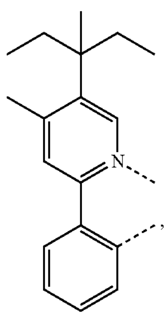
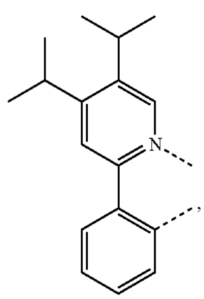


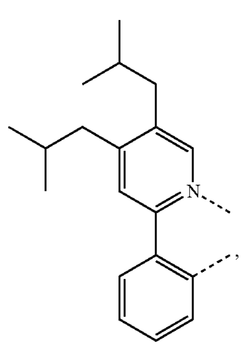
L_{B44}



L_{B45}



L_{B46}



L_{B47}

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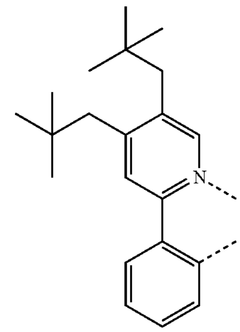
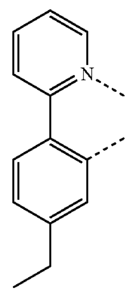
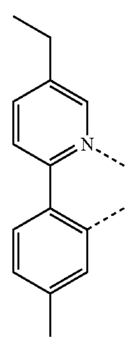


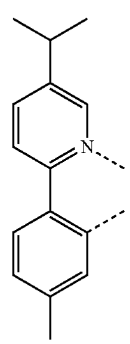
L_{B48}



L_{B49}



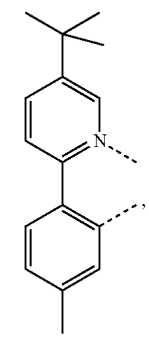
L_{B50}



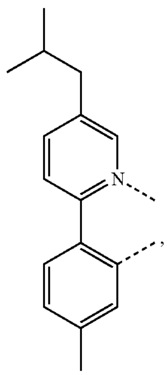
L_{B51}

-continued

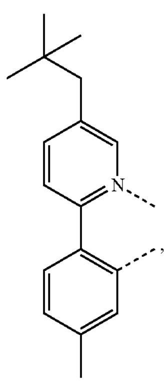
L_{B52}



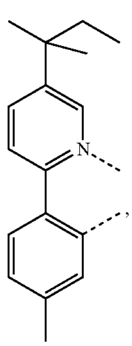
L_{B53}



L_{B54}



L_{B55}



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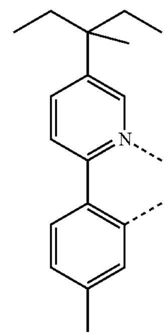
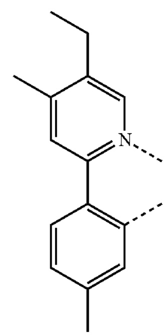
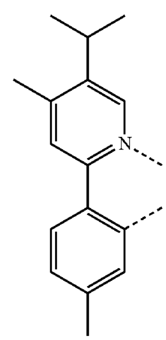
L_{B56}



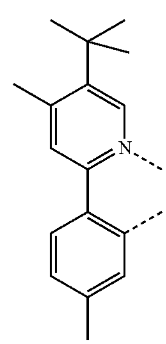
L_{B57}



L_{B58}



L_{B59}



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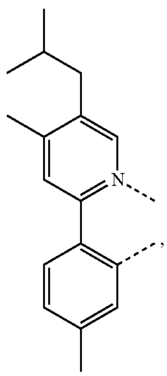
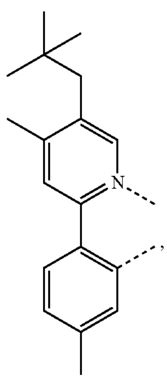
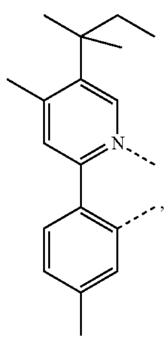
L_{B60}



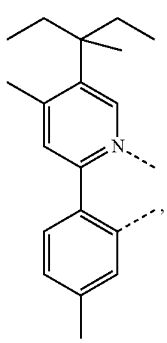
L_{B61}



L_{B62}



L_{B63}



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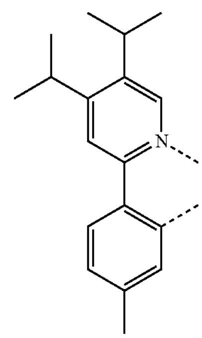
L_{B64}



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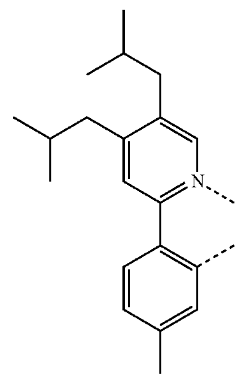

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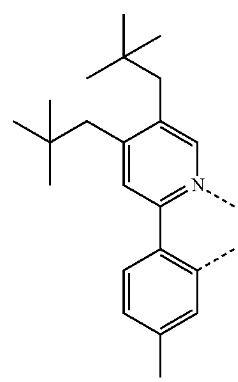

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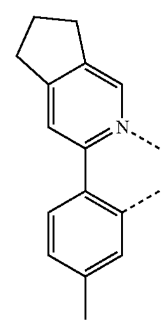

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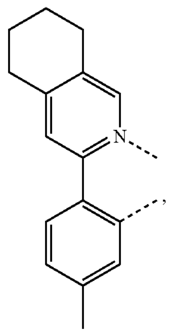

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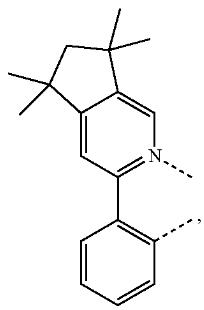

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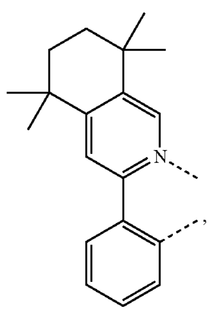

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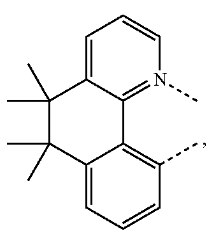

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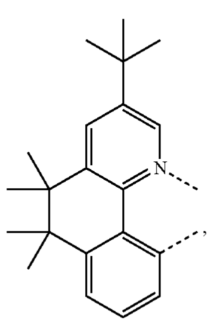

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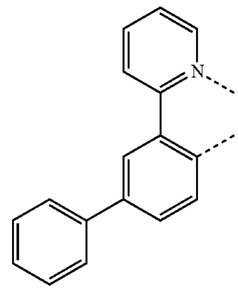

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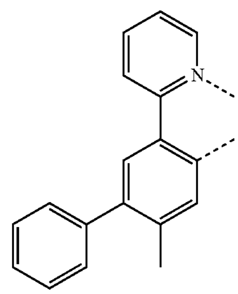

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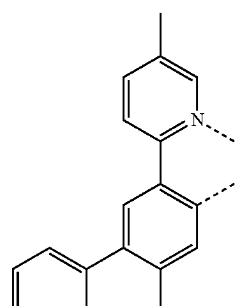

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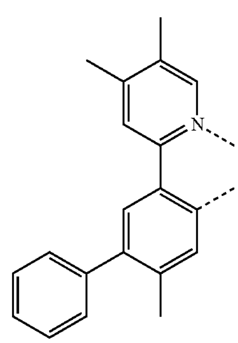

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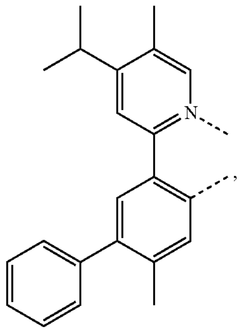

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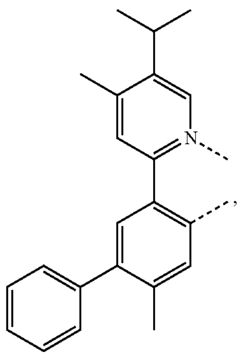

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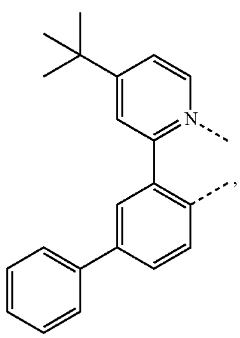

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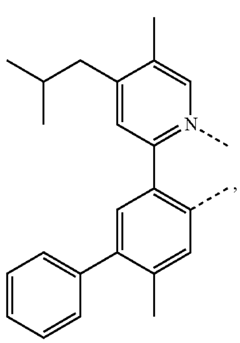

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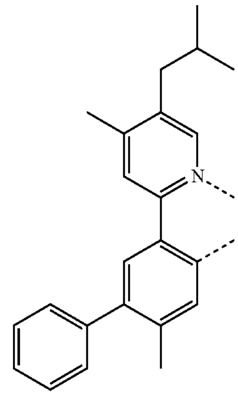

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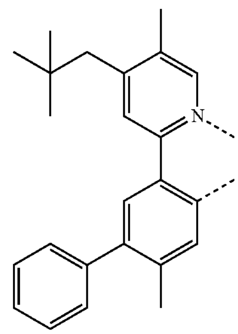

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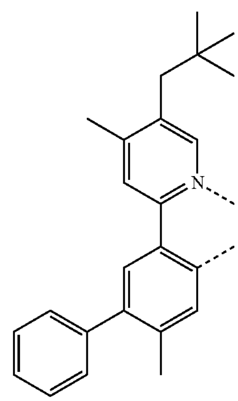

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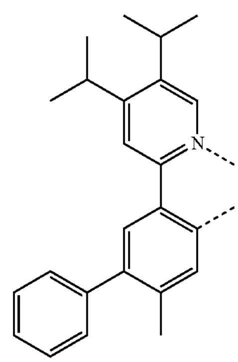

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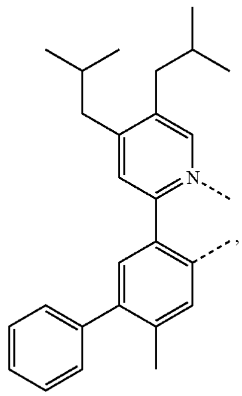

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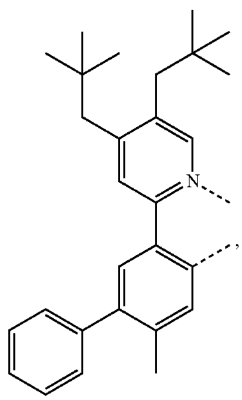

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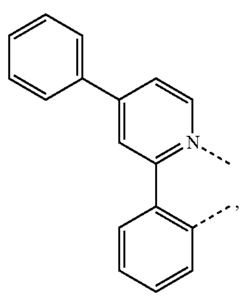

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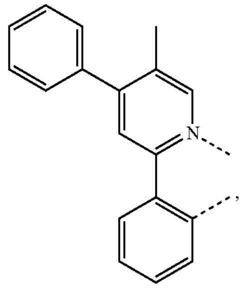

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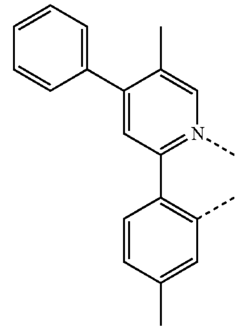

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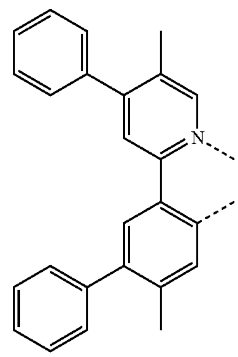

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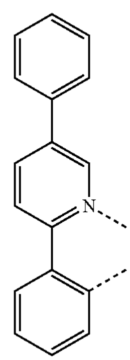

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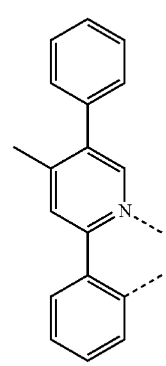

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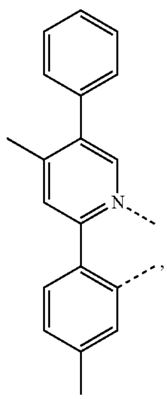

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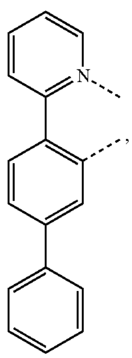

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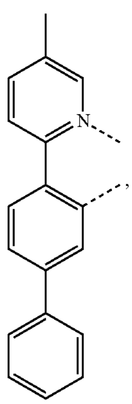

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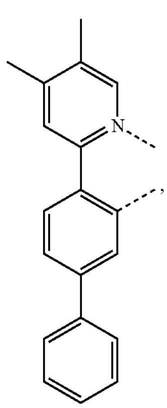

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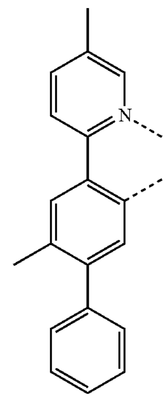

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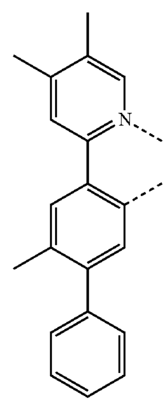

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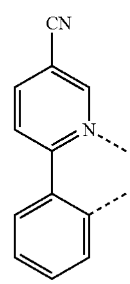

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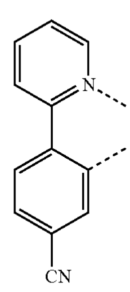

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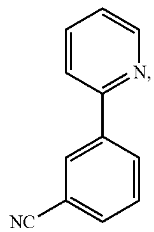

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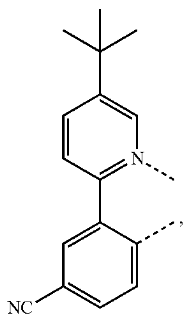

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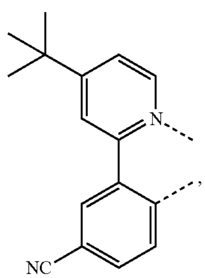

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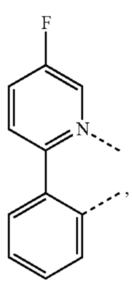

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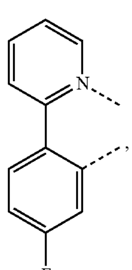

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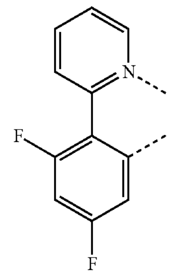

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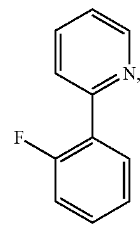

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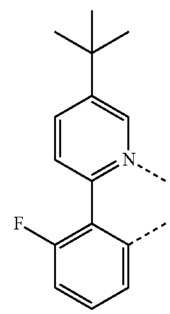

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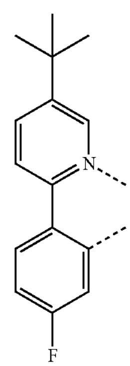

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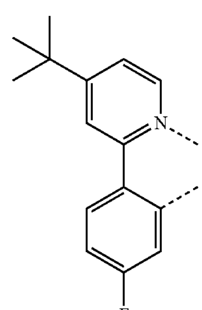

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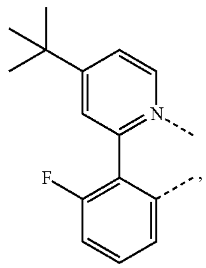

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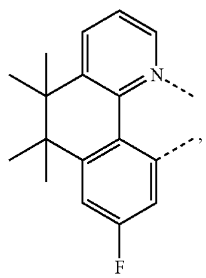

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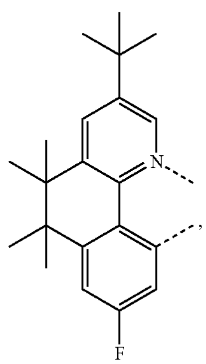

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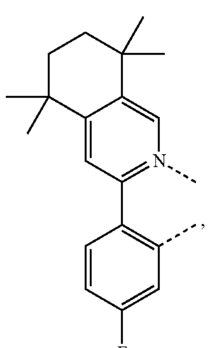

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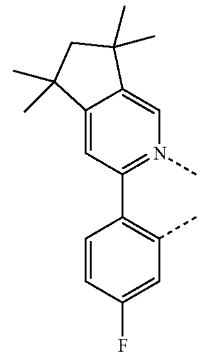

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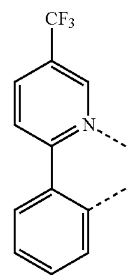

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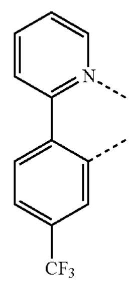

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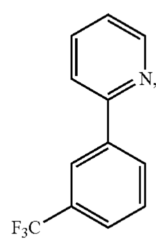

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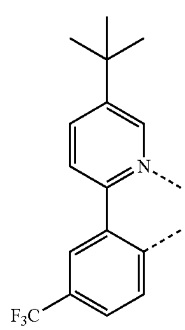

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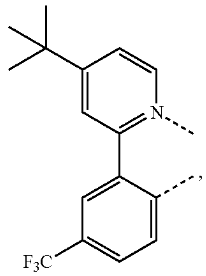

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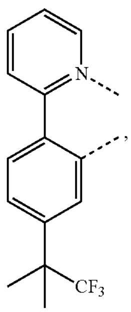

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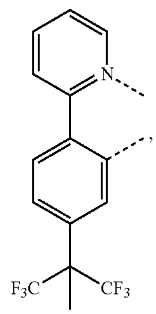

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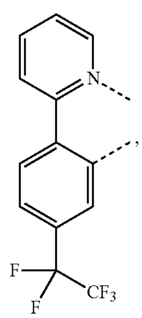

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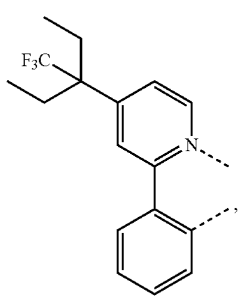

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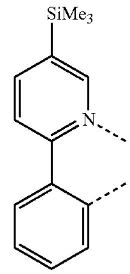

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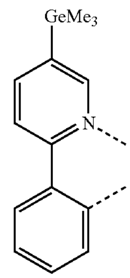

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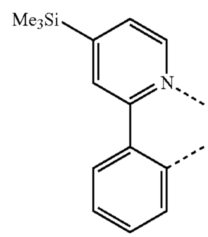

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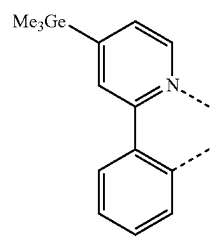

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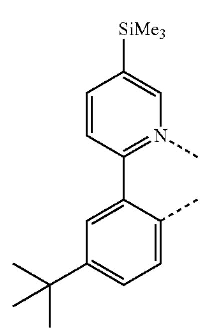

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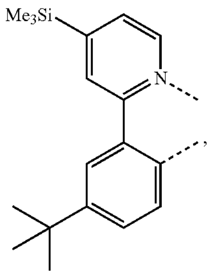

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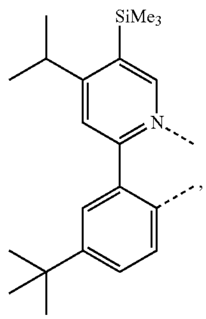

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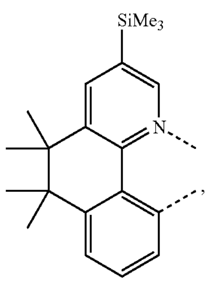

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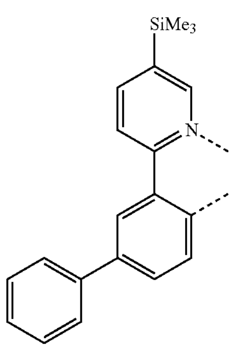

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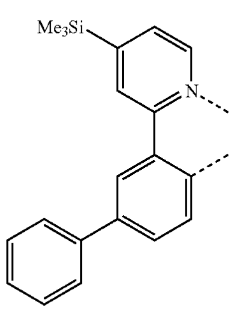

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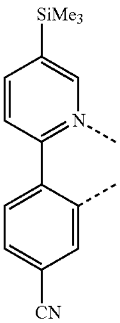

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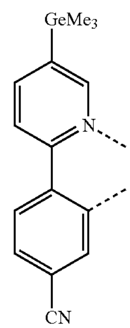

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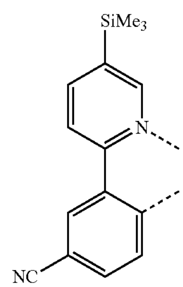

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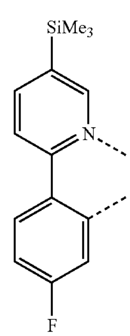

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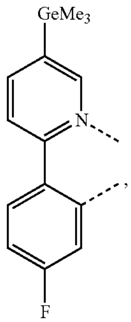

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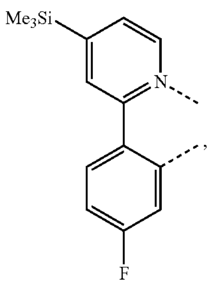

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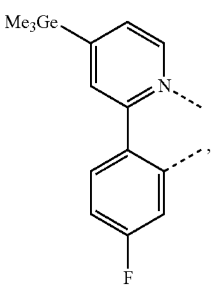

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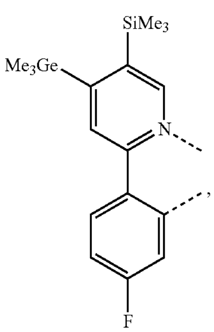

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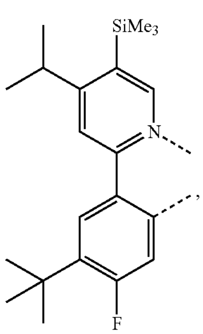

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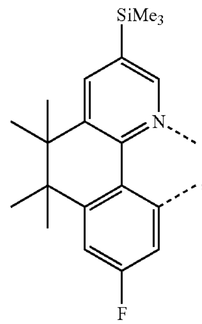

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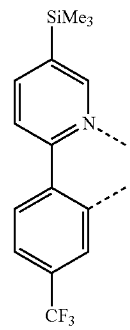

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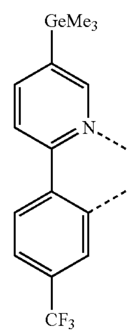

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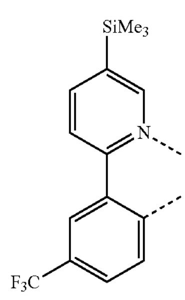

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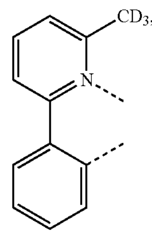

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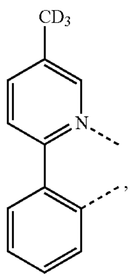

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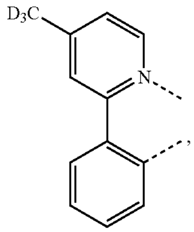

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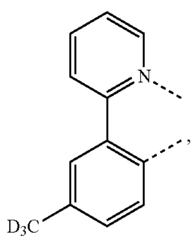

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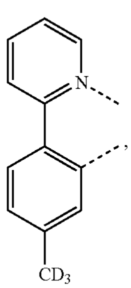

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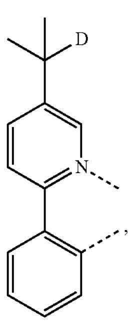

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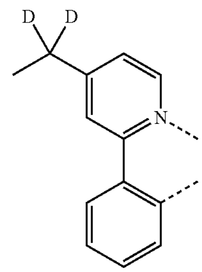

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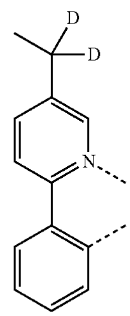

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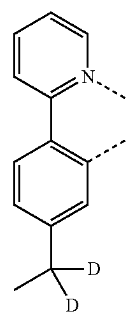

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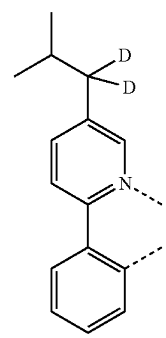

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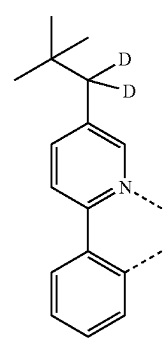

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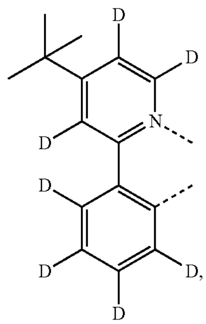

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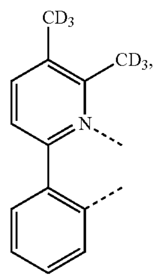

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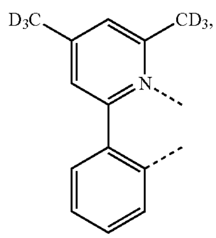

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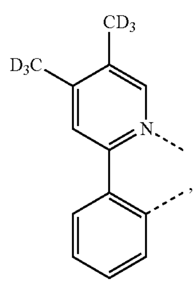

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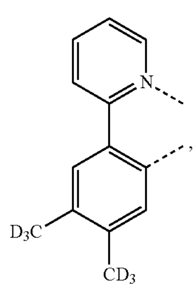

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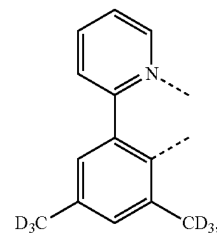

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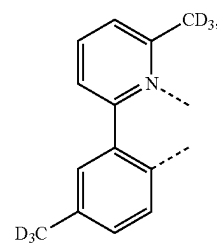

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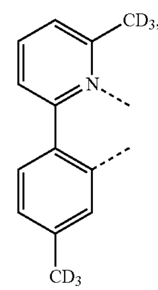

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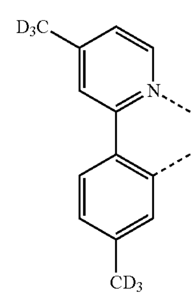

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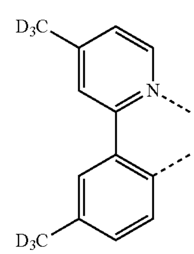

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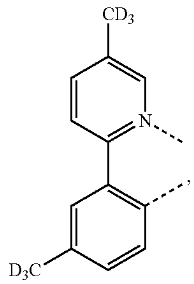

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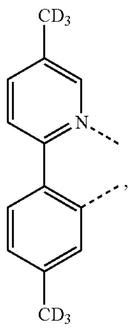

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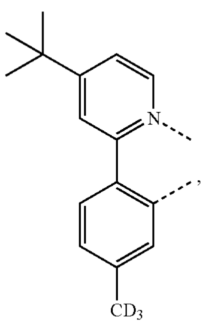

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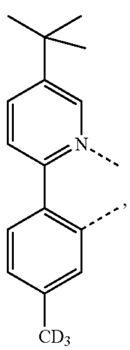

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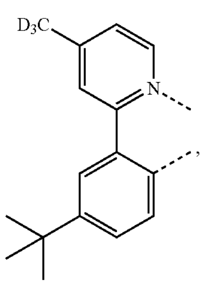

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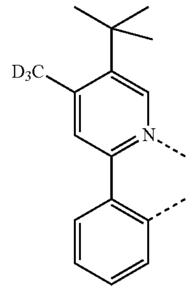

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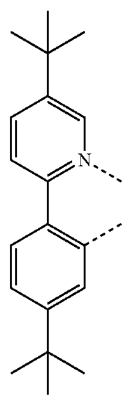

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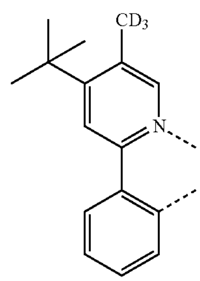

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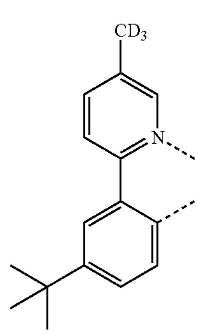

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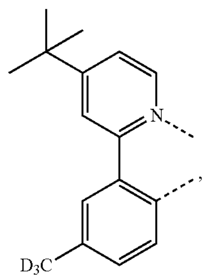

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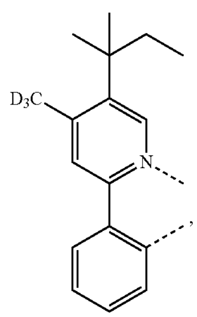

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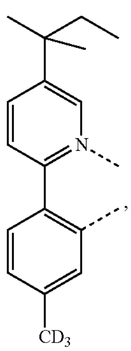

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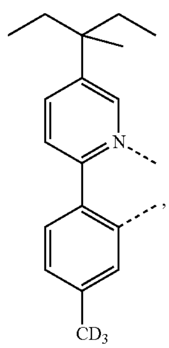

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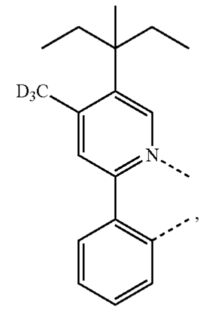

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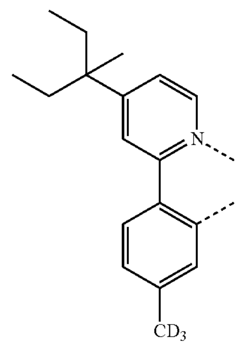

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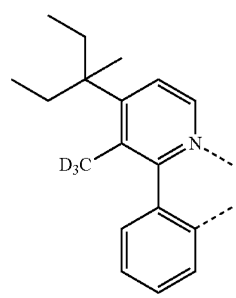

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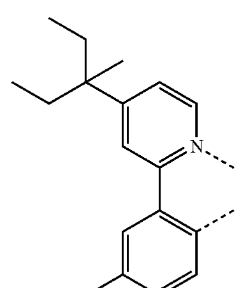

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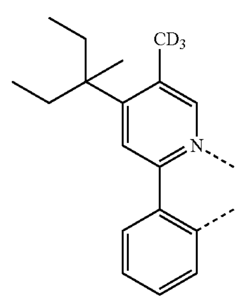

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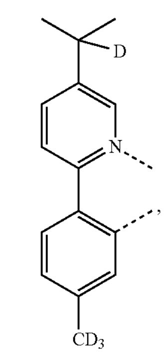

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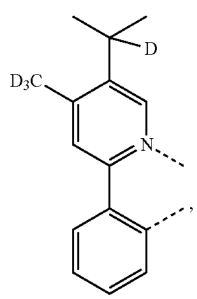

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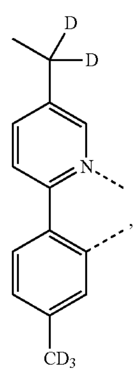

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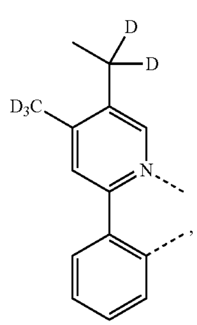

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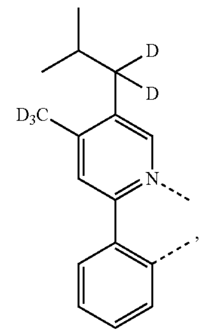

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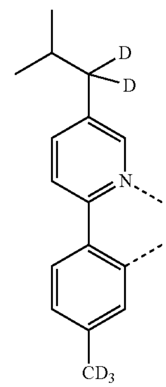

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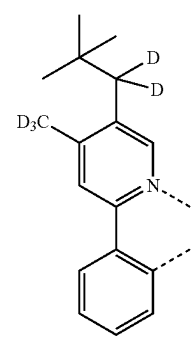

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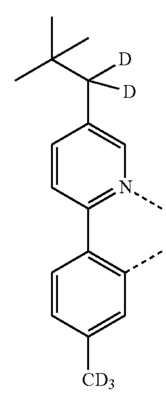

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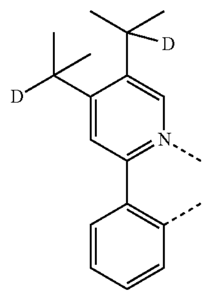
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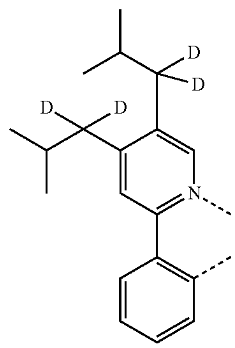
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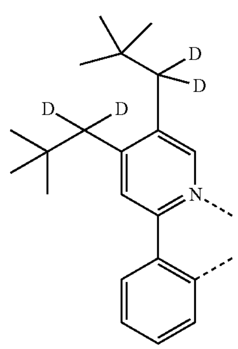
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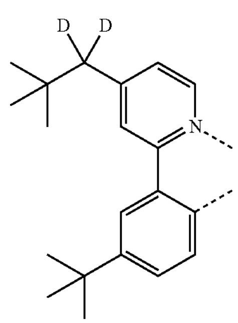
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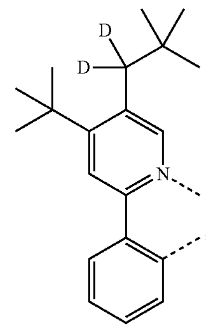
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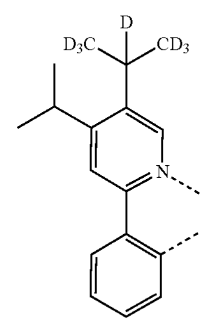
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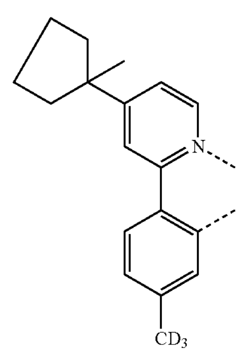
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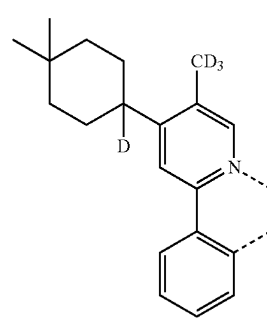
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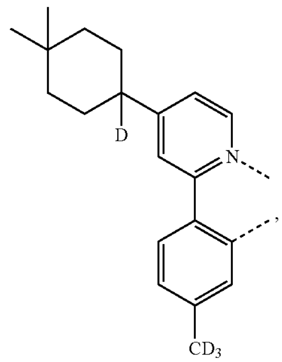

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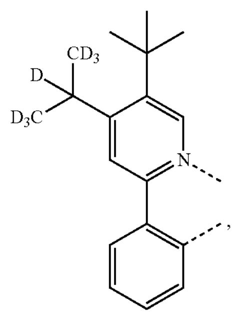

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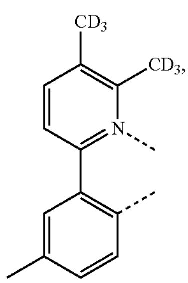

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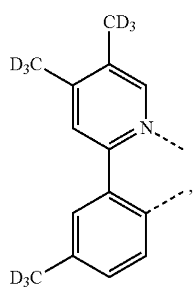

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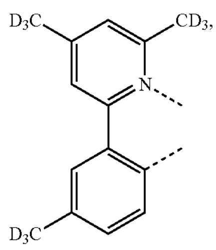

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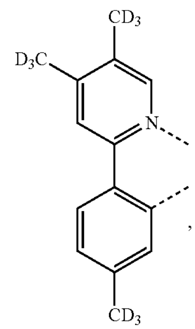

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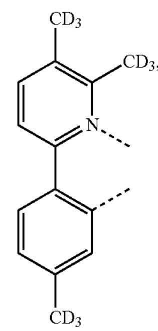

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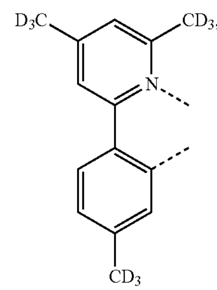

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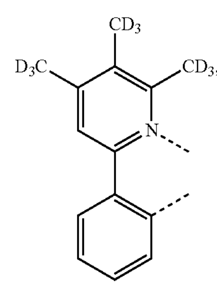

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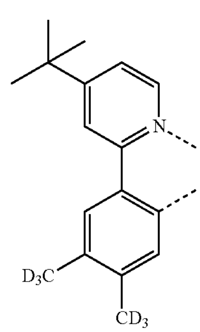

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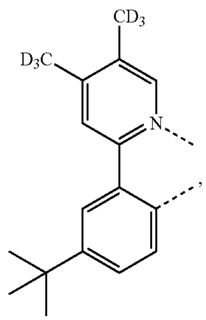

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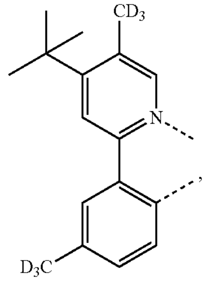

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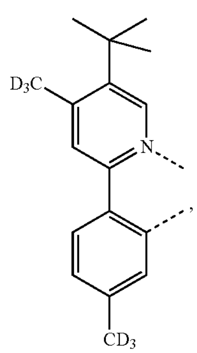

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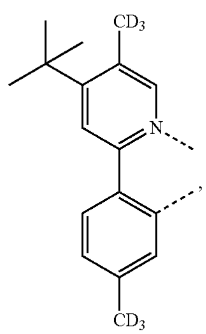

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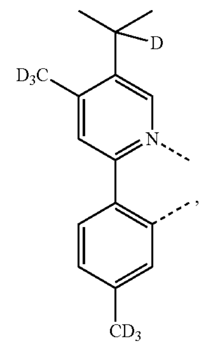

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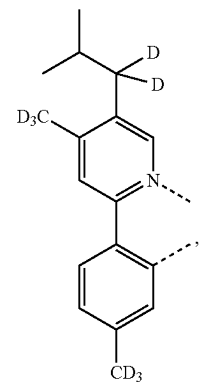

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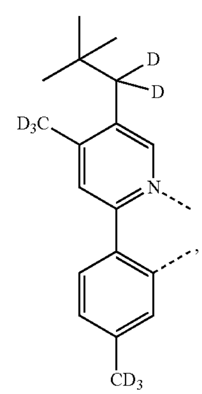

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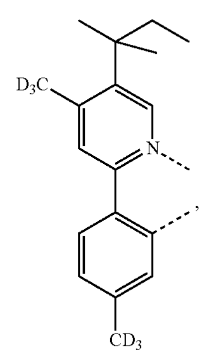

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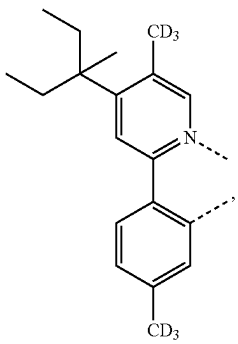

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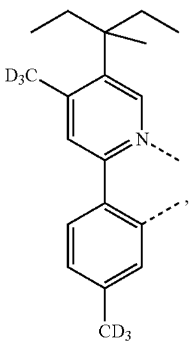

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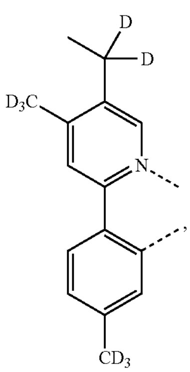

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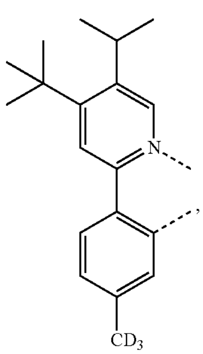

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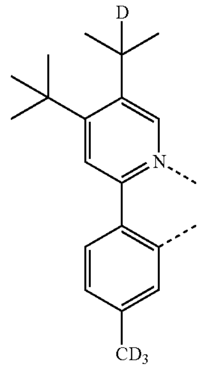

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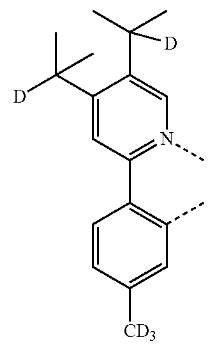

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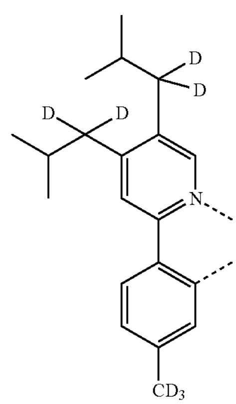

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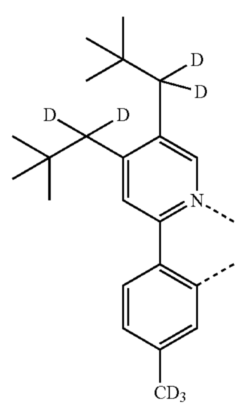

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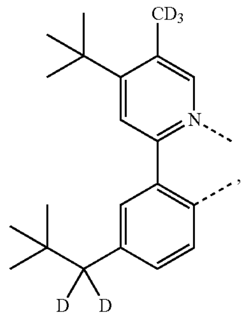

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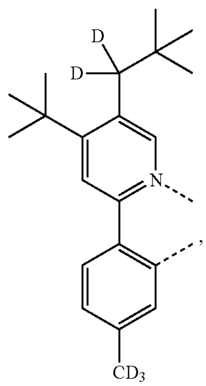

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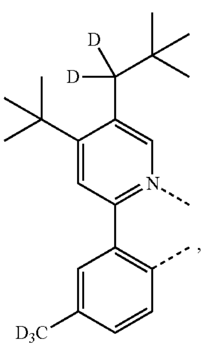

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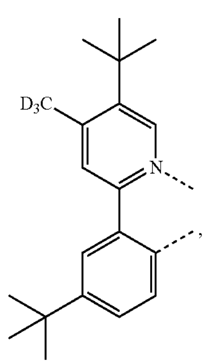

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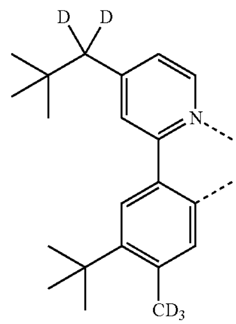

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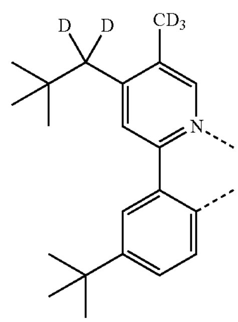

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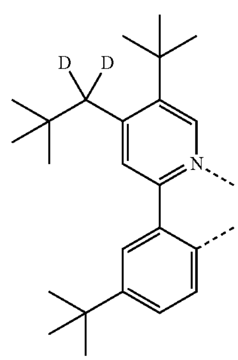

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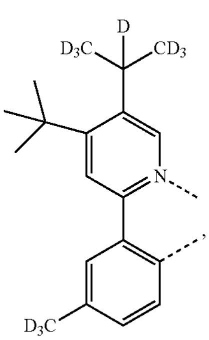

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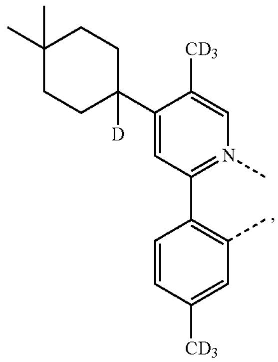

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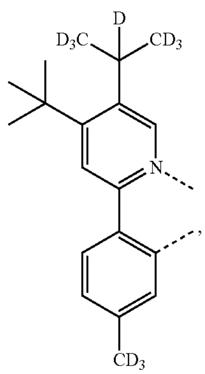

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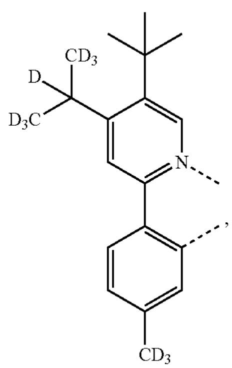

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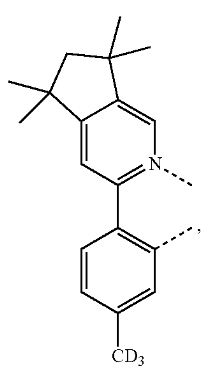

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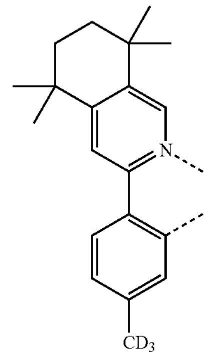

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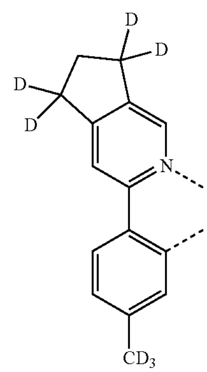

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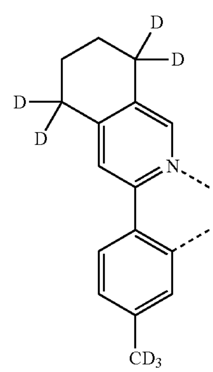

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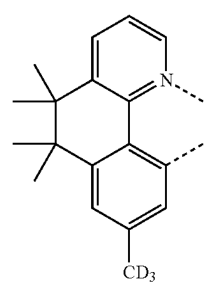

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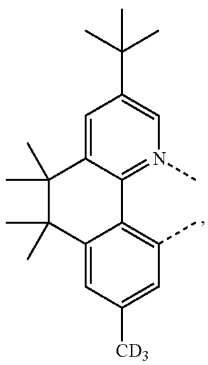

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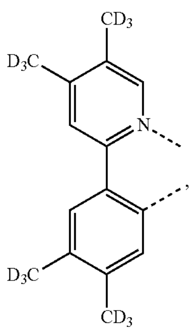

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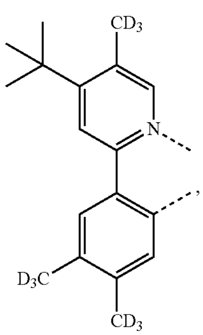

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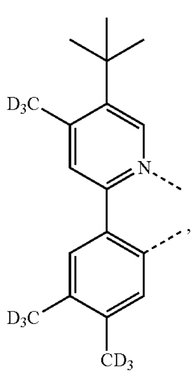

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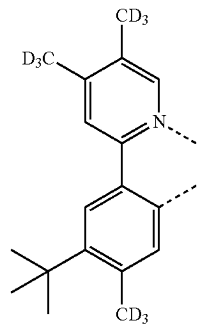

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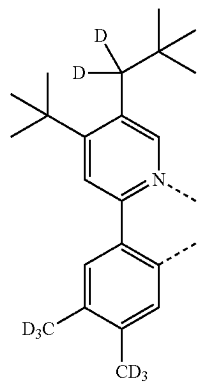

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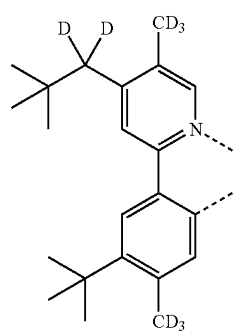

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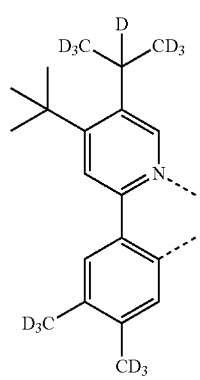

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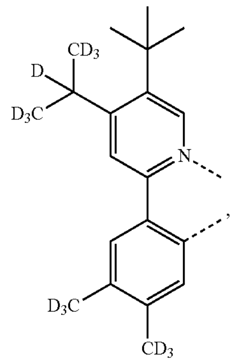

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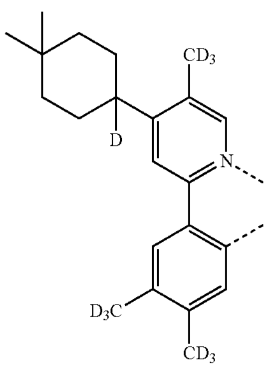

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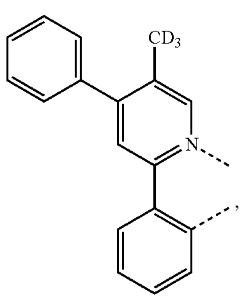

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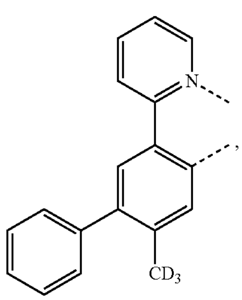

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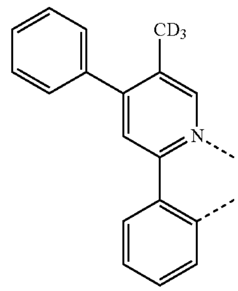

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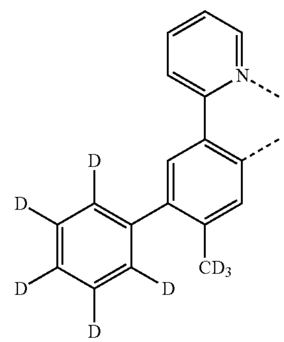

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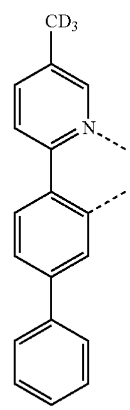

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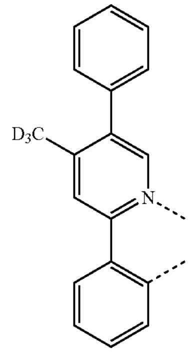

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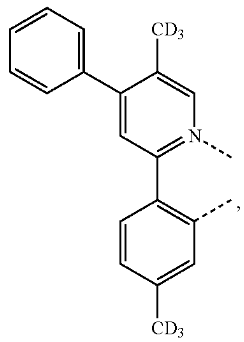

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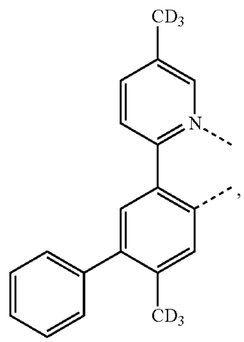

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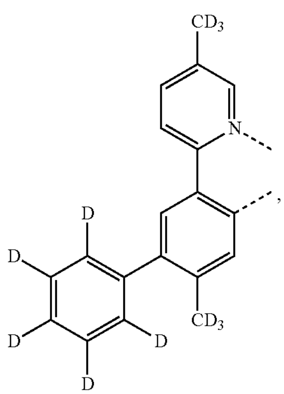

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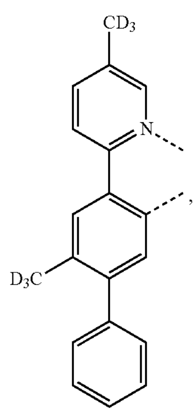

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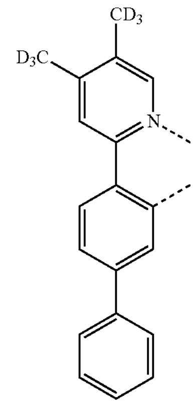

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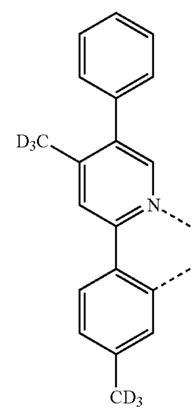

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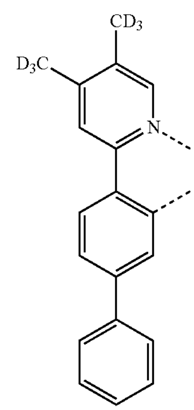

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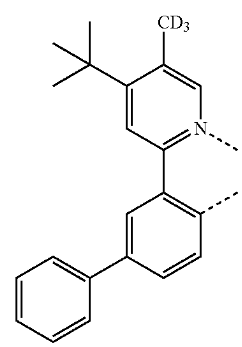

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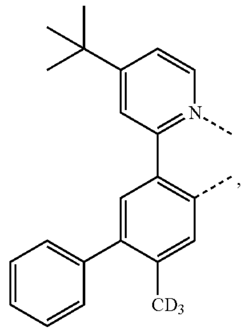

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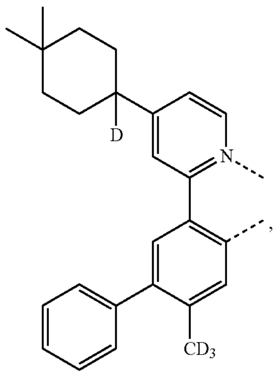

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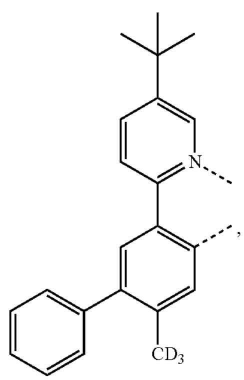

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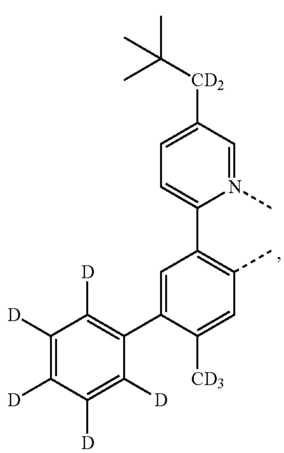

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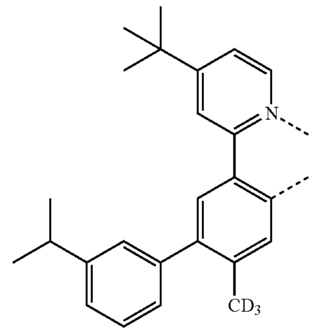

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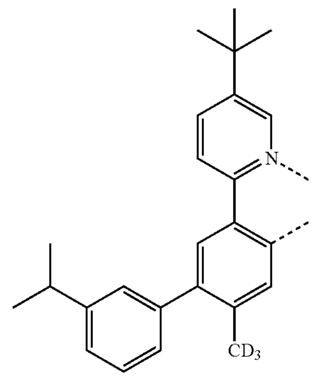

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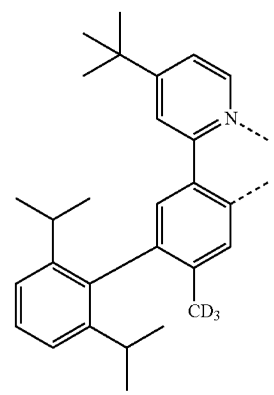

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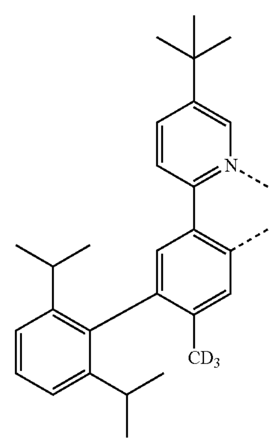

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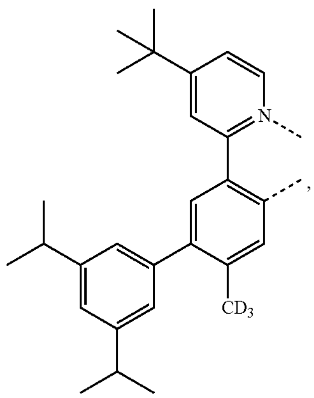

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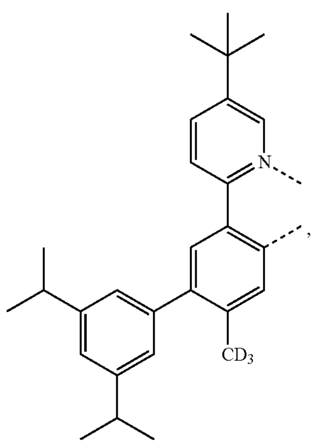

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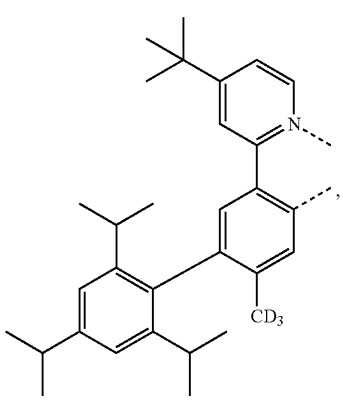

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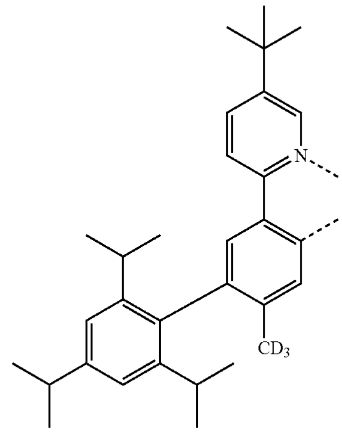

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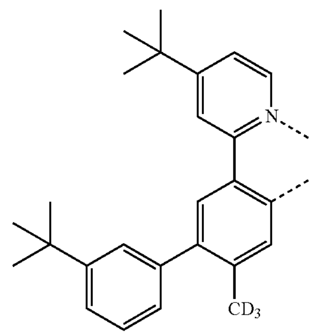

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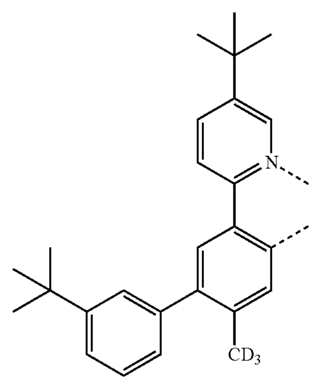

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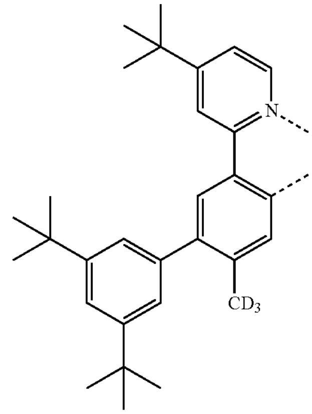

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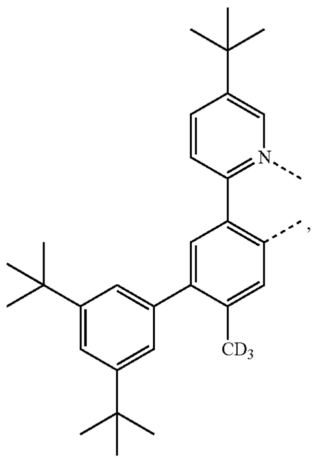

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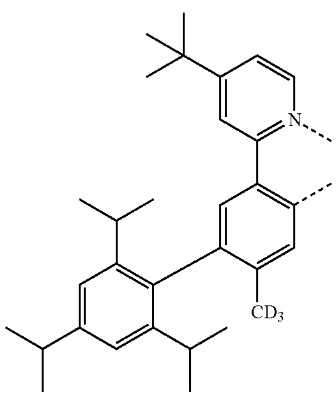

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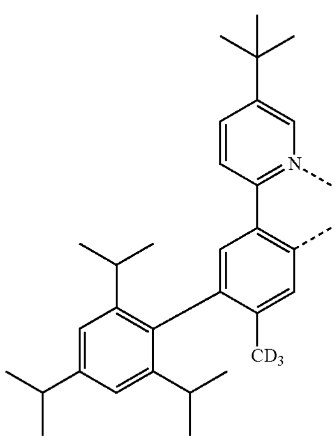

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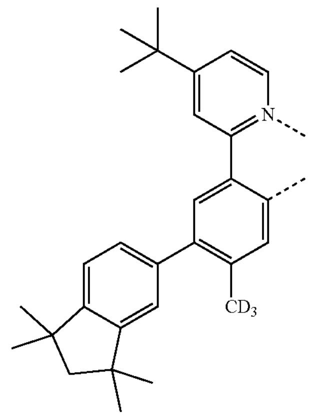

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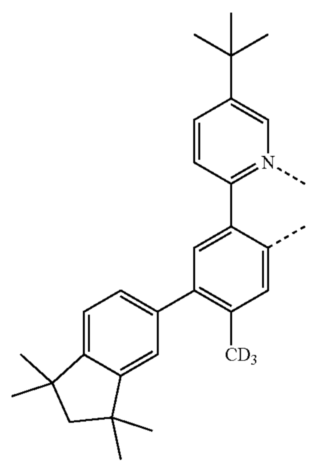

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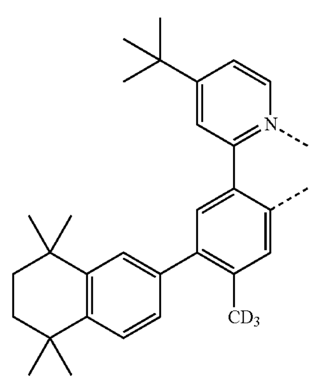

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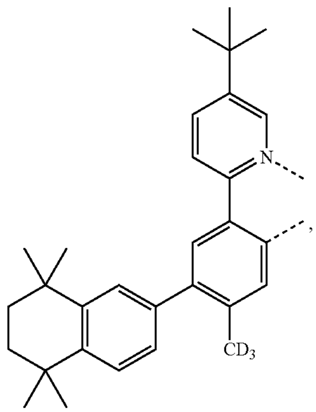

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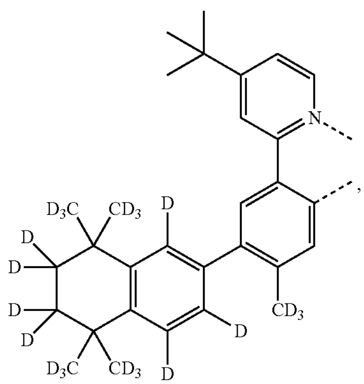

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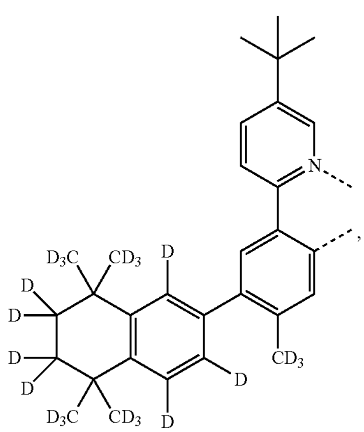

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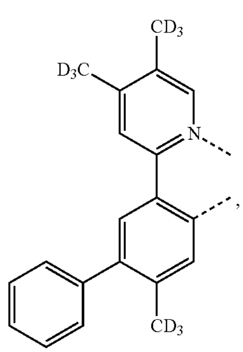

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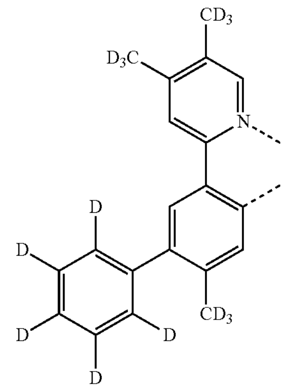

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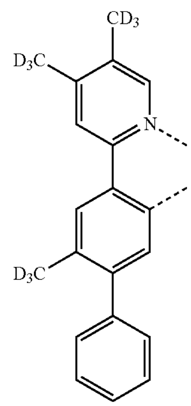

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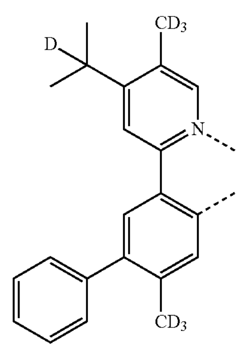

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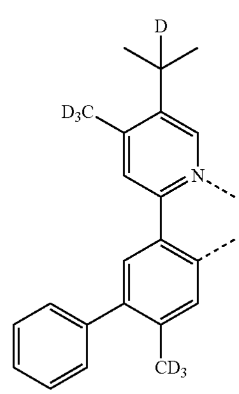

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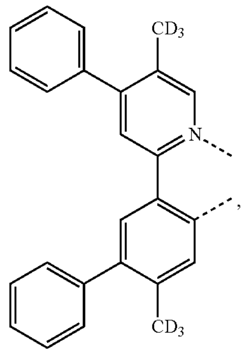

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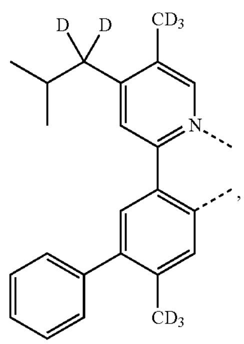

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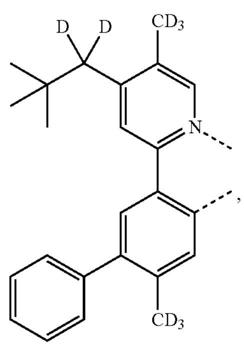

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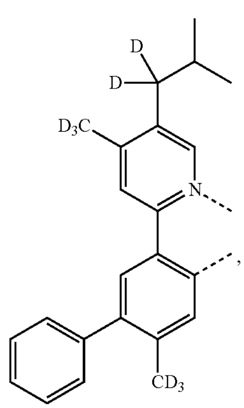

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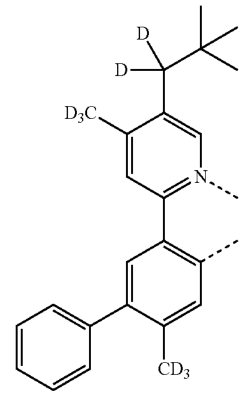

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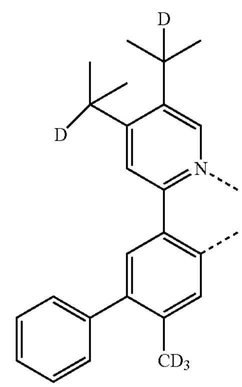

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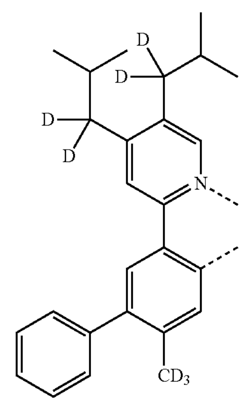

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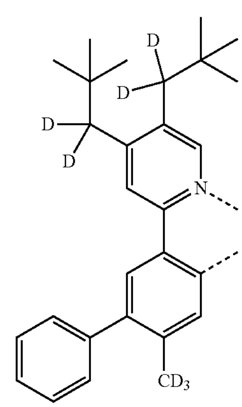

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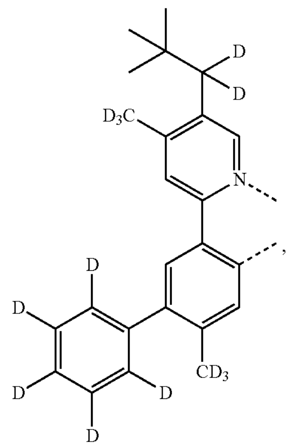

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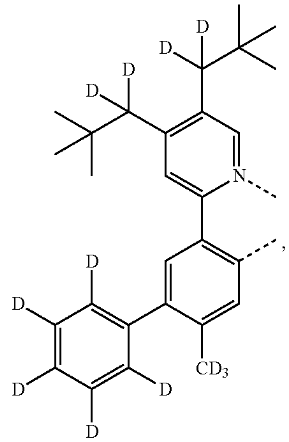

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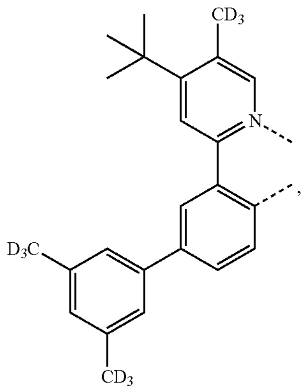

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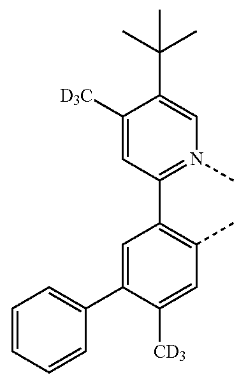

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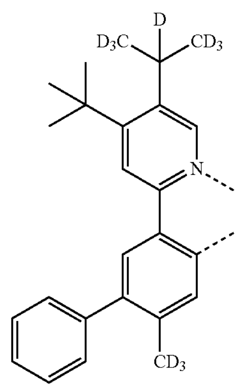

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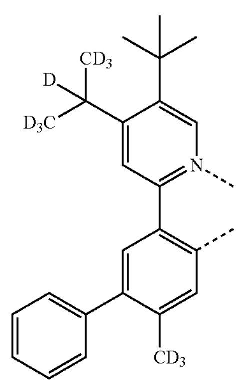

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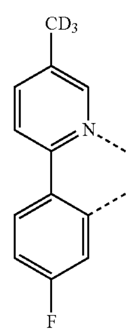

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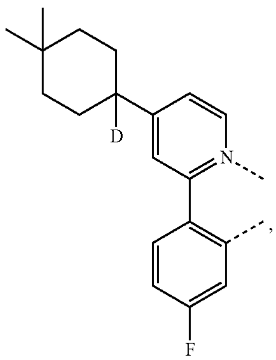

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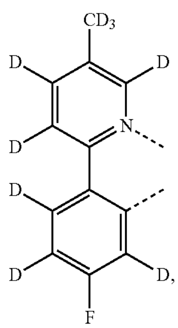

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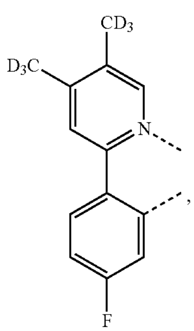

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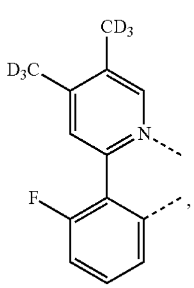

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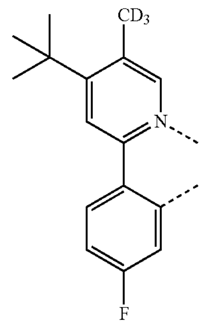

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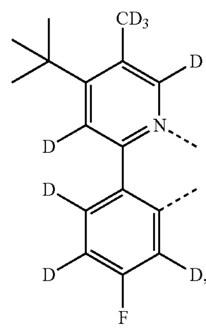

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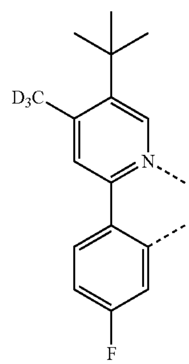

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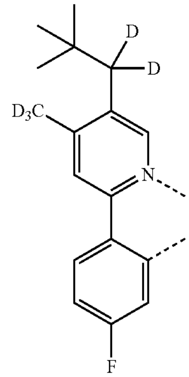

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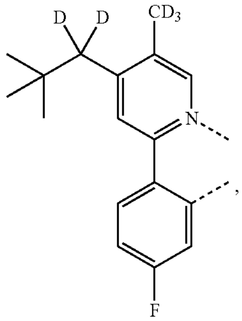

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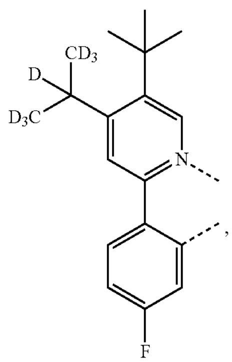

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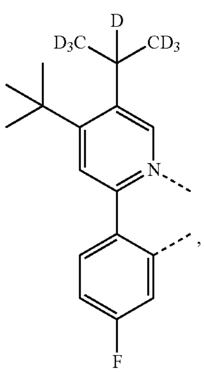

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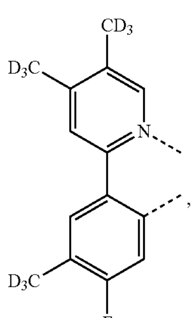

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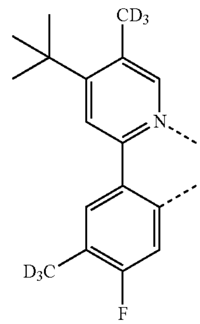

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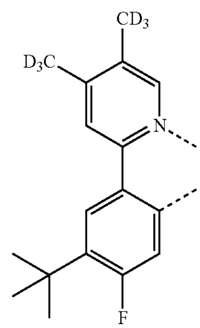

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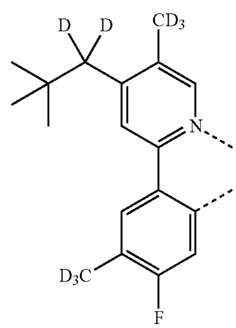

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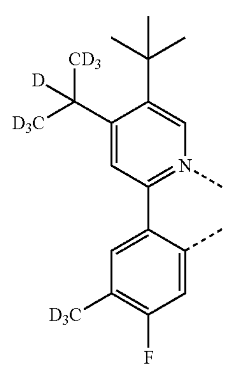

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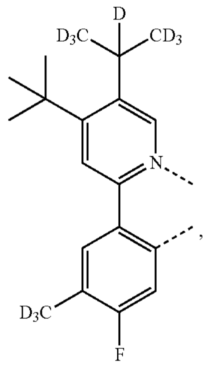

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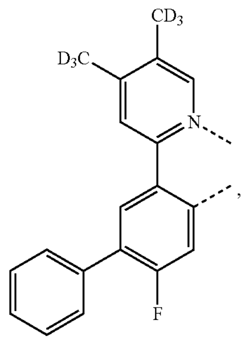

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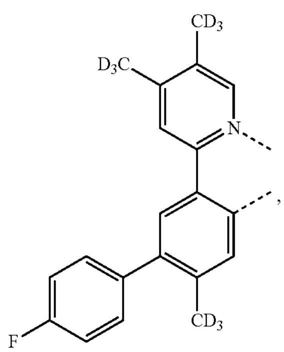

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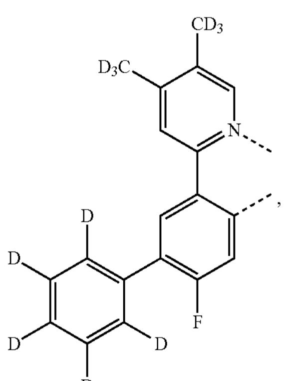

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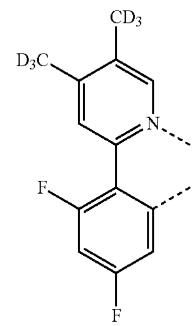

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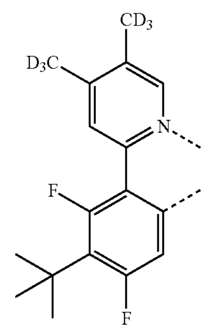

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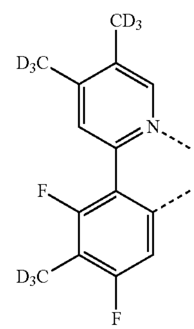

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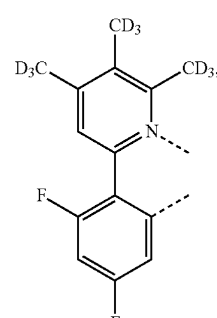

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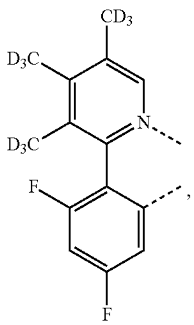

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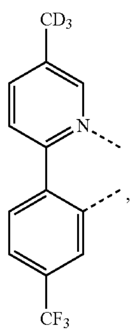

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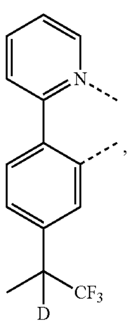

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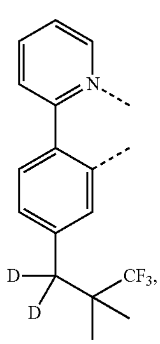

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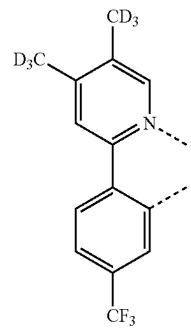

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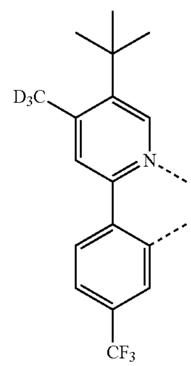

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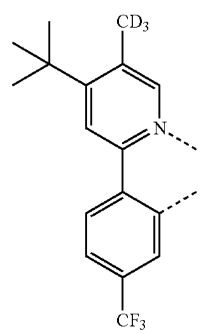

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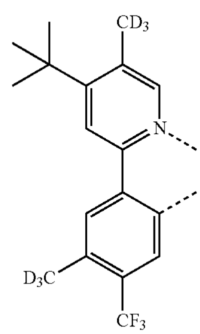

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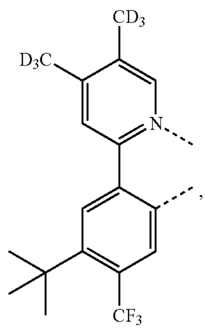

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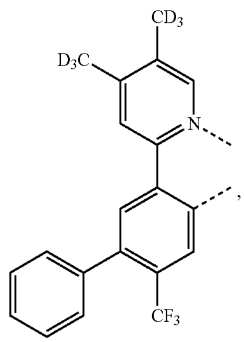

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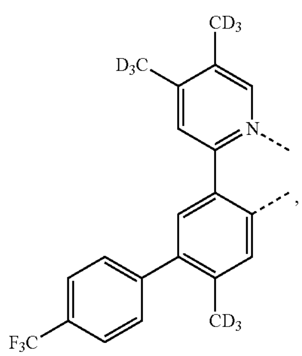

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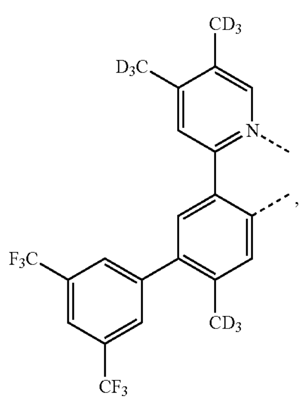

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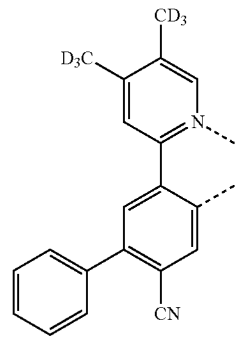

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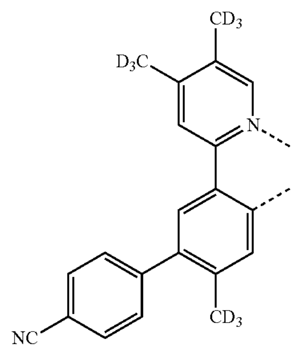

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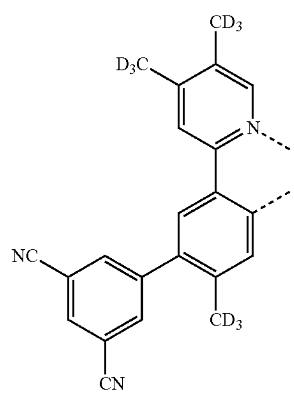

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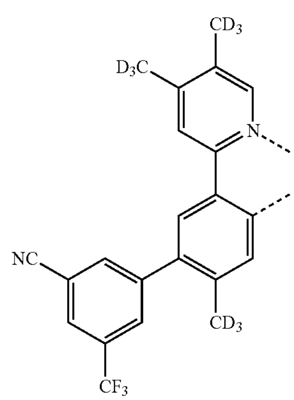

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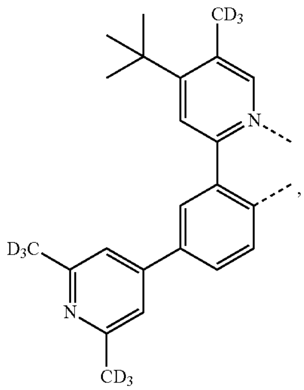

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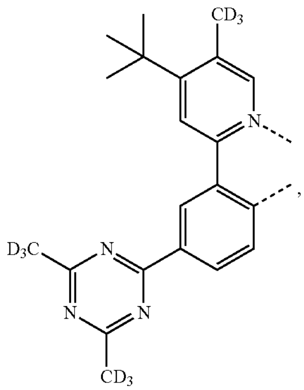

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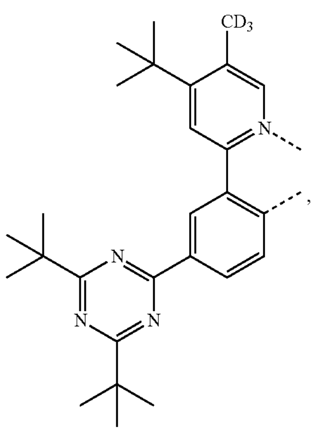

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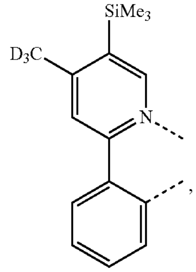

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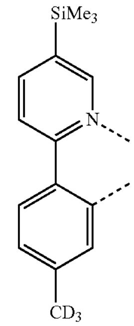

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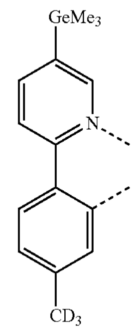

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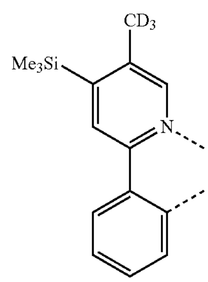

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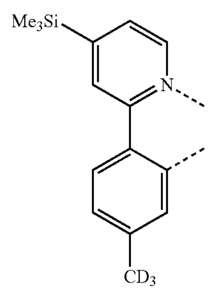

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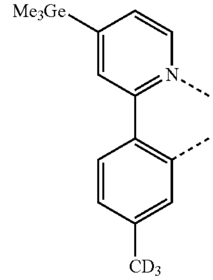

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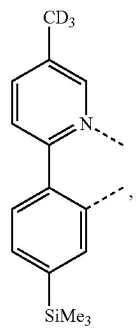

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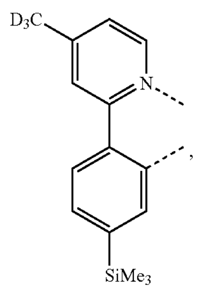

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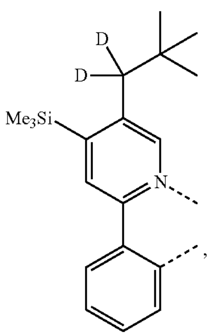

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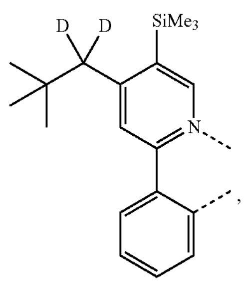

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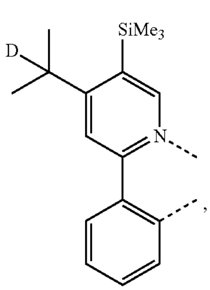

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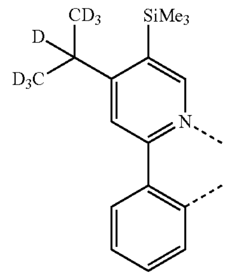

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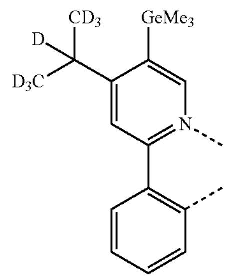

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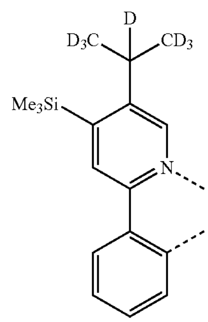

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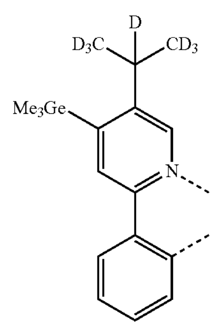

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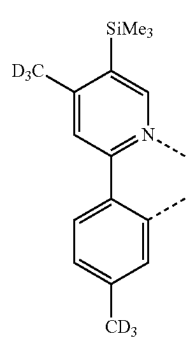

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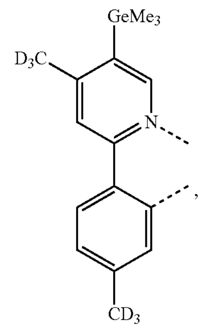

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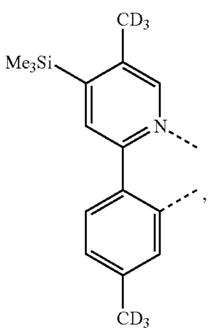

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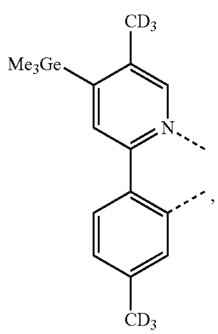

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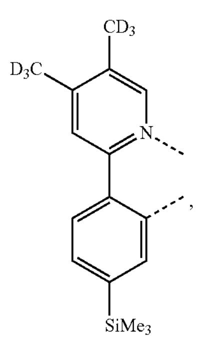

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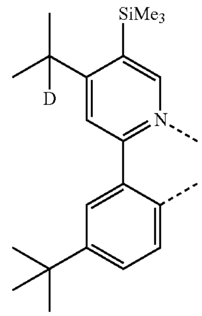

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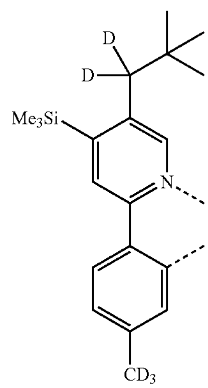

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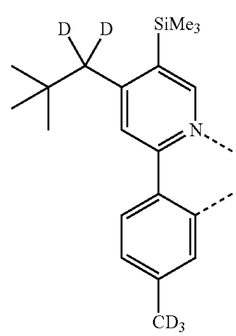

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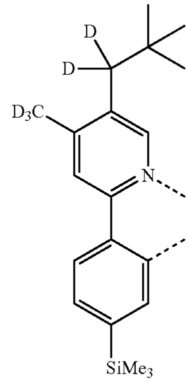

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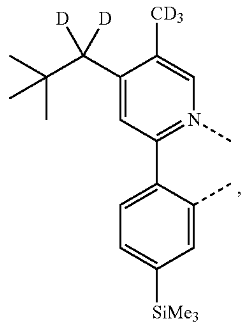

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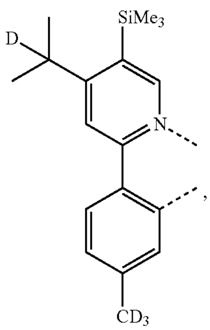

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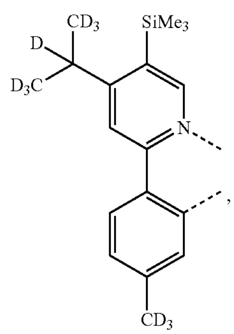

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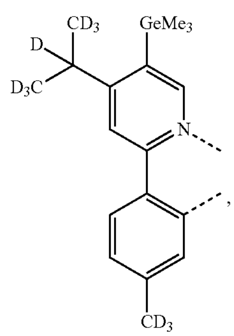

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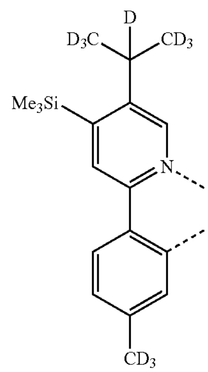

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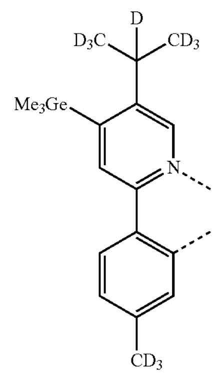

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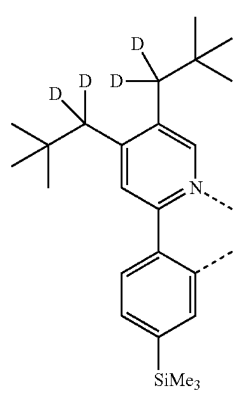

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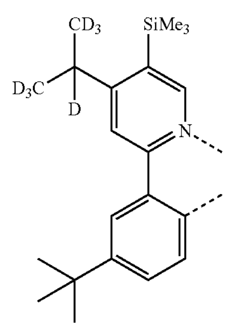

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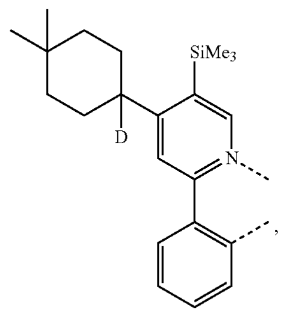

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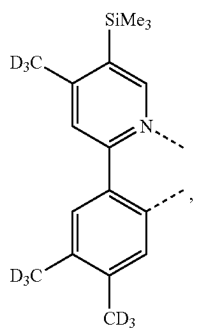

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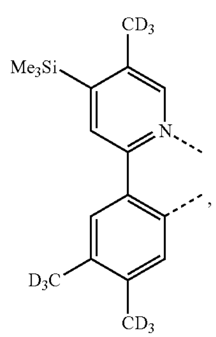

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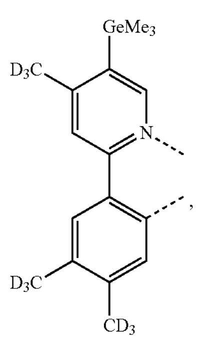

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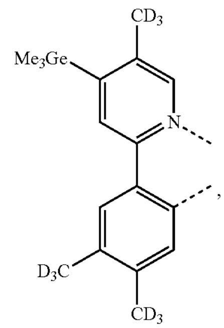

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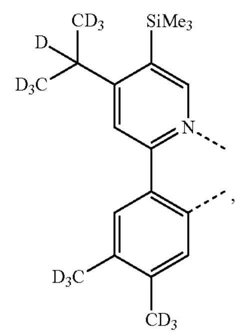

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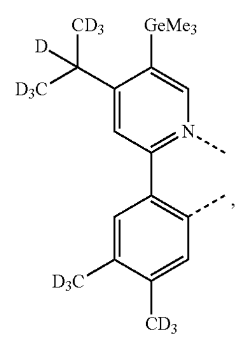

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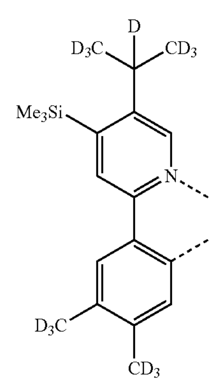

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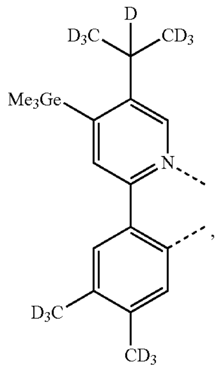

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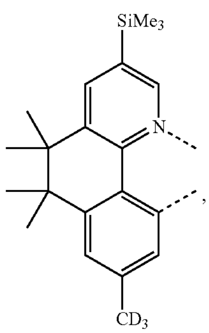

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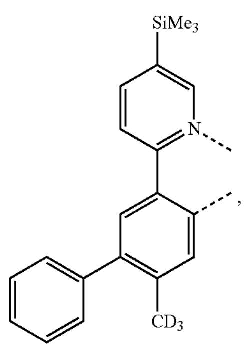

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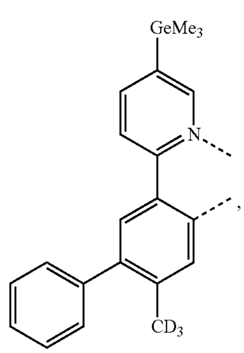

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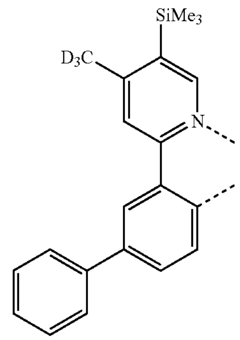

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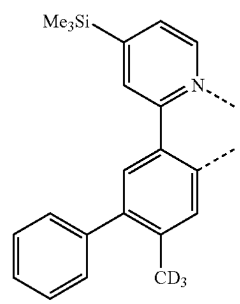

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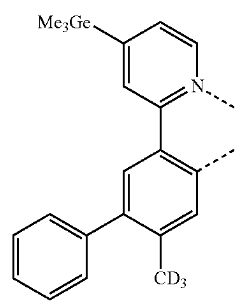

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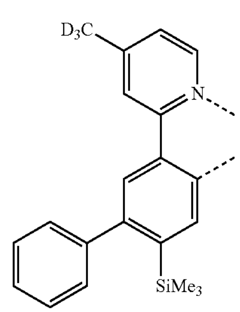

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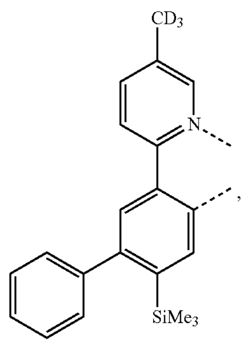

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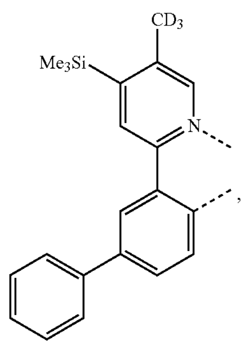

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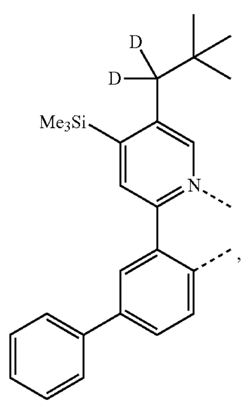

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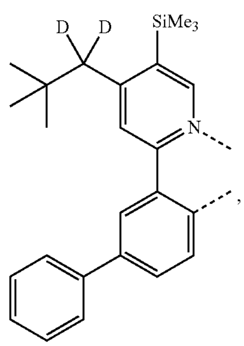

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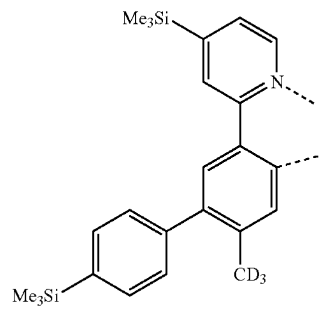

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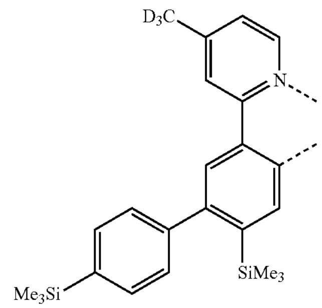

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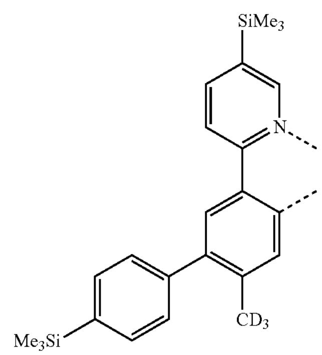

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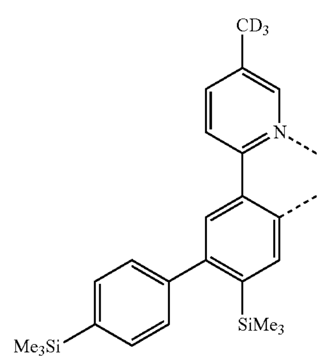

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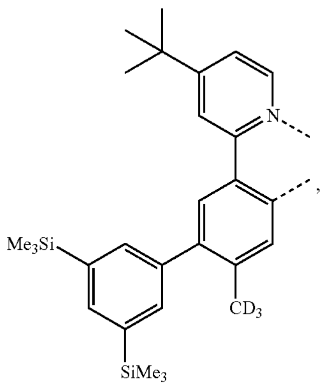

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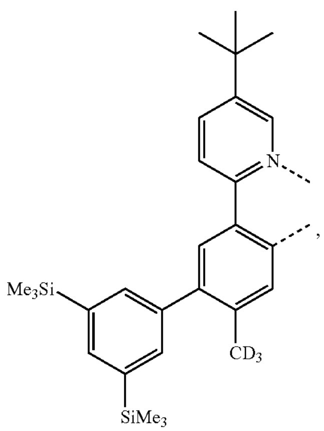

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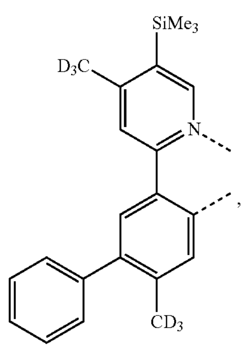

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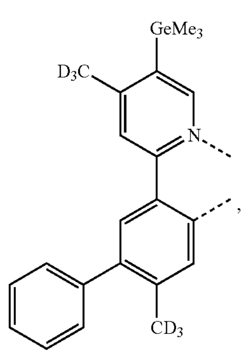

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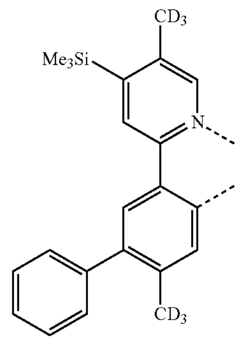

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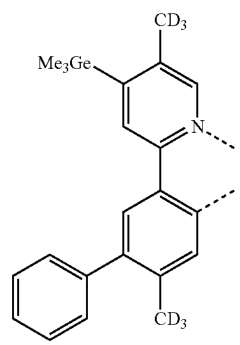

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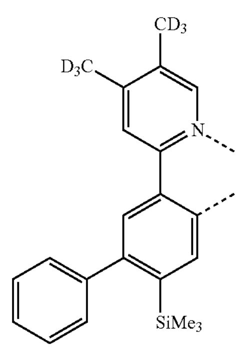

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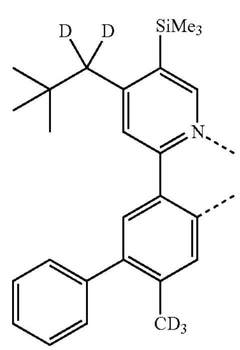

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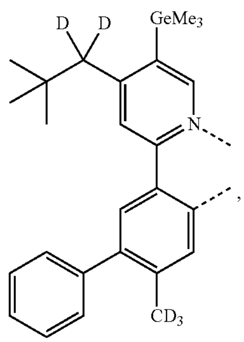

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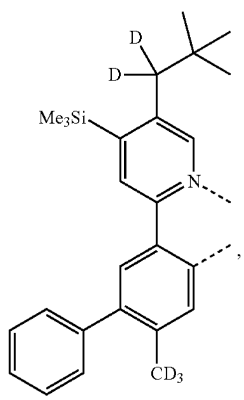

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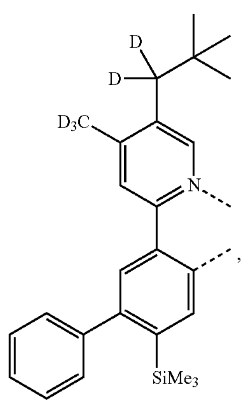

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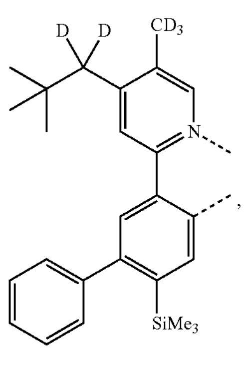

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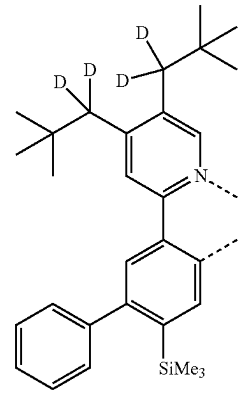

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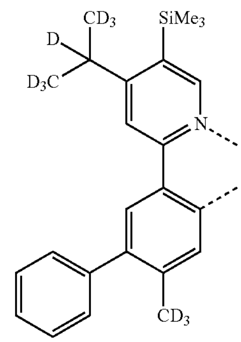

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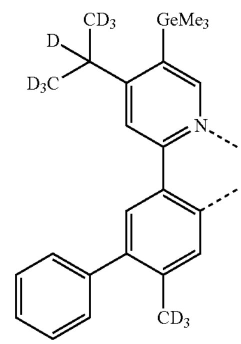

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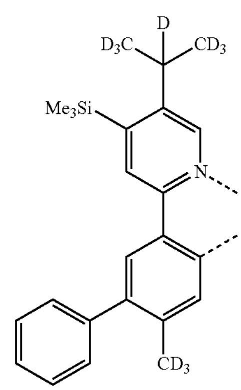

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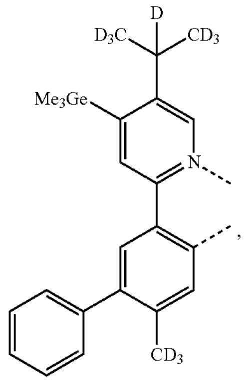

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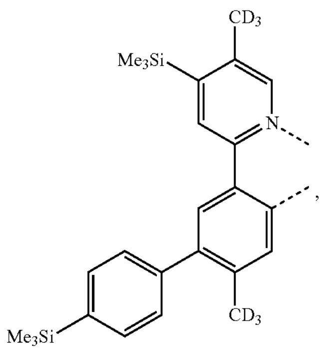

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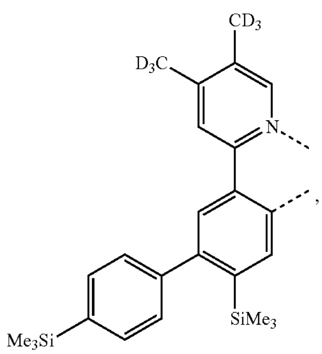

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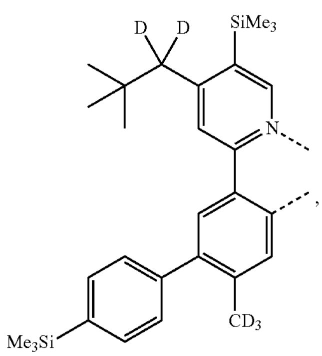

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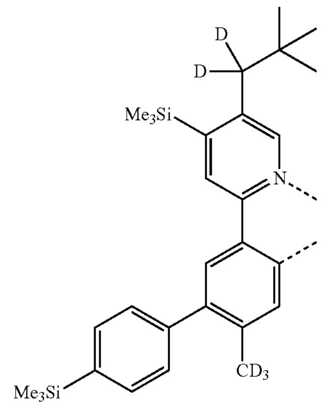

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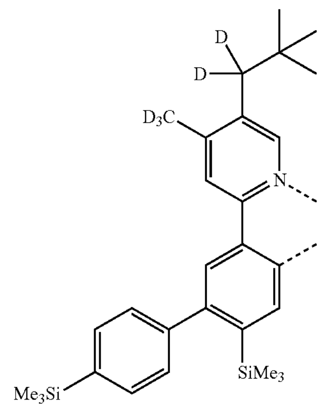

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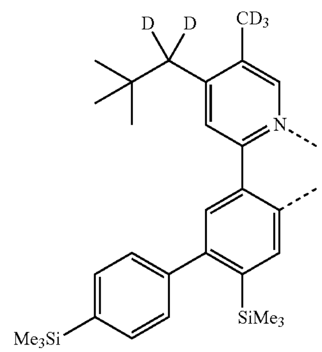

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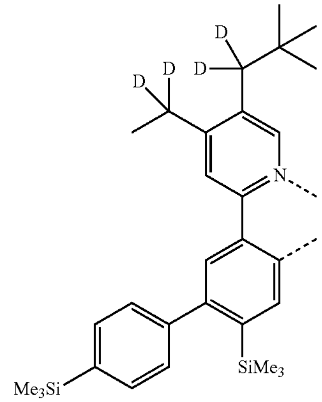

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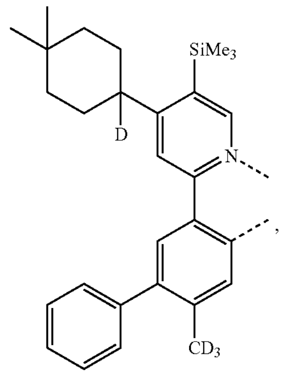

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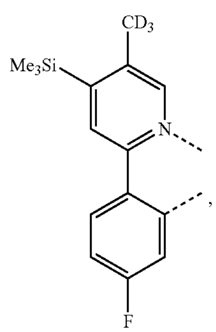

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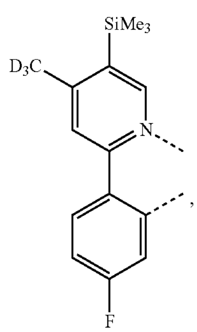

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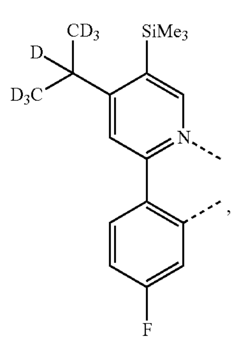

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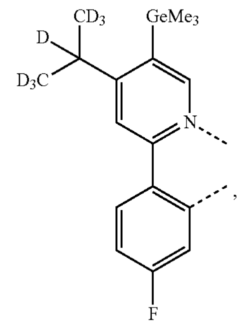

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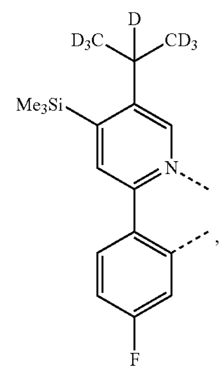

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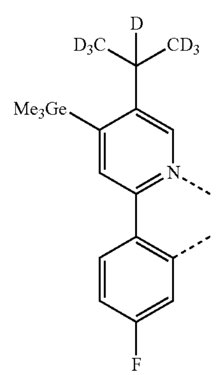

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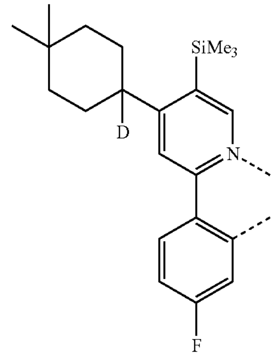

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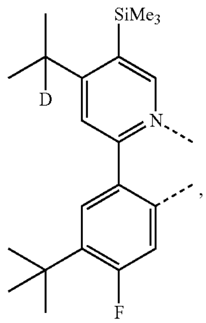

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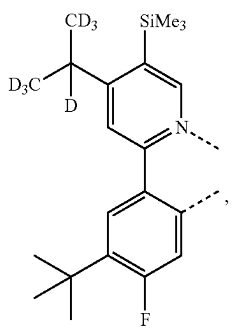

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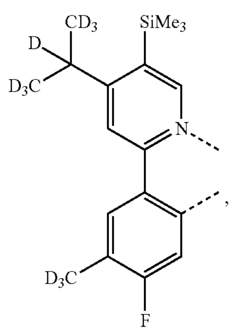

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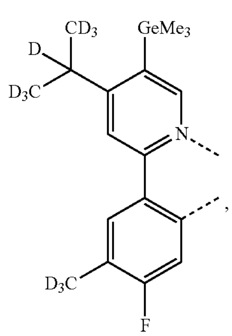

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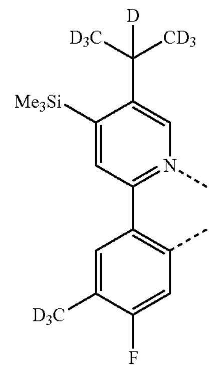

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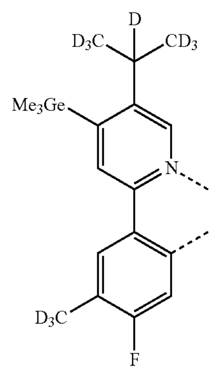

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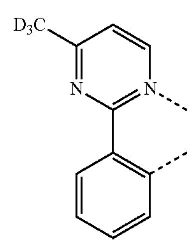

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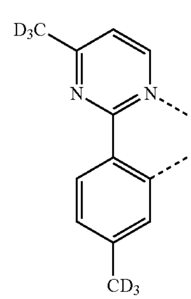

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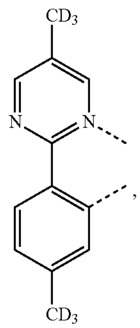

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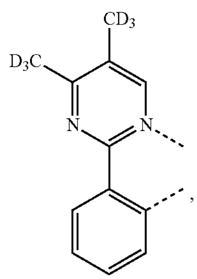

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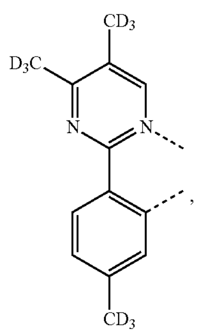

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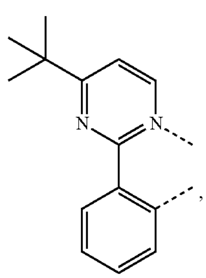

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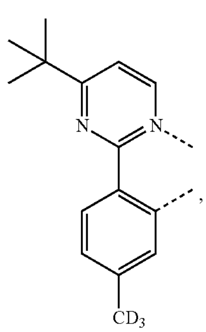

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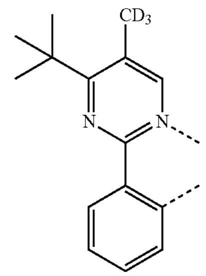

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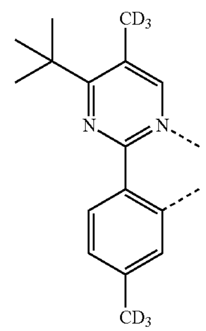

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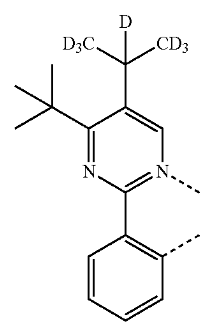

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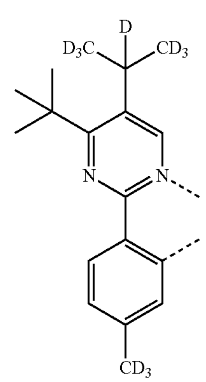

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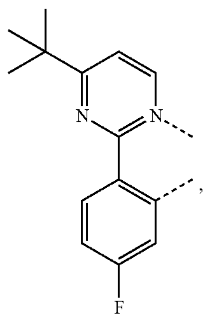

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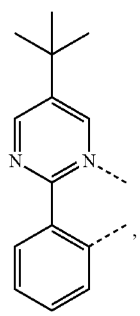

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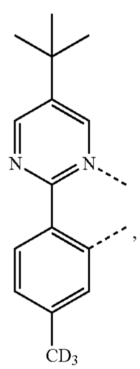

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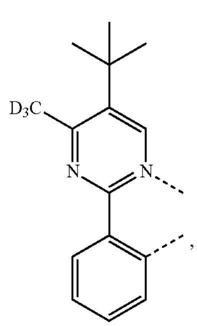

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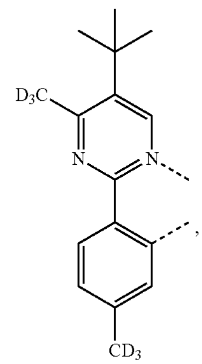

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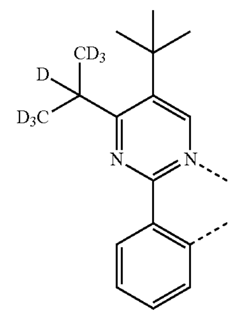

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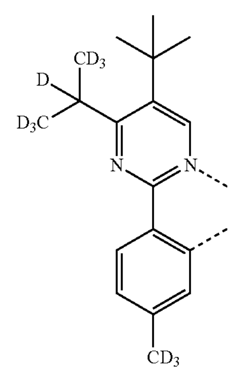

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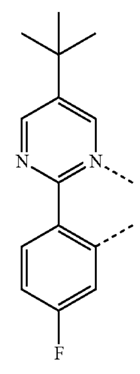

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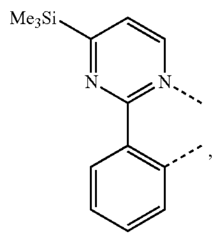
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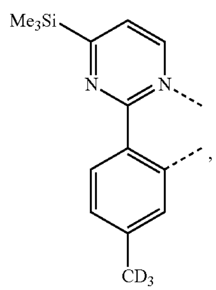
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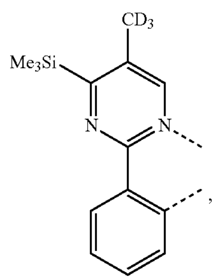
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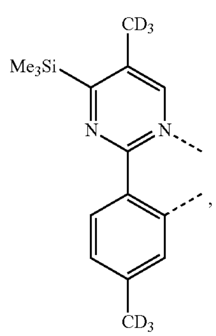
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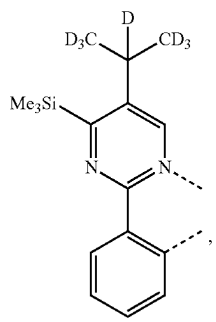
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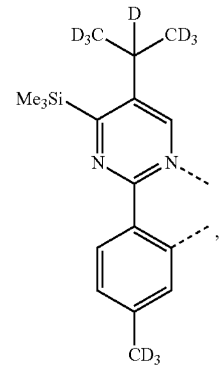
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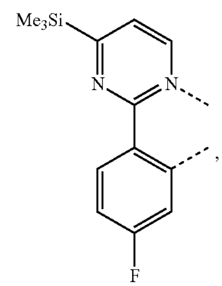
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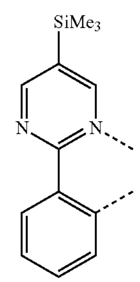
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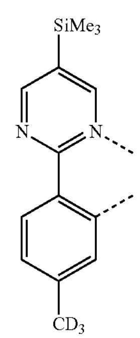
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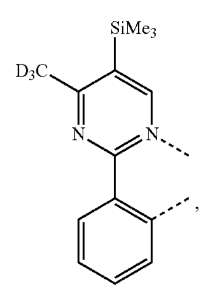
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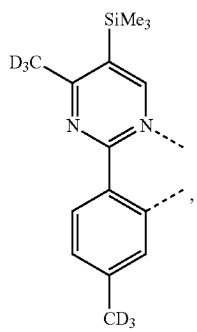

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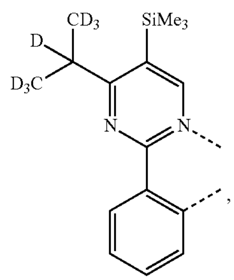

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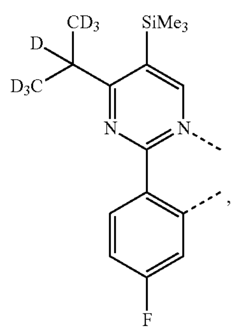

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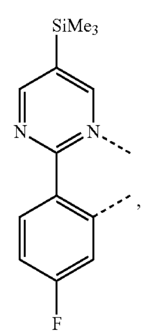

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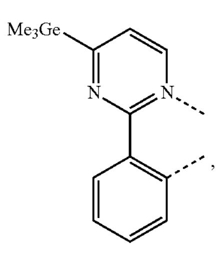

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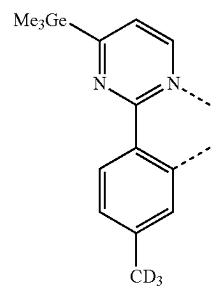

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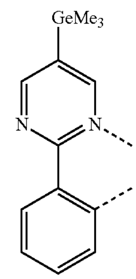

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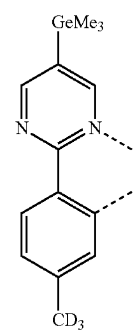

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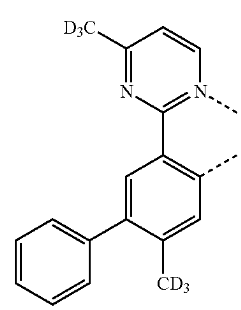

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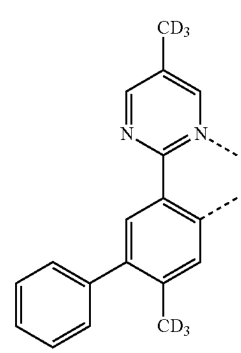

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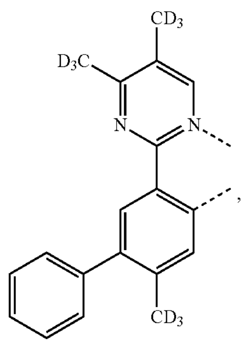

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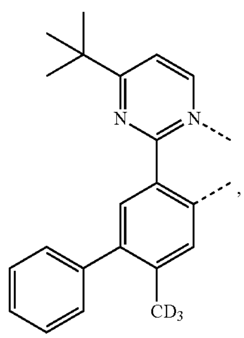

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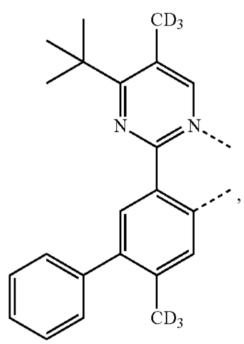

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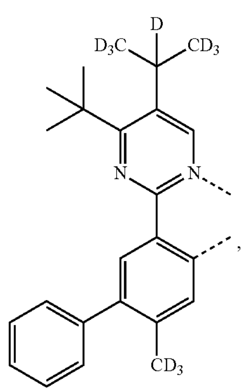

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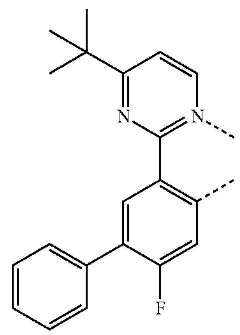

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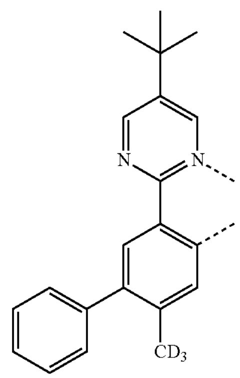

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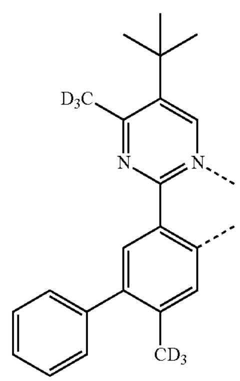

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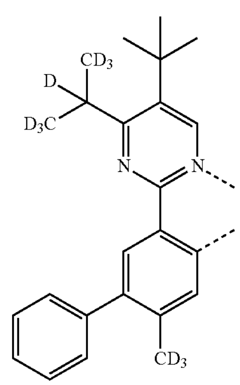

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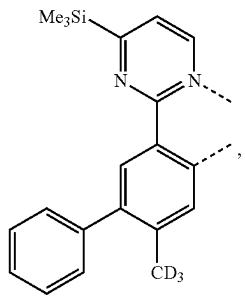

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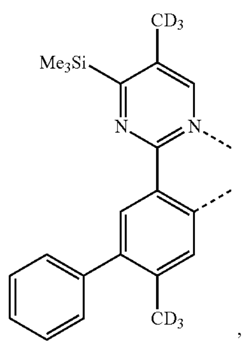

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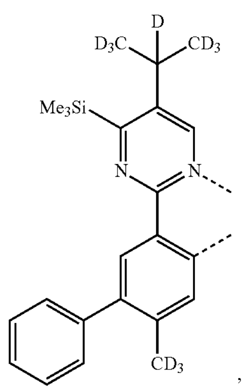

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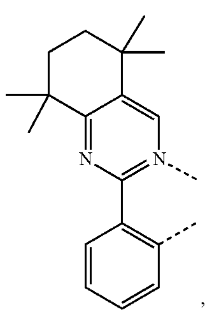

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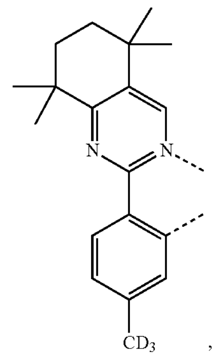

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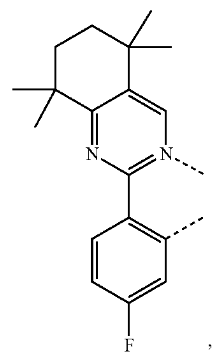

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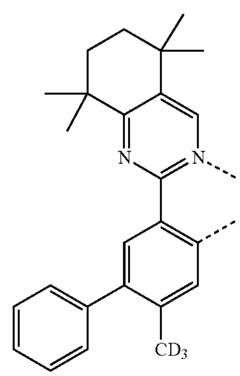

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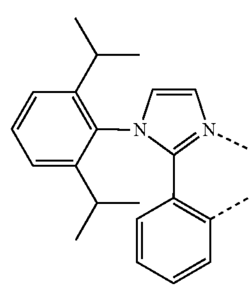

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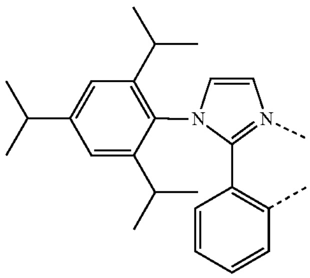

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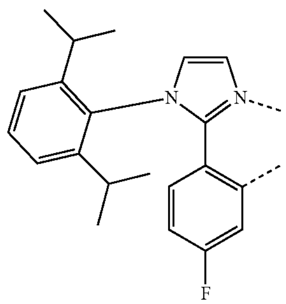

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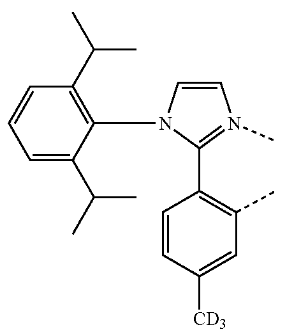

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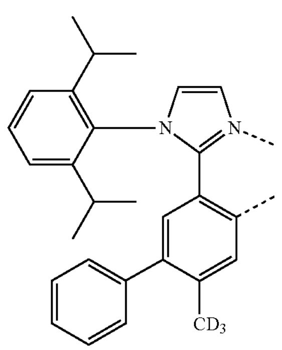

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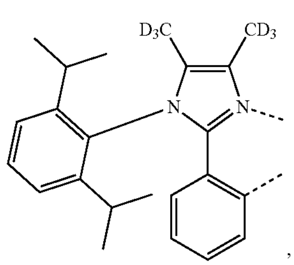

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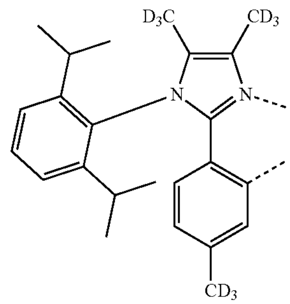

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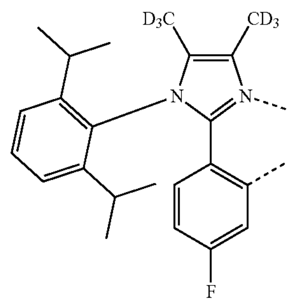

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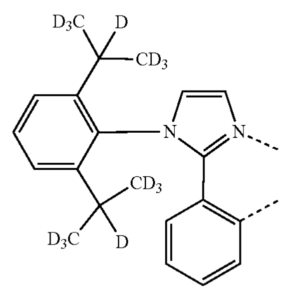

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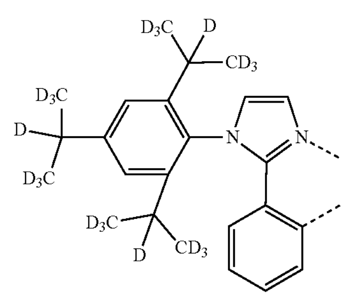

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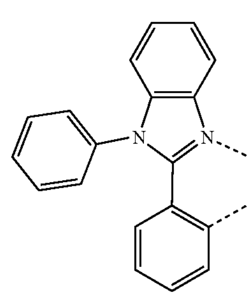

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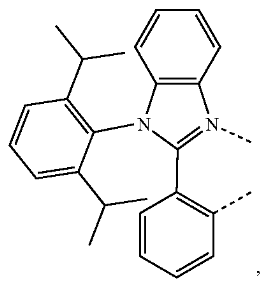

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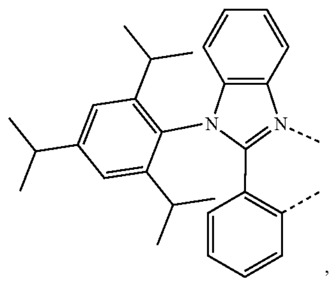

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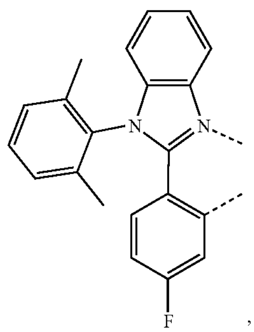

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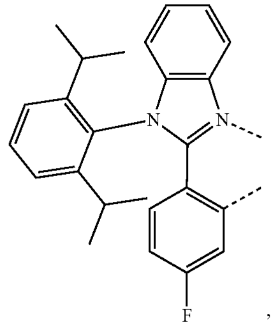

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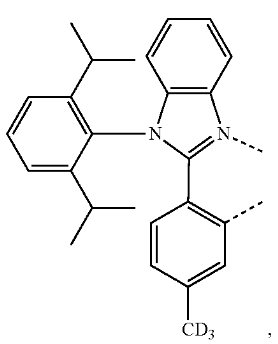

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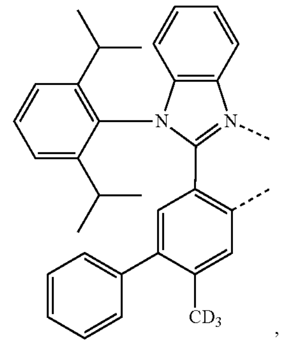

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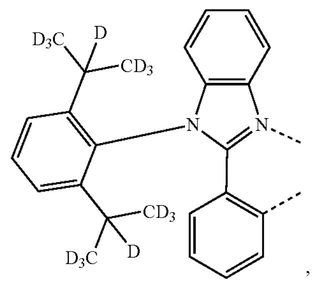

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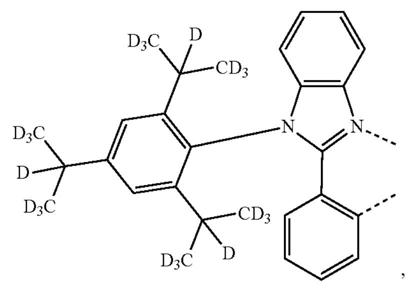

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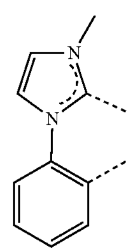

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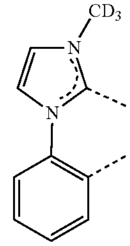

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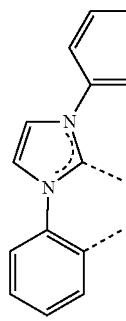

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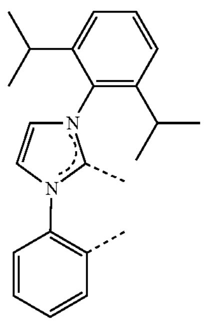

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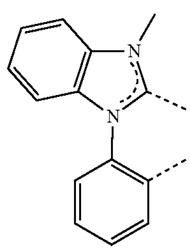

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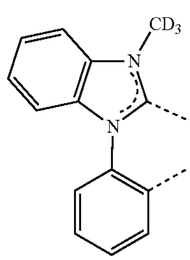

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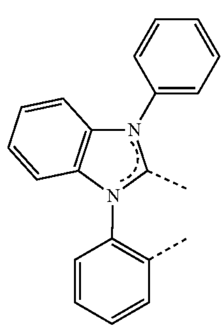

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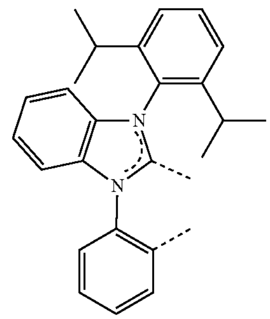

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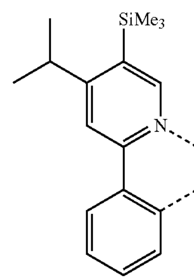
L_{B531}



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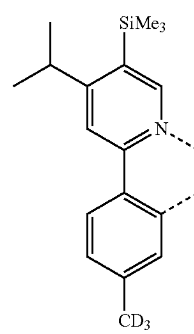

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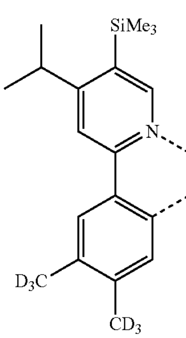

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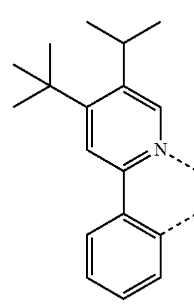

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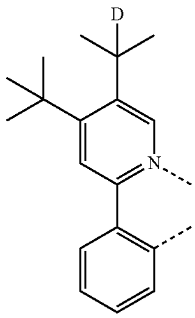

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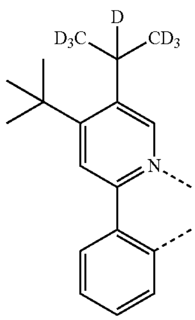

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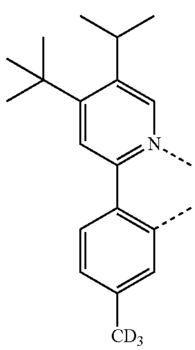

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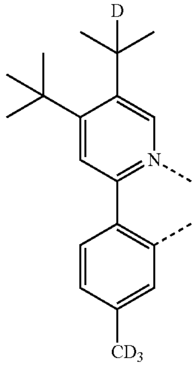

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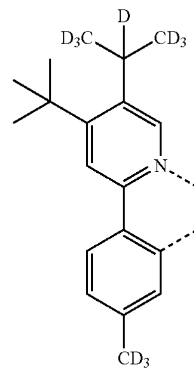
L_{B539}



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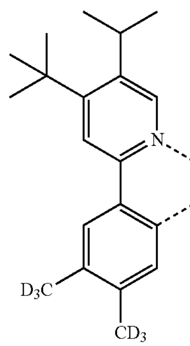
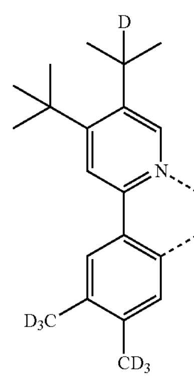
L_{B540}



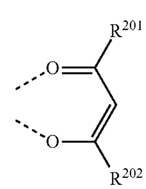
, and

L_{B541}



;

[0315] wherein each L_{Cj-I} has a structure based on formula



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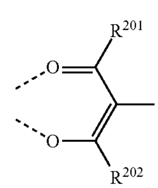
and

[0316] each L_{Cj-II} has a structure based on formula



,

wherein for each L_{Cj} in $L_{Cj\text{-}I}$ and $L_{Cj\text{-}II}$, R^{201} and R^{202} are defined in the following LIST 11:

L_{Cj}	R^{201}	R^{202}	L_{Cj}	R^{201}	R^{202}	L_{Cj}	R^{201}	R^{202}	L_{Cj}	R^{201}	R^{202}
L_{C1}	R^{D1}	R^{D1}	L_{C193}	R^{D1}	R^{D3}	L_{C385}	R^{D17}	R^{D40}	L_{C577}	R^{D143}	R^{D120}
L_{C2}	R^{D2}	R^{D2}	L_{C194}	R^{D1}	R^{D4}	L_{C386}	R^{D17}	R^{D41}	L_{C578}	R^{D143}	R^{D133}
L_{C3}	R^{D3}	R^{D3}	L_{C195}	R^{D1}	R^{D5}	L_{C387}	R^{D17}	R^{D42}	L_{C579}	R^{D143}	R^{D134}
L_{C4}	R^{D4}	R^{D4}	L_{C196}	R^{D1}	R^{D9}	L_{C388}	R^{D17}	R^{D43}	L_{C580}	R^{D143}	R^{D135}
L_{C5}	R^{D5}	R^{D5}	L_{C197}	R^{D1}	R^{D10}	L_{C389}	R^{D17}	R^{D48}	L_{C581}	R^{D143}	R^{D136}
L_{C6}	R^{D6}	R^{D6}	L_{C198}	R^{D1}	R^{D17}	L_{C390}	R^{D17}	R^{D49}	L_{C582}	R^{D143}	R^{D144}
L_{C7}	R^{D7}	R^{D7}	L_{C199}	R^{D1}	R^{D18}	L_{C391}	R^{D17}	R^{D50}	L_{C583}	R^{D143}	R^{D145}
L_{C8}	R^{D8}	R^{D8}	L_{C200}	R^{D1}	R^{D20}	L_{C392}	R^{D17}	R^{D54}	L_{C584}	R^{D143}	R^{D146}
L_{C9}	R^{D9}	R^{D9}	L_{C201}	R^{D1}	R^{D22}	L_{C393}	R^{D17}	R^{D55}	L_{C585}	R^{D143}	R^{D147}
L_{C10}	R^{D10}	R^{D10}	L_{C202}	R^{D1}	R^{D37}	L_{C394}	R^{D17}	R^{D58}	L_{C586}	R^{D143}	R^{D149}
L_{C11}	R^{D11}	R^{D11}	L_{C203}	R^{D1}	R^{D40}	L_{C395}	R^{D17}	R^{D59}	L_{C587}	R^{D143}	R^{D151}
L_{C12}	R^{D12}	R^{D12}	L_{C204}	R^{D1}	R^{D41}	L_{C396}	R^{D17}	R^{D78}	L_{C588}	R^{D143}	R^{D154}
L_{C13}	R^{D13}	R^{D13}	L_{C205}	R^{D1}	R^{D42}	L_{C397}	R^{D17}	R^{D79}	L_{C589}	R^{D143}	R^{D155}
L_{C14}	R^{D14}	R^{D14}	L_{C206}	R^{D1}	R^{D43}	L_{C398}	R^{D17}	R^{D81}	L_{C590}	R^{D143}	R^{D161}
L_{C15}	R^{D15}	R^{D15}	L_{C207}	R^{D1}	R^{D48}	L_{C399}	R^{D17}	R^{D87}	L_{C591}	R^{D143}	R^{D175}
L_{C16}	R^{D16}	R^{D16}	L_{C208}	R^{D1}	R^{D49}	L_{C400}	R^{D17}	R^{D88}	L_{C592}	R^{D144}	R^{D3}
L_{C17}	R^{D17}	R^{D17}	L_{C209}	R^{D1}	R^{D50}	L_{C401}	R^{D17}	R^{D89}	L_{C593}	R^{D144}	R^{D5}
L_{C18}	R^{D18}	R^{D18}	L_{C210}	R^{D1}	R^{D54}	L_{C402}	R^{D17}	R^{D93}	L_{C594}	R^{D144}	R^{D17}
L_{C19}	R^{D19}	R^{D19}	L_{C211}	R^{D1}	R^{D55}	L_{C403}	R^{D17}	R^{D116}	L_{C595}	R^{D144}	R^{D18}
L_{C20}	R^{D20}	R^{D20}	L_{C212}	R^{D1}	R^{D58}	L_{C404}	R^{D17}	R^{D117}	L_{C596}	R^{D144}	R^{D20}
L_{C21}	R^{D21}	R^{D21}	L_{C213}	R^{D1}	R^{D59}	L_{C405}	R^{D17}	R^{D118}	L_{C597}	R^{D144}	R^{D22}
L_{C22}	R^{D22}	R^{D22}	L_{C214}	R^{D1}	R^{D78}	L_{C406}	R^{D17}	R^{D119}	L_{C598}	R^{D144}	R^{D37}
L_{C23}	R^{D23}	R^{D23}	L_{C215}	R^{D1}	R^{D79}	L_{C407}	R^{D17}	R^{D120}	L_{C599}	R^{D144}	R^{D40}
L_{C24}	R^{D24}	R^{D24}	L_{C216}	R^{D1}	R^{D81}	L_{C408}	R^{D17}	R^{D133}	L_{C600}	R^{D144}	R^{D41}
L_{C25}	R^{D25}	R^{D25}	L_{C217}	R^{D1}	R^{D87}	L_{C409}	R^{D17}	R^{D134}	L_{C601}	R^{D144}	R^{D42}
L_{C26}	R^{D26}	R^{D26}	L_{C218}	R^{D1}	R^{D88}	L_{C410}	R^{D17}	R^{D135}	L_{C602}	R^{D144}	R^{D43}
L_{C27}	R^{D27}	R^{D27}	L_{C219}	R^{D1}	R^{D89}	L_{C411}	R^{D17}	R^{D136}	L_{C603}	R^{D144}	R^{D48}
L_{C28}	R^{D28}	R^{D28}	L_{C220}	R^{D1}	R^{D93}	L_{C412}	R^{D17}	R^{D143}	L_{C604}	R^{D144}	R^{D49}
L_{C29}	R^{D29}	R^{D29}	L_{C221}	R^{D1}	R^{D116}	L_{C413}	R^{D17}	R^{D144}	L_{C605}	R^{D144}	R^{D54}
L_{C30}	R^{D30}	R^{D30}	L_{C222}	R^{D1}	R^{D117}	L_{C414}	R^{D17}	R^{D145}	L_{C606}	R^{D144}	R^{D58}
L_{C31}	R^{D31}	R^{D31}	L_{C223}	R^{D1}	R^{D118}	L_{C415}	R^{D17}	R^{D146}	L_{C607}	R^{D144}	R^{D59}
L_{C32}	R^{D32}	R^{D32}	L_{C224}	R^{D1}	R^{D119}	L_{C416}	R^{D17}	R^{D147}	L_{C608}	R^{D144}	R^{D78}
L_{C33}	R^{D33}	R^{D33}	L_{C225}	R^{D1}	R^{D120}	L_{C417}	R^{D17}	R^{D149}	L_{C609}	R^{D144}	R^{D79}
L_{C34}	R^{D34}	R^{D34}	L_{C226}	R^{D1}	R^{D133}	L_{C418}	R^{D17}	R^{D151}	L_{C610}	R^{D144}	R^{D81}
L_{C35}	R^{D35}	R^{D35}	L_{C227}	R^{D1}	R^{D134}	L_{C419}	R^{D17}	R^{D154}	L_{C611}	R^{D144}	R^{D87}
L_{C36}	R^{D36}	R^{D36}	L_{C228}	R^{D1}	R^{D135}	L_{C420}	R^{D17}	R^{D155}	L_{C612}	R^{D144}	R^{D88}
L_{C37}	R^{D37}	R^{D37}	L_{C229}	R^{D1}	R^{D136}	L_{C421}	R^{D17}	R^{D161}	L_{C613}	R^{D144}	R^{D89}
L_{C38}	R^{D38}	R^{D38}	L_{C230}	R^{D1}	R^{D143}	L_{C422}	R^{D17}	R^{D175}	L_{C614}	R^{D144}	R^{D93}
L_{C39}	R^{D39}	R^{D39}	L_{C231}	R^{D1}	R^{D144}	L_{C423}	R^{D50}	R^{D3}	L_{C615}	R^{D144}	R^{D116}
L_{C40}	R^{D40}	R^{D40}	L_{C232}	R^{D1}	R^{D145}	L_{C424}	R^{D50}	R^{D5}	L_{C616}	R^{D144}	R^{D117}
L_{C41}	R^{D41}	R^{D41}	L_{C233}	R^{D1}	R^{D146}	L_{C425}	R^{D50}	R^{D18}	L_{C617}	R^{D144}	R^{D118}
L_{C42}	R^{D42}	R^{D42}	L_{C234}	R^{D1}	R^{D147}	L_{C426}	R^{D50}	R^{D20}	L_{C618}	R^{D144}	R^{D119}
L_{C43}	R^{D43}	R^{D43}	L_{C235}	R^{D1}	R^{D149}	L_{C427}	R^{D50}	R^{D22}	L_{C619}	R^{D144}	R^{D120}
L_{C44}	R^{D44}	R^{D44}	L_{C236}	R^{D1}	R^{D151}	L_{C428}	R^{D50}	R^{D37}	L_{C620}	R^{D144}	R^{D133}
L_{C45}	R^{D45}	R^{D45}	L_{C237}	R^{D1}	R^{D154}	L_{C429}	R^{D50}	R^{D40}	L_{C621}	R^{D144}	R^{D134}
L_{C46}	R^{D46}	R^{D46}	L_{C238}	R^{D1}	R^{D155}	L_{C430}	R^{D50}	R^{D41}	L_{C622}	R^{D144}	R^{D135}
L_{C47}	R^{D47}	R^{D47}	L_{C239}	R^{D1}	R^{D161}	L_{C431}	R^{D50}	R^{D42}	L_{C623}	R^{D144}	R^{D136}
L_{C48}	R^{D48}	R^{D48}	L_{C240}	R^{D1}	R^{D175}	L_{C432}	R^{D50}	R^{D43}	L_{C624}	R^{D144}	R^{D145}
L_{C49}	R^{D49}	R^{D49}	L_{C241}	R^{D4}	R^{D3}	L_{C433}	R^{D50}	R^{D48}	L_{C625}	R^{D144}	R^{D146}
L_{C50}	R^{D50}	R^{D50}	L_{C242}	R^{D4}	R^{D5}	L_{C434}	R^{D50}	R^{D49}	L_{C626}	R^{D144}	R^{D147}
L_{C51}	R^{D51}	R^{D51}	L_{C243}	R^{D4}	R^{D9}	L_{C435}	R^{D50}	R^{D54}	L_{C627}	R^{D144}	R^{D149}
L_{C52}	R^{D52}	R^{D52}	L_{C244}	R^{D4}	R^{D10}	L_{C436}	R^{D50}	R^{D55}	L_{C628}	R^{D144}	R^{D151}
L_{C53}	R^{D53}	R^{D53}	L_{C245}	R^{D4}	R^{D17}	L_{C437}	R^{D50}	R^{D58}	L_{C629}	R^{D144}	R^{D154}
L_{C54}	R^{D54}	R^{D54}	L_{C246}	R^{D4}	R^{D18}	L_{C438}	R^{D50}	R^{D59}	L_{C630}	R^{D144}	R^{D155}
L_{C55}	R^{D55}	R^{D55}	L_{C247}	R^{D4}	R^{D20}	L_{C439}	R^{D50}	R^{D78}	L_{C631}	R^{D144}	R^{D161}
L_{C56}	R^{D56}	R^{D56}	L_{C248}	R^{D4}	R^{D22}	L_{C440}	R^{D50}	R^{D79}	L_{C632}	R^{D144}	R^{D175}
L_{C57}	R^{D57}	R^{D57}	L_{C249}	R^{D4}	R^{D37}	L_{C441}	R^{D50}	R^{D81}	L_{C633}	R^{D145}	R^{D3}
L_{C58}	R^{D58}	R^{D58}	L_{C250}	R^{D4}	R^{D40}	L_{C442}	R^{D50}	R^{D87}	L_{C634}	R^{D145}	R^{D5}
L_{C59}	R^{D59}	R^{D59}	L_{C251}	R^{D4}	R^{D41}	L_{C443}	R^{D50}	R^{D88}	L_{C635}	R^{D145}	R^{D17}
L_{C60}	R^{D60}	R^{D60}	L_{C252}	R^{D4}	R^{D42}	L_{C444}	R^{D50}	R^{D89}	L_{C636}	R^{D145}	R^{D18}
L_{C61}	R^{D61}	R^{D61}	L_{C253}	R^{D4}	R^{D43}	L_{C445}	R^{D50}	R^{D93}	L_{C637}	R^{D145}	R^{D20}
L_{C62}	R^{D62}	R^{D62}	L_{C254}	R^{D4}	R^{D48}	L_{C446}	R^{D50}	R^{D116}	L_{C638}	R^{D145}	R^{D22}
L_{C63}	R^{D63}	R^{D63}	L_{C255}	R^{D4}	R^{D49}	L_{C447}	R^{D50}	R^{D117}	L_{C639}	R^{D145}	R^{D37}
L_{C64}	R^{D64}	R^{D64}	L_{C256}	R^{D4}	R^{D50}	L_{C448}	R^{D50}	R^{D118}	L_{C640}	R^{D145}	R^{D40}
L_{C65}	R^{D65}	R^{D65}	L_{C257}	R^{D4}	R^{D54}	L_{C449}	R^{D50}	R^{D119}	L_{C641}	R^{D145}	R^{D41}
L_{C66}	R^{D66}	R^{D66}	L_{C258}	R^{D4}	R^{D55}	L_{C450}	R^{D50}	R^{D120}	L_{C642}	R^{D145}	R^{D42}
L_{C67}	R^{D67}	R^{D67}	L_{C259}	R^{D4}	R^{D58}	L_{C451}	R^{D50}	R^{D133}	L_{C643}	R^{D145}	R^{D43}
L_{C68}	R^{D68}	R^{D68}	L_{C260}	R^{D4}	R^{D59}	L_{C452}	R^{D50}	R^{D134}	L_{C644}	R^{D145}	R^{D48}
L_{C69}	R^{D69}	R^{D69}	L_{C261}	R^{D4}	R^{D78}	L_{C453}	R^{D50}	R^{D135}	L_{C645}	R^{D145}	R^{D49}
L_{C70}	R^{D70}	R^{D70}	L_{C262}	R^{D4}	R^{D79}	L_{C454}	R^{D50}	R^{D136}	L_{C646}	R^{D145}	R^{D54}
L_{C71}	R^{D71}	R^{D71}	L_{C263}	R^{D4}	R^{D81}	L_{C455}	R^{D50}	R^{D143}	L_{C647}	R^{D145}	R^{D58}
L_{C72}	R^{D72}	R^{D72}	L_{C264}	R^{D4}	R^{D87}	L_{C456}	R^{D50}	R^{D144}	L_{C648}	R^{D145}	R^{D59}
L_{C73}	R^{D73}	R^{D73}	L_{C265}	R^{D4}	R^{D88}	L_{C457}	R^{D50}	R^{D145}	L_{C649}	R^{D145}	R^{D78}

-continued

L_{Cj}	R^{201}	R^{202}	L_{Cj}	R^{201}	R^{202}	L_{Cj}	R^{201}	R^{202}	L_{Cj}	R^{201}	R^{202}
L_{C74}	R^{D74}	R^{D74}	L_{C266}	R^{D4}	R^{D89}	L_{C458}	R^{D50}	R^{D146}	L_{C650}	R^{D145}	R^{D79}
L_{C75}	R^{D75}	R^{D75}	L_{C267}	R^{D4}	R^{D93}	L_{C459}	R^{D50}	R^{D147}	L_{C651}	R^{D145}	R^{D81}
L_{C76}	R^{D76}	R^{D76}	L_{C268}	R^{D4}	R^{D116}	L_{C460}	R^{D50}	R^{D149}	L_{C652}	R^{D145}	R^{D87}
L_{C77}	R^{D77}	R^{D77}	L_{C269}	R^{D4}	R^{D117}	L_{C461}	R^{D50}	R^{D151}	L_{C653}	R^{D145}	R^{D88}
L_{C78}	R^{D78}	R^{D78}	L_{C270}	R^{D4}	R^{D118}	L_{C462}	R^{D50}	R^{D154}	L_{C654}	R^{D145}	R^{D89}
L_{C79}	R^{D79}	R^{D79}	L_{C271}	R^{D4}	R^{D119}	L_{C463}	R^{D50}	R^{D155}	L_{C655}	R^{D145}	R^{D93}
L_{C80}	R^{D80}	R^{D80}	L_{C272}	R^{D4}	R^{D120}	L_{C464}	R^{D50}	R^{D161}	L_{C656}	R^{D145}	R^{D116}
L_{C81}	R^{D81}	R^{D81}	L_{C273}	R^{D4}	R^{D133}	L_{C465}	R^{D50}	R^{D175}	L_{C657}	R^{D145}	R^{D117}
L_{C82}	R^{D82}	R^{D82}	L_{C274}	R^{D4}	R^{D134}	L_{C466}	R^{D55}	R^{D3}	L_{C658}	R^{D145}	R^{D118}
L_{C83}	R^{D83}	R^{D83}	L_{C275}	R^{D4}	R^{D135}	L_{C467}	R^{D55}	R^{D5}	L_{C659}	R^{D145}	R^{D119}
L_{C84}	R^{D84}	R^{D84}	L_{C276}	R^{D4}	R^{D136}	L_{C468}	R^{D55}	R^{D18}	L_{C660}	R^{D145}	R^{D120}
L_{C85}	R^{D85}	R^{D85}	L_{C277}	R^{D4}	R^{D143}	L_{C469}	R^{D55}	R^{D20}	L_{C661}	R^{D145}	R^{D133}
L_{C86}	R^{D86}	R^{D86}	L_{C278}	R^{D4}	R^{D144}	L_{C470}	R^{D55}	R^{D22}	L_{C662}	R^{D145}	R^{D134}
L_{C87}	R^{D87}	R^{D87}	L_{C279}	R^{D4}	R^{D145}	L_{C471}	R^{D55}	R^{D37}	L_{C663}	R^{D145}	R^{D135}
L_{C88}	R^{D88}	R^{D88}	L_{C280}	R^{D4}	R^{D146}	L_{C472}	R^{D55}	R^{D40}	L_{C664}	R^{D145}	R^{D136}
L_{C89}	R^{D89}	R^{D89}	L_{C281}	R^{D4}	R^{D147}	L_{C473}	R^{D55}	R^{D41}	L_{C665}	R^{D145}	R^{D146}
L_{C90}	R^{D90}	R^{D90}	L_{C282}	R^{D4}	R^{D149}	L_{C474}	R^{D55}	R^{D42}	L_{C666}	R^{D145}	R^{D147}
L_{C91}	R^{D91}	R^{D91}	L_{C283}	R^{D4}	R^{D151}	L_{C475}	R^{D55}	R^{D43}	L_{C667}	R^{D145}	R^{D149}
L_{C92}	R^{D92}	R^{D92}	L_{C284}	R^{D4}	R^{D154}	L_{C476}	R^{D55}	R^{D48}	L_{C668}	R^{D145}	R^{D151}
L_{C93}	R^{D93}	R^{D93}	L_{C285}	R^{D4}	R^{D155}	L_{C477}	R^{D55}	R^{D49}	L_{C669}	R^{D145}	R^{D154}
L_{C94}	R^{D94}	R^{D94}	L_{C286}	R^{D4}	R^{D161}	L_{C478}	R^{D55}	R^{D54}	L_{C670}	R^{D145}	R^{D155}
L_{C95}	R^{D95}	R^{D95}	L_{C287}	R^{D4}	R^{D175}	L_{C479}	R^{D55}	R^{D58}	L_{C671}	R^{D145}	R^{D161}
L_{C96}	R^{D96}	R^{D96}	L_{C288}	R^{D9}	R^{D3}	L_{C480}	R^{D55}	R^{D59}	L_{C672}	R^{D145}	R^{D175}
L_{C97}	R^{D97}	R^{D97}	L_{C289}	R^{D9}	R^{D5}	L_{C481}	R^{D55}	R^{D78}	L_{C673}	R^{D146}	R^{D3}
L_{C98}	R^{D98}	R^{D98}	L_{C290}	R^{D9}	R^{D10}	L_{C482}	R^{D55}	R^{D79}	L_{C674}	R^{D146}	R^{D5}
L_{C99}	R^{D99}	R^{D99}	L_{C291}	R^{D9}	R^{D17}	L_{C483}	R^{D55}	R^{D81}	L_{C675}	R^{D146}	R^{D17}
L_{C100}	R^{D100}	R^{D100}	L_{C292}	R^{D9}	R^{D18}	L_{C484}	R^{D55}	R^{D87}	L_{C676}	R^{D146}	R^{D18}
L_{C101}	R^{D101}	R^{D101}	L_{C293}	R^{D9}	R^{D20}	L_{C485}	R^{D55}	R^{D88}	L_{C677}	R^{D146}	R^{D20}
L_{C102}	R^{D102}	R^{D102}	L_{C294}	R^{D9}	R^{D22}	L_{C486}	R^{D55}	R^{D89}	L_{C678}	R^{D146}	R^{D22}
L_{C103}	R^{D103}	R^{D103}	L_{C295}	R^{D9}	R^{D37}	L_{C487}	R^{D55}	R^{D93}	L_{C679}	R^{D146}	R^{D37}
L_{C104}	R^{D104}	R^{D104}	L_{C296}	R^{D9}	R^{D40}	L_{C488}	R^{D55}	R^{D116}	L_{C680}	R^{D146}	R^{D40}
L_{C105}	R^{D105}	R^{D105}	L_{C297}	R^{D9}	R^{D41}	L_{C489}	R^{D55}	R^{D117}	L_{C681}	R^{D146}	R^{D41}
L_{C106}	R^{D106}	R^{D106}	L_{C298}	R^{D9}	R^{D42}	L_{C490}	R^{D55}	R^{D118}	L_{C682}	R^{D146}	R^{D42}
L_{C107}	R^{D107}	R^{D107}	L_{C299}	R^{D9}	R^{D43}	L_{C491}	R^{D55}	R^{D119}	L_{C683}	R^{D146}	R^{D43}
L_{C108}	R^{D108}	R^{D108}	L_{C300}	R^{D9}	R^{D48}	L_{C492}	R^{D55}	R^{D120}	L_{C684}	R^{D146}	R^{D48}
L_{C109}	R^{D109}	R^{D109}	L_{C301}	R^{D9}	R^{D49}	L_{C493}	R^{D55}	R^{D133}	L_{C685}	R^{D146}	R^{D49}
L_{C110}	R^{D110}	R^{D110}	L_{C302}	R^{D9}	R^{D50}	L_{C494}	R^{D55}	R^{D134}	L_{C686}	R^{D146}	R^{D54}
L_{C111}	R^{D111}	R^{D111}	L_{C303}	R^{D9}	R^{D54}	L_{C495}	R^{D55}	R^{D135}	L_{C687}	R^{D146}	R^{D58}
L_{C112}	R^{D112}	R^{D112}	L_{C304}	R^{D9}	R^{D55}	L_{C496}	R^{D55}	R^{D136}	L_{C688}	R^{D146}	R^{D59}
L_{C113}	R^{D113}	R^{D113}	L_{C305}	R^{D9}	R^{D58}	L_{C497}	R^{D55}	R^{D143}	L_{C689}	R^{D146}	R^{D78}
L_{C114}	R^{D114}	R^{D114}	L_{C306}	R^{D9}	R^{D59}	L_{C498}	R^{D55}	R^{D144}	L_{C690}	R^{D146}	R^{D79}
L_{C115}	R^{D115}	R^{D115}	L_{C307}	R^{D9}	R^{D78}	L_{C499}	R^{D55}	R^{D145}	L_{C691}	R^{D146}	R^{D81}
L_{C116}	R^{D116}	R^{D116}	L_{C308}	R^{D9}	R^{D79}	L_{C500}	R^{D55}	R^{D146}	L_{C692}	R^{D146}	R^{D87}
L_{C117}	R^{D117}	R^{D117}	L_{C309}	R^{D9}	R^{D81}	L_{C501}	R^{D55}	R^{D147}	L_{C693}	R^{D146}	R^{D88}
L_{C118}	R^{D118}	R^{D118}	L_{C310}	R^{D9}	R^{D87}	L_{C502}	R^{D55}	R^{D149}	L_{C694}	R^{D146}	R^{D89}
L_{C119}	R^{D119}	R^{D119}	L_{C311}	R^{D9}	R^{D88}	L_{C503}	R^{D55}	R^{D151}	L_{C695}	R^{D146}	R^{D93}
L_{C120}	R^{D120}	R^{D120}	L_{C312}	R^{D9}	R^{D89}	L_{C504}	R^{D55}	R^{D154}	L_{C696}	R^{D146}	R^{D117}
L_{C121}	R^{D121}	R^{D121}	L_{C313}	R^{D9}	R^{D93}	L_{C505}	R^{D55}	R^{D155}	L_{C697}	R^{D146}	R^{D118}
L_{C122}	R^{D122}	R^{D122}	L_{C314}	R^{D9}	R^{D116}	L_{C506}	R^{D55}	R^{D161}	L_{C698}	R^{D146}	R^{D119}
L_{C123}	R^{D123}	R^{D123}	L_{C315}	R^{D9}	R^{D117}	L_{C507}	R^{D55}	R^{D175}	L_{C699}	R^{D146}	R^{D120}
L_{C124}	R^{D124}	R^{D124}	L_{C316}	R^{D9}	R^{D118}	L_{C508}	R^{D116}	R^{D3}	L_{C700}	R^{D146}	R^{D133}
L_{C125}	R^{D125}	R^{D125}	L_{C317}	R^{D9}	R^{D119}	L_{C509}	R^{D116}	R^{D5}	L_{C701}	R^{D146}	R^{D134}
L_{C126}	R^{D126}	R^{D126}	L_{C318}	R^{D9}	R^{D120}	L_{C510}	R^{D116}	R^{D17}	L_{C702}	R^{D146}	R^{D135}
L_{C127}	R^{D127}	R^{D127}	L_{C319}	R^{D9}	R^{D133}	L_{C511}	R^{D116}	R^{D18}	L_{C703}	R^{D146}	R^{D136}
L_{C128}	R^{D128}	R^{D128}	L_{C320}	R^{D9}	R^{D134}	L_{C512}	R^{D116}	R^{D20}	L_{C704}	R^{D146}	R^{D146}
L_{C129}	R^{D129}	R^{D129}	L_{C321}	R^{D9}	R^{D135}	L_{C513}	R^{D116}	R^{D22}	L_{C705}	R^{D146}	R^{D147}
L_{C130}	R^{D130}	R^{D130}	L_{C322}	R^{D9}	R^{D136}	L_{C514}	R^{D116}	R^{D37}	L_{C706}	R^{D146}	R^{D149}
L_{C131}	R^{D131}	R^{D131}	L_{C323}	R^{D9}	R^{D143}	L_{C515}	R^{D116}	R^{D40}	L_{C707}	R^{D146}	R^{D151}
L_{C132}	R^{D132}	R^{D132}	L_{C324}	R^{D9}	R^{D144}	L_{C516}	R^{D116}	R^{D41}	L_{C708}	R^{D146}	R^{D154}
L_{C133}	R^{D133}	R^{D133}	L_{C325}	R^{D9}	R^{D145}	L_{C517}	R^{D116}	R^{D42}	L_{C709}	R^{D146}	R^{D155}
L_{C134}	R^{D134}	R^{D134}	L_{C326}	R^{D9}	R^{D146}	L_{C518}	R^{D116}	R^{D43}	L_{C710}	R^{D146}	R^{D161}
L_{C135}	R^{D135}	R^{D135}	L_{C327}	R^{D9}	R^{D147}	L_{C519}	R^{D116}	R^{D48}	L_{C711}	R^{D146}	R^{D175}
L_{C136}	R^{D136}	R^{D136}	L_{C328}	R^{D9}	R^{D149}	L_{C520}	R^{D116}	R^{D49}	L_{C712}	R^{D133}	R^{D3}
L_{C137}	R^{D137}	R^{D137}	L_{C329}	R^{D9}	R^{D151}	L_{C521}	R^{D116}	R^{D54}	L_{C713}	R^{D133}	R^{D5}
L_{C138}	R^{D138}	R^{D138}	L_{C330}	R^{D9}	R^{D154}	L_{C522}	R^{D116}	R^{D58}	L_{C714}	R^{D133}	R^{D3}
L_{C139}	R^{D139}	R^{D139}	L_{C331}	R^{D9}	R^{D155}	L_{C523}	R^{D116}	R^{D59}	L_{C715}	R^{D133}	R^{D18}
L_{C140}	R^{D140}	R^{D140}	L_{C332}	R^{D9}	R^{D161}	L_{C524}	R^{D116}	R^{D78}	L_{C716}	R^{D133}	R^{D20}
L_{C141}	R^{D141}	R^{D141}	L_{C333}	R^{D9}	R^{D175}	L_{C525}	R^{D116}	R^{D79}	L_{C717}	R^{D133}	R^{D22}
L_{C142}	R^{D142}	R^{D142}	L_{C334}	R^{D10}	R^{D3}	L_{C526}	R^{D116}	R^{D81}	L_{C718}	R^{D133}	R^{D37}
L_{C143}	R^{D143}	R^{D143}	L_{C335}	R^{D10}	R^{D5}	L_{C527}	R^{D116}	R^{D87}	L_{C719}	R^{D133}	R^{D40}
L_{C144}	R^{D144}	R^{D144}	L_{C336}	R^{D10}	R^{D17}	L_{C528}	R^{D116}	R^{D88}	L_{C720}	R^{D133}	R^{D41}
L_{C145}	R^{D145}	R^{D145}	L_{C337}	R^{D10}	R^{D18}	L_{C529}	R^{D116}	R^{D89}	L_{C721}	R^{D133}	R^{D42}
L_{C146}	R^{D146}	R^{D146}	L_{C338}	R^{D10}	R^{D20}	L_{C530}	R^{D116}	R^{D93}	L_{C722}	R^{D133}	R^{D43}
L_{C147}	R^{D147}	R^{D147}	L_{C339}	R^{D10}	R^{D22}	L_{C531}	R^{D116}	R^{D117}	L_{C723}	R^{D133}	R^{D48}

-continued

L_{Cj}	R^{201}	R^{202}	L_{Cj}	R^{201}	R^{202}	L_{Cj}	R^{201}	R^{202}	L_{Cj}	R^{201}	R^{202}
L_{C148}	R^{D148}	R^{D148}	L_{C340}	R^{D10}	R^{D37}	L_{C532}	R^{D116}	R^{D118}	L_{C724}	R^{D133}	R^{D49}
L_{C149}	R^{D149}	R^{D149}	L_{C341}	R^{D10}	R^{D40}	L_{C533}	R^{D116}	R^{D119}	L_{C725}	R^{D133}	R^{D54}
L_{C150}	R^{D150}	R^{D150}	L_{C342}	R^{D10}	R^{D41}	L_{C534}	R^{D116}	R^{D120}	L_{C726}	R^{D133}	R^{D58}
L_{C151}	R^{D151}	R^{D151}	L_{C343}	R^{D10}	R^{D42}	L_{C535}	R^{D116}	R^{D133}	L_{C727}	R^{D133}	R^{D59}
L_{C152}	R^{D152}	R^{D152}	L_{C344}	R^{D10}	R^{D43}	L_{C536}	R^{D116}	R^{D134}	L_{C728}	R^{D133}	R^{D78}
L_{C153}	R^{D153}	R^{D153}	L_{C345}	R^{D10}	R^{D48}	L_{C537}	R^{D116}	R^{D135}	L_{C729}	R^{D133}	R^{D79}
L_{C154}	R^{D154}	R^{D154}	L_{C346}	R^{D10}	R^{D49}	L_{C538}	R^{D116}	R^{D136}	L_{C730}	R^{D133}	R^{D81}
L_{C155}	R^{D155}	R^{D155}	L_{C347}	R^{D10}	R^{D50}	L_{C539}	R^{D116}	R^{D143}	L_{C731}	R^{D133}	R^{D87}
L_{C156}	R^{D156}	R^{D156}	L_{C348}	R^{D10}	R^{D54}	L_{C540}	R^{D116}	R^{D144}	L_{C732}	R^{D133}	R^{D88}
L_{C157}	R^{D157}	R^{D157}	L_{C349}	R^{D10}	R^{D55}	L_{C541}	R^{D116}	R^{D145}	L_{C733}	R^{D133}	R^{D89}
L_{C158}	R^{D158}	R^{D158}	L_{C350}	R^{D10}	R^{D58}	L_{C542}	R^{D116}	R^{D146}	L_{C734}	R^{D133}	R^{D93}
L_{C159}	R^{D159}	R^{D159}	L_{C351}	R^{D10}	R^{D59}	L_{C543}	R^{D116}	R^{D147}	L_{C735}	R^{D133}	R^{D117}
L_{C160}	R^{D160}	R^{D160}	L_{C352}	R^{D10}	R^{D78}	L_{C544}	R^{D116}	R^{D149}	L_{C736}	R^{D133}	R^{D118}
L_{C161}	R^{D161}	R^{D161}	L_{C353}	R^{D10}	R^{D79}	L_{C545}	R^{D116}	R^{D151}	L_{C737}	R^{D133}	R^{D119}
L_{C162}	R^{D162}	R^{D162}	L_{C354}	R^{D10}	R^{D81}	L_{C546}	R^{D116}	R^{D154}	L_{C738}	R^{D133}	R^{D120}
L_{C163}	R^{D163}	R^{D163}	L_{C355}	R^{D10}	R^{D87}	L_{C547}	R^{D116}	R^{D155}	L_{C739}	R^{D133}	R^{D133}
L_{C164}	R^{D164}	R^{D164}	L_{C356}	R^{D10}	R^{D88}	L_{C548}	R^{D116}	R^{D161}	L_{C740}	R^{D133}	R^{D134}
L_{C165}	R^{D165}	R^{D165}	L_{C357}	R^{D10}	R^{D89}	L_{C549}	R^{D116}	R^{D175}	L_{C741}	R^{D133}	R^{D135}
L_{C166}	R^{D166}	R^{D166}	L_{C358}	R^{D10}	R^{D93}	L_{C550}	R^{D143}	R^{D3}	L_{C742}	R^{D133}	R^{D136}
L_{C167}	R^{D167}	R^{D167}	L_{C359}	R^{D10}	R^{D116}	L_{C551}	R^{D143}	R^{D5}	L_{C743}	R^{D133}	R^{D146}
L_{C168}	R^{D168}	R^{D168}	L_{C360}	R^{D10}	R^{D117}	L_{C552}	R^{D143}	R^{D17}	L_{C744}	R^{D133}	R^{D147}
L_{C169}	R^{D169}	R^{D169}	L_{C361}	R^{D10}	R^{D118}	L_{C553}	R^{D143}	R^{D18}	L_{C745}	R^{D133}	R^{D149}
L_{C170}	R^{D170}	R^{D170}	L_{C362}	R^{D10}	R^{D119}	L_{C554}	R^{D143}	R^{D20}	L_{C746}	R^{D133}	R^{D151}
L_{C171}	R^{D171}	R^{D171}	L_{C363}	R^{D10}	R^{D120}	L_{C555}	R^{D143}	R^{D22}	L_{C747}	R^{D133}	R^{D154}
L_{C172}	R^{D172}	R^{D172}	L_{C364}	R^{D10}	R^{D133}	L_{C556}	R^{D143}	R^{D37}	L_{C748}	R^{D133}	R^{D155}
L_{C173}	R^{D173}	R^{D173}	L_{C365}	R^{D10}	R^{D134}	L_{C557}	R^{D143}	R^{D40}	L_{C749}	R^{D133}	R^{D161}
L_{C174}	R^{D174}	R^{D174}	L_{C366}	R^{D10}	R^{D135}	L_{C558}	R^{D143}	R^{D41}	L_{C750}	R^{D133}	R^{D175}
L_{C175}	R^{D175}	R^{D175}	L_{C367}	R^{D10}	R^{D136}	L_{C559}	R^{D143}	R^{D42}	L_{C751}	R^{D175}	R^{D3}
L_{C176}	R^{D176}	R^{D176}	L_{C368}	R^{D10}	R^{D143}	L_{C560}	R^{D143}	R^{D43}	L_{C752}	R^{D175}	R^{D5}
L_{C177}	R^{D177}	R^{D177}	L_{C369}	R^{D10}	R^{D144}	L_{C561}	R^{D143}	R^{D48}	L_{C753}	R^{D175}	R^{D18}
L_{C178}	R^{D178}	R^{D178}	L_{C370}	R^{D10}	R^{D145}	L_{C562}	R^{D143}	R^{D49}	L_{C754}	R^{D175}	R^{D20}
L_{C179}	R^{D179}	R^{D179}	L_{C371}	R^{D10}	R^{D146}	L_{C563}	R^{D143}	R^{D54}	L_{C755}	R^{D175}	R^{D22}
L_{C180}	R^{D180}	R^{D180}	L_{C372}	R^{D10}	R^{D147}	L_{C564}	R^{D143}	R^{D58}	L_{C756}	R^{D175}	R^{D37}
L_{C181}	R^{D181}	R^{D181}	L_{C373}	R^{D10}	R^{D149}	L_{C565}	R^{D143}	R^{D59}	L_{C757}	R^{D175}	R^{D40}
L_{C182}	R^{D182}	R^{D182}	L_{C374}	R^{D10}	R^{D151}	L_{C566}	R^{D143}	R^{D78}	L_{C758}	R^{D175}	R^{D41}
L_{C183}	R^{D183}	R^{D183}	L_{C375}	R^{D10}	R^{D154}	L_{C567}	R^{D143}	R^{D79}	L_{C759}	R^{D175}	R^{D42}
L_{C184}	R^{D184}	R^{D184}	L_{C376}	R^{D10}	R^{D155}	L_{C568}	R^{D143}	R^{D81}	L_{C760}	R^{D175}	R^{D43}
L_{C185}	R^{D185}	R^{D185}	L_{C377}	R^{D10}	R^{D161}	L_{C569}	R^{D143}	R^{D87}	L_{C761}	R^{D175}	R^{D48}
L_{C186}	R^{D186}	R^{D186}	L_{C378}	R^{D10}	R^{D175}	L_{C570}	R^{D143}	R^{D88}	L_{C762}	R^{D175}	R^{D49}
L_{C187}	R^{D187}	R^{D187}	L_{C379}	R^{D17}	R^{D3}	L_{C571}	R^{D143}	R^{D89}	L_{C763}	R^{D175}	R^{D54}
L_{C188}	R^{D188}	R^{D188}	L_{C380}	R^{D17}	R^{D5}	L_{C572}	R^{D143}	R^{D93}	L_{C764}	R^{D175}	R^{D58}
L_{C189}	R^{D189}	R^{D189}	L_{C381}	R^{D17}	R^{D18}	L_{C573}	R^{D143}	R^{D116}	L_{C765}	R^{D175}	R^{D59}
L_{C190}	R^{D190}	R^{D190}	L_{C382}	R^{D17}	R^{D20}	L_{C574}	R^{D143}	R^{D117}	L_{C766}	R^{D175}	R^{D78}
L_{C191}	R^{D191}	R^{D191}	L_{C383}	R^{D17}	R^{D22}	L_{C575}	R^{D143}	R^{D118}	L_{C767}	R^{D175}	R^{D79}
L_{C192}	R^{D192}	R^{D192}	L_{C384}	R^{D17}	R^{D37}	L_{C576}	R^{D143}	R^{D119}	L_{C768}	R^{D175}	R^{D81}
L_{C769}	R^{D193}	R^{D193}	L_{C877}	R^{D1}	R^{D193}	L_{C985}	R^{D4}	R^{D193}	L_{C1093}	R^{D9}	R^{D193}
L_{C770}	R^{D194}	R^{D194}	L_{C878}	R^{D1}	R^{D194}	L_{C986}	R^{D4}	R^{D194}	L_{C1094}	R^{D9}	R^{D194}
L_{C771}	R^{D195}	R^{D195}	L_{C879}	R^{D1}	R^{D195}	L_{C987}	R^{D4}	R^{D195}	L_{C1095}	R^{D9}	R^{D195}
L_{C772}	R^{D196}	R^{D196}	L_{C880}	R^{D1}	R^{D196}	L_{C988}	R^{D4}	R^{D196}	L_{C1096}	R^{D9}	R^{D196}
L_{C773}	R^{D197}	R^{D197}	L_{C881}	R^{D1}	R^{D197}	L_{C989}	R^{D4}	R^{D197}	L_{C1097}	R^{D9}	R^{D197}
L_{C774}	R^{D198}	R^{D198}	L_{C882}	R^{D1}	R^{D198}	L_{C990}	R^{D4}	R^{D198}	L_{C1098}	R^{D9}	R^{D198}
L_{C775}	R^{D199}	R^{D199}	L_{C883}	R^{D1}	R^{D199}	L_{C991}	R^{D4}	R^{D199}	L_{C1099}	R^{D9}	R^{D199}
L_{C776}	R^{D200}	R^{D200}	L_{C884}	R^{D1}	R^{D200}	L_{C992}	R^{D4}	R^{D200}	L_{C1100}	R^{D9}	R^{D200}
L_{C777}	R^{D201}	R^{D201}	L_{C885}	R^{D1}	R^{D201}	L_{C993}	R^{D4}	R^{D201}	L_{C1101}	R^{D9}	R^{D201}
L_{C778}	R^{D202}	R^{D202}	L_{C886}	R^{D1}	R^{D202}	L_{C994}	R^{D4}	R^{D202}	L_{C1102}	R^{D9}	R^{D202}
L_{C779}	R^{D203}	R^{D203}	L_{C887}	R^{D1}	R^{D203}	L_{C995}	R^{D4}	R^{D203}	L_{C1103}	R^{D9}	R^{D203}
L_{C780}	R^{D204}	R^{D204}	L_{C888}	R^{D1}	R^{D204}	L_{C996}	R^{D4}	R^{D204}	L_{C1104}	R^{D9}	R^{D204}
L_{C781}	R^{D205}	R^{D205}	L_{C889}	R^{D1}	R^{D205}	L_{C997}	R^{D4}	R^{D205}	L_{C1105}	R^{D9}	R^{D205}
L_{C782}	R^{D206}	R^{D206}	L_{C890}	R^{D1}	R^{D206}	L_{C998}	R^{D4}	R^{D206}	L_{C1106}	R^{D9}	R^{D206}
L_{C783}	R^{D207}	R^{D207}	L_{C891}	R^{D1}	R^{D207}	L_{C999}	R^{D4}	R^{D207}	L_{C1107}	R^{D9}	R^{D207}
L_{C784}	R^{D208}	R^{D208}	L_{C892}	R^{D1}	R^{D208}	L_{C1000}	R^{D4}	R^{D208}	L_{C1108}	R^{D9}	R^{D208}
L_{C785}	R^{D209}	R^{D209}	L_{C893}	R^{D1}	R^{D209}	L_{C1001}	R^{D4}	R^{D209}	L_{C1109}	R^{D9}	R^{D209}
L_{C786}	R^{D210}	R^{D210}	L_{C894}	R^{D1}	R^{D210}	L_{C1002}	R^{D4}	R^{D210}	L_{C1110}	R^{D9}	R^{D210}
L_{C787}	R^{D211}	R^{D211}	L_{C895}	R^{D1}	R^{D211}	L_{C1003}	R^{D4}	R^{D211}	L_{C1111}	R^{D9}	R^{D211}
L_{C788}	R^{D212}	R^{D212}	L_{C896}	R^{D1}	R^{D212}	L_{C1004}	R^{D4}	R^{D212}	L_{C1112}	R^{D9}	R^{D212}
L_{C789}	R^{D213}	R^{D213}	L_{C897}	R^{D1}	R^{D213}	L_{C1005}	R^{D4}	R^{D213}	L_{C1113}	R^{D9}	R^{D213}
L_{C790}	R^{D214}	R^{D214}	L_{C898}	R^{D1}	R^{D214}	L_{C1006}	R^{D4}	R^{D214}	L_{C1114}	R^{D9}	R^{D214}
L_{C791}	R^{D215}	R^{D215}	L_{C899}	R^{D1}	R^{D215}	L_{C1007}	R^{D4}	R^{D215}	L_{C1115}	R^{D9}	R^{D215}
L_{C792}	R^{D216}	R^{D216}	L_{C900}	R^{D1}	R^{D216}	L_{C1008}	R^{D4}	R^{D216}	L_{C1116}	R^{D9}	R^{D216}
L_{C793}	R^{D217}	R^{D217}	L_{C901}	R^{D1}	R^{D217}	L_{C1009}	R^{D4}	R^{D217}	L_{C1117}	R^{D9}	R^{D217}
L_{C794}	R^{D218}	R^{D218}	L_{C902}	R^{D1}	R^{D218}	L_{C1010}	R^{D4}	R^{D218}	L_{C1118}	R^{D9}	R^{D218}
L_{C795}	R^{D219}	R^{D219}	L_{C903}	R^{D1}	R^{D219}	L_{C1011}	R^{D4}	R^{D219}	L_{C1119}	R^{D9}	R^{D219}
L_{C796}	R^{D220}	R^{D220}	L_{C904}	R^{D1}	R^{D220}	L_{C1012}	R^{D4}	R^{D220}	L_{C1120}	R^{D9}	R^{D220}
L_{C797}	R^{D221}	R^{D221}	L_{C905}	R^{D1}	R^{D221}	L_{C1013}	R^{D4}	R^{D221}	L_{C1121}	R^{D9}	R^{D221}

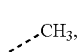
-continued

L_{Cj}	R^{201}	R^{202}	L_{Cj}	R^{201}	R^{202}	L_{Cj}	R^{201}	R^{202}	L_{Cj}	R^{201}	R^{202}
L_{C798}	R^{D222}	R^{D222}	L_{C906}	R^{D1}	R^{D222}	L_{C1014}	R^{D4}	R^{D222}	L_{C1122}	R^{D9}	R^{D222}
L_{C799}	R^{D223}	R^{D223}	L_{C907}	R^{D1}	R^{D223}	L_{C1015}	R^{D4}	R^{D223}	L_{C1123}	R^{D9}	R^{D223}
L_{C800}	R^{D224}	R^{D224}	L_{C908}	R^{D1}	R^{D224}	L_{C1016}	R^{D4}	R^{D224}	L_{C1124}	R^{D9}	R^{D224}
L_{C801}	R^{D225}	R^{D225}	L_{C909}	R^{D1}	R^{D225}	L_{C1017}	R^{D4}	R^{D225}	L_{C1125}	R^{D9}	R^{D225}
L_{C802}	R^{D226}	R^{D226}	L_{C910}	R^{D1}	R^{D226}	L_{C1018}	R^{D4}	R^{D226}	L_{C1126}	R^{D9}	R^{D226}
L_{C803}	R^{D227}	R^{D227}	L_{C911}	R^{D1}	R^{D227}	L_{C1019}	R^{D4}	R^{D227}	L_{C1127}	R^{D9}	R^{D227}
L_{C804}	R^{D228}	R^{D228}	L_{C912}	R^{D1}	R^{D228}	L_{C1020}	R^{D4}	R^{D228}	L_{C1128}	R^{D9}	R^{D228}
L_{C805}	R^{D229}	R^{D229}	L_{C913}	R^{D1}	R^{D229}	L_{C1021}	R^{D4}	R^{D229}	L_{C1129}	R^{D9}	R^{D229}
L_{C806}	R^{D230}	R^{D230}	L_{C914}	R^{D1}	R^{D230}	L_{C1022}	R^{D4}	R^{D230}	L_{C1130}	R^{D9}	R^{D230}
L_{C807}	R^{D231}	R^{D231}	L_{C915}	R^{D1}	R^{D231}	L_{C1023}	R^{D4}	R^{D231}	L_{C1131}	R^{D9}	R^{D231}
L_{C808}	R^{D232}	R^{D232}	L_{C916}	R^{D1}	R^{D232}	L_{C1024}	R^{D4}	R^{D232}	L_{C1132}	R^{D9}	R^{D232}
L_{C809}	R^{D233}	R^{D233}	L_{C917}	R^{D1}	R^{D233}	L_{C1025}	R^{D4}	R^{D233}	L_{C1133}	R^{D9}	R^{D233}
L_{C810}	R^{D234}	R^{D234}	L_{C918}	R^{D1}	R^{D234}	L_{C1026}	R^{D4}	R^{D234}	L_{C1134}	R^{D9}	R^{D234}
L_{C811}	R^{D235}	R^{D235}	L_{C919}	R^{D1}	R^{D235}	L_{C1027}	R^{D4}	R^{D235}	L_{C1135}	R^{D9}	R^{D235}
L_{C812}	R^{D236}	R^{D236}	L_{C920}	R^{D1}	R^{D236}	L_{C1028}	R^{D4}	R^{D236}	L_{C1136}	R^{D9}	R^{D236}
L_{C813}	R^{D237}	R^{D237}	L_{C921}	R^{D1}	R^{D237}	L_{C1029}	R^{D4}	R^{D237}	L_{C1137}	R^{D9}	R^{D237}
L_{C814}	R^{D238}	R^{D238}	L_{C922}	R^{D1}	R^{D238}	L_{C1030}	R^{D4}	R^{D238}	L_{C1138}	R^{D9}	R^{D238}
L_{C815}	R^{D239}	R^{D239}	L_{C923}	R^{D1}	R^{D239}	L_{C1031}	R^{D4}	R^{D239}	L_{C1139}	R^{D9}	R^{D239}
L_{C816}	R^{D240}	R^{D240}	L_{C924}	R^{D1}	R^{D240}	L_{C1032}	R^{D4}	R^{D240}	L_{C1140}	R^{D9}	R^{D240}
L_{C817}	R^{D241}	R^{D241}	L_{C925}	R^{D1}	R^{D241}	L_{C1033}	R^{D4}	R^{D241}	L_{C1141}	R^{D9}	R^{D241}
L_{C818}	R^{D242}	R^{D242}	L_{C926}	R^{D1}	R^{D242}	L_{C1034}	R^{D4}	R^{D242}	L_{C1142}	R^{D9}	R^{D242}
L_{C819}	R^{D243}	R^{D243}	L_{C927}	R^{D1}	R^{D243}	L_{C1035}	R^{D4}	R^{D243}	L_{C1143}	R^{D9}	R^{D243}
L_{C820}	R^{D244}	R^{D244}	L_{C928}	R^{D1}	R^{D244}	L_{C1036}	R^{D4}	R^{D244}	L_{C1144}	R^{D9}	R^{D244}
L_{C821}	R^{D245}	R^{D245}	L_{C929}	R^{D1}	R^{D245}	L_{C1037}	R^{D4}	R^{D245}	L_{C1145}	R^{D9}	R^{D245}
L_{C822}	R^{D246}	R^{D246}	L_{C930}	R^{D1}	R^{D246}	L_{C1038}	R^{D4}	R^{D246}	L_{C1146}	R^{D9}	R^{D246}
L_{C823}	R^{D17}	R^{D193}	L_{C931}	R^{D50}	R^{D193}	L_{C1039}	R^{D145}	R^{D193}	L_{C1147}	R^{D168}	R^{D193}
L_{C824}	R^{D17}	R^{D194}	L_{C932}	R^{D50}	R^{D194}	L_{C1040}	R^{D145}	R^{D194}	L_{C1148}	R^{D168}	R^{D194}
L_{C825}	R^{D17}	R^{D195}	L_{C933}	R^{D50}	R^{D195}	L_{C1041}	R^{D145}	R^{D195}	L_{C1149}	R^{D168}	R^{D195}
L_{C826}	R^{D17}	R^{D196}	L_{C934}	R^{D50}	R^{D196}	L_{C1042}	R^{D145}	R^{D196}	L_{C1150}	R^{D168}	R^{D196}
L_{C827}	R^{D17}	R^{D197}	L_{C935}	R^{D50}	R^{D197}	L_{C1043}	R^{D145}	R^{D197}	L_{C1151}	R^{D168}	R^{D197}
L_{C828}	R^{D17}	R^{D198}	L_{C936}	R^{D50}	R^{D198}	L_{C1044}	R^{D145}	R^{D198}	L_{C1152}	R^{D168}	R^{D198}
L_{C829}	R^{D17}	R^{D199}	L_{C937}	R^{D50}	R^{D199}	L_{C1045}	R^{D145}	R^{D199}	L_{C1153}	R^{D168}	R^{D199}
L_{C830}	R^{D17}	R^{D200}	L_{C938}	R^{D50}	R^{D200}	L_{C1046}	R^{D145}	R^{D200}	L_{C1154}	R^{D168}	R^{D200}
L_{C831}	R^{D17}	R^{D201}	L_{C939}	R^{D50}	R^{D201}	L_{C1047}	R^{D145}	R^{D201}	L_{C1155}	R^{D168}	R^{D201}
L_{C832}	R^{D17}	R^{D202}	L_{C940}	R^{D50}	R^{D202}	L_{C1048}	R^{D145}	R^{D202}	L_{C1156}	R^{D168}	R^{D202}
L_{C833}	R^{D17}	R^{D203}	L_{C941}	R^{D50}	R^{D203}	L_{C1049}	R^{D145}	R^{D203}	L_{C1157}	R^{D168}	R^{D203}
L_{C834}	R^{D17}	R^{D204}	L_{C942}	R^{D50}	R^{D204}	L_{C1050}	R^{D145}	R^{D204}	L_{C1158}	R^{D168}	R^{D204}
L_{C835}	R^{D17}	R^{D205}	L_{C943}	R^{D50}	R^{D205}	L_{C1051}	R^{D145}	R^{D205}	L_{C1159}	R^{D168}	R^{D205}
L_{C836}	R^{D17}	R^{D206}	L_{C944}	R^{D50}	R^{D206}	L_{C1052}	R^{D145}	R^{D206}	L_{C1160}	R^{D168}	R^{D206}
L_{C837}	R^{D17}	R^{D207}	L_{C945}	R^{D50}	R^{D207}	L_{C1053}	R^{D145}	R^{D207}	L_{C1161}	R^{D168}	R^{D207}
L_{C838}	R^{D17}	R^{D208}	L_{C946}	R^{D50}	R^{D208}	L_{C1054}	R^{D145}	R^{D208}	L_{C1162}	R^{D168}	R^{D208}
L_{C839}	R^{D17}	R^{D209}	L_{C947}	R^{D50}	R^{D209}	L_{C1055}	R^{D145}	R^{D209}	L_{C1163}	R^{D168}	R^{D209}
L_{C840}	R^{D17}	R^{D210}	L_{C948}	R^{D50}	R^{D210}	L_{C1056}	R^{D145}	R^{D210}	L_{C1164}	R^{D168}	R^{D210}
L_{C841}	R^{D17}	R^{D211}	L_{C949}	R^{D50}	R^{D211}	L_{C1057}	R^{D145}	R^{D211}	L_{C1165}	R^{D168}	R^{D211}
L_{C842}	R^{D17}	R^{D212}	L_{C950}	R^{D50}	R^{D212}	L_{C1058}	R^{D145}	R^{D212}	L_{C1166}	R^{D168}	R^{D212}
L_{C843}	R^{D17}	R^{D213}	L_{C951}	R^{D50}	R^{D213}	L_{C1059}	R^{D145}	R^{D213}	L_{C1167}	R^{D168}	R^{D213}
L_{C844}	R^{D17}	R^{D214}	L_{C952}	R^{D50}	R^{D214}	L_{C1060}	R^{D145}	R^{D214}	L_{C1168}	R^{D168}	R^{D214}
L_{C845}	R^{D17}	R^{D215}	L_{C953}	R^{D50}	R^{D215}	L_{C1061}	R^{D145}	R^{D215}	L_{C1169}	R^{D168}	R^{D215}
L_{C846}	R^{D17}	R^{D216}	L_{C954}	R^{D50}	R^{D216}	L_{C1062}	R^{D145}	R^{D216}	L_{C1170}	R^{D168}	R^{D216}
L_{C847}	R^{D17}	R^{D217}	L_{C955}	R^{D50}	R^{D217}	L_{C1063}	R^{D145}	R^{D217}	L_{C1171}	R^{D168}	R^{D217}
L_{C848}	R^{D17}	R^{D218}	L_{C956}	R^{D50}	R^{D218}	L_{C1064}	R^{D145}	R^{D218}	L_{C1172}	R^{D168}	R^{D218}
L_{C849}	R^{D17}	R^{D219}	L_{C957}	R^{D50}	R^{D219}	L_{C1065}	R^{D145}	R^{D219}	L_{C1173}	R^{D168}	R^{D219}
L_{C850}	R^{D17}	R^{D220}	L_{C958}	R^{D50}	R^{D220}	L_{C1066}	R^{D145}	R^{D220}	L_{C1174}	R^{D168}	R^{D220}
L_{C851}	R^{D17}	R^{D221}	L_{C959}	R^{D50}	R^{D221}	L_{C1067}	R^{D145}	R^{D221}	L_{C1175}	R^{D168}	R^{D221}
L_{C852}	R^{D17}	R^{D222}	L_{C960}	R^{D50}	R^{D222}	L_{C1068}	R^{D145}	R^{D222}	L_{C1176}	R^{D168}	R^{D222}
L_{C853}	R^{D17}	R^{D223}	L_{C961}	R^{D50}	R^{D223}	L_{C1069}	R^{D145}	R^{D223}	L_{C1177}	R^{D168}	R^{D223}
L_{C854}	R^{D17}	R^{D224}	L_{C962}	R^{D50}	R^{D224}	L_{C1070}	R^{D145}	R^{D224}	L_{C1178}	R^{D168}	R^{D224}
L_{C855}	R^{D17}	R^{D225}	L_{C963}	R^{D50}	R^{D225}	L_{C1071}	R^{D145}	R^{D225}	L_{C1179}	R^{D168}	R^{D225}
L_{C856}	R^{D17}	R^{D226}	L_{C964}	R^{D50}	R^{D226}	L_{C1072}	R^{D145}	R^{D226}	L_{C1180}	R^{D168}	R^{D226}
L_{C857}	R^{D17}	R^{D227}	L_{C965}	R^{D50}	R^{D227}	L_{C1073}	R^{D145}	R^{D227}	L_{C1181}	R^{D168}	R^{D227}
L_{C858}	R^{D17}	R^{D228}	L_{C966}	R^{D50}	R^{D228}	L_{C1074}	R^{D145}	R^{D228}	L_{C1182}	R^{D168}	R^{D228}
L_{C859}	R^{D17}	R^{D229}	L_{C967}	R^{D50}	R^{D229}	L_{C1075}	R^{D145}	R^{D229}	L_{C1183}	R^{D168}	R^{D229}
L_{C860}	R^{D17}	R^{D230}	L_{C968}	R^{D50}	R^{D230}	L_{C1076}	R^{D145}	R^{D230}	L_{C1184}	R^{D168}	R^{D230}
L_{C861}	R^{D17}	R^{D231}	L_{C969}	R^{D50}	R^{D231}	L_{C1077}	R^{D145}	R^{D231}	L_{C1185}	R^{D168}	R^{D231}
L_{C862}	R^{D17}	R^{D232}	L_{C970}	R^{D50}	R^{D232}	L_{C1078}	R^{D145}	R^{D232}	L_{C1186}	R^{D168}	R^{D232}
L_{C863}	R^{D17}	R^{D233}	L_{C971}	R^{D50}	R^{D233}	L_{C1079}	R^{D145}	R^{D233}	L_{C1187}	R^{D168}	R^{D233}
L_{C864}	R^{D17}	R^{D234}	L_{C972}	R^{D50}	R^{D234}	L_{C1080}	R^{D145}	R^{D234}	L_{C1188}	R^{D168}	R^{D234}
L_{C865}	R^{D17}	R^{D235}	L_{C973}	R^{D50}	R^{D235}	L_{C1081}	R^{D145}	R^{D235}	L_{C1189}	R^{D168}	R^{D235}
L_{C866}	R^{D17}	R^{D236}	L_{C974}	R^{D50}	R^{D236}	L_{C1082}	R^{D145}	R^{D236}	L_{C1190}	R^{D168}	R^{D236}
L_{C867}	R^{D17}	R^{D237}	L_{C975}	R^{D50}	R^{D237}	L_{C1083}	R^{D145}	R^{D237}	L_{C1191}	R^{D168}	R^{D237}
L_{C868}	R^{D17}	R^{D238}	L_{C976}	R^{D50}	R^{D238}	L_{C1084}	R^{D145}	R^{D238}	L_{C1192}	R^{D168}	R^{D238}
L_{C869}	R^{D17}	R^{D239}	L_{C977}	R^{D50}	R^{D239}	L_{C1085}	R^{D145}	R^{D239}	L_{C1193}	R^{D168}	R^{D239}
L_{C870}	R^{D17}	R^{D240}	L_{C978}	R^{D50}	R^{D240}	L_{C1086}	R^{D145}	R^{D240}	L_{C1194}	R^{D168}	R^{D240}
L_{C871}	R^{D17}	R^{D241}	L_{C979}	R^{D50}	R^{D241}	L_{C1087}	R^{D145}	R^{D241}	L_{C1195}	R^{D168}	R^{D241}

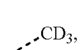
-continued

L_{Cj}	R^{201}	R^{202}	L_{Cj}	R^{201}	R^{202}	L_{Cj}	R^{201}	R^{202}	L_{Cj}	R^{201}	R^{202}
L_{C872}	R^{D17}	R^{D242}	L_{C980}	R^{D50}	R^{D242}	L_{C1088}	R^{D145}	R^{D242}	L_{C1196}	R^{D168}	R^{D242}
L_{C873}	R^{D17}	R^{D243}	L_{C981}	R^{D50}	R^{D243}	L_{C1089}	R^{D145}	R^{D243}	L_{C1197}	R^{D168}	R^{D243}
L_{C874}	R^{D17}	R^{D244}	L_{C982}	R^{D50}	R^{D244}	L_{C1090}	R^{D145}	R^{D244}	L_{C1198}	R^{D168}	R^{D244}
L_{C875}	R^{D17}	R^{D245}	L_{C983}	R^{D50}	R^{D245}	L_{C1091}	R^{D145}	R^{D245}	L_{C1199}	R^{D168}	R^{D245}
L_{C876}	R^{D17}	R^{D246}	L_{C984}	R^{D50}	R^{D246}	L_{C1092}	R^{D145}	R^{D246}	L_{C1200}	R^{D168}	R^{D246}
L_{C1201}	R^{D10}	R^{D193}	L_{C1255}	R^{D55}	R^{D193}	L_{C1309}	R^{D37}	R^{D193}	L_{C1363}	R^{D143}	R^{D193}
L_{C1202}	R^{D10}	R^{D194}	L_{C1256}	R^{D55}	R^{D194}	L_{C1310}	R^{D37}	R^{D194}	L_{C1364}	R^{D143}	R^{D194}
L_{C1203}	R^{D10}	R^{D195}	L_{C1257}	R^{D55}	R^{D195}	L_{C1311}	R^{D37}	R^{D195}	L_{C1365}	R^{D143}	R^{D195}
L_{C1204}	R^{D10}	R^{D196}	L_{C1258}	R^{D55}	R^{D196}	L_{C1312}	R^{D37}	R^{D196}	L_{C1366}	R^{D143}	R^{D196}
L_{C1205}	R^{D10}	R^{D197}	L_{C1259}	R^{D55}	R^{D197}	L_{C1313}	R^{D37}	R^{D197}	L_{C1367}	R^{D143}	R^{D197}
L_{C1206}	R^{D10}	R^{D198}	L_{C1260}	R^{D55}	R^{D198}	L_{C1314}	R^{D37}	R^{D198}	L_{C1368}	R^{D143}	R^{D198}
L_{C1207}	R^{D10}	R^{D199}	L_{C1261}	R^{D55}	R^{D199}	L_{C1315}	R^{D37}	R^{D199}	L_{C1369}	R^{D143}	R^{D199}
L_{C1208}	R^{D10}	R^{D200}	L_{C1262}	R^{D55}	R^{D200}	L_{C1316}	R^{D37}	R^{D200}	L_{C1370}	R^{D143}	R^{D200}
L_{C1209}	R^{D10}	R^{D201}	L_{C1263}	R^{D55}	R^{D201}	L_{C1317}	R^{D37}	R^{D201}	L_{C1371}	R^{D143}	R^{D201}
L_{C1210}	R^{D10}	R^{D202}	L_{C1264}	R^{D55}	R^{D202}	L_{C1318}	R^{D37}	R^{D202}	L_{C1372}	R^{D143}	R^{D202}
L_{C1211}	R^{D10}	R^{D203}	L_{C1265}	R^{D55}	R^{D203}	L_{C1319}	R^{D37}	R^{D203}	L_{C1373}	R^{D143}	R^{D203}
L_{C1212}	R^{D10}	R^{D204}	L_{C1266}	R^{D55}	R^{D204}	L_{C1320}	R^{D37}	R^{D204}	L_{C1374}	R^{D143}	R^{D204}
L_{C1213}	R^{D10}	R^{D205}	L_{C1267}	R^{D55}	R^{D205}	L_{C1321}	R^{D37}	R^{D205}	L_{C1375}	R^{D143}	R^{D205}
L_{C1214}	R^{D10}	R^{D206}	L_{C1268}	R^{D55}	R^{D206}	L_{C1322}	R^{D37}	R^{D206}	L_{C1376}	R^{D143}	R^{D206}
L_{C1215}	R^{D10}	R^{D207}	L_{C1269}	R^{D55}	R^{D207}	L_{C1323}	R^{D37}	R^{D207}	L_{C1377}	R^{D143}	R^{D207}
L_{C1216}	R^{D10}	R^{D208}	L_{C1270}	R^{D55}	R^{D208}	L_{C1324}	R^{D37}	R^{D208}	L_{C1378}	R^{D143}	R^{D208}
L_{C1217}	R^{D10}	R^{D209}	L_{C1271}	R^{D55}	R^{D209}	L_{C1325}	R^{D37}	R^{D209}	L_{C1379}	R^{D143}	R^{D209}
L_{C1218}	R^{D10}	R^{D210}	L_{C1272}	R^{D55}	R^{D210}	L_{C1326}	R^{D37}	R^{D210}	L_{C1380}	R^{D143}	R^{D210}
L_{C1219}	R^{D10}	R^{D211}	L_{C1273}	R^{D55}	R^{D211}	L_{C1327}	R^{D37}	R^{D211}	L_{C1381}	R^{D143}	R^{D211}
L_{C1220}	R^{D10}	R^{D212}	L_{C1274}	R^{D55}	R^{D212}	L_{C1328}	R^{D37}	R^{D212}	L_{C1382}	R^{D143}	R^{D212}
L_{C1221}	R^{D10}	R^{D213}	L_{C1275}	R^{D55}	R^{D213}	L_{C1329}	R^{D37}	R^{D213}	L_{C1383}	R^{D143}	R^{D213}
L_{C1222}	R^{D10}	R^{D214}	L_{C1276}	R^{D55}	R^{D214}	L_{C1330}	R^{D37}	R^{D214}	L_{C1384}	R^{D143}	R^{D214}
L_{C1223}	R^{D10}	R^{D215}	L_{C1277}	R^{D55}	R^{D215}	L_{C1331}	R^{D37}	R^{D215}	L_{C1385}	R^{D143}	R^{D215}
L_{C1224}	R^{D10}	R^{D216}	L_{C1278}	R^{D55}	R^{D216}	L_{C1332}	R^{D37}	R^{D216}	L_{C1386}	R^{D143}	R^{D216}
L_{C1225}	R^{D10}	R^{D217}	L_{C1279}	R^{D55}	R^{D217}	L_{C1333}	R^{D37}	R^{D217}	L_{C1387}	R^{D143}	R^{D217}
L_{C1226}	R^{D10}	R^{D218}	L_{C1280}	R^{D55}	R^{D218}	L_{C1334}	R^{D37}	R^{D218}	L_{C1388}	R^{D143}	R^{D218}
L_{C1227}	R^{D10}	R^{D219}	L_{C1281}	R^{D55}	R^{D219}	L_{C1335}	R^{D37}	R^{D219}	L_{C1389}	R^{D143}	R^{D219}
L_{C1228}	R^{D10}	R^{D220}	L_{C1282}	R^{D55}	R^{D220}	L_{C1336}	R^{D37}	R^{D220}	L_{C1390}	R^{D143}	R^{D220}
L_{C1229}	R^{D10}	R^{D221}	L_{C1283}	R^{D55}	R^{D221}	L_{C1337}	R^{D37}	R^{D221}	L_{C1391}	R^{D143}	R^{D221}
L_{C1230}	R^{D10}	R^{D222}	L_{C1284}	R^{D55}	R^{D222}	L_{C1338}	R^{D37}	R^{D222}	L_{C1392}	R^{D143}	R^{D222}
L_{C1231}	R^{D10}	R^{D223}	L_{C1285}	R^{D55}	R^{D223}	L_{C1339}	R^{D37}	R^{D223}	L_{C1393}	R^{D143}	R^{D223}
L_{C1232}	R^{D10}	R^{D224}	L_{C1286}	R^{D55}	R^{D224}	L_{C1340}	R^{D37}	R^{D224}	L_{C1394}	R^{D143}	R^{D224}
L_{C1233}	R^{D10}	R^{D225}	L_{C1287}	R^{D55}	R^{D225}	L_{C1341}	R^{D37}	R^{D225}	L_{C1395}	R^{D143}	R^{D225}
L_{C1234}	R^{D10}	R^{D226}	L_{C1288}	R^{D55}	R^{D226}	L_{C1342}	R^{D37}	R^{D226}	L_{C1396}	R^{D143}	R^{D226}
L_{C1235}	R^{D10}	R^{D227}	L_{C1289}	R^{D55}	R^{D227}	L_{C1343}	R^{D37}	R^{D227}	L_{C1397}	R^{D143}	R^{D227}
L_{C1236}	R^{D10}	R^{D228}	L_{C1290}	R^{D55}	R^{D228}	L_{C1344}	R^{D37}	R^{D228}	L_{C1398}	R^{D143}	R^{D228}
L_{C1237}	R^{D10}	R^{D229}	L_{C1291}	R^{D55}	R^{D229}	L_{C1345}	R^{D37}	R^{D229}	L_{C1399}	R^{D143}	R^{D229}
L_{C1238}	R^{D10}	R^{D230}	L_{C1292}	R^{D55}	R^{D230}	L_{C1346}	R^{D37}	R^{D230}	L_{C1400}	R^{D143}	R^{D230}
L_{C1239}	R^{D10}	R^{D231}	L_{C1293}	R^{D55}	R^{D231}	L_{C1347}	R^{D37}	R^{D231}	L_{C1401}	R^{D143}	R^{D231}
L_{C1240}	R^{D10}	R^{D232}	L_{C1294}	R^{D55}	R^{D232}	L_{C1348}	R^{D37}	R^{D232}	L_{C1402}	R^{D143}	R^{D232}
L_{C1241}	R^{D10}	R^{D233}	L_{C1295}	R^{D55}	R^{D233}	L_{C1349}	R^{D37}	R^{D233}	L_{C1403}	R^{D143}	R^{D233}
L_{C1242}	R^{D10}	R^{D234}	L_{C1296}	R^{D55}	R^{D234}	L_{C1350}	R^{D37}	R^{D234}	L_{C1404}	R^{D143}	R^{D234}
L_{C1243}	R^{D10}	R^{D235}	L_{C1297}	R^{D55}	R^{D235}	L_{C1351}	R^{D37}	R^{D235}	L_{C1405}	R^{D143}	R^{D235}
L_{C1244}	R^{D10}	R^{D236}	L_{C1298}	R^{D55}	R^{D236}	L_{C1352}	R^{D37}	R^{D236}	L_{C1406}	R^{D143}	R^{D236}
L_{C1245}	R^{D10}	R^{D237}	L_{C1299}	R^{D55}	R^{D237}	L_{C1353}	R^{D37}	R^{D237}	L_{C1407}	R^{D143}	R^{D237}
L_{C1246}	R^{D10}	R^{D238}	L_{C1300}	R^{D55}	R^{D238}	L_{C1354}	R^{D37}	R^{D238}	L_{C1408}	R^{D143}	R^{D238}
L_{C1247}	R^{D10}	R^{D239}	L_{C1301}	R^{D55}	R^{D239}	L_{C1355}	R^{D37}	R^{D239}	L_{C1409}	R^{D143}	R^{D239}
L_{C1248}	R^{D10}	R^{D240}	L_{C1302}	R^{D55}	R^{D240}	L_{C1356}	R^{D37}	R^{D240}	L_{C1410}	R^{D143}	R^{D240}
L_{C1249}	R^{D10}	R^{D241}	L_{C1303}	R^{D55}	R^{D241}	L_{C1357}	R^{D37}	R^{D241}	L_{C1411}	R^{D143}	R^{D241}
L_{C1250}	R^{D10}	R^{D242}	L_{C1304}	R^{D55}	R^{D242}	L_{C1358}	R^{D37}	R^{D242}	L_{C1412}	R^{D143}	R^{D242}
L_{C1251}	R^{D10}	R^{D243}	L_{C1305}	R^{D55}	R^{D243}	L_{C1359}	R^{D37}	R^{D243}	L_{C1413}	R^{D143}	R^{D243}
L_{C1252}	R^{D10}	R^{D244}	L_{C1306}	R^{D55}	R^{D244}	L_{C1360}	R^{D37}	R^{D244}	L_{C1414}	R^{D143}	R^{D244}
L_{C1253}	R^{D10}	R^{D245}	L_{C1307}	R^{D55}	R^{D245}	L_{C1361}	R^{D37}	R^{D245}	L_{C1415}	R^{D143}	R^{D245}
L_{C1254}	R^{D10}	R^{D246}	L_{C1308}	R^{D55}	R^{D246}	L_{C1362}	R^{D37}	R^{D246}	L_{C1416}	R^{D143}	R^{D246}

[0317] wherein R^{D1} to R^{D246} have the structures defined in the following LIST 12:



R^{D1}



R^{D2}

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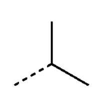
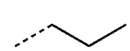
R^{D3}



R^{D4}



R^{D5}

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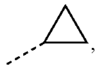
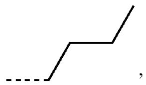
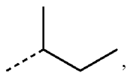
R^{D6}



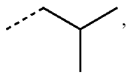
R^{D7}



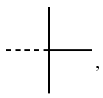
R^{D8}



R^{D9}



R^{D10}



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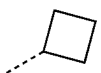

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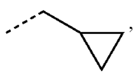

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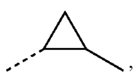

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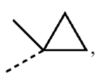

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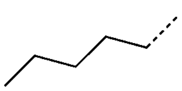

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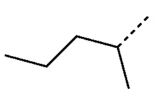

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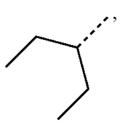

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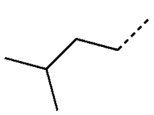

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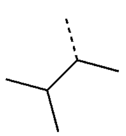

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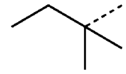
R^{D20}



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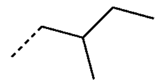

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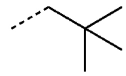

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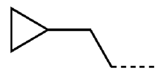

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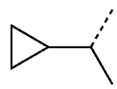

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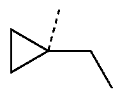

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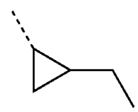

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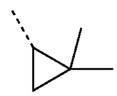

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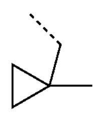

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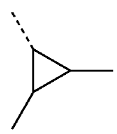

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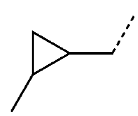
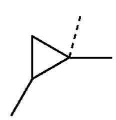
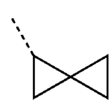


R^{D31}



R^{D32}



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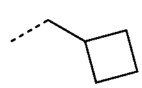

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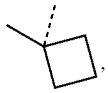
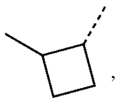
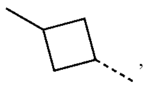
R^{D34}



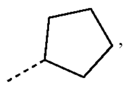
R^{D35}



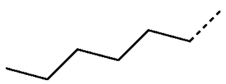
R^{D36}



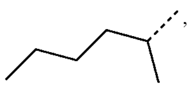
R^{D37}



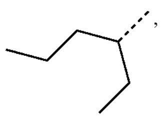
R^{D38}



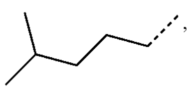
R^{D39}



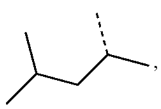
R^{D40}



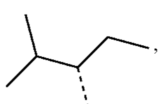
R^{D41}



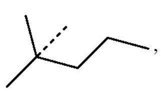
R^{D42}



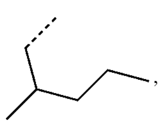
R^{D43}



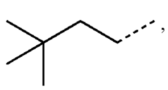
R^{D44}



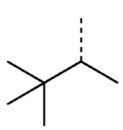
R^{D45}



R^{D46}



R^{D47}



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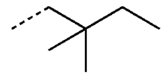
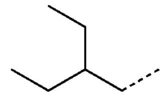
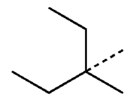
R^{D48}



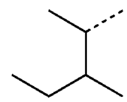
R^{D49}



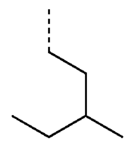
R^{D50}



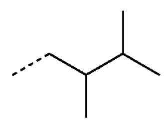
R^{D51}



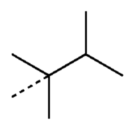
R^{D52}



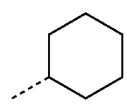
R^{D53}



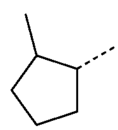
R^{D54}



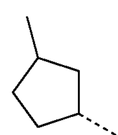
R^{D55}



R^{D56}

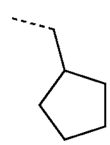


R^{D57}



R^{D58}

R^{D59}



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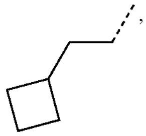
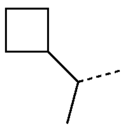
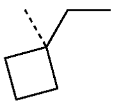
R^{D60}



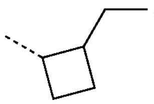
R^{D61}



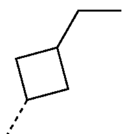
R^{D62}



R^{D63}

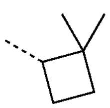


R^{D64}

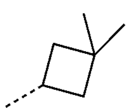


R^{D65}

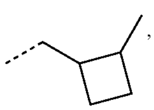
R^{D66}



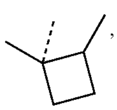
R^{D67}



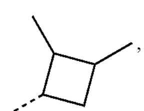
R^{D68}



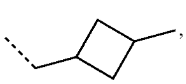
R^{D69}



R^{D70}



R^{D71}



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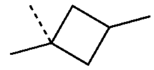
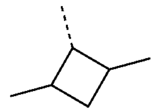
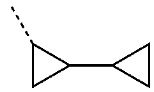
R^{D72}



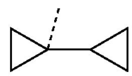
R^{D73}



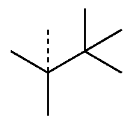
R^{D74}



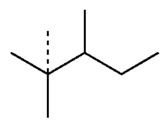
R^{D75}



R^{D76}



R^{D77}



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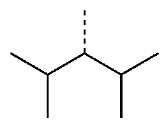

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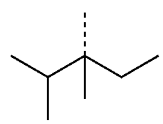

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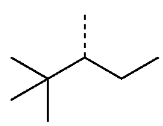

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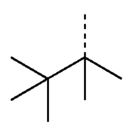

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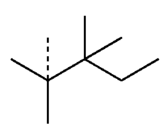

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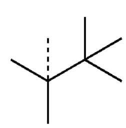

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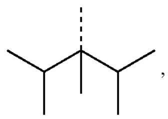

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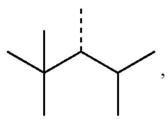

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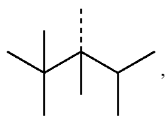

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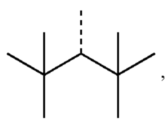

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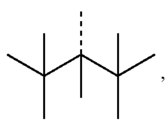

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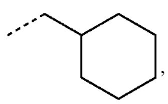

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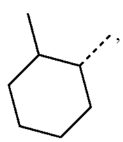

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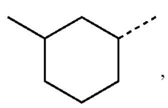

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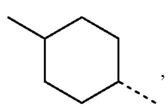

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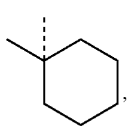

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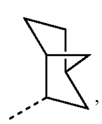

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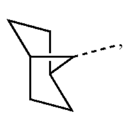

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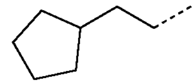

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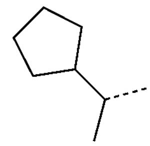

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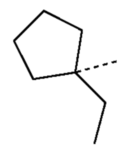

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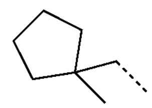

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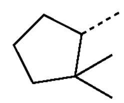

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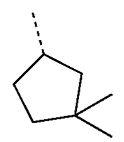

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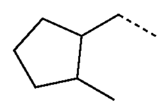

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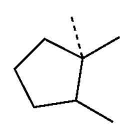

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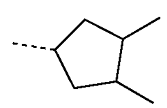

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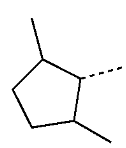

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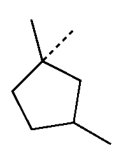

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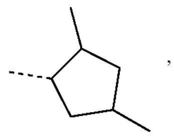

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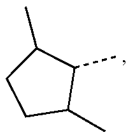

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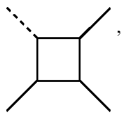

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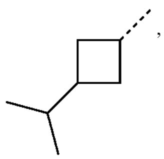

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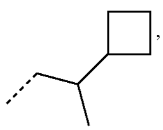

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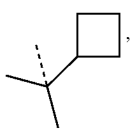

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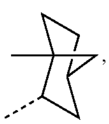

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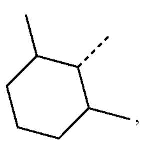

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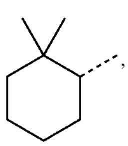

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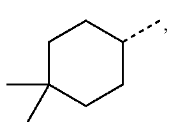

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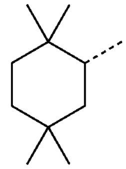

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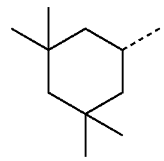

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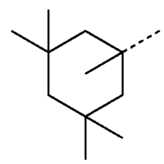

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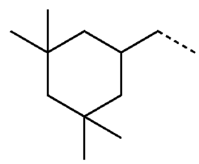

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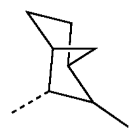

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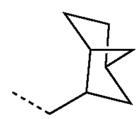

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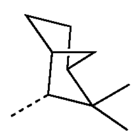

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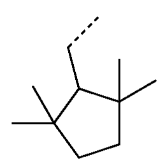

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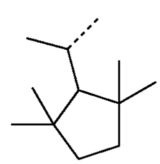

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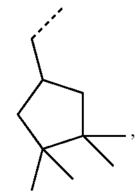

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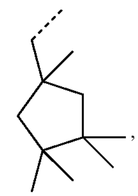

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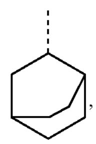

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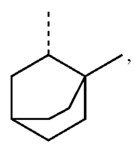

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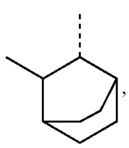

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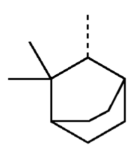

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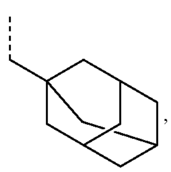

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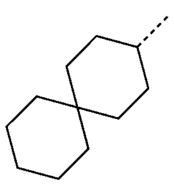

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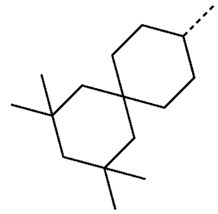

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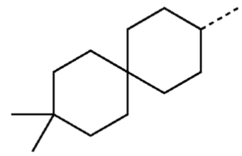

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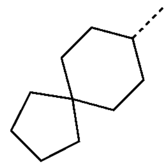

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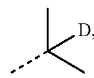

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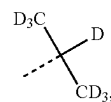

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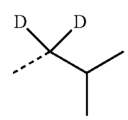

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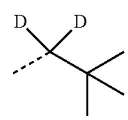

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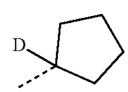

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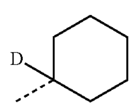

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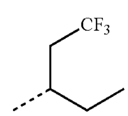

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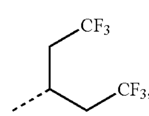

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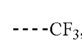

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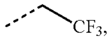
R^{D146}



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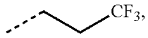

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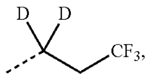

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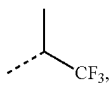

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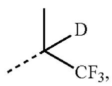

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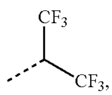

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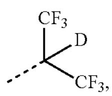

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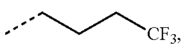

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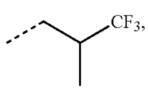

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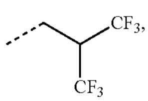

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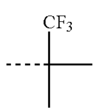

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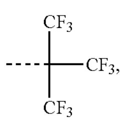

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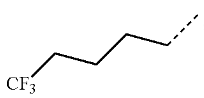

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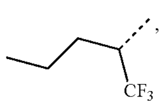

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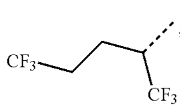

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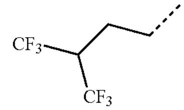

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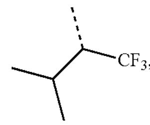

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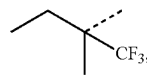

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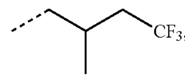

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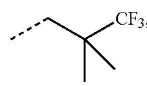

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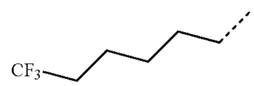

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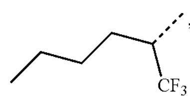

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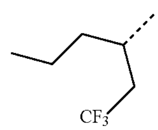

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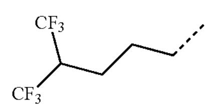

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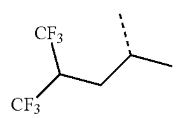

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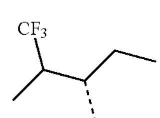

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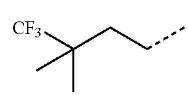

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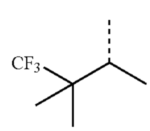

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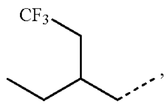

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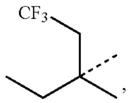

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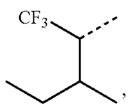

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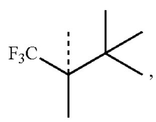

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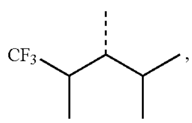

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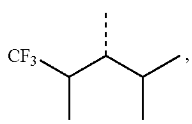

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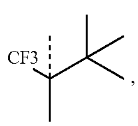

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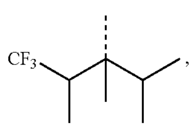

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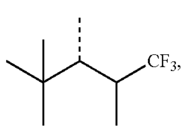

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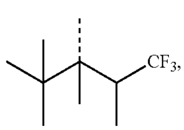

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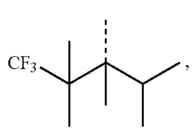

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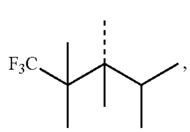

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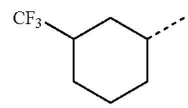

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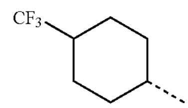

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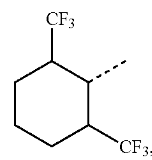

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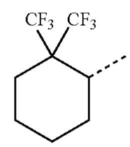

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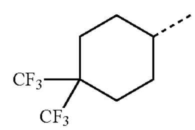

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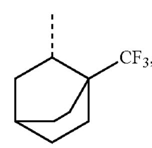

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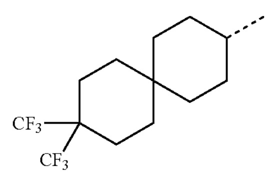

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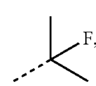

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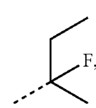

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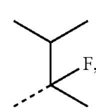

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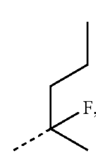

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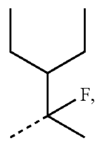
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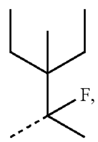
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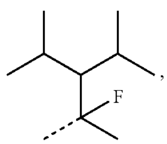
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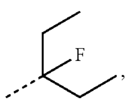
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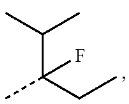
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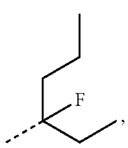
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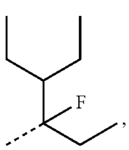
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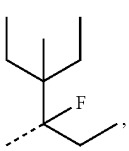
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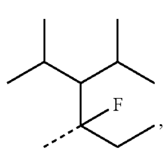
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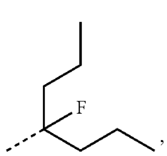
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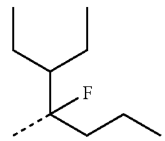
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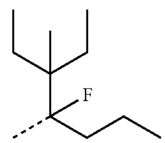
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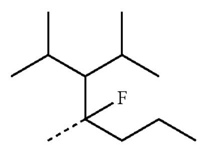
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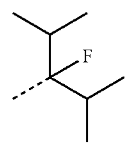
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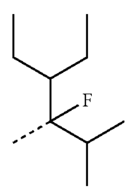
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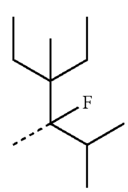
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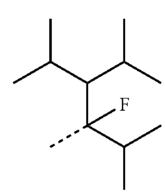
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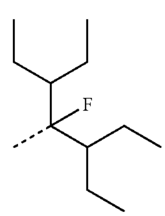
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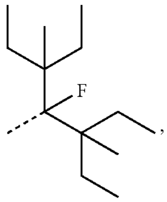

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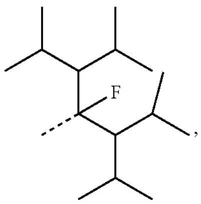

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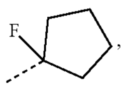

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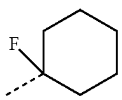

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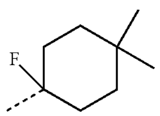

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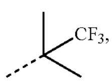

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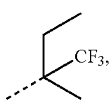
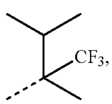
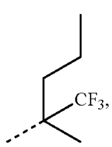


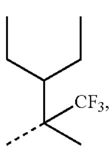
R^{D222}



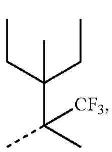
R^{D223}



R^{D224}



R^{D225}



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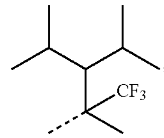
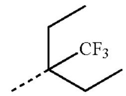
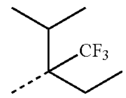
R^{D226}



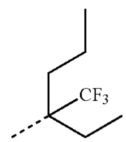
R^{D227}



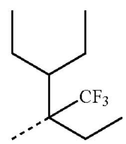
R^{D228}



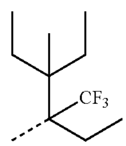
R^{D229}



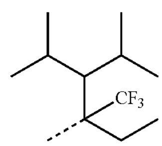
R^{D230}



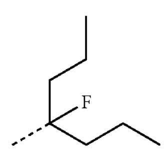
R^{D231}



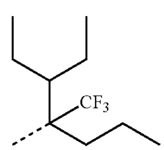
R^{D232}



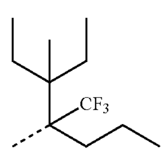
R^{D233}



R^{D234}



R^{D235}



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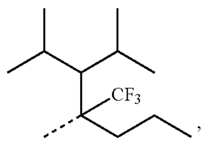
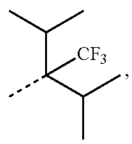
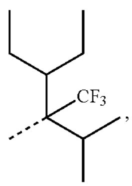
R^{D236}



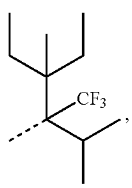
R^{D237}



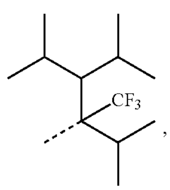
R^{D238}



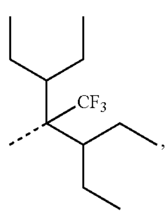
R^{D239}



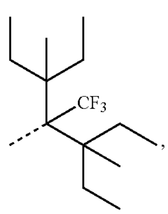
R^{D240}



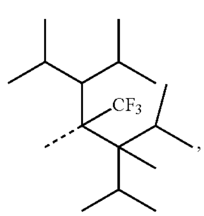
R^{D241}



R^{D242}



R^{D243}



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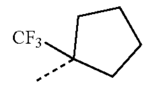
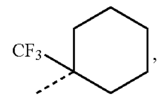
R^{D244}

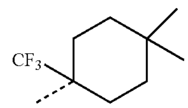


R^{D245}



and

R^{D246}



[0318] In some embodiments, the compound is selected from the group consisting of only those compounds whose L_{Bk} corresponds to one of the following: L_{B1}, L_{B30}, L_{B31}, L_{B109}, L_{B110}, L_{B112}, L_{B113}, L_{B114}, L_{B125}, L_{B127}, L_{B138}, L_{B140}, L_{B149}, L_{B150}, L_{B170}, L_{B171}, L_{B172}, L_{B174}, L_{B208}, L_{B241}, L_{B312}, L_{B315}, L_{B356}, L_{B367}, L_{B371}, L_{B382}, L_{B439}, L_{B440}, L_{B455}, L_{B456}, L_{B457}, L_{B458}, L_{B461}, L_{B462}, L_{B463}, L_{B469}, and L_{B476}.

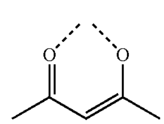
[0319] In some embodiments, the compound is selected from the group consisting of only those compounds whose L_{Bk} corresponds to one of the following: L_{B1}, L_{B30}, L_{B31}, L_{B125}, L_{B138}, L_{B171}, L_{B172}, L_{B356}, L_{B357}, L_{B367}, L_{B371}, L_{B382}, L_{B455}, and L_{B456}.

[0320] In some embodiments, the compound is selected from the group consisting of only those compounds having L_{Cj-I} or L_{Cj-II} ligand whose corresponding R^{201} and R^{202} are defined to be one of the following structures: R^{D1}, R^{D3}, R^{D4}, R^{D5}, R^{D9}, R^{D10}, R^{D17}, R^{D18}, R^{D20}, R^{D22}, R^{D37}, R^{D40}, R^{D41}, R^{D42}, R^{D43}, R^{D48}, R^{D49}, R^{D50}, R^{D54}, R^{D55}, R^{D58}, R^{D59}, R^{D78}, R^{D79}, R^{D81}, R^{D87}, R^{D88}, R^{D89}, R^{D93}, R^{D116}, R^{D117}, R^{D118}, R^{D119}, R^{D120}, R^{D133}, R^{D134}, R^{D135}, R^{D136}, R^{D143}, R^{D144}, R^{D145}, R^{D146}, R^{D147}, R^{D149}, R^{D151}, R^{D154}, R^{D155}, R^{D161}, R^{D175}, R^{D190}, R^{D193}, R^{D200}, R^{D201}, R^{D206}, R^{D210}, R^{D214}, R^{D215}, R^{D216}, R^{D218}, R^{D219}, R^{D220}, R^{D227}, R^{D237}, R^{D241}, R^{D242}, R^{D245} and R^{D246}.

[0321] In some embodiments, the compound is selected from the group consisting of only those compounds having L_{Cj-I} or L_{Cj-II} ligand whose corresponding R^{201} and R^{202} are defined to be one of selected from the following structures: R^{D1}, R^{D3}, $R^{D4}R^{D5}$, R^{D9}, R^{D10}, R^{D17}, R^{D22}, R^{D43}, R^{D50}, R^{D78}, R^{D116}, R^{D118}, R^{D133}, R^{D134}, R^{D135}, R^{D136}, R^{D143}, R^{D144}, R^{D145}, R^{D146}, R^{D149}, R^{D151}, R^{D154}, R^{D155}, R^{D190}, R^{D193}, R^{D200}, R^{D201}, R^{D206}, R^{D210}, R^{D214}, R^{D215}, R^{D216}, R^{D218}, R^{D219}, R^{D220}, R^{D227}, R^{D237}, R^{D241}, R^{D242}, R^{D245} and R^{D246}.

[0322] In some embodiments, the compound is selected from the group consisting of only those compounds having one of the structures of the following LIST 13 for the L_{Cj-I} ligand:

L_{C1-I}



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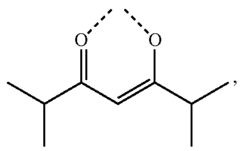
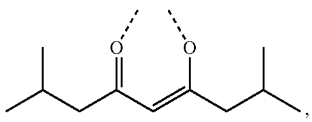
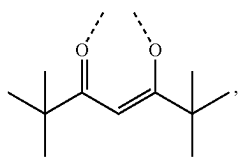
$L_{C4\text{-}I}$



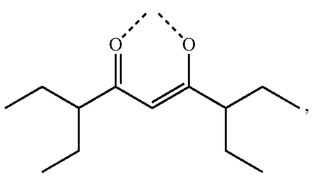
$L_{C9\text{-}I}$



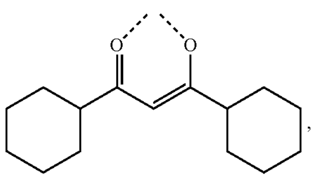
$L_{C10\text{-}I}$



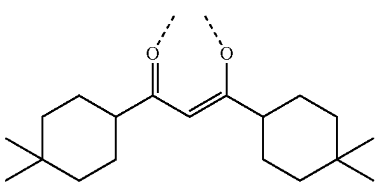
$L_{C17\text{-}I}$



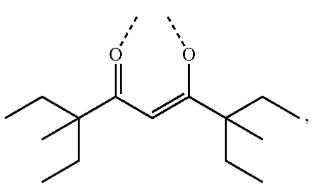
$L_{C55\text{-}I}$



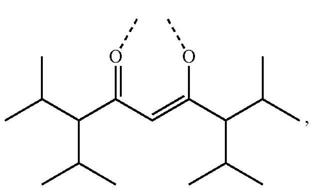
$L_{C116\text{-}I}$



$L_{C50\text{-}I}$



$L_{C78\text{-}I}$



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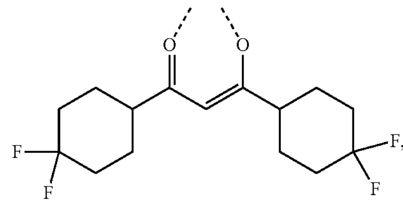
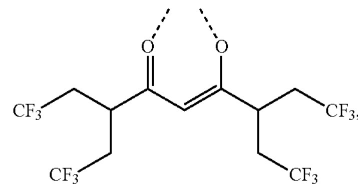
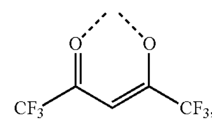
$L_{C190\text{-}I}$



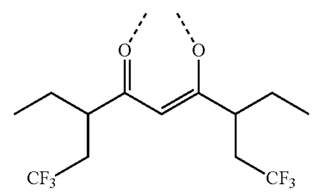
$L_{C144\text{-}I}$



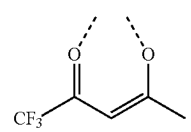
$L_{C145\text{-}I}$



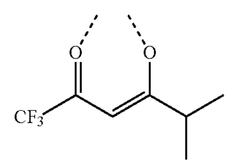
$L_{C143\text{-}I}$



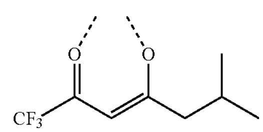
$L_{C232\text{-}I}$



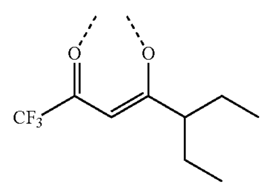
$L_{C279\text{-}I}$



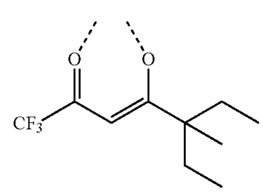
$L_{C325\text{-}I}$



$L_{C414\text{-}I}$



$L_{C457\text{-}I}$



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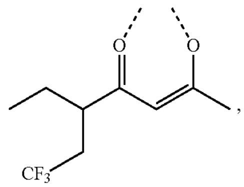
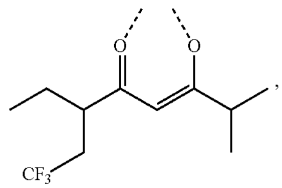
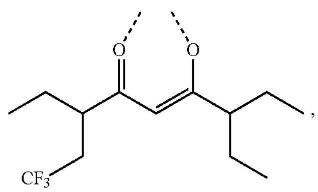


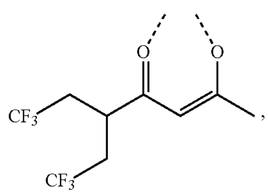
$L_{C230\text{-}I}$



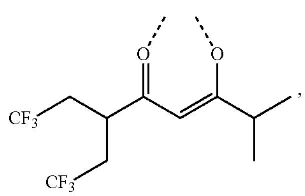
$L_{C277\text{-}I}$



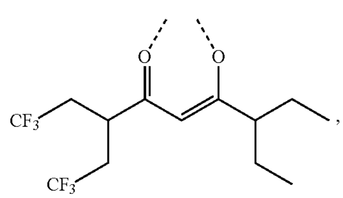
$L_{C412\text{-}I}$



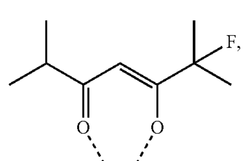
$L_{C231\text{-}I}$



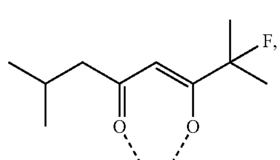
$L_{C278\text{-}I}$



$L_{C413\text{-}I}$



$L_{C985\text{-}I}$



$L_{C1093\text{-}I}$

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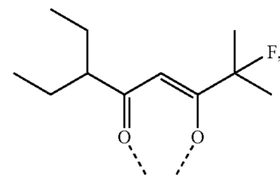
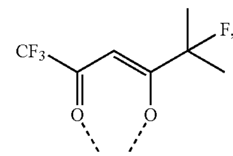
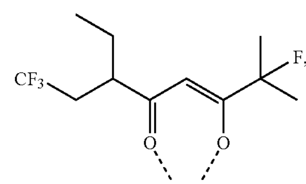


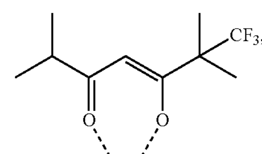
$L_{C823\text{-}I}$



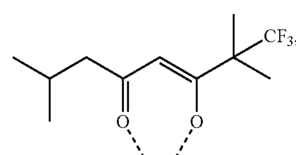
$L_{C1039\text{-}I}$



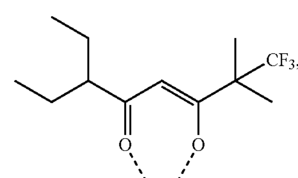
$L_{C1147\text{-}I}$



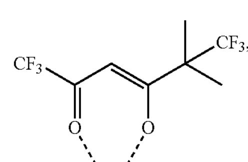
$L_{C1012\text{-}I}$



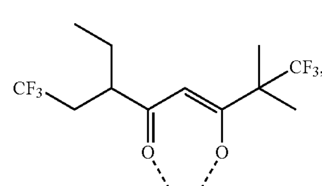
$L_{C1120\text{-}I}$



$L_{C850\text{-}I}$



$L_{C1066\text{-}I}$



$L_{C1174\text{-}I}$

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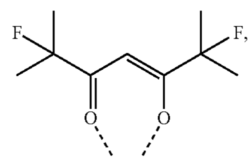
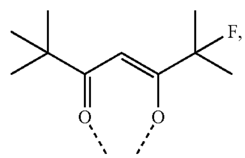
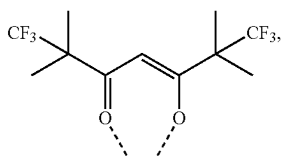
$L_{C769\text{-}I}$



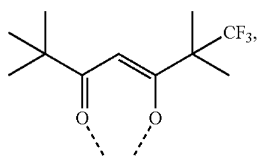
$L_{C1201\text{-}I}$



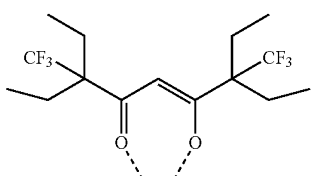
$L_{C796\text{-}I}$



$L_{C1228\text{-}I}$

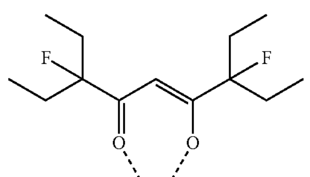


$L_{C803\text{-}I}$



, and

$L_{C776\text{-}I}$



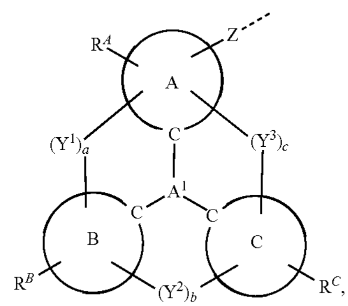
[0323] In some embodiments, the compound has a formula selected from the group consisting of $Ir(L_A)_3$, $Ir(L_A)_2(L_B)$, $Ir(L_A)(L_B)_2$, $Ir(L_A)_2(L_C)$, and $Ir(L_A)(L_B)(L_C)$. In some embodiments, L_A is selected from the group consisting of the structures of LIST 2, LIST 4, and LIST 7, L_B is selected from the group consisting of the structures of LIST 8, LIST 9, and LIST 10 (L_{Bk}), and L_C is selected from the group consisting of the structures of $L_{Cj\text{-}I}$ and $L_{Cj\text{-}II}$ defined herein.

[0324] In some embodiments, L_A is selected from the group consisting of the structures of LIST 2 and L_B is selected from the group consisting of the structures of L_{Bk}. In some embodiments, L_A is selected from the group consisting of the structures of LIST 4 and L_B is selected from the group consisting of the structures of L_{Bk}. In some embodiments, L_A is selected from LIST 7 defined herein, and L_B is selected from the group consisting of the structures of L_{Bk} wherein k is an integer from 1 to 541. In some embodiments, L_A is selected from LIST 7 defined herein, and L_C is selected from the group consisting of the structures of $L_{Cj\text{-}I}$ and $L_{Cj\text{-}II}$ wherein j is an integer from 1 to 1416.

[0325] In some embodiments, the compound can be $Ir(L_A)_3$, $Ir(L_A)_2(L_B)$, $Ir(L_A)(L_B)_2$, $Ir(L_A)_2(L_C)$, $Ir(L_A)(L_C)_2$, or $Ir(L_A)(L_B)(L_C)$. In some of these embodiments, the compound can be $Ir(L_{A1})_3$ consisting of the compounds of $Ir(L_{A1})_3$ to $Ir(L_{A241})_3$, $Ir(L_{Ai})_2(L_B)$, $Ir(L_{Ai})(L_B)_2$, $Ir(L_A)_2(L_{Bk})$, $Ir(L_A)(L_{Bk})_2$, $Ir(L_{A1})_2(L_{Bk})$ consisting of the compounds from $Ir(L_{A1})_2(L_{B1})$ to $Ir(L_{A241})_2$ (L_{B541}), $Ir(L_{Ai})(L_{Bk})_2$ consisting of the compounds from $Ir(L_{A1})(L_{B1})_2$ to $Ir(L_{A238})(L_{B541})_2$, $Ir(L_{Ai})_2(L_{Cj\text{-}I})$, consisting of the compounds of $Ir(L_{A1})_2(L_{C1\text{-}I})$ to $Ir(L_{A241})_2(L_{C1416\text{-}II})$, $Ir(L_{Ai})_2(L_{Cj\text{-}II})$, consisting of the compounds of $Ir(L_{A1})_2(L_{C1\text{-}II})$ to $Ir(L_{A241})_2(L_{C1416\text{-}II})$, $Ir(L_{Ai})(L_{Bk})(L_{Cj\text{-}I})$ consisting of the compounds of $Ir(L_{A1})(L_{B1})(L_{C1\text{-}I})$ to $Ir(L_{A241})(L_{B541})(L_{C1416\text{-}I})$, or $Ir(L_{Ai})(L_{Bk})(L_{Cj\text{-}II})$ consisting of the compounds of $Ir(L_{A1})(L_{B1})(L_{Cj\text{-}II})$ to $Ir(L_{A241})(L_{B541})(L_{C1416\text{-}II})$, wherein L_{Ai}, L_{Bk}, $L_{Cj\text{-}II}$ and $L_{Cj\text{-}II}$ are all defined herein.

What is claimed is:

1. A compound having a first ligand L_A comprising a structure of Formula I:



wherein:

L_A is coordinated to a metal M selected from the group consisting of Ir, Rh, Re, Ru, Os, Pt, Pd, Ag, Au, and Cu;

each of moiety A, moiety B, and moiety C is independently a monocyclic ring or a polycyclic fused ring system, wherein the monocyclic ring or each ring of the polycyclic fused ring system is independently a 5-membered to 10-membered carbocyclic or heterocyclic ring;

A^1 is selected from the group consisting of B, N, P, P═O, P═S, Al, Ga, SiR", GeR", and SnR";

each of a, b, and c is independently 0 or 1, where 0 represents absent and 1 represents present;

a+b+c is 2 or 3;

each of Y^1, Y^2, and Y^3 is independently selected from the group consisting of a direct bond, BR, BRR', NR, PR, P(O)R, O, S, Se, C═O, C═S, C═Se, C═NR, C═CRR', S═O, SO_2, CR, CRR', SiRR', and GeRR';

Z is a direct bond to metal M, or Z is a direct bond or an organic linker to a monocyclic ring or a polycyclic fused ring system that is coordinated to M;

each of R^A, R^B, and R^C independently represents mono to the maximum allowable substitution;

each R, R', R", R^A, R^B, and R^C is independently hydrogen or a substituent selected from the group consisting of deuterium, halogen, alkyl, cycloalkyl, heteroalkyl, heterocycloalkyl, arylalkyl, alkoxy, aryloxy, amino, silyl, germyl, boryl, alkenyl, cycloalkenyl, heteroalkenyl, alkynyl, aryl, heteroaryl, acyl, carboxylic acid, ether, ester, nitrile, isonitrile, sulfanyl, sulfinyl, sulfonyl, phosphino, selenyl, and combinations thereof;

M can be coordinated to other ligands;

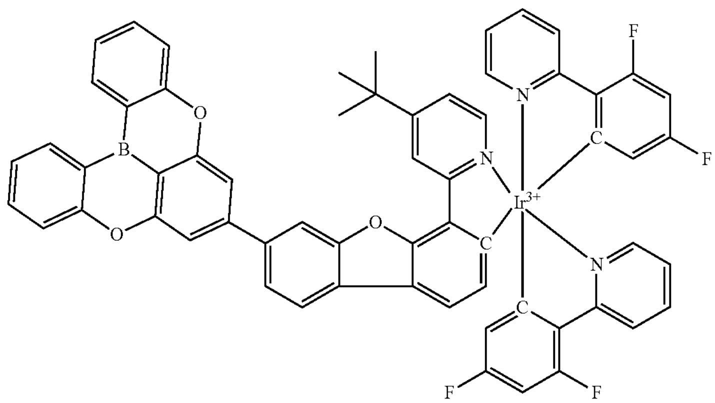
L_A can be a monodentate, bidentate, tridentate, tetradentate, pentadentate ligand, or hexadentate ligand; and

wherein any two substituents can be joined or fused to form a ring, with the proviso that

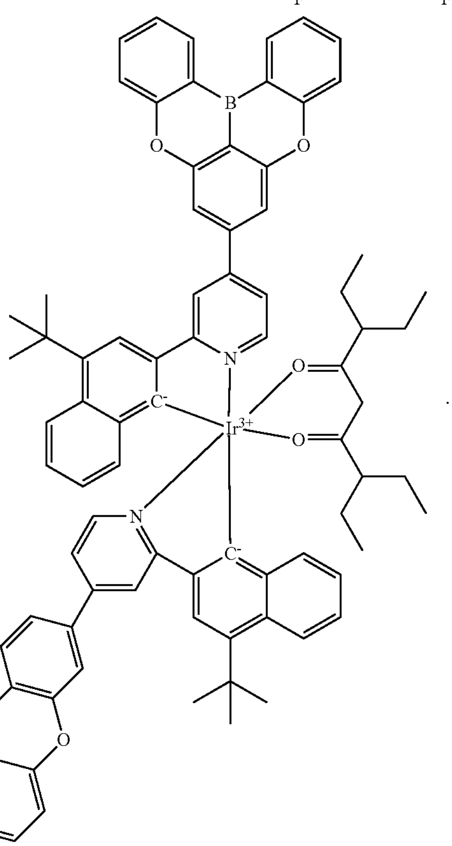
(1) if M is Pt, then L_A is a monodentate ligand;

(2) if Formula I is part of a bidentate ligand and ring A is a monocyclic ring, then Z is not a direct bond to the metal; and

(3) the compound is not



or

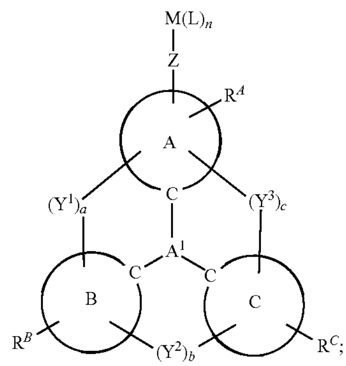


.

2. The compound of claim 1, wherein each of moiety A, moiety B, and moiety C is independently selected from the group consisting of the following Cyclic Moiety List: benzene, pyridine, pyrimidine, pyridazine, pyrazine, triazine, imidazole, pyrazole, pyrrole, oxazole, furan, thiophene, thiazole, triazole, naphthalene, quinoline, isoquinoline, quinazoline, benzofuran, aza-benzofuran, benzoxazole, aza-benzoxazole, benzothiophene, aza-benzothiophene, benzothiazole, aza-benzothiazole, benzoselenophene, aza-benzoselenophene, indene, aza-indene, indole, aza-indole, benzimidazole, aza-benzimidazole, benzobenzimidazole, aza-benzobenzimidazole, carbazole, aza-carbazole, dibenzofuran, aza-dibenzofuran, phenanthro[3,2-b]benzofuran, dibenzothiophene, aza-dibenzothiophene, quinoxaline, phthalazine, phenanthrene, aza-phenanthrene, anthracene, aza-anthracene, phenanthridine, fluorene, and aza-fluorene; and/or wherein A^1 is selected from the group consisting of B, N, P, Al, and Ga; and/or wherein Y^1 is selected from the

group consisting of direct bond, O, S, and Se; and/or wherein Y^2 is selected from the group consisting of direct bond, O, S, and Se; and/or wherein Y^3 is selected from the group consisting of direct bond, O, S, and Se; and/or wherein metal M is Ir, Pt, or Pd; and/or wherein Z is a direct bond.

3. The compound of claim **1**, wherein the compound comprises a structure of Formula IIA:



wherein:

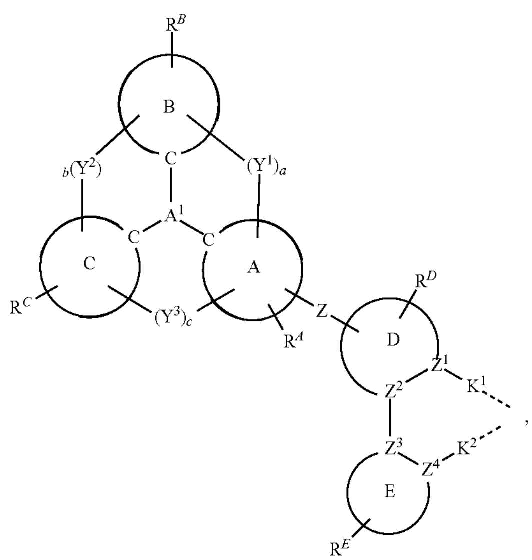
X^1 and X^2 are each independently C or N;

n is an integer from 1 to 5;

each L is independently a monodentate, bidentate, tridentate, tetradentate, or pentadentate ligand; and

each L can be the same or different.

4. The compound of claim **1**, wherein L_A comprises a structure of Formula IIIA:



wherein:

each of moiety D and moiety E is independently a monocyclic ring or a polycyclic fused ring system, wherein the monocyclic ring or each ring of the polycyclic fused ring system is independently a 5-membered to 10-membered carbocyclic or heterocyclic ring;

each of X^1, X^2, and Z^1 to Z^4 is independently C or N;

K^1 and K^2 are each independently a direct bond or selected from the group consisting of a direct bond, O, S, $N(R^\alpha)$, $P(R^\alpha)$, $B(R^\alpha)$, $C(R^\alpha)(R^\beta)$, and $Si(R^\alpha)(R^\beta)$;

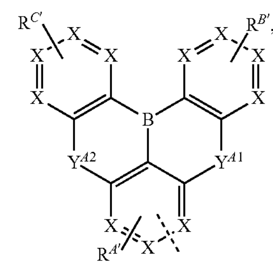
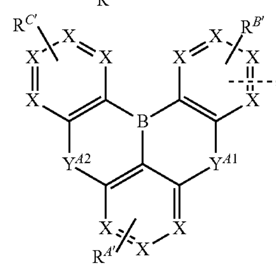
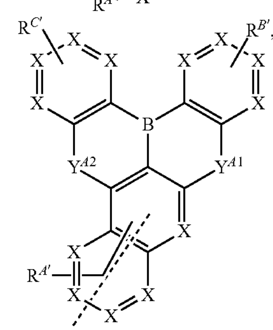
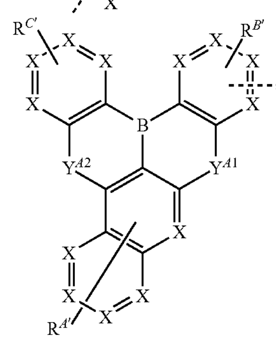
R^D and R^E each independently represent mono to the maximum allowable substitution;

each R^α, R^β, R^D, and R^E is independently hydrogen or a substituent selected from the group consisting of deuterium, halogen, alkyl, cycloalkyl, heteroalkyl, heterocycloalkyl, arylalkyl, alkoxy, aryloxy, amino, silyl, germyl, boryl, alkenyl, cycloalkenyl, heteroalkenyl, alkynyl, aryl, heteroaryl, acyl, carboxylic acid, ether, ester, nitrile, isonitrile, sulfanyl, sulfinyl, sulfonyl, phosphino, selenyl, and combinations thereof;

any two substituents can be joined or fused to form a ring; and

L_A is coordinated to M through the indicated dashed lines.

5. The compound of claim **1**, wherein the ligand L_A is selected from the group consisting of:

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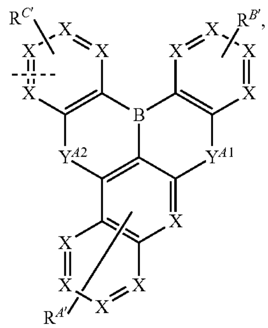

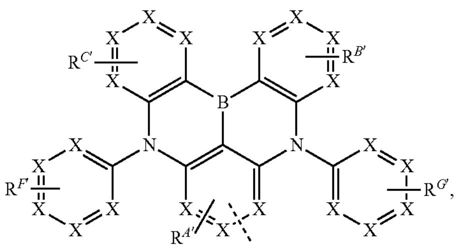

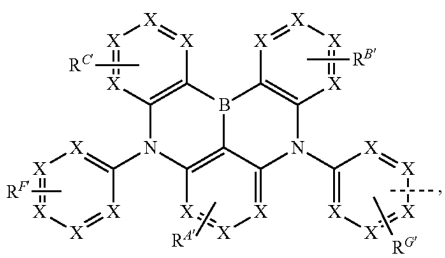

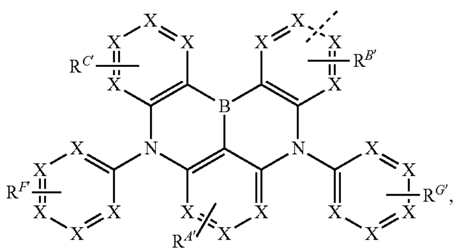

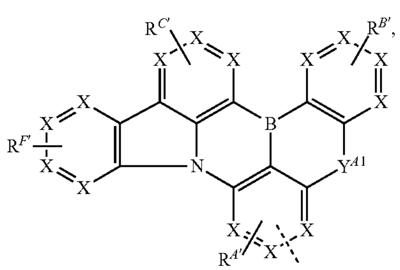

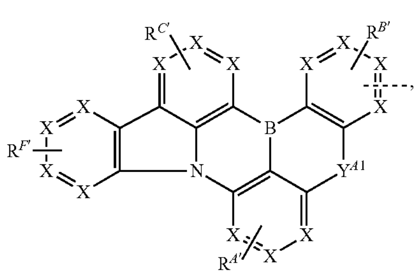

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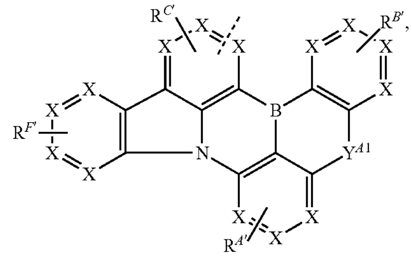

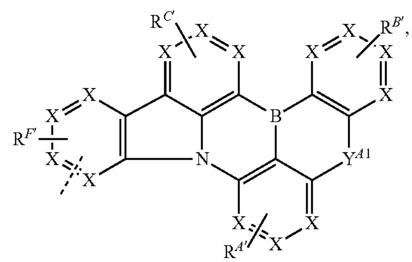

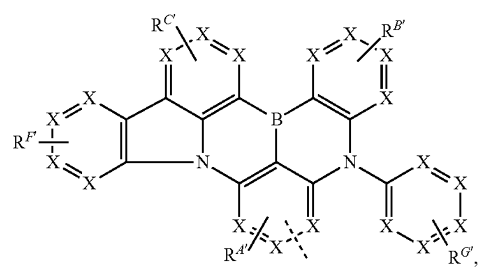

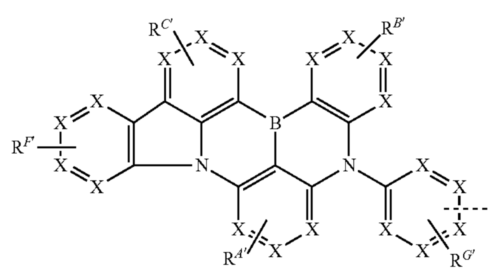

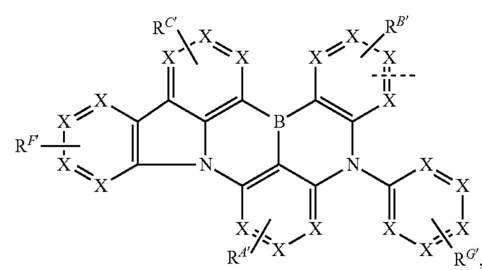

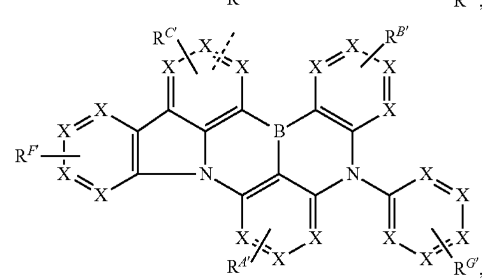

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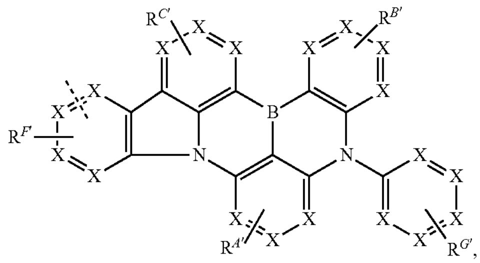

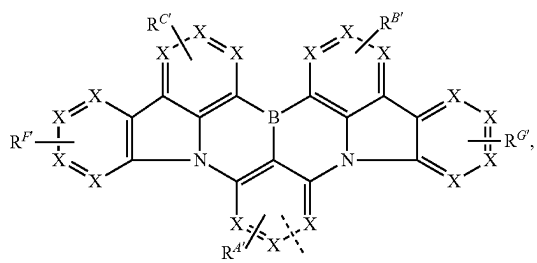

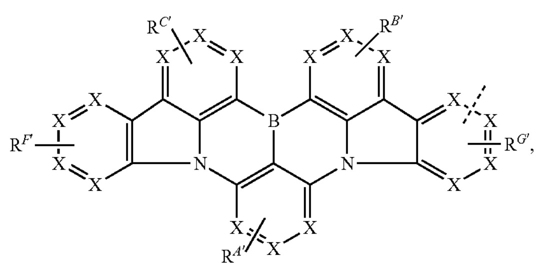

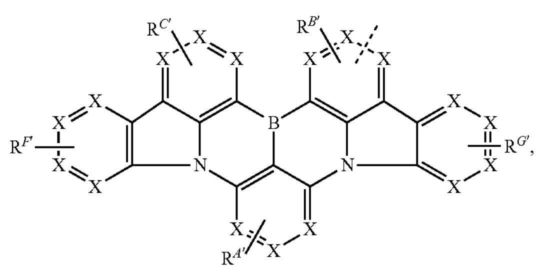

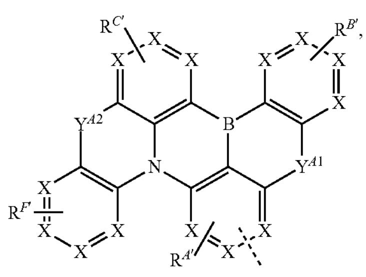

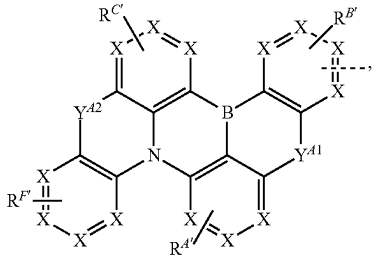

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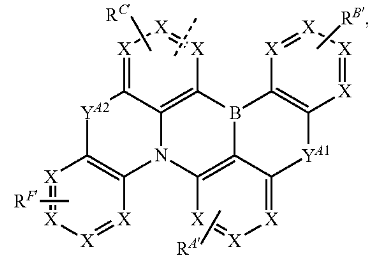

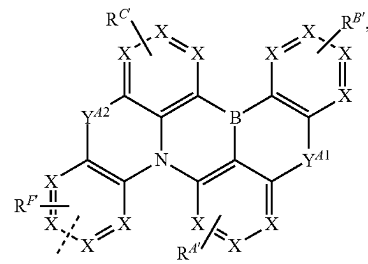

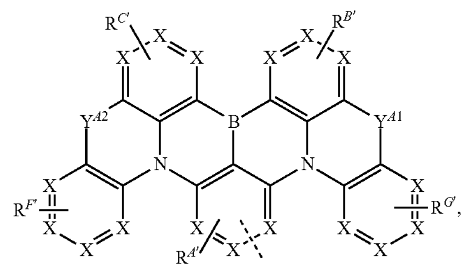

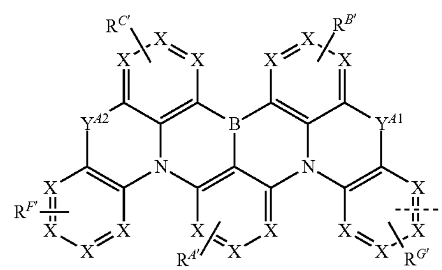

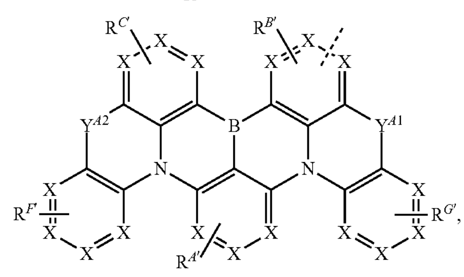

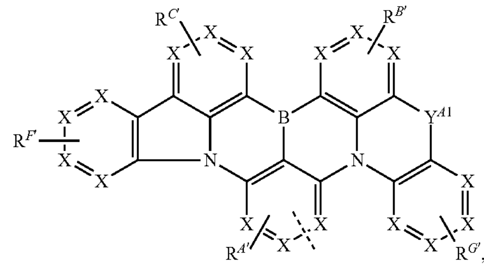

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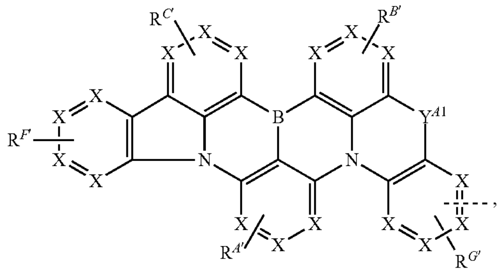

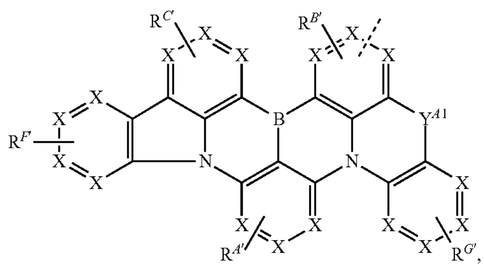

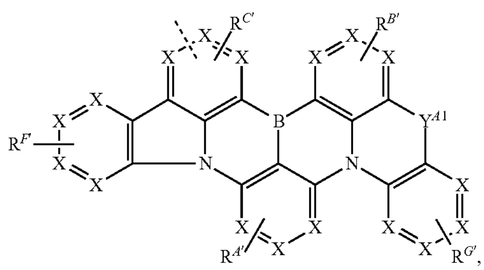

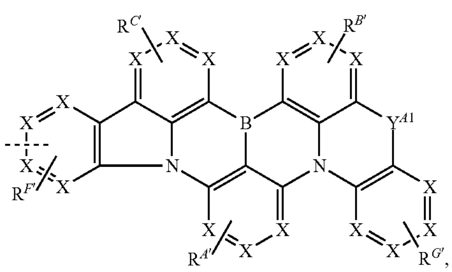

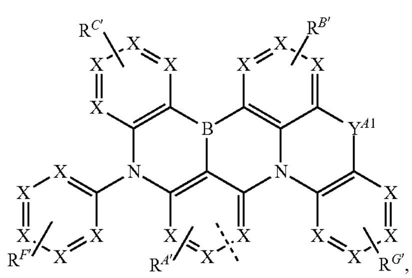

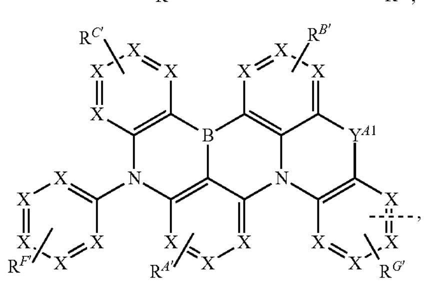

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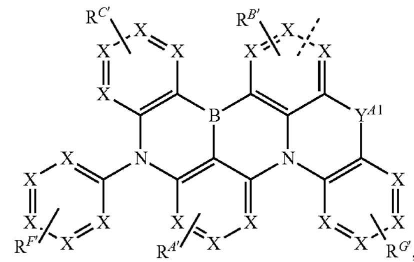

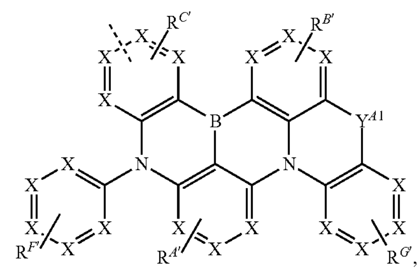

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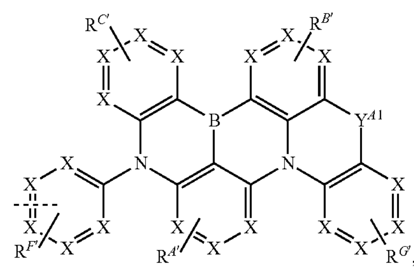


wherein

for each occurrence, X is independently C or N;

each of Y^{A1} and Y^{A2} is independently selected from the group consisting of a direct bond, BR, BRR', NR, PR, P(O)R, O, S, Se, C═O, C═S, C═Se, C═NR, C═CRR', S═O, SO_2, CR, CRR', SiRR', and GeRR';

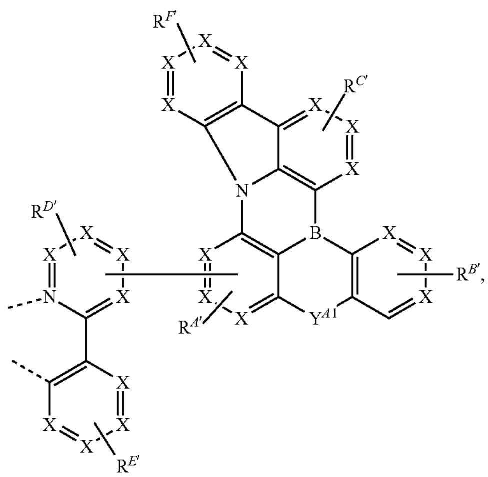
- - - - represents chelation to the metal;

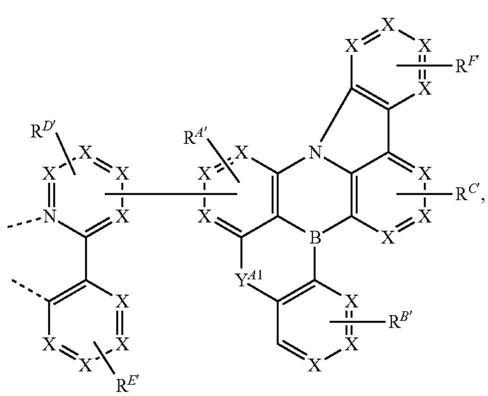
each of $R^{A'}$, $R^{B'}$, $R^{C'}$, $R^{F'}$, and $R^{G'}$ independently represents mono to the maximum allowable substitution;

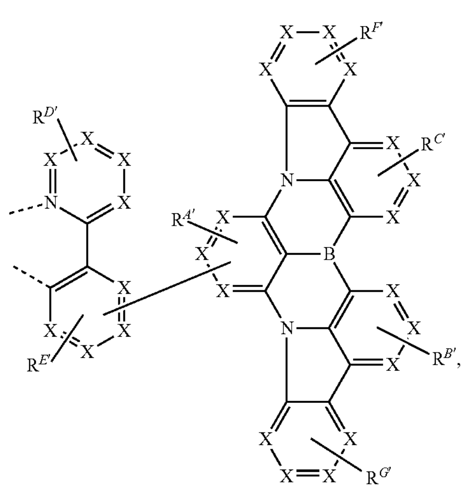
each R, R', $R^{A'}$, $R^{B'}$, $R^{C'}$, $R^{F'}$, and $R^{G'}$ is independently hydrogen or a substituent selected from the group consisting of deuterium, halogen, alkyl, cycloalkyl, heteroalkyl, heterocycloalkyl, arylalkyl, alkoxy, aryloxy, amino, silyl, germyl, boryl, alkenyl, cycloalkenyl, heteroalkenyl, alkynyl, aryl, heteroaryl, acyl, carboxylic acid, ether, ester, nitrile, isonitrile, sulfanyl, sulfinyl, sulfonyl, phosphino, selenyl, and combinations thereof; and

any two substituents can be joined or fused to form a ring.

6. The compound of claim 1, wherein the ligand L_A is selected from the group consisting of:







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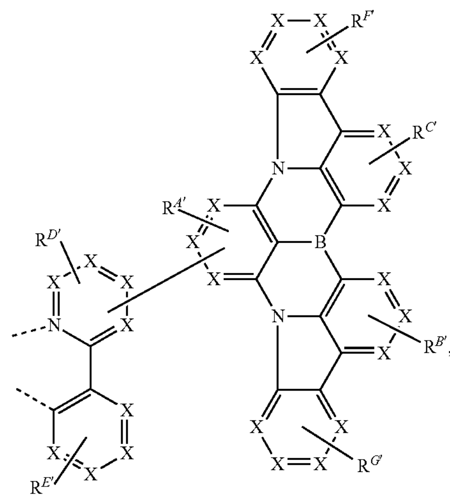

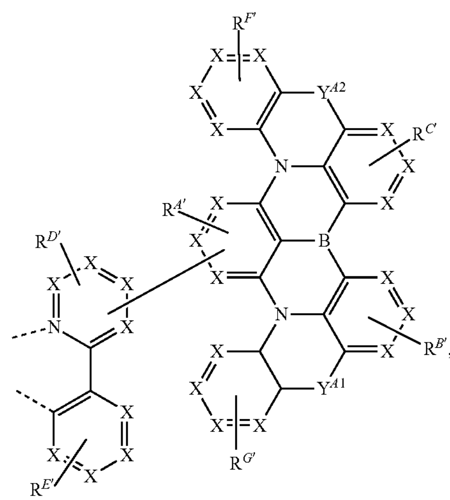

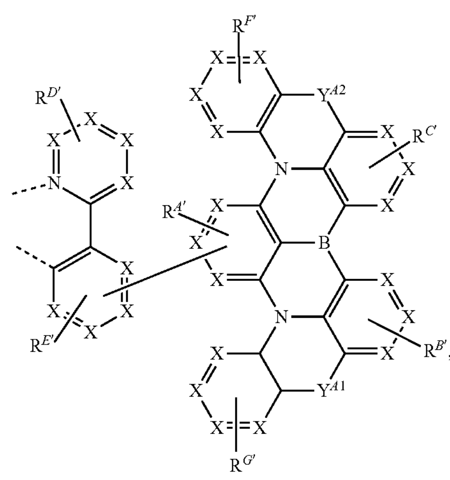

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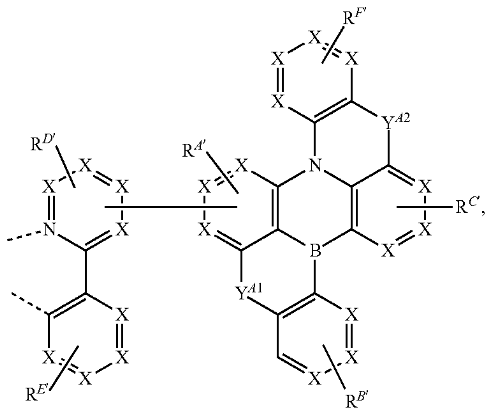

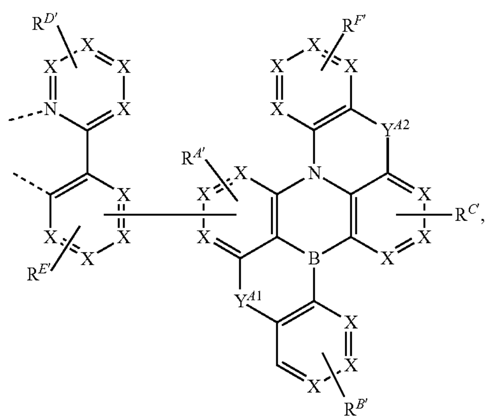

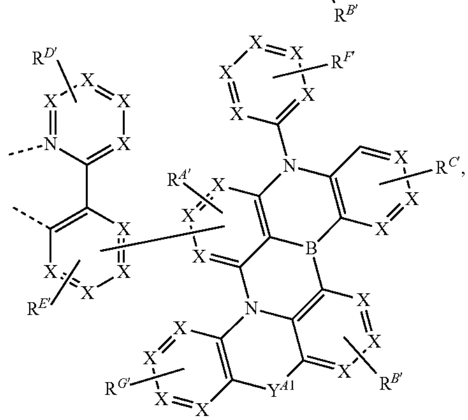

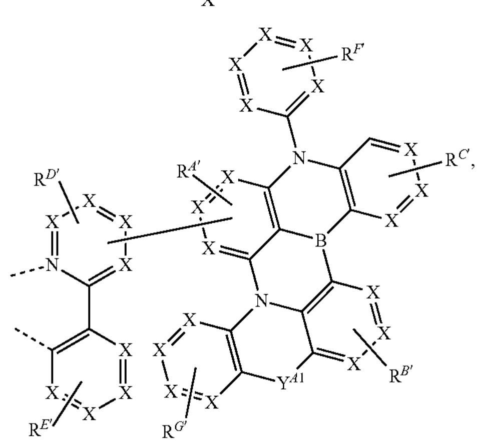

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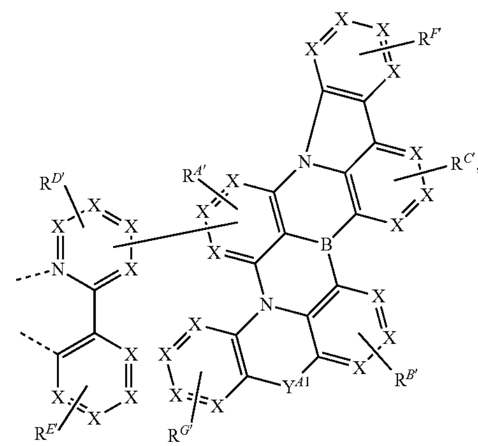

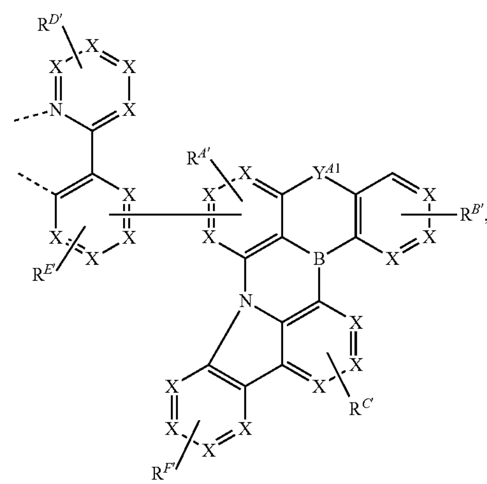

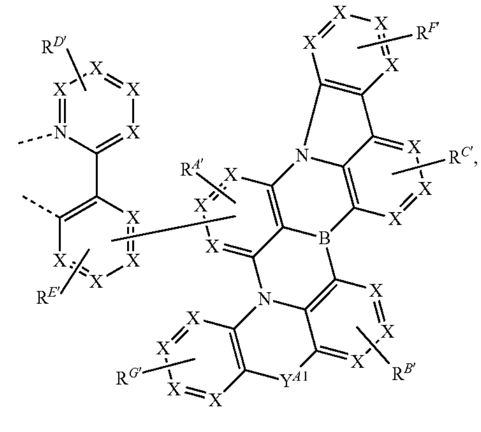

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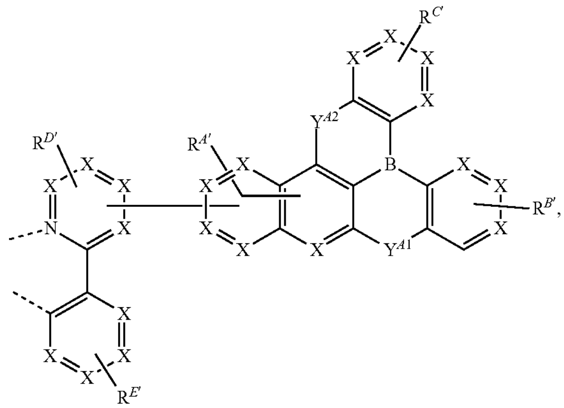

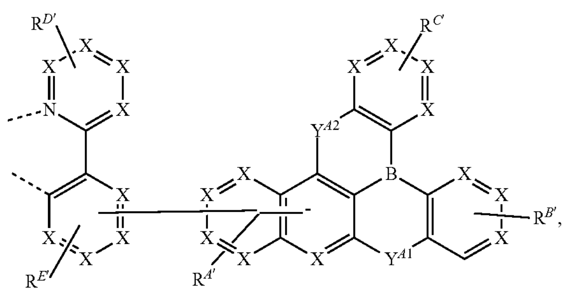

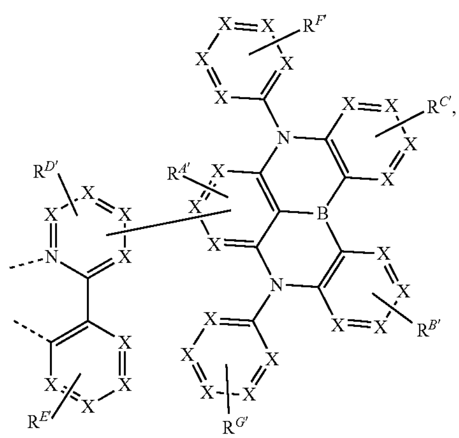

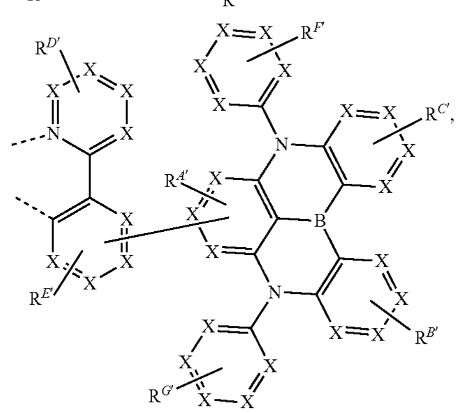

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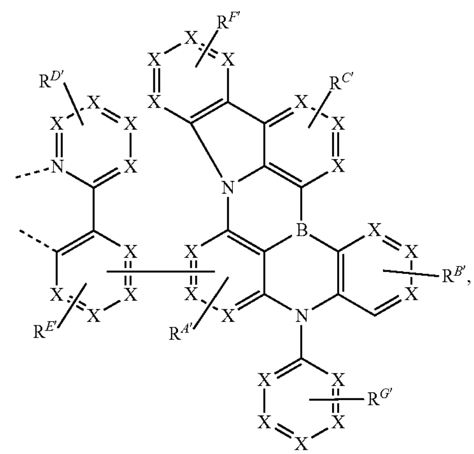

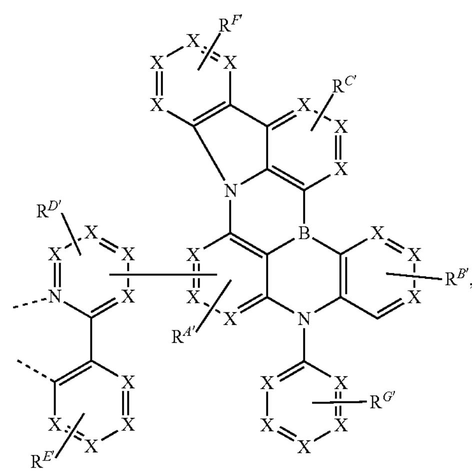

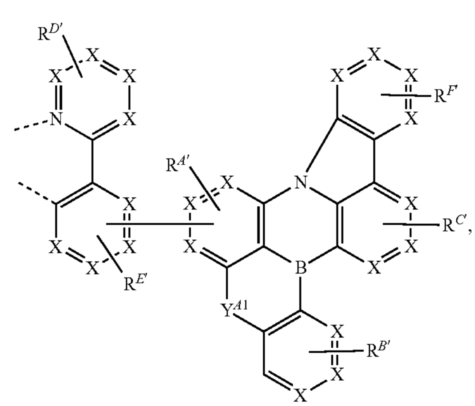

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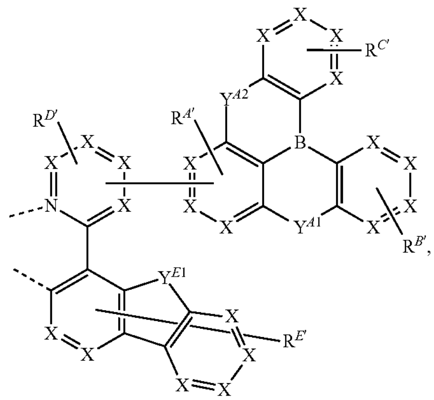

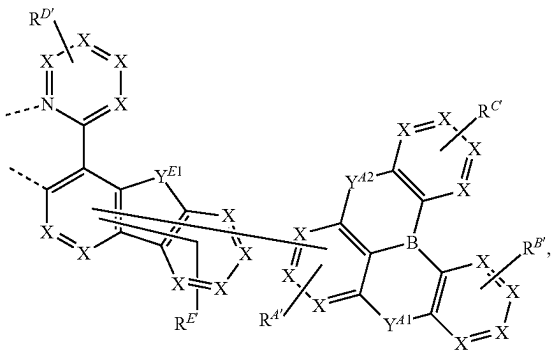

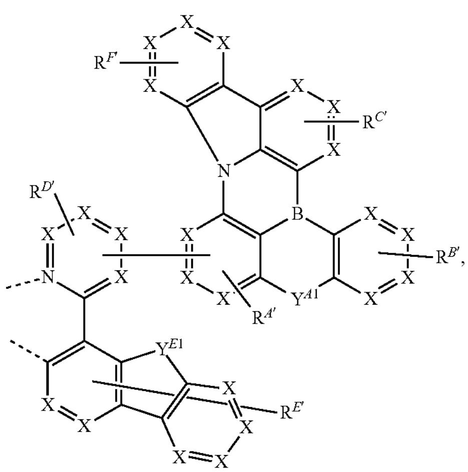

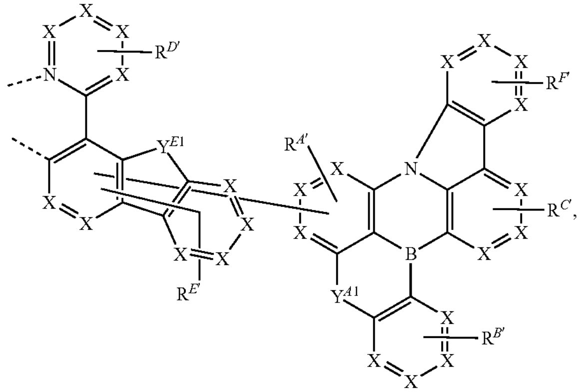

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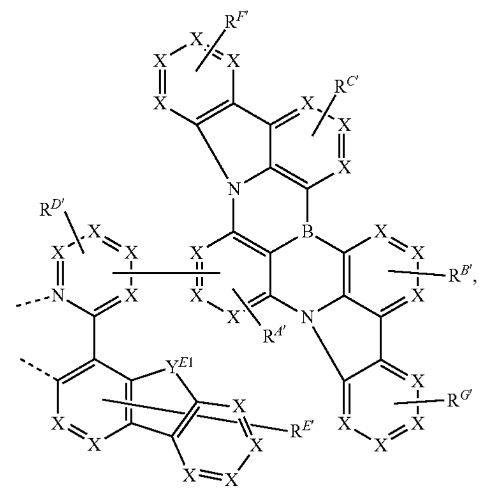

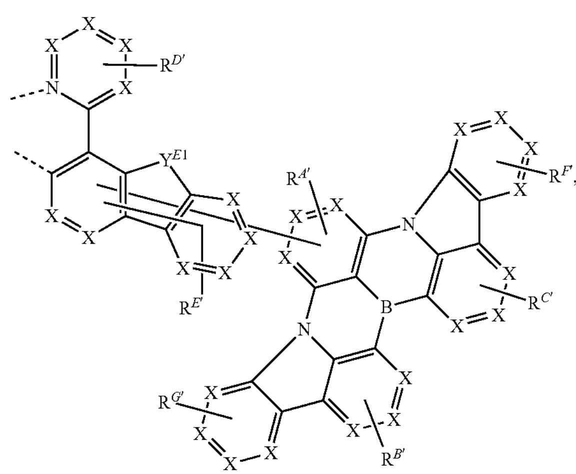

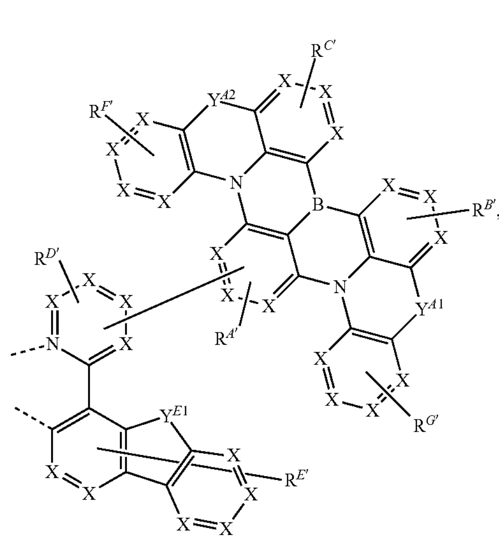

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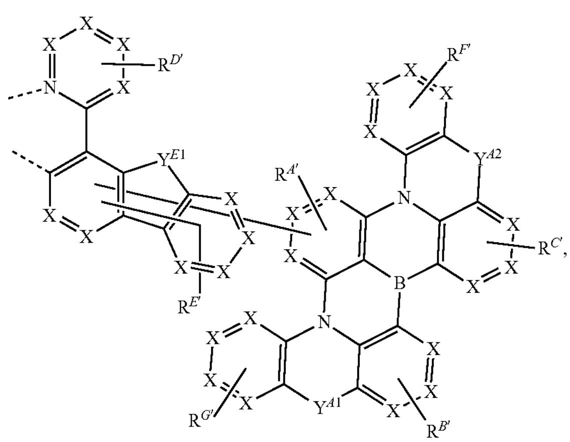

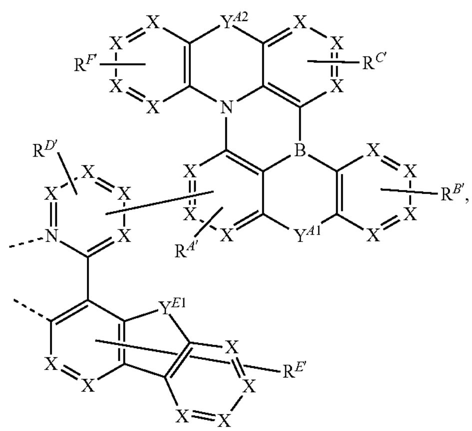

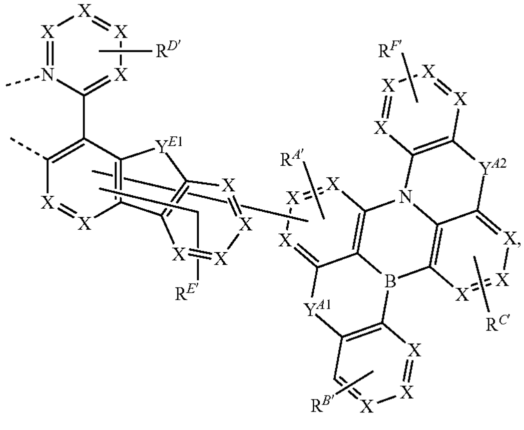

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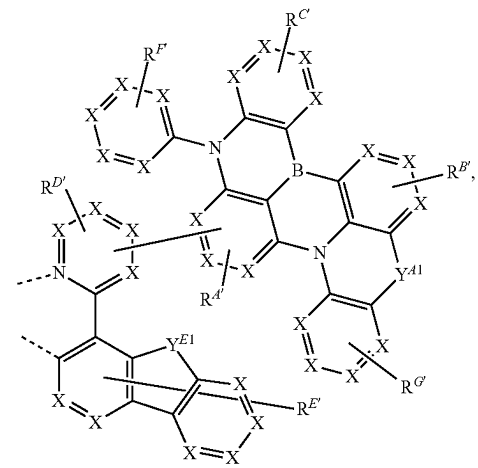

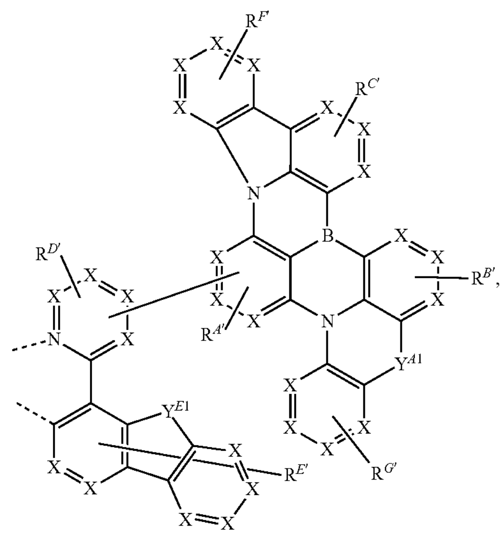

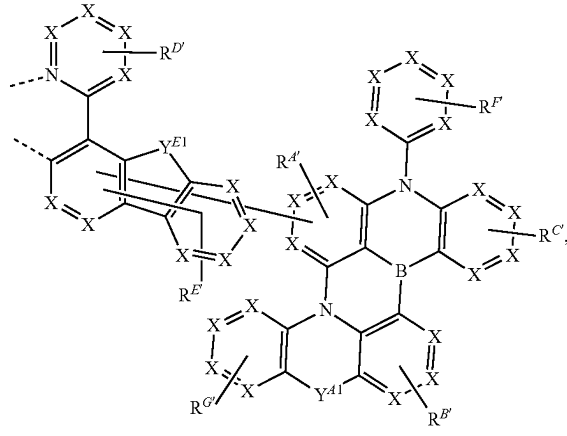

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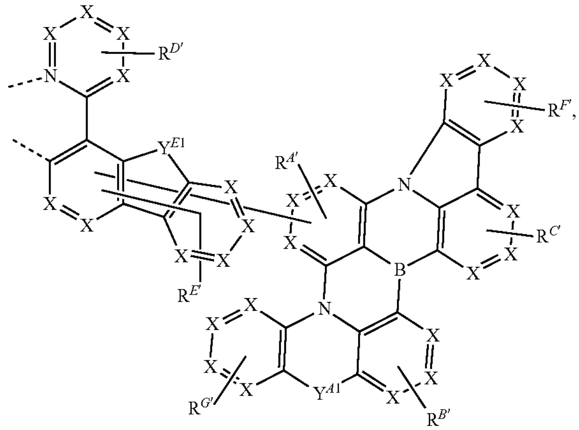

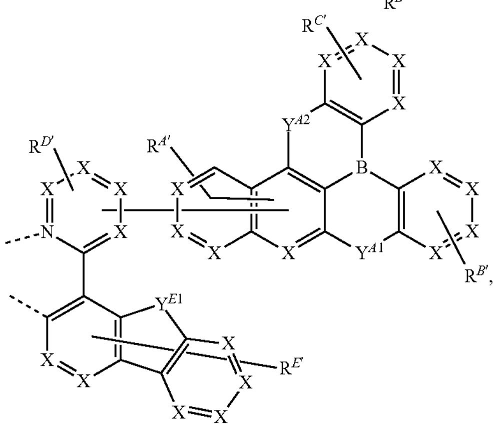

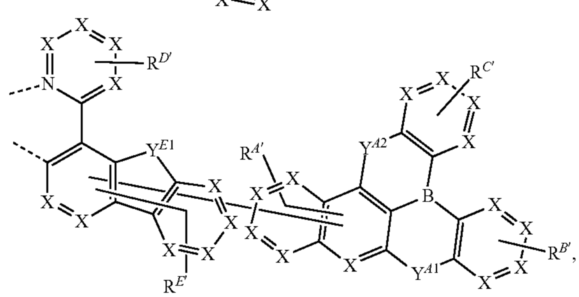

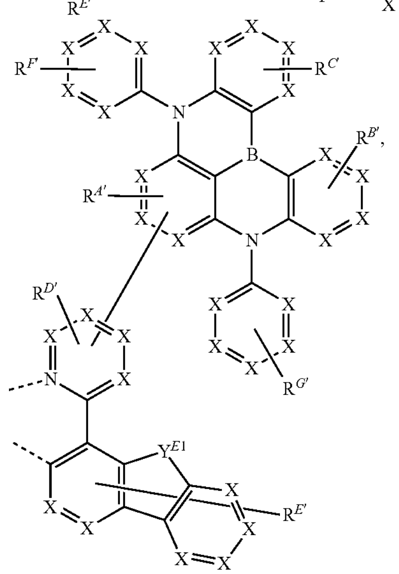

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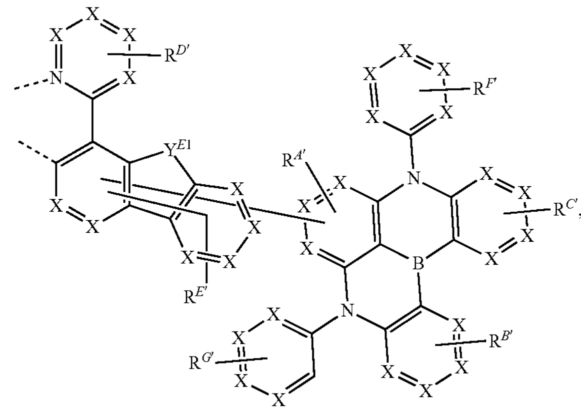

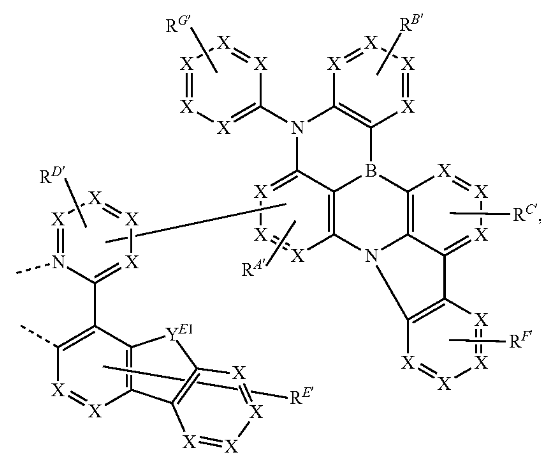

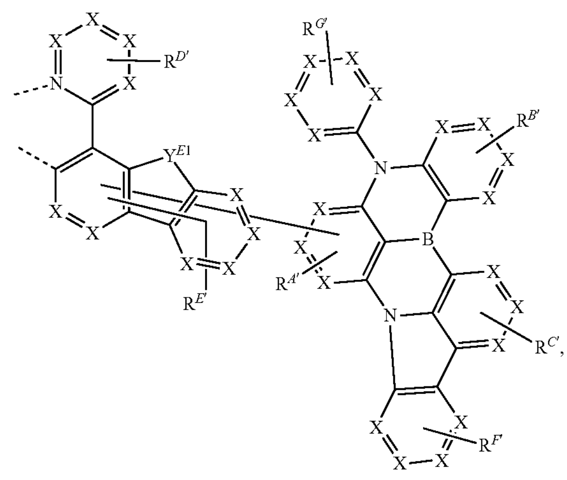

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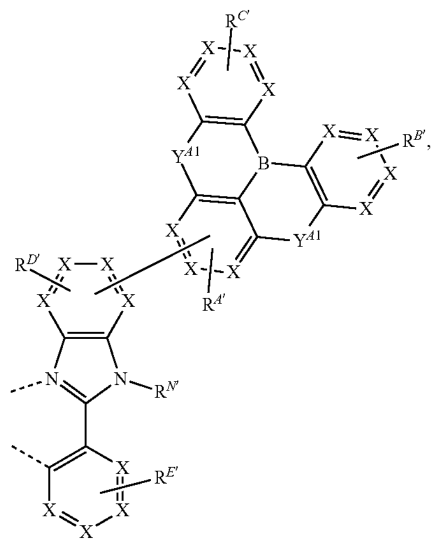

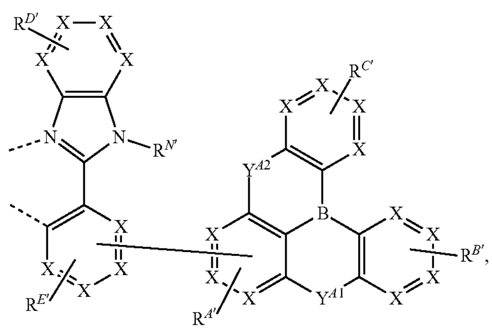

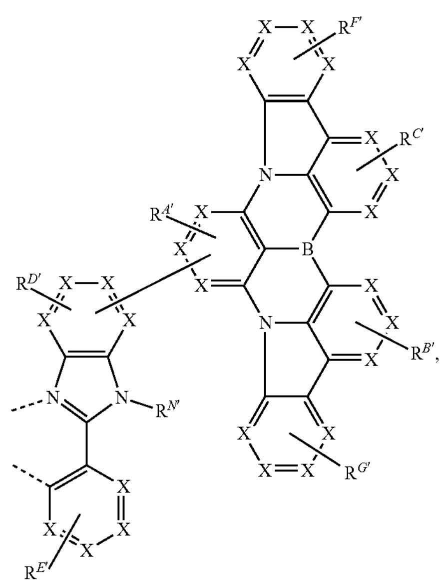

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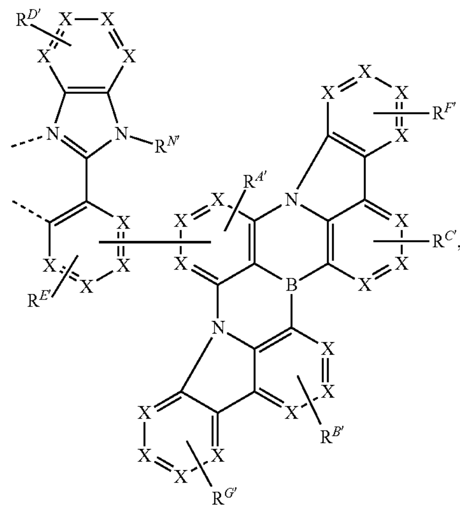

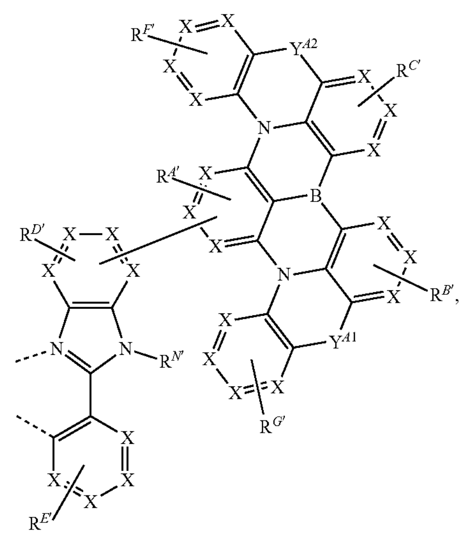

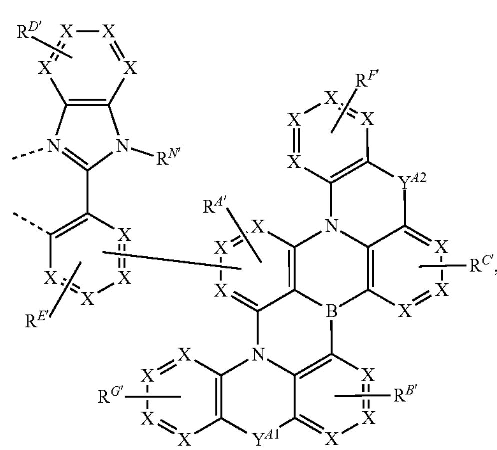

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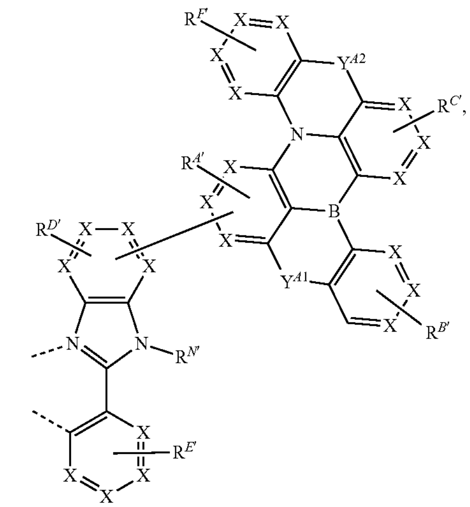

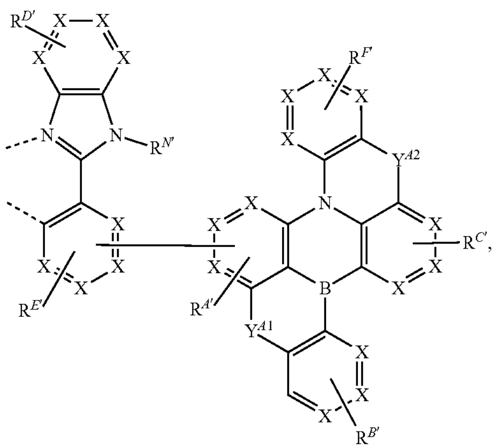

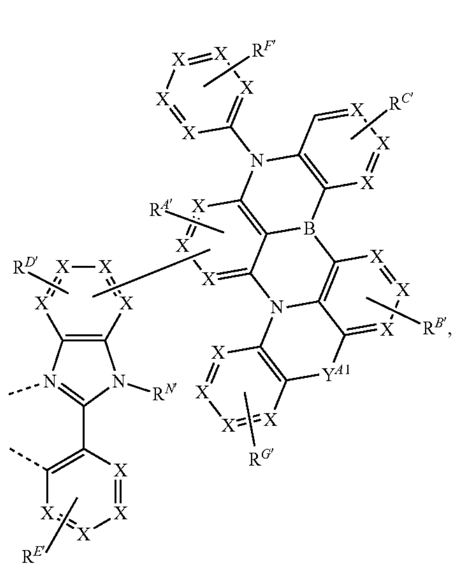

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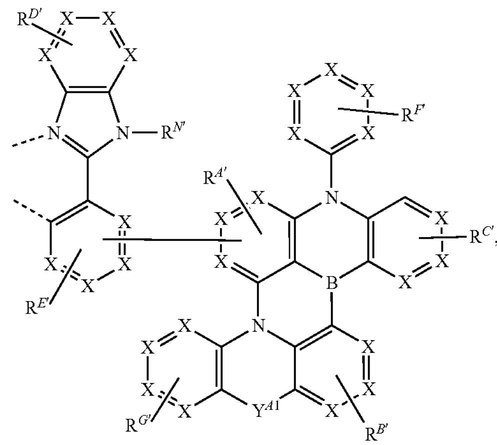

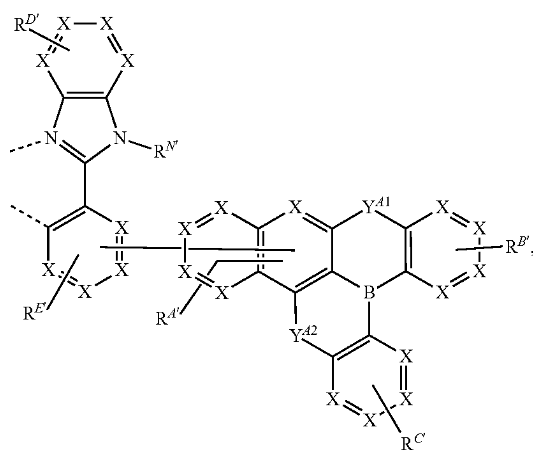

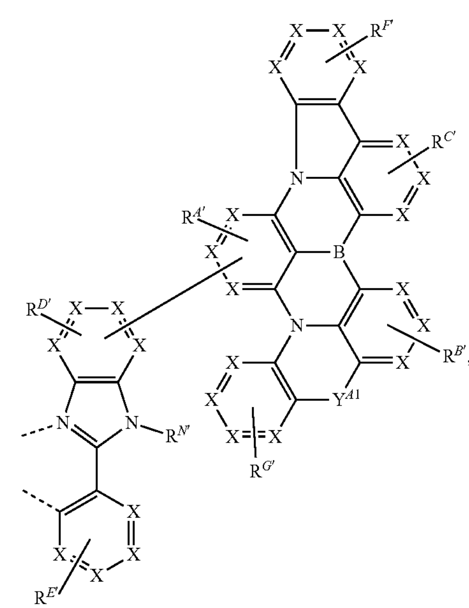

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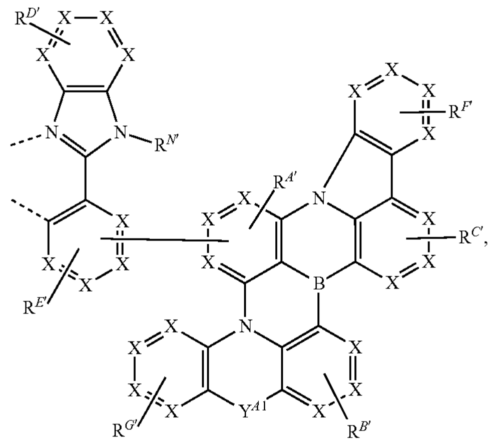

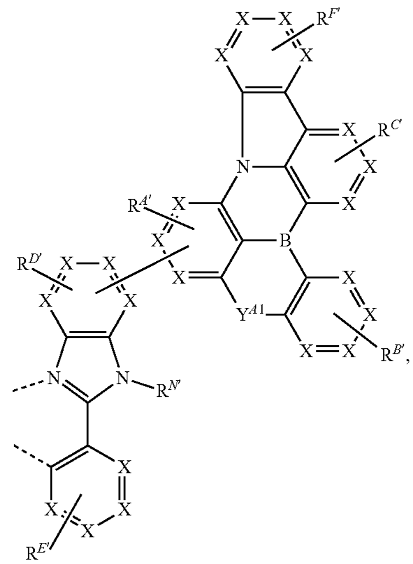

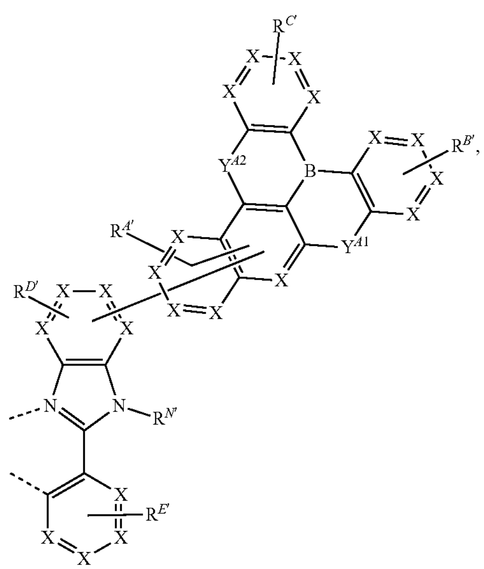

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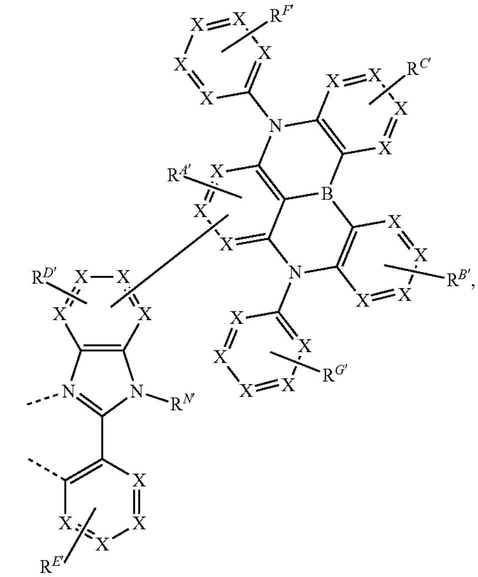

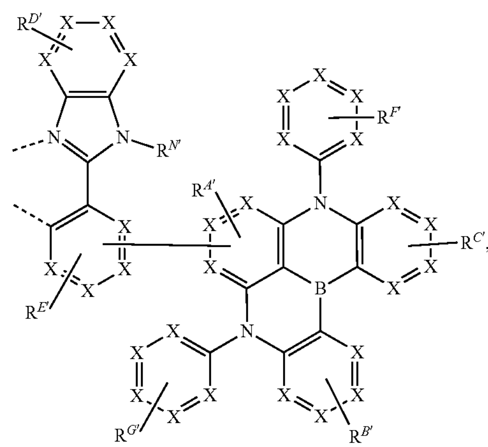

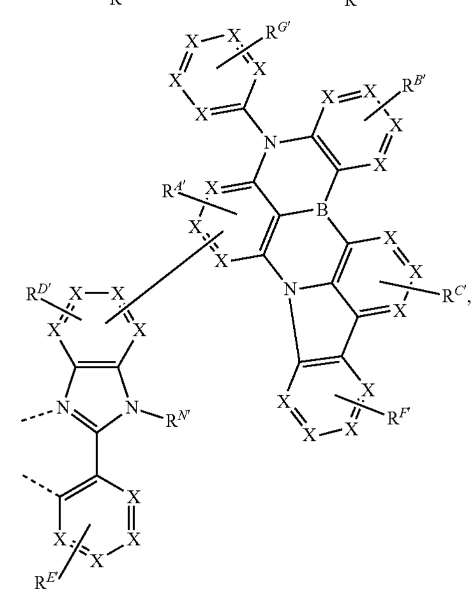

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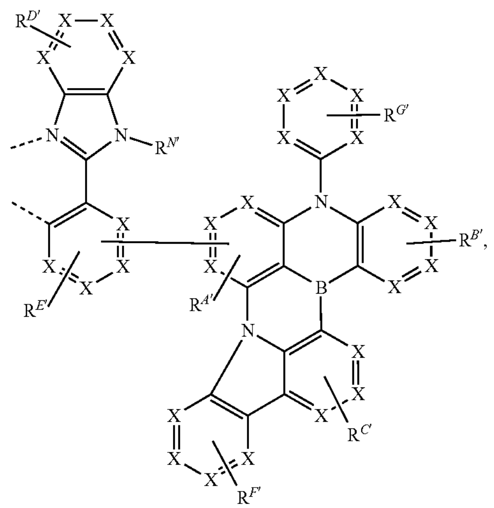

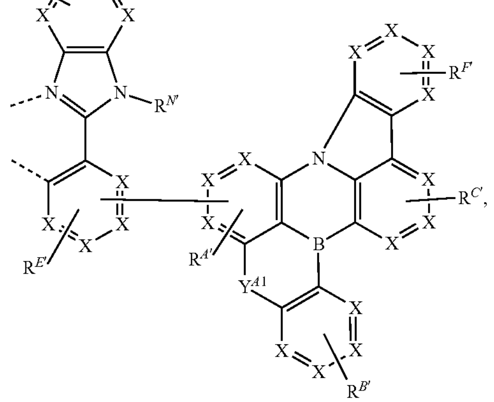

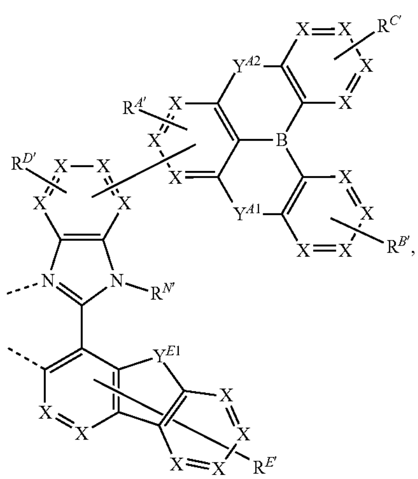

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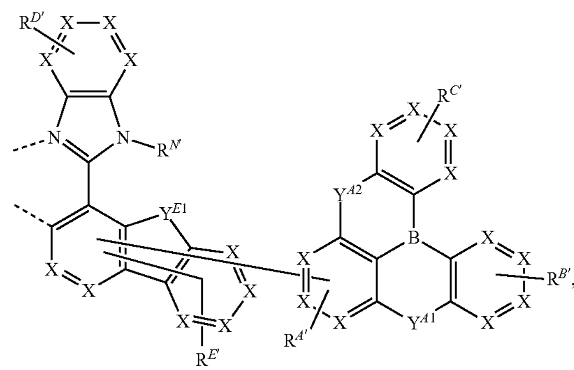

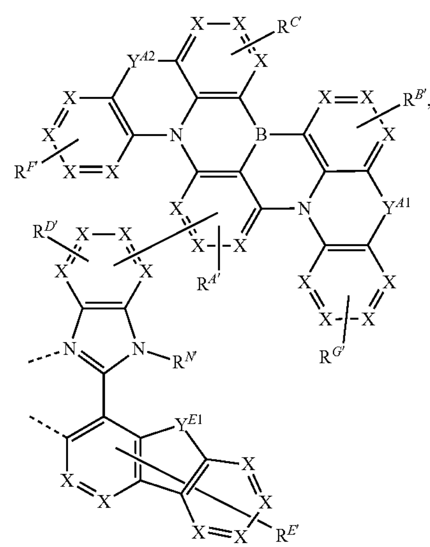

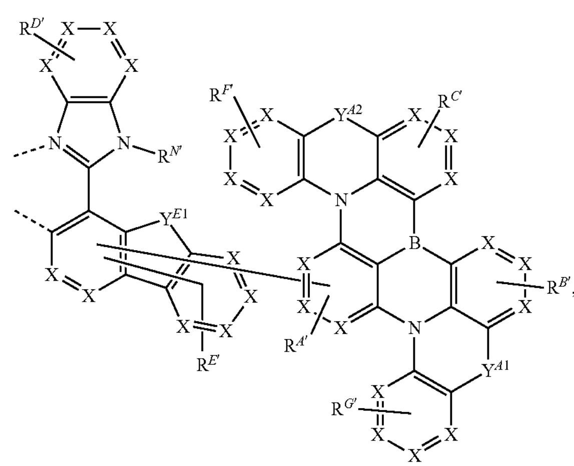

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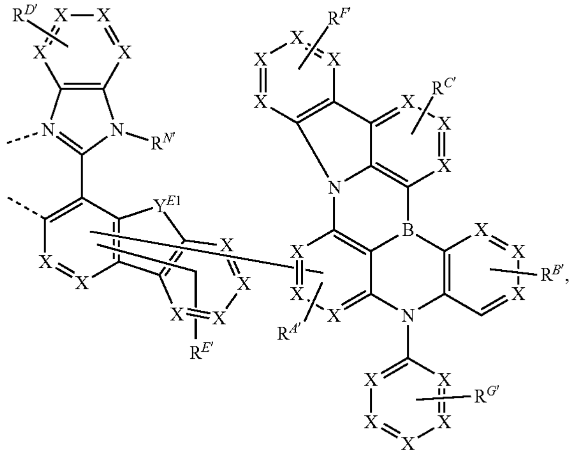

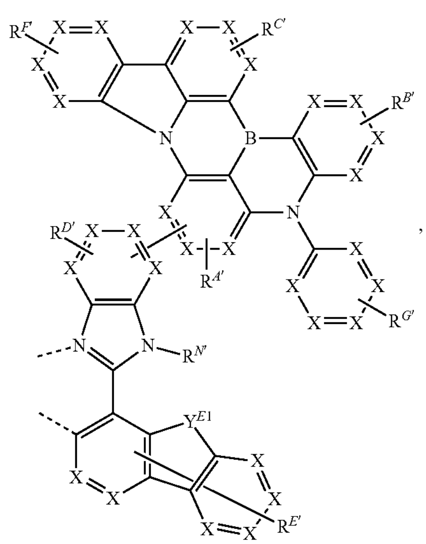

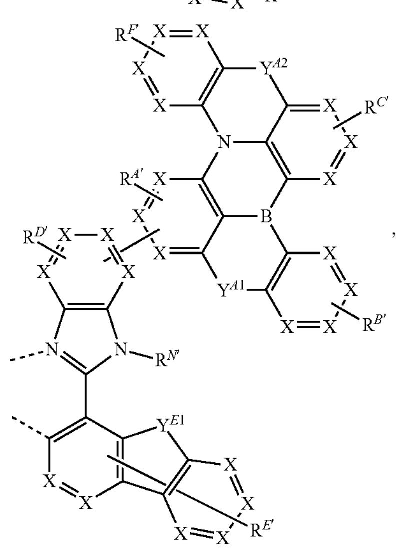

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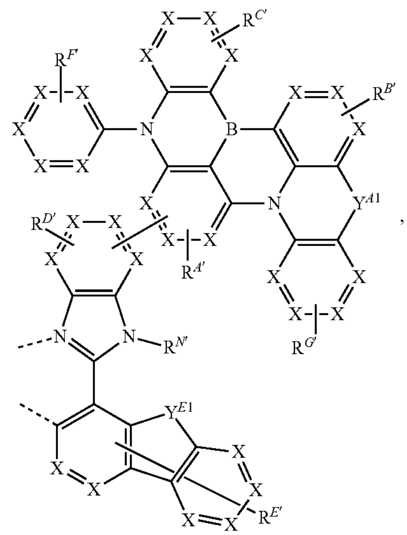

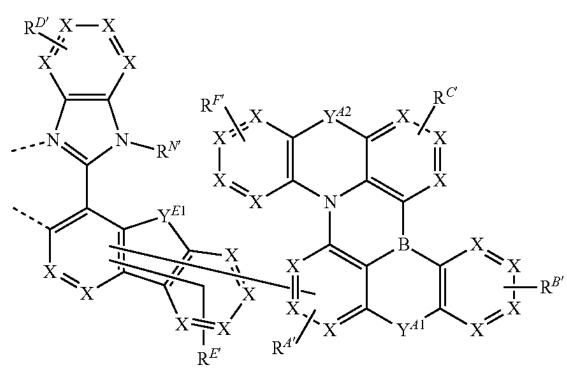

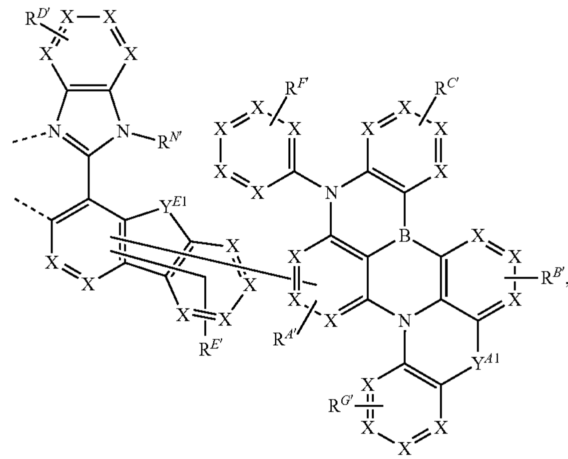

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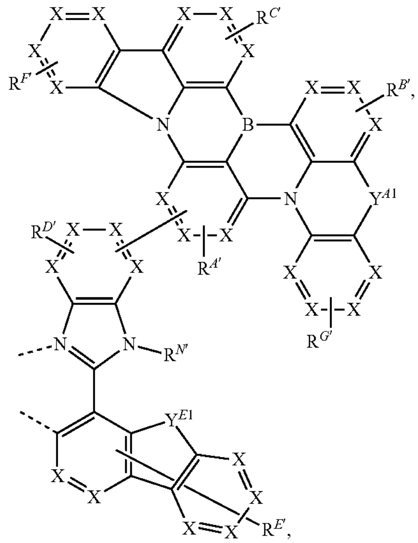

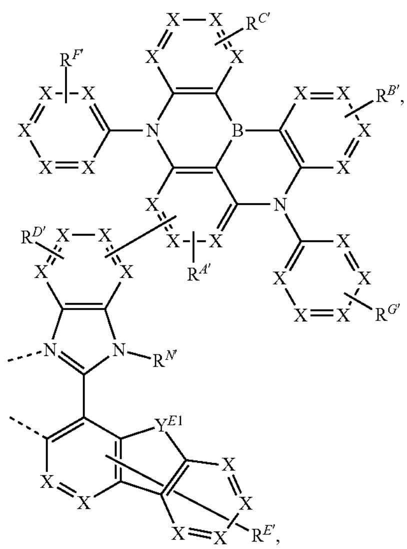

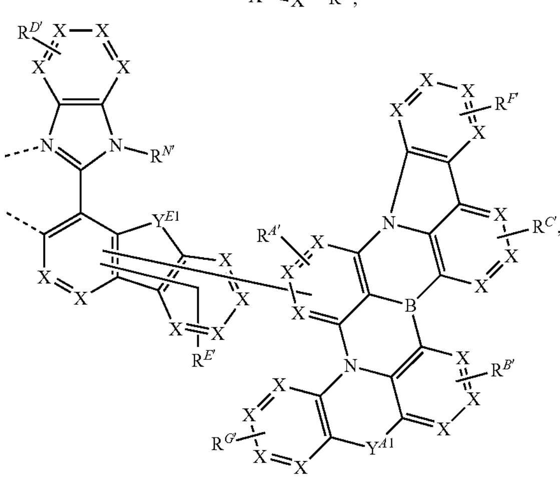

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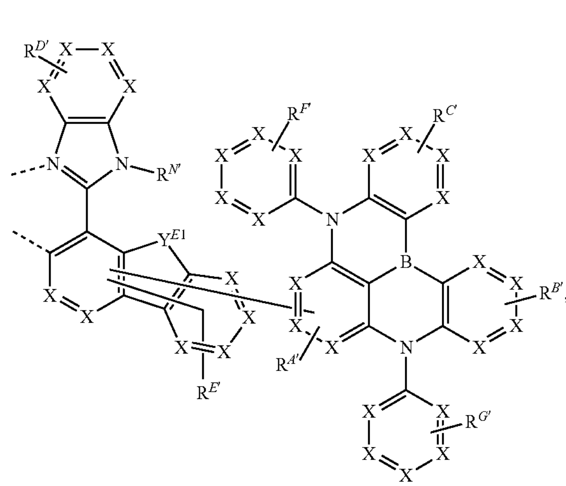

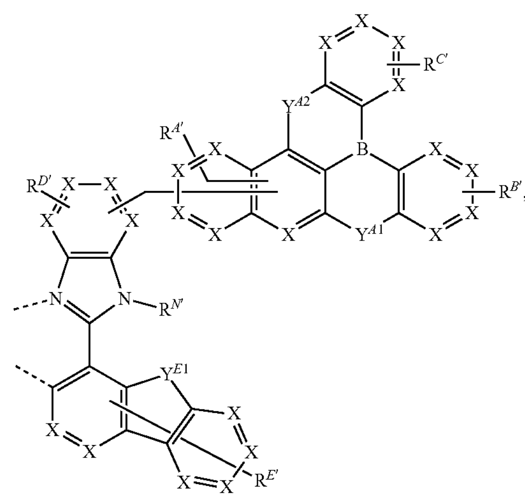

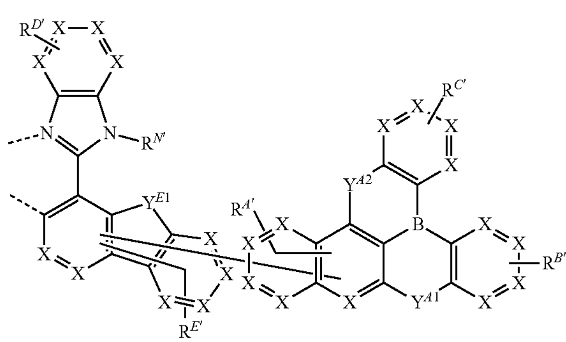

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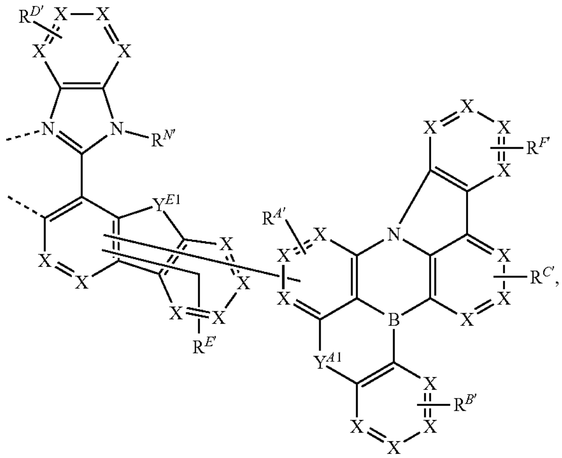

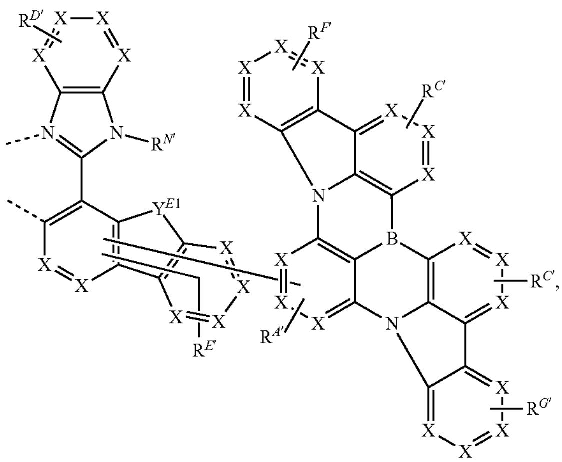

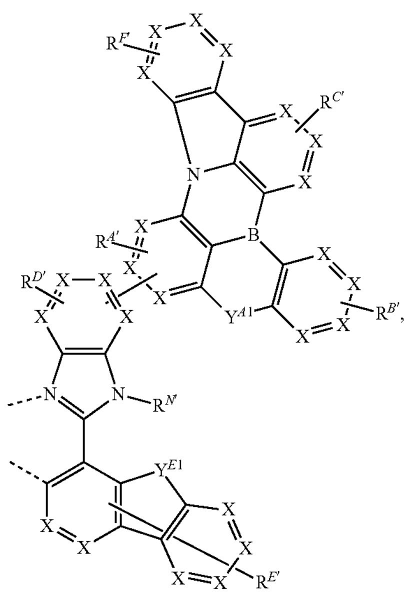

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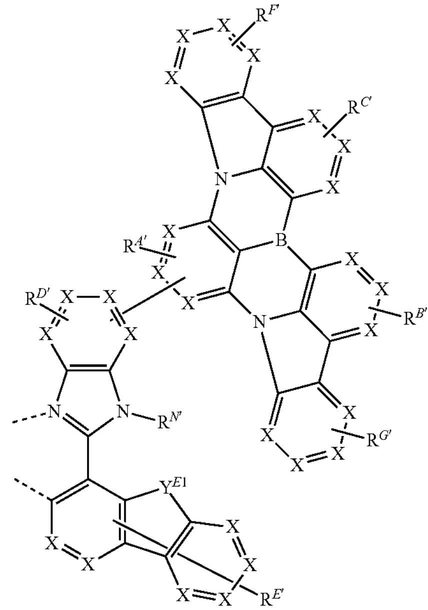

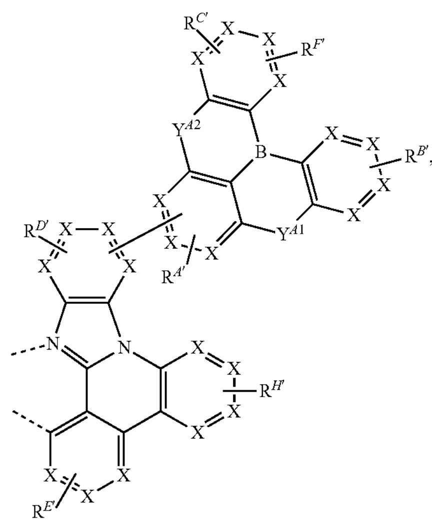

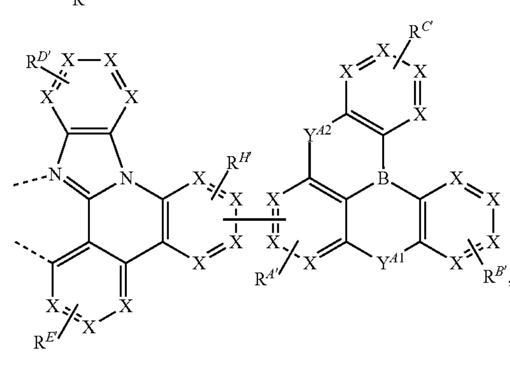

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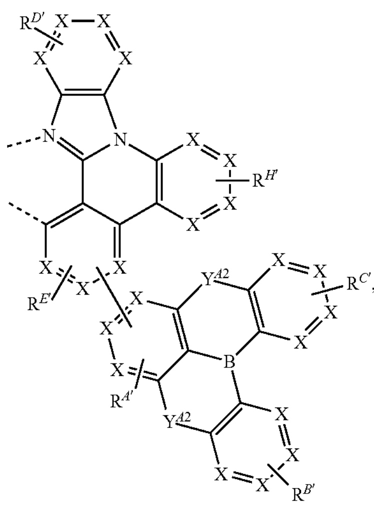

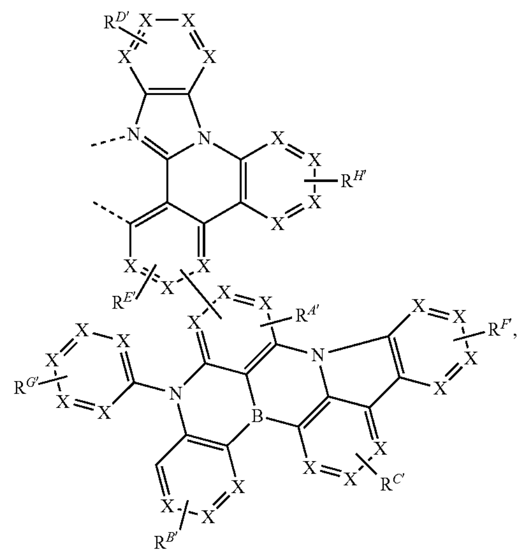

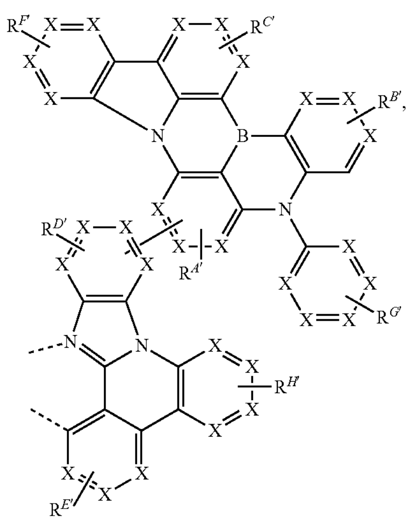

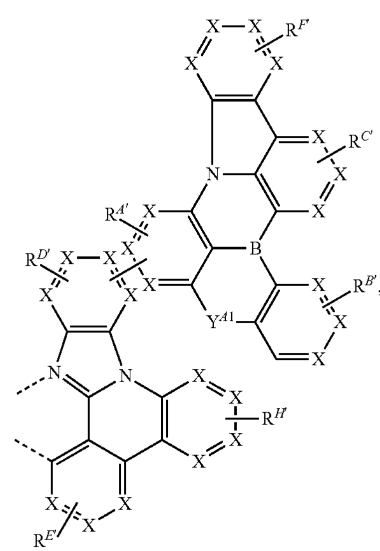

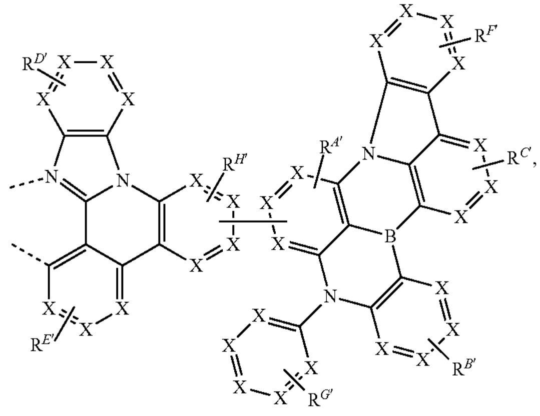

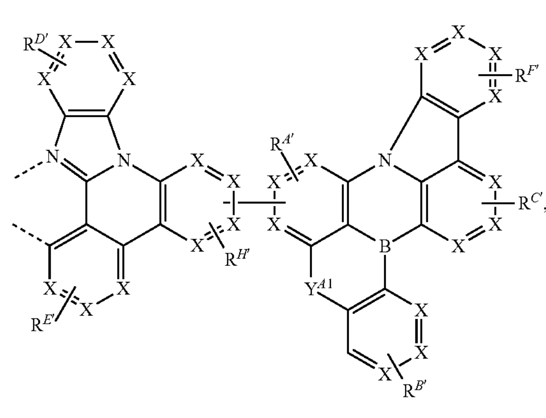

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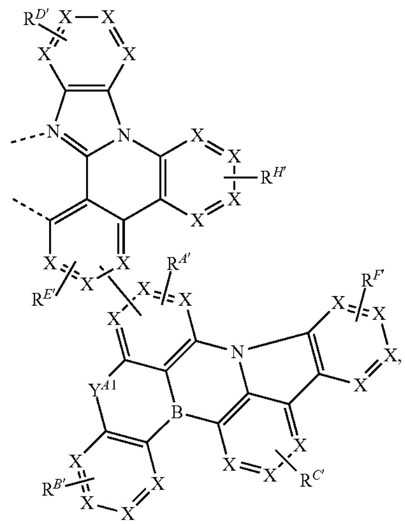

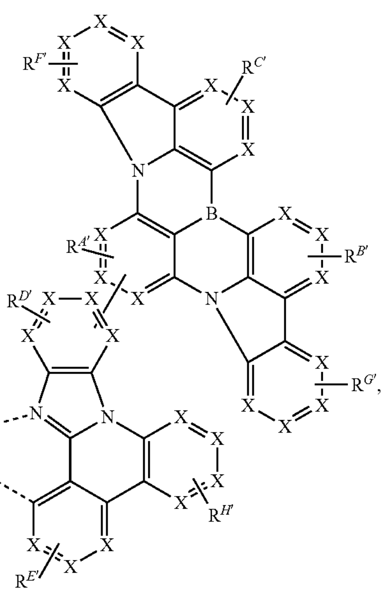

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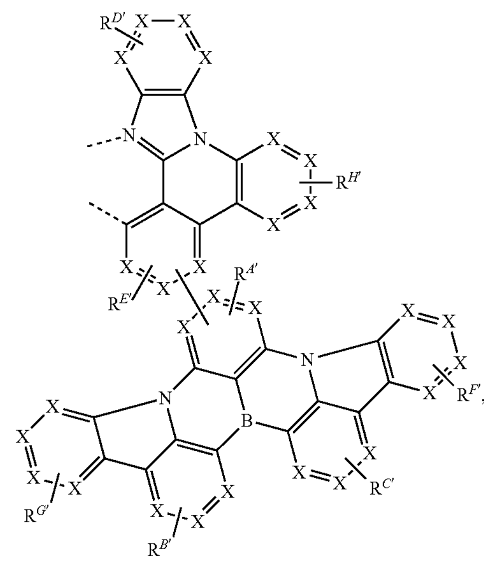

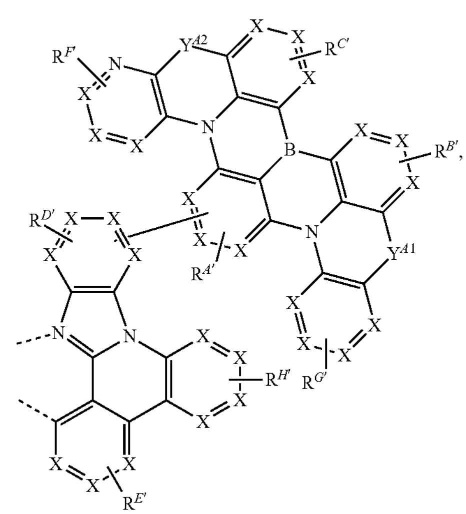

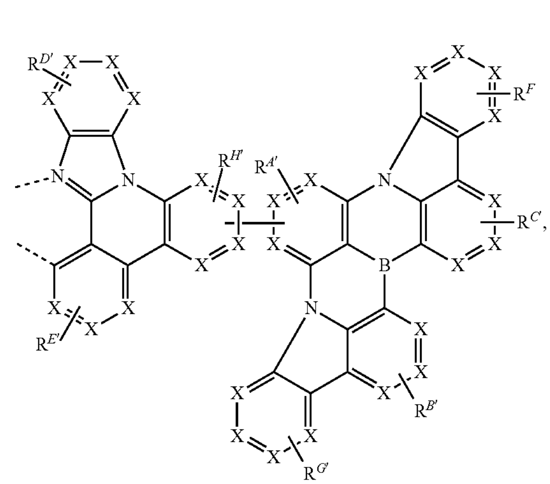

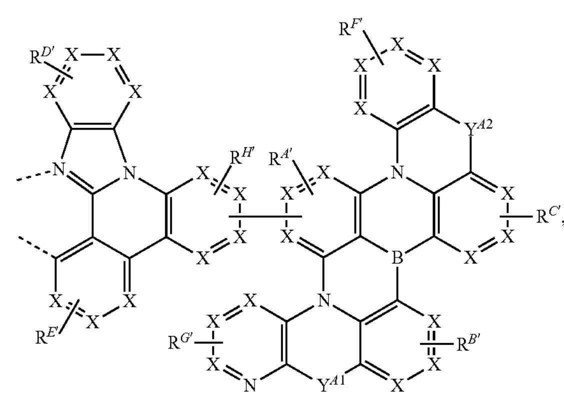

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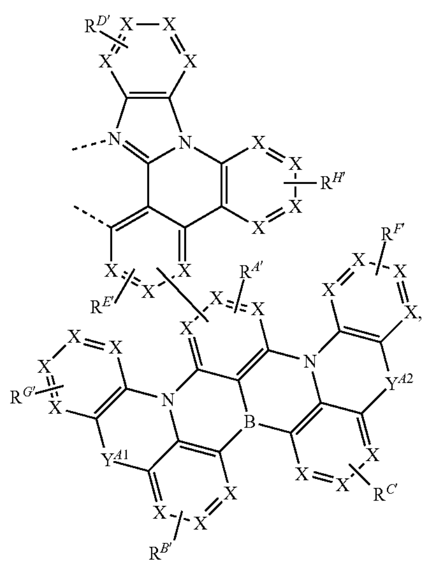

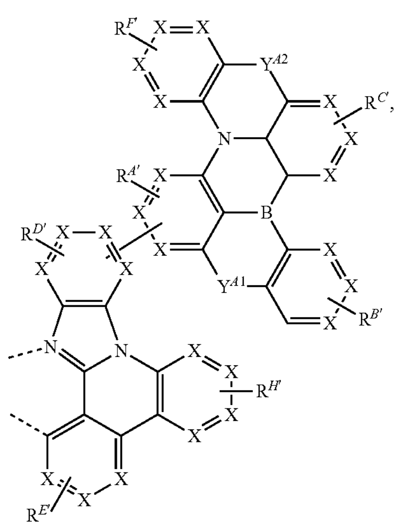

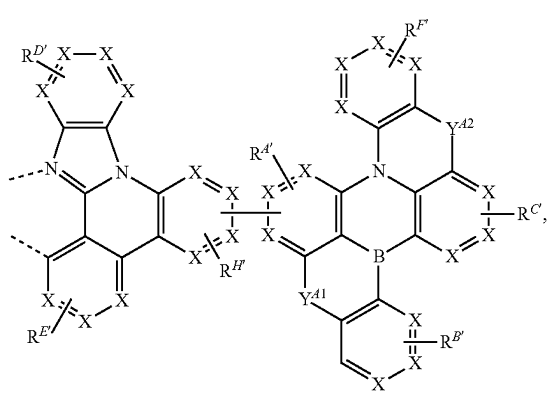

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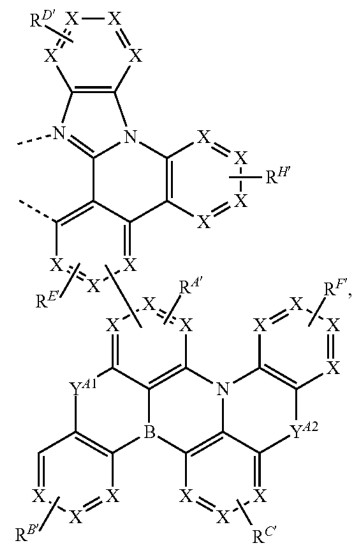

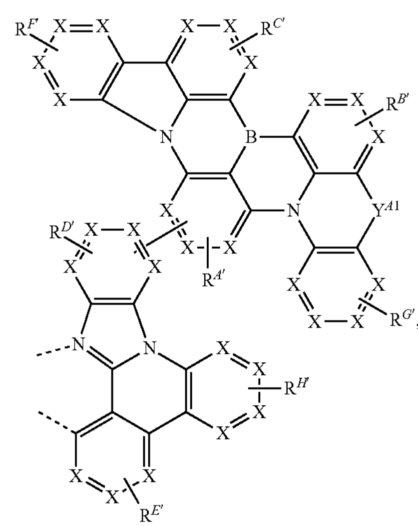

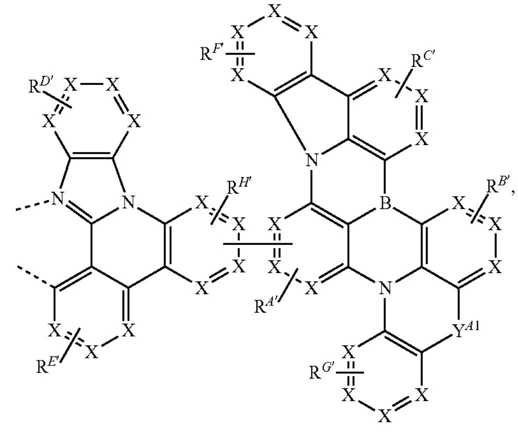

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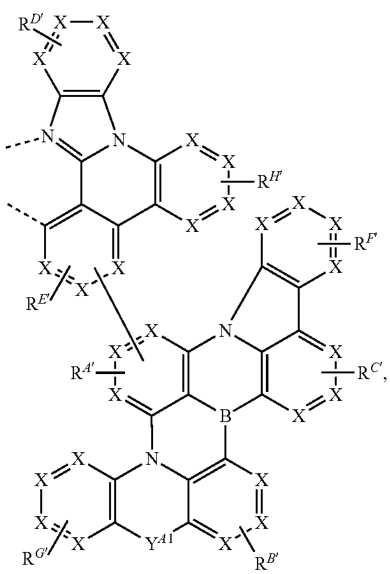

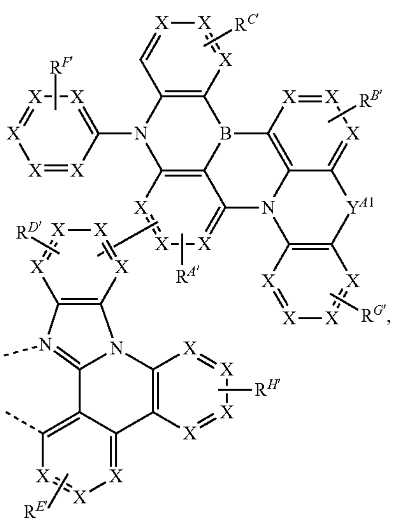

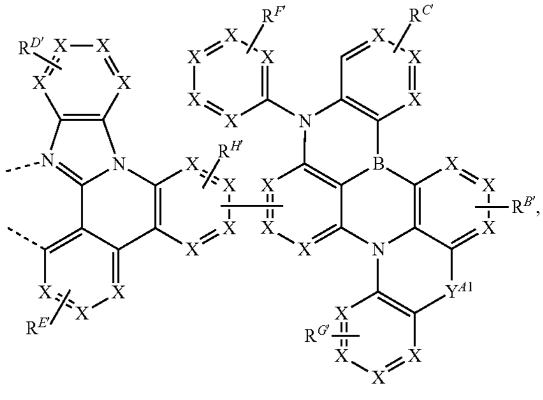

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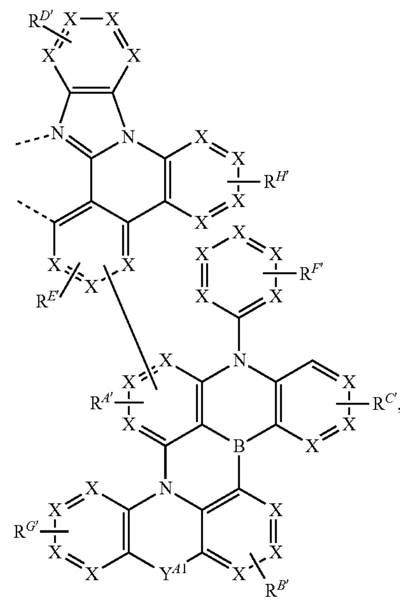

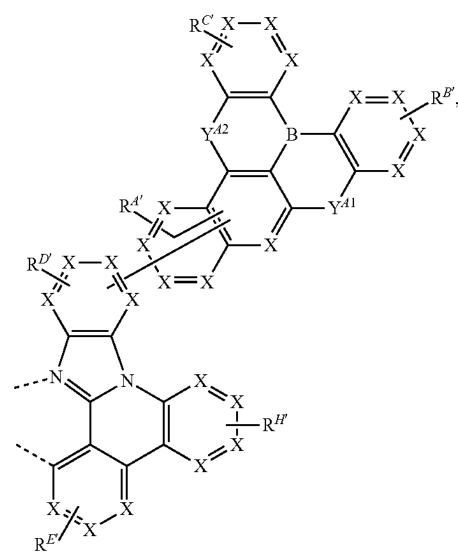

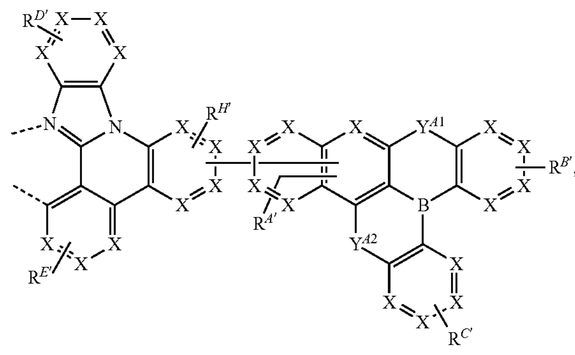

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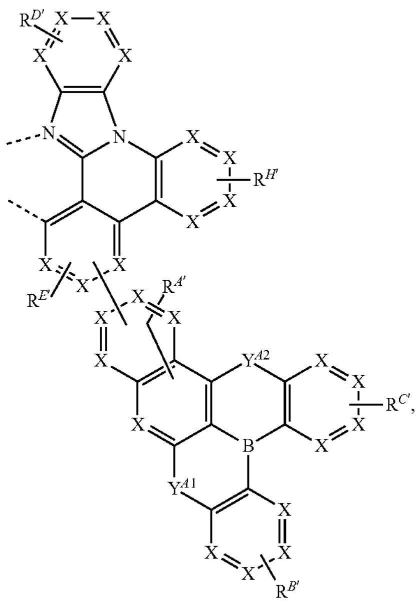

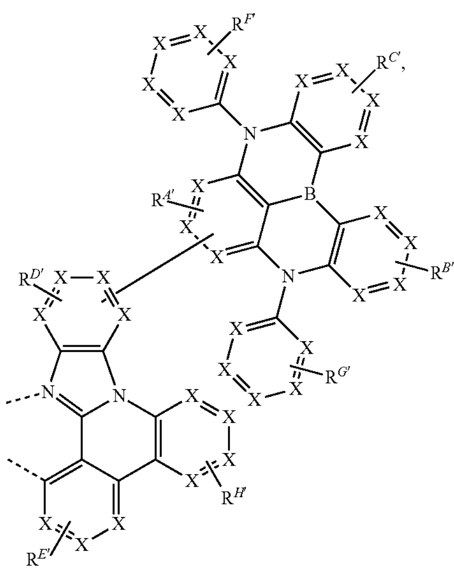

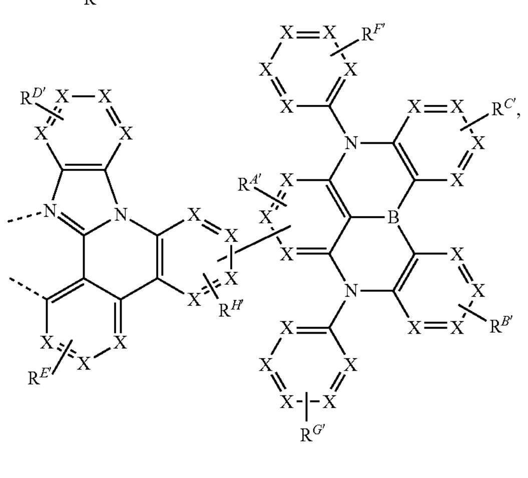

and

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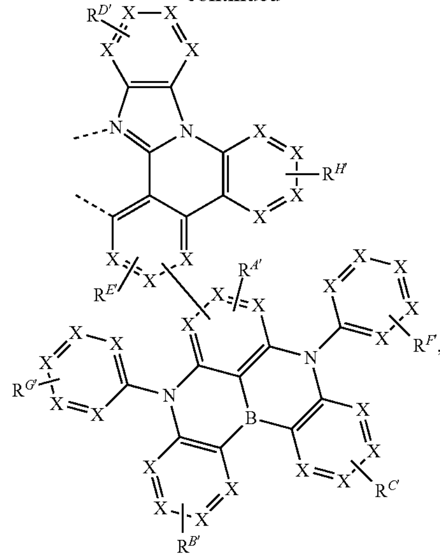


wherein

for each occurrence, X is independently C or N;

each of Y^{A1}, Y^{A2} and Y^{E1} is independently selected from the group consisting of a direct bond, BR, BRR', NR, PR, P(O)R, O, S, Se, C═O, C═S, C═Se, C═NR, C═CRR', S═O, SO_2, CR, CRR', SiRR', and GeRR';

each of the two - - - - represents chelation to the metal;

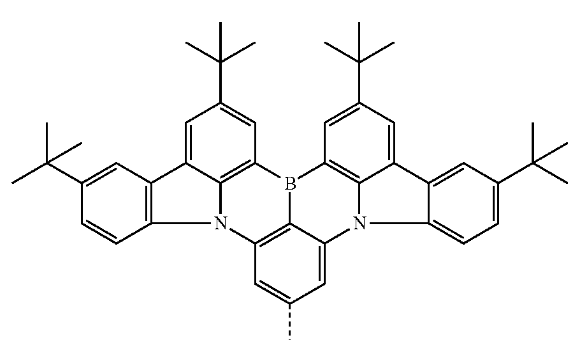
each of $R^{A'}$, $R^{B'}$, $R^{C'}$, $R^{D'}$, $R^{E'}$, $R^{F'}$, $R^{G'}$, and $R^{H'}$ independently represents mono to the maximum allowable substitution;

each R, R', $R^{A'}$, $R^{B'}$, $R^{C'}$, $R^{D'}$, $R^{E'}$, $R^{F'}$, $R^{G'}$, $R^{H'}$, and $R^{N'}$ is independently hydrogen or a substituent selected from the group consisting of deuterium, halogen, alkyl, cycloalkyl, heteroalkyl, heterocycloalkyl, arylalkyl, alkoxy, aryloxy, amino, silyl, germyl, boryl, alkenyl, cycloalkenyl, heteroalkenyl, alkynyl, aryl, heteroaryl, acyl, carboxylic acid, ether, ester, nitrile, isonitrile, sulfanyl, sulfinyl, sulfonyl, phosphino, selenyl, and combinations thereof; and

any two substituents can be joined or fused to form a ring.

7. The compound of claim **1**, wherein the ligand L_A is selected from L_{AAi}, wherein i is an integer from 1 to 15, and each L_{AAi} is defined in the following LIST 5:

L_{AA1}



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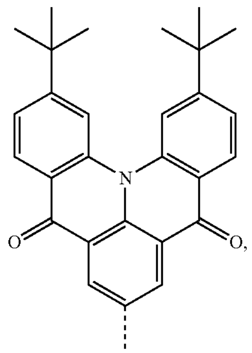
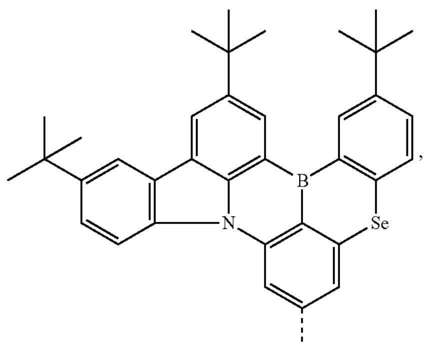
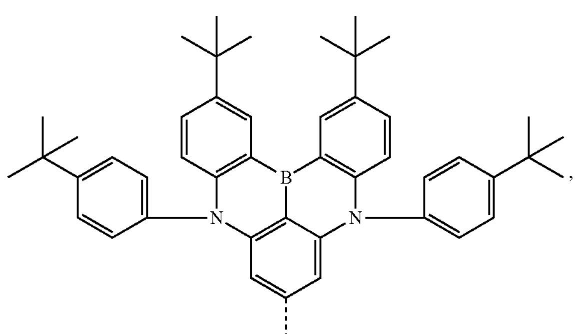
L_{AA2}



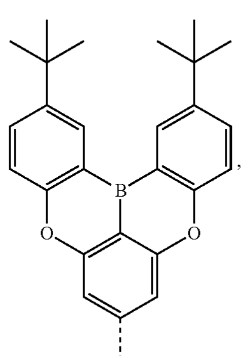
L_{AA3}



L_{AA4}



L_{AA5}



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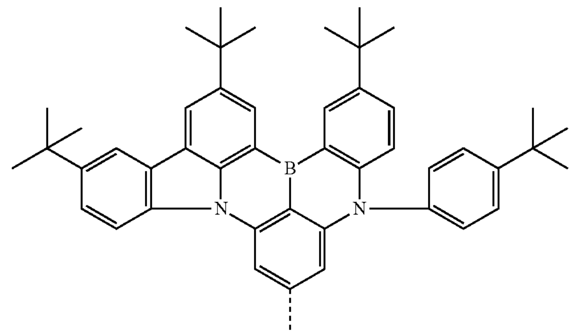
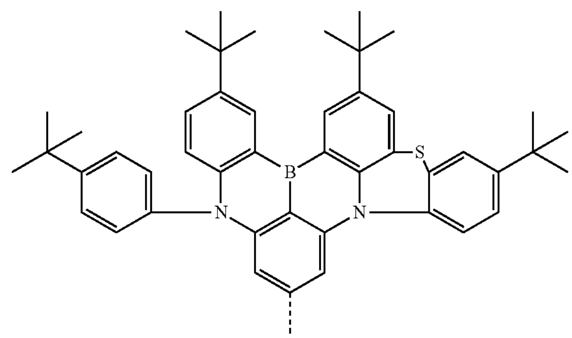
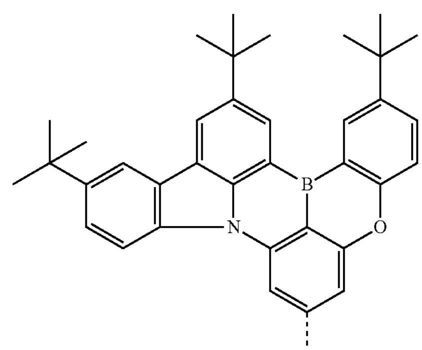
L_{AA6}



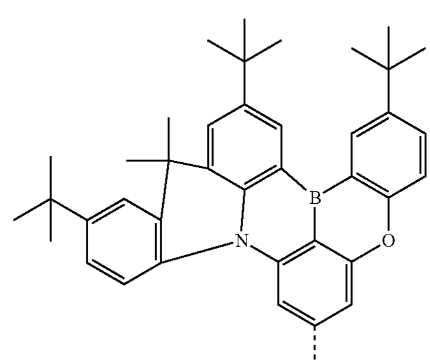
L_{AA7}



L_{AA8}



L_{AA9}



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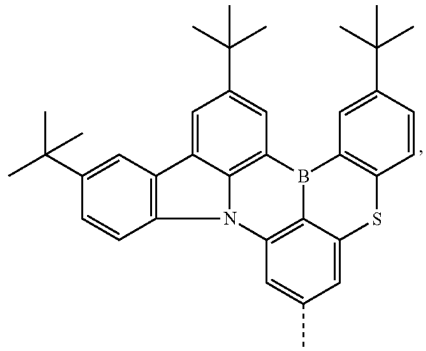
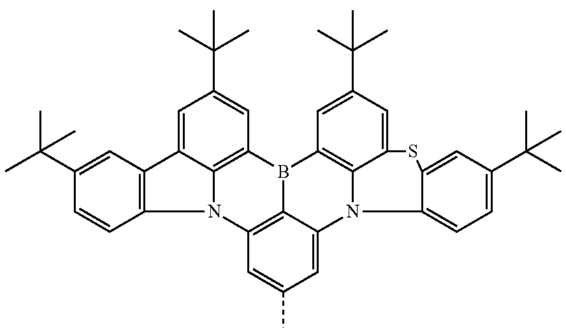
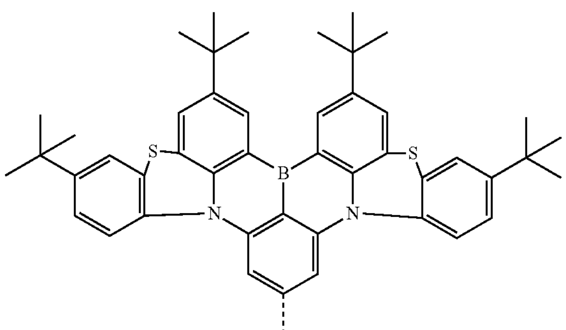
L_{AA10}



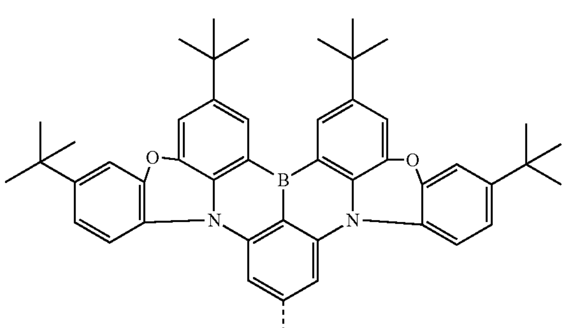
L_{AA11}



L_{AA12}



L_{AA13}



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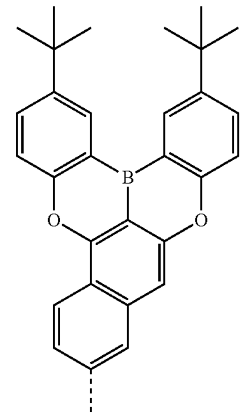
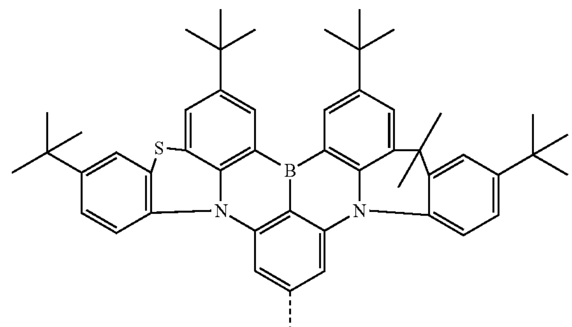
L_{AA14}



and

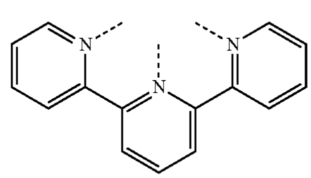
L_{AA15}



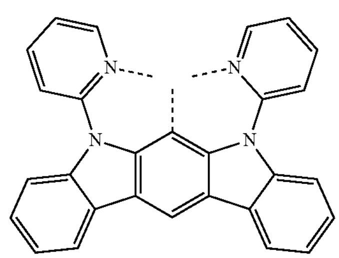
8. The compound of claim **7**, wherein the compound has a Formula $Ir(L_A)(L_{B'})(L_C)$ or $Pt(L_A)(L_{B'})$, wherein $L_{B'}$ is a tridentate ligand, L_C is a bidentate ligand.

9. The compound of claim **8**, wherein $L_{B'}$ is selected from $L_{BBi'}$, wherein i' is an integer from 1 to 53, and each $L_{BBi'}$ is defined in the following LIST 6:

L_{BB1}



L_{BB2}



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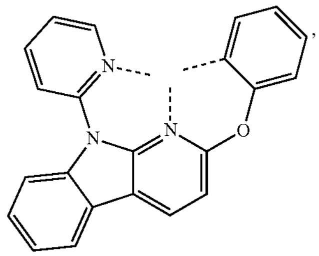
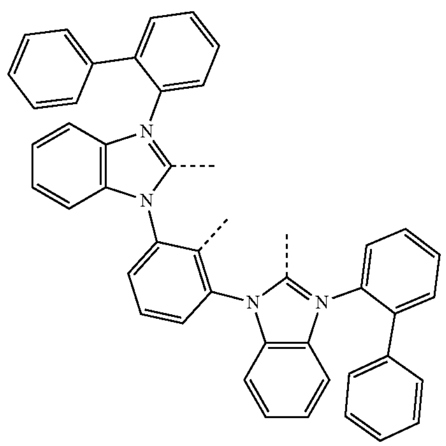
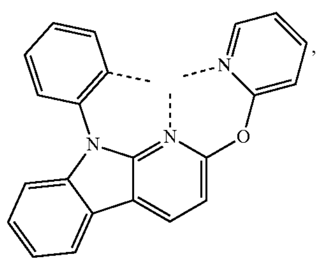
L_{BB3}



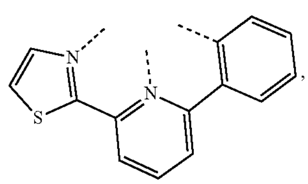
L_{BB4}



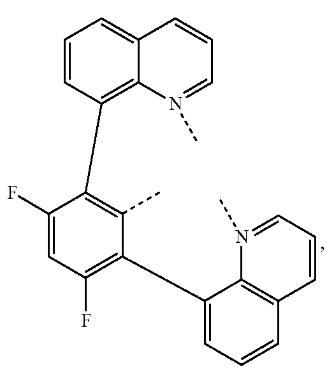
L_{BB5}



L_{BB6}



L_{BB7}



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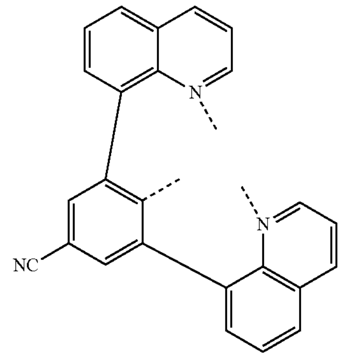
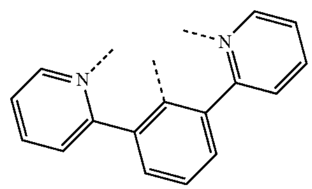
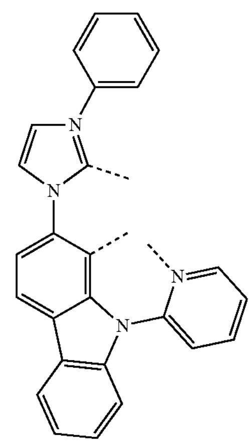
L_{BB8}



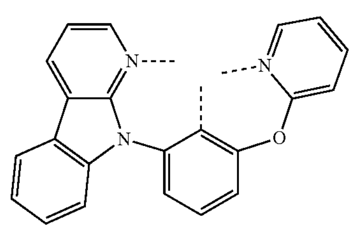
L_{BB9}



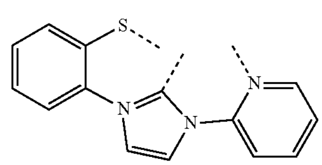
L_{BB10}



L_{BB11}



L_{BB12}



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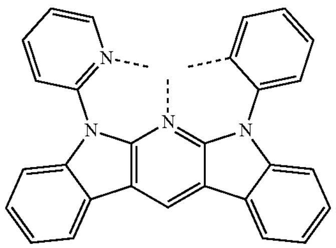
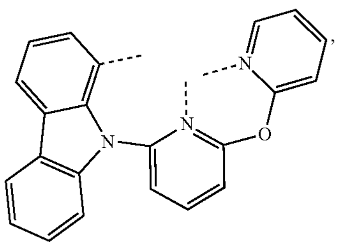
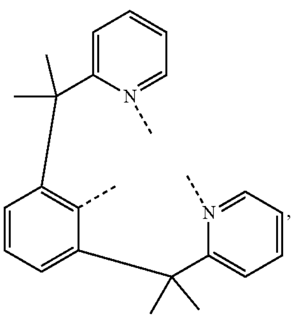
L_{BB13}



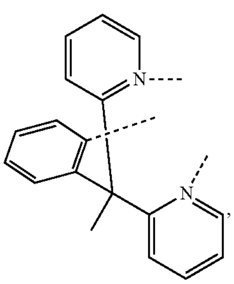
L_{BB14}



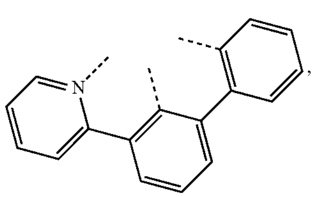
L_{BB15}



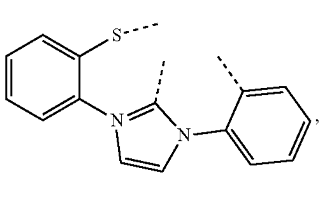
L_{BB16}



L_{BB17}



L_{BB18}



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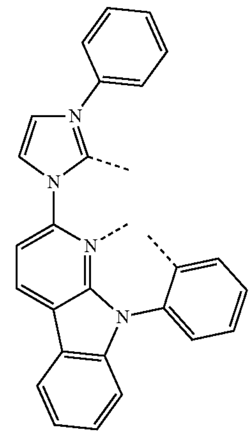
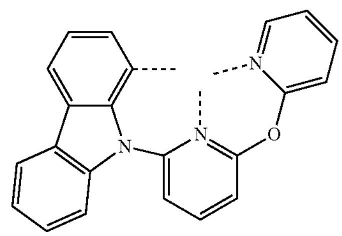
L_{BB19}



L_{BB20}



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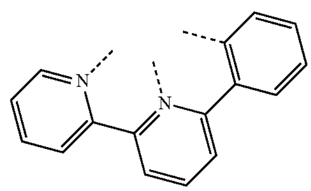
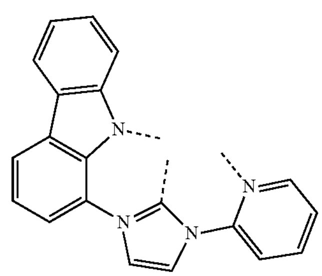
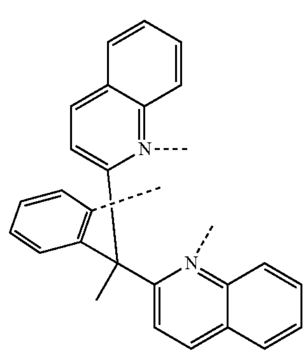


L_{BB22}



L_{BB23}



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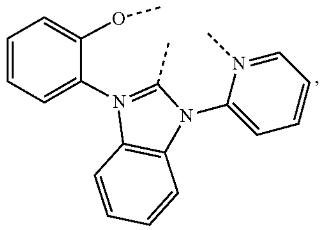
L_{BB24}



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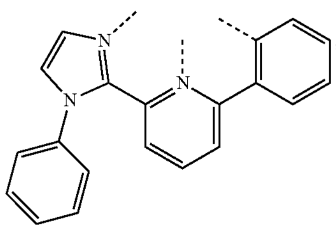
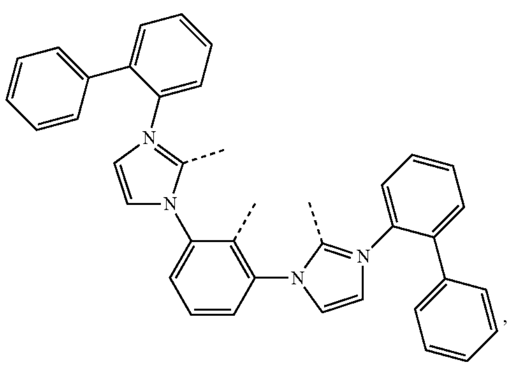
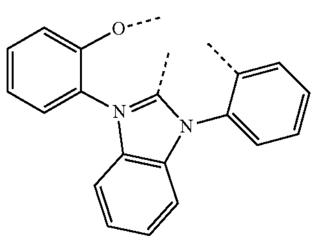


L_{BB26}



L_{BB27}



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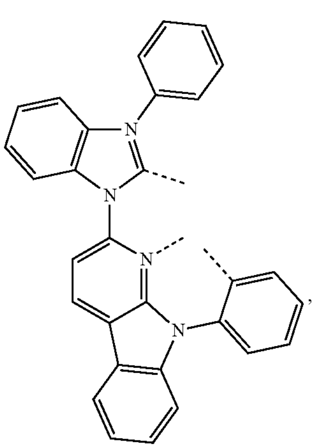

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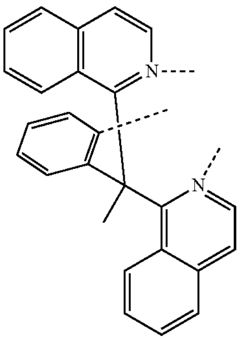

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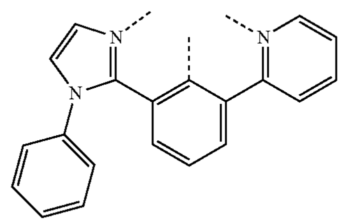

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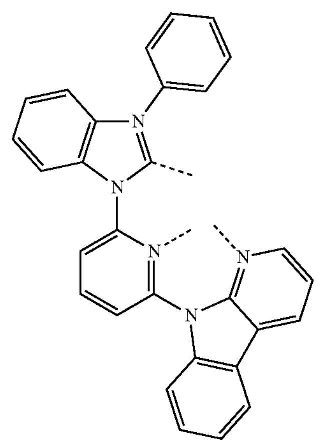

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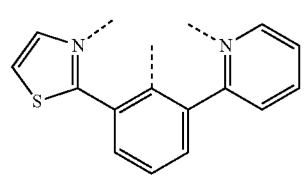

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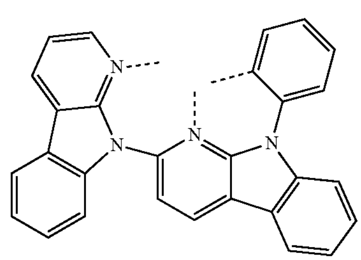

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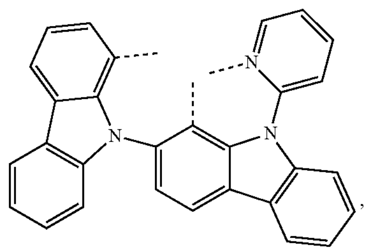
L_{BB34}



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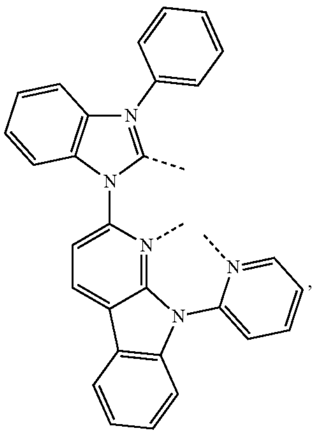

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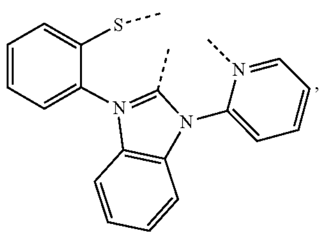

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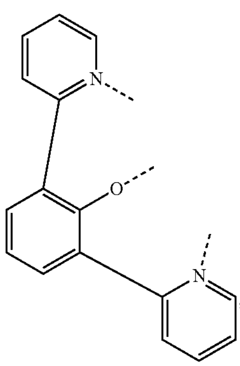

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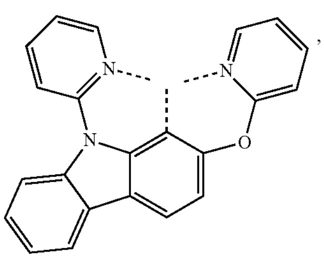

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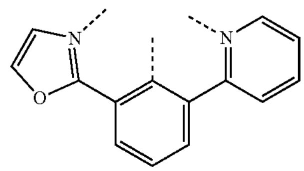

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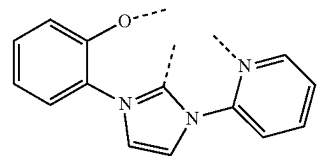

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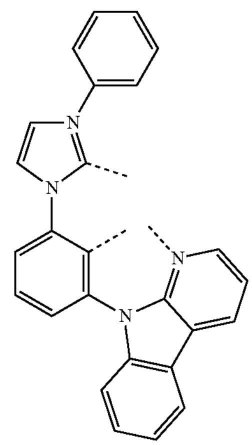
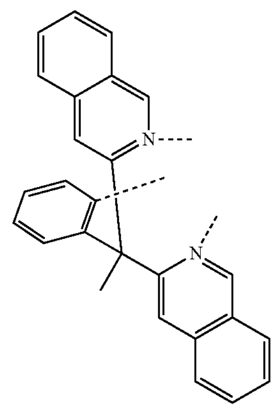
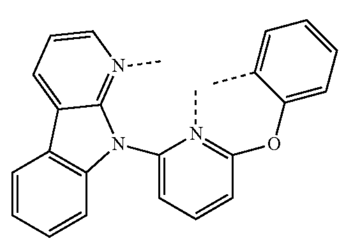


L_{BB42}



L_{BB43}



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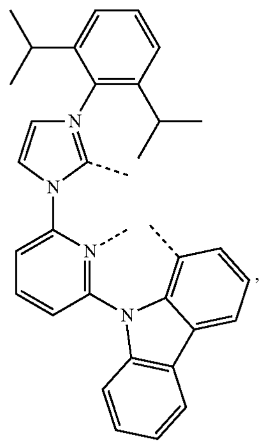
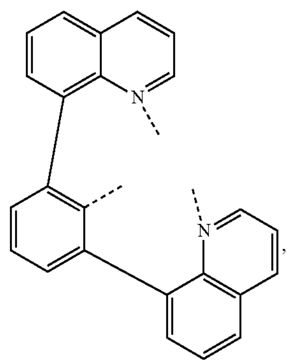
L_{BB44}



L_{BB45}



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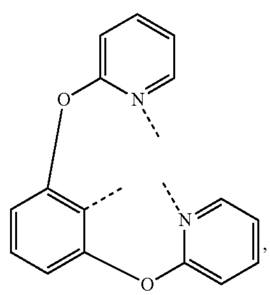
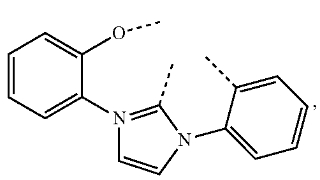
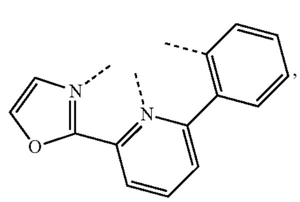


L_{BB47}



L_{BB48}



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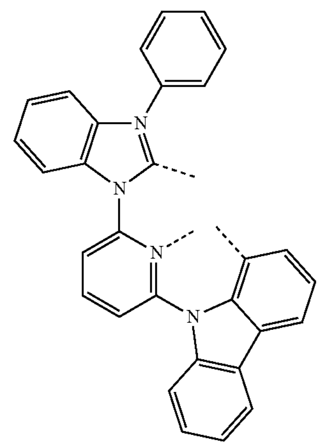
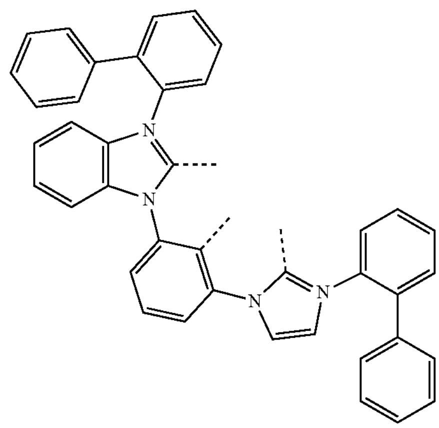
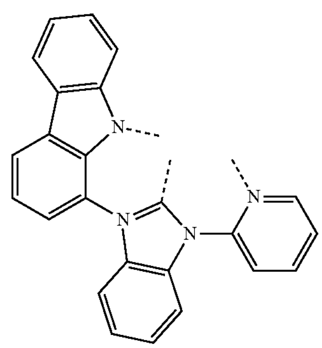
L_{BB49}



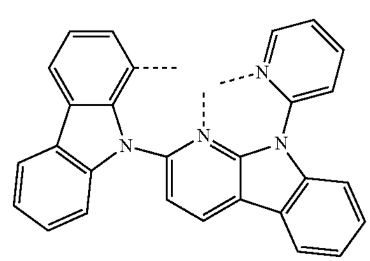
L_{BB50}



L_{BB51}



L_{BB52}



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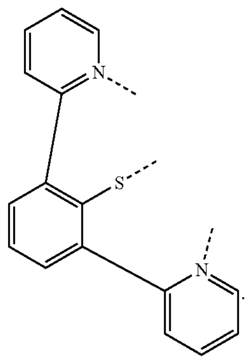
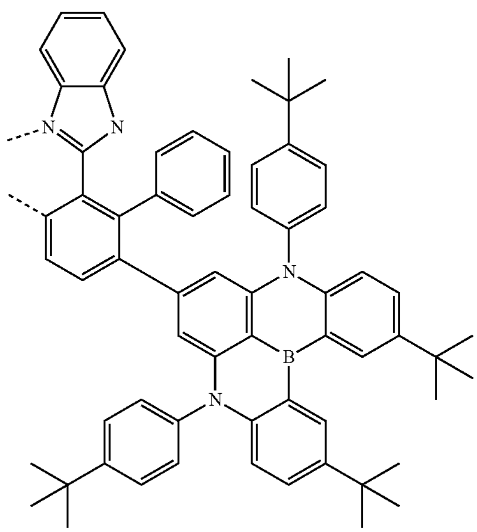
L_{BB53}

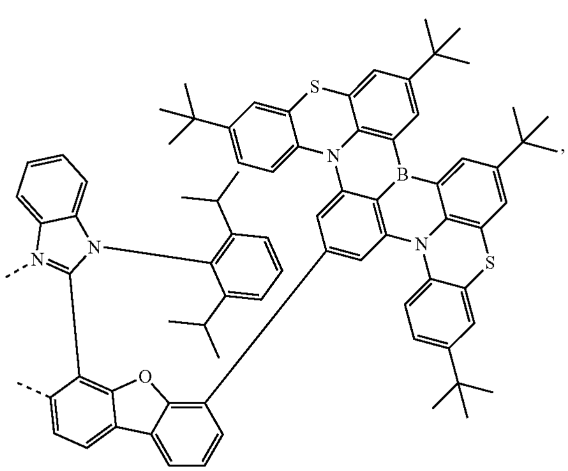


10. The compound of claim **1**, wherein the ligand L_A is selected from $L_{Ai''}$, wherein i" is an integer from 1 to 241, and each $L_{Ai''}$ is defined in the following LIST 7:

L_{A1}



L_{A2}



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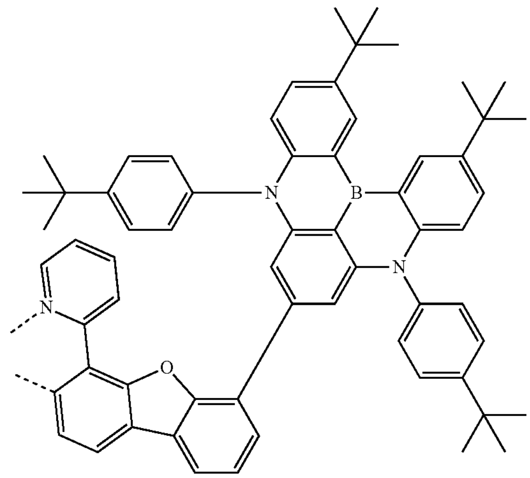
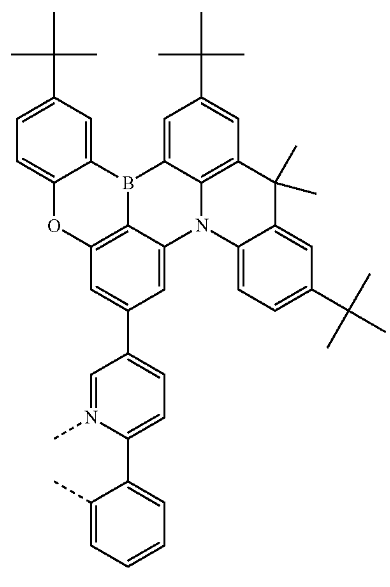
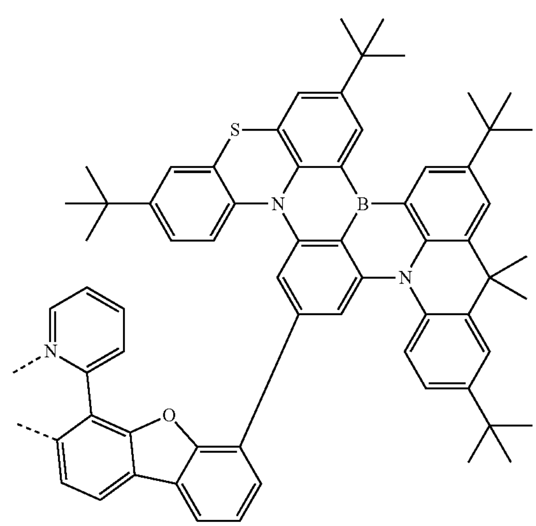
L_{A3}



L_{A4}



L_{A5}



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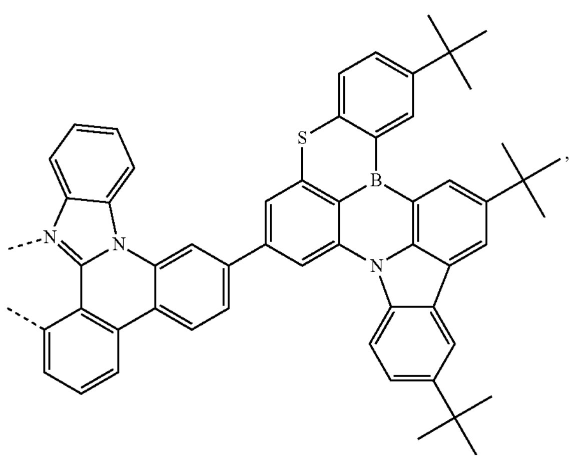
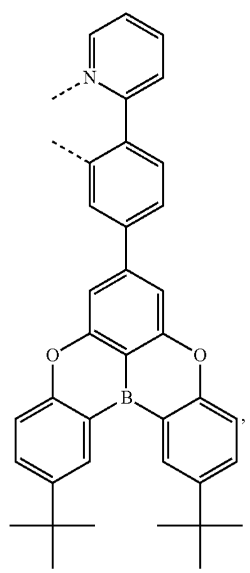
L_{A6}



L_{A7}



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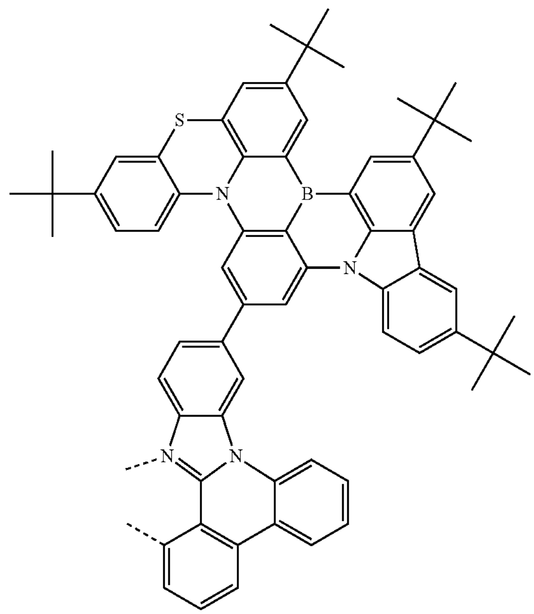
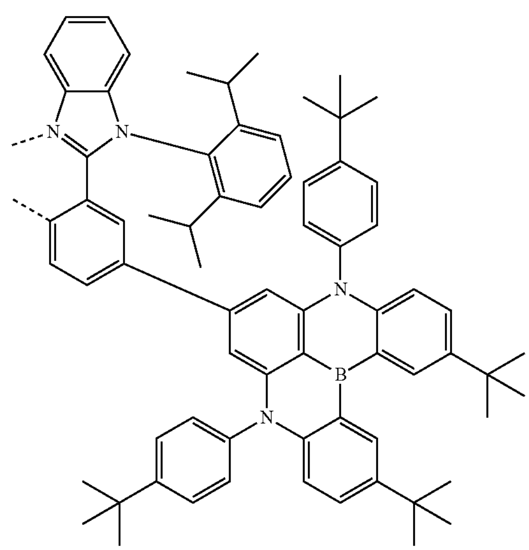
L_{A8}



L_{A9}



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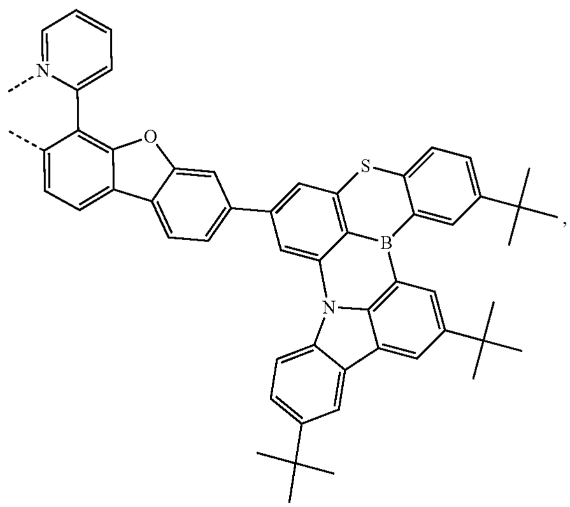
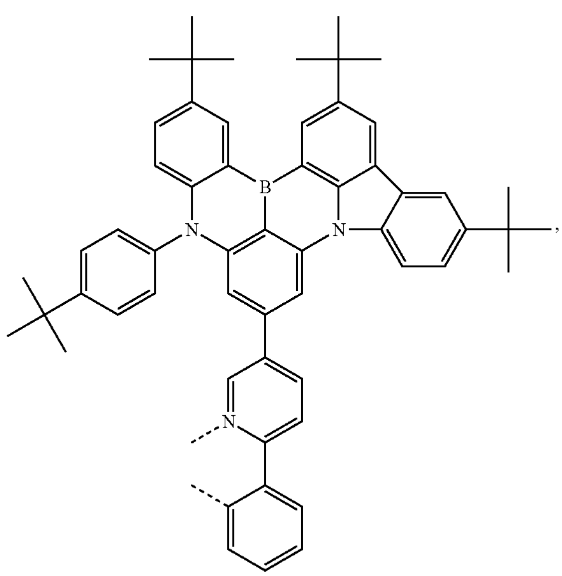
L$_{A10}$



L$_{A11}$



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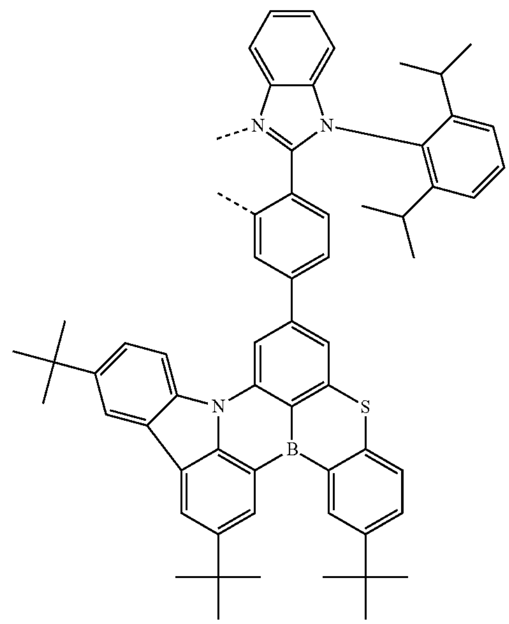
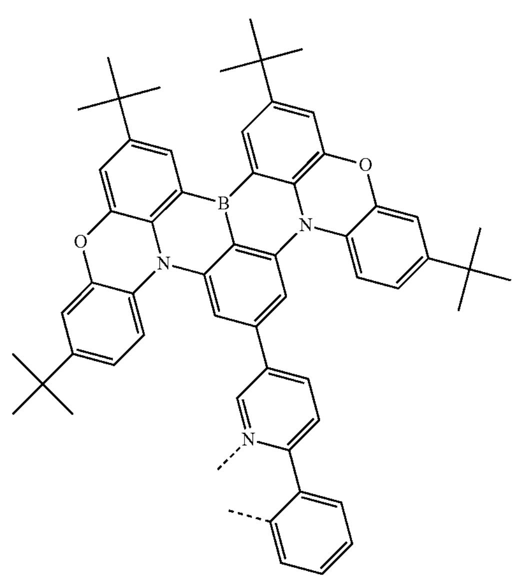
L$_{A12}$



L$_{A13}$



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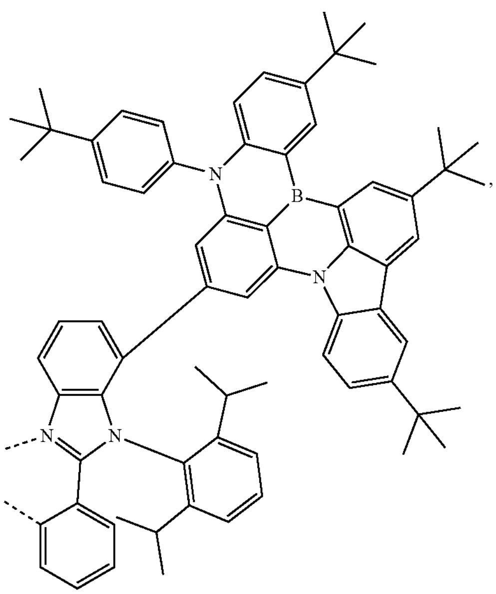
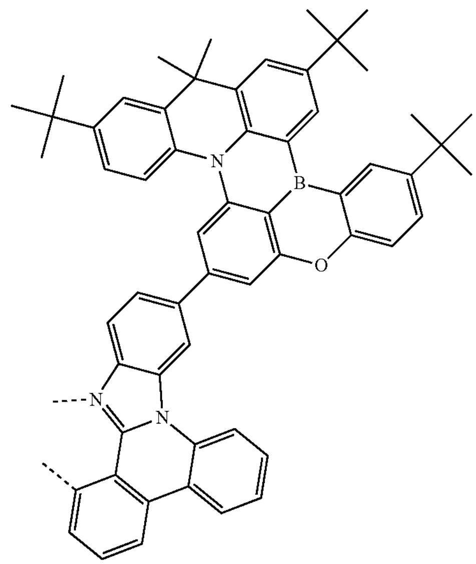
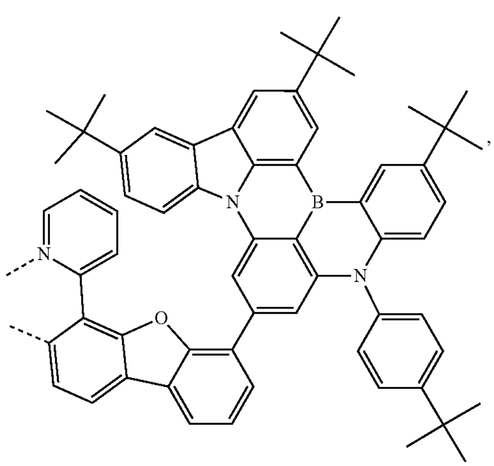
L_{A14}



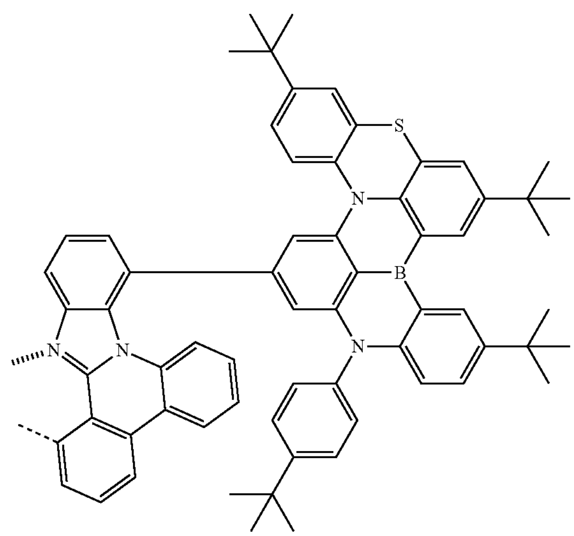
L_{A16}



L_{A15}



L_{A17}



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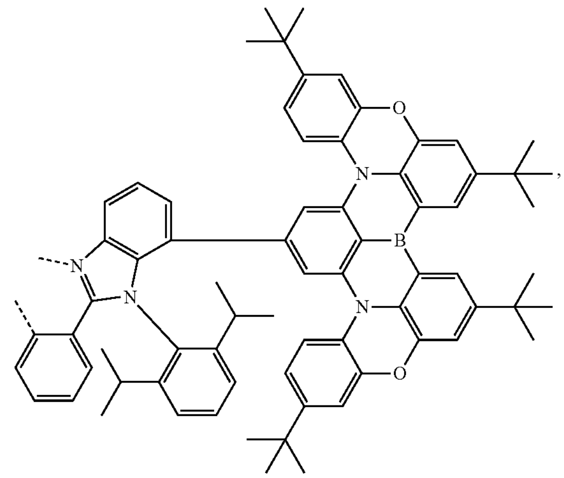
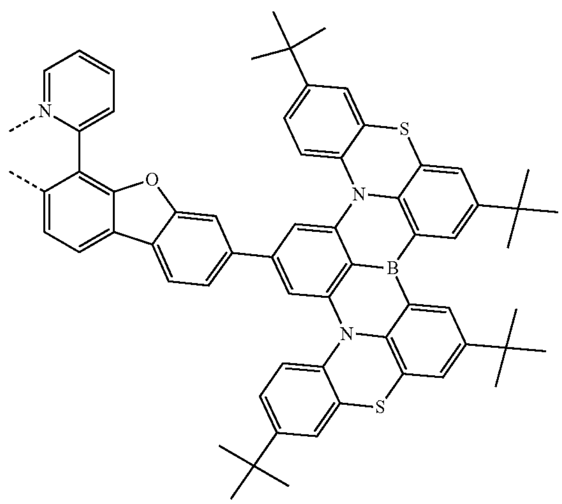
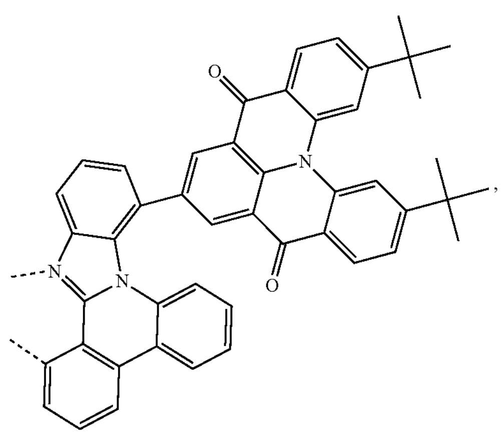
L_{A18}



L_{A19}



L_{A20}



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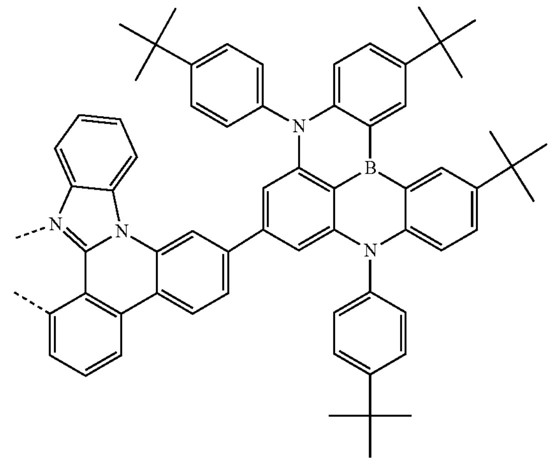
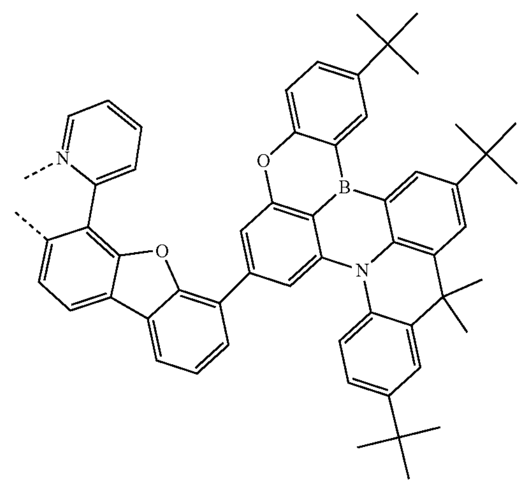
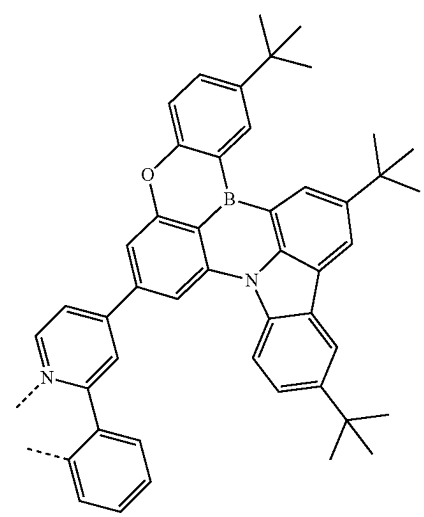
L_{A21}



L_{A22}



L_{A23}



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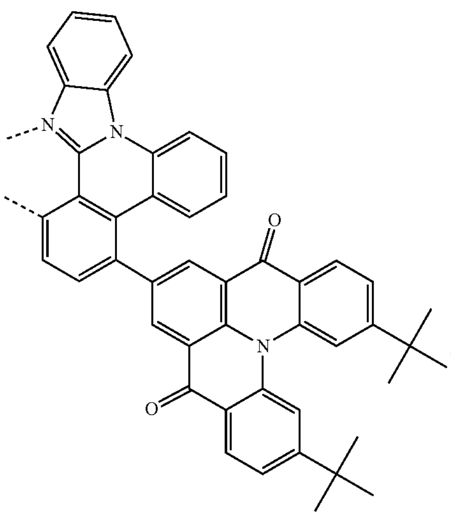
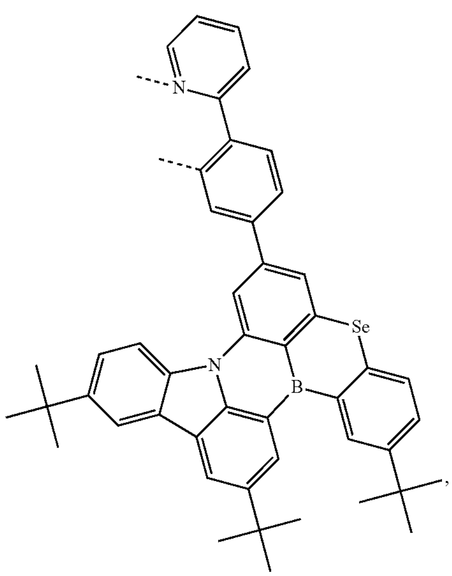
L$_{A24}$



L$_{A25}$



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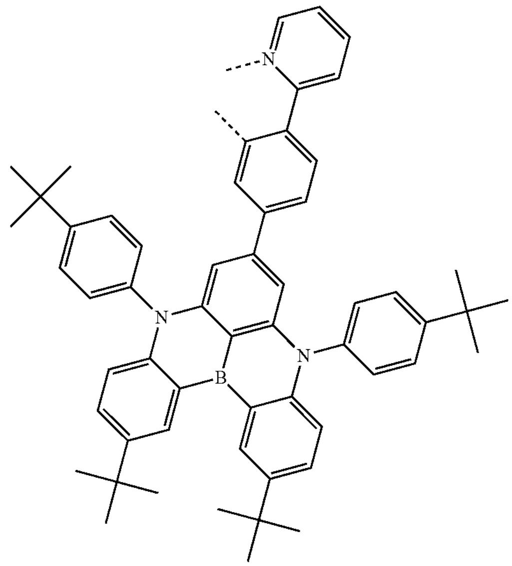

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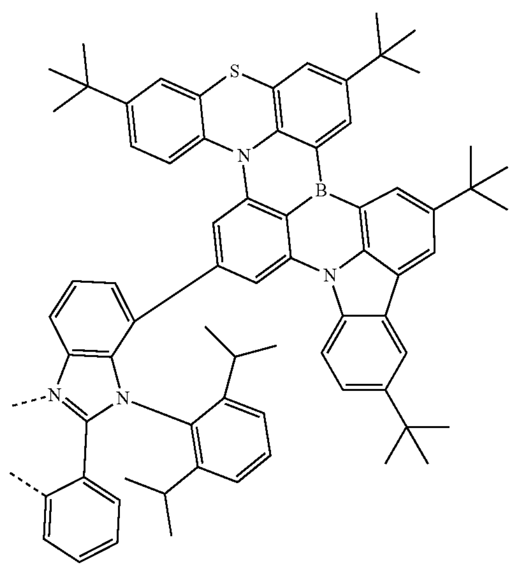

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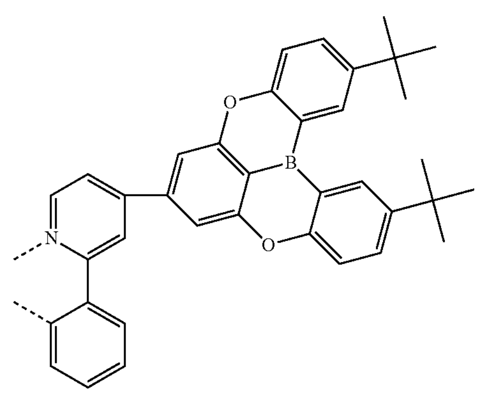

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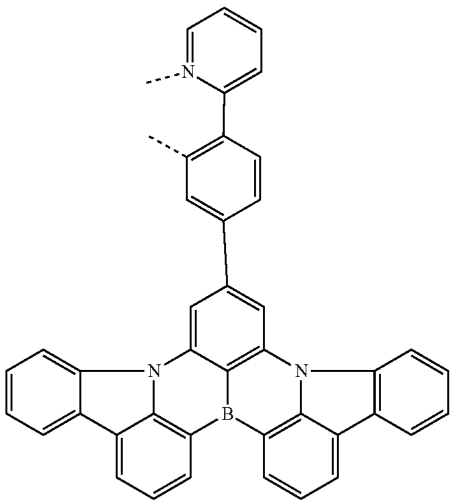
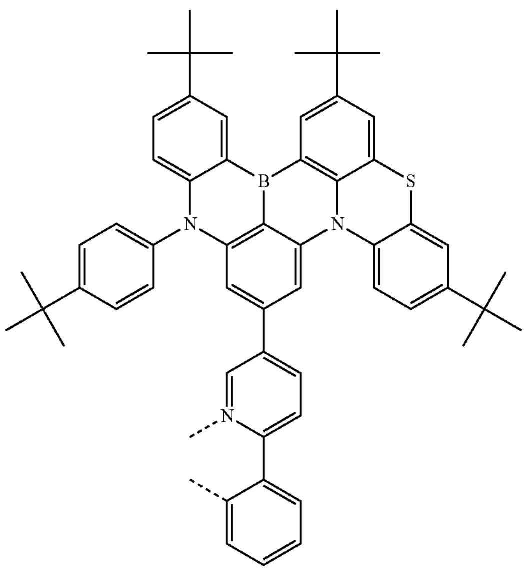
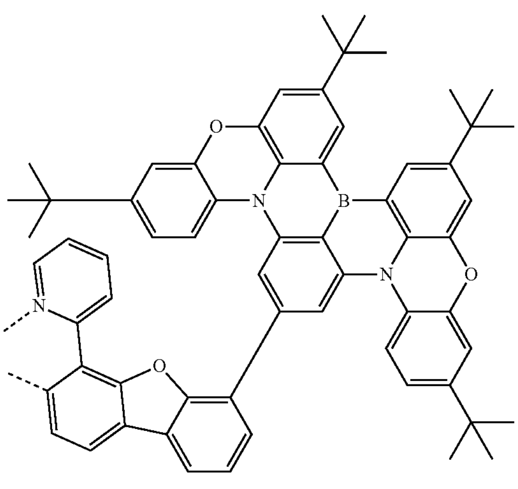
L_{A29}



L_{A30}



L_{A31}



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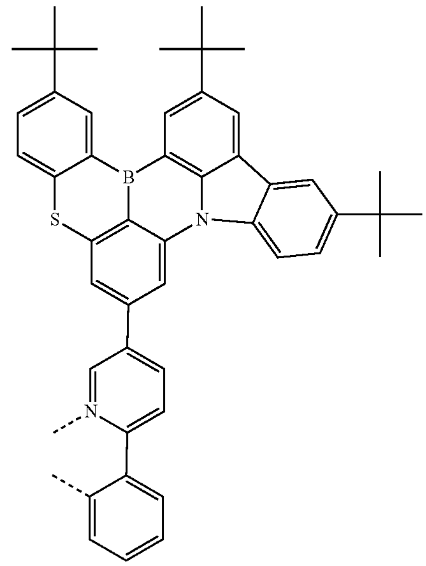
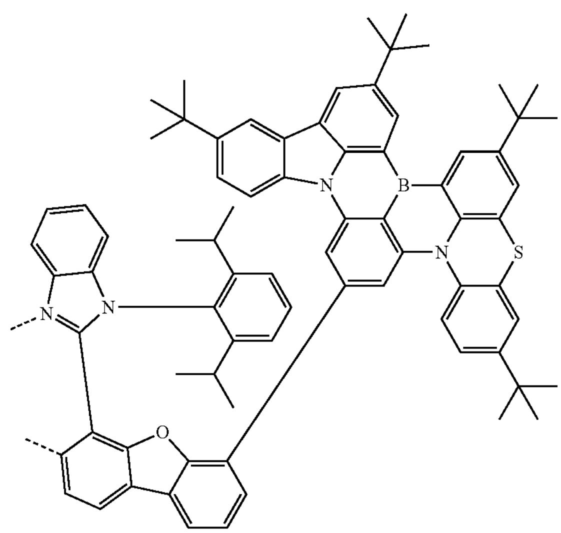
L_{A32}



L_{A33}



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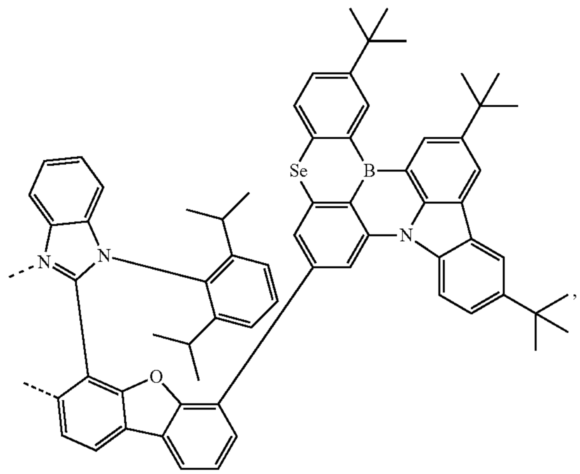
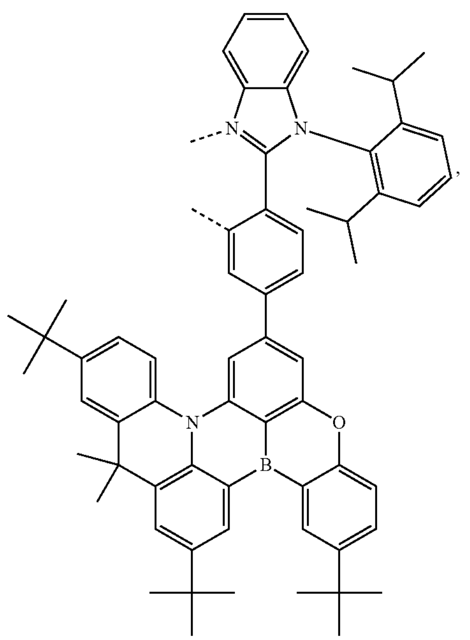
L_{A34}



L_{A35}



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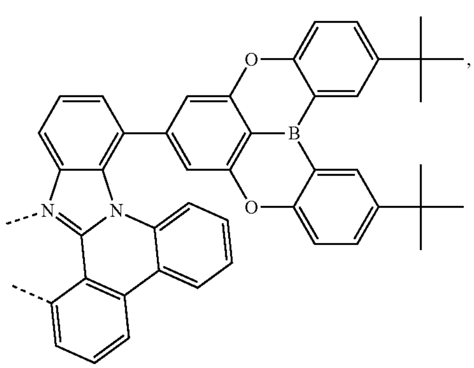

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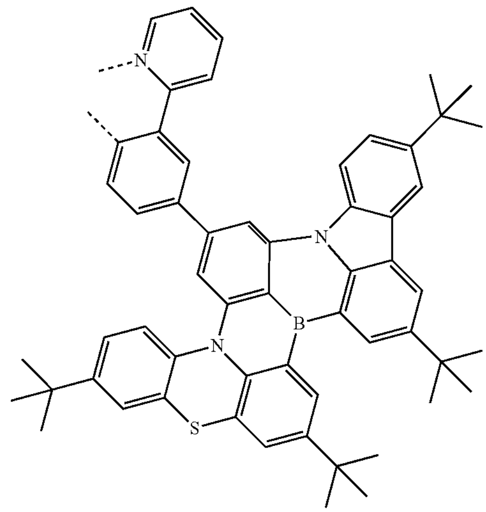
L_{A37}



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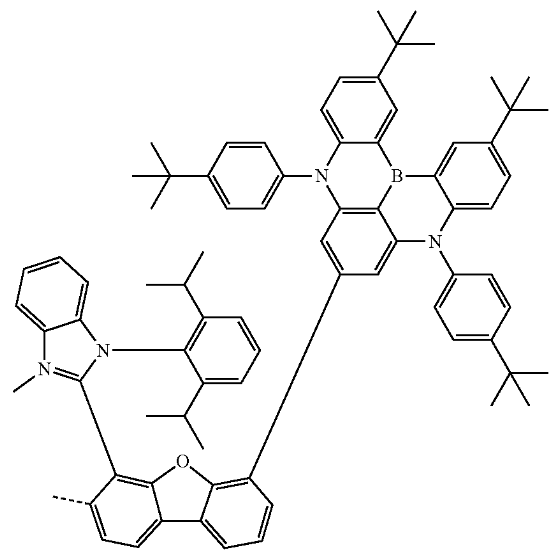
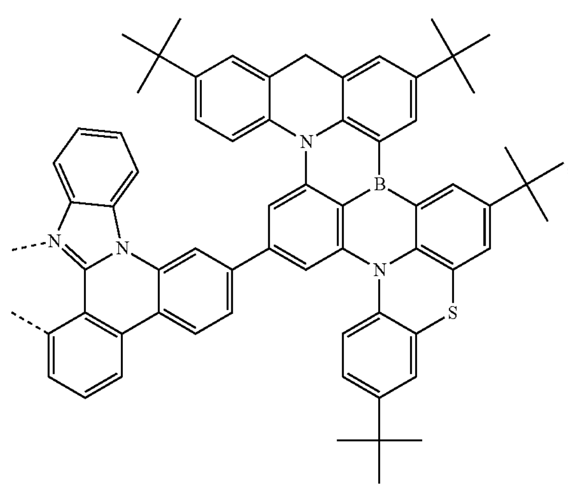


L_{A39}



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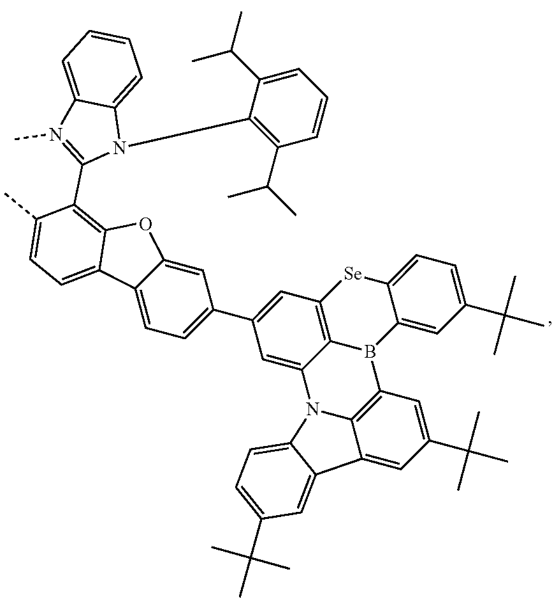
L$_{A40}$



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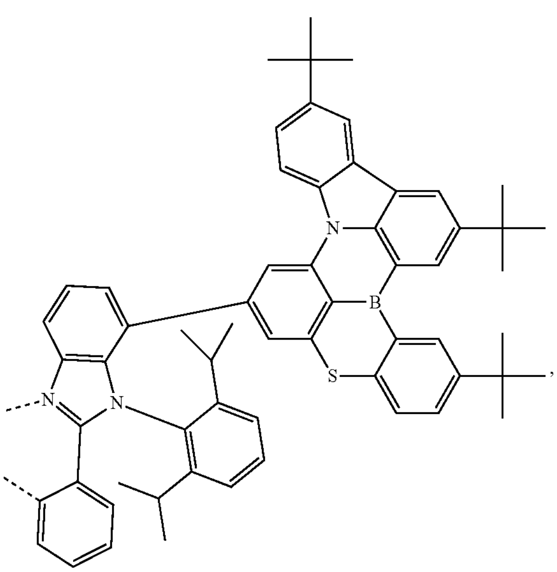

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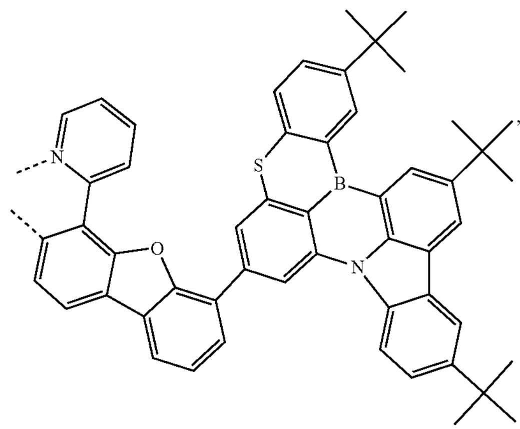

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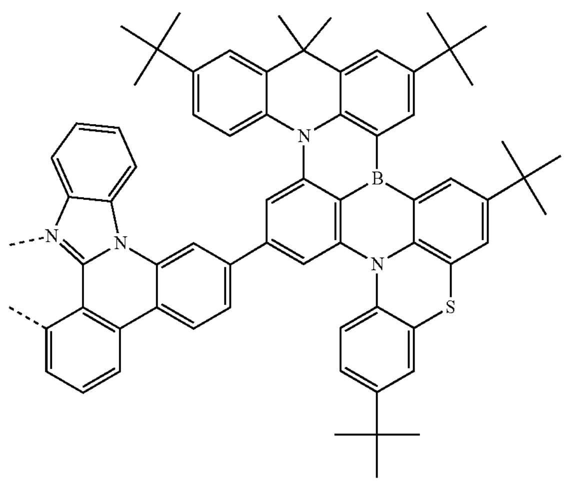

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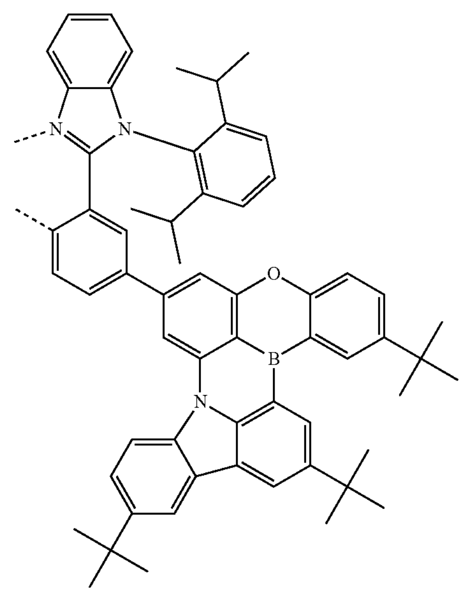

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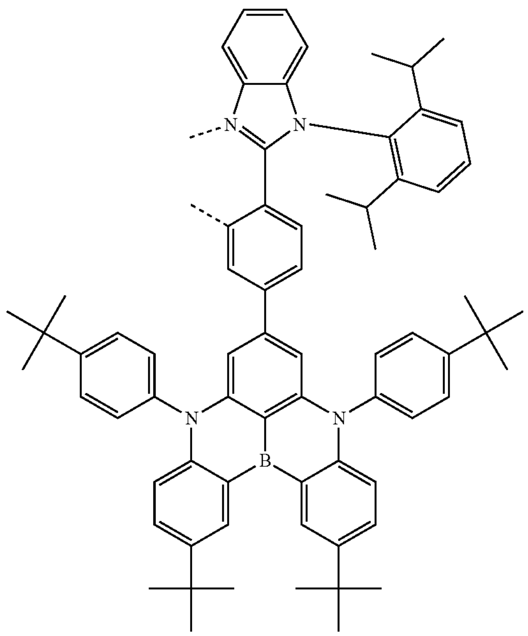

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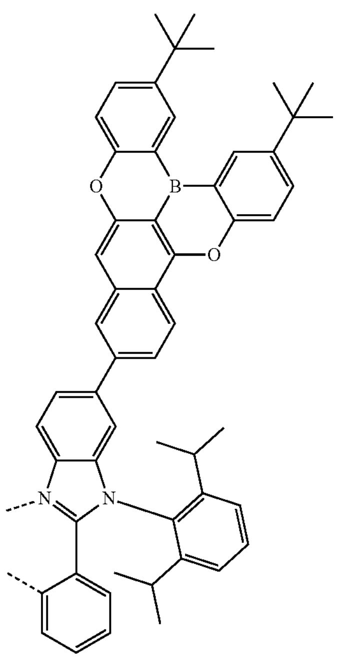

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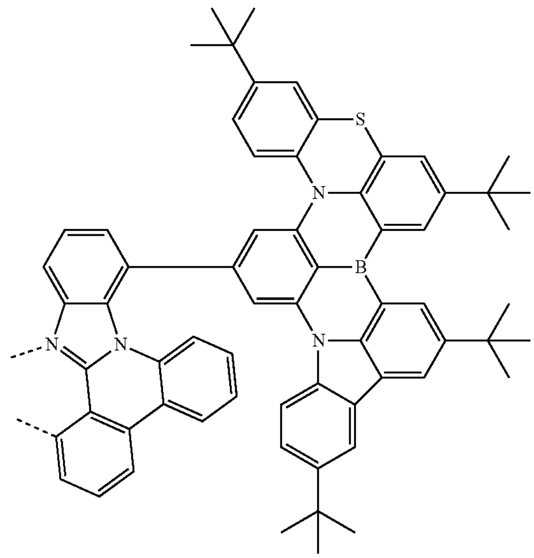

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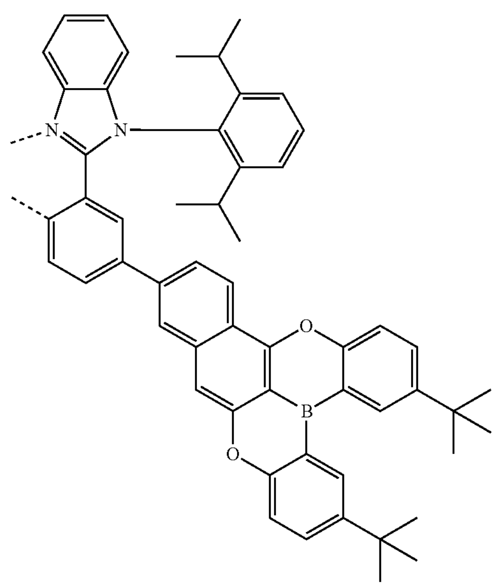

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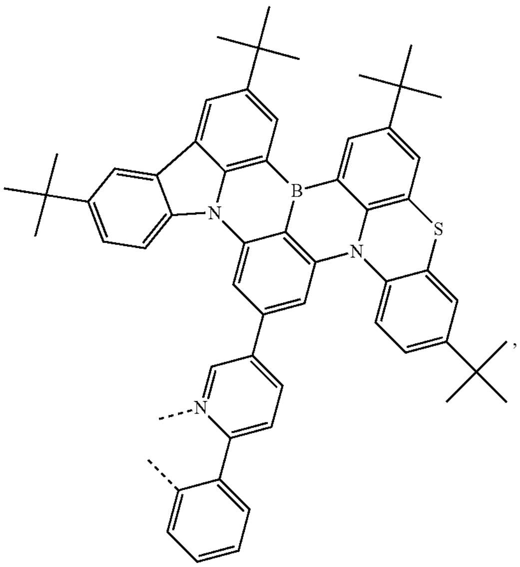

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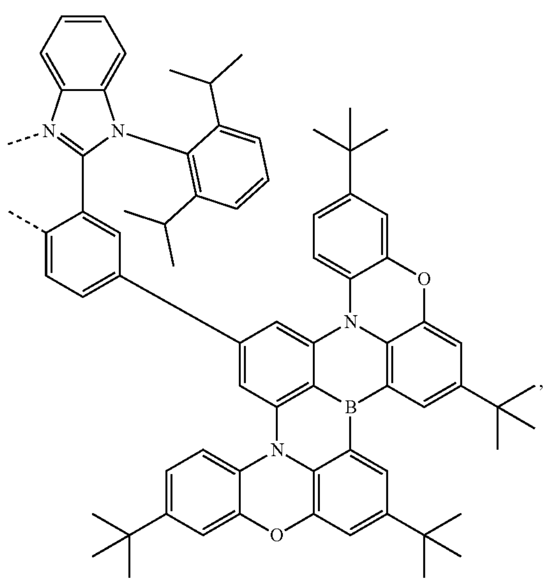

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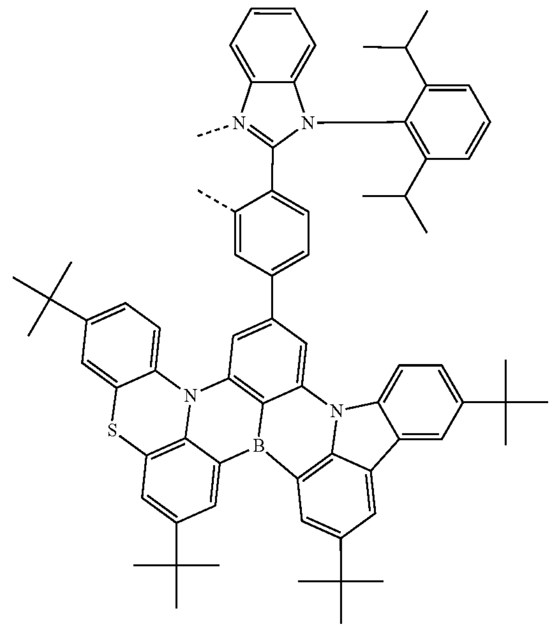

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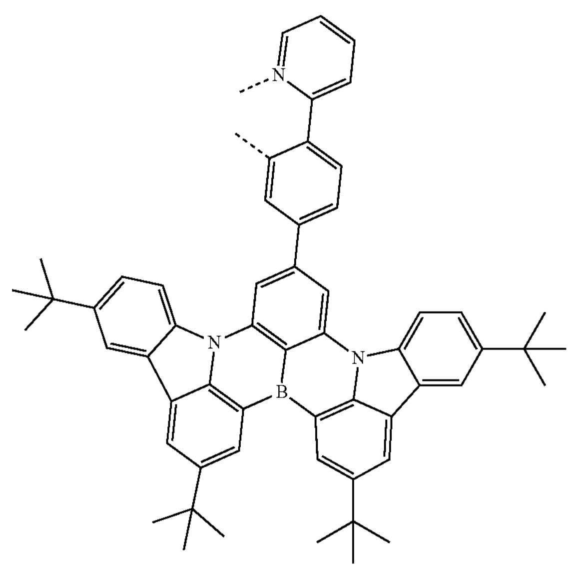

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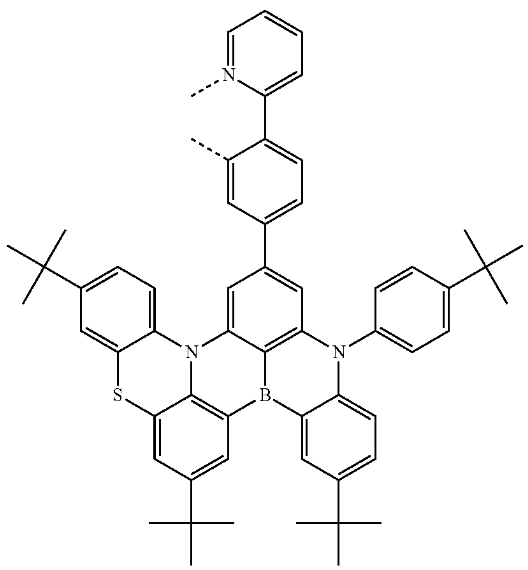

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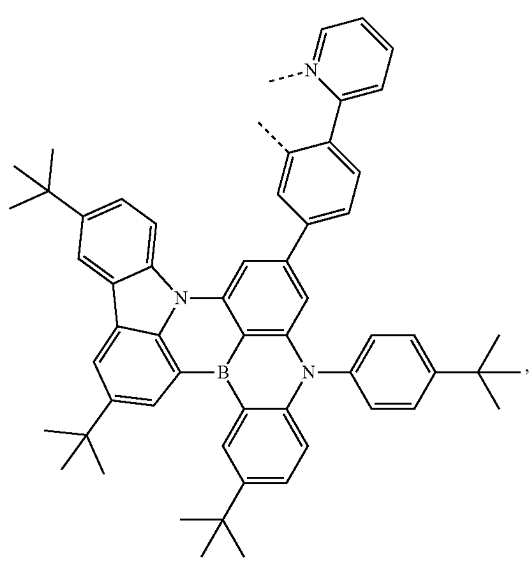

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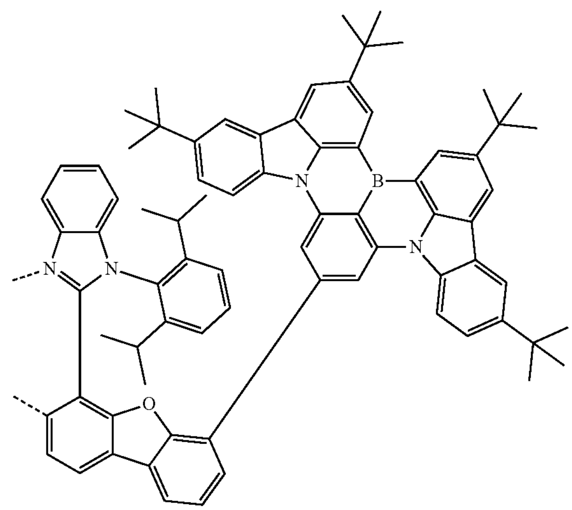

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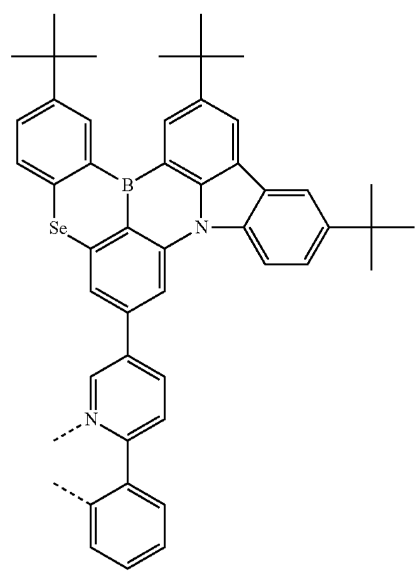

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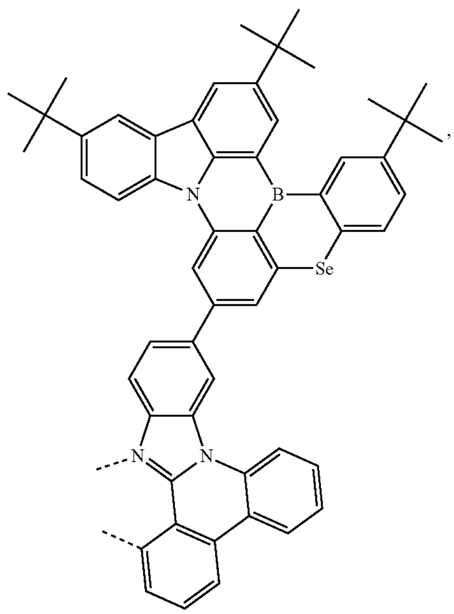

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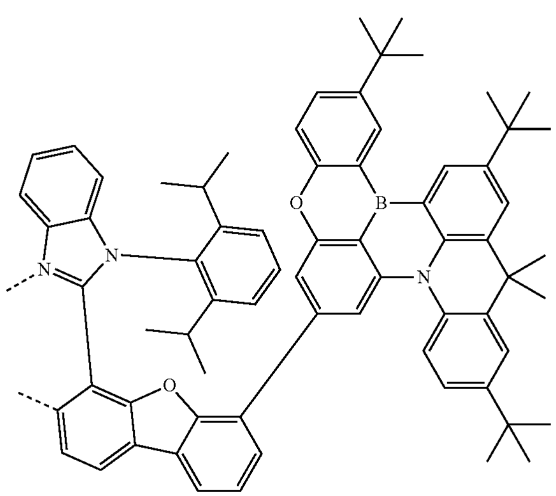

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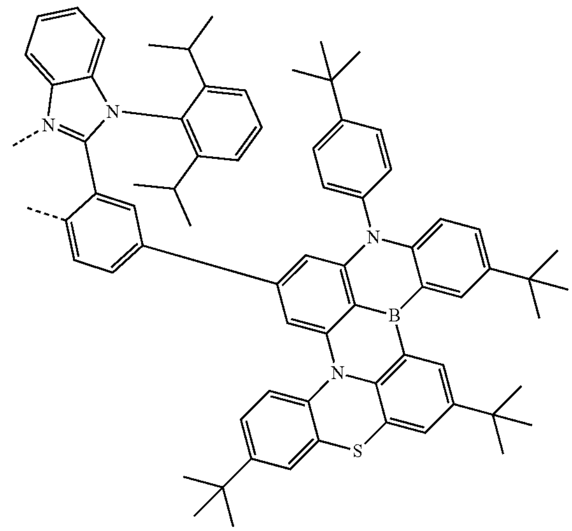

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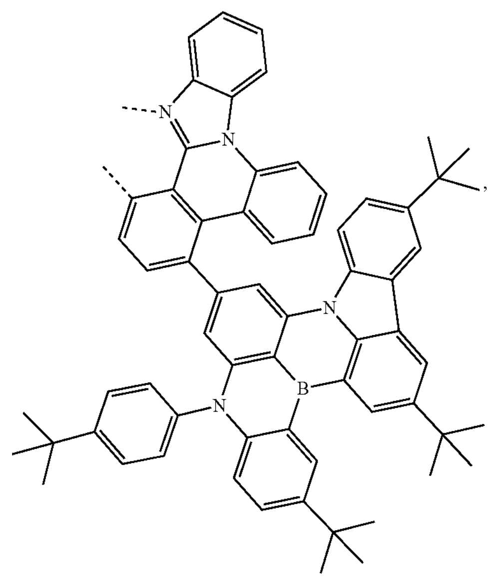

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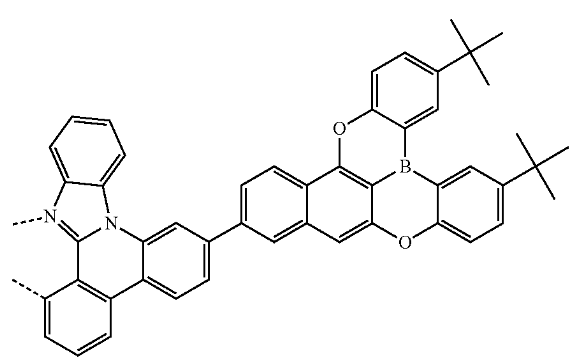

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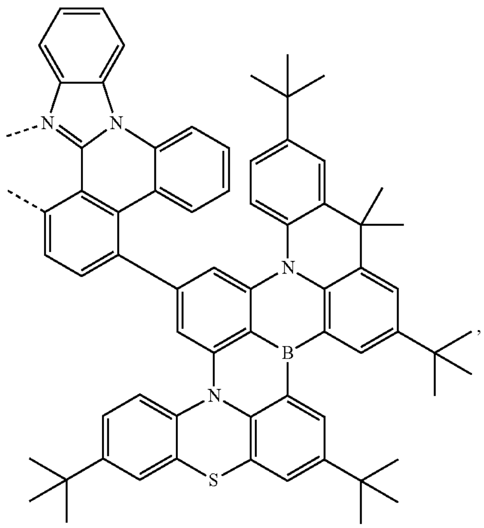

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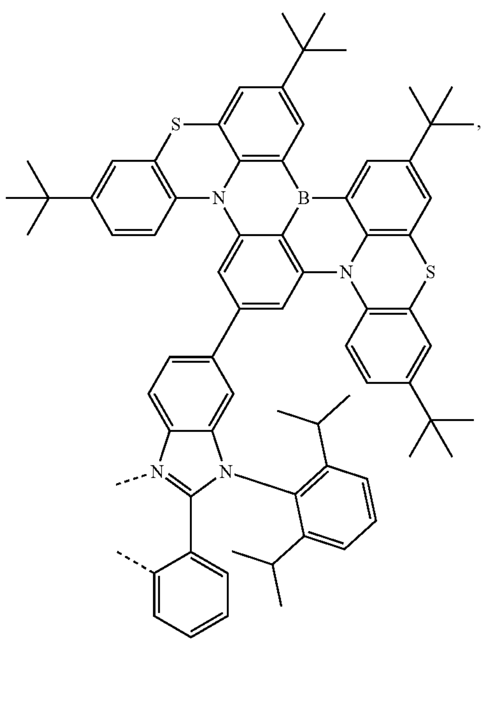

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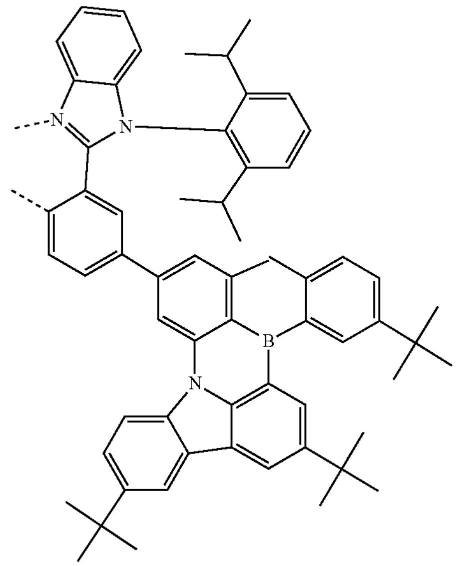

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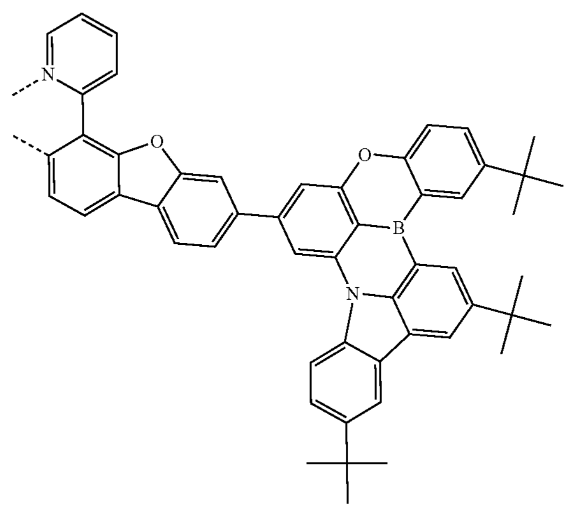

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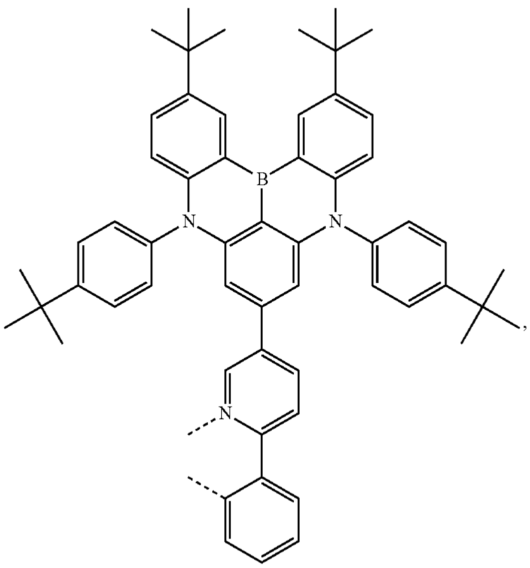

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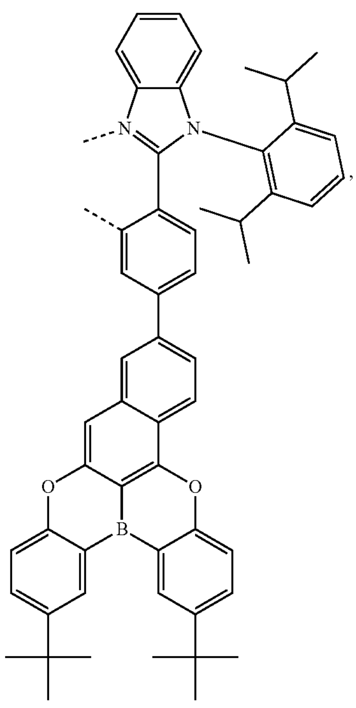

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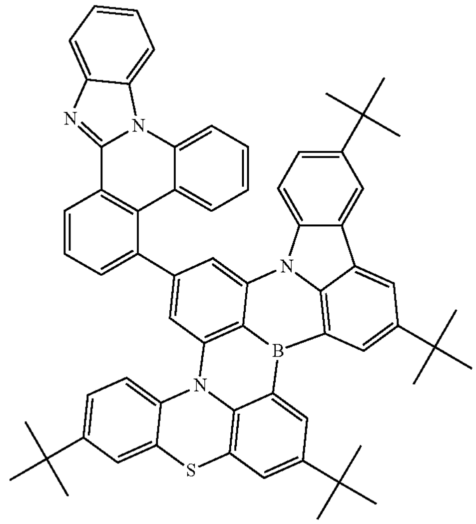

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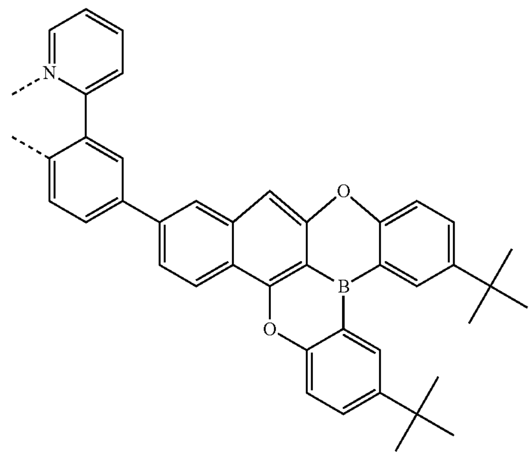

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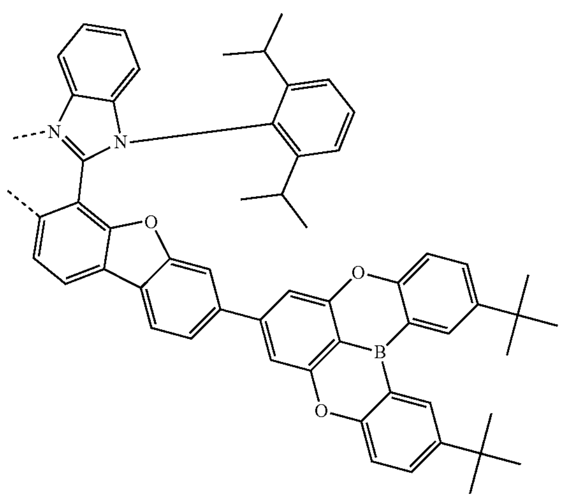

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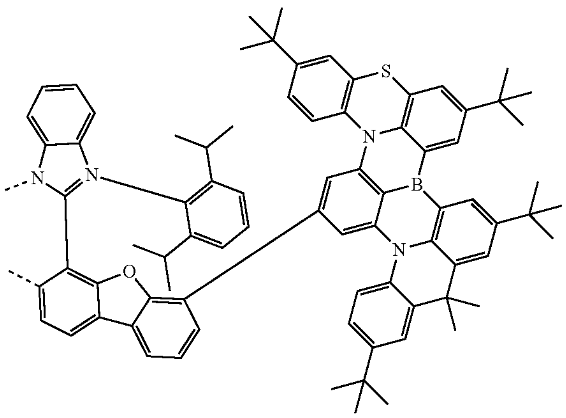

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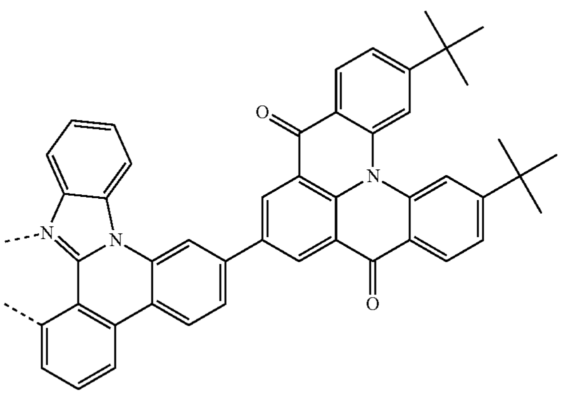

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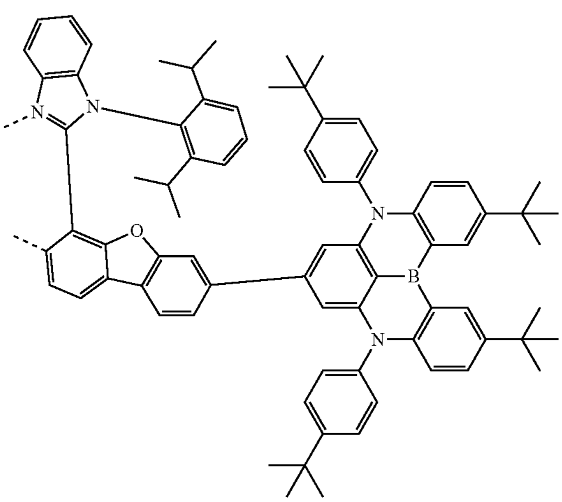

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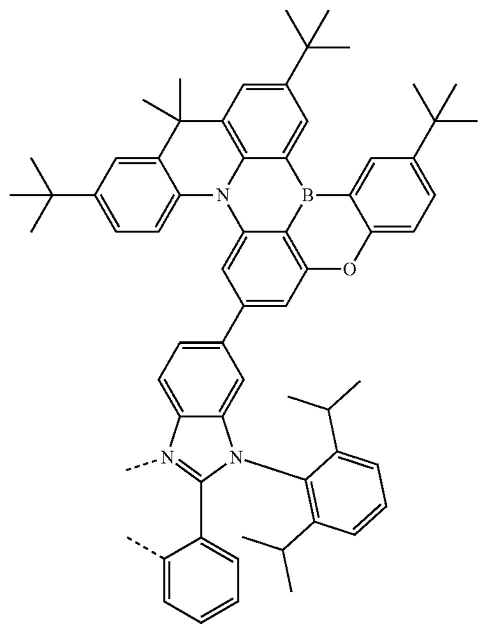

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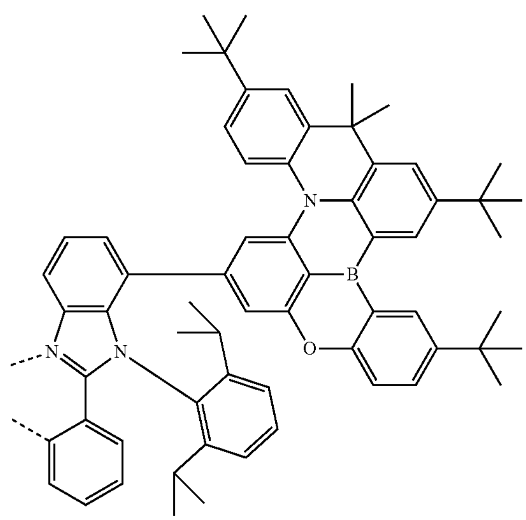

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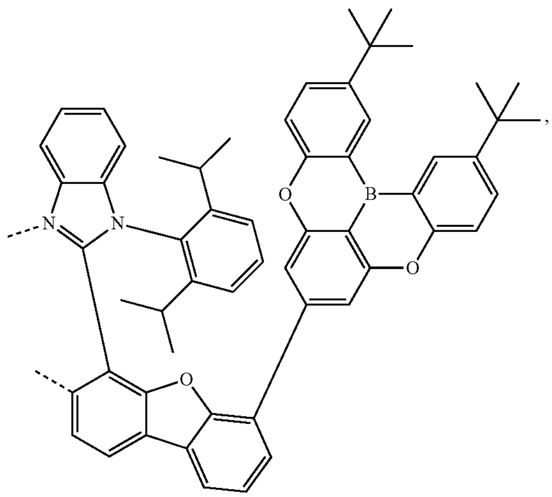

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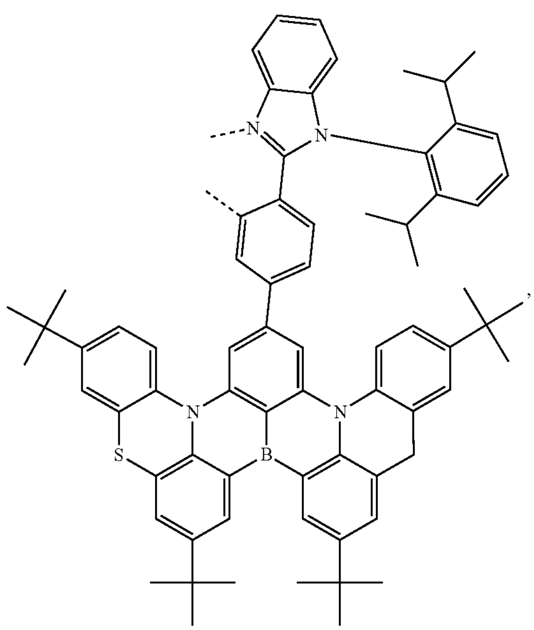

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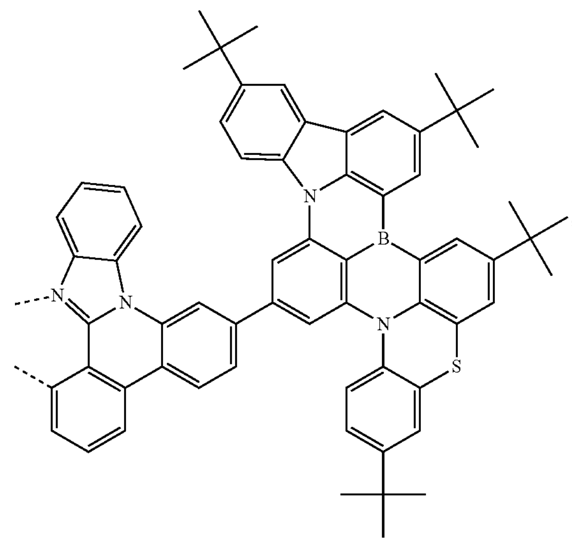

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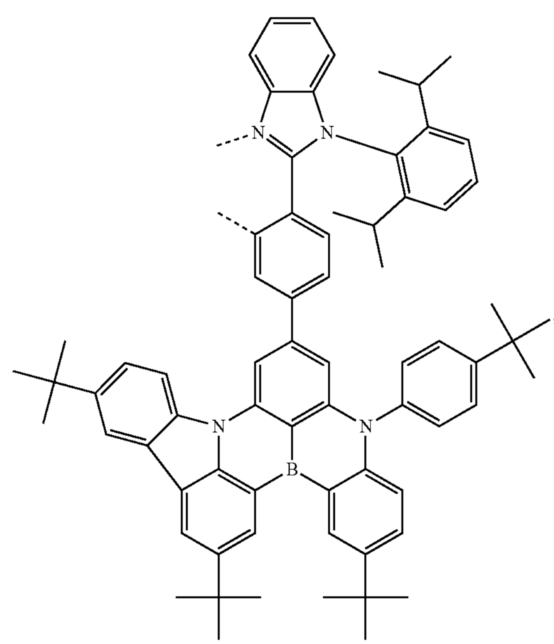

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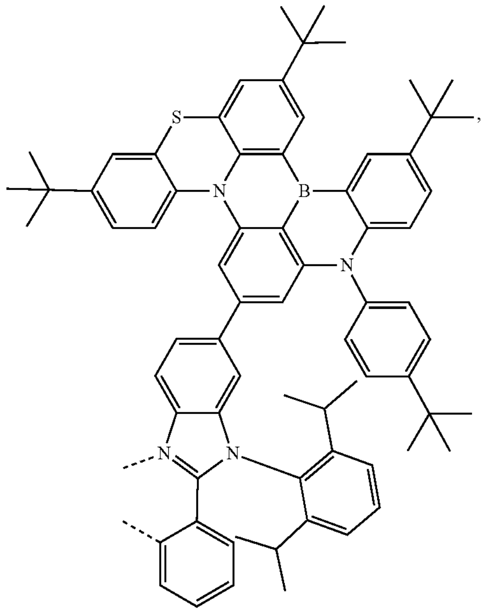

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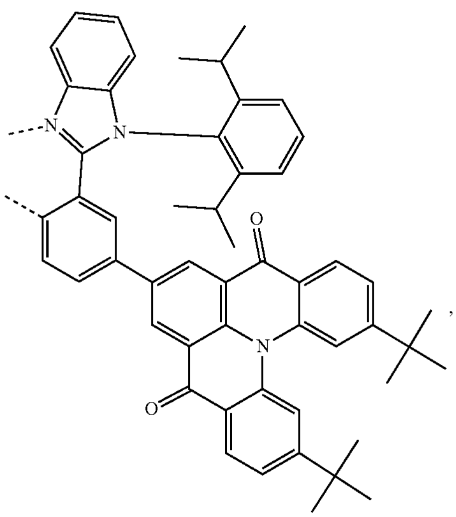

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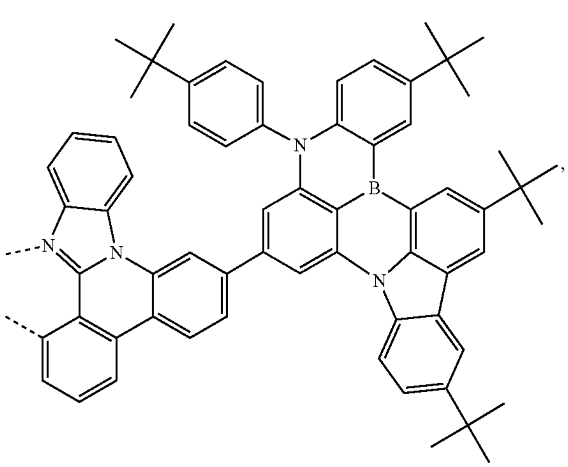

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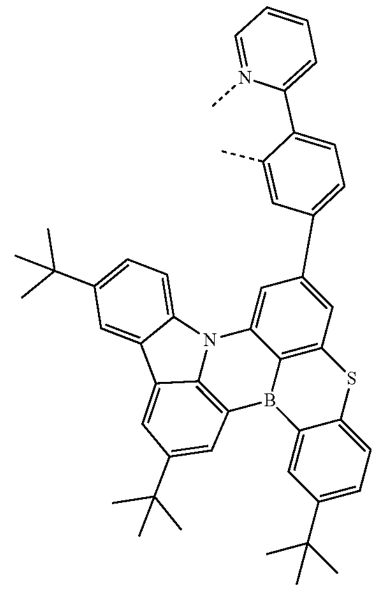

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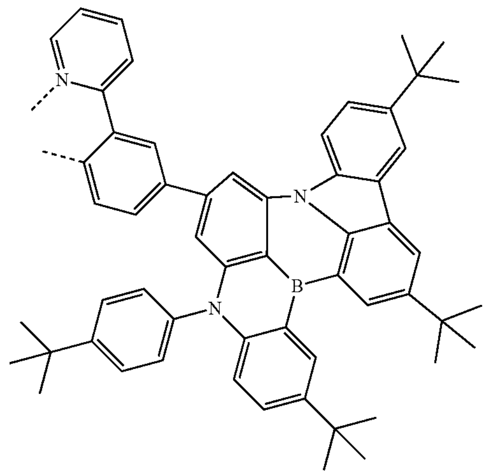

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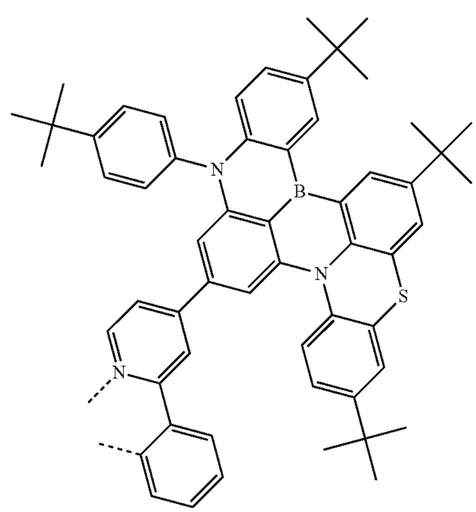

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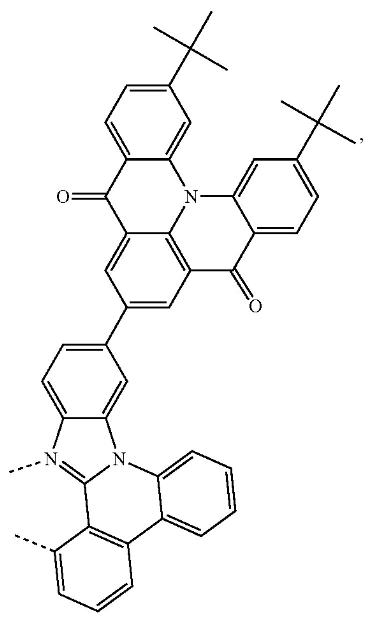

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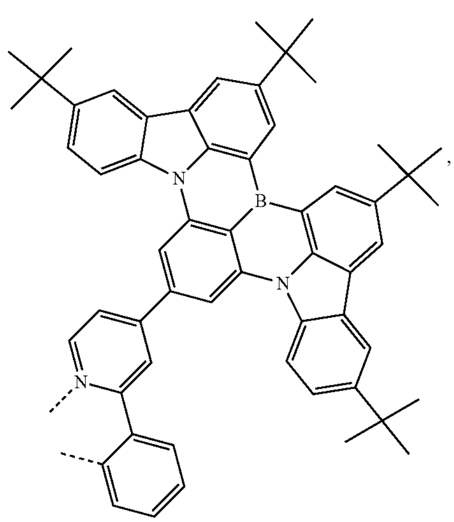

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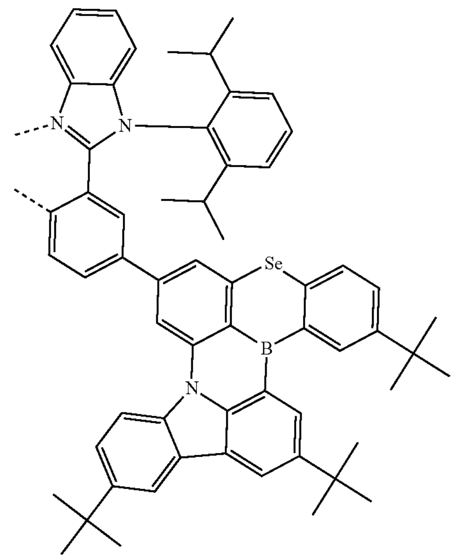

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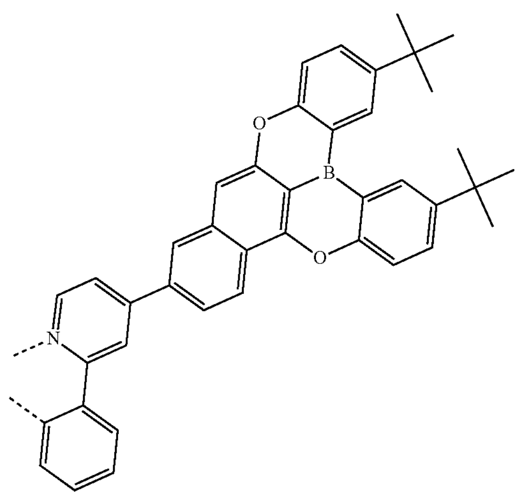

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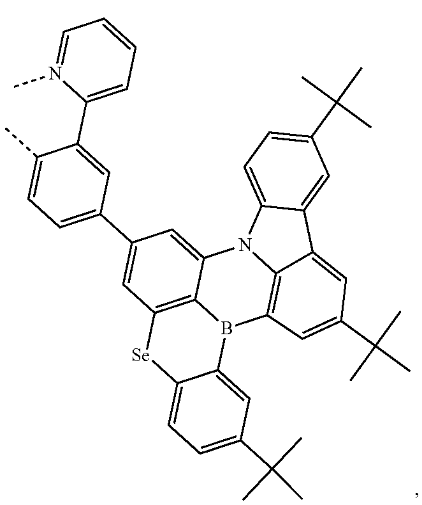

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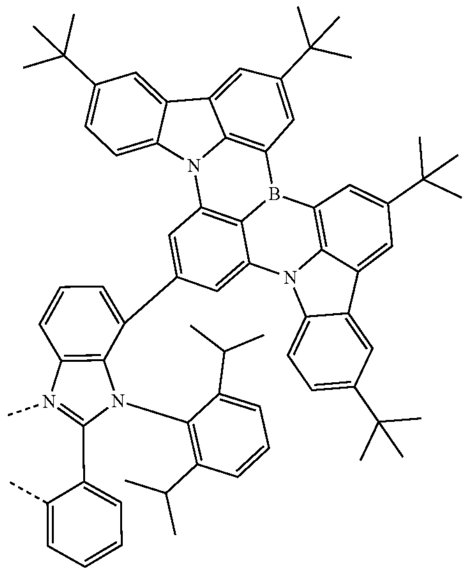

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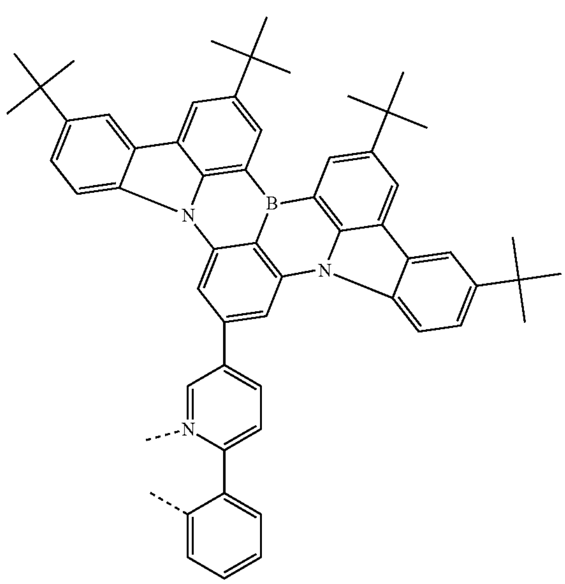

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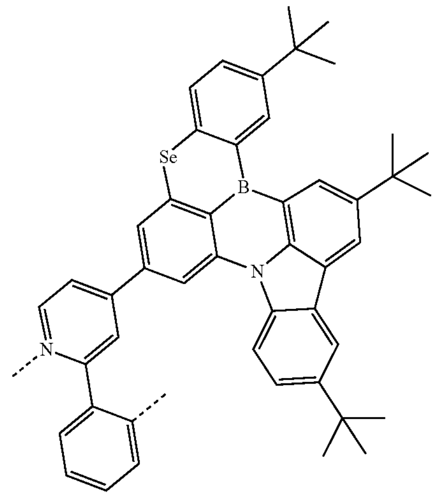

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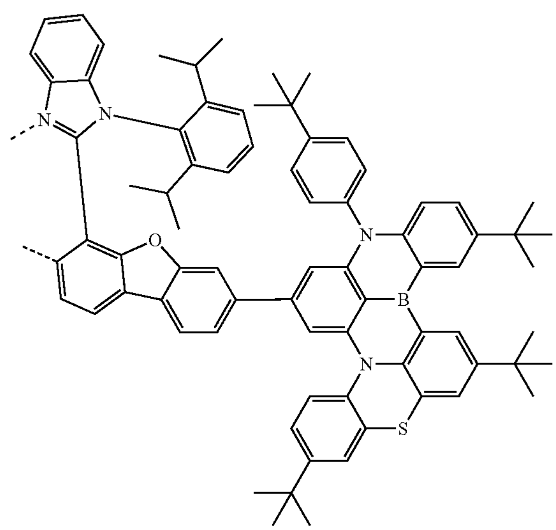

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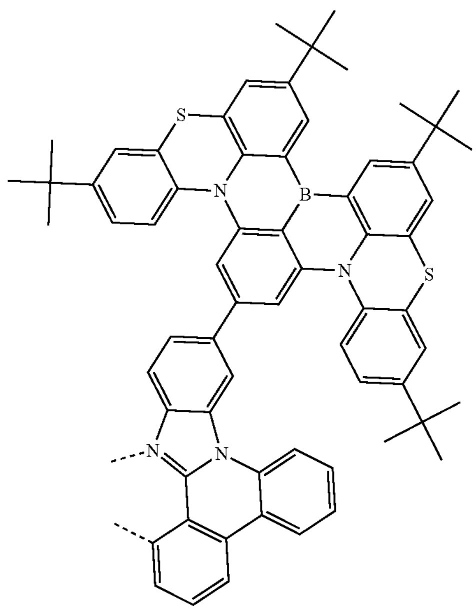

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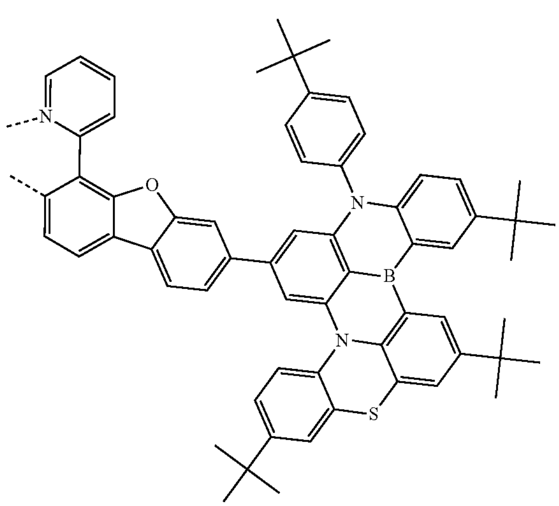

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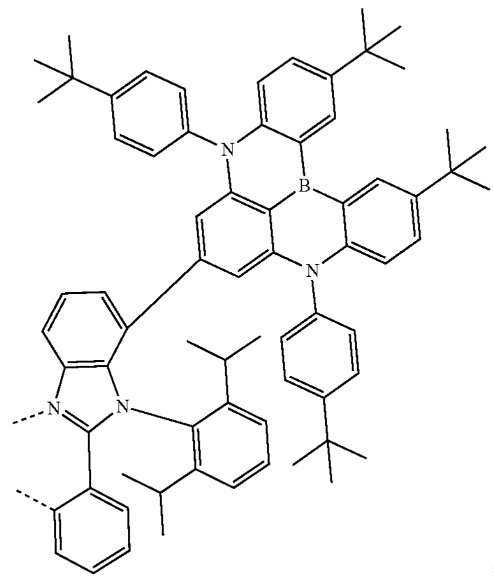

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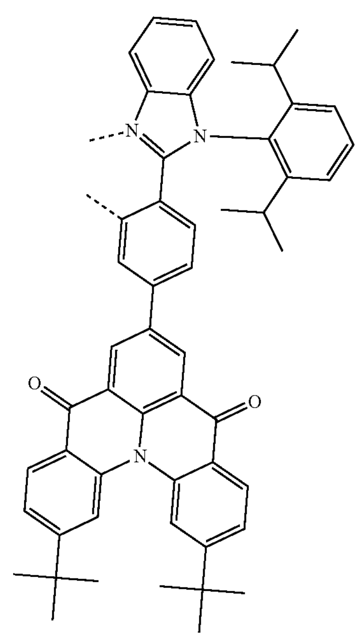

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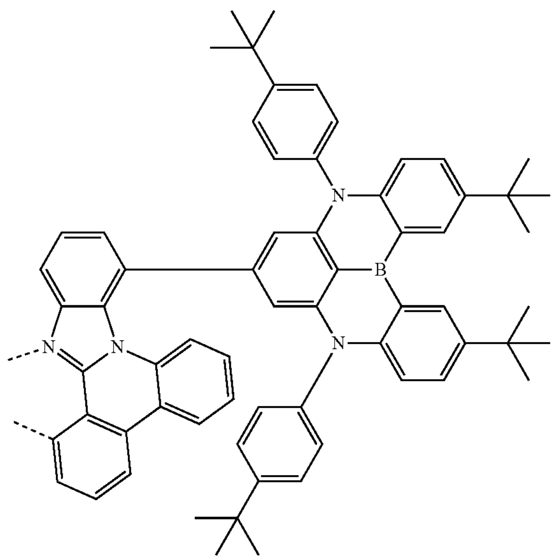

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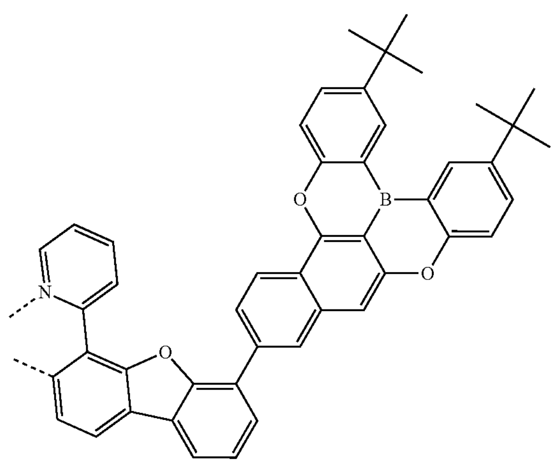

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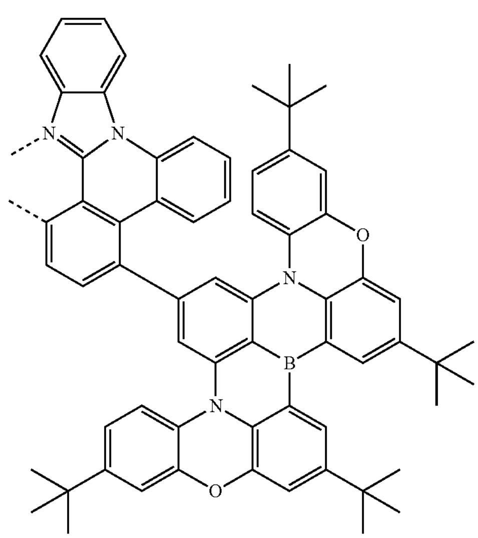

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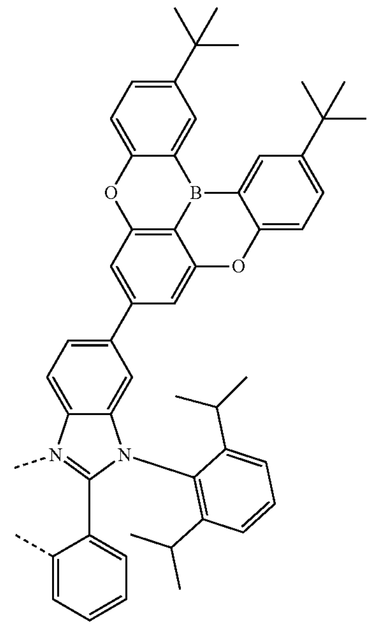

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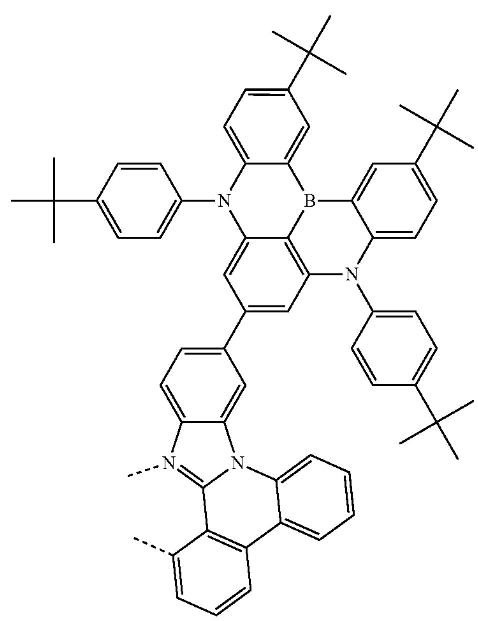

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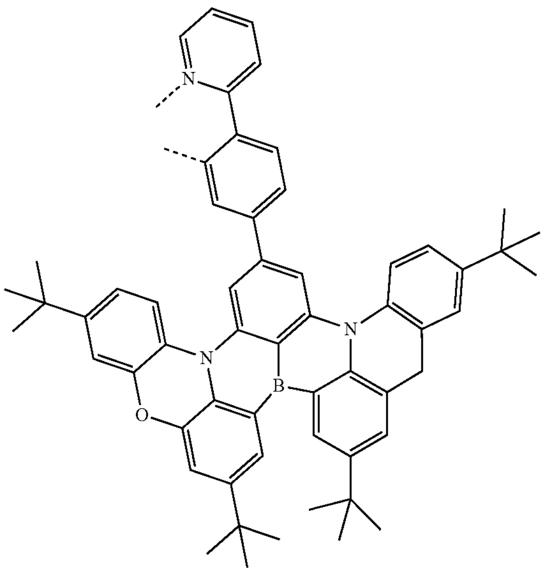

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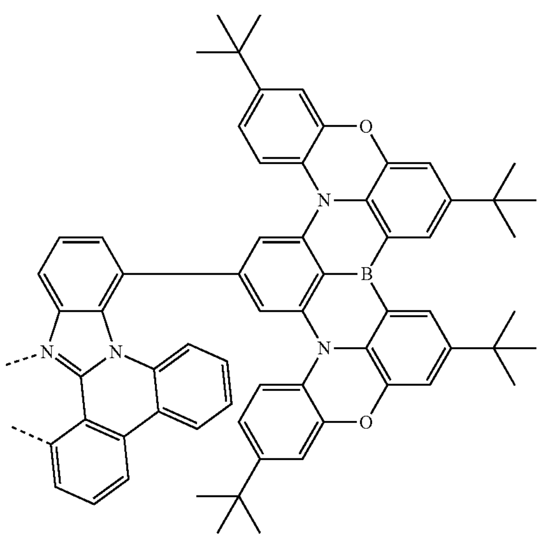

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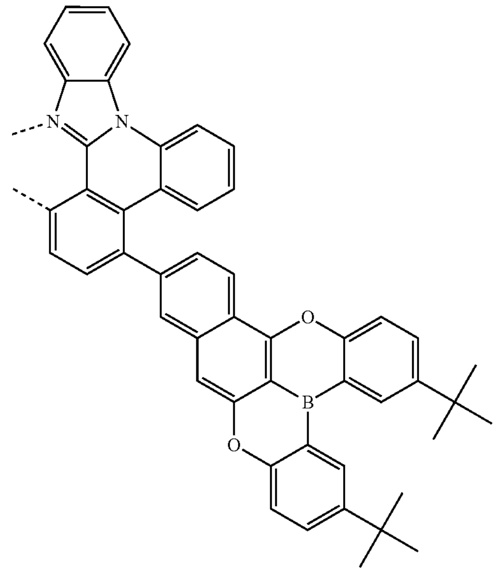

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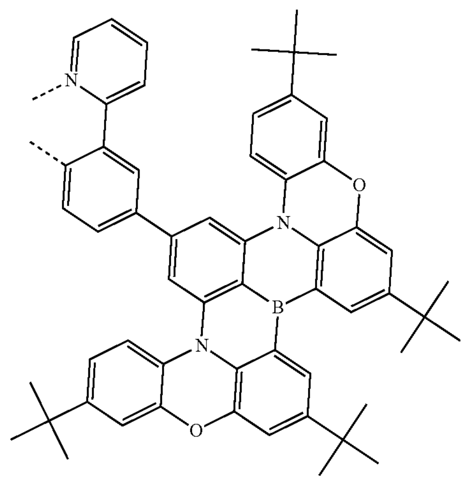

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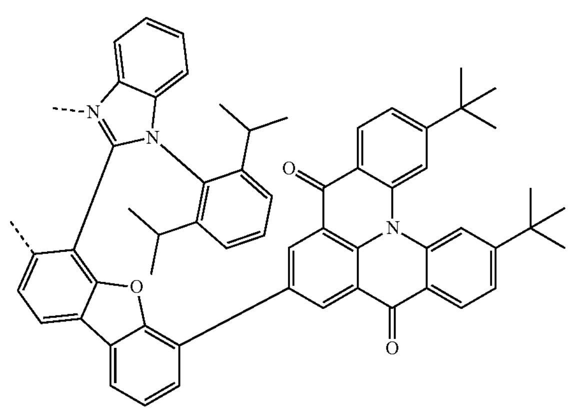

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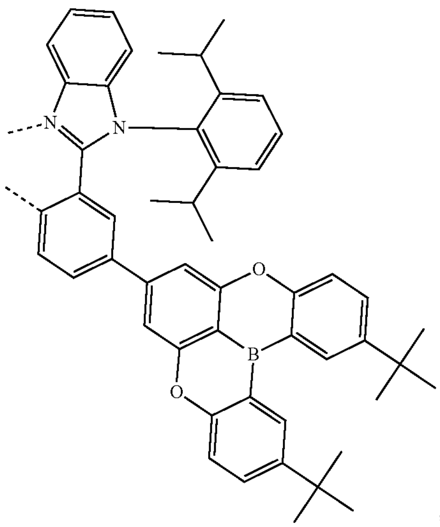

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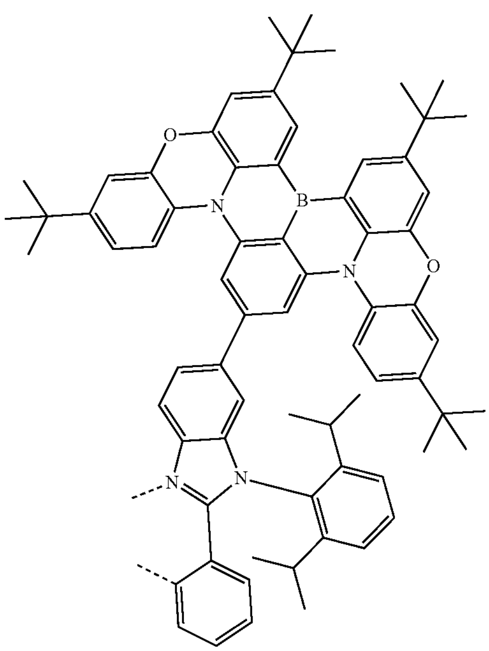

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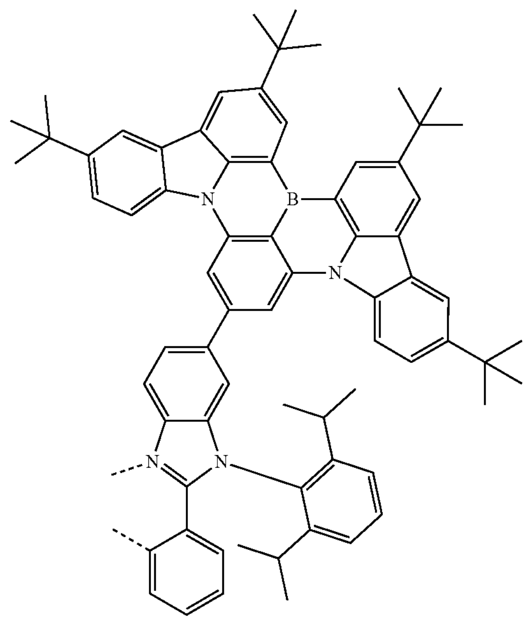

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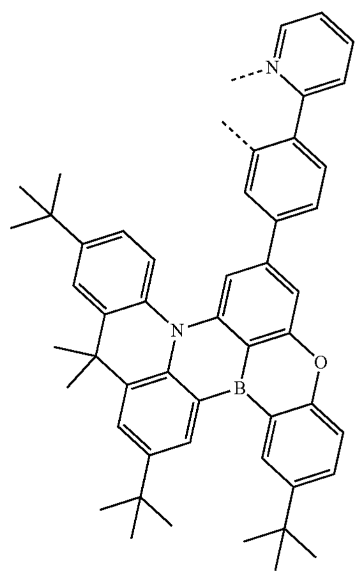

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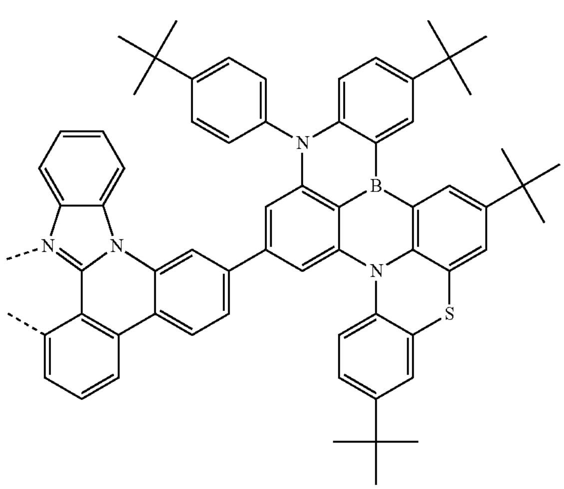

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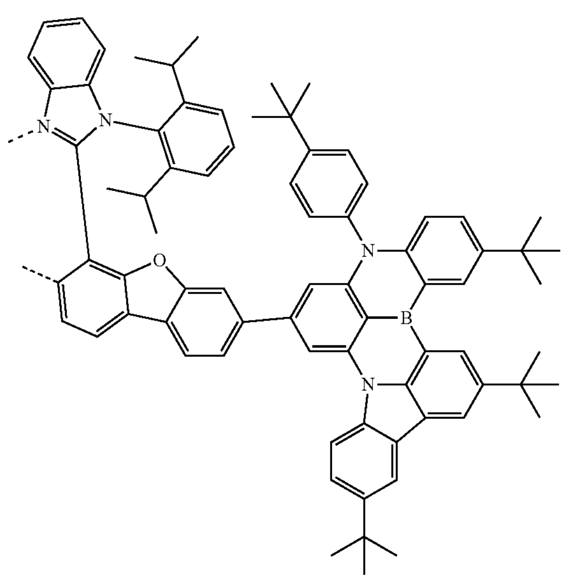

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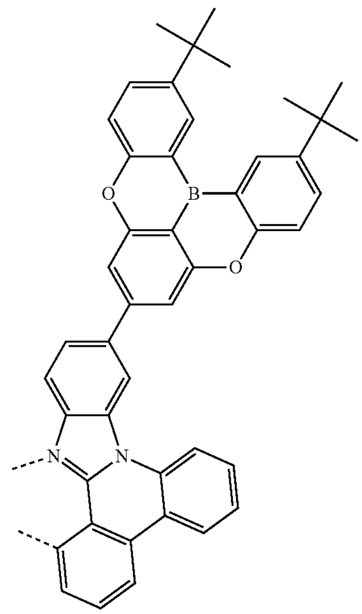

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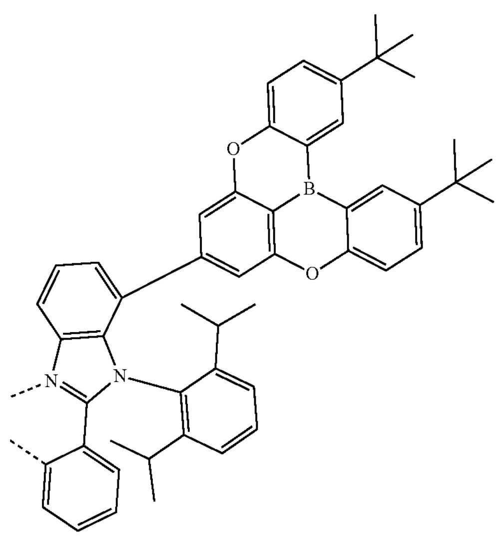

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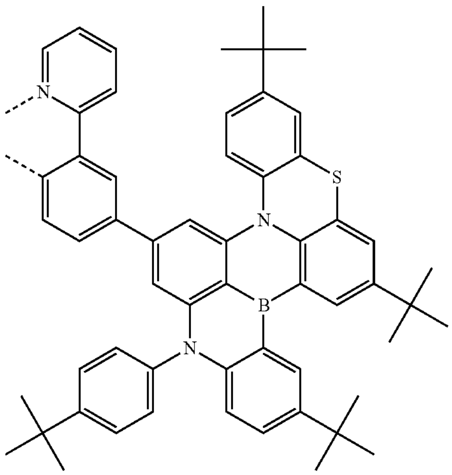

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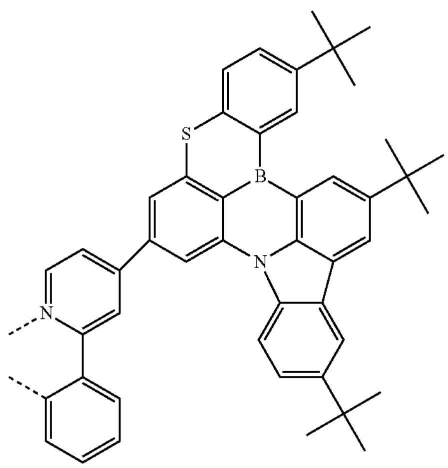

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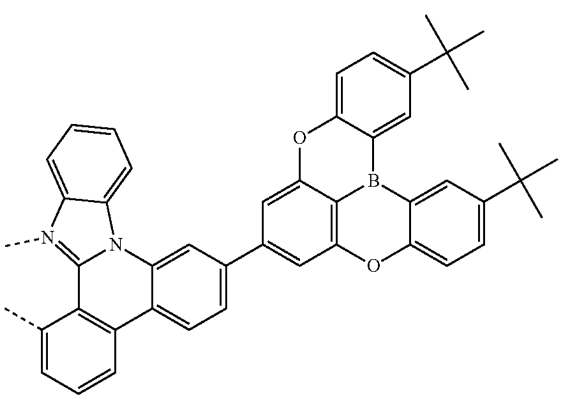

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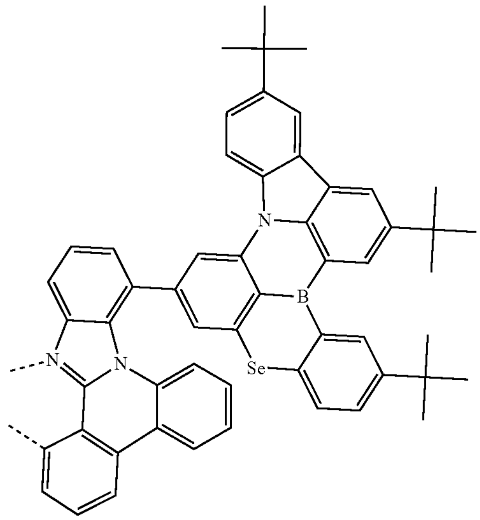

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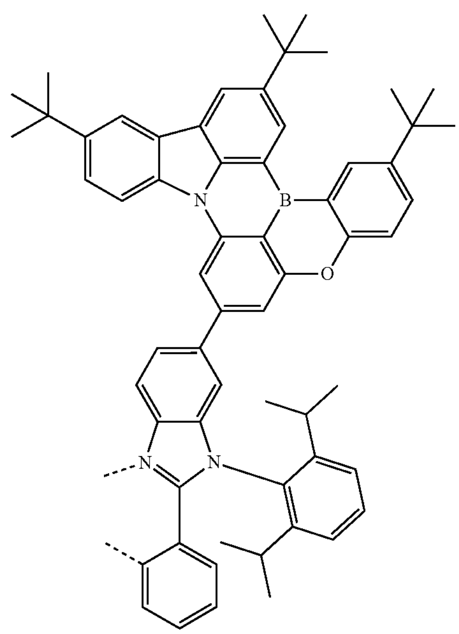

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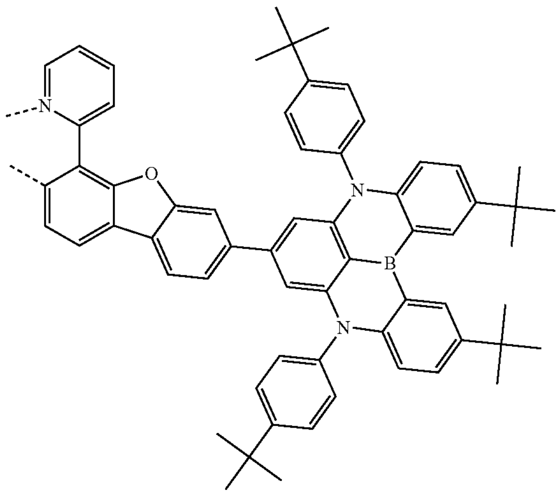

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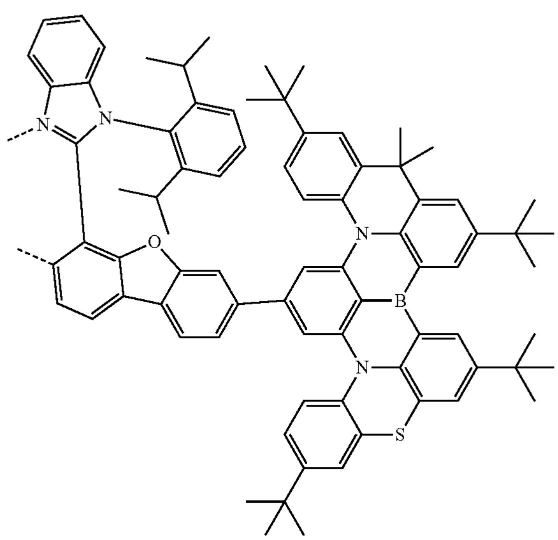

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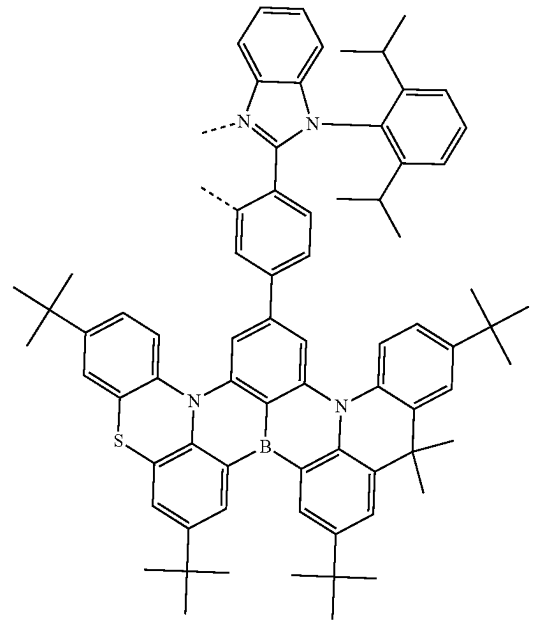

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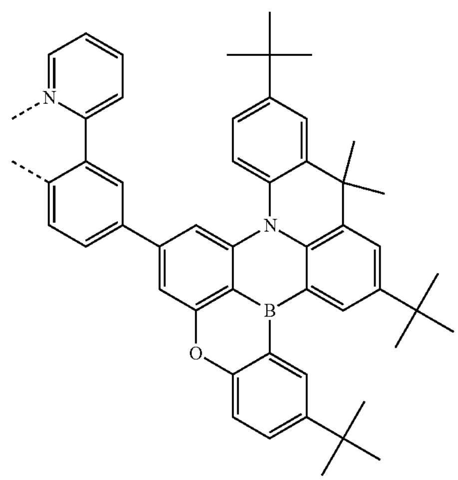

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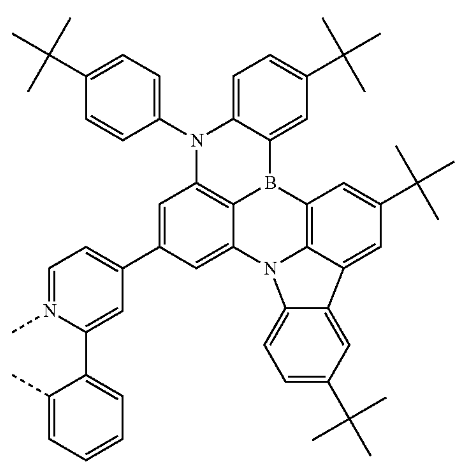

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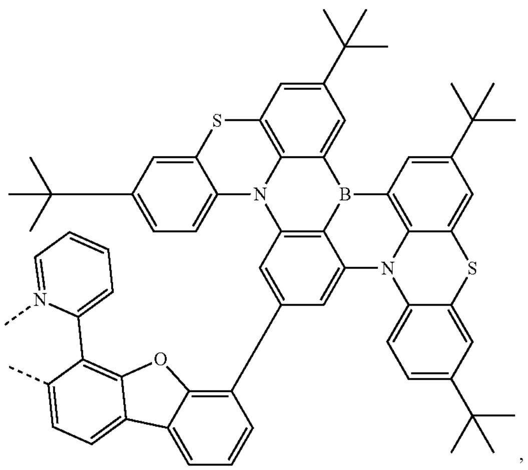

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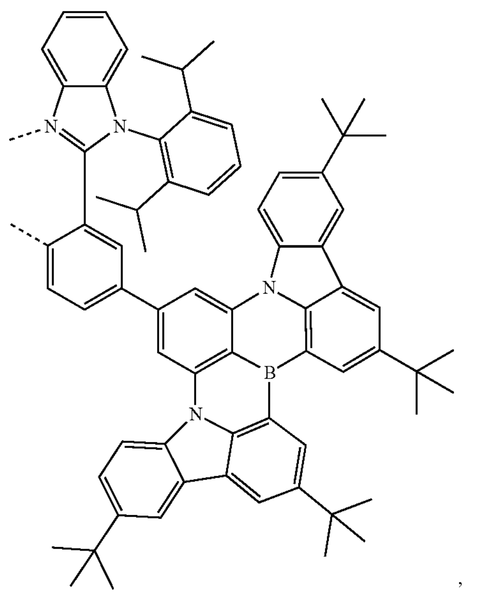

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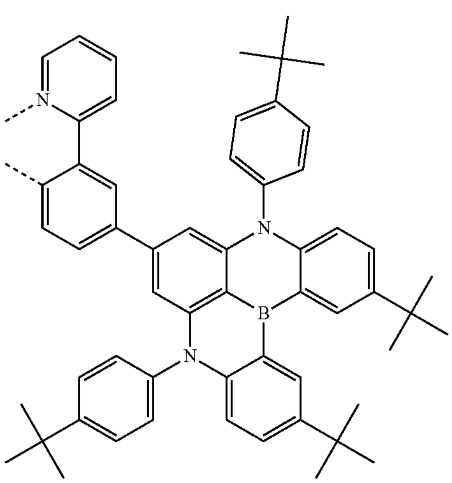

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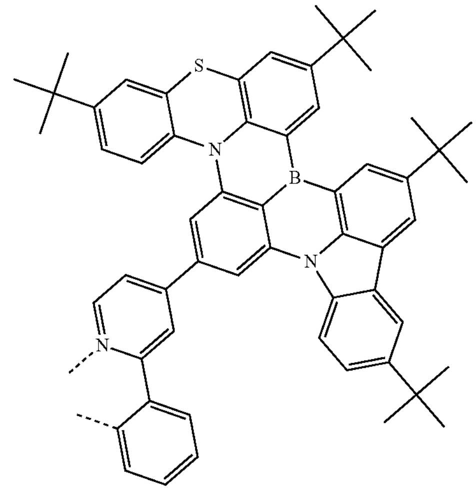

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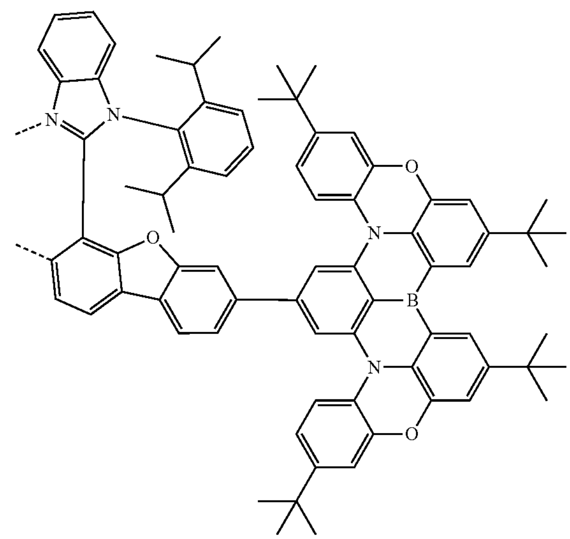

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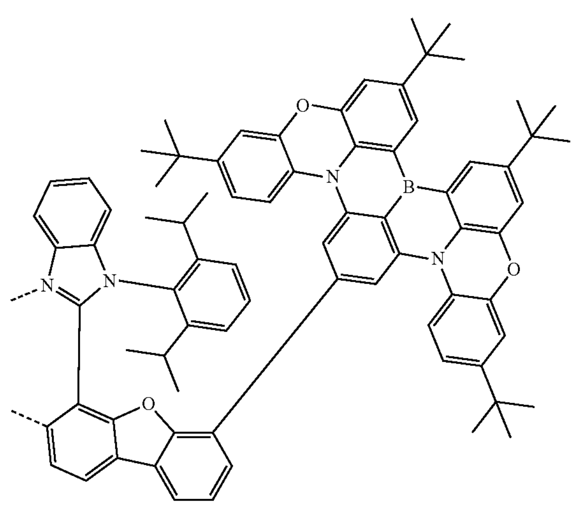

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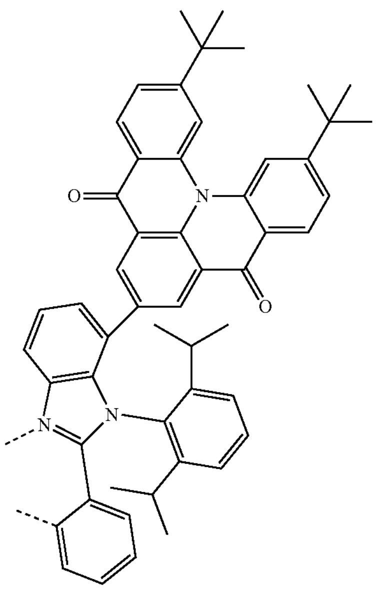

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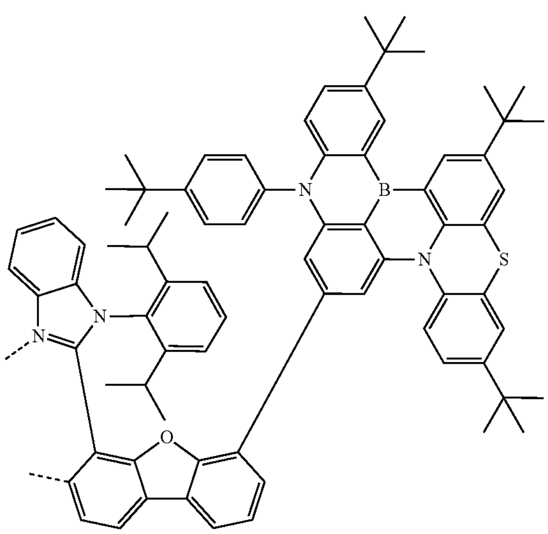

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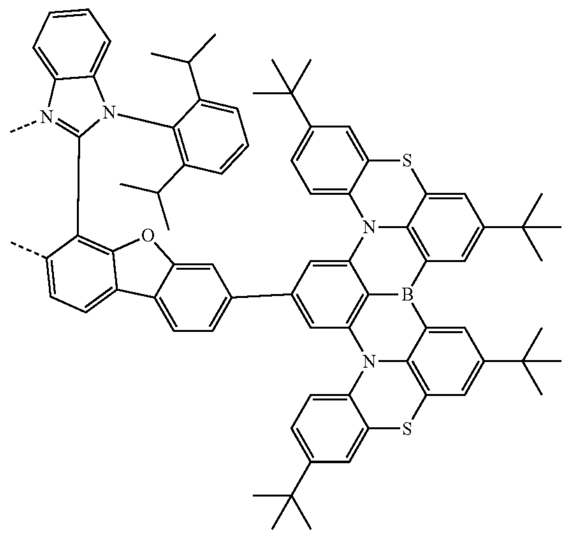

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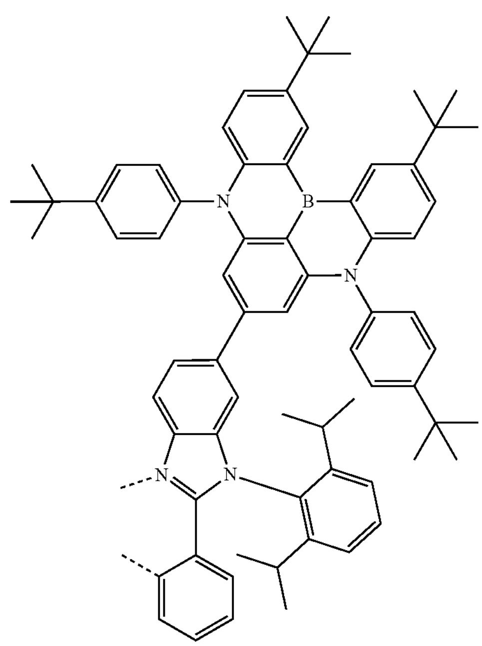

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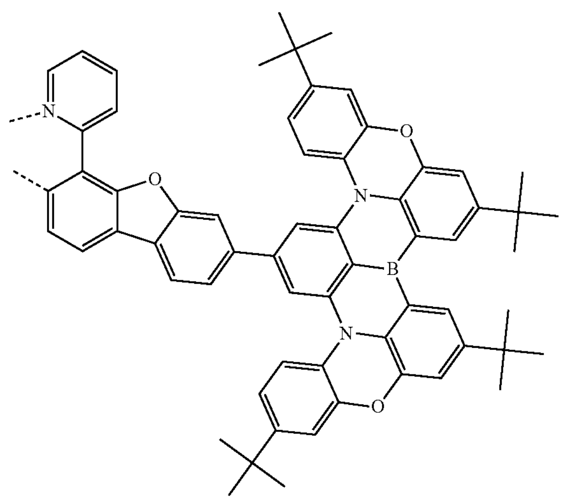

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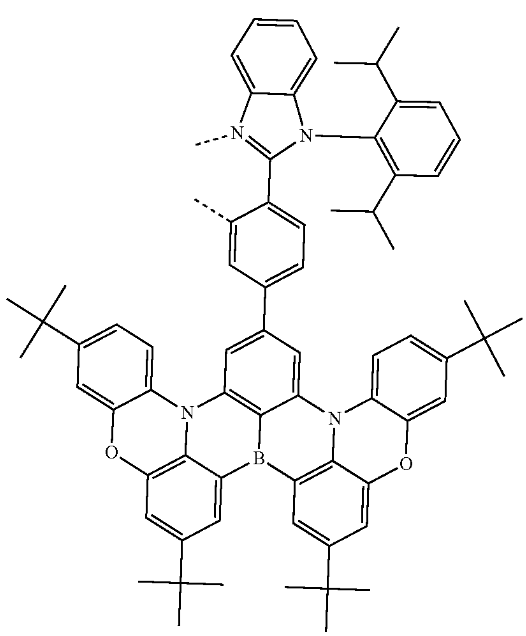

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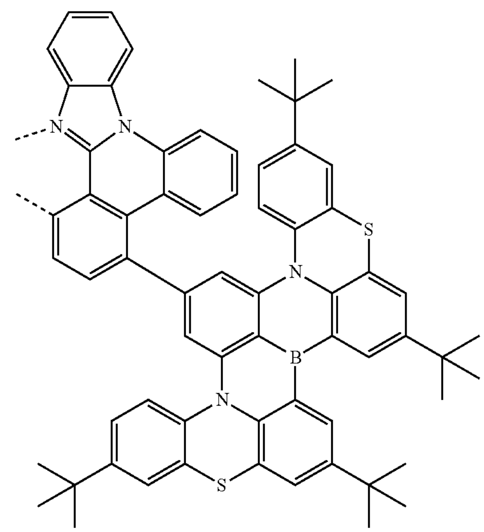

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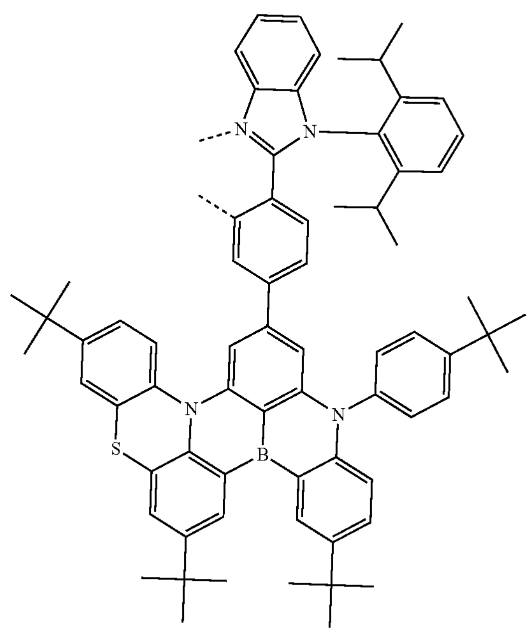

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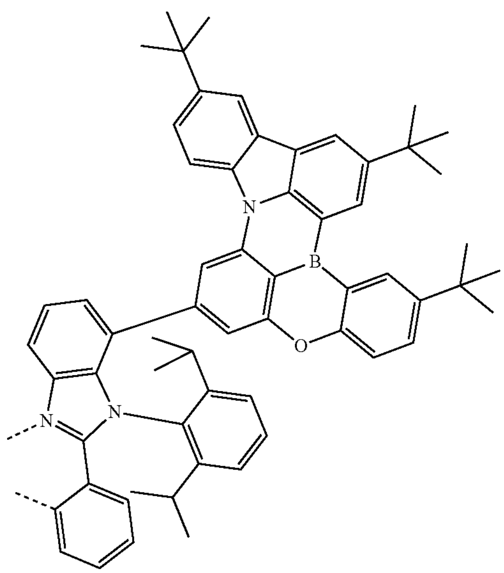

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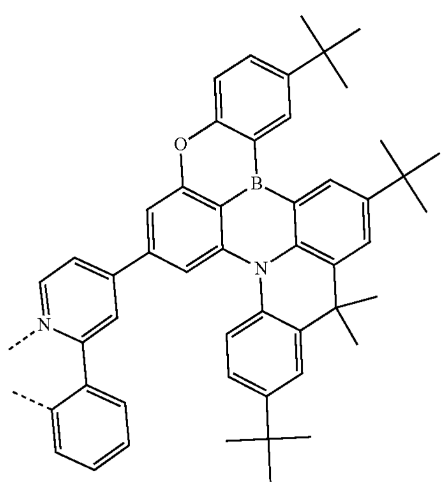

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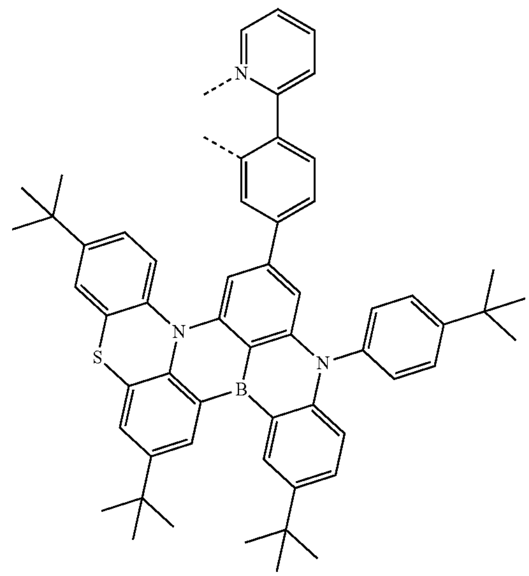

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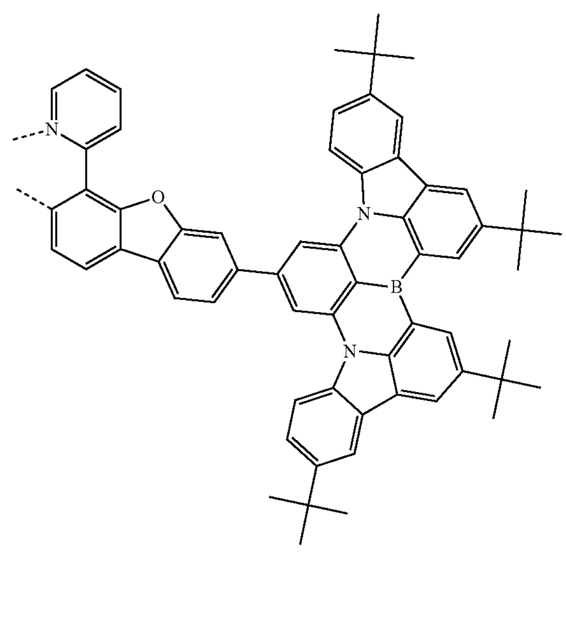

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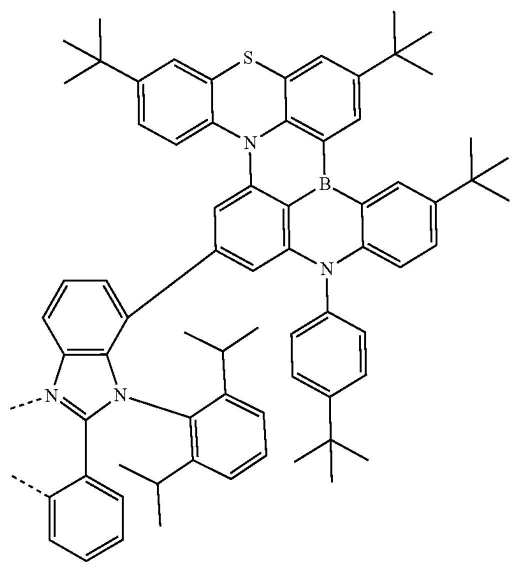

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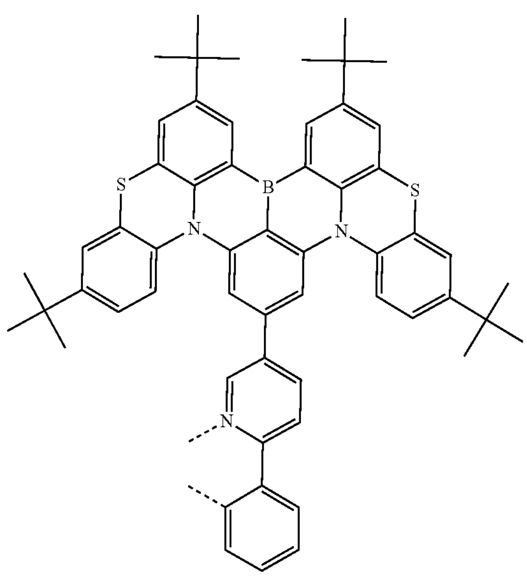

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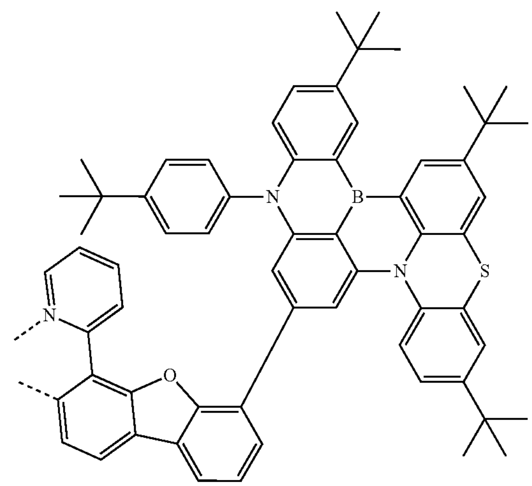

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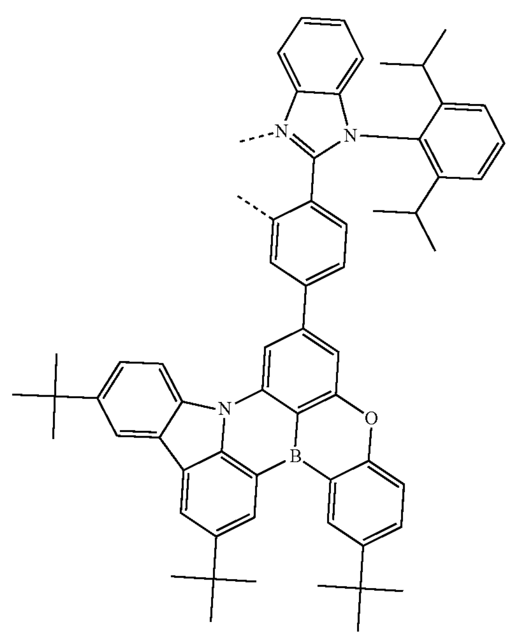

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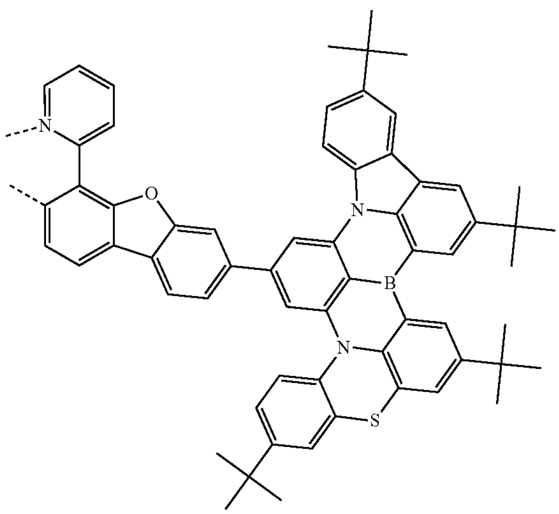

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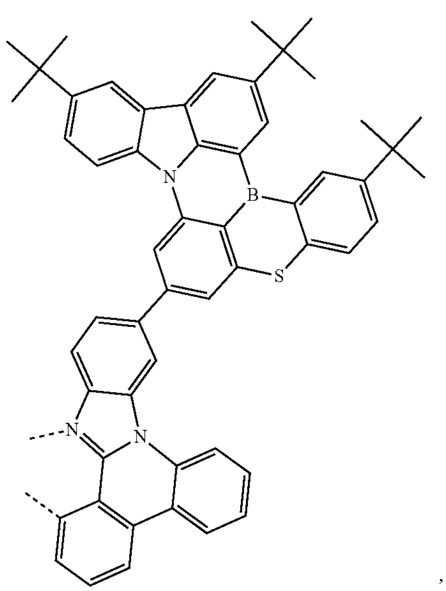

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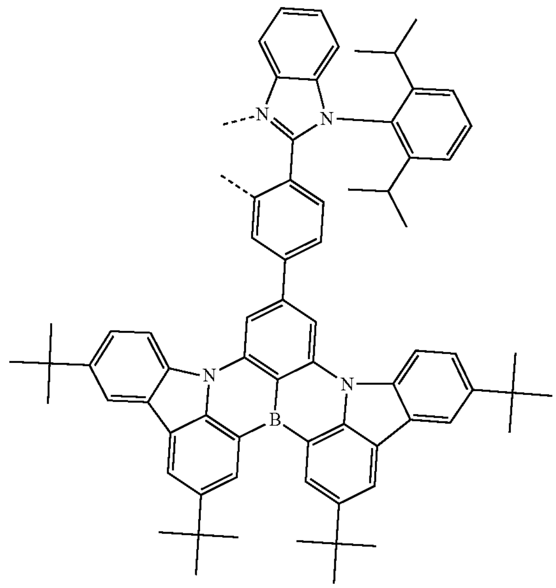

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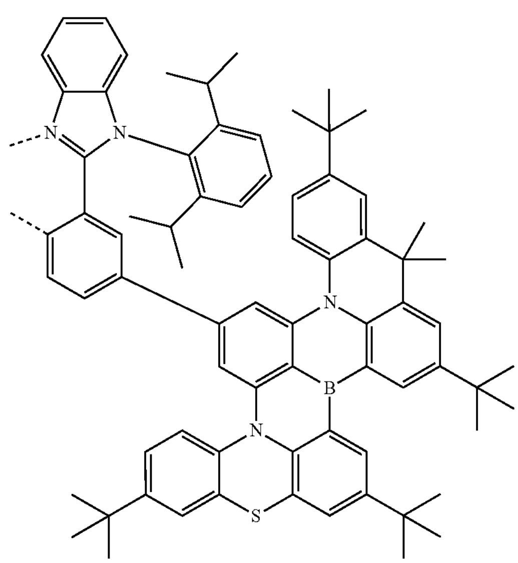

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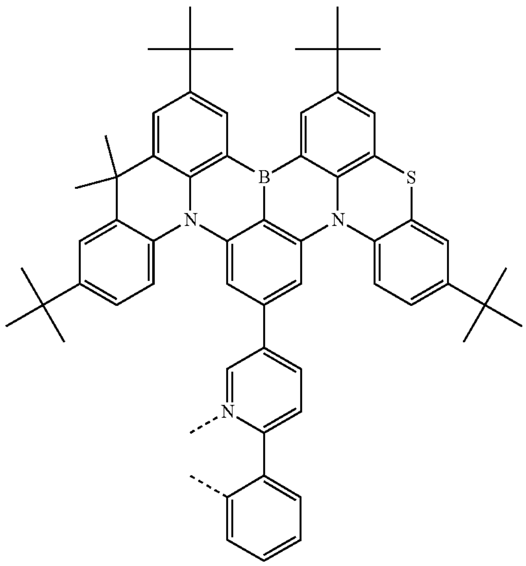

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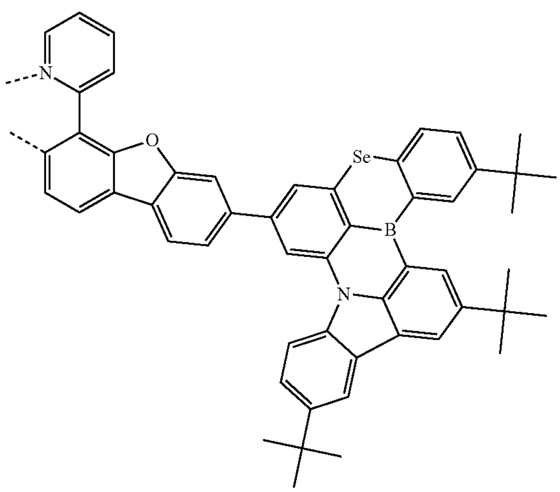

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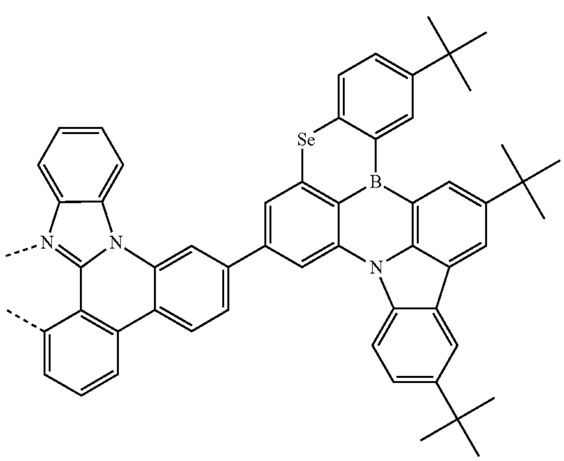

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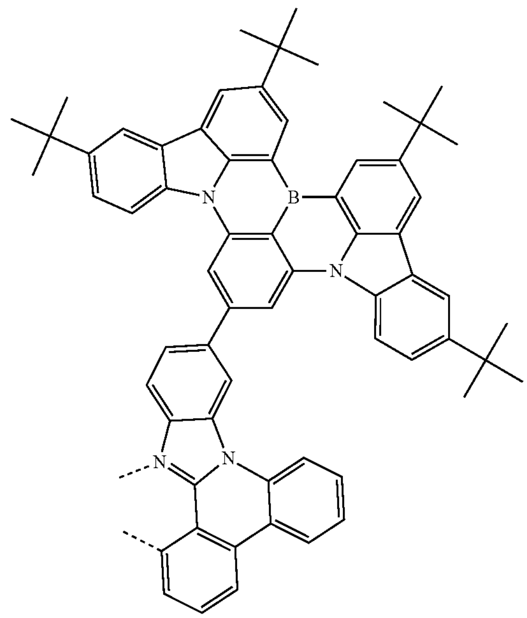

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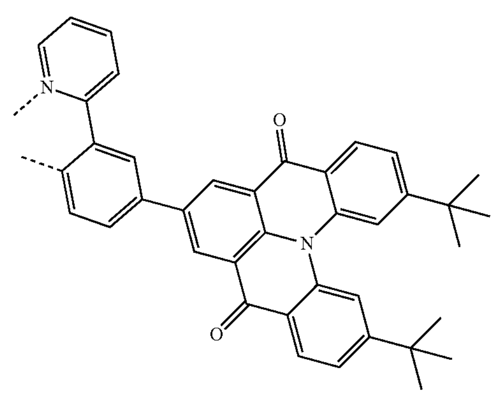

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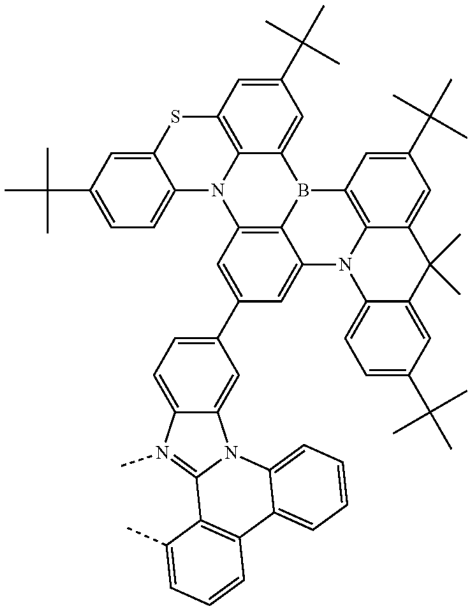

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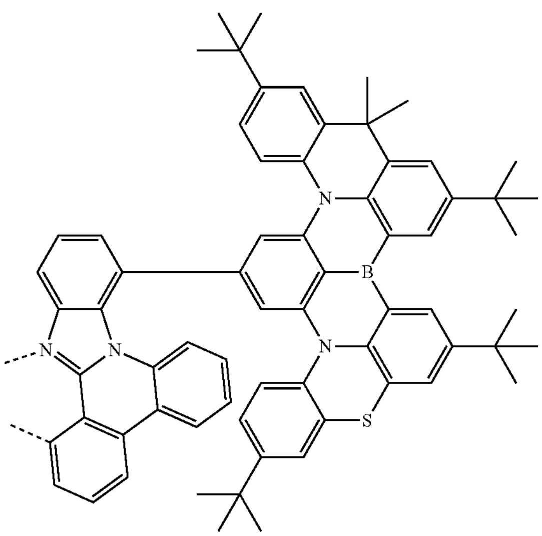

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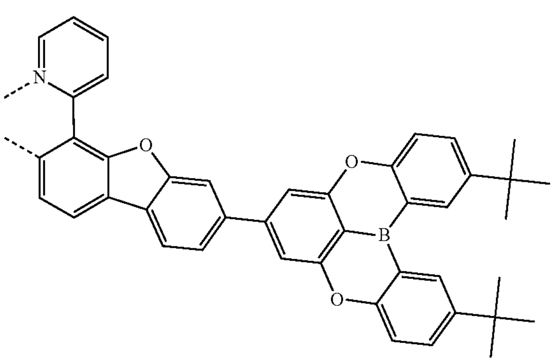

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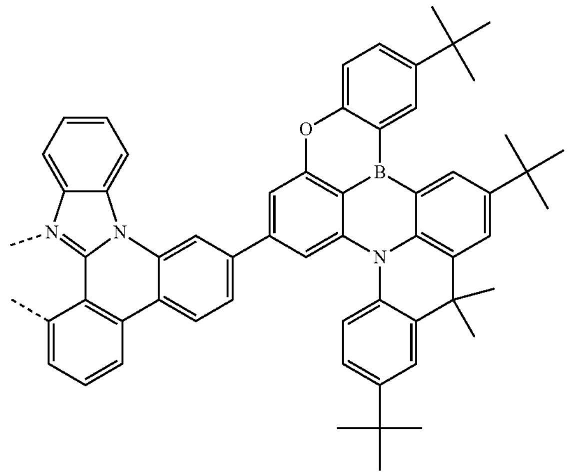

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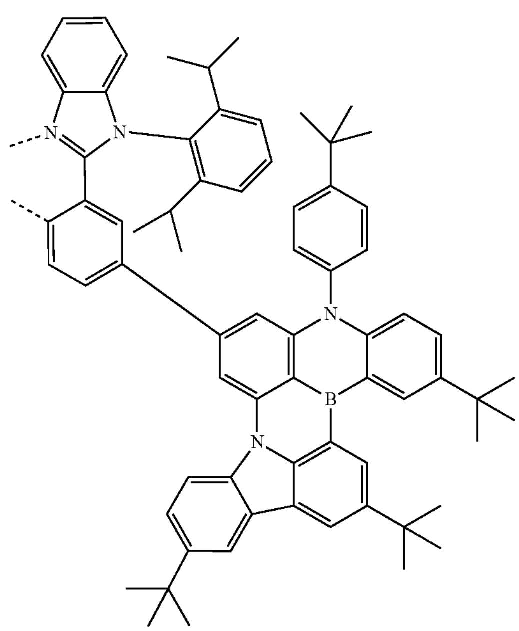

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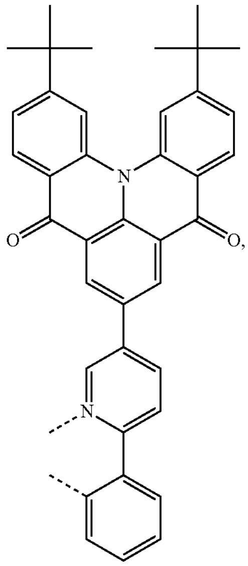

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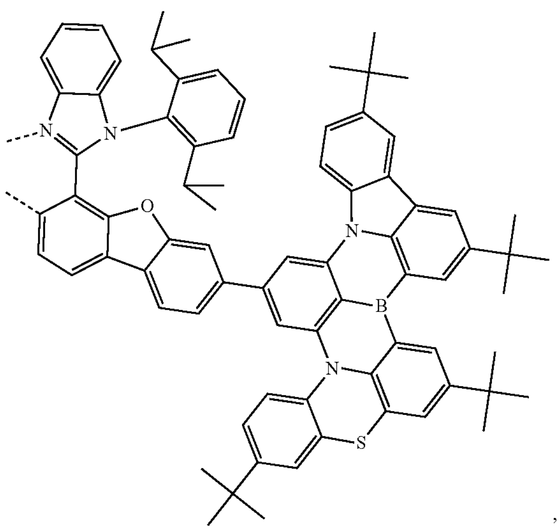

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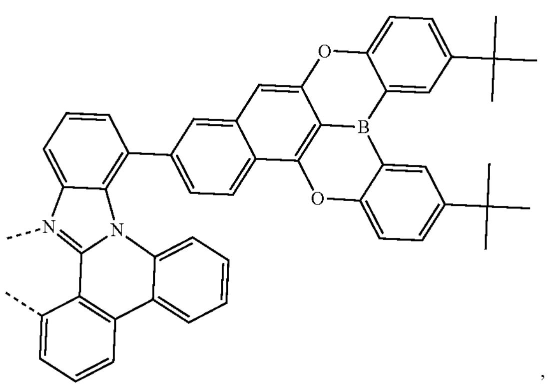

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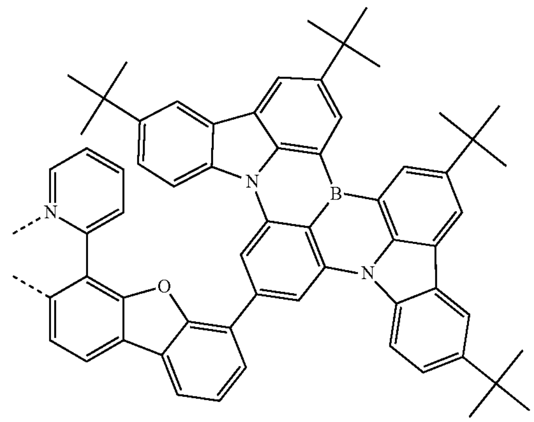

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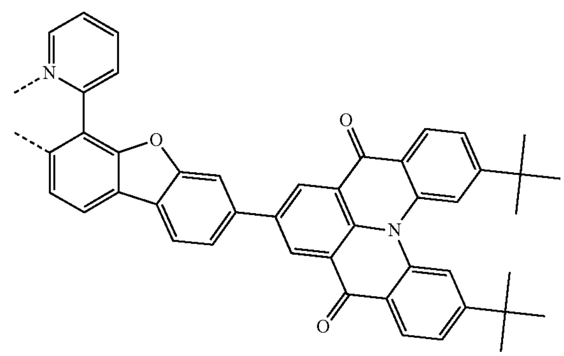

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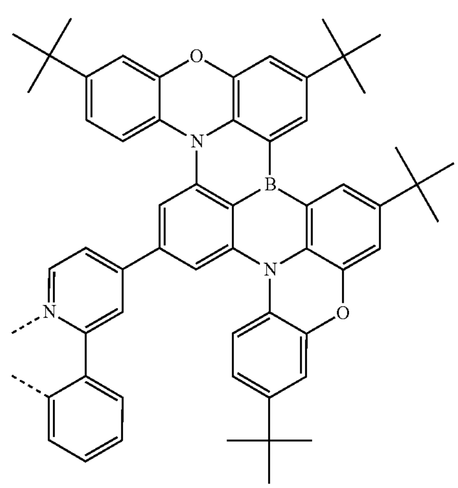

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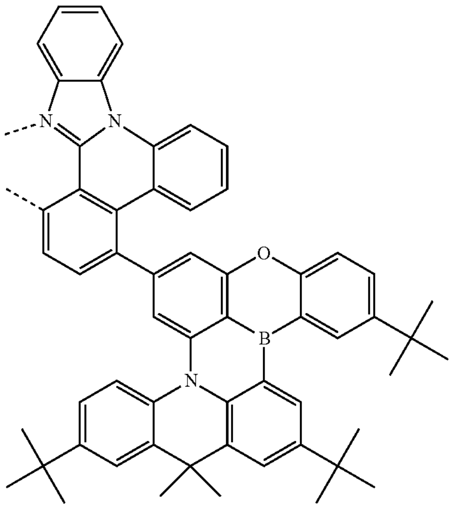

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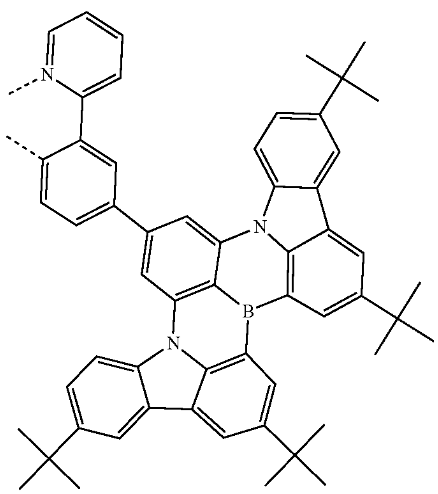

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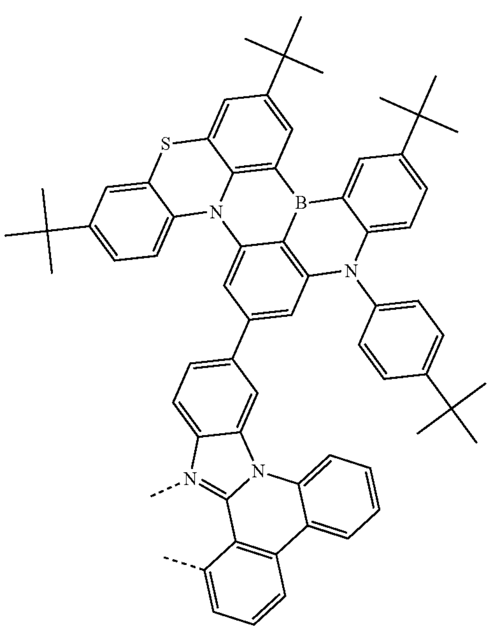

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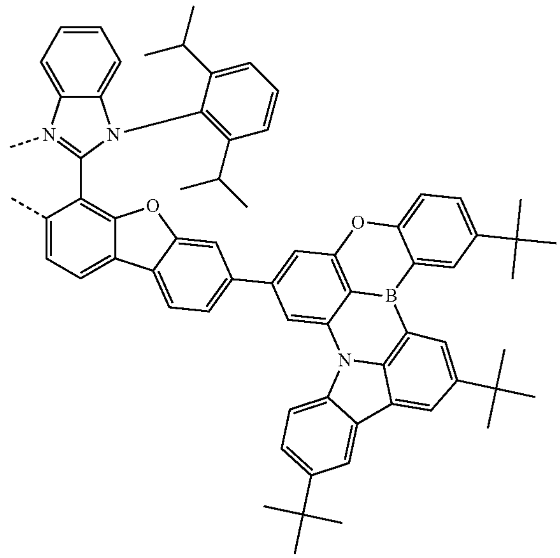

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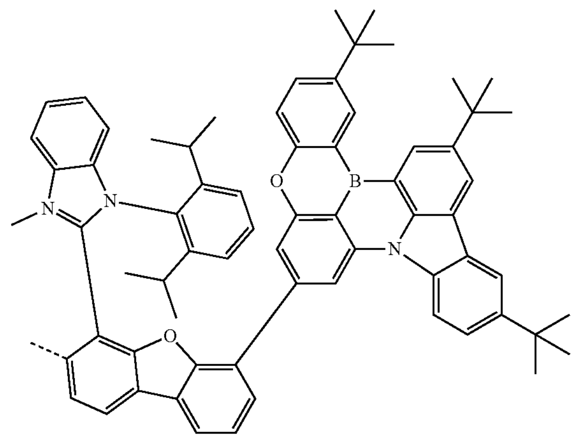

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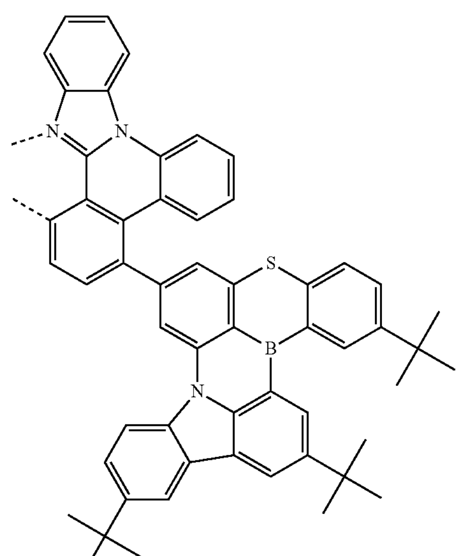

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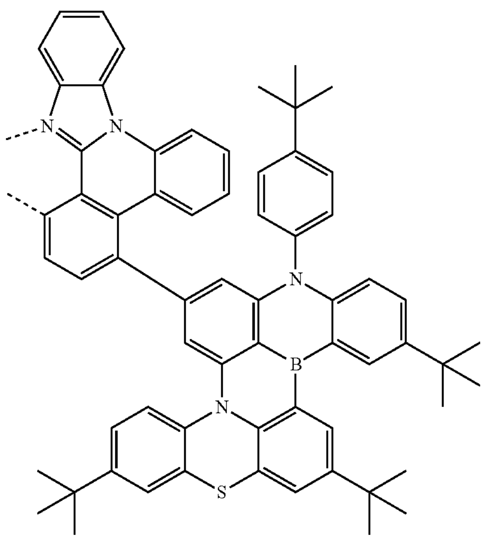

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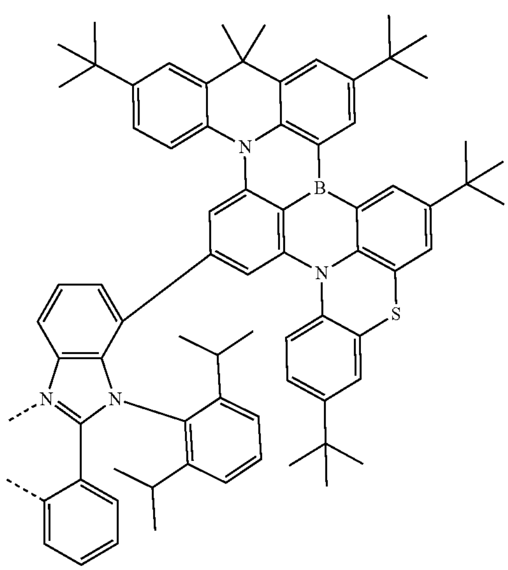

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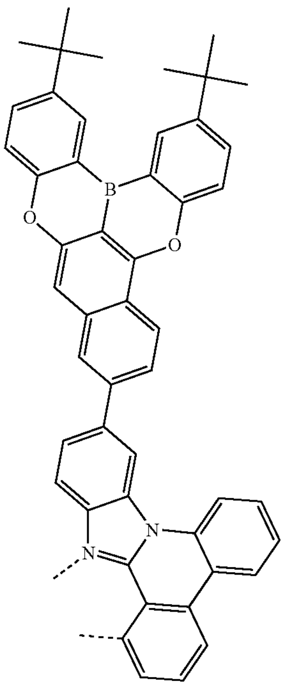

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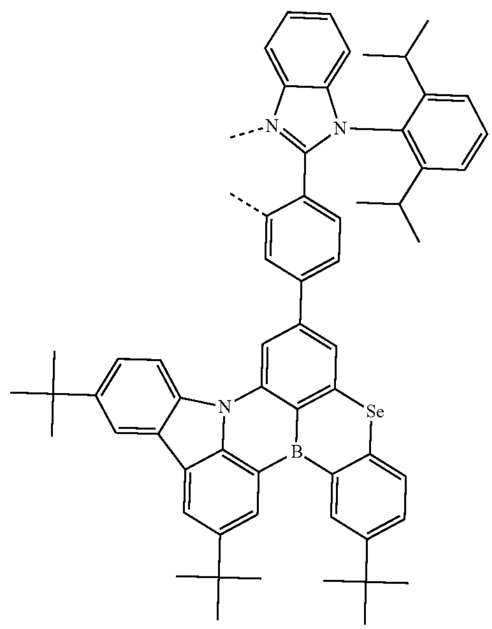

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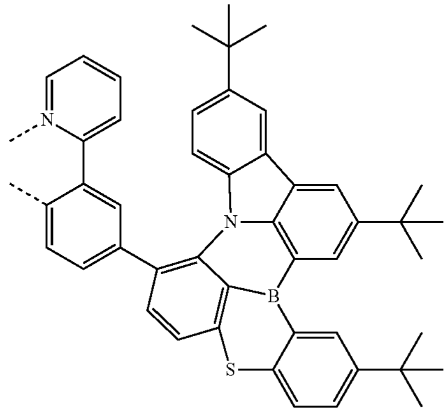

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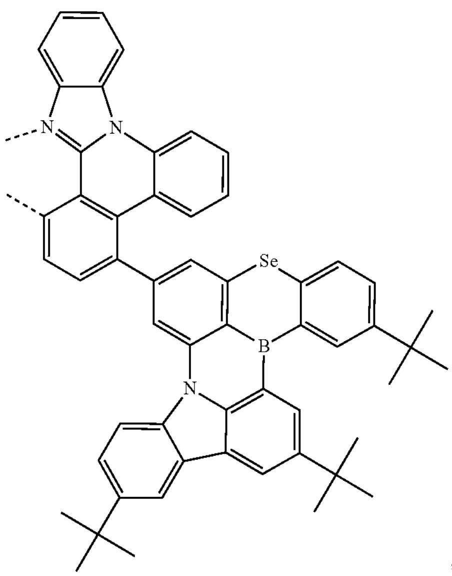

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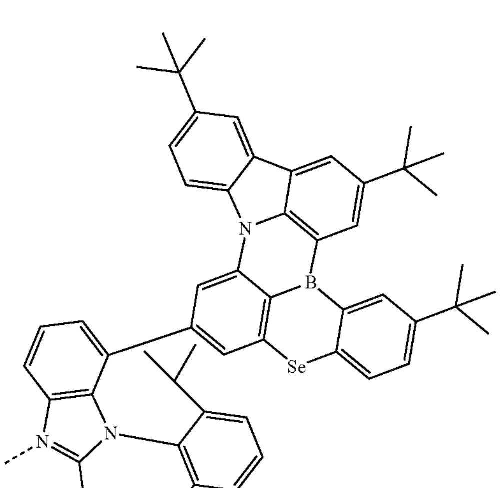

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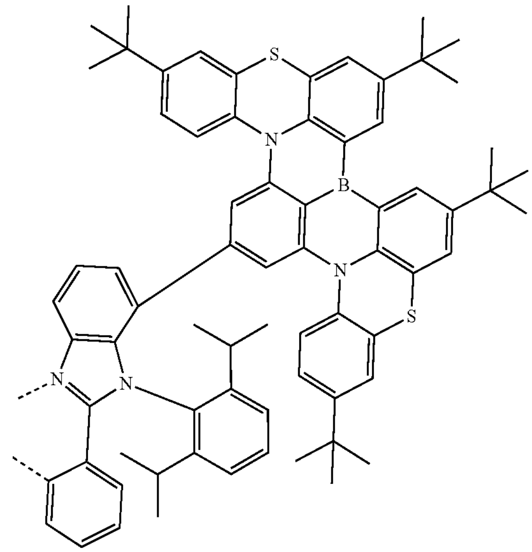

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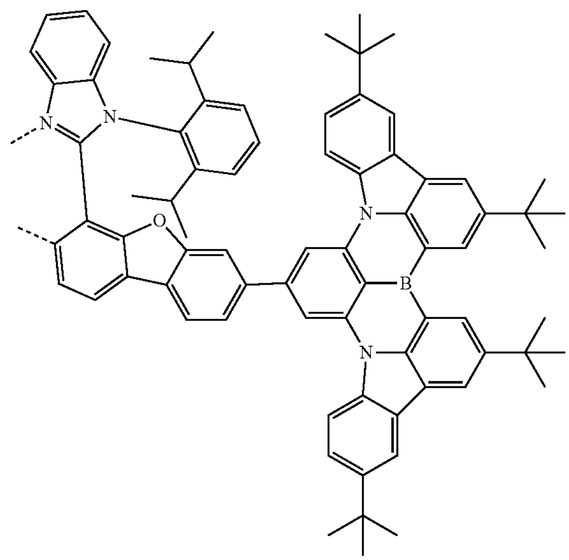

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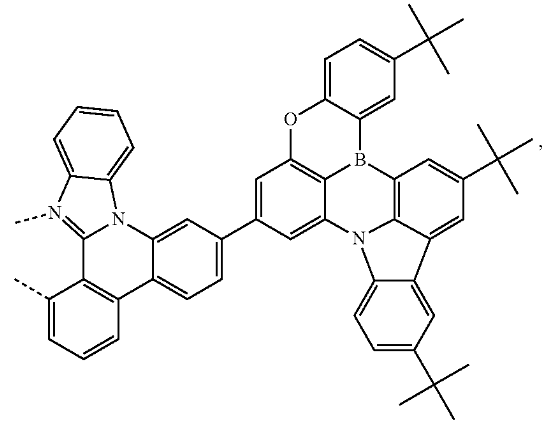

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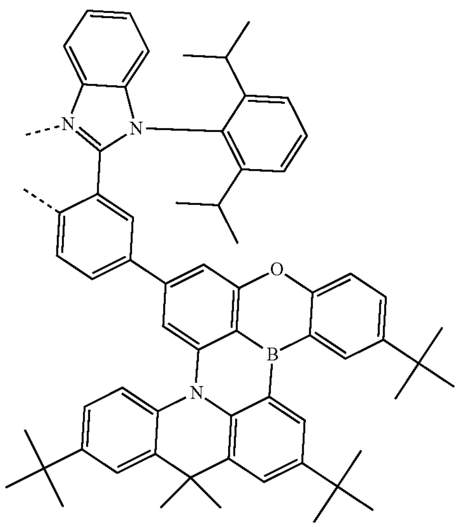

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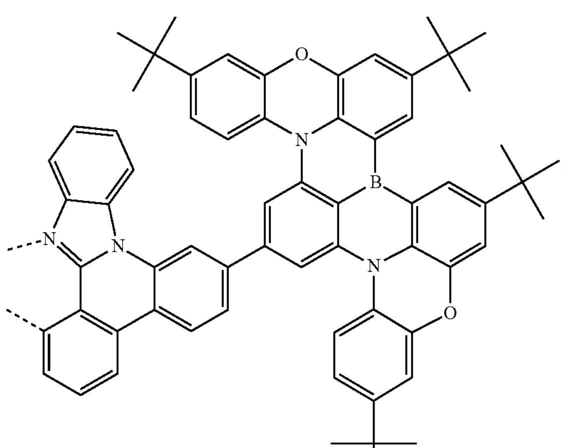

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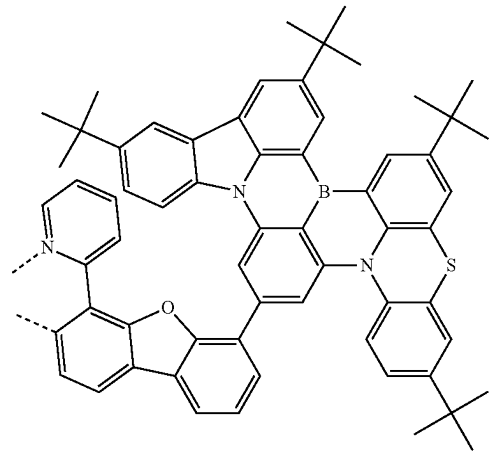

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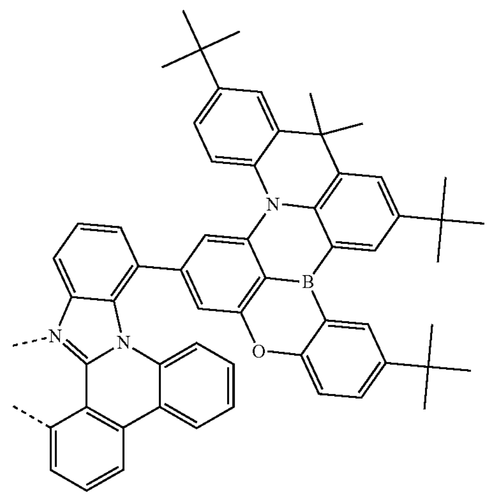

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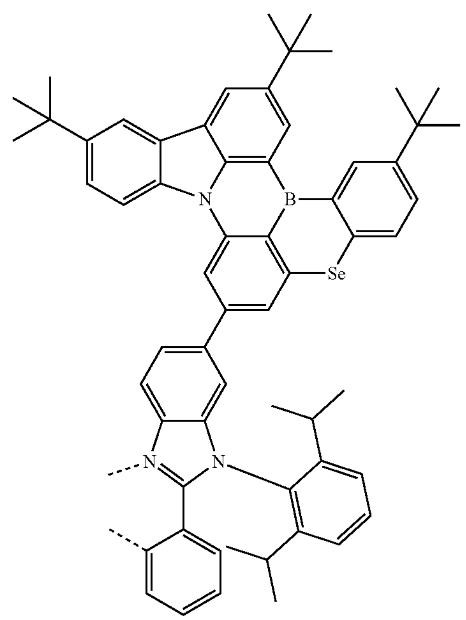

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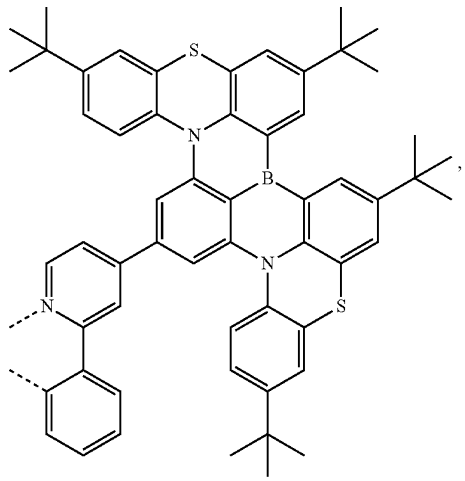

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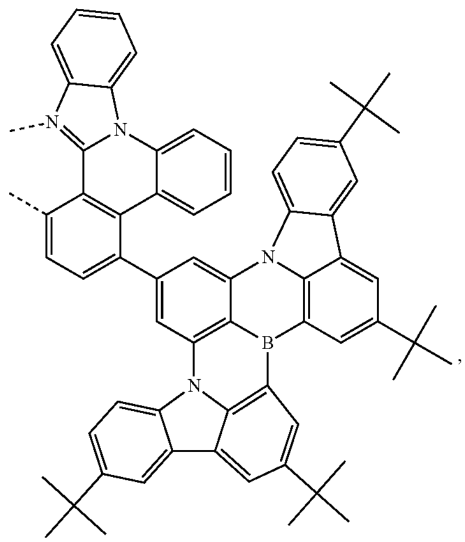

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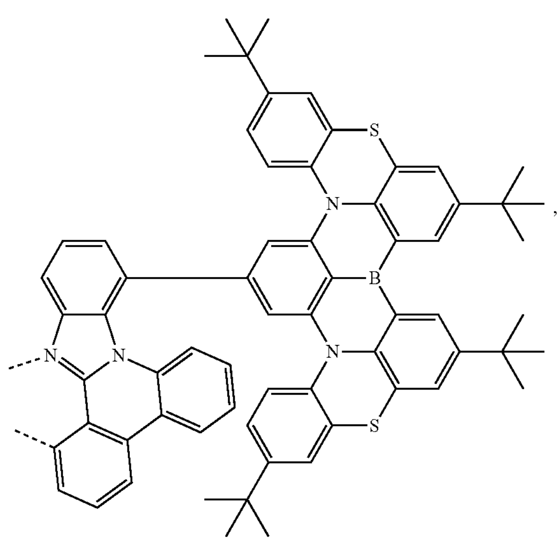

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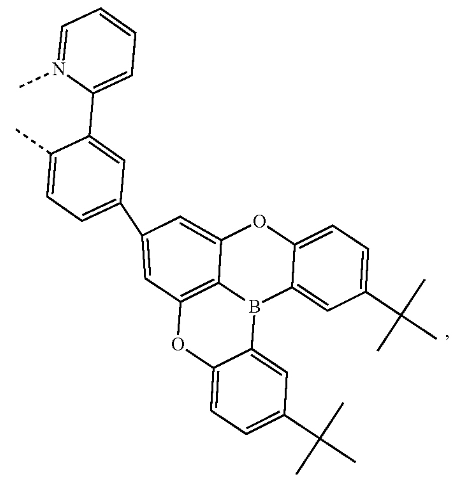

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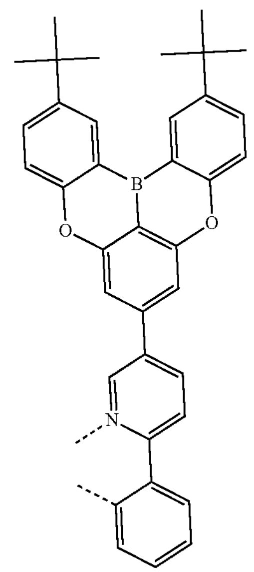

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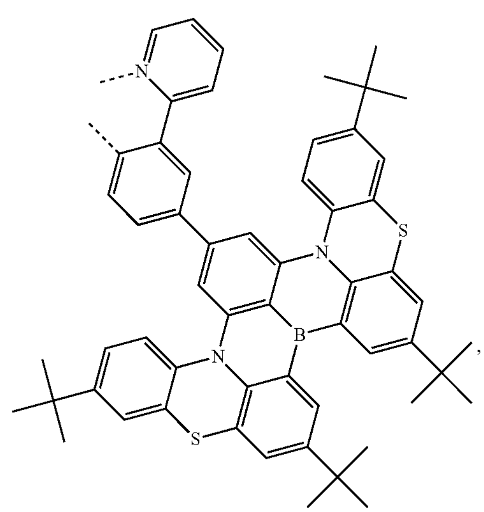

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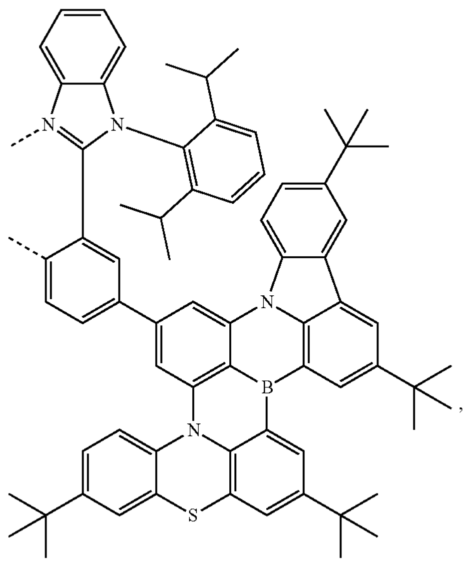

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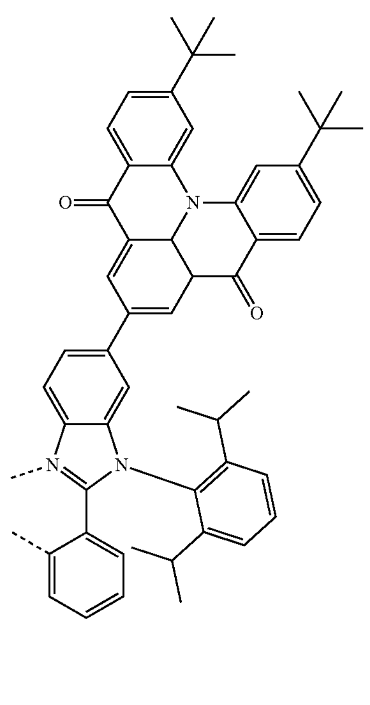

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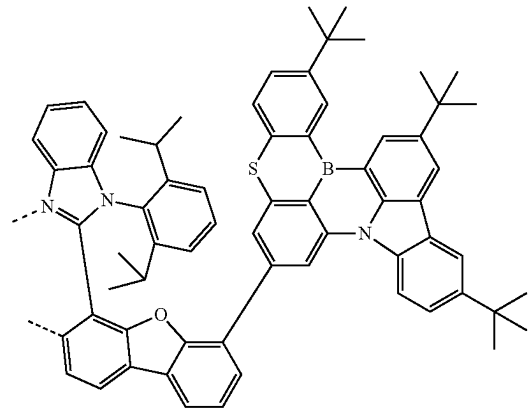

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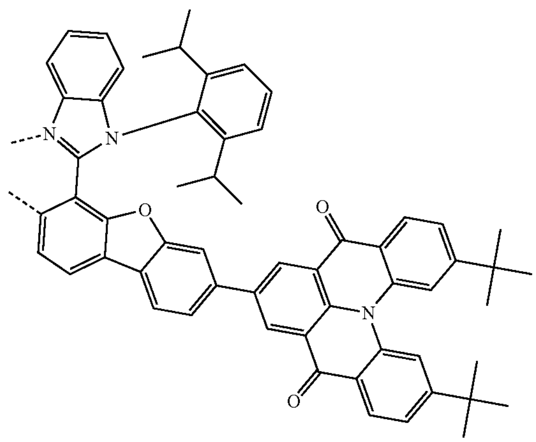

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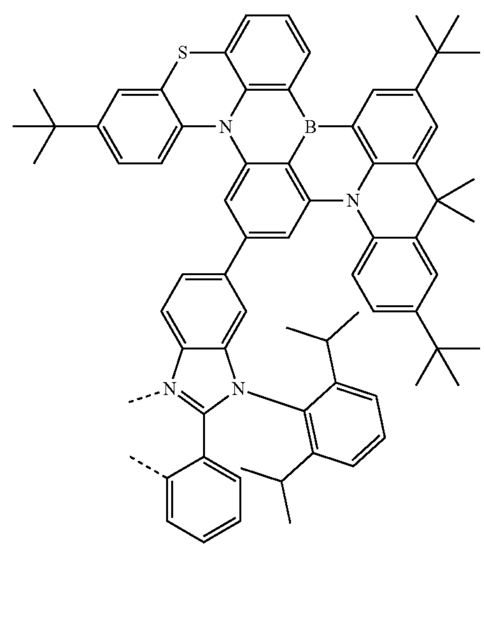

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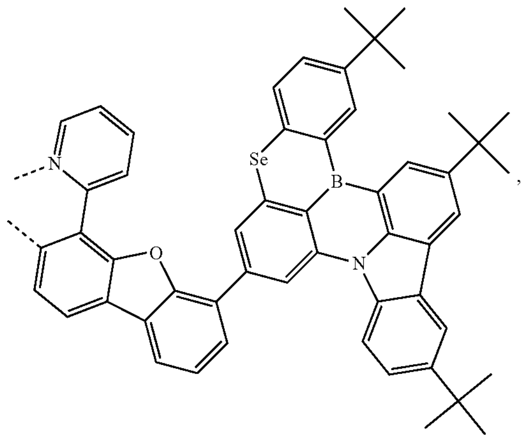
L_{A201}



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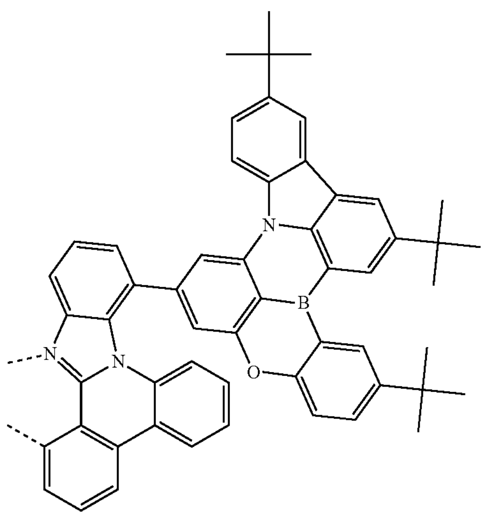

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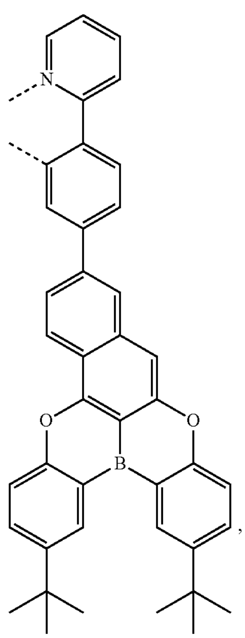

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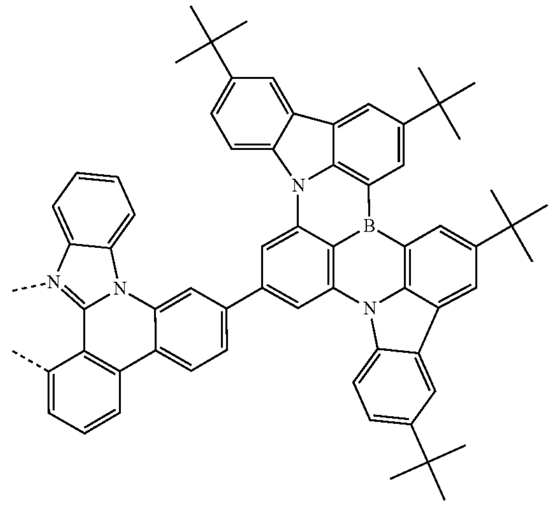

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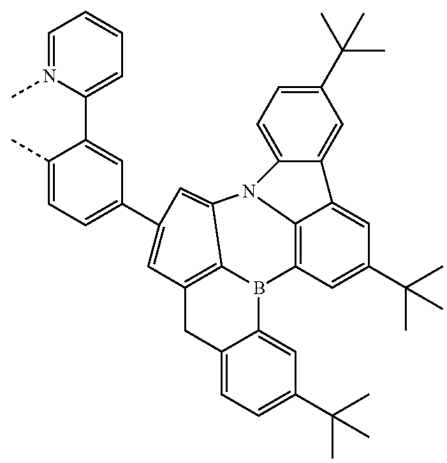

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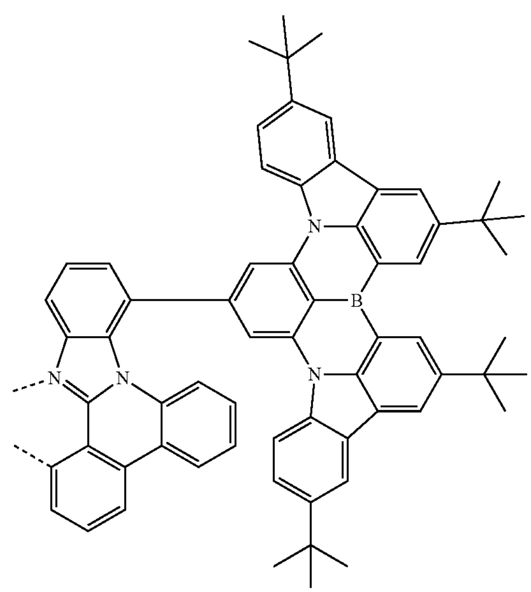

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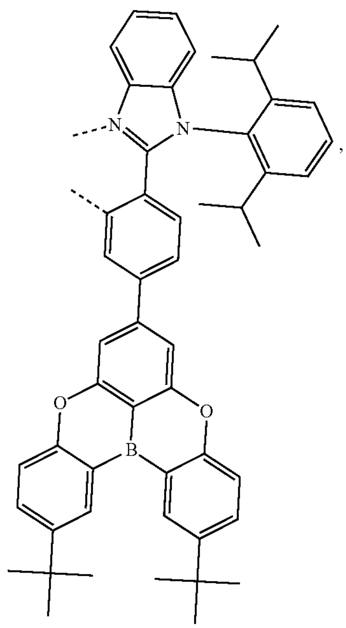

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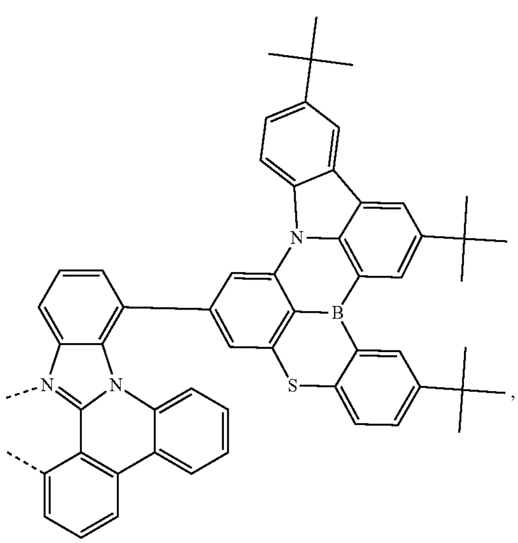

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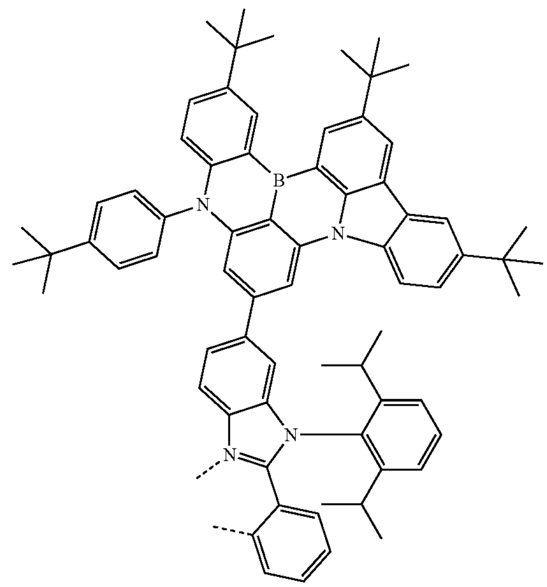

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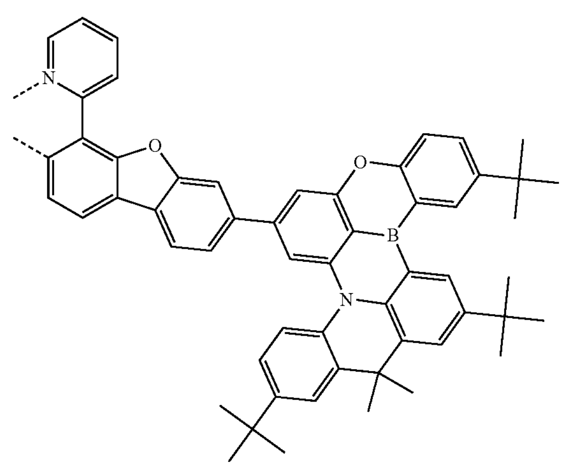

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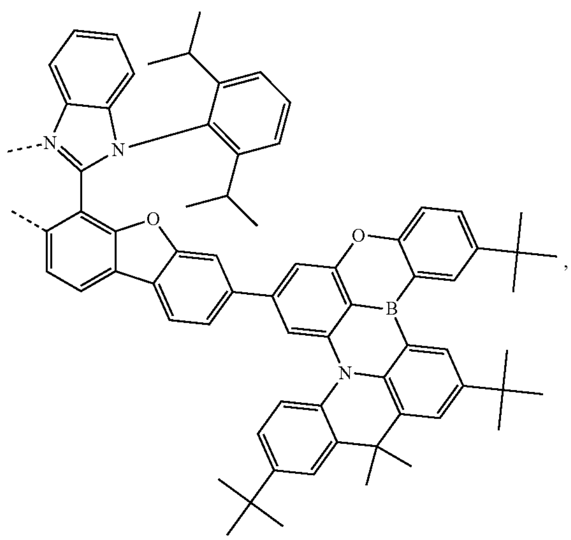

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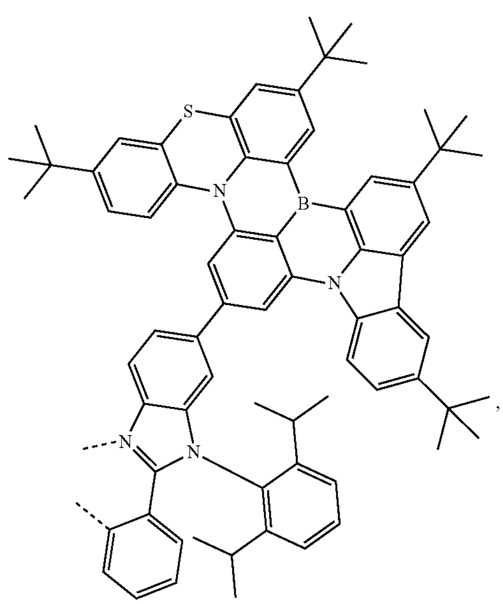

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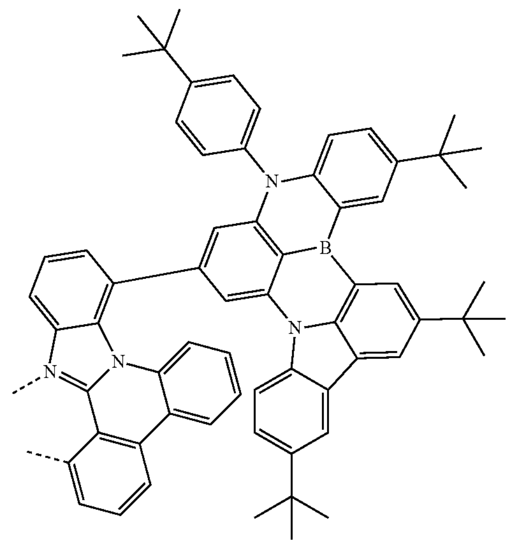
L$_{A213}$



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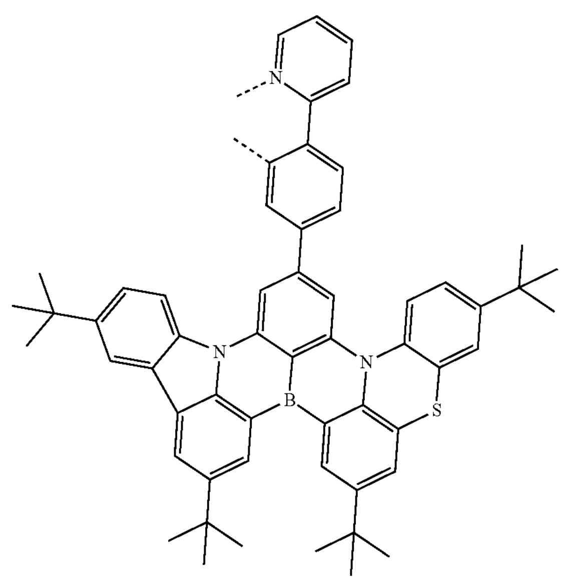

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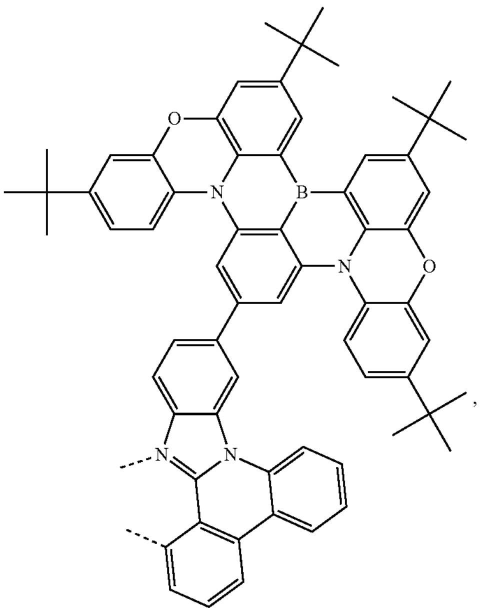

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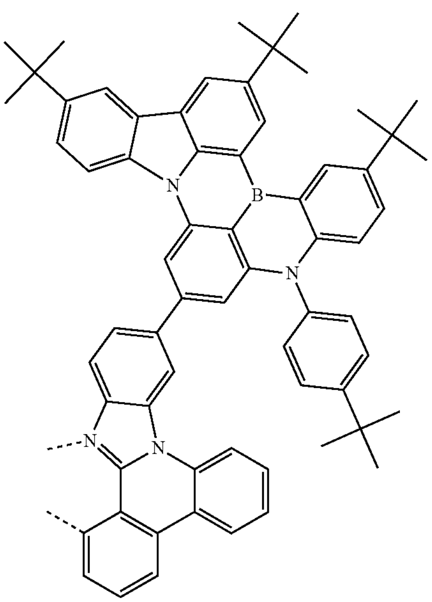

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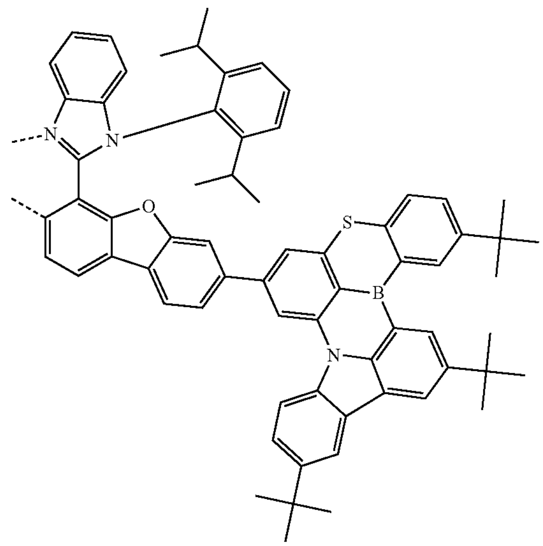

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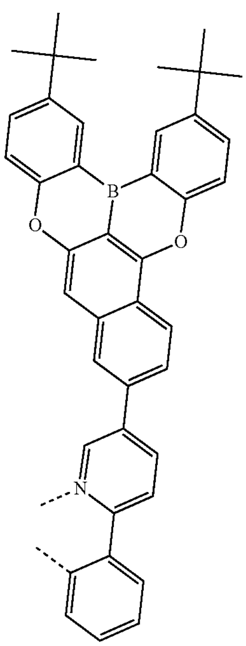

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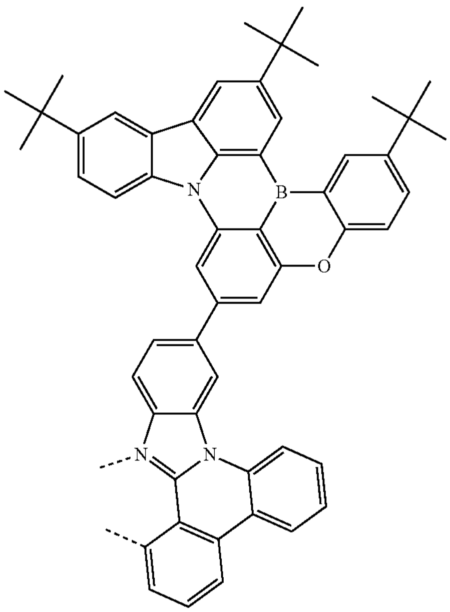

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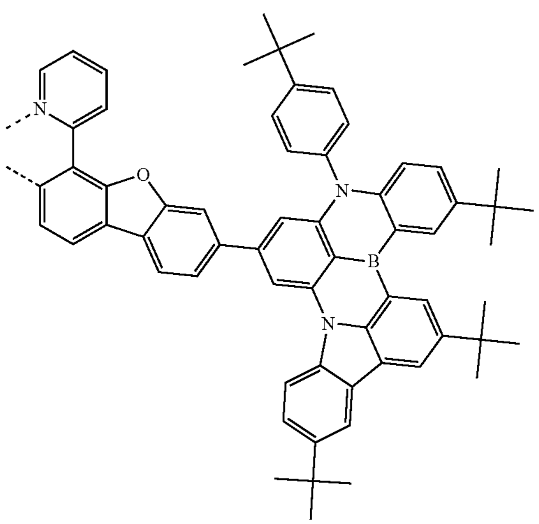

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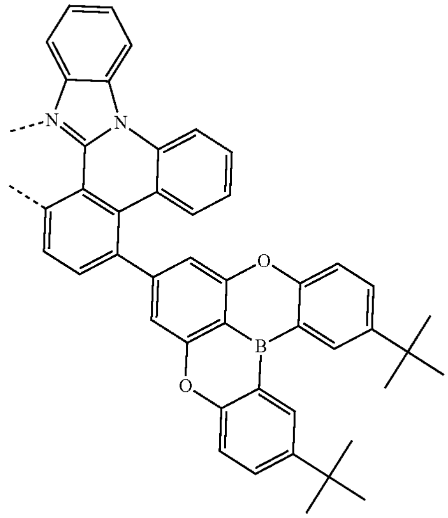

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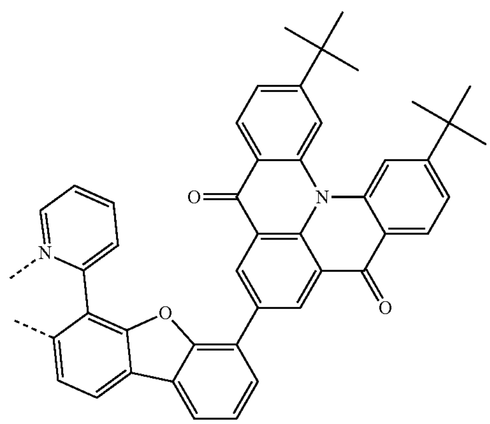

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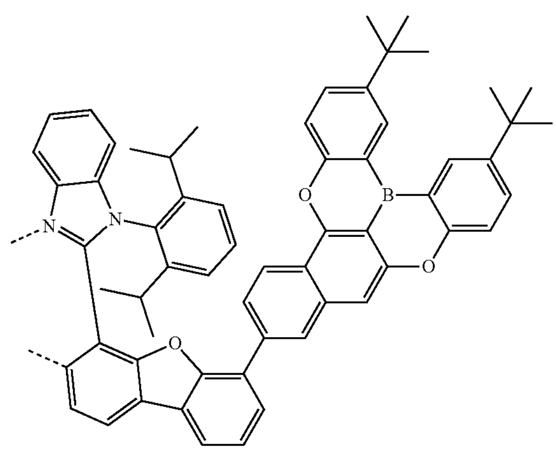

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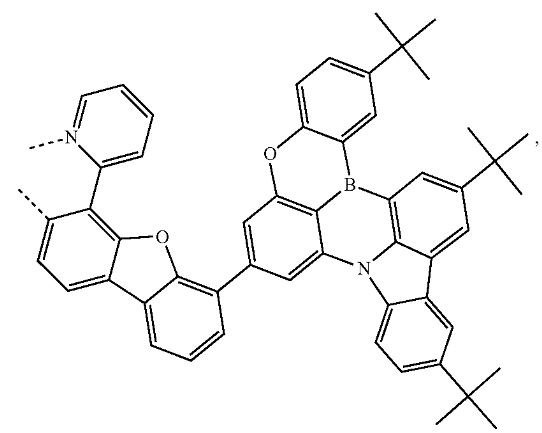

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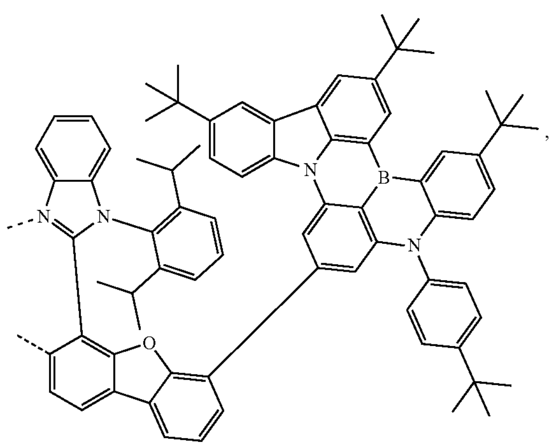

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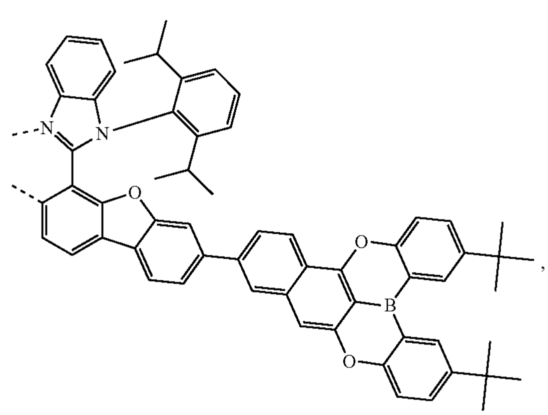

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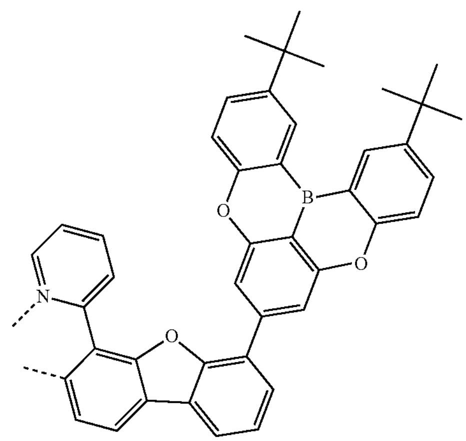

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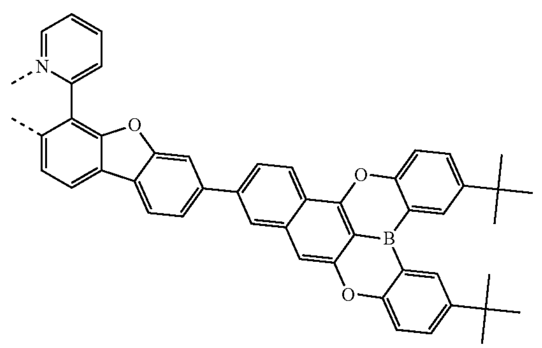

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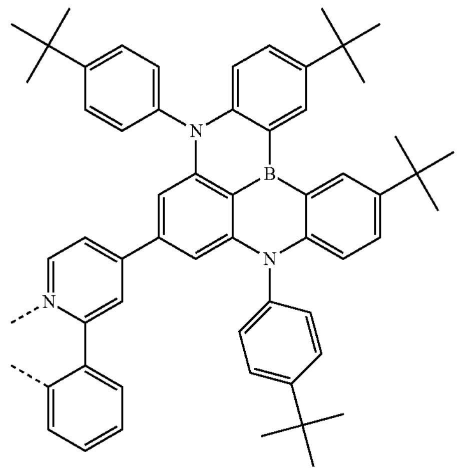

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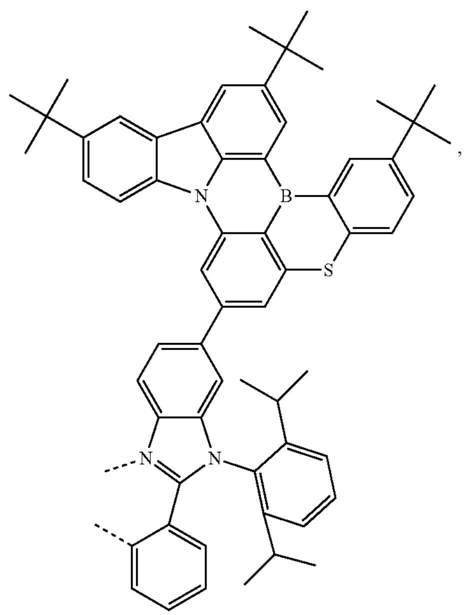
L_{A230}



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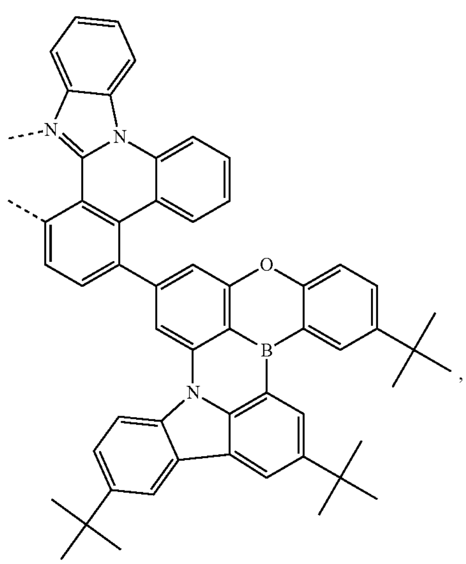

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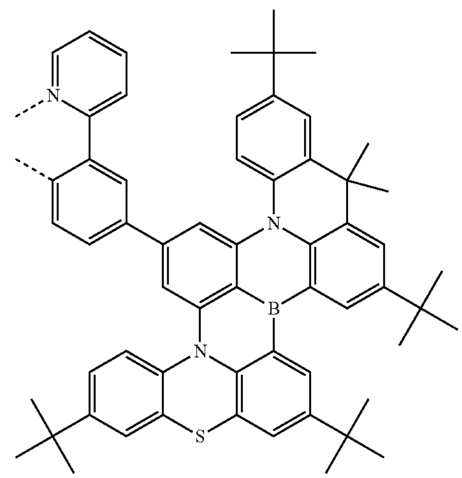

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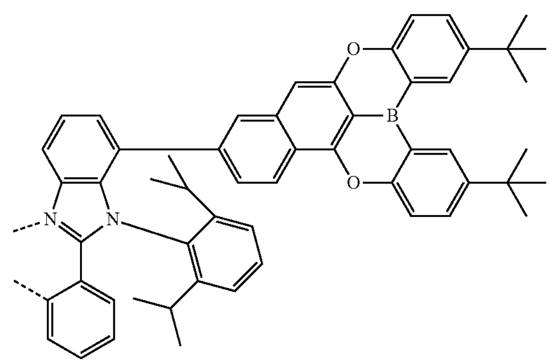

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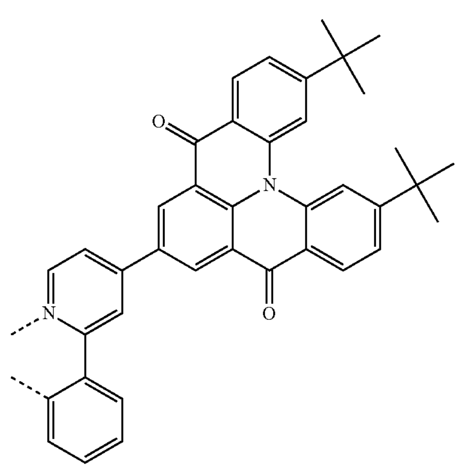

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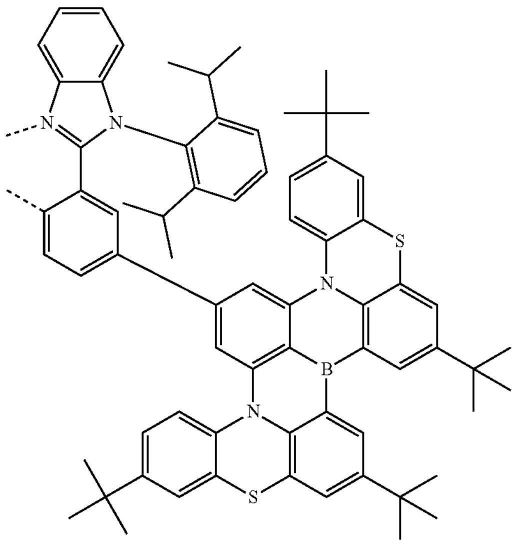

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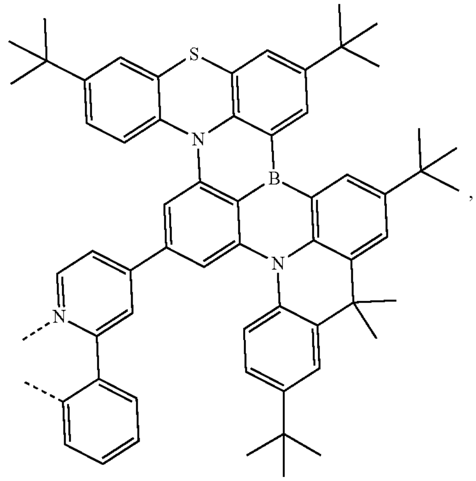

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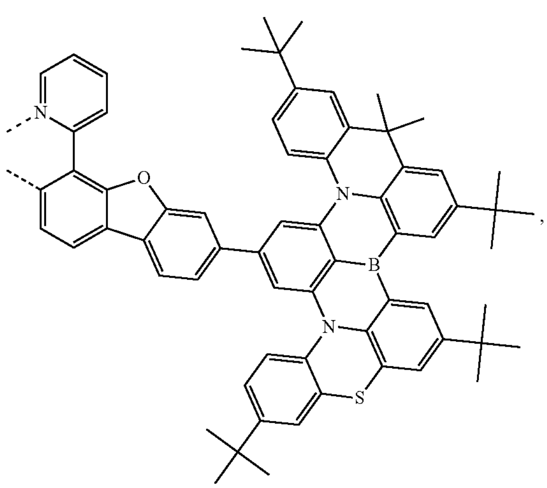

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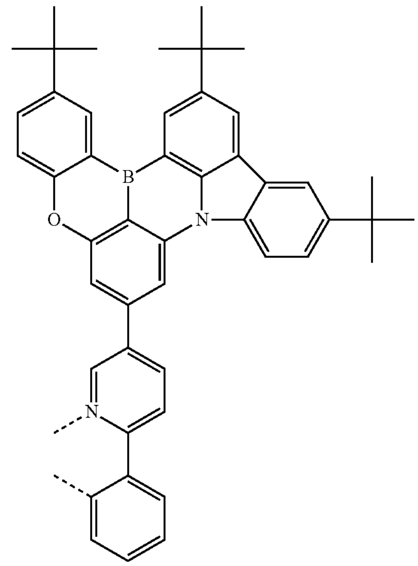

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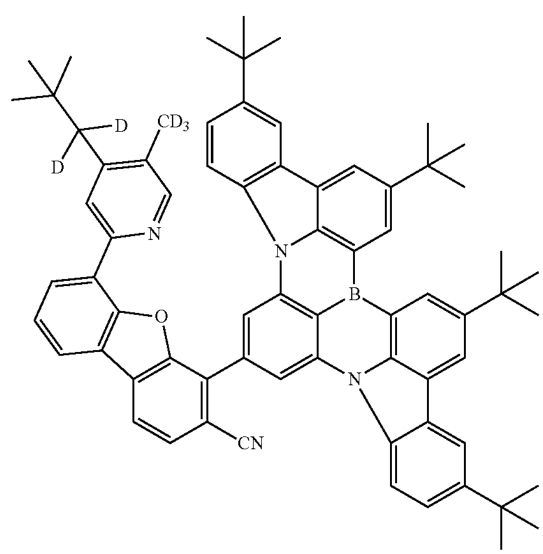

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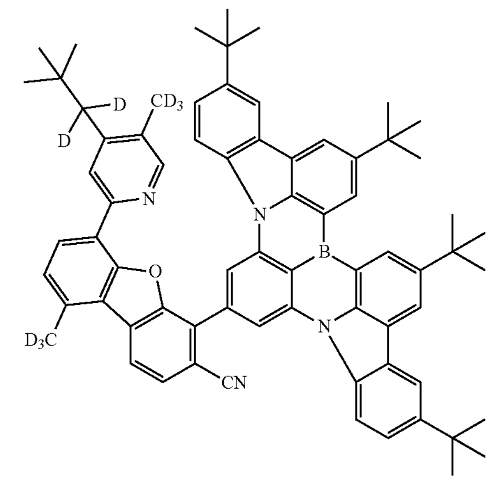
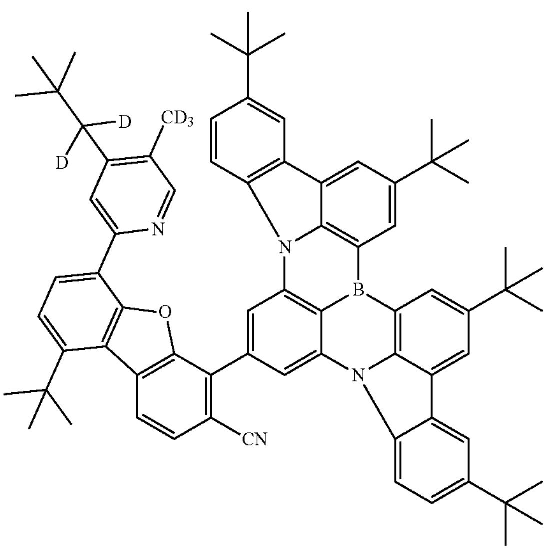
L_{A240}



and

L_{A241}



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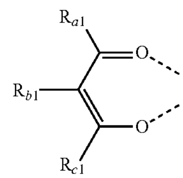
11. The compound of claim **1**, wherein the compound has a formula of $M(L_A)_p(L_B)_q(L_C)_r$ wherein L_B and L_C are each a bidentate ligand; and wherein p is 1, 2, or 3; q is 0, 1, or 2; r is 0, 1, or 2; and p+q+r is the oxidation state of the metal M.

12. The compound of claim **11**, wherein the compound has a formula selected from the group consisting of $Ir(L_A)_3$, $Ir(L_A)(L_B)_2$, $Ir(L_A)_2(L_B)$, $Ir(L_A)_2(L_C)$, and $Ir(L_A)(L_B)(L_C)$; and wherein L_A, L_B, and L_C are different from each other; or a formula of $Pt(L_A)(L_{B'})$.

13. The compound of claim **11**, wherein L_B and L_C are each independently selected from the group consisting of:



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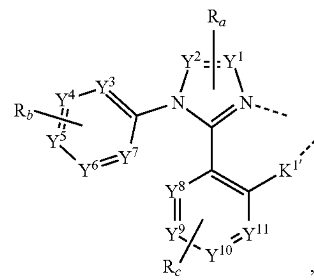

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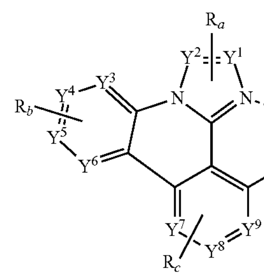

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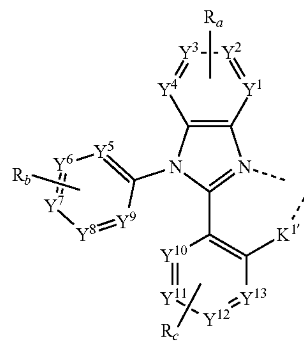

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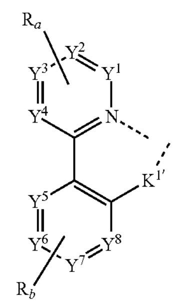

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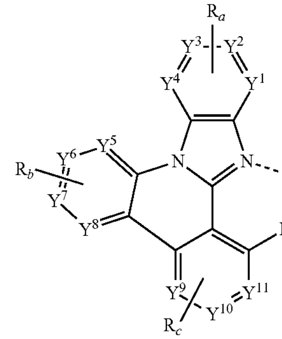

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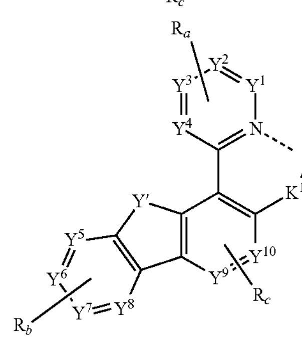

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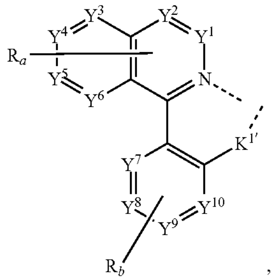
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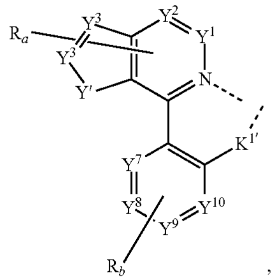
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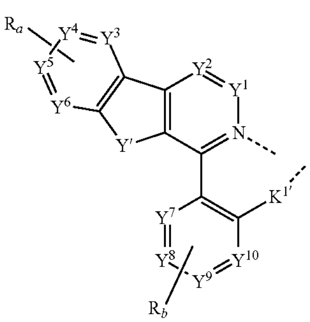
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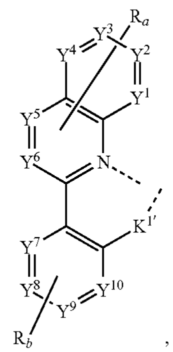
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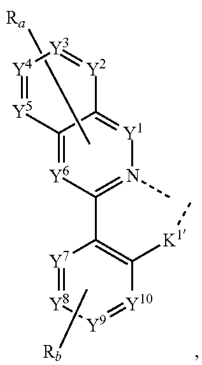
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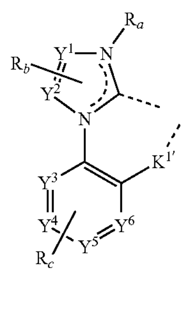
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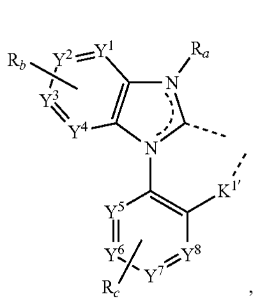
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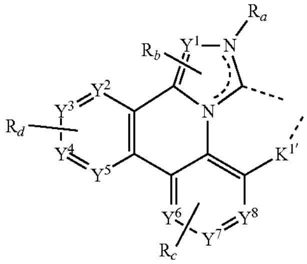
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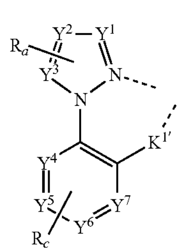
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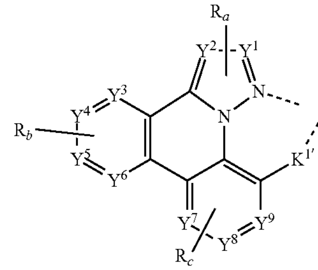
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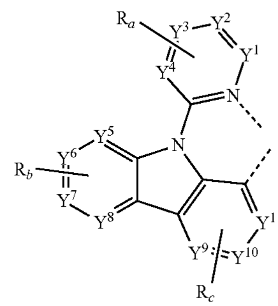
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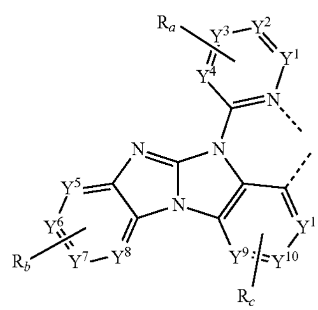
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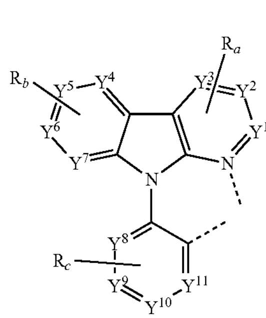
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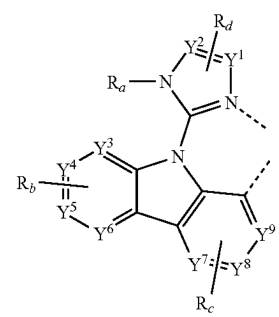
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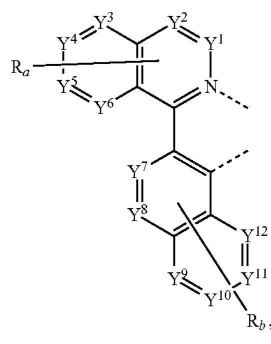
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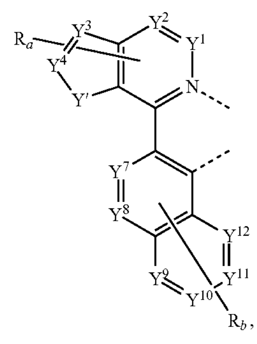
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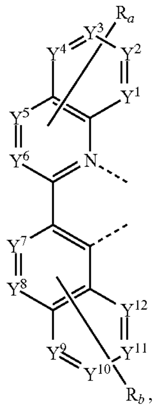

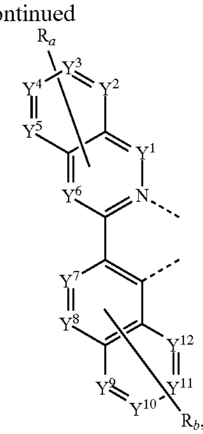

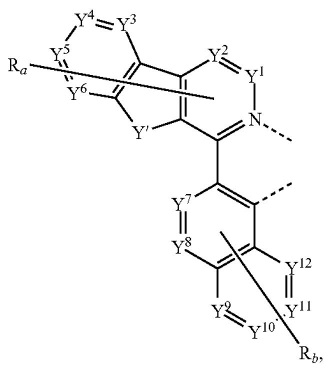

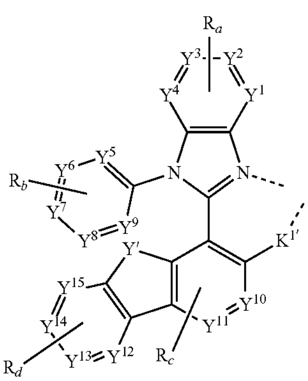

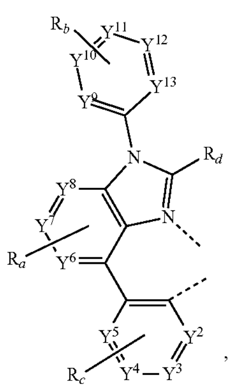

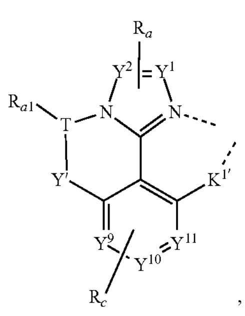

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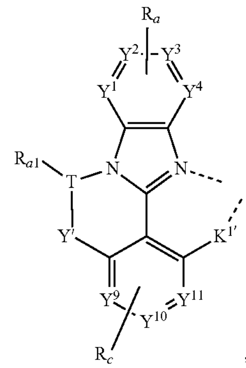

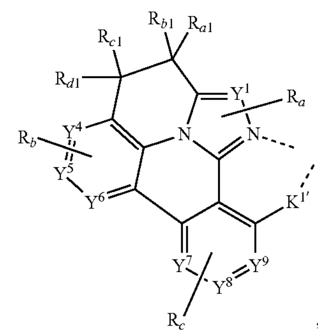

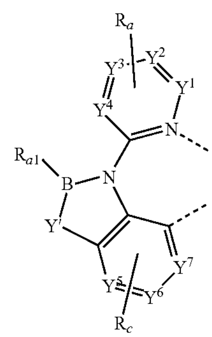

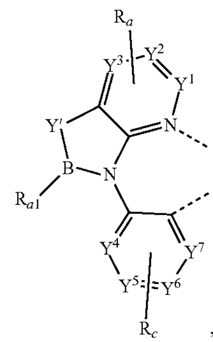

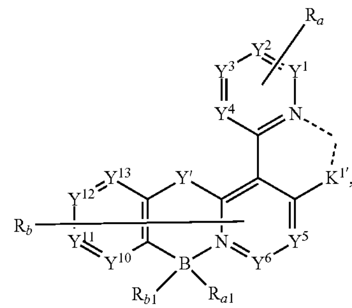

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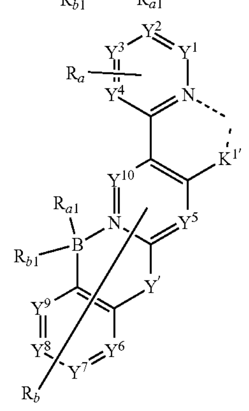


wherein:

T is selected from the group consisting of B, Al, Ga, and In;

$K^{1'}$ is selected from the group consisting of a single bond, O, S, NR_e, PR_e, BR_e, CR_eR_f, and SiR_eR_f;

each of Y^1 to Y^{13} is independently selected from the group consisting of C and N;

Y^1 is selected from the group consisting of BR_e, BR_eR_f, NR_e, PR_e, $P(O)R_e$, O, S, Se, C=O, C=S, C=Se, $C{=}NR_e$, $C{=}CR_eR_f$, S=O, SO_2, CR_eR_f, SiR_eR_f, and GeR_eR_f;

R_e and R_f can be fused or joined to form a ring;

each R_a, R_b, R_c, and R_d independently represents from mono to the maximum allowed number of substitutions, or no substitution;

each of R_{a1}, R_{b1}, R_{c1}, R_{d1}, R_a, R_b, R_c, R_d, R_e, and R_f is independently a hydrogen or a subsituent selected from the group consisting of deuterium, halide, alkyl, cycloalkyl, heteroalkyl, arylalkyl, alkoxy, aryloxy, amino, silyl, germyl, boryl, alkenyl, cycloalkenyl, heteroalkenyl, alkynyl, aryl, heteroaryl, acyl, carbonyl, carboxylic acid, ester, nitrile, isonitrile, sulfanyl, selenyl, sulfinyl, sulfonyl, phosphino, and combinations thereof; and

any two substituents of R_{a1}, R_{b1}, R_{c1}, R_{d1}, R_a, R_b, R_c, and R_d can be fused or joined to form a ring or form a multidentate ligand.

14. The compound of claim **11**, wherein L_A can be selected from L_{Ai}, wherein i is an integer from 1 to 241; and L_B is selected from L_{Bk}, wherein k is an integer from 1 to 541, wherein:

when the compound has formula $Ir(L_{Ai})_3$, the compound is selected from the group consisting of $Ir(L_{A1})_3$ to $Ir(L_{A241})_3$;

when the compound has formula $Ir(L_{Ai})(L_{Bk})_2$, the compound is selected from the group consisting of $Ir(L_{A1})(L_{B1})_2$ to $Ir(L_{A241})(L_{B541})_2$;

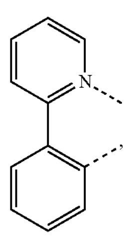
when the compound has formula $Ir(L_{Ai})_2(L_{Bk})$, the compound is selected from the group consisting of $Ir(L_{A1})_2(L_{B1})$ to $Ir(L_{A241})_2(L_{B541})$;

when the compound has formula $Ir(L_{Ai})_2(L_{Cj-I})$, j is an integer from 1 to 1416, wherein the compound is selected from the group consisting of $Ir(L_{A1})_2(L_{Cj-I})$ to $Ir(L_{A241})_2(L_{C1416-I})$; and

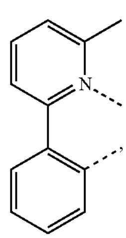
when the compound has formula $Ir(L_{Ai})_2(L_{Cj-II})$, j is an integer from 1 to 1416, wherein the compound is selected from the group consisting of $Ir(L_{A1})_2(L_{C1-II})$ to $Ir(L_{A241})_2(L_{C1416-II})$;

wherein k is an integer from 1 to 541, and each L_{Bk} has the structure defined as follows:

L_{B1}



L_{B2}



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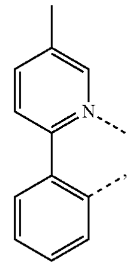
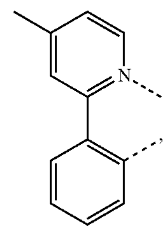
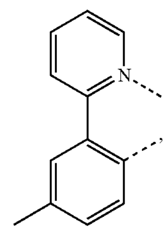
L_{B3}



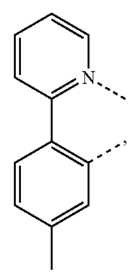
L_{B4}



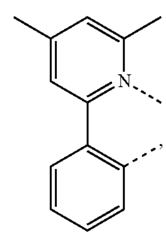
L_{B5}



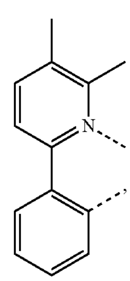
L_{B6}



L_{B7}



L_{B8}



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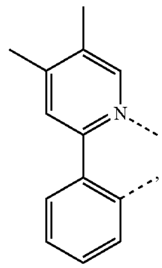
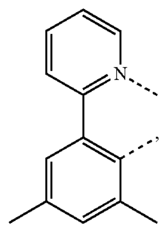
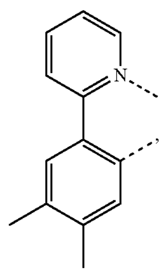
L_{B9}



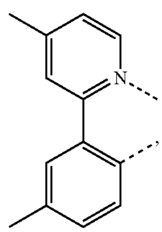
L_{B10}



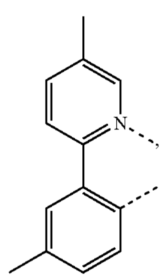
L_{B11}



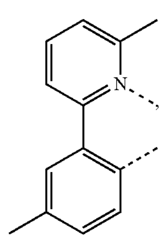
L_{B12}



L_{B13}



L_{B14}



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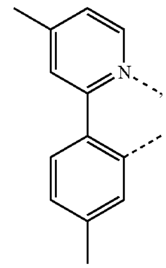
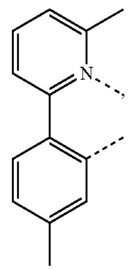
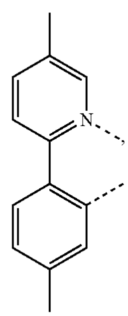
L_{B15}



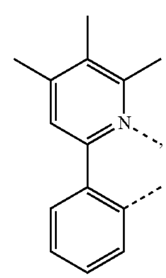
L_{B16}



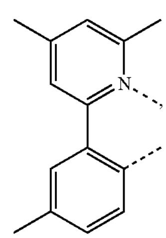
L_{B17}



L_{B18}



L_{B19}



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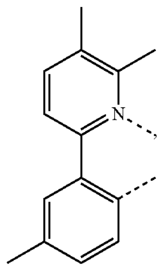
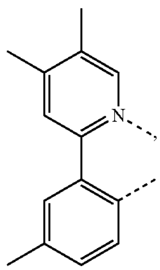

L_{B20}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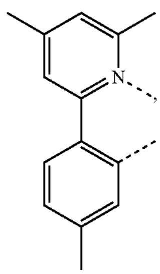
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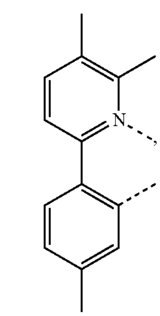
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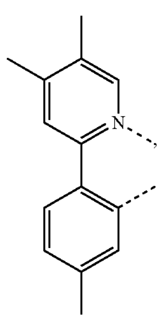
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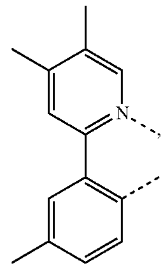
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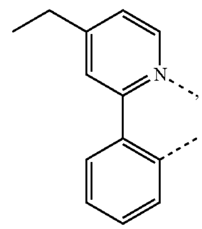
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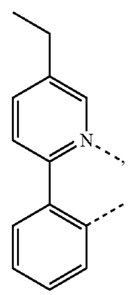
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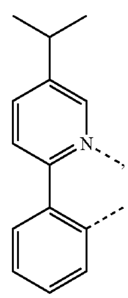
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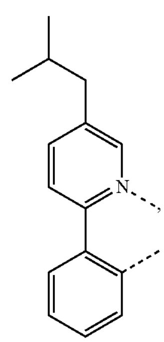
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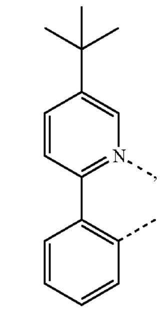
L_{B30}



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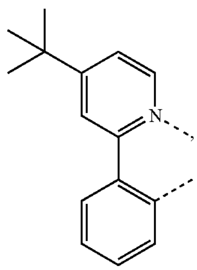
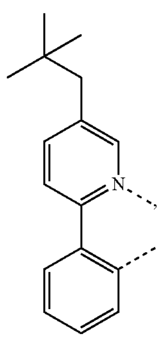


L_{B32}



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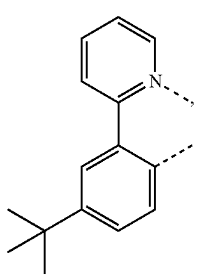

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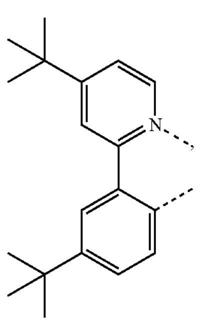

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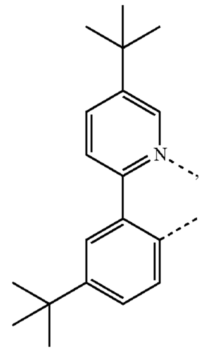
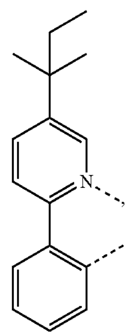
L_{B35}



L_{B36}



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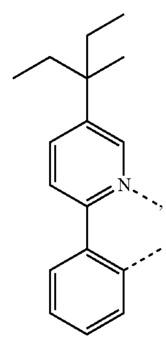
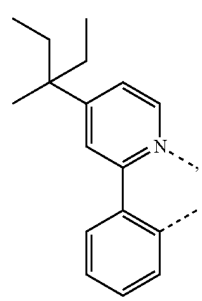


L_{B38}



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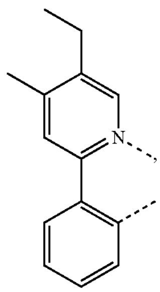
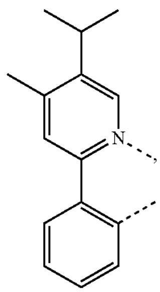
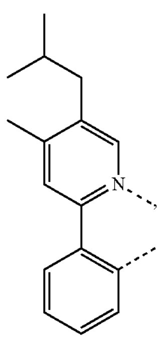
L_{B39}



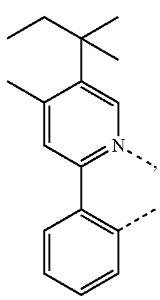
L_{B40}



L_{B41}



L_{B42}



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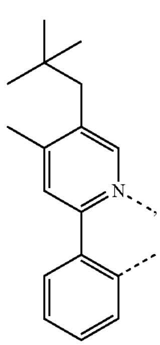

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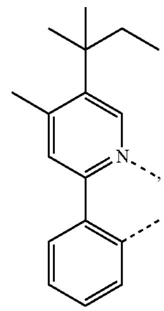

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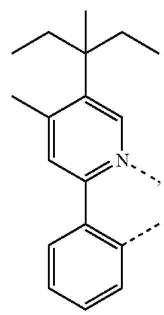

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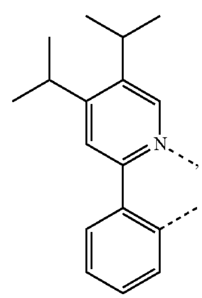

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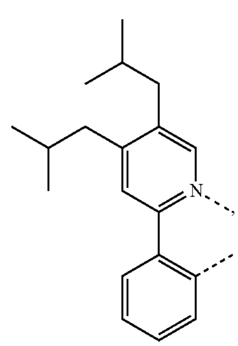

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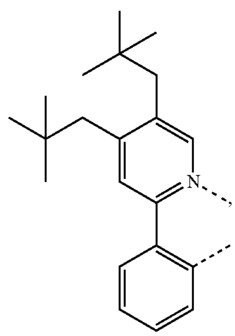

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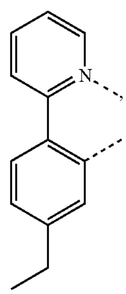

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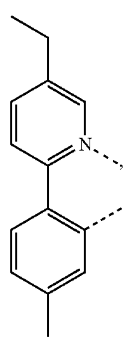

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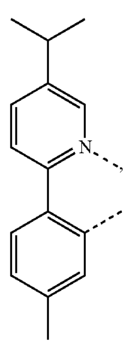

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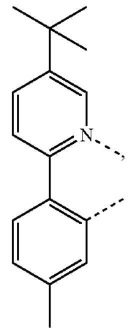

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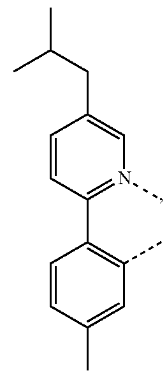

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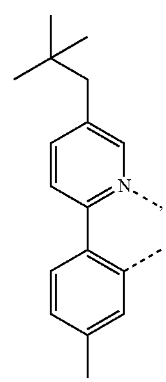

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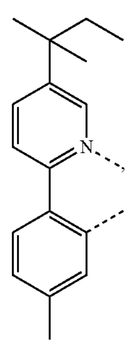

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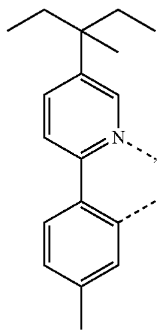

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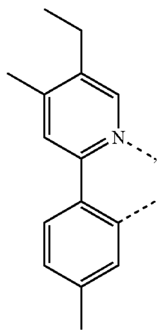

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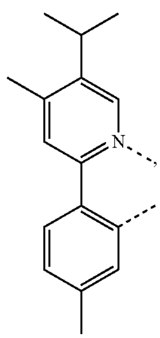

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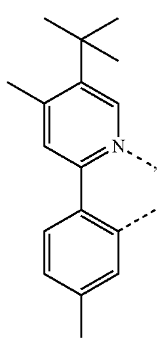

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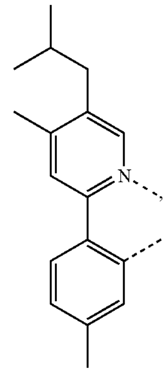

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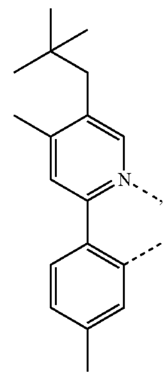

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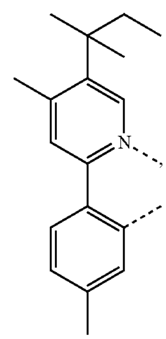

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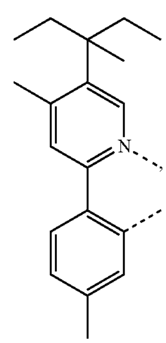

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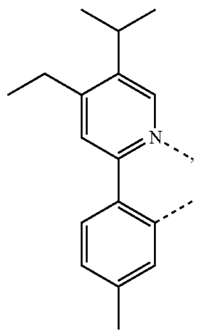

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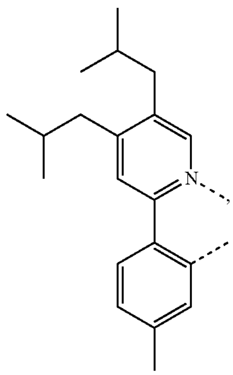

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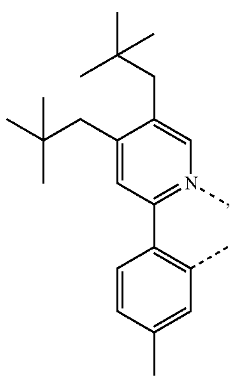

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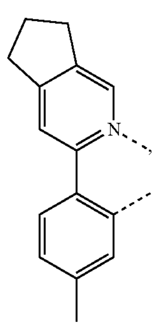

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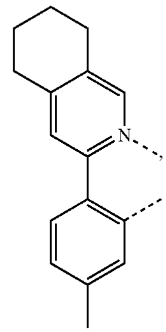

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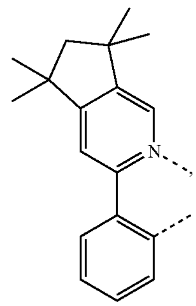

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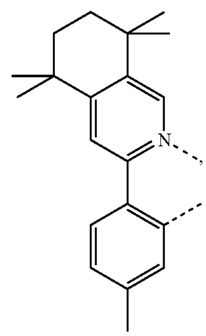

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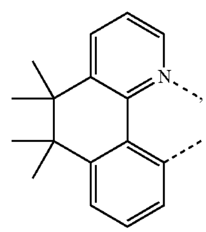

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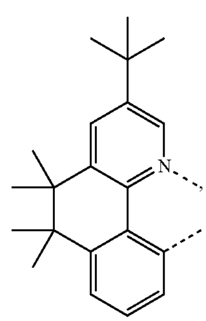

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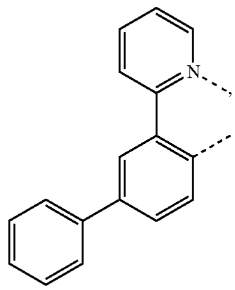
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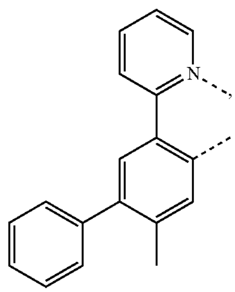
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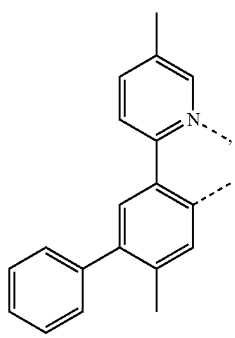
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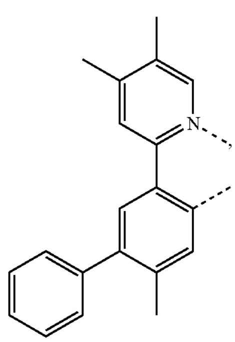
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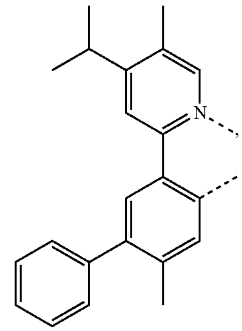
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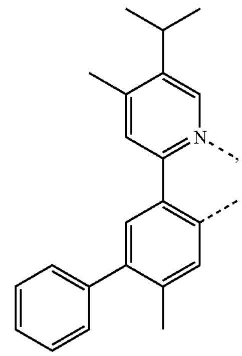
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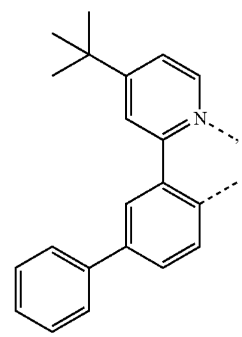
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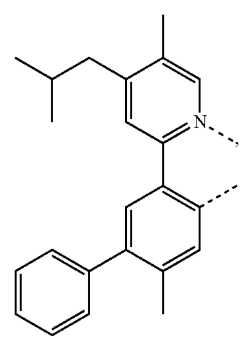
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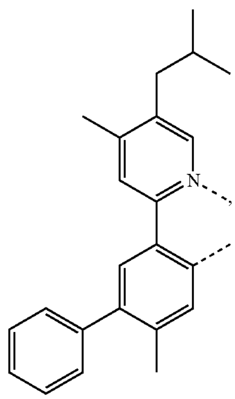

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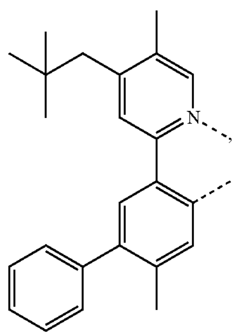

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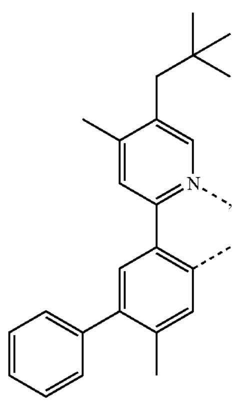

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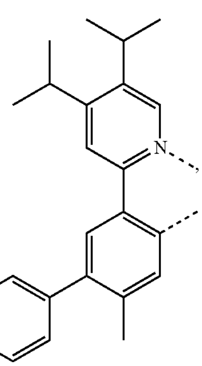

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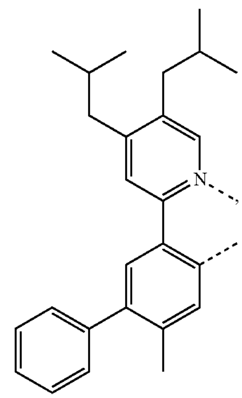

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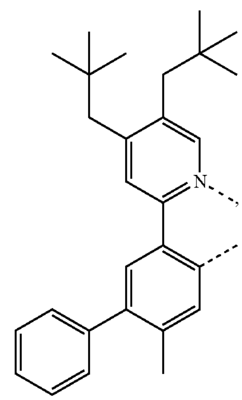

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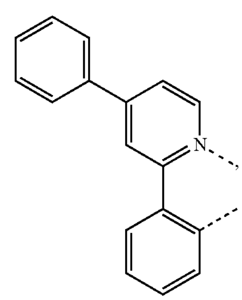

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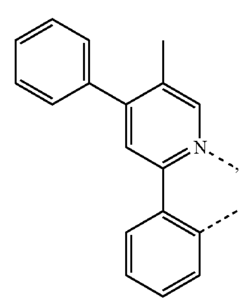

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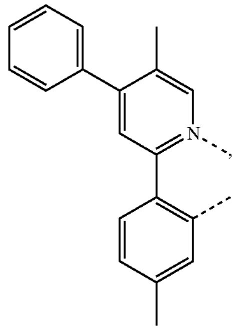

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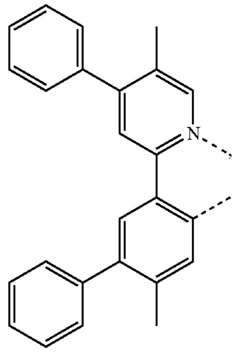

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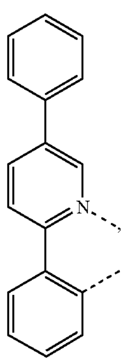

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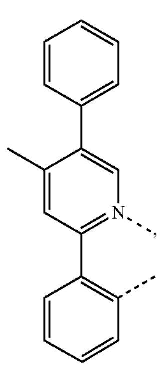

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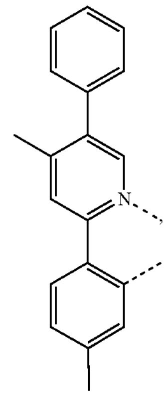

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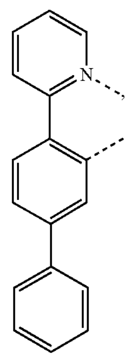

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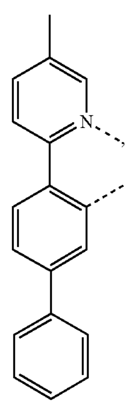

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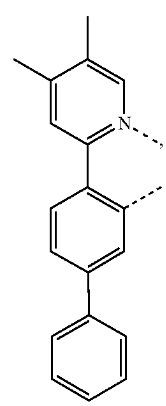

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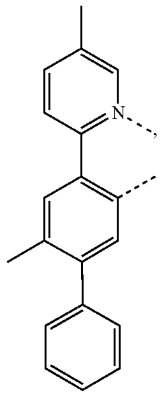

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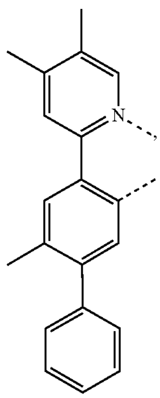

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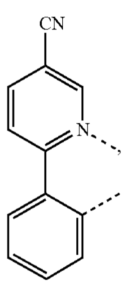

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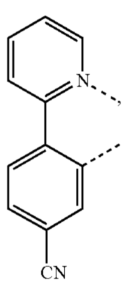

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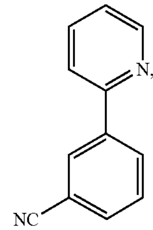

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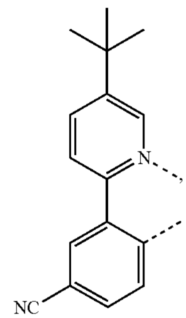

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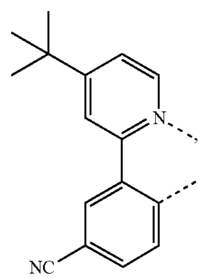

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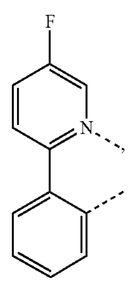

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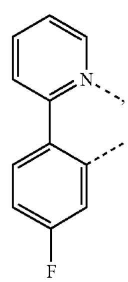

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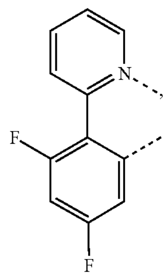

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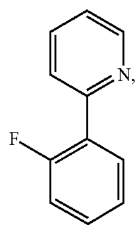

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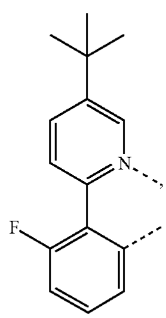

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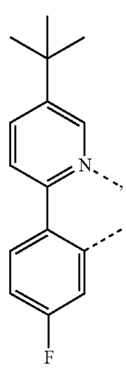

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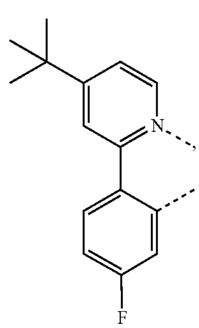

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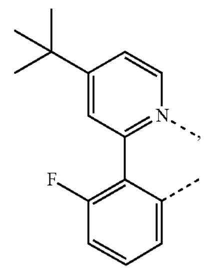

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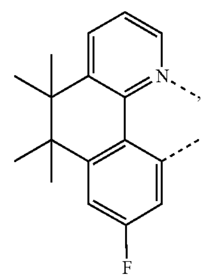

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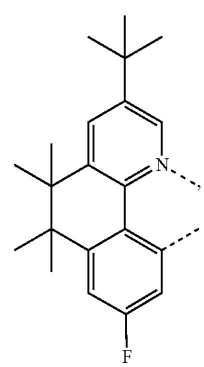

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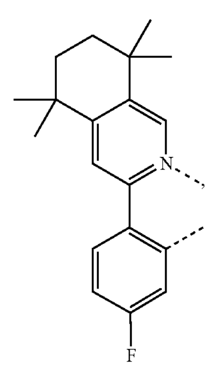

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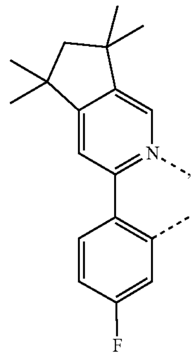

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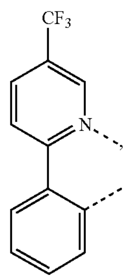

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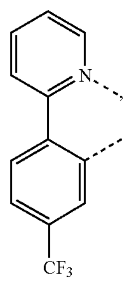

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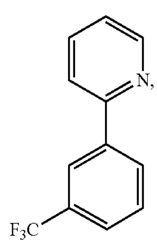

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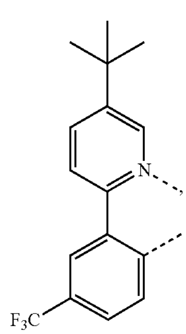

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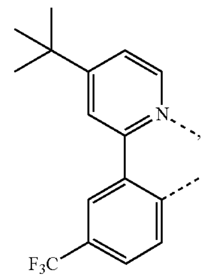

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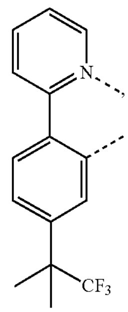

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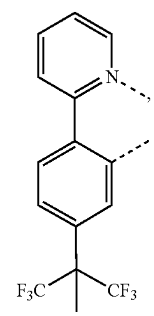

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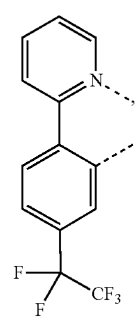

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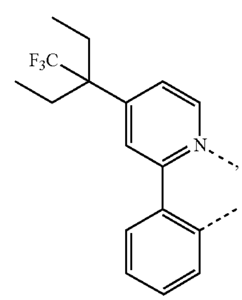

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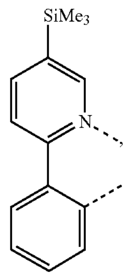
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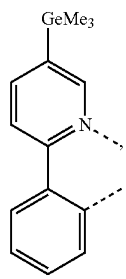
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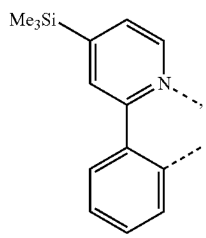
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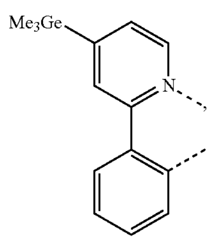
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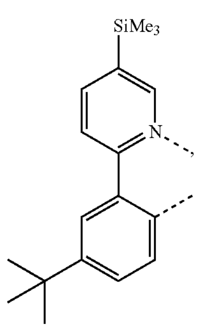
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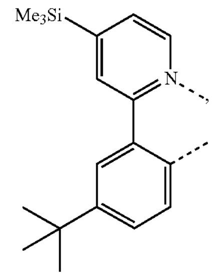
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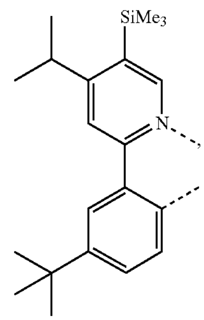
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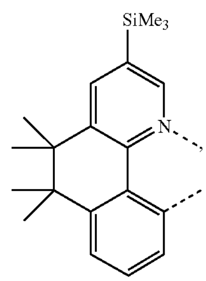
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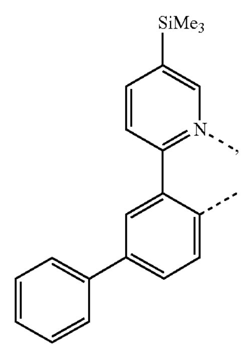
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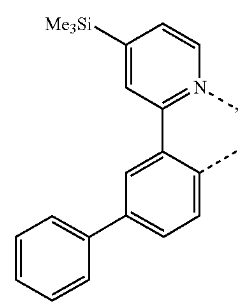
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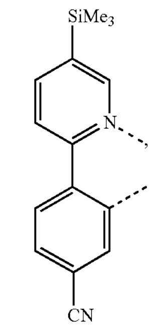

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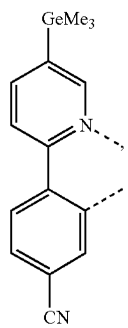

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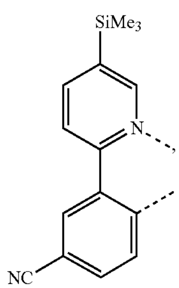

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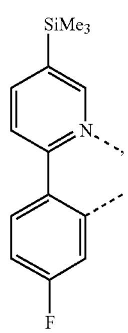

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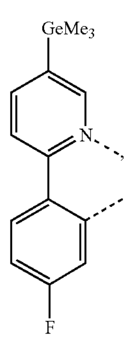

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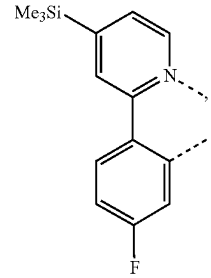

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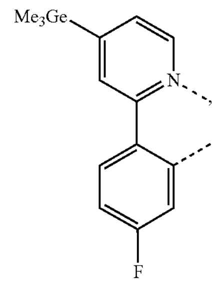

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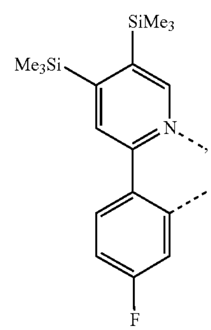

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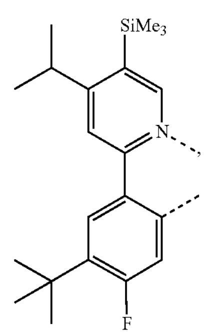

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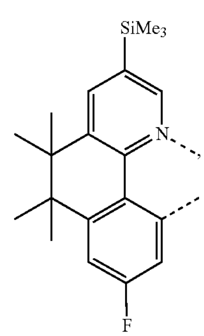

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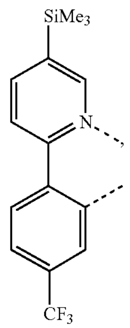

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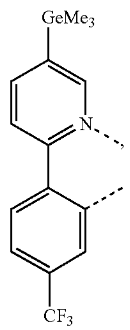

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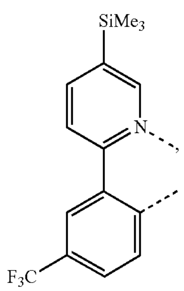

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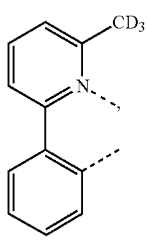

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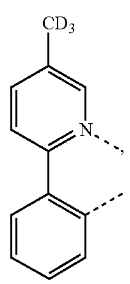

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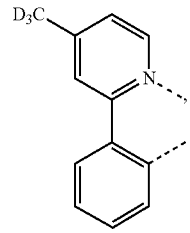

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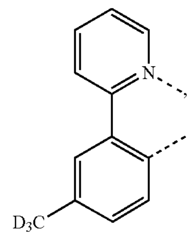

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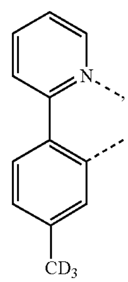

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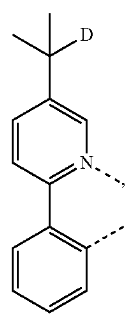

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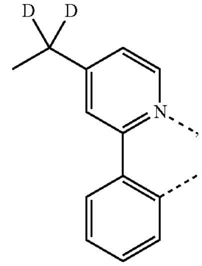

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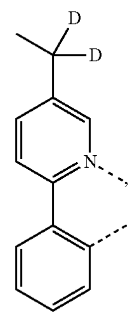

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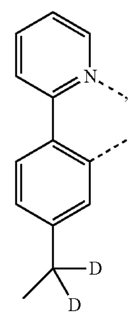

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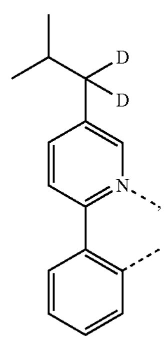

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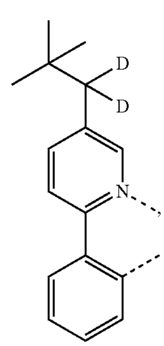

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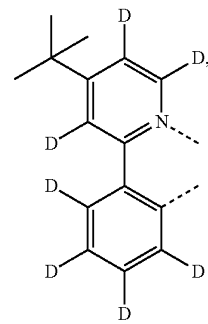

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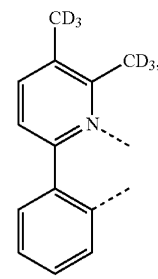

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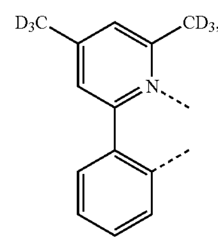

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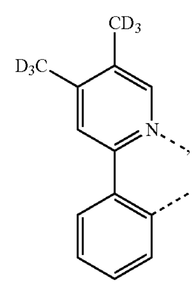

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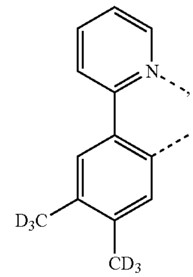

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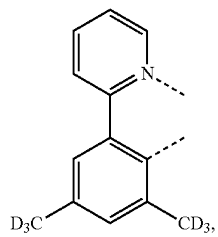

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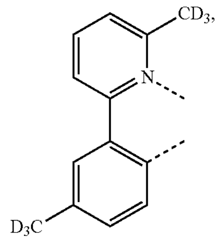

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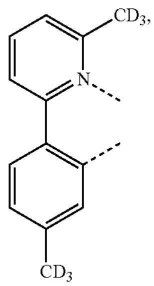

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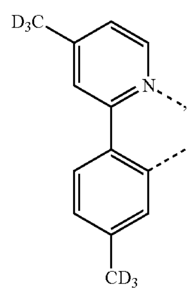

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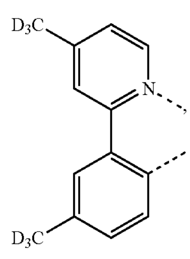

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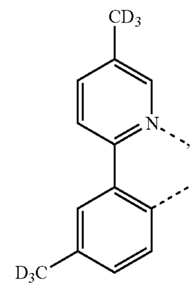

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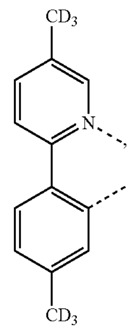

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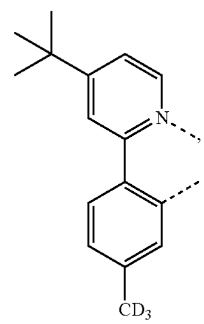

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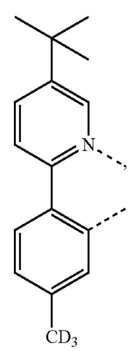

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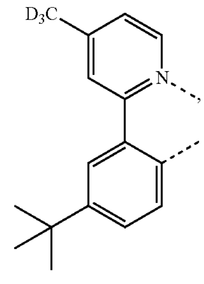

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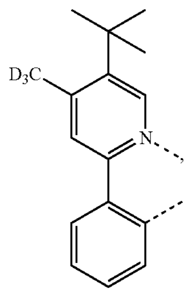

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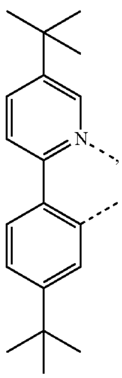

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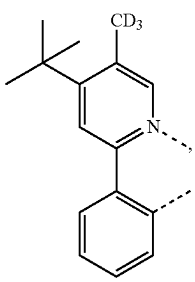

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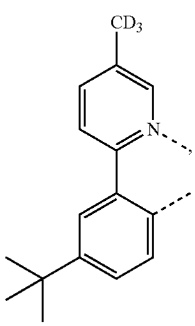

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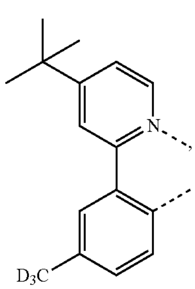

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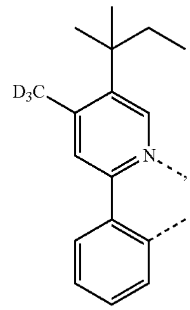

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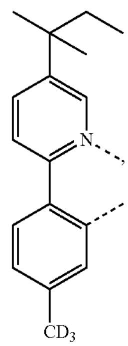

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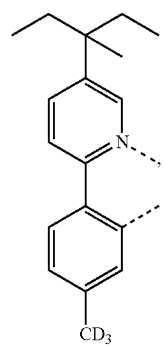

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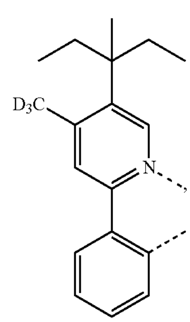

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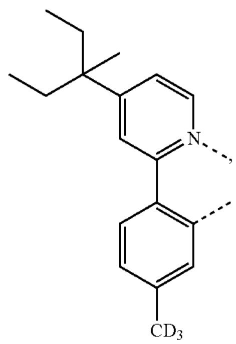

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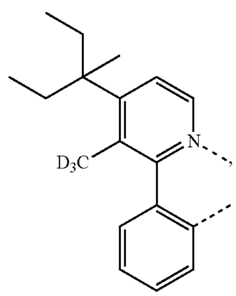

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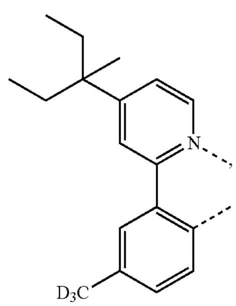

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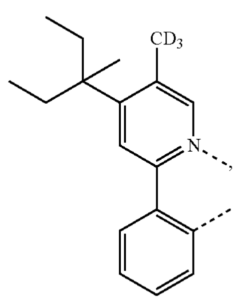

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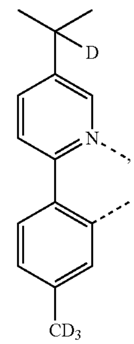

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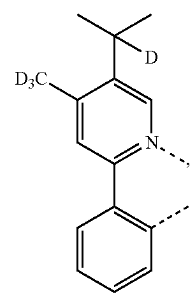

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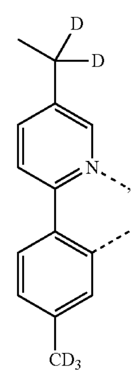

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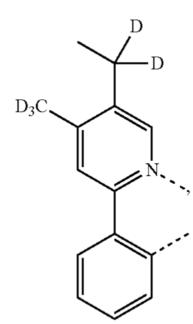

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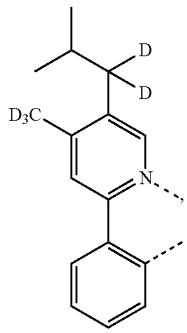

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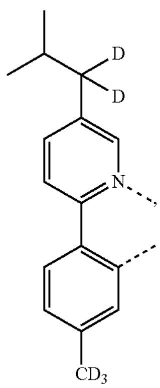

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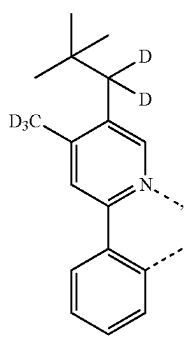

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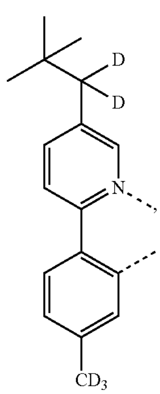

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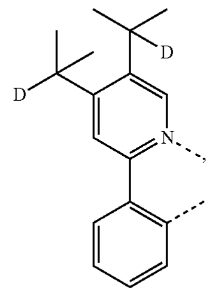

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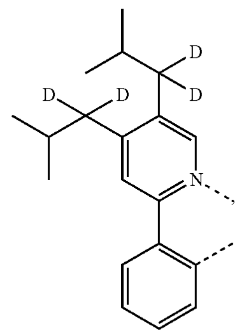

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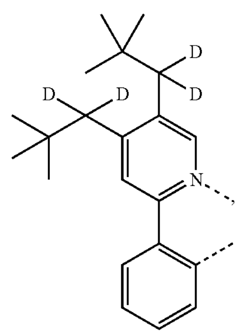

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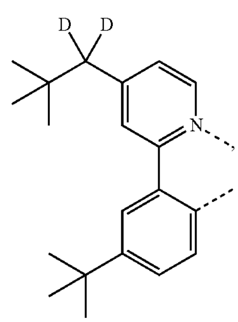

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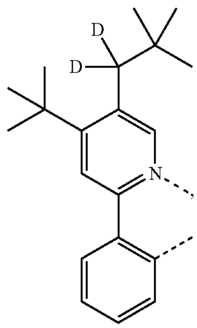

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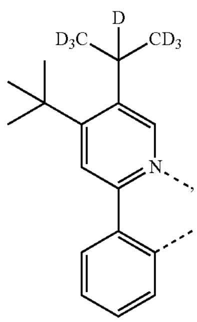

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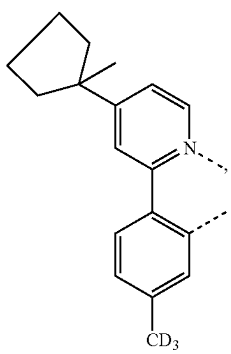

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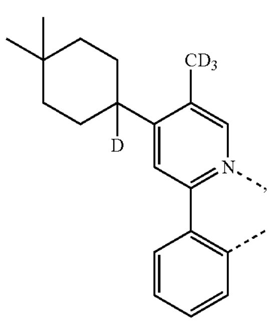

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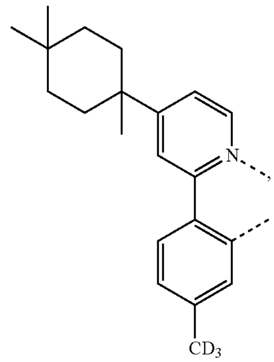

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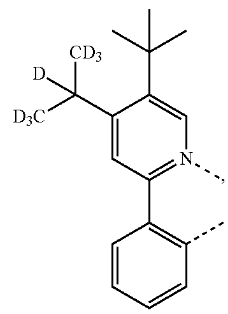

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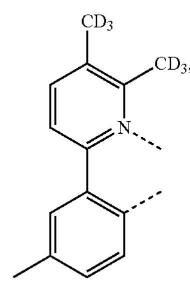

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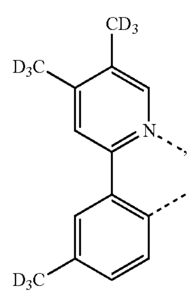

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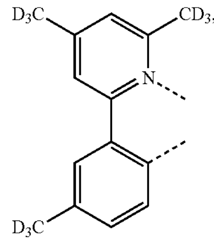

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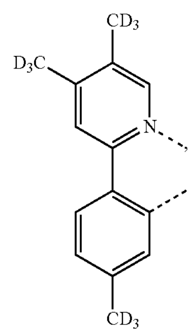

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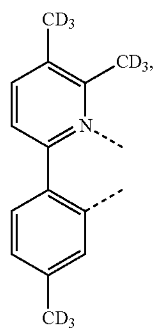

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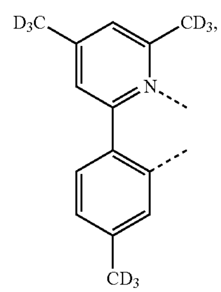

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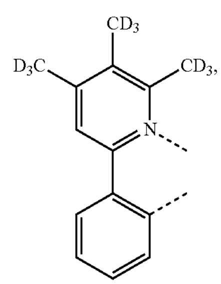

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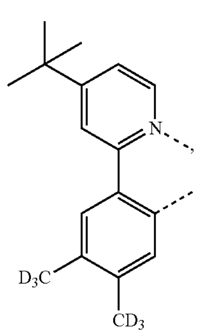

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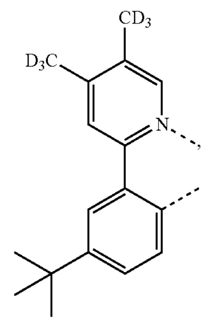

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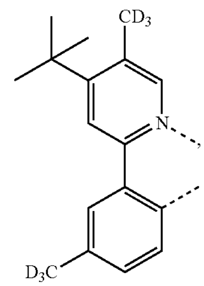

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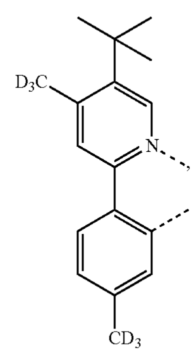

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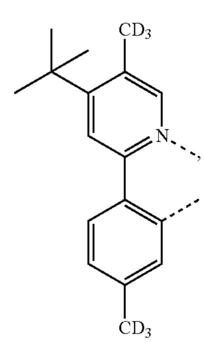

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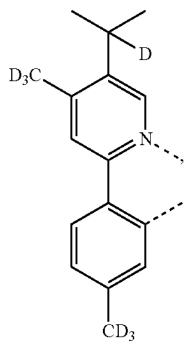

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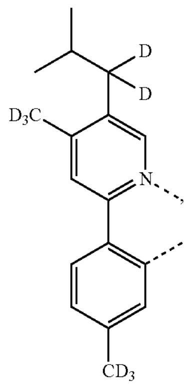

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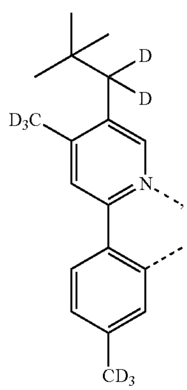

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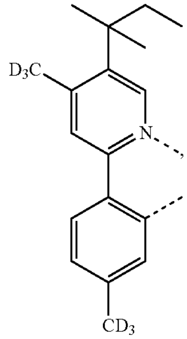

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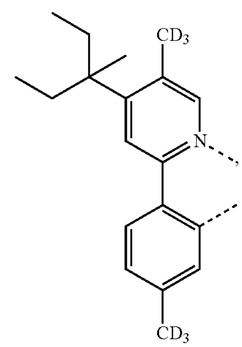

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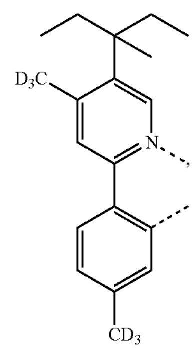

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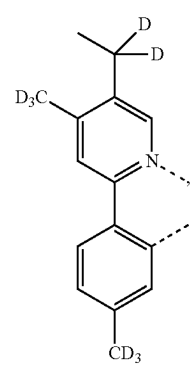

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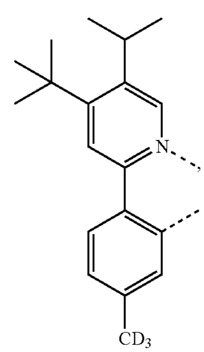

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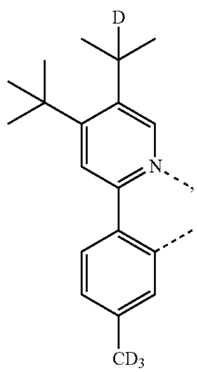

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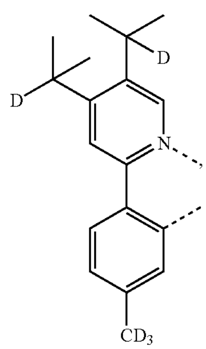

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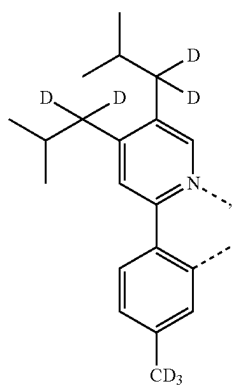

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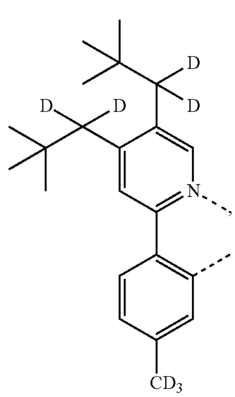

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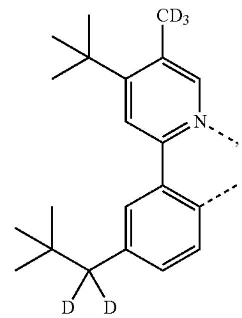

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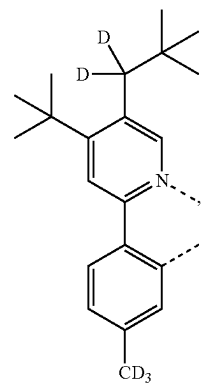

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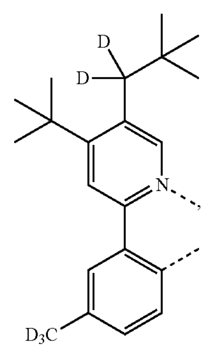

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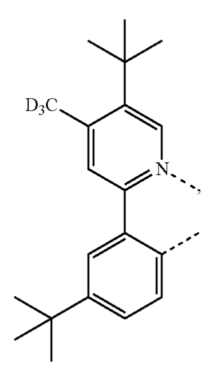

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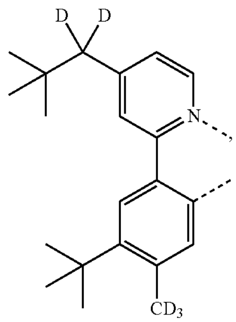

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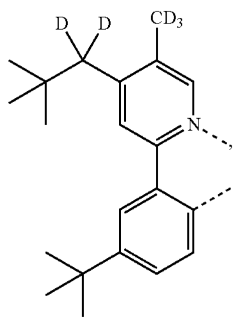

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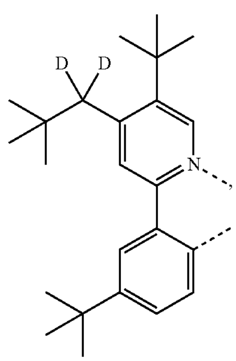

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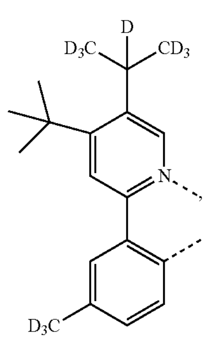

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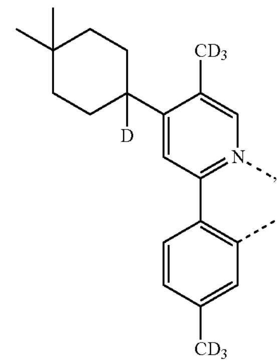

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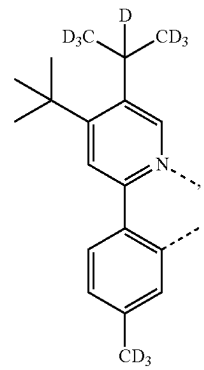

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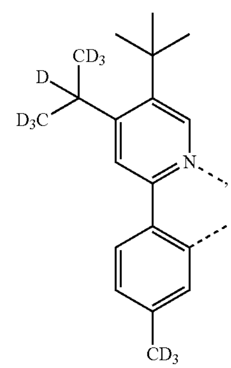

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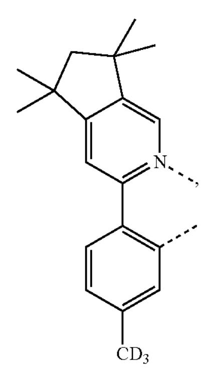

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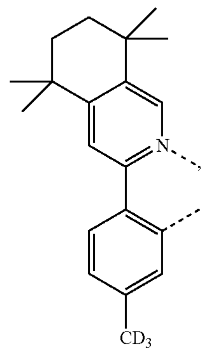

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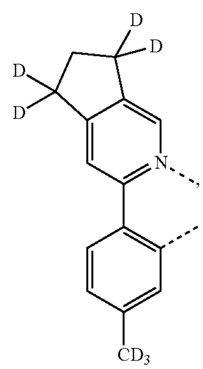

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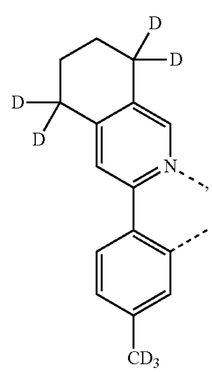

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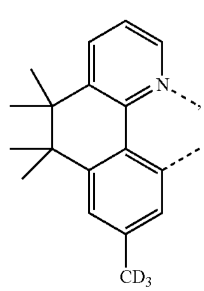

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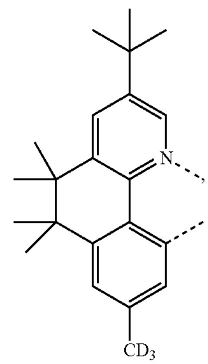

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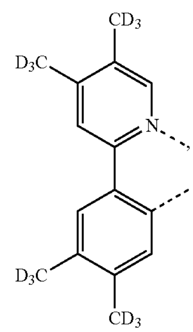

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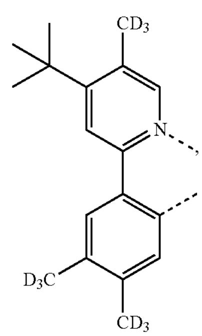

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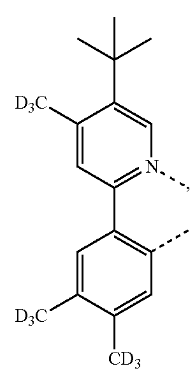

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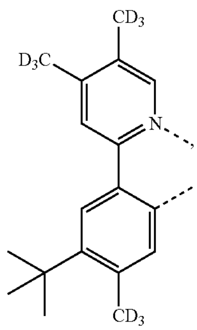

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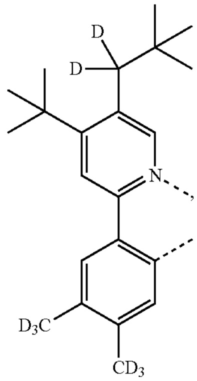

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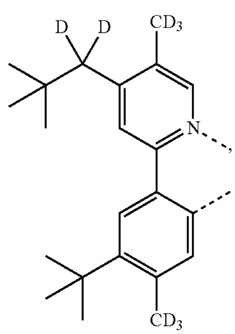

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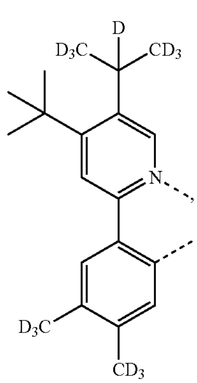

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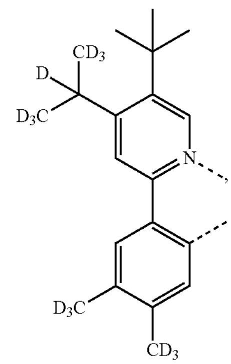

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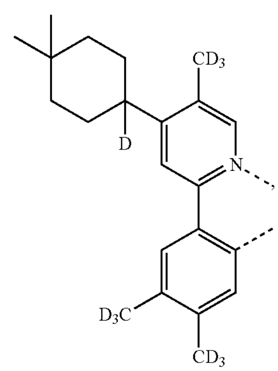

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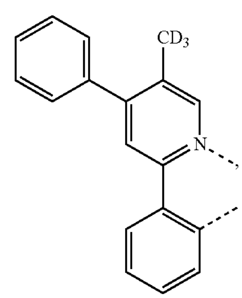

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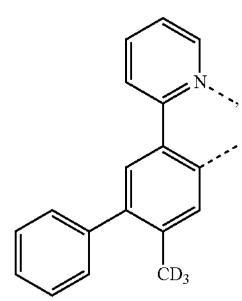

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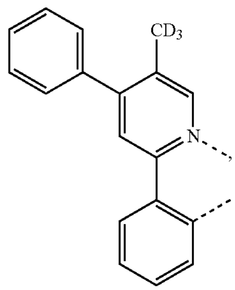

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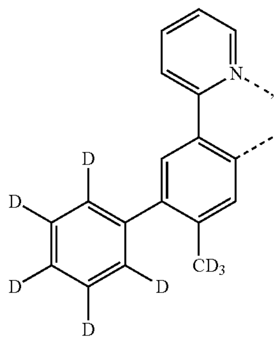

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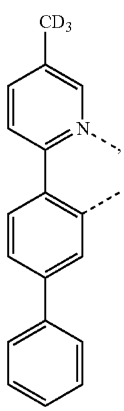

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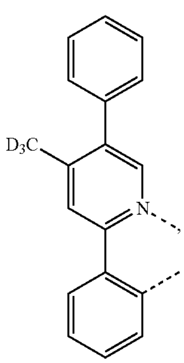

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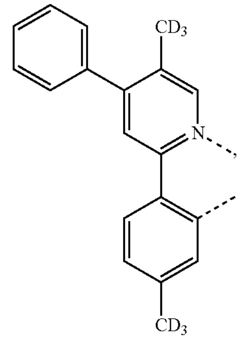

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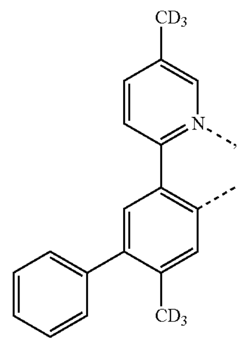

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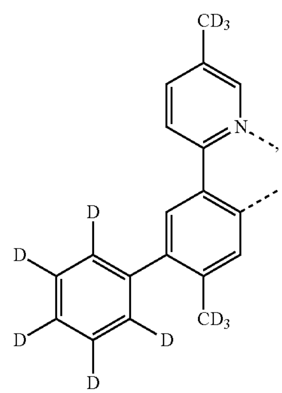

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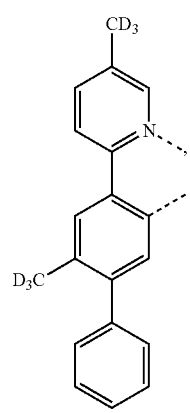

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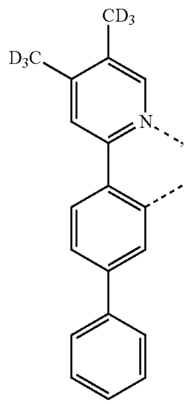

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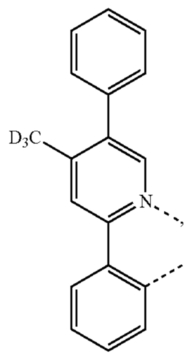

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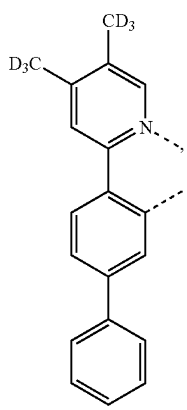

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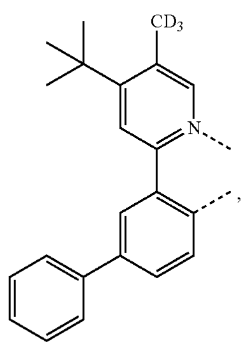

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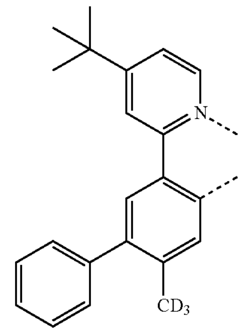

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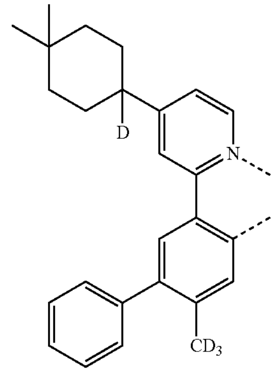

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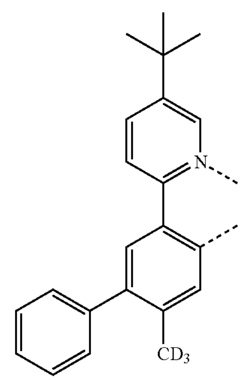

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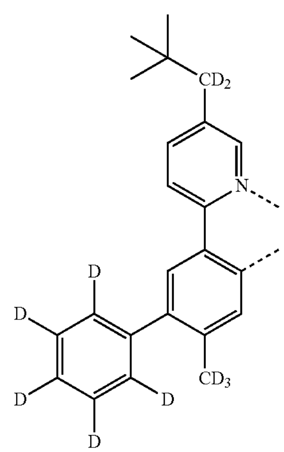

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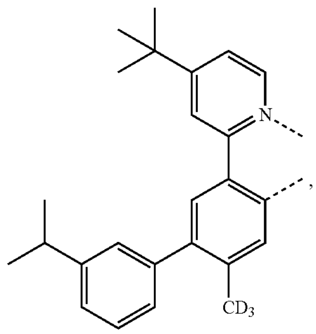

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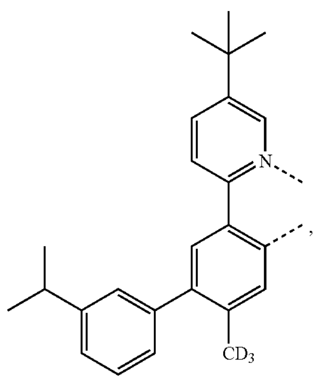

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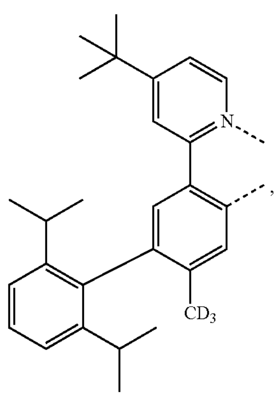

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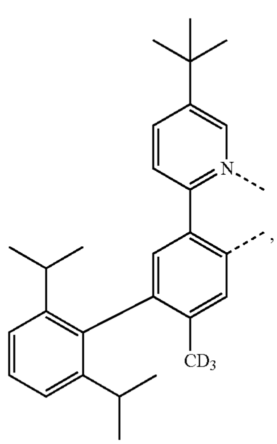

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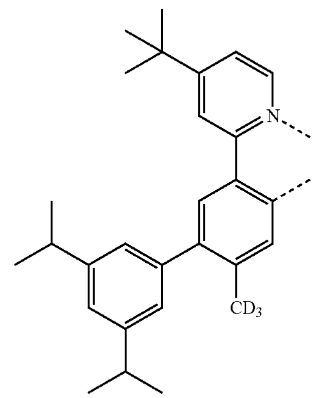

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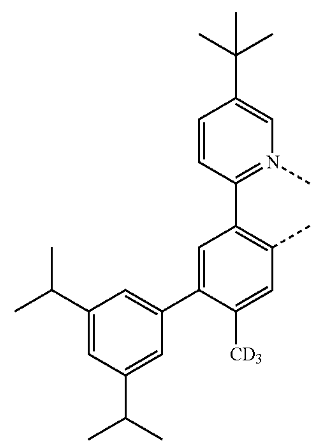

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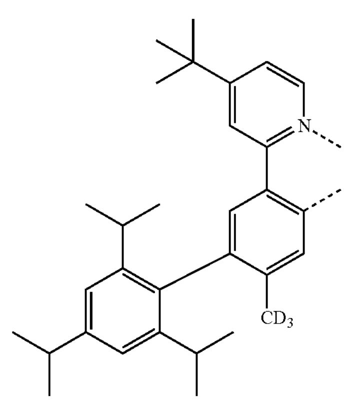

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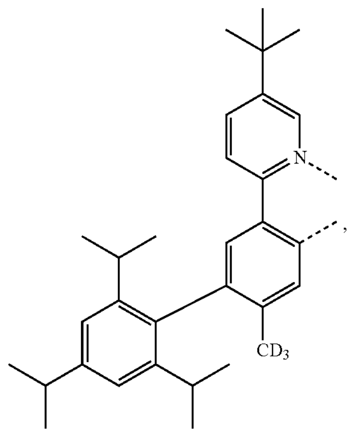

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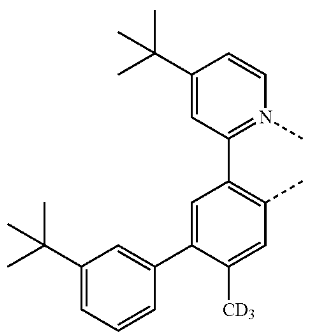

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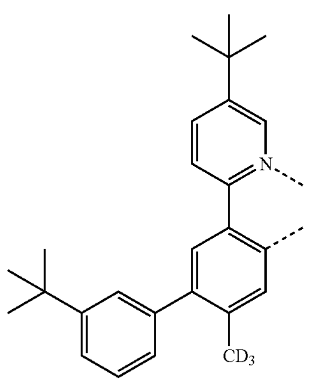

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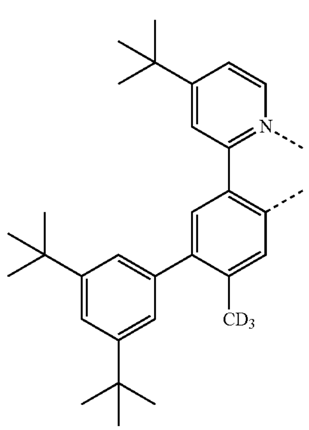

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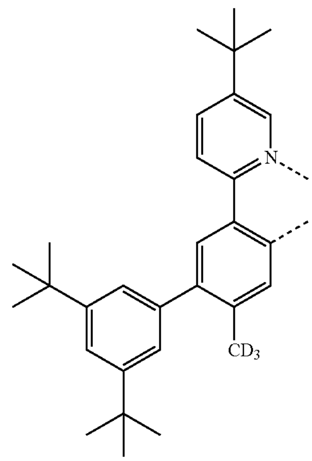

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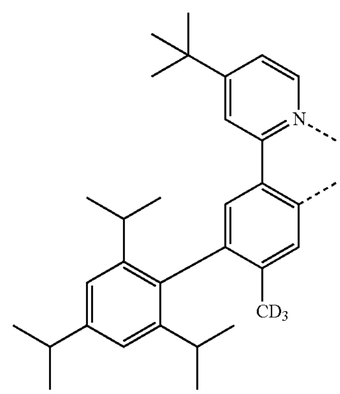

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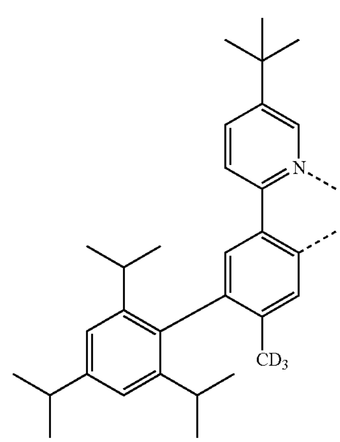

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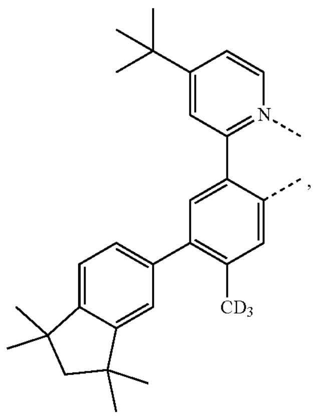

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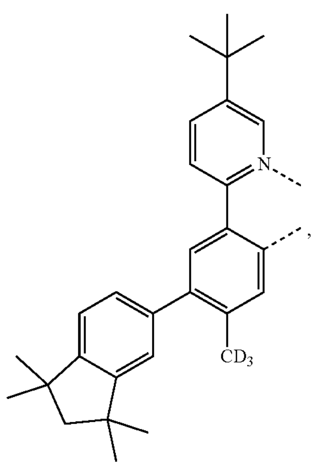

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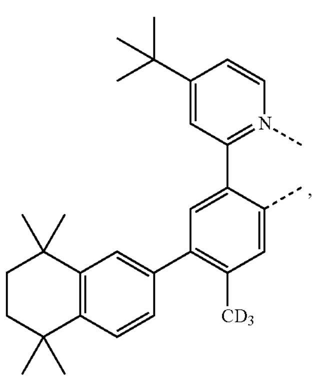

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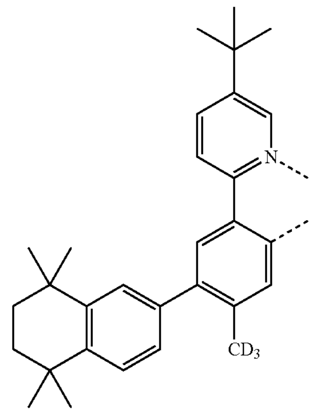

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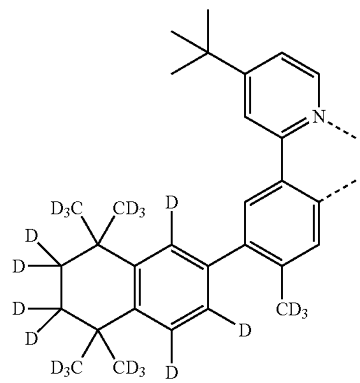

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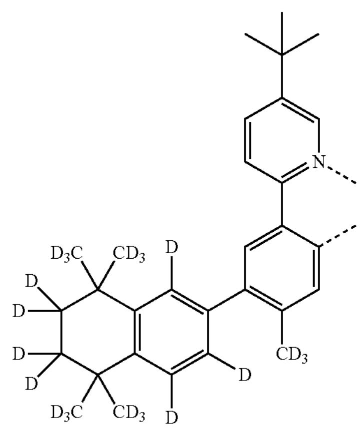

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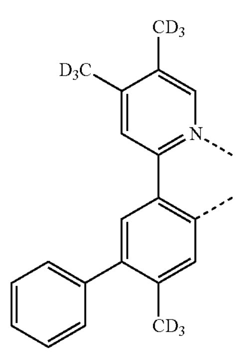

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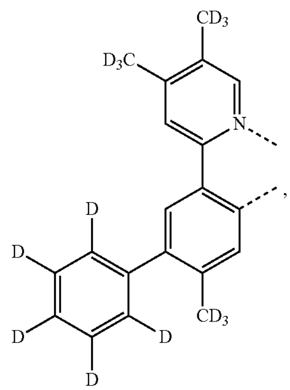

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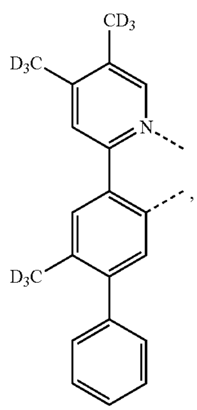

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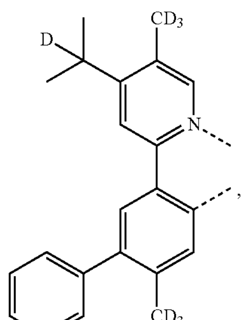

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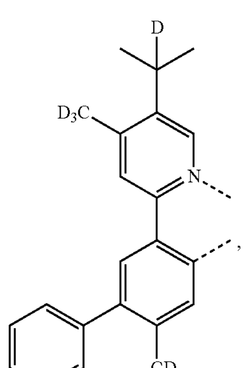

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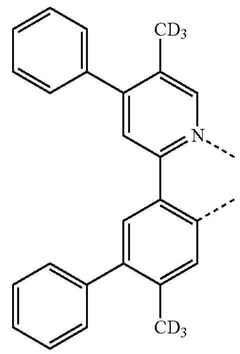

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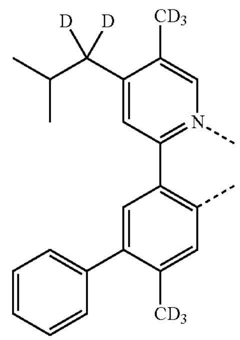

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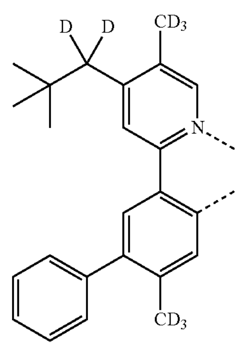

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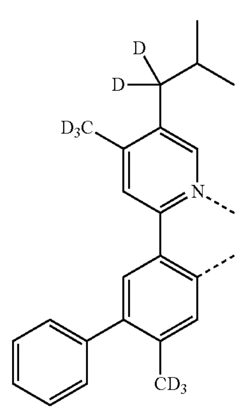

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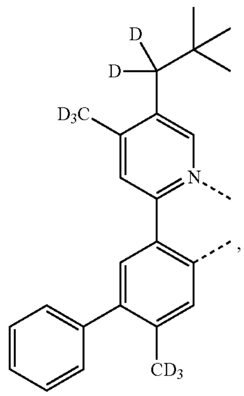

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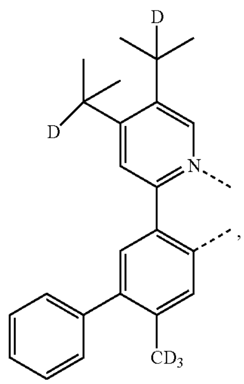

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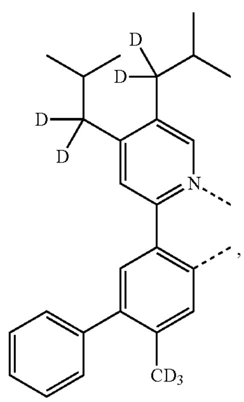

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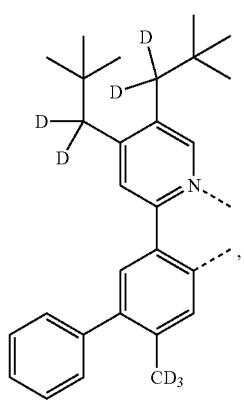

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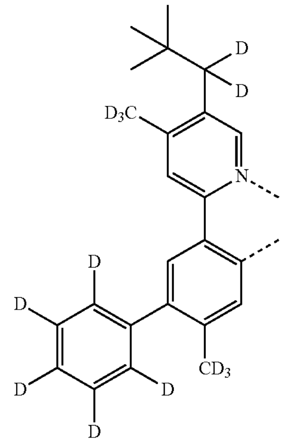

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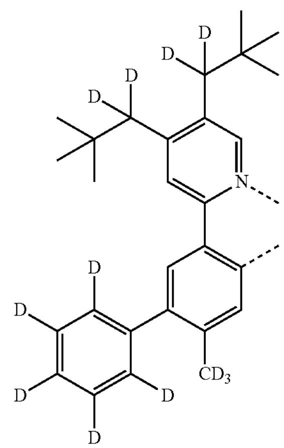

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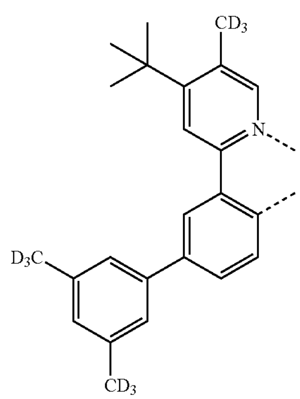

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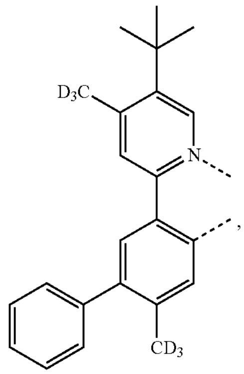

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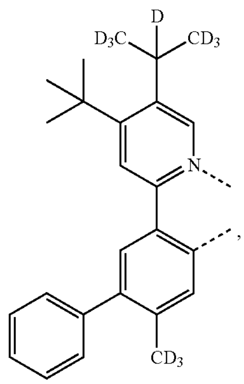

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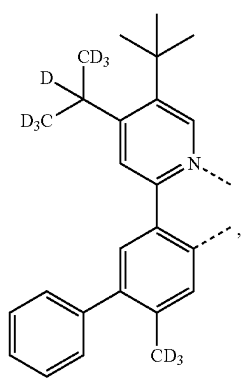

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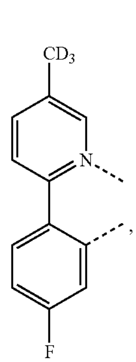

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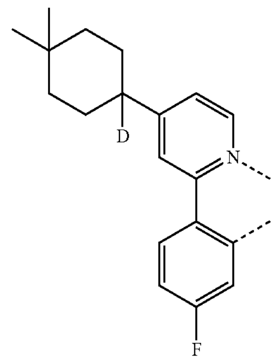

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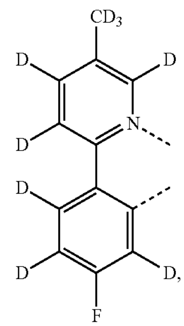

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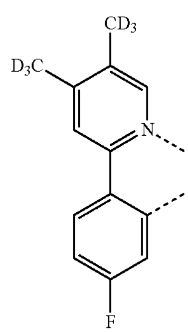

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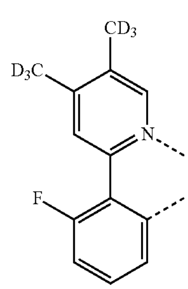

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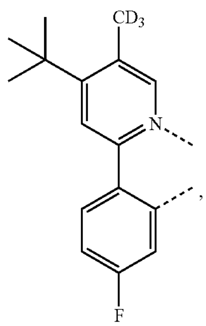

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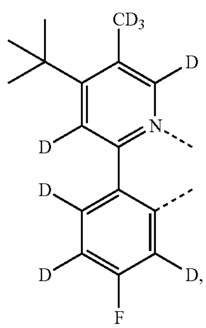

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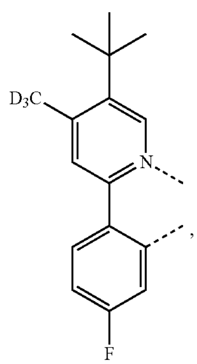

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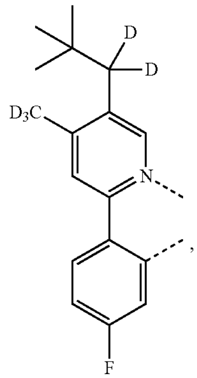

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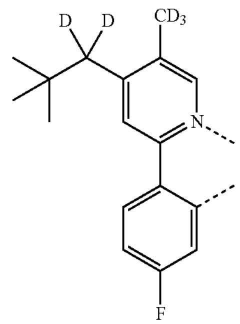

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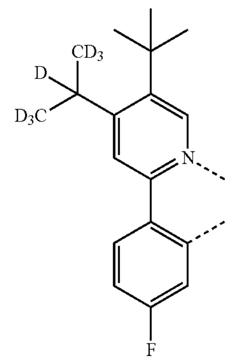

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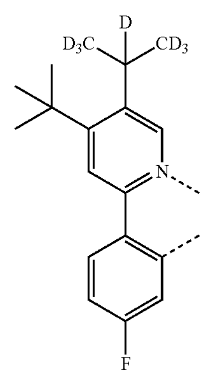

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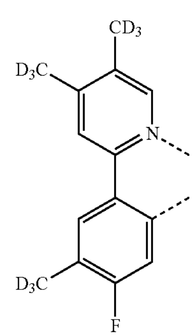

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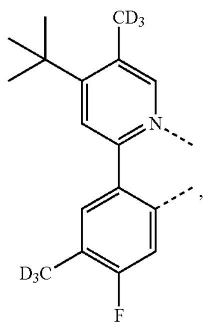

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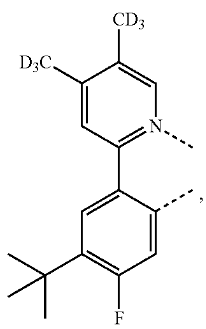

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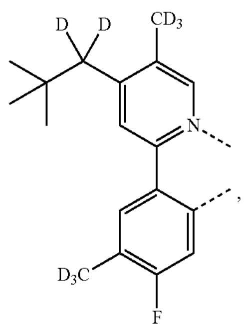

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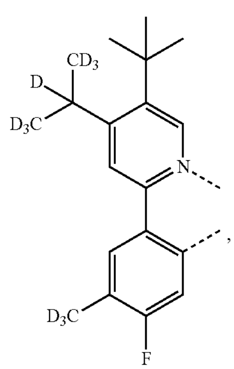

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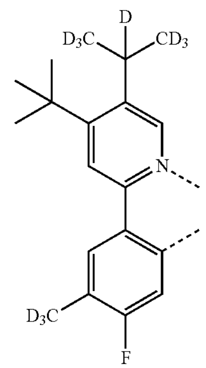

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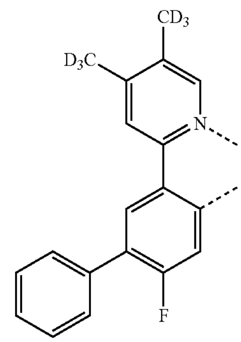

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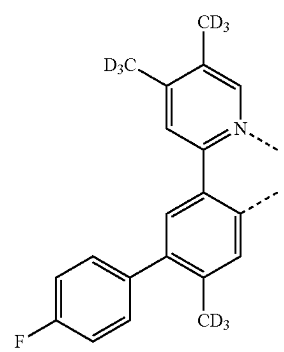

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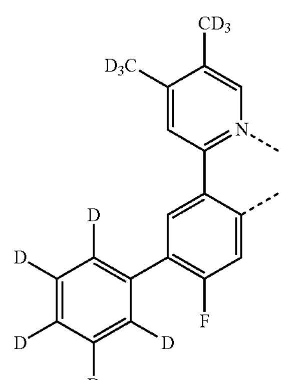

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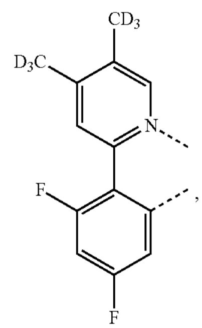

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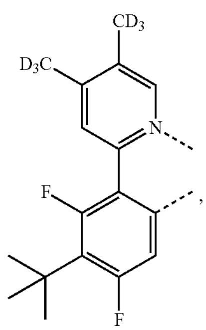

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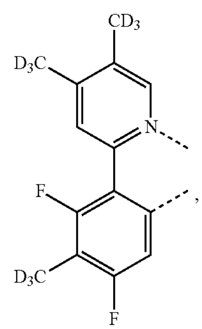

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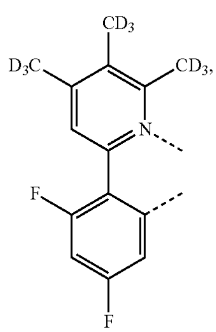

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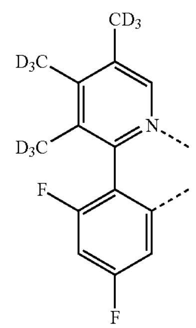

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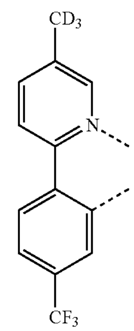

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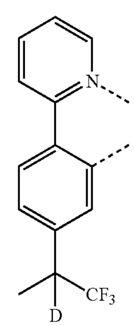

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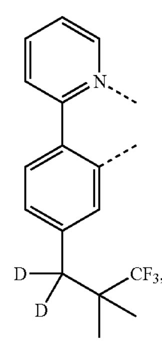

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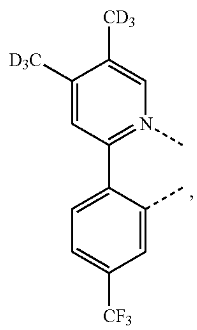

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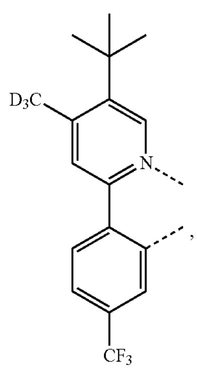

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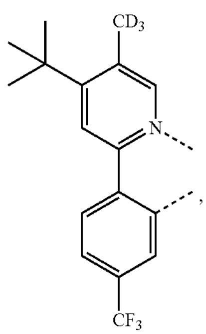

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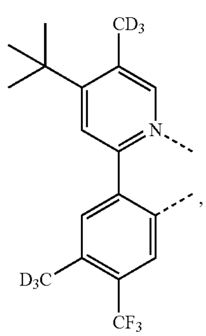

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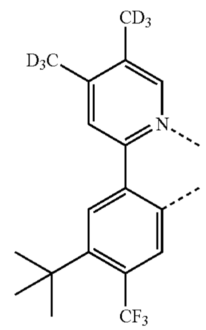

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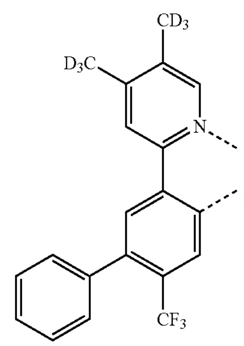

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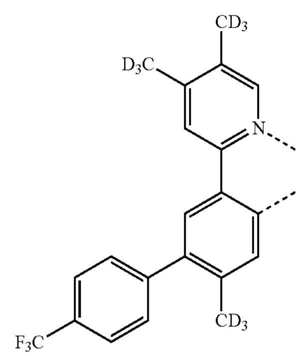

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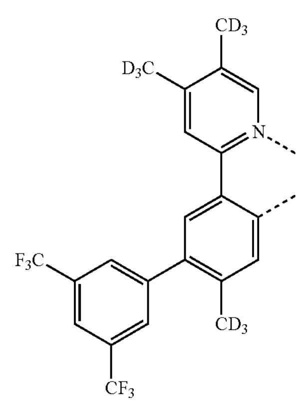

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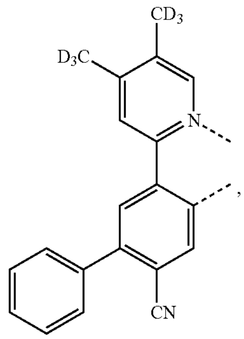

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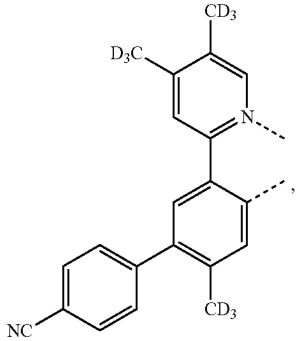

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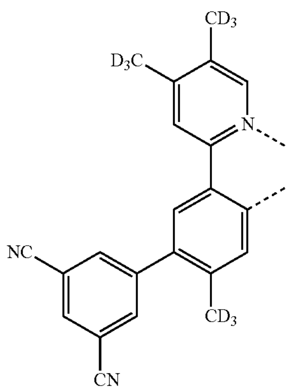

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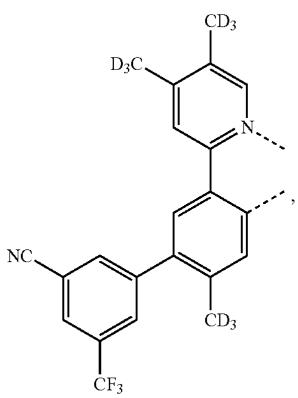

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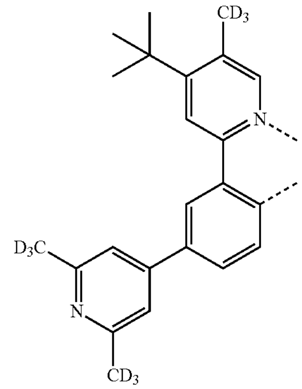

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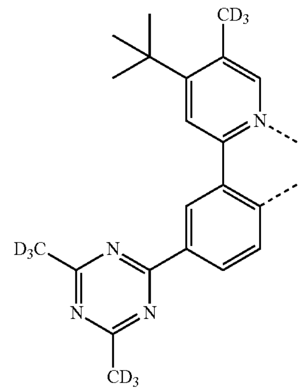

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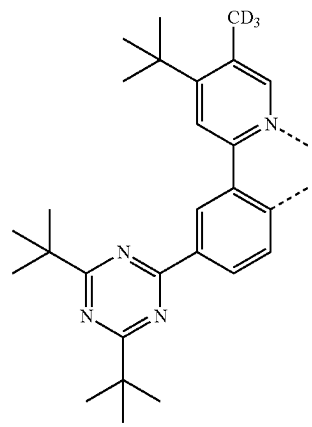

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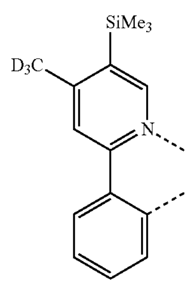

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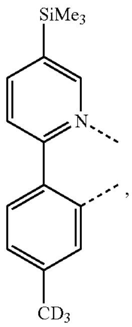

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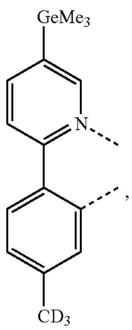

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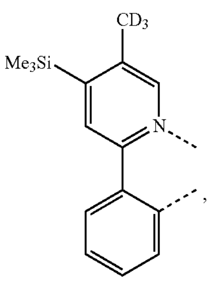

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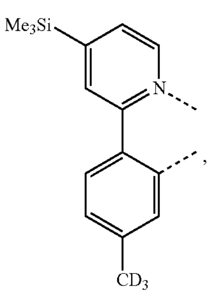

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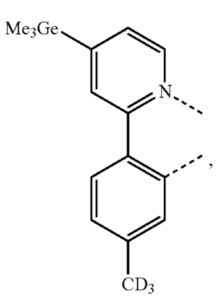

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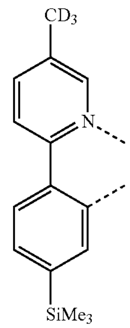

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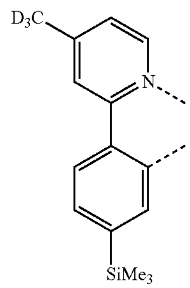

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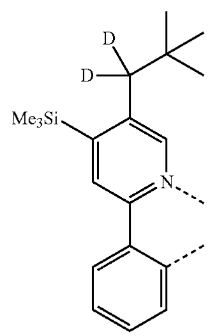

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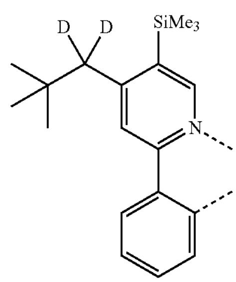

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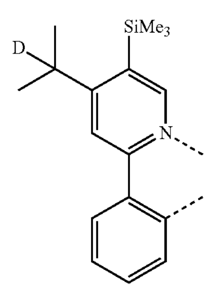

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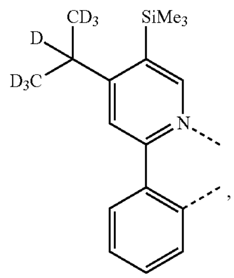

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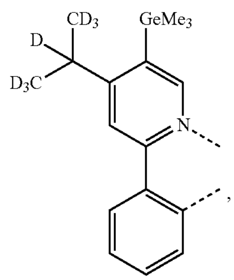

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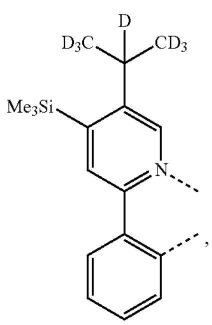

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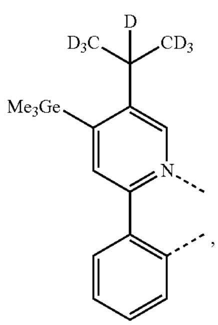

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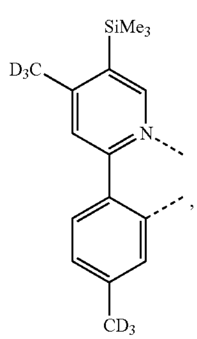

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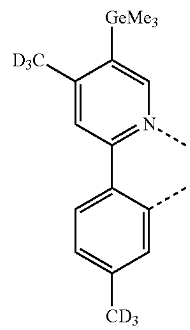

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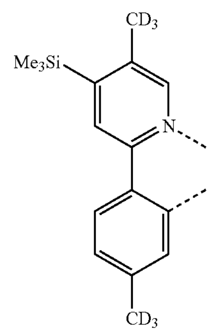

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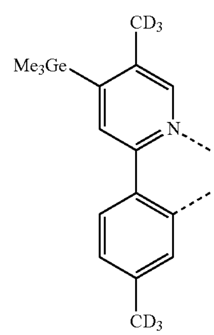

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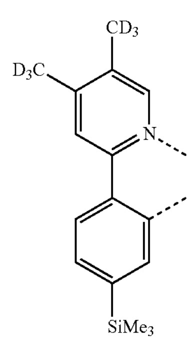

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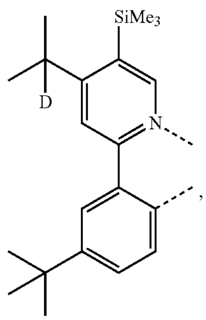

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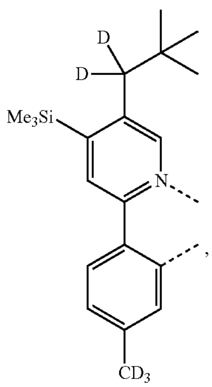

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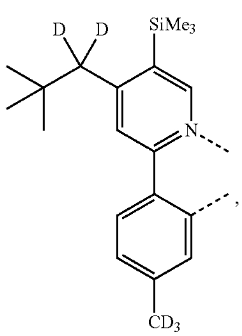

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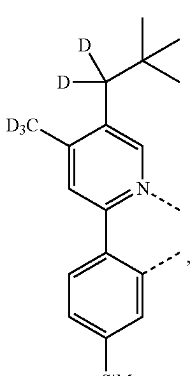

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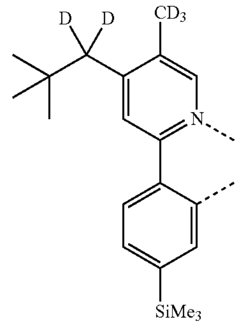

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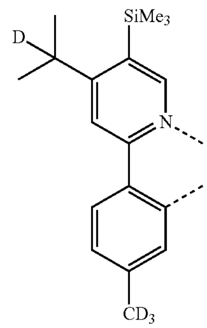

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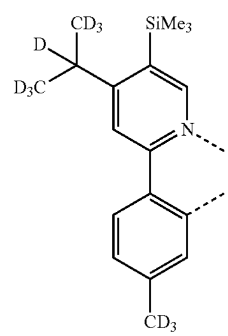

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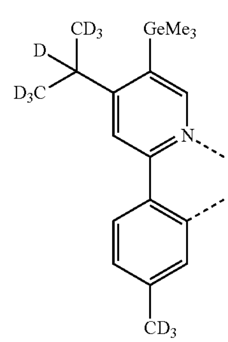

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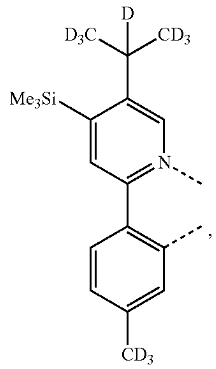

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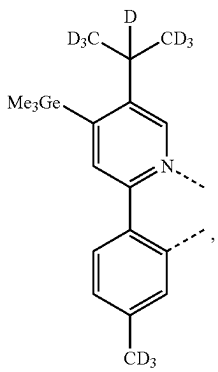

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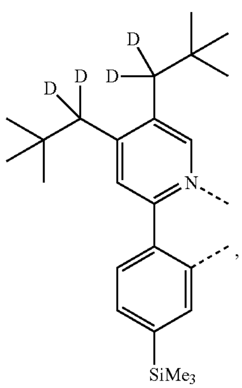

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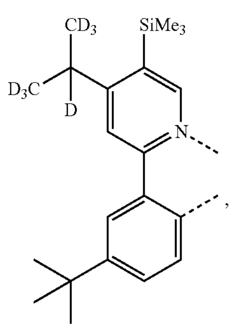

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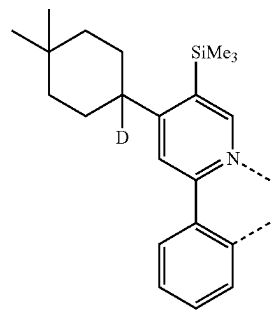

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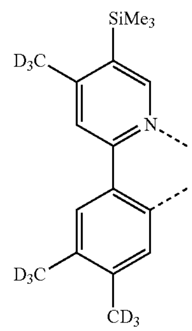

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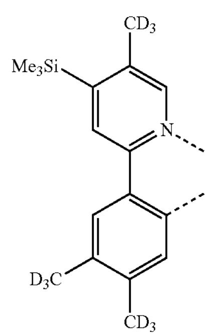

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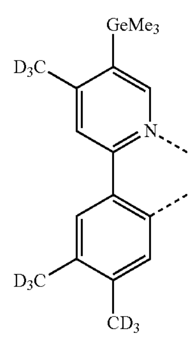

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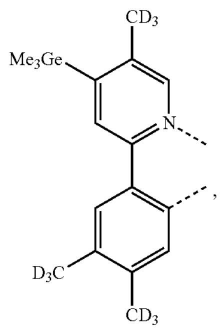

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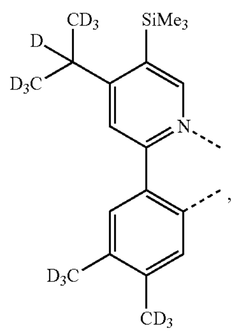

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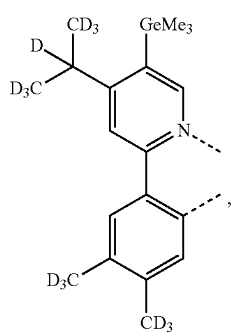

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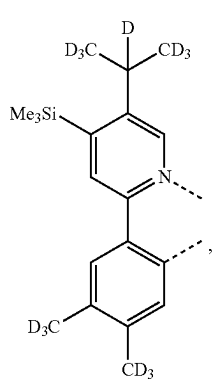

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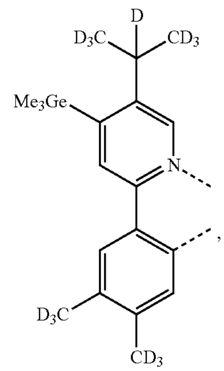

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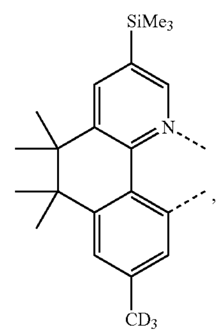

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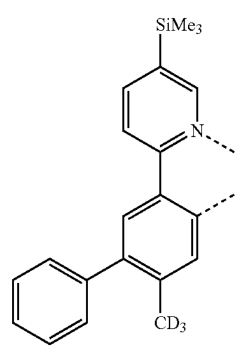

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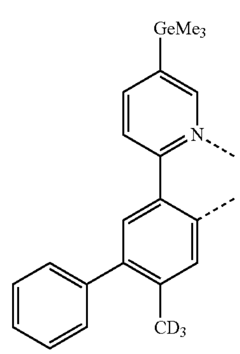

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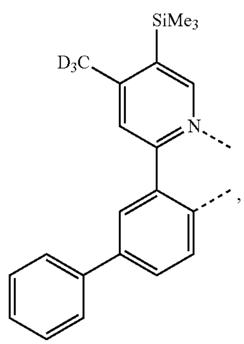

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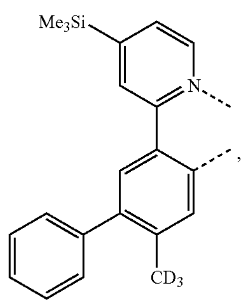

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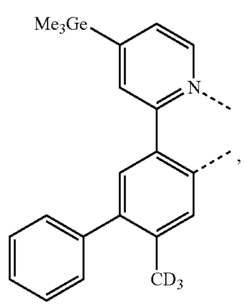

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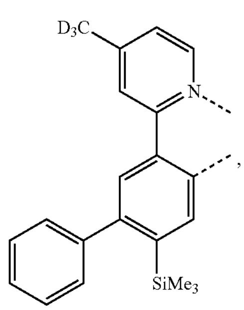

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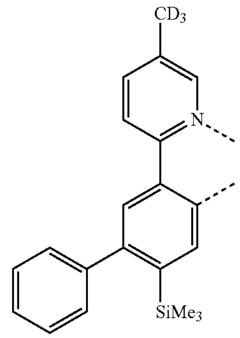

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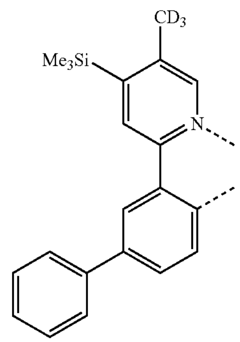

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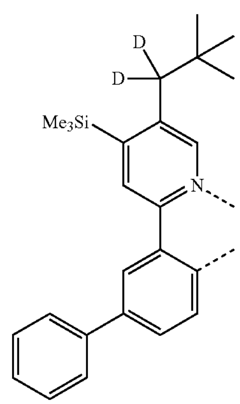

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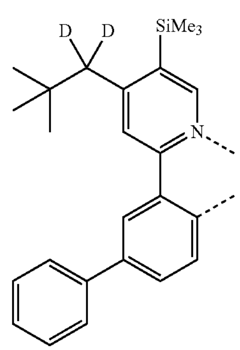

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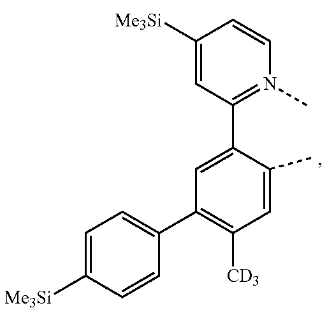

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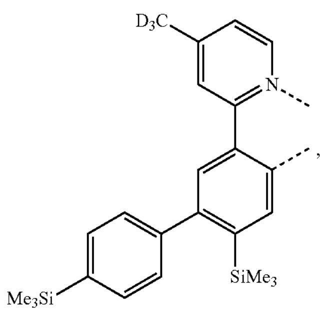

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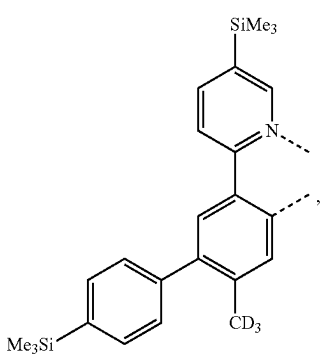

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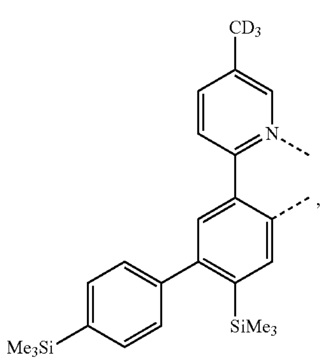

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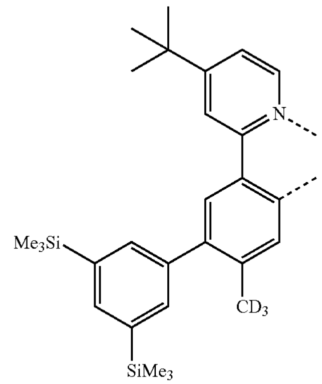

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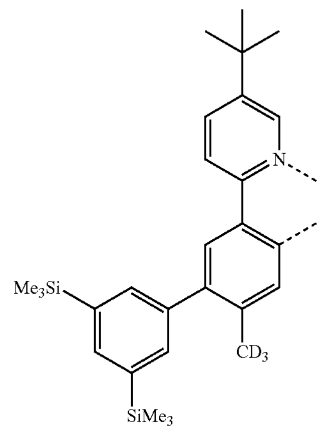

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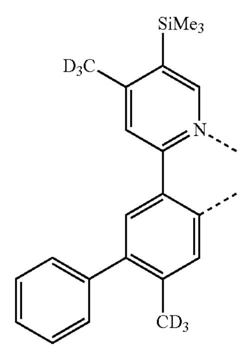

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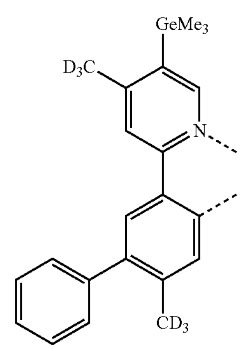

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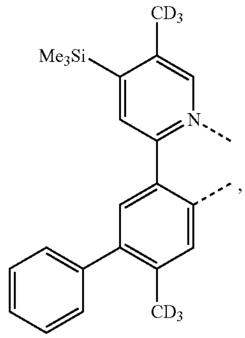

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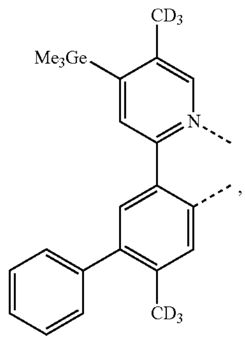

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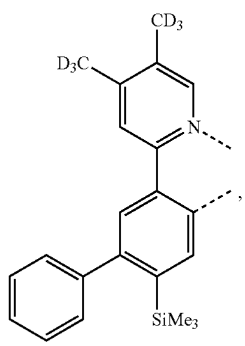

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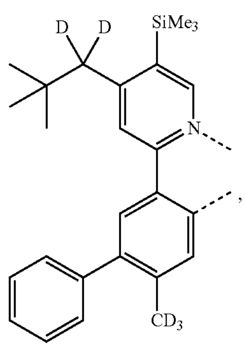

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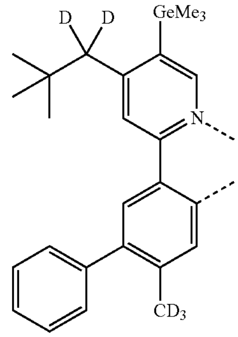

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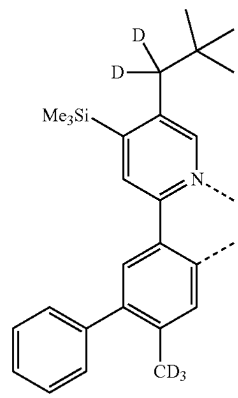

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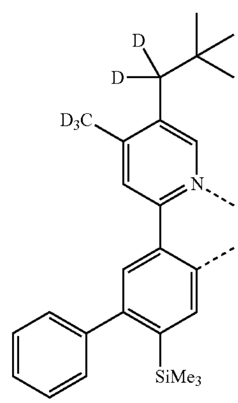

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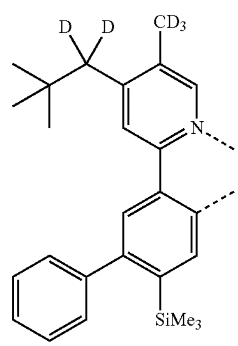

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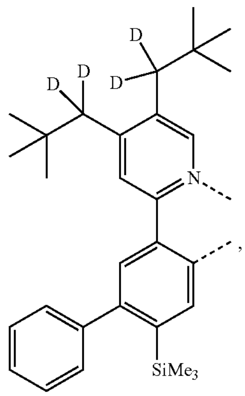

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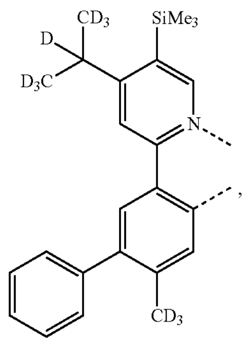

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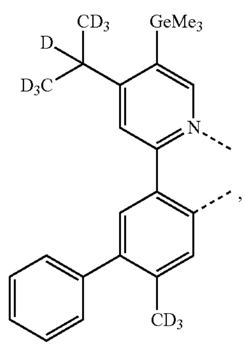

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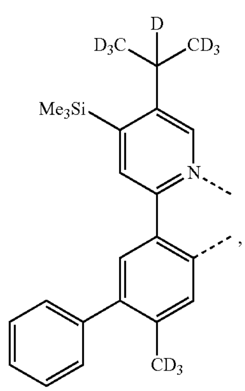

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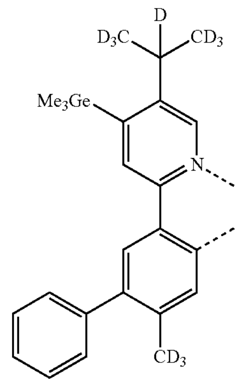

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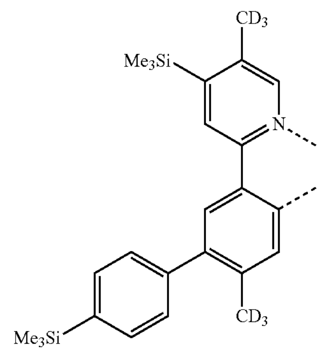

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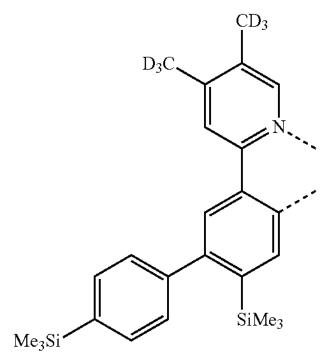

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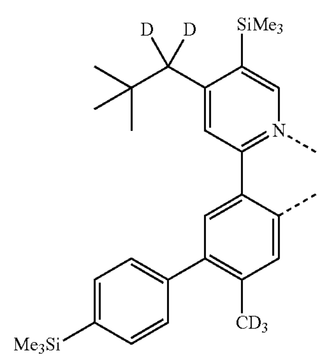

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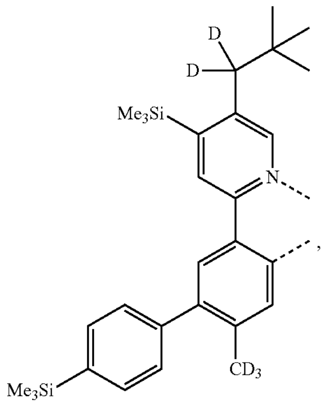

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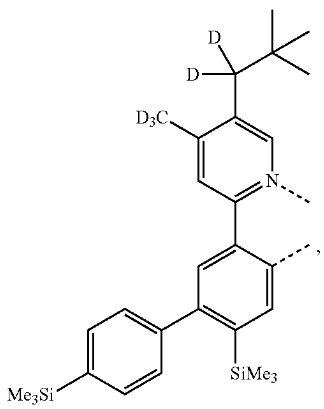

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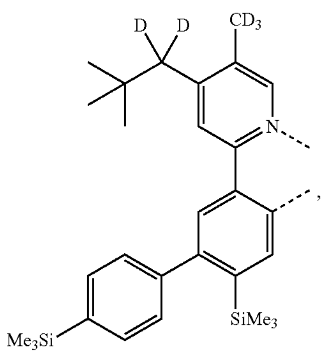

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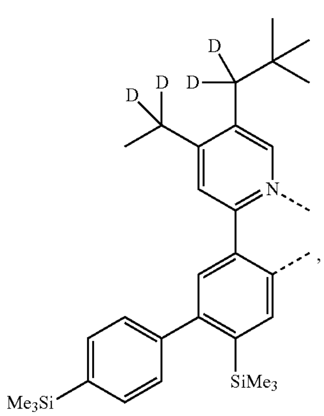

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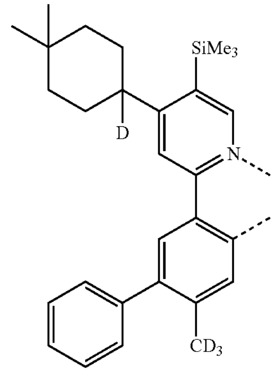

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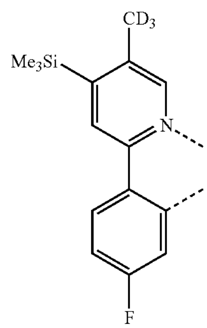

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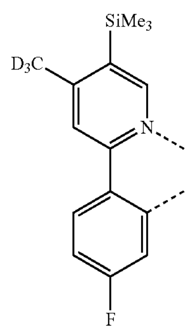

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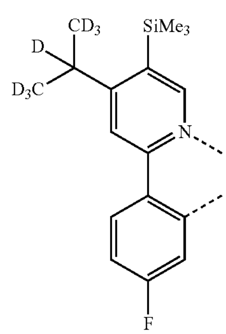

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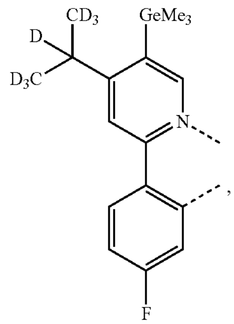

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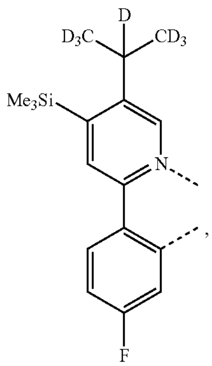

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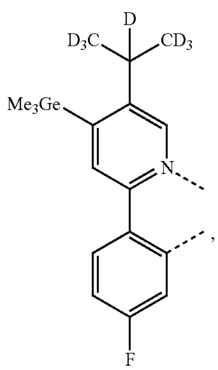

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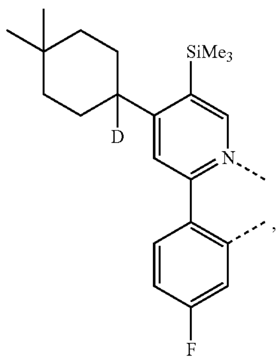

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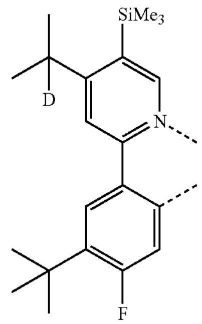

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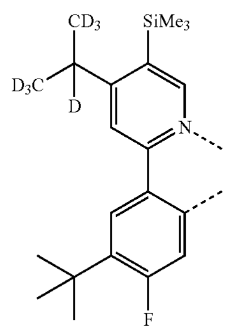

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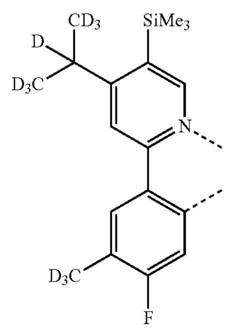

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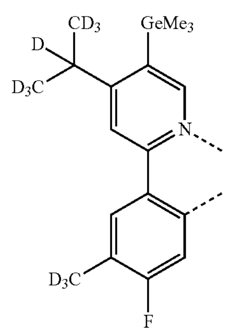

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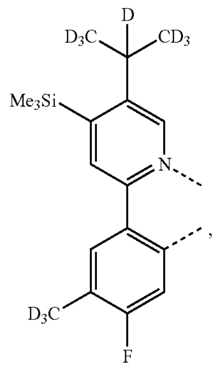

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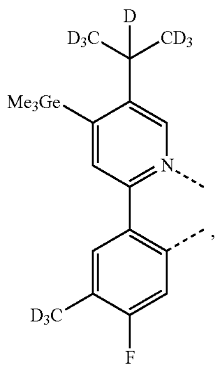

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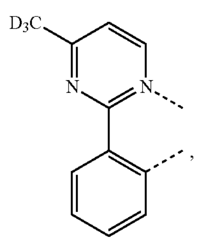

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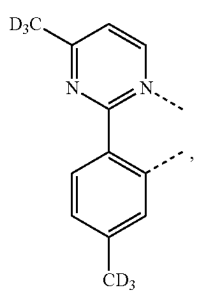

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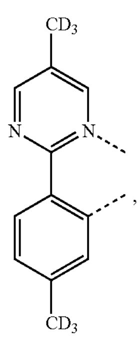

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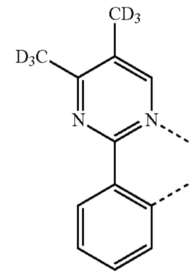

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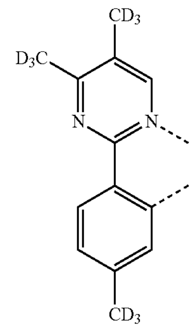

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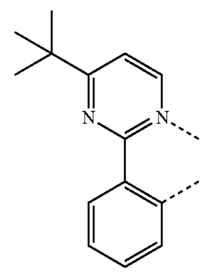

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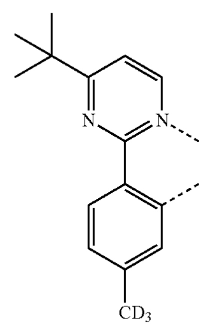

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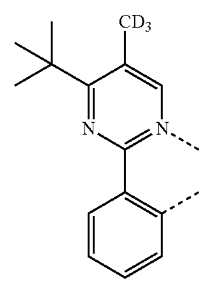

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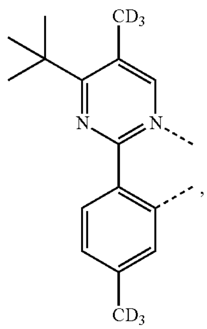

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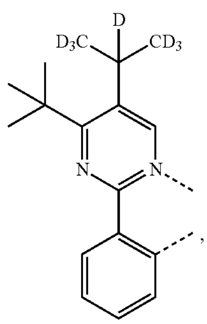

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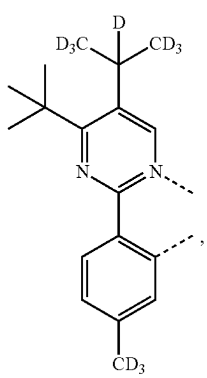

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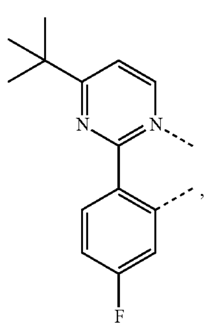

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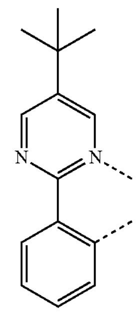

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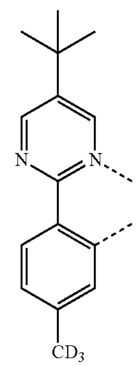

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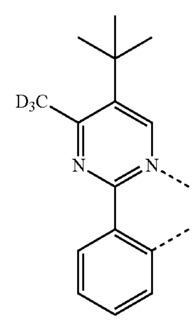

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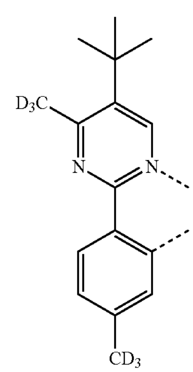

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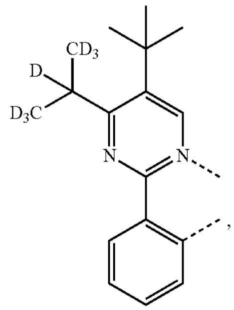

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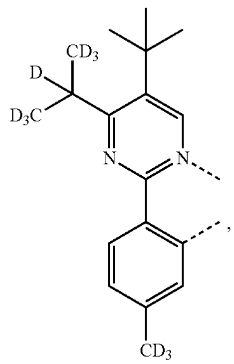

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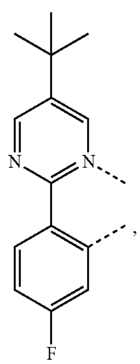

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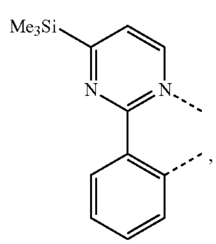

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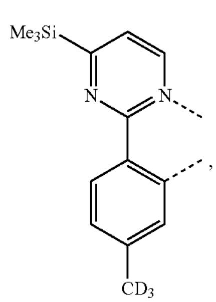

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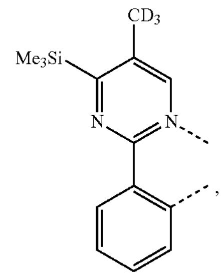

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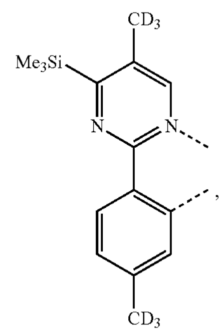

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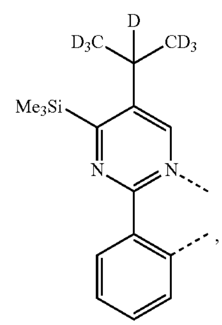

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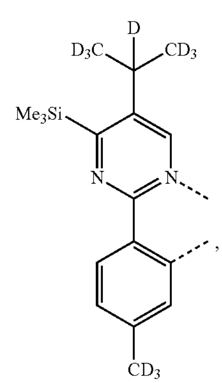

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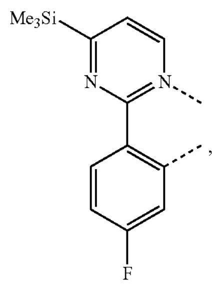

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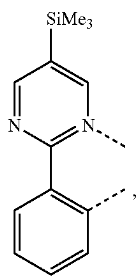

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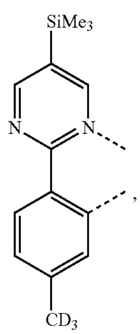

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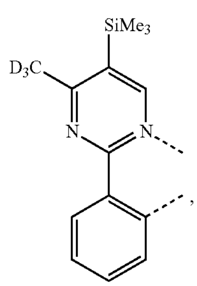

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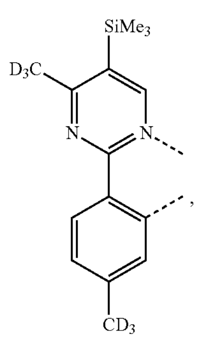

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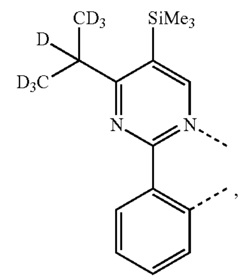

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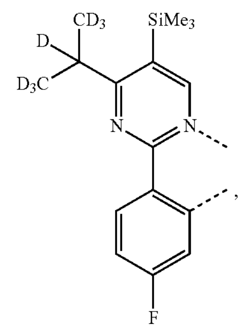

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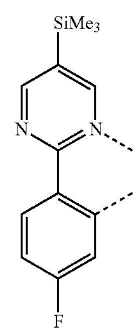

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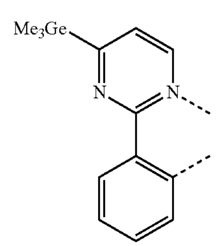

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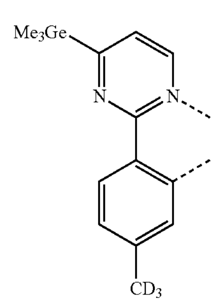

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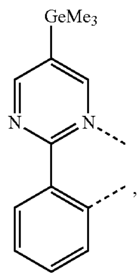

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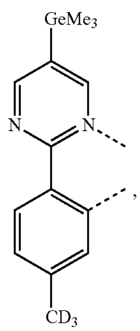

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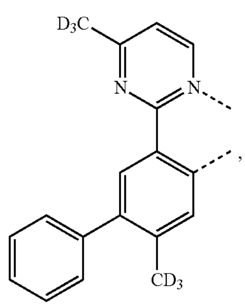

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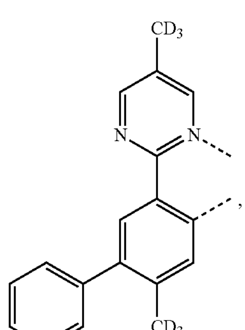

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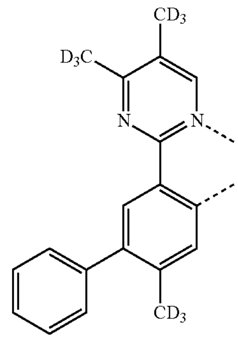

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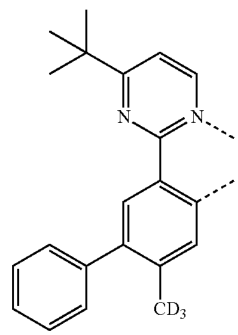

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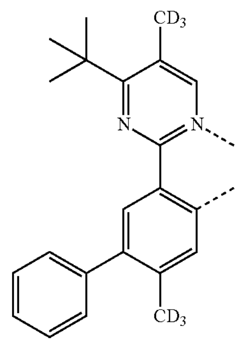

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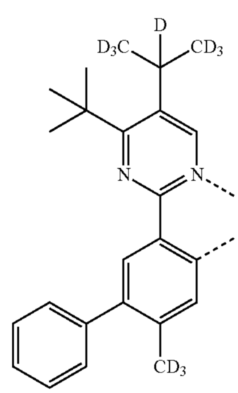

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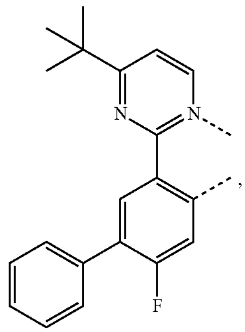
L_{B493}



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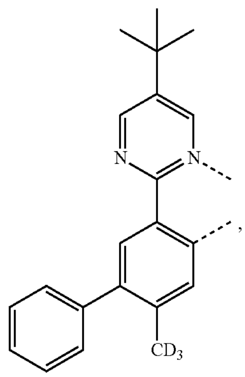

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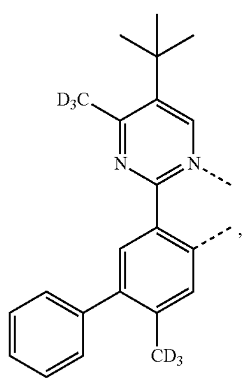

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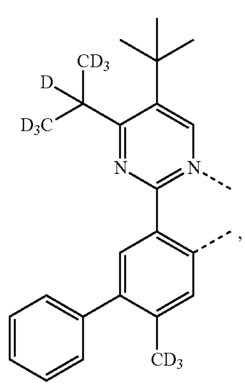

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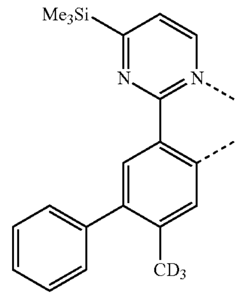

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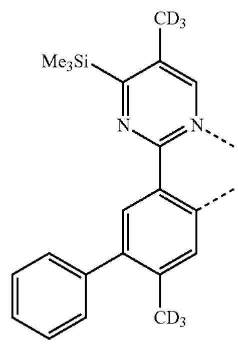

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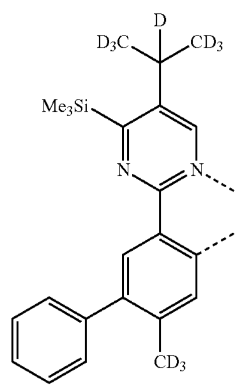

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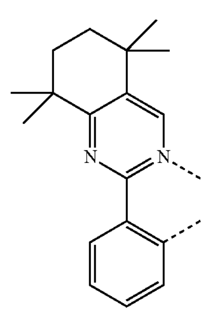

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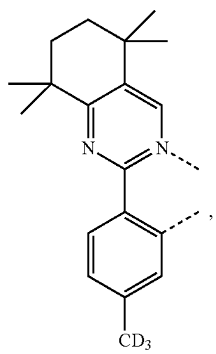

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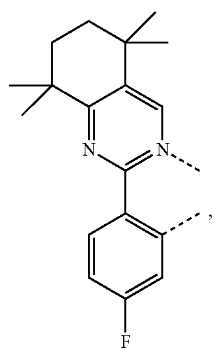

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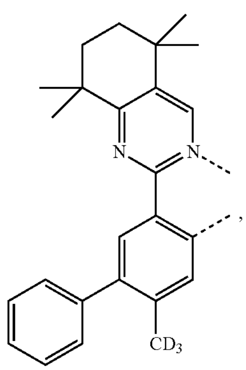

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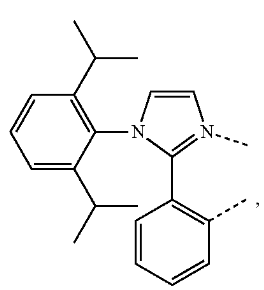

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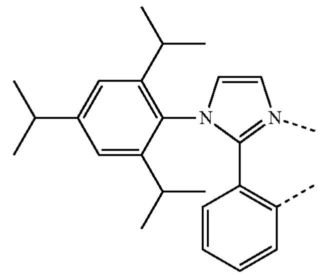

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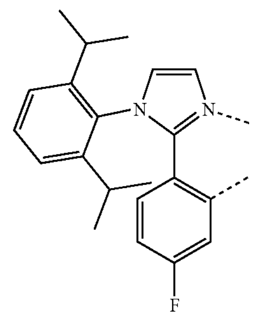

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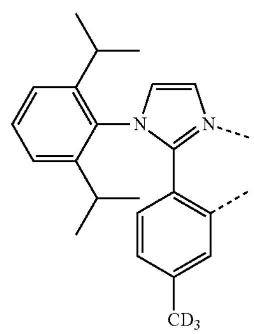

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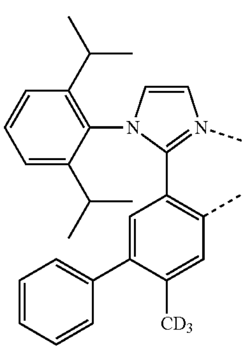

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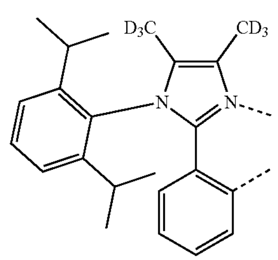

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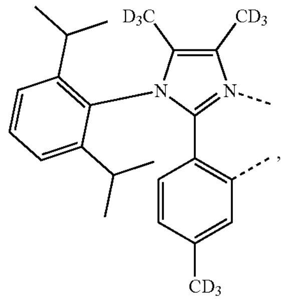

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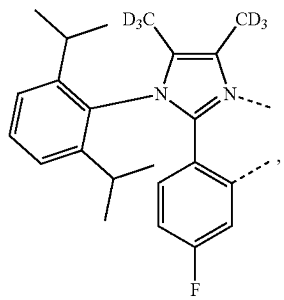

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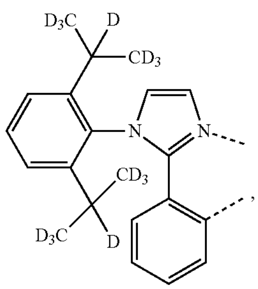

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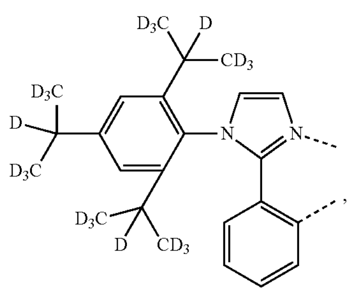

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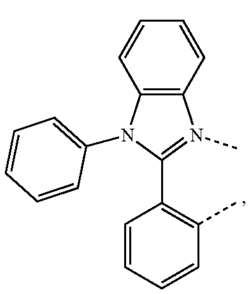

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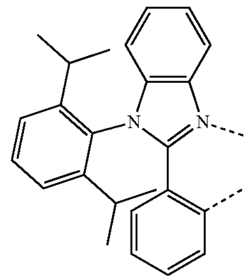

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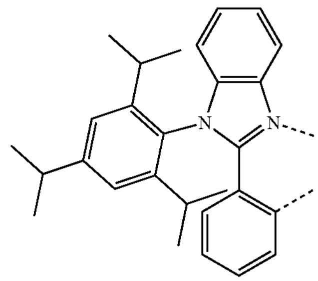

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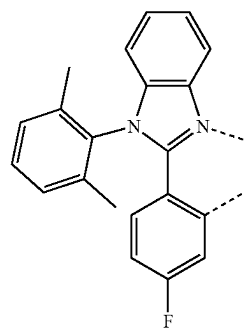

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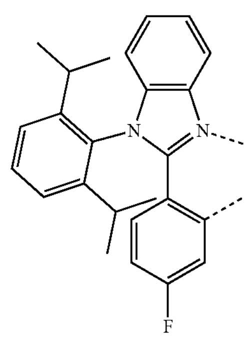

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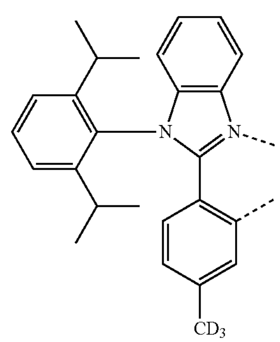

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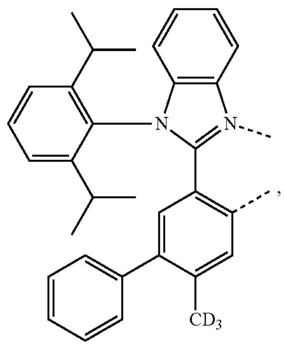

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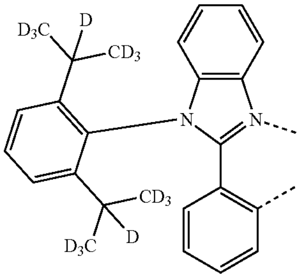

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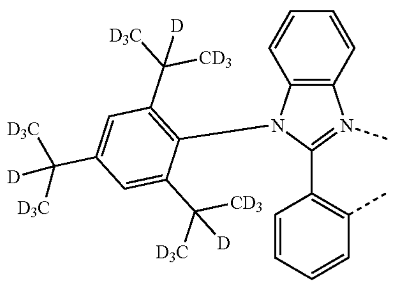

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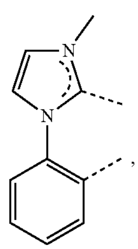

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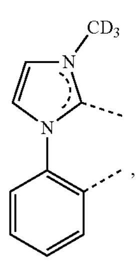

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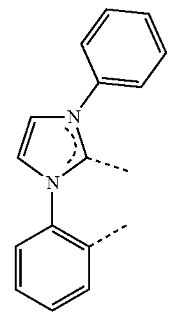

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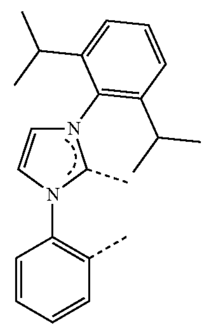

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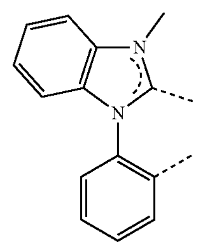

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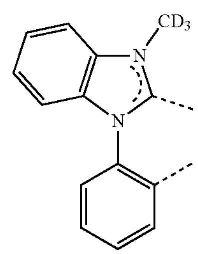

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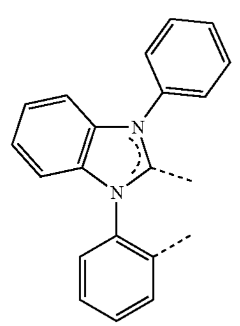

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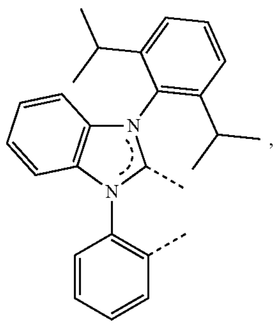

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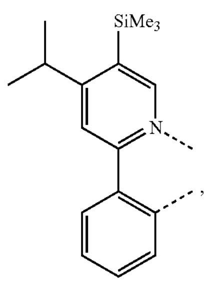

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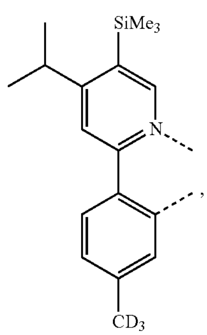

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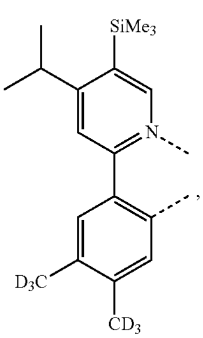

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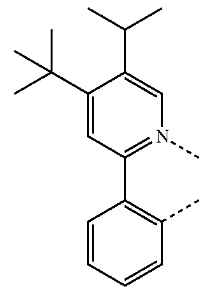

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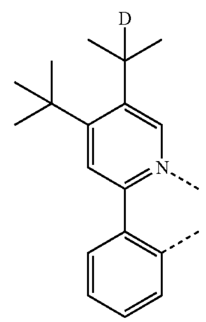

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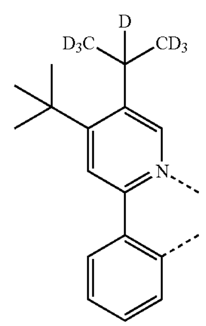

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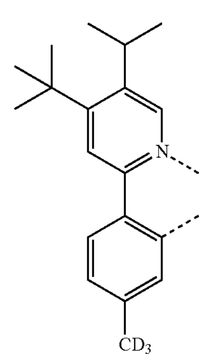

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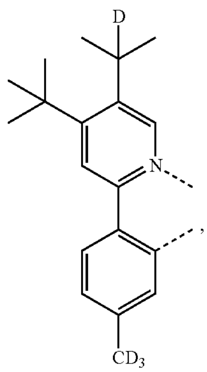
L_{B538}



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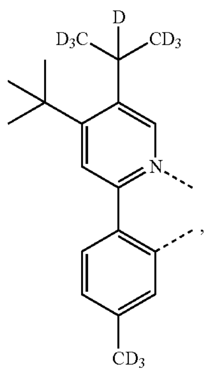

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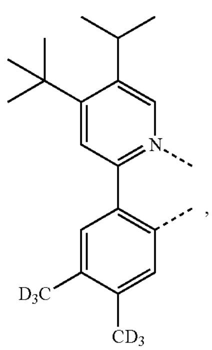

and

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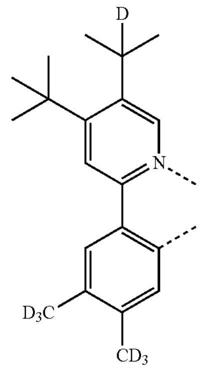
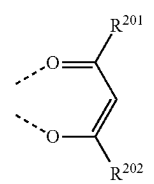
L_{B541}

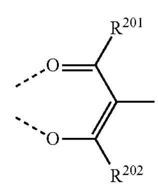


wherein each R has a structure based on formula



and

each has a structure based on formula



wherein for each L_{Cj} in $L_{Cj\text{-}I}$ and $L_{Cj\text{-}II}$, R^{201} and R^{202} are defined in the following LIST 11:

L_{Cj}	R^{201}	R^{202}	L_{Cj}	R^{201}	R^{202}	L_{Cj}	R^{201}	R^{202}	L_{Cj}	R^{201}	R^{202}
L_{C1}	R^{D1}	R^{D1}	L_{C193}	R^{D1}	R^{D3}	L_{C385}	R^{D17}	R^{D40}	L_{C577}	R^{D143}	R^{D120}
L_{C2}	R^{D2}	R^{D2}	L_{C194}	R^{D1}	R^{D4}	L_{C386}	R^{D17}	R^{D41}	L_{C578}	R^{D143}	R^{D133}
L_{C3}	R^{D3}	R^{D3}	L_{C195}	R^{D1}	R^{D5}	L_{C387}	R^{D17}	R^{D42}	L_{C579}	R^{D143}	R^{D134}
L_{C4}	R^{D4}	R^{D4}	L_{C196}	R^{D1}	R^{D9}	L_{C388}	R^{D17}	R^{D43}	L_{C580}	R^{D143}	R^{D135}
L_{C5}	R^{D5}	R^{D5}	L_{C197}	R^{D1}	R^{D10}	L_{C389}	R^{D17}	R^{D48}	L_{C581}	R^{D143}	R^{D136}
L_{C6}	R^{D6}	R^{D6}	L_{C198}	R^{D1}	R^{D17}	L_{C390}	R^{D17}	R^{D49}	L_{C582}	R^{D143}	R^{D144}
L_{C7}	R^{D7}	R^{D7}	L_{C199}	R^{D1}	R^{D18}	L_{C391}	R^{D17}	R^{D50}	L_{C583}	R^{D143}	R^{D145}
L_{C8}	R^{D8}	R^{D8}	L_{C200}	R^{D1}	R^{D20}	L_{C392}	R^{D17}	R^{D54}	L_{C584}	R^{D143}	R^{D146}
L_{C9}	R^{D9}	R^{D9}	L_{C201}	R^{D1}	R^{D22}	L_{C393}	R^{D17}	R^{D55}	L_{C585}	R^{D143}	R^{D147}
L_{C10}	R^{D10}	R^{D10}	L_{C202}	R^{D1}	R^{D37}	L_{C394}	R^{D17}	R^{D58}	L_{C586}	R^{D143}	R^{D149}
L_{C11}	R^{D11}	R^{D11}	L_{C203}	R^{D1}	R^{D40}	L_{C395}	R^{D17}	R^{D59}	L_{C587}	R^{D143}	R^{D151}
L_{C12}	R^{D12}	R^{D12}	L_{C204}	R^{D1}	R^{D41}	L_{C396}	R^{D17}	R^{D78}	L_{C588}	R^{D143}	R^{D154}
L_{C13}	R^{D13}	R^{D13}	L_{C205}	R^{D1}	R^{D42}	L_{C397}	R^{D17}	R^{D79}	L_{C589}	R^{D143}	R^{D155}
L_{C14}	R^{D14}	R^{D14}	L_{C206}	R^{D1}	R^{D43}	L_{C398}	R^{D17}	R^{D81}	L_{C590}	R^{D143}	R^{D161}
L_{C15}	R^{D15}	R^{D15}	L_{C207}	R^{D1}	R^{D48}	L_{C399}	R^{D17}	R^{D87}	L_{C591}	R^{D143}	R^{D175}
L_{C16}	R^{D16}	R^{D16}	L_{C208}	R^{D1}	R^{D49}	L_{C400}	R^{D17}	R^{D88}	L_{C592}	R^{D144}	R^{D3}
L_{C17}	R^{D17}	R^{D17}	L_{C209}	R^{D1}	R^{D50}	L_{C401}	R^{D17}	R^{D89}	L_{C593}	R^{D144}	R^{D5}
L_{C18}	R^{D18}	R^{D18}	L_{C210}	R^{D1}	R^{D54}	L_{C402}	R^{D17}	R^{D93}	L_{C594}	R^{D144}	R^{D17}

-continued

L_{Cj}	R^{201}	R^{202}	L_{Cj}	R^{201}	R^{202}	L_{Cj}	R^{201}	R^{202}	L_{Cj}	R^{201}	R^{202}
L_{C19}	R^{D19}	R^{D19}	L_{C211}	R^{D1}	R^{D55}	L_{C403}	R^{D17}	R^{D116}	L_{C595}	R^{D144}	R^{D18}
L_{C20}	R^{D20}	R^{D20}	L_{C212}	R^{D1}	R^{D58}	L_{C404}	R^{D17}	R^{D117}	L_{C596}	R^{D144}	R^{D20}
L_{C21}	R^{D21}	R^{D21}	L_{C213}	R^{D1}	R^{D59}	L_{C405}	R^{D17}	R^{D118}	L_{C597}	R^{D144}	R^{D22}
L_{C22}	R^{D22}	R^{D22}	L_{C214}	R^{D1}	R^{D78}	L_{C406}	R^{D17}	R^{D119}	L_{C598}	R^{D144}	R^{D37}
L_{C23}	R^{D23}	R^{D23}	L_{C215}	R^{D1}	R^{D79}	L_{C407}	R^{D17}	R^{D120}	L_{C599}	R^{D144}	R^{D40}
L_{C24}	R^{D24}	R^{D24}	L_{C216}	R^{D1}	R^{D81}	L_{C408}	R^{D17}	R^{D133}	L_{C600}	R^{D144}	R^{D41}
L_{C25}	R^{D25}	R^{D25}	L_{C217}	R^{D1}	R^{D87}	L_{C409}	R^{D17}	R^{D134}	L_{C601}	R^{D144}	R^{D42}
L_{C26}	R^{D26}	R^{D26}	L_{C218}	R^{D1}	R^{D88}	L_{C410}	R^{D17}	R^{D135}	L_{C602}	R^{D144}	R^{D43}
L_{C27}	R^{D27}	R^{D27}	L_{C219}	R^{D1}	R^{D89}	L_{C411}	R^{D17}	R^{D136}	L_{C603}	R^{D144}	R^{D48}
L_{C28}	R^{D28}	R^{D28}	L_{C220}	R^{D1}	R^{D93}	L_{C412}	R^{D17}	R^{D143}	L_{C604}	R^{D144}	R^{D49}
L_{C29}	R^{D29}	R^{D29}	L_{C221}	R^{D1}	R^{D116}	L_{C413}	R^{D17}	R^{D144}	L_{C605}	R^{D144}	R^{D54}
L_{C30}	R^{D30}	R^{D30}	L_{C222}	R^{D1}	R^{D117}	L_{C414}	R^{D17}	R^{D145}	L_{C606}	R^{D144}	R^{D58}
L_{C31}	R^{D31}	R^{D31}	L_{C223}	R^{D1}	R^{D118}	L_{C415}	R^{D17}	R^{D146}	L_{C607}	R^{D144}	R^{D59}
L_{C32}	R^{D32}	R^{D32}	L_{C224}	R^{D1}	R^{D119}	L_{C416}	R^{D17}	R^{D147}	L_{C608}	R^{D144}	R^{D78}
L_{C33}	R^{D33}	R^{D33}	L_{C225}	R^{D1}	R^{D120}	L_{C417}	R^{D17}	R^{D149}	L_{C609}	R^{D144}	R^{D79}
L_{C34}	R^{D34}	R^{D34}	L_{C226}	R^{D1}	R^{D133}	L_{C418}	R^{D17}	R^{D151}	L_{C610}	R^{D144}	R^{D81}
L_{C35}	R^{D35}	R^{D35}	L_{C227}	R^{D1}	R^{D134}	L_{C419}	R^{D17}	R^{D154}	L_{C611}	R^{D144}	R^{D87}
L_{C36}	R^{D36}	R^{D36}	L_{C228}	R^{D1}	R^{D135}	L_{C420}	R^{D17}	R^{D155}	L_{C612}	R^{D144}	R^{D88}
L_{C37}	R^{D37}	R^{D37}	L_{C229}	R^{D1}	R^{D136}	L_{C421}	R^{D17}	R^{D161}	L_{C613}	R^{D144}	R^{D89}
L_{C38}	R^{D38}	R^{D38}	L_{C230}	R^{D1}	R^{D143}	L_{C422}	R^{D17}	R^{D175}	L_{C614}	R^{D144}	R^{D93}
L_{C39}	R^{D39}	R^{D39}	L_{C231}	R^{D1}	R^{D144}	L_{C423}	R^{D50}	R^{D3}	L_{C615}	R^{D144}	R^{D116}
L_{C40}	R^{D40}	R^{D40}	L_{C232}	R^{D1}	R^{D145}	L_{C424}	R^{D50}	R^{D5}	L_{C616}	R^{D144}	R^{D117}
L_{C41}	R^{D41}	R^{D41}	L_{C233}	R^{D1}	R^{D146}	L_{C425}	R^{D50}	R^{D18}	L_{C617}	R^{D144}	R^{D118}
L_{C42}	R^{D42}	R^{D42}	L_{C234}	R^{D1}	R^{D147}	L_{C426}	R^{D50}	R^{D20}	L_{C618}	R^{D144}	R^{D119}
L_{C43}	R^{D43}	R^{D43}	L_{C235}	R^{D1}	R^{D149}	L_{C427}	R^{D50}	R^{D22}	L_{C619}	R^{D144}	R^{D120}
L_{C44}	R^{D44}	R^{D44}	L_{C236}	R^{D1}	R^{D151}	L_{C428}	R^{D50}	R^{D37}	L_{C620}	R^{D144}	R^{D133}
L_{C45}	R^{D45}	R^{D45}	L_{C237}	R^{D1}	R^{D154}	L_{C429}	R^{D50}	R^{D40}	L_{C621}	R^{D144}	R^{D134}
L_{C46}	R^{D46}	R^{D46}	L_{C238}	R^{D1}	R^{D155}	L_{C430}	R^{D50}	R^{D41}	L_{C622}	R^{D144}	R^{D135}
L_{C47}	R^{D47}	R^{D47}	L_{C239}	R^{D1}	R^{D161}	L_{C431}	R^{D50}	R^{D42}	L_{C623}	R^{D144}	R^{D136}
L_{C48}	R^{D48}	R^{D48}	L_{C240}	R^{D1}	R^{D175}	L_{C432}	R^{D50}	R^{D43}	L_{C624}	R^{D144}	R^{D145}
L_{C49}	R^{D49}	R^{D49}	L_{C241}	R^{D4}	R^{D3}	L_{C433}	R^{D50}	R^{D48}	L_{C625}	R^{D144}	R^{D146}
L_{C50}	R^{D50}	R^{D50}	L_{C242}	R^{D4}	R^{D5}	L_{C434}	R^{D50}	R^{D49}	L_{C626}	R^{D144}	R^{D147}
L_{C51}	R^{D51}	R^{D51}	L_{C243}	R^{D4}	R^{D9}	L_{C435}	R^{D50}	R^{D54}	L_{C627}	R^{D144}	R^{D149}
L_{C52}	R^{D52}	R^{D52}	L_{C244}	R^{D4}	R^{D10}	L_{C436}	R^{D50}	R^{D55}	L_{C628}	R^{D144}	R^{D151}
L_{C53}	R^{D53}	R^{D53}	L_{C245}	R^{D4}	R^{D17}	L_{C437}	R^{D50}	R^{D58}	L_{C629}	R^{D144}	R^{D154}
L_{C54}	R^{D54}	R^{D54}	L_{C246}	R^{D4}	R^{D18}	L_{C438}	R^{D50}	R^{D59}	L_{C630}	R^{D144}	R^{D155}
L_{C55}	R^{D55}	R^{D55}	L_{C247}	R^{D4}	R^{D20}	L_{C439}	R^{D50}	R^{D78}	L_{C631}	R^{D144}	R^{D161}
L_{C56}	R^{D56}	R^{D56}	L_{C248}	R^{D4}	R^{D22}	L_{C440}	R^{D50}	R^{D79}	L_{C632}	R^{D144}	R^{D175}
L_{C57}	R^{D57}	R^{D57}	L_{C249}	R^{D4}	R^{D37}	L_{C441}	R^{D50}	R^{D81}	L_{C633}	R^{D145}	R^{D3}
L_{C58}	R^{D58}	R^{D58}	L_{C250}	R^{D4}	R^{D40}	L_{C442}	R^{D50}	R^{D87}	L_{C634}	R^{D145}	R^{D5}
L_{C59}	R^{D59}	R^{D59}	L_{C251}	R^{D4}	R^{D41}	L_{C443}	R^{D50}	R^{D88}	L_{C635}	R^{D145}	R^{D17}
L_{C60}	R^{D60}	R^{D60}	L_{C252}	R^{D4}	R^{D42}	L_{C444}	R^{D50}	R^{D89}	L_{C636}	R^{D145}	R^{D18}
L_{C61}	R^{D61}	R^{D61}	L_{C253}	R^{D4}	R^{D43}	L_{C445}	R^{D50}	R^{D93}	L_{C637}	R^{D145}	R^{D20}
L_{C62}	R^{D62}	R^{D62}	L_{C254}	R^{D4}	R^{D48}	L_{C446}	R^{D50}	R^{D116}	L_{C638}	R^{D145}	R^{D22}
L_{C63}	R^{D63}	R^{D63}	L_{C255}	R^{D4}	R^{D49}	L_{C447}	R^{D50}	R^{D117}	L_{C639}	R^{D145}	R^{D37}
L_{C64}	R^{D64}	R^{D64}	L_{C256}	R^{D4}	R^{D50}	L_{C448}	R^{D50}	R^{D118}	L_{C640}	R^{D145}	R^{D40}
L_{C65}	R^{D65}	R^{D65}	L_{C257}	R^{D4}	R^{D54}	L_{C449}	R^{D50}	R^{D119}	L_{C641}	R^{D145}	R^{D41}
L_{C66}	R^{D66}	R^{D66}	L_{C258}	R^{D4}	R^{D55}	L_{C450}	R^{D50}	R^{D120}	L_{C642}	R^{D145}	R^{D42}
L_{C67}	R^{D67}	R^{D67}	L_{C259}	R^{D4}	R^{D58}	L_{C451}	R^{D50}	R^{D133}	L_{C643}	R^{D145}	R^{D43}
L_{C68}	R^{D68}	R^{D68}	L_{C260}	R^{D4}	R^{D59}	L_{C452}	R^{D50}	R^{D134}	L_{C644}	R^{D145}	R^{D48}
L_{C69}	R^{D69}	R^{D69}	L_{C261}	R^{D4}	R^{D78}	L_{C453}	R^{D50}	R^{D135}	L_{C645}	R^{D145}	R^{D49}
L_{C70}	R^{D70}	R^{D70}	L_{C262}	R^{D4}	R^{D79}	L_{C454}	R^{D50}	R^{D136}	L_{C646}	R^{D145}	R^{D54}
L_{C71}	R^{D71}	R^{D71}	L_{C263}	R^{D4}	R^{D81}	L_{C455}	R^{D50}	R^{D143}	L_{C647}	R^{D145}	R^{D58}
L_{C72}	R^{D72}	R^{D72}	L_{C264}	R^{D4}	R^{D87}	L_{C456}	R^{D50}	R^{D144}	L_{C648}	R^{D145}	R^{D59}
L_{C73}	R^{D73}	R^{D73}	L_{C265}	R^{D4}	R^{D88}	L_{C457}	R^{D50}	R^{D145}	L_{C649}	R^{D145}	R^{D78}
L_{C74}	R^{D74}	R^{D74}	L_{C266}	R^{D4}	R^{D89}	L_{C458}	R^{D50}	R^{D146}	L_{C650}	R^{D145}	R^{D79}
L_{C75}	R^{D75}	R^{D75}	L_{C267}	R^{D4}	R^{D93}	L_{C459}	R^{D50}	R^{D147}	L_{C651}	R^{D145}	R^{D81}
L_{C76}	R^{D76}	R^{D76}	L_{C268}	R^{D4}	R^{D116}	L_{C460}	R^{D50}	R^{D149}	L_{C652}	R^{D145}	R^{D87}
L_{C77}	R^{D77}	R^{D77}	L_{C269}	R^{D4}	R^{D117}	L_{C461}	R^{D50}	R^{D151}	L_{C653}	R^{D145}	R^{D88}
L_{C78}	R^{D78}	R^{D78}	L_{C270}	R^{D4}	R^{D118}	L_{C462}	R^{D50}	R^{D154}	L_{C654}	R^{D145}	R^{D89}
L_{C79}	R^{D79}	R^{D79}	L_{C271}	R^{D4}	R^{D119}	L_{C463}	R^{D50}	R^{D155}	L_{C655}	R^{D145}	R^{D93}
L_{C80}	R^{D80}	R^{D80}	L_{C272}	R^{D4}	R^{D120}	L_{C464}	R^{D50}	R^{D161}	L_{C656}	R^{D145}	R^{D116}
L_{C81}	R^{D81}	R^{D81}	L_{C273}	R^{D4}	R^{D133}	L_{C465}	R^{D50}	R^{D175}	L_{C657}	R^{D145}	R^{D117}
L_{C82}	R^{D82}	R^{D82}	L_{C274}	R^{D4}	R^{D134}	L_{C466}	R^{D55}	R^{D3}	L_{C658}	R^{D145}	R^{D118}
L_{C83}	R^{D83}	R^{D83}	L_{C275}	R^{D4}	R^{D135}	L_{C467}	R^{D55}	R^{D5}	L_{C659}	R^{D145}	R^{D119}
L_{C84}	R^{D84}	R^{D84}	L_{C276}	R^{D4}	R^{D136}	L_{C468}	R^{D55}	R^{D18}	L_{C660}	R^{D145}	R^{D120}
L_{C85}	R^{D85}	R^{D85}	L_{C277}	R^{D4}	R^{D143}	L_{C469}	R^{D55}	R^{D20}	L_{C661}	R^{D145}	R^{D133}
L_{C86}	R^{D86}	R^{D86}	L_{C278}	R^{D4}	R^{D144}	L_{C470}	R^{D55}	R^{D22}	L_{C662}	R^{D145}	R^{D134}
L_{C87}	R^{D87}	R^{D87}	L_{C279}	R^{D4}	R^{D145}	L_{C471}	R^{D55}	R^{D37}	L_{C663}	R^{D145}	R^{D135}
L_{C88}	R^{D88}	R^{D88}	L_{C280}	R^{D4}	R^{D146}	L_{C472}	R^{D55}	R^{D40}	L_{C664}	R^{D145}	R^{D136}
L_{C89}	R^{D89}	R^{D89}	L_{C281}	R^{D4}	R^{D147}	L_{C473}	R^{D55}	R^{D41}	L_{C665}	R^{D145}	R^{D146}
L_{C90}	R^{D90}	R^{D90}	L_{C282}	R^{D4}	R^{D149}	L_{C474}	R^{D55}	R^{D42}	L_{C666}	R^{D145}	R^{D147}
L_{C91}	R^{D91}	R^{D91}	L_{C283}	R^{D4}	R^{D151}	L_{C475}	R^{D55}	R^{D43}	L_{C667}	R^{D145}	R^{D149}
L_{C92}	R^{D92}	R^{D92}	L_{C284}	R^{D4}	R^{D154}	L_{C476}	R^{D55}	R^{D48}	L_{C668}	R^{D145}	R^{D151}

-continued

L_{Cj}	R^{201}	R^{202}	L_{Cj}	R^{201}	R^{202}	L_{Cj}	R^{201}	R^{202}	L_{Cj}	R^{201}	R^{202}
L_{C93}	R^{D93}	R^{D93}	L_{C285}	R^{D4}	R^{D155}	L_{C477}	R^{D55}	R^{D49}	L_{C669}	R^{D145}	R^{D154}
L_{C94}	R^{D94}	R^{D94}	L_{C286}	R^{D4}	R^{D161}	L_{C478}	R^{D55}	R^{D54}	L_{C670}	R^{D145}	R^{D155}
L_{C95}	R^{D95}	R^{D95}	L_{C287}	R^{D4}	R^{D175}	L_{C479}	R^{D55}	R^{D58}	L_{C671}	R^{D145}	R^{D161}
L_{C96}	R^{D96}	R^{D96}	L_{C288}	R^{D9}	R^{D3}	L_{C480}	R^{D55}	R^{D59}	L_{C672}	R^{D145}	R^{D175}
L_{C97}	R^{D97}	R^{D97}	L_{C289}	R^{D9}	R^{D5}	L_{C481}	R^{D55}	R^{D78}	L_{C673}	R^{D146}	R^{D3}
L_{C98}	R^{D98}	R^{D98}	L_{C290}	R^{D9}	R^{D10}	L_{C482}	R^{D55}	R^{D79}	L_{C674}	R^{D146}	R^{D5}
L_{C99}	R^{D99}	R^{D99}	L_{C291}	R^{D9}	R^{D17}	L_{C483}	R^{D55}	R^{D81}	L_{C675}	R^{D146}	R^{D17}
L_{C100}	R^{D100}	R^{D100}	L_{C292}	R^{D9}	R^{D18}	L_{C484}	R^{D55}	R^{D87}	L_{C676}	R^{D146}	R^{D18}
L_{C101}	R^{D101}	R^{D101}	L_{C293}	R^{D9}	R^{D20}	L_{C485}	R^{D55}	R^{D88}	L_{C677}	R^{D146}	R^{D20}
L_{C102}	R^{D102}	R^{D102}	L_{C294}	R^{D9}	R^{D22}	L_{C486}	R^{D55}	R^{D89}	L_{C678}	R^{D146}	R^{D22}
L_{C103}	R^{D103}	R^{D103}	L_{C295}	R^{D9}	R^{D37}	L_{C487}	R^{D55}	R^{D93}	L_{C679}	R^{D146}	R^{D37}
L_{C104}	R^{D104}	R^{D104}	L_{C296}	R^{D9}	R^{D40}	L_{C488}	R^{D55}	R^{D116}	L_{C680}	R^{D146}	R^{D40}
L_{C105}	R^{D105}	R^{D105}	L_{C297}	R^{D9}	R^{D41}	L_{C489}	R^{D55}	R^{D117}	L_{C681}	R^{D146}	R^{D41}
L_{C106}	R^{D106}	R^{D106}	L_{C298}	R^{D9}	R^{D42}	L_{C490}	R^{D55}	R^{D118}	L_{C682}	R^{D146}	R^{D42}
L_{C107}	R^{D107}	R^{D107}	L_{C299}	R^{D9}	R^{D43}	L_{C491}	R^{D55}	R^{D119}	L_{C683}	R^{D146}	R^{D43}
L_{C108}	R^{D108}	R^{D108}	L_{C300}	R^{D9}	R^{D48}	L_{C492}	R^{D55}	R^{D120}	L_{C684}	R^{D146}	R^{D48}
L_{C109}	R^{D109}	R^{D109}	L_{C301}	R^{D9}	R^{D49}	L_{C493}	R^{D55}	R^{D133}	L_{C685}	R^{D146}	R^{D49}
L_{C110}	R^{D110}	R^{D110}	L_{C302}	R^{D9}	R^{D50}	L_{C494}	R^{D55}	R^{D134}	L_{C686}	R^{D146}	R^{D54}
L_{C111}	R^{D111}	R^{D111}	L_{C303}	R^{D9}	R^{D54}	L_{C495}	R^{D55}	R^{D135}	L_{C687}	R^{D146}	R^{D58}
L_{C112}	R^{D112}	R^{D112}	L_{C304}	R^{D9}	R^{D55}	L_{C496}	R^{D55}	R^{D136}	L_{C688}	R^{D146}	R^{D59}
L_{C113}	R^{D113}	R^{D113}	L_{C305}	R^{D9}	R^{D58}	L_{C497}	R^{D55}	R^{D143}	L_{C689}	R^{D146}	R^{D78}
L_{C114}	R^{D114}	R^{D114}	L_{C306}	R^{D9}	R^{D59}	L_{C498}	R^{D55}	R^{D144}	L_{C690}	R^{D146}	R^{D79}
L_{C115}	R^{D115}	R^{D115}	L_{C307}	R^{D9}	R^{D78}	L_{C499}	R^{D55}	R^{D145}	L_{C691}	R^{D146}	R^{D81}
L_{C116}	R^{D116}	R^{D116}	L_{C308}	R^{D9}	R^{D79}	L_{C500}	R^{D55}	R^{D146}	L_{C692}	R^{D146}	R^{D87}
L_{C117}	R^{D117}	R^{D117}	L_{C309}	R^{D9}	R^{D81}	L_{C501}	R^{D55}	R^{D147}	L_{C693}	R^{D146}	R^{D88}
L_{C118}	R^{D118}	R^{D118}	L_{C310}	R^{D9}	R^{D87}	L_{C502}	R^{D55}	R^{D149}	L_{C694}	R^{D146}	R^{D89}
L_{C119}	R^{D119}	R^{D119}	L_{C311}	R^{D9}	R^{D88}	L_{C503}	R^{D55}	R^{D151}	L_{C695}	R^{D146}	R^{D93}
L_{C120}	R^{D120}	R^{D120}	L_{C312}	R^{D9}	R^{D89}	L_{C504}	R^{D55}	R^{D154}	L_{C696}	R^{D146}	R^{D117}
L_{C121}	R^{D121}	R^{D121}	L_{C313}	R^{D9}	R^{D93}	L_{C505}	R^{D55}	R^{D155}	L_{C697}	R^{D146}	R^{D118}
L_{C122}	R^{D122}	R^{D122}	L_{C314}	R^{D9}	R^{D116}	L_{C506}	R^{D55}	R^{D161}	L_{C698}	R^{D146}	R^{D119}
L_{C123}	R^{D123}	R^{D123}	L_{C315}	R^{D9}	R^{D117}	L_{C507}	R^{D55}	R^{D175}	L_{C699}	R^{D146}	R^{D120}
L_{C124}	R^{D124}	R^{D124}	L_{C316}	R^{D9}	R^{D118}	L_{C508}	R^{D116}	R^{D3}	L_{C700}	R^{D146}	R^{D133}
L_{C125}	R^{D125}	R^{D125}	L_{C317}	R^{D9}	R^{D119}	L_{C509}	R^{D116}	R^{D5}	L_{C701}	R^{D146}	R^{D134}
L_{C126}	R^{D126}	R^{D126}	L_{C318}	R^{D9}	R^{D120}	L_{C510}	R^{D116}	R^{D17}	L_{C702}	R^{D146}	R^{D135}
L_{C127}	R^{D127}	R^{D127}	L_{C319}	R^{D9}	R^{D133}	L_{C511}	R^{D116}	R^{D18}	L_{C703}	R^{D146}	R^{D136}
L_{C128}	R^{D128}	R^{D128}	L_{C320}	R^{D9}	R^{D134}	L_{C512}	R^{D116}	R^{D20}	L_{C704}	R^{D146}	R^{D146}
L_{C129}	R^{D129}	R^{D129}	L_{C321}	R^{D9}	R^{D135}	L_{C513}	R^{D116}	R^{D22}	L_{C705}	R^{D146}	R^{D147}
L_{C130}	R^{D130}	R^{D130}	L_{C322}	R^{D9}	R^{D136}	L_{C514}	R^{D116}	R^{D37}	L_{C706}	R^{D146}	R^{D149}
L_{C131}	R^{D131}	R^{D131}	L_{C323}	R^{D9}	R^{D143}	L_{C515}	R^{D116}	R^{D40}	L_{C707}	R^{D146}	R^{D151}
L_{C132}	R^{D132}	R^{D132}	L_{C324}	R^{D9}	R^{D144}	L_{C516}	R^{D116}	R^{D41}	L_{C708}	R^{D146}	R^{D154}
L_{C133}	R^{D133}	R^{D133}	L_{C325}	R^{D9}	R^{D145}	L_{C517}	R^{D116}	R^{D42}	L_{C709}	R^{D146}	R^{D155}
L_{C134}	R^{D134}	R^{D134}	L_{C326}	R^{D9}	R^{D146}	L_{C518}	R^{D116}	R^{D43}	L_{C710}	R^{D146}	R^{D161}
L_{C135}	R^{D135}	R^{D135}	L_{C327}	R^{D9}	R^{D147}	L_{C519}	R^{D116}	R^{D48}	L_{C711}	R^{D146}	R^{D175}
L_{C136}	R^{D136}	R^{D136}	L_{C328}	R^{D9}	R^{D149}	L_{C520}	R^{D116}	R^{D49}	L_{C712}	R^{D133}	R^{D3}
L_{C137}	R^{D137}	R^{D137}	L_{C329}	R^{D9}	R^{D151}	L_{C521}	R^{D116}	R^{D54}	L_{C713}	R^{D133}	R^{D5}
L_{C138}	R^{D138}	R^{D138}	L_{C330}	R^{D9}	R^{D154}	L_{C522}	R^{D116}	R^{D58}	L_{C714}	R^{D133}	R^{D3}
L_{C139}	R^{D139}	R^{D139}	L_{C331}	R^{D9}	R^{D155}	L_{C523}	R^{D116}	R^{D59}	L_{C715}	R^{D133}	R^{D18}
L_{C140}	R^{D140}	R^{D140}	L_{C332}	R^{D9}	R^{D161}	L_{C524}	R^{D116}	R^{D78}	L_{C716}	R^{D133}	R^{D20}
L_{C141}	R^{D141}	R^{D141}	L_{C333}	R^{D9}	R^{D175}	L_{C525}	R^{D116}	R^{D79}	L_{C717}	R^{D133}	R^{D22}
L_{C142}	R^{D142}	R^{D142}	L_{C334}	R^{D10}	R^{D3}	L_{C526}	R^{D116}	R^{D81}	L_{C718}	R^{D133}	R^{D37}
L_{C143}	R^{D143}	R^{D143}	L_{C335}	R^{D10}	R^{D5}	L_{C527}	R^{D116}	R^{D87}	L_{C719}	R^{D133}	R^{D40}
L_{C144}	R^{D144}	R^{D144}	L_{C336}	R^{D10}	R^{D17}	L_{C528}	R^{D116}	R^{D88}	L_{C720}	R^{D133}	R^{D41}
L_{C145}	R^{D145}	R^{D145}	L_{C337}	R^{D10}	R^{D18}	L_{C529}	R^{D116}	R^{D89}	L_{C721}	R^{D133}	R^{D42}
L_{C146}	R^{D146}	R^{D146}	L_{C338}	R^{D10}	R^{D20}	L_{C530}	R^{D116}	R^{D93}	L_{C722}	R^{D133}	R^{D43}
L_{C147}	R^{D147}	R^{D147}	L_{C339}	R^{D10}	R^{D22}	L_{C531}	R^{D116}	R^{D117}	L_{C723}	R^{D133}	R^{D48}
L_{C148}	R^{D148}	R^{D148}	L_{C340}	R^{D10}	R^{D37}	L_{C532}	R^{D116}	R^{D118}	L_{C724}	R^{D133}	R^{D49}
L_{C149}	R^{D149}	R^{D149}	L_{C341}	R^{D10}	R^{D40}	L_{C533}	R^{D116}	R^{D119}	L_{C725}	R^{D133}	R^{D54}
L_{C150}	R^{D150}	R^{D150}	L_{C342}	R^{D10}	R^{D41}	L_{C534}	R^{D116}	R^{D120}	L_{C726}	R^{D133}	R^{D58}
L_{C151}	R^{D151}	R^{D151}	L_{C343}	R^{D10}	R^{D42}	L_{C535}	R^{D116}	R^{D133}	L_{C727}	R^{D133}	R^{D59}
L_{C152}	R^{D152}	R^{D152}	L_{C344}	R^{D10}	R^{D43}	L_{C536}	R^{D116}	R^{D134}	L_{C728}	R^{D133}	R^{D78}
L_{C153}	R^{D153}	R^{D153}	L_{C345}	R^{D10}	R^{D48}	L_{C537}	R^{D116}	R^{D135}	L_{C729}	R^{D133}	R^{D79}
L_{C154}	R^{D154}	R^{D154}	L_{C346}	R^{D10}	R^{D49}	L_{C538}	R^{D116}	R^{D136}	L_{C730}	R^{D133}	R^{D81}
L_{C155}	R^{D155}	R^{D155}	L_{C347}	R^{D10}	R^{D50}	L_{C539}	R^{D116}	R^{D143}	L_{C731}	R^{D133}	R^{D87}
L_{C156}	R^{D156}	R^{D156}	L_{C348}	R^{D10}	R^{D54}	L_{C540}	R^{D116}	R^{D144}	L_{C732}	R^{D133}	R^{D88}
L_{C157}	R^{D157}	R^{D157}	L_{C349}	R^{D10}	R^{D55}	L_{C541}	R^{D116}	R^{D145}	L_{C733}	R^{D133}	R^{D89}
L_{C158}	R^{D158}	R^{D158}	L_{C350}	R^{D10}	R^{D58}	L_{C542}	R^{D116}	R^{D146}	L_{C734}	R^{D133}	R^{D93}
L_{C159}	R^{D159}	R^{D159}	L_{C351}	R^{D10}	R^{D59}	L_{C543}	R^{D116}	R^{D147}	L_{C735}	R^{D133}	R^{D117}
L_{C160}	R^{D160}	R^{D160}	L_{C352}	R^{D10}	R^{D78}	L_{C544}	R^{D116}	R^{D149}	L_{C736}	R^{D133}	R^{D118}
L_{C161}	R^{D161}	R^{D161}	L_{C353}	R^{D10}	R^{D79}	L_{C545}	R^{D116}	R^{D151}	L_{C737}	R^{D133}	R^{D119}
L_{C162}	R^{D162}	R^{D162}	L_{C354}	R^{D10}	R^{D81}	L_{C546}	R^{D116}	R^{D154}	L_{C738}	R^{D133}	R^{D120}
L_{C163}	R^{D163}	R^{D163}	L_{C355}	R^{D10}	R^{D87}	L_{C547}	R^{D116}	R^{D155}	L_{C739}	R^{D133}	R^{D133}
L_{C164}	R^{D164}	R^{D164}	L_{C356}	R^{D10}	R^{D88}	L_{C548}	R^{D116}	R^{D161}	L_{C740}	R^{D133}	R^{D134}
L_{C165}	R^{D165}	R^{D165}	L_{C357}	R^{D10}	R^{D89}	L_{C549}	R^{D116}	R^{D175}	L_{C741}	R^{D133}	R^{D135}
L_{C166}	R^{D166}	R^{D166}	L_{C358}	R^{D10}	R^{D93}	L_{C550}	R^{D143}	R^{D3}	L_{C742}	R^{D133}	R^{D136}

-continued

L_{Cj}	R^{201}	R^{202}	L_{Cj}	R^{201}	R^{202}	L_{Cj}	R^{201}	R^{202}	L_{Cj}	R^{201}	R^{202}
L_{C167}	R^{D167}	R^{D167}	L_{C359}	R^{D10}	R^{D116}	L_{C551}	R^{D143}	R^{D5}	L_{C743}	R^{D133}	R^{D146}
L_{C168}	R^{D168}	R^{D168}	L_{C360}	R^{D10}	R^{D117}	L_{C552}	R^{D143}	R^{D17}	L_{C744}	R^{D133}	R^{D147}
L_{C169}	R^{D169}	R^{D169}	L_{C361}	R^{D10}	R^{D118}	L_{C553}	R^{D143}	R^{D18}	L_{C745}	R^{D133}	R^{D149}
L_{C170}	R^{D170}	R^{D170}	L_{C362}	R^{D10}	R^{D119}	L_{C554}	R^{D143}	R^{D20}	L_{C746}	R^{D133}	R^{D151}
L_{C171}	R^{D171}	R^{D171}	L_{C363}	R^{D10}	R^{D120}	L_{C555}	R^{D143}	R^{D22}	L_{C747}	R^{D133}	R^{D154}
L_{C172}	R^{D172}	R^{D172}	L_{C364}	R^{D10}	R^{D133}	L_{C556}	R^{D143}	R^{D37}	L_{C748}	R^{D133}	R^{D155}
L_{C173}	R^{D173}	R^{D173}	L_{C365}	R^{D10}	R^{D134}	L_{C557}	R^{D143}	R^{D40}	L_{C749}	R^{D133}	R^{D161}
L_{C174}	R^{D174}	R^{D174}	L_{C366}	R^{D10}	R^{D135}	L_{C558}	R^{D143}	R^{D41}	L_{C750}	R^{D133}	R^{D175}
L_{C175}	R^{D175}	R^{D175}	L_{C367}	R^{D10}	R^{D136}	L_{C559}	R^{D143}	R^{D42}	L_{C751}	R^{D175}	R^{D3}
L_{C176}	R^{D176}	R^{D176}	L_{C368}	R^{D10}	R^{D143}	L_{C560}	R^{D143}	R^{D43}	L_{C752}	R^{D175}	R^{D5}
L_{C177}	R^{D177}	R^{D177}	L_{C369}	R^{D10}	R^{D144}	L_{C561}	R^{D143}	R^{D48}	L_{C753}	R^{D175}	R^{D18}
L_{C178}	R^{D178}	R^{D178}	L_{C370}	R^{D10}	R^{D145}	L_{C562}	R^{D143}	R^{D49}	L_{C754}	R^{D175}	R^{D20}
L_{C179}	R^{D179}	R^{D179}	L_{C371}	R^{D10}	R^{D146}	L_{C563}	R^{D143}	R^{D54}	L_{C755}	R^{D175}	R^{D22}
L_{C180}	R^{D180}	R^{D180}	L_{C372}	R^{D10}	R^{D147}	L_{C564}	R^{D143}	R^{D58}	L_{C756}	R^{D175}	R^{D37}
L_{C181}	R^{D181}	R^{D181}	L_{C373}	R^{D10}	R^{D149}	L_{C565}	R^{D143}	R^{D59}	L_{C757}	R^{D175}	R^{D40}
L_{C182}	R^{D182}	R^{D182}	L_{C374}	R^{D10}	R^{D151}	L_{C566}	R^{D143}	R^{D78}	L_{C758}	R^{D175}	R^{D41}
L_{C183}	R^{D183}	R^{D183}	L_{C375}	R^{D10}	R^{D154}	L_{C567}	R^{D143}	R^{D79}	L_{C759}	R^{D175}	R^{D42}
L_{C184}	R^{D184}	R^{D184}	L_{C376}	R^{D10}	R^{D155}	L_{C568}	R^{D143}	R^{D81}	L_{C760}	R^{D175}	R^{D43}
L_{C185}	R^{D185}	R^{D185}	L_{C377}	R^{D10}	R^{D161}	L_{C569}	R^{D143}	R^{D87}	L_{C761}	R^{D175}	R^{D48}
L_{C186}	R^{D186}	R^{D186}	L_{C378}	R^{D10}	R^{D175}	L_{C570}	R^{D143}	R^{D88}	L_{C762}	R^{D175}	R^{D49}
L_{C187}	R^{D187}	R^{D187}	L_{C379}	R^{D17}	R^{D3}	L_{C571}	R^{D143}	R^{D89}	L_{C763}	R^{D175}	R^{D54}
L_{C188}	R^{D188}	R^{D188}	L_{C380}	R^{D17}	R^{D5}	L_{C572}	R^{D143}	R^{D93}	L_{C764}	R^{D175}	R^{D58}
L_{C189}	R^{D189}	R^{D189}	L_{C381}	R^{D17}	R^{D18}	L_{C573}	R^{D143}	R^{D116}	L_{C765}	R^{D175}	R^{D59}
L_{C190}	R^{D190}	R^{D190}	L_{C382}	R^{D17}	R^{D20}	L_{C574}	R^{D143}	R^{D117}	L_{C766}	R^{D175}	R^{D78}
L_{C191}	R^{D191}	R^{D191}	L_{C383}	R^{D17}	R^{D22}	L_{C575}	R^{D143}	R^{D118}	L_{C767}	R^{D175}	R^{D79}
L_{C192}	R^{D192}	R^{D192}	L_{C384}	R^{D17}	R^{D37}	L_{C576}	R^{D143}	R^{D119}	L_{C768}	R^{D175}	R^{D81}
L_{C769}	R^{D193}	R^{D193}	L_{C877}	R^{D1}	R^{D193}	L_{C985}	R^{D4}	R^{D193}	L_{C1093}	R^{D9}	R^{D193}
L_{C770}	R^{D194}	R^{D194}	L_{C878}	R^{D1}	R^{D194}	L_{C986}	R^{D4}	R^{D194}	L_{C1094}	R^{D9}	R^{D194}
L_{C771}	R^{D195}	R^{D195}	L_{C879}	R^{D1}	R^{D195}	L_{C987}	R^{D4}	R^{D195}	L_{C1095}	R^{D9}	R^{D195}
L_{C772}	R^{D196}	R^{D196}	L_{C880}	R^{D1}	R^{D196}	L_{C988}	R^{D4}	R^{D196}	L_{C1096}	R^{D9}	R^{D196}
L_{C773}	R^{D197}	R^{D197}	L_{C881}	R^{D1}	R^{D197}	L_{C989}	R^{D4}	R^{D197}	L_{C1097}	R^{D9}	R^{D197}
L_{C774}	R^{D198}	R^{D198}	L_{C882}	R^{D1}	R^{D198}	L_{C990}	R^{D4}	R^{D198}	L_{C1098}	R^{D9}	R^{D198}
L_{C775}	R^{D199}	R^{D199}	L_{C883}	R^{D1}	R^{D199}	L_{C991}	R^{D4}	R^{D199}	L_{C1099}	R^{D9}	R^{D199}
L_{C776}	R^{D200}	R^{D200}	L_{C884}	R^{D1}	R^{D200}	L_{C992}	R^{D4}	R^{D200}	L_{C1100}	R^{D9}	R^{D200}
L_{C777}	R^{D201}	R^{D201}	L_{C885}	R^{D1}	R^{D201}	L_{C993}	R^{D4}	R^{D201}	L_{C1101}	R^{D9}	R^{D201}
L_{C778}	R^{D202}	R^{D202}	L_{C886}	R^{D1}	R^{D202}	L_{C994}	R^{D4}	R^{D202}	L_{C1102}	R^{D9}	R^{D202}
L_{C779}	R^{D203}	R^{D203}	L_{C887}	R^{D1}	R^{D203}	L_{C995}	R^{D4}	R^{D203}	L_{C1103}	R^{D9}	R^{D203}
L_{C780}	R^{D204}	R^{D204}	L_{C888}	R^{D1}	R^{D204}	L_{C996}	R^{D4}	R^{D204}	L_{C1104}	R^{D9}	R^{D204}
L_{C781}	R^{D205}	R^{D205}	L_{C889}	R^{D1}	R^{D205}	L_{C997}	R^{D4}	R^{D205}	L_{C1105}	R^{D9}	R^{D205}
L_{C782}	R^{D206}	R^{D206}	L_{C890}	R^{D1}	R^{D206}	L_{C998}	R^{D4}	R^{D206}	L_{C1106}	R^{D9}	R^{D206}
L_{C783}	R^{D207}	R^{D207}	L_{C891}	R^{D1}	R^{D207}	L_{C999}	R^{D4}	R^{D207}	L_{C1107}	R^{D9}	R^{D207}
L_{C784}	R^{D208}	R^{D208}	L_{C892}	R^{D1}	R^{D208}	L_{C1000}	R^{D4}	R^{D208}	L_{C1108}	R^{D9}	R^{D208}
L_{C785}	R^{D209}	R^{D209}	L_{C893}	R^{D1}	R^{D209}	L_{C1001}	R^{D4}	R^{D209}	L_{C1109}	R^{D9}	R^{D209}
L_{C786}	R^{D210}	R^{D210}	L_{C894}	R^{D1}	R^{D210}	L_{C1002}	R^{D4}	R^{D210}	L_{C1110}	R^{D9}	R^{D210}
L_{C787}	R^{D211}	R^{D211}	L_{C895}	R^{D1}	R^{D211}	L_{C1003}	R^{D4}	R^{D211}	L_{C1111}	R^{D9}	R^{D211}
L_{C788}	R^{D212}	R^{D212}	L_{C896}	R^{D1}	R^{D212}	L_{C1004}	R^{D4}	R^{D212}	L_{C1112}	R^{D9}	R^{D212}
L_{C789}	R^{D213}	R^{D213}	L_{C897}	R^{D1}	R^{D213}	L_{C1005}	R^{D4}	R^{D213}	L_{C1113}	R^{D9}	R^{D213}
L_{C790}	R^{D214}	R^{D214}	L_{C898}	R^{D1}	R^{D214}	L_{C1006}	R^{D4}	R^{D214}	L_{C1114}	R^{D9}	R^{D214}
L_{C791}	R^{D215}	R^{D215}	L_{C899}	R^{D1}	R^{D215}	L_{C1007}	R^{D4}	R^{D215}	L_{C1115}	R^{D9}	R^{D215}
L_{C792}	R^{D216}	R^{D216}	L_{C900}	R^{D1}	R^{D216}	L_{C1008}	R^{D4}	R^{D216}	L_{C1116}	R^{D9}	R^{D216}
L_{C793}	R^{D217}	R^{D217}	L_{C901}	R^{D1}	R^{D217}	L_{C1009}	R^{D4}	R^{D217}	L_{C1117}	R^{D9}	R^{D217}
L_{C794}	R^{D218}	R^{D218}	L_{C902}	R^{D1}	R^{D218}	L_{C1010}	R^{D4}	R^{D218}	L_{C1118}	R^{D9}	R^{D218}
L_{C795}	R^{D219}	R^{D219}	L_{C903}	R^{D1}	R^{D219}	L_{C1011}	R^{D4}	R^{D219}	L_{C1119}	R^{D9}	R^{D219}
L_{C796}	R^{D220}	R^{D220}	L_{C904}	R^{D1}	R^{D220}	L_{C1012}	R^{D4}	R^{D220}	L_{C1120}	R^{D9}	R^{D220}
L_{C797}	R^{D221}	R^{D221}	L_{C905}	R^{D1}	R^{D221}	L_{C1013}	R^{D4}	R^{D221}	L_{C1121}	R^{D9}	R^{D221}
L_{C798}	R^{D222}	R^{D222}	L_{C906}	R^{D1}	R^{D222}	L_{C1014}	R^{D4}	R^{D222}	L_{C1122}	R^{D9}	R^{D222}
L_{C799}	R^{D223}	R^{D223}	L_{C907}	R^{D1}	R^{D223}	L_{C1015}	R^{D4}	R^{D223}	L_{C1123}	R^{D9}	R^{D223}
L_{C800}	R^{D224}	R^{D224}	L_{C908}	R^{D1}	R^{D224}	L_{C1016}	R^{D4}	R^{D224}	L_{C1124}	R^{D9}	R^{D224}
L_{C801}	R^{D225}	R^{D225}	L_{C909}	R^{D1}	R^{D225}	L_{C1017}	R^{D4}	R^{D225}	L_{C1125}	R^{D9}	R^{D225}
L_{C802}	R^{D226}	R^{D226}	L_{C910}	R^{D1}	R^{D226}	L_{C1018}	R^{D4}	R^{D226}	L_{C1126}	R^{D9}	R^{D226}
L_{C803}	R^{D227}	R^{D227}	L_{C911}	R^{D1}	R^{D227}	L_{C1019}	R^{D4}	R^{D227}	L_{C1127}	R^{D9}	R^{D227}
L_{C804}	R^{D228}	R^{D228}	L_{C912}	R^{D1}	R^{D228}	L_{C1020}	R^{D4}	R^{D228}	L_{C1128}	R^{D9}	R^{D228}
L_{C805}	R^{D229}	R^{D229}	L_{C913}	R^{D1}	R^{D229}	L_{C1021}	R^{D4}	R^{D229}	L_{C1129}	R^{D9}	R^{D229}
L_{C806}	R^{D230}	R^{D230}	L_{C914}	R^{D1}	R^{D230}	L_{C1022}	R^{D4}	R^{D230}	L_{C1130}	R^{D9}	R^{D230}
L_{C807}	R^{D231}	R^{D231}	L_{C915}	R^{D1}	R^{D231}	L_{C1023}	R^{D4}	R^{D231}	L_{C1131}	R^{D9}	R^{D231}
L_{C808}	R^{D232}	R^{D232}	L_{C916}	R^{D1}	R^{D232}	L_{C1024}	R^{D4}	R^{D232}	L_{C1132}	R^{D9}	R^{D232}
L_{C809}	R^{D233}	R^{D233}	L_{C917}	R^{D1}	R^{D233}	L_{C1025}	R^{D4}	R^{D233}	L_{C1133}	R^{D9}	R^{D233}
L_{C810}	R^{D234}	R^{D234}	L_{C918}	R^{D1}	R^{D234}	L_{C1026}	R^{D4}	R^{D234}	L_{C1134}	R^{D9}	R^{D234}
L_{C811}	R^{D235}	R^{D235}	L_{C919}	R^{D1}	R^{D235}	L_{C1027}	R^{D4}	R^{D235}	L_{C1135}	R^{D9}	R^{D235}
L_{C812}	R^{D236}	R^{D236}	L_{C920}	R^{D1}	R^{D236}	L_{C1028}	R^{D4}	R^{D236}	L_{C1136}	R^{D9}	R^{D236}
L_{C813}	R^{D237}	R^{D237}	L_{C921}	R^{D1}	R^{D237}	L_{C1029}	R^{D4}	R^{D237}	L_{C1137}	R^{D9}	R^{D237}
L_{C814}	R^{D238}	R^{D238}	L_{C922}	R^{D1}	R^{D238}	L_{C1030}	R^{D4}	R^{D238}	L_{C1138}	R^{D9}	R^{D238}
L_{C815}	R^{D239}	R^{D239}	L_{C923}	R^{D1}	R^{D239}	L_{C1031}	R^{D4}	R^{D239}	L_{C1139}	R^{D9}	R^{D239}
L_{C816}	R^{D240}	R^{D240}	L_{C924}	R^{D1}	R^{D240}	L_{C1032}	R^{D4}	R^{D240}	L_{C1140}	R^{D9}	R^{D240}

-continued

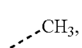
L_{Cj}	R^{201}	R^{202}	L_{Cj}	R^{201}	R^{202}	L_{Cj}	R^{201}	R^{202}	L_{Cj}	R^{201}	R^{202}
L_{C817}	R^{D241}	R^{D241}	L_{C925}	R^{D1}	R^{D241}	L_{C1033}	R^{D4}	R^{D241}	L_{C1141}	R^{D9}	R^{D241}
L_{C818}	R^{D242}	R^{D242}	L_{C926}	R^{D1}	R^{D242}	L_{C1034}	R^{D4}	R^{D242}	L_{C1142}	R^{D9}	R^{D242}
L_{C819}	R^{D243}	R^{D243}	L_{C927}	R^{D1}	R^{D243}	L_{C1035}	R^{D4}	R^{D243}	L_{C1143}	R^{D9}	R^{D243}
L_{C820}	R^{D244}	R^{D244}	L_{C928}	R^{D1}	R^{D244}	L_{C1036}	R^{D4}	R^{D244}	L_{C1144}	R^{D9}	R^{D244}
L_{C821}	R^{D245}	R^{D245}	L_{C929}	R^{D1}	R^{D245}	L_{C1037}	R^{D4}	R^{D245}	L_{C1145}	R^{D9}	R^{D245}
L_{C822}	R^{D246}	R^{D246}	L_{C930}	R^{D1}	R^{D246}	L_{C1038}	R^{D4}	R^{D246}	L_{C1146}	R^{D9}	R^{D246}
L_{C823}	R^{D17}	R^{D193}	L_{C931}	R^{D50}	R^{D193}	L_{C1039}	R^{D145}	R^{D193}	L_{C1147}	R^{D168}	R^{D193}
L_{C824}	R^{D17}	R^{D194}	L_{C932}	R^{D50}	R^{D194}	L_{C1040}	R^{D145}	R^{D194}	L_{C1148}	R^{D168}	R^{D194}
L_{C825}	R^{D17}	R^{D195}	L_{C933}	R^{D50}	R^{D195}	L_{C1041}	R^{D145}	R^{D195}	L_{C1149}	R^{D168}	R^{D195}
L_{C826}	R^{D17}	R^{D196}	L_{C934}	R^{D50}	R^{D196}	L_{C1042}	R^{D145}	R^{D196}	L_{C1150}	R^{D168}	R^{D196}
L_{C827}	R^{D17}	R^{D197}	L_{C935}	R^{D50}	R^{D197}	L_{C1043}	R^{D145}	R^{D197}	L_{C1151}	R^{D168}	R^{D197}
L_{C828}	R^{D17}	R^{D198}	L_{C936}	R^{D50}	R^{D198}	L_{C1044}	R^{D145}	R^{D198}	L_{C1152}	R^{D168}	R^{D198}
L_{C829}	R^{D17}	R^{D199}	L_{C937}	R^{D50}	R^{D199}	L_{C1045}	R^{D145}	R^{D199}	L_{C1153}	R^{D168}	R^{D199}
L_{C830}	R^{D17}	R^{D200}	L_{C938}	R^{D50}	R^{D200}	L_{C1046}	R^{D145}	R^{D200}	L_{C1154}	R^{D168}	R^{D200}
L_{C831}	R^{D17}	R^{D201}	L_{C939}	R^{D50}	R^{D201}	L_{C1047}	R^{D145}	R^{D201}	L_{C1155}	R^{D168}	R^{D201}
L_{C832}	R^{D17}	R^{D202}	L_{C940}	R^{D50}	R^{D202}	L_{C1048}	R^{D145}	R^{D202}	L_{C1156}	R^{D168}	R^{D202}
L_{C833}	R^{D17}	R^{D203}	L_{C941}	R^{D50}	R^{D203}	L_{C1049}	R^{D145}	R^{D203}	L_{C1157}	R^{D168}	R^{D203}
L_{C834}	R^{D17}	R^{D204}	L_{C942}	R^{D50}	R^{D204}	L_{C1050}	R^{D145}	R^{D204}	L_{C1158}	R^{D168}	R^{D204}
L_{C835}	R^{D17}	R^{D205}	L_{C943}	R^{D50}	R^{D205}	L_{C1051}	R^{D145}	R^{D205}	L_{C1159}	R^{D168}	R^{D205}
L_{C836}	R^{D17}	R^{D206}	L_{C944}	R^{D50}	R^{D206}	L_{C1052}	R^{D145}	R^{D206}	L_{C1160}	R^{D168}	R^{D206}
L_{C837}	R^{D17}	R^{D207}	L_{C945}	R^{D50}	R^{D207}	L_{C1053}	R^{D145}	R^{D207}	L_{C1161}	R^{D168}	R^{D207}
L_{C838}	R^{D17}	R^{D208}	L_{C946}	R^{D50}	R^{D208}	L_{C1054}	R^{D145}	R^{D208}	L_{C1162}	R^{D168}	R^{D208}
L_{C839}	R^{D17}	R^{D209}	L_{C947}	R^{D50}	R^{D209}	L_{C1055}	R^{D145}	R^{D209}	L_{C1163}	R^{D168}	R^{D209}
L_{C840}	R^{D17}	R^{D210}	L_{C948}	R^{D50}	R^{D210}	L_{C1056}	R^{D145}	R^{D210}	L_{C1164}	R^{D168}	R^{D210}
L_{C841}	R^{D17}	R^{D211}	L_{C949}	R^{D50}	R^{D211}	L_{C1057}	R^{D145}	R^{D211}	L_{C1165}	R^{D168}	R^{D211}
L_{C842}	R^{D17}	R^{D212}	L_{C950}	R^{D50}	R^{D212}	L_{C1058}	R^{D145}	R^{D212}	L_{C1166}	R^{D168}	R^{D212}
L_{C843}	R^{D17}	R^{D213}	L_{C951}	R^{D50}	R^{D213}	L_{C1059}	R^{D145}	R^{D213}	L_{C1167}	R^{D168}	R^{D213}
L_{C844}	R^{D17}	R^{D214}	L_{C952}	R^{D50}	R^{D214}	L_{C1060}	R^{D145}	R^{D214}	L_{C1168}	R^{D168}	R^{D214}
L_{C845}	R^{D17}	R^{D215}	L_{C953}	R^{D50}	R^{D215}	L_{C1061}	R^{D145}	R^{D215}	L_{C1169}	R^{D168}	R^{D215}
L_{C846}	R^{D17}	R^{D216}	L_{C954}	R^{D50}	R^{D216}	L_{C1062}	R^{D145}	R^{D216}	L_{C1170}	R^{D168}	R^{D216}
L_{C847}	R^{D17}	R^{D217}	L_{C955}	R^{D50}	R^{D217}	L_{C1063}	R^{D145}	R^{D217}	L_{C1171}	R^{D168}	R^{D217}
L_{C848}	R^{D17}	R^{D218}	L_{C956}	R^{D50}	R^{D218}	L_{C1064}	R^{D145}	R^{D218}	L_{C1172}	R^{D168}	R^{D218}
L_{C849}	R^{D17}	R^{D219}	L_{C957}	R^{D50}	R^{D219}	L_{C1065}	R^{D145}	R^{D219}	L_{C1173}	R^{D168}	R^{D219}
L_{C850}	R^{D17}	R^{D220}	L_{C958}	R^{D50}	R^{D220}	L_{C1066}	R^{D145}	R^{D220}	L_{C1174}	R^{D168}	R^{D220}
L_{C851}	R^{D17}	R^{D221}	L_{C959}	R^{D50}	R^{D221}	L_{C1067}	R^{D145}	R^{D221}	L_{C1175}	R^{D168}	R^{D221}
L_{C852}	R^{D17}	R^{D222}	L_{C960}	R^{D50}	R^{D222}	L_{C1068}	R^{D145}	R^{D222}	L_{C1176}	R^{D168}	R^{D222}
L_{C853}	R^{D17}	R^{D223}	L_{C961}	R^{D50}	R^{D223}	L_{C1069}	R^{D145}	R^{D223}	L_{C1177}	R^{D168}	R^{D223}
L_{C854}	R^{D17}	R^{D224}	L_{C962}	R^{D50}	R^{D224}	L_{C1070}	R^{D145}	R^{D224}	L_{C1178}	R^{D168}	R^{D224}
L_{C855}	R^{D17}	R^{D225}	L_{C963}	R^{D50}	R^{D225}	L_{C1071}	R^{D145}	R^{D225}	L_{C1179}	R^{D168}	R^{D225}
L_{C856}	R^{D17}	R^{D226}	L_{C964}	R^{D50}	R^{D226}	L_{C1072}	R^{D145}	R^{D226}	L_{C1180}	R^{D168}	R^{D226}
L_{C857}	R^{D17}	R^{D227}	L_{C965}	R^{D50}	R^{D227}	L_{C1073}	R^{D145}	R^{D227}	L_{C1181}	R^{D168}	R^{D227}
L_{C858}	R^{D17}	R^{D228}	L_{C966}	R^{D50}	R^{D228}	L_{C1074}	R^{D145}	R^{D228}	L_{C1182}	R^{D168}	R^{D228}
L_{C859}	R^{D17}	R^{D229}	L_{C967}	R^{D50}	R^{D229}	L_{C1075}	R^{D145}	R^{D229}	L_{C1183}	R^{D168}	R^{D229}
L_{C860}	R^{D17}	R^{D230}	L_{C968}	R^{D50}	R^{D230}	L_{C1076}	R^{D145}	R^{D230}	L_{C1184}	R^{D168}	R^{D230}
L_{C861}	R^{D17}	R^{D231}	L_{C969}	R^{D50}	R^{D231}	L_{C1077}	R^{D145}	R^{D231}	L_{C1185}	R^{D168}	R^{D231}
L_{C862}	R^{D17}	R^{D232}	L_{C970}	R^{D50}	R^{D232}	L_{C1078}	R^{D145}	R^{D232}	L_{C1186}	R^{D168}	R^{D232}
L_{C863}	R^{D17}	R^{D233}	L_{C971}	R^{D50}	R^{D233}	L_{C1079}	R^{D145}	R^{D233}	L_{C1187}	R^{D168}	R^{D233}
L_{C864}	R^{D17}	R^{D234}	L_{C972}	R^{D50}	R^{D234}	L_{C1080}	R^{D145}	R^{D234}	L_{C1188}	R^{D168}	R^{D234}
L_{C865}	R^{D17}	R^{D235}	L_{C973}	R^{D50}	R^{D235}	L_{C1081}	R^{D145}	R^{D235}	L_{C1189}	R^{D168}	R^{D235}
L_{C866}	R^{D17}	R^{D236}	L_{C974}	R^{D50}	R^{D236}	L_{C1082}	R^{D145}	R^{D236}	L_{C1190}	R^{D168}	R^{D236}
L_{C867}	R^{D17}	R^{D237}	L_{C975}	R^{D50}	R^{D237}	L_{C1083}	R^{D145}	R^{D237}	L_{C1191}	R^{D168}	R^{D237}
L_{C868}	R^{D17}	R^{D238}	L_{C976}	R^{D50}	R^{D238}	L_{C1084}	R^{D145}	R^{D238}	L_{C1192}	R^{D168}	R^{D238}
L_{C869}	R^{D17}	R^{D239}	L_{C977}	R^{D50}	R^{D239}	L_{C1085}	R^{D145}	R^{D239}	L_{C1193}	R^{D168}	R^{D239}
L_{C870}	R^{D17}	R^{D240}	L_{C978}	R^{D50}	R^{D240}	L_{C1086}	R^{D145}	R^{D240}	L_{C1194}	R^{D168}	R^{D240}
L_{C871}	R^{D17}	R^{D241}	L_{C979}	R^{D50}	R^{D241}	L_{C1087}	R^{D145}	R^{D241}	L_{C1195}	R^{D168}	R^{D241}
L_{C872}	R^{D17}	R^{D242}	L_{C980}	R^{D50}	R^{D242}	L_{C1088}	R^{D145}	R^{D242}	L_{C1196}	R^{D168}	R^{D242}
L_{C873}	R^{D17}	R^{D243}	L_{C981}	R^{D50}	R^{D243}	L_{C1089}	R^{D145}	R^{D243}	L_{C1197}	R^{D168}	R^{D243}
L_{C874}	R^{D17}	R^{D244}	L_{C982}	R^{D50}	R^{D244}	L_{C1090}	R^{D145}	R^{D244}	L_{C1198}	R^{D168}	R^{D244}
L_{C875}	R^{D17}	R^{D245}	L_{C983}	R^{D50}	R^{D245}	L_{C1091}	R^{D145}	R^{D245}	L_{C1199}	R^{D168}	R^{D245}
L_{C876}	R^{D17}	R^{D246}	L_{C984}	R^{D50}	R^{D246}	L_{C1092}	R^{D145}	R^{D246}	L_{C1200}	R^{D168}	R^{D246}
L_{C1201}	R^{D10}	R^{D193}	L_{C1255}	R^{D55}	R^{D193}	L_{C1309}	R^{D37}	R^{D193}	L_{C1363}	R^{D143}	R^{D193}
L_{C1202}	R^{D10}	R^{D194}	L_{C1256}	R^{D55}	R^{D194}	L_{C1310}	R^{D37}	R^{D194}	L_{C1364}	R^{D143}	R^{D194}
L_{C1203}	R^{D10}	R^{D195}	L_{C1257}	R^{D55}	R^{D195}	L_{C1311}	R^{D37}	R^{D195}	L_{C1365}	R^{D143}	R^{D195}
L_{C1204}	R^{D10}	R^{D196}	L_{C1258}	R^{D55}	R^{D196}	L_{C1312}	R^{D37}	R^{D196}	L_{C1366}	R^{D143}	R^{D196}
L_{C1205}	R^{D10}	R^{D197}	L_{C1259}	R^{D55}	R^{D197}	L_{C1313}	R^{D37}	R^{D197}	L_{C1367}	R^{D143}	R^{D197}
L_{C1206}	R^{D10}	R^{D198}	L_{C1260}	R^{D55}	R^{D198}	L_{C1314}	R^{D37}	R^{D198}	L_{C1368}	R^{D143}	R^{D198}
L_{C1207}	R^{D10}	R^{D199}	L_{C1261}	R^{D55}	R^{D199}	L_{C1315}	R^{D37}	R^{D199}	L_{C1369}	R^{D143}	R^{D199}
L_{C1208}	R^{D10}	R^{D200}	L_{C1262}	R^{D55}	R^{D200}	L_{C1316}	R^{D37}	R^{D200}	L_{C1370}	R^{D143}	R^{D200}
L_{C1209}	R^{D10}	R^{D201}	L_{C1263}	R^{D55}	R^{D201}	L_{C1317}	R^{D37}	R^{D201}	L_{C1371}	R^{D143}	R^{D201}
L_{C1210}	R^{D10}	R^{D202}	L_{C1264}	R^{D55}	R^{D202}	L_{C1318}	R^{D37}	R^{D202}	L_{C1372}	R^{D143}	R^{D202}
L_{C1211}	R^{D10}	R^{D203}	L_{C1265}	R^{D55}	R^{D203}	L_{C1319}	R^{D37}	R^{D203}	L_{C1373}	R^{D143}	R^{D203}
L_{C1212}	R^{D10}	R^{D204}	L_{C1266}	R^{D55}	R^{D204}	L_{C1320}	R^{D37}	R^{D204}	L_{C1374}	R^{D143}	R^{D204}
L_{C1213}	R^{D10}	R^{D205}	L_{C1267}	R^{D55}	R^{D205}	L_{C1321}	R^{D37}	R^{D205}	L_{C1375}	R^{D143}	R^{D205}
L_{C1214}	R^{D10}	R^{D206}	L_{C1268}	R^{D55}	R^{D206}	L_{C1322}	R^{D37}	R^{D206}	L_{C1376}	R^{D143}	R^{D206}

-continued

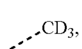
L_{Cj}	R^{201}	R^{202}	L_{Cj}	R^{201}	R^{202}	L_{Cj}	R^{201}	R^{202}	L_{Cj}	R^{201}	R^{202}
L_{C1215}	R^{D10}	R^{D207}	L_{C1269}	R^{D55}	R^{D207}	L_{C1323}	R^{D37}	R^{D207}	L_{C1377}	R^{D143}	R^{D207}
L_{C1216}	R^{D10}	R^{D208}	L_{C1270}	R^{D55}	R^{D208}	L_{C1324}	R^{D37}	R^{D208}	L_{C1378}	R^{D143}	R^{D208}
L_{C1217}	R^{D10}	R^{D209}	L_{C1271}	R^{D55}	R^{D209}	L_{C1325}	R^{D37}	R^{D209}	L_{C1379}	R^{D143}	R^{D209}
L_{C1218}	R^{D10}	R^{D210}	L_{C1272}	R^{D55}	R^{D210}	L_{C1326}	R^{D37}	R^{D210}	L_{C1380}	R^{D143}	R^{D210}
L_{C1219}	R^{D10}	R^{D211}	L_{C1273}	R^{D55}	R^{D211}	L_{C1327}	R^{D37}	R^{D211}	L_{C1381}	R^{D143}	R^{D211}
L_{C1220}	R^{D10}	R^{D212}	L_{C1274}	R^{D55}	R^{D212}	L_{C1328}	R^{D37}	R^{D212}	L_{C1382}	R^{D143}	R^{D212}
L_{C1221}	R^{D10}	R^{D213}	L_{C1275}	R^{D55}	R^{D213}	L_{C1329}	R^{D37}	R^{D213}	L_{C1383}	R^{D143}	R^{D213}
L_{C1222}	R^{D10}	R^{D214}	L_{C1276}	R^{D55}	R^{D214}	L_{C1330}	R^{D37}	R^{D214}	L_{C1384}	R^{D143}	R^{D214}
L_{C1223}	R^{D10}	R^{D215}	L_{C1277}	R^{D55}	R^{D215}	L_{C1331}	R^{D37}	R^{D215}	L_{C1385}	R^{D143}	R^{D215}
L_{C1224}	R^{D10}	R^{D216}	L_{C1278}	R^{D55}	R^{D216}	L_{C1332}	R^{D37}	R^{D216}	L_{C1386}	R^{D143}	R^{D216}
L_{C1225}	R^{D10}	R^{D217}	L_{C1279}	R^{D55}	R^{D217}	L_{C1333}	R^{D37}	R^{D217}	L_{C1387}	R^{D143}	R^{D217}
L_{C1226}	R^{D10}	R^{D218}	L_{C1280}	R^{D55}	R^{D218}	L_{C1334}	R^{D37}	R^{D218}	L_{C1388}	R^{D143}	R^{D218}
L_{C1227}	R^{D10}	R^{D219}	L_{C1281}	R^{D55}	R^{D219}	L_{C1335}	R^{D37}	R^{D219}	L_{C1389}	R^{D143}	R^{D219}
L_{C1228}	R^{D10}	R^{D220}	L_{C1282}	R^{D55}	R^{D220}	L_{C1336}	R^{D37}	R^{D220}	L_{C1390}	R^{D143}	R^{D220}
L_{C1229}	R^{D10}	R^{D221}	L_{C1283}	R^{D55}	R^{D221}	L_{C1337}	R^{D37}	R^{D221}	L_{C1391}	R^{D143}	R^{D221}
L_{C1230}	R^{D10}	R^{D222}	L_{C1284}	R^{D55}	R^{D222}	L_{C1338}	R^{D37}	R^{D222}	L_{C1392}	R^{D143}	R^{D222}
L_{C1231}	R^{D10}	R^{D223}	L_{C1285}	R^{D55}	R^{D223}	L_{C1339}	R^{D37}	R^{D223}	L_{C1393}	R^{D143}	R^{D223}
L_{C1232}	R^{D10}	R^{D224}	L_{C1286}	R^{D55}	R^{D224}	L_{C1340}	R^{D37}	R^{D224}	L_{C1394}	R^{D143}	R^{D224}
L_{C1233}	R^{D10}	R^{D225}	L_{C1287}	R^{D55}	R^{D225}	L_{C1341}	R^{D37}	R^{D225}	L_{C1395}	R^{D143}	R^{D225}
L_{C1234}	R^{D10}	R^{D226}	L_{C1288}	R^{D55}	R^{D226}	L_{C1342}	R^{D37}	R^{D226}	L_{C1396}	R^{D143}	R^{D226}
L_{C1235}	R^{D10}	R^{D227}	L_{C1289}	R^{D55}	R^{D227}	L_{C1343}	R^{D37}	R^{D227}	L_{C1397}	R^{D143}	R^{D227}
L_{C1236}	R^{D10}	R^{D228}	L_{C1290}	R^{D55}	R^{D228}	L_{C1344}	R^{D37}	R^{D228}	L_{C1398}	R^{D143}	R^{D228}
L_{C1237}	R^{D10}	R^{D229}	L_{C1291}	R^{D55}	R^{D229}	L_{C1345}	R^{D37}	R^{D229}	L_{C1399}	R^{D143}	R^{D229}
L_{C1238}	R^{D10}	R^{D230}	L_{C1292}	R^{D55}	R^{D230}	L_{C1346}	R^{D37}	R^{D230}	L_{C1400}	R^{D143}	R^{D230}
L_{C1239}	R^{D10}	R^{D231}	L_{C1293}	R^{D55}	R^{D231}	L_{C1347}	R^{D37}	R^{D231}	L_{C1401}	R^{D143}	R^{D231}
L_{C1240}	R^{D10}	R^{D232}	L_{C1294}	R^{D55}	R^{D232}	L_{C1348}	R^{D37}	R^{D232}	L_{C1402}	R^{D143}	R^{D232}
L_{C1241}	R^{D10}	R^{D233}	L_{C1295}	R^{D55}	R^{D233}	L_{C1349}	R^{D37}	R^{D233}	L_{C1403}	R^{D143}	R^{D233}
L_{C1242}	R^{D10}	R^{D234}	L_{C1296}	R^{D55}	R^{D234}	L_{C1350}	R^{D37}	R^{D234}	L_{C1404}	R^{D143}	R^{D234}
L_{C1243}	R^{D10}	R^{D235}	L_{C1297}	R^{D55}	R^{D235}	L_{C1351}	R^{D37}	R^{D235}	L_{C1405}	R^{D143}	R^{D235}
L_{C1244}	R^{D10}	R^{D236}	L_{C1298}	R^{D55}	R^{D236}	L_{C1352}	R^{D37}	R^{D236}	L_{C1406}	R^{D143}	R^{D236}
L_{C1245}	R^{D10}	R^{D237}	L_{C1299}	R^{D55}	R^{D237}	L_{C1353}	R^{D37}	R^{D237}	L_{C1407}	R^{D143}	R^{D237}
L_{C1246}	R^{D10}	R^{D238}	L_{C1300}	R^{D55}	R^{D238}	L_{C1354}	R^{D37}	R^{D238}	L_{C1408}	R^{D143}	R^{D238}
L_{C1247}	R^{D10}	R^{D239}	L_{C1301}	R^{D55}	R^{D239}	L_{C1355}	R^{D37}	R^{D239}	L_{C1409}	R^{D143}	R^{D239}
L_{C1248}	R^{D10}	R^{D240}	L_{C1302}	R^{D55}	R^{D240}	L_{C1356}	R^{D37}	R^{D240}	L_{C1410}	R^{D143}	R^{D240}
L_{C1249}	R^{D10}	R^{D241}	L_{C1303}	R^{D55}	R^{D241}	L_{C1357}	R^{D37}	R^{D241}	L_{C1411}	R^{D143}	R^{D241}
L_{C1250}	R^{D10}	R^{D242}	L_{C1304}	R^{D55}	R^{D242}	L_{C1358}	R^{D37}	R^{D242}	L_{C1412}	R^{D143}	R^{D242}
L_{C1251}	R^{D10}	R^{D243}	L_{C1305}	R^{D55}	R^{D243}	L_{C1359}	R^{D37}	R^{D243}	L_{C1413}	R^{D143}	R^{D243}
L_{C1252}	R^{D10}	R^{D244}	L_{C1306}	R^{D55}	R^{D244}	L_{C1360}	R^{D37}	R^{D244}	L_{C1414}	R^{D143}	R^{D244}
L_{C1253}	R^{D10}	R^{D245}	L_{C1307}	R^{D55}	R^{D245}	L_{C1361}	R^{D37}	R^{D245}	L_{C1415}	R^{D143}	R^{D245}
L_{C1254}	R^{D10}	R^{D246}	L_{C1308}	R^{D55}	R^{D246}	L_{C1362}	R^{D37}	R^{D246}	L_{C1416}	R^{D143}	R^{D246}

wherein R^{D1} to R^{D246} have the structures defined in the following LIST 12:

R^{D1}

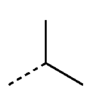


R^{D2}

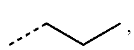


R^{D3}

R^{D4}



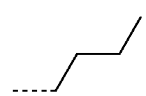
R^{D5}



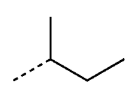
R^{D6}

-continued

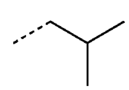
R^{D7}



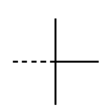
R^{D8}



R^{D9}

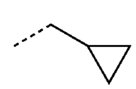


R^{D10}



R^{D11}

R^{D12}



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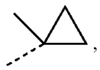
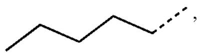
R^{D13}

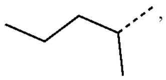
R^{D14}



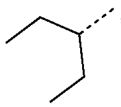
R^{D15}



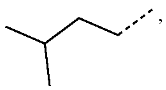
R^{D16}



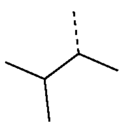
R^{D17}



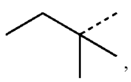
R^{D18}



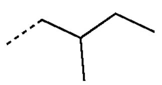
R^{D19}



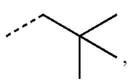
R^{D20}



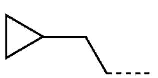
R^{D21}



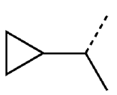
R^{D22}



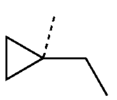
R^{D23}



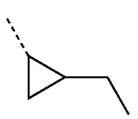
R^{D24}



R^{D25}



R^{D26}



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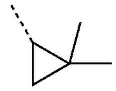
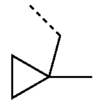
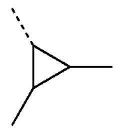
R^{D27}



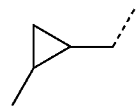
R^{D28}



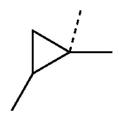
R^{D29}



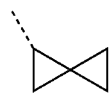
R^{D30}



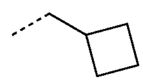
R^{D31}



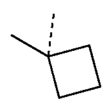
R^{D32}



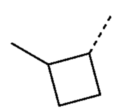
R^{D33}



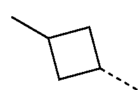
R^{D34}



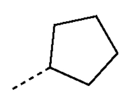
R^{D35}



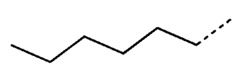
R^{D36}



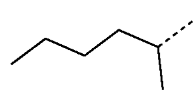
R^{D37}



R^{D38}



R^{D39}



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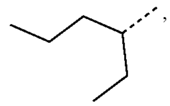
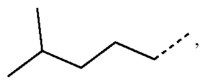
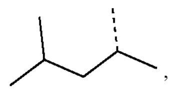
R^{D40}



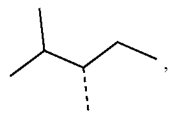
R^{D41}



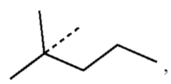
R^{D42}



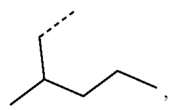
R^{D43}



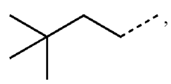
R^{D44}



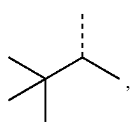
R^{D45}



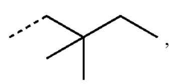
R^{D46}



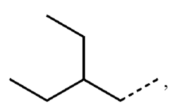
R^{D47}



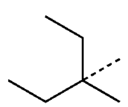
R^{D48}



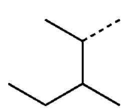
R^{D49}



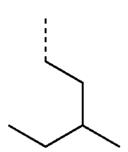
R^{D50}



R^{D51}



R^{D52}



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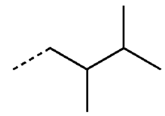
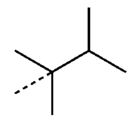
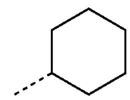
R^{D53}



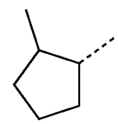
R^{D54}



R^{D55}



R^{D56}



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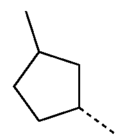

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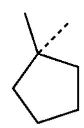

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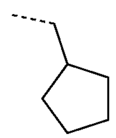

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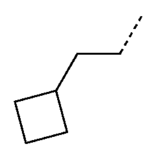

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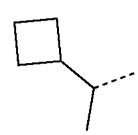

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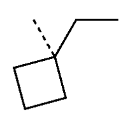

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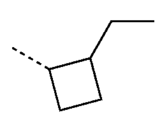

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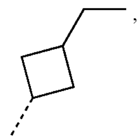
R^{D64}



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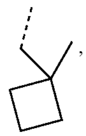

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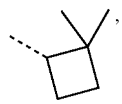

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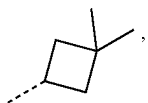

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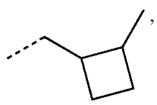

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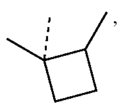

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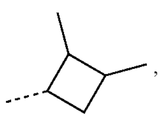

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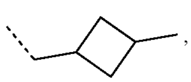

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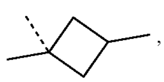

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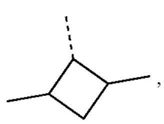
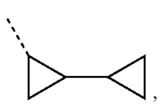


R^{D74}



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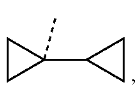

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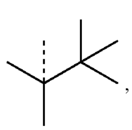

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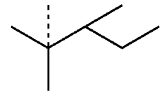
R^{D77}



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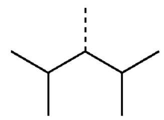

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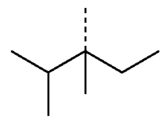

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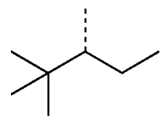

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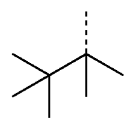

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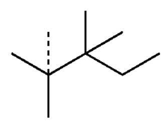

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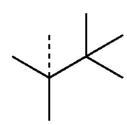

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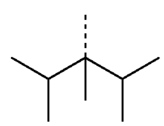

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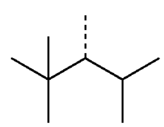

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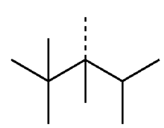

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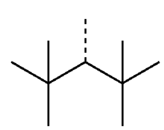

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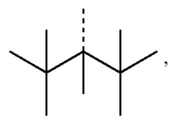
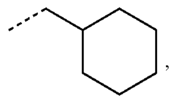
R^{D88}



R^{D89}



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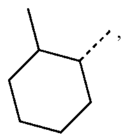

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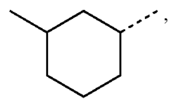

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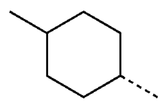

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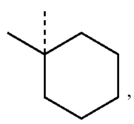
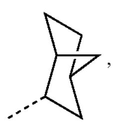
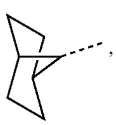


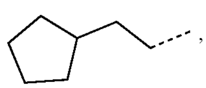
R^{D94}



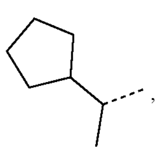
R^{D95}



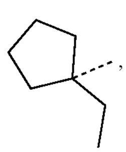
R^{D96}



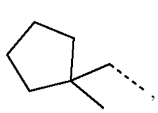
R^{D97}



R^{D98}



R^{D99}



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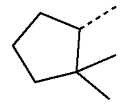
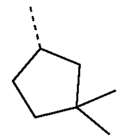
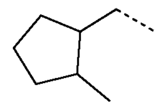
R^{D100}



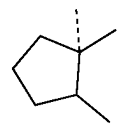
R^{D101}



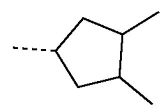
R^{D102}



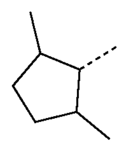
R^{D103}



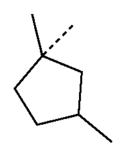
R^{D104}



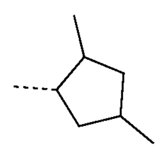
R^{D105}



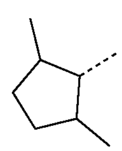
R^{D106}



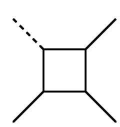
R^{D107}



R^{D108}



R^{D109}



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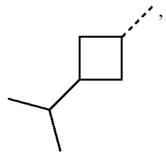
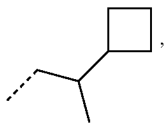
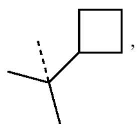
R^{D110}



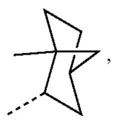
R^{D111}



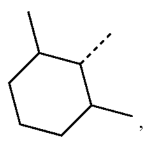
R^{D112}



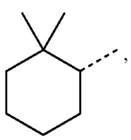
R^{D113}



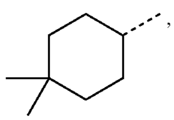
R^{D114}



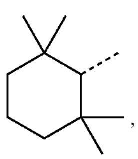
R^{D115}



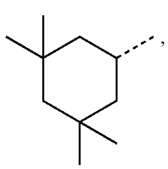
R^{D116}



R^{D117}



R^{D118}



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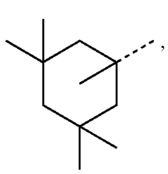

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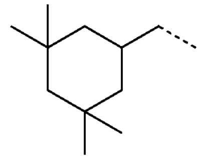
R^{D120}



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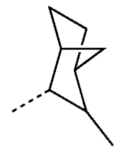
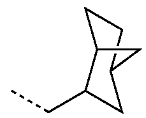
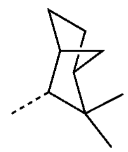


R^{D122}



R^{D123}



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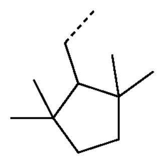
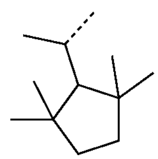
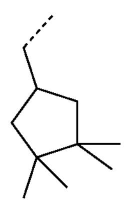


R^{D125}



R^{D126}



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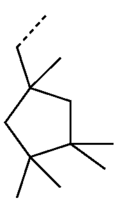

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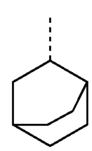

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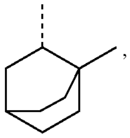

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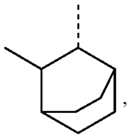

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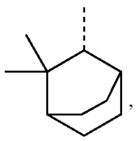

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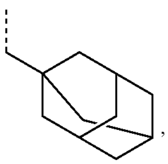

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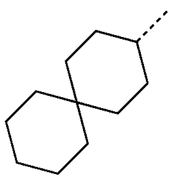

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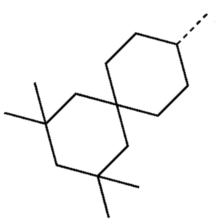

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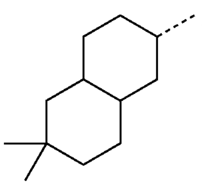

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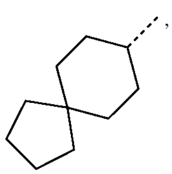

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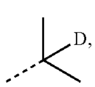

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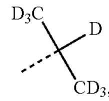

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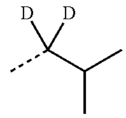

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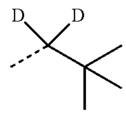

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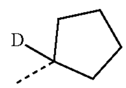

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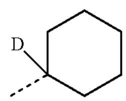

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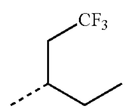

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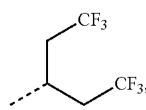

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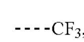

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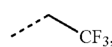

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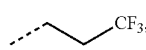

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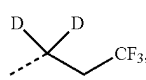

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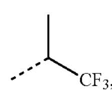

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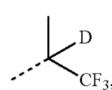

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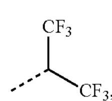

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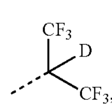

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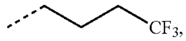

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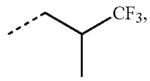

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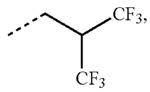

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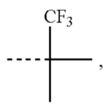

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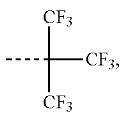

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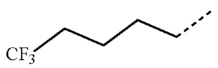

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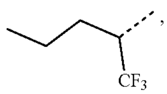

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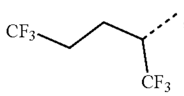

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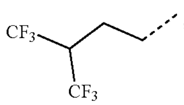

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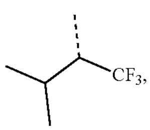

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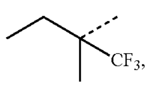

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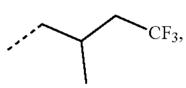

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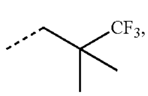

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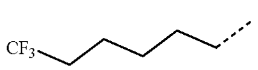

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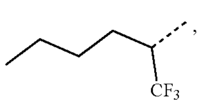

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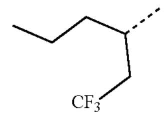

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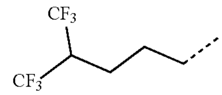

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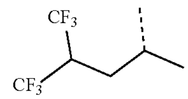

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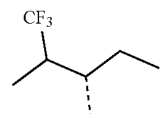

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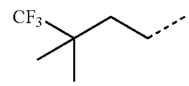

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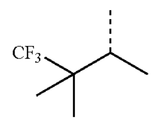

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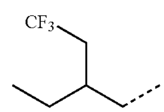

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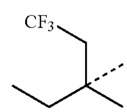

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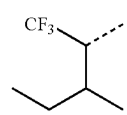

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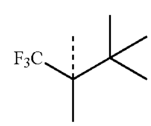

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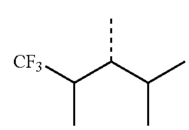

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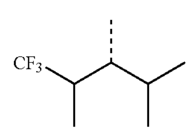

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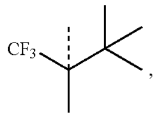

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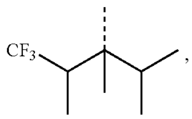

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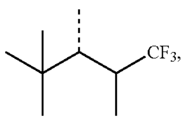

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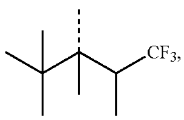

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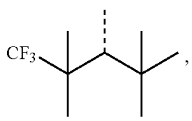

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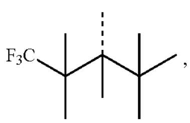

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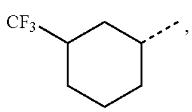

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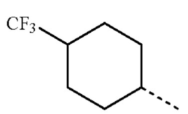

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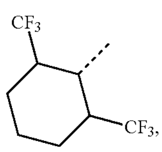

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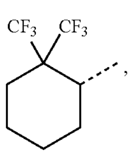

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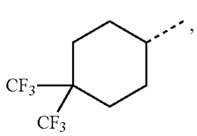

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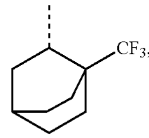

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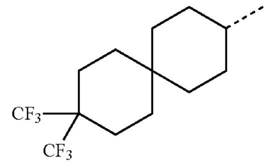

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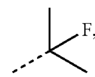

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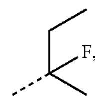

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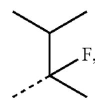

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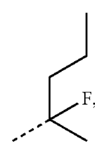

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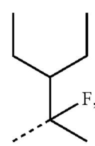

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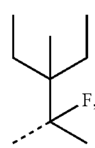

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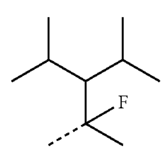

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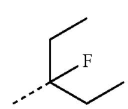

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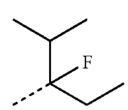

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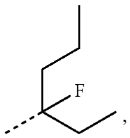

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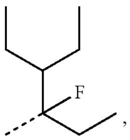

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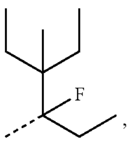

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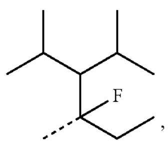

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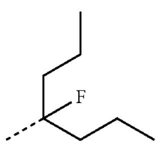

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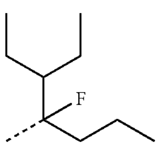

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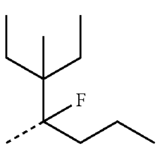

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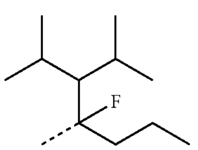

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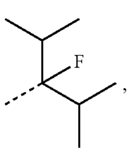

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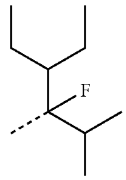

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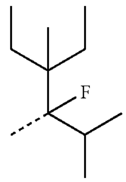

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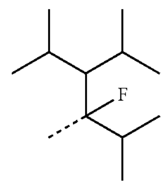

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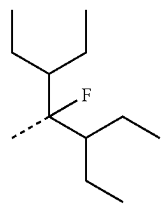

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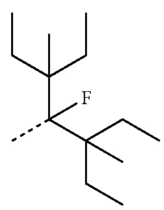

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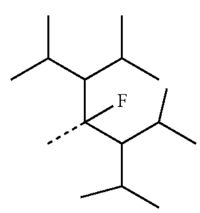

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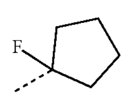

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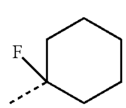

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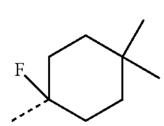

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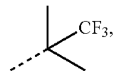

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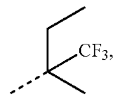

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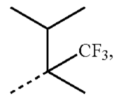

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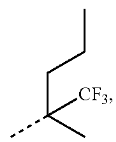

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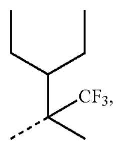

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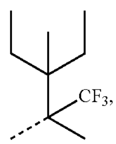

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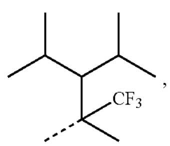

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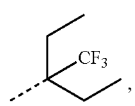

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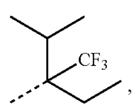

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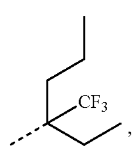

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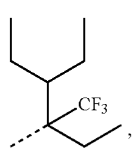

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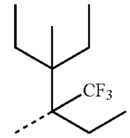

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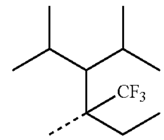

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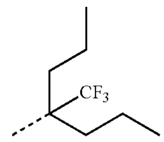
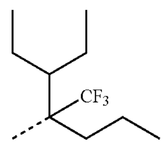


R^{D234}



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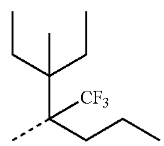

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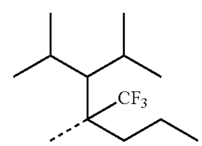

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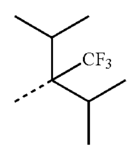

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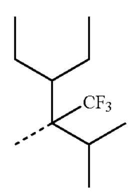

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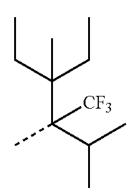

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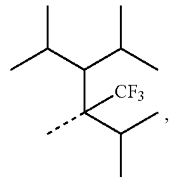

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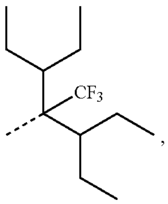

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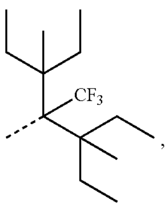

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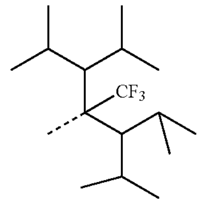
R^{D243}



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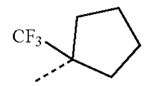

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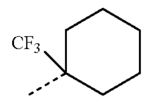

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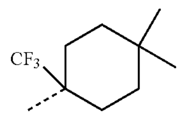
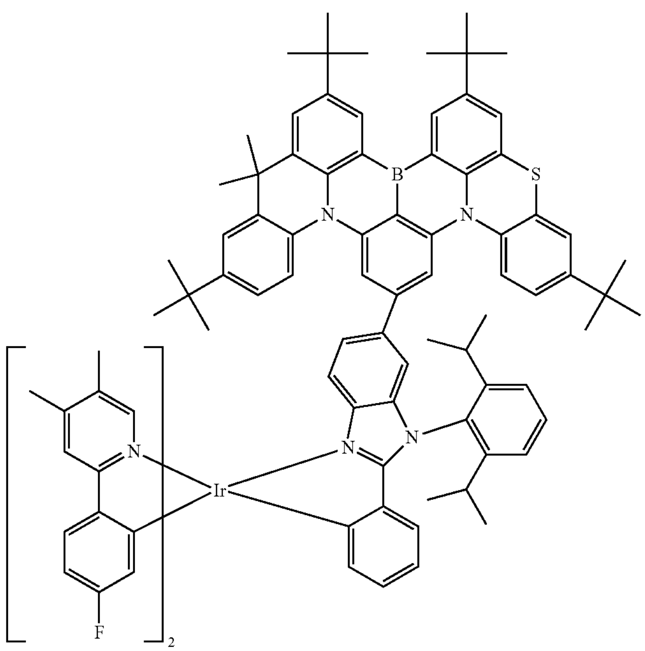
R^{D246}



15. The compound of claim **1**, wherein the compound is selected from the group consisting of:



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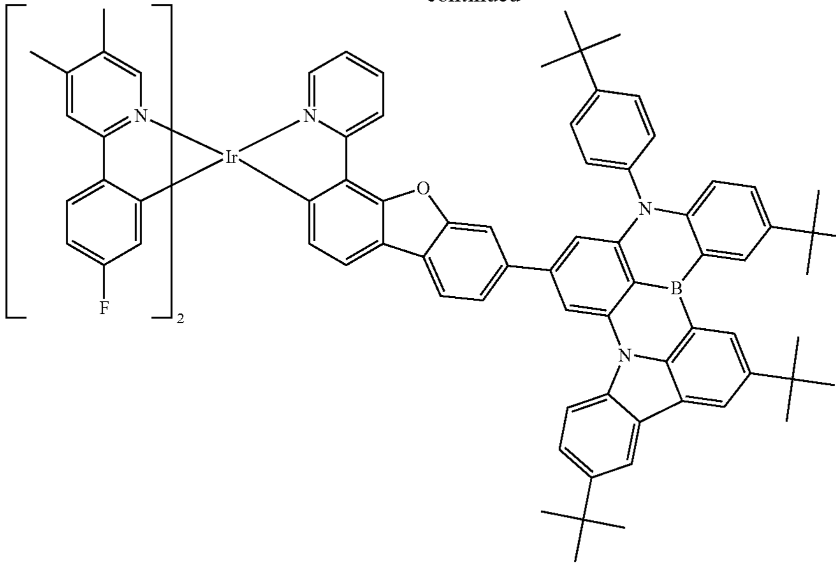
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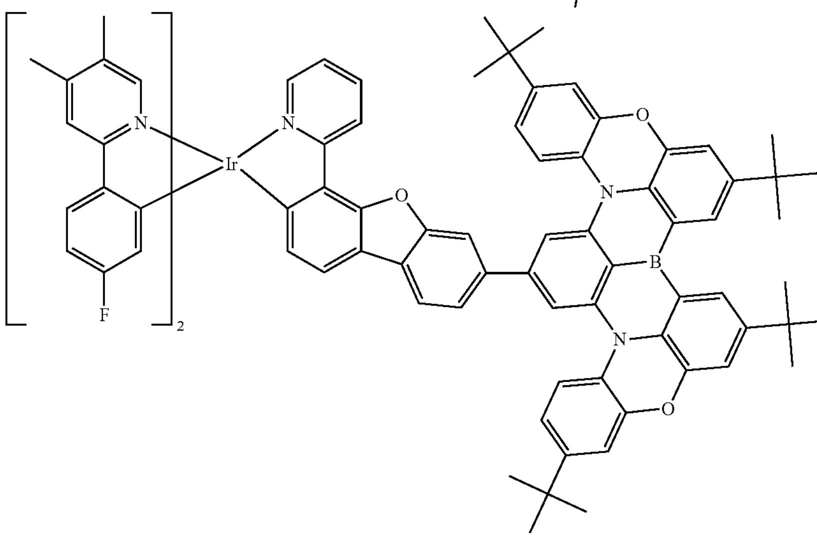
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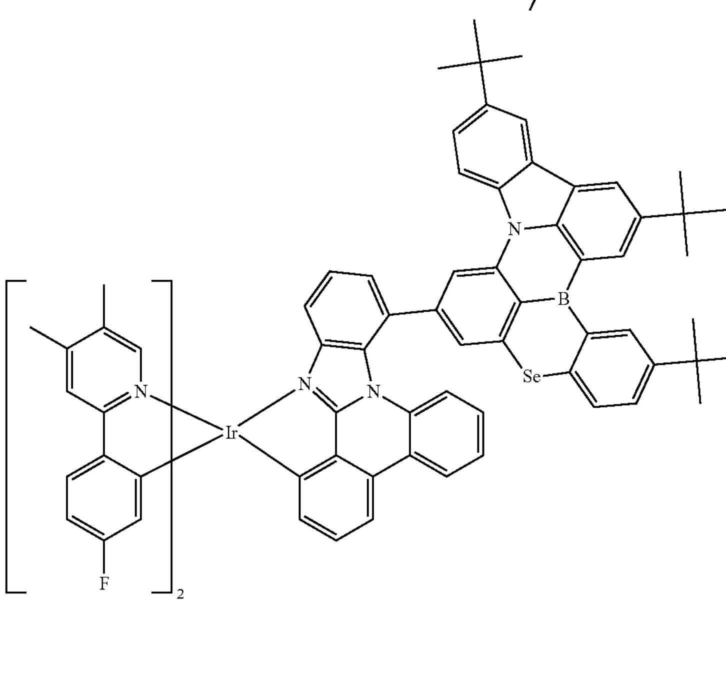
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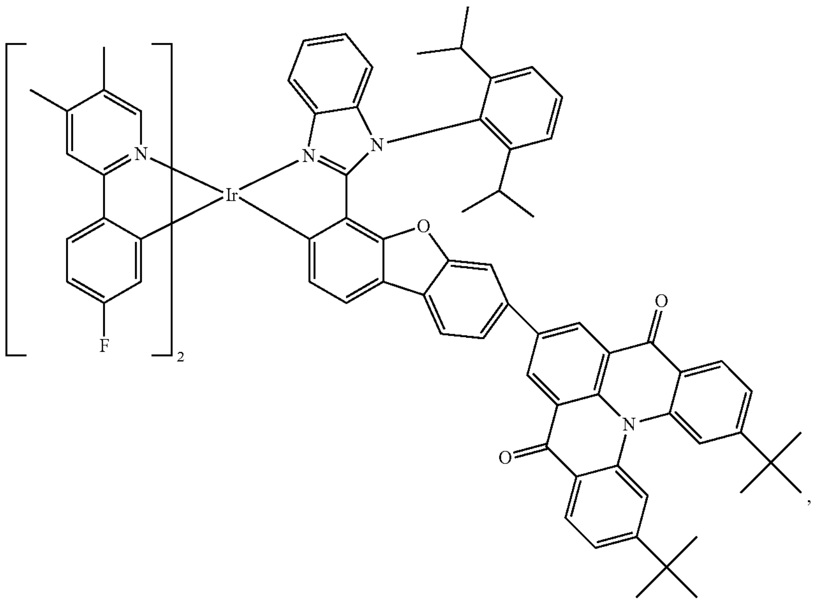

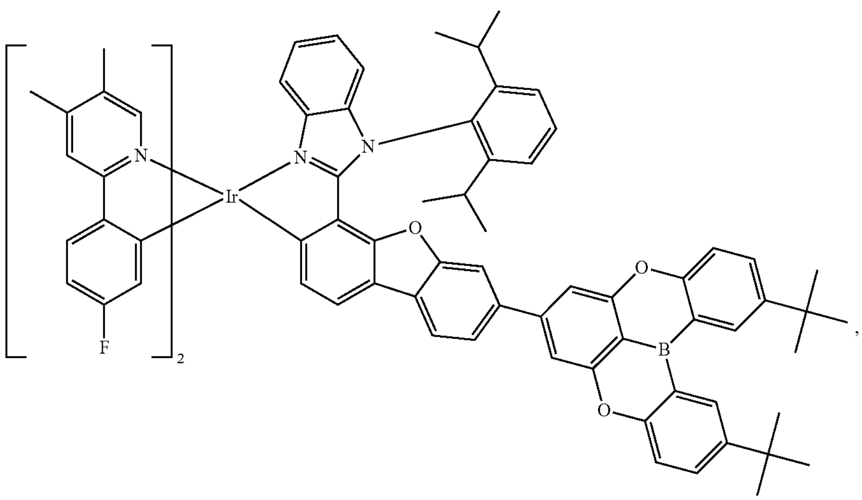

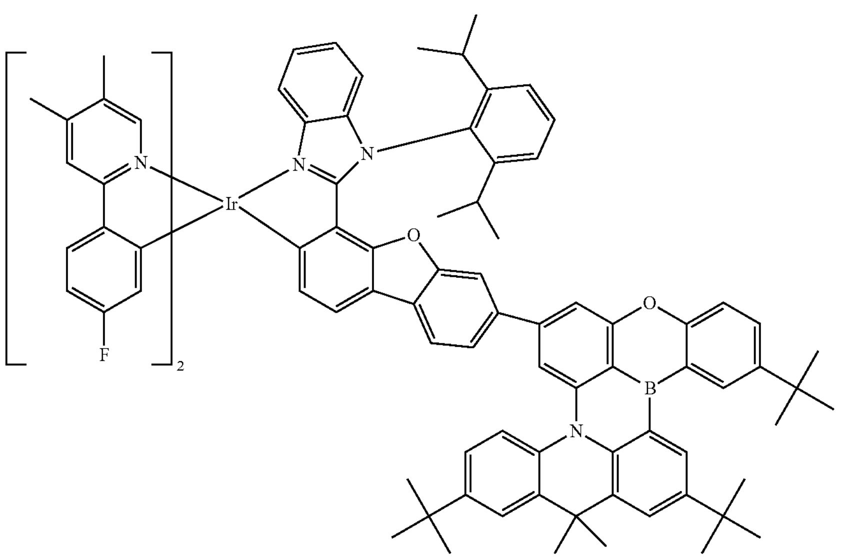

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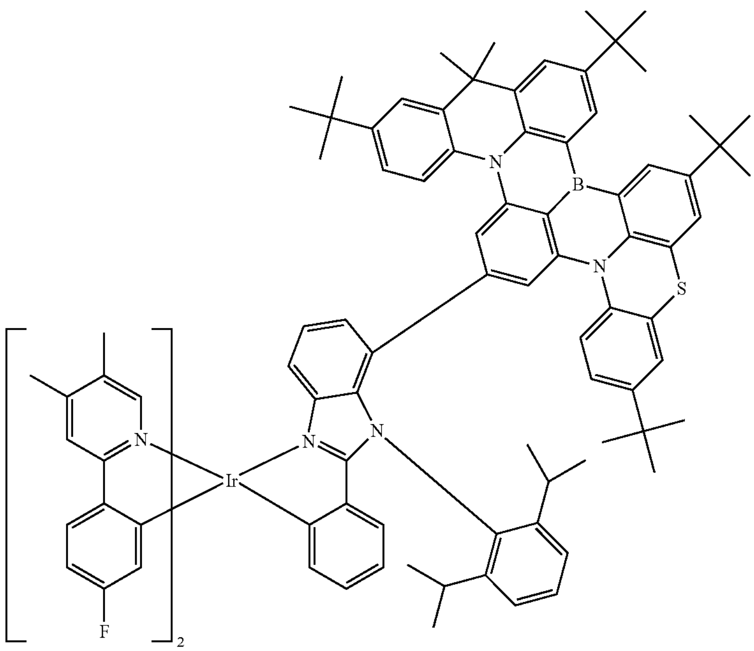

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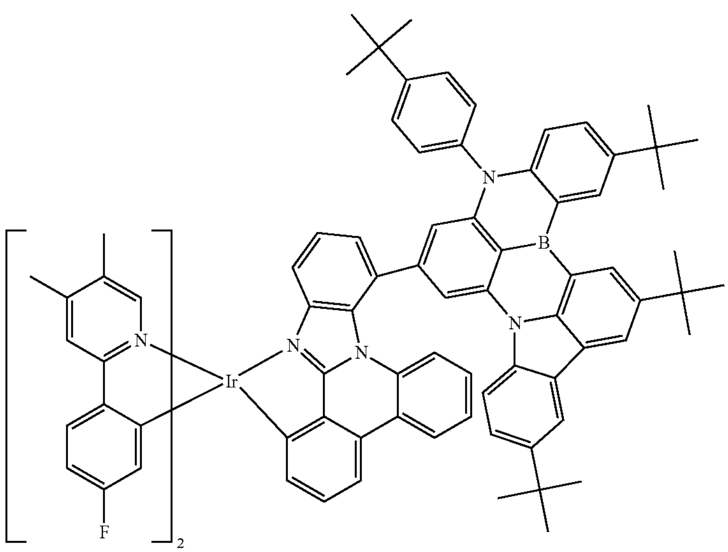

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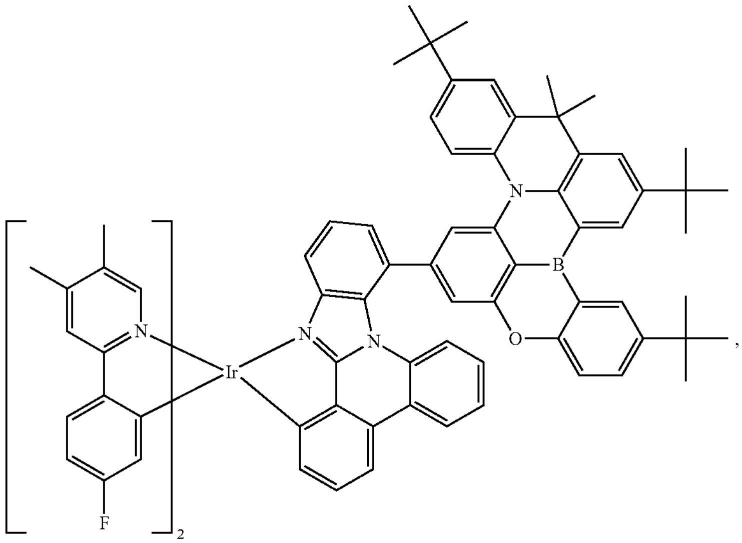

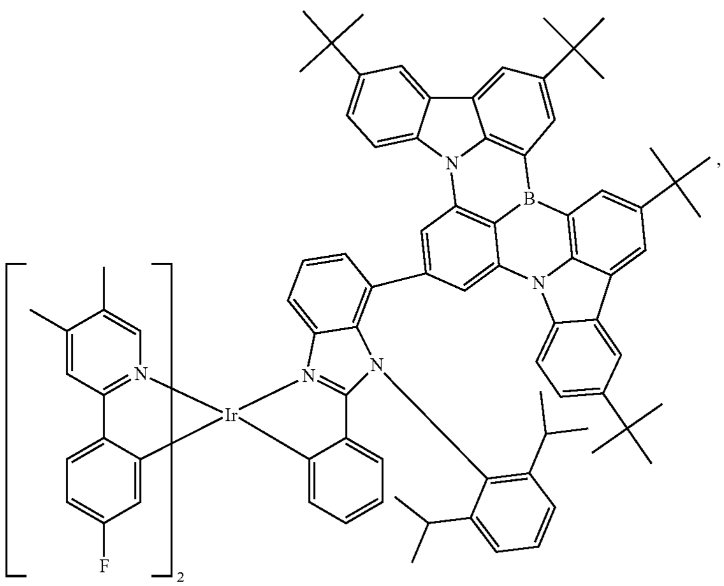

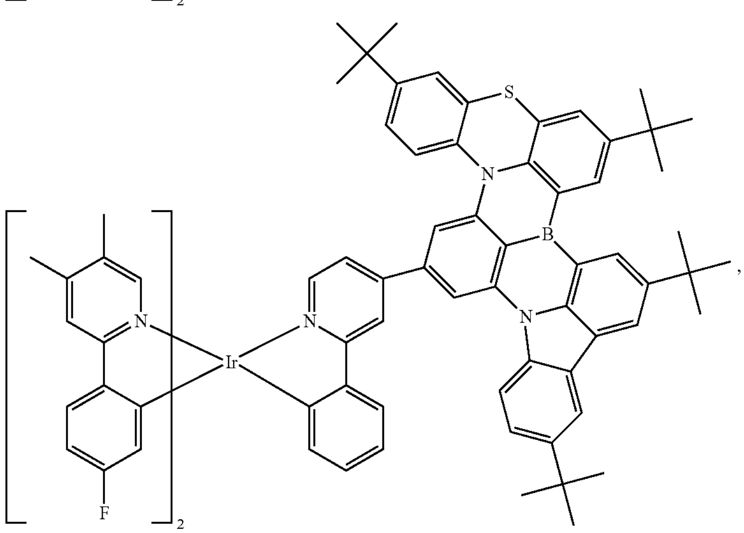

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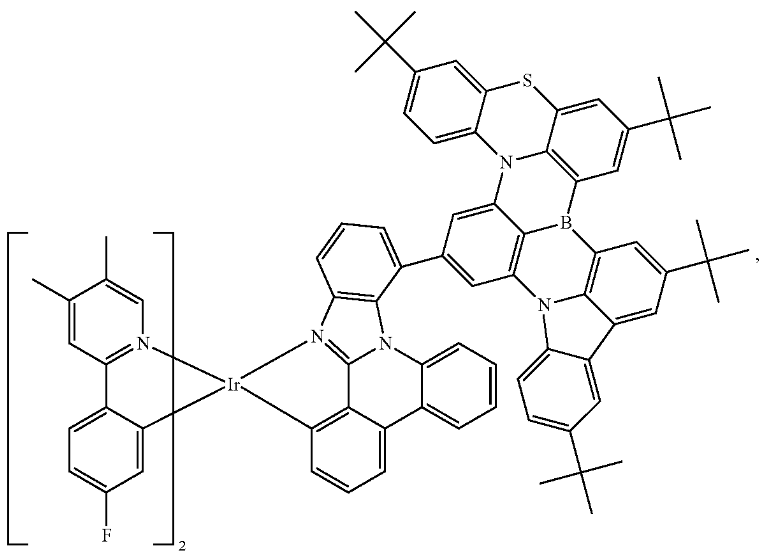

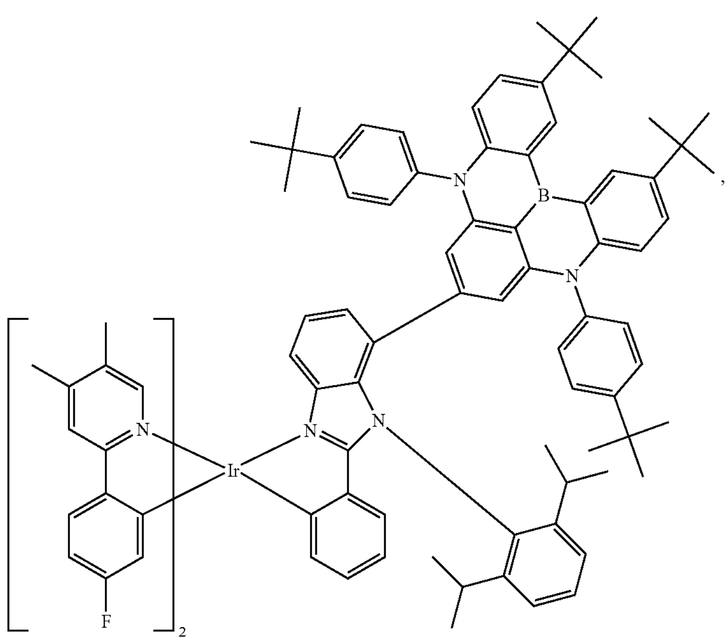

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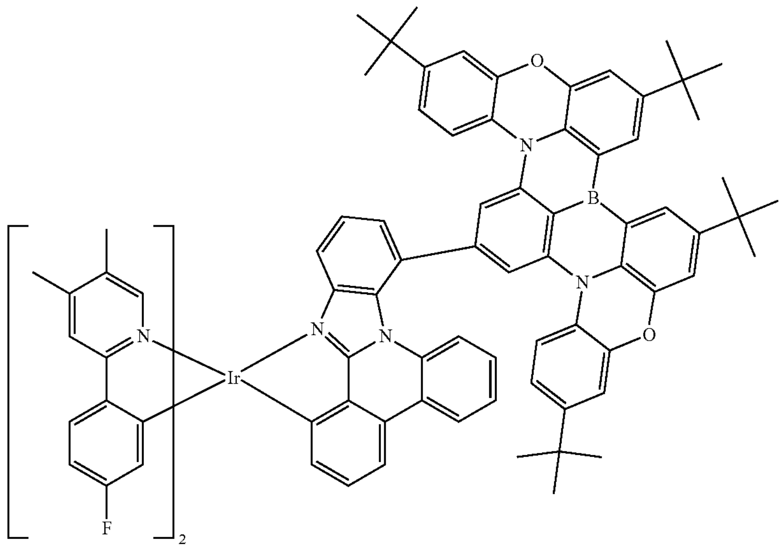
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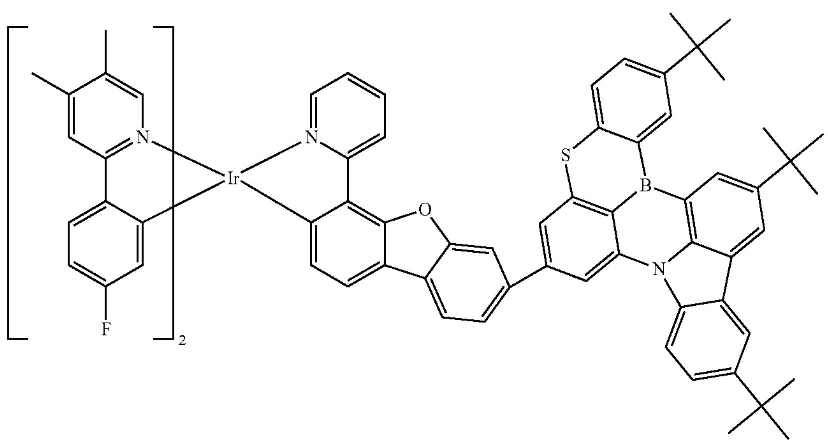
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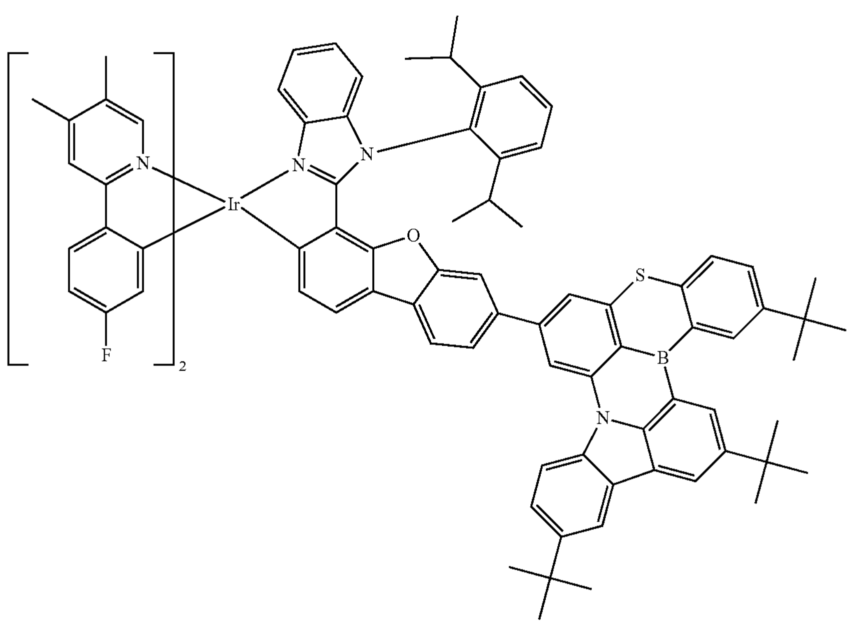
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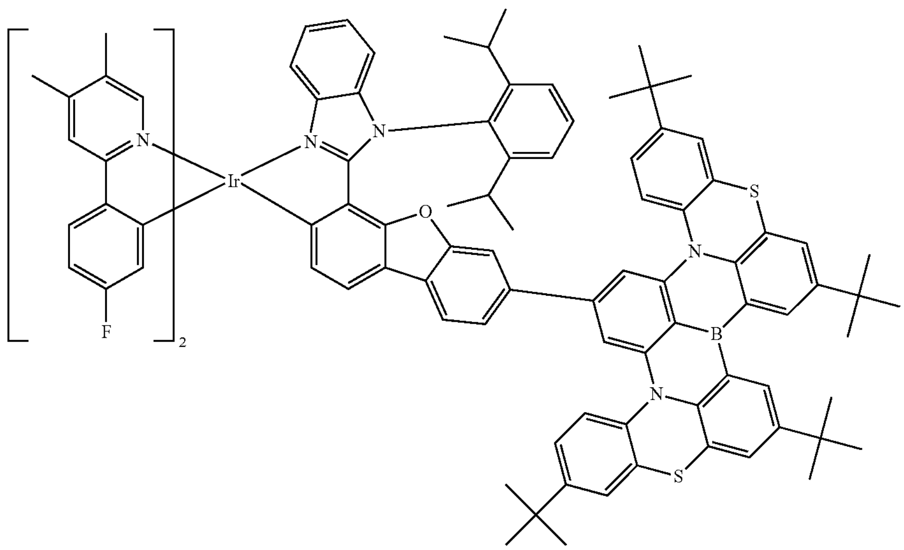

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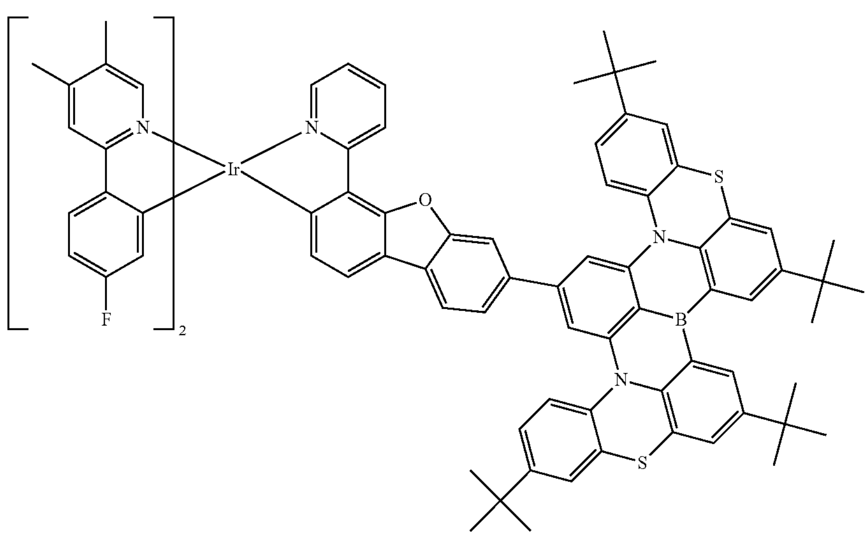

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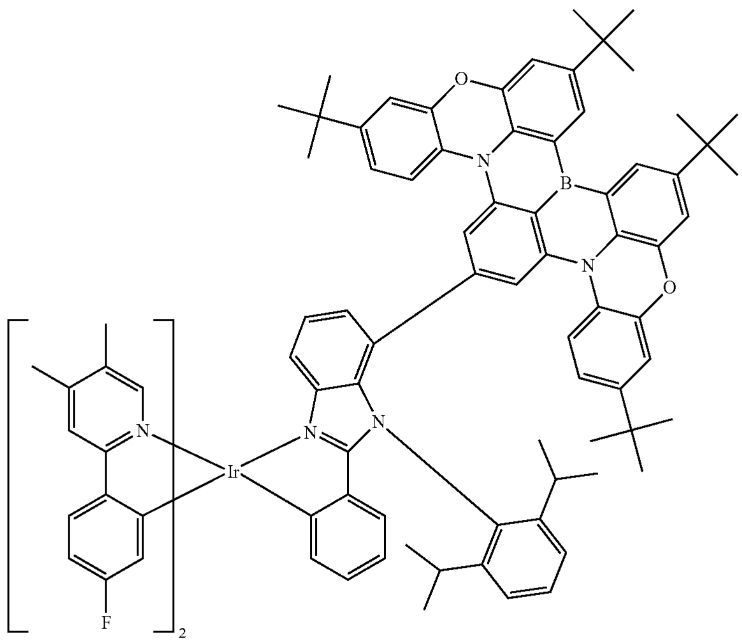

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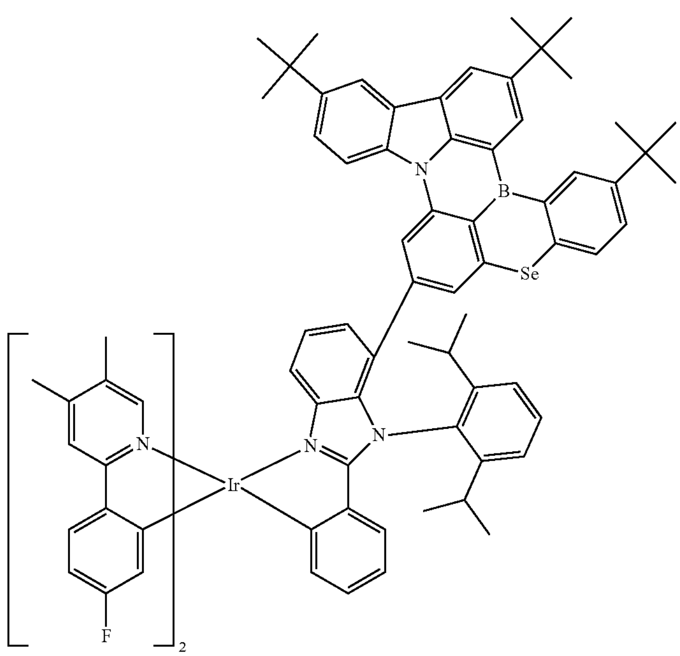

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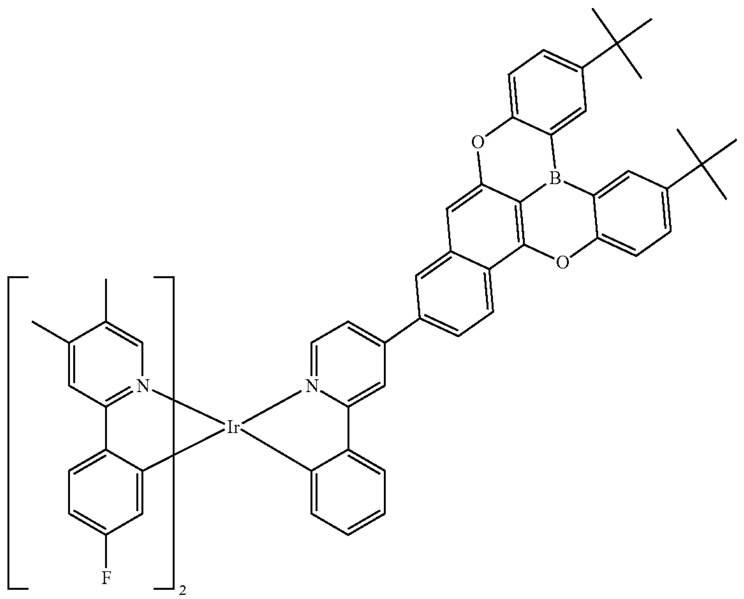

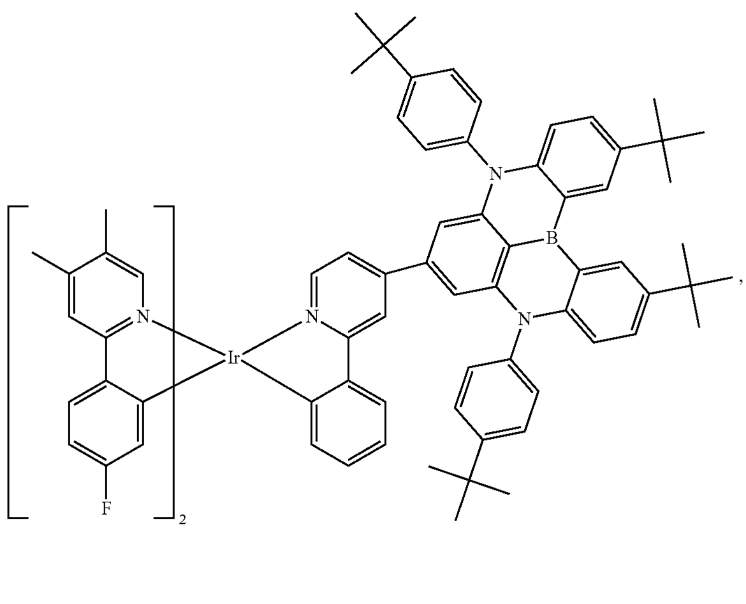

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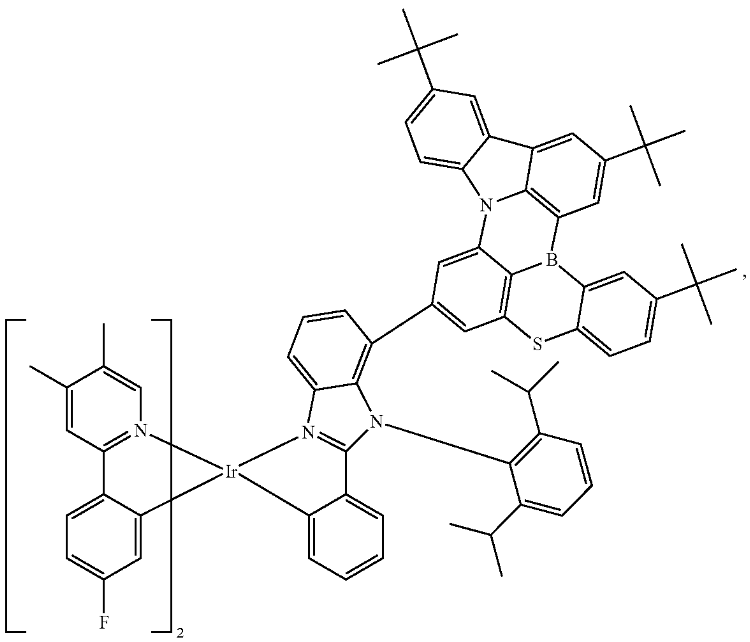

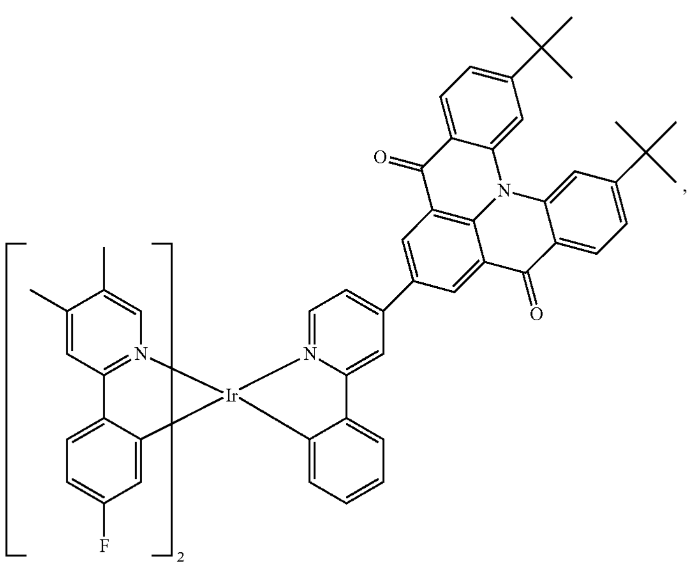

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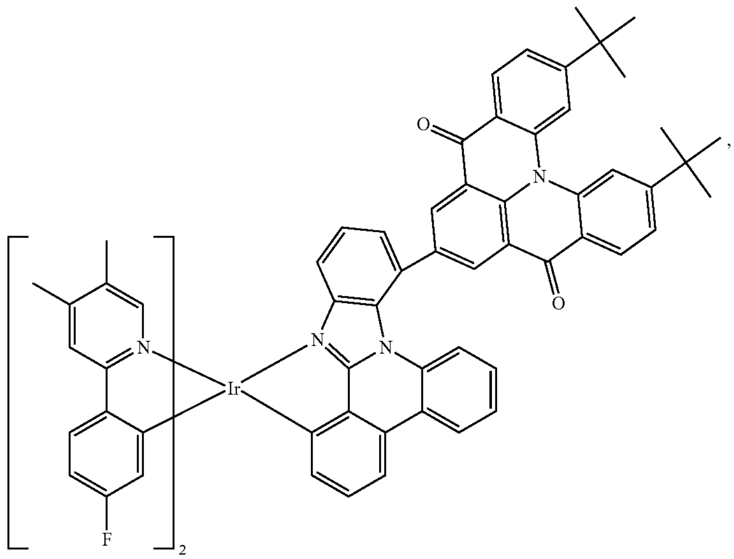

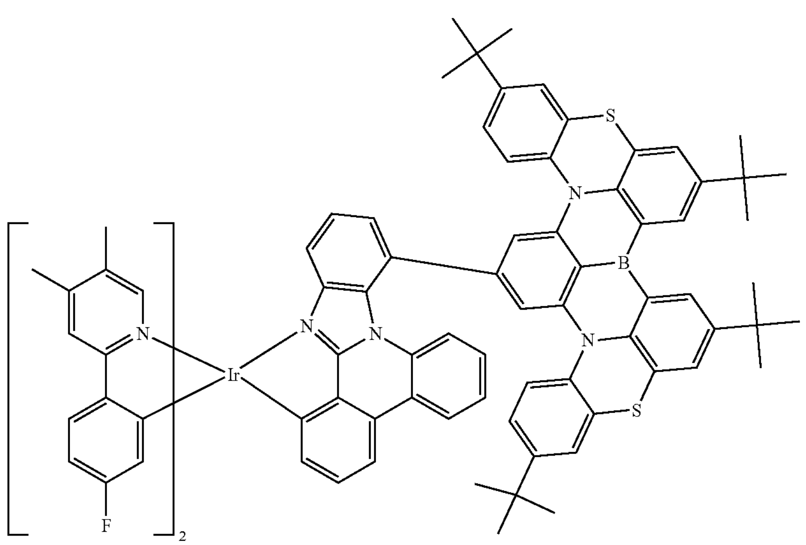

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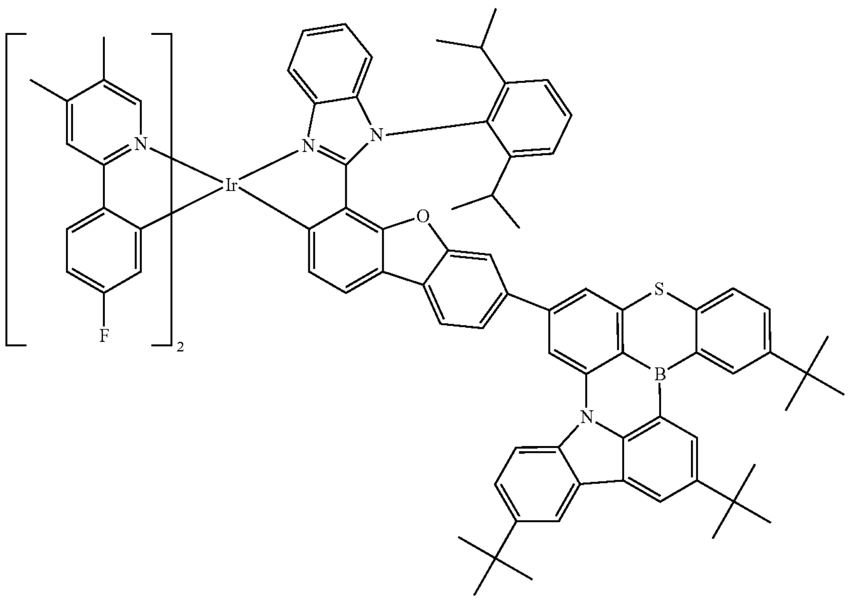

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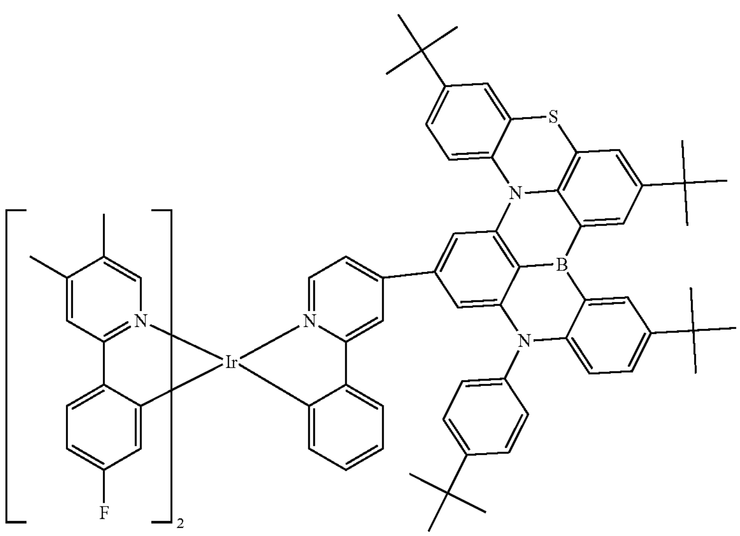

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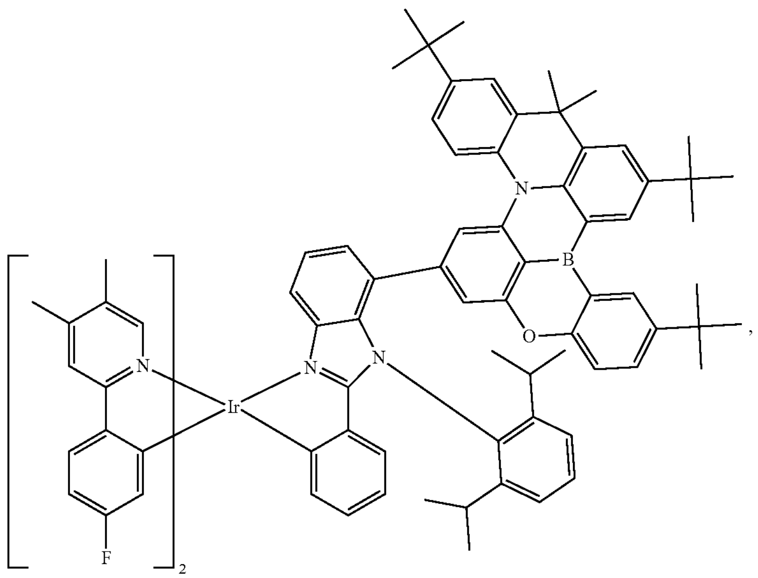

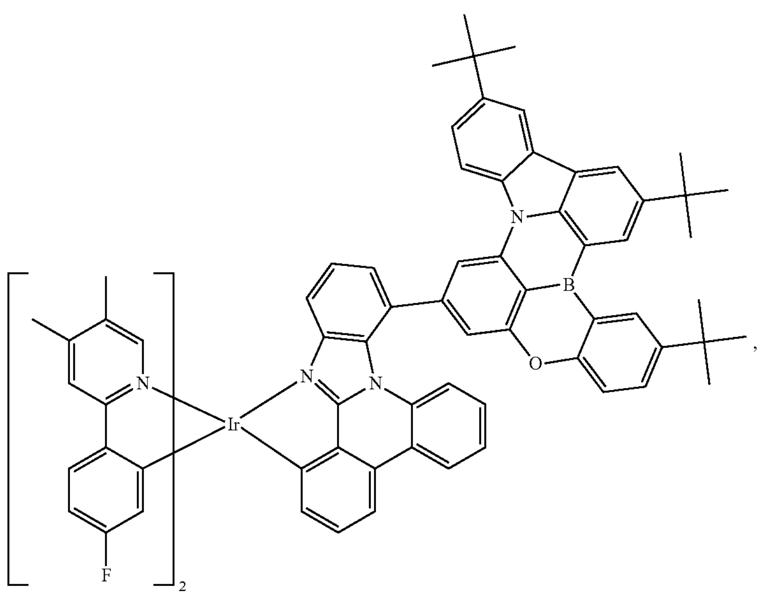

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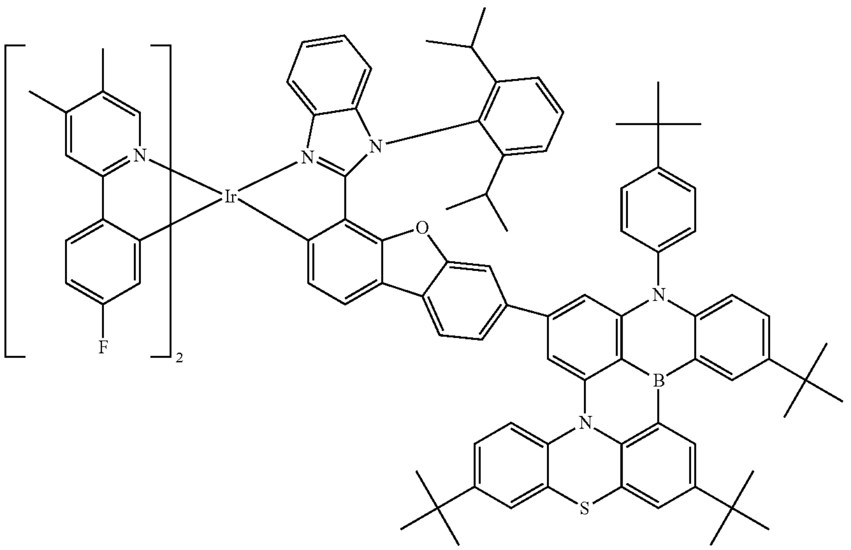

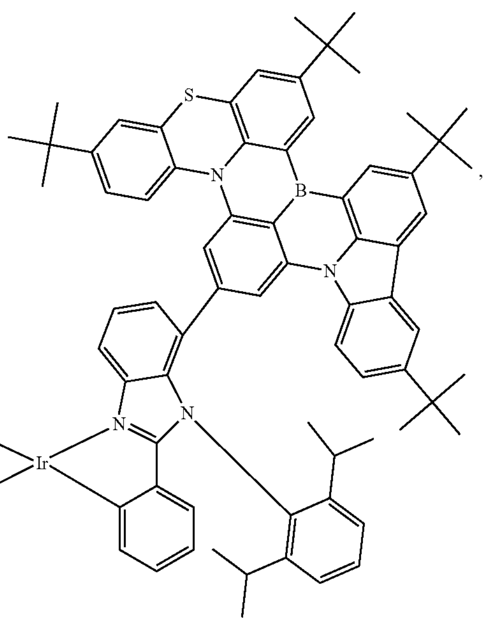

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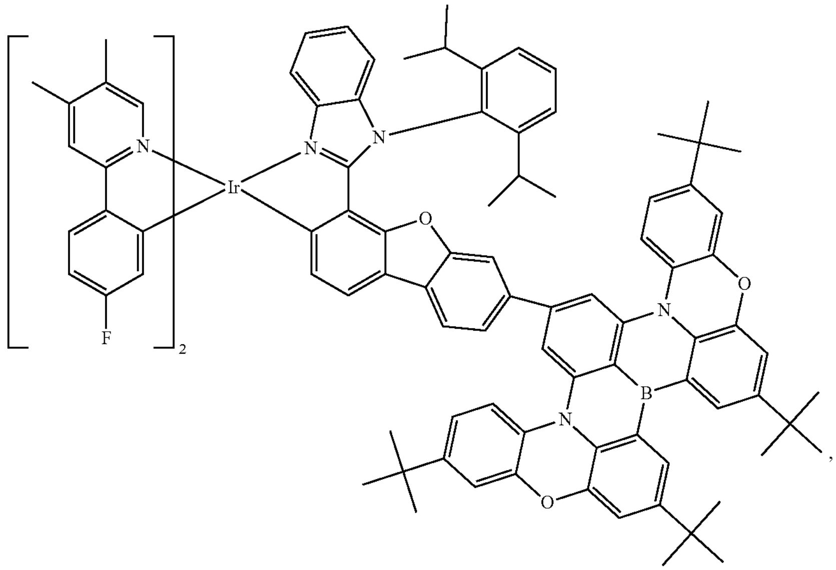

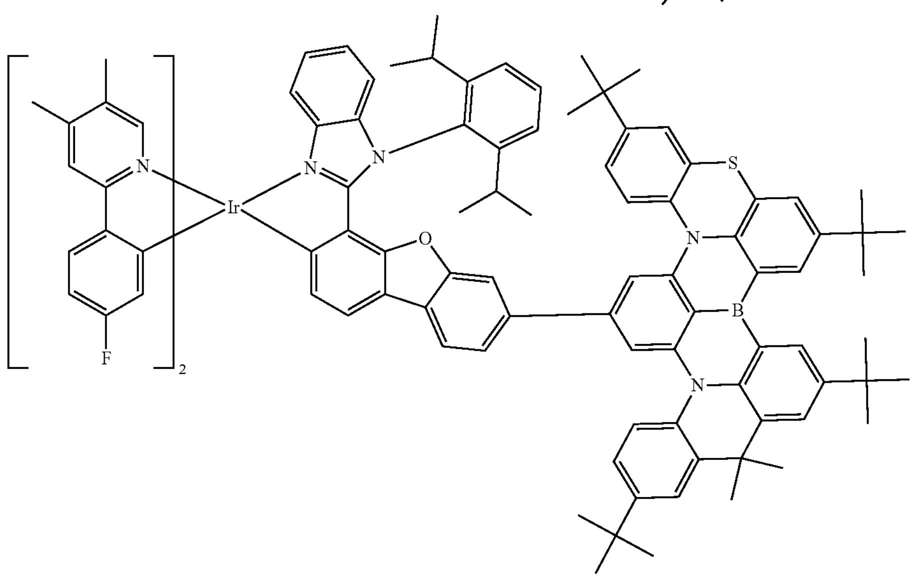

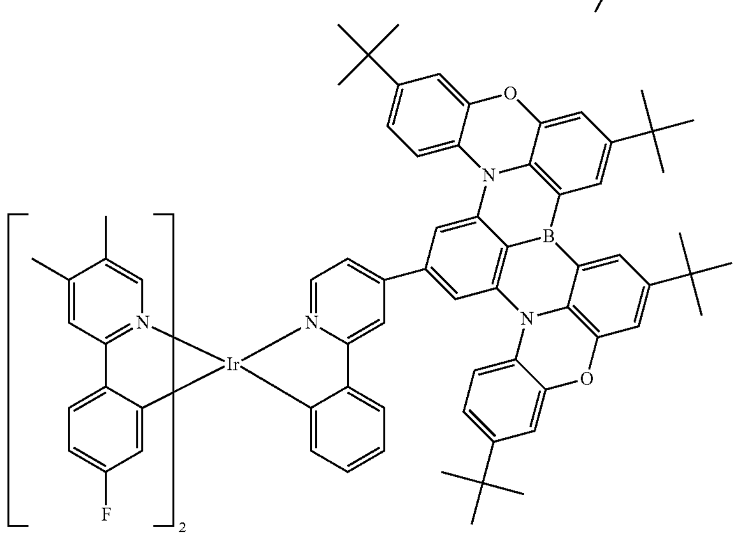

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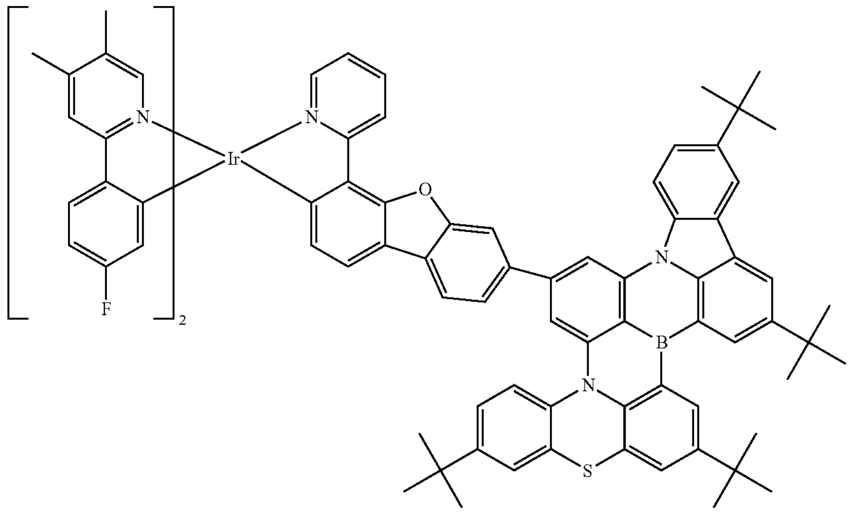

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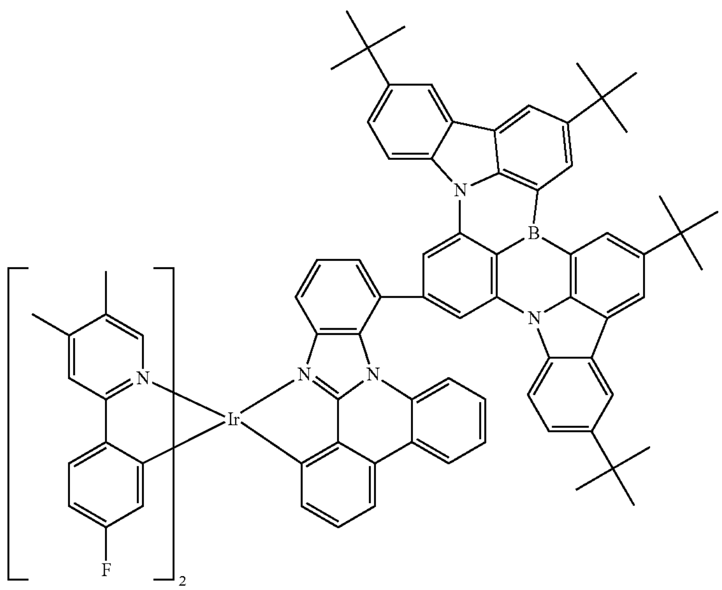

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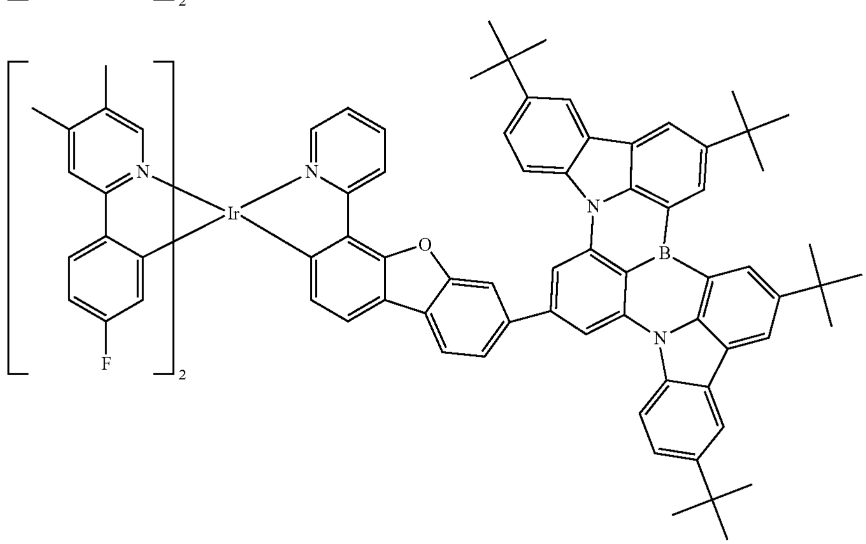

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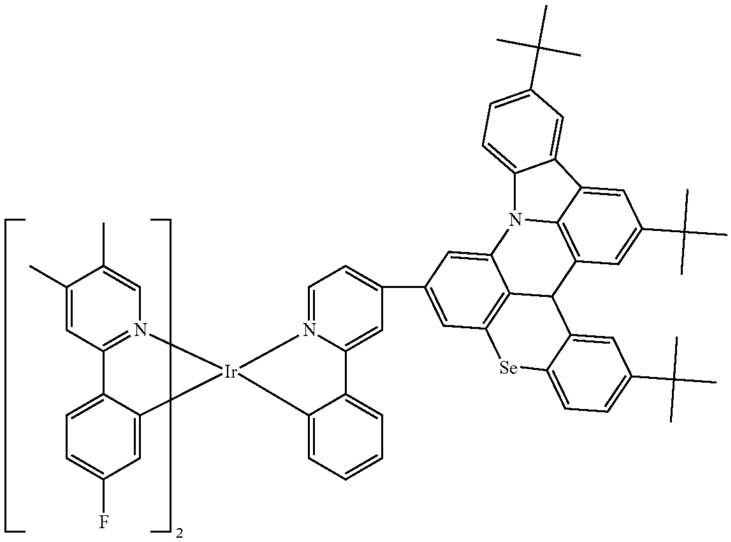

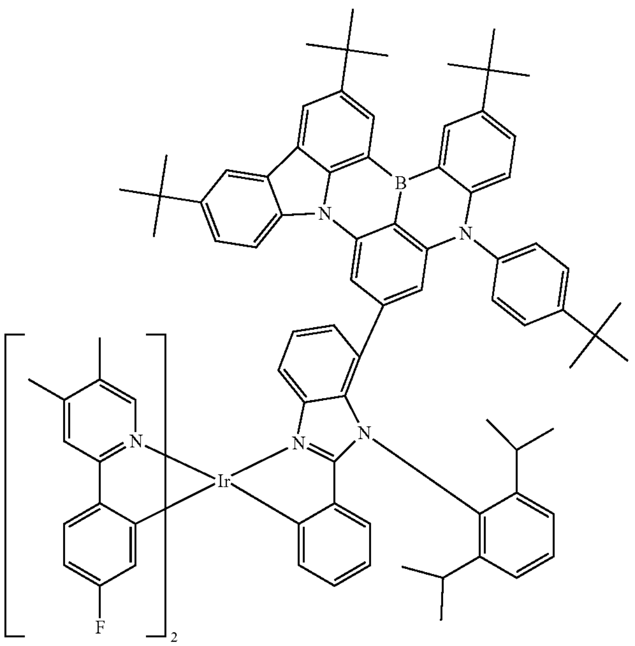

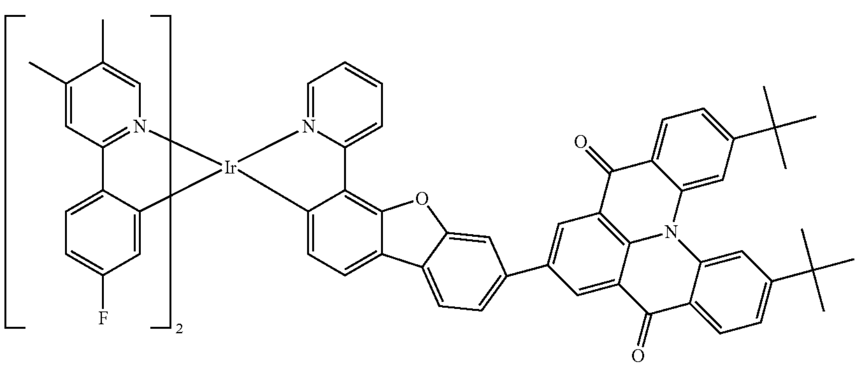

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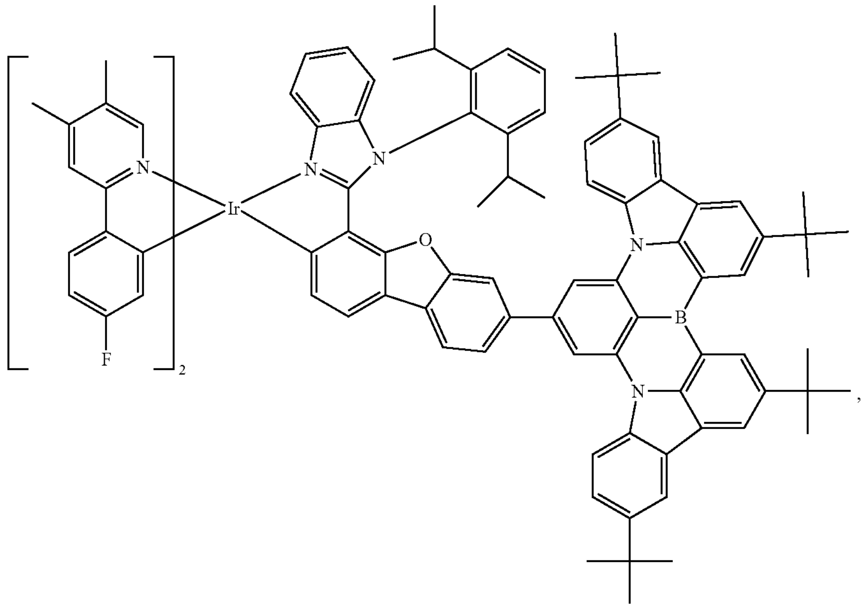

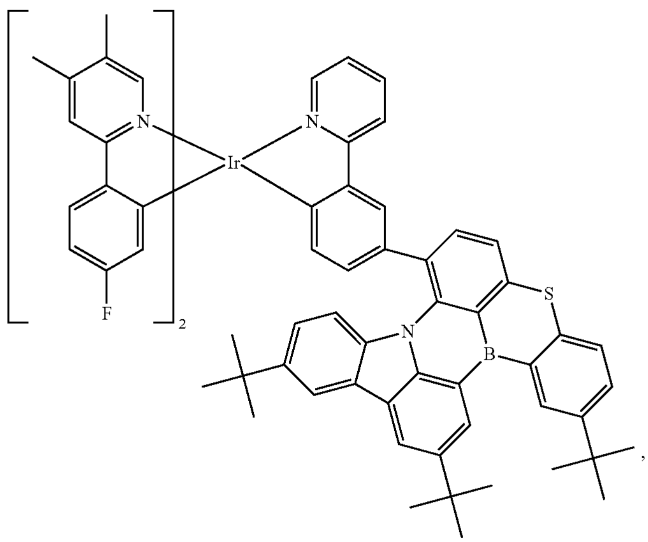

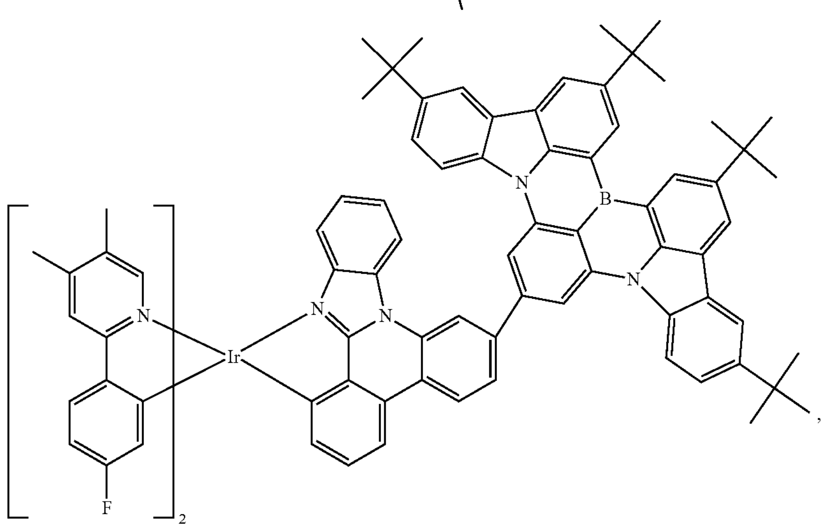

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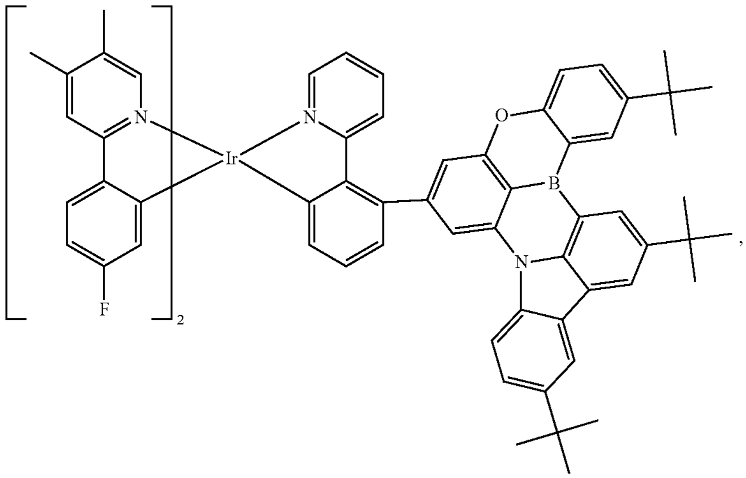

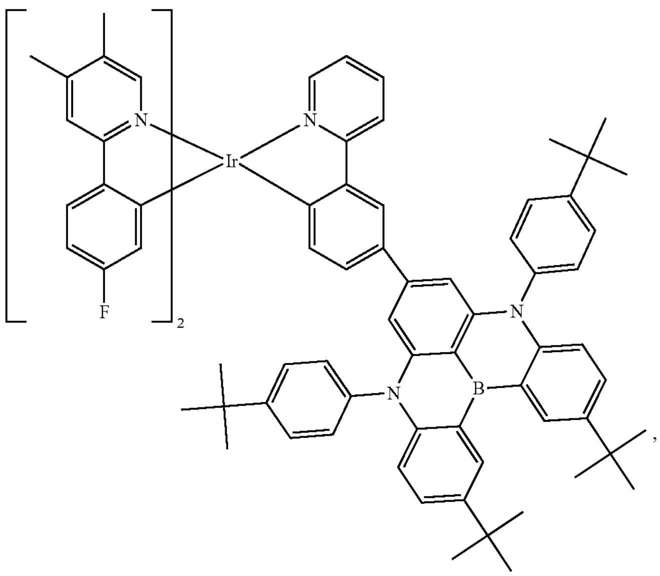

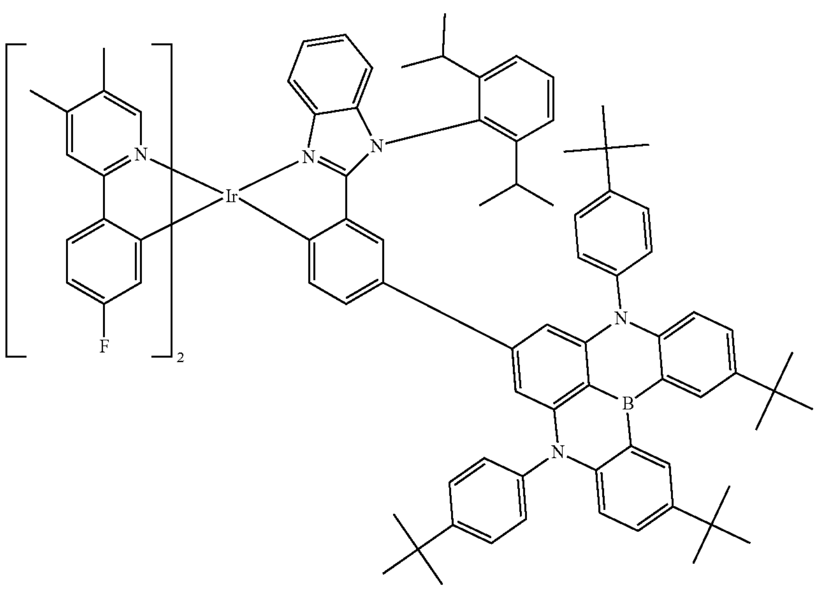

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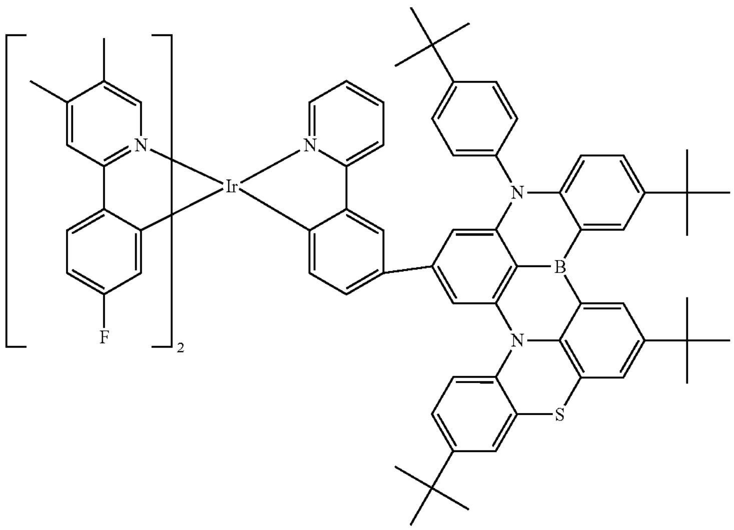

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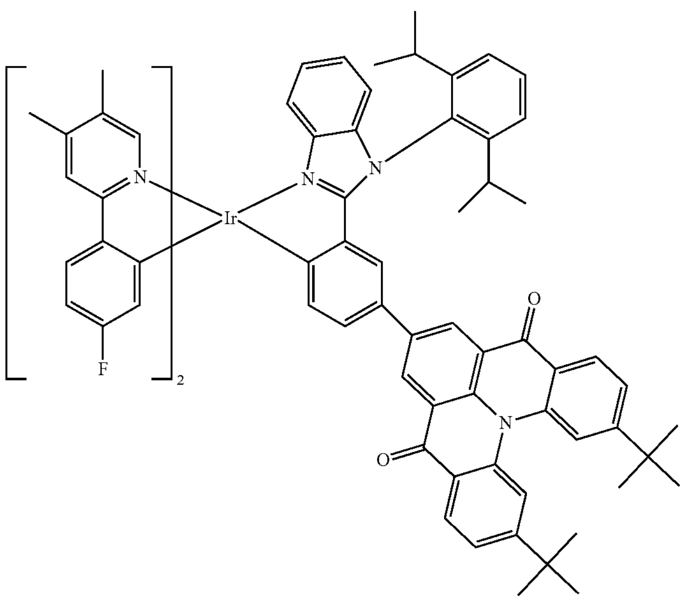

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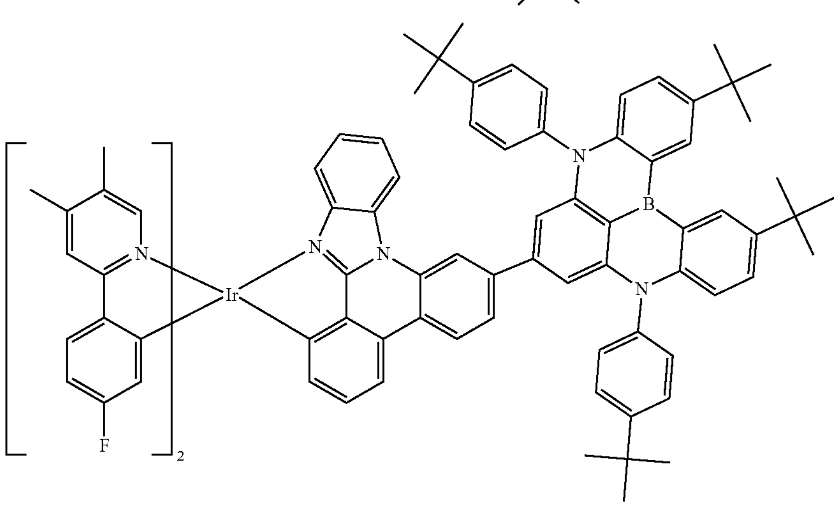

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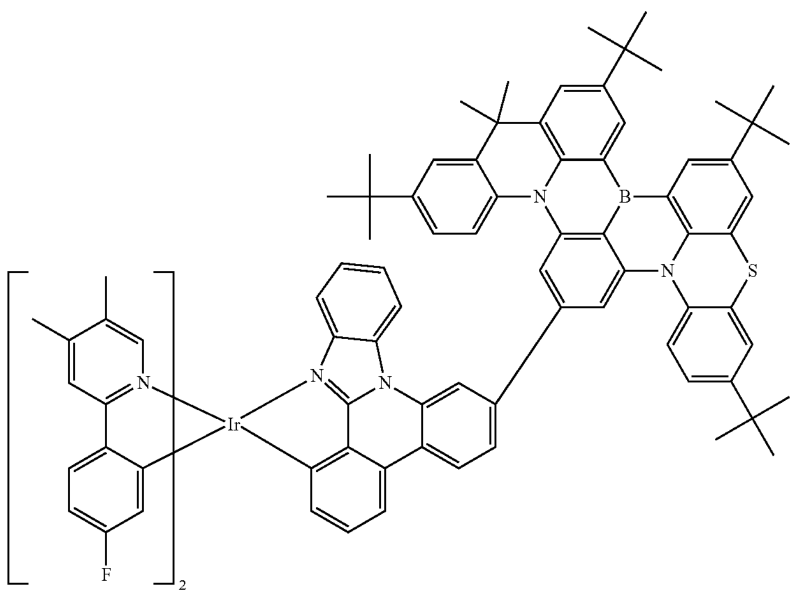

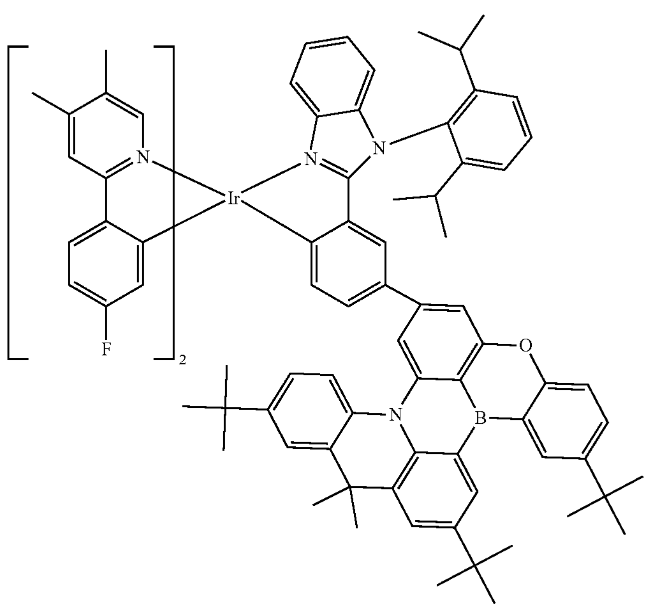

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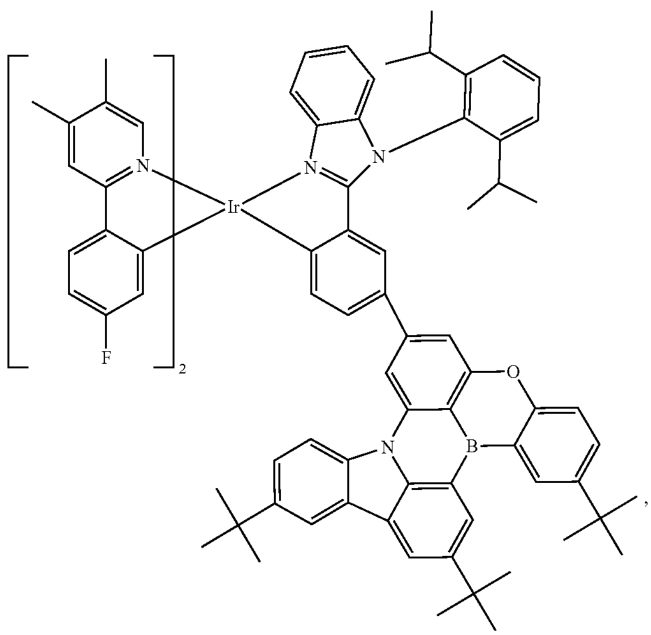

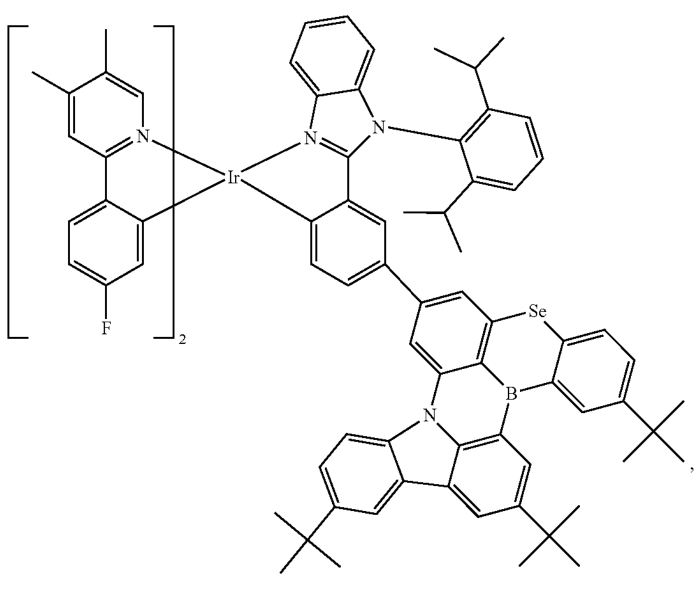

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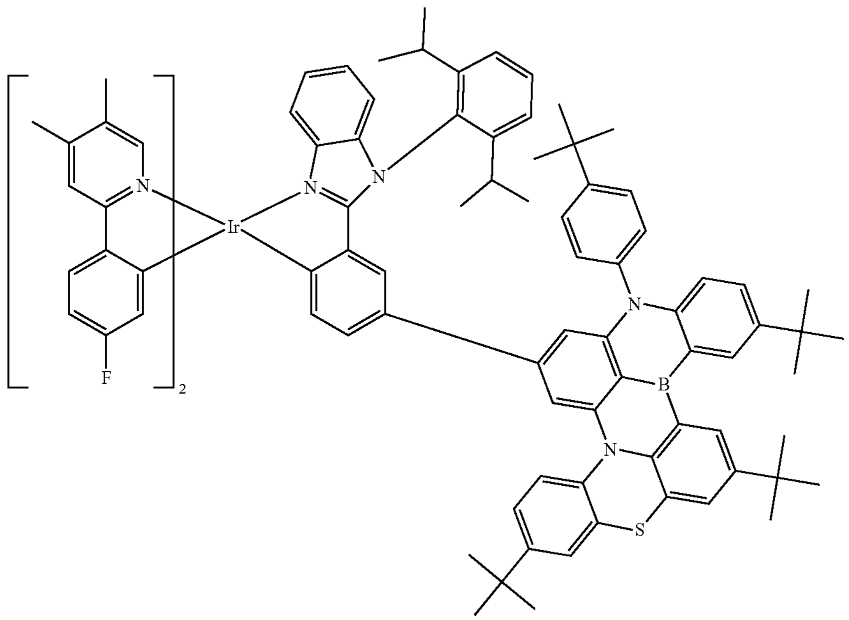

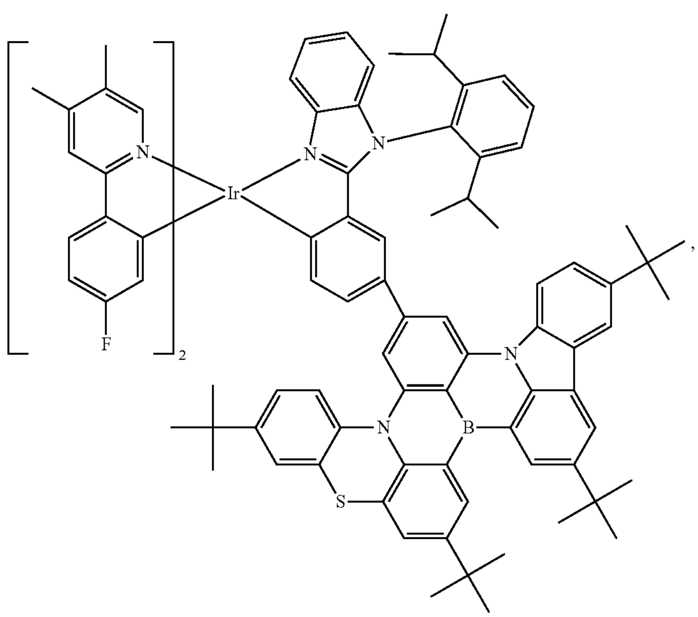

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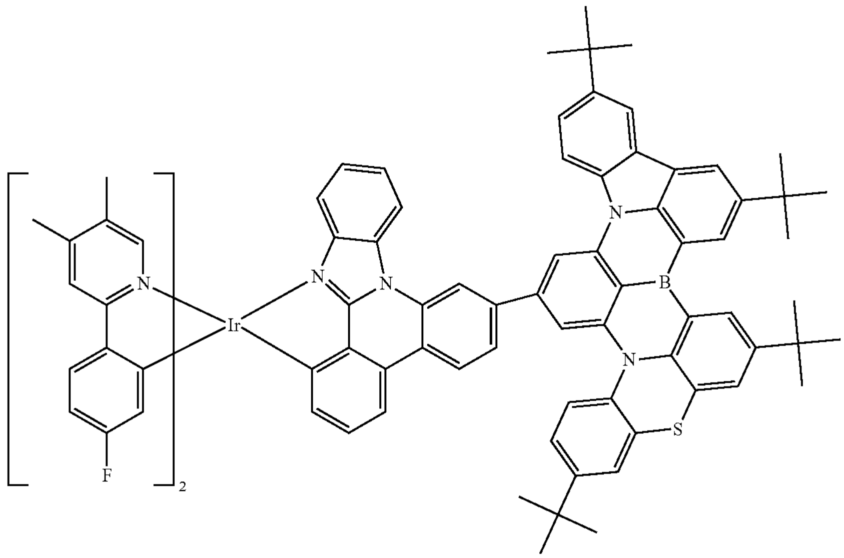

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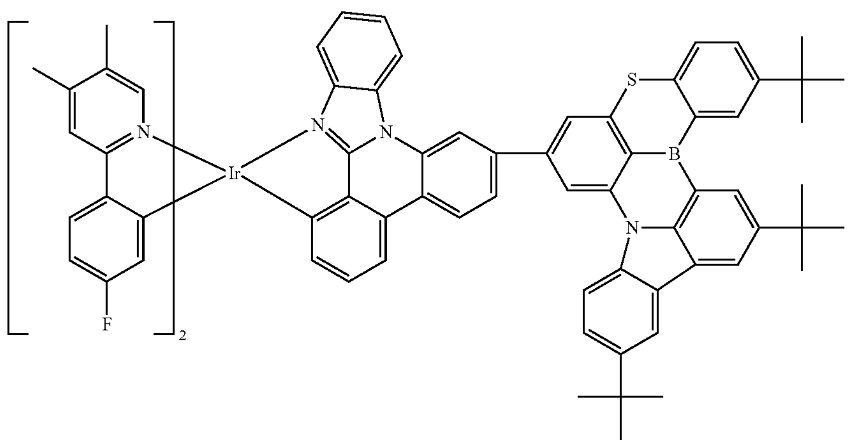

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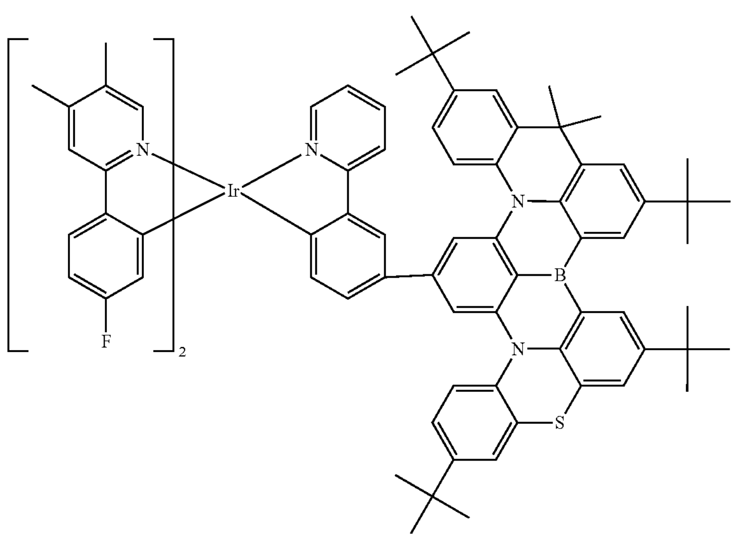

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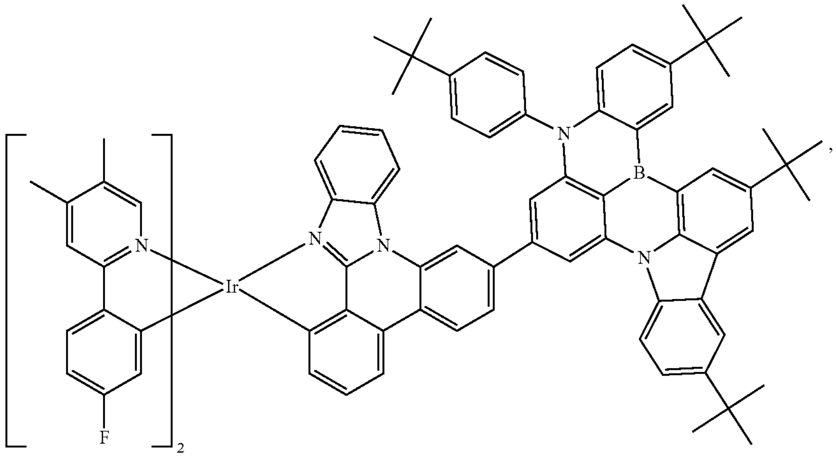

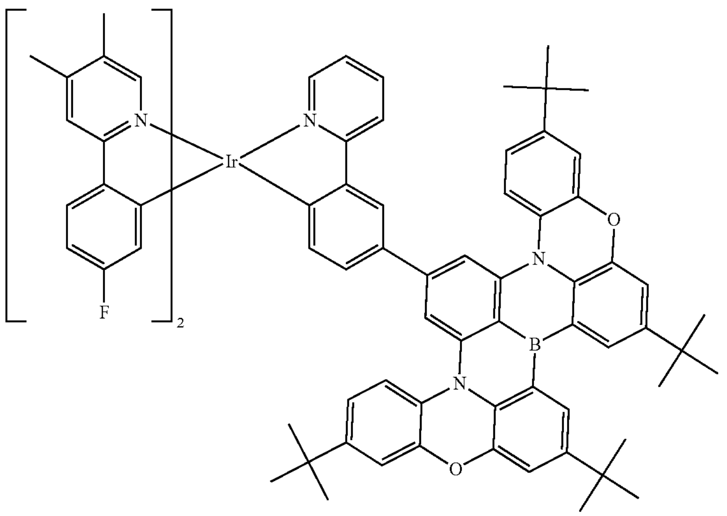

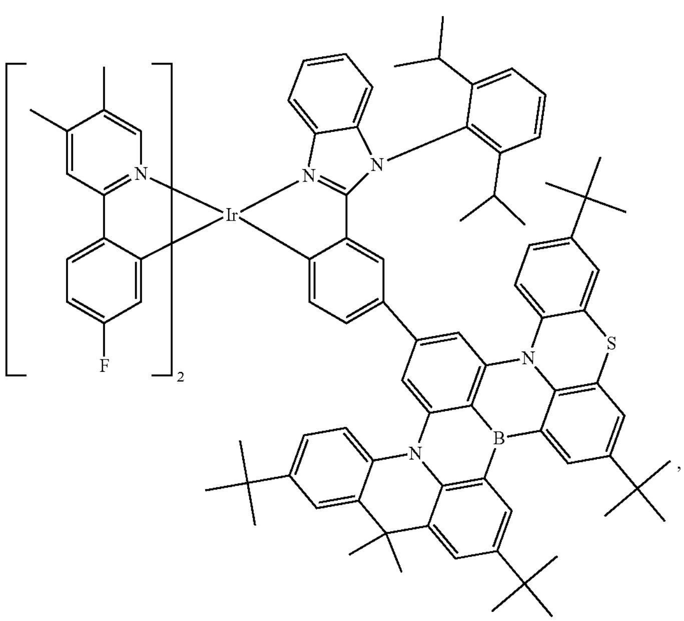

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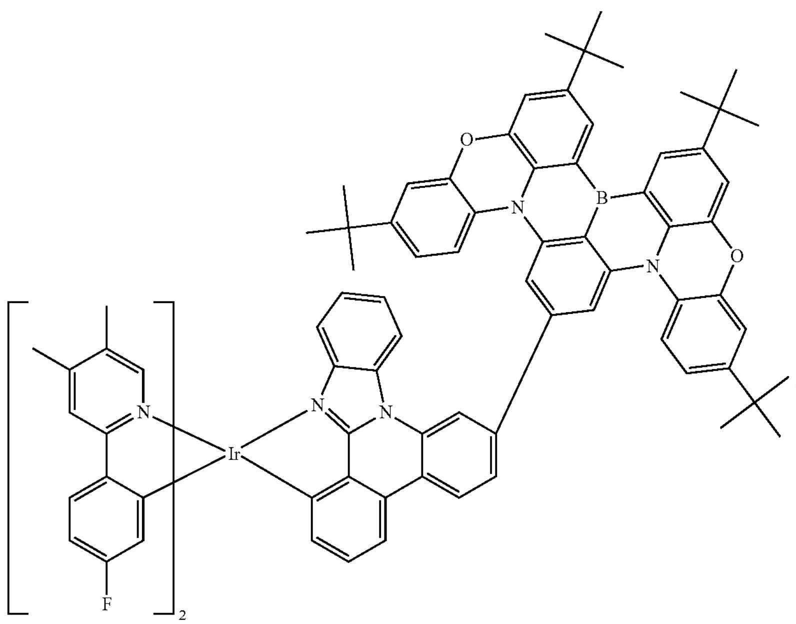

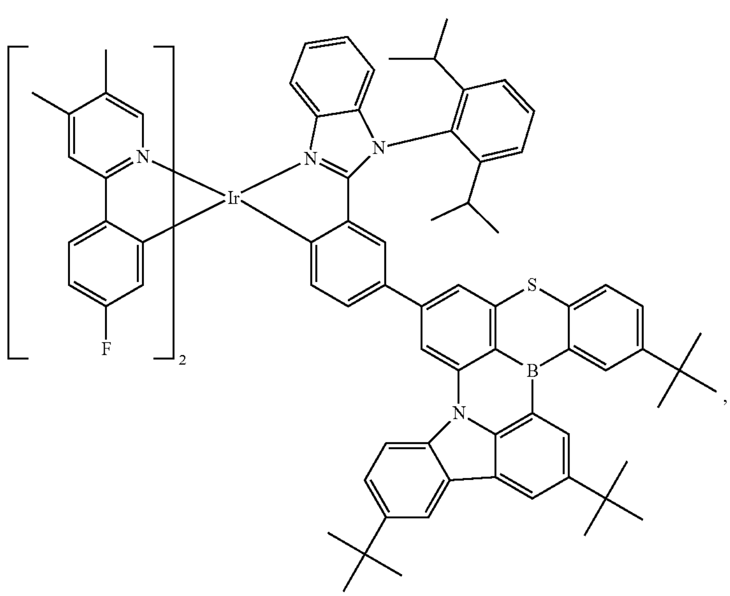

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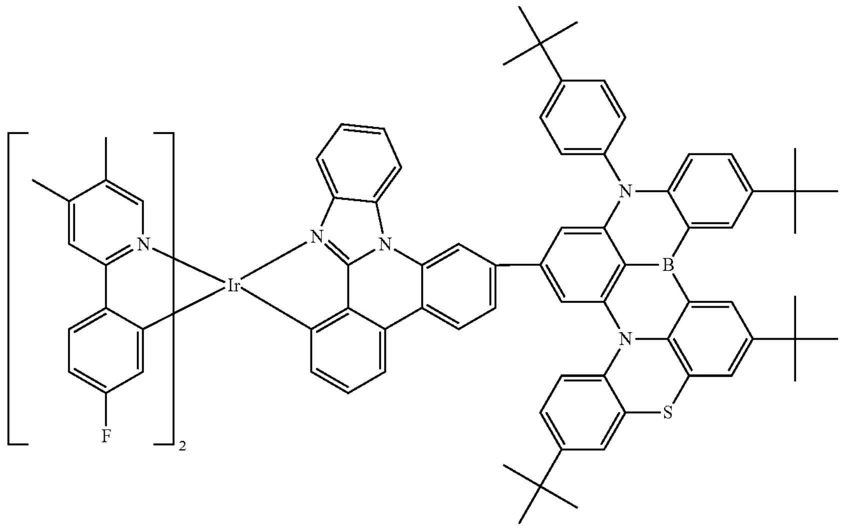

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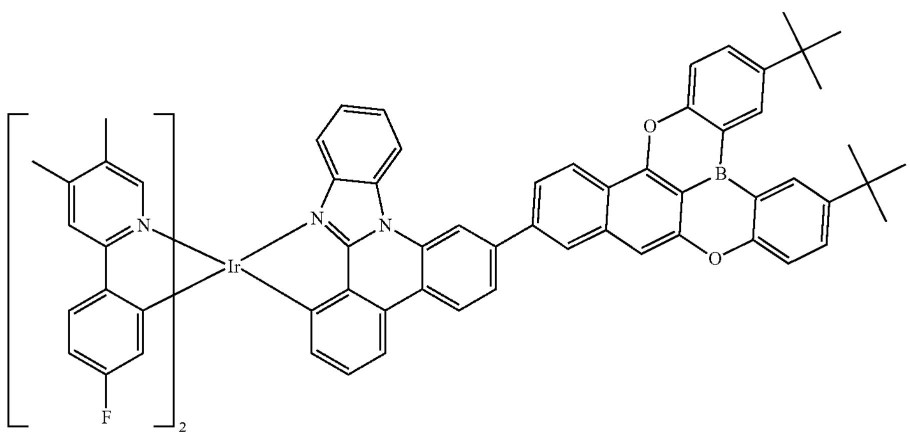

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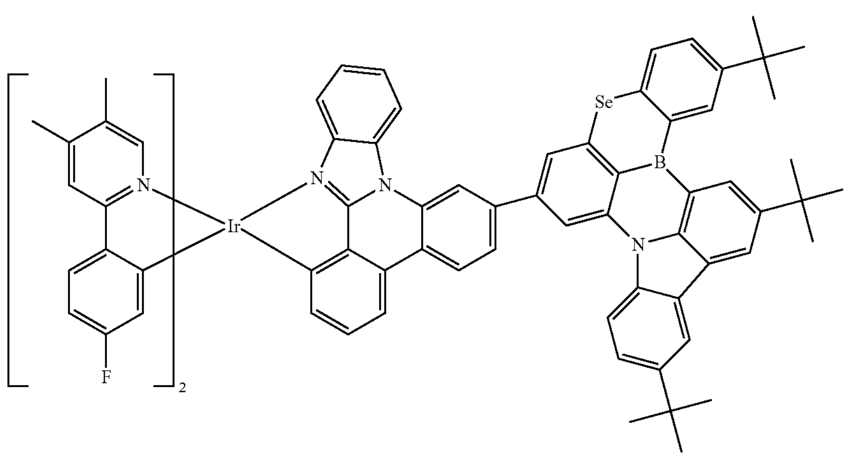

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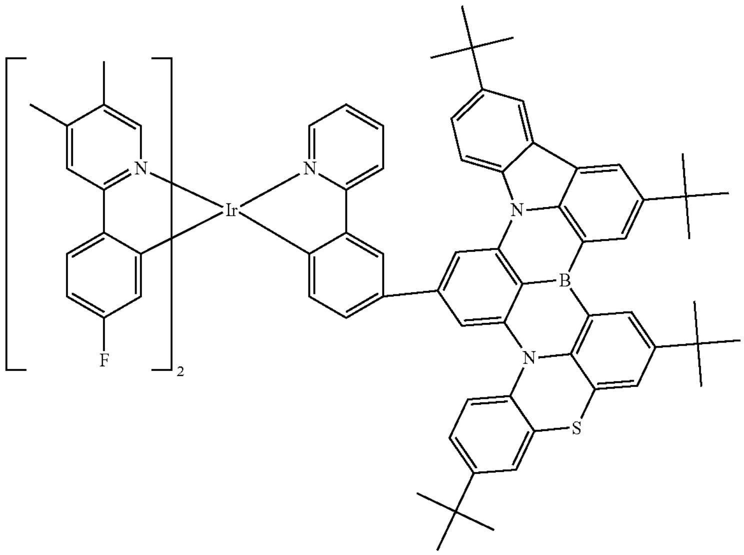

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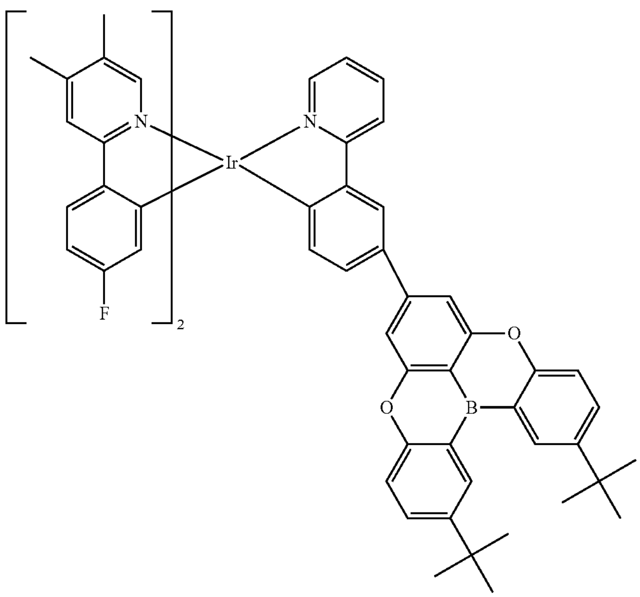

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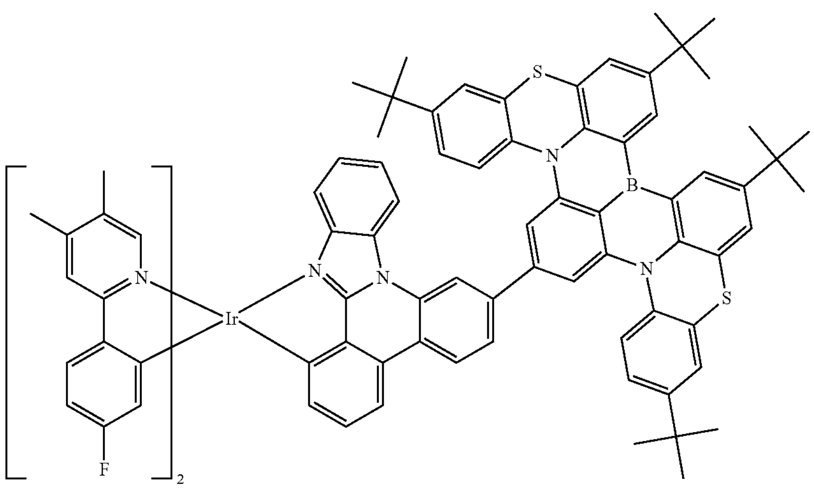

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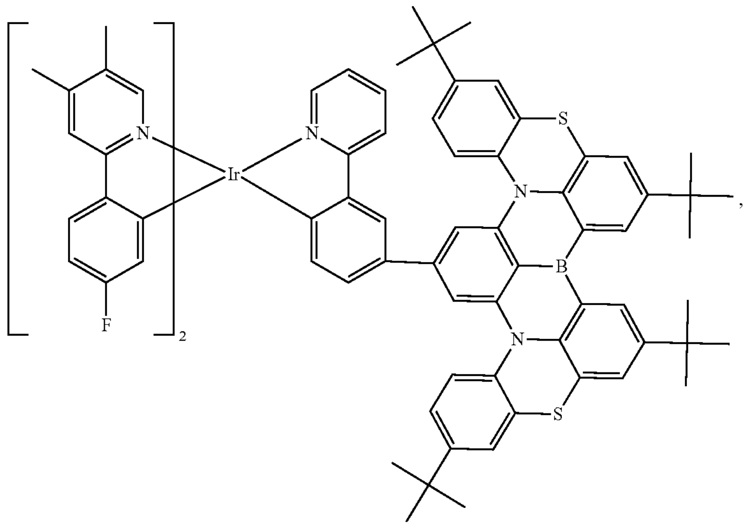

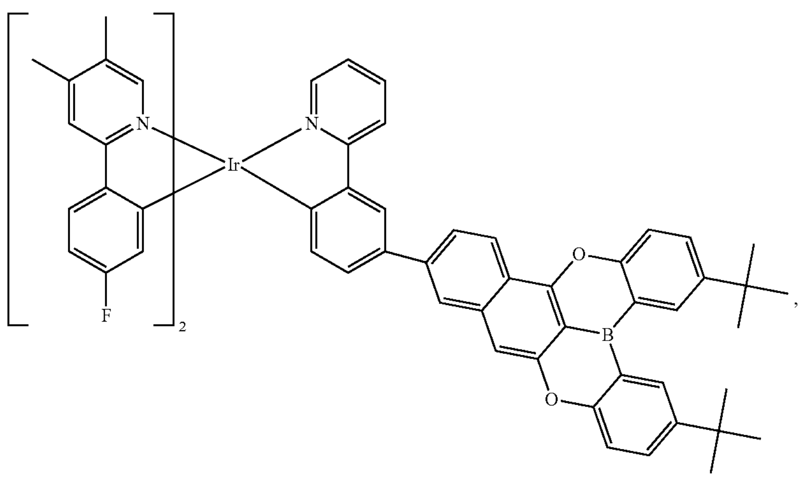

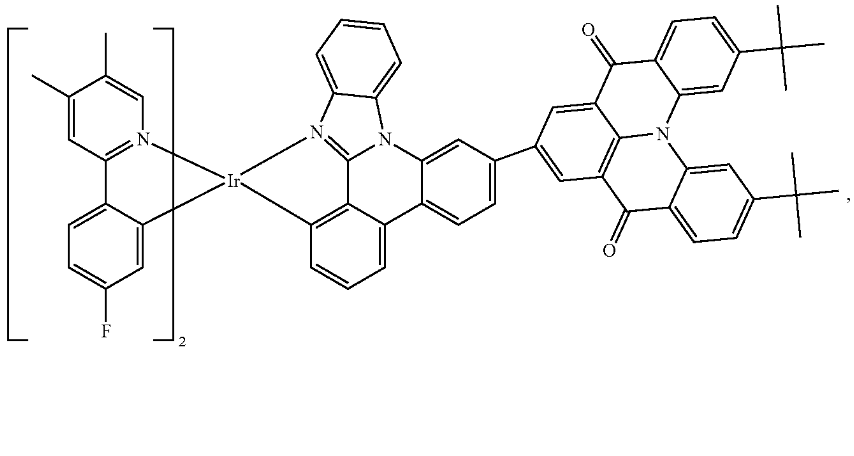

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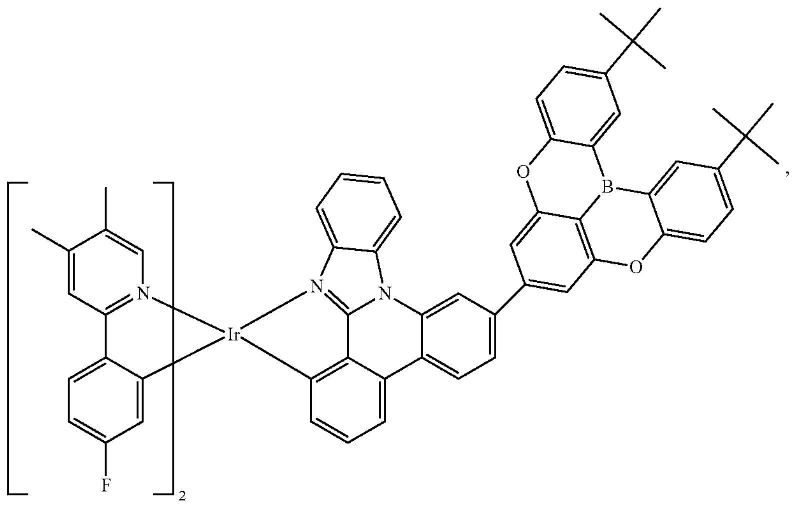

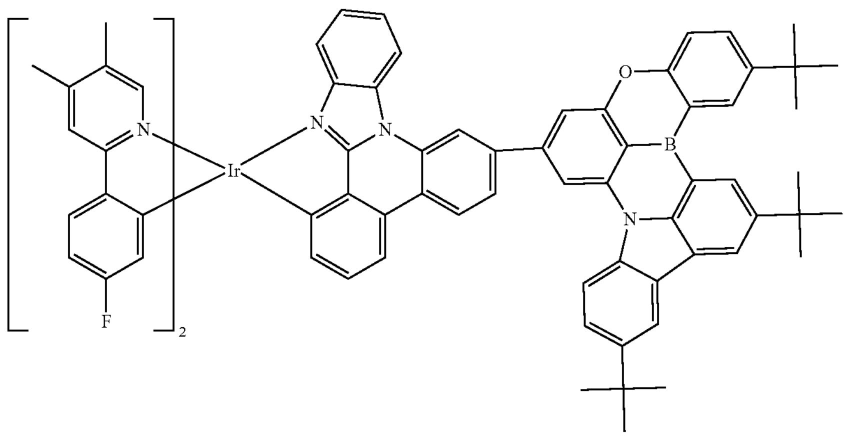

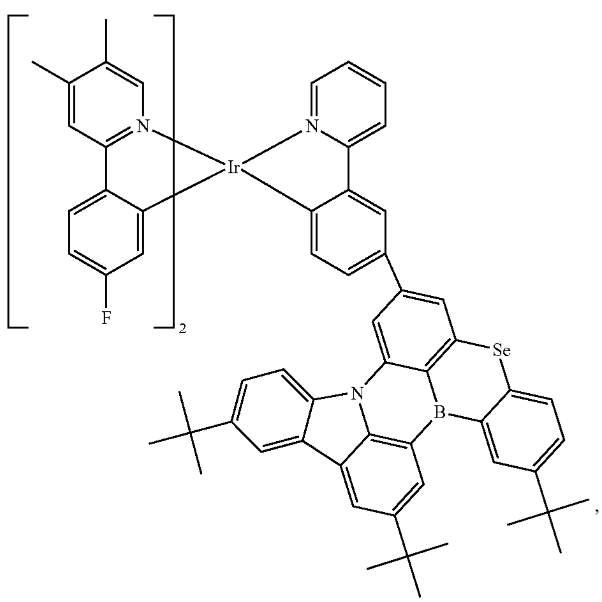

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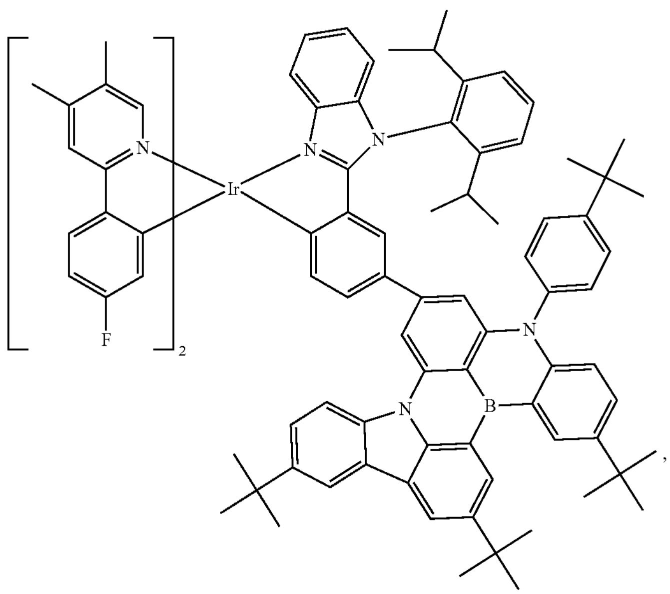

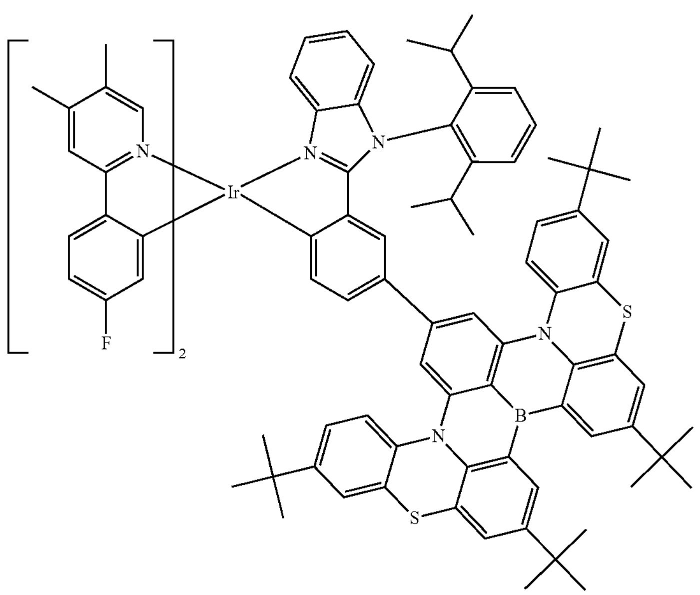

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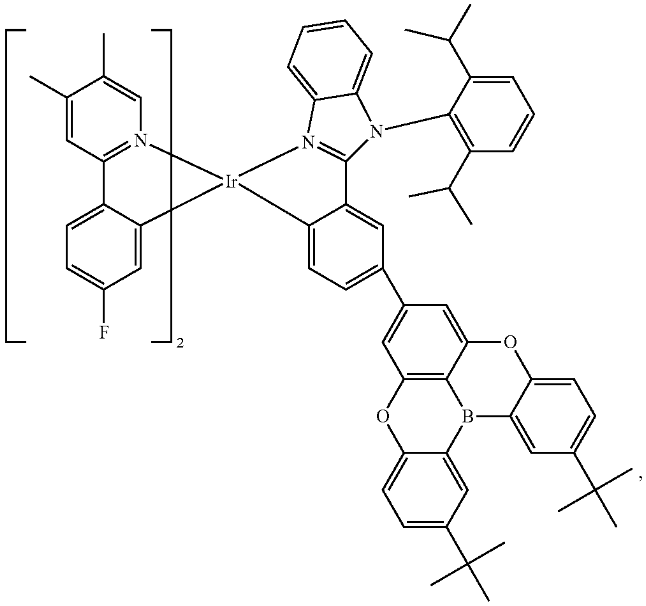

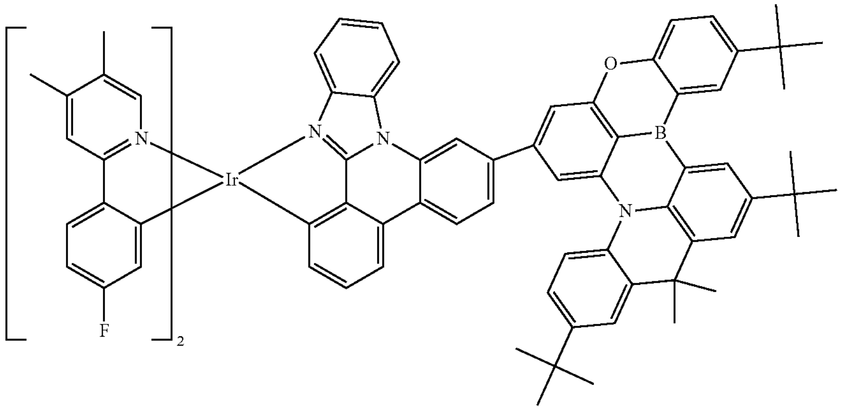

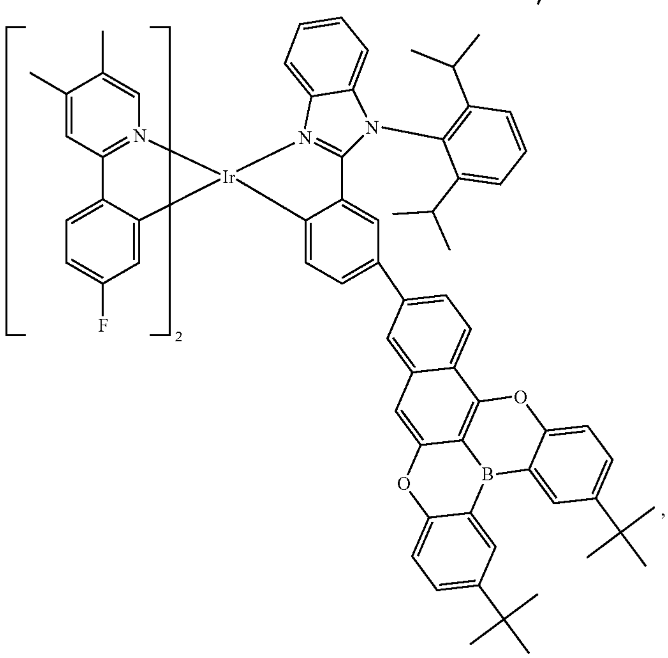

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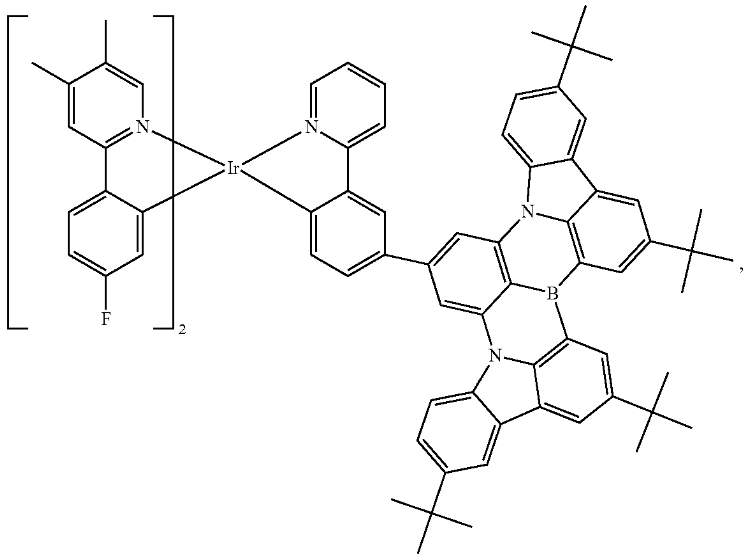

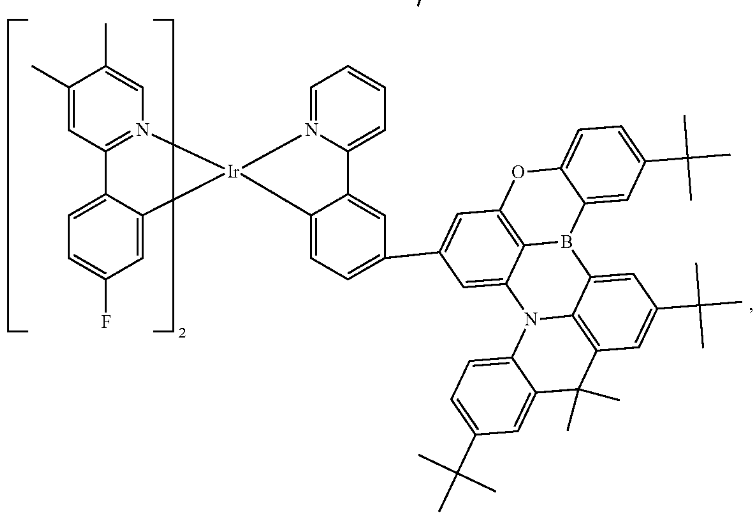

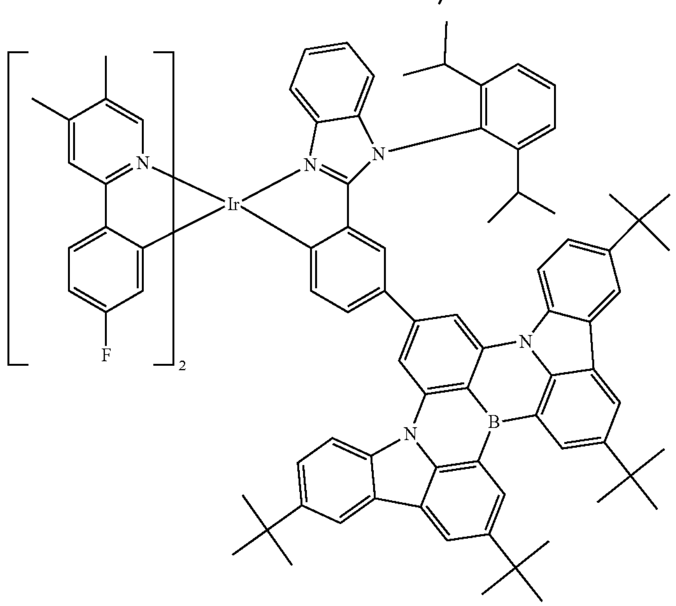

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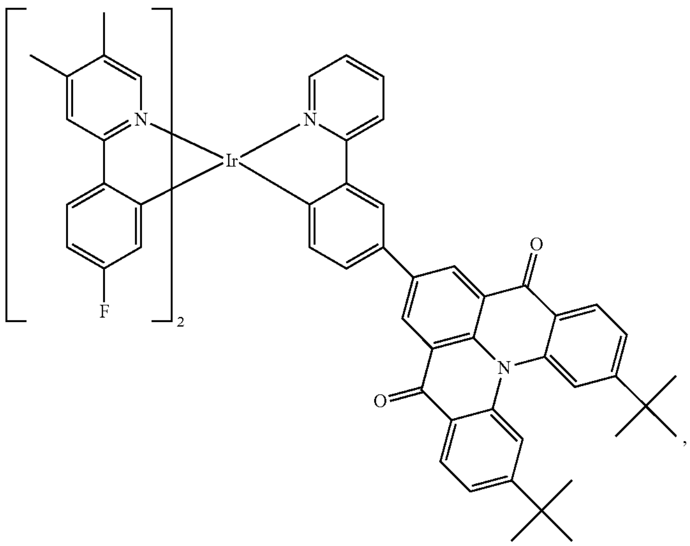

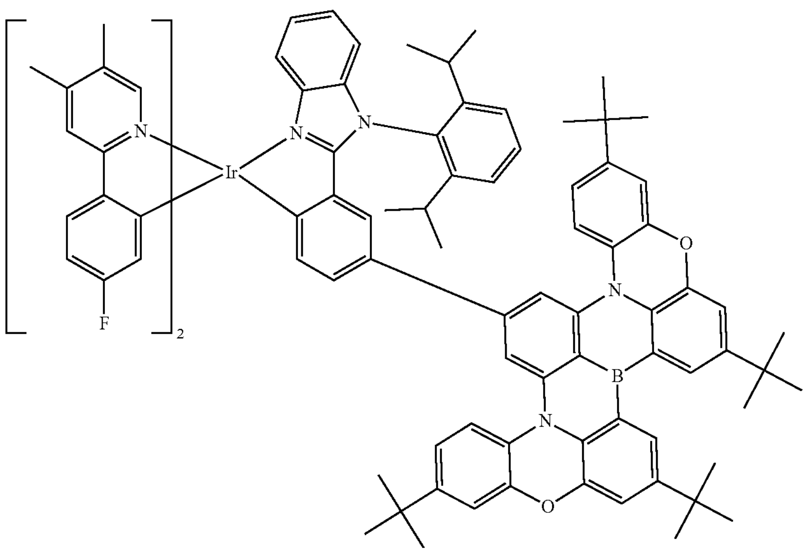

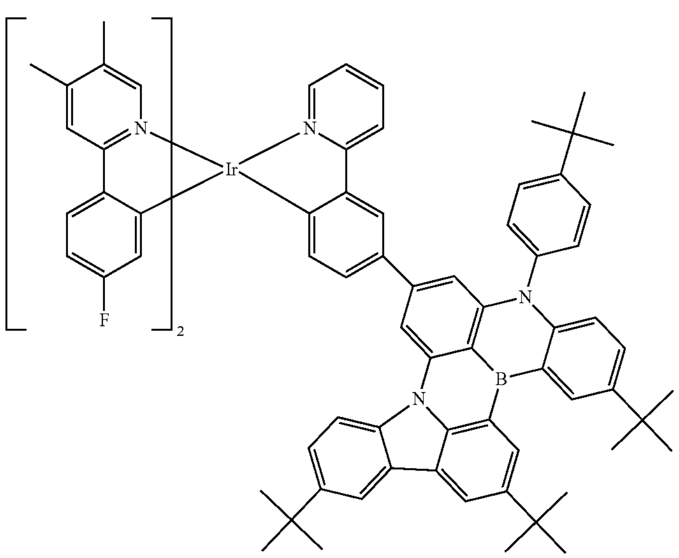

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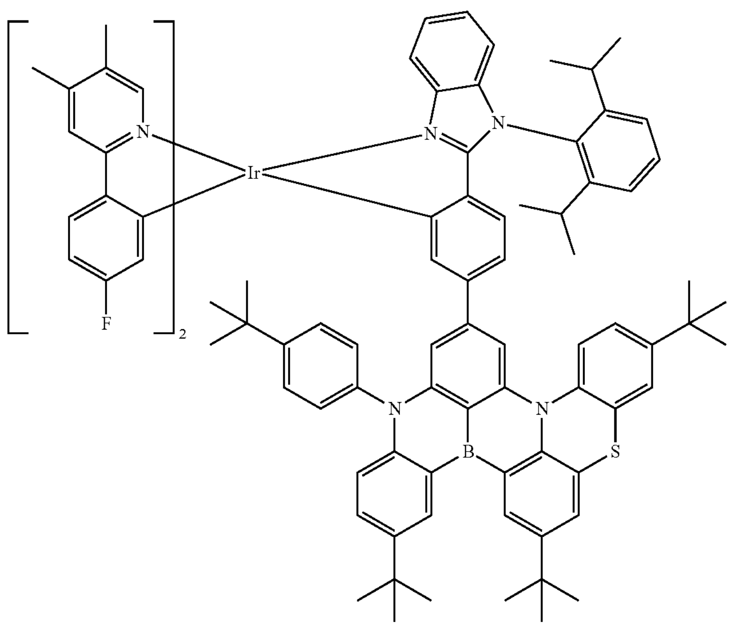

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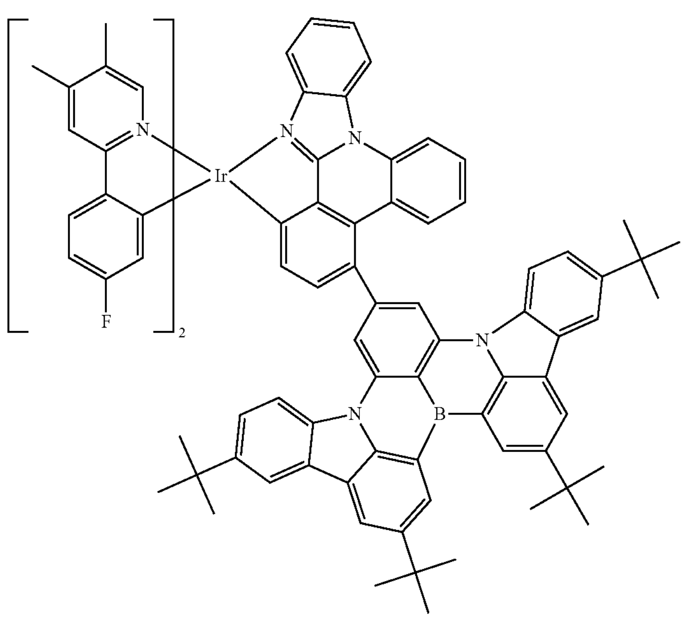

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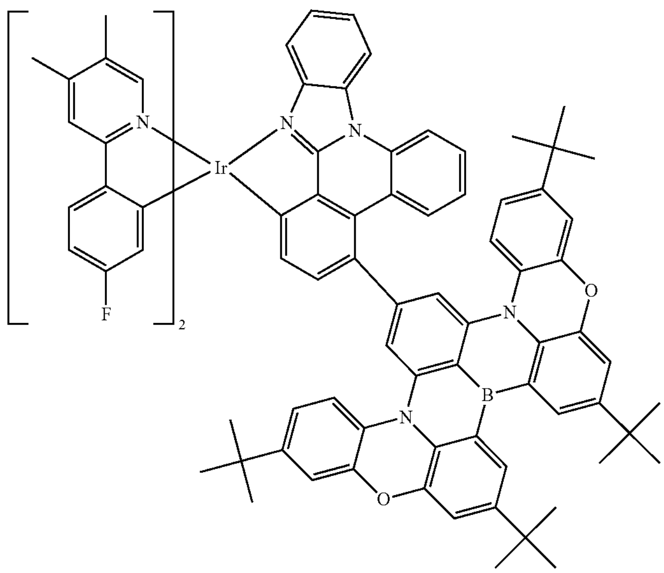

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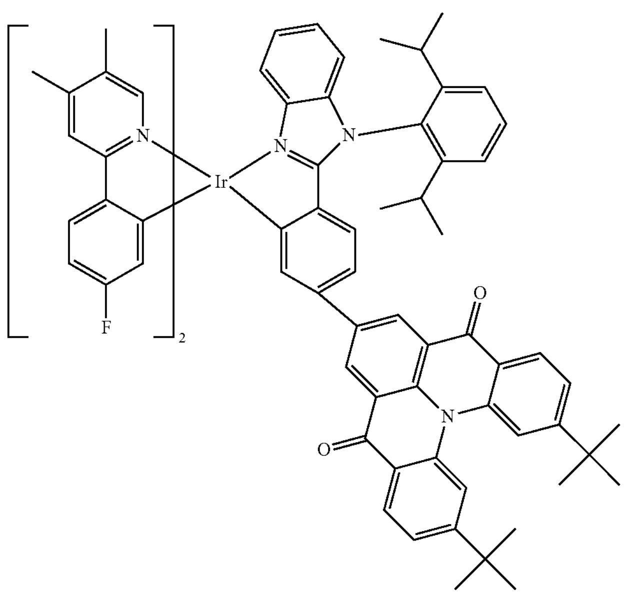

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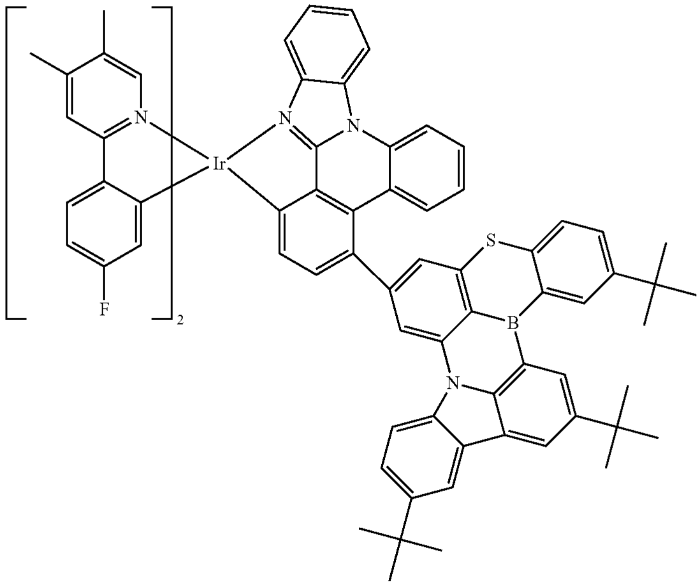

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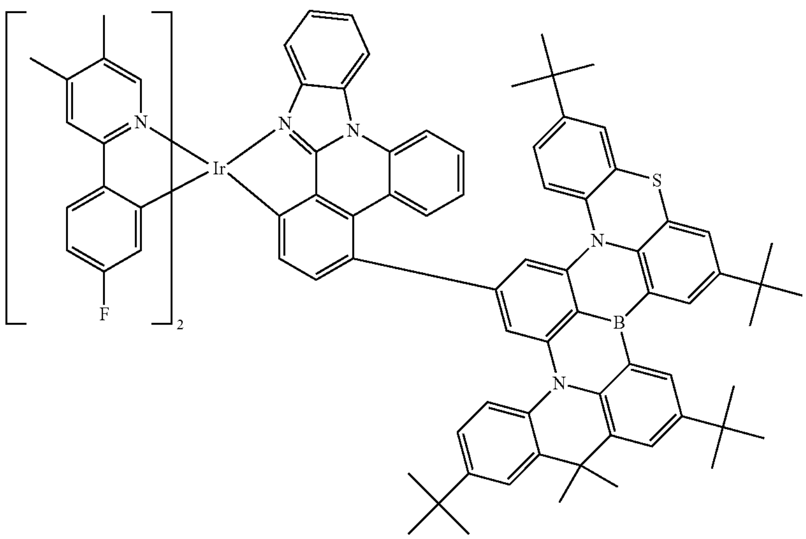

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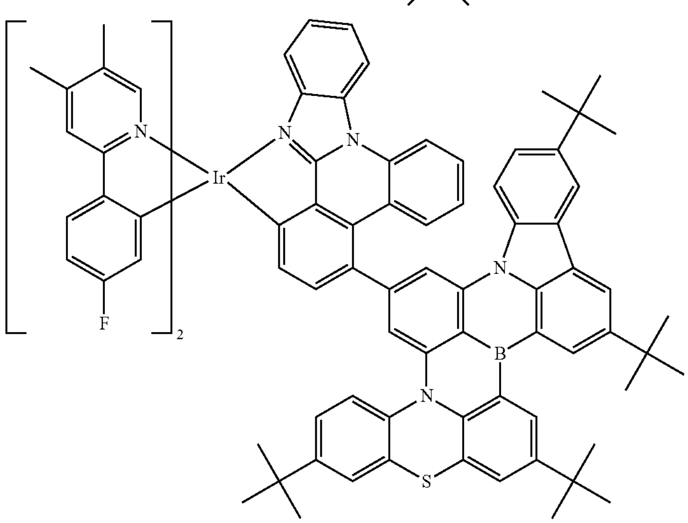

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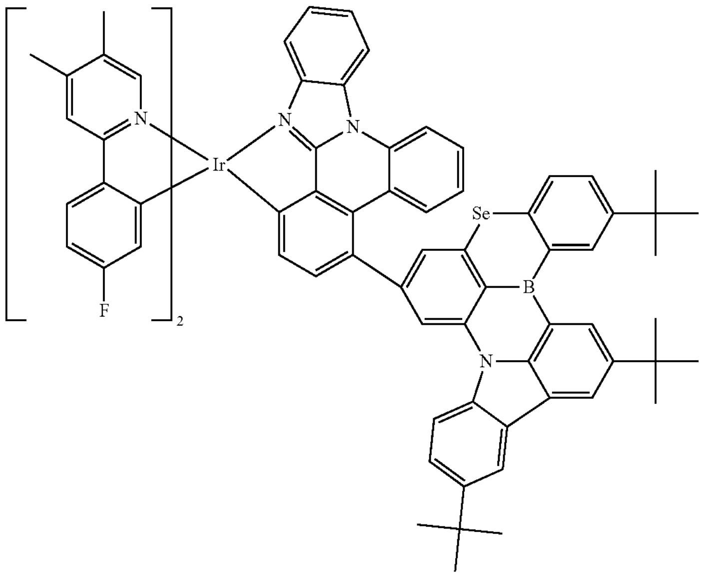

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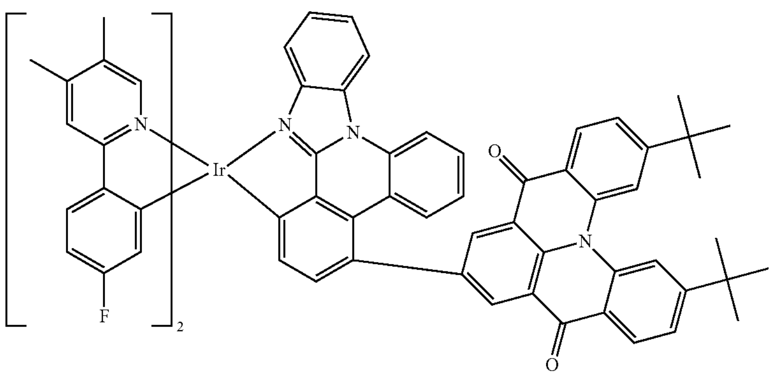

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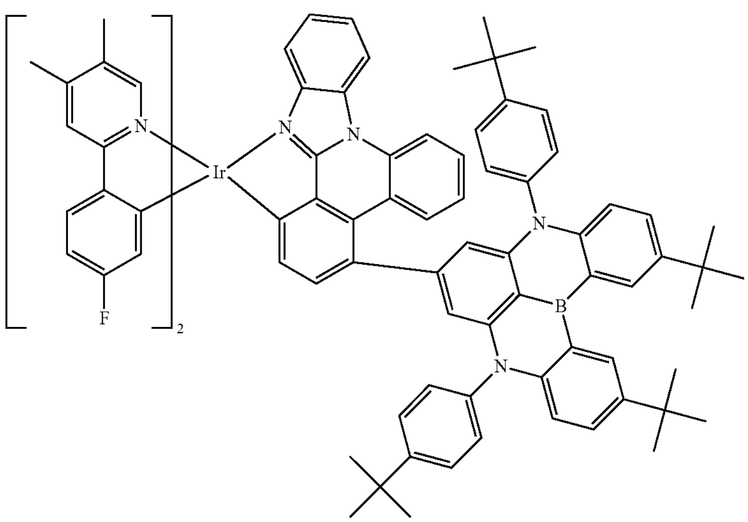

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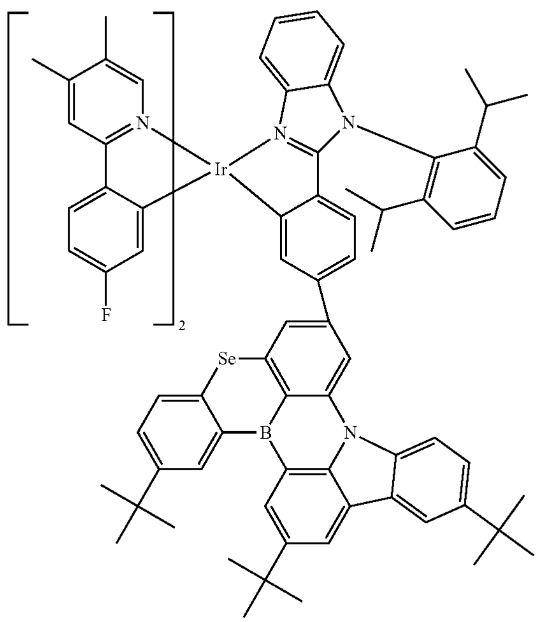

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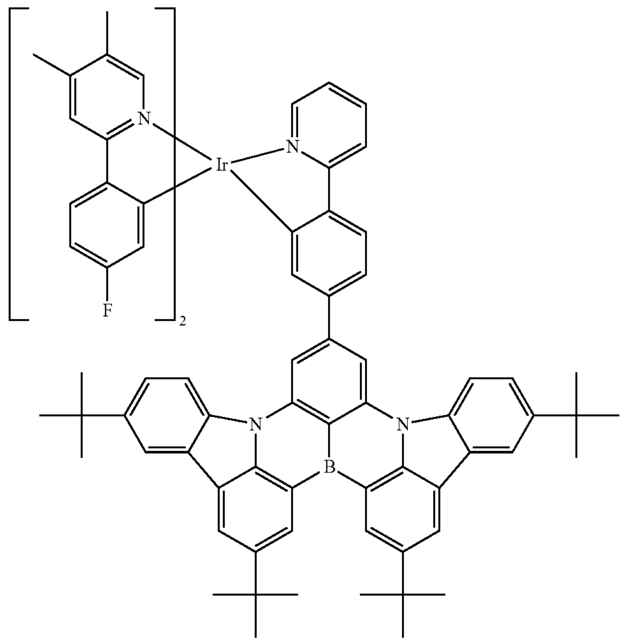

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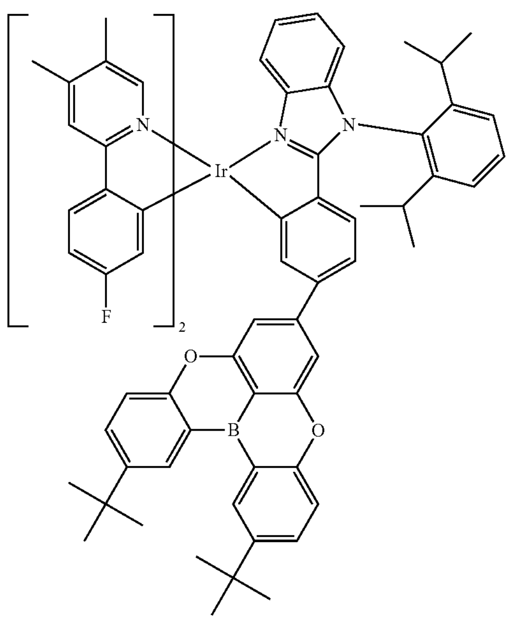

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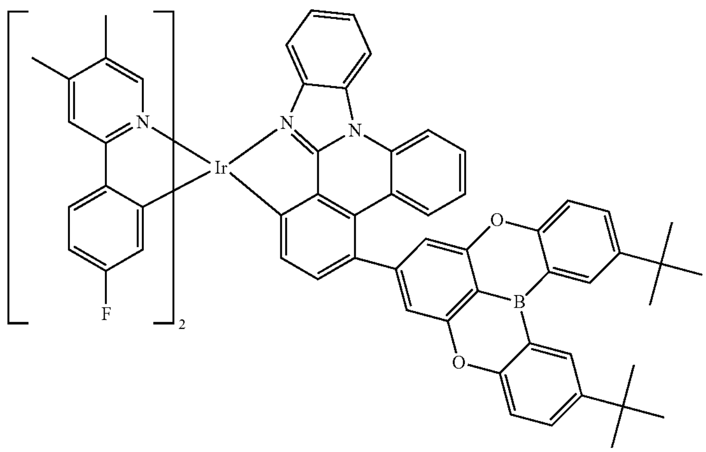

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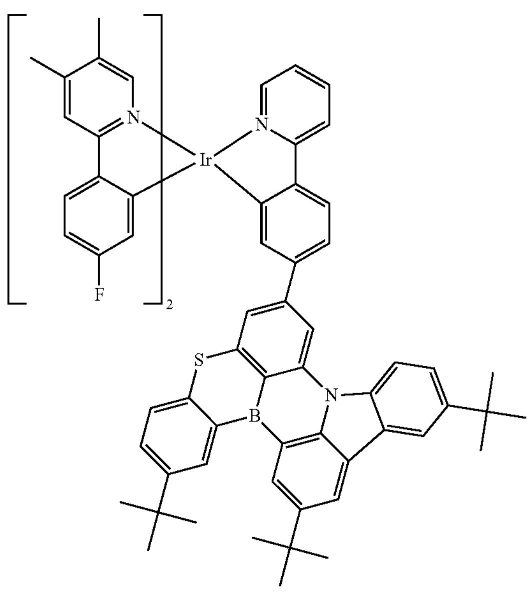

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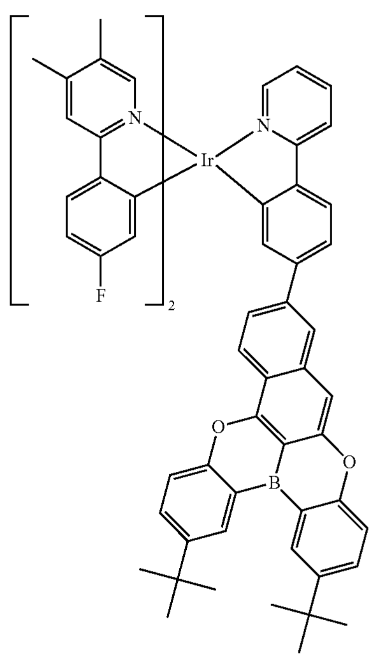

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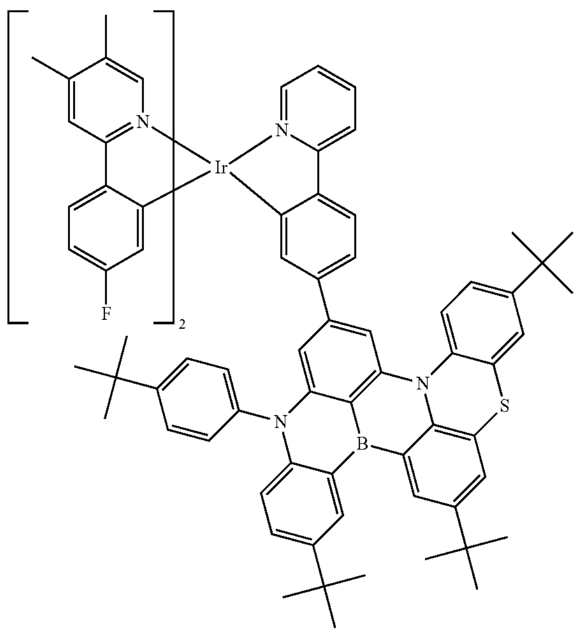
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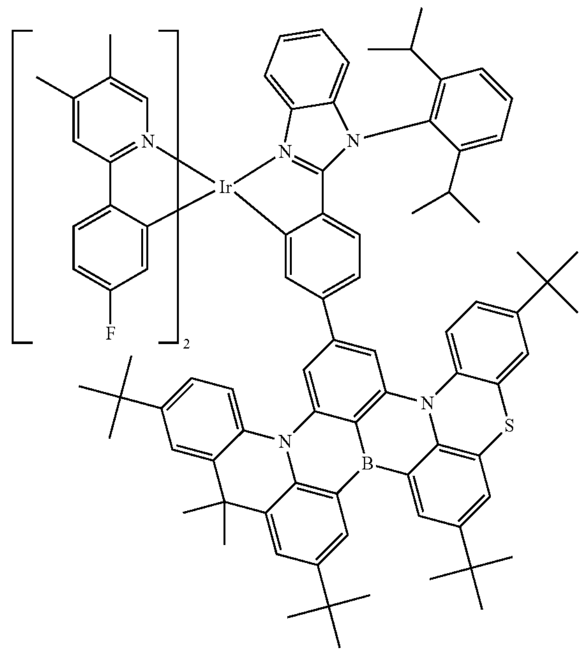
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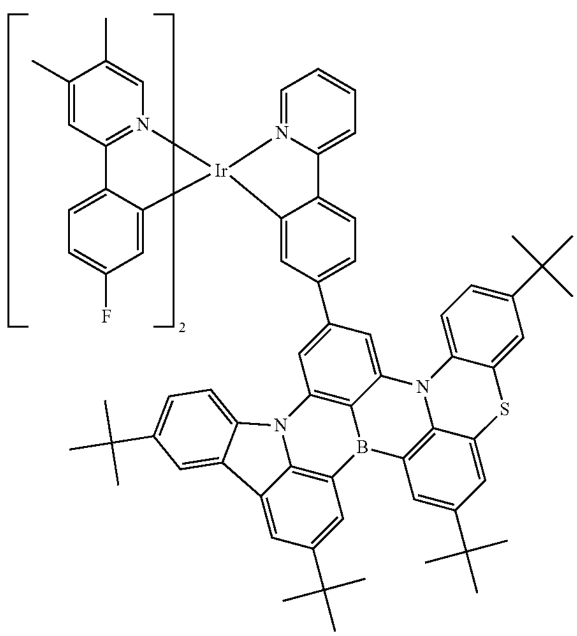
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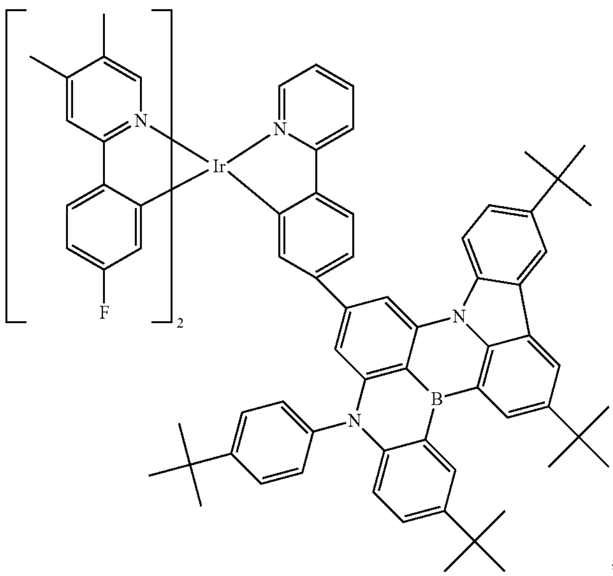

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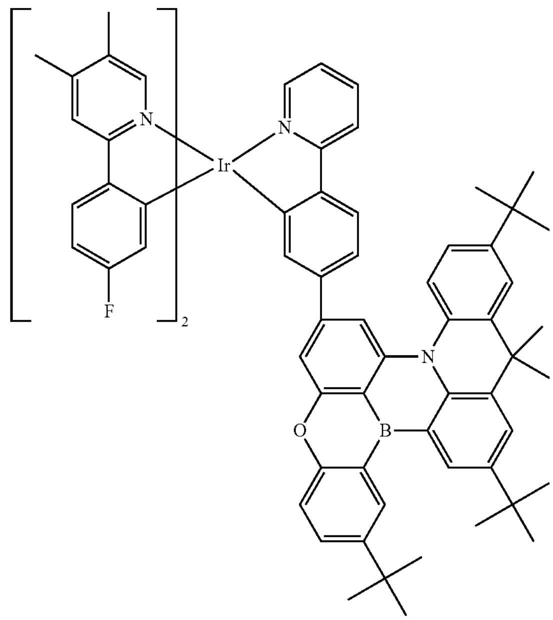

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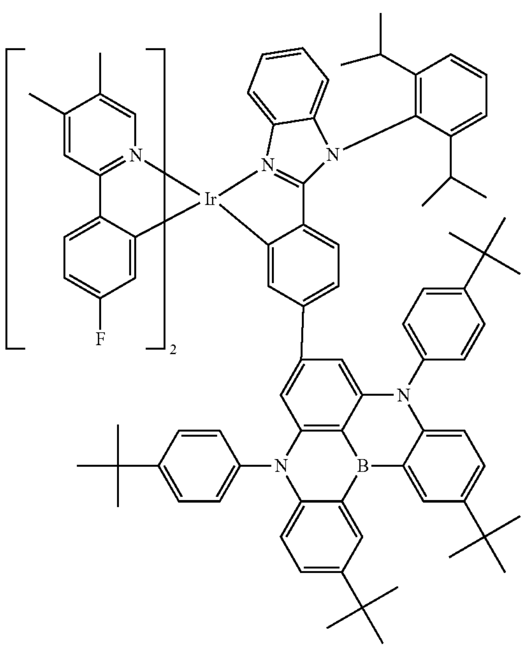

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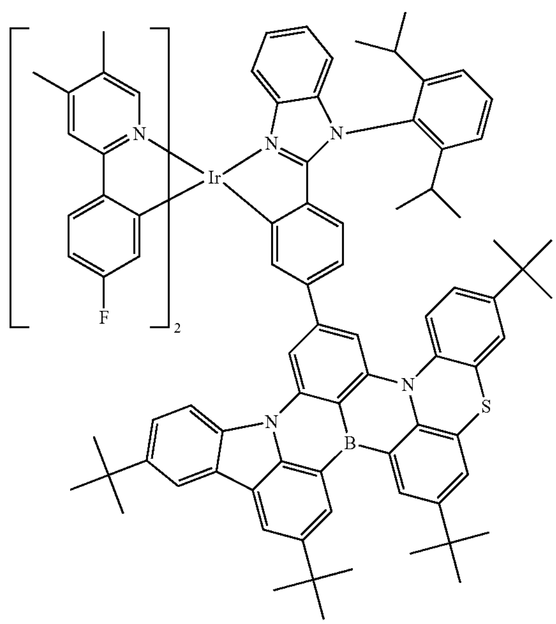

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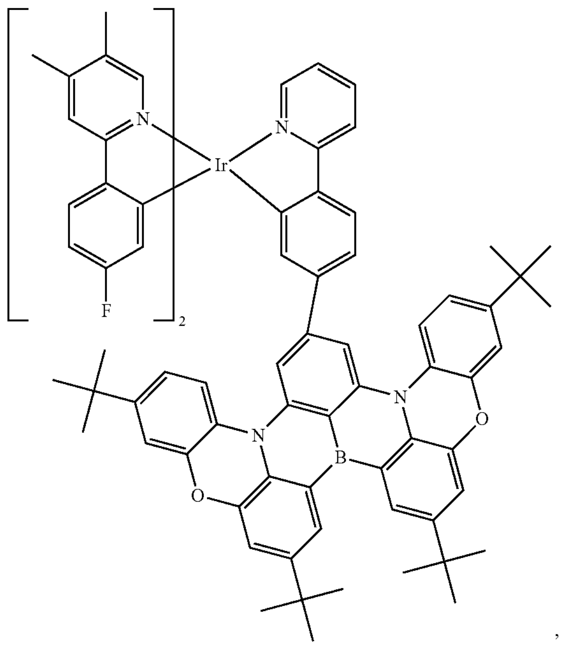

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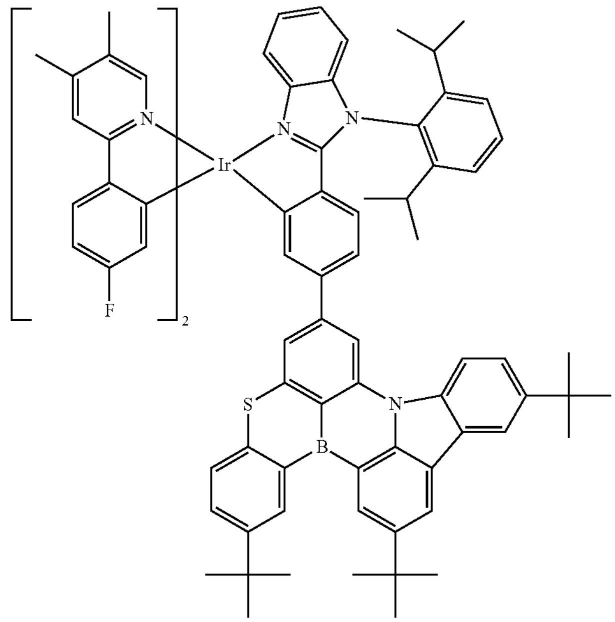

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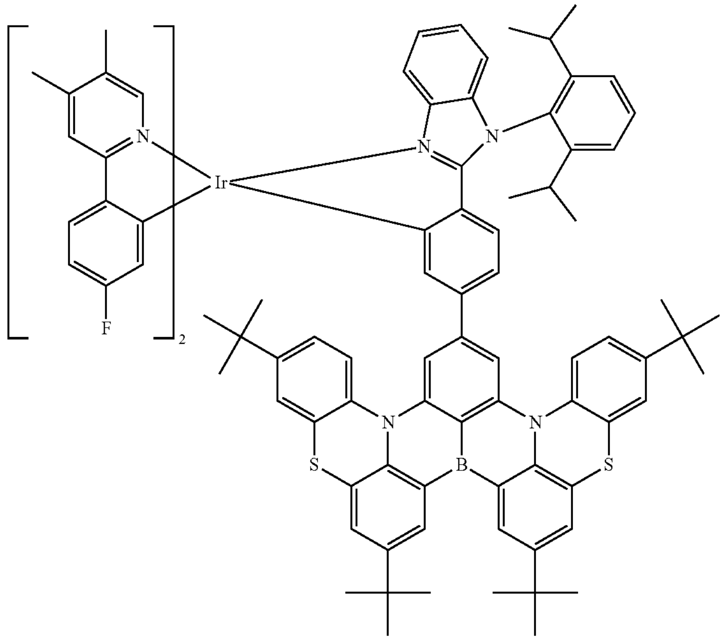

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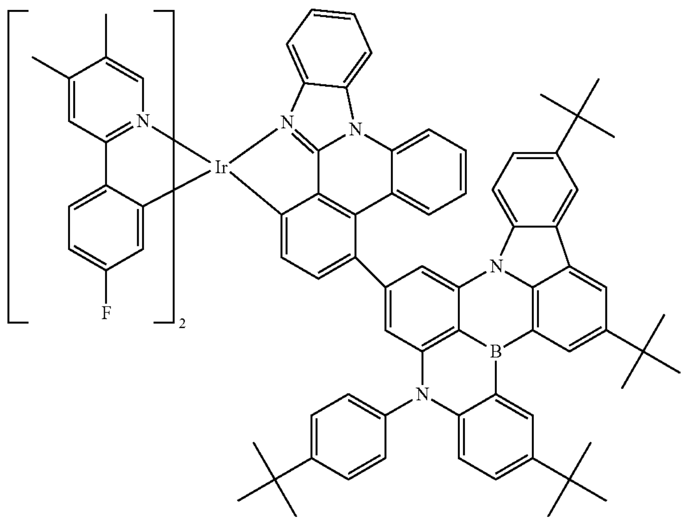

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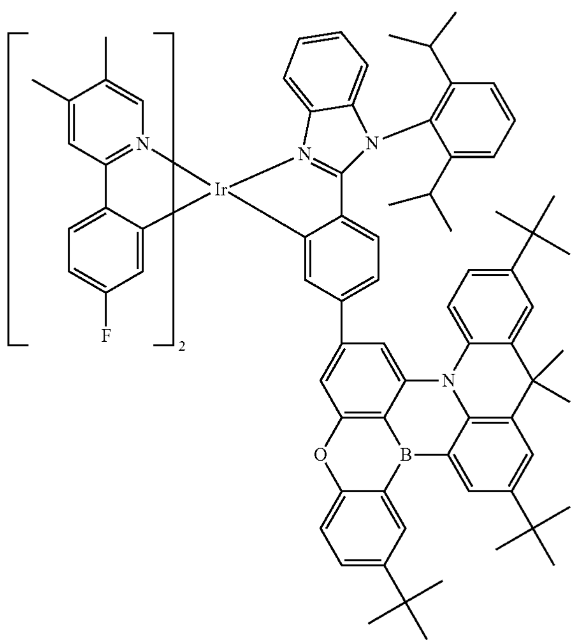

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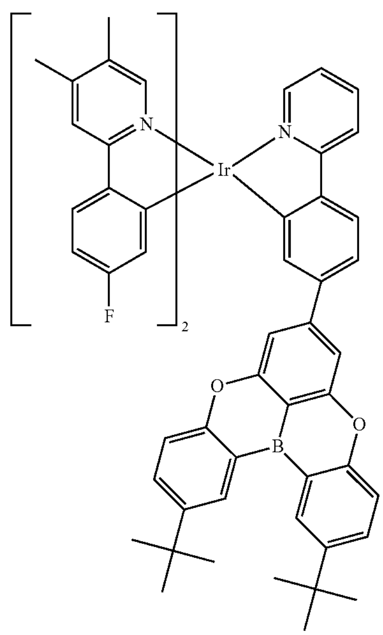

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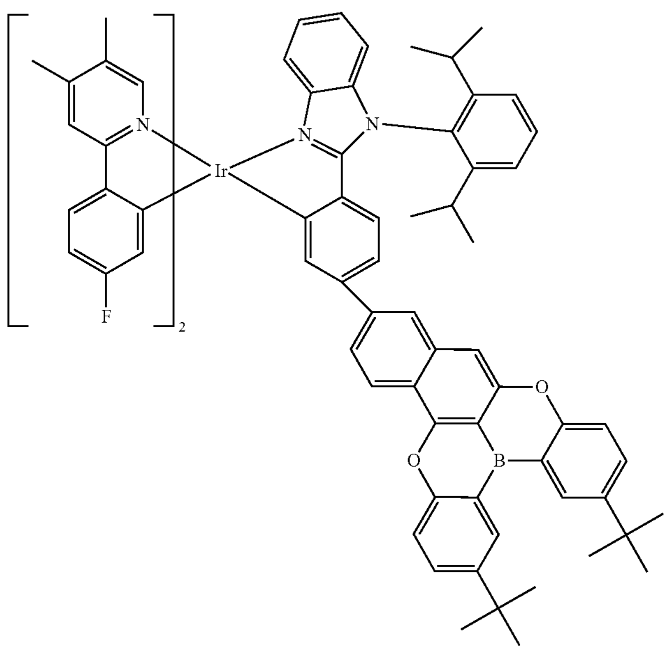

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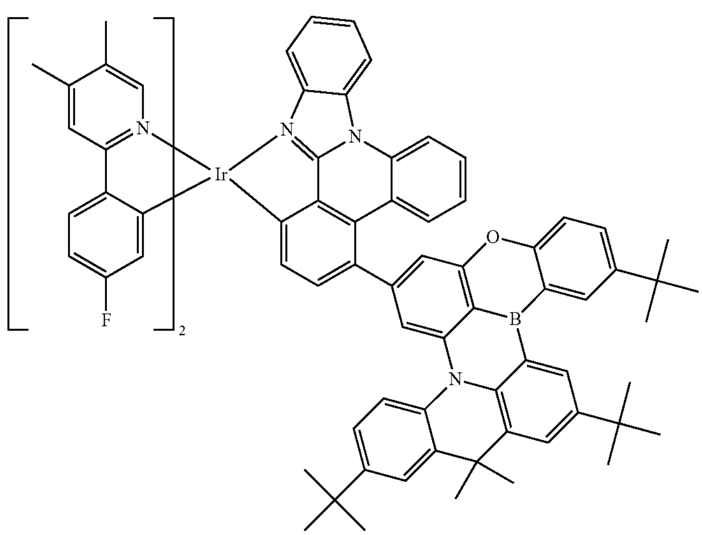

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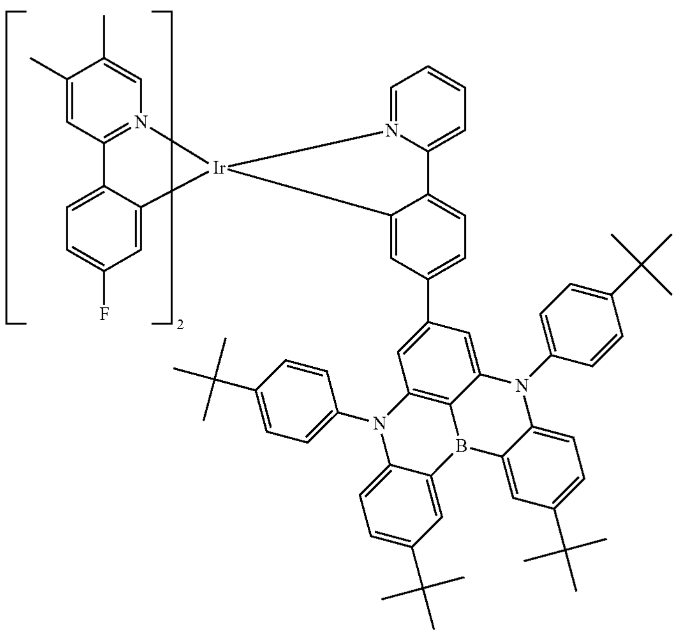

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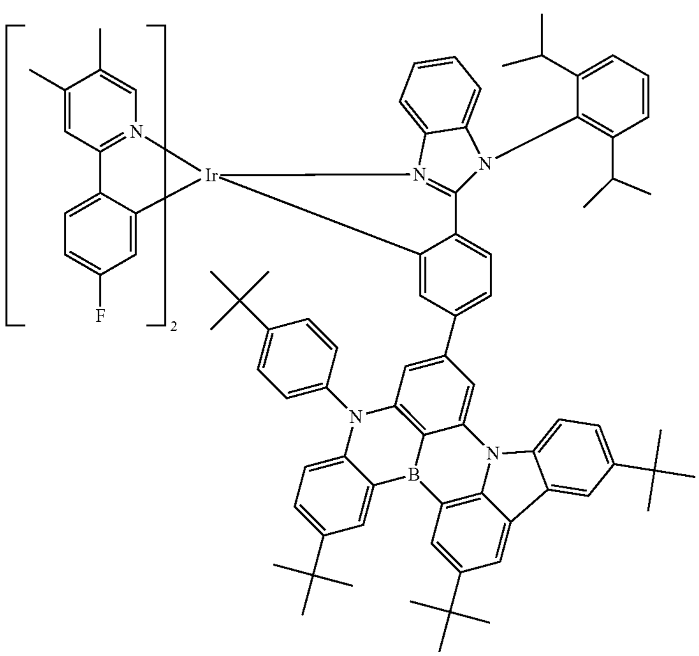

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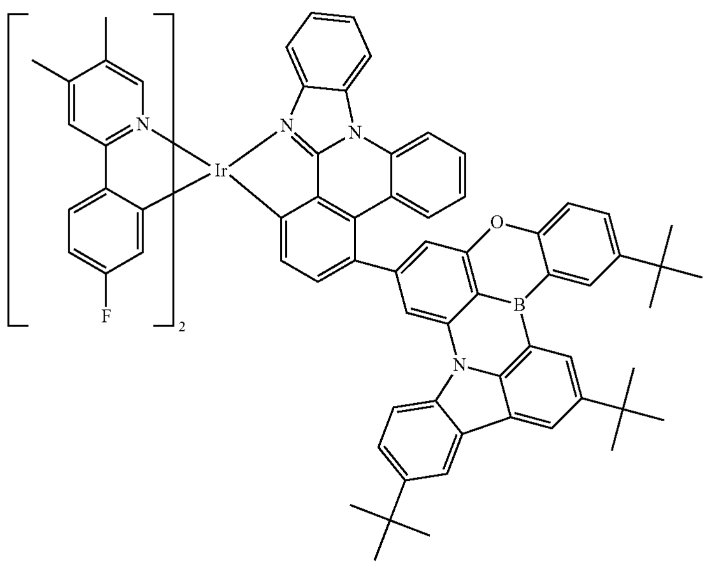

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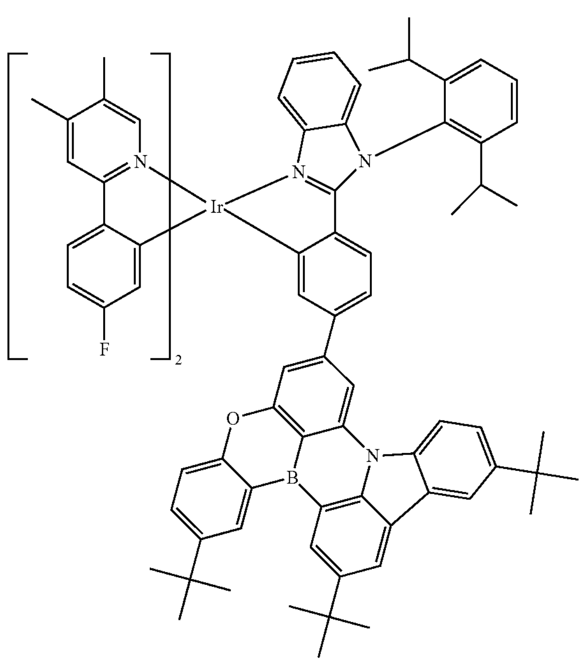

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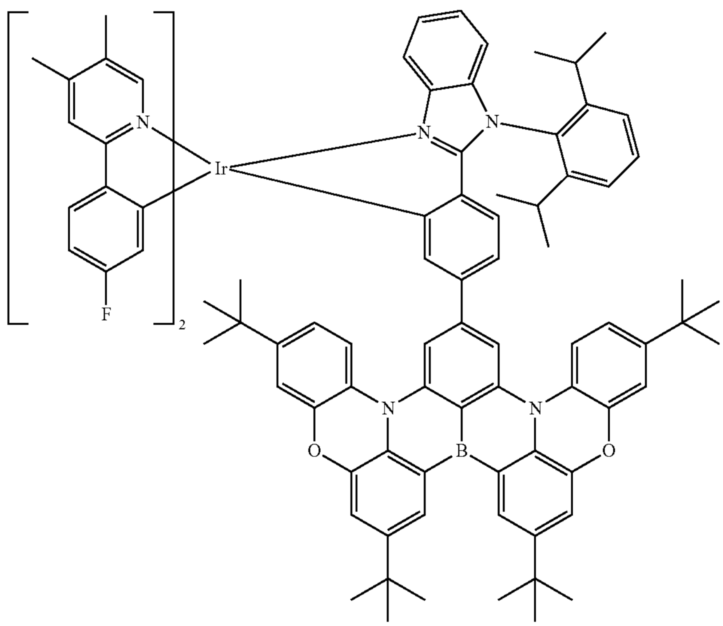

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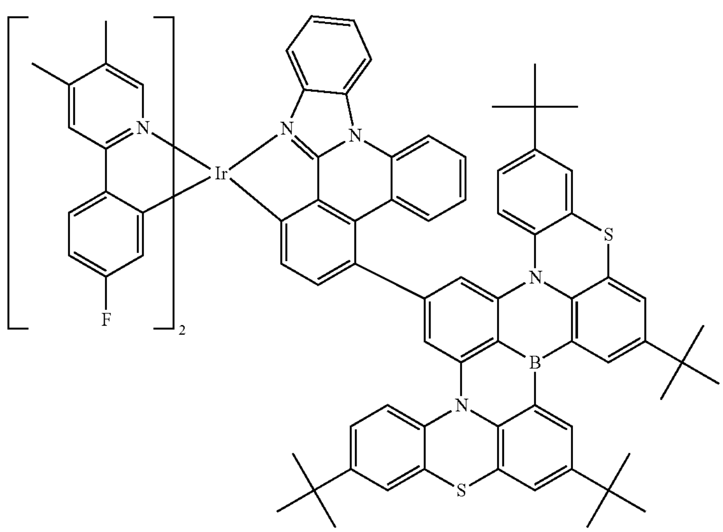

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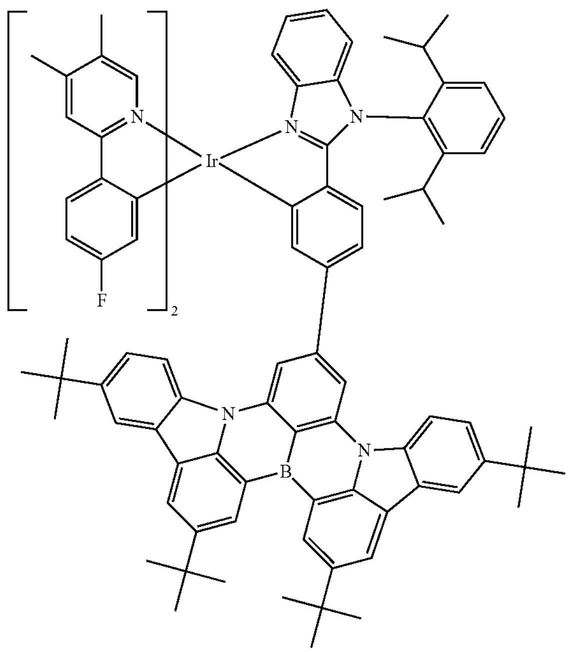

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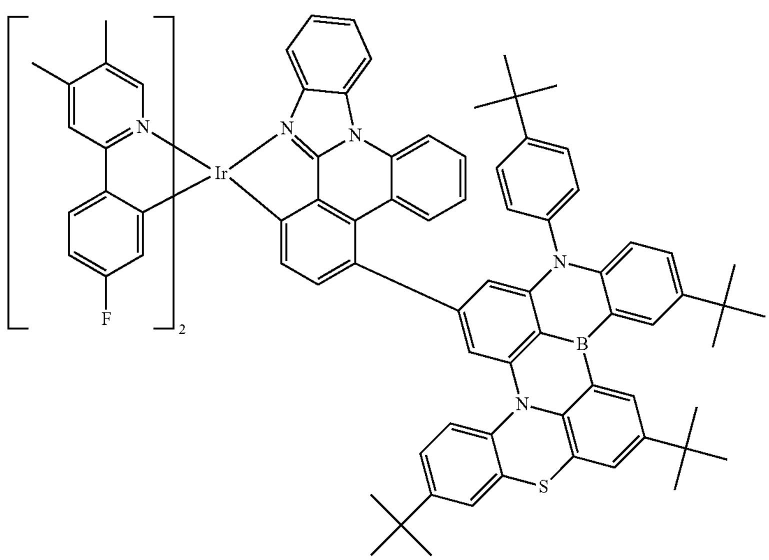

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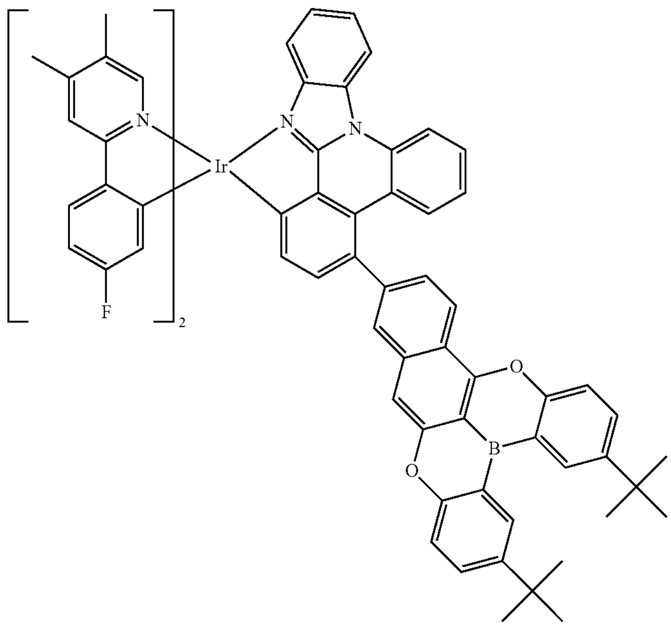

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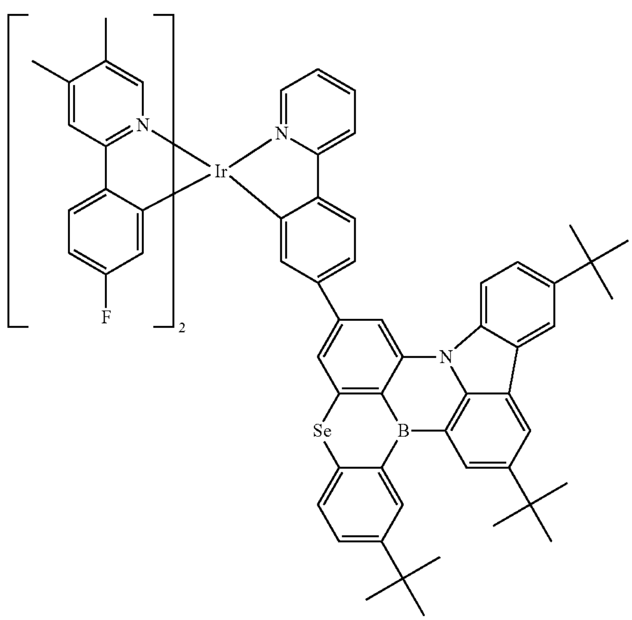

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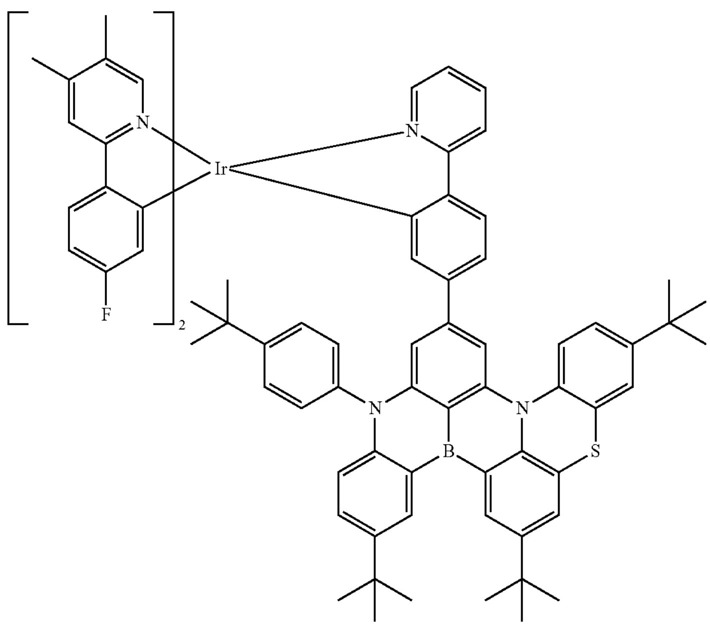

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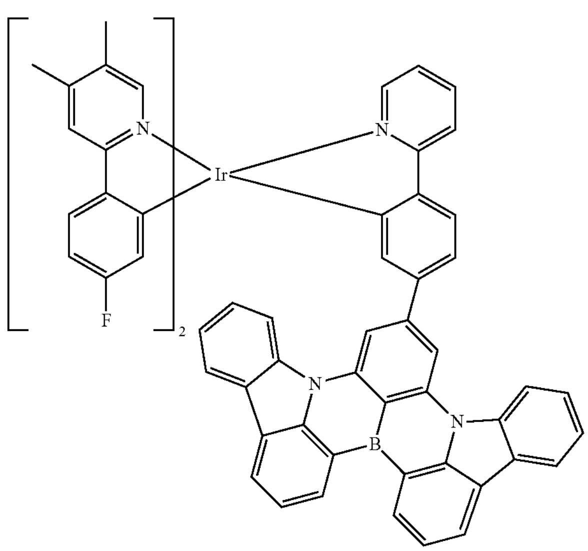

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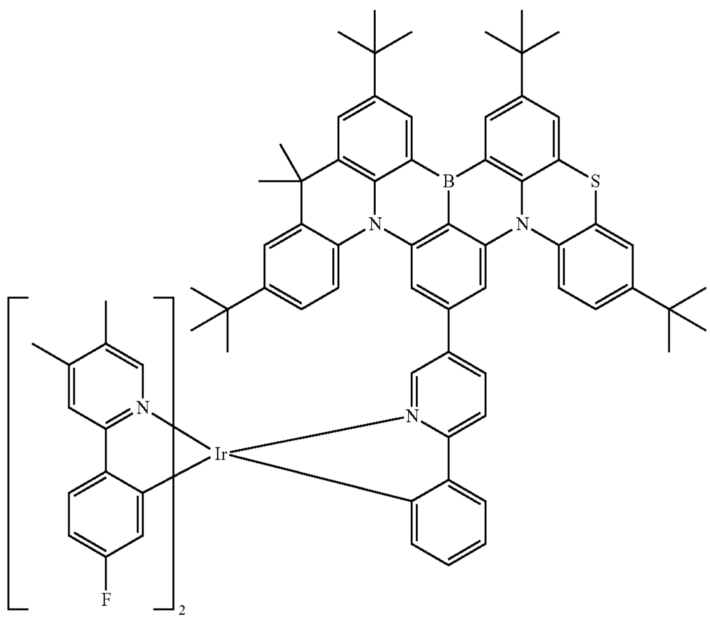

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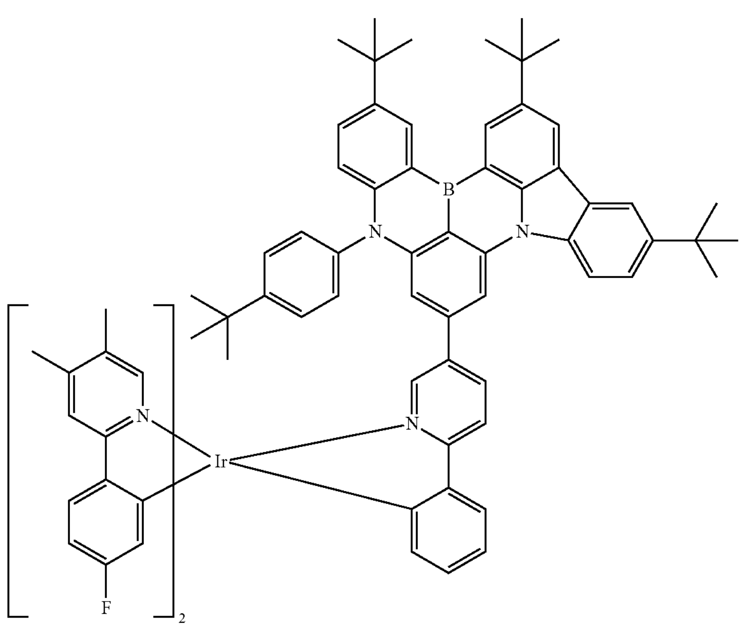

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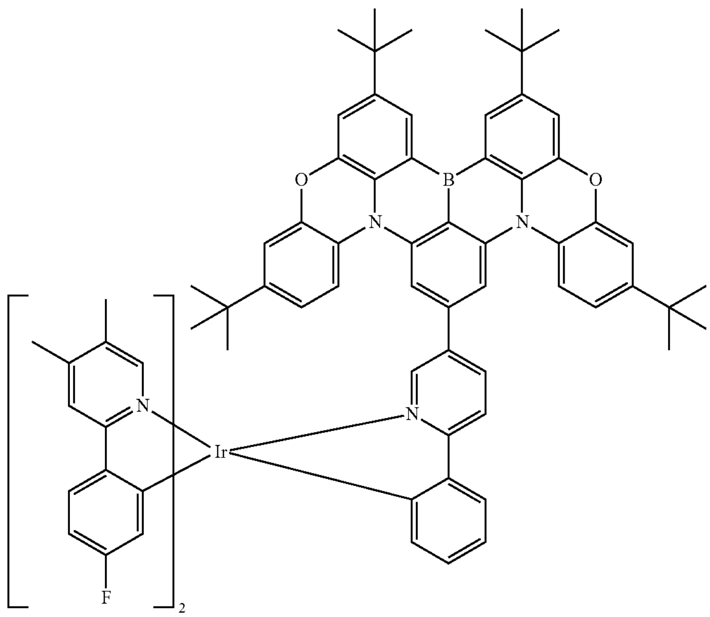

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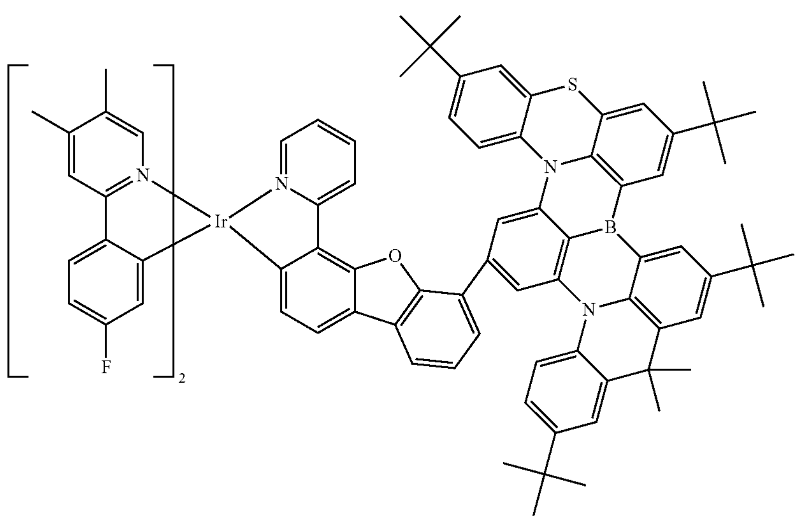

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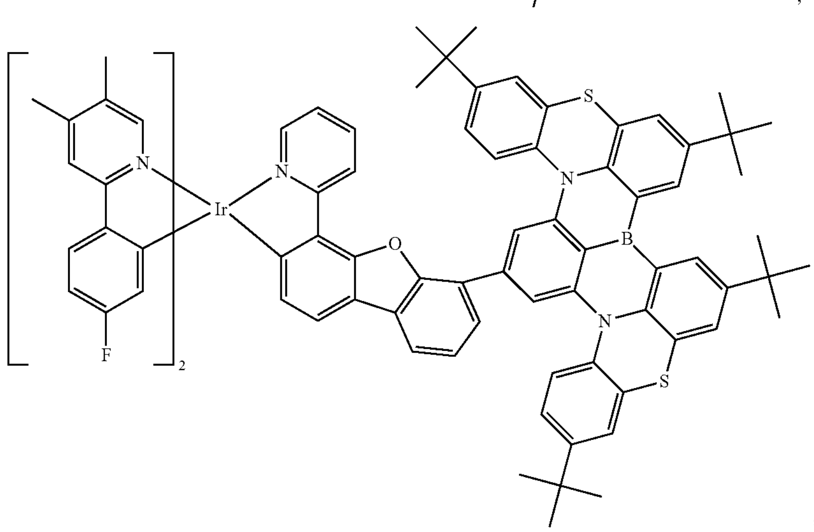

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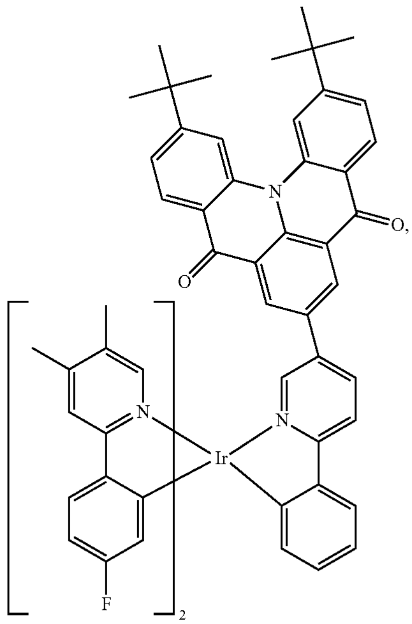

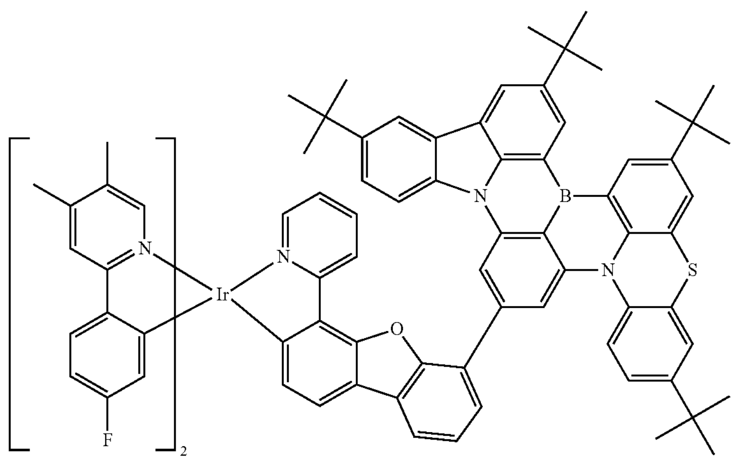
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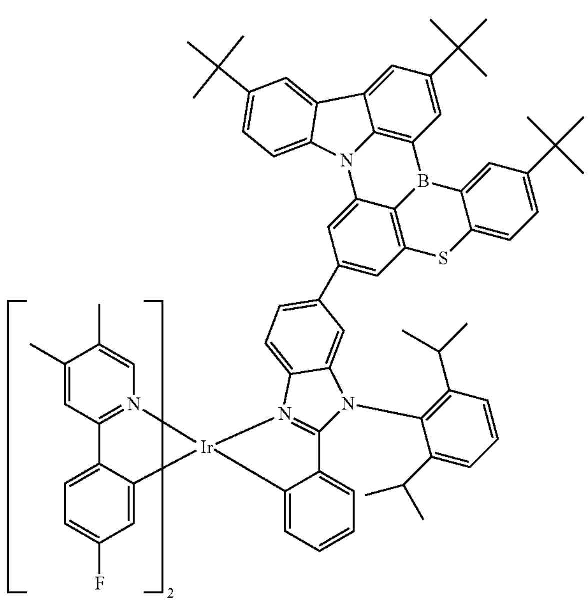
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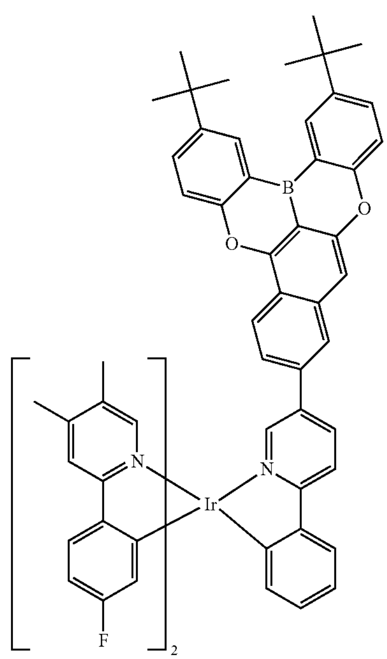
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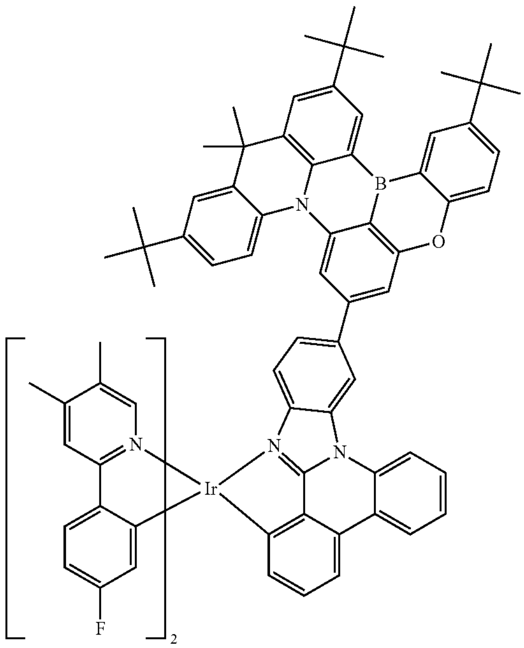

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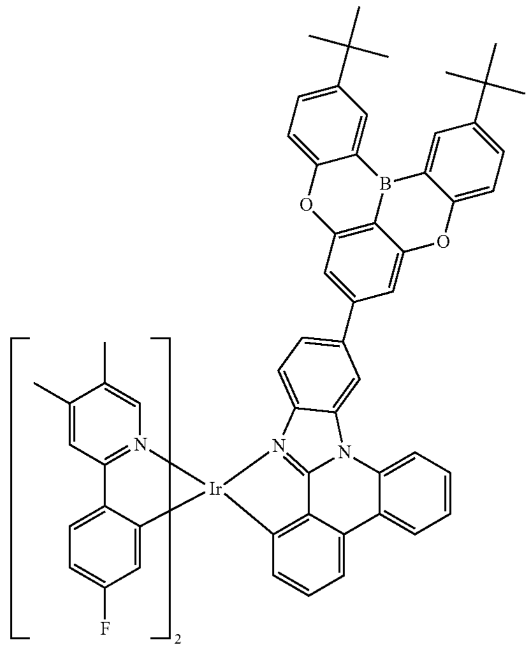

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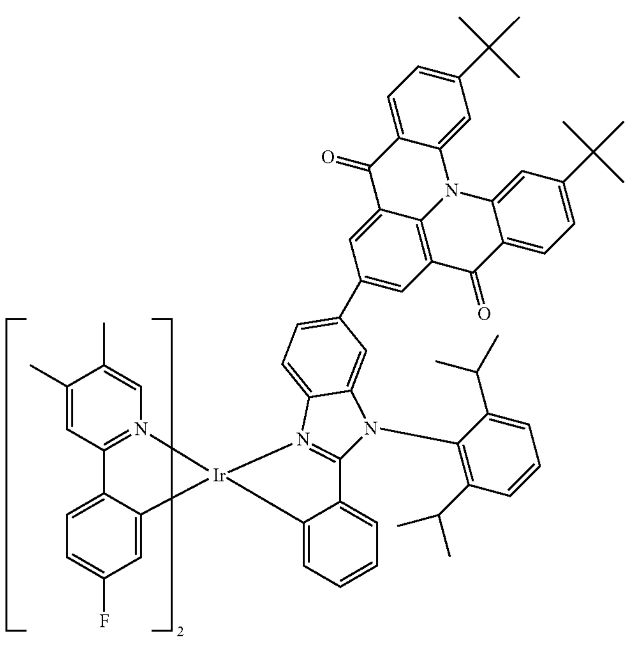

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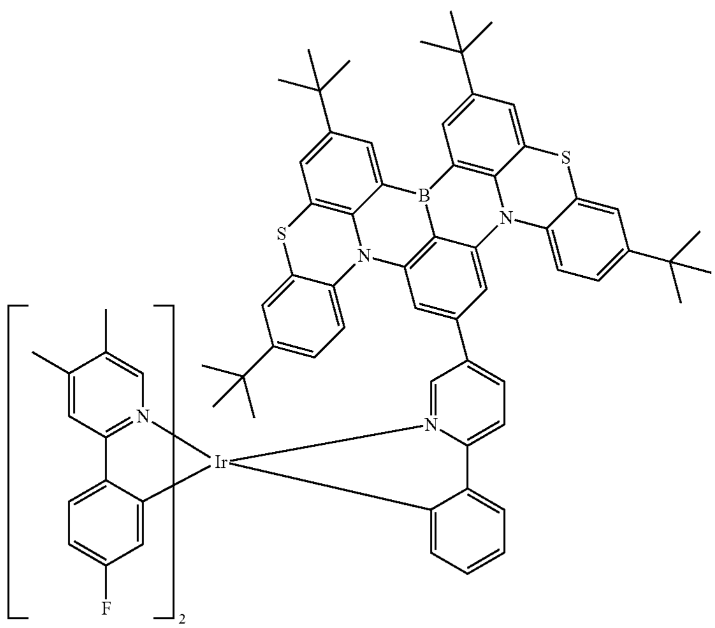

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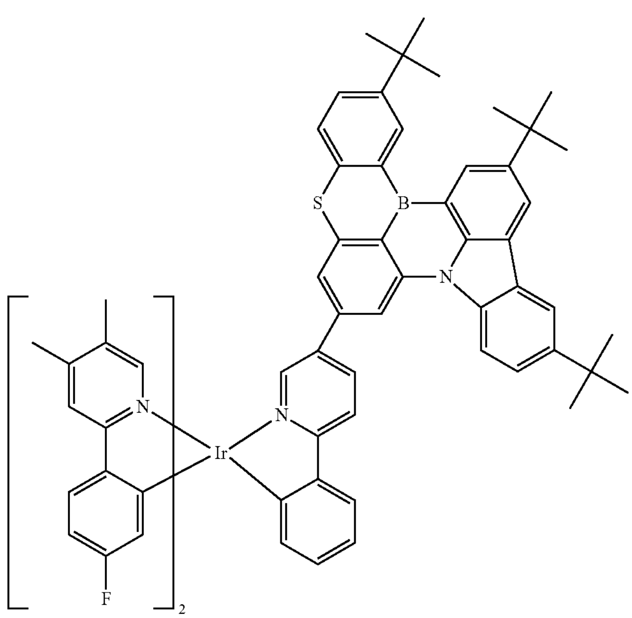

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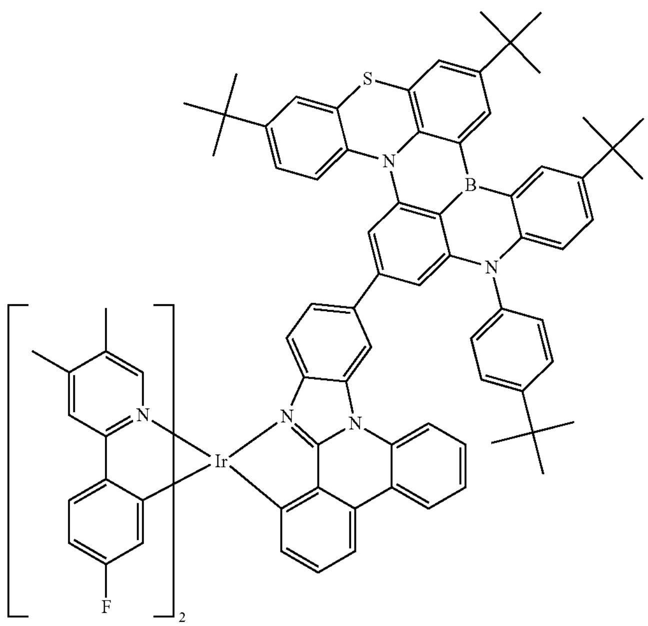

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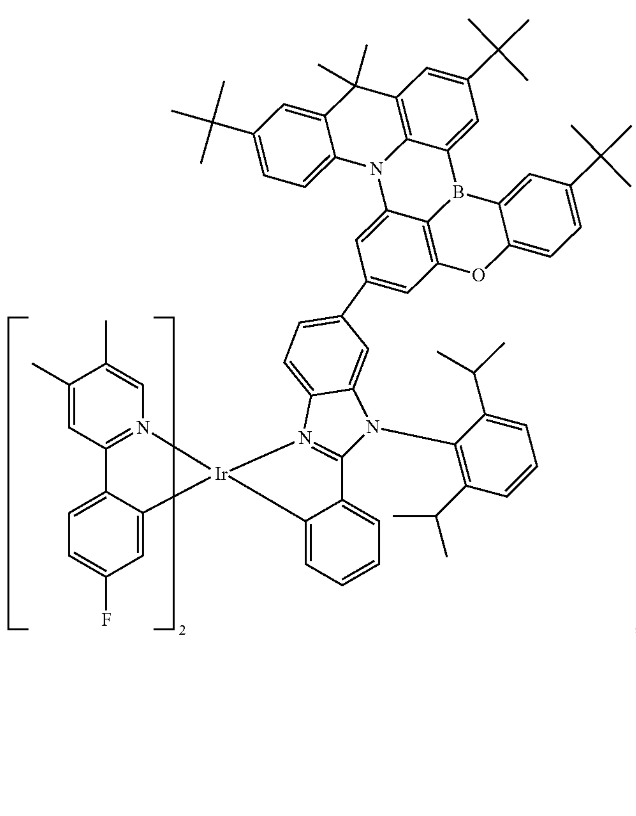

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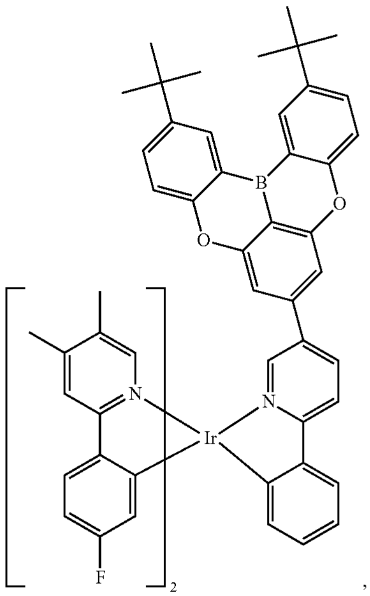
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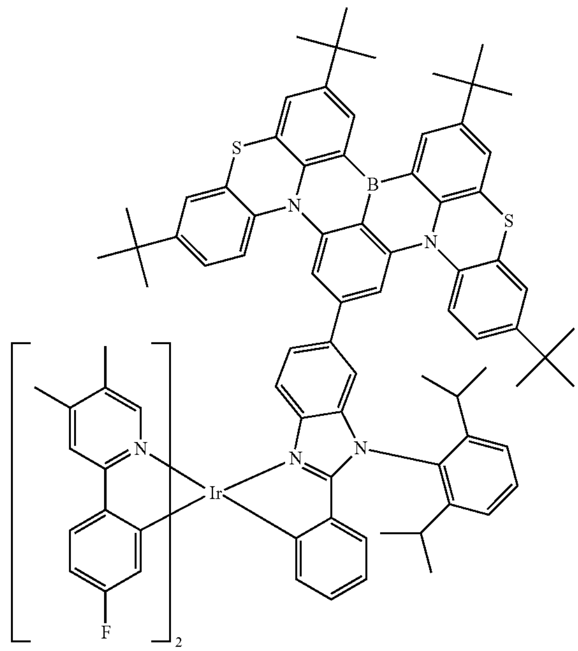
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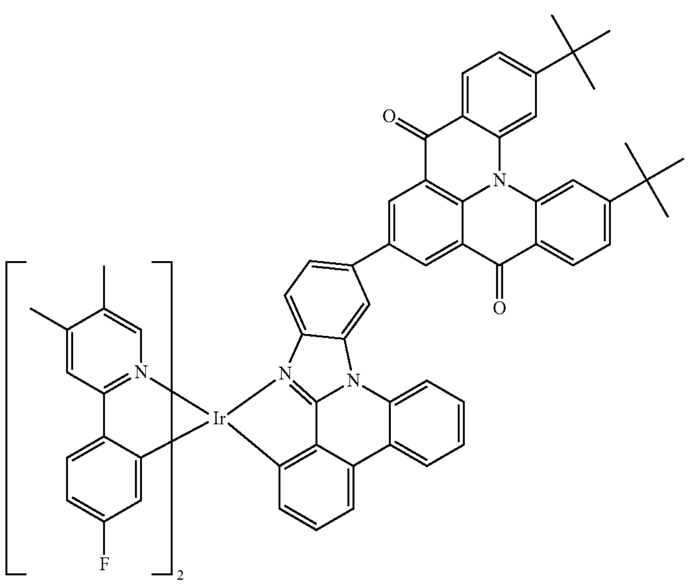
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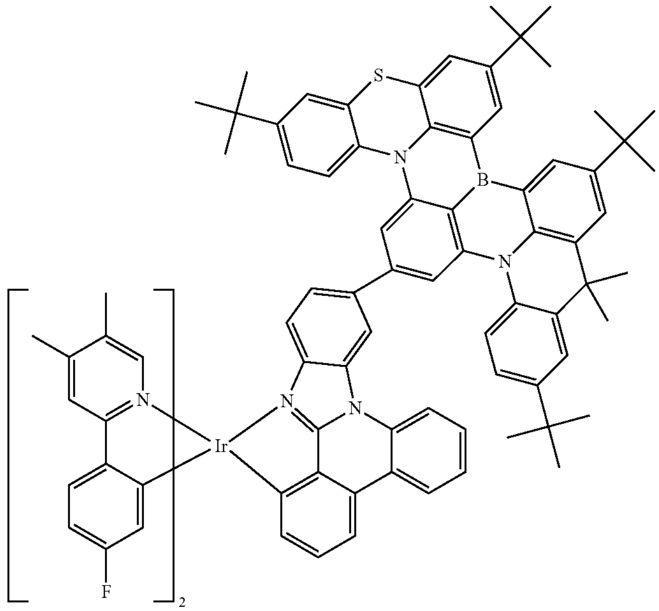

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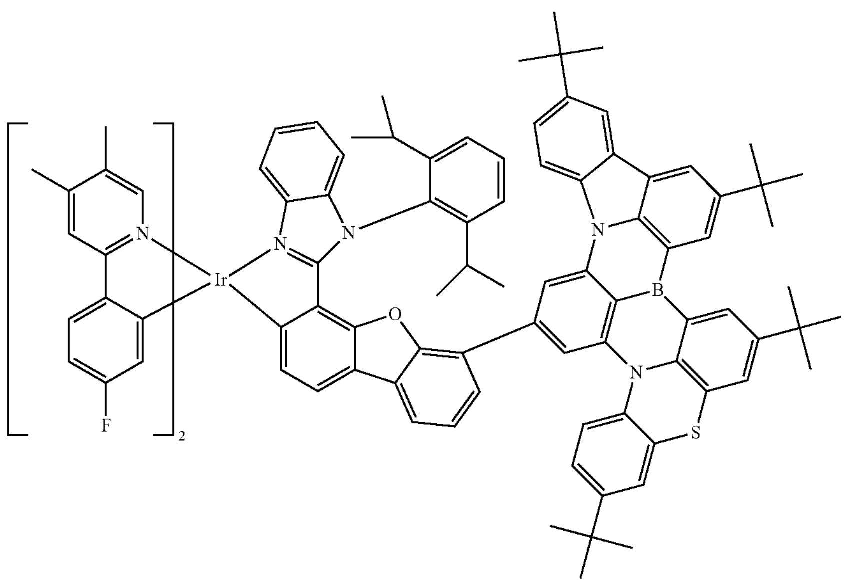

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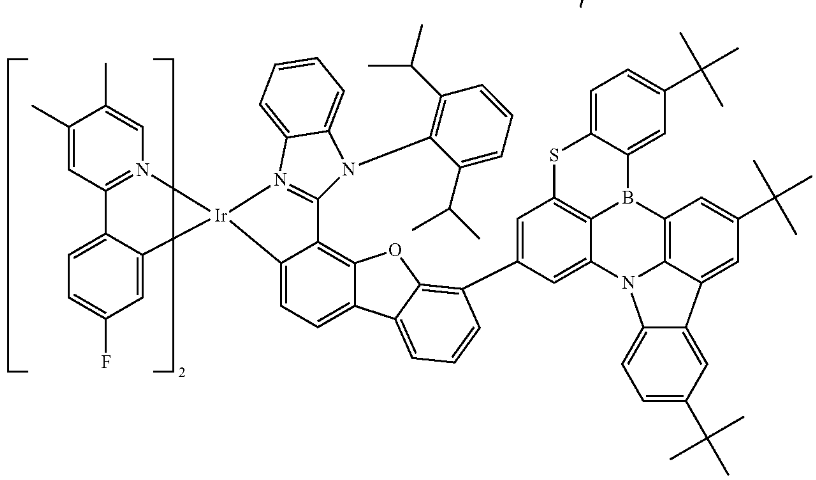

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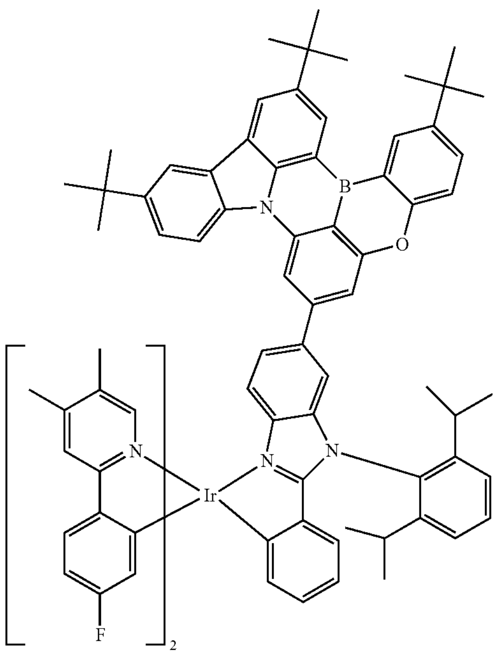

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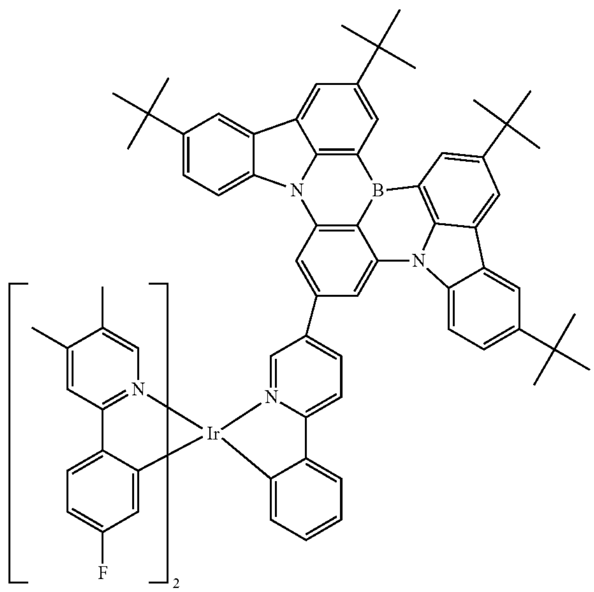

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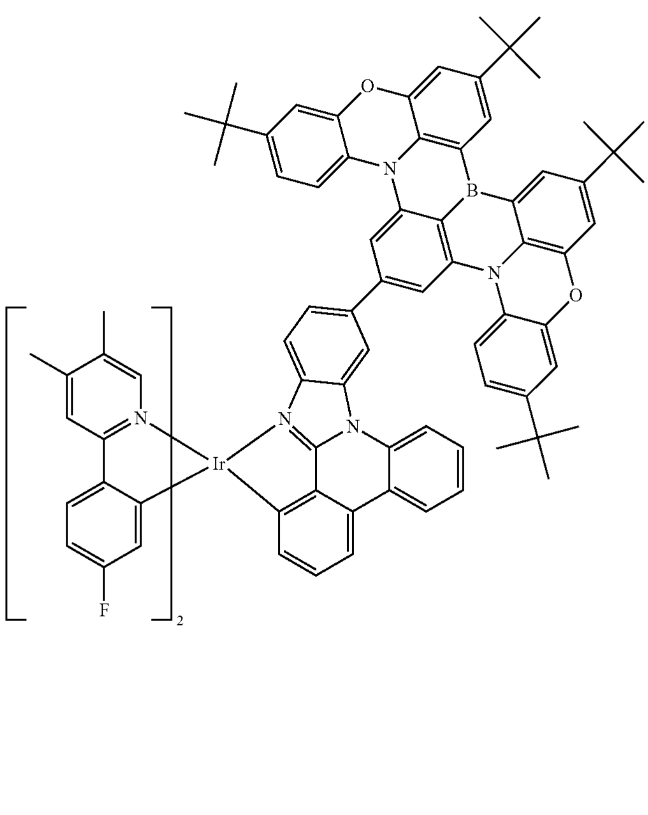

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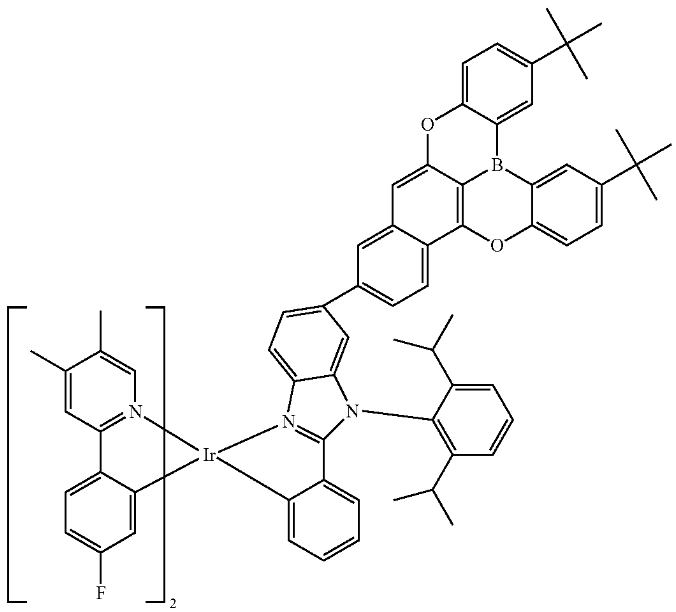

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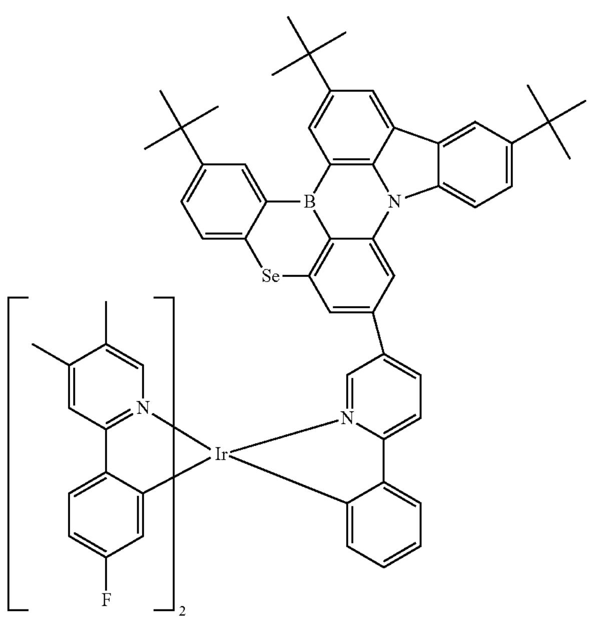

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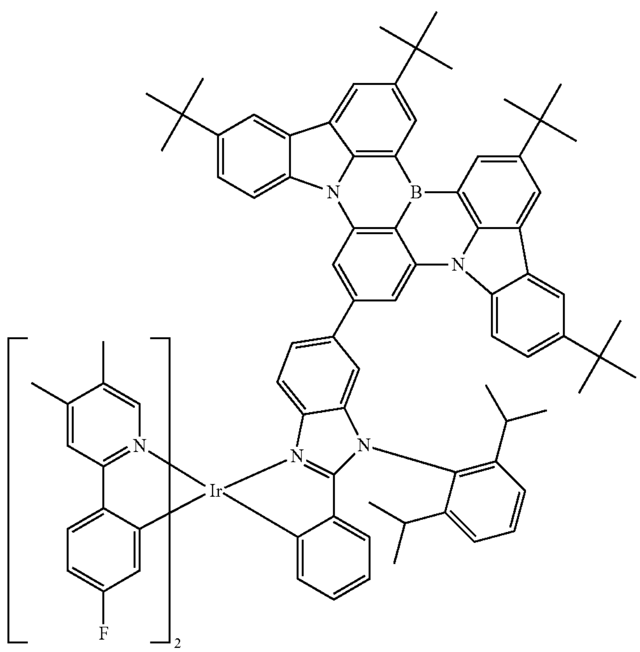

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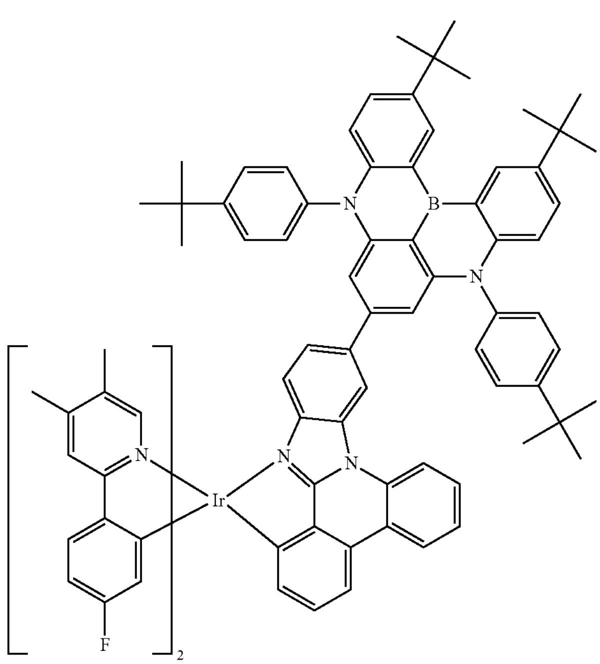

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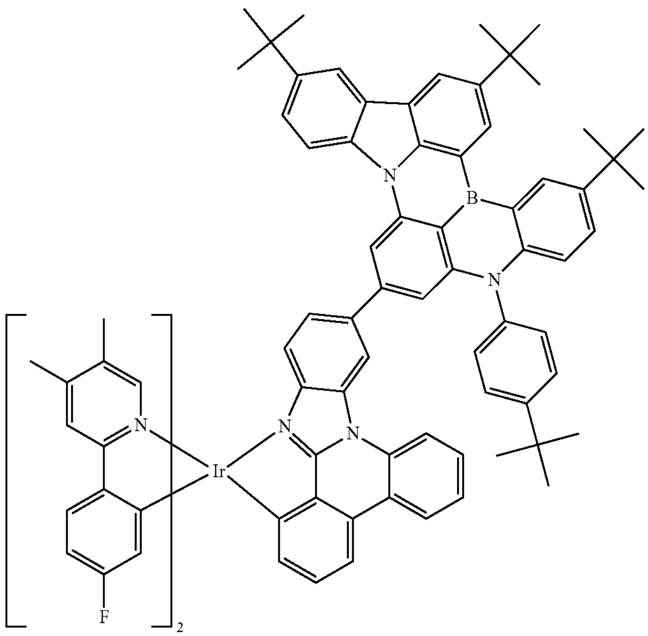

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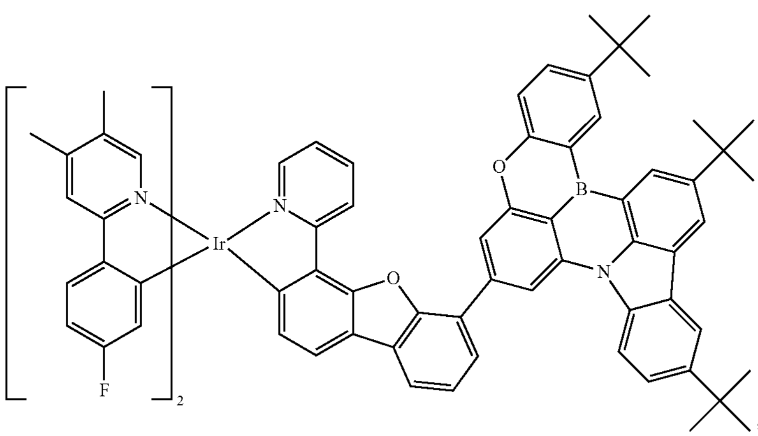

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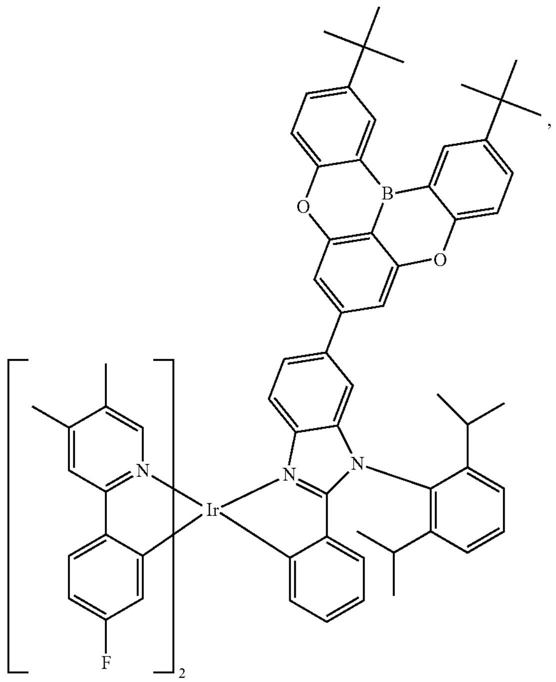

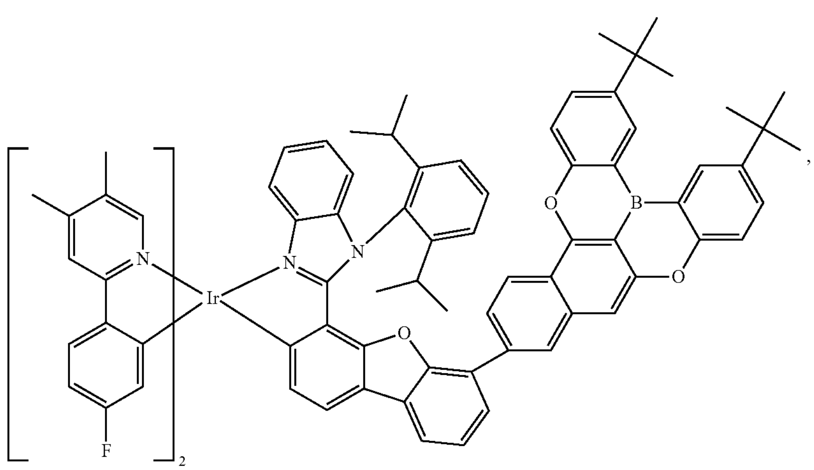

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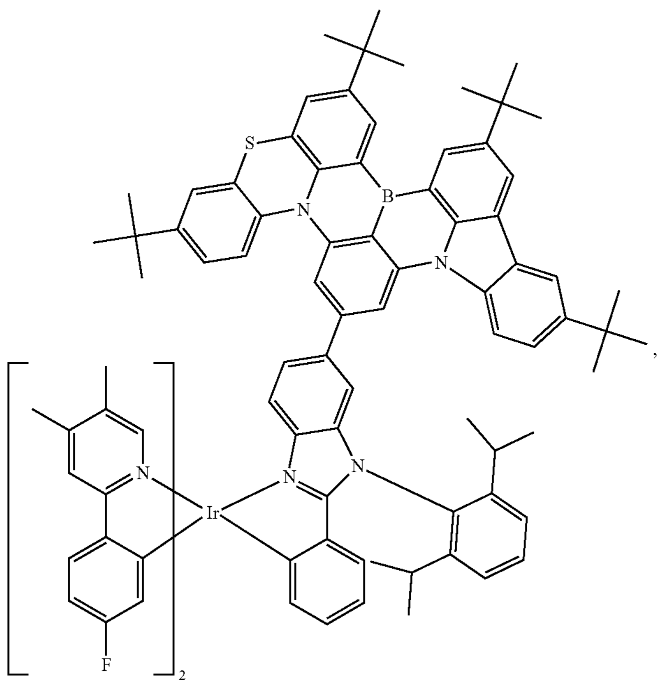

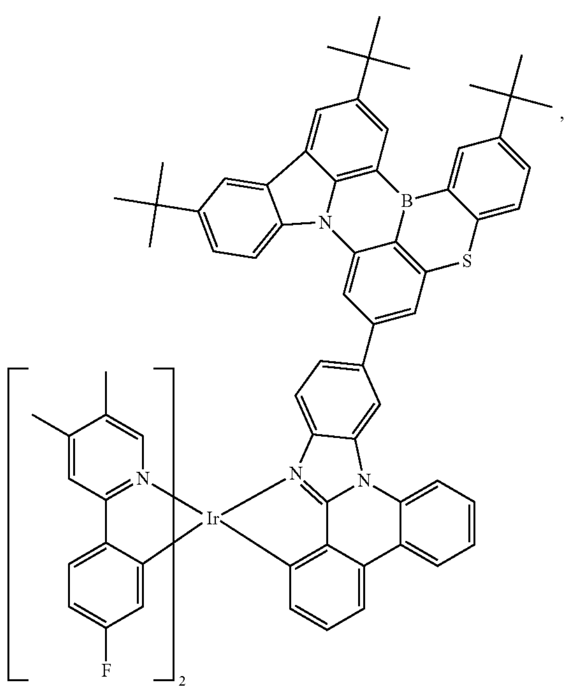

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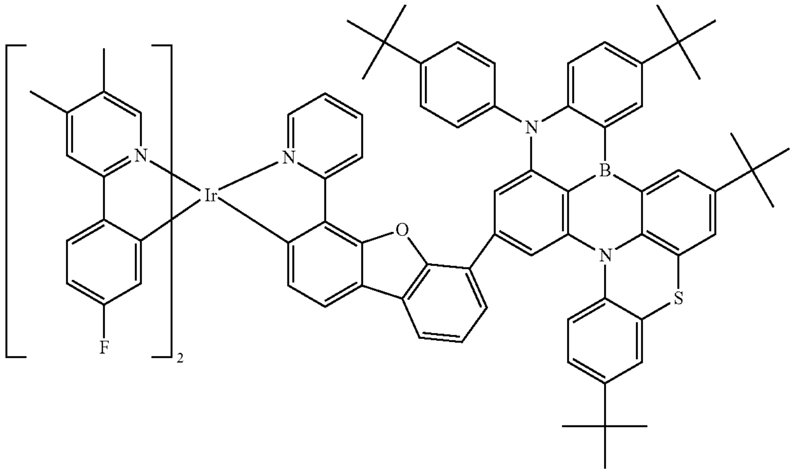

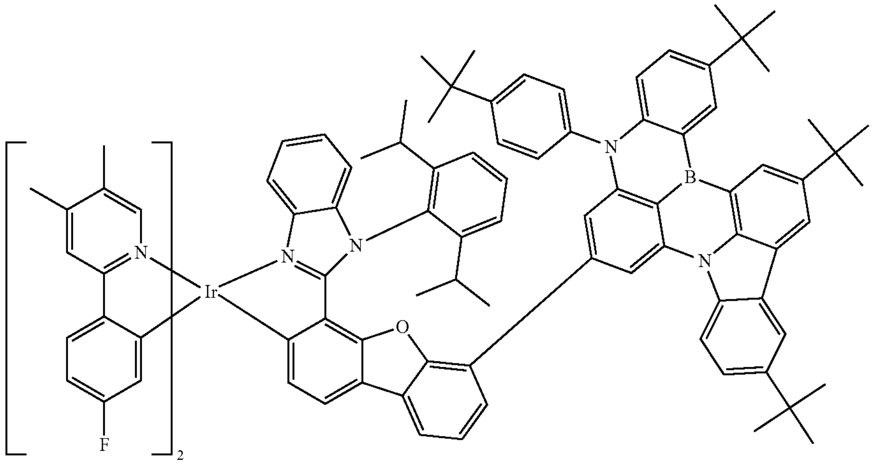

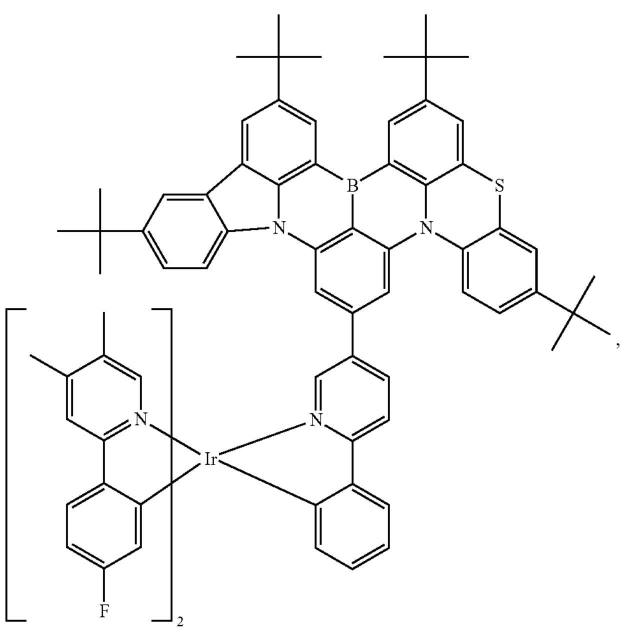

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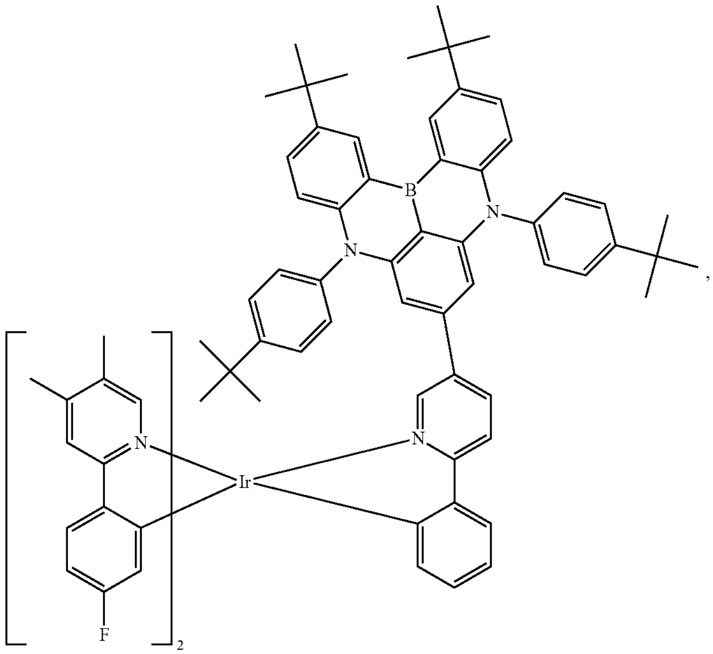

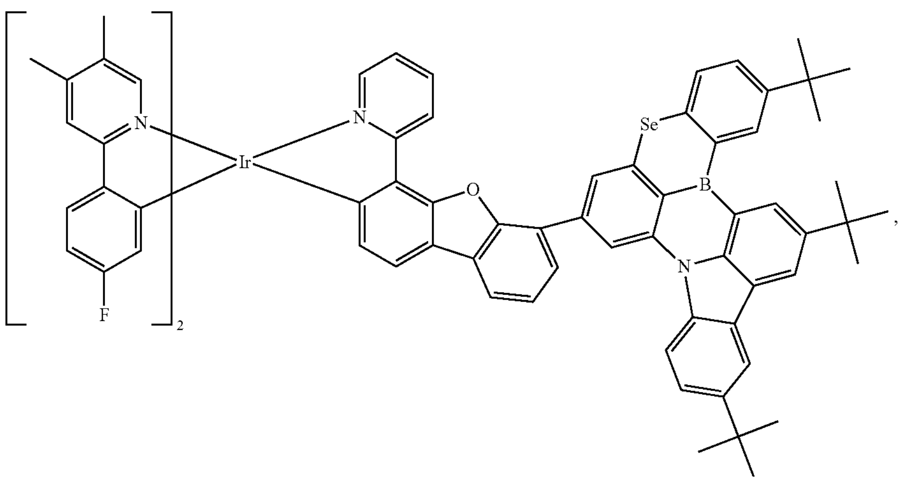

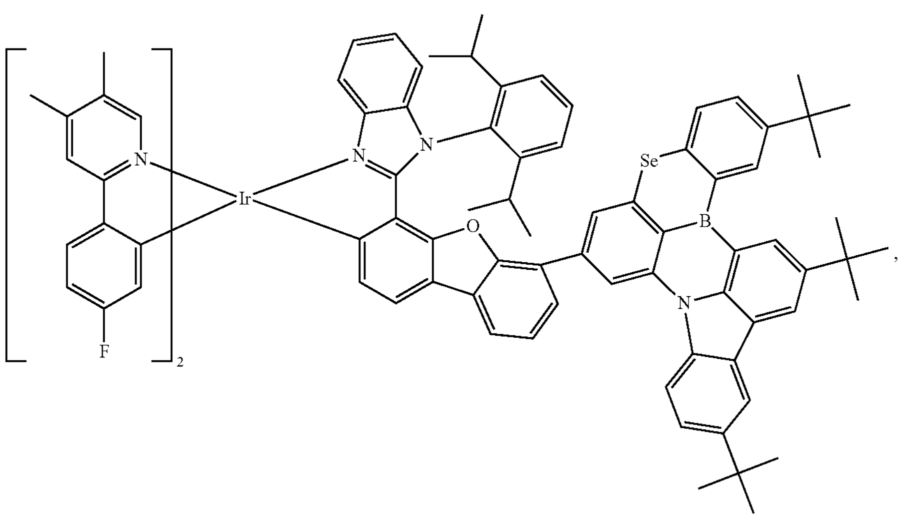

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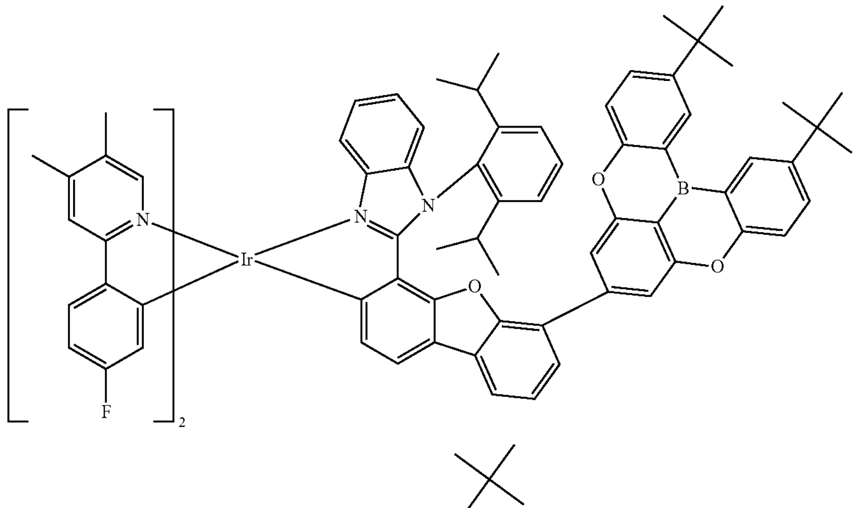

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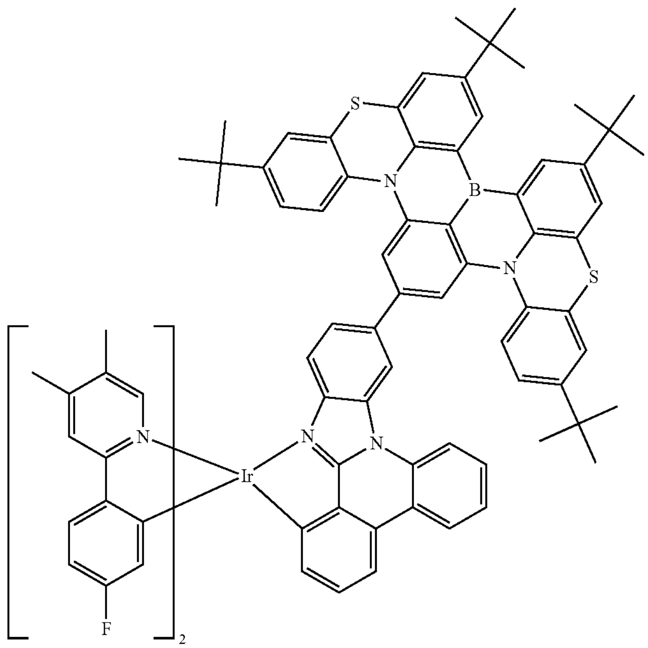

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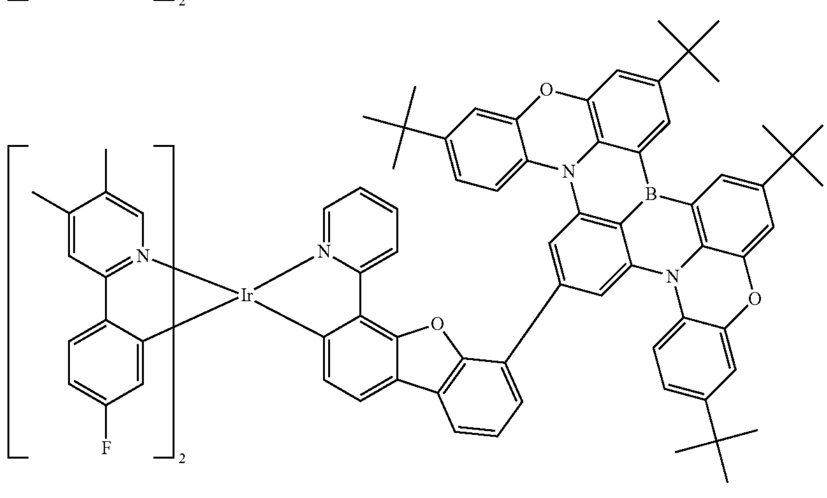

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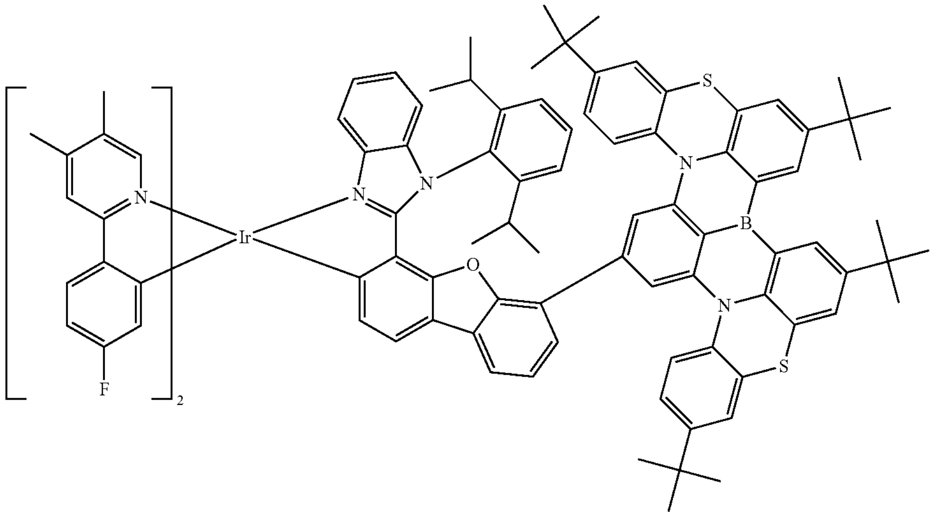

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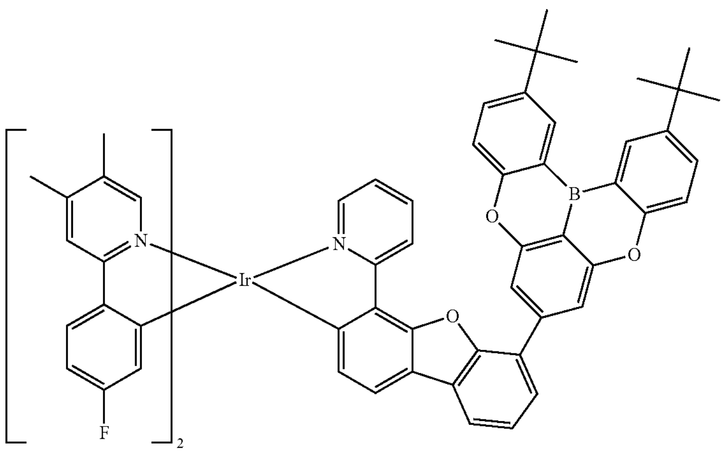

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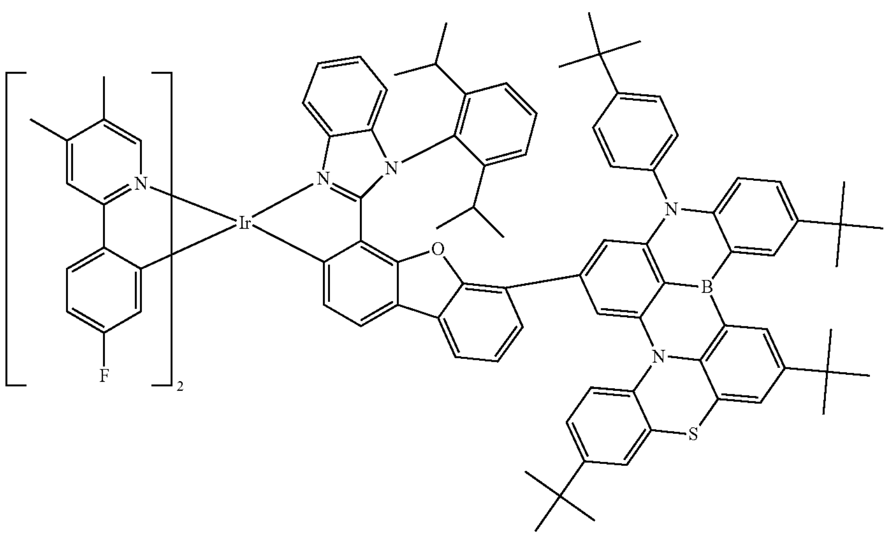

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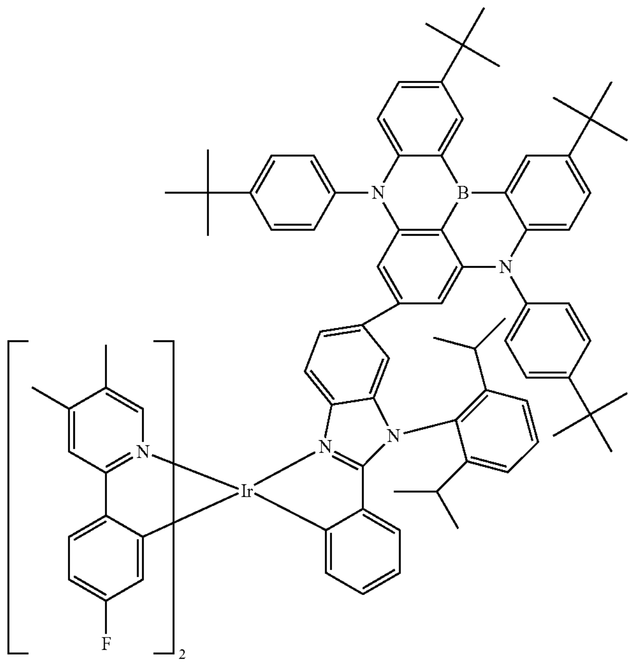

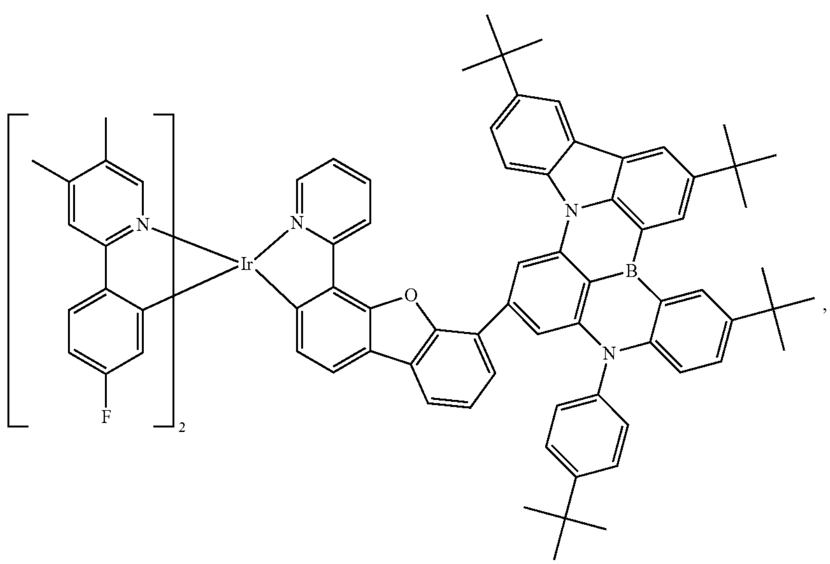

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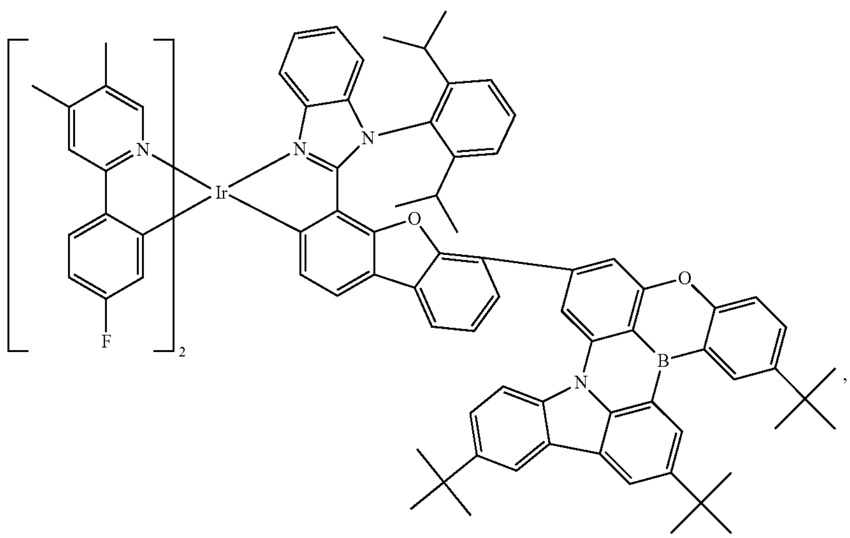

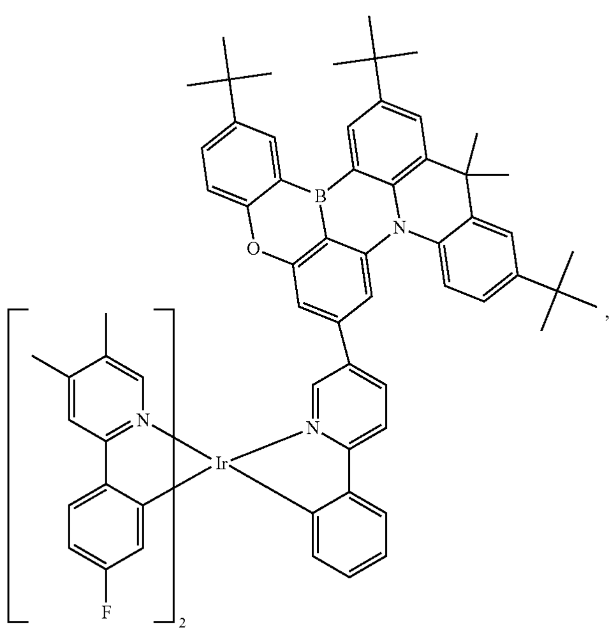

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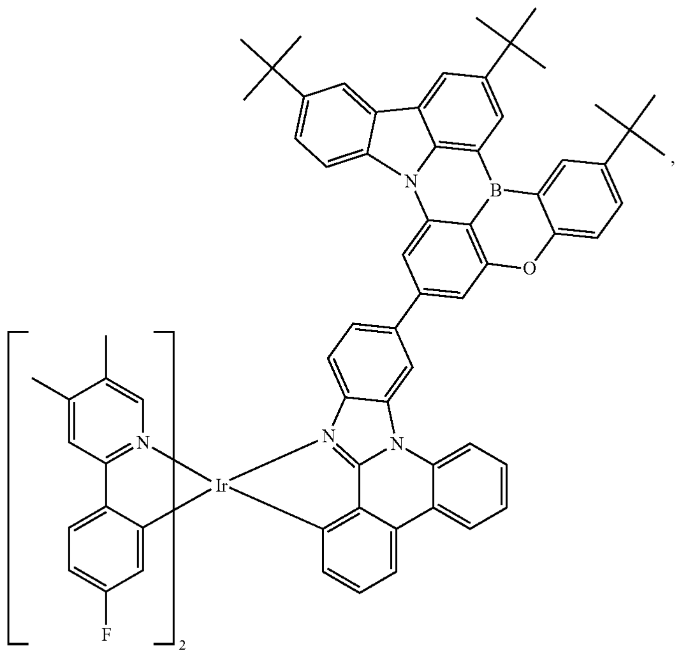

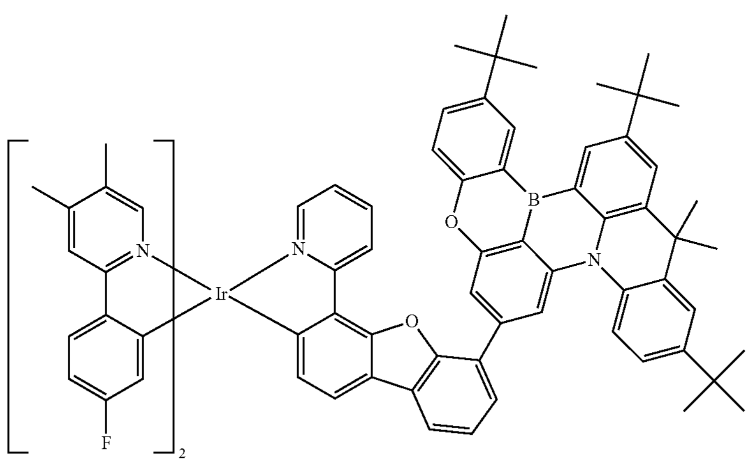

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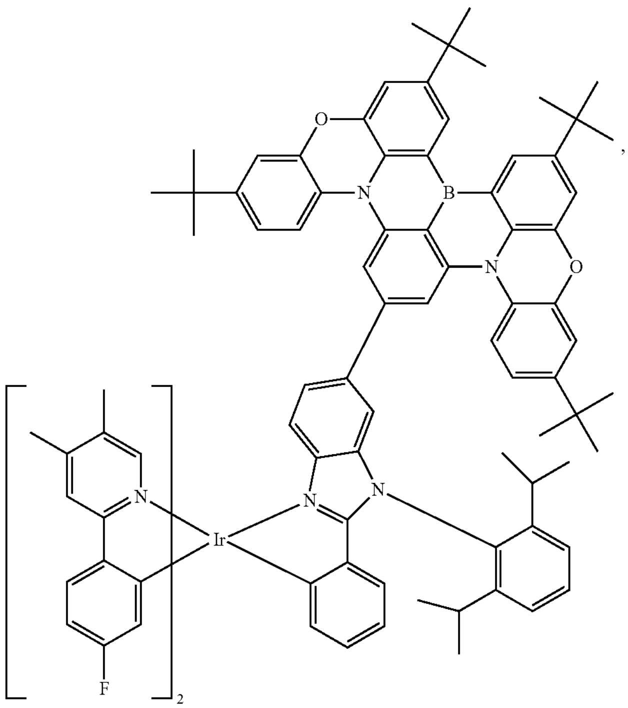

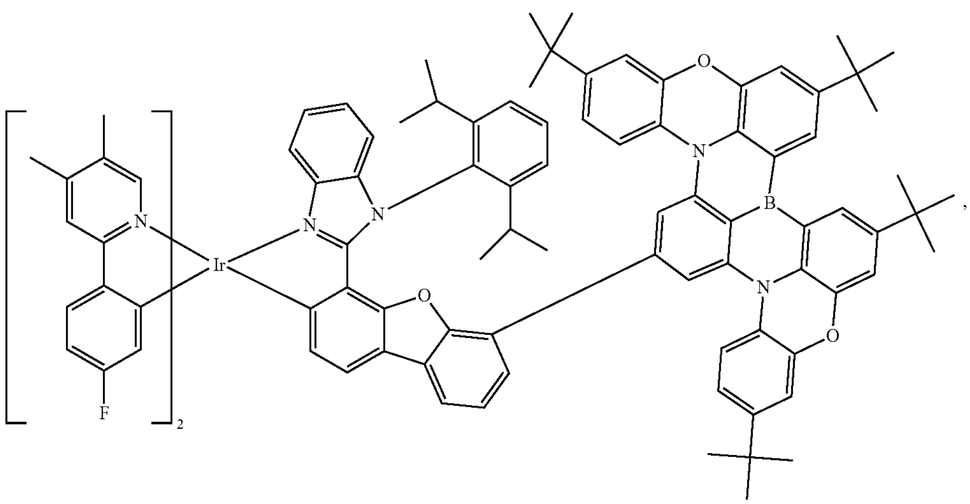

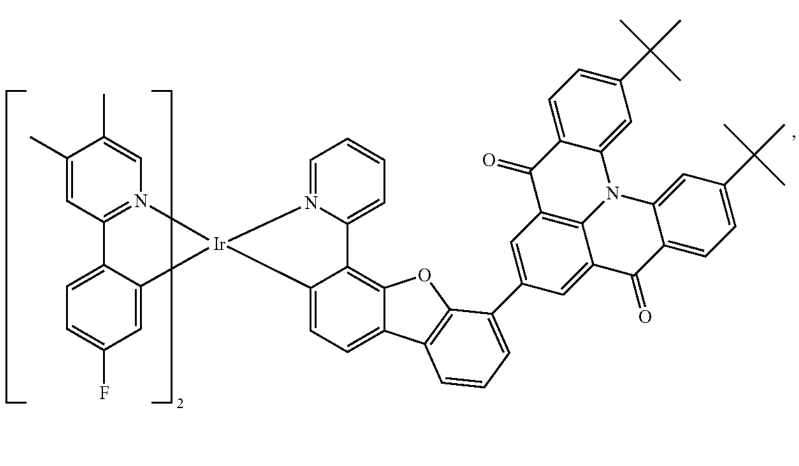

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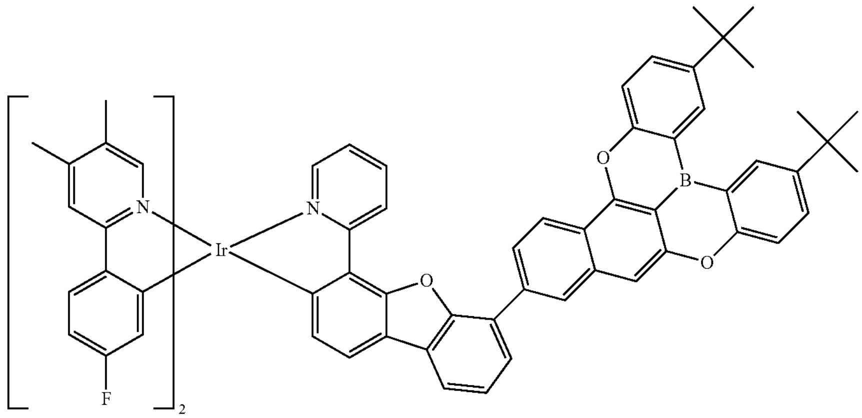

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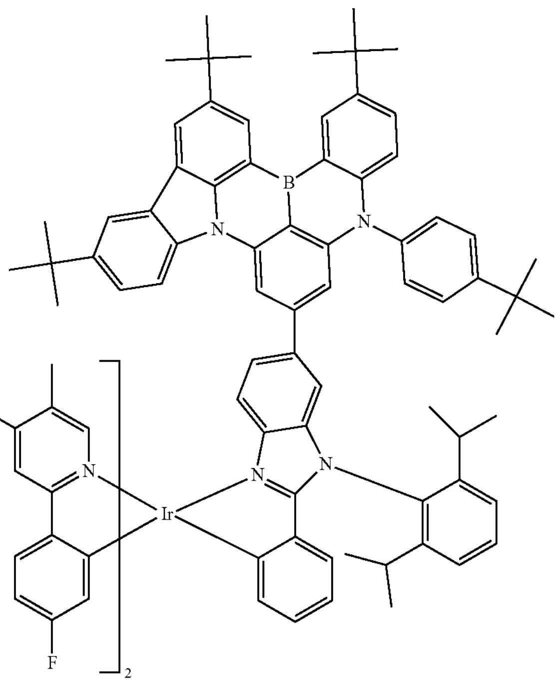

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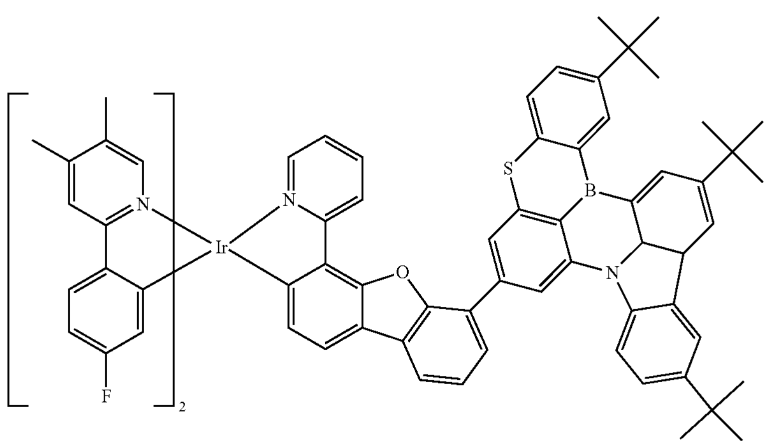

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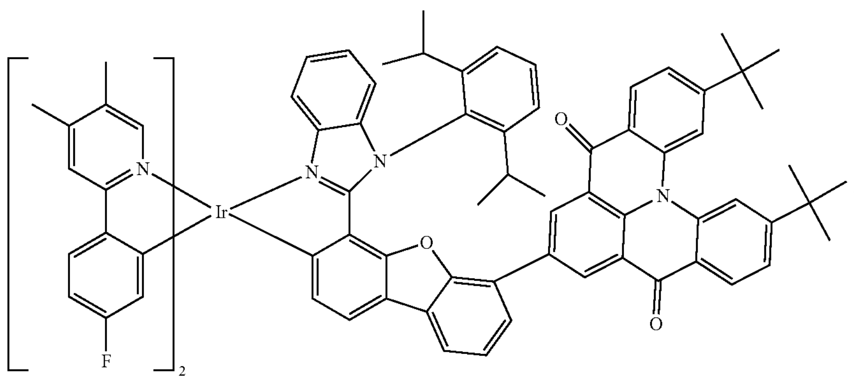

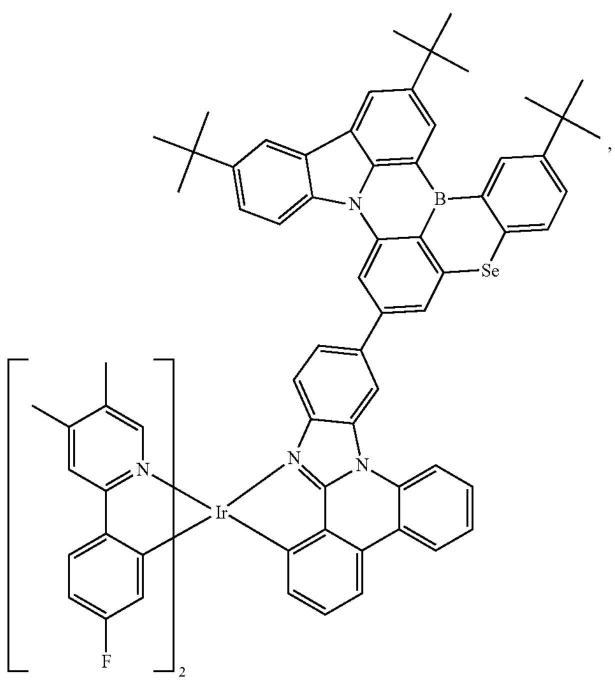

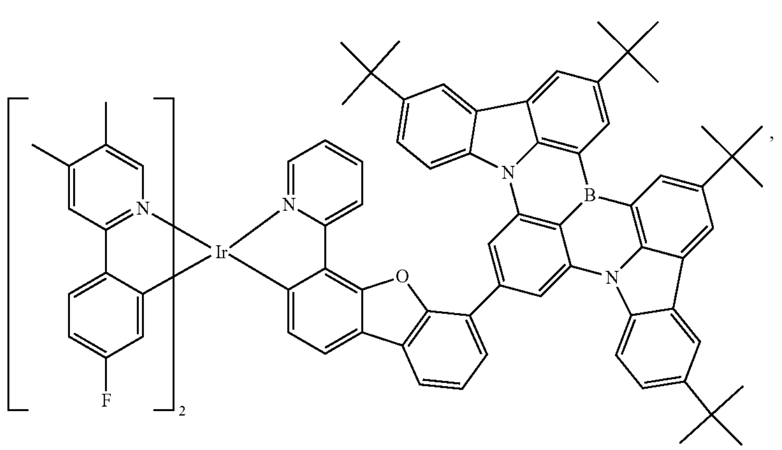

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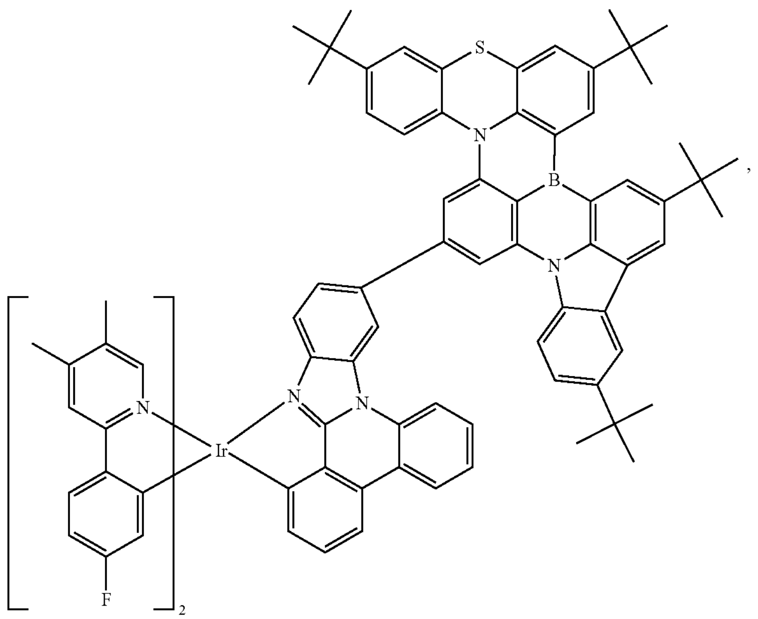

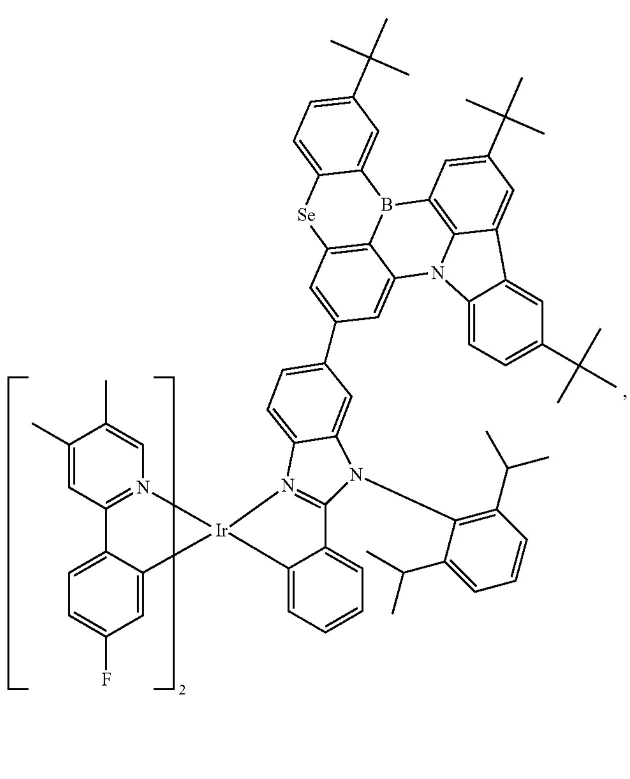

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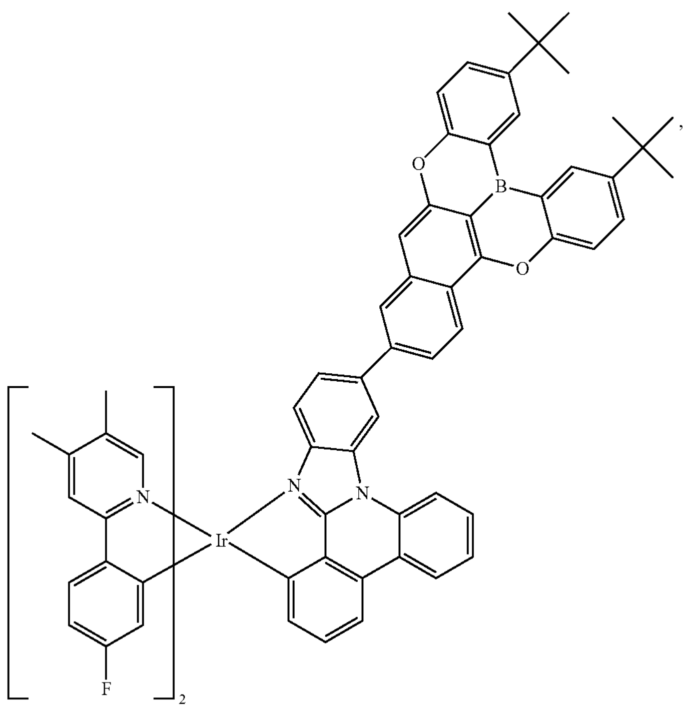

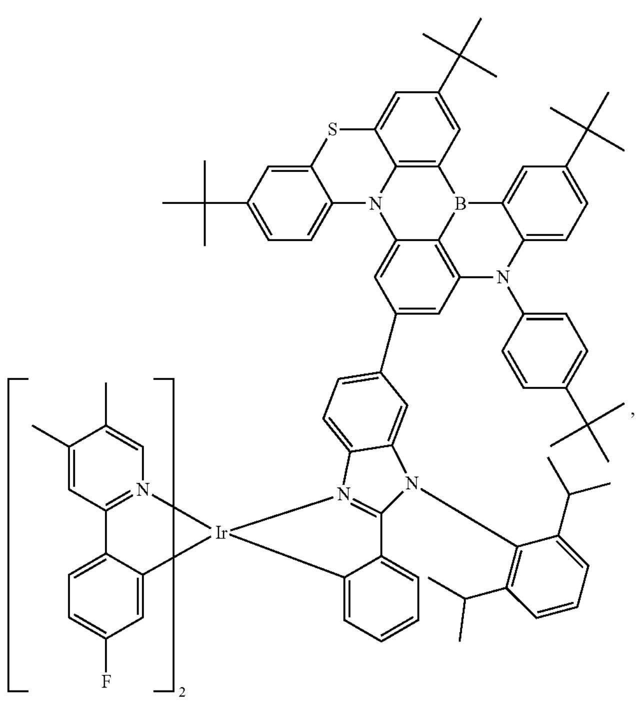

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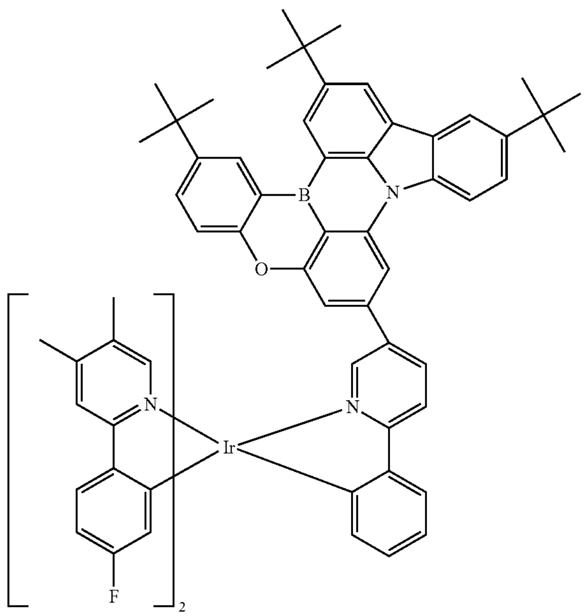

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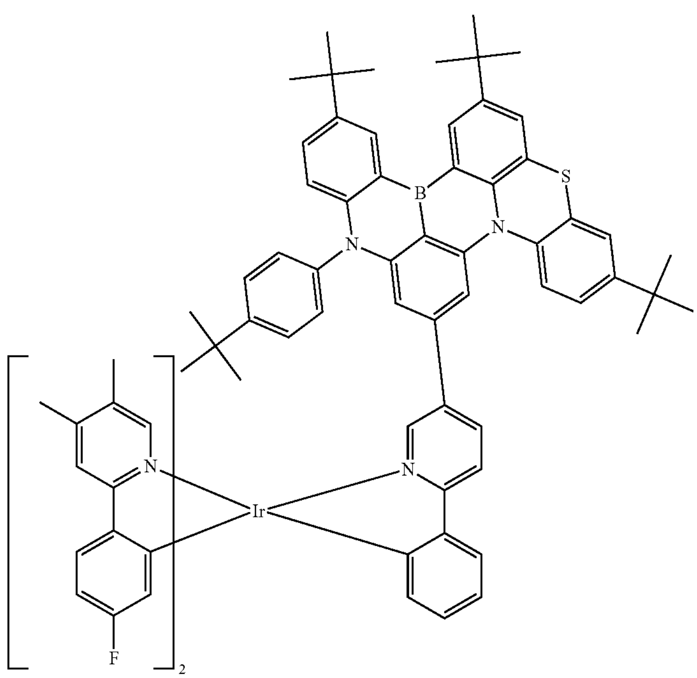

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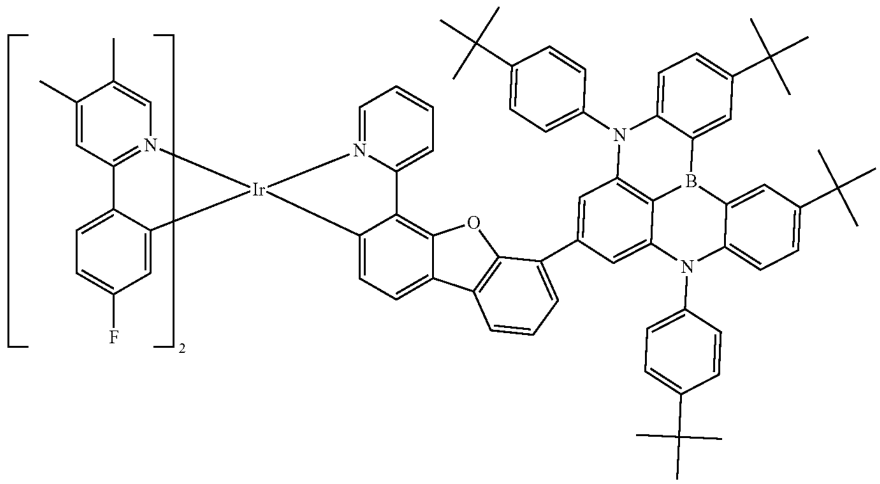

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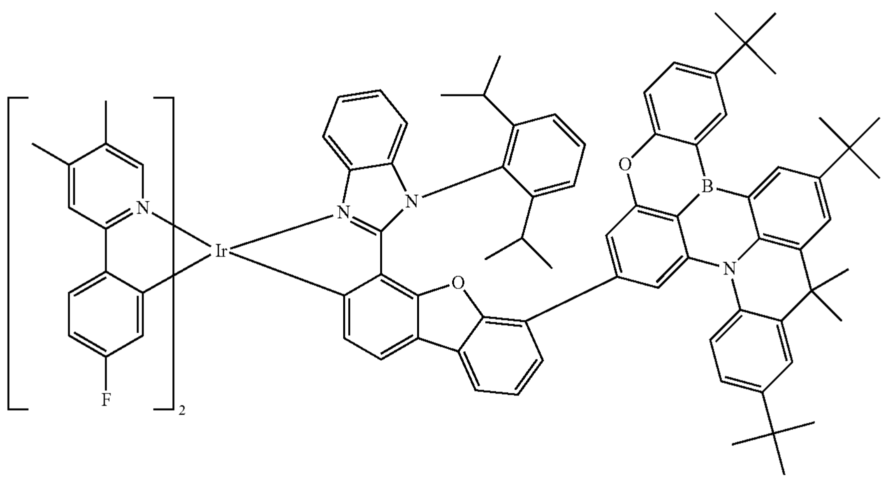

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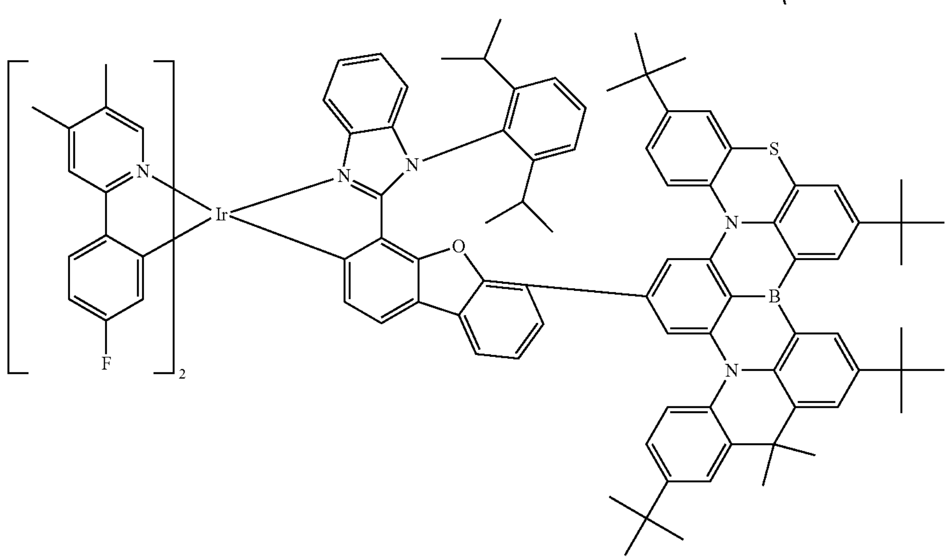

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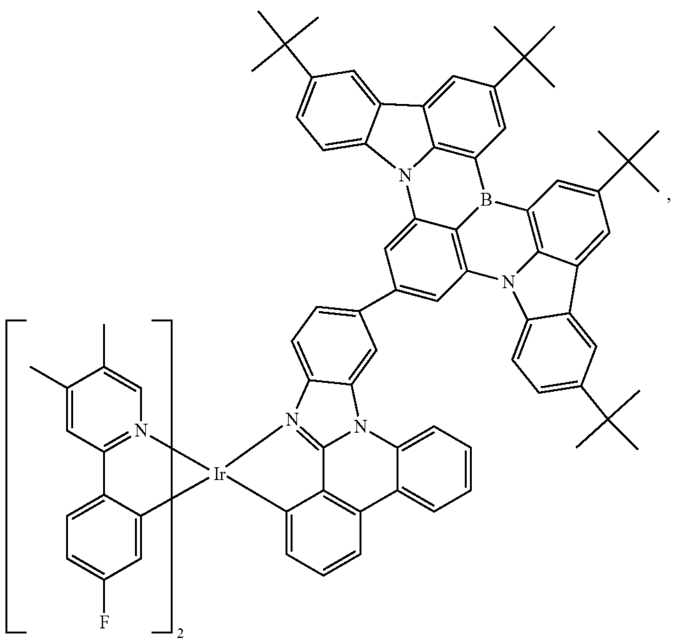

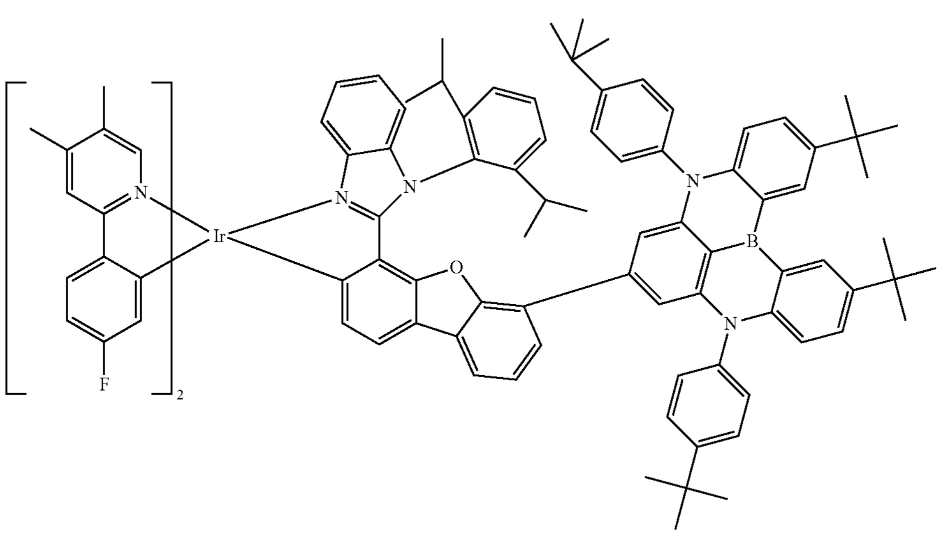

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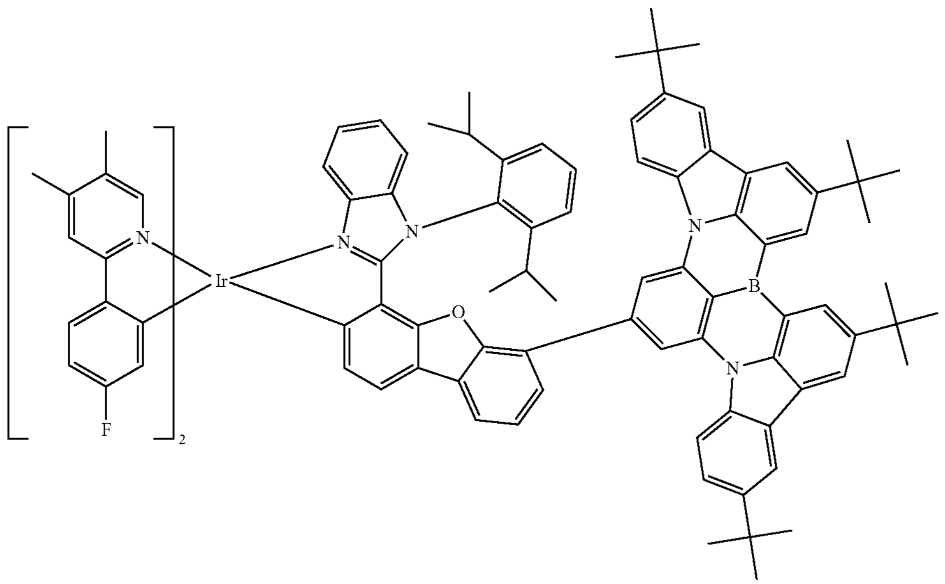

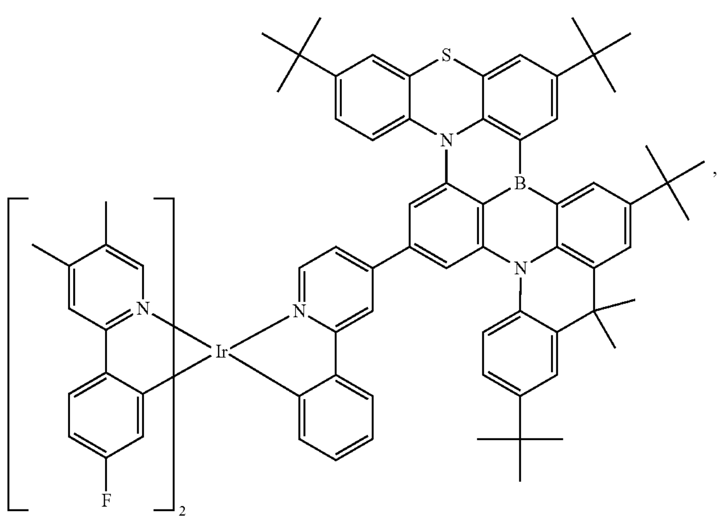

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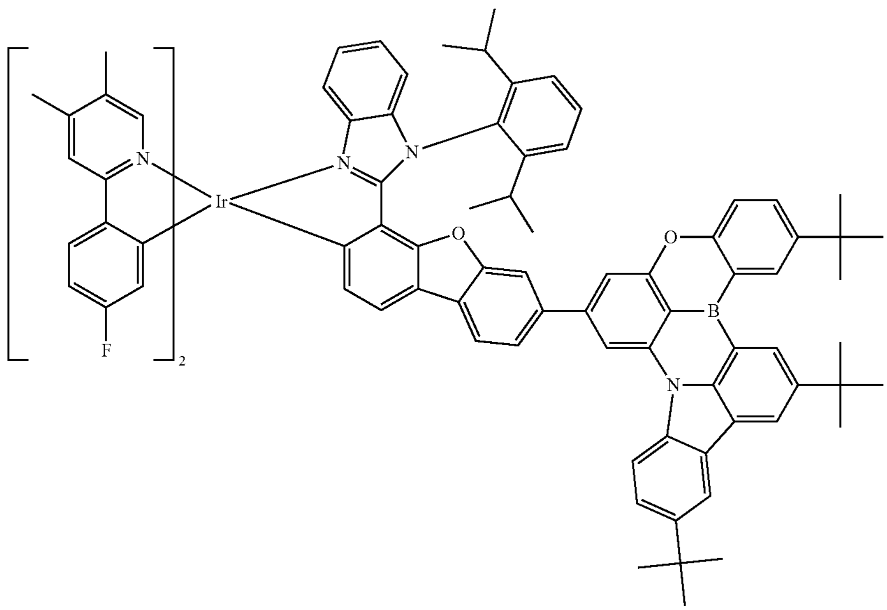

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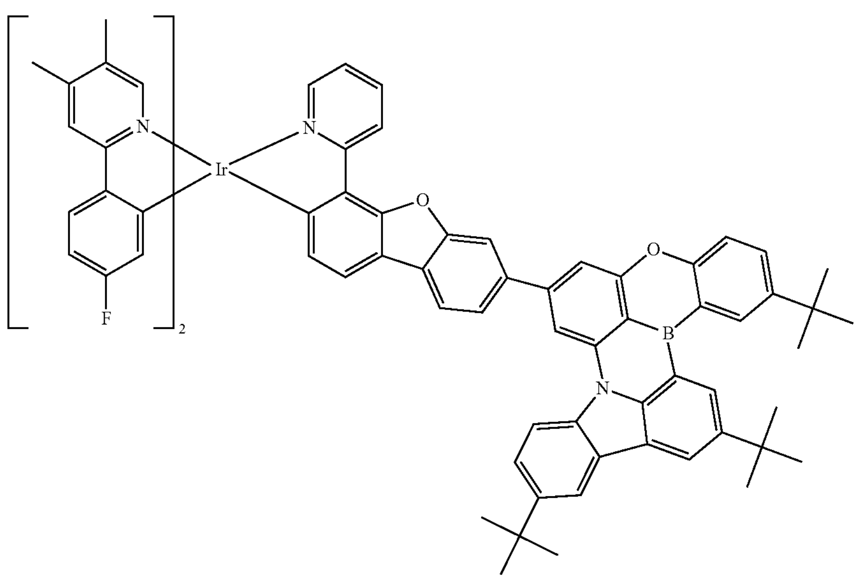

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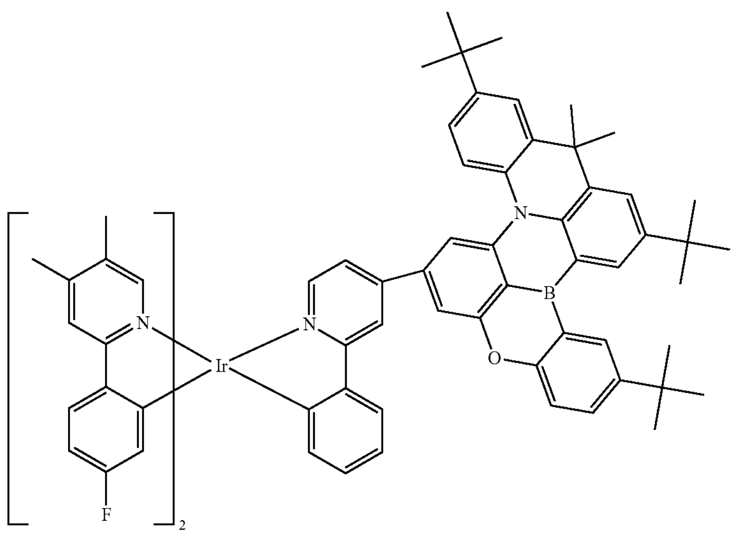

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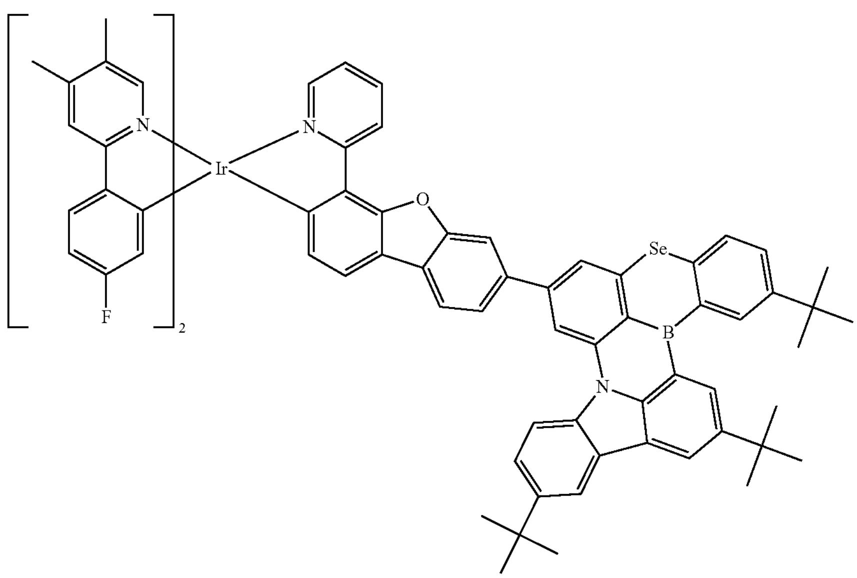

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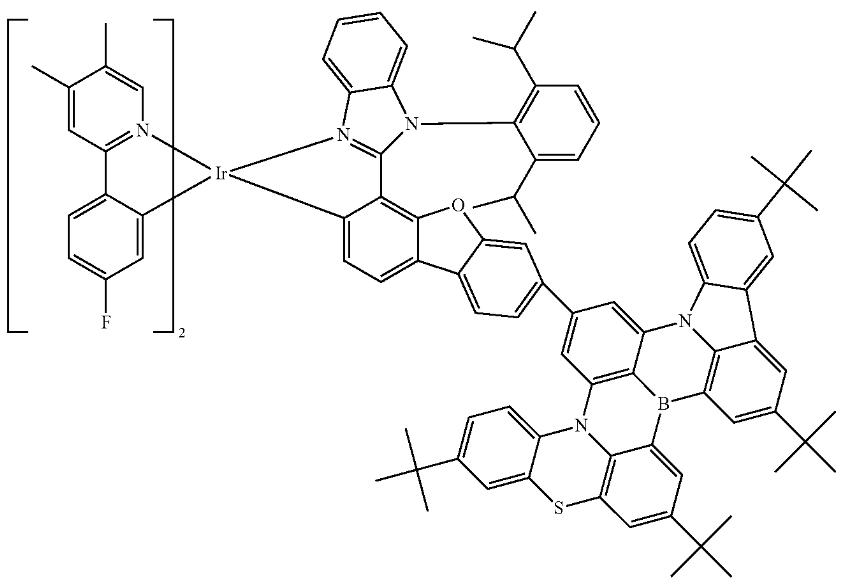

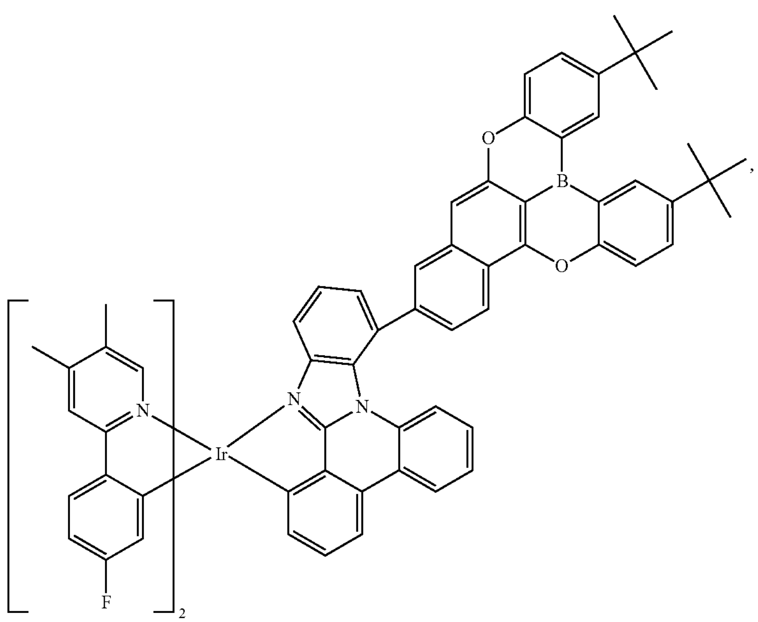

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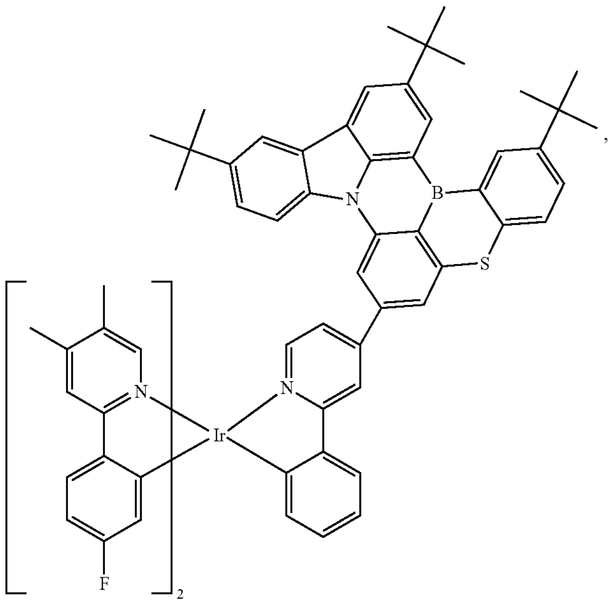

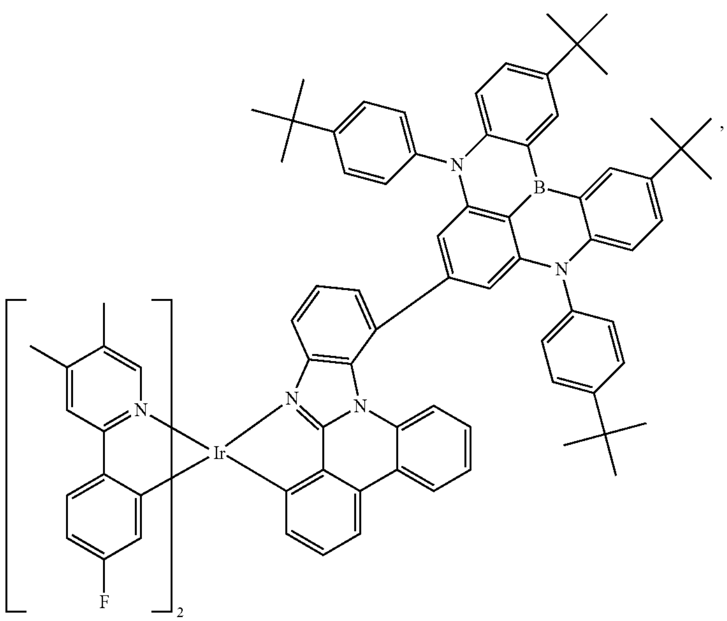

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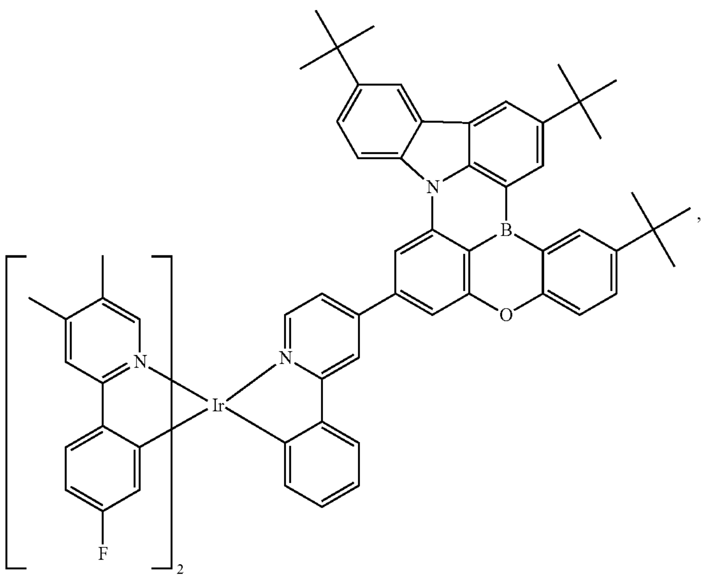

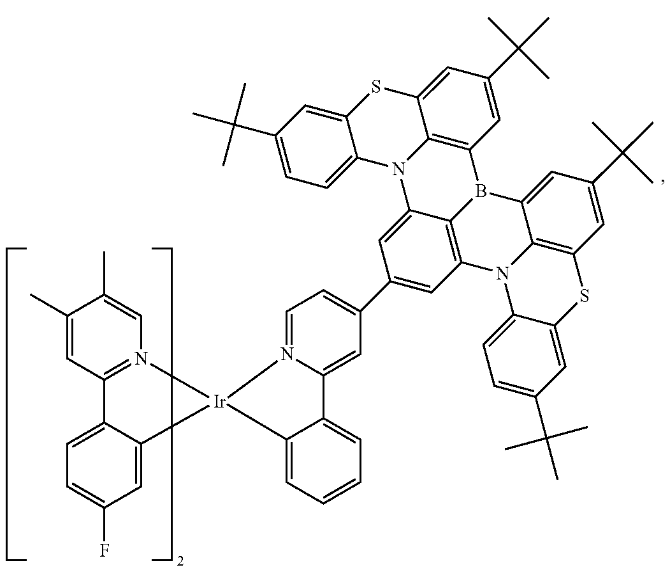

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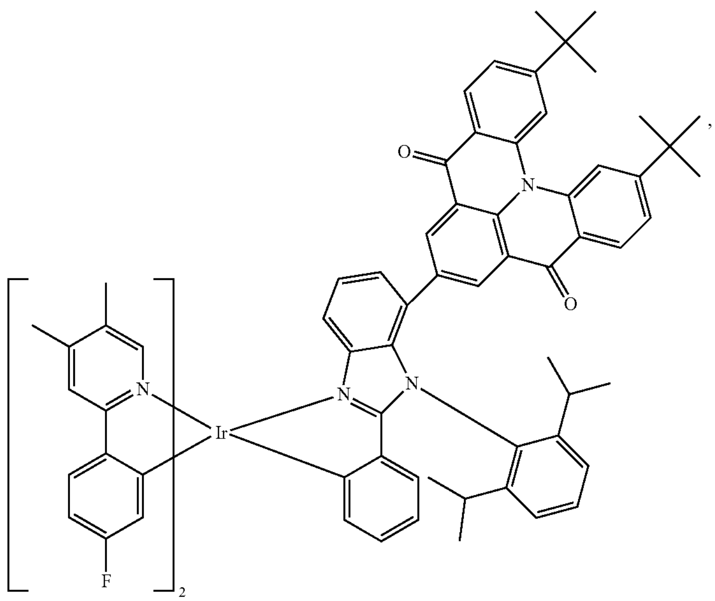

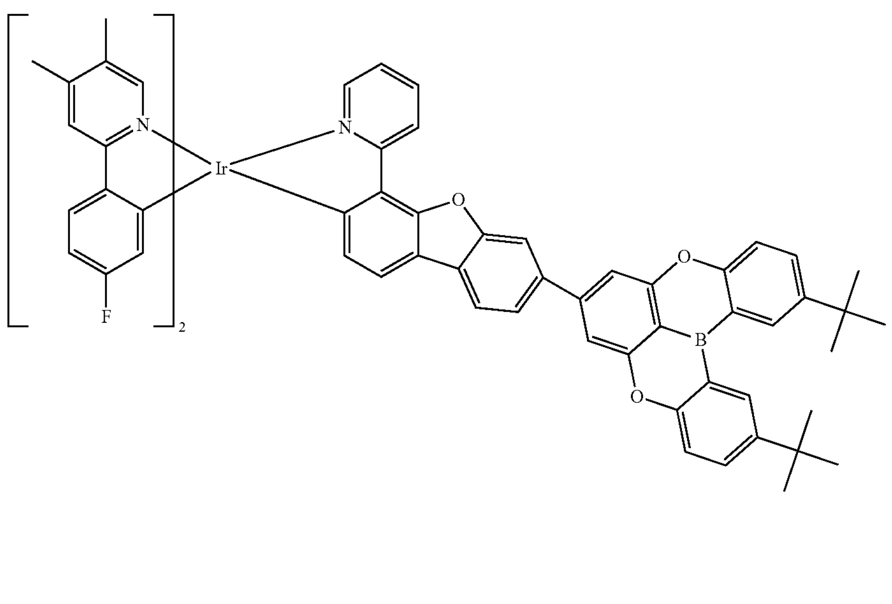

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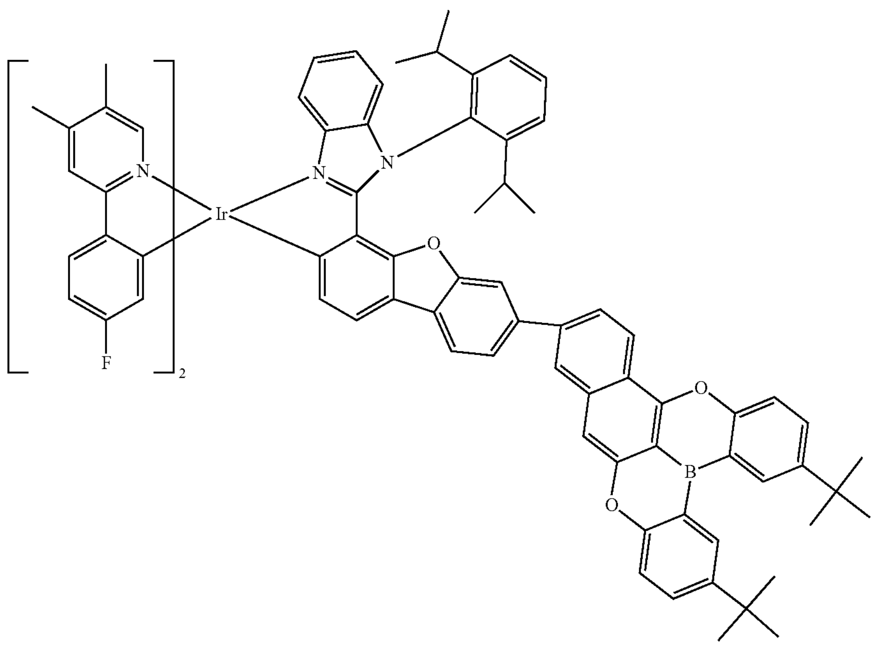

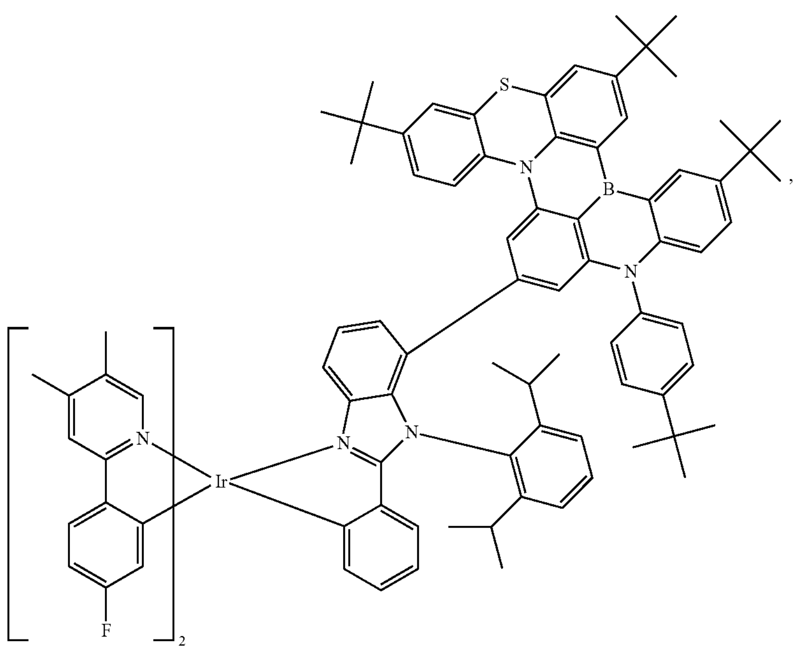

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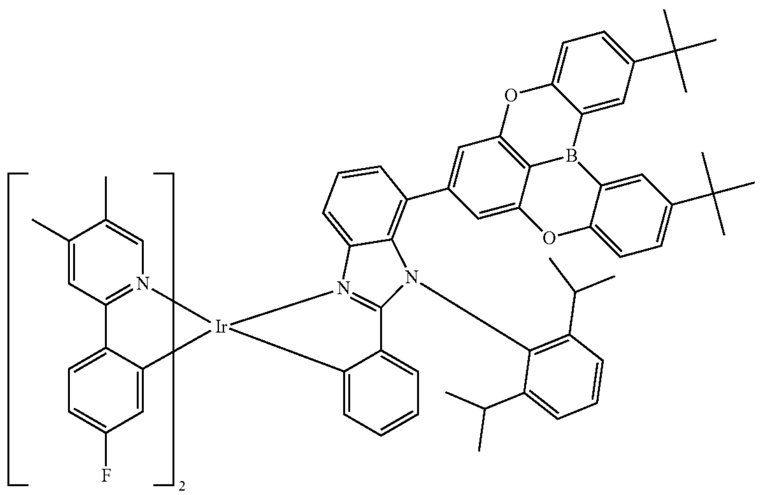

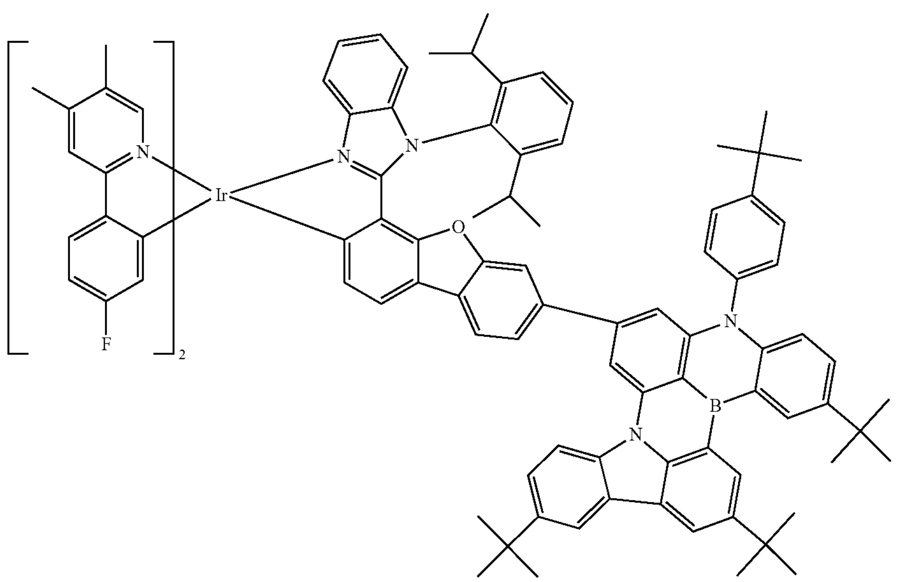

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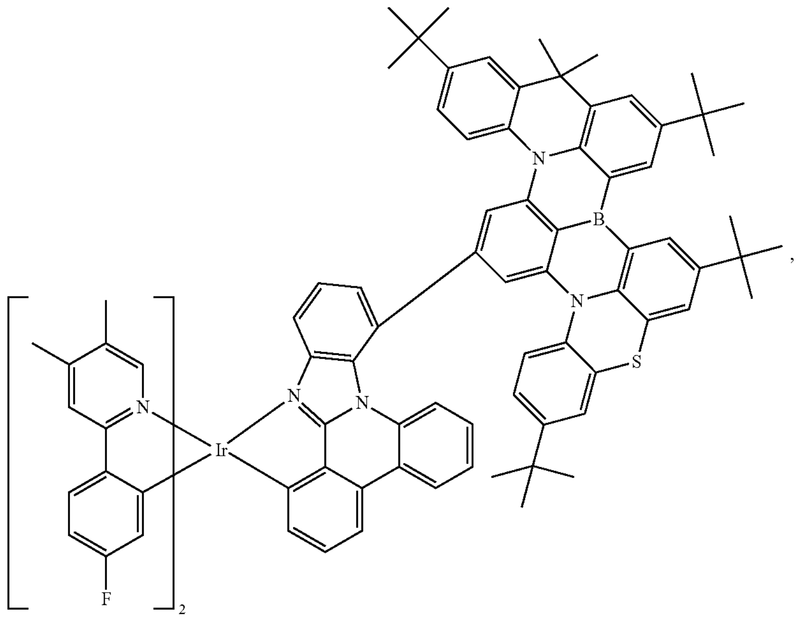

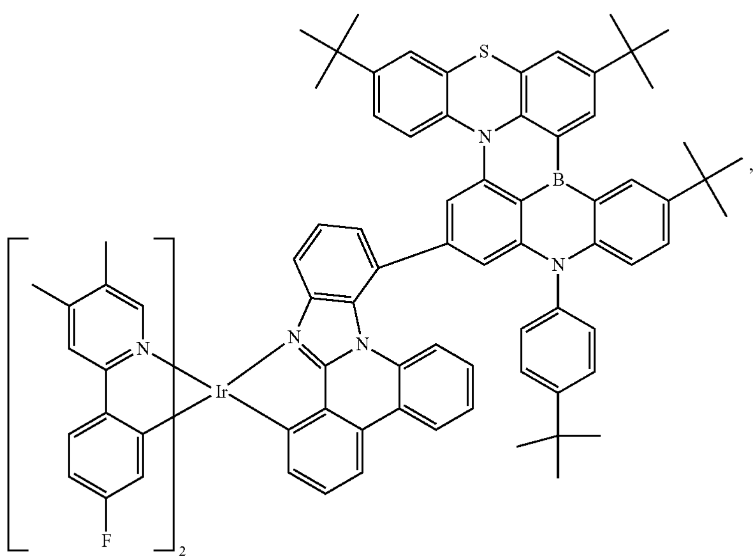

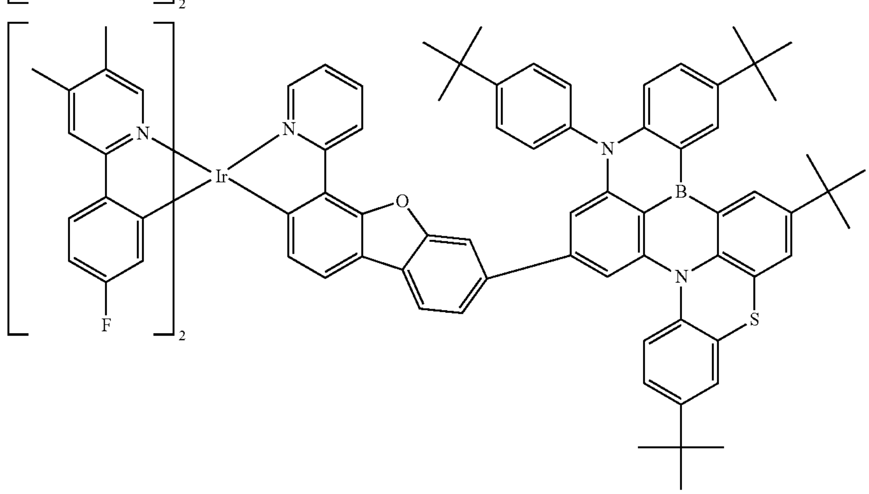

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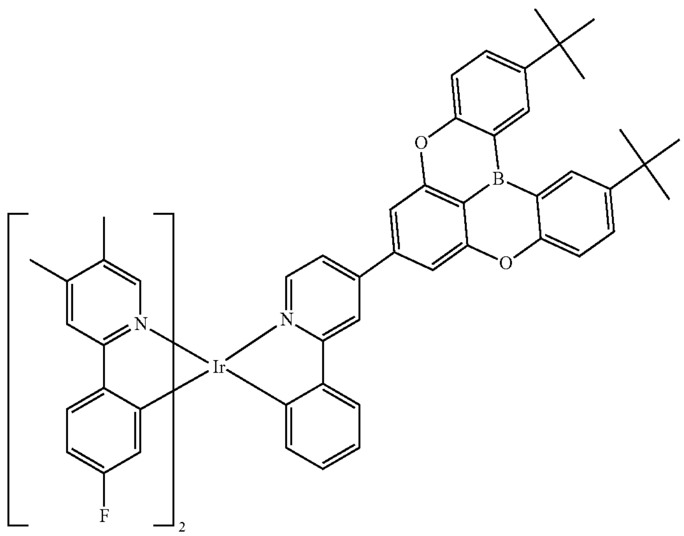

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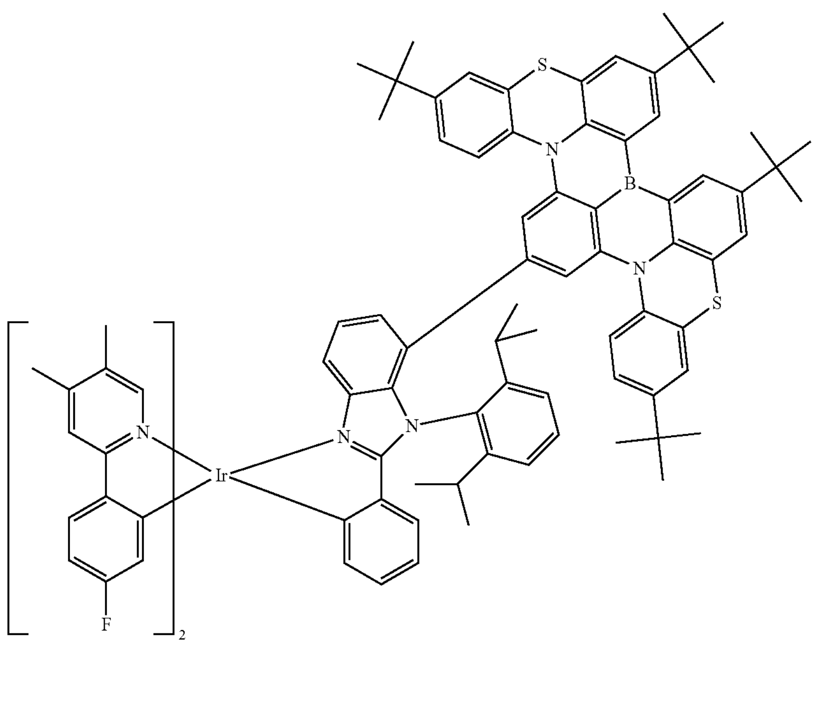

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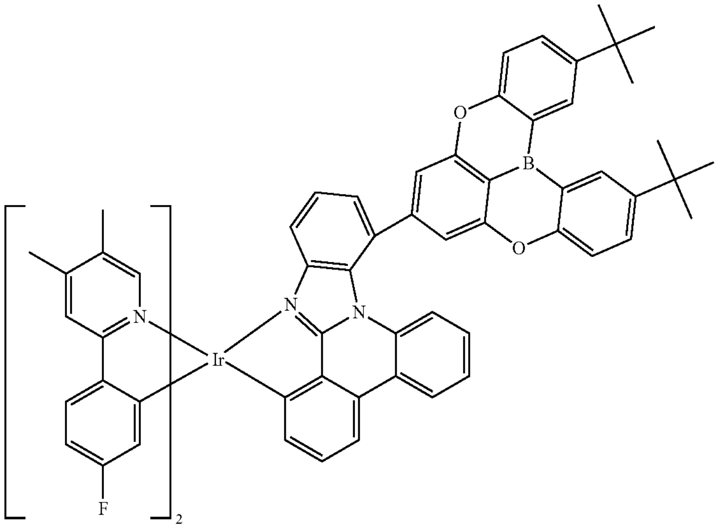

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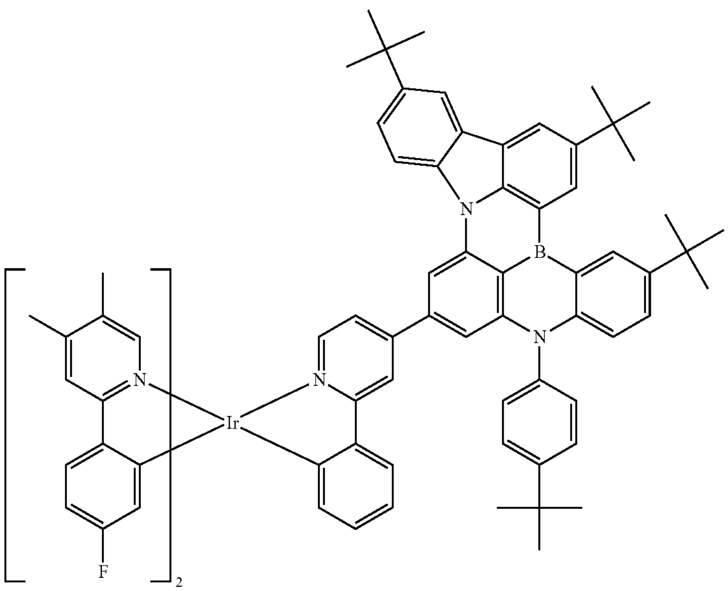

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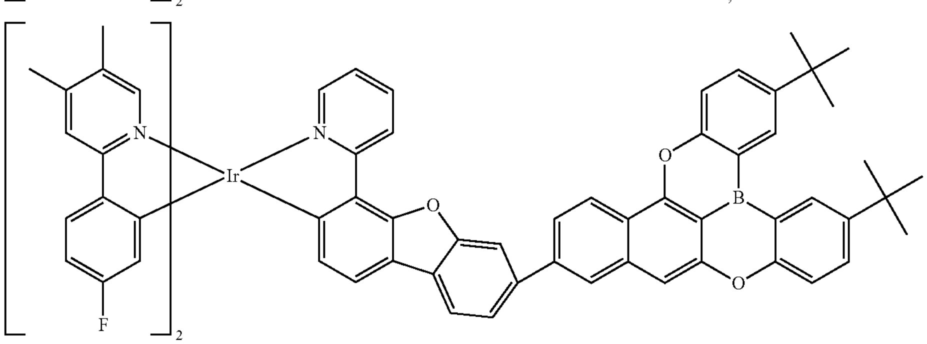

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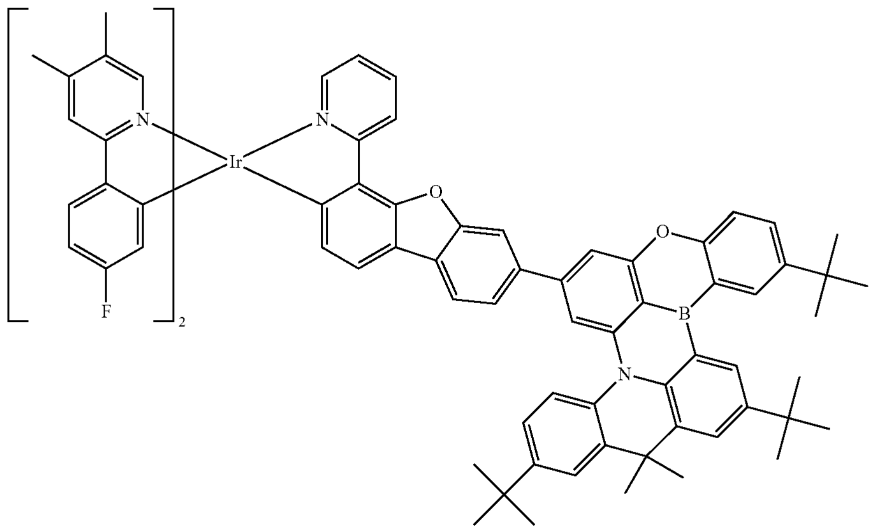

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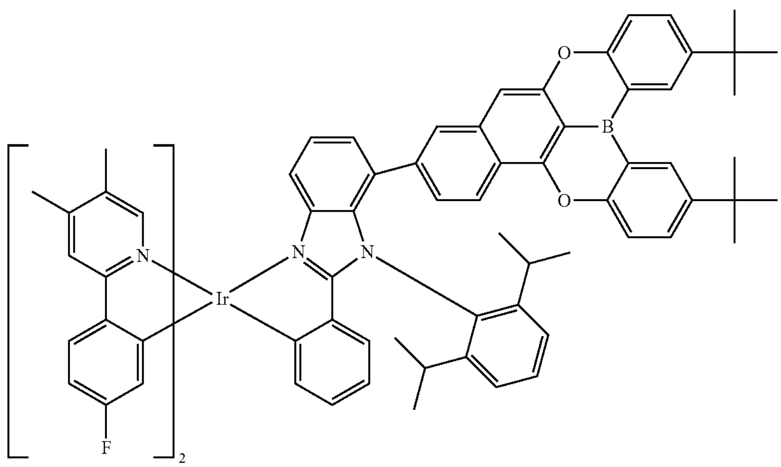

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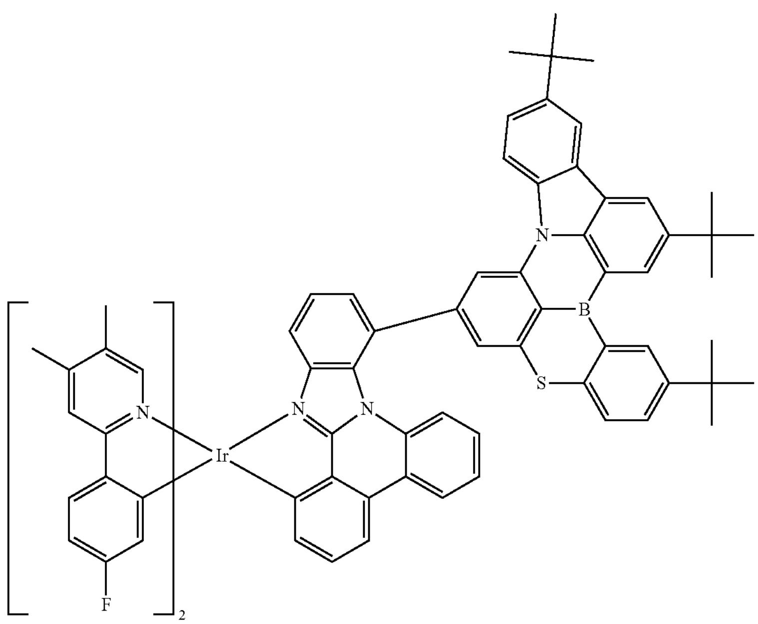

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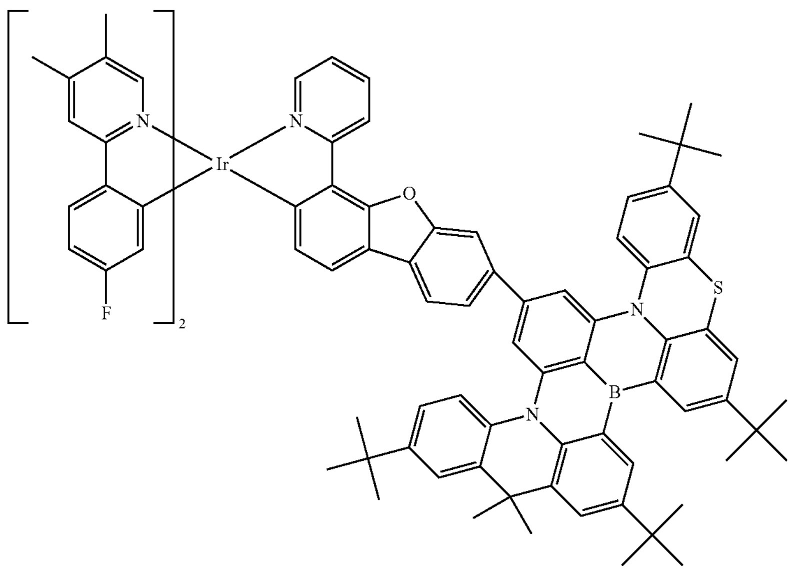

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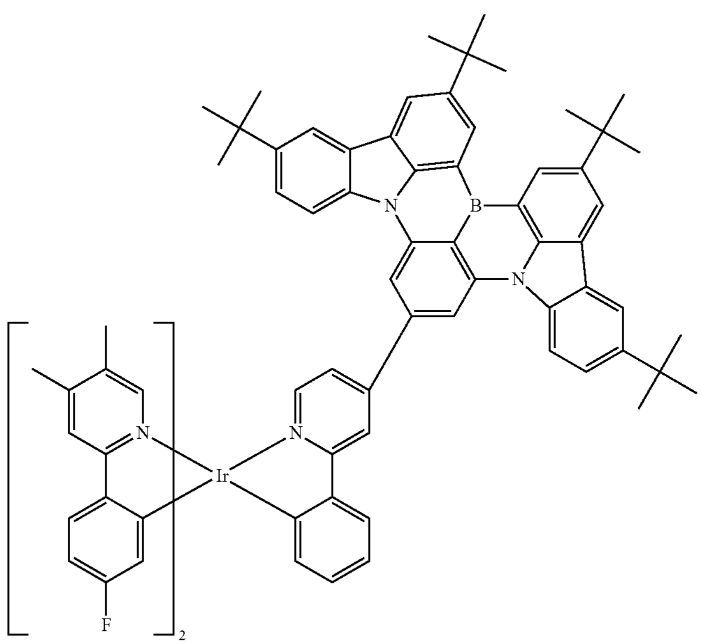

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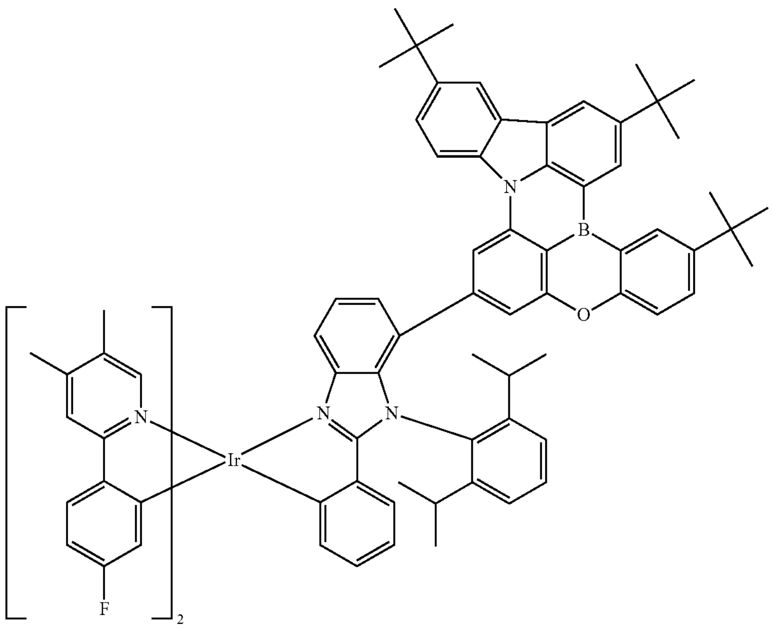

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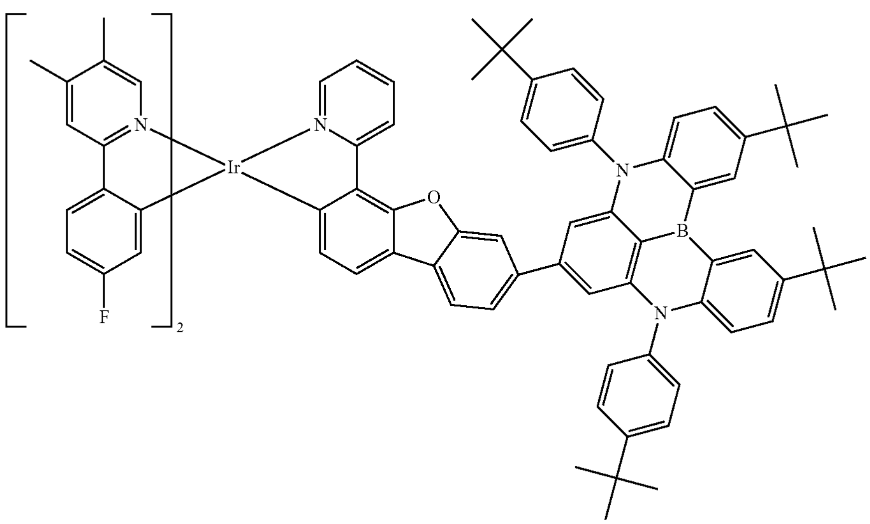

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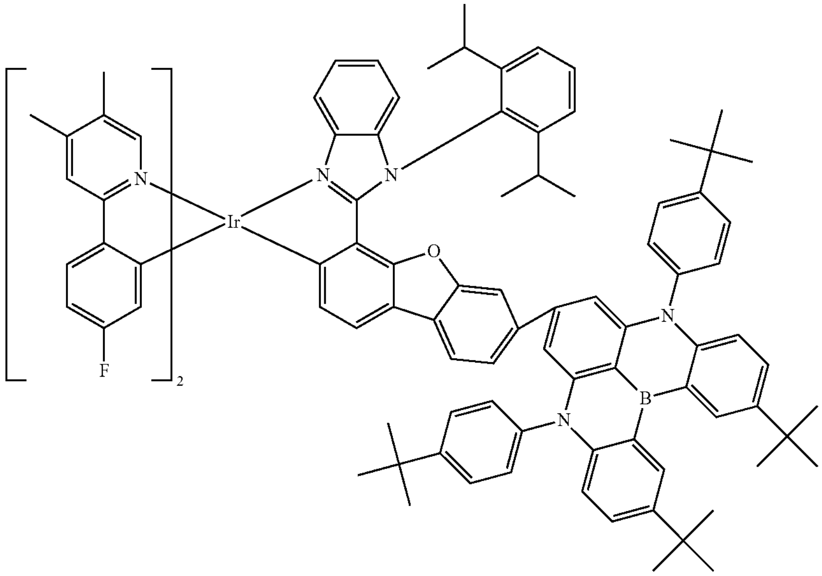

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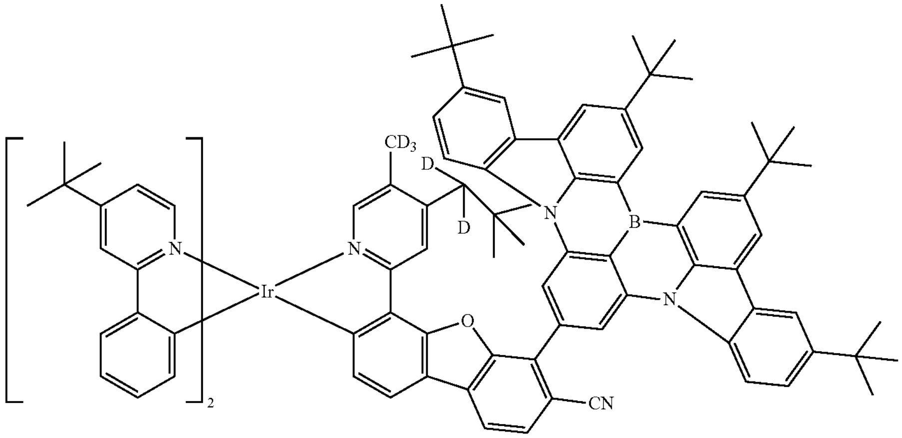

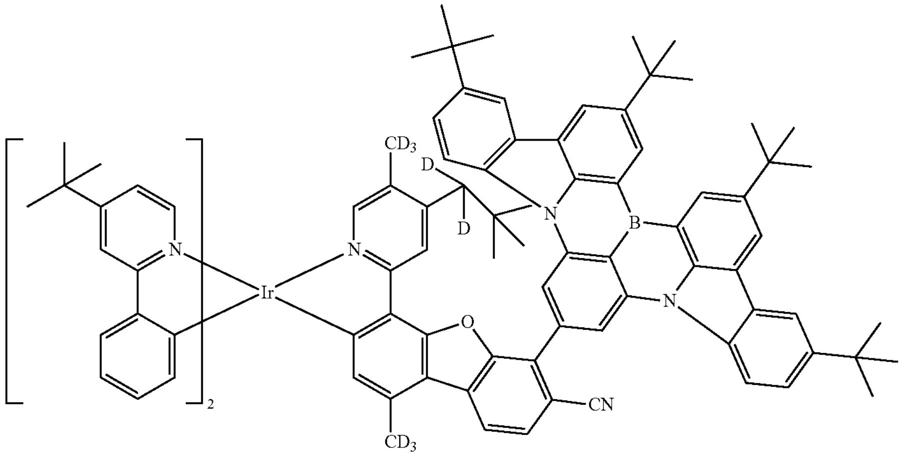

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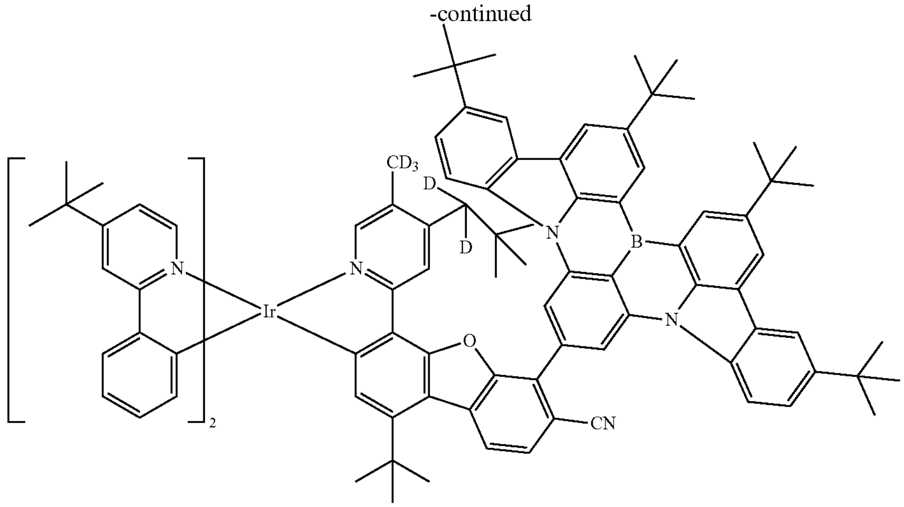
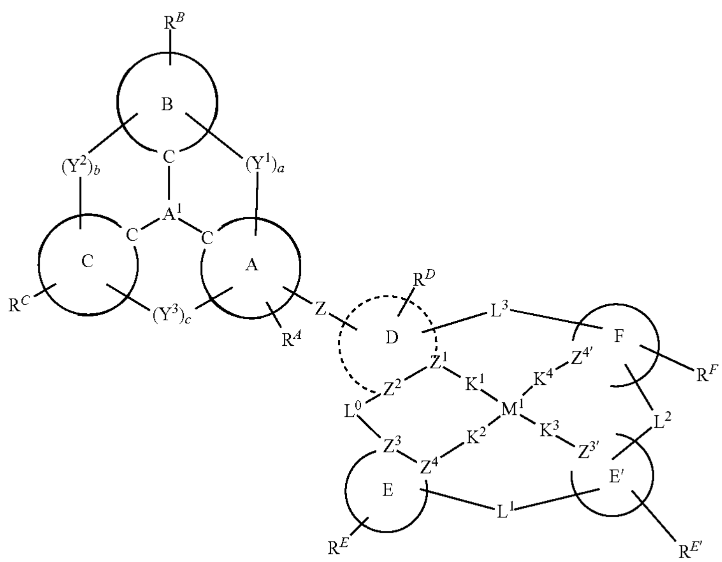


16. The compound of claim 11, wherein the compound has a structure of Formula IVA:



wherein:

M^1 is Pd or Pt;

each of the moieties E, E', F, and, when present, D are each independently monocyclic or polycyclic ring structure, wherein the monocyclic ring or each ring of the polycyclic fused ring system is independently a 5-membered to 10-membered carbocyclic or heterocyclic ring;

when M^1 is Pt, moiety D can be present or absent;

if moiety D is absent, Z is chelated to metal M^1;

Z is a direct bond or an organic linker;

each of Z^1, Z^2, Z^3, Z^4, $Z^{3'}$ and $Z^{4'}$ are each independently C or N;

K^1, K^2, K^3, and K^4 are each independently selected from the group consisting of a direct bond, O, and S, wherein at least two of them are direct bonds;

L^0, L^1, L^2, and L^3 are each independently absent or selected from the group consisting of a direct bond, BR, BRR', NR, PR, P(O)R, O, S, Se, C═O, C═S, C═Se, C═NR, C═CRR', S═O, SO_2, CR, CRR', SiRR', GeRR', alkylene, cycloalkyl, aryl, cycloalkylene, arylene, heteroarylene, and combinations thereof, wherein at least one of L^1 and L^2 is present;

R^D, R^E, $R^{E'}$, and R^F each independently represents zero, mono, or up to a maximum allowed number of substitutions;

each of R, R', R^D, R^E, $R^{E'}$, and R^F is independently a hydrogen or a substituent selected from the group consisting of deuterium, halide, alkyl, cycloalkyl, heteroalkyl, arylalkyl, alkoxy, aryloxy, amino, silyl, germyl, boryl, alkenyl, cycloalkenyl, heteroalkenyl, alkynyl, aryl, heteroaryl, acyl, carbonyl, carboxylic acid, ester, nitrile, isonitrile, sulfanyl, selenyl, sulfinyl, sulfonyl, phosphino, and combinations thereof; and two adjacent R^A, R^B, R^C, R^D, R^E, $R^{E'}$, and R^F can be joined or fused together to form a ring; with the proviso that if M^1 is Pt, then L^0 and L^3 are both absent and moiety D is optionally present.

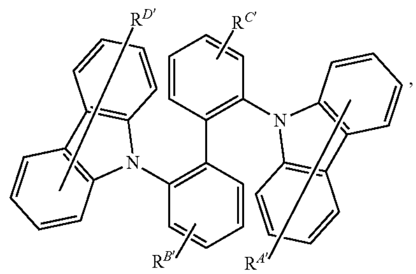
17. An organic light emitting device (OLED) comprising:

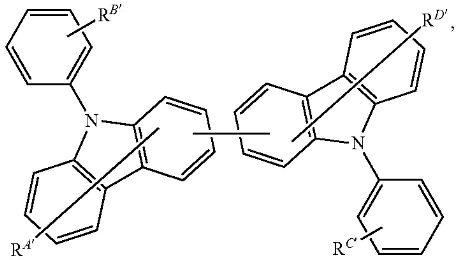
an anode;

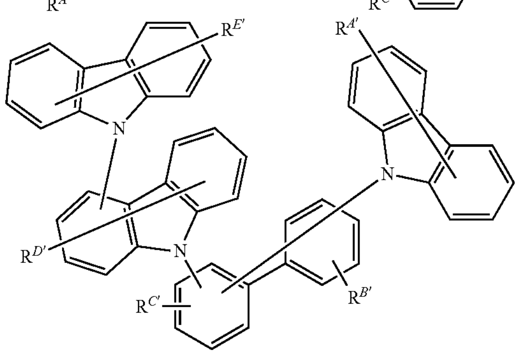
a cathode; and

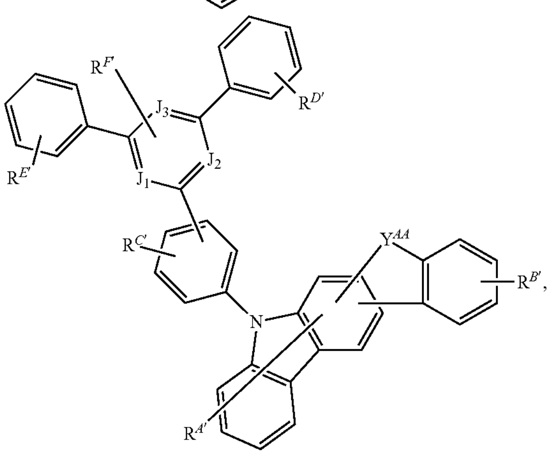
an organic layer disposed between the anode and the cathode, wherein the organic layer comprises a compound according to claim **1**.

18. The OLED of claim **17**, wherein the organic layer further comprises a host, wherein the host is selected from the group consisting of:









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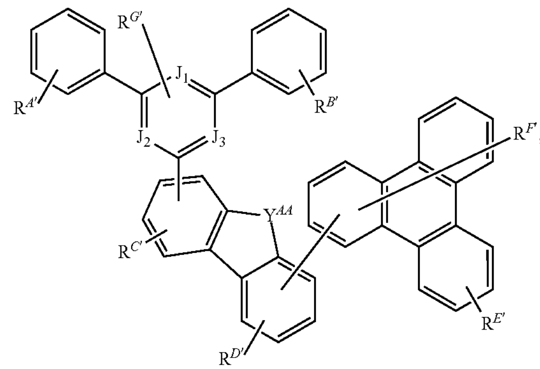

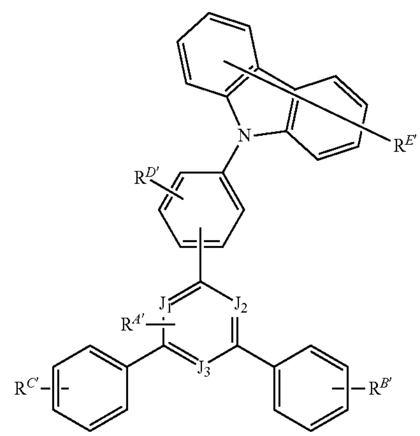

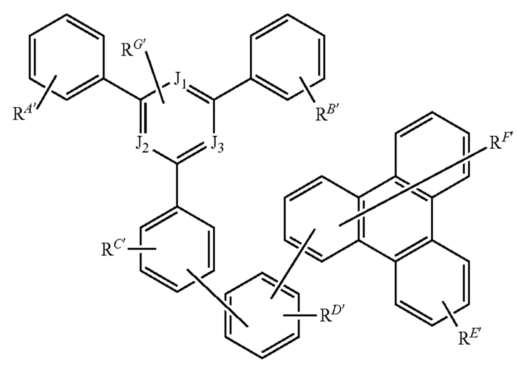

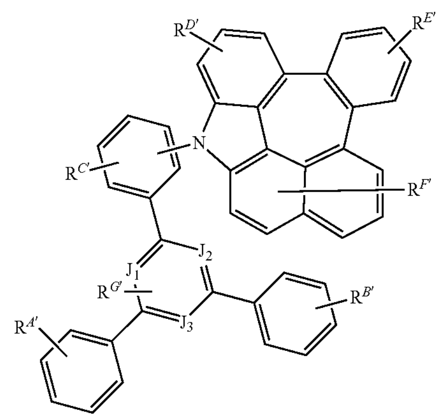

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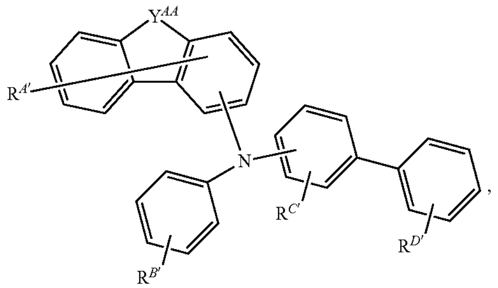

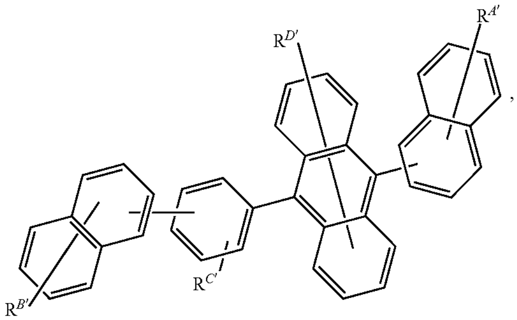

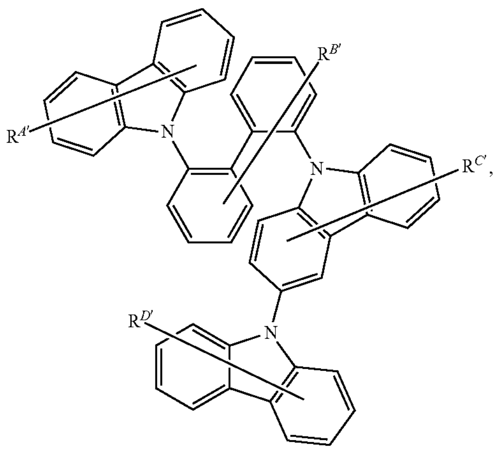

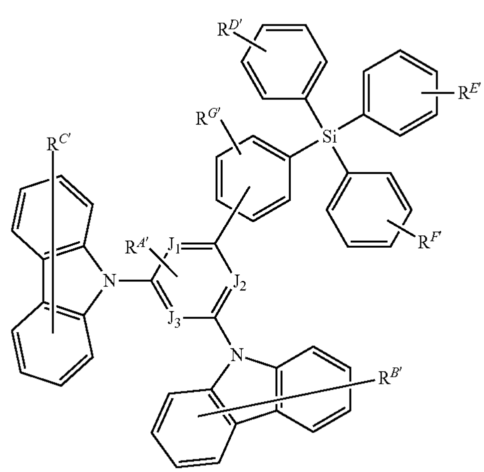

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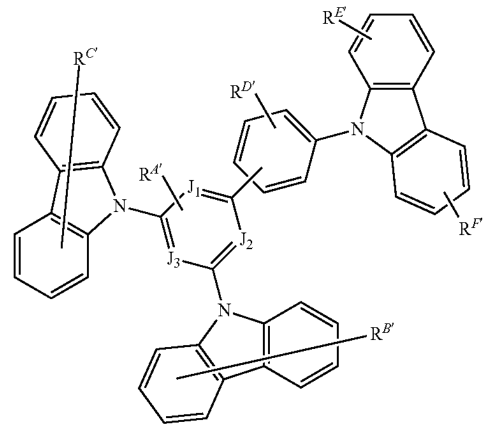

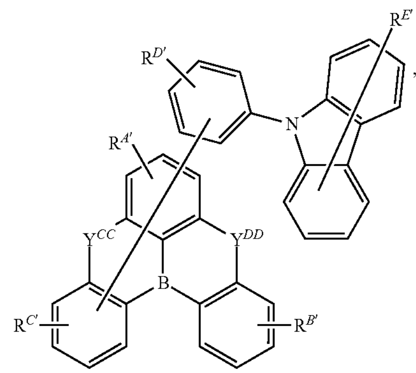

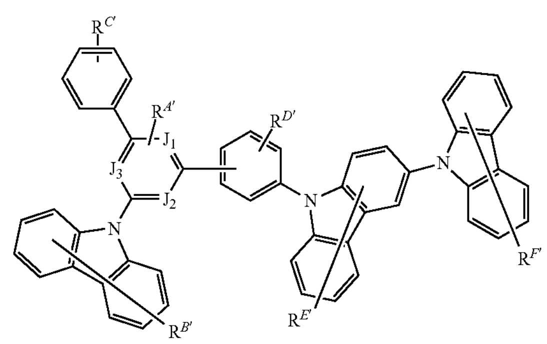

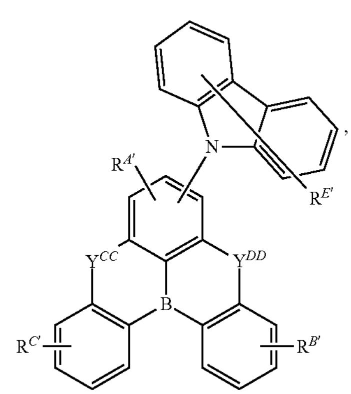

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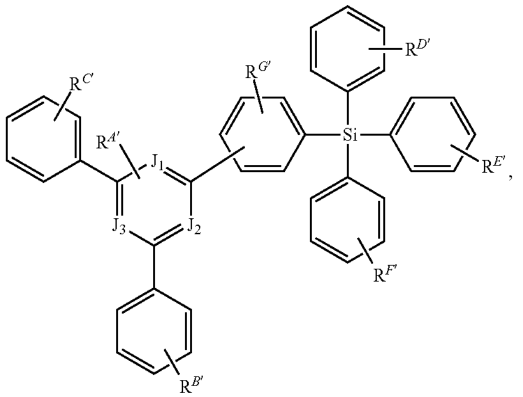

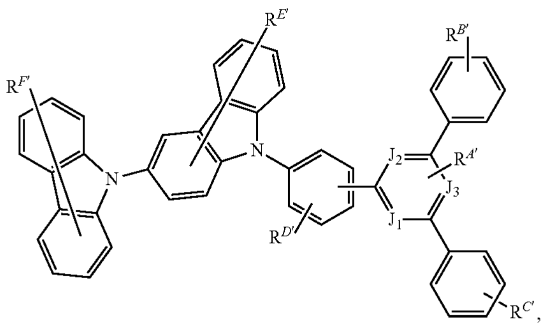

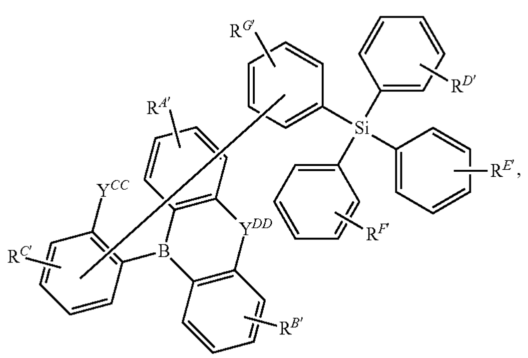

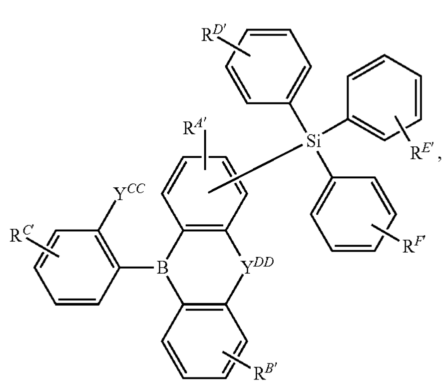

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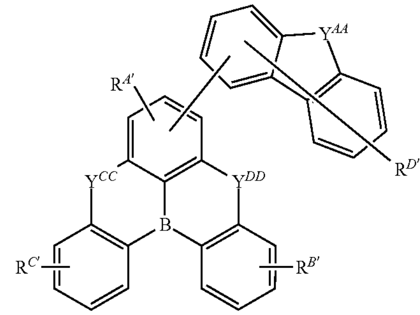

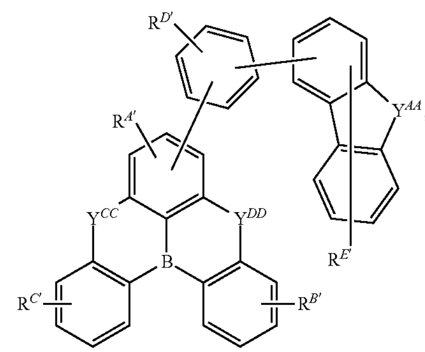

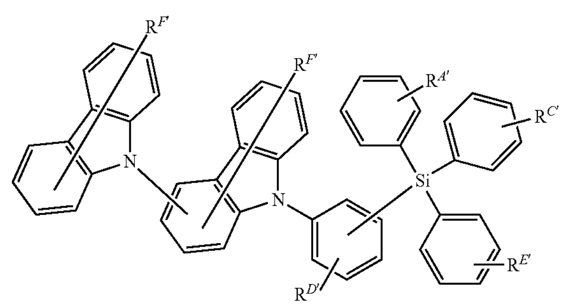

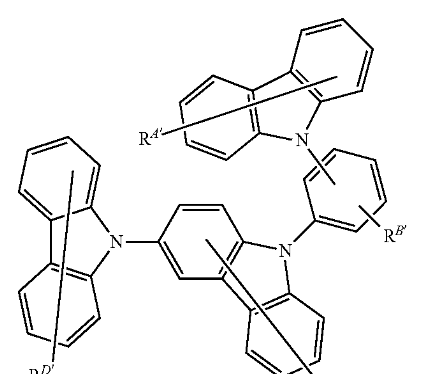

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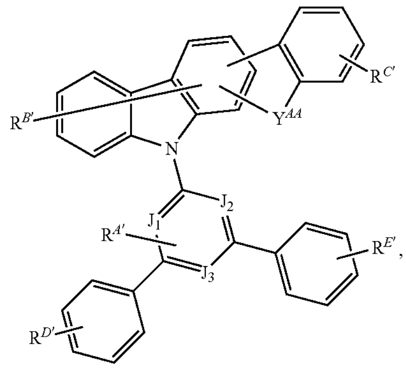

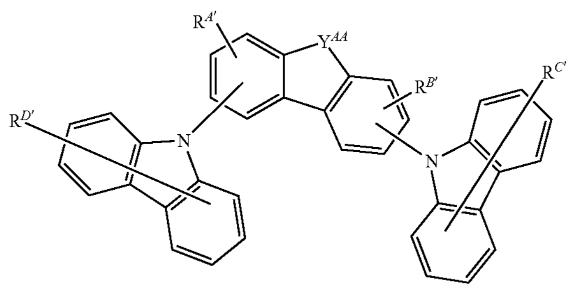

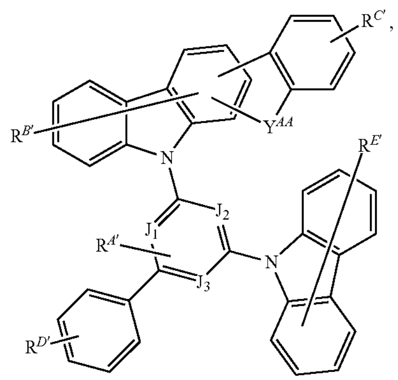

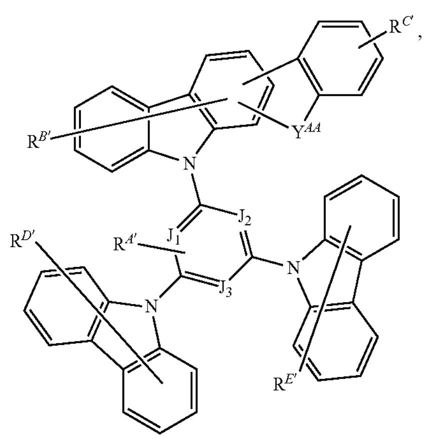

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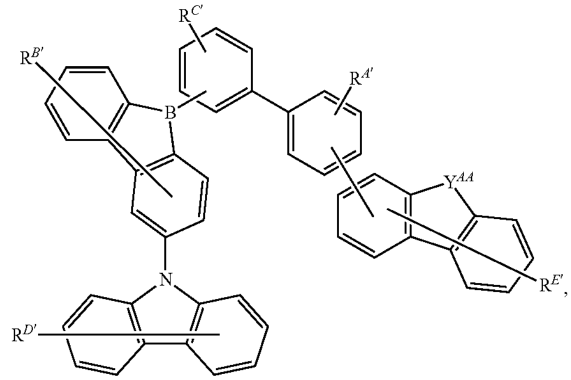

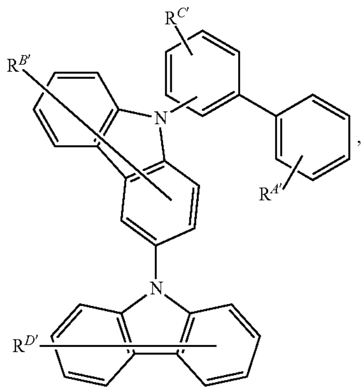

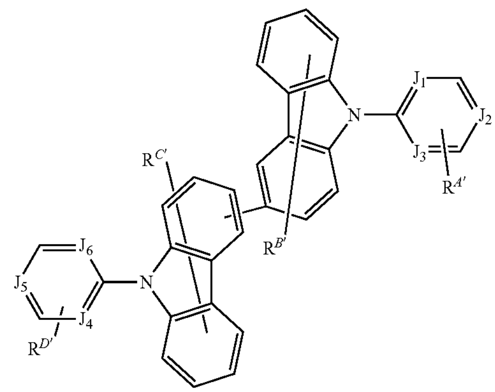

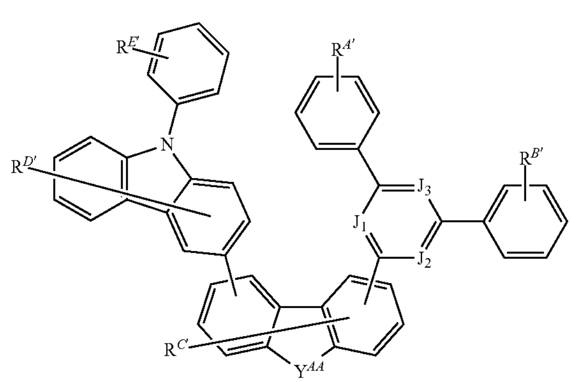

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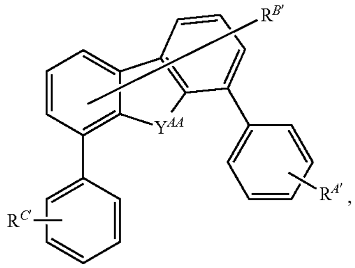

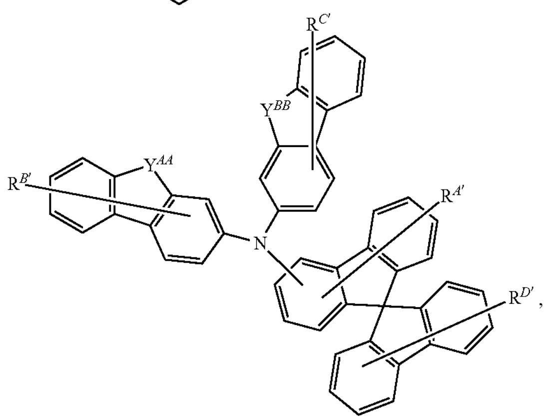

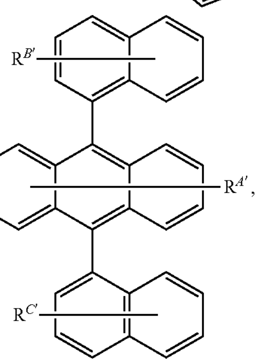

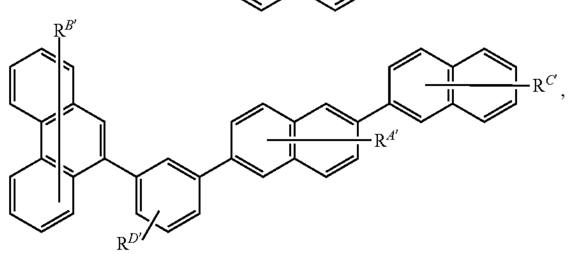

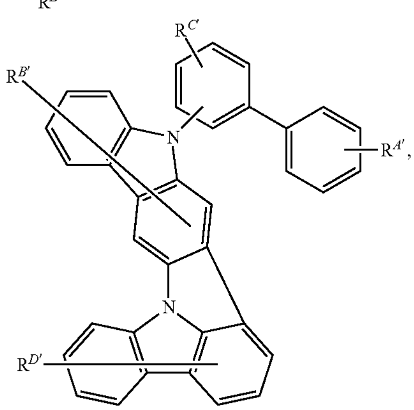

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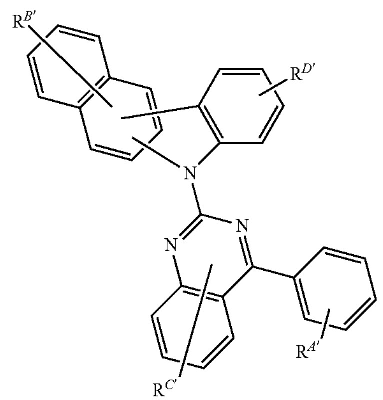

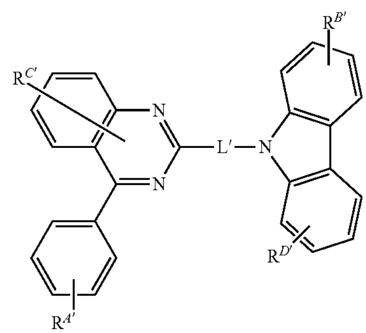

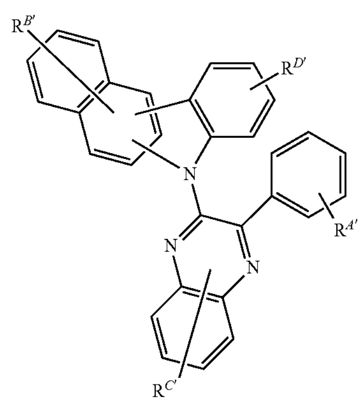

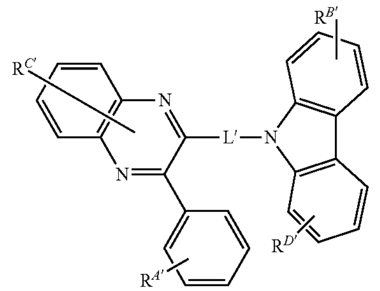

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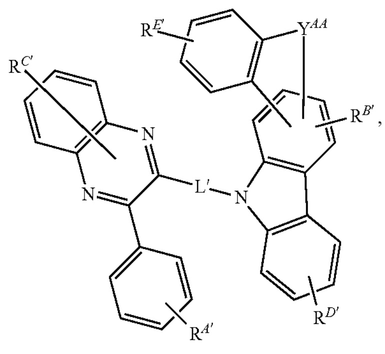

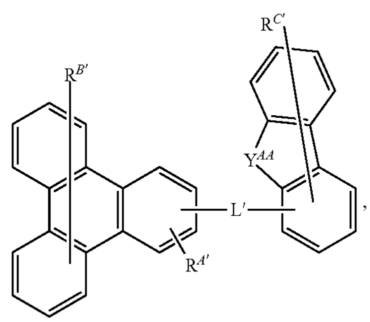

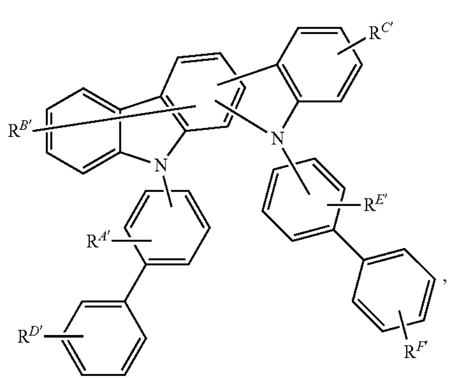

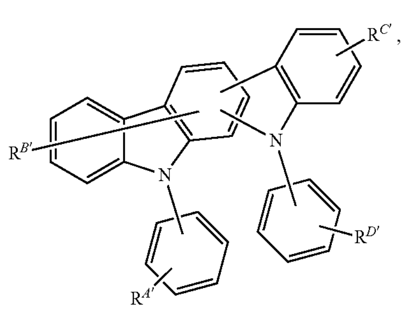

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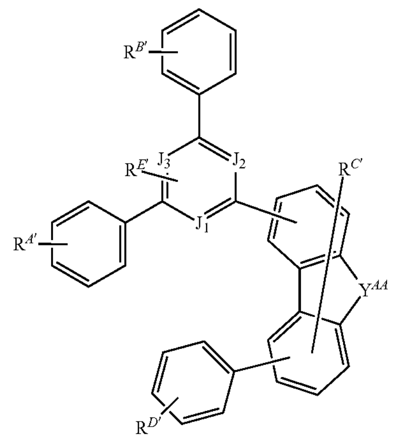

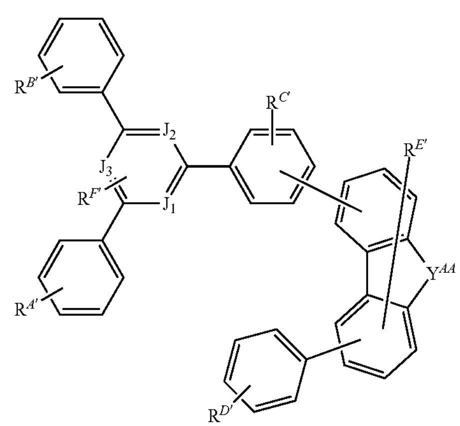

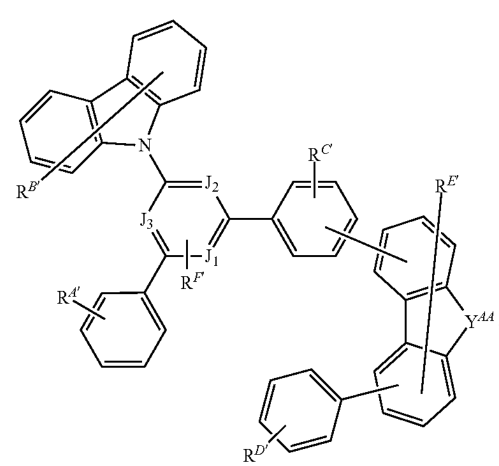

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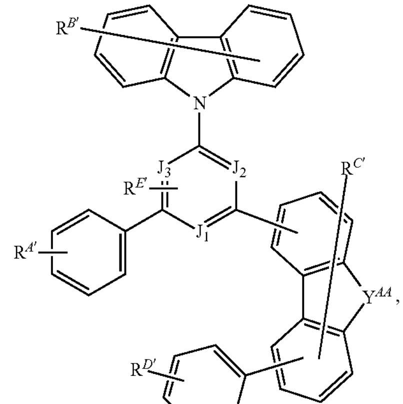

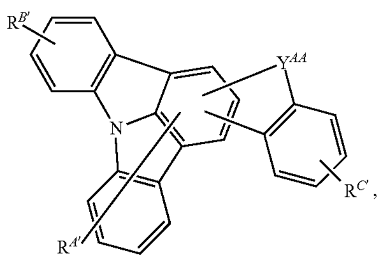

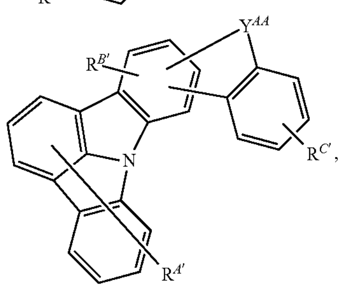

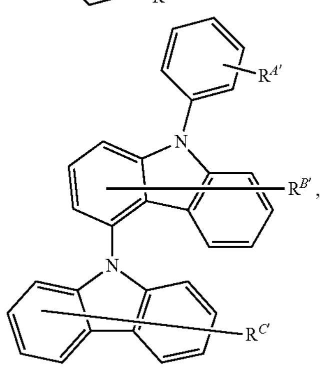

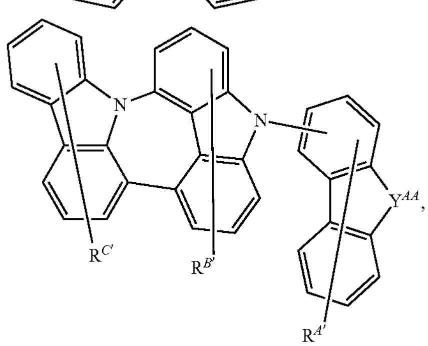

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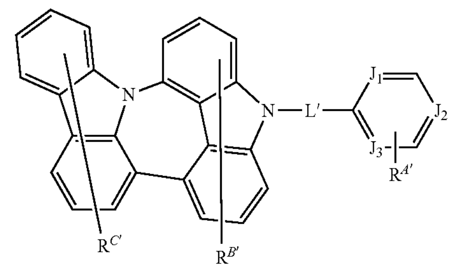

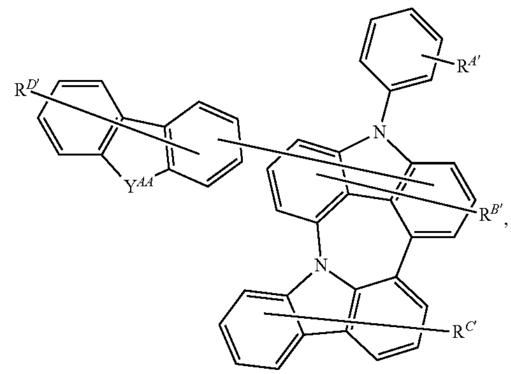

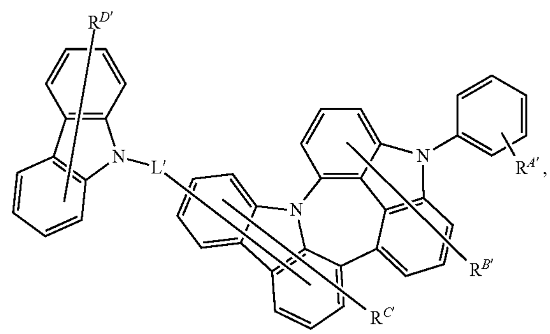

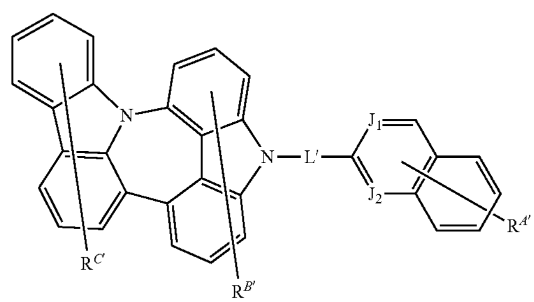

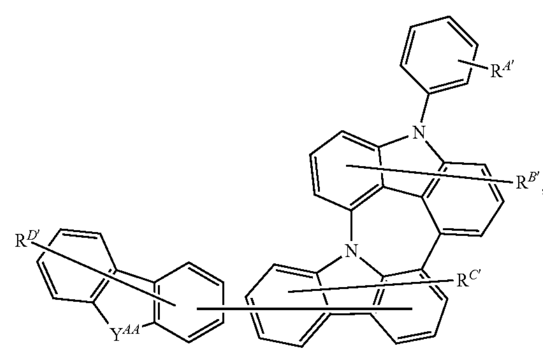

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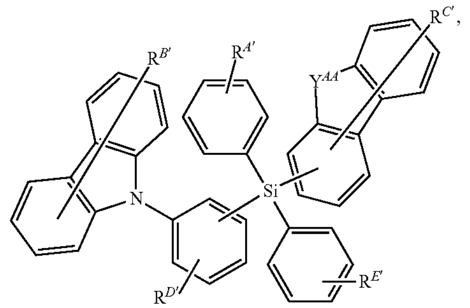

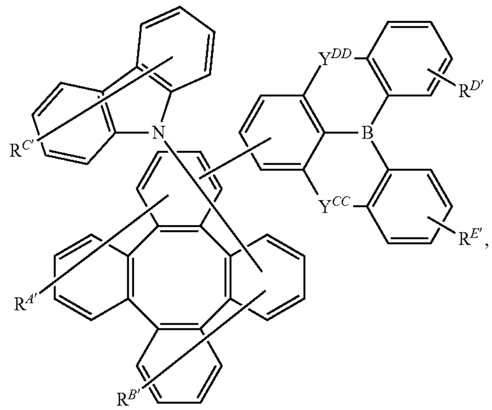

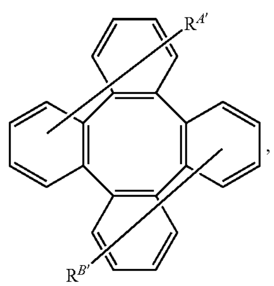

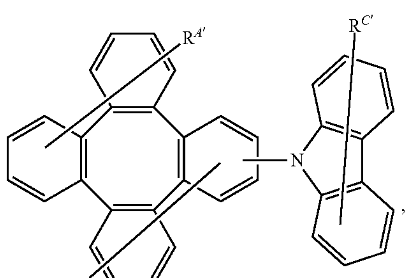

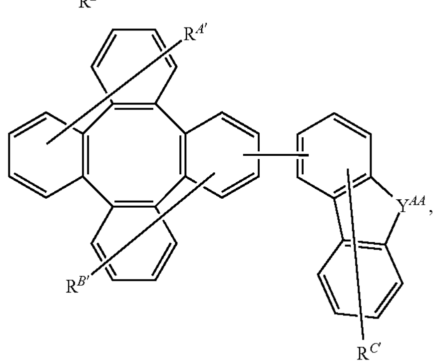

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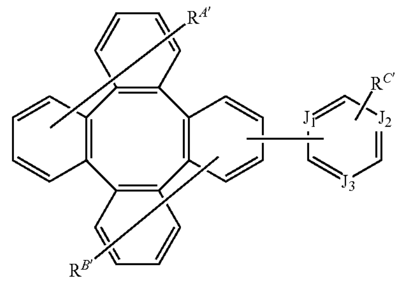

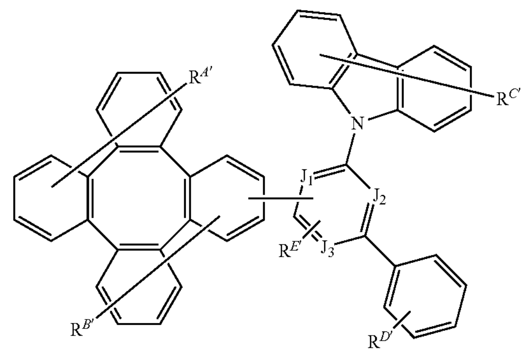

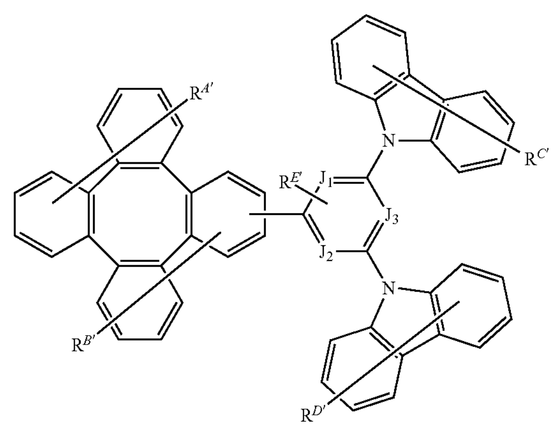

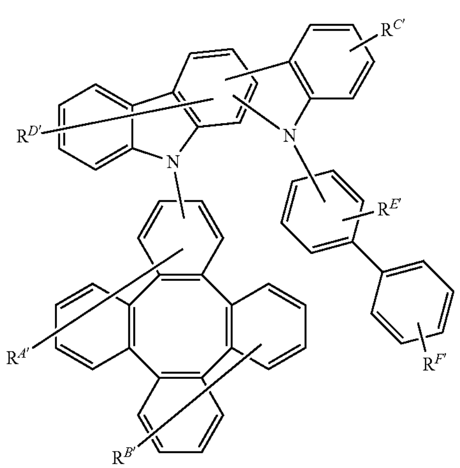

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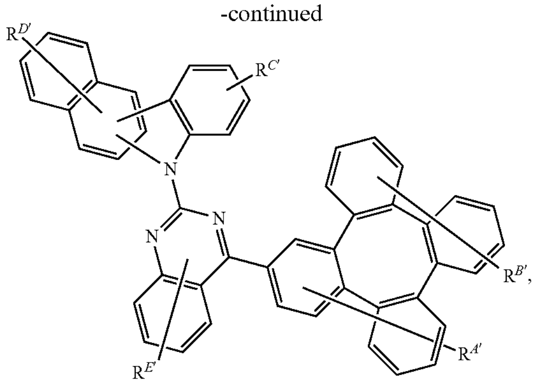

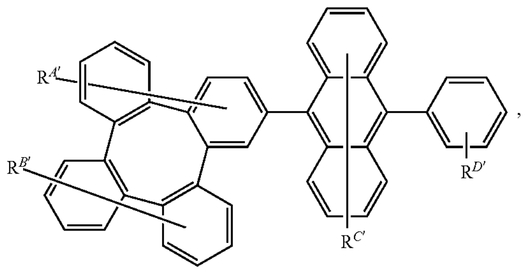

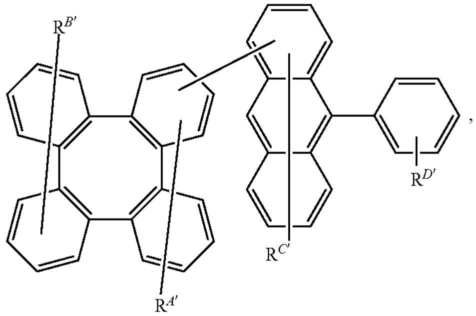

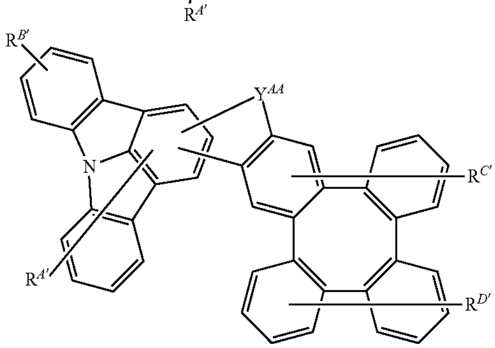

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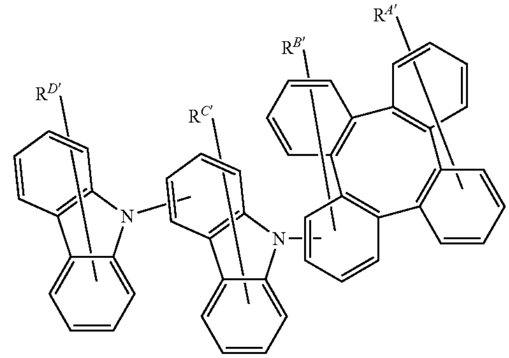

wherein:

each of J_1 to J_6 is independently C or N;

L' is a direct bond or an organic linker;

each Y^{AA}, Y^{BB}, Y^{CC}, and Y^{DD} is independently selected from the group consisting of absent a bond, direct bond, O, S, Se, CRR', SiRR', GeRR', NR, BR, BRR';

each of $R^{A'}$, $R^{B'}$, $R^{C'}$, $R^{D'}$, $R^{E'}$, $R^{F'}$, and $R^{G'}$ independently represents mono, up to the maximum substitutions, or no substitutions;

each R, R', $R^{A'}$, $R^{B'}$, $R^{C'}$, $R^{D'}$, $R^{E'}$, $R^{F'}$, and $R^{G'}$ is independently a hydrogen or a substituent selected from the group consisting of the General Substituents as defined herein; any two substituents can be joined or fused to form a ring; and where possible, each unsubstituted aromatic carbon atom is optionally replaced with N to form an aza-substituted ring.

19. The OLED of claim **17**, wherein the compound is a sensitizer, and the OLED further comprises an acceptor selected from the group consisting of a fluorescent emitter, a delayed fluorescence emitter, and combination thereof.

20. A consumer product comprising an organic light-emitting device comprising:

an anode;

a cathode; and

an organic layer disposed between the anode and the cathode, wherein the organic layer comprises a compound according to claim **1**.

* * * * *